The Certified Six Sigma Black Belt Handbook

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The Certified Six Sigma Black Belt Handbook

Third Edition

T. M. Kubiak and Donald W. Benbow

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For Frank Surgot, my lifelong friend:
We met as children and grew up together. As the years have passed, I have come to consider you a friend beyond all friends. You are the brother I never had.
—T. M. Kubiak

For my grandchildren Sarah, Emily, Dana, Josiah, Regan, Alec, Marah, and Liam, and to my great-granddaughter, Satori.
—Donald W. Benbow

Problems breed problems, and the lack of a disciplined method of openly attacking them breeds more problems.

—Phillip Crosby

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Preface to the Third Edition

In the spirit of customer–supplier relationships, we are pleased to provide our readers with the third edition of *The Certified Six Sigma Black Belt Handbook*. The handbook has been updated to reflect the most recent Six Sigma Black Belt Body of Knowledge released in 2015.

As with all ASQ certification–based handbooks, the primary audience for this work is the individual who plans to prepare to sit for the Six Sigma Black Belt certification examination. Therefore, the book assumes the necessary background and experience in quality and Six Sigma. Concepts are dealt with briefly, but facilitated with practical examples. We have intentionally avoided theoretical discussion unless such a discussion was necessary to communicate a concept. As always, readers are encouraged to use additional sources when they are seeking much deeper levels of discussion. Most of the citations provided in the references section will be helpful in this regard.

A secondary audience for the handbook is the quality and Six Sigma professional who would like a relevant Six Sigma reference book. With this audience in mind, we have greatly expanded the appendices section:

- Although the Body of Knowledge was updated in 2015, we have elected to keep both the 2001 and 2007 bodies of knowledge so that readers can compare changes and perhaps offer recommendations for future changes.

- All tables were developed using a combination of Microsoft Excel and Minitab 15. As such, the reader may find some differences between our tables and those published in other sources. Note that many statistical tables were produced years ago either by hand or using rudimentary calculators. These tables have been handed down from author to author and have remained largely unchanged. Our approach was to revert to the formulas and algorithms that produced the tables and then re-develop them using statistical software.

- Additional exercises have been included particularly to address the more difficult concepts.

- Additional tables and figures have been added to clarify concepts. Tables, in particular, are used to help collect concepts and to make comparisons and contrasts more easily accomplished.

- Additional content has been added between the DMAIC parts of the book (that is, Parts IV–VII) to help smooth the transition between phases and to better relate the underlying concepts of the DMAIC methodology.
- Content that was removed in the 2015 Body of Knowledge was retained and placed in the last chapter of the book for preservation of knowledge purposes.
- Additional content has been added that is not part of the body of knowledge because of direct relevance. Such content has been identified as not being in the body of knowledge.
- Additional content has been added to ensure that the Black Belt is fully trained in concepts taught to the Green Belt. Inconsistencies between the two bodies of knowledge were noted. For example, the Green Belt body of knowledge may discuss material not covered in the Black Belt body of knowledge. The additional content is meant to address such gaps.
- The glossary has been updated where appropriate.
- An acronym glossary is now included in the Appendix.
- The CD now includes both Excel and Minitab files for most exercises. This will permit our readers to work the examples more easily and, hopefully, facilitate the learning process. However, this book is not intended to serve as a learning tool for Excel or Minitab.
- The supplemental and simulated exam problems are now fully worked so that readers can more readily follow the problem-solving process.

We are confident readers will find the above improvements useful.

As you might expect, the number of the chapters and sections follows the same method used in the SSBB Body of Knowledge. This has made for some awkward placement of discussions (for example, the normal distribution is referred to several times before it is defined). In some cases, redundancy of discussion exists. However, where possible, we have tried to reference the main content in the handbook and refer the reader there for the primary discussion.

After the first edition was published, we received several comments from readers stating that their answers did not completely agree with those given in the examples. In many instances, we found that discrepancies could be attributed to the use of computers with different bits, the number of significant digits accounted for by the software used, the sequence in which the arithmetic was performed, and the propagation of errors due to rounding or truncation. Therefore, we urge the reader to carefully consider the above points as the examples are worked. Also, it should be noted that due to software updates and improvements by Minitab, the results shown in charts and graphs in the third edition may differ slightly with the second edition. These differences are minor and can be ignored.

However, we do recognize that errors occasionally occur. We ask that readers recommend suggestions, additions, corrections, or deletions, as well as seek out any corrections that may have been found and published, by emailing authors@asq.org.

Finally, the enclosed CD contains supplementary problems covering each chapter and a simulated exam that has problems distributed among chapters according to the scheme published in the Body of Knowledge. It is suggested that the reader study a particular chapter, repeating any calculations independently, and then do the supplementary problems for that chapter. After attaining success with all chapters, the reader may complete the simulated exam to confirm mastery of the entire Six Sigma Black Belt Body of Knowledge.

—The Authors

Preface to the Second Edition

In the spirit of customer-supplier relationships, we are pleased to provide our readers with the second edition of *The Certified Six Sigma Black Belt Handbook*. The handbook has been updated to reflect the most recent Six Sigma Black Belt Body of Knowledge, released in 2007.

As with all ASQ certification–based handbooks, the primary audience for this work is the individual who plans to prepare to sit for the Six Sigma Black Belt certification examination. Therefore, the book assumes the individual has the necessary background and experience in quality and Six Sigma. Concepts are dealt with briefly but facilitated with practical examples. We have intentionally avoided theoretical discussion unless such a discussion was necessary to communicate a concept. As always, readers are encouraged to use additional sources when seeking much deeper levels of discussion. Most of the citations provided in the references will be helpful in this regard.

A secondary audience for the handbook is the quality and Six Sigma professional who would like a relevant Six Sigma reference book. With this audience in mind, we have greatly expanded the appendices section:

- Although the Body of Knowledge was updated in 2007, we have elected to keep the 2001 Body of Knowledge so that readers can compare changes and perhaps offer recommendations for future Bodies of Knowledge.
- All tables were developed using a combination of Microsoft Excel and Minitab 15. Thus, the reader may find some differences between our tables and those published in other sources. Appendices 29–33 are examples of where such differences might occur. Note that years ago many statistical tables were produced either by hand or by using rudimentary calculators. These tables have been handed down from author to author and have remained largely unchanged. Our approach was to revert to the formulas and algorithms that produced the tables and then redevelop them using statistical software.
- The table for control constants has been expanded to now include virtually all control constants. To the best of our knowledge, this handbook is probably the only reference source that includes this information.
- Tables for both cumulative and noncumulative forms of the most useful distributions are now present—for example, binomial, Poisson, and normal.

- Additional alpha values in tables have been included. For example, large alpha values for the left side of the *F*-distribution now exist. Thus, it will no longer be necessary to use the well-known conversion property of the distribution to obtain critical *F* values associated with higher alpha values. Though the conversion formula is straightforward, everyone seems to get it wrong. We expect our readers will appreciate this.
- The glossary has grown significantly. Most notable is the inclusion of more terms relating to lean.
- A second glossary has been added as well. This short glossary is limited to the most common Japanese terms used by quality and Six Sigma professionals.

We are confident that readers will find the above additions useful.

As you might expect, chapter and section numbering follows the same method used in the Six Sigma Black Belt Body of Knowledge. This has made for some awkward placement of discussions (for example, the normal distribution is referred to several times before it is defined), and in some cases, redundancy of discussion exists. However, where possible, we have tried to reference the main content in the handbook and refer the reader there for the primary discussion.

After the first edition was published, we received several comments from readers who stated that their answers did not completely agree with those given in the examples. In many instances, we found that discrepancies could be attributed to the following: use of computers with different bits, the number of significant digits accounted for by the software used, the sequence in which the arithmetic was performed, and the propagation of errors due to rounding or truncation. Therefore, we urge the reader to carefully consider the above points as the examples are worked. However, we do recognize that errors occasionally occur and thus have established a SharePoint site that will permit readers to recommend suggestions, additions, corrections, or deletions, as well as to seek out any corrections that may have been found and published. The SharePoint site address is http://asqgroups.asq.org/cssbbhandbook/.

Finally, the enclosed CD contains supplementary problems covering each chapter and a simulated exam that has problems distributed among chapters according to the scheme published in the Body of Knowledge. It is suggested that the reader study a particular chapter, repeating any calculations independently, and then do the supplementary problems for that chapter. After attaining success with all chapters, the reader may complete the simulated exam to confirm mastery of the entire Six Sigma Black Belt Body of Knowledge.

—The Authors

Preface to the First Edition

We decided to number chapters and sections by the same method used in the Body of Knowledge (BOK) specified for the Certified Six Sigma Black Belt examination. This made for some awkward placement (the normal distribution is referred to several times before it is defined), and in some cases, redundancy. We thought the ease of access for readers, who might be struggling with some particular point in the BOK, would more than balance these disadvantages.

The enclosed CD contains supplementary problems covering each chapter and a simulated exam that has problems distributed among chapters according to the scheme published in the Body of Knowledge. It is suggested that the reader study a particular chapter, repeating any calculations independently, and then do the supplementary problems for that chapter. After attaining success with all chapters, the reader may complete the simulated exam to confirm mastery of the entire Six Sigma Black Belt Body of Knowledge.

—The Authors

Acknowledgments

This was a difficult project and we could not have completed it successfully without much needed support.

Special thanks go to our wives, Darlene and Jane, for their continued patience and support.

Thanks go to the reviewers of our manuscript. Their input brought clarity to us. We, in turn, hope that we have successfully captured that clarity and transferred it to the book you are now holding.

We would like to express our deepest appreciation to Minitab Inc. for providing us with the use of Minitab 17 and for permission to use several examples from both the Minitab and Quality Companion software programs. This software was instrumental in creating and verifying examples used throughout the book.

In addition, we would like to thank the Quality Press management staff, Matt Meinholz and Paul O'Mara, for their outstanding support and exceptional patience while we completed this project.

Finally, we would like to thank the staff of New Paradigm for applying their finely tuned project management, copyediting, and typesetting skills to this project. Their support has allowed us to produce a final product suitable for the ASQ Quality Press family of publications.

—The Authors

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Part I

Organization-Wide Planning and Deployment

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Chapter 1

Organization-Wide Considerations

FUNDAMENTALS OF SIX SIGMA AND LEAN METHODOLOGIES

> Define and describe the value, foundations, philosophy, history, and goals of these approaches, and describe the integration and complementary relationship between them. (Understand)
>
> **Body of Knowledge I.A.1**

Most of the techniques found in the Six Sigma toolbox have been available for some time thanks to the groundbreaking work of many professionals in the quality sciences. Some of the material in the following pages is found in ASQ's biographies.

Walter A. Shewhart successfully brought together the disciplines of statistics, engineering, and economics and became known as the father of modern quality control. The lasting and tangible evidence of that union, and for which he is most widely known, is the control chart, a simple but highly effective tool that represented an initial step toward what Shewhart called "the formulation of a scientific basis for securing economic control." Shewhart was concerned that statistical theory should serve the needs of industry. While the literature of the day discussed the stochastic nature of both biological and technical systems, and spoke of the possibility of applying statistical methodology to these systems, Shewhart actually showed how it was to be done; in that respect, the field of quality control can claim a genuine pioneer in Shewhart. His monumental work *Economic Control of Quality of Manufactured Product*, published in 1931, is regarded as a complete and thorough exposition of the basic principles of quality control. Most of Shewhart's professional career was spent as an engineer at Western Electric from 1918 to 1924, and at Bell Telephone Laboratories, where he served in several capacities as a member of the technical staff from 1925 until his retirement in 1956. He also lectured on quality control and applied statistics at the University of London, Stevens Institute of Technology, the graduate school of the US Department of Agriculture, and in India. He was a member of the visiting committee at Harvard's Department of Social Relations, an honorary professor at Rutgers, and a member of the advisory committee of the Princeton mathematics department. Called on frequently as a consultant, Shewhart served the War Department, the United Nations, and the government of India, and he was active with the National Research Council and the International Statistical Institute.

W. Edwards Deming, well known for his role as adviser, consultant, author, and teacher to some of the most influential businessmen, corporations, and scientific pioneers of quality control, is the most widely known proponent of statistical quality control. He has been described variously as a national folk hero in Japan, where he was influential in the spectacular rise of Japanese industry after World War II, as a curmudgeon, as the high prophet of quality control, as an imperious old man, and as founder of the third wave of the Industrial Revolution.

In 1942, while at the Bureau of the Census, Deming was retained as a consultant to the secretary of war and was asked by W. Allen Wallis, a statistician at Stanford University, for ideas on ways to aid the war effort. Deming suggested a short course in Shewhart's methods to teach the basics of applied statistics to engineers and others. The idea was adopted quickly and enthusiastically, and the first course was held in the summer of 1942. Among the instructors were Deming and Ralph Wareham, and the attendees included Holbrook Working and Eugene L. Grant of Stanford. The courses were repeated many times, often with Deming as instructor. The influence of the courses on the individuals who formed the core of the statistical quality control movement in the United States and who founded the American Society for Quality (ASQ) is well known.

Because of his work at the USDA and his expertise in statistics, Deming was sent to Japan in 1946 by the Economic and Scientific Section of the War Department to study agricultural production and related problems in the war-damaged nation. He returned to Japan in 1948 to conduct more studies for the occupation forces. During these trips, Deming made contact with Japanese statisticians and developed a lasting admiration and fondness for the Japanese people. Deming convinced Kenichi Koyanagi, one of the founding members of the Union of Japanese Scientists and Engineers (JUSE), of the potential of statistical methods in the rebuilding of Japanese industry. Koyanagi, in turn, suggested the idea to JUSE, which invited Deming to teach courses in statistical methods to Japanese industry. Under the auspices of the Supreme Commander of the Allied Powers, Deming arrived in Japan to teach in June 1950. He returned five times as teacher and consultant to Japanese industry.

Deming successfully influenced a new group of managers who had risen to the top in Japanese business after the war. They were hungry for new ideas to help them correct serious and persistent quality problems. Their interest was in contrast to behavior in the United States, where management was abandoning the precepts learned in the wartime quality control courses. Deming gave his Japanese students not only statistical theory, but also confidence. "I told [Japanese industrialists] Japanese quality could be the best in the world, instead of the worst," he said. Still, many were skeptical. "I was the only man in Japan who believed that Japanese industry could do that." Deming made his prophetic statement that the Japanese could capture world market within five years if they followed his advice. "They beat my prediction. I had said it would need five years. It took four." In recognition of Deming's efforts in Japan, JUSE created the Deming

Prize in 1951. He was awarded the Second Order Medal of the Sacred Treasure by Emperor Hirohito in 1960.

During the period of his activities in Japan, Deming pursued a similar mission in the United States. However, it has taken the United States much longer to pay attention to his teachings. In 1946 he became a professor of statistics in the Graduate School of Business Administration at New York University, and he began the consulting practice that he maintained until his death in 1993.

Deming developed a list of 14 points in which he emphasized the need for change in management structure and attitudes. As stated in his book *Out of the Crisis* (1986), these 14 points are as follows:

1. Create constancy of purpose for improvement of product and service.
2. Adopt a new philosophy.
3. Cease dependence on inspection to achieve quality.
4. End the practice of awarding business on the basis of price tag alone. Instead, minimize total cost by working with a single supplier.
5. Improve constantly and forever every process for planning, production, and service.
6. Institute training on the job.
7. Adopt and institute leadership.
8. Drive out fear.
9. Break down barriers between staff areas.
10. Eliminate slogans, exhortations, and targets for the workforce.
11. Eliminate numerical quotas for the workforce and numerical goals for management.
12. Remove barriers that rob people of pride of workmanship. Eliminate the annual rating or merit system.
13. Institute a vigorous program of education and self-improvement for everyone.
14. Put everybody in the company to work to accomplish the transformation.

> *It is most important that top management be quality-minded. In the absence of sincere manifestation of interest at the top, little will happen below.*

The only thing that distinguishes these comments from those heard commonly today is that they were written in 1945 by *Joseph M. Juran*. Juran joined the Bell System as an engineer in 1924; two years later, he was one of three people assigned to a new department formed to carry out what is known today as *statistical quality control*. Juran prepared what may have been the first text on statistical quality control—and perhaps the ancestor of today's widely used *Western Electric Statistical Quality Control Handbook.*

Juran was on the faculty at New York University, where he served for a time as head of the Department of Industrial Engineering, and his arrangement with NYU allowed him to spend a good deal of time on matters not connected with teaching. One of these was the writing and editing of the *Quality Control Handbook.* First published in 1951, the *Handbook* has grown from 15 chapters to 52, paralleling and aiding the growth of the quality field.

During the late 1940s, Juran began to develop what may have been the most influential course on the subject of quality since the eight-day seminars conducted during World War II. Juran's "Managing for Quality" has been taught to more than 100,000 people in over 40 countries. For almost 30 years, Juran offered the course through the American Management Association. Given his longstanding interests, Juran's association with AMA was not surprising. His writings in *Industrial Quality Control* during the 1950s and 1960s also reflect his interest in management. As a contributing author and editor for 16 years of "Management's Corner," Juran frequently emphasized the role of management in quality. He saw early on the broadening role of quality, and urged quality professionals to prepare for it.

Juran developed the Juran trilogy: three managerial processes—quality planning, quality control, and quality improvement—for use in managing for quality. Juran wrote hundreds of papers and 12 books, including *Juran's Quality Control Handbook* (2010), early editions of *Juran's Quality Planning & Analysis for Enterprise Quality* (with Gryna 2007), and *Juran on Leadership for Quality* (2003). His approach to quality improvement includes the following points:

- Create awareness of the need and opportunity for improvement
- Mandate quality improvement; make it a part of every job description
- Create the infrastructure: establish a quality council, select projects for improvement, appoint teams, provide facilitators
- Provide training in how to improve quality
- Review progress regularly
- Give recognition to the winning teams
- Propagandize the results
- Revise the reward system to enforce the rate of improvement
- Maintain momentum by enlarging the business plan to include goals for quality improvement

Philip B. Crosby was a legend in the discipline of quality. A noted quality professional, consultant, and author, he is widely recognized for promoting the concept of "zero defects" and for defining quality as *conformance to requirements*. He founded Philip Crosby Associates, Inc. (PCA), teaching management how to establish a preventive culture to get things done right the first time. Crosby's first book, *Quality Is Free*, has been credited with playing a large part in initiating the quality revolution in the United States and Europe. He published a total of 13 books, including *The Absolutes of Leadership* in 1996.

Crosby's 14 steps to quality improvement are as follows:

1. Make it clear that management is committed to quality.
2. Form quality improvement teams with representatives from each department.
3. Determine how to measure where current and potential quality problems lie.
4. Evaluate the cost of quality and explain its use as a management tool.
5. Raise the quality awareness and personal concern of all employees.
6. Take formal actions to correct problems identified through previous steps.
7. Establish a committee for the zero defects program.
8. Train all employees to actively carry out their part of the quality improvement program.
9. Hold a "zero defects day" to let all employees realize that there has been a change.
10. Encourage individuals to establish improvement goals for themselves and their groups.
11. Encourage employees to communicate to management the obstacles they face in attaining their improvement goals.
12. Recognize and appreciate those who participate.
13. Establish quality councils to communicate on a regular basis.
14. Do it all over again to emphasize that the quality improvement program never ends.

The name *Armand V. Feigenbaum* and the term "total quality control" are virtually synonymous. During his career, the precepts of total quality control were carefully laid out and tirelessly promulgated in the United States and around the world. Feigenbaum's ideas are contained in his now famous book *Total Quality Control*, first published in 1951 under the title *Quality Control: Principles, Practice, and Administration*, and based on his earlier articles and program installations in the field.

The book has been translated into more than a score of languages, including Japanese, Chinese, French, and Spanish. Feigenbaum is president and CEO of General Systems Co. in Pittsfield, Massachusetts, an engineering firm that designs and installs operational systems for corporations in the United States, Europe, Asia, and Latin America. Prior to assuming his responsibilities at General Systems Co., Feigenbaum was the manager of worldwide manufacturing operations and quality control for the General Electric Co.

Feigenbaum is recognized as an innovator in the area of quality cost management. His was the first text to characterize quality costs as the costs of prevention, appraisal, and internal and external failure. He lists three steps to quality:

1. Quality leadership
2. Modern quality technology
3. Organizational commitment

The career of *Kaoru Ishikawa* in some ways parallels the economic history of contemporary Japan. Ishikawa, like Japan as a whole, learned the basics of statistical quality control developed by Americans. But just as Japan's economic accomplishments are not limited to imitating foreign products, so the country's quality achievements—and Ishikawa's in particular—go well beyond the efficient application of imported ideas. Perhaps Ishikawa's most important contribution has been his key role in the development of a specifically Japanese quality strategy. The hallmark of the Japanese approach is broad involvement in quality, not only top to bottom within the organization, but also start to finish in the product life cycle. The bottom-up approach is best exemplified by the *quality circle*. As a member of the editorial board of *Quality Control for the Foreman*, as chief executive director of Quality Control Circle Headquarters at the Union of Japanese Scientists and Engineers (JUSE), and as editor of JUSE's two books on quality circles (*QC Circle Koryo* and *How to Operate QC Circle Activities*), Ishikawa played a major role in the growth of quality circles. One of Ishikawa's early achievements contributed to the success of quality circles. The cause-and-effect diagram—often called the *Ishikawa diagram* and perhaps the achievement for which he is best known—has provided a powerful tool that can easily be used by non-specialists to analyze and solve problems. Although the quality circle was developed in Japan, it spread to more than 50 countries, a development Ishikawa never foresaw.

The following points summarize Ishikawa's philosophy:

- Quality first—not short-term profit first.
- Consumer orientation—not producer orientation. Think from the standpoint of the other party.
- The next process is your customer—breaking down the barrier of sectionalism.
- Using facts and data to make presentations—utilization of statistical methods.
- Respect for humanity as a management philosophy—full participatory management.
- Cross-functional management.

Genichi Taguchi taught that any departure from the nominal or target value for a characteristic represents a loss to society. He also popularized the use of fractional factorial experiments and stressed the concept of robustness. Genichi Taguchi is also known for developing a methodology to improve quality and reduce costs, known in the United States as the "Taguchi methods." He also developed the quality loss function.

Taguchi worked for the Institute of Statistical Mathematics from 1948 to 1950 and gained recognition for his contributions to industrial experiments dealing with the production of penicillin. He was hired by the Electrical Communication

Laboratory (ECL) in 1950, even as statistical quality control was gaining popularity in leading Japanese companies.

During this time, both ECL and Bell Laboratories were developing crossbar and telephone switching systems. Working on the project for ECL provided Taguchi with plenty of opportunity for experimentation and data analysis. Six years later, the ECL systems project was completed, around the same time Bell Labs completed its version. Nippon AT&T awarded a contract to ECL, however, for its superior production.

Taguchi also found time to write *Experimental Design and Life Test Analysis* and *Design of Experiments for Engineers*. In 1960 the latter book helped earn him Japan's Deming Prize for his contributions to quality engineering. Two years later, after he had earned his doctorate in science, Taguchi wrote a second edition of *Design of Experiments* that introduced industrial research on the signal-to-noise ratio. He left ECL but maintained his relationship in a consulting capacity.

After joining the associate research staff of the Japanese Standards Association, Taguchi founded the Quality Research Group. Since 1963 the group has met monthly to discuss industry applications. Aoyama Gakuin University in Japan invited Taguchi to teach in 1965, and he stayed on for 17 years and helped develop the university's engineering department.

By the early 1980s Taguchi was making a name for himself in the United States. A supplier had introduced Taguchi's methods to Ford Motor Co., and he was invited to provide seminars to Ford executives in 1982.

Dorian Shainin worked more than 60 years to improve the professional approach to industrial problem solving. He is best known for the "Shainin techniques," practical tools he developed to help manufacturers solve problems, including problems that had been considered unsolvable.

Through his work with more than 900 organizations, Shainin developed a discipline called *statistical engineering*. He specialized in creating strategies to enable engineers to "talk to the parts" and solve "unsolvable" problems. The discipline has been used successfully for product development, quality improvement, analytical problem solving, manufacturing cost reduction, product reliability, product liability prevention, and research and development.

In the early 1960s Shainin served Grumman Aerospace as a reliability consultant for the lunar module of NASA's Apollo project. The lunar module prototype components and systems had been empirically tested using the Shainin multiple environment overstress probe testing system to be statistically sure that even the weakest failure mode had a statistical margin of safety. NASA initially awarded that contract to Grumman because no other aerospace competitive proposal demonstrated that safety ability.

Shainin wrote more than 100 articles and was the author or coauthor of several books, including *Managing Manpower in the Industrial Environment*, *Tool Engineers Handbook*, *Quality Control Handbook*, *New Decision-Making Tools for Managers*, *Quality Control for Plastics Engineers*, *Statistics in Action*, and the *Manufacturing, Planning, and Estimating Handbook.*

In addition to these noted individuals, Toyota Motor Company has been recognized as the leader in developing the concept of lean manufacturing systems. Various approaches to quality have been in vogue over the years, as shown in Table 1.1.

Table 1.1 Some approaches to quality over the years.

Quality approach	Approximate time frame	Short description
Quality circles	1979–1981	Quality improvement or self-improvement study groups composed of a small number of employees (10 or fewer) and their supervisor. Quality circles originated in Japan, where they are called *quality control circles*.
Statistical process control (SPC)	Mid-1980s	The application of statistical techniques to control a process. Also called *statistical quality control*.
ISO 9000	1987–present	A set of international standards on quality management and quality assurance (ISO 9001) developed to help companies effectively document the quality system elements to beimplemented to maintain an efficient quality system. The standards, initially published in 1987, are not specific to any particular industry, product, or service. The standards were developed by the International Organization for Standardization (ISO), a specialized international agency for standardization composed of the national standards bodies of 91 countries. These standards have undergone major revisions over the years, with the latest revision occurring in 2015.
Reengineering	1996–1997	A breakthrough approach involving the restructuring of an entire organization and its processes.
Benchmarking	1988–1996	An improvement process in which a company measures its performance against that of best-in-class companies, determines how those companies achieved their performance levels, and uses the information to improve its own performance. The subjects that can be benchmarked include strategies, operations, processes, and procedures.
Balanced scorecard	1990s–present	A management concept that helps managers at all levels monitor their results in their key areas.
Baldrige Award criteria	1990s–present	An award established by the U.S. Congress in 1987 to raise awareness of quality management and recognize U.S. companies that have implemented successful quality management systems. Two awards may be given annually in each of six categories: manufacturing company, service company, small business, education, health care, and nonprofit. The award is named after the late secretary of commerce Malcolm Baldrige, a proponent of quality management. The U.S. Commerce Department's National Institute of Standards and Technology manages the award, and ASQ administers it.

Continued

Table 1.1 *Continued.*

Quality approach	Approximate time frame	Short description
Six Sigma	1995–present	As described elsewhere is this chapter.
Lean	2000–present	As described elsewhere is this chapter.
Lean Six Sigma	2002–present	This approach combines the individual concepts of lean and Six Sigma and recognizes that both are necessary to effectively drive sustained improvement.

A wide range of companies has found that when the Six Sigma philosophy is fully embraced, the enterprise thrives. What is this Six Sigma philosophy? Several definitions have been proposed, with the following common threads:

- Use of teams that are assigned well-defined projects that have direct impact on the organization's bottom line.
- Training in statistical thinking at all levels, and providing key people with extensive training in advanced statistics and project management. These key people are designated "Black Belts."
- Emphasis on the DMAIC approach to problem solving: define, measure, analyze, improve, and control.
- A management environment that supports these initiatives as a business strategy.

The literature is replete with examples of projects that have returned high dollar amounts to the organizations involved. Black Belts are often required to manage four projects per year for a total of $500,000–$5,000,000 in contributions to the company's bottom line.

Opinions on the definition of Six Sigma differ:

- *Philosophy*. The philosophical perspective views all work as processes that can be defined, measured, analyzed, improved, and controlled (DMAIC). Processes require inputs and produce outputs. If you control the inputs, you will control the outputs. This is generally expressed as the $Y = f(X)$ concept.
- *Set of tools*. Six Sigma as a set of tools includes all the qualitative and quantitative techniques used by the Six Sigma expert to drive process improvement. A few such tools include statistical process control (SPC), control charts, failure mode and effects analysis, and process mapping. Six Sigma professionals do not totally agree as to exactly which tools constitute the set.

- *Methodology*. The methodological view of Six Sigma recognizes the underlying and rigorous approach known as DMAIC. DMAIC defines the steps a Six Sigma practitioner is expected to follow, starting with identifying the problem and ending with implementing long-lasting solutions. While DMAIC is not the only Six Sigma methodology in use, it is certainly the most widely adopted and recognized.
- *Metrics*. In simple terms, Six Sigma quality performance means 3.4 defects per million opportunities (accounting for a 1.5-sigma shift in the mean).

In the first edition of this book, we used the following to define Six Sigma:

> *Six Sigma is a fact-based, data-driven philosophy of improvement that values defect prevention over defect detection. It drives customer satisfaction and bottom-line results by reducing variation and waste, thereby promoting a competitive advantage. It applies anywhere variation and waste exist, and every employee should be involved.*

However, going forward, we combine the definitions of lean and Six Sigma and offer a definition for Lean Six Sigma. This is discussed in detail later in this section.

The term "lean thinking" refers to the use of ideas originally employed in lean manufacturing to improve functions in all departments of an enterprise.

The National Institute of Standards and Technology (NIST), through its Manufacturing Extension Partnership, defines *lean* as follows:

> *A systematic approach to identifying and eliminating waste (non-value-added activities) through continuous improvement by flowing the product at the pull of the customer in pursuit of perfection.*

ASQ defines the term *non-value-added* as follows:

> *A process step or function that is not required for the direct achievement of process output. This step or function is identified and examined for potential elimination.*

This represents a shift in focus for manufacturing engineering, which has traditionally studied ways to improve value-added functions and activities (for example, how can this process run faster and more precisely?). Lean thinking doesn't ignore the valued-added activities, but it does shine the spotlight on waste. A discussion of various categories of waste is provided in the waste analysis section of Chapter 23.

Lean manufacturing seeks to eliminate or reduce these wastes by use of the following:

- *Teamwork* with well-informed, cross-trained employees who participate in the decisions that impact their function
- *Clean*, organized, and well-marked work spaces
- *Flow systems* instead of batch and queue (that is, reduce batch size toward its ultimate ideal—one)
- *Pull systems* instead of push systems (that is, replenish what the customer has consumed)

- *Reduced lead times* through more efficient processing, setups, and scheduling

The history of lean thinking may be traced to Eli Whitney, who is credited with spreading the concept of part interchangeability. Henry Ford, who went to great lengths to reduce cycle times, furthered the idea of lean thinking, and later the Toyota Production System (TPS) packaged most of the tools and concepts now known as *lean manufacturing*.

Lean and Six Sigma have the same general purpose of providing the customer with the best possible quality, cost, delivery, and a newer attribute—nimbleness. There is a great deal of overlap, and disciples of both disagree as to which techniques belong where. Six Sigma Black Belts need to know a lot about lean (witness the appearance of lean topics in the Body of Knowledge for Black Belt certification).

The most successful users of implementations have begun with the lean approach, making the workplace as efficient and effective as possible, reducing the (now) eight wastes, and using value stream maps to improve understanding and throughput. When process problems remain, the more technical Six Sigma statistical tools may be applied. One thing both methodologies have in common is that both require strong management support to make them the standard way of doing business.

Some organizations have responded to this dichotomy of approaches by forming a Lean Six Sigma problem-solving team with specialists in the various aspects of each discipline, but with each member cognizant of the others' fields. Task forces from this team are formed and reshaped depending on the problem at hand.

Given the earlier discussion, we believe a combined definition is required and proffer the following:

> *Lean Six Sigma is a fact-based, data-driven philosophy of improvement that values defect prevention over defect detection. It drives customer satisfaction and bottom-line results by reducing variation, waste, and cycle time while promoting the use of work standardization and flow, thereby creating a competitive advantage. It applies anywhere variation and waste exist, and every employee should be involved.*

SIX SIGMA, LEAN, AND CONTINUOUS IMPROVEMENT METHODOLOGIES

> Describe when to use six sigma instead of other problem-solving approaches, and describe the importance of aligning six sigma objectives and organizational goals. Describe screening criteria and how such criteria can be used for the selection of six sigma projects, lean initiatives, and other continuous improvement methods. (Apply)
>
> **Body of Knowledge I.A.2**

The demarcation between Six Sigma and lean has blurred. We are hearing about terms such as "Lean Six Sigma" with greater frequency because process improvement requires aspects of both approaches to attain positive results. Six Sigma focuses on reducing process variation and enhancing process control, whereas lean—also known as *lean manufacturing*—drives out waste (non-value-added) and promotes work standardization and flow. Six Sigma practitioners should be well versed in both. More details of what is sometimes referred to as *lean thinking* are given in Chapters 25 and 26.

The two initiatives approach their common purpose from slightly different angles:

- Lean focuses on waste reduction, whereas Six Sigma emphasizes variation reduction.
- Lean achieves its goals by using less technical tools such as kaizen, workplace organization, and visual controls, whereas Six Sigma tends to use statistical data analysis, design of experiments, and hypothesis tests.

The most successful implementations of lean and Six Sigma have an oversight group with top management representation and support. This group defines and prioritizes problems and establishes teams to solve them. The oversight group is responsible for maintaining a systemic approach. It also provides the training, support, recognition, and rewards for teams. The following are examples of problems that would be assigned to teams:

- A number of customers of an accounting firm have complained about the amount of time the firm takes to perform an audit. The oversight group forms a team consisting of three auditors (one of them a lead auditor), two cost accountants, and two representatives from the firm's top customers. The oversight group asks the team to determine whether the lead time is indeed inordinate and to propose measures that will reduce it. The team begins by benchmarking (see Chapter 4) a customer's internal audit process. After allowing for differences between internal and external audits, the team concludes that the lead time should be shortened. The team next uses the material discussed in Chapter 14 to construct a value stream map, which displays work in progress, cycle times, and communication channels. A careful study of the map data shows several areas where lead time can be decreased.

- A team has been formed to reduce cycle times on an appliance assembly line. The team consists of the 12 workers on the line (six from each of the two shifts) as well as the two shift coaches and the line supervisor. Although this makes a large team, it helps ensure that everyone's creative energy is tapped. The team decides to start a job-rotation process in which each assembler will work one station for a month and then move on to the next station. After three months, the workers universally dislike this procedure, but they agree to continue through at least one complete rotation. At the end of nine months, or one and a half rotations, the team acknowledges that the rotation system has helped improve standard work (see Chapter 25) because each person better understands what the next person needs. They are also better equipped to accommodate absences and the training of new people. The resulting reduction in cycle times surprises everyone.

- A team has been charged with improving the operation of a shuttle brazer. Automotive radiators are loaded on this machine and shuttled through a series of gas-fired torches to braze the connections. The operator can adjust the shuttle speed, wait time, gas pressure, torch angle, and torch height. There is a tendency to adjust one or more of these settings to produce leak-free joints, but no one seems to know the best settings. The team decides to conduct a full-factorial 25 designed experiment with four replications (see Chapter 24) during a planned plant shutdown.

- A company is plagued with failure to meet deadlines for software projects. A team is formed to study and improve the design/code/test process. The team splits into three subteams, one for each phase. The design subteam discovers that this crucial phase endures excess variation in the form of customer needs. This occurs because customers change the requirements and because sometimes the software package is designed to serve multiple customers whose needs aren't known until late in the design phase. The subteam helps the designers develop a generic Gantt chart (see Chapter 12) for the design phase itself. It also establishes a better process for determining potential customer needs (see Chapter 10). The design group decides to develop configurable software packages that permit the user to specify the functions needed.

  The coding subteam finds that those responsible for writing the actual code are often involved with multiple projects, leading to tension between project managers. This results in spurts of activity and concentration being spent on several projects with the resulting inefficiencies. The subteam collaborates with the project manager to establish a format for prioritization matrices (see Chapter 13), which provide better guidance for coders.

  The testing subteam determines that there is poor communication between designers and testers regarding critical functions, especially those that appeared late in the design phase. After discussions with those involved, it is decided that for each project a representative of the testing group should be an ex officio member of the design group.

Organizational goals must be consistent with the long-term strategies of the enterprise. One technique for developing such strategies is called *hoshin planning*. In this process, a company develops up to four vision statements that indicate where the company should be in the next five years. Company goals, projects, and relevant work plans are developed on the basis of these vision statements. Periodic audits are then conducted to monitor progress. Further details of the hoshin planning system are discussed later in this chapter.

Once Six Sigma projects have had some success, there will usually be more project ideas than it is possible to undertake at one time. Some sort of project proposal format may be needed, along with an associated process for project selection. It is common to require that project proposals include precise statements of the problem definition and some preliminary measures of the seriousness of the problem, including its impact on the goals of the enterprise.

A project selection group, including Master Black Belts, Black Belts, organizational champions, and key executive supporters, establishes a set of criteria for project selection and team assignments. In some companies the project selection group assigns some projects to Six Sigma teams and other projects to teams using other methodologies. For example, problems involving extensive data analysis and improvements using designed experiments would likely be assigned to a Six Sigma team, whereas a process improvement not involving these techniques might be assigned to a lean manufacturing team employing kaizen tools. Details on kaizen new product design should follow the Design for Six Sigma (DFSS) guidelines as detailed in Chapters 31 and 32.

The project selection criteria always have as key elements the furthering of organizational goals. One key to gauging both the performance and the health of an organization and its processes lies with its selection and use of metrics. These are usually converted to financial terms such as return on investment, cost reduction, and increases in sales and/or profit. Other things being approximately equal, projects with the greatest contributions to the bottom line receive the highest priority. More details on project metrics are covered in Chapter 11.

RELATIONSHIPS AMONG BUSINESS SYSTEMS AND PROCESSES

> Describe the interactive relationships among business systems, processes, and internal and external stakeholders, and the impact those relationships have on business systems. (Understand)
>
> **Body of Knowledge I.A.3**

Processes

Once a product or service has been designed, the next step is to develop a process to produce it. In some fields the process is referred to as a set of procedures or routines. Regardless of the name used, process design is a critical step for all stakeholders:

- The *customer* needs a process that provides a quality output in an acceptable time frame and at an acceptable cost.
- The *producer* needs a process that uses reasonably priced inputs and that reliably generates output that meets or exceeds customer expectations.
- The *marketplace* requires a quality product at a competitive price as an output from a process that is agile enough to meet changing customer requirements.

- *Suppliers* need a dependable market for their products that are the inputs to the process.

These stakeholders generate sometimes conflicting forces on the process, as shown in Figure 1.1.

A process is a series of steps designed to produce products and/or services. A process is often diagrammed with a flowchart depicting inputs, the path that material or information follows, and outputs. An example of a process flowchart is shown in Figure 1.2. Understanding and improving processes is a key part of every Six Sigma project.

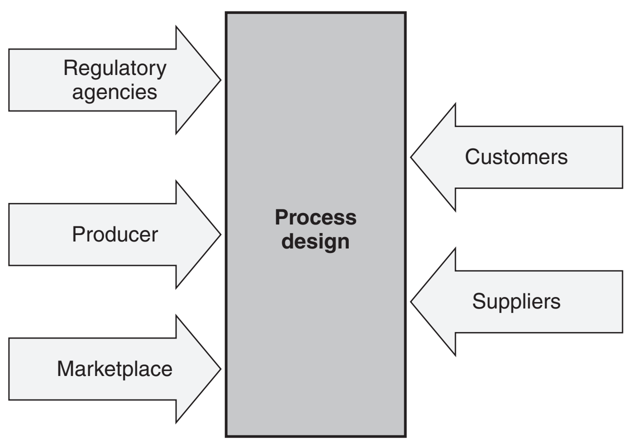

Figure 1.1 Some of the forces impacting process design.

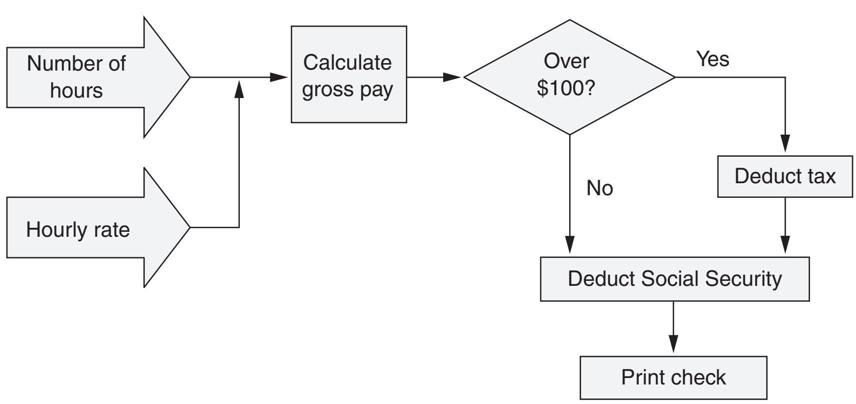

Figure 1.2 Example of a process flowchart.

The basic strategy of Six Sigma is contained in DMAIC. These steps constitute the cycle Six Sigma practitioners use to manage problem-solving projects. The individual parts of the DMAIC cycle are explained in Chapters 10–30.

Business Systems

A business system is designed to implement a process or, more commonly, a set of processes. The business system must coordinate the often conflicting forces on the design and implementation of its processes. This means that the business system must consider the goals and objectives of the enterprise as it provides resources and guidance. Business systems make certain that process inputs are in the right place at the right time so that each step of the process has the resources it needs. Perhaps most importantly, a business system must have as its goal the continual improvement of its processes, products, and services. To this end, the business system is responsible for collecting and analyzing data from the process and other sources that will help in the continual incremental improvement of process outputs. Figure 1.3 illustrates relationships among systems, processes, subprocesses, and steps. Note that each part of a system can be broken into a series of processes, each of which may have subprocesses. The subprocesses may be further broken into steps. In addition, the business system imposes performance expectations on the process and its designers in terms of safety, cost, quality, and delivery. The performance of a process, in turn, is a determining factor in the success of its business system. The Six Sigma Black Belt is often called on to help teams improve process

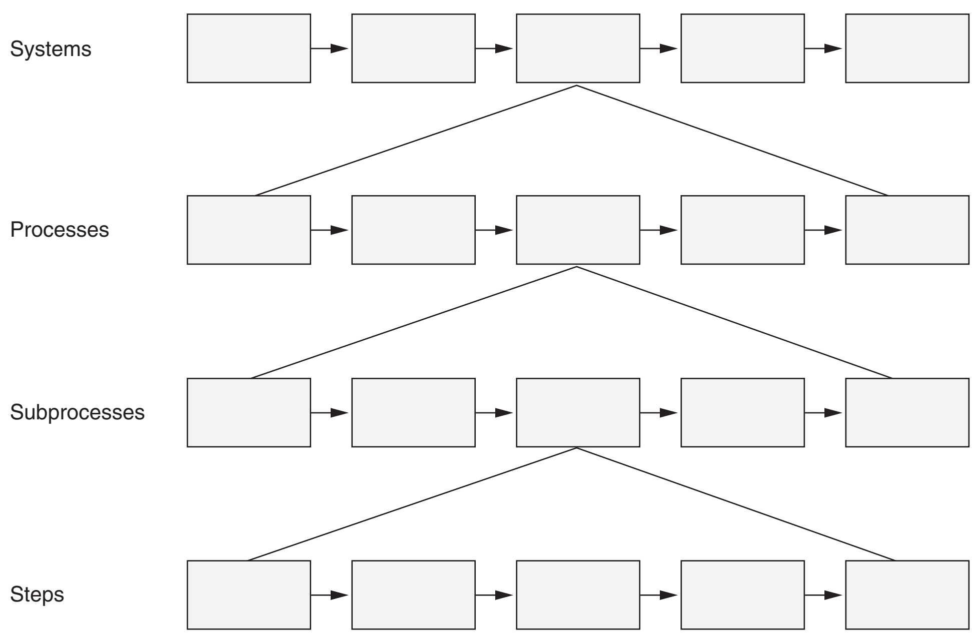

Figure 1.3 Relationships among systems, processes, subprocesses, and steps.

performance. Much of this book is devoted to providing tools that can be used in response to such requests.

STRATEGIC PLANNING AND DEPLOYMENT FOR INITIATIVES

> Define the importance of strategic planning for six sigma projects and lean initiatives. Demonstrate how hoshin kanri (X-matrix), portfolio analysis, and other tools can be used in support of strategic deployment of these projects. Use feasibility studies, SWOT analysis (strengths, weaknesses, opportunities, and threats), PEST analysis (political, economic, social, and technological) and contingency planning and business continuity planning to enhance strategic planning and deployment. (Apply)
>
> **Body of Knowledge I.A.4**

In general, a strategic planning process must do three things:

- Study the current state (that is, the business environment in which we operate)
- Envision the ideal future state (that is, where we'd like to be in three to five years)
- Plan the path (that is, how we'll get from here to there)

Strategic planning must be a priority of top management. For strategic planning to be successful, senior management must be actively engaged in the process. It is not something that can be delegated. Further, successful strategic planning must create a line of sight from strategic to tactical to operational activities. This does not mean that individuals executing activities at the lowest level of the organization must understand the details of the strategic plan. However, it does mean that their actions are traceable back to the strategic plan itself.

Various strategic planning models have been proposed, as described in the following subsections.

Porter's Five Forces

In 1979, Michael Porter listed five forces that affect the success of an enterprise:

- *The bargaining power of customers.* This force represents the ability and buying power of your customers to drive prices down.
- *The bargaining power of suppliers.* This force represents the ability and power of your suppliers to drive up prices of their products.

- *The threat of new entrants.* This force represents the ease with which new competitors can enter the market and drive prices down.
- *The threat of substitute products.* This force represents the extent to which different products or services can be substituted for your own.
- *The intensity of competitive rivalry.* This force represents the strength of the competition in your industry.

An organization should analyze each of these forces as it forms its strategic plan. It should also continually monitor each force because such forces can be highly dynamic, and changes in any one of these forces can require a change in strategy.

Portfolio Analysis

Portfolio analysis is the process of determining which products to produce. As a strategic planning tool, it can be used to add to a current product line or to develop an entirely new family of products. The strategic advantage of this approach is that the product family will have common modules that will make manufacturing and inventory more efficient and permit "mass customization." The steps in portfolio analysis are as follows:

1. Study the basic physical/chemical/biological principles involved (for example, how does powder coating work at the molecular level?).
2. Use the basic principles studied in step 1 to outline a family of products (for example, powder coating spray guns, curing systems, turnkey systems).
3. List the modules required for each product.
4. Form a matrix with products composing the rows and function modules composing the columns (see Figure 1.4).
5. Study the matrix to determine which products to build.

Steps 3–5 are repeated in an iterative fashion, adjusting modules, adding or deleting products, and redesigning modules until a final combination is determined.

Hoshin Kanri

One translation of the word *hoshin* refers to a shining arrow such as found on a compass. The word *kanri* may be translated as *management*. Hoshin kanri, then, provides tools for stating objectives for the organization and managing their implementation. Hoshin kanri is a very tightly controlled planning and execution protocol that requires goal setting, beginning at the highest level. Each lower level of the organization is required to form objectives and goals that support those of the next higher level.

Following are the basic steps in the hoshin kanri process at each level of the hierarchy:

	Module						
Product	**A**	**B**	**C**	**D**	**E**	**F**	**G**
Systems for custom auto shops	Y	Y		Y		Y	Y
Systems for education and training	Y	Y		Y			
Home systems	Y	Y		Y		Y	
Commercial turnkey systems	Y	Y	Y	Y	Y	Y	
Systems for coating wood	Y		Y	Y	Y	Y	Y
Systems for using thermosets	Y	Y	Y		Y		Y
Systems for curing other finishes		Y		Y	Y	Y	
Stand-alone spray guns		Y		Y			

Figure 1.4 Example of a product family matrix.

- Statement of the organization's goals (occasionally referred to as *objectives*)
- Formulation of strategies that support achievement of the goals
- Development of time-based tactics that support achievement of each strategy
- Development of metrics that measure the progress toward achieving each strategy
- Assignment of accountability for strategies and tactics
- Regular reviews to assess the actual performance of the plan
- Establishment of recovery plans, or the adjustment of strategies and tactics as necessary

A key advantage of an effectively implemented hoshin planning process is that strategies are deployed and aligned both horizontally and vertically throughout an organization.

Figure 1.5, from Kubiak (2012), represents the underlying principles and concepts upon which hoshin kanri is built. Most authors agree that the process focuses on the development of both long-term and mid-term plans, while some authors suggest a focus on short-term plans as well. Long-term plans are usually two to five years, mid-term plans are from one to two or three years, and short-term plans are generally one year long and are commonly called the *annual* or *operational plan*.

There are various ways to display the goals and objectives for hoshin kanri analysis.

Kubiak (2012) uses Table 1.2 as a way to illustrate a typical format.

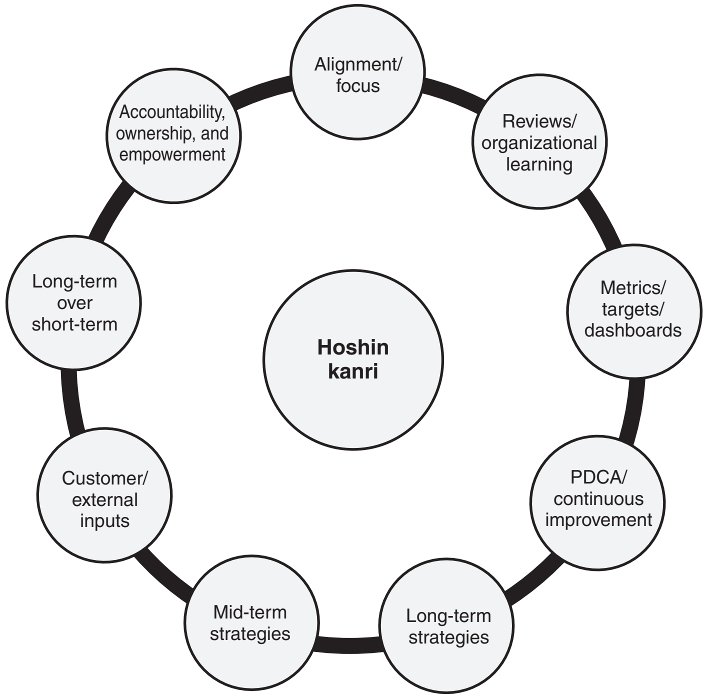

Figure 1.5 Fundamental principles and concepts of hoshin kanri.

Table 1.2 Example of a typical hoshin kanri strategy deployment form.

G&O category	Goals and objectives	Strategies	Primary measure	Counter-measure	Results	Strategy owner
Customer	Reduce customer complaints from 60% to 15%	1.1 Introduce “No questions asked return policy”	% complaints received during returns	% returns	25% reduction to date	R. U. Done
		1.2 . . .				
Suppliers	Reduce PPM at receiving inspection from 500K to 300K	2.1 Have suppliers transition from hand to wave soldering	% reduction in PPM at receiving inspection	% reduction in PPM at source inspection	60% reduction achieved to date	I. M. Fini
		2.2 . . .				
Employees	89% satisfied	3.1 . . .				
		3.2 . . .				

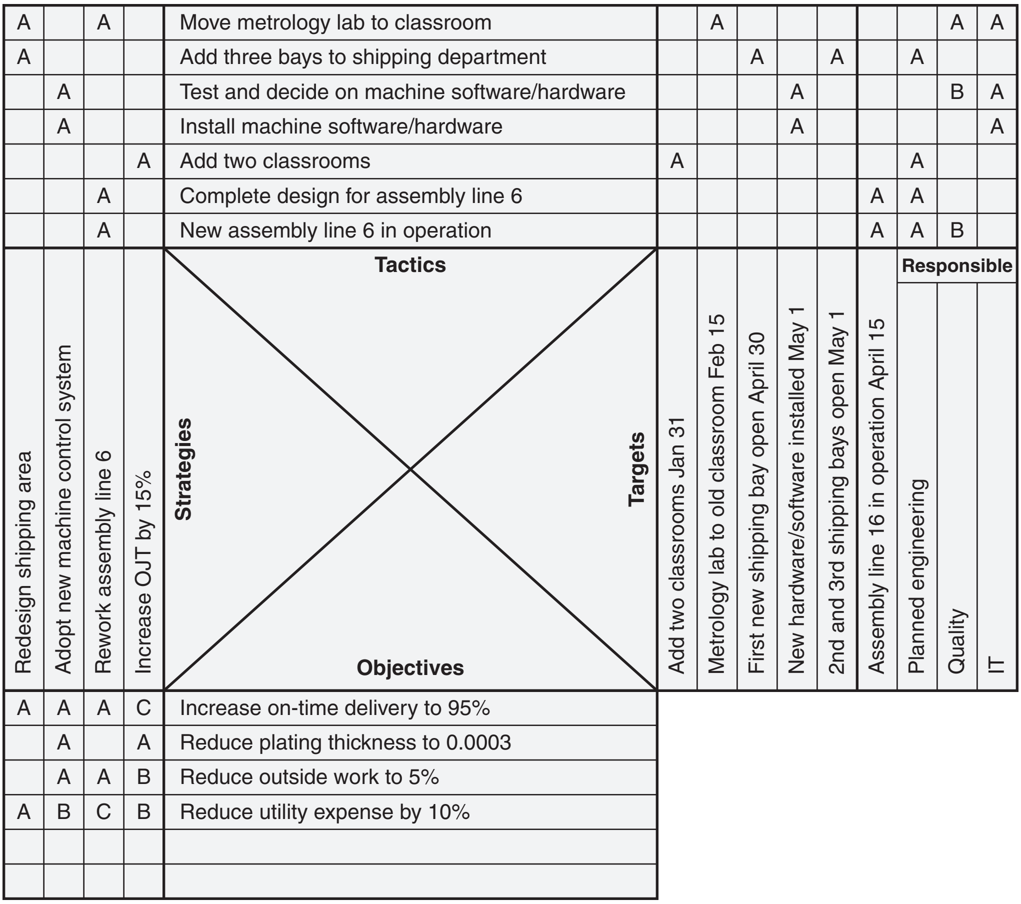

Figure 1.6 X-matrix example 1.

The hoshin kanri X-matrix, also call a *policy deployment matrix*, summarizes the strategic plan linking objectives with the design for achieving them. The X-matrix is used in various ways. Examples are shown in Figures 1.6 and 1.7. The X-matrix is constructed by starting with the objectives and moving clockwise around the diagram, next listing strategies for attaining the objectives. The box representing the intersection of a particular objective row and the column for a particular strategy contains a symbol representing the extent to which that strategy supports that objective. In these figures the letter A represents the strongest support. The next step is to list the tactics section, with tactical support of strategies again shown with letters. The final section represents targets and assigns responsibilities.

Feasibility Study

A *feasibility study* is a preliminary investigation to determine the viability of a project, business venture, or the like. Such a study may look at the feasibility from the

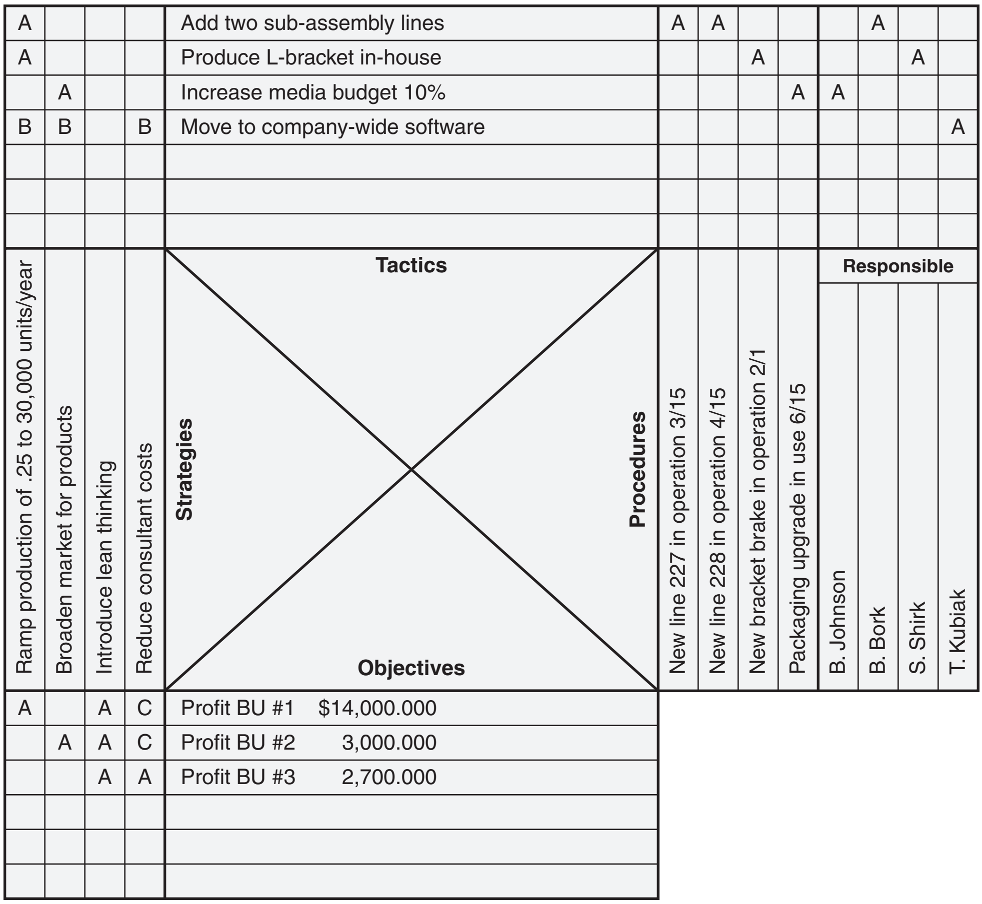

Figure 1.7 X-matrix example 2.

perspective of cost, technology, culture, legality, and so forth. At the conclusion of a feasibility study, management usually determines whether to proceed with the project or business venture.

Risk Analysis

When analyzing risk, we often deal with the concept of *expected profit*. Expected profit is defined as follows:

$$\text{Expected profit} = \Sigma_{i=1}^{n} \ (\text{Profit}_i \times \text{Probability}_i)$$

where

n = Total number of outcomes

Risk analysis for real-life problems tends to be less precise primarily because the probabilities are usually not known and must be estimated. However, the overall concept is similar.

EXAMPLE 1.1

A gambler is considering whether to bet $1.00 on red at a roulette table. If the ball falls into a red cell, the gambler will receive a $1.00 profit. Otherwise, the gambler will lose the $1.00 bet. The wheel has 38 cells, and 18 are red.

Assuming a fair wheel, the probability of winning is 18/38 ≈ 0.474, and the probability of losing is 20/38 ≈ 0.526. This risk analysis is best represented in a table similar to Table 1.3.

Table 1.3 Data for Example 1.1.

Outcome	Profit	Probability	Profit × probability ($M)
Win	$1.00	0.474	$0.474
Lose	($1.00)	0.526	($0.526)
		Expected profit =	($0.052)

In this case the gambler can expect to lose an average of about a nickel (–$0.052) for each $1.00 bet.

EXAMPLE 1.2

A proposed Six Sigma project is aimed at improving quality enough to attract one or two new customers. The project will cost $3 million (M). Previous experience indicates that the probability of getting customer A is between 60% and 70%, and the probability of getting customer B is between 10% and 20%. The probability of getting both A and B is between 5% and 10%. One way to analyze this problem is to make two tables, one for the worst case and the other for the best case, as indicated in Table 1.4.

Table 1.4 Data for Example 1.2.

	Worst case			Best case		
Outcome	Profit ($M)	Probability	Profit × probability ($M)	Profit ($M)	Probability	Profit × probability ($M)
A only	2.00	0.60	1.20	2.00	0.70	1.40
B only	2.00	0.10	0.20	2.00	0.20	0.40
A and B	7.00	0.05	0.35	7.00	0.10	0.70
None	(3.00)	0.25	(0.75)	(3.00)	0.00	0.00
		Expected profit =	1.00		Expected profit =	2.50

Assuming that the data are correct, the project will improve profit of the enterprise by between $1M and $2.5M.

SWOT

A *SWOT* (strengths, weaknesses, opportunities, and threats) analysis is an effective strategic planning tool applicable to a business or project objective.

Strengths and weaknesses are identified with respect to the internal capabilities of an organization. They provide an introspective view. Strengths and weaknesses should not be considered opposites. This can be readily seen in Figure 1.7.

On the other hand, opportunities and threats look outside the organization. Quite simply, we are trying to identify opportunities for the organization and threats to the organization.

When analyzing a SWOT, the question to ask is, "How can the organization leverage its strengths or improve its weaknesses to take advantage of the opportunities while mitigating or eliminating the threats?"

EXAMPLE 1.3

A private higher-education institution has experienced declining enrollment for four consecutive years, particularly from the two major cities in the state where it is located. This declining enrollment has impacted profitability. The president is feeling pressure from the board of directors, which wants answers as to why the enrollment levels are declining. The industrial engineering department has proposed a Six Sigma project aimed at improving the enrollment rate for students from the two major cities. As the department prepares to make probability estimates, it produces the SWOT analysis shown in Figure 1.8.

	Strengths	**Weaknesses**
Internal	• High academic entrance requirements • Strong general education program • High percentage of full-time faculty • New endowments for faculty chairs • Fundraising goals are being met each year • Well-designed internship and study abroad opportunities	• Print communications for prospective students are bland and generic • Recruitment staff is too passive in contacting prospective students • Uneven implementation of assessment and program review methods • Student-centered customer focus is not recognized throughout the institution
	Opportunities	**Threats**
External	• Metropolitan area location has population advantages • Area corporations offer tuition reimbursements • Reputable business program with a global studies focus is not marketed as well as it could be	• The number of traditional-age prospective students is expected to decline • With the cost of education escalating, community colleges have recently become viable competitors • Public universities have undertaken significant marketing efforts to position themselves as comparable in experience to private colleges

Figure 1.8 SWOT analysis for Example 1.3.

PEST

An analysis similar to SWOT is the *PEST analysis*, which brings together four environmental scanning perspectives that serve as useful input into an organization's strategic planning process:

- *Political*. This perspective looks at political stability, threats of wars, regulatory issues, tax considerations, labor problems, laws, and so forth.
- *Economic*. This perspective looks at exchange rates, market conditions, unemployment rates, inflation factors, interest rates, and so forth.
- *Social*. This perspective looks at education considerations, demographics, health factors, cultural implications, and so forth.
- *Technological*. This perspective looks at new technology developments, rate of technological change, cost impact of technology, and so forth.

EXAMPLE 1.4

A nonprofit organization seeks to improve quality of life for citizens in a neighborhood by upgrading storm sewers. Its PEST analysis would look similar to Figure 1.9.

Political	Economic
• The proposal will be competing with other projects for public money • The neighborhood must have adequate representation on the governing bodies • Other neighborhoods will be improved as a result of the proposal, so they should support it • The current situation allows storm water to infiltrate the sanitary sewer system, which impacts all users	• An increase in real estate taxes for two years will be necessary to cover the costs of the upgrade • Environmental costs will be associated with increased stream and river pollution that occurs from ponded storm water
Social	**Technological**
• People in the affected neighborhood tend to feel disenfranchised and mistreated by the government decision makers, which may impact attitudes toward others • Lack of an adequate storm sewer system has led to an increasing trend of gastrointestinal illnesses as the population increases and more demand is placed on the existing sewer system	• A study should be commissioned to determine which approach will best solve the storm water problem • New pipe-TV techniques are now available

Figure 1.9 PEST analysis for Example 1.4.

Unintended Consequences

A system may be thought of as the set of processes that make up an enterprise. When improvements are proposed, it is important to take a systems approach. This means that consideration should be given to the effect the proposed changes will have on other processes within the system, and therefore on the enterprise as a whole. Operating a system at less than its best mode is called *suboptimization*. Changes in a system may optimize individual processes but suboptimize the system as a whole. It is important to put measurement systems in place that reflect unintended consequences of proposed actions.

EXAMPLE 1.5

The resources invested in improving process A might be more profitably invested in process B.

EXAMPLE 1.6

The improvement of the throughput rate for process A exceeds the ability of process B, thus creating a bottleneck at process B. As result, the work-in-progress increases dramatically.

EXAMPLE 1.7

A distribution center loads its trucks in a manner that minimizes its work. However, this method requires the receiving organization to expend more time, energy, resources, and dollars to unload the trucks. A different loading arrangement might be more expensive for the distribution center, but it would significantly reduce costs for the entire system.

Contingency Planning

Tools and techniques discussed earlier in this chapter enable long-range planning, but a crisis that occurs in minutes can change everything if the organization isn't prepared. A contingency plan (also called a *plan B*) can be divided into three topics:

1. Crisis management
   - Natural disasters
   - Terrorism
   - On-the-job injuries/accidents, and so on

- Positive events such as a huge order that can't be met with current resources

2. Business continuity
   - Loss of key personnel
   - Sudden change in customer/supplier/marketplace
   - Strike by bargaining unit personnel
3. Asset security
   - Loss of intellectual property or other assets
   - Loss of data
   - Mismanagement (neglect, theft)
   - Product recall

A contingency planning team should address each of these areas using the following steps:

1. Use a cause-and-effect diagram (Chapter 23) for each of the areas to generate lists of events that could cause disruption of business.
2. Prioritize the items. One method:
   (a) Give each item a score of 1 to 10 to reflect the impact on the business, using 10 for the most impactful
   (b) Give each item a score of 1 to 10 to reflect the probability of occurrence, using 10 as the most probable
   (c) Multiply the two scores to provide a priority index
3. Beginning with the highest-priority items, develop a plan for each that will address the following:
   - *Reaction*. How to protect people and property during the event. Include activities like evacuation, accounting for all personnel, and equipment damage assessment.
   - *Continuity*. How to maintain the business after the event. Include activities like contracting out some work, temporary warehousing, logistics, and security measures. Insurance against some events may be appropriate.

Contingency planning may motivate other activities in the area of prevention and mitigation of certain events.

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Chapter 2

Leadership

ROLES AND RESPONSIBILITIES

> Describe the responsibilities of executive leadership, champions, sponsors, process owners, master black belts, black belts, and green belts in driving six sigma and lean initiatives. Describe how each group influences project deployment in terms of providing or managing resources, enabling changes in organizational structure, and supporting communications about the purpose and deployment of the initiatives. (Understand)
>
> **Body of Knowledge I.B.1**

The definition and role of leadership have undergone major shifts in recent years. The leadership model that is most effective in the deployment of Six Sigma envisions the leader as a problem solver. The leader's job is to implement systems that identify and solve problems that impede the effectiveness of the processes. This requires two steps:

- Allocate resources to support team-based problem identification and solution
- Allocate resources to install corrections and ensure that the problems do not recur

This concept of leadership implies a good understanding of team dynamics and Six Sigma problem-solving techniques. Deployment of a culture-changing initiative such as the adoption of Six Sigma rarely succeeds without engaged, visible, and active senior-level management involvement. Initiatives starting in the rank and file seldom gain the critical mass necessary to sustain and fuel their own existence.

Enterprises with successful Six Sigma programs have found it useful to delineate roles and responsibilities for people involved in project activity. Although titles vary somewhat from company to company, the following list represents the general thinking regarding each role. The definitions provided can be found in Appendix 34.

Black Belts

Black Belts work full time on Six Sigma projects. These projects are usually prioritized on the basis of their potential financial impact on the enterprise. Individuals designated as Black Belts must be thoroughly trained in statistical methods and be proficient at working with teams to implement project success. Breyfogle (2003) suggests that the number of Black Belts equal about 1% of the number of employees in the organization.

> *Black Belt (BB)—A Six Sigma role associated with an individual typically assigned full-time to train and mentor Green Belts as well as lead improvement projects using specified methodologies such as DMAIC; define, measure, analyze, design, and verify (DMADV); and design for Six Sigma (DFSS).*

Master Black Belts

Master Black Belts have advanced knowledge in statistics and other fields and provide technical support to the Black Belts.

> *Master Black Belt (MBB)—A Six Sigma role associated with an individual typically assigned full-time to train and mentor Black Belts as well as lead the strategy to ensure that the improvement projects chartered are the right strategic projects for the organization. Master Black Belts are usually the authorizing body to certify Green Belts and Black Belts.*

Green Belts

A Green Belt works under the direction of a Black Belt, assisting with all phases of project operation. Green Belts typically are less adept at statistics and other problem-solving techniques.

> *Green Belt (GB)—A Six Sigma role associated with an individual who retains his or her regular position within the firm but is trained in the tools, methods, and skills necessary to conduct Six Sigma improvement projects either individually or as part of larger teams.*

Champion

A champion is typically a top-level manager who is familiar with the benefits of Six Sigma strategies and provides support for the program.

> *Champion—A Six Sigma role associated with a senior manager who ensures that his or her projects are aligned with the organization's strategic goals and priorities, provides the Six Sigma team with resources, removes organizational barriers for the team, participates in project tollgate reviews, and essentially serves as the team's backer. A champion is also known as a* sponsor.

Executive

Successful implementations of Six Sigma have unwavering support from the company-level executives. Executives demonstrate their support for Six Sigma through their communications and actions.

Process Owner

Process owners should be at a sufficiently high level in the organization to make decisions regarding process changes. It is only natural that managers responsible for a particular process frequently have a vested interest in keeping things as they are. They should be involved in any discussion of change. In most cases, they are willing to support changes but need to see evidence that recommended improvements are for the long-term good of the enterprise. A team member with a "show me" attitude can make a very positive contribution to the team. Process owners should be provided with opportunities for training at least to the Green Belt level.

> *Process owner—A Six Sigma role associated with an individual who coordinates the various functions and work activities at all levels of a process, has the authority or ability to make changes in the process as required, and manages the entire process cycle so as to ensure performance effectiveness.*

Organizational goals must be consistent with the long-term strategies of the enterprise. One technique for developing such strategies is called *hoshin planning*, which was discussed in more detail in Chapter 1. In this process, a company develops up to four vision statements that indicate where the company should be in the next five years. Company goals, projects, and relevant work plans are developed on the basis of these vision statements. Periodic audits are then conducted to monitor progress.

ORGANIZATIONAL ROADBLOCKS AND CHANGE MANAGEMENT

> Describe how an organization's structure and culture can impact six sigma projects. Identify common causes of six sigma failures, including lack of management support and lack of resources. Apply change management techniques, including stakeholder analysis, readiness assessments, and communication plans to overcome barriers and drive organization-wide change. (Apply)
>
> **Body of Knowledge I.B.2**

Leaders must understand that change, like any movement, generates friction and resistance. Awareness of this resistance and the ability to mitigate it are important skills. Many organizations have a structure and culture that were formed when an important goal was for the organization to continue running the way it always had. A highly centralized organizational structure tends to have more of this inertia than a decentralized one. Such a structure and culture tend to resist any change. An unintended consequence is that fundamental improvements are difficult to achieve. Some symptoms of such a system are listed here, with possible remedies:

- *Symptom*: A system that requires many signatures for expenditures discourages or delays improvements, particularly when signatories are busy or travel frequently.

  Remedies:

  – A multiple-signature e-mail procedure can be instituted.

  – Teams can be empowered with budgets.

- *Symptom*: Adherence to an "if it ain't broke, don't fix it" philosophy serves as a barrier to change.

  Remedies:

  – One company encourages employees to stop the assembly line when they see a problem or an opportunity for improvement. A wall clock reflects the number of minutes the line was stopped during the shift. They try to have 30–40 minutes of downtime per shift, because only when the line is down are they solving problems and making improvements.

  – Everyone must understand that an important part of his or her job is to make improvements. Incentives for these improvements must be provided.

- *Symptom*: Managers are not properly trained as change agents.

  Remedy: All personnel with management responsibilities need to understand the basics of change management.

Project champions are often the ones best able to break roadblocks. In many situations, the motto "No champion, no project" is valid. Typically, the champion is at an executive level of the organization. Sometimes, the champion is the process owner.

There was a time when a manager's job was to keep things running the way they always did, that is, to prevent change. Today it is understood that change is critical to an enterprise, and the management of changing processes has become a science of its own. Following are some common errors that managers need to avoid:

- Inadequate communication about coming changes. Management may assume that a single memo or meeting is appropriate, but people need to have time and an opportunity to digest and react to changes.
- Assigning the change-management function to people without the preparation or resources to execute it.
- Improper or inadequate explanation of the change. Employees tend to fear and resist change that they do not understand.

For best results, a team with executive leadership should be charged with leading the change and advising top management of necessary actions. Effective communication is essential to success. The following steps are typical in the management of change:

1. Communicate the need for change. Use data regarding market share, competition, and prospects for expansion. Benchmarking (see Chapter 4) is a useful tool.
2. Communicate a view of a future state with a successful change.
3. Establish and communicate near-term, intermediate-term, and long-term goals and metrics.
4. Identify and use forces for change; identify and reduce barriers to change.
5. Communicate early successes and recognize those responsible.
6. Lock in improvements.
7. Establish the need for being a nimble, changing organization.

Stakeholder Analysis

The success of any proposed change depends on the buy-in by impacted stakeholders. Internal and external stakeholders should be involved in the problem-solving and decision-making processes to the extent possible. At a minimum, they need to be in the communication loop from the beginning. One way to identify stakeholders is to use the SIPOC format illustrated in Figure 2.1. Additional information on stakeholder analysis can be found in Chapter 3 and the SIPOC can be found in Chapter 10.

Through interviews, surveys, group meetings, and any other ways deemed feasible, the stakeholders' attitudes toward the proposed change should be assessed and analyzed. A format for summarizing results is shown in Figure 2.2.

Readiness Assessment

Understanding the causes of resistance to change and taking steps to address them are crucial to the success of the proposed change. Timing is, of course, an important element. Determining organizational readiness for change involves

Suppliers	Inputs	Process	Outputs	Customers
Raw material suppliers	Routing dept.	Quality dept.	Regulatory agencies	Regional representatives
Maintenance dept.	Accounting	Reliability dept.	Shipping dept.	Retail outlets
Lube/coolant dept.	Information technology	Design engineering	Packaging	Plumbers
Incoming inspection	Programming	Bargaining unit	Final inspection	State regulators
Receiving				Commercial and residential users

Figure 2.1 SIPOC model used to identify stakeholders for a proposed change in a manufacturing process.

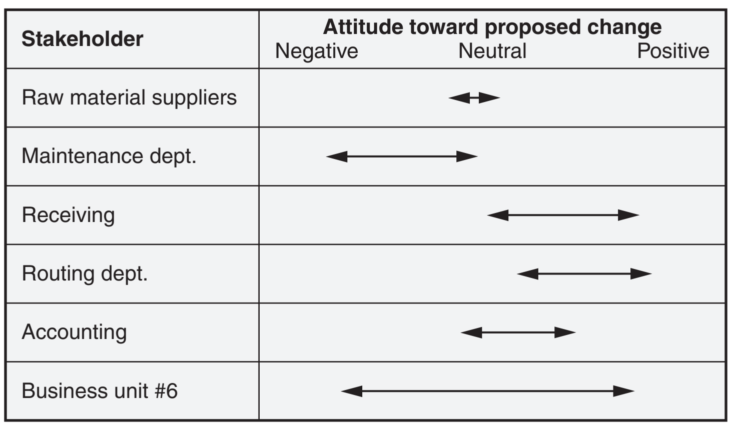

Figure 2.2 Form for displaying stakeholders' attitudes.

assessment of several traits. This assessment tends to be subjective, and the traits will vary depending on the situation. Figure 2.3 displays an example of a readiness assessment. The readiness assessment provides guidance regarding communication and information decisions.

Communication Plans

The Six Sigma Black Belt should develop the ability to effectively communicate the strategic need for change. Communication and transparency are so critical to the success of a proposed change that they require a formal plan. Key issues are listed below:

Trait	Score (1–10)
Alignment with organizational goals	9
Awareness of the need for change	3
Commitment to change	4
Openness to change	6
Willingness to be involved in change	5
Willingness to enable change	3

Figure 2.3 Example of a readiness assessment.

1. All stakeholders should be informed of the need for the change at the earliest possible time—perhaps even before the details are in place. This may involve meetings, articles in newsletters, the use of social media—any method that can convey the message.
2. Involvement in determining the need and the requisite change to meet the need aids communication. Involvement, even through representatives, can help.
3. Continuous communication of the status throughout the change cycle is key to maintaining support.

Additional information on communication plans can be found in Chapter 7.

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Part II

Organizational Process Management and Measures

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Chapter 3

Impact on Stakeholders

> Describe the impact six sigma projects can have on customers, suppliers, and other stakeholders. (Understand)
>
> **Body of Knowledge II.A**

This chapter will address stakeholders from two perspectives: (1) the impact projects can have on stakeholders, and (2) the impact stakeholders can have on projects.

Tague (2005) defines a stakeholder as "anyone with an interest or right in an issue, or anyone who can affect or be affected by an action or change. Stakeholders may be individuals, groups, internal or external to the organization." Notice that this definition may include customers and suppliers as well. To help identify stakeholders related to a project, she recommends asking the following set of eight questions:

- Who might receive benefits?
- Who might experience negative effects?
- Who might have to change behavior?
- Who has goals that align with these goals?
- Who has goals that conflict with these goals?
- Who has responsibility for action or decision?
- Who has resources or skills that are important to this issue?
- Who has expectations for this issue or action?

A form similar to the one depicted in Table 3.1 can be used to capture relevant information about the stakeholders. Let's review the form:

- *Stakeholder.* The list provided above is an excellent starting point for identifying stakeholders. There are two types of stakeholders: primary and secondary. *Primary stakeholders* are directly affected by the project. *Secondary stakeholders* are involved in implementing, funding, monitoring, and so on, and are considered intermediaries. Consider identifying each stakeholder as primary or secondary. An alternate

Table 3.1 A simple form for completing a stakeholder analysis.

Stakeholder	Project impact	Level of influence	Level of importance	Current attitude	Action plan

stakeholder analysis form is shown in Figure 3.1. Notice that this form allows for the categorization of stakeholders.

- *Project impact*. Clarify the impact of the project on the stakeholder. This establishes the relationship between the stakeholder and the project.
- *Level of influence*. Rate the level of influence the stakeholder has on the project. Consider a scale from 1 to 5, or low, medium, and high. Low and high works well, too.
- *Level of importance*. Rate the level of importance each stakeholder has in the project. Consider a scale from 1 to 5, or low, medium, and high. Low and high works well, too.
- *Current attitude*. Rate the attitude the stakeholder has toward the project. Consider a scale from 1 to 5, or low, medium, and high. Low and high works well, too.
- *Action plan*. Outline different strategies for dealing with each stakeholder. The strategies should focus on reducing opposition and increasing support.

Once Table 3.1 has been completed, the rating from the level of influence and importance columns can be transferred to Table 3.2. Table 3.2 provides the added benefit of classifying stakeholders:

- *High influence/high importance*. Collaborate with these stakeholders.
- *High influence/low importance*. Work with these stakeholders to involve them and increase their level of interest. These stakeholders are capable of sabotaging plans and escalating problems and are considered high-risk.
- *Low influence/high importance*. Protect and defend these stakeholders. Work to give them a voice and help increase their level of influence.
- *Low influence/low importance*. Monitor these stakeholders, but don't spend resources on them.

Whenever possible, involve stakeholders with high influence; otherwise, neutralize their influence.

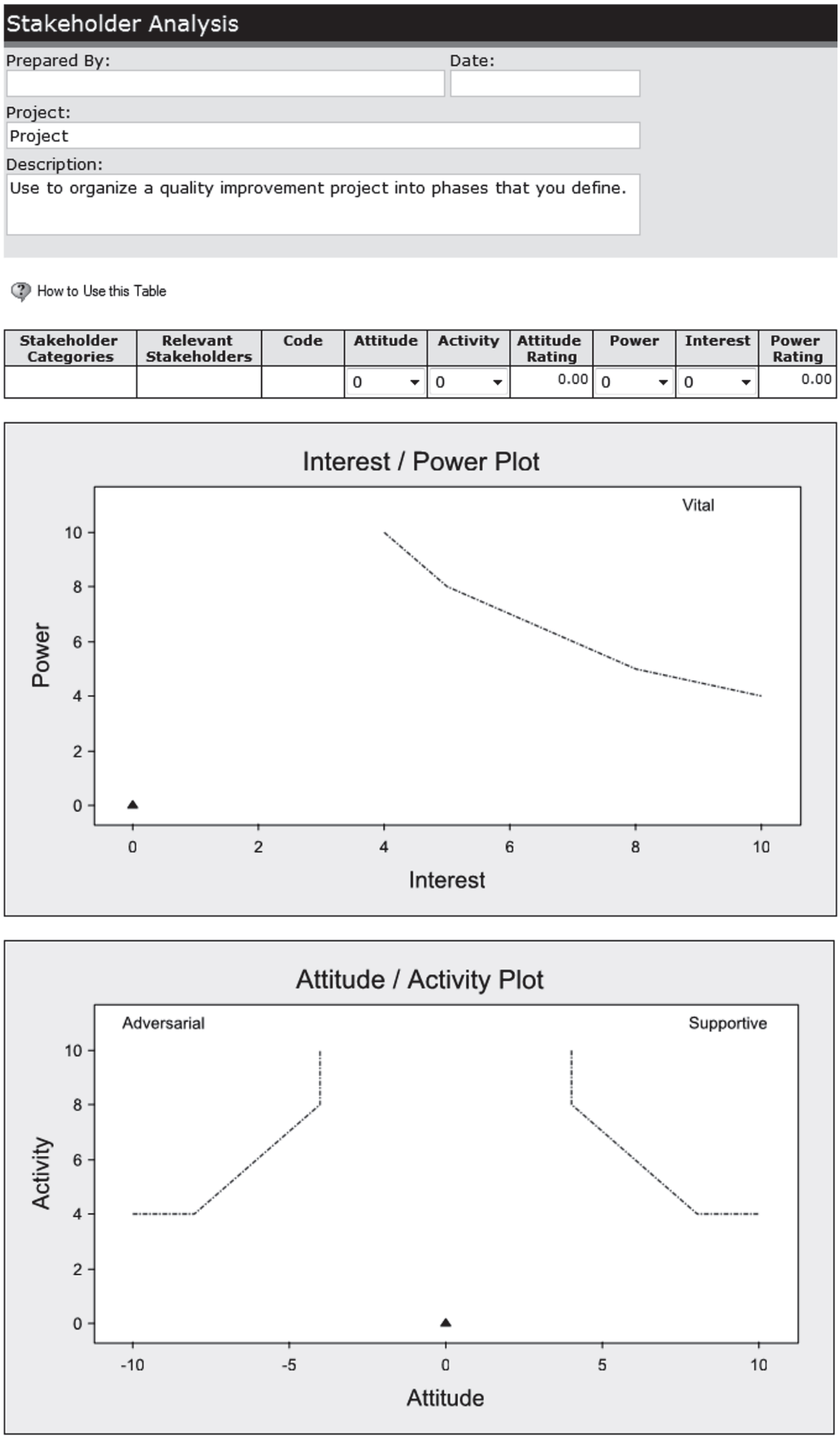

Figure 3.1 Alternate example of a stakeholder analysis.

Source: Courtesy of Minitab Quality Companion 3 software, Minitab Inc.

Table 3.2 Example of an influence-importance stakeholder table.

		Influence	
		High	**Low**
Importance	**High**	Collaborate	Protect and defend
	Low	High risk	Do not spend resources

Source: Adapted from Tague (2005).

Table 3.3 Example of an impact-cooperation stakeholder table.

		Potential to impact	
		High	**Low**
Potential for cooperation	**High**	Mixed blessing	Supportive
	Low	Nonsupportive	Marginal

Source: Adapted from Collins and Huge (1993).

Andersen (2007) suggests another means of classifying stakeholders. His method provides additional insight and is depicted in Table 3.3. In this table, stakeholders are classified as to their potential impact on the organization versus their potential for cooperation with the organization. Again, this requires categorizing each stakeholder as either low or high. Once complete, stakeholders can be charted and viewed as:

- *High impact/high cooperation*. These stakeholders are considered a "mixed blessing" and should be handled through cooperation.
- *High impact/low cooperation*. These individuals are considered "nonsupportive." Minimize dependency on this type of stakeholder through a defensive strategy.
- *Low impact/high cooperation*. These stakeholders are considered "supportive" and should be involved in relevant discussions and decisions.
- *Low impact/low cooperation*. These stakeholders are considered "marginal" and should be monitored. Like Tague's low influence/low importance, don't spend resources on them.

Black Belts as well as Master Black Belts may find that a significant portion of their time is dedicated to engaging project stakeholders. As Covey (1989) would say in his fifth habit, "Seek first to understand, then to be understood." The stakeholder analysis is a highly structured method dedicated to this purpose. However, if done properly, it can be a very sensitive document. If left open to public consumption within the organization, it can be damaging, hurtful, or perceived as offensive. Use this tool effectively, but use it with discretion.

Chapter 4

Benchmarking

Define and distinguish between various types of benchmarking, e.g., best practices, competitive, collaborative, breakthrough. Select measures and performance goals for projects resulting from benchmarking activities. (Apply)

Body of Knowledge II.B

Companies living in the "realm of the cool," "next big thing," or "social convention" environments are trying to translate a program with diminished authenticity to fit into their standardized box and have little chance of equaling the performance of the innovators.

Benchmarking the early successes of Six Sigma requires an understanding of the environment the companies lived in. It requires being able to honestly comprehend the factors that enabled the program to succeed. Cloning a MBB, copying job descriptions, and requiring a certain number of projects per year will not replicate results.

—Mike Carnell

Improving processes and products is often aided by comparing the current state with outstanding processes or products. In some cases, the comparison will be with other divisions within the same company. In other cases, external comparisons are more appropriate. The use of these comparisons is called *benchmarking*. The *benchmark* is the recognized or accepted measurement value that establishes the best performance for a given process.

Benchmarking often assists a Six Sigma team in setting targets and finding new ways to achieve them. It is an especially helpful technique when a quality improvement team has run out of new ideas. Consider the following benchmarking example.

EXAMPLE 4.1

A payroll department team charged with making improvements in the quality and efficiency of the department's work seeks out a company known for the quality of its payroll work. With that company's agreement, the team studies the payroll function, looking for ideas it can use in its home department.

The information for benchmarking may come from various sources, including publications, professional meetings, university research, customer feedback, site visits, and analysis of competitors' products. A downside of benchmarking within a particular industry is that it tends to put one in second place, at best.

Benchmarking helps teams strive toward excellence while reducing the tendency to feel that locally generated ideas are the only ones worth considering. Moreover, it is important because it provides an organization with the opportunity to see what level of process performance is possible. Seeing a gap in process performance between what is and what could be helps an organization determine its desired rate of improvement and provides the ability to set meaningful intermediate stretch goals or targets. Benchmarking is useful for driving breakthrough improvement over continuous improvement.

Types of Benchmarking

What are the types of benchmarking? There are many different answers to this question depending on the source. For instance, according to the APQC (2014-3), the following five types of benchmarking exist:

- *Functional benchmarking.* The act of comparing the process or performance of a business function (for example, finance, IT, or human resources) in two or more organizations.
- *Generic benchmarking.* The act of comparing the process or performance of two or more companies irrespective of their industries.
- *Internal benchmarking.* The act of conducting performance or process benchmarking within an organization by comparing similar business units or business processes.
- *Process benchmarking.* The act of measuring and comparing discrete process performance and functionality against other organizations.
- *Strategic benchmarking.* The systematic process of evaluating alternatives, implementing strategies, and improving performance by understanding and adapting successful strategies from external partners who participate in an ongoing strategic alliance.

Camp (1995), the recognized father of benchmarking, has defined the following four types:

- *Internal benchmarking.* The process of comparing your own processes to similar ones within your own organization
- *Competitive benchmarking.* The process of comparing your own processes to similar ones of competitors within your own industry
- *Noncompetitive benchmarking.* The process of comparing your own processes to similar ones of noncompetitors outside your industry
- *Generic process benchmarking.* Similar to noncompetitive benchmarking, but seeks to identify who does each process absolutely the best

Westcott (2014) identifies four types as follows:

- *Internal benchmarking.* This compares a process in one function with that in another function, or compares the same process across locations. Data are easy to collect; however, the focus is limited and possibly biased.
- *Competitive benchmarking.* This type compares direct competitors at the local, regional, national, or even global level. Thus, legal concerns may be a consideration, and ethical issues become very complicated at global levels.
- *Functional benchmarking.* This compares processes to other organizations with similar processes in the same function, but outside the industry.
- *Generic benchmarking.* Finding organizations that have best-in-class processes and approaches from which one may learn and translate to improvements at one's own organization.

If that's not complicated enough, the 2015 release of the Black Belt Body of Knowledge section shown in the shaded box at the beginning of this chapter identified the following, which we, as the authors, are obligated to address:

- *Best practice benchmarking.* Most authors who are authorities on the subject of benchmarking do not consider this a specific type of benchmarking probably because there is not universal agreement on what constitutes "best practices." APQC (2014-3) places the concept of "best practice" in perspective when it writes, "there is no single 'best practice' because best is not the same for everyone. Every organization is different in some way—missions, cultures, environments, and technologies. What is meant by 'best' are practices that have produced superior results. Best practices are then adapted to fit a particular organization."
- *Competitive benchmarking.* A recognized type of benchmarking, competitive benchmarking forces organizations to take an external perspective. However, focusing on industry practices may limit opportunities to achieve high levels of performance, particularly if a given industry is not known for its quality.

- *Collaborative benchmarking.* Although generally not considered a type of benchmarking since, by definition, all benchmarking is collaborative, this refers to the cooperation between various organizations to achieve benchmarking results. "Collaborative benchmarking" may permit access to specific benchmarking partners that may not exist with the other types of benchmarking.
- *Breakthrough benchmarking.* Although generally not considered a type of benchmarking, the concept of "breakthrough" is often associated with benchmarking. The simple reason for this is that benchmarking makes organizations aware of what's possible and, more specifically, achievable. Consequently, organizations that act through projects based on their effective use of benchmarking have been able to achieve significant (that is, breakthrough) levels of improvement. These organizations now understand that routinely setting low improvement goals of, say, 10% is trivial and likely to be considered managerially irresponsible in today's highly competitive economic environment.

Let's take a moment to review and analyze the benchmarking types that have been offered by the various sources and see what we can conclude:

- There are differences between the definitions even for similar terminology. For example, APQC's and Westcott's generic benchmarking focus more on having two excellent organizations partnering than determining what can be benchmarked, while Camp focuses on determining who is absolutely the best at a process and then seeking out that organization to benchmark.
- When partnering is involved, benchmarking must be collaborative.
- Ultimately, functional benchmarking must decompose into process benchmarking.
- Strategic benchmarking simply appears to encompass the basic concepts of strategic planning, competitive analysis, SWOT, and PEST when access to partners is permitted.

In Kubiak and Benbow (2009), we identified four benchmarking types. However, we have reviewed our previous categorization with regard to those above and have settled on the following at this time:

- *Internal benchmarking.* Compares similar processes within the same company. However, the search for exceptional performance is often limited by the company's culture, norms, and history.
- *Competitive benchmarking.* A recognized type of benchmarking, competitive benchmarking forces organizations to take an external perspective. However, focusing on industry practices may limit

opportunities to achieve high levels of performance, particularly if a given industry is not known for its quality.

- *Process benchmarking.* Compares similar processes typically outside the organization's industry, which provides ample opportunities to seek out benchmarking partners. Process benchmarking includes functional, generic, and noncompetitive benchmarking.

The above three benchmarking types should be done at the process level. Further, the concept of benchmarking is not an organizational luxury or fad, but an essential aspect of the strategic planning process. When integrated into this process, it can force organizations to wake up to reality and take dramatic steps to obtain breakthrough achievements or even simply survive.

Benchmarking Methodologies

There are probably as many benchmarking methodologies as there are organizations. Several are given in Table 4.1. Since insight into the commonality among the methodologies isn't obvious at the first step, the sub-steps have been identified for the APQC methodology.

A key takeaway from Table 4.1 is that benchmarking requires the use of a strict methodology. It is not a haphazard activity, but one that must be planned and funded. Organizations that approach it any other way will likely fail.

Dangers of Benchmarking

If you missed the opening quote in this chapter, now is a good time to reread it. We'll wait.

When it was first introduced in the early 1980s, organizations were enamored with benchmarking. It seems as though every organization had a benchmarking specialist, money was being spent, and, of course, trips to Japan were being planned and executed.

More importantly, successful processes in one organization were being studied and copied into others. These copied processes rarely succeeded. Why? Common reasons include:

- Not piloted
- No time
- No money
- No manpower
- Incompatible system

However, one of the most forgotten considerations when transplanting processes is the culture, norms, and history of the receiving organization. Consider the following example.

Table 4.1 Comparison of benchmarking methodologies.

Westcott (2014)	Camp (1993)	Munro (2008)	APQC (2014-TBD)
1. Review, refine, and define existing process to be benchmarked 2. Determine what to benchmark 3. Form a benchmarking team 4. Identify benchmark partners 5. Collect and analyze benchmarking information 6. Evaluate organization's performance versus benchmark partner 7. Determine how upgrading practice will impact the organizations involved 8. Establish new strategic targets 9. Implement improvements and a system to monitor progress 10. Do it all over again	1. Determine what to benchmark 2. Determine who to benchmark against 3. Exercise all data sources that will help you do this kind of work 4. Analyze the gap between what you do and what the sum of the practices would say you should look like 5. Revise your internal performance measurements and goals	1. Flowchart the current process 2. Identify the area to be improved 3. Brainstorm ideas 4. Investigate how others (internal and external) perform similar processes 5. Develop plans for application of ideas 6. Pilot test ideas 7. Initiate the new process 8. Evaluate the new process	1. Opportunity identification • Identify strategic business opportunity • Business case for change 2. Preliminary scope • Business process "current state" • Scope benchmarking study 3. Secondary research • Define research scope • Define information resources • Collect, sort, and report 4. Primary research • Define research scope • Select target companies • Conduct data capture activities 5. Normalize data • Sort and compile data for comparison • Identify best practices and enablers 6. Gap analysis • Compare current performance to best practices • Assess adaptability of practices and enablers 7. Implementation planning • Select opportunities • Conduct cost–benefit analysis • Final report

Continued

Table 4.1 *Continued.*

Westcott (2014)	Camp (1993)	Munro (2008)	APQC (2014-TBD)
			8. Executive support • Present findings and recommendations • Commit to change and provide resources 9. Implementation management • Implementation work plan with contingencies • Change management • Pilot site/full implementation 10. Process management • Measure/monitor • Recalibrate/recycle

EXAMPLE 4.2

Consider a five-step machining process that must be accomplished on five different machines. Machine shop A has the benchmark machine process. It is family-owned, staffed by all family members, located in the backwoods of Arkansas, and nonunion. Machine shop B, located in Brooklyn, New York, has just reopened after a two-week machinist union wildcat strike. Do the culture, norms, and history of machine shop B have an impact on the successful transplantation of the process used by machine shop A?

Another danger associated with benchmarking is that organizations become exposed to potential breaches of confidentiality. Such breaches may occur by conduct of their partners or their own employees. Trade secrets and process patents may be involved, and certain restrictions may need to be established and respected. Further, when competitive benchmarking is being considered, legal aspects may need to be taken into account to prevent any improprieties or the appearance of improprieties.

At the outset of benchmarking, organizations recognized the need to establish a code of conduct. One such code is provided in Figure 4.1. Notice that it is long and thorough. One portion of the code deals with not wasting a partner's time. This inclusion was because during the early days of benchmarking, many organizations simply window-shopped. Their intent was to take without giving. It is

Preamble—To guide benchmarking encounters and enhance the professionalism and effectiveness of benchmarking, many organizations have adopted this common Code of Conduct. All organizations are encouraged to abide by this Code of Conduct. Adherence to these principles will contribute to efficient, effective, and ethical benchmarking.

1. *Principle of Legality*.
   - If there is any potential question on the legality of an issue, don't do it.
   - Avoid discussions or actions that could lead to or imply an interest in restraint of trade, market and/or customer allocation schemes, price fixing, dealing arrangements, bid rigging, or bribery. Don't discuss costs with competitors if costs are an element of pricing.
   - Refrain from the acquisition of trade secrets by any means that could be interpreted as improper, including the breach or inducement of a breach of any duty to maintain secrecy. Do not disclose or use any trade secret that may have been obtained through improper means or that was disclosed by another in violation of a duty to maintain secrecy or limit its use. Do not, as a consultant or client, extend one benchmarking effort's findings to another company without first obtaining permission from the parties of the first effort.
2. *Principle of Exchange*.
   - Be willing to provide the same type and level of information that you request from your benchmarking partner to your benchmarking partner.
   - Communicate fully and early in the relationship to clarify expectations, avoid misunderstanding, and establish mutual interest in the benchmarking exchange. Be honest and complete.
3. *Principle of Confidentiality*.
   - Treat benchmarking interchanges as confidential to the individuals and companies involved. Information must not be communicated outside the partnering organizations without the prior consent of the benchmarking partner who shared the information.
   - A company's participation in a study is confidential and should not be communicated externally without its prior permission.
4. *Principle of Use*.
   - Use information obtained through benchmarking only for purposes of formulating improvement of operations or processes within the companies participating in the benchmarking effort.
   - The use or communication of a benchmarking partner's name with the data obtained or practices observed requires the prior permission of that partner.
   - Do not use benchmarking as a means to market or sell.
5. *Principle of First-Party Contact*.
   - Initiate benchmarking contacts, whenever possible, through a benchmarking contact designated by the partner company.
   - Respect the corporate culture of partner companies and work within mutually agreed-on procedures.
   - Obtain mutual agreement with the designated benchmarking contact on any hand-off of communication or responsibility to other parties.
6. *Principle of Third-Party Contact*.
   - Obtain an individual's permission before providing his or her name in response to a contact request.
   - Avoid communicating a contact's name in an open forum without the contact's permission.
7. *Principle of Preparation*.
   - Demonstrate commitment to the efficiency and effectiveness of benchmarking by completing preparatory work prior to making an initial benchmarking contact, and follow a benchmarking process.
   - Make the most of your benchmarking partner's time by being fully prepared for each exchange.
   - Help your benchmarking partners prepare by providing them with an interview guide or questionnaire and agenda prior to benchmarking visits.

Figure 4.1 Benchmarking code of conduct.
Source: Westcott et al. (2014).

only when partners cooperate and collaborate do they achieve the full benefits of benchmarking.

Selecting Measures and Performance Goals

While the actual measures for a project are selected based on the process being improved, benchmarking can be a valuable tool in helping an organization recognize and set the desired levels of improvement to be achieved.

Historically, many organizations have set mundane process improvement goals. Usually, they accomplish this by reviewing the annual performance data and selecting one of the following methods:

- Guessing
- Guessing (without looking at the annual performance data)
- Always using a 10% improvement
- Using the previous annual goal
- Using what the boss provides
- Using what corporate provides
- Using some other equally sophisticated method

Benchmarking, however, allows organizations to set breakthrough improvement goals or otherwise meaningful and defensible targets and goals. Figure 4.2 illustrates the relationship that exists between goals, targets, and benchmarks.

Organizations aim for benchmarks as the ultimate measure of success. However, many organizations do not realize that benchmarks are dynamic. More specifically, those organizations setting the benchmarks are usually in engaged in a

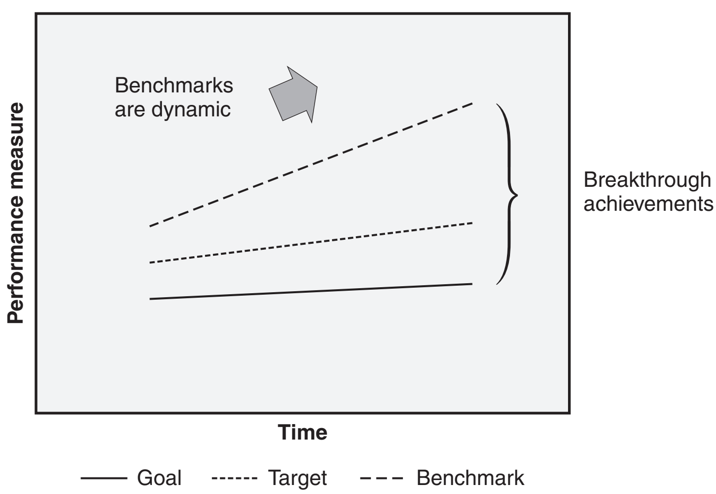

Figure 4.2 Selecting performance levels for projects resulting from benchmarking.

continual process of improving them. Thus, any given organization shooting for a benchmark needs to aim higher, or they will forever be chasing the benchmark.

A *target*, in the context of benchmarking, is some intermediary goal that lies between the goal and the benchmark. Occasionally, a target is known as a *stretch goal*. A target is useful when the prospect of achieving the benchmark is infeasible, perhaps due to cost, manpower, or other limitations.

EXAMPLE 4.3

Company A's widget product line is operating at 65% yield, though it has taken years to achieve that level due to attempts to increase the yield through an annual project goal of 10% per year. The company has recently implemented a benchmarking initiative and discovered from a noncompetitive partner (who is in the same industry) that a yield of 98% is quite feasible. Such a benchmark represents an increase of almost 50%. The director of production is reluctant to accept the benchmark as the project goal. However, he agrees to the project goal of a 30% increase (goal yield of 85%), a target increase of 40% (target yield of 91%), and the benchmark of 50% increase (benchmark yield of 98%).

Chapter 5

Business Measures

Many of the Lean Six Sigma tools deal with analyzing numerical values. The metrics the organization chooses reflects its business philosophy and, largely, determines its success. Unfortunately, many organizations use too many metrics, often lagging in nature, which may drown them in data and cause them to be unable to change with customer needs. This section lists some tools that will help in the selection and use of business performance measures.

PERFORMANCE MEASURES

> Define and describe balanced scorecard, key performance indicators (KPIs), customer loyalty metrics, and leading and lagging indicators. Explain how to create a line of sight from performance measures to organizational strategies. (Analyze)
>
> **Body of Knowledge II.C.1**

Balanced Scorecard

The *balanced scorecard*, a term coined by Kaplan and Norton (1996) in the early 1990s, separates organizational metrics into four perspectives:

- *Financial perspective.* This provides the organization with insight into how well its goals, objectives, and strategies are contributing to shareholder value and the bottom line. It provides shareholders with a direct line of sight into the health and well-being of the organization. Common metrics used for this perspective are given in the next section.
- *Customer perspective.* This perspective defines an organization's value proposition and measures how effective the organization is in creating value for its customers through its goals, objectives, strategies, and processes. The customer perspective is not easily obtained. Multiple approaches and listening posts are frequently required, and must be compared and reconciled to ensure a correct understanding of

the customer's stated and unstated needs and expectations of the organization's products and services, and the strength of its customer relationships.

- *Internal business processes perspective*. Internal business processes include all organizational processes designed to create and deliver the customer's value proposition. Organizations that perform these processes well generally have high levels of customer satisfaction and strong customer relationships since efficient and effective internal processes can be considered predictors of the metrics included in the customer perspective.
- *Learning and growth perspective*. Generally, this perspective includes the capabilities and skills of an organization and how it is focused and channeled to support the internal processes used to create customer value. The development and effective use of human resources and information systems, as well as the organizational culture, are key aspects of this perspective.

Key Performance Indicators

Key performance indicators (KPIs) are both financial and nonfinancial metrics that reflect an organization's *key business drivers* (KBDs), also known as *critical success factors* (CSFs). Organizations will benefit from the use of KPIs that are defined with the following criteria in mind:

- Quantitative and measurable
- Goal based
- Strategy based
- Time bounded

Another way of defining KPIs is in terms of the SMART acronym:

- *Specific*. KPIs should be laser focused and process based.
- *Measurable*. KPIs must be quantitative and easily determined.
- *Achievable yet aggressive*. KPIs may be set with regard to benchmark levels, yet remain obtainable. Sometimes this part of the acronym is defined as "actionable." All KPIs should be actionable, yet they must remain within reach. KPIs set at levels too high for a given set of resources can adversely impact employee morale and, subsequently, organizational performance.
- *Relevant*. KPIs should be linked to the organization's strategies, goals, and objectives. If they are not, disregard them and select others.
- *Time bounded*. KPI levels should reflect a specific period; they should never be open-ended and never be longer than the duration of the

strategy. Time-bounded KPIs can be measured. When an appropriate time frame is chosen, KPIs can create a sense of urgency and employee focus.

Some organizations have developed and embedded their KPIs within the framework of the balanced scorecard. Common examples include the following:

- Financial
  - Return on investment
  - Return on capital
  - Return on equity
  - Economic value added
- Customer
  - Customer satisfaction levels
  - Retention rates
  - Referral rates
  - Quality
  - On-time delivery rates
- Internal business processes
  - Defect rates
  - Cycle time
  - Throughput rates
  - Quality
  - On-time delivery rates
- Learning and growth
  - Employee satisfaction
  - Employee turnover rate
  - Absentee rate
  - Percentage of internal promotions

These examples should be considered a starter set and subjected to the criteria identified earlier to ensure their appropriateness. Also, note that the metrics in these perspectives are not necessarily mutually exclusive and may very well overlap. For example, quality and on-time delivery rates appear in both the customer and internal business process perspectives. Customers judge an organization by these metrics and make determinations regarding the continuance of a business relationship with a supplier. On the other hand, these same metrics may be used by a supplier as a predictor of the customer satisfaction levels that might be achieved because of its internal business processes.

Organizational metrics do not end with KPIs. Lower-level metrics down to the individual process level may be required to support KPIs and to provide deep, meaningful, and actionable insight in order for organizations to effectively conduct and run their businesses.

Financial Impact of Customer Loyalty

Real profits are generated by loyal customers—not just satisfied customers.

—Rafael Aguayo

Customer loyalty is a term used to describe the behavior of customers—in particular, customers who exhibit a high level of satisfaction, conduct repeat business, or provide referrals and testimonials. It results from an organization's processes, practices, and efforts designed to deliver its services or products in ways that drive such behavior.

Many organizations recognize that it is far easier and less costly to retain customers than to attract new ones. Hence, customer loyalty drives customer retention and, by extension, has a financial impact. However, customer loyalty can be a double-edged sword. Some organizations strive blindly to retain all their customers, including those that are unprofitable. Chapter 10 discusses the concept of customer segmentation. In one context of segmentation, customers may be classified as profitable or unprofitable. Conceptually, the idea of unprofitable customers is difficult to understand. Moreover, severing ties with this customer segment may be even more difficult, as it may be seen as highly unusual or simply "not done." Enlightened organizations understand this distinction well and act accordingly. Such organizations have gained insight into the cost of maintaining customer relationships through extensive data collection and analysis. This insight has given rise to such terms as "relationship costs" and "relationship revenue."

Organizations must be careful not to confuse loyal customers with tolerant customers. Customers have a zone of tolerance whereby a single bad experience or even a series of bad experiences spread over a sufficiently long time frame may strain the customer–supplier relationship yet keep it intact. However, the cumulative effects of such experiences may push a customer beyond his or her zone of tolerance, resulting in customer defection or a reduction in business. At this point, a customer may deem that it is no longer worth maintaining the business relationship. Consider your own banking relationship for the moment. How many times would it take for your bank to provide bad service before you are willing to forsake convenience and take your business elsewhere?

Organizations, especially those in a niche market or those that may be a single/sole source provider, should have a deep understanding of what drives their customer satisfaction levels and relationships. Limited competition or the cost of changing suppliers or service providers may be all that is keeping customers loyal. Given the right opportunity, these "loyal" customers might readily defect.

Common Customer Loyalty Metrics

There are numerous so-called customer loyalty metrics used by organizations. However, when scrutinized, we find that they are not always a direct measure of customer behavior. Consequently, we offer the following basic customer loyalty metrics as a starting point:

- *Customer referrals.* Customers are generally reluctant to refer individuals or organizations unless their confidence is high that the individuals or organization will provide a superior product or service in the same manner that they themselves received. "Customer referrals" is an excellent measure of customer loyalty, particularly when it can be expressed as a rate.
- *Customer abandonment rate.* This is essentially the complement of the customer retention rate. Some organizations prefer the abandonment rate because it focuses effort on customer issues and tends to have a stronger internal impact, whereas the use of the retention rate can sometimes lull organizations into a false sense of security. It is sometimes called *customer churn* or the *customer churn rate.*
- *Customer retention rate.* This metric has applicability when customers have an active role in staying with an organization.

  For example, membership dues with the American Society for Quality must be actively paid each year to remain a member. There is no automatic renewal process. Members elect to stay as members because they see value in retaining themselves as members. ASQ has retained them as members due to a concerted effort as part of a strategy that is in place to generate membership value. Because the strategy works, it influences the member (customer) retention rate.

  However, caution is required when using the retention rate as a customer loyalty metric, particularly when:

  – No real alternative to leaving the organization exists; such is the case of monopolies (for example, utilities) or sole/single source suppliers.

  – The cost or difficulty of leaving the organization is very high; consider the sole/single source supplier, or just the matter of moving your financials from one bank to another. This may sound simple, but suppose you have several integrated checking and savings accounts that you transfer funds between and use for all your bill paying. The difficulty involved in switching banks could be very high and quite time-consuming.

Organizations using the retention rate metric with customers facing these conditions have a retention rate that is probably overstated, and should be concerned about possible future mass customer defections when the conditions change.

- *Customer renewal rate.* The renewal rate is a useful metric when customers must actively engage with the organization to demonstrate their loyalty. If we continue with the ASQ membership example from above, members must actively renew their membership each year. ASQ makes it easy for members to do so by offering a variety of ways (for example, online, mail, call-in).
- *Repeat customers.* When we talk about repeat customers, we are talking about the time between purchases of the product or service relative to the durability of the product or need for the service. Examples on the product side include the purchase of an automobile or refrigerator. Generally, such purchases happen infrequently and there is a lot of competition for the customer's attention. To be a repeat customer with long spans of time between purchases makes a very positive statement about the customer experience. This metric is particularly useful if it can be expressed as a rate.
- *Share of wallet.* This refers to customers extending their loyalty to the purchase of other products and services within the same organization based on their experience with a given product or service. However, when the customer cannot sufficiently distinguish supplying organizations (that is, suppliers of products and services) apart, this extension of loyalty is sometimes conducted as a matter of convenience on the part of the customer rather than being earned on the part of the receiving organization.

Lagging and Leading Indicators

There are two types of indicators:

- *Lagging.* This indicator follows an event. It is reported after the fact.
- *Leading.* This indicator predicts or infers a future event.

Depending on how it is viewed, an indicator can be both lagging and leading. Probably the best way to understand the concept of lagging and leading indicators is through examples.

EXAMPLE 5.1

In January, the sales department of an organization visited 15 customers, conducted eight public demonstrations, released 10 proposals, and booked six orders that are all expected to hit in the fourth quarter of the year. Consequently, customer visits, demonstrations, proposals, and orders are considered leading indicators for sales. Sales, when reported after the fact, is considered a lagging indicator.

EXAMPLE 5.2

Consider a stoplight. The yellow light is both a leading indicator of the red light and lagging indicator of the green light.

If we apply the definitions from above, we will see that several of the financial measures given in the next section, such as market share, revenue growth, and margin, are all lagging indicators. In general, many of the indicators reported on balance sheets are lagging indicators since they are reported after the fact.

Creating Line of Sight from Measures to Strategies

The details of this subsection were addressed in Chapter 1. However, it is important to keep in mind that an organization must have an effective strategic planning methodology that aligns the organization both horizontally and vertically while decomposing the plan downward from strategic to tactical to operational plans and into projects.

At each level, measures or metrics are developed to ensure accountability, and responsibility is maintained. Concurrently, the strategic, tactical, operational, and project plans are integrated with other plans as necessary, such as the capital plan and strategic human resource plan.

When the above occurs, line of sight must be maintained. See Chapter 1 for specific details.

FINANCIAL MEASURES

Define and use revenue growth, market share, margin, net present value (NPV), return on investment (ROI), and cost-benefit analysis (CBA). Explain the difference between hard cost measures (from profit and loss statements) and soft cost benefits of cost avoidance and reduction. (Apply)

Body of Knowledge II.C.2

Organizations have come under severe financial scrutiny in recent years due to the Sarbanes–Oxley Act. Consequently, it is important that Black Belts have a solid understanding of the various financial measures that may be used to cost-justify projects, and higher-level business performance metrics that may weigh upon project selection.

When preparing a project proposal or report, it is generally expected to state the project's benefits in financial terms. Thus, it is often necessary to conduct a detailed financial analysis. This section will address some of the key tools needed to support such an analysis and will discuss some of the advantages and disadvantages of each. Remember, as with any analysis, it is critical to enumerate all assumptions on which it is based since it may be necessary to revisit them from time to time. Also, whenever possible, work with the organization's financial department. This helps to instill confidence and credibility into the numbers.

Furthermore, some benefits cannot be stated financially. For example, improved morale and improved team-building skills. However, these benefits are important as they may be used to break ties between projects with equal financial benefits. Therefore, they should be identified in the report, but recognize that the financial aspects usually dominate the benefit section of the report.

Common Financial Measures

Revenue growth is the projected increase in income that will result from the project. Revenue growth may be stated in dollars per year or as a percentage per year.

An organization's *market share* of a particular product or service is that percentage of the dollar value that is sold relative to the total dollar value sold by all organizations in a given market. A project's goal may be to increase this percentage. During a market slowdown, market share is often watched closely because an increase in market share, even with a drop in sales, can produce an increase in sales when the slowdown ends.

Conceptually, *margin* refers to the difference between income and cost. However, there are multiple financial concepts of margin, each of which is used to measure the efficiency and effectiveness of an organization's financial capabilities and finesse:

- *Percent profit margin*. This metric provides an overall sense of how efficiently an organization converts sales into profits. It is computed as follows:

  Percent profit margin

  $$= \left(\frac{\text{Sales of a product or service} - \text{Cost of a product or service}}{\text{Sales of a product or service}} \right)(100)$$

  $$= \left(\frac{\text{Net profit}}{\text{Net sales}} \right)(100)$$

- *Percent gross margin*. This metric measures the efficiency of producing sales. It is computed as follows:

  $$\text{Percent gross margin} = \left(\frac{\text{Gross profit}}{\text{Net sales}} \right)(100)$$

- *Percent operating margin*. This metric measures how effectively an organization controls costs not related to the production and sales of a product or service. It is computed as:

  $$\text{Percent operating margin} = \left(\frac{\text{Operating profit}}{\text{Net sales}}\right)(100)$$

- *Percent net margin*. This metric measures the effectiveness of an organization at generating net profit from sales. It is computed as:

  $$\text{Percent net margin} = \left(\frac{\text{Net profit}}{\text{Net sales}}\right)(100)$$

  Notice that it is computed the same as the percent profit margin. Generally, when we speak of percent profit margin, we are speaking of the percent net margin.

To better understand some of the above equations, we must define some of the component terms such as:

- Net sales = Revenue (from sales)
- Gross profit = Net sales – Cost of goods sold
- Cost of goods sold = Cost of raw goods + Cost of labor + Cost of inventory
- Net profit = Net income
  = Net earnings
  = Bottom line
  = Total revenue – Total expenses
- Operating expenses = Cost of goods sold + General and administrative expenses + Depreciation + Other expenses
- Operating profit = Operating income = Earnings
  = Revenue (from sales) – Operating expenses

Net Present Value

Discounted cash flow techniques were developed to take into account the time value of money and to improve the accuracy of cash/capital project evaluations. One such method, *net present value* (NPV), consists of:

- Finding the present value (PV) of each cash flow, discounted at an appropriate percentage rate (determined by the cost of capital to the organization and the projected risk of the flow—the higher the risk, the higher the discount rate required).
- Computing the project's NPV by adding the discounted net cash flows.

- Determining whether the project is a candidate for acceptance, when the NPV is positive. If two projects are positive, the one with a higher NPV is a more likely choice.

The PV of an amount to be received in the future is given by the formula

$$PV = F\,(1 + i)^{-n}$$

where

PV = Present value

F = Amount to be received n periods from now

i = Annual interest rate expressed as a decimal

EXAMPLE 5.3

In four years, $2500 will be available. What is the PV of that money, assuming an annual interest rate of 8%?

$$PV = 2500(1 + 0.08)^{-4}$$

$$PV = 1837.57$$

Therefore, $1837.57 invested at 8% compounded annually will be worth $2500 after four years.

EXAMPLE 5.4

Suppose a project requiring an initial investment of $4000 pays $1000 every six months for two years beginning one year from the date of the investment. What is the NPV?

$$\text{Present value of the first payment} = PV_1 = \$1000(1 + 0.08)^{-1} = \$925.93$$

$$\text{Present value of the second payment} = PV_2 = \$1000(1 + 0.08)^{-1.5} = \$890.97$$

$$\text{Present value of the third payment} = PV_3 = \$1000(1 + 0.08)^{-2} = \$857.34$$

$$\text{Present value of the fourth payment} = PV_4 = \$1000(1 + 0.08)^{-2.5} = \$824.97$$

$$\text{Present value of the fifth payment} = PV_5 = \$1000(1 + 0.08)^{-3} = \$793.83$$

$$\text{Total present value of the income} = \$4293.04$$

Since the initial investment is $4000, the net present value is $293.04.

Return on Investment

One of the most easily understood financial numbers is the return on investment (ROI):

$$\text{ROI} = \frac{\text{Income}}{\text{Cost}}(100\%)$$

where *income* includes money that will be made due to the project dollars saved, and costs avoided, and *cost* is the amount of money required to implement the project.

In basic terms, the ROI metric measures the effectiveness of an organization's ability to use its resources to generate income.

Return on investment is used both as an estimating tool to determine the potential return on an investment contemplated as well as a measure of actual payoff upon project completion. ROI tends to be more useful for evaluating the payoff potential of shorter-term projects inasmuch as it typically ignores the time value of money.

Cost–Benefit Analysis (CBA)

Top management must approve funding for projects that require significant financial resources. Therefore, before a proposed project is initiated, a cost–benefit analysis using projected revenues, costs, and net cash flows is performed to determine the project's financial feasibility. This analysis is based on estimates of the project's benefits and costs and the timing thereof. The results of the analysis are a key factor in deciding which projects will be funded. Upon project completion, accounting data are analyzed to verify the project's financial impact on the organization. In other words, management must confirm whether the project added to or detracted from the company's financial well-being.

In today's competitive global environment, resource allocation is complicated by the fact that business needs and opportunities are greater and that improvements are often more difficult to achieve. Many of the easier projects have already been done. Therefore, top management requires that project benefits and costs be evaluated so that projects can be correlated to overall revenues, costs, customer satisfaction, market share, and other criteria. These factors are analyzed to maximize business returns and to limit risks, costs, and exposures. Therefore, quality improvement projects are considered business investments—as are all cash or capital investments—in which benefits must exceed costs. Quality managers are additionally challenged in that many of these benefits and costs are not easily quantifiable, as is normally expected by executives, accountants, and financial officers.

A *cost–benefit* (also known as a *benefit–cost*) *analysis* attempts to evaluate the benefits and costs associated with a project or initiative. Two types of benefits can be evaluated: direct or indirect. Direct benefits are easily measured and include items such as increased production, higher quality, increased sales, reduced delivery costs, higher reliability, decreased deficiencies, and lower warranty costs.

Indirect benefits are more difficult to measure and include such items as improved quality of work life, increased internal customer satisfaction, and better-trained employees. Similarly, two types of costs can be evaluated. Direct costs include equipment, labor, training, machinery, and salaries. Indirect costs include downtime, opportunity costs, pollution, and displaced workers.

The simple formula for computing a benefit–cost (occasionally referred to as cost–benefit, which is the reciprocal) ratio is:

$$\frac{\sum \text{NPV of all benefits anticipated}}{\sum \text{NPV of all costs anticipated}}$$

If the formula above yields a value greater than 1, the project is generally worth considering, but it may not be as attractive as other projects. However, in practice, organizations may set thresholds such that the ratio is above a lower limit.

Typically, a cost–benefit analysis is conducted on the basis of direct costs and direct benefits to:

- Assist in the performance of a *what-if analysis*
- Compare the benefits, costs, and risks of competing projects
- Allocate resources among multiple projects
- Justify expenditures for equipment, products, or people

Because many indirect benefits and costs are difficult to quantify, and time is needed to identify such costs, some worthwhile quality projects or initiatives could fail to obtain the necessary funding.

Quality professionals are expected to help their organizations profit from quality investments, so no project should receive automatic approval and resource allocation. To be competitive, companies must find ways to increase profitability while providing customers with high levels of satisfaction. The challenge for improvement projects, programs, and initiatives is to find the link between critical business performance measurements, such as profitability, and quality investments. Although some quality benefits and costs are difficult to measure or are unknowable, the quality manager still must help determine and measure quality's effect on the organization's finances.

EXAMPLE 5.5

The NPV costs to conduct a project are estimated to be \$90,000. The NPV benefits or savings due to the project are estimated at \$855,000. Compute the benefit-to-cost ratio.

Using the above formula, we have:

$$\frac{\sum \text{NPV of all benefits anticipated}}{\sum \text{NPV of all costs anticipated}} = \frac{\$855{,}000}{\$90{,}000} = 9.5$$

This means \$9.50 in benefits for every dollar expended.

Cost of Quality

Note: This subject area has been removed from the 2015 Body of Knowledge. However, we have elected to retain this content in the handbook.

The standard categories of quality costs are as follows:

- *Appraisal costs.* Expenses involved in the inspection process
- *Prevention costs.* Costs of all activities whose purpose is to prevent failures
- *Internal failure costs.* Costs incurred when a failure occurs in-house
- *External failure costs.* Costs incurred when a failure occurs while the customer owns the product

The total cost of quality is the sum of these four amounts. Cost-of-quality calculations have been successfully used to justify proposed improvements. However, it is essential that representatives from the accounting function be involved to ensure that appropriate conventions are used in such proposals.

It is the bias of most quality professionals that if businesses spent more on prevention, they could reduce the total cost of quality because failure costs and, in many cases, appraisal costs could be reduced.

The traditional way to sketch the relationship between quality costs and the quality level is illustrated in Figure 5.1. It indicates that as the quality level

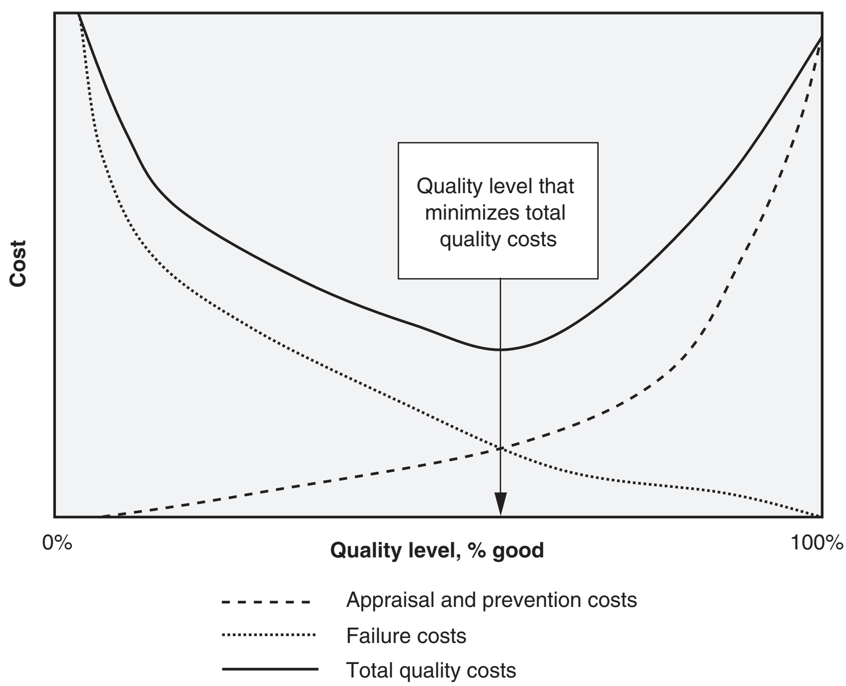

Figure 5.1 Traditional quality cost curves.

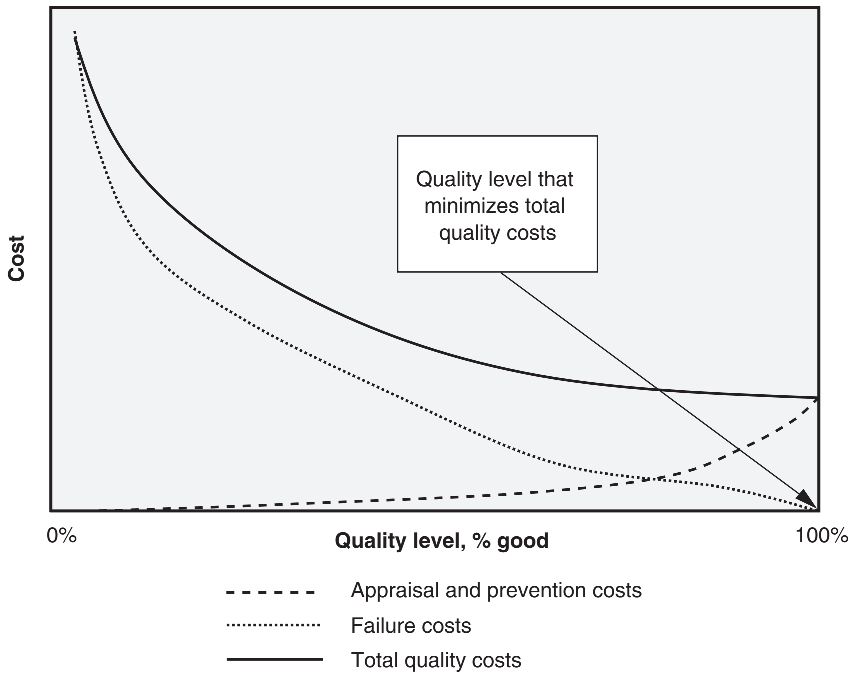

Figure 5.2 Modern quality cost curves.

approaches 100%, appraisal and prevention costs approach infinity. The optimum total quality cost level is where the total quality cost curve is minimized. This optimum point is located directly above the intersection of the appraisal/prevention and failure curves.

Figure 5.2 demonstrates a different path of thought with regard to quality costs. It illustrates that if it were possible to achieve perfect quality (that is, 100% quality level), the total quality cost curve would be minimized, with finite appraisal/prevention and failure costs.

Defining Hard versus Soft Dollars

The discussion of what constitutes hard and soft dollar savings is a long-standing debate. It would seem to be a simple issue, but so many organizations have defined them differently over the years that confusion abounds. Ashenbaum (2006) noted that, "Generally speaking, cost savings are understood as tangible bottom-line reductions resulting in saved money that could be removed from budgets or reinvested back into the business. There also needs to be a prior baseline or standard cost for the purchased product or service so that these savings could be measured against a prior time period's spend." (Note: In the context in which Ashenbaum offered this definition, cost savings are synonymous with hard dollar savings.) Based on this definition, let's extract the key characteristics of *cost (hard) dollar savings*:

- A prior baseline of spending must be established, typically 12 months.
- The dollars must have been planned and in the budget. This goes along with the baseline requirement above.
- The cost savings must affect the bottom line (that is, the profit and loss statement or balance sheet).

Unexpectedly, this definition also settles the question regarding whether spent or reinvested cost savings were ever savings at all. Some organizations tend to empty the savings pot as soon as it's filled. Then they wonder or even complain about the lack of savings from the Lean Six Sigma strategic initiative. Once a project books its hard savings, these savings exist regardless of what the organization chooses to do with them thereafter.

So, what are soft dollar savings? *Soft dollar savings* are *cost avoidance savings* and are, by exclusion, everything that is not hard dollar savings. This concept is illustrated in Figure 5.3. This figure makes the simple point that if it isn't hard dollar savings, then it is soft dollar savings. However, Ashenbaum offers some additional clarity on the subject:

- *Avoidance is a cost reduction that does not lower the cost of products/services when compared against historical results, but rather minimizes or avoids entirely the negative impact to the bottom line that a price increase would have caused.*
- *When there is an increase in output-capacity without increasing resource expenditure, in general, the cost avoidance savings are the amount that would have been spent to handle the increased volume/output.*
- *Avoidances include process improvements that do not immediately reduce costs or assets but provide benefits through improved process efficiency, employee productivity, customer satisfaction, improved competitiveness, etc. Over time, cost avoidance often becomes cost savings.*

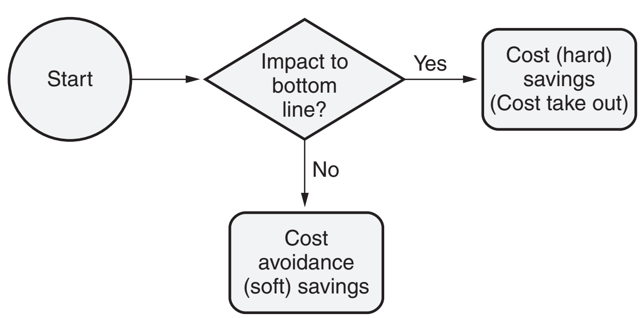

Figure 5.3 Determining hard versus soft cost savings.

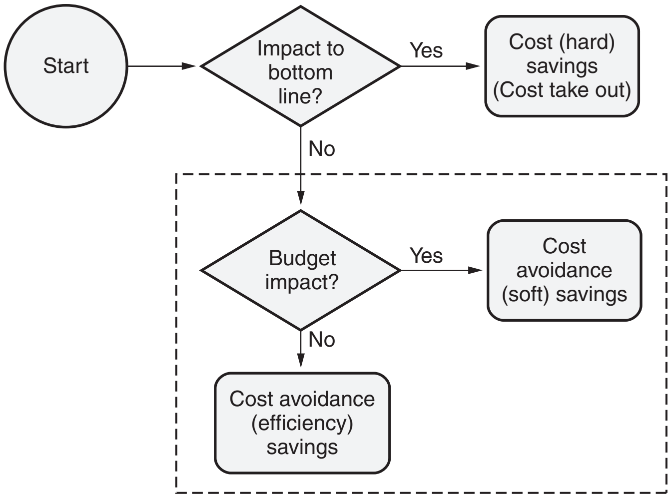

Figure 5.4 Expanding the definition of cost avoidance to include efficiency savings.

If we consider Ashenbaum's clarifications of soft dollars, we can expand Figure 5.3, resulting in Figure 5.4. Notice that "cost avoidance" now depends on whether it has a budget impact or not. Let's consider the following examples:

- *Cost avoidance (budget impacting).* This type of cost avoidance eliminates or reduces items in the budget marked for future spending. For example, three engineers are budgeted for hire in the fourth quarter. The cost of the three engineers is not in the baseline, nor has any spending for these engineers occurred. Eliminating these planned expenses has an impact on the future budget and is thus cost avoidance.
- *Cost avoidance (non–budget impacting).* This type of cost avoidance results from productivity or efficiencies gained in a process (that is, reduction of non-value-added activities) without a head count reduction. For example, the process cycle time that involved two workers was reduced by 10 percent. Assuming the 10 percent amounts to two hours per worker per week, the two workers save four hours per week. These four hours are allocated to other tasks.

Another way to impact the bottom line is through revenue or top-line growth. How might a Lean Six Sigma project affect such growth? Consider a project that identifies and removes a production bottleneck or constraint. As a direct consequence of removing that constraint, the organization is able to accept new customer orders (something it wasn't able to do previously) or work off backlogged orders. In this situation, revenue growth is achieved, and the reason is the constraint removal as an outcome of the project. It would seem that "revenue growth" would be a desirable category to which to attribute the dollar impact of Lean Six Sigma project dollars. This is illustrated in Figure 5.5.

Some organizations further refine cost savings into the third category of working capital/cash flow. Projects falling into this category might improve internal

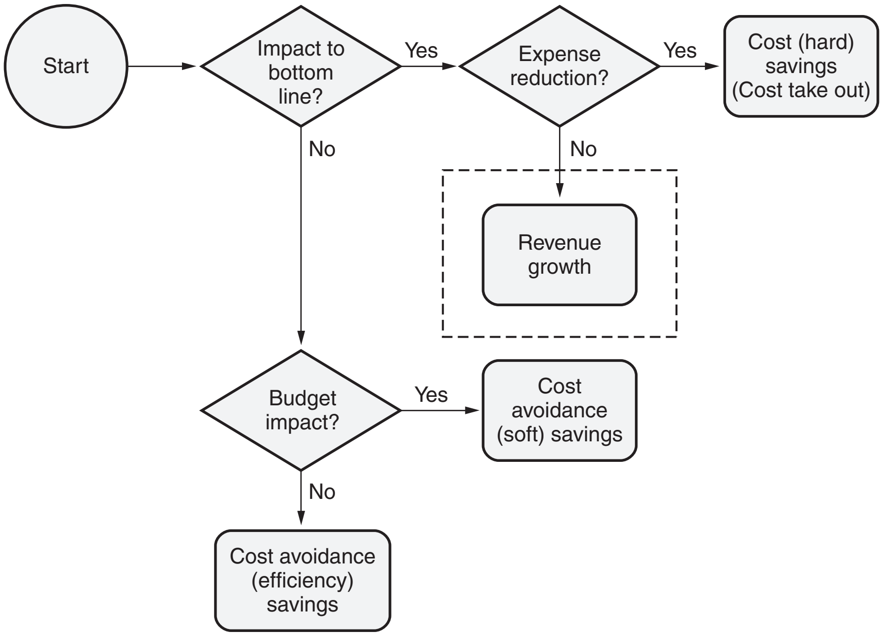

Figure 5.5 Expanding the definition of cost savings to include revenue growth.

invoicing processes, resulting in lower accounts receivable, or reduce inventory requirements through increased process efficiency and reduced claims. Working capital/cash flow is reflected in Figure 5.6.

In summary, two major categories of savings have been identified, each of which can be further refined:

- Hard savings (cost savings)
  - Cost take out
  - Revenue growth
  - Working capital/cash flow
- Soft savings (cost avoidance)
  - Cost avoidance (budget impacting)
  - Cost avoidance (non–budget impacting)

At one time or another, several of the pioneers in Lean Six Sigma, such as Allied-Signal (later Honeywell International) and GE, favored categories of savings similar to those above.

False Savings

One of the most troubling and abused areas of savings is cost avoidance (non–budget impacting). This area of savings is typically generated from productivity or efficiency gains. Many organizations will save a small slice of time in

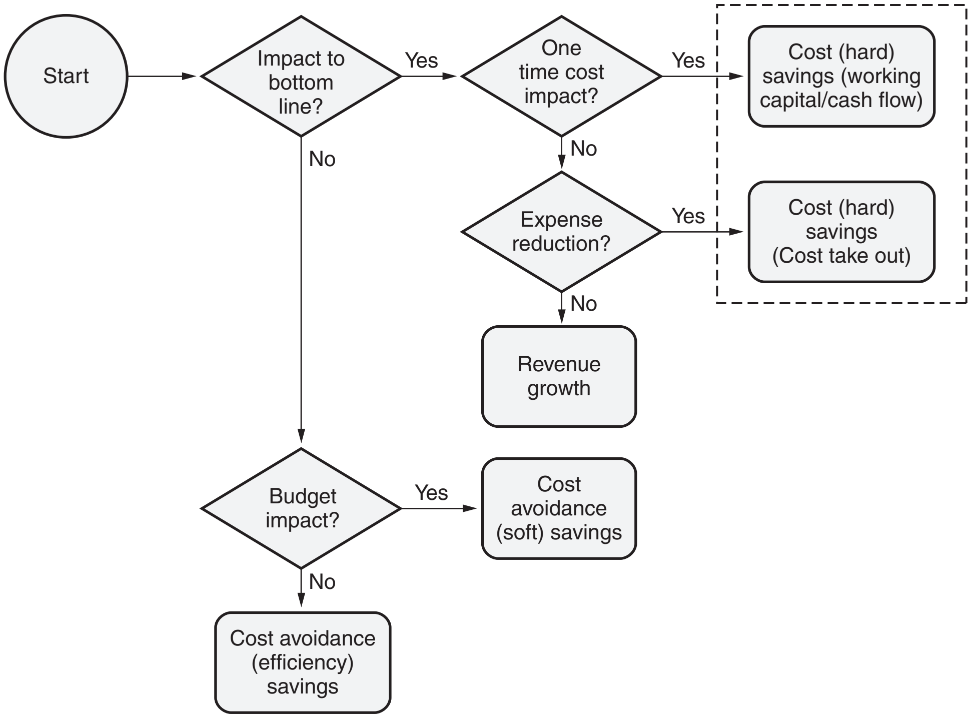

Figure 5.6 Expanding the definition of cost savings to include working capital.

an employee workday, multiply that by the number of workers performing the same job and extend that over a year. Voilà! Enormous savings are generated and booked. This approach is common in many transaction-based organizations such as large banks and call centers where a few seconds may be saved on a transaction or calling up a computer screen with customer data. These transactions may be performed hundreds of thousands of times per day across multiple individuals.

On the surface, this approach seems fine and is generally accepted. However, the fallacy lies with the ability to aggregate these miniscule time slices into a sufficiently meaningful size that will accommodate other work. If this can't be done, the "time saved" will simply be absorbed into adjacent work activities. Then they are not even paper savings. Furthermore, the ability to measure such time savings might be questionable. Consider trying to measure five or 10 seconds saved from a two- to three-minute transaction. In many cases, we are dealing at the noise level. Yet, these types of savings are reported all the time. Although they may be categorized as cost avoidance due to productivity or efficiency gains, are they really that if the gains cannot actually be used or even measured reliably?

Identifying Project Savings

Table 5.1 provides a starting point for categorizing improvement in business metrics to the various categories of savings at the outset of a project. However,

Table 5.1 Examples of savings by category.

Business metric	Savings categories				
	Cost (hard) savings			Cost (soft) avoidance	
	Cost take out (bottom line)	Revenue growth (top line)	Working capital/ cash flow	Cost avoidance (budget)	Cost avoidance (non-budget)
Labor					
The amount of labor required to perform a process decreases					
Shift premium decreases or is eliminated					
Overtime decreases					
Less rework occurs					
Material					
Less material is consumed (for example, solvents, office supplies, and so on)					
Less raw materials are purchased					
Less material is stocked in inventory					
Shelf life is extended/ expirations are reduced					
Reduced lead time in acquiring parts					
Reduced price of raw materials					
Increased inventory turns					
Inventory carrying costs are reduced					
Capacity					
Machine cycle time decreases					
Available capacity increases and can be used					
Overhead					
Travel expenses are reduced or eliminated					
Reduction of floor space					

Continued

Table 5.1 *Continued.*

Business metric	Savings categories				
	Cost (hard) savings			Cost (soft) avoidance	
	Cost take out (bottom line)	Revenue growth (top line)	Working capital/ cash flow	Cost avoidance (budget)	Cost avoidance (non-budget)
Improvement in building and grounds					
Reduction in expenses and supplies					
Fewer utilities are used (for example, electric, gas, water)					
Equipment					
Capital requirements are reduced or eliminated					
General and miscellaneous					
Less scrap is produced					
Customer penalties are decreased or eliminated					
Planned future costs are reduced or eliminated					
Less office supplies consumed					
Customer					
Improvement in customer satisfaction					
Improvement in voice of the customer					
Reduced warranty claims					
Increased OEM acceptance rates					
Process					
Increased yields					
Improvement in voice of the process					
Revenue growth					
Increased capacity					
Removal of process bottlenecks					

at the beginning of a project our knowledge of potential savings is limited. Some categories of savings by business metric may hold throughout the project duration, with only minor increases or decreases. Others that were initially projected may never materialize. Still others that were unforeseen may surface near the project's end and yield healthy returns.

It is important to remember that any given project may have a wide variety of savings. Therefore, it is crucial to involve the financial representative to ensure that potential savings are not overlooked and to provide or enhance the credibility of the savings.

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Part III

Team Management

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Chapter 6

Team Formation

If management is split about teams, implementing them won't work, plain and simple.

—Deborah Harrington-Mackin

This chapter will address the various aspects of forming a team, including team types, constraints placed on teams, team roles and responsibilities, criteria for selecting team members, and factors that make teams successful.

TEAM TYPES AND CONSTRAINTS

> Define and describe various teams, including virtual, cross-functional, and self-directed. Determine what team type will work best for a given set of constraints, e.g., geography, technology availability, staff schedules, time zones. (Apply)
>
> **Body of Knowledge III.A.1**

The two general types of teams are *formal* and *informal*. *Formal teams* have a specific goal or goals linked to the organization's plans. The goal is referred to as the *mission* or *statement of purpose* of the team. The formal team may also have a charter that includes its mission statement, a listing of team members, and a statement of support from management. *Informal teams* usually have none of these documents and may have a more fluid membership depending on their needs.

Virtual teams are made up of people in different locations who may never meet in person. Instead, the team may meet using teleconferencing facilities, or they may conduct all communication through written words via e-mail. These teams are used when expertise is separated geographically or temporally. When virtual teams work, they have significant benefits such as:

- Reduced travel expenses
- Reduced office equipment requirements
- Increased flexibility

Consider some of the big consultancies such as Accenture or KPMG. Although their consultants may be traveling for one purpose, they are able to participate in meetings wherever they may be. Essentially, their office is wherever they happen to be at the moment.

Process improvement teams are formed to find and propose changes in specific processes. These teams are usually *cross-functional*, involving representation from various groups likely to be impacted by the changes.

Self-directed work teams (SDWT) and *work group teams* usually have a broader and more ongoing mission involving day-to-day operations, own a work process, and are generally colocated. They typically are in a position to make decisions about safety, quality, maintenance, scheduling, personnel, and so forth.

According to Westcott (2014), "Self-managed work teams, however, require more up-front planning, support structures, and nurturing. The transition will usually take a significant time period, and needs consistent support from management." SDWTs often require a lot of training to be successful, and members of the team must be engaged and want to be a member of such a team. The application of an SDWT cannot simply be thrust on a work group and be successful without the proper preparation and infrastructure.

There are different types of constraints that affect the efficiency and effectiveness of each type of team. Considering them when forming a team will go a long way toward making the team successful. Let's look at each constraint type briefly:

- *Geography.* Geography might include different sites that are relatively close or widely dispersed, but definitely not colocated.
- *Technology availability.* The technology available may differ vastly depending on the organization's resources and culture. Such technologies include teleconferencing and web meetings, among others. Choosing the right technology or technologies depends on the needs of the team over its life cycle.
- *Staff schedules.* Team members may have vastly different schedules. Some may work different shifts, or even the same shifts but in different time zones.
- *Time zones.* Today's virtual team members may operate in widely different time zones. Some members may be at work while others may have completed their workday and are participating in meetings from home. In today's global organizations, it might be possible to have 12- to 16-hour time differences between team members. However, such operating conditions are not particularly conducive to running an efficient team and, if possible, should be avoided when selecting team members.

Table 6.1 illustrates which team type is affected by which constraint. The self-directed work team is not affected by the constraints since members of this type of team are generally colocated and work the same shift. See the definition of self-directed work team given above.

On the other hand, both cross-functional and virtual teams can be significantly affected by each of the constraints. Members of these teams can be located

Table 6.1 Determining which team type is affected by which constraint.

Constraint	Self-directed work team	Cross-functional team	Virtual team
Geography		X	X
Technology availability		X	X
Staff schedules		X	X
Time zones		X	X

anywhere in any time zone working on any shift with access to incompatible technologies. It is for this reason that team member selection and team success factors addressed later in this chapter be given additional consideration when putting together these types of teams.

TEAM ROLES AND RESPONSIBILITIES

> Define and describe various team roles and responsibilities for leader, facilitator, coach, and individual member. (Understand)
>
> **Body of Knowledge III.A.2**

Teams seem to work best when members understand their assigned roles and the team dynamics. Some standard team roles together with typical duties include:

- *Champion*. The role of the champion was discussed in detail in Chapter 1. This individual does not generally attend regular team meetings.
- *Team leader*. This is commonly the Black Belt. The role of the Black Belt was discussed in detail in Chapter 1:
  - Chairs team meetings and maintains team focus on the goal.
  - Monitors progress toward the goal and communicates this to the organization.
  - Manages administrative and record-keeping details.
  - Establishes and follows up on action assignments for individual team members.
- *Facilitator*. This may be the team leader or a distinct role depending on the skill level required:

- Makes certain all members have an opportunity for input and that a full discussion of issues occurs. In some cases, the facilitator chairs the meeting.
- Helps the team leader keep the team on track.
- Summarizes progress using visual aids as appropriate.
- Provides methods for reaching decisions.
- Mitigates nonproductive behavior.
- Helps resolve conflicts.

• *Scribe/recorder*. This may be a temporary or rotating position:

- Maintains and publishes meeting minutes. May communicate action assignment reminders to individuals.
- May use visual aids to record discussion points so all can see and react.

• *Timekeeper*. This role is used when a timed agenda is used:

- Keeps track of the time.
- Notifies the team leader or facilitator of the time remaining on each agenda item.

• *Coach*. In the case of a Lean Six Sigma team, the coach is likely to be the Master Black Belt:

- Works with the team leader and facilitator to move the team toward the objective.
- Helps provide resources for completion of team member assignments.
- Evaluates the team's progress.

• *Team member*:

- Participates in team meetings.
- Communicates ideas and expertise.
- Listens openly to all ideas.
- Completes action assignments as scheduled.

It is not uncommon that some individuals find the role of "team member" to be troublesome. More specifically, they find that they do not know how to behave in this role. As this information becomes known, it is imperative that the team leader takes immediate action. Such action might include "team (member) training." This training would provide instruction regarding how to serve on a team, what is expected of a team member, and so on. Such training may be formal or informal depending on the extent of the need.

TEAM MEMBER SELECTION CRITERIA

> Describe various factors that influence the selection of team members, including the ability to influence, openness to change, required skills sets, subject matter expertise, and availability. (Apply)
>
> **Body of Knowledge III.A.3**

Various criteria are often used by organizations to select team members. Such criteria are important because their use is foundational to making a team successful.

Typical critical include:

- *Ability to influence.* The ability to influence is important in that this skill helps keep the team moving forward by building consensus on decisions.
- *Openness to change.* Team members that hunker down and dig into a position even when presented with facts that contradict their position can create great frustration for the team. Team members that are open to change create harmony and keep the team moving forward.
- *Possessing the required skill set.* Teams usually require a variety of skill sets. Possessing a required skill set fills a void and keeps the team moving forward.
- *Being a subject matter expert.* A subject matter expert is an individual who is intimately familiar with the detailed workings of the process under investigation. This individual will bring facts and knowledge to the team when needed.
- *Being available.* Teams cannot be staffed simply by whoever is available. Unbelievably, many organizations do this. A team must select the right individual, one who has the ability to influence, is open to change, and possesses the right skill set. This individual may or may not be a subject matter expert. Alternately, teams are often staffed with members who delegate their attendance to subordinates who are not in a position to make decisions. Such actions stall teams or even cause them to fail.

  Another condition that should be considered when addressing the topic of "availability" is the constraint set identified in Table 6.1. Ideally, we would like to minimize the impact of those constraints when we select team members. For example, given two equally qualified team members, it would be more desirable to choose one for the team who is colocated with other team members than one who is located 12 time zones away.

- *Know how to participate as a team member.* Surprisingly, team members often do not know how to do this. It is the job of the team leader to recognize this deficiency early and to obtain the necessary training for the team.

In some situations, team composition will change over the life cycle of the project. For example, design engineers may be involved in the concept phase, and production personnel would be brought in for the preproduction/launch phase. Some team members may be in one, some, or all phases depending on the nature of the project. Of course, care must be exercised to ensure that team members are brought in at the appropriate times. Waiting too long to involve production personnel may result in an elegant yet non-buildable product.

A team must be provided with the resources needed to complete team activities and assignments. This includes ensuring that team member participation is viewed as part of a team member's job and that adequate time is allotted to completing team assignments.

Finally, teams should be staffed with actively involved members who can make decisions on behalf of the organizations they represent. If teams members cannot make decisions on behalf of their organizations, then other members should be chosen or the team should not be launched.

TEAM SUCCESS FACTORS

> Identify and describe the elements necessary for successful teams, e.g., management support, clear goals, ground rules, timelines. (Apply)
>
> **Body of Knowledge III.A.4**

Using teams to accomplish Lean Six Sigma projects has become commonplace. Team members and team dynamics bring resources to bear on a project that people working individually would not be able to produce.

The best teams with the best intentions will perform suboptimally, however, without careful preparation. There seems to be no standard list of preparation steps, but the following items should be performed at a minimum:

- *Team purpose and goals*. The team should have a clear purpose and a set of goals. These should be directly related to the project charter. Teams should not be asked to define their own purpose, although intermediate goals and timetables can be team generated. Teams often flounder when purposes are too vague. If team members have been selected to represent groups or supply technical knowledge, these roles should be announced at the initial or kickoff meeting.

The initiation of the team should be in the form of a written document from an individual at the executive level of the organization. A separate document from each member's immediate supervisor is also very helpful in providing up-front recognition that the team member's participation is important for the success of the team.

- *Team-building training*. Without an understanding of how a team works and the individual behavior that advances team progress, the team will often get caught in personality concerns and turf wars.
- *Team meetings*. A schedule of team meetings should be published early, and team members should be asked to commit to attending all meetings. Additional subgroup meetings may be scheduled, but full team meetings should be held as originally scheduled.
- *Management support*. Teams can succeed only if management wants them to. If clear management support is not in place, the wisdom of launching the team is in question. Team members must know that they have the authority to gather data, ask difficult questions, and, in general, think differently.
- *Sponsorship*. All teams should have a sponsor who has a vested interest in the project. The sponsor reviews the team's progress, provides resources, and removes organizational roadblocks.
- *Team ground rules and norms*. The establishment of team norms is a helpful technique that is often initiated at the first team meeting. Team norms provide clear guidelines regarding what the team will and will not tolerate, and often define the consequences of violating the norms. Examples of team norms include:
  - Being on time. A consequence of being late may be to put a dollar into a kitty that will be used at the team's discretion later.
  - Holding one conversation.
  - Demonstrating civility and courtesy to all members.
  - Accomplishing assigned tasks on time.
  - Providing advance notice of not being able to attend a meeting.
  - Participating in each meeting.
  - Following a prepared agenda.
  - Pulling the team back on track when it strays from the agenda topic.
  - Three knock rule (to be discussed in Chapter 8).

The wider availability of higher-performing lower-cost technology has greatly facilitated the increase in the use of virtual teams. These teams require somewhat more attention to communication and documentation activities and, in general, logistics considerations.

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Chapter 7

Team Facilitation

Leaders don't invent motivation in their followers, they unlock it.

—John W. Gardner

The chapter will address various ways to motivate and demotivate team members, the stages of team development, a powerful technique for facilitating team communication, and a well-known leadership model.

MOTIVATIONAL TECHNIQUES

> Describe and apply techniques to motivate team members. Identify factors that can demotivate team members and describe techniques to overcome them. (Apply)
>
> **Body of Knowledge III.B.1**

Demotivating Factors

There are many factors either working independently or in combination with others that can demotivate team members. The following are some of the more significant ones:

- *Constantly changing scope.* Generally, the source of this issue is leadership. In this case, it would be the project sponsor. When project scope changes constantly, it is likely due to the lack of a signed project charter by the project sponsor. Revisit the project charter with the sponsor and the team and work the issue until agreement is achieved and the charter is signed. If necessary, use the project coach for assistance.
- *Unclear problem statement.* Hopefully, this is just a matter of poor writing. However, unclear problem statements can occur when the pain point is not defined. In some cases, this points to "pet" projects. When this occurs, engage your project coach. Lean Six Sigma team resources should not conduct projects such as these.

- *Unclear scope.* Work with the project sponsor to clarify the scope. Don't let this issue linger. In fact, projects with unclear scopes should never launch.
- *Insufficient resources.* This insufficiency may be in terms of dollars and/or manpower. A quick way of dealing with this issue is to adjust the project plan to reflect the constraints (that is, dollars and/or manpower) and present the plan to the project sponsor. If the adjusted plan is not acceptable to the sponsor, it is the sponsor's responsibility to secure the appropriate resources so that the plan can be modified accordingly.
- *Failure to achieve goals.* Decompose the goals into smaller, intermediate, and more achievable milestones. Celebrate these milestones when they're accomplished.
- *Team members' supervisors preempt team's work.* This can and does create significant problems for teams in many organizations. In addition, it sends a mixed message to the organization regarding the importance of the work of the team. Furthermore, it creates a dilemma for the team member. As soon as these issues occur, the team leader must work through the sponsor who will, in turn, work with the appropriate supervisors to resolve it.
- *Team members who aren't team players.* Occasionally, there will be a time when the application of motivation techniques fails to improve the behavior of a team member. When this occurs, it is usually best to have a private conversation with the team member with the express intent of notifying them that their time with the team is limited. Further, contingent on a turnaround in behavior, their participation on the team will be terminated. This drastic step will usually yield the desired result for most team members. However, there are those stubborn few for which it does not work. In this case, the team leader must be prepared to replace the team member. Removing a team member actually takes a courageous leader. Many team leaders would be reluctant to take this necessary action due to the high level of conflict involved. Consequently, the entire morale and productivity of the team suffers.

Motivation Techniques

In many enterprises people feel like anonymous cogs in a giant machine. They become unwilling to invest much of themselves in a project. An important part of team leadership is generating and maintaining a high level of motivation toward project completion. The following techniques have proved successful.

Recognition. Individuals like to be recognized for their unique contributions. Some forms of recognition include the following:

- Letters of appreciation sent to individuals and placed in personnel files
- Public expressions of appreciation via meetings, newsletters, and so forth
- Positive feedback and comments from management
- Inclusion in the performance appraisal, which is important, particularly because this type of recognition tells the individual that the project was not separate from their "regular" job and was important to the organization. Also, it can be a means of reinforcing a Lean Six Sigma deployment.

Ideally, this type of recognition should be tailored to the individual to be most effective. For example, shy individuals would probably not appreciate public recognition. The responsibility lies with the individual providing the recognition to determine the best type to deliver.

Rewards. Rewards fall into two categories: monetary and nonmonetary.

Monetary rewards are effective, especially when considerable personal time or sacrifice is involved. However, particular care must be taken with this type of reward. Frequently, the individual giving the reward delivers the same dollar amount to each team member without considering the amount of personal time, sacrifice, or results achieved by each team member.

This is the easy route for the giver, but it may generate hard feelings within the team because all the members know the degree to which each team member participated. One way around this issue is to provide a set monetary limit for the entire team and allow the members to determine how the money is spent or divided.

Monetary rewards can include:

- Bonuses based on milestones met
- Percentage of the project savings
- Fixed or scale-based dollar amounts

EXAMPLE 7.1

Consider a vacation as an award. Awarding a vacation meant for one to a married individual is not appropriate. Similarly, a vacation to an individual who has no vacation left in the time frame the vacation must be used is not appropriate. Also, giving a vacation usually requires the recipient to pay taxes on it as earned income. Again, not appropriate. In most cases, the organization should "gross up the reward" and pay the taxes on behalf of the recipient.

When monetary rewards are involved, it is recommended that an objective means of determining the reward or, at least, an agreement is made in advance. Otherwise, recipients may tend to second-guess the reward structure. This can set morale on a downward spiral.

Nonmonetary rewards include tokens of appreciation such as trophies, gifts, apparel, vacations, and so on. Nonmonetary rewards, like monetary rewards, must be thought out in advance. This is explained in Example 7.1.

Relationships within the Team

During team meetings, facilitators and other team members can enhance motivation by setting an example of civility and informality, relaxing outside roles, and exhibiting a willingness to learn from one another. Simply, they can serve as role models.

Additionally, it is helpful if one of the team ground rules is to "leave your stripes at the door." This translates to a level playing field during the meeting because all titles and ranks are left outside the meeting room. All team members become equals.

Celebrating the achievement of a project milestone may enhance teamwork. Some team leaders have found that the adage "celebrate early, celebrate often" is highly effective.

In addition, some authorities suggest that participation and motivation are improved simply by referring to team sessions as workshops rather than meetings.

Westcott (2014) notes that, "An effective team leader can provide an environment in which team members feel motivated. This can be achieved by applying the *six R's*:

1. *Reinforce*. Identify and positively reinforce work done well.
2. *Request information*. Discuss team members' views. Is anything preventing expected performance?
3. *Resources*. Identify needed resources, the lack of which could impede quality performance.
4. *Responsibility*. Customers make paydays possible; all employees have a responsibility to the customers, internal and external.
5. *Role*. Be a role model. Don't just tell; demonstrate how to do it. Observe learners' performance. Together, critique the approach and work out an improved method.
6. *Repeat*. Apply the above principles regularly and repetitively."

Above all, one of the most important motivating factors is *communication*, which will be discussed in a later section in this chapter.

TEAM STAGES OF DEVELOPMENT

> Identify and describe the classic stages of team development: forming, storming, norming, performing, and adjourning. (Apply)
>
> **Body of Knowledge III.B.2**

Teams are said to go through several growth stages:

- *Forming*. Members struggle to understand the team goal and its meaning for them individually.

  Each new team is a new experience for an individual, with new members with new expectations and probably in a new environment. This creates difficulty for members, and they proceed cautiously at the outset.

- *Storming*. Members express their own opinions and ideas, often disagreeing with others.

  Team members are thinking based on individual experiences. Collaboration has yet to occur. Members may want to redefine the team's purpose and goals, and some will test the leader's authority.

- *Norming*. Members begin to understand the need to operate as a team rather than as a group of individuals.

  Team members are beginning to think in terms of a shared purpose and team goals. Interpersonal conflicts are becoming less frequent and work toward goals is beginning to proceed. Team members are starting to communicate.

- *Performing*. Team members work together to reach their common goal.

  Team members are working in a smooth and continuous manner. The team is cohesive and communication is open, honest, and flowing. Members understand one another and recognize each other's strengths and weaknesses.

These four stages are considered traditional. Recently, however, two additional stages have been added:

- *Adjourning*. A final team meeting is held, during which management decisions regarding the project are discussed, and other loose ends are tied up. This is a critical stage. Formal closeout of teams is important in that it generates a feeling of accomplishment. All too often, team

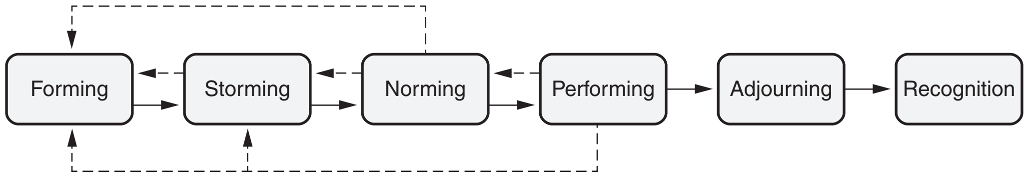

Figure 7.1 Team stages.

members are left hanging, not knowing whether their role on a team is complete or if they can reallocate their precious time.

- *Recognition*. The team's contribution is acknowledged. The techniques of reward and recognition addressed above are employed.

In Figure 7.1, the horizontal arrows indicate the standard stages of team development. The dashed arrows show alternate paths taken by teams in some circumstances. Without effective leadership, a team can regress from norming or performing into a previous stage. If a team leader or facilitator observes signs of regressing, he or she should remind the group of the goals and agenda and the need to press forward. Team members accustomed to working with one another on similar projects may be able to skip the storming stage and possibly the norming stage. There have also been examples of teams that go through the other stages and omit the performing stage.

TEAM COMMUNICATION

> Describe and explain the elements of an effective communication plan, e.g., audience identification, message type, medium, frequency. (Apply)
>
> **Body of Knowledge III.B.3**

Lack of adequate communication is one of the most frequently noted causes of team failure. Serious effort toward improving communication should be made at each stage of team development. In some situations, such as large projects or those with geographical barriers, it is necessary to develop a formal communication plan.

It was noted in Chapter 6 that at the time a team is initiated, a clear statement of the goals and purpose of the team is critical. If team members have been selected to represent groups or supply technical knowledge, these roles should be announced at the initial or kickoff meeting. The team's initiation should be in the form of a written document from an individual at the executive level of the organization. A separate document from each member's immediate supervisor is also helpful in recognizing that the team member's participation is important for the success of the team.

Ongoing communication throughout the duration of a project is essential. Such communication includes:

- Announcements of team meetings—including time, location, agenda, and such—must be made as early as possible.
- Minutes or a summary of the meeting should be provided immediately after each session.
- Disposition of action items.
- Routine status reports and presentations to management.
- Feedback to the team from management.
- Feedback to the team from the team leader, coach, or sponsor.
- Collection of routine status reports from the team members to the team leader.
- Final reports and closeout memos.

One particularly useful document for facilitating communication is the communications plan shown in Table 7.1. This document helps direct the flow of communication both within and outside of the team.

A communications plan defines who will deliver, what will be communicated, to whom, how often, and by what means. Communications could include status reports, presentations, newsletters, and so on. Methods may be either formal or informal. In all cases, though, the communications plan must reflect the needs of the audience.

Communicating within the Team

Communication within the team should take place on a regular basis. All communication within the team should be open, honest, and with no fear of retribution. In general, team members should be aware of any project problems or issues

Table 7.1 Constructing a communications plan.

Who delivers	What message	To whom	By what method	How often

before anyone outside of the team. No team member should ever be blindsided because project information bypassed him or her.

In addition, team members need to be apprised of their own performance. Too often, this valuable activity is ignored either because it is uncomfortable for the project leader or because human capital resources are limited and the project leader is reluctant to alienate a scarce resource.

As noted previously, lack of adequate communication is one of the most frequently noted causes of team failure. Serious effort toward communication improvement should be made at each stage of team development.

In some situations, such as large projects or those with wide geographical boundaries, the development of a formal communications plan is necessary. This is true particularly in the case of virtual teams.

Many teams operating in today's large organizations may never be face to face or, for that matter, ever see one another. Fortunately, technology facilitates communication.

However, careful planning for communication is still required. Accommodations for extreme time zone differences might be necessary, and technologies may be new to some sites and localities. Therefore, it is important to test any form of technical communication before operating it in a live environment.

With the advent of technology that permits round-the-clock, worldwide contact, the use of virtual teams has become a reality. As noted previously, virtual team members may never see one another. This makes the communication process that much more difficult.

If the communication process is conducted via audio only, team members have no ability to read body language. They are limited to verbal cues and voice

EXAMPLE 7.2

Table 7.2 provides an example of how a communication plan may be used by a team. Notice that it is used for communicating both within and outside of the team.

Table 7.2 Example of a communications plan.

Who delivers	What message	To whom	By what method	How often
Team leader	Progress report	Project sponsor	Face-to-face presentation	Second Tuesday of each month
Team leader	Progress report	Project sponsor subordinates	E-mail	Second Wednesday of each month
Team scribe	Team meeting minutes	Team members	E-mail	Within 24 hours after each team meeting
Coach	Team evaluation	Team members	Face to face	Quarterly

intonations. Consequently, some team members may find it difficult to function as a virtual team member. Therefore, team communication must be carefully planned, monitored, and adjusted as necessary.

Communication outside the Team

As with communication within the team, communication outside the team should also take place on a regular basis. Such communications typically include status reports and presentations. Status reports should be brief. Generally, a bullet format is sufficient when grouped under a series of headings. Typical headings include:

- Accomplishments since last report
- Problems, issues, or concerns
- Actions being taken
- 30-day plans

A recipient of a status report in the above format might conclude that the project leader is indeed making progress, has incurred problems but has recognized them (some leaders don't), has taken mitigating action (therefore, I do not need to intervene), and is proceeding forward once more.

The recipient reads the status report and completes it with a feeling of confidence in the project leader. Poorly written status reports invite criticism and, occasionally, unwarranted scrutiny and intrusion.

Presentations are another format commonly used to communicate outside the team, and are usually delivered to higher levels of the organization. The project leader may request a spot on the sponsoring executive's staff meeting agenda or may be invited to attend.

While the above bullet format is acceptable, more detail is required. Remember, the presentation will be interactive and usually face to face. Therefore, messages need to be clear and crisp. There is no room for ambiguity.

Furthermore, the presenter should expect interruptions and questions out of order and not related to the intended delivery sequence. Hence, the presenter's motto is "Be prepared!" As with poorly written status reports, a poorly delivered presentation can be a disaster that makes recovery difficult and often brings unwanted help.

TEAM LEADERSHIP MODELS

> Describe and select appropriate leadership approaches (e.g., direct, coach, support, delegate) to ensure team success. (Apply)
>
> **Body of Knowledge III.B.4**

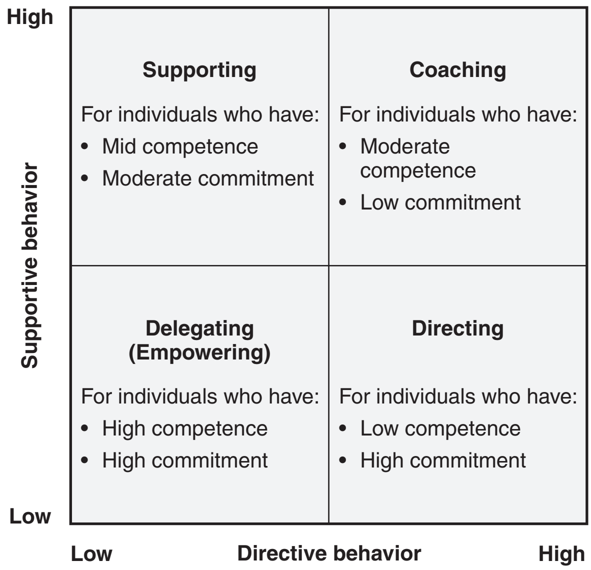

Figure 7.2 Situational leadership model.
Source: Adapted from Blanchard et al. (1999).

Blanchard et al. (1999) developed what is described as a *situational leadership* model. This model is illustrated in Figure 7.2. In this model the leadership style is categorized by quadrant according to the leader's degree of supportive and directive behavior. *Supportive behavior* involves two-way communication and focuses on emotional and social support, while *directive behavior* establishes a clear path to the goal by specifying what is to be done, how it is to be done, and who is going to do it.

Each leadership style now will be addressed:

- *Directing.* This style is characterized by high directive and low supportive behavior. It is suited for individuals who possess low competence and high commitment. Such individuals are often described as the *enthusiastic beginner.* They are new to a task and motivated about it.
- *Coaching.* This style is characterized by high directive and high supportive behavior. It is suited for individuals who possess moderate competence and low commitment. Such individuals are often described as the *disillusioned learner.* They are new to a task, but not motivated about it.
- *Supporting.* This style is characterized by low directive and high supportive behavior. It is suited for individuals who possess mid competence and moderate commitment. Such individuals are often described as the *capable, but cautious, performer.* They know their task, but are not motivated about it.

- *Delegating (Empowering).* This style is characterized by low directive and low supportive behavior. It is suited for individuals who possess high competence and high commitment. Such individuals are often described as the *self-reliant achiever*. They are proficient in their task and are highly motivated about it.

It is important to note that leadership styles are highly dynamic and depend on the competency or commitment level of an individual at any given time. For example, an individual who is competent at a specific task may vary in their commitment toward that task over time depending on their motivation. Similarly, an individual who is committed to their task will likely demonstrate an increase in competency as they gain proficiency over time. Consequently, leaders should recognize these dynamics and adjust their styles accordingly.

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Chapter 8

Team Dynamics

There are no problems we cannot solve together and very few we can solve by ourselves.

—Lyndon Johnson

This chapter will address conflict management, group behavior and dynamics, various ways to motivate and demotivate team members, the stages of team development, a powerful technique for facilitating team communication, and a well-known leadership model.

GROUP BEHAVIORS

> Identify and use various conflict resolution techniques (e.g., coaching, mentoring, intervention) to overcome negative group dynamics, including dominant and reluctant participants, groupthink, rushing to finish, and digressions. (Evaluate)
>
> **Body of Knowledge III.C.1**

Once teams have been formed, they must be built, because a true team is more than a collection of individuals. A team begins to take on a life of its own that is greater than the sum of its parts.

The building of a team begins with well-trained members who understand the roles and responsibilities of team members and how they may differ from their roles outside the team. For example, a first-line supervisor might find himself deferring to members of his group in a team session more than he would outside team meetings. Teams are built with understanding and trust in fellow team members; therefore, an early step in the building of a team should be informal introductions of each member.

Various devices are used for this process, such as "Tell your name, a non–job related interest, and what you hope the team accomplishes." This last request sometimes uncovers so-called hidden agendas. These are goals individuals have in addition to the team's stated goals and purposes. Whether or not such agendas are verbalized, team members must recognize their existence and importance. The most successful teams are able to suppress the hidden agendas in favor of team progress on the stated ones.

The best model for team leadership is the coach who strives to motivate all members to contribute their best. Productive teams occur when team coaches facilitate progress while recognizing and dealing with obstacles. Team leaders or team facilitators, or both, may perform the coaching function. Table 8.1 contains common team obstacles and associated solutions.

Coaching

Coaching is a process by which a more experienced individual helps enhance the existing skills and capabilities that reside in a less experienced individual. Coaching is about listening, observing, and providing constructive, practical, and meaningful feedback.

During training, coaching helps the trainee translate the theoretical learning into applied learning while also helping the trainee develop confidence in their newly developing knowledge and skills.

Post training, coaches help projects stay on track and advance toward completion in a timely manner.

Coaches provide guidance and direction on how to navigate organizational barriers, select and use the proper tools and techniques, prepare for tollgate reviews, discover solutions on their own, provide intervention where needed, and generally serve as a sounding board for project-related issues. They are particularly useful in providing guidance regarding team conflict and facilitation issues given their considerable experience running project teams as both a Black Belt and Master Black Belt.

In some cases the coach may find that the individual is unsuited to work a specific project and consequently will have the project assigned to another Belt or request that the project champion have the project canceled if it no longer appears viable.

Mentoring

While coaching focuses on the individual as it relates to Lean Six Sigma, mentoring focuses on the individual from the career perspective. Mentors are usually experienced individuals (not necessarily in Lean Six Sigma) who have in-depth knowledge about the organization as well as the individual (that is, *mentee*). Usually, they come from within the organization, though not necessarily the same department as their mentee. Their role is to help provide guidance, wisdom, and a possible road map to career advancement.

Like the coach, the mentor also helps the individual navigate the organization. It would be conceivable that both a coach and a mentor would support an individual Black Belt concurrently. The coach would focus on the individual's role as a Belt and the mentor would focus on the individual's overall career. The mentor would be of great help to the individual in finding a suitable role elsewhere.

From the above, we see that the mentor is generally involved with the Black Belt from a career standpoint. Therefore, Black Belts might consider limiting their

Table 8.1 Negative group dynamics—team obstacles and solutions.

Obstacle	Solutions
• A person or group dominates the discussion	• Go around the team, asking each person for one comment or reaction. • Ask dominating people to summarize their positions or proposals and e-mail them to all team members. • If the dominating people tend to react negatively to the suggestions of others, ask them for their ideas first or adopt the "no judgments allowed" rule from the brainstorming technique. • Speak to the dominating people between team meetings, requesting their assistance and cooperation in making sure all voices are heard.
• A person or group is reluctant to participate	• Be sure to welcome and express appreciation for every comment or contribution. • Form small subgroups that report to the full team. • Make action assignments for each person, with brief reports at the beginning of the next team meeting. • Speak to the reluctant people between team meetings, requesting their assistance and cooperation in making sure all voices are heard.
• A tendency exists to accept opinions without data	• Emphasize the importance of basing decisions on facts from the first meeting onward. • Raise questions such as: – Are there data that can support that? – How do we know that? – How could we verify that? – Who could collect some data on that?
• Groupthink: emphasis on consensus building has influenced a team to seek consensus too early or members are more concerned with avoiding a conflict	• Provide full opportunity for the expression of all views. • Seek out dissenting views. • Ask individuals to play devil's advocate by opposing early consensus. • Be sure a preliminary written conclusion includes dissenting views. • Ensure that the team ground rules call for multiple alternatives to be examined.
• Team members begin to air old disputes	• Make sure ground rules state the need for a fresh start, keeping history in the past. • Have a printed agenda, possibly with scheduled times for agenda items. A facilitator may say, "It's 3:10; let's move on to item number 2." • Assign team members the job of collecting data regarding the issue in dispute.

Continued

Table 8.1 *Continued.*

Obstacle	Solutions
• The team is floundering because it has lost sight of its goals and objectives	• Make sure goals and objectives are clear and well understood at the first team meeting. • Start team meetings by revisiting goals and objectives and offering a reminder as to where the team is in its journey and which objective is next. • Use graphics based on PERT and Gantt charts to help keep teams focused. • Bring in an outside voice, which often generates new ideas and approaches. • Consider using a facilitator.
• The team is rushing to meet its milestones or designated accomplishments without the benefit of a thorough study or analysis Note: When the above occurs, the team risks: • Suboptimizing • Unintended consequences (that is, solving one problem but creating others • Missing root causes	• Encourage the team to study a far-ranging list of approaches and solutions. Use divergent thinking tools such as brainstorming and cause-and-effect diagrams to broaden the perspective. • When a change is studied, be sure it is clearly communicated to all who might be impacted. Actively seek input regarding possible unintended consequences. • If a root cause is "turned off," the problem should go away. Try toggling the root cause on and off and observe whether the problem is toggling also.
• The team becomes troubled over the issue of attribution (that is, who should get credit for an accomplishment)	• If an idea comes from outside the team, that should be acknowledged. • If an idea was advanced before the team was formed, that should be acknowledged. • Often, the source of ideas and concepts developed during team sessions is not known, but if one person or group was responsible, that should be acknowledged.
• The team digresses too far from its goals and objectives	• Assign an individual or group to follow through on a side topic. • Set a specific time as a deadline beyond which no effort will be spent on the topic.

use of the mentor to discussing project-specific issues such as conflicts and negative group behavior and dynamics.

Conflict Management

Conflict is inevitable, and every team must be prepared to deal with it.

Bens (2011) notes that facilitating conflict has two steps:

1. *Venting emotions.* Individuals will not begin to resolve issues until they have vented their emotions. This step involves making people feel they have been heard and helping diffuse any pent-up emotions.
2. *Resolving issues.* It is important that a facilitator stay calm, neutral, assertive, and in control of the situation. Reference to the team norms is helpful, and a continued emphasis on effective listening is necessary. Slow things down and focus on a structured approach (for example, force-field analysis) for moving forward. Capture key points on a flip chart or other visual media and revisit when there is a need to regain control. Create closure by testing agreed-upon points and issues. The following are the five basic approaches for resolving issues:
   - *Avoid.* Ignore the issue with the hope that it will go away. Employ this technique when the issue cannot be resolved profitably.
   - *Accommodate.* This involves asking people to try to get along. In truth, one person will ultimately have to give in to another. Employ this technique when keeping the peace is more important than finding a solution.
   - *Compromise.* This occurs when views are polarized and each must meet somewhere in the middle. Compromise is not a consensus. Consensus-building will be discussed later in this chapter.
   - *Compete.* This creates a win/lose situation and is the least desirable means of attempting to resolve conflict.
   - *Collaborate.* This creates a win/win situation and encourages the participants to seek solutions that they can live with. This approach is based on data, fact, and trust. Participants must be committed to finding a consensus-based solution that works.

Intervention by Coach

The coach will want to attend team meetings from time to time to make evaluations and observations. Above all, the coach will want to determine whether the team is making progress or is heading for a state of imminent failure.

If failure is close or imminent, the coach should take immediate action to prevent it. This form of action is called *intervention*. Specifically, an intervention is an action taken by a leader or a facilitator to support the effective functioning of a team or work group.

An intervention might take the form of replacing the team leader, replacing team members, adding a professional facilitator to the team, providing additional training to specific members, or whatever the coach decides is appropriate.

Regardless, the coach should discuss his or her evaluations and observations with the team frequently as well as follow up with additional evaluations and observations at later meetings to determine whether behavior changes have occurred.

Intervention by Team Leader or Facilitator

Table 8.2 provides a summary of conditions when intervention might be necessary and criteria for determining when to actually intervene.

Table 8.2 Deciding to intervene requires significant forethought when the proper conditions are at hand.

Conditions for intervening	Criteria for deciding to intervene
• A participant is not listening or paying attention or is distracting other participants • One or more side conversations are taking place • Participants are interrupting one another • Personal attacks, discourtesy, or incivility is occurring • The discussion is drifting off track • Violation of team norms or ground rules occurred • Violation of organization policy has occurred	1. Is this a serious problem? Consider whether this problem is a single serious problem or a series of recurring issues. 2. Will this problem go away by itself? 3. How much time will an intervention take, and do we have that much time? 4. How much of a disruption will the intervention cause, and can we recover from it? 5. Will the intervention impact team member relationships and/or the flow of the meeting? 6. Can the intervention hurt the climate of the meeting? 7. Will anyone's self-esteem be damaged? 8. What's the chance that the intervention will work? 9. Do I know these people well enough to do this? 10. Do I have enough credibility to do this? 11. Is it appropriate enough to intervene, given their level of openness and trust? 12. What will happen if I do nothing (that is, take no action)? Alternately, should I take action after the meeting?

Source: Adapted from Bens (1999).

Notice that the left column of Table 8.2 is bulleted. This is because the order of the conditions does not matter. The right column is numbered because it represents a serial thinking process.

In essence, the team leader or facilitator witnesses a possible situation described by the left column and mentally assesses that situation by asking the series of questions outlined in the right column. Based on the answers to those questions, a decision is made and the facilitator acts to intervene or not. Of course, this whole assessment takes place in seconds.

In Chapter 6 we discussed the concept of team norms and ground rules. Some teams gloss over this action swiftly. However, when done correctly, it makes team facilitation and, particularly, intervention much easier.

More often than not, situations requiring interventions are simply violations of team norms or ground rules. To ease the burden of facilitation, it is useful to post the norms and ground rules at each meeting. All that is required on the part of the facilitator is a minor interruption and a reference to the posting. Team members quickly understand and become accustomed to them. After a while, they begin to facilitate themselves.

In Chapter 6 one of the team norms was the "three knock rule." This rule refers to a team member knocking three times on the table to interrupt the meeting for the sole purpose of gaining everyone's attention. With everyone's attention in hand, the team member explains that a team norm has been violated, such as too many side conversations, straying too far from the topic, and so on.

Essentially, the "three knock rule" shares the authority and role of the facilitator with all the team members. Once team members understand how this rule works, they will make adequate use of it, and meetings will begin to run smoothly.

MEETING MANAGEMENT

> Select and use various meeting management techniques, including using agendas, starting on time, requiring pre-work by attendees, and ensuring that the right people and resources are available. (Apply)
>
> **Body of Knowledge III.C.2**

Time is perhaps the most critical resource for any team. Team meeting time must be treated as the rare and valuable commodity that it is. It is up to the team leader and the facilitator to make every minute count, although every team member has this responsibility as well. Some practices that have proved useful follow:

- *Form agenda committee.* This committee will generate the meeting agenda in advance of the scheduled meeting and will be responsible for getting the resources called for by each agenda item. For smaller teams, the team leader often prepares the agenda.

- *Publish timed meeting agendas.* A timed meeting agenda is simply a meeting agenda with time limits on each item. Assign a timekeeper and stick to the schedule as closely as possible. Leave five to ten minutes as a buffer at the end of the meeting and also for conducting an end-of-meeting process check.

  At times, there will be agenda items that require more discussion time than allotted. There are two ways to handle these items. The first way is to discuss within the time allotted and carry over the discussion into additional meetings until the issue is resolved. The second way is to spin off a sub-team and let the sub-team address the issue. Once the sub-team has completed its task, it will report to the main team with its resolution.
- *Publish project timeline*. Use a Gantt chart (see Chapter 12) timeline displaying milestones and dates, updating as needed. This visual provides constant reinforcement of the milestones and due dates and retains a sense of urgency.
- *Publish actions assigned*. In some cases, the team leader will want to review progress on these assignments between meetings.
- *Publish reminders of assigned actions*. Intermediate reminders between meetings are helpful to ensure that actions are completed on time.
- *Maintain "parking lot" list*. This list contains important items that may not be relevant at the current time but should not be forgotten, or may not be relevant to the team at all, but should be forwarded to another authority for appropriate action or investigation.
- *Starting on time*. This may seem trivial, but it is of great importance—so much so that it should be incorporated into the team's norms. Lateness can be a great source of frustration and distraction for other members who are on time. Furthermore, some leaders have a propensity to backtrack for the benefit of late arrivals. This wastes time.
- *Require pre-work*. Coming to a meeting with pre-work complete is like going to class with your homework complete. For example, if the meeting is to discuss a procedure, the pre-work should be to have each member review the procedure beforehand and come to the meeting prepared to discuss it. If necessary, the pre-work might include the member doing any related investigatory work that might be required to answer any of his or her own questions as well.
- *Ensuring that the right people and resources are available*. Occasionally, a meeting might require a special attendee, require a move to a special location for a particular meeting, and so on. Any unusual conditions or differences from normal meeting situations need to be given special attention and planned for in advance to ensure that meetings are executed smoothly and conducted effectively and efficiently.

- *Assign prescribed roles*. This approach was covered in Chapter 6. Recall that the special roles include team leader, facilitator, scribe, timekeeper, and coach.
- *Create the team norms*. The use of team norms was addressed in Chapter 6. Team norms describe the behavior we want team members to follow.
- *Conduct in-meeting process checks*. Conduct periodic process checks during the meeting. Such checks include determining whether the team is making process and whether it is moving at the right speed. It also provides the opportunity to ensure that all team members are moving forward together.
- *Conduct end-of-meeting process check*. A few minutes are reserved at the end of each meeting and are included on the agenda to conduct a formal process check. This process check focuses on how well the meeting went and how it could be improved. Lessons learned from this activity are made effective with the next team meeting.

Each of the above techniques works well to keep teams on track. Collectively, they keep teams effective, efficient, and moving forward. None of the techniques are difficult to learn or apply. Apply a few at each meeting and before you know it, your team will be running smoothly and achieving its goals. When achieved, celebrate them!

TEAM DECISION-MAKING METHODS

> Define, select, and use various tools (e.g., consensus, nominal group technique, multi-voting) for decision-making. (Apply)
>
> **Body of Knowledge III.C.3**

Several useful tools are available for helping teams make decisions. These include:

- Consensus building
- Nominal group technique (NGT)
- Multivoting
- Force-field analysis

Consensus-Building

Leadership by consensus views process as being as important as product. That is, the impact of the process by which decisions are made will have a far-reaching effect on the quality of the product.”

—Richard Wynn and Charles W. Guditus

It is important to recognize that a consensus is not a compromise. Consensus creates a situation or outcome that all the participants can live with. If no situation or outcome can be found, discussion continues.

Bens (1999) provides an interesting take on consensus when she states, "Working toward consensus forces dissenters to collaborate. If they refuse and consensus isn't reached because of specific individuals, the minutes should reflect who blocked progress." She goes on to state that "Consensus isn't designed to make people happy or leave them in 100% agreement. Its goal is to create an outcome that represents the best feasible course of action given the circumstances."

Consensus requires a skilled facilitator who can, according to Bens (1999):

- Integrate and summarize complex ideas to the satisfaction of the group members
- Gain buy-in to the purpose of the meeting from all members
- Link people's ideas together so that they can see how they are saying the same thing
- Record ideas on a flip chart that so that group members can see how they have contributed
- Draw out everyone's input into a clear goal and objective for the group's activities

While attempting to build consensus, the facilitator must simultaneously navigate the harsh waters of conflict management. It is for this reason that it is sometimes best that the roles of team leader and facilitator are split.

Nominal Group Technique (NGT)

The *nominal group technique* (NGT) is used to prioritize a list of items. The technique begins with all team members writing their ideas down on a slip of paper without any discussion. After a period of time or when it appears that members have run out of ideas, the facilitator selects one idea from each team member and captures it on a flip chart. This process continues until all ideas have been collected. During this selection period, there is no discussion and ideas are not judged. When all of the ideas have been captured, like ideas can be consolidated with agreement from the contributing members.

Now it is time to prioritize the ideas. Each idea is designated with a letter, and each team member writes one letter on a slip of paper while assigning a priority to it. For example, if there are four items, each team member writes the letters A, B, C, and D on separate pieces of paper and then ranks each of the items, using a higher number for a higher ranking. The highest-ranking item would be marked "4," the next highest would be marked "3," and so on. The totals are compiled for each item and represent the team's consensus. Items with the highest totals are the priority items to be worked on by the team. If necessary, the process can be repeated for ties.

EXAMPLE 8.1

Table 8.3 provides an example of the nominal group technique with four team members. Review the scoring for all team members. Notice that team members 1 and 4 ranked item "A" high, while team members 2, 3, and 4 all ranked item "D" the highest. The resulting priority for the team, in order from highest to lowest, will be D, C, A, and B. Hopefully, this ranking will not cause dissension problems with team member 1.

Table 8.3 Example of the nominal group technique.

Item	Team member 1	Team member 2	Team member 3	Team member 4	Total
A	4	1	1	3	**9**
B	1	2	2	1	**6**
C	3	3	3	2	**11**
D	2	4	4	4	**14**

Multivoting

Multivoting is a variation of NGT in which each team member has 100 points to allocate to the items on the list as he or she feels appropriate, assigning the largest number of points to the highest-ranking item. Points assigned to each item are totaled and ranked accordingly. Items with the most points are worked on first.

Although we can define a single top priority in Example 8.2, it is important to look at the underlying votes to validate the results. Only one of the four members demonstrated real conviction. As a team leader, would the vote given in this example convince you to move forward?

EXAMPLE 8.2

This example provides the team with the same items to prioritize as in the previous example. Only this time, the members will have 100 points to allocate among the four items. The results are given in Table 8.4. Upon reviewing the results, we notice that team member 1 allocated all 100 points to item "A," indicating a strong feeling toward it as a top priority. Team member 2 spread his points not quite evenly among items "B," "C," and "D," with some emphasis placed on item "D." Team member 3 felt somewhat strongly about item "B," but not so strongly as to allocate all of them. In fact, she did allocate some points to items "C" and "D," evenly. Team member 4 felt moderately split between items "C" and "D." Still feeling undecided, he voted for item "B." Going into

Continued

Continued

this round of voting, only team member 1 felt decisive enough to establish a priority. Consequently, the vote shown in the last column of Table 8.4 indicates item "D," followed by a tie of items "A" and "B," and then item "C." The range of the total votes varied 10 points (95–105), which doesn't provide much discrimination between the items. When such closeness of the totals occurs, it is generally an indication that more discussion regarding the items is required.

Table 8.4 Example of the multivoting technique.

Item	Team member 1	Team member 2	Team member 3	Team member 4	Total
A	100	0	0		**100**
B	0	30	50	20	**100**
C	0	30	25	40	**95**
D	0	40	25	40	**105**

EXAMPLE 8.3

Suppose Example 8.2 had 20 items on the list instead of four. In this case, we could provide each member with as many as one-third of the votes each, which is six or seven on the low end, to as many as one-half, which is 10 on the high end.

An alternative form of multi-voting used when the list of items is long is to provide each team member a number of votes that is approximately one-third to one-half the number of items on the list.

Force-Field Analysis

A well known, but little used, quality tool that is quite helpful in driving organizational change is the force-field analysis. In addition, it can be quite beneficial in helping complete the "action plan" column of the stakeholder analysis form (Table 3.1).

Siebels (2004) defines force-field analysis as a "technique for analyzing the forces that aid or hinder an organization in reaching an objective." Such an objective could be cultural change.

Figure 8.1 illustrates the general format of a force-field analysis. The desired change is listed at the top in simple terms. Driving forces are listed on the left side of the vertical line. Driving forces are those forces that aid in achieving the objective. Similarly, restraining forces are placed on the right side of the line. Restraining forces are those forces that hinder or oppose the objective.

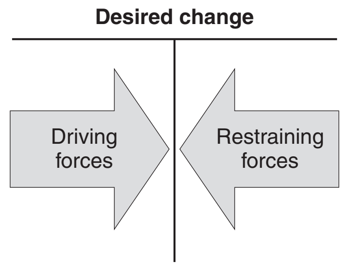

Figure 8.1 A force-field analysis diagram.

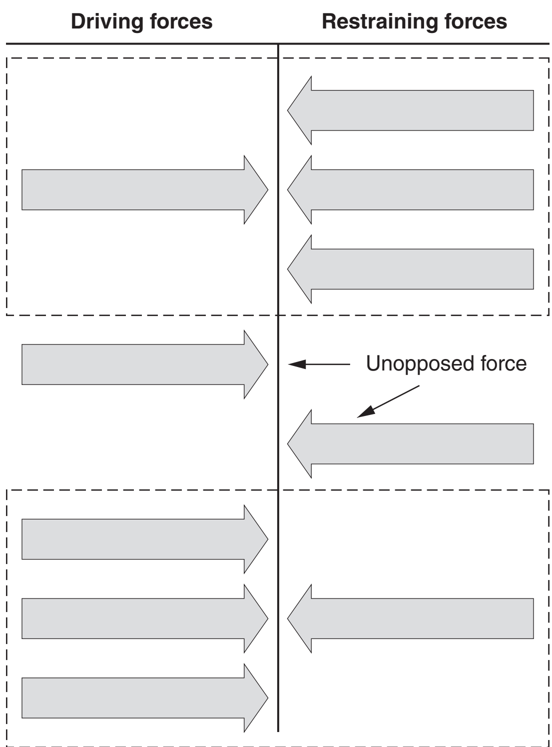

Figure 8.2 An example of many-to-one forces.

It is important to note that driving and restraining forces may be many-to-one in either direction, or even unopposed, as shown in Figure 8.2. When the unopposed force is a restraining force and is active, the reader may need to resort to the techniques discussed in Chapter 3. If these turn out to be ineffective, the sponsor may need to rethink the viability of the project.

Cultural maturity and readiness assessments can contain a wealth of information regarding both the driving and restraining forces. In addition, consider the use of brainstorming sessions. These can be conducted candidly and will likely surface additional and insightful information.

A force-field analysis can be done for change at any level, and is useful any time that there are two opposing sides to an issue that need to be considered. Identifying which forces can and cannot be changed will help reduce wasted energy. Focusing energy on removing or reducing forces that are restraining change, or ensuring that the driving forces are maintained or increased, will be more productive.

EXAMPLE 8.4

Figure 8.3 illustrates an example of losing weight provided by Tague (2005). As you review this example, notice that the driving and restraining forces are not necessarily or intended to be opposites. Also, notice that the relative strength of each force is not stated. This may be considered a drawback of the force-field analysis tool.

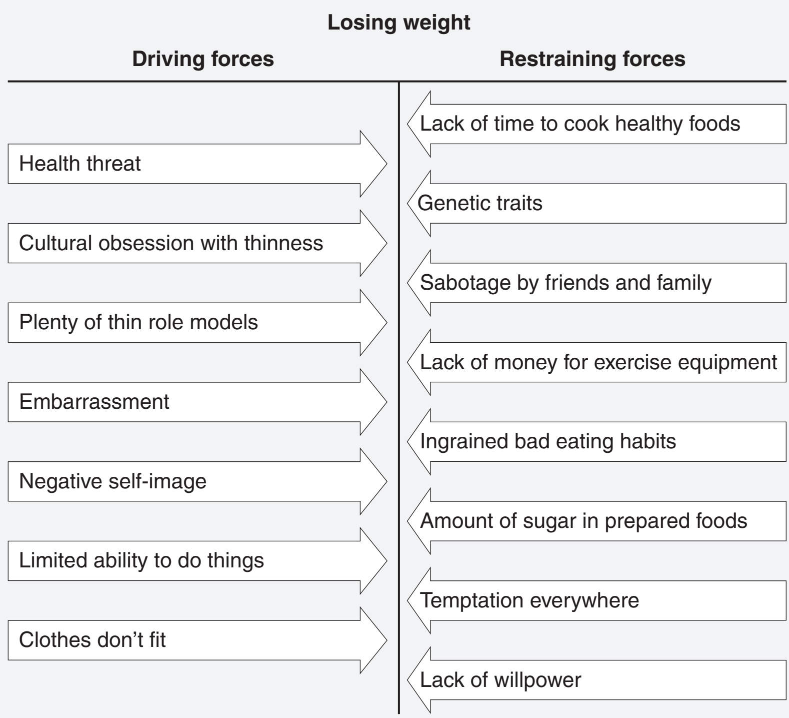

Figure 8.3 A force-field analysis diagram for Example 8.4.
Source: Adapted from Tague (2005).

Chapter 9
Team Training

Management will recognize the need for education and training when they realize that people are an asset and not an expense.

—William Scherkenbach

The terms "training" and "education" are often used interchangeably. However, *education* focuses on broadening an individual's knowledge base and expands thinking processes. Further, it helps employees understand concepts and accept increased job responsibilities, and prepares them for future jobs and leadership roles. Simply, education helps individuals learn how to think.

Training is considered a subset of education that focuses on increasing proficiency in a skill. Therefore, training typically refers to skill-based instruction and addresses the specific skills employees need to perform a current job or task. Such skills can be highly diverse. For example, they might include learning how to apply quality tools, implementing customer service principles, running a new piece of equipment, or facilitating a team. Training may take place in either a formal or informal setting, such as a classroom or on-the-job training. Training is appropriate when skills are lacking, but also when the individual is willing to learn. A primary attribute of training is that it produces measureable performance outcomes, whereas education alone does not.

NEEDS ASSESSMENT

> Identify the steps involved to implement an effective training curriculum: identify skills gaps, develop learning objectives, prepare a training plan, and develop training materials. (Understand)
>
> **Body of Knowledge III.D.1**

The Importance of a Training Needs Analysis

Successful organizations understand and acknowledge the importance their human assets play in achieving their success. However, the degree of success depends on the appropriate level and relevance of their human assets' education and training. Remember, an organization's strategies are executed through its people. If the current education and training of the people cannot support the strategies, the organization must act on this gap via the training needs analysis.

A *training need* is essentially a performance gap between what people know or are able to do and what they should know or be able to do to perform their work competently. Therefore, a *training needs analysis* is a diagnostic method to identify the gap between current performance and desired performance. Organizational training needs stem from the strategic planning process, while individual training needs stem from both strategic planning and individual performance. A needs analysis is also known as a *needs assessment* or *training requirements analysis*.

A needs analysis uses a variety of tools and techniques to identify the training needs for a specific population, target group, or team. Through a systematic process, relevant data are collected and used to formulate a training plan for employees. In addition to identifying training needs for individual employees, a needs analysis also investigates how such training can support the organization's strategies.

A training needs analysis effort produces facts about the target audience, training needs, learning styles, and so on. However, a good needs analysis does more than merely gather information. It makes a case for training. To that end, a training needs analysis must be well executed to garner top-down support.

Defining the Extent and Nature of the Job

Before any type of training needs or performance gap analysis can be conducted, it is critical that the organization be aware of the extent and nature of the jobs associated with each target group involved. To achieve this awareness, the following three aspects of each job must be considered in detail:

- *Job analysis.* Pearn and Kandola (1993) describe a job analysis as a "systematic procedure for obtaining detailed information about a job, task, or role that will be performed or is currently being performed." However, a more traditional definition describes a *job analysis* as a process for collecting information about duties, responsibilities, skills, and outcomes. Also, a description of the work environment may be included if it is unique or extraordinary in some manner. An example of such an environment might be a manufacturing clean room.
- *Task analysis.* The purpose of the *task analysis* is to identify the specific tasks required to perform a particular job. Often, this is accomplished by direct observation of the individual performing the job. It is important that all tasks be identified regardless of whether or not they are observable at the time the task analysis is being performed. An example of a task might be "prepare minutes for all management meetings."
- *Skills analysis.* The purpose of the *skills analysis* is to identify the specific skills required to perform the specific tasks identified by the task analysis. In many instances, it is likely that multiple skills will be required to perform a specific task. Examples of skills associated with

the task example might be "ability to listen effectively" and "ability to write clearly."

Once completed, the above three aspects of each job will be used to draft a job description and a job specification. *Job descriptions*, according to Rae (1997) "are statements of the outline of the whole job and show the duties and responsibilities involved in the job." The job description also "details the skills, knowledge, and attitudes which are required by the individual in order to carry out the duties involved in the job." A subset of the job description, the *job specification*, according to Rae (1997), defines "what the job holder should do and be capable of doing." Bee and Bee (1994) add that the job specification also includes carrying out the job to a prescribed standard. The greater the detail of the job specification, the easier it will be to determine performance gaps.

The Importance of Training Plans

When skills training is required, it is crucial to have a training plan. Every organization faces important decisions about the wide variety of organizational needs that might be supported, including new hire orientation; technical and process training; quality training; certification training; team, interpersonal, and communication skills training; diversity training; and management and leadership programs. Consequently, training dollars represent a significant budget item.

In spite of their organizational importance, training expenditures are often vulnerable to cost-reduction pressures. When senior leaders do not view training as a strategic, cost-justified investment, business downturns can result in training budget cutbacks. Therefore, employee training should reinforce strategic plans with direct line of sight.

Training plans provide the road map for meeting critical training requirements. Without a plan, important organizational training requirements may go unnoticed, may be poorly addressed, or not addressed at all. In fact, many organizations with strategic plans that have training implications still leave such training to happenstance.

In large organizations, various business units, sites, functional groups, and departments may have individual training plans. All training plans, though, should be a result of a strategic business planning process and should provide a mechanism for ensuring that training and education build the organization's capabilities as well as enable employees to a make a positive contribution to the organization's success.

Components of an Effective Training Plan

Bee and Bee (1994) suggest that a viable training plan should comprise three critical components:

- *Statement of policy.* This constitutes a statement of direction for the time period covered by the training plan. For example, there may be a specific focus of training such as qualification or certification training, leadership or first-line supervisory training, interpersonal skills training versus professional or technical skills training, and so on.

- *Training budget.* This would include all costs associated with developing and executing the training plans. Also, it is important to specify how training is paid.
- *Operational plan.* This covers all aspects of timing and non-financial resources. For example:
  - Identify the need
  - State objectives and outcomes
  - Determine content and sequence of instruction, if designed internally, or select from qualified external suppliers
  - Identify the target groups
  - Develop schedules for all aspects of training, such as development, delivery, evaluation, and so on
  - Specify resource requirements such as instructors, facilities, equipment, materials
  - Establish measurements
  - Select training delivery methods, if delivered by internal instructor as assessed, or select from qualified external suppliers
  - Train instructors
  - Deliver training
  - Evaluate outcomes
  - Modify training as needed
  - Conduct classes on lessons learned
  - Add documentation to the organization's knowledge database

Remember, executing the training plan should be treated as any other project. That is, it should be managed properly using the appropriate project management tools and techniques. Therefore, planning for training should closely resemble the planning done for any other project.

Applying the Training Plan

In simplest terms, one of the key deliverables of a training plan will be a matrix similar to that shown in Table 9.1. This table illustrates for each identified skill an associated level of competency. The "entry" level competency is the minimal level of competence required to be accepted into the training program, curriculum, or course. For example, in the Lean Six Sigma context, an entry-level skill might be basic math or algebra. Both are prerequisites to Lean Six Sigma training. Also, the nomenclature used to describe the competency levels in the table is not overly important. These are simply an ordinal categorization that indicates an increase in skill level as one moves from left to right across the table. Many different categorizations are in use today. Find one that works for your organization and stick with it.

As organizations mature and their training plans are refined, Table 9.1 will likely evolve into a more comprehensive matrix similar to that shown in Table 9.2. Let's briefly interpret this table. There are four curriculums, 10 courses, nine

Terminology is subject to change, but reflects increasing levels of competency

Table 9.1 Example of a mastery grid.

	Competency level				
Skill	**Entry**	**Beginner**	**Practitioner**	**Expert**	**Master**
A					
B					
C					
D					
E					

Required for acceptance into the program

Narrative, but preferably measurable, descriptions of skill proficiencies achieved

Table 9.2 Example of a training plan in matrix summary form.

				Competency level				
Curriculum	**Course**	**Skill**	**Target group**	**Entry**	**Beginner**	**Practitioner**	**Expert**	**Master**
I	101	A	1, 6		1, 6			
III	102	A	3			3		
I	103	C	1			1		
I, II	201	D	1, 2, 4			1, 2, 4		
I, II, III	202	E	1, 2, 3, 4			1, 2, 3, 4		
I, II	203	F	1, 2, 4		1, 2, 4			
II	301	G	4, 5			4, 5		
II, III	302	H	3, 4, 5			3, 4, 5		
II	303	I	4				4	
IV	401	J	4					4

skills, and six target groups. Target group 1 is required to take all of curriculum 1, while target groups 2, 3, 4, and 6 are required to take specific courses from curriculum 1. Similarly, target group 4 is required to take all of curriculum 2, while target groups 2, 3, and 5 are required to take specific courses from curriculum 2.

DELIVERY

> Describe various techniques used to deliver effective training, including adult learning theory, soft skills, and modes of learning. (Understand)
>
> **Body of Knowledge III.D.2**

I hear, I forget.
I see, I remember.
I do, I understand.

—Chinese proverb

Note: Some of the following material on training delivery is extracted from Westcott (2006).

Traditionally, training delivery methods have included live instructors, printed handouts, and media aids (for example, whiteboards, chalkboards, flip charts, overhead projection of slides and transparencies, or computerized projection). Lectures, discussions, role-plays, simple paper-based games, case studies, workbooks, video, audio, and self-directed learning with printed materials are often used. Lectures and role-playing date back to classical Greece, more than 2000 years ago. Lectures were the primary method of instruction, but creative instructors now use many other tools to enhance learning.

Self-Directed Learning (SDL)

Increasingly, newer technologies for delivering self-directed learning (SDL), also called *learner-controlled instruction* (LCI), are becoming available. *Self-directed learning* requires students to work without an instructor and at their own pace. This permits adult learners to build toward achieving a desired competency level in a needed knowledge or skill. Self-directed learning works well when there is only one person to train at any given time, when the learners are geographically distant, when training needs to occur at multiple sites simultaneously, when consistency of skill execution is a necessity, and when refresher training is needed.

Lectures and Presentations

A *lecture* is a one-dimensional, oral transmission of information to students by an instructor, either in person or via electronic media. It is the most frequently used manner of conveying material to students and often is used in combination with other tools. Lectures are one type of presentation, but presentations can also include other oral, visual, and multimedia formats.

Lectures and presentations offer the advantage of being relatively easy to develop, present, and revise. However, it is usually difficult to master knowledge and skills by this means. Furthermore, this delivery method results in low retention rates.

Discussion Format

The *discussion* format allows participants freedom to present views, opinions, and ideas in an unrestrained environment. Roundtables and panel discussions are variations. These methods normally work best with a knowledgeable and experienced discussion leader. The Socratic method of repeated questioning has been successfully used to elicit and share the accumulated knowledge and experience of course members.

However, the discussion format requires a skilled facilitator in order to execute it successfully. Also, some participants may be reluctant to speak up in this environment.

Experiential Training

The focus of this form of training is on experiencing the effects typically encountered in a real-life situation. The *experiential* approach employs many types of structured experiences (for example, games, simulations, role-plays) to facilitate learning. Typically, participants assume roles within a stated scenario designed to surface one or more learning points. They then "play out" the scenario to a conclusion under the eye of a skilled facilitator. No outcome is prescribed, but a well-designed scenario will tend to produce a predictable range of responses highlighting the training content. Experiential training is used more frequently with soft skill training.

Experiential training permits faster learning than on the job, and allows mistakes to occur without penalties. However, it requires a great deal of planning and can be costly to provide.

Coaching

Used on a one-to-one basis, or for a small group or team, and conducted at the workplace, *coaching* is a teaching, learning, counseling relationship designed to develop job-related knowledge and skills and improve performance. It requires a continuous flow of instructions, comments, explanations, and suggestions from coach to employee, with the coach demonstrating, listening, questioning, relating learner's experiences, assisting, motivating, encouraging, and rewarding performance. The coaching option is used primarily to teach complex skills and operations—when the number of trainees is small and the training is needed infrequently—and is a follow-on to other forms of employee training. It has the advantage of being flexible and adaptable. Its disadvantage is that its success depends on the competence of the coach.

Case Studies

Case studies can be developed in-house or purchased from a variety of sources. However, a form of mini case study, the *critical incident*, may often be of greater value than a larger, more inclusive case study. It is carefully crafted, usually presented in a succinct written form, and is designed to engender learning about how to respond to a specific event, situation, or condition.

The critical incident is frequently derived from real situations within the organization for which the training is designed. A critical incident might consist of just a few sentences describing a specific type of event or condition without reference to names or locations. Critical incidents are usually "sprinkled" throughout the training to illustrate specific learning points. See Kubiak (2012) for a detailed discussion on the advantages and disadvantages of the critical incident approach.

Workbooks

Workbooks contain a collection of discussion topics, exercises, extra readings, and other reference material. They may be used to supplement other materials provided during classroom presentations. Kubiak (2012) notes that workbooks can serve several purposes:

- As a learning tool
- To demonstrate knowledge or skills
- As an on-the-job training tool

Workbooks do have some drawbacks: they require self-motivated participants to use them effectively, poorly designed materials may be boring, well-designed materials may be expensive, and participants may become "stuck" and lose interest.

Instructional Games, Simulations, and Role-Playing

Instructional games are activities designed to augment other training methods in order to focus the participants' learning toward specific training outcomes. One example: "In-box" exercises usually present each participant with a variety of papers (for example, reports, charts, tables) that resemble what a manager may find in an in-box. The participant analyzes the documents and takes what they consider appropriate action for each. Responses are either discussed in class or evaluated by the instructor off-line, with instructional notes affixed to the documents.

Simulations resemble instructional games but also include correspondence between some aspect of the game and reality. *Role-playing* consists of spontaneously performing in an assigned scenario.

Experience has shown that adult learners prefer to practice rather than just listen to lectures and that they prefer to interact with other learners when in a group learning environment.

Remote Learning

Correspondence schools and open learning programs at educational institutions are types of *remote learning*. Students can earn degrees in many fields without ever stepping into a classroom. This self-directed learning style consists of an instructor mailing (that is, through traditional or electronic mail), a course syllabus, and a list of assignments to students. Students complete the required work and send it to the instructor, who then returns the graded assignment.

Remote learning is especially popular with nontraditional students who would not be able to attend classes in person. This type of learning requires a tremendous amount of self-discipline.

Distance Learning

One popular form of *distance learning* is where students in more than one location are scheduled and linked together electronically (for example, by cable, satellite TV, Internet, Skype, FaceTime) to enable one instructor to simultaneously teach students in multiple locations. The connection provides face-to-face video between the students and the instructor and audio from the instructor to the students. Audio interaction from the students to the instructor and between students is usually muted and under the control of the instructor.

Computer-Based Instructional Techniques

Although much training continues to be conducted using the traditional techniques of lecture, discussion, and on-the-job training, technology-based approaches are rapidly overtaking the traditional methods. Computers, electronic communications, and the Internet have created dramatic new opportunities to provide learning. New products and techniques are appearing nearly every day.

Computer-based instructional techniques have the added advantages of being able to be widely distributed with ease, updated with little difficulty, and offered with more activity options such as simulations and games. However, learners must be motivated to use this method of delivery, and not everyone adapts well to it.

Job Aids

Job aids are virtually any type of media that can substitute for formal training and/or provide reinforcement or reference after training. Often, providing a job aid can be the simplest and most cost-effective way to assist workers in doing their jobs. A few types of job aids are:

- Procedure manual
- Laminated checklist hung at a workstation

- Instructions printed on the back of a form providing details for completing the form
- Pertinent information for each critical step described on a shop work order or traveler

Overall, jobs aids can be cost-effective, available when and where they are needed, and easily kept current, and diminish the need for individuals to rely on memory. However, job aids are not a substitute for expertise or performance, and do not replace good judgment.

On-the-Job Training (OJT)

A significant amount of skills training occurs through on-the-job training (OJT). The learner is usually an employee who has been newly hired, transferred, or promoted into a position and lacks the knowledge and skill to perform some specific job component. A more experienced employee (for example, peer or supervisor) is usually assigned to work with the trainee. Specific tasks are demonstrated and performed in the actual work environment to help the trainee gain the knowledge or skills previously lacking.

Because it is common OJT practice for an employee who is familiar with the job to train another employee, without objectives or a planned system, inconsistencies can result in the way employees perform their duties. In addition, the lack of structure can cause bad habits to be included in the training. Another problem occurs when a relatively inexperienced employee is responsible for training others. The employee might know how to do the job, but, as anyone who has done training knows, it is much more difficult to teach someone to do a job than to do it yourself.

On-the-job training is highly regarded as a learning method. It matches the day-to-day realities of the organization and can be delivered just in time (JIT). However, it can be hampered by physical constraints of the job site, it may disrupt work flow, and can create potential safety issues.

Blended Learning

Although each learning method has been discussed separately, it must be made clear that these methods can be mixed in any manner necessary to accommodate the needs of the target group. This mixing is called *blended learning*.

Blended learning accelerates learning and job application, provides the course leader with more options, and costs considerably less than a formal classroom setting. However, it requires more coordination, and tracking the progress of the learner is more difficult.

Adult Learning Theory

Training has been defined as a subset of education that focuses on increasing proficiency in a skill. It is a "push" strategy that pushes knowledge, skills, and attitudes toward employees to enhance their job performance. By contrast, learning is a "pull" strategy by which the employee pulls knowledge, skills, and attitudes to enhance their job performance.

Olson and Hergenhahn (1997) state that "*learning* refers to a change in behavior potentiality, and *performance* refers to the translation of this potentiality into behavior." Rothwell (2008) states this more simply as "learning gives individuals the potential to get results. But performance is the actual realization of that potential."

Assumptions About Adult Learners

The following assumptions about adult learners are helpful in understanding how to design curricula, deliver material, and more effectively transfer knowledge:

- They learn for different reasons:
  - To build social networks
  - To meet expectations such as those that might be imposed by those in authority
  - To advance in their careers
  - To seek stimulation
  - To help others
  - To learn for the sake of learning

  At the outset of the training, the instructor should consider addressing the above topics with the learners to help gain a feel for the individual motivations for participating in the learning activity.
- They prefer to focus on learning that centers on key ideas, principles, and experiences rather than on general learning topics.
- They prefer to learn in an environment that is both physically and psychologically comfortable.
- They retain only about eight percent of material when training is conducted off the job. This percentage can be increased by providing the learning with memorable instruction, appealing to their physical senses, relating new concepts to existing knowledge and understanding, or helping them solve problems faced on the job.
- They face barriers to learning such as scheduling difficulties, money, management support, and so on. The best way to learn about them is to simply ask.

Table 9.3 provides a summary of training preferences of adult learners as seen by Tolbize (2008). Preferences are further categorized by hard and soft skills. Notice that all generation types cite similar learning preferences for hard skills.

Table 9.3 Carnes's training preferences of adult learners by generation.

Generation	Time frame	Training preferences
Baby boomers	1946–1964	• Transfer learning to jobs • Formal classrooms • Mixed reception on computers and Internet use
Gen X	1965–1979	• Personal interaction such as coaching and mentoring • Interactions with manager and peers • e-learning
Gen Y	1980–1999	• Virtual learning • Threaded discussions • Social media

Source: Compiled from Carnes (2011).

Characteristics That Affect Adult Learners

Trainers need to be aware of any characteristics of adult learners that might impede their ability to learn or the trainer's ability to transfer knowledge and skills. Several key characteristics to be mindful of include hearing, seeing, memory, and specific learning disabilities. All of these just mentioned may be age-related in nature. In addition, specific learning disabilities can affect anyone of any age. For example, consider dyslexia as one type of learning disability.

EVALUATION

Describe various techniques to evaluate training, including evaluation planning, feedback surveys, pre-training and post-training testing. (Understand)

Body of Knowledge III.D.3

Management has a tendency to believe that when people have completed training, they are "trained."

—Howard Gitlow and Shelly Gitlow

Evaluating and Selecting Training Materials and Resources

Although some organizations may do well at conducting a training needs analysis and developing a training plan, most organizations invoke minimal effort when it comes to evaluating and selecting training materials and resources. This occurs for one or more of the following reasons:

- They simply don't know how.
- It was overlooked and wasn't planned into the budget.
- It takes time and effort, and after all, it's the employee's responsibility to learn.
- If purchased, they assume that the high cost of training materials makes them high caliber.
- If developed in-house, "we only do good work, so why do we need to evaluate it?"
- They may use existing materials with some modifications due to cost constraints.

Evaluating and selecting training materials and resources does take considerable planning and effort.

To help organizations more effectively evaluate and select training materials and resources, the following checklist has been developed:

- *Language*
  - Is the language at the appropriate level for the target group?
  - Is the spelling correct?
  - Is the grammar correct?
  - Are idioms, slang, or excessive jargon used?
- *Accuracy*
  - Is the material technically accurate?
  - Is the material current?
  - Is the math correct?
- *Format*
  - Is the material consistent and uniform in design and presentation format?
  - Is the format easy to follow?
  - Is the material too dense?
  - Is there a mixture of words, illustrations, and graphics?
  - Is the material sufficiently modularized both in content and in time?
- *Progression*
  - Is an adequate foundation presented?
  - Does the material flow logically?

- *Infrastructure*
  - Does the material stand alone?
  - Are supplementary materials required?
  - Is an instructor required, and if so, how many?
- *Content*
  - Is the content unambiguous?
  - Is the material presented without bias?
  - Does the content support the learning objectives and outcomes?
  - Is the material consistent with other material that either exists or is under consideration?
- *Development and outcomes*
  - Does the material address different learning styles?
  - Does the material address different generational preferences?
  - Will the material meet the objectives established by the training plan?
  - Will the material permit the target group to achieve the desired level of competency?
  - What target groups are suitable for the training materials?
  - Do the training materials require solution manuals, and if so, are they provided?
- *Responsibilities and ownership*
  - Who is responsible for updating the material (if the material is purchased)?
  - How often is the material updated (assuming the material is purchased and the supplier is responsible for updates)?
  - How are errors in the materials handled (assuming the material is purchased and the supplier is responsible for updates)?
  - Who owns the training materials (if delivered by a third party)?
  - Is there a one-time purchasing fee or recurring licensing fees (if purchased from a third party)?
- *Delivery*
  - Do instructors require certification?
  - Are the instructors knowledgeable in the subject matter?

  - Is the material available for delivery in different mediums (that is, classroom, electronic, and so on)?
  - Is the supplier accredited (assuming a third-party trainer and accreditation is required)?
  - Do the training materials require the training to take place in a special environment or require special facilities, tools, audiovisual equipment, computers, or software?
  - How much notification is required if delivered by a third party?
- *Evaluations*
  - Who evaluates the training materials and resources, and what are their qualifications for doing so?
  - How are students evaluated against training objectives?
  - How are the instructors evaluated?
  - Has the material been tested previously, and if so, what were the results?
- *Duration*
  - What is the course duration?
  - Is the course contiguous, periodic, or ad hoc (for example, three days in a row, the first week of every month for four months, or as time permits, respectively)?

A weighted criteria approach can be used for evaluating and selecting training material. Further, it would provide a more objective basis to the evaluation process that would facilitate meaningful discussions among the evaluators. Evaluators would now have a quantitative basis for justifying their decisions.

Consider the above checklist to be a starting point. Evaluating and selecting training materials and resources can take a significant amount of time and effort. Consequently, it should be incorporated into the training plan schedule and budget. Also, remember that not everyone is capable of evaluating training material. This little tidbit was included in the checklist and should not be overlooked: "Who is qualified to evaluate the training material?" should be addressed early on and not left to whomever is available.

Data Collection Methods

Numerous methods exist for collecting data for evaluating training. These include:

- Follow-up survey and questionnaires
- Observations on the job
- Follow-up interviews

- Follow-focus groups
- Testing (pre- and post-training)
- Performance monitoring
- Review of operational data (for example, scrap, rework)

In addition to the above, consider the use of operational data such as scrap and rework data or whatever is relevant within your organization. An excellent starting point for determining the data appropriate for any given job is the job description. Though not containing the hard data itself, job descriptions might serve as a pointer to the root source.

Part IV

Define

The purpose of this phase is provide a compelling business case appropriately scoped, complete with SMART goals, and linked to a strategic plan. Common tools used in this phase are shown in Table IV.1.

As all Lean Six Sigma professionals know, DMAIC is the de facto methodology for improving processes using Lean Six Sigma. The reason for this is quite simple. The methodology is:

- Easy to understand
- Logical
- Complete

Further, it is sufficiently general to encompass virtually all improvement situations whether they are business or personal in nature.

When Black Belts first learn about DMAIC, they, for the most part, see it depicted in a linear manner as shown in the figure above. Unfortunately, this is an illusion that leads inexperienced Black Belts to plow ahead from phase to phase, steamrolling weak champions/sponsors, particularly if the Black Belts lack the support of a Master Black Belt coach. When this occurs, projects either fail or results are not sustained.

Table IV.1 Common tools used in *define* phase.

5 whys	*Data collection plan*	*Prioritization matrix*
Activity network diagrams	*Failure mode and effects analysis (FMEA)*	Process decision program chart
Advanced quality planning	*Flowchart/process mapping (as is)*	Project charter
Affinity diagrams	Focus groups	*Project management*
Auditing	*Force-field analysis*	Project scope
Benchmarking	*Gantt chart*	*Project tracking*
Brainstorming	*Interrelationship digraphs*	Quality function deployment (QFD)
Cause-and-effect diagrams	Kano model	*Run charts*
Check sheets	Matrix diagrams	Sampling
Communication plan	*Meeting minutes*	Stakeholder analysis
Control charts	*Multivoting*	*Suppliers–inputs–process–outputs–customers (SIPOC)*
Critical-to-quality (CTQ) tree	*Nominal group technique*	*Tollgate review*
Customer feedback	*Pareto charts*	*Tree diagrams*
Customer identification	*Program evaluation and review technique (PERT)*	*Y=f (X)*
Customer interviews		
Data collection		

Note: Tools shown in *italic* are used in more than one phase.

Figure IV.1 depicts DMAIC in the real world. As we all know, tollgates occur after each phase. Diligent application of the proper tools and, above all, significant preparation are required to pass them. Passing them is not a given. Notice that up to and including the *analyze* phase, the failure of a tollgate could set the project back one or two phases. This usually occurs when knowledge contrary to expectations is gained in a phase. In this case, the project team may be forced to retreat all the way back to *define* in order to restate the problem. Beyond the analyze phase, a failed tollgate usually results in correcting or performing additional work within the phase. However, at any phase, a project may be terminated. Generally, this will happen when projects are no longer synchronized with strategy, champions/sponsors have lost interest or transferred, or the potential savings are less than expected.

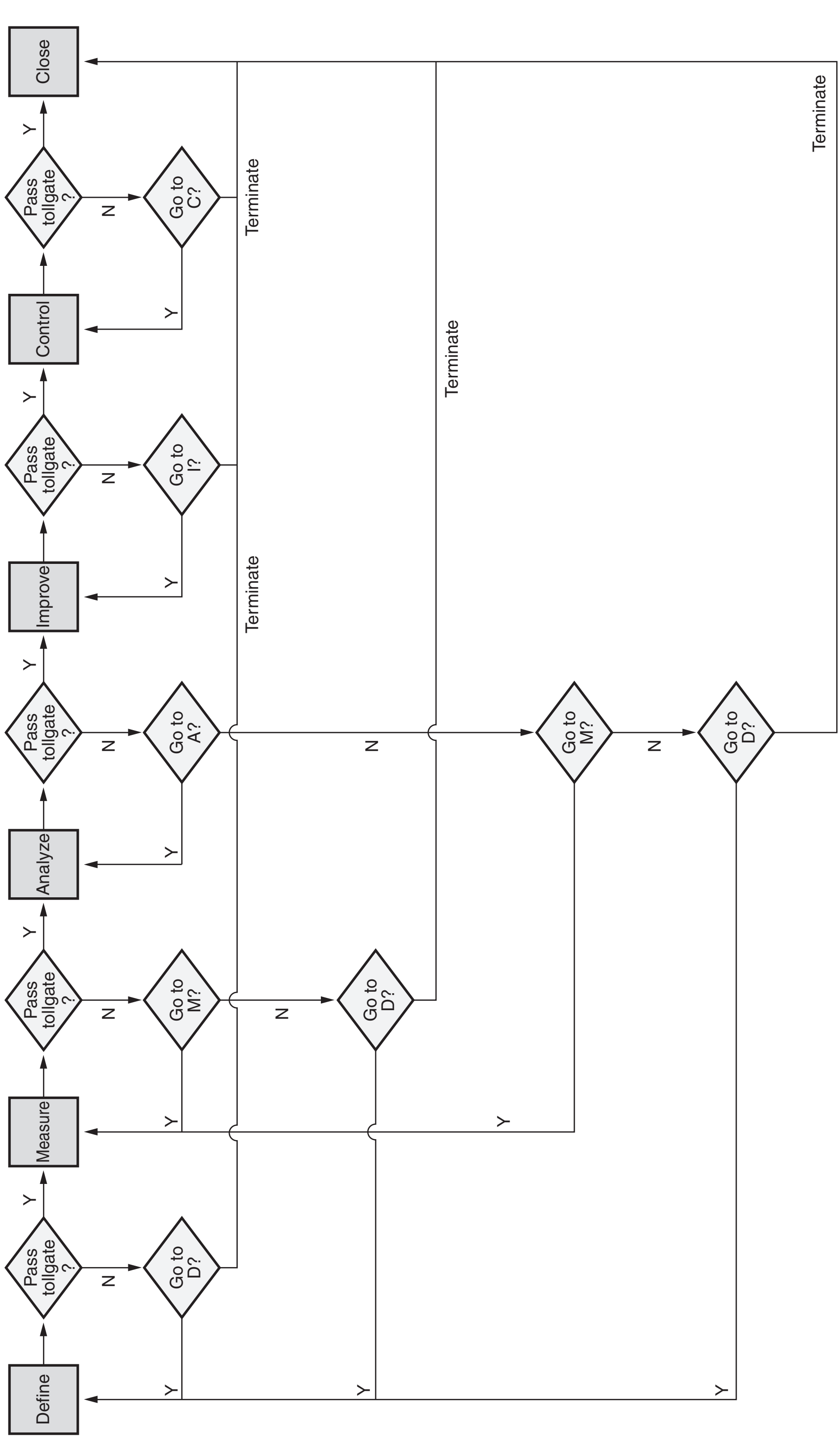

Figure IV.1 DMAIC in the real world.

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Chapter 10

Voice of the Customer

The voice of the customer guides world-class leaders' every action and decision.

—Yu Sang Chang, George Labovitz, and Victor Rosansky

In contrast to natural process limits, specification limits are customer determined or derived from customer requirements and are used to define acceptable levels of process performance. Said to be the "*voice of the customer*" (VOC), specification limits may be one-sided (that is, upper or lower) or two-sided. The difference between the upper and lower specification limits is known as the *tolerance*.

In some cases, customers provide specifications for products or services explicitly. This is often the situation when the customer is the Department of Defense or any of the military branches.

In others, customers express requirements in value terms, the components that influence the buy decision, such as price, product quality, innovation, service quality, company image, and reputation.

In still other cases, customers may spotlight only their needs or wants, thus leaving it up to the organization to translate them into internal specifications.

Tools such as quality function deployment, critical-to-quality analysis (also known as *customer requirements tree analysis*), and so on, often help with the last two situations.

CUSTOMER IDENTIFICATION

> Identify and segment customers and show how a project will impact both internal and external customers. (Apply)
>
> **Body of Knowledge IV.A.1**

An early step in any project is to seek the voice of the customer (VOC). The internal and external customers of a project include all those who are impacted by the project. For most projects, the customers can be grouped into categories and/or segments, with segments being a further breakdown within categories. This categorization and/or segmentation is driven by the similarity of customer requirements and often includes the following:

- Internal and external to the organization
- Customer demographics:
  - Age groups, especially for consumer goods
  - Geographical location, including climate, language and ethnic issues, and shipping considerations
- Industry types (for example, the project might impact customers in construction, agriculture, and stationary equipment industries)

When possible, a list of customers, or stakeholders in general, within a category and/or segment should be constructed. When a project team proposes changes of any type, all stakeholders, internal and external, must be consulted, or at a minimum, the stakeholders' concerns must be represented. This is usually accomplished best through the stakeholder analysis discussed in Chapter 3.

It is easy to underestimate the value of understanding and providing for customers' needs. For without customers, we have nothing!

CUSTOMER DATA COLLECTION

> Identify and select appropriate data collection methods (e.g., surveys, focus groups, interviews, observations) to gather voice of the customer data. Ensure the data collection methods used are reviewed for validity and reliability. (Analyze)
>
> **Body of Knowledge IV.A.2**

Data Validity and Reliability

Because data errors are generally hard to detect, data validity and reliability considerations should be at the forefront of any data collection activity.

Data validity refers to the general correctness of a data set. *Data validation*, on the other hand, refers to the process of validating data against a set of documented acceptance criteria, usually as the data are entered.

Common data validation checks include numeric versus alpha character checks, range checks, logic checks, spell checks, comparison checks, lookup checks, and so on.

When the data collection instrument is a survey, pre-testing can help reduce data validity issues that might arise, by eliminating confusing questions.

EXAMPLE 10.1

The data field for the number of items sold per day is three characters long. The data are validated upon entry to ensure that a numeric value is entered between 0 (zero) and 999, inclusive.

If the number of items sold on a given day is 345, but is entered as 258, the data could only be validated as numeric. We could not validate 345 versus 258 since it would not be possible to distinguish one valid numeric entry from another.

However, if sales per day never exceeds 500 and an entry of 572 is entered, it would be possible to validate an entry exceeding a maximum value.

Data reliability refers to the overall consistency, stability, or dependability of a data set. In the context of data, reliability can be thought of as the extent to which data are reproducible.

Reliability in data collection instruments such as surveys, for example, means they will produce the same results regardless of when administered. Reliability problems in surveys generally occur when respondents:

- Do not understand the question
- Are asked about something they don't recall
- Are asked about something that is not relevant to them
- Are asked compound questions (that is, questions containing "and" or "or")

Reliability problems generally call for revision and improvement of the data collection instrument.

EXAMPLE 10.2

The true weight of an individual is 150 pounds. The individual weighs himself on a scale (our data collection instrument) daily. Each day the scale reads 150 pounds. The scale is said to be both valid and reliable. It is valid because it gives the correct weight and it is reliable because it gives the same weight consistently.

One day, the individual, who still weighs 150 pounds, gets on the scale and it reads 170 pounds. Each day thereafter, the scale reads 145 pounds. Although the scale is now biased, it is still reliable because it is consistent, stable, and dependable. However, it is not valid anymore because it no longer gives the correct weight.

One day, the individual, who still weighs 150 pounds, gets on the scale and it reads 170 pounds. Each day thereafter, the scale reads a different value than 150 pounds. The scale is no longer valid or reliable. It is not reliable because it is not consistent, stable, or dependable in providing the weight.

There is a key aspect about data validity and reliability that should be noted. That is, you can have a data collection instrument that is reliable but not valid, but you cannot have a valid instrument that is not reliable.

Regardless of the data collection instrument, the concepts of data validity and reliability must be considered. Furthermore, it is often useful to not only validate each data collection instrument unto itself, but also validate each against the others. For example, does the survey instrument provide data similar to that which the telephone or focus group data provides? If it doesn't, is there a logical reason for it? The ability to cross-validate data across multiple listening posts can provide increased assurance of the validity of critical customer issues and complaints and can create a sense of urgency in otherwise complacent organizations.

Designing a Data Collection System

Many organizations collect customer data but lack a systematic approach for doing so. Important data are often collected independently via many different departments, but never shared, collated, reconciled, or analyzed to draw meaningful conclusions that drive action.

When designing a customer data collection system, the following key considerations should be at the forefront regardless of the data collection instrument used:

- *Collect data as objectively and consistently as possible.* Essentially, minimize measurement error. This concept should permeate all data collection mechanisms such as surveys, data collection check sheets, and so on. Consider using scripts for interviews to ensure that the same questions are asked the same way each time.

- *Collect data at the right level of granularity.* Consider what type of analysis will be done, what questions you are trying to answer, or the type of requirements you are trying to discover. If data are collected at too high a level, information is lost. Data collected at too low a level is meaningless. Fortunately, it often can be summarized to an appropriate higher level for analysis. However, this creates additional work and consumes time.

- *Consider independent sources of data collection.* Multiple data collection mechanisms or *listening posts* should be used, as shown in Figure 10.1. This permits data regarding each customer category or segment to be validated. If the data cannot be validated, this is generally a signal that improvement in the mechanisms (for example, revised questions) is required.

- *Use multiple media for collecting data.* Different customer categories or segments may favor different media for providing feedback, as shown in Figure 10.2. The key is to determine how to best align listening posts with the most appropriate media.

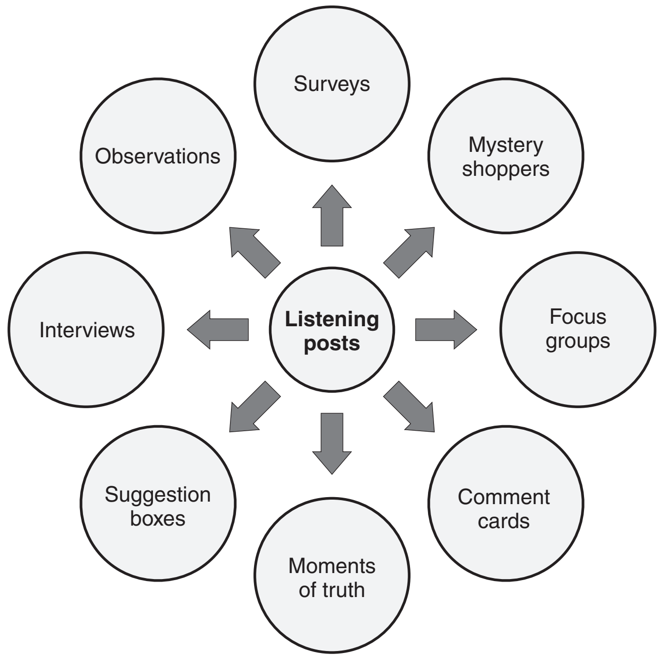

Figure 10.1 Examples of common listening posts.

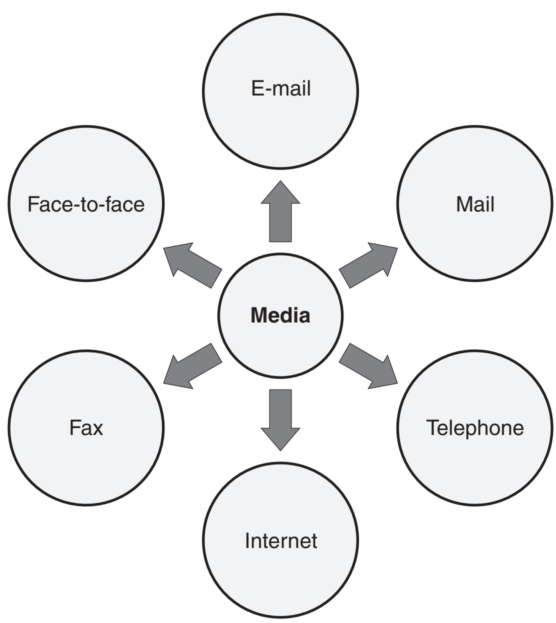

Figure 10.2 Examples of media used for listening posts.

- *Make it easy for the customer.* Customers must have easy access to the organization to provide feedback. Similarly to using multiple media, this bullet speaks to the ease with which the customer has access to the organization.

  For example, suppose an organization uses an automated telephone system for collecting customer feedback. Customers may become frustrated with attempting to determine the correct routing path to take to provide the feedback. Consequently, they decide that their input and time is not worth the effort. As a result, the organization loses a source of valuable information regarding the performance of its products and services. Similarly, surveys that are too long or inappropriately designed may also cause customers to abandon participation.

The above points focus mainly on collecting customer data from the organization's own internal sources of information. Such data should include both internal and external customers. Table 10.1 identifies these multiples sources of customer data. Notice that there are many internal sources of customer data. Actually, many organizations are data rich with regard to such information but often fail to tap into it or even recognize it as an important source.

Unfortunately, the linear, project-by-project approach limits an organization's ability to fully understand and exploit customer data. Consequently, all customer data must be viewed in its entirety in order for meaningful conclusions to surface. Results from each data source should be compared to determine patterns of reinforcement or contradiction for each customer category or segment for each requirement, need, or want. Only now can an organization begin to translate requirements, needs, or wants into specifications. An added value of this analysis is the insight gained into the effectiveness of the listening posts. Listening posts that are performing poorly can be improved, modified, or even eliminated.

Table 10.1 Sources of customer data.

Internal	**External**
Complaints	Research (that is, magazines, newspapers, trade journals)
Claim resolutions	Public information (that is, customers' and competitors' annual reports)
Warranty and guarantee usage	Advertising media (that is, television, radio, websites)
Service records	Industry market research
Customer-contact employees	Customers of competitors
Listening post data	Industry conferences and forums
Market research	
Transaction data	
Defection data	

CUSTOMER REQUIREMENTS

Define, select, and apply appropriate tools to determine customer needs and requirements, including critical-to-X (CTX when 'X' can be quality, cost, safety, etc.), CTQ tree, quality function deployment (QFD), supplier, input, process, output, customer (SIPOC) and Kano model. (Analyze)

Body of Knowledge IV.A.3

When the customer says, "That ain't it," then that, in fact, ain't it.

—John Guaspari

The best customer data collection and analysis is useless unless there is a system to use the data to effect changes. This system should study each item of customer feedback to determine which processes, product, and/or services will be impacted. The volume and/or urgency of customer concerns will help determine the requirements that customers deem critical. Some type of criticality rating system should be applied to the feedback data (for example, a phone call from a customer is rated higher than a response to a questionnaire).

For example, suppose that analysis of customer data identifies six areas where product quality is compromised. With the help of the criticality rating scale, perhaps two of the six would be deemed important enough to motivate immediate improvement projects.

Critical to X (CTx) Requirements

The concept behind *critical to x* (CTx), where *x* is a variable, is simply that *x* is an area or areas of impact on the customer. Critical customer requirements are usually expressed as expectations or needs, but not necessarily in quantifiable terms. Process CTx's represent measurable product or process performance characteristics and act to quantify the critical customer requirements. Let's look at some of the more familiar "*x*'s":

- *Critical-to-quality (CTQ).* CTQs may include the physical dimensions of height, width, depth, and weight. They may even include the electrical characteristics of a product, such as impedance. They describe the requirements of quality in general terms but lack the specificity to be measurable. However, CTQs can be translated into measurable terms or critical requirements through the inclusion of customer-defined specifications. For example, the customer may specify that the weight of the product be between 15 and 20 pounds. Products outside these specifications will not meet the customer's critical requirements.

- *Critical-to-cost (CTC).* CTCs are similar to CTQs but deal exclusively with the impact of cost on the customer. However, CTCs and CTQs may be similar, yet stated by the customer for different reasons. For example, the weight of the product may also serve as a CTC. Again, the customer might require the weight to be between 15 and 20 pounds. However, the specification is necessary to achieve an impact on cost. Products heavier than 20 pounds may require more power consumption, thus increasing the cost.
- *Critical-to-safety (CTS).* CTSs are stated customer needs regarding the safety of the product or process. Once more, consider the previous example of product weight. Product weight is the CTS variable. Though identical to the CTQ and the CTC, it is identified by the customer because any product heavier than 20 pounds has been shown to introduce a high rate of back injuries. Product weights less than 15 pounds have been shown to introduce a high rate of back injuries as well since the lower weight induces the operator to pick up and transport multiple products. So far, we have seen that CTx's may be identical across categories, but for different reasons.
- *Critical-to-process (CTP).* CTPs are typically key process input variables. In terms of the process, we might think of CTPs as the critical X's in the $Y = f(X)$ equation. We may know and understand what the X's are (for example, temperature, pressure, humidity), but not necessarily their specific setting or level. Once the settings are determined or specified, they can be set to achieve a consistent Y.
- *Critical-to-delivery (CTD).* CTDs represent those customers with stated needs regarding delivery. Typically, one thinks of late deliveries. However, delivering too early may be a problem for some customers, as it represents excess inventory requiring payment before the need for such inventory arises. CTDs are translated into critical customer requirements through the quantification of these impact areas. For example, the customer may specify that products be received no earlier than two days prior to the delivery date and no later than one day after the delivery date.

Projects aligned with CTx's, and subsequently with critical customer requirements, have the biggest impact on the customer and often on the business directly. A useful tool for ensuring this alignment or translation from critical customer requirement to CTx is the tree diagram—an example of which is shown in Figure 10.3.

Critical-to-Quality Flow-Down

Customer satisfaction generally falls into three dimensions:

- Quality
- Delivery
- Cost

Cost and delivery are easy to quantify. Customers are willing to pay *x* per item and expect its delivery *y* days after they place the order. Quantifying quality characteristics presents more of a challenge.

One of the tools devised to help is called the *critical-to-quality* (CTQ) *flow-down*. Its purpose is to start with the high-level strategic goal of customer satisfaction and determine how this goal "flows down" into measurable goals. The nomenclature for the various levels is illustrated in Figure 10.4.

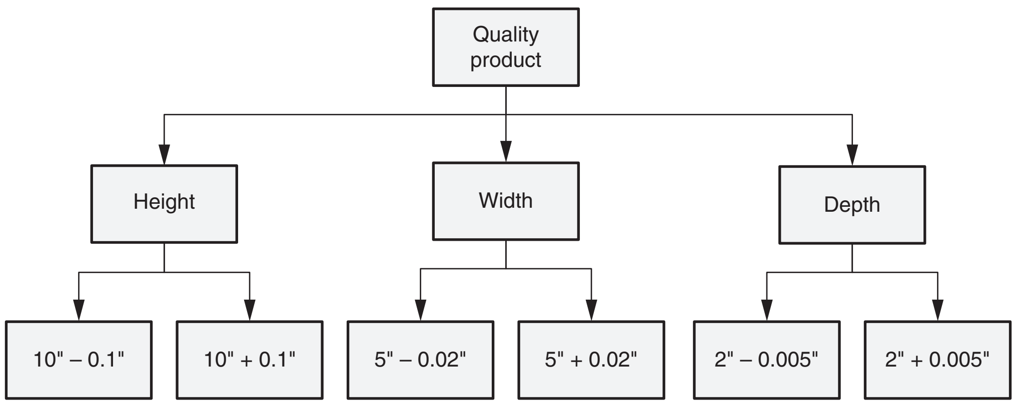

Figure 10.3 Example of a CTQ tree diagram.

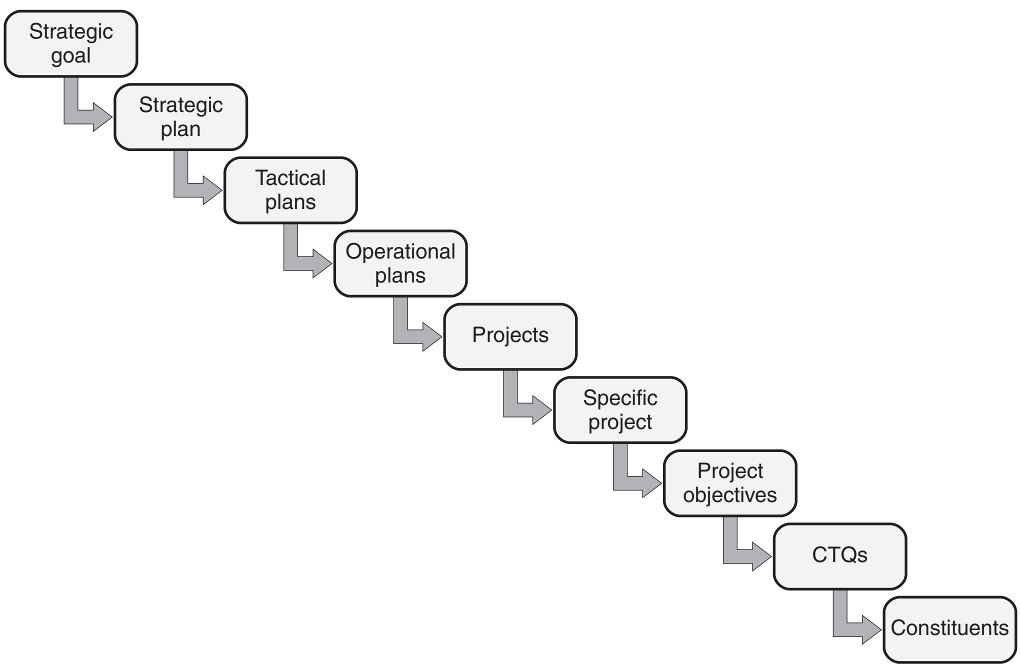

Figure 10.4 CTQ flow-down.

EXAMPLE 10.3

High on Mid America Landscaping's list of strategic goals is customer satisfaction. A survey of past customers raised the need to establish a project team to ensure that customers understand the nature of the plants and trees they are purchasing. The team identifies a number of constituent parts to the problem. These constituents will guide the team. The CTQ flow-down diagram is shown in Figure 10.5.

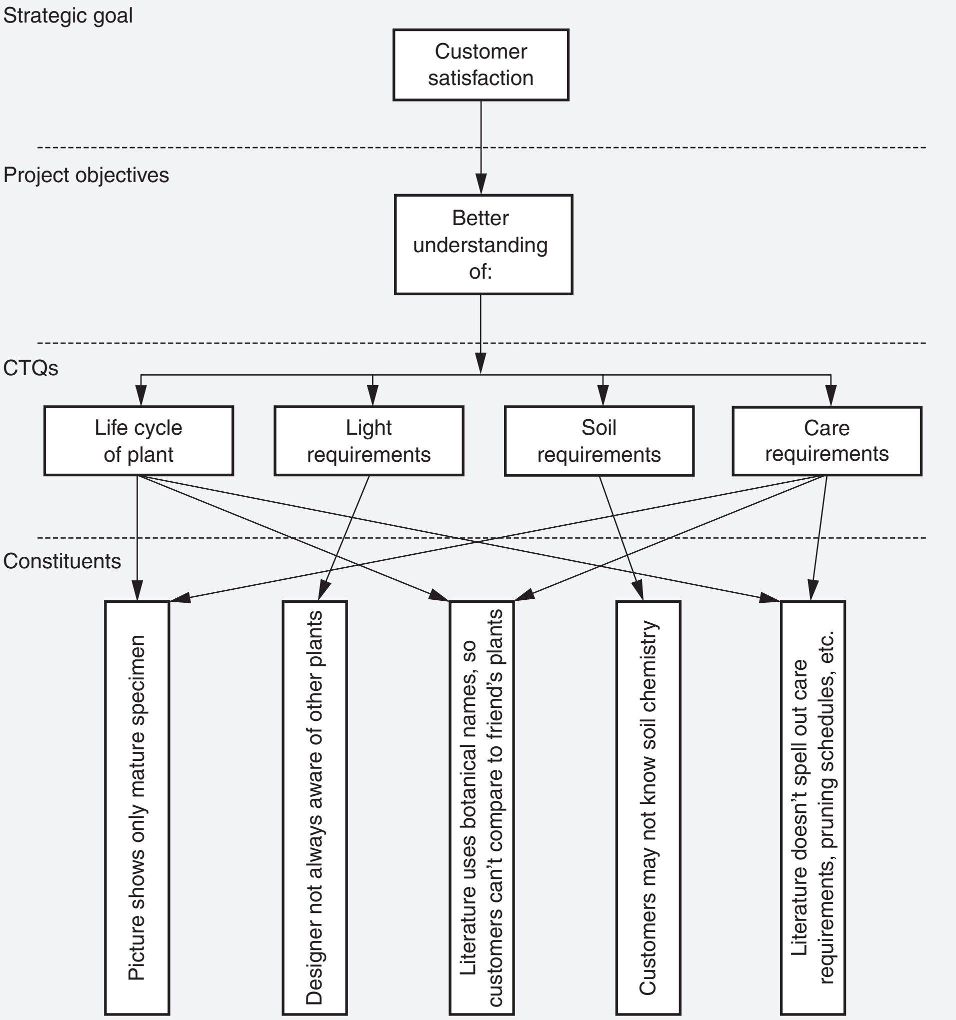

Figure 10.5 Example of a CTQ flow-down.

Quality Function Deployment

Quality function deployment (QFD) provides a process for planning new or redesigned products and services. The input to the process is the VOC. The QFD process requires that a team discover the needs and desires of its customer and study the organization's response to these needs and desires. The QFD matrix helps illustrate the linkage between the VOC and the resulting technical requirements.

A QFD matrix consists of several parts. There is no standard format matrix or key for the symbols, but the example shown in Figure 10.6 is typical. A map of the various parts of Figure 10.6 is shown in Figure 10.7.

Let's look at each of area of Figure 10.7 individually:

- *Area 1: Customer requirements.* This area contains the customer requirements, which are developed from analysis of the VOC and often include a scale reflecting the importance of the individual entries.
- *Area 2: Technical requirements.* This area contains the technical requirements that are established in response to the customer requirements. The symbols on the top line in this section indicate whether lower (↓) or higher (↑) is better. A circle indicates that target is best.
- *Area 3: Relationship matrix.* This area displays the strength of the relationship between the technical requirements and the customer requirements. Various symbols can be used here. The most common are shown in Figure 10.6.
- *Area 4: Comparison with competition.* This area plots the comparison with the competition for the customer requirements and is not always shown on QFD matrices.
- *Area 5: Action plans.* This area contains notes or plans concerning improvement activities.
- *Area 6: Comparison with competition.* This area, like area 4, is not always shown on QFD matrices. It plots the comparison with the competition for the technical requirements.
- *Area 7: Target values.* This area contains a list of the target values for each of the technical requirements.
- *Area 8: Co-relationships.* This area shows the co-relationships between the technical requirements. A positive co-relationship indicates that both technical requirements can be improved at the same time. A negative co-relationship indicates that improving one of the technical requirements will worsen the other.

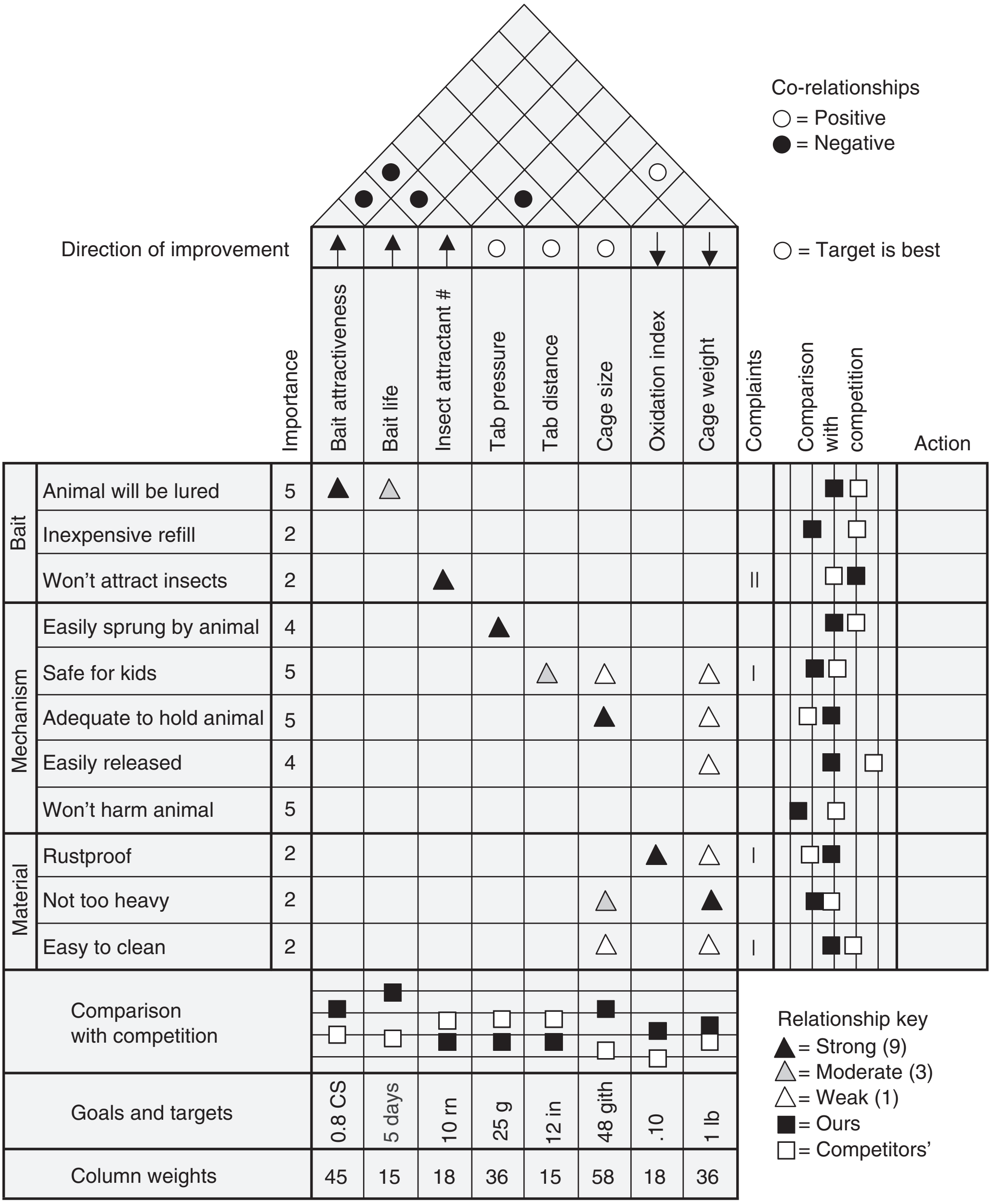

Figure 10.6 Example of a QFD matrix for an animal trap.

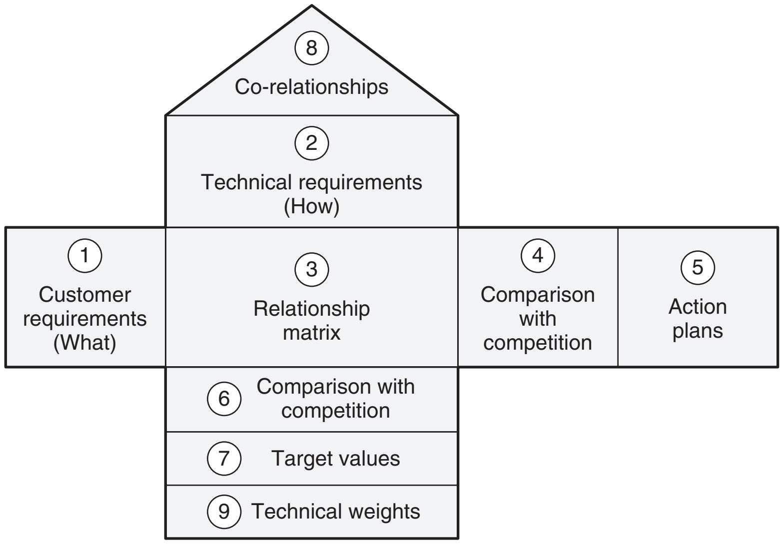

Figure 10.7 The structure of the QFD matrix (sometimes called the house of quality).

- *Area 9: Technical weights.* This area provides the column weights for each of the technical requirements. The column weights indicate the importance of the technical requirements in meeting customer requirements. The value in the "column weights" row is obtained by multiplying the value in the "importance" column in the customer requirements section by values assigned to the symbols in the relationship matrix. These assigned values are arbitrary; in the example, a strong relationship was assigned a 9, moderate 3, and weak 1.

The completed matrix can provide a database for product development, serve as a basis for planning product or process improvements, and suggest opportunities for new or revised product or process introductions.

The customer requirements section is sometimes called the "What," while the technical requirements section is referred to as the "How." This "what versus how" concept can be applied through a series of matrices intended to decompose the technical requirements to lower levels that can eventually be assigned as tasks.

In addition, the basic QFD product-planning matrix can be followed with similar matrices for planning the parts that make up the product and for planning the processes that will produce the parts. This concept is illustrated in Figure 10.8.

If a matrix has more than 25 customer voice lines, it tends to become unmanageable. In such a situation, a convergent tool such as the affinity diagram (see Chapter 13) may be used to condense the list.

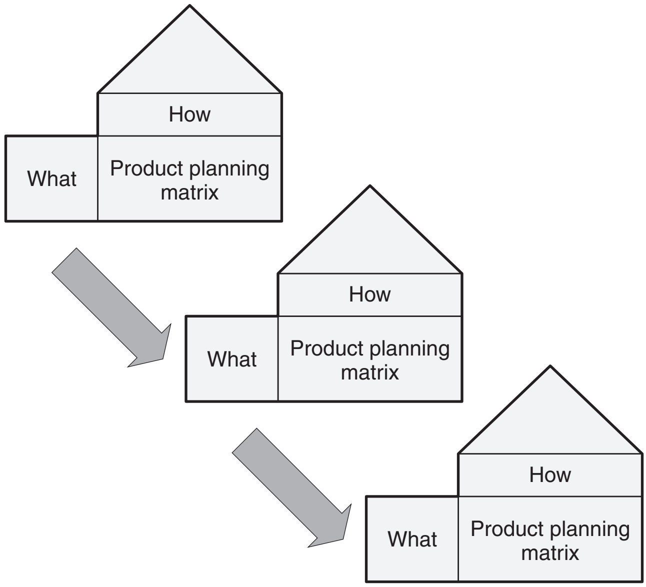

Figure 10.8 Using QFD for product planning and beyond.

Suppliers–Inputs–Process–Outputs–Customers (SIPOC)

The *SIPOC* is another useful tool in the study of processes. The acronym stands for the key elements of a process:

- *Suppliers.* These are the internal and external providers of resources, materials, knowledge, and services.
- *Inputs.* These are the resources, material, services, and knowledge that feed that process. In the context of $Y = f(X)$, these are the X's.
- *Process.* This is the step or sequence of steps that is being investigated, analyzed, and improved.
- *Outputs.* These are the products and services produced by the process. In the context of $Y = f(X)$, these are the Y's.
- *Customers.* These organizations or users receive the final outputs of the process.

Developing a SIPOC constitutes completing the form shown in Figure 10.9. An alternate SIPOC form is shown in Figure 10.10. Note that this form allows specific requirements to be captured on both the inputs and outputs.

A key benefit of the SIPOC is that it is much easier to complete than either a process map or a value stream map. However, SIPOCs can be used as a basis for constructing detailed process maps and value stream maps at future dates.

Furthermore, SIPOCs help identify the voice of the customer as well as provide quick oversight into the initial X's and Y's.

Suppliers	Inputs	Process	Outputs	Customers

Figure 10.9 Example of a common SIPOC form.

SIPOC	
Prepared by:	Date:
Project:	
Project	

Suppliers	Inputs		Process	Outputs		Customers
	Description	Requirements		Description	Requirements	

Figure 10.10 Alternate example of a SIPOC form.
Source: Courtesy of Minitab Quality Companion 3 software, Minitab Inc.

Kano Model

Noriaki Kano's model of customer satisfaction identifies several types of requirements that impact customer satisfaction. They are illustrated in Figure 10.11 and described below:

- *Must-be requirements.* These are the requirements the customer assumes will be there, and when they aren't, the customer will be very dissatisfied. Such requirements are generally unspoken. Note that the line labeled "must-be" in Figure 10.11 is on the dissatisfied side when the requirement is missing. However, as the requirement becomes fulfilled, it has little effect on customer satisfaction.

  Consider, for example, a shirt with a buttonhole that is sewed shut. Customers do not expect buttonholes to be sewed shut. When this occurs, customers are very dissatisfied.

  "Must-be" requirements are also known as *basic* requirements, *expected* requirements, *must-have* requirements, and a whole host of other names.

- *One-dimensional requirements.* These are the requirements in which the customer expects a certain level of fulfillment, and anything exceeding that level increases satisfaction. Such requirements are generally spoken and determined by classic research methods.

  Consider, for example, a computer connection that may be advertised as having a certain data transfer rate. If the rate in use is less than the

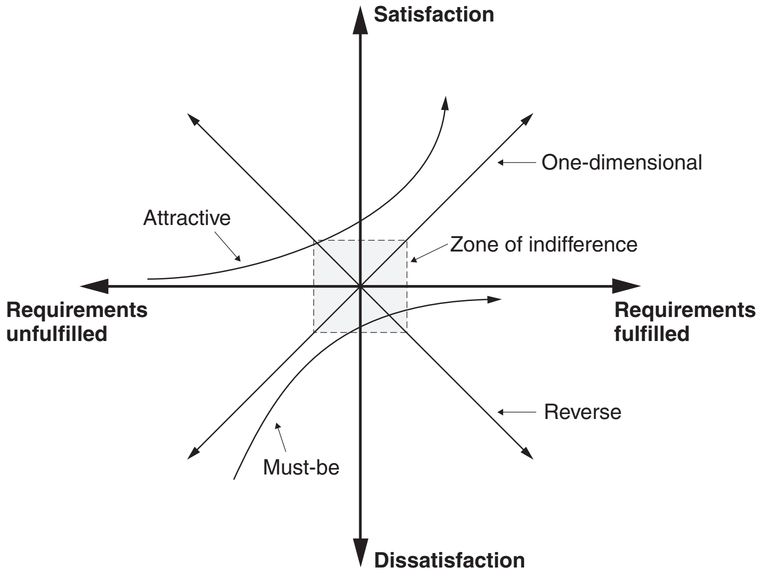

Figure 10.11 Kano model for customer satisfaction.

advertised rate, the customer is dissatisfied. However, transfer rates above the advertised rate will increase the level of satisfaction.

One-dimensional requirements are also known as *performance* requirements, *normal* requirements, *linear* requirements, and *satisfiers*. They are called linear requirements because satisfaction tends to increase linearly with the increase in the fulfillments of requirements.

- *Attractive requirements*. These are the requirements that get to the heart of the customer. When present, they increase satisfaction, but their absence does not cause dissatisfaction, because the customer does not expect them. Consequently, these are unspoken.

  Consider, for example, a set of kitchen cupboards that arrives with an extra corbel because the designer thought it would add a nice touch.

  Attractive requirements are also known as *excitement* requirements, *exciters*, *exciting needs*, and *delighters*. Again, the reader will likely find additional names for attractive requirements in the literature beyond those mentioned here.

- *Indifferent requirements.* These requirements do not cause satisfaction if present, nor do they cause dissatisfaction if absent. This could refer to features the customer didn't expect or features the customer rarely uses. Indifferent requirements are shown as a zone of indifference in Figure 10.11.

Consider, for example, the smartphones of today that are loaded with many features of which most owners of the phones will only use a few.

Indifferent requirements are also known as *indifference* requirements or *neutral* requirements.

- *Reverse requirements.* These are the requirements that refer to features the customer doesn't want and whose presence causes dissatisfaction. Occurrences of these tend to be rare. Such requirements are most often unspoken before the product is developed, but customers are usually very vocal after exposure to them.

  Consider, for example, the initial introduction of seat belts in automobiles. Most owners disliked them. Consequently, they developed a workaround, which was to buckle the belts and tuck them between the seat and back cushions. This prevented any buzzers from sounding or lights from flashing.

  Reverse requirements are also known as *reversal* requirements.

An awareness of these features of customer satisfaction (and dissatisfaction) provides insight and guidance for the organization's goal of ever-improving customer satisfaction.

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Chapter 11

Business Case and Project Charter

The *project charter*, though a simple living document, requires significant preparation time (see Figure 11.1). It serves as a formal contract between the champion and Black Belt, and helps the team stay on track with the goals of the project and organization. There are typically six key elements of a meaningful charter document:

- Business case
- Problem statement

Project identification related information	
Business case	Problem statement
Goal statement	Project scope
Project plan	Project team

Figure 11.1 A basic form of a charter document.

- Goal statement
- Project scope
- Project plan
- Project team

Figure 11.2 illustrates an alternate form of a project charter. Note how this form emphasizes baseline and goal performance and provides the opportunity to link the project directly to customer satisfaction.

Each element of the project charter will now be addressed.

BUSINESS CASE

> Describe business case justification used to support projects. (Understand)
>
> **Body of Knowledge IV.B.1**

The business case identifies the dollars to be saved and establishes how the project aligns to the organization's strategies. It also states how the organization will fare better when the project reaches its goals. Although the business case generally appears first in a project charter, it is often easier to write the business case once the problem statement, goal statement, and project scope are defined.

EXAMPLE 11.1

The completion of this project will reduce the rework of product B on production line A by 57% and latent defects by more than 82%. Consequently, we expect to reduce in-process inventory by 35%, achieve on-time delivery to our top three customers, and eliminate their product complaints. Hard dollars savings are expected to be $2.75M in the first 12 months and efficiency savings are projected to be $65K. Most of the hard dollars savings are from eliminating a $1.9M capital expenditure, while the balance is from eliminating two hiring requisitions that were planned and opened for quality engineers to address these quality problems. Furthermore, the increased production yield will help advance strategy D of entering the emerging market in the southern region.

Project Charter

Project Authorization

Organization:	Champion:	Process Owner:

Project:	Project #:
Project	

Problem Statement:

Project Objective:

Estimated Defect Level:	Initial Goal:	Estimated Benefits:
Approval Date:	**Champion Signature:**	**Process Owner Signature:**
Estimated Completion Date:	**Project Leader:**	**Financial Analyst:**

Project Team

Name	Role	Comments	Phone

Project Definition and Scoping

Metrics (unit of measure):

Critical to Satisfaction (linkage to customer):

Defect Definition (include opportunity):

Scope of Project:

Figure 11.2 Example of a project charter document.

Source: Courtesy of Minitab Quality Companion 3 software, Minitab Inc.

PROBLEM STATEMENT

> Develop a project problem statement and evaluate it in relation to baseline performance and improvement goals. (Evaluate)
>
> **Body of Knowledge IV.B.2**

The problem statement identifies what is not working, or where the pain point lies, and should be a concise explanation of a current state that adversely impacts the organization. If the pain point cannot be identified, it is unlikely that a meaningful project can be defined. This critical statement is often glossed over.

In many ways, the problem definition phase is the most important phase of the DMAIC cycle. If this phase is not done thoroughly, teams may move on to subsequent phases only to stall and cycle back through *define*. This phase should be emphasized, and teams should not move forward until the sponsor signs off on it.

In the goal section, we'll learn how to make meaningful goal statements so that baseline performance can be established.

EXAMPLE 11.2

Consider the following problem statements:

- The increased reject rate for product X during the second quarter of this fiscal year has caused us to lose 2% of our market share to competitor B.
- The rising cost of product B since the beginning of this year has caused sales to drop 19% over the third quarter of last year.

If you can't find the pain in the statement, then the problem statement is not well written. In both of the examples above, the pain is well defined. In the first example, the pain is that competitors are taking some of the market share. In the second example, the high cost of product B is making the company lose its competitiveness.

Notice the specificity of the wording of the problem statements. In particular, they are worded such that they are directly linked to determining baseline performance. In the first example, we can determine both the reject rate and the market share at the beginning of the second quarter. In the second example, we can determine the cost of product B at the beginning of the year and the sales in the third quarter of the previous year.

PROJECT SCOPE

> Develop and review project boundaries to ensure that the project has value to the customer. (Analyze)
>
> **Body of Knowledge IV.B.3**

The project scope specifies the boundaries of the project. The most common scope limitations are budget, time, authority, and resources. Additional aspects of the scope are addressed below.

Lean Six Sigma projects sometimes suffer from a disagreement between the project team members and the sponsor regarding project boundaries. The process of defining scope, of course, can result in problems of extremes:

- Project definitions with scopes that are too broad may lead a team into a morass of connecting issues and associated problems beyond the team's resources. For example, "Improve customer satisfaction" with a complex product or service.
- Project boundaries that are set too narrow could restrict teams from finding root causes. For example, "Improve customer satisfaction by reducing variation in plating thickness" implies a restriction from looking at machining processes, which could be the root cause of the customer problems.

The tendency is to err on the side of making the scope too broad rather than too narrow. Extra attention, effort, and time may be needed to ensure a proper scope. Don't shortcut the process.

Several tools are available to assist in setting a project scope. These include:

- *Pareto charts* to help in the prioritizing processes and sometimes in support of project justification
- *Cause-and-effect diagrams* to broaden the thinking within set categories
- *Affinity diagrams* to show linkages between the project and other projects or processes
- *Process maps* to provide visualization and perhaps illuminate obvious project boundaries and relationships.

Generally, there are two methods useful in defining projects' scopes. These are the:

- Dimensional method
- Functional method

EXAMPLE 11.3

The project scope will entail process A, employee category B, in site C, who produce products D, E, and F for customer G. All other employee categories will be excluded.

Dimensional Method

This approach identifies an initial eight possible dimensions for characterizing a project scope:

- *Process.* Processes receive input from other processes and feed output to other processes. Tightly bound the process or subprocess associated with the project.
- *Demographics.* This would consider factors such as employee categorizations, gender, age, education, job level, and so on.
- *Relationships.* This would include such entities as suppliers, customers, contract personnel, and so on.
- *Organizational*. Which business units, divisions, sites, or departments are included?
- *Systems*. Which manual or computerized systems are included? For example, those using system A.
- *Geographical.* Which country, region, or site is included?
- *Customer.* Which segmentation or category will be considered?
- *Combinations.* Any combination of the aforementioned.

Note: This list is subject to expansion. Readers are encouraged to submit ideas to authors@asq.org.

The advantage of this method is that it communicates the scope in straightforward and easy-to-understand terms.

Unfortunately, this method does not focus in on process inputs as well as the functional method.

In addition to defining what is in scope, it is often useful to define what is out of scope. Although one would seem to define the other, experience has shown that what is out of scope is frequently overlooked or not understood fully unless explicitly stated. Many sponsors, team members, and other stakeholders often fail to make this important connection. For example, process B is in scope, while processes A and C are out of scope.

Unwieldy scopes are one of the most frequent reasons cited for the demise of projects. When the scope is too large to be completed within the project plan time frame, or additional resources are not available, there may be a reluctance to go back and re-scope the project. When it is too small, projected savings may be overstated. Note: The project charter should be considered a living document and adjusted as information is gained and learning takes place.

Functional Method

The functional method is rooted more in logic and functional relationships between input and output variables than the dimensional method. The intent is to isolate critical input variables. Other names for this method include *process mapping decomposition*, x–y *diagrams*, Y = f(X), and *big* Y *exercises*.

This method is best illustrated by an example provided by Lynch, et al. (2003). The example deals with the issue of poor electric motor reliability. Figure 11.3 portrays the functional $Y = f(X)$ breakdown. Each of the X's should be supported by operational data and may be supported by subject matter expert opinions.

The breakdown continues, in this example, until the smallest meaningful part with continuous data is identified. Keep in mind that another Y from the brush reliability equation could have been identified as well. This would have yielded an additional, but concurrent, project.

The disadvantage of using this approach lies in the difficulty of explaining it to senior leaders and other levels of management.

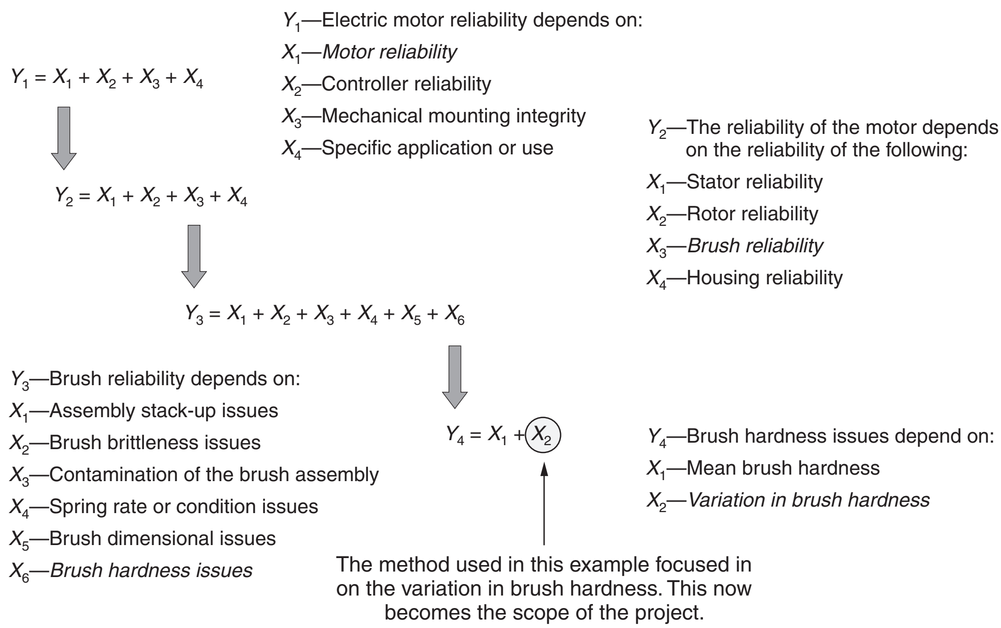

Figure 11.3 Example of project scoping using the cause-and-effect method–$Y = f(X)$ format.
Source: Lynch et al. (2003).

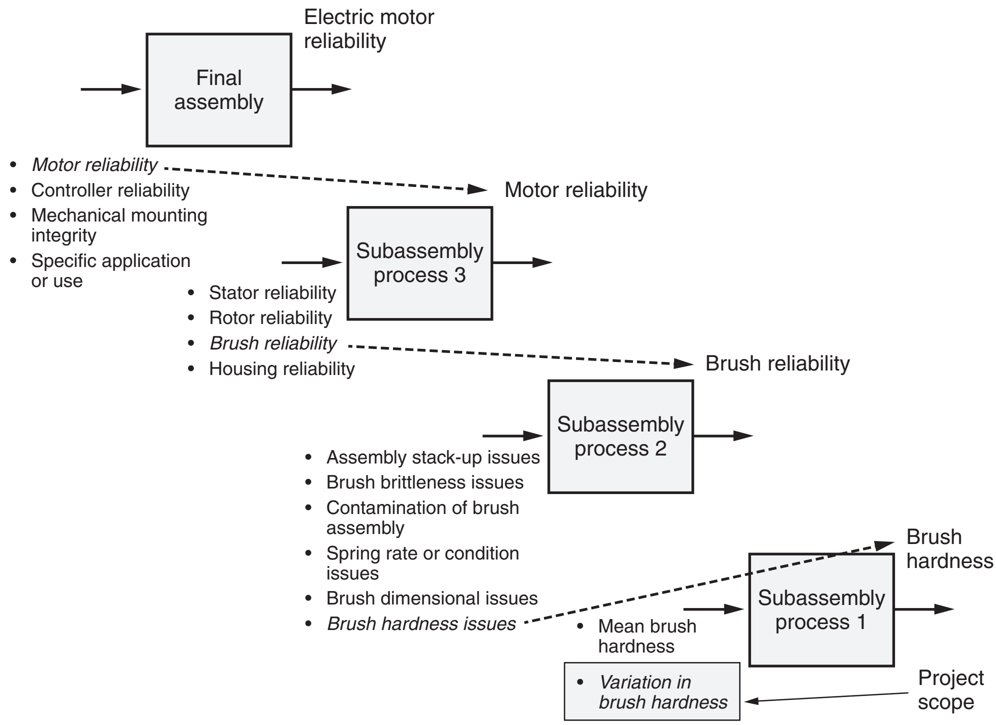

Figure 11.4 Example of project scoping using the cause-and-effect method–process mapping decomposition format.

Figure 11.4 is the same as Figure 11.3 but uses a process mapping decomposition breakdown format to isolate critical input variables. This format may be more appealing to senior leaders and management in general.

GOALS AND OBJECTIVES

> Identify SMART (specific, measurable, actionable, relevant and time bound) goals and objectives on the basis of the project's problem statement and scope. (Analyze)
>
> **Body of Knowledge IV.B.4**

The *goal statement* identifies the project's objectives and targets and how the success of the project will be measured.

Goal statements must define at least one success (that is, primary) measure. However, baseline performance must be established for each primary measure. Frequently, primary measures are accompanied by secondary or countermeasures to ensure that when primary measures are improved, other measures (that is, secondary measures) do not suffer in the process. Likewise, baseline performance must be established for each secondary measure used.

Remember, goals should be SMART. SMART was discussed in Chapter 5, but will be recapped briefly here:

- *Specific*. This is not the place to be generic or philosophic. Nail down the goal.
- *Measurable*. Unless the team has measurable goals, they won't know whether they are making progress or whether they have succeeded.
- *Achievable*, yet aggressive. This is a judgment call, and experience with project planning and execution will help in meeting this requirement.
- *Relevant*. The goal must be specifically linked to the strategic goals of the enterprise.
- *Time bounded*. The goal must make sense in the time frame that the team has been given to work within.

Remember, all processes can be measured in terms of quality (that is, defects) and cycle time. The cost of the process flows from these two metrics. Reducing cost without focusing on reducing defects and cycle time is a recipe for disaster.

Project Plan

The project charter sets forth a tentative schedule for the project timing after a thorough discussion and subsequent agreement with the project champion. This information is needed to facilitate the project pipeline creation and portfolio management process. Executives need to be aware that project time and completion dates are not easily manipulated since they may be part of an overall schedule that is being managed closely along with critical resources.

EXAMPLE 11.4

Consider the following SMART goal statements:

- Reduce the reject rate for product X from its baseline of 45% to 5% within six months. Note: This is an absolute percentage reduction. The relative percentage reduction would be 88.9%. It is important to be clear when stating percentage changes.
- Reduce the cycle time of product B from 90 days to less than 10 days 95% of the time within three months.

When a project plan is included, it becomes possible to manage the project from a portfolio perspective, and particularly with regard to cost, schedule, and resources.

Generally, a Gantt chart depicting the DMAIC phases and noting the resources required by phase will be sufficient. Gantt charts will be addressed in Chapter 12. However, at a minimum, an estimated start and completion date is required.

Project Team

The project team members should be identified along with their expected time contributions to the project. Project team members are knowledgeable and valuable resources who will likely be in demand for other projects as well.

Knowing the total demand requirements placed on team members is essential to ensure they are not overloaded and that their "regular" jobs do not suffer from their contributions to projects.

Charter Housekeeping Considerations

Furthermore, it is prudent to have charters replete with the appropriate signatures of authorization, not the least of which are the sponsor and financial representatives. The inclusion of the financial approval lends credibility and support to the proposed financial gains.

The project sponsor has the ultimate accountability for developing the project charter. He or she will likely seek counsel from a Black Belt or Master Black Belt. However, in the end, senior leaders will look to the project sponsor for the success or failure of the project.

PROJECT PERFORMANCE MEASUREMENTS

> Identify and evaluate performance measurements (e.g., cost, revenue, delivery, schedule, customer satisfaction) that connect critical elements of the process to key outputs. (Analyze)
>
> **Body of Knowledge IV.B.5**

In the Body of Knowledge subtopics IV.B.1 through IV.B.4 above, the topic of the Six Sigma project charter was addressed. Now we must turn our attention to determining those measures associated with a project in the context of project management.

Projects exist for producing results. Portny (2010) states that three components must be present for an activity to be defined as a *project*. These components are depicted in Figure 11.5. Resources include both the financial component and the human capital component. These components define boundaries or constraints

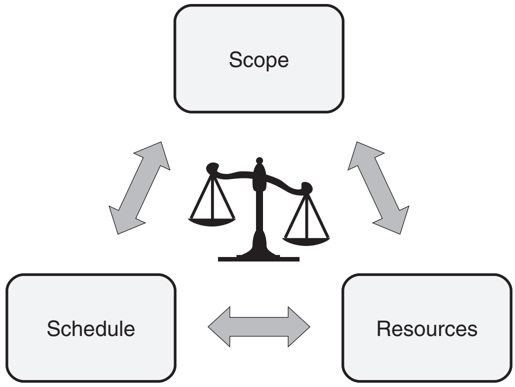

Figure 11.5 The components of any project.
Source: Adapted from Portny (2010).

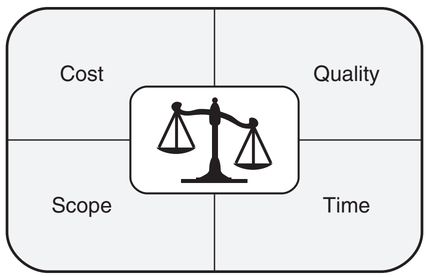

Figure 11.6 The balanced quadrant.
Source: Adapted from Williams (2008).

for the project. Without any one of them, the concept of a project evaporates. For example, consider a project without a scope. What would be accomplished? How about a project without a schedule? When would it be complete? Without any resources, who would do the project and how would it be paid for?

Williams (2008) offers what she calls the *balanced quadrant* concept depicted in Figure 11.6. Each quadrant represents one aspect of a project. Managing a project requires maintaining a balance among the four quadrants. Consider the following examples:

- An increase in scope results in an increase in time and cost
- An increase in quality results in an increase in cost and perhaps time
- A decrease in cost results in a decrease in scope, time, and/or quality

In most cases, the project manager or team leader cannot make balancing decisions alone. Often, the internal and external stakeholders must be consulted, particularly when cost is a consideration.

The above discussion raises a key point about Lean Six Sigma projects. That is, such projects are not managed by the DMAIC methodology alone. Like all significant projects, Lean Six Sigma projects must be managed using project management techniques. This includes using the following project performance measurements when appropriate:

- Cost
- Revenue
- Delivery (that is, on-time delivery)
- Schedule
- Customer satisfaction

Of the five above, cost and schedule are required at a minimum since they can be used to form the basis of the cost performance index (CPI) and schedule performance index (SPI). These two indices are fundamental project management indices. See Kubiak (2012). Revenue, delivery, and customer satisfaction, among others, may be needed depending on the specifics of the project.

A word of caution is in order with regard to managing Lean Six Sigma projects using project management techniques. The general concepts and techniques of project management work well. However, one must remember that these techniques were developed for projects where the plan, from start to finish, is well known. This is not the case for Lean Six Sigma projects.

Although the DMAIC methodology itself is rigorous, it is built upon the premise that we don't know the solution in advance and that we follow the data. Consequently, it is difficult to plan, resource, and cost out each phase with a high degree of precision. This does not mean we should abandon these planning tools and techniques; just recognize their limitations.

PROJECT CHARTER REVIEW

> Explain the importance of having periodic project charter reviews with stakeholders. (Understand)
>
> **Body of Knowledge IV.B.6**

Stakeholders should be actively engaged in the review of Lean Six Sigma projects.

On one hand, champions must be engaged in a specific set of reviews related to the projects for which they are responsible. Recall that the role of the champions was addressed in Chapter 1. This set of reviews is known as *tollgates*. Tollgates will be addressed in detail in Chapter 12.

In addition to tollgates, many organizations instigate an integrated and routine set of reviews across all active projects. This set of reviews would include executive management, other senior leaders, middle management, and, of course, interested stakeholders other than champions.

If a large number of projects are under way, half of the projects might be conducted one month and the other half the next month and rotated on this cycle thereafter. This permits stakeholders to view all of the projects over a relatively short time frame. Projects not performing to plan may receive additional support and resources to rectify them.

Involving stakeholders in the review process helps ensure cooperation and buy-in and/or, at least, surfaces earlier an awareness of resistance that some stakeholders may harbor. When this awareness becomes known, the techniques of Chapter 3 can be applied.

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Chapter 12

Project Management (PM) Tools

Identify and use the following PM tools to track projects and document their progress. (Evaluate)

Body of Knowledge IV.C.1

This chapter will address a variety of basic project management tools, including Gantt charts, tollgate reviews, work breakdown structures, and the RACI matrix. When used fully, these tools provide a Black Belt with the capability to efficiently and effectively manage a project.

Gantt Charts

A *Gantt chart* is a type of bar chart used in process/project planning and control to display planned work and finished work in relation to time. It is also called a *milestone chart*.

The Gantt chart provides an excellent visualization of time-based progress. However, to put together a meaningful Gantt chart, the approach shown in Example 12.1 is often necessary for large projects.

EXAMPLE 12.1

A team uses a mailed survey to obtain information on customer preferences. The team identifies the following activities necessary for project completion:

A. Construct survey questionnaire

B. Decide whether to buy or build software to analyze the data

C. Print and mail questionnaire

D. Buy or build software

E. Test software

Continued

Continued

F. Enter data into database

G. Use software to analyze results

H. Interpret results and write final report

As the key in Figure 12.1 indicates, the numbers below the line in each activity box provide data regarding start and finish times. *Slack time* for an activity is defined as:

Slack time = Latest start time – Earliest start time

Program evaluation and review technique (PERT) and critical path method (CPM) have essentially merged in current software packages. The *critical path* is the path from start to finish that requires the most time. In Figure 12.1, there are just two paths:

- Path ACEGH requires 10 + 5 + 10 + 10 + 15 = 50 days
- Path BDFGH requires 5 + 15 + 10 + 10 + 15 = 55 days

Therefore, BDFGH is the critical path.

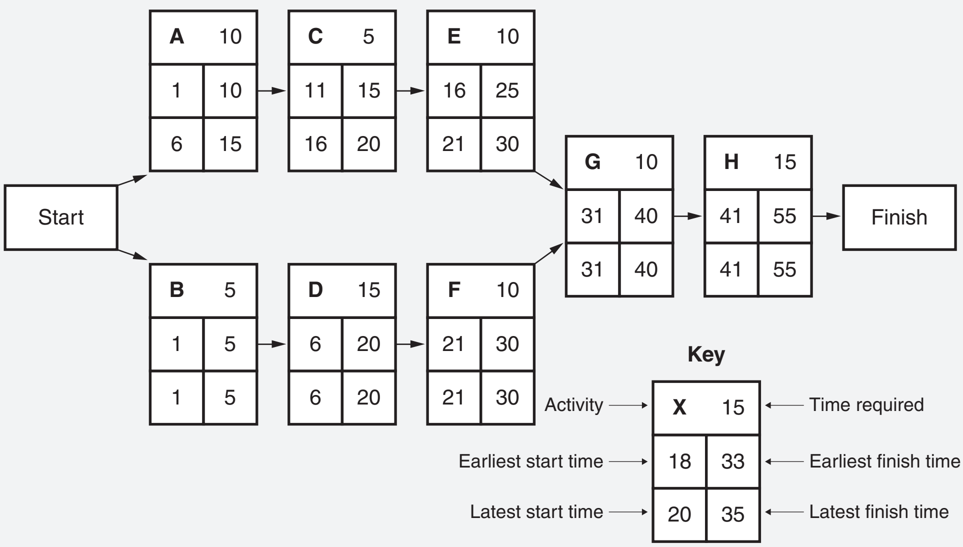

Figure 12.1 Activity network diagram.

Software packages are available to identify and calculate the critical path for projects with multiple paths. If activities on the critical path are delayed, the entire project will be delayed. Critical path time is the time required to complete the project. One way to complete the project in less time is to decrease the time for at least one of the activities. This is usually accomplished by putting more resources into one or more activities on the critical path. This is sometimes

EXAMPLE 12.2

Using the data from Example 12.1, explain the Gantt chart depicted in Figure 12.2.

There are two important things to note in this example. The first is that the Gantt chart shown in Figure 12.2 uses five-day work weeks although the weeks are numbered in seven-day increments.

The second is that the first day of work on any activity is always available as a full day of work. For example, suppose the earliest start time is day 1, the earliest finish time is day 10, and the activity consumes 10 days. There are 10 days available for the activity because day 1 is available for work.

Task A: Using Block A and the legend in Figure 12.1, we know that task A can start as early as day 1 and as late as day 6. However, it can finish as early as day 10 and as late as day 15. Task A will consume 10 days to complete. The slack time for this task is five days. We will start task A at time 1. This is depicted in Figure 12.2 on line A.

Task B: This is independent of any other task as shown in Figure 12.1. It can start as early as day 1 and as late as day 1. However, it can finish as early as day 5 and as late as day 5. Task B is highly constrained. Also, it consumes 5 days to complete. There is no slack time for this task. We are forced to start task B at day 1. This is depicted in Figure 12.2 on line B.

Task C: This task must follow task A and precede task E from Figure 12.1. It can start as early as day 11 and as late as day 16. However, it must finish as early as day 15 and as late as day 20. Task C will consume 5 days to complete. The slack time for this task is five days. We will start this task immediately following the completion of task A at day 11. This is depicted in Figure 12.2 on line C.

Task D: This task must follow task B and precede task F from Figure 12.1. It can start as early as day 6 and as late as day 6. However, it must finish as early as day 20 and as late as day 20. Task D is highly constrained. Also, it will consume 15 days to complete. There is no slack time for this task. We are forced to start task D at day 6. This is depicted in Figure 12.2 on line D.

Task E: This task must follow task C and precede task G from Figure 12.1 It can start as early as day 16 and as late as day 21. However, it must finish as early as day 25 and as

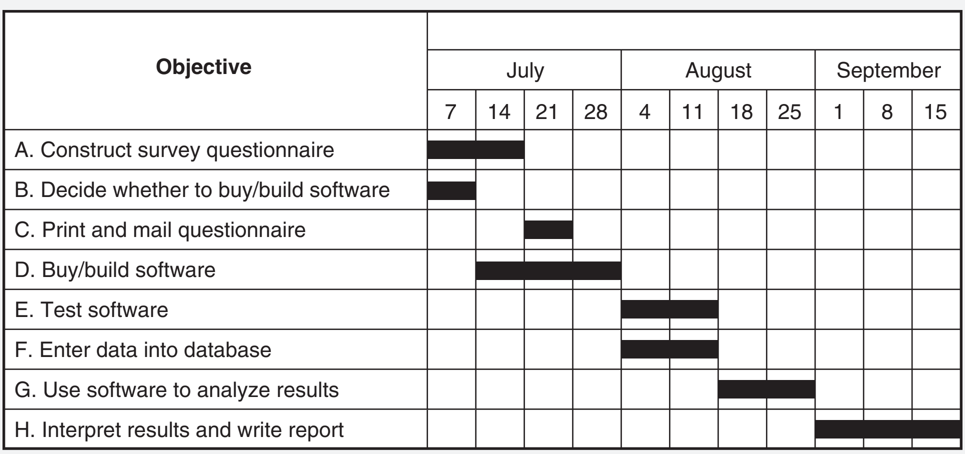

Figure 12.2 Example of a Gantt chart.

Continued

Continued

late as day 30. Task E will consume 10 days to complete. The slack time for this task is five days. We can start this task immediately following the completion of task C at day 16, but we will delay it five days and start it at the same time as task F on day 21. This is depicted in Figure 12.2 on line E.

Task F: This task must follow task D and precede task G from Figure 12.1. It can start as early as day 21 and as late as day 21. However, it must finish as early as day 30 and as late as day 30. Task F is highly constrained. Also, it will consume 10 days to complete. There is no slack time for this task. We are forced to start task F at day 21. This is depicted in Figure 12.2 on line F.

Task G.: This task cannot start until both tasks E and F are completed since both of these tasks are predecessors from Figure 12.1. It can start as early as day 31 and as late as day 31. However, it must finish as early as day 40 and as late as day 40. Task G is highly constrained. Also, it will consume 10 days to complete. There is no slack time for this task. We are forced to start task G at day 31. This is depicted in Figure 12.2 on line G.

Task H: This task cannot start until task G is complete as shown in Figure 12.1. It can start as early as day 41 and as late as day 41. However, it must finish as early as day 55 and as late as day 55. Task H is highly constrained. Also, it will consume 15 days to complete. There is no slack time for this task. We are forced to start task H at day 41. This is depicted in Figure 12.2 on line H.

referred to as "crashing" the project. The information from the PERT/CPM analysis can be used to construct a Gantt chart, as shown in Figure 12.2.

The charts shown in Figures 12.1 and 12.2 have the advantage of being easy to understand while conveying essential data, and therefore they are often used for presentations to executive groups in addition to their use as tracking devices. Additional charts and diagrams can be produced for each of the activities listed. These graphical tools are also useful for tracking and evaluating the project at various phases and at final management reviews. *Storyboards* are sometimes used to convey project information involving changes that are easier to draw or photograph than to explain in words. A common application of storyboards is for before-and-after pictures, often called *current state* and *future state*, for a proposed project. This approach is appropriate for facility or product redesign projects. Storyboards for Six Sigma projects are often formatted into five sections labeled *define*, *measure*, *analyze*, *improve*, and *control* (DMAIC), with charts, figures, and documents illustrating the activity in each area.

Tollgate Reviews

Tollgate reviews are the bonds that hold the DMAIC framework together. They are a necessary part of the DMAIC infrastructure. Tollgates serve as meaningful checkpoints, facilitate communications, and ensure sponsor participation. They exist as a check and balance to ensure that teams are ready to transition from one phase to the next in an orderly manner. Tollgates focus projects on moving forward.

The *tollgate review* is a formal review process conducted by a champion who asks a series of focused questions aimed at ensuring that the team has performed diligently during a phase. The result of a tollgate is a "go" or "no-go" decision. The "go" decision allows the team to move forward to the next phase. If it is in the last phase, the "go" decision" brings about project closure. If the decision is "no-go," the team must remain in the phase, retreat to an earlier phase (in some instances), or perhaps the project is terminated or suspended. The concept of "go/no-go" decisions was illustrated in Figure IV.1.

Effective tollgate reviews are usually characterized by an objective evaluation of the work performed by the team during the phase, and the willingness to identify and resolve problems during the reviews. Poor preparation, incomplete documentation, champions desiring to change the scope, and champions delegating replacements typically characterize poor tollgate reviews.

Many thoughts exist regarding the purpose of tollgates, who should attend them, what questions should be asked, and what the exit criteria are from the tollgate phase. Table 12.1 provides a summary of one viewpoint. The "questions to address" are minimal in the table, but they are critical.

Notice that the first question asked is "Is the project consistent with the goals of the organization?" Project alignment to the strategy and subsequently the goals and objectives of the organization are critical. This question appears in every phase because of the highly dynamic nature of organizations. When the answer is "no," it is incumbent upon the champion to shut the project down and allow the project resources to be allocated elsewhere. However, notice that there is something particularly unique about this question. Unlike the others in the table, this question should be both asked and answered by the champion for it is simply a rhetorical question posed to the team. The champion is in the position to have direct access to the proper resources to be able to answer this question. The champion should come prepared at every tollgate to address this critical question.

Table 12.2 provides an additional set of questions the champion might ask during tollgate reviews. Of course, common to each phase are questions regarding budget and schedule. Also included is a question regarding what obstacles the team might have encountered and how the champion may be able to help. Recall that one job of the champion is the removal of organizational barriers the team might face.

Tollgate reviews, when executed well, provide real value to the Lean Six Sigma deployment. They keep projects focused and on track, and executives and champions engaged.

Work Breakdown Structure (WBS)

A *work breakdown structure* is a hierarchical decomposition of the work content for a given project. The WBS permits the project manager to systematically break the work into smaller, more manageable sizes that can be planned, executed, and tracked more efficiently. Figure 12.3 represents a generalized work breakdown structure. The project is first decomposed into small modules as shown in level 2. This level is subsequently broken down further into even smaller modules of work as shown in level 3. This process continues down to the level "n." This level

Table 12.1 A brief summary of the tollgate process.

Component	Define	Measure	Analyze	Improve	Control
Purpose	Provide a compelling business case appropriately scoped, complete with SMART goals, and linked to a hoshin plan	Collect process performance data for primary and secondary metrics to gain insight and understanding into root causes and to establish performance baselines	Analyze and establish optimal performance settings for each X and verify root causes	Identify and implement process improvement solutions	Establish and deploy a control plan to ensure that gains in performance are maintained.
Participants required	• Sponsor • Process owner • MBB/BB coach • Deployment champion • Project team • Finance partner	• Sponsor • Process owner • MBB/BB coach • Deployment champion • Project team	• Sponsor • Process owner • MBB/BB coach • Deployment champion • Project team • Finance partner	• Sponsor • Process owner • MBB/BB coach • Deployment champion • Project team	• Sponsor • Process owner • MBB/BB coach • Deployment champion • Project team • Finance partner
Questions to address	• Is this project consistent with the goals of the organization? • Do we have the right level of engagement from stakeholders and business partners?	• Is this project still consistent with the goals of the organization? • Do we have the right level of engagement from stakeholders and business partners?	• Is this project still consistent with the goals of the organization? • Do we have the right level of engagement from stakeholders and business partners?	• Is this project still consistent with the goals of the organization? • Do we have the right level of engagement from stakeholders and business partners?	• Is this project still consistent with the goals of the organization? • Have resources been allocated to move into replication?

Continued

Table 12.1 *Continued.*

Component	Define	Measure	Analyze	Improve	Control
Questions to address (continued)	• Have resources been allocated to move to the next phase? • Do conflicts exist with other projects or activities?	• Have resources been allocated to move to the next phase?	• Have resources been allocated to move to the next phase? • What are the market and timing dependencies?	• Have resources been allocated to move to the next phase?	• Is the replication schedule and plan appropriate? • What are the market and timing dependencies? • Have responsibilities identified in the control plan been transferred to appropriate parties?
Exit criteria	• Sponsor approval • Finance approval • Funding approval	• Sponsor approval	• Sponsor approval • Finance approval	• Sponsor approval	• Sponsor approval • Finance approval • Hand-off to process owner

Table 12.2 Additional questions to be asked by the champion during tollgate reviews.

Phase	Questions
Define	• Who are the stakeholders? • What are the primary and secondary measures? • Are you facing any barriers or obstacles I can help with? • Are we on schedule/budget?
Measure	• What is the process capability? • What data still remain to be collected? • What did the FMEA show? • What questions still remain to be answered about the process? • Are you facing any barriers or obstacles I can help with? • Are we on schedule/budget?
Analyze	• What are the critical X's? • What are the root causes? • Are you facing any barriers or obstacles I can help with? • Are we on schedule/budget?
Improve	• What is the process capability of the revised process? • What criteria will be used to select the optimum solution from the set of solutions? • Will we run a pilot? • Are you facing any barriers or obstacles I can help with? • Are we on schedule/budget?
Control	• Is the process documented? • Is the control plan completed? • Has the process been transitioned back to the process owner? • What lessons have we learned? • Do we know where we might be able to replicate our new process across the organization? • Are you facing any barriers or obstacles I can help with? • Are we on schedule/budget?

Source: Adapted and compiled from Phillips and Stone (2002).

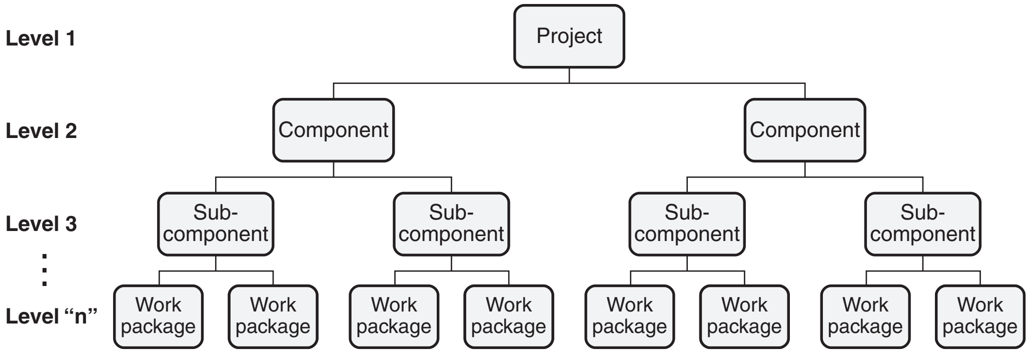

Figure 12.3 A generalized work breakdown structure.

represents the smallest work contents that must be tracked in order to manage the project successfully. At level "n," the work packages (also called *tasks*) can be assigned to individuals for execution. Time, material, and expenses are collected at this level and can be summarized upward to any level desired. Thus, considerable thought must be given to the development of the WBS.

Since the WBS is developed to assign and track work, the work packages should appear on the project schedule or Gantt chart. It should be noted that some of the integrated off-the-shelf software available permits the development and construction of a whole WBS and the subsequent translation into a project schedule, thus giving the project leader increased visibility into project performance.

RACI Model (Responsible, Accountable, Consulted and Informed)

In addition to the communication plan shown in Table 7.1, another effective tool for facilitating communication is the RACI matrix. This is depicted in Figure 12.4. RACI stands for *responsibility*, *accountability*, *consultation*, *inform*. See Price and Works, "Balancing Roles and Responsibilities in Lean Six Sigma." Let's look at the meaning of the acronym in detail:

- *Responsibility.* Individuals who actively participate in an activity. Notice that more than one individual at a time may be held responsible
- *Accountability.* The individual ultimately accountable for results. Only one individual may be accountable at a time. This will become evident in the example given below.
- *Consulted.* Individuals who must be consulted before a decision is made. Sometimes this appears as *consultation* or *counselor.*
- *Informed.* Individuals who must be informed about a decision because they are affected. These individuals do not need to take part in the decision-making process.

Task → / Player ↓	Group A	Group B	Group C	Group D	Group E	Group F	Group G
Player 1							
Player 2							
Player 3							
Player 4							
Player 5							
Player 6							
Player 7							

Figure 12.4 Example of a common RACI matrix form.

The RACI matrix is usefully defined in detail by a team or committee, and in some cases, by an individual (for example, project manager). Frequently, a RACI is included to provide clarification. Also, it is important to keep in mind that there should be single points of accountability with sufficient authority to make decisions.

EXAMPLE 12.3

An organization is in the initial stages of beginning its Lean Six Sigma deployment. Due to significant miscommunications and misunderstandings, the deployment incurred multiple setbacks. Consequently, the deployment team identified the set of stakeholders and tasks shown in Figure 12.5.

Figure 12.5 depicts one possible RACI matrix the deployment team felt would work for their organization. Notice that only one stakeholder is accountable in each column.

Because of the popularity and usefulness of the RACI matrix, it has evolved over the years, and numerous variants of it can be found. These include: RASCI (sometimes RASIC), RACI-VS, RACIO (sometimes CAIRO), and DACI, among others. Interested readers can easily look these variations up on the Internet.

Continued

Continued

Task → / Player ↓	Set policy	Identify projects	Select projects	Work projects	Achieve results	Maintain gains	Coach and mentor
Executive	A	C	I, C	I	I	I	
Sponsor		C	C	I	A	I	
Process owner		C	C	I, C	I	A, R	
MBB	C	R	R, C	C	I	I	A
BB		R	C	A, R	R	R	
GB		C		R	R		
Quality council	R	A	A	I	I	I	

Figure 12.5 RACI matrix for Example 12.3.

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Chapter 13

Analytical Tools

> Identify and use the following analytical tools throughout the DMAIC cycle. (Apply)
>
> **Body of Knowledge IV.D**

There is no doubt that there is no right improvement tool for all operations and/or situations. The company that hangs its reputation on one is doomed to eventual extinction.

—Ernst & Young

This chapter will address what is commonly known as the *seven management and planning tools*. Although the body of knowledge calls for only five of the seven tools, all seven will be included for completeness purposes. The two excluded from the body of knowledge are the *process decision program chart* and the *interrelationship digraph*. These will not be part of the examination.

As you read this chapter, remember that although tools are often fun to use, they are still tools and are a means to an end; not an end unto themselves. This is something Black Belts frequently forget, particularly when they are in training or being driven by a Master Black Belt who has lost sight of the big picture.

Process Decision Program Chart (PDPC)

The *process decision program chart* (PDPC) is a tool that identifies all events that can go wrong and the appropriate countermeasures for these events. It graphically represents all sequences that lead to a desirable effect.

The PDPC is based on a tree diagram (discussed below) and is often used when a plan is large and complex, must be on schedule, is being implemented for the first time, and the cost of failing is high.

Tague (2005) suggests the following procedure for developing a PDPC:

1. Obtain a high-level tree diagram of the proposed plan. This diagram should show the second level of main activities and the third level of broadly defined tasks.
2. Brainstorm what could go wrong for each third-level task.

EXAMPLE 13.1

This example is taken from Tague (2005).

A medical group is planning to improve the care of patients with chronic illnesses such as diabetes and asthma through a new chronic illness management program (CIMP). They have defined four main elements and, for each of these elements, key components. The information is laid out in the process decision program chart of Figure 13.1.

Dotted lines represent sections of the chart that have been omitted. Only some of the potential problems and countermeasures identified by the planning team are shown on the chart. Thinking about these helped the team create a better program. For example, one of the possible problems with patients' goal setting is backsliding. The team liked the idea of each patient having a buddy or sponsor and will add that to the program design. Other areas of the chart helped them plan better rollout, such as arranging for all staff to visit a clinic with a CIMP program in place. Still other areas allowed them to plan in advance for problems, such as training the CIMP nurses how to counsel patients who choose inappropriate goals.

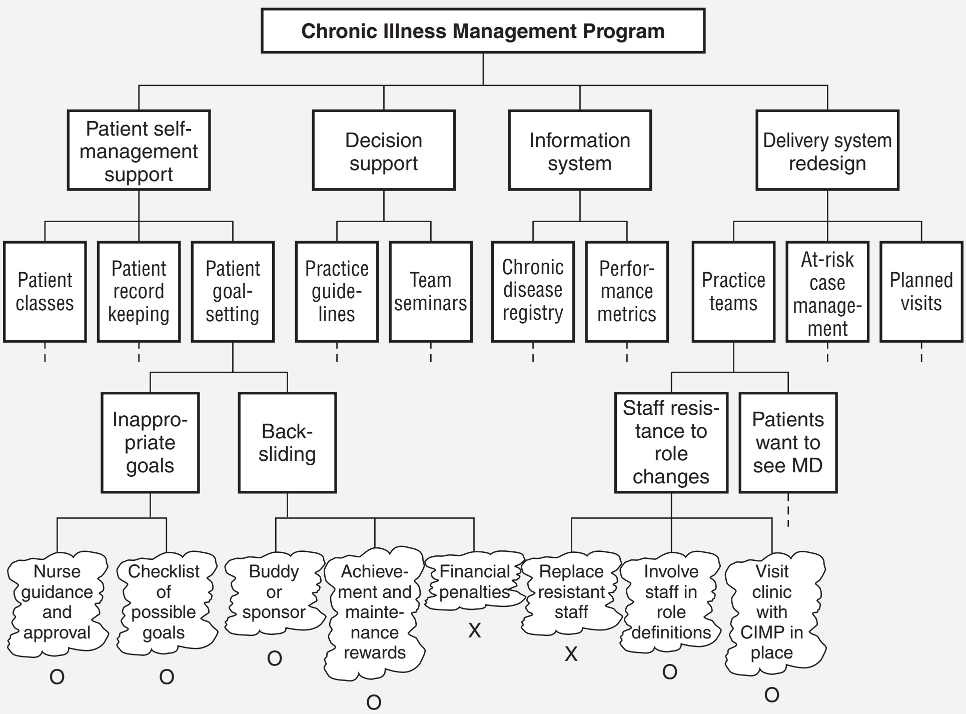

Figure 13.1 Process decision program chart example.

3. Eliminate any problems that are improbable or whose consequences are insignificant. Show the problems as the fourth level on the diagram.
4. Brainstorm countermeasures for each problem. Such countermeasures might either prevent or remedy the problem. Show countermeasures at the fifth level by outlining in clouds or jagged lines as a convention.
5. Assess the practicality of each countermeasure in terms of cost, time, ease of implementation, and effectiveness. Use an "X" to depict those countermeasures that are impractical and an "O" for those that are practical.

Interrelationship Digraph (ID)

The *interrelationship digraph* (ID) is a tool that displays the cause-and-effect relationships between ideas or factors in a complex situation. It identifies meaningful categories from a mass of ideas and is useful when relationships are difficult to determine. The ID is also known as a *relations diagram*, *circle diagram*, *or hand-off map*.

The basic procedure for developing the ID follows:

1. State the issue or problem to be explored.
2. Brainstorm the factors related to the issue or problem and list them individually on sticky notes.
3. Arrange the notes in a circle on a sheet of easel paper or something similar.
4. Start with the note at the 12 o'clock position, and moving clockwise, compare it with every other note (factor). Ask whether the current factor causes or influences the next factor, and so on. If it does, draw an arrow from the current factor to the one being compared and move on.
5. When all factors have been compared, count the number of incoming and outgoing arrows for each factor and designate as I/O (for example 3/1). The factors that have the most outgoing arrows are the basic causes or drivers. The factors that have the most incoming arrows are the outcomes or final effects.
6. Begin improvement activities on the key drivers.

EXAMPLE 13.2

Consider Figure 13.2. A team brainstormed the factors shown in this figure, identified the causal relationships, and counted the number of incoming and outgoing arrows for each factor. Notice that the number of incoming and outgoing arrows have been identified below each factor in Figure 13.2.

Since this ID is relatively small, we can readily determine the drivers and the outcomes. The drivers have the largest number of outgoing arrows:

1. Poor scheduling practices has seven outgoing arrows.
2. Late order from the customer has five outgoing arrows.

The resulting outcome has the largest number of incoming arrows:

1. Poor scheduling of trucker has four incoming arrows.

Now that two top drivers have been identified, the team can begin to formulate plans for its improvement efforts.

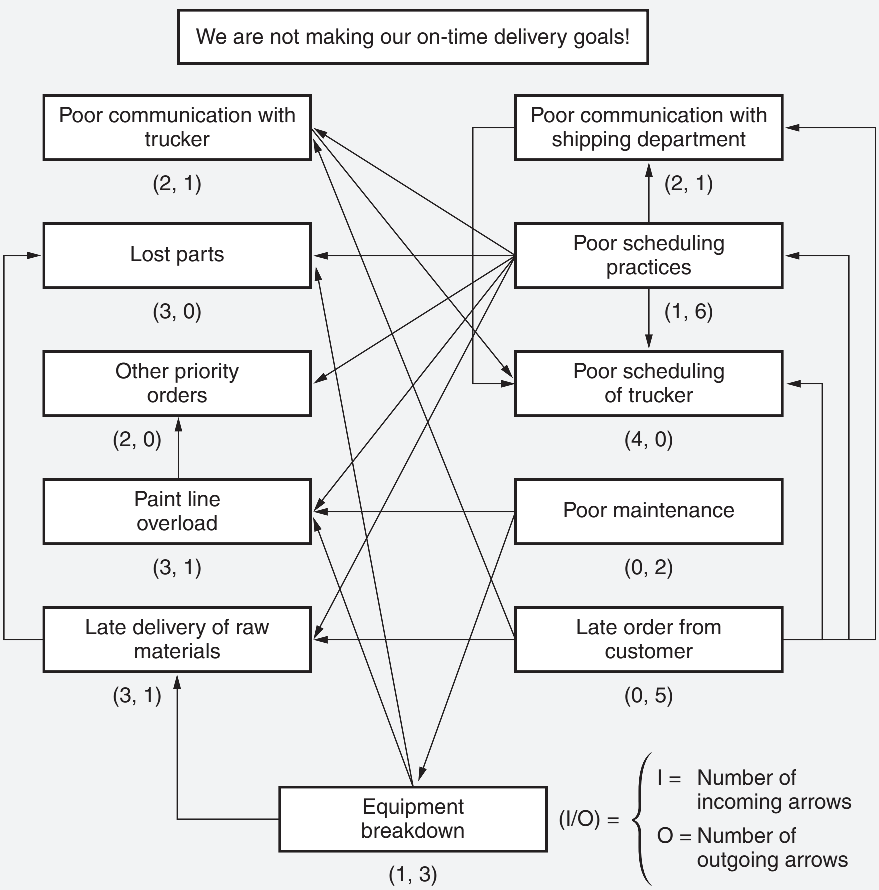

Figure 13.2 Example of an interrelationship digraph.

Affinity Diagrams

The *affinity diagram* is a tool used to organize information and help achieve order out of the chaos that can develop in a brainstorming session. Large amounts of data, concepts, and ideas are grouped based on their natural relationship to one another. It is more a creative process than a logical process. The affinity diagram is also known as the *KJ method*,

The affinity diagram is useful when there are numerous ideas, bordering on chaos; the issues are large and difficult to grasp, and group consensus is necessary. See Tague (2005).

The basic procedure for developing the affinity diagram follows:

1. Conduct a brainstorming session and collect ideas on a sticky note or card.
2. Post ideas randomly on a large work surface or sheet of easel paper.
3. Gather all participants around the notes.
4. Without talking, participants should look for related ideas and gather them into logical groups. Also, participants may move notes that other participants have moved. If notes appear to belong to more than one group, replicate a note for each group. It is OK to have outliers or lone ideas.
5. At this point, participants may talk. Review all the notes within a group and look for a theme of the group. This theme may be already on a note. If not, capture this theme on a note and place it at the top of the group. Do this for all the groups. This is an important step as it is necessary to reach consensus on the groupings.
6. If appropriate, combine groups into supergroups.

The affinity diagram is a highly valued technique because it is evolutionary in nature. According to Tague (2005), it allows "groupings to happen" as well as allows a group to "move beyond habitual thinking and preconceived categories." That's why it's important to avoid placing the notes in any order or developing any category headings in advance. See Example 13.3.

EXAMPLE 13.3

This example is taken from Tague (2005).

> *The ZZ-400 manufacturing team used an affinity diagram to organize its list of potential performance indicators. Table 13.1 shows the list the team members brainstormed. Because the team works a shift schedule and members could not meet to do the affinity diagram together, they modified the procedure.*
>
> *They wrote each idea on a sticky note and put all the notes randomly on a rarely used door. Over several days, everyone reviewed the notes in their spare time and moved the notes into related groups. Some people reviewed the evolving pattern several times. After a few days, the natural grouping shown in Figure 13.3 had emerged.*

Notice that one of the notes, "Safety," has become part of the heading for its group. The rest of the heading was added after the grouping emerged. Five broad areas of performance were identified: product quality, maintenance, manufacturing cost, safety and environmental, and volume.

Also, notice that "Maintenance costs" was placed into two groups, which are circled in Figure 13.3.

Table 13.1 Affinity diagram for Example 13.3.

Possible performance measures			
% purity	% trace metals	Maintenance costs	Number of emergency jobs
Pounds produced	Environmental accidents	Material costs	Overtime costs
Number of pump seal failures	Viscosity	C_{pk} values	Safety
Days since last lost-time	% rework or reject	Hours downtime	% uptime
Number of OSHA recordables	Number of customer returns	Customer complaints	Overtime total hours worked
$ per pounds produced	Raw material utilization	Yield	Utility cost
ppm water	Color	Service factor	Time between turnarounds
Hours worked per employee	Pounds of waste	Housekeeping score	% capacity filled

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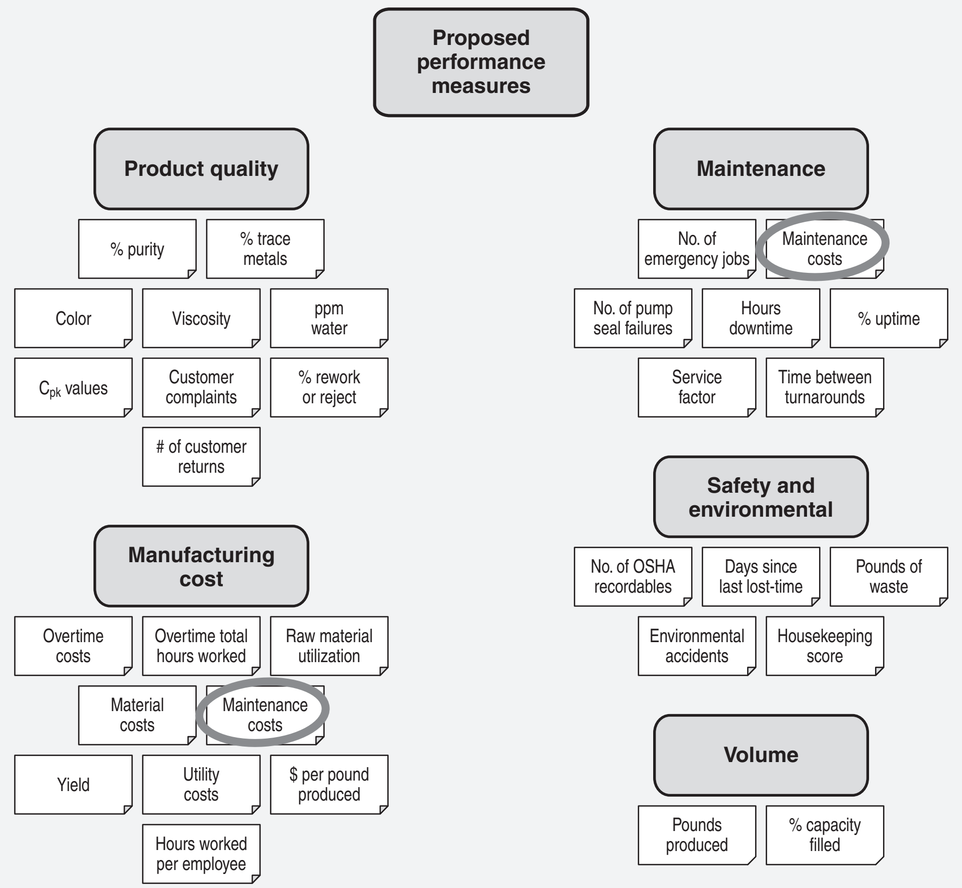

Figure 13.3 Example of an affinity diagram.

Tree Diagrams

The *tree diagram* is a tool that depicts the hierarchy of tasks and subtasks needed to complete an objective. The finished diagram resembles a tree. Tree diagrams may be depicted either vertically top-down or horizontally left-to-right. When depicted top-down, tasks move from general (top) to specific (down), and when depicted left-to-right, tasks move from left (general) to right (specific).

Tree diagrams may be used in a wide variety of applications, including critical-to-quality tree, decision tree, fault tree analysis, Gozinto chart, organization chart, process decision program chart, five whys, work breakdown structure (WBS), and so on. Figure 13.4 illustrates the general structure of a tree diagram using the top-down format. "Tasks" were chosen in this figure. Had we chosen a "system" as the top tier, the second would have been subsystems, the third tier components, and the fourth tier parts.

The basic procedure for developing the tree diagram follows:

1. State the issue or problem to be explored. Write it on a note. This serves as the top of the tree.

2. Depending on the nature of the issue or problem, different types of questions must be asked to generate subsequent levels of the tree. For example, for each item at each level, Tague (2005) suggests:

   – If the top item of the tree is a WBS, plan, or goal, ask questions like "What tasks must be done to accomplish this?" or "How can this be accomplished?"

   – If the top item of the tree is a root cause analysis, ask questions like "What causes this?" or "Why does this happen?"

   – If the tree diagram is to serve as a Gozinto chart, ask a question like "What are the components?"

   Of course, the wording at each level might have to change to reflect the context of the level. For example, "components" might be replaced by "systems," "subsystems," or "parts."

   Brainstorm each item at each level until complete. When done properly, this technique could lead to very large tree diagrams.

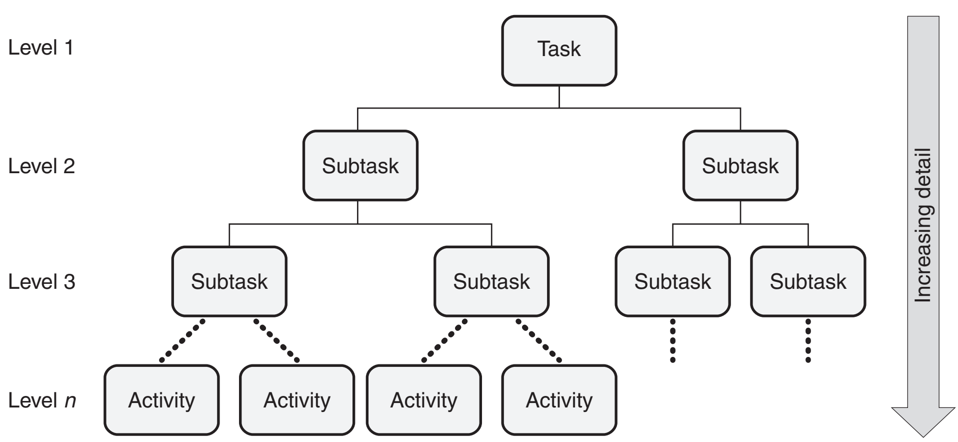

Figure 13.4 Example of the general structure of a tree diagram.

3. Tague (2005) suggests:

   Do a "necessary and sufficient" check. Are all the items at this level necessary for the one the level above? If all the items at this level were present or accomplished, would they be sufficient *for the one on the level above?*

4. Stop the tree diagram when it no longer makes sense to divide the lowest level. For example, the root cause is evident, a specific action can be carried out, or a part is a single entity.
5. Draw connecting lines from each item on a given level to the appropriate supporting items on the next level down.

EXAMPLE 13.4

Consider the example given in Figure 13.5.

Notice that the tree diagram completes at different levels. Also, notice that the best setting is not known for the viscosity of the paint. Consequently, it leads to runs, which, in turn, leads to tables being rejected.

This example has led to the discovery of a root cause. The next step would be to establish a project that would determine the optimal viscosity setting so that this root cause could be eliminated. A detailed review of this tree diagram would reveal other potential actions as well (for example, find the missing clamping instructions or create a new one, revise the tubing specification, and so on).

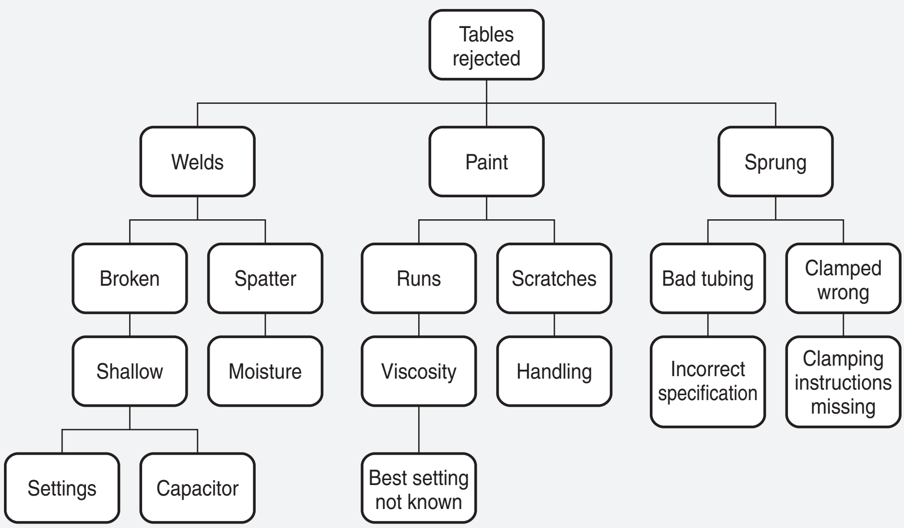

Figure 13.5 Tree diagram for Example 13.4.

Matrix Diagram

The *matrix diagram* is a tool that identifies the relationships that exist between groups of data. It can be also used to identify the strength of those relationships as well. The matrix diagram does not use a singular format. Instead, there are six different forms, including the C, L, T, X, Y, and roof-shaped format. The different shapes that can be selected depend on the number of variables to be compared. This is represented in Table 13.2.

Strengths of relationships between variables can be depicted using different symbols, as shown in Table 13.3, and are used in exactly the same manner as described in Chapter 10 when we discussed quality functional deployment. Any choice of symbols can be used as long as they are sufficiently distinct. In some cases, users may only need to use two symbols depending on the comparison to be made. For example: high and low, and large and small.

The basic procedure for developing the matrix diagram follows:

1. Determine what type of variables or groups of items are to be compared.
2. Select the appropriate type of matrix: C, L, T, X, Y, or roof-shape depending on the number of variables to be compared.
3. Complete the matrix headings using the items from each group as row and column headings.
4. Select the appropriate symbols for the matrix.

Table 13.2 Matrix diagram shapes by number of variables or groups compared.

Matrix shape	Number of variables
C	3
L	2
T	3
X	4
Y	3
Roof	1

Table 13.3 Relationship symbols and weights for matrix diagrams.

Relationship	Symbol	Weight
Weak	▯	1
Medium	○	3
Strong	⊙	9

5. Complete each cell in the matrix by doing the appropriate comparisons and applying the necessary symbols.
6. Review the matrix for patterns and draw conclusions.

EXAMPLE 13.5

This example is taken from Tague (2005).

Consider Figures 13.6 through 13.11.

- C-shape. *Think of C meaning, "cube." Because this matrix is three-dimensional, it is difficult to draw and infrequently used. If it is important to compare three groups simultaneously, consider using a three-dimensional model or computer software that can provide a clear visual image.*

  This figure shows one point on a C-shaped matrix relating products, customers, and manufacturing locations. Zig Company's model B is made at the Mississippi plant. (See Figure 13.6.)

- L-shape. *This L-shaped matrix summarizes customers' requirements. The team placed numbers in the boxes to show numerical specifications and used check marks to show choice of packaging. The L-shaped matrix actually forms an upside-down L. This is the most basic and most common matrix format. (See Figure 13.7.)*

- T-shape. *This T-shaped matrix relates product models (group A) to their manufacturing locations (group B) and to their customers (group C).*

  Examining the matrix in different ways reveals different information. For example, concentrating on model A, we see that it is produced in large volume

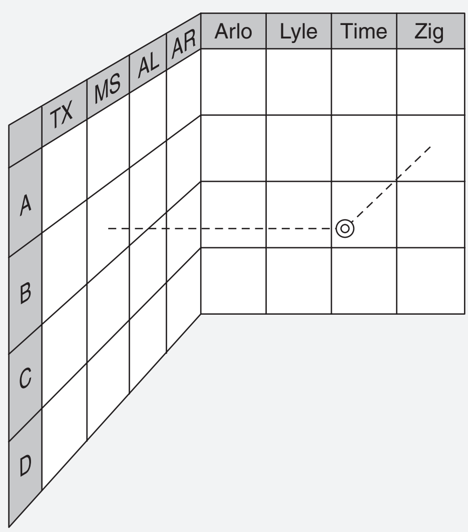

Figure 13.6 C-shaped matrix example.

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Customer Requirements

	Customer D	Customer M	Customer R	Customer T
Purity %	> 99.2	> 99.2	> 99.4	> 99.0
Trace metals (ppm)	< 5	—	< 10	< 25
Water (ppm)	< 10	< 5	< 10	—
Viscosity (cp)	20–35	20–30	10–50	15–35
Color	< 10	< 10	< 15	< 10
Drum		✔		
Truck	✔			✔
Railcar			✔	

Figure 13.7 L-shaped matrix example.

Products—Customers—Manufacturing Locations

Texas plant	●		○	○
Mississippi plant		●		○
Alabama plant	○			●
Arkansas plant		○	●	
● Large volume ○ Small volume	Model A	Model B	Model C	Model D
Zig Corp.		●		
Arlo Co.	○	○	○	●
Lyle Co.			○	○
Time Inc.	●			●

Figure 13.8 T-shaped matrix example.

at the Texas plant and in small volume at the Alabama plant. Time Inc. is the major customer for model A, while Arlo Co. buys a small amount. If we choose to focus on the customer rows, we learn that only one customer, Arlo, buys all four models. Zig buys just one. Time makes large purchases of A and D, while Lyle is a relatively minor customer (see Figure 13.8).

- *X-shape. This figure extends the T-shaped matrix example into an X-shaped matrix by including the relationships of freight lines with the manufacturing sites they serve and the customers who use them. Each axis of the matrix is related to the two adjacent ones, but not to the one across. Thus, the product*

Continued

models are related to the plant sites and to the customers, but not to the freight lines.

A lot of information can be contained in an X-shaped matrix. In this one, we can observe that Red Lines and Zip Inc., which seem to be minor carriers based on volume, are the only carriers that serve Lyle Co. Lyle doesn't buy much, but it and Arlo are the only customers for model C. Model D is made at three locations, while the other models are made at two. (See Figure 13.9.)

- Y-shape. *This Y-shaped matrix shows the relationships between customer requirements, internal process metrics, and the departments involved. Symbols show the strength of the relationships: primary relationships, such as the manufacturing department's responsibility for production capacity; secondary relationships, such as the link between product availability and inventory levels; minor relationships, such as the distribution department's responsibility for order lead time; and no relationship, such as between the purchasing department and on-time delivery.*

  The matrix tells an interesting story about on-time delivery. The distribution department is assigned primary responsibility for that customer requirement. The two metrics most strongly related to on-time delivery are inventory levels and order lead time. Of the two, distribution has only a weak relationship with order lead time and none with inventory levels. Perhaps the responsibility for on-time delivery needs to be reconsidered. (See Figure 13.10.)

Manufacturing Sites—Products—Customers—Freight Lines

○		●	○	Texas plant	●		○	○
	○	●	●	Mississippi plant		●		○
		●	●	Alabama plant	○			●
○	○		○	Arkansas plant		○	●	
Red Lines	Zip Inc.	World-wide	Trans South		Model A	Model B	Model C	Model D
		●	○	Zig Corp.		●		
			●	Arlo Co.	○	○	○	●
○	○			Lyle Co.			○	○
	○	●		Time Inc.	●			●

● Large volume
○ Small volume

Figure 13.9 X-shaped matrix example.

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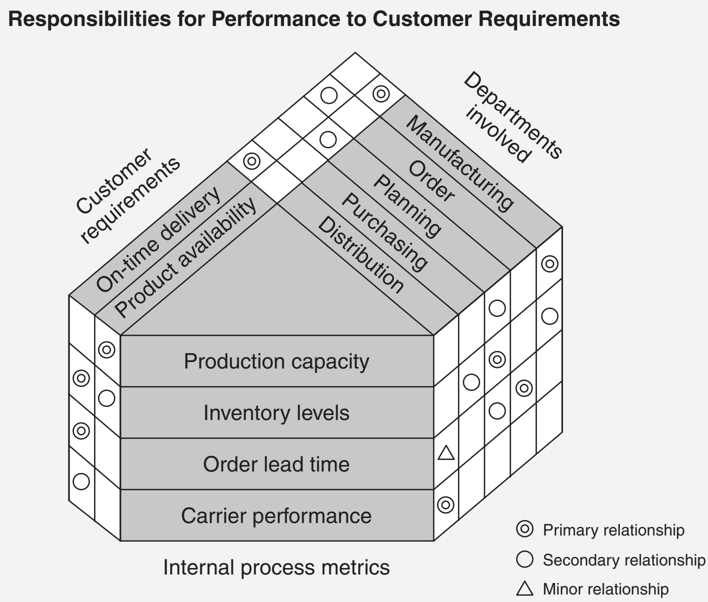

Figure 13.10 Y-shaped matrix example.

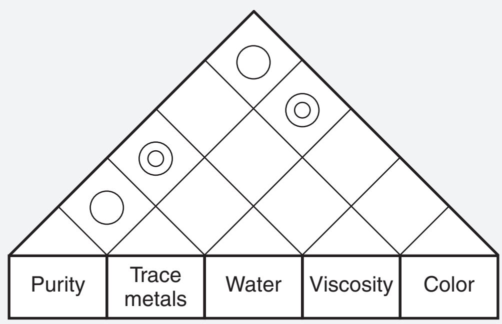

Figure 13.11 Roof-shaped matrix example.

- Roof-shape. *The roof-shaped matrix is used with an L- or T-shaped matrix to show one group of items relating to itself. It is most commonly used with a house of quality, where it forms the "roof" of the "house." For example, a strong relationship links color and trace metals, while viscosity is unrelated to any of the other requirements.* (See Figure 13.11.)

The matrix diagram is a highly useful tool. The key lies with selecting the most appropriate one for conveying the proper message.

Prioritization Matrix

The *prioritization matrix* is a tool used to choose between several options that have many useful benefits, but where not all of them are of equal value. The choices are prioritized according to known weighted criteria and then narrowed down to the most desirable or effective one(s) to accomplish the task or problem at hand.

The basic procedure for developing the prioritization matrix follows:

1. Identify the goal and clarify the options available.
2. Identify the criteria the decision must meet.
3. Develop the relative importance for each criterion (that is, weighting). This relative importance is a weighting in 0.05 increments between zero and one (that is, $0 < \text{weighting} < 1$). There are many ways to obtain agreement on these weights. Multivoting is one such method.
4. Create an L-matrix. List the relative importance weightings, criteria, and options as shown in Figure 13.12.
5. Rank each option within a criterion. If there are "n" options, rank the lowest option as "1" and the highest option as "n." Do this for each criterion. As always, there are many ways to reach an agreement on the ranks. Select one that works best for your team. If there is a tie between

Goal					
Relative importance	Weight 1	Weight 2	Weight 3	Weight 4	= 1.00
	Criteria				
Options	**Criteria 1**	**Criteria 2**	**Criteria 3**	**Criteria 4**	**Total**
Option 1					
Option 2					
Option 3					
Option 4					

Figure 13.12 Example of the general structure of a prioritization matrix.

options, use the standard rank averaging procedure you would use with any statistical data. For example, if two rank for third out of four options, then they have ranks of third and fourth and rank values of 2 and 1, respectively. The average rank value is 1.5. Thus, both would be assigned a rank value of 1.5.

6. When all criteria (that is, all columns) have been ranked, create another prioritization matrix with blank cells and cross-multiply the relative importance or weighting in each column by the rank for each option to create a new row value. Do this for all options. When complete, sum across to create a row total.
7. Select the option with the highest row total.
8. Conduct a sanity check to ensure the solution makes sense.

EXAMPLE 13.6

Consider the example given in Figure 13.13.

In the example, the options are four software packages: A, B, C, and D. The team determines by consensus the criteria against which the options will be measured and the relative importance (that is, weighting) of each criterion.

The criteria and their relative importance follows:

Determine the most suitable software package					
Relative importance	0.25	0.30	0.40	0.05	1.00
	Criteria				
Options	**Compatibility**	**Cost**	**Ease of use**	**Training time**	**Total**
Package A	4	1.5	3	3	N/A
Package B	1	4	2	2	N/A
Package C	3	1.5	4	4	N/A
Package D	2	3	1	1	N/A

Figure 13.13 First step of the prioritization matrix for Example 13.6.

Continued

Continued

Compatibility: 0.20

Cost: 0.30

Ease of use: 0.40

Training time: 0.05

Total time: 1.00

Each option is ranked for each criterion, with 1 being the lowest and 4 being the highest. In the example in Figure 13.13, packages A and C are tied for third place for cost. Thus, the two ranks are third and fourth with rank values of 2 and 1, respectively. Therefore, each option receives a rank value of 1.5. Package B has the lowest cost (that is, highest rank), so it is assigned a rank value of 4. Package D has the next lowest cost, so it is assigned a rank value of 3. Packages A and C have the same cost, so each is assigned a rank value of 1.5.

The next step, as shown in Figure 13.14, is to multiply each of the option values by the relative importance (that is, weightings) and calculate the row totals. (Note the calculations shown in Figure 13.14. The "4" and the "1" are obtained from the corresponding cells in Figure 13.13.) The option with the highest total represents the team's consensus and therefore its software package selection. From Figure 13.12, we can see that the team will select software package C.

Determine the most suitable software package					
Relative importance	0.25	0.30	0.40	0.05	1.00
	Criteria				
Options	**Compatibility**	**Cost**	**Ease of use**	**Training time**	**Total**
Package A	1.00	0.45	1.20	0.15	2.80
Package B	0.25	1.20	0.80	0.10	2.35
Package C	0.75	0.45	1.60	0.20	3.00
Package D	0.50	0.90	0.40	0.05	1.85

(0.25) × 4

(0.25) × 1

Figure 13.14 Second step of the prioritization matrix for Example 13.6.

For other methods of determining the best solution among several options using the prioritization matrix, see Tague (2005).

Activity Network Diagram

The *activity network diagram* (AND) is a tool used to illustrate a sequence of events or activities (nodes) and the interconnectivity of such nodes. It is used for scheduling and especially for determining the critical path through nodes. It is also known as an *arrow diagram*. We touched on the activity network diagram in the Chapter 12 introduction to Gantt charts.

The basic procedure for developing the activity network diagram (or arrow diagram) follows:

Note: The following is an excerpt from Tague (2005).

1. *List all the necessary tasks in the project or process. One convenient method is to write each task on the top half of a card or sticky note. Across the middle of the card, draw a horizontal arrow pointing right.*
2. *Determine the correct sequence of the tasks. Do this by asking three questions for each task:*
   - *Which tasks must happen before this one can begin?*
   - *Which tasks can be done at the same time as this one?*
   - *Which tasks should happen immediately after this one?*

   It can be useful to create a table with four columns—prior tasks, this task, simultaneous tasks, and following tasks.
3. *Diagram the network of tasks. If you are using notes or cards, arrange them in sequence on a large piece of paper. Time should flow from left to right and concurrent tasks should be vertically aligned. Leave space between the cards.*
4. *Between each two tasks, draw circles for "events." An event marks the beginning or end of a task. Thus, events are nodes that separate tasks.*
5. *Look for three common problem situations and redraw them using "dummies" or extra events. A dummy is an arrow drawn with dotted lines used to separate tasks that would otherwise start and stop with the same events or to show logical sequence. Dummies are not real tasks.*

   Problem situations:

   - *Two simultaneous tasks start and end at the same events.*

     Solution: *Use a dummy and an extra event to separate them. In Figure 13.15, event 2 and the dummy between 2 and 3 have been added to separate tasks A and B.*

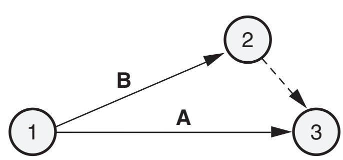

Figure 13.15 Dummy separating simultaneous tasks.

- *Task C cannot start until both tasks A and B are complete; a fourth task, D, cannot start until A is complete, but need not wait for B. (See Figure 13.16.)*

  Solution: *Use a dummy between the end of task A and the beginning of task C.*

- *A second task can be started before part of a first task is done.*

  Solution: *Add an extra event where the second task can begin, and use multiple arrows to break the first task into two subtasks. In Figure 13.17, event 2 was added, splitting task A.*

6. *When the network is correct, label all events in sequence with event numbers in the circles. It can be useful to label all tasks in sequence, using letters.*
7. *Determine task times—the best estimate of the time that each task should require. Use one measuring unit (hours, days, or weeks) throughout, for consistency. Write the time on each task's arrow.*
8. *Determine the "critical path"—the longest path from the beginning to the end of the project. Mark the critical path with a heavy line or color. Calculate the length of the critical path: the sum of all the task times on the path.*

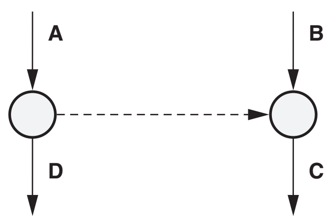

Figure 13.16 Dummy keeping sequence correct.

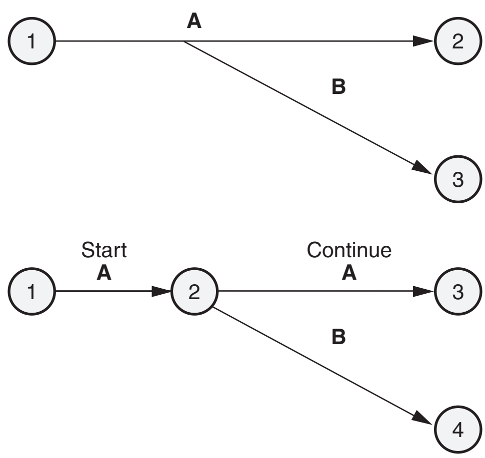

Figure 13.17 Using an extra event.

9. Calculate the earliest times each task can start and finish, based on how long preceding tasks take. These are called earliest start (ES) and earliest finish (EF). Start with the first task, where ES = 0, and work forward. Draw a square divided into four quadrants, as in Figure 13.18. Write the ES in the top left box and the EF in the top right.

   For each task:

   - Earliest start (ES) = The largest EF of the tasks leading into this one
   - Earliest finish (EF) = ES + Task time for this task

   Note: Using an ES = 0 creates a problem directly with transferring tasks to a Gantt chart since there is no time zero or day zero. It is for this reason that we pointed out the two important notes at the beginning of Example 12.2.

10. Calculate the latest times each task can start and finish without upsetting the project schedule, based on how long later tasks will take. These are called latest start (LS) and latest finish (LF). Start from the last task, where the latest finish is the project deadline, and work backwards. Write the LS in the lower left box and the LF in the lower right box.

    - Latest finish (LF) = The smallest LS of all tasks immediately following this one
    - Latest start (LS) = LF – Task time for this task

11. Calculate slack times for each task and for the entire project.

    - Total slack is the time a job could be postponed without delaying the project schedule.
      - Total slack = LS – ES = LF – EF

    - Free slack is the time a task could be postponed without affecting the early start of any job following it.
      - Free slack = The earliest ES of all tasks immediately following this one – EF

Figure 13.19 shows a schematic way to remember which numbers to subtract.

ES Earliest start	**EF** Earliest finish
LS Latest start	**LF** Latest finish

Figure 13.18 Arrow diagram time box.

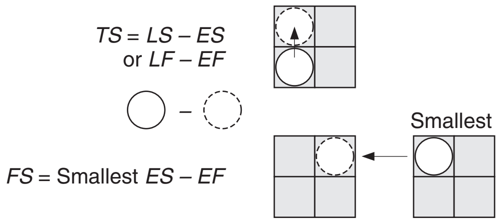

Figure 13.19 Remembering slack calculations.

EXAMPLE 13.7

Tasks. *Figure 13.20 shows an arrow diagram for a benchmarking project. There are 14 tasks, shown as solid arrows. The number on the arrow is the task time in days, based on the experience and judgment of the group planning the project.*

Events. *There are 15 events, represented by the circled numbers. The events mark the beginning and ending times for the tasks. They will serve as milestones for monitoring progress throughout the project.*

Dummies. *Tasks E and F show the first kind of problem situation, where two tasks can occur simultaneously. Developing a detailed plan and conducting preliminary research can be done at the same time. Dummy 6-7 separates them. Task 10-11 is also a dummy to separate the simultaneous tasks J (develop questions) and K (identify current state).*

The second kind of problem situation is illustrated around tasks H, I, J, K, and L. Task L (analyzing public data) cannot start until both tasks H (identifying key practices and measures) and I (collecting public data) are done. Task J (develop questions) and task K (identify current state) can start as soon as task H is complete. Task 8-9 is a dummy to separate the starting time of tasks J and K from that of task L.

The third kind of problem situation is illustrated around tasks M, N, and O. Originally, tasks N and O were one task—develop new process. Task M (visiting benchmark partners) can begin at the same time, but the visits need to be completed before final decisions can be made about the new process. To show this, subtask O (finalize new process) and extra event 13 were created. A dummy was also needed because tasks M and N are simultaneous. Now the network shows that the visits must be completed before the process can be finalized.

Critical path. *The longest path, marked in bold, is the critical path. Its tasks and their times are:*

A	B	C	D	F	I	L	N	O	P	Total
5	2	10	2	4	10	6	12	2	4	57

The team was surprised that the external visits, which members assumed would be a scheduling bottleneck, were not on the critical path—if they can schedule visits with a three-week lead time, as task G assumes. The entire project will require 57 days if all tasks on the critical path proceed as scheduled.

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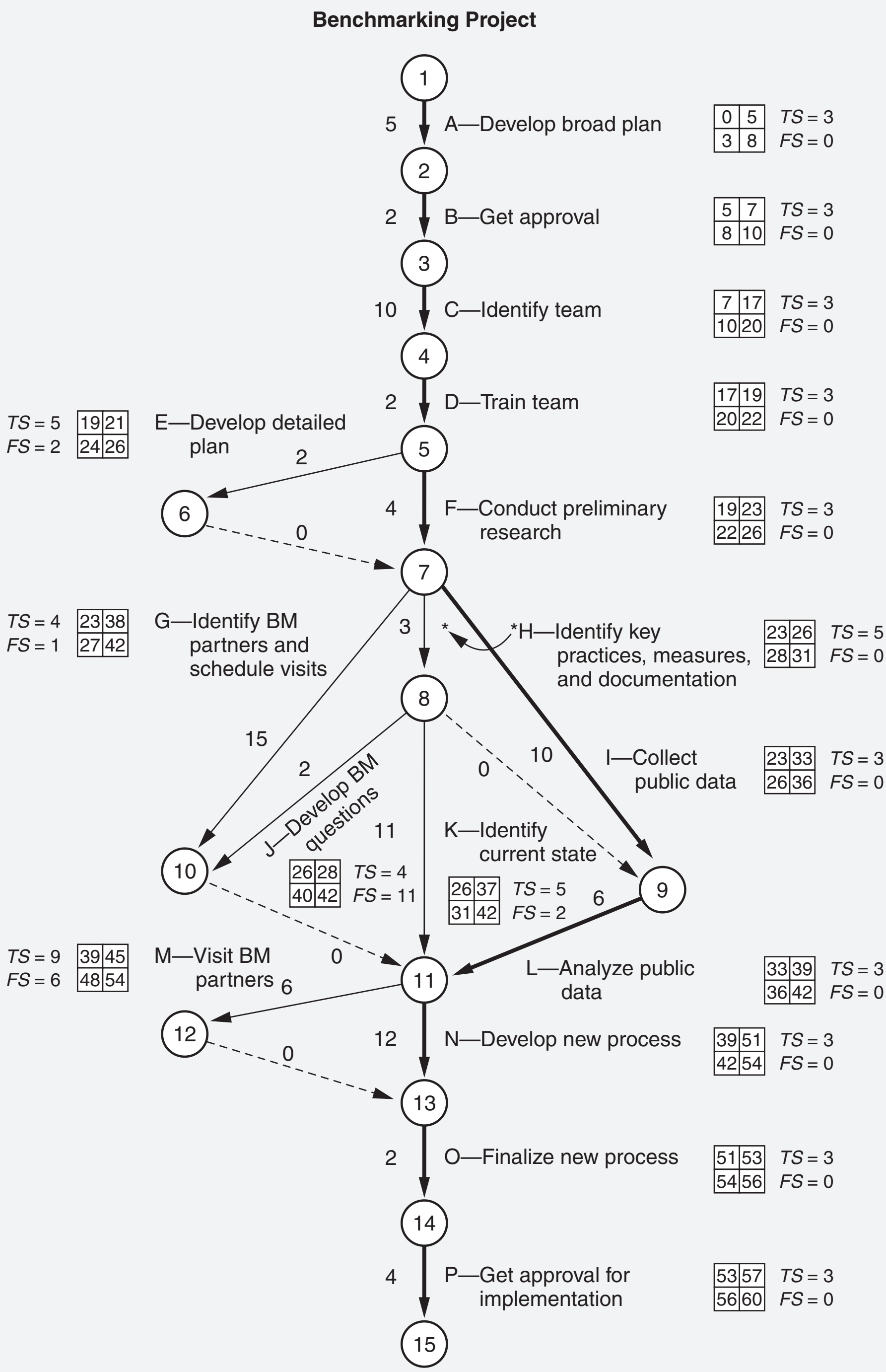

Figure 13.20 Arrow diagram example.

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Earliest and latest start and finish times. *The earliest times are calculated forward from the start. The ES is the largest EF of the tasks just before the one being calculated. The ES for the first task is zero. (Sic. See Table 13.4.)*

The latest times are calculated backward from the end. LF is the smallest LS of the tasks just after the one being calculated. The project has a 12-week deadline, or 60 working days, so 60 is used as the LF for the last task. (Sic. See Table 13.5.) All these times are shown beside each task in the four-box grid.

Slack times. *The slack times for each task are shown beside each grid. Because 60 days have been allotted for the project and the critical path adds up to 57 days, there is slack time for all the tasks. For each of the tasks on the critical path, the total slack time is three, the slack time for the entire project. The tasks not on the critical path have up to nine days total slack.*

Surprisingly, there is free slack of six days for visiting the benchmark partners, so six days of scheduling problems with the partners can be tolerated without delaying later tasks. There is total slack of nine days for the visits, so if the project is on schedule up to that task, nine days' delay in making visits will not affect the project completion date.

Note: See Table 13.16 for slack time calculations. Tables 13.4 and 13.5 were partially completed in Tague (2005). In addition, Table 13.6 was not part of the original example. However, we included it to help demonstrate how slack times are calculated.

Table 13.4 Calculation of earliest start and earliest finish times for Example 13.7.

Task	ES +	Task time =	EF
A	0	5	5
B	5	2	7
C	7	10	17
D	17	2	19
E	19	2	21
F	19	4	23
G	23	15	38
H	23	3	26
I	23	10	33
J	26	2	28
K	26	11	37
L	33	6	39
M	39	6	45
N	39	12	51
O	51	2	53
P	53	4	57

Source: Adapted from Tague (2005).

Continued

Table 13.5 Calculation of latest start and latest finish times for Example 13.7.

Task	LF –	Task time =	LS
P	60	4	56
O	56	2	54
N	54	12	42
M	54	6	48
L	42	6	36
K	42	11	31
J	42	2	40
I	36	10	26
H	31	3	28
G	42	15	27
F	26	4	22
E	26	2	24
D	22	2	20
C	20	10	10
B	10	2	8
A	8	5	3

Source: Adapted from Tague (2005).

Continued

Continued

Table 13.6 Calculation of earliest start and earliest finish times for Example 13.7.

Task	ES	LS	EF	LF	TS = LS – ES	FS
A	0	3	5	8	3	5 – 5 = 0
B	5	8	7	10	3	7 – 7 = 0
C	7	10	17	20	3	17 – 17 = 0
D	17	20	19	22	3	19 – 19 = 0
E	19	24	21	26	5	21 – 19 = 2
F	19	22	23	26	3	23 – 23 = 0
G	23	27	38	42	4	39 – 38 = 1
H	23	28	26	31	5	26 – 26 = 0
I	23	26	33	36	3	33 – 33 = 0
J	26	40	28	42	4	39 – 28 = 11
K	26	31	37	42	5	39 – 37 = 2
L	33	36	39	42	3	39 – 39 = 0
M	39	48	45	54	9	54 – 48 = 6
N	39	42	51	54	3	51 – 51 = 0
O	51	54	53	56	3	53 – 53 = 0
P	53	56	57	60	3	0

Source: Adapted from Tague (2005).

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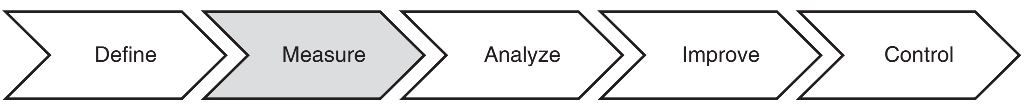
Part V

Measure



The purpose of this phase is to collect process performance data for primary and secondary metrics to gain insight and understanding into root causes and to establish performance baselines. Common tools used in this phase are shown in Table V.1.

Table V.1 Common tools used in *measure* phase.

Basic statistics	*Histograms*	*Project tracking*
Brainstorming	*Hypothesis testing*	Regression
Cause-and-effect diagrams	*Measurement systems analysis (MSA)*	*Run charts*
Check sheets	*Meeting minutes*	*Scatter diagrams*
Circle diagrams	Operational definitions	Spaghetti diagrams
Correlation	*Pareto charts*	*Statistical process control (SPC)*
Data collection	Probability	*Suppliers–inputs–process–outputs–customers (SIPOC)*
Data collection plan	*Process capability analysis*	Taguchi loss function
Failure mode and effects analysis (FMEA)	Process flow metrics	*Tollgate review*
Flowcharts	*Process maps*	*Value stream maps*
Gage R&R	*Process sigma*	
Graphical methods	*Project management*	

Note: Tools shown in *italic* are used in more than one phase.

Chapter 14

Process Characteristics

This chapter will address three key topics dealing with process characteristics: (1) a variety of process flow metrics that can be used to gauge the health of a process, (2) how these process flow metrics relate to the concept of the hidden factory that many organizations know all too well, and (3) a host of visualization tools used to analyze process flow.

PROCESS FLOW METRICS

> Identify and use process flow metrics (e.g., work in progress (WIP), work in queue (WIQ), touch time, takt time, cycle time, throughput) to determine constraints. Describe the impact that "hidden factories" can have on process flow metrics. (Analyze)
>
> **Body of Knowledge V.A.1**

Consider any business process such as generating an invoice, a government process such as serving a subpoena, a manufacturing process, or a design process. The following metrics may be used to evaluate flow in the process:

- *Work in progress* (WIP). This is the all the material that has been input into the process, but that has not reached the output or finished stage. This includes the material being processed, waiting to be processed, or stored as inventory at each step. Work in progress is also known as *work in process*.

  In the case of a business process such as invoice processing, WIP would be the number of invoices that have started the process but that have not yet finished it. In general, the larger the WIP, the longer the time required to complete the process. WIP, although considered an asset by some accounting systems, is in many ways a liability because it requires space, environmental control, labeling, record keeping, and other upkeep expenses.

• *Work in queue* (WIQ). This is the material waiting to be processed by one or more steps in the process and is one component of WIP. It is important to segment the WIQ by step to help identify process bottlenecks or constraints. Also, it is desirable to maintain WIQ in "first in, first out" (FIFO) order depending on the shelf life of the inventory.

• *Touch time*. This is the time that a unit of product is actually being worked on at any step in the process. Again, it may be useful to segment the total touch time by step so that processes with large touch times may be readily identified for improvement purposes. Touch time is generally considered value-added. See the note below.

• *Takt time*. This is the rate in time per unit at which the process must complete units in order to achieve the customer demand. The takt time is computed as:

$$\text{Takt time} = \frac{\text{Time available}}{\text{Number of units to be produced}}$$

Takt is similar to the word "drumbeat" in German, referring to the constant rate of output.

Occasionally the takt time (the reciprocal of the takt rate) is confused with the takt rate, which is the customer demand stated in units per time and computed as:

$$\text{Takt rate} = \frac{\text{Number of units to be produced}}{\text{Time available}}$$

• *Cycle time*. This is the time required to complete one unit from the beginning of the process to the end of the process. The average cycle time is computed over the completion of multiple units and thus cycles. If the cycle is greater than the takt time, the customer demand will not be met without a modification to the process. Such a modification might include an increase of manpower.

• *Throughput*. This is the amount of output that passes through the process in a specified period of time. For example, a takt time of 10 minutes per subpoena is equivalent to a throughput rate of six subpoenas per hour.

EXAMPLE 14.1

Forty-eight subpoenas are to be served in 480 minutes. Therefore:

$$\text{Takt time} = \frac{480 \text{ minutes}}{48 \text{ subpoenas}} = 10 \text{ minutes/subpoena}$$

$$\text{Takt rate} = \frac{48 \text{ subpoenas}}{480 \text{ minutes}} = 0.1 \text{ subpoenas/minute}$$

The following two definitions are not part of the body of knowledge, but are relevant to this subject matter and thus are included for completeness purposes:

• *Value-added activity.* An activity that adds value to a product or service. In the case of a product, the value-added work changes the form, fit, or function of a product.

A special note of clarification is required. In some instances, a valued-added activity is defined as an activity for which the customer is willing to pay. However, customers sometimes pay for activities that do not add value. Consequently, we cannot define value-added activities as such. We must define them independently of the customer, as we have done above.

By definition, all other activities that are not value-added are non-value-added and should be scrutinized for possible reduction, simplification, or elimination. Note that, based on the discussion above, touch time may not necessarily be the same as value-added time.

• *Setup time.* This is the time from the last unit on one job to the first good unit of the next job, and includes the time it takes to remove all the dies and tooling from the previous job and replace them with the dies and tooling for the next job. Setup time is also known as *changeover time*, *turnover time*, or *prep time*. Although it usually refers to manufacturing machinery, setup time may be used in other types of businesses, such as the time it takes to change a band's equipment on stage.

The above metrics, along with quality metrics, are useful in determining the health of a process. Both sets of metrics combined with improved design, manufacturing, and producibility techniques, in conjunction with Lean Six Sigma methodologies, will generally result in moving the direction of the metrics as follows:

- Reducing work in progress
- Reducing work in queue
- Reducing touch time
- Reducing takt time
- Reducing cycle time
- Increasing throughput
- Increasing value-added activity
- Reducing setup time

Inevitably, there will be a time when improving one metric will come at the expense of another. When this occurs, it will be necessary to optimize the process according to some criterion. Usually, that criterion is cost.

Therefore, it is valuable to understand the cost dimension associated with the various elements of a process, particularly when it comes time to make trade-off decisions or otherwise optimize the process, for example, the cost associated with touch time, the inventory in progress, inventory in queue, setup time, and so on.

Hidden Factory

Google the definition of the "hidden factory" and you'll see definitions all over the map. Many will indicate that the "hidden factory" relates to work that is not explicitly shown on any report, not documented on any process, or clocked on any time card.

This seems to indicate that organizations are almost completely ignorant about it. The above definitions paint a picture of "hear no evil," "see no evil," "speak no evil." Nothing can be farther from the truth.

Let's clarify the concept of the "hidden factory." The *hidden factory* constitutes the collection of activities in a process that generate waste. Consequently, the hidden factory loses time, money, and effort. Note: The concept of waste will be discussed in detail in Chapter 23.

Most of today's organizations that are sufficiently cognizant of what constitutes the hidden factory have, to some extent, attempted to track components of waste such as scrap and rework. Some organizations have tried to implement cost of quality systems, but many organizations are limited in their progress by their accounting systems. Often, it takes creative approaches to gimmick their systems to be able to track something simple such as the labor and material costs associated with a work group dedicated to rework and repair.

In the previous subsection we looked at a variety of process flow metrics. When the hidden factory exists (as it usually does), or is particularly large, these metrics are typically inflated. Consider the following examples of when defects occur along the process and rework must take place:

- Work in progress is increased because of defective raw material, and defective product occurs at various stages.
- Work in queue is increased for the same reason as work in progress.
- Cycle time is increased as defective products are being channeled off for rework and repair.
- Throughput is reduced because of rejected product due to defects.
- Setup time may increase because the time to produce the first good unit of the next lot takes longer because of defects being produced during the setup process.

The above are examples of what happens to the process flow metrics with just one example of the seven wastes (that is, defects). When influenced by other waste factors as well, the hidden factory can have an enormous effect on the bottom line of an organization. Fortunately, if used properly, the process flow metrics can help ferret out these waste factors in advance of serious trouble.

PROCESS ANALYSIS TOOLS

> Select, use, and evaluate various tools, e.g., value stream maps, process maps, work instructions, flowcharts, spaghetti diagrams, circle diagrams, gemba walk. (Evaluate)
>
> **Body of Knowledge V.A.2**

Flowcharts

A *flowchart* is a pictorial or graphical representation of a process and is used to help understand, communicate, document, improve, and/or manipulate a process off-line. An example of a common form of a flowchart is shown in Figure 14.1. The flowchart is also known as a *process flowchart* or *flow diagram,* among a variety of other names.

Numerous symbols can be used when developing flowcharts. These are depicted in Figure 14.2. Flowcharting has been used extensively with computer programming, as evident by some of the symbols shown in this figure. Although dated, virtually anything can be flowcharted through the collected use of these 28 symbols. Twenty-eight is a lot of symbols to use, however, and it is common to use a reduced set of symbols, at least in the early stages of development, primarily for convenience purposes. This reduced set is shown with dashed lines in Figure 14.2.

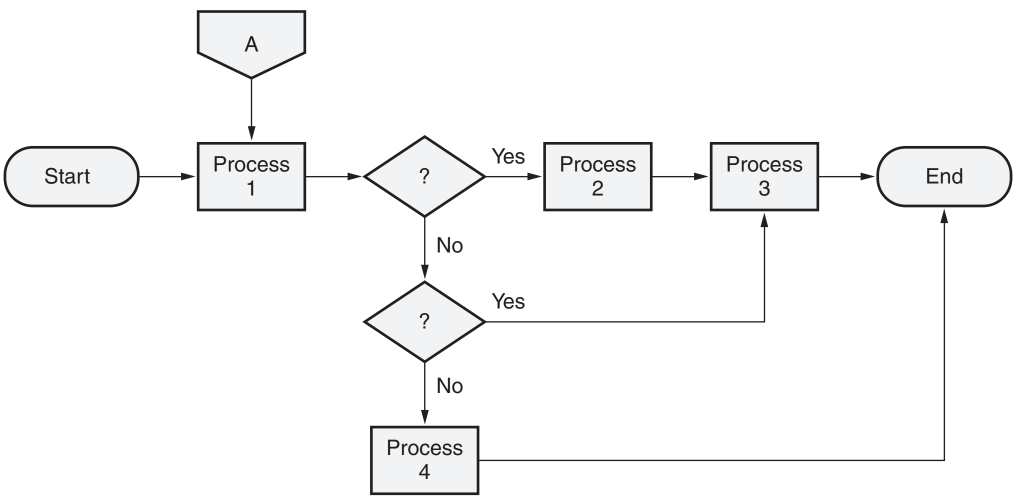


Figure 14.1 Example of a common flowchart.

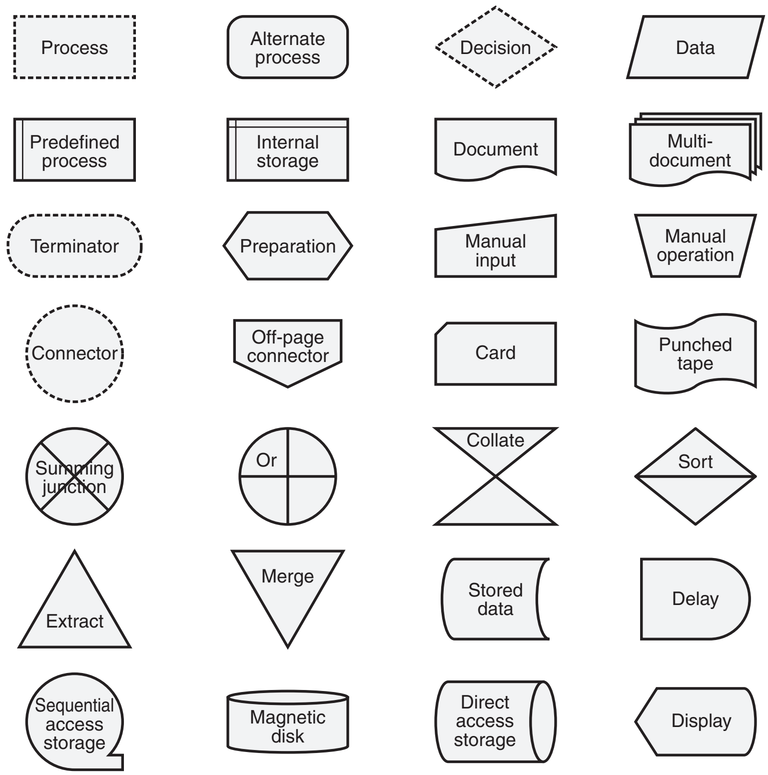


Figure 14.2 Common flowchart symbols available in PowerPoint .

Tague (2005) suggests the following process for developing flowcharts:

1. Define the process to be flowcharted
2. Define the boundaries of the process
3. Determine the level of detail to be included
4. Brainstorm the activities that take place
5. Arrange the activities in the proper sequence
6. Draw arrows to show the flow of the process
7. Review the process with others involved to seek agreement

Various types of flowcharts exist. One of the most common and most useful variants is the swim lane flowchart depicted in Figure 14.3. This particular flowchart can be oriented either vertically or horizontally. Notice that processes are segmented by functional group, department, or other logical organizational unit. This is particularly valuable in that it provides visual clarity into the overall circuitous nature of the process. For example, look at both the marketing and finance departments. Notice that the process oscillates between these two departments

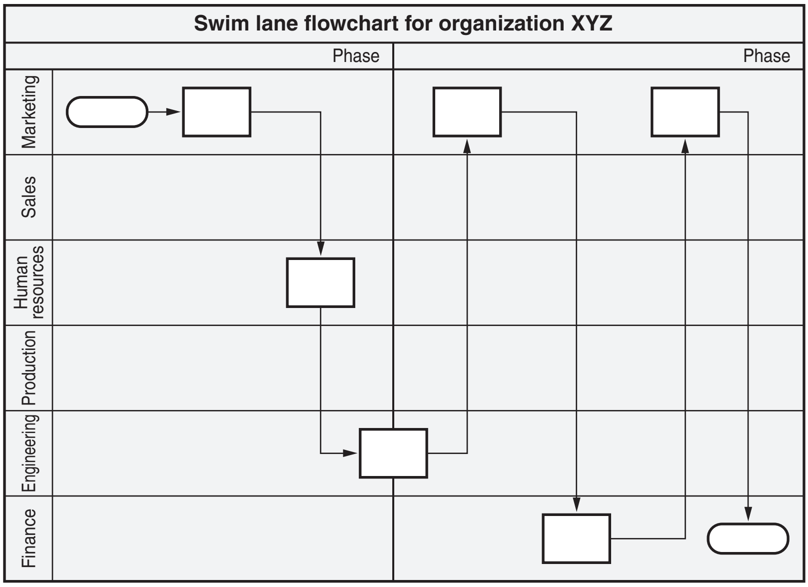


Figure 14.3 Example of a swim lane flowchart.

before being terminated in the finance department. This raises questions regarding process flow and surfaces opportunities for improvement.

Process Maps

Although process maps use the same symbols as flowcharts, they reveal much more about a process than flowcharts do. Process maps provide the Black Belt with more guidance because they address the key foundational concept of Lean Six Sigma: the concept of $Y = f(X)$, or simply, outputs are a function of inputs.

If the Lean Six Sigma project should take on more of a lean focus, the process map can evolve into a *value stream map* (VSM). However, the value stream map differs from the process map in that the symbols used are relevant to lean.

Categorizing and Classifying Process Input Variables

Kubiak (2007) describes the basic anatomy of a process map, as shown in Figure 14.4. *Input variables*, our list of X's, flow into a transform function that we call a *process*. Flowing from the process is one or more desired *outputs*.

Notice that the inputs appear below the flow line, while outputs appear above the flow line. This is simply a matter of aesthetics. Otherwise, inputs and outputs appearing on the same line would significantly clutter the map, particularly when multiple process blocks appear adjacent to one another.

Notice the list of input variables. They should be familiar as they represent the typical categories (also known as the *six M's*) of the main bones in a fishbone diagram. (Note: Some authors include "management" as another category and call

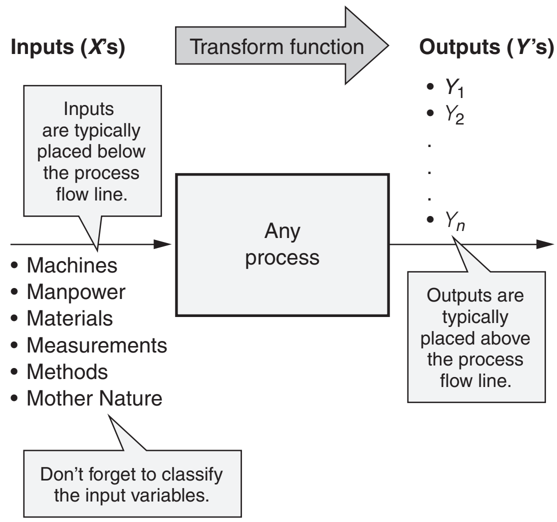


Figure 14.4 Demonstrating the $Y = f(X)$ concept.

the list the *seven M*'s.). Using the six M's as a structured approach provides some assurance that few inputs will be overlooked. Table 14.1 identifies each of the M's along with alternative terminology that may be used in service- or transaction-based industries, for example. Commentary and insight about each "M" have been included.

One important aspect of input variables not visible in Figure 14.4 is their classification. Classifying input variables helps Lean Six Sigma practitioners focus on those inputs that are controllable and guides practitioners away from spending time and energy on those that are not.

An example of a common classification scheme is shown in Table 14.2. Properly classifying input variables requires recognizing the need to fully understand a process in relation to the culture and business needs of the organization.

Let's assume that a technician represents the manpower required to perform a given process. What classification or variable designation from Table 14.2 should a Black Belt assign it?

- If the team determines that the skill and experience level of the technician can be assigned appropriately to the needs of the process, then the team would classify "technician" as a controllable input variable. It would appear in the process input list as "technician (C)." The input variable is followed by the classification designation in parentheses.
- If the team determines that management will never invest in training technicians and that the technician assigned to the process is fixed, the team may consider classifying "technician" as a noise input variable. If this were the case, then it would appear in the process input list as "technician (N)."

Table 14.1 The six M's: useful categories when thinking about input variables.

Input (*X*)	Alternate terminology	Comment
Machines	Equipment	Machines or equipment need not be costly or even high-tech. Don't overlook the basics. For example, service and/or transactional-based processes might include the use of simple devices such as a stapler or highlighter. When considering input variables of this nature, it is often helpful to consider developing a list of "tools required."
Manpower	People	Human resources may take various forms such as skilled technicians, engineers, or administrative and clerical personnel. Even highly automated processes may occasionally call for human support when preventive or corrective maintenance or actions are required.
Materials	Materials	Materials may include raw materials or even intermediate subassemblies. Materials are often consumed or transformed during the execution of a process.
Measurements	Measurements	Always remember to ensure the measurement system is capable. If a measurement system is assumed capable when, in fact, it is not, erroneous results may occur. These include overlooking true critical input variables or concluding some variables are critical when they are not.
Methods	Processes	Processes come in all shapes and sizes. They may be well defined or very loosely defined. In the context of a manufacturing environment, there is a category of processes known as "special processes."
Mother Nature	Environment	Variables falling into this category can be associated with either an internal or external environment. This is an important distinction, particularly when determining whether such variables are "noise" variables. For example, temperature and humidity would likely be considered controllable variables when the underlying process takes place in a clean room environment. However, if another process is conducted outdoors, temperature and humidity might very likely be considered noise variables since they may be impossible or too costly to control.

This example illustrates the need to revisit processes on a periodic basis since organizations continually change and evolve. New and insightful management may recognize the need to grow and improve the skill level of its technicians.

Hence, input variables previously classified as noise may now be considered controllable. Likewise, the opposite may become true.

When counseling teams, I have generally found it more effective to focus on addressing whether an input variable can be controlled than trying to determine whether it will ever be controlled. This avoids the complications of having a team second-guess management.

Table 14.2 Classifying input variables.

Variable designation	Type of variable	Comment
C	Controllable	These are variables over which the process owner has control (regardless of whether such control may ever be exercised). More specifically, characteristics or values of controllable variables can be set or manipulated in a manner that drives one or more output variables (Y's) in the desired direction.
N	Noise	Noise variables are those that either can not be controlled or perhaps are too expensive to control. It is important that such variables be defined so that the Green Belt team or project Black Belt knows which variables should not be addressed. Attempts to control "noise" variables often result in frustrated teams and/or failed projects.
SOP	Standard operating procedure	A standard operating procedure is a unique and predefined way of performing a process. For example, it may be an instruction document for assembling a bicycle or preparing an expense report. Just because an input variable is defined as an SOP, no inference should be drawn regarding the quality of the process encompassed by the procedure. SOP variables are a subset of controllable variables. Designating an input variable as SOP does not exclude it from the process owner's control. However, it does suggest that minimal variation is probably associated with it.
X	Critical	Critical input variables are a subset of the set of controllable variables. These are variables that have been determined to have a significant impact on one or more output variables (Y's). Significance may be demonstrated through statistical tools such as design of experiments, regression, and so on. Early in the DMAIC phase and particularly before the completion of *analyze*, use of this designation should be considered, at best, tentative. Only after the analyze phase has been completed will we have the knowledge to state with some degree of certainty that a variable is a "critical input." Be careful not to confuse this classification of variable with the "X" used in the equation $Y = f(X)$.

Gaining Insights from the Process Map

Figure 14.4 demonstrated the basic architecture and components of the process map. If we expand Figure 14.4 so that it represents a series of linked processes or even subprocesses, we obtain something similar to what is presented in Figure 14.5.

Though still simple in nature, Figure 14.5 depicts a representation of processes we are likely to encounter in any organization. For the sake of convenience, inputs

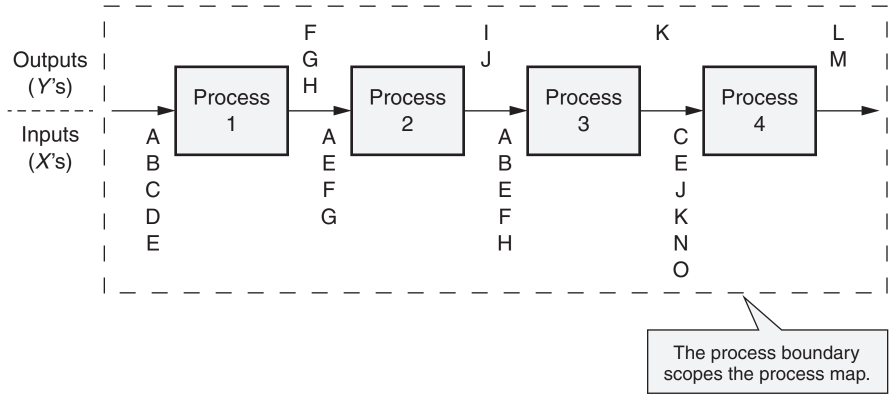


Figure 14.5 Analyzing inputs and outputs.

and outputs in Figure 14.5 have been defined simply as letters. Since input variable classifications are not the focus of this section, they have been omitted.

Upon reviewing Figure 14.5, several immediate observations can be made, including:

- Processes should have boundaries. Define them.
- Identical inputs may be required at different processes.
- Multiple process outputs may occur.
- Outputs of one process may become inputs of another process.
- Some outputs may or may not be linked to processes outside of the process boundaries.

We can now extend the thinking captured in the above bullets for each of the variables in Figure 14.5. These can be summarized easily in Table 14.3. Note the comments for each variable in Table 14.3. Notice that some of the comments are essentially a statement of fact, while others demand action, seek information, or require further investigation.

A well-developed process map serves as an effective communication tool and is a constant reminder of where a team should and should not focus its time and energy.

Value Stream Maps

According to Manos and Vincent (2012), a value stream is "all the actions (both value-added and non-value-added currently required to bring a product through the main flows essential to every product: (1) the production flow from raw material into the arms of the customer, and (2) the design flow from concept to launch."

Table 14.3 Identifying potential issues with inputs and outputs.

Variable	Comment
A	Input to processes 1, 2, and 3. However, it is not an output from any of the processes within the scope of the project at hand. If it is output from another process outside the scope, the project team should ascertain its source to ensure its availability when required.
B	Input to processes 1 and 3. However, it is not an output from any of the processes within the scope of the project at hand. If it is output from another process outside the scope, the project team should ascertain its source to ensure its availability when required.
C	Input to processes 1 and 4. However, it is not an output from any of the processes within the scope of the project at hand. If it is output from another process outside the scope, the project team should ascertain its source to ensure its availability when required.
D	Input to process 1. However, it is not an output from any of the processes within the scope of the project at hand. If it is output from another process outside the scope, the project team should ascertain its source to ensure its availability when required.
E	Input to processes 1, 2, 3, and 4. However, it is not an output from any of the processes within the scope of the project at hand. If it is output from another process outside the scope, the project team should ascertain its source to ensure its availability when required.
F	Output from process 1 and input to both processes 2 and 3.
G	Output from process 1 and input to process 2.
H	Output from process 1 and input to process 3.
I	Output from process 2. This variable is no longer found in the remainder of the process map. It might be an extraneous output that is no longer needed, and process 2 was never changed to eliminate its production. Alternately, it could be used in another process beyond the scope of the project at hand. Either way, action is required.
J	Output from process 2 and input to process 4.
K	Output from process 3 and input to process 4.
L	Output from process 4. It may or may not be used in another process.
M	Output from process 4. It may or may not be used in another process.
N	Input to process 4. However, it is not an output from any of the processes within the scope of the project at hand. If it is output from another process outside the scope, the project team should ascertain its source to ensure its availability when required.
O	Input to process 4. However, it is not an output from any of the processes within the scope of the project at hand. If it is output from another process outside the scope, the project team should ascertain its source to ensure its availability when required.

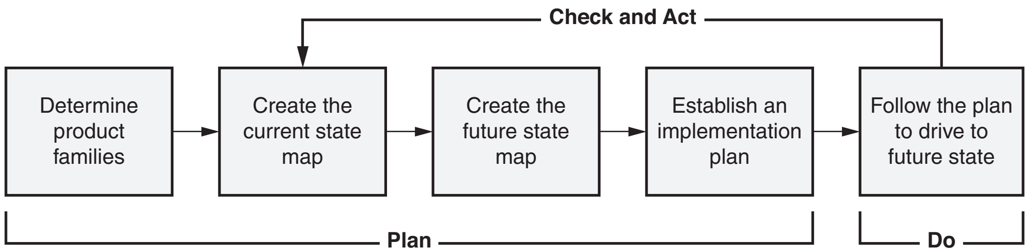


Figure 14.6 The process of value stream mapping.
Source: Manos and Vincent (2012).

They go on to say that "Value stream mapping is simply an illustration of the sequential activities that take place within a value stream. At each step of the map the practitioner evaluates whether value is being created and whether one or more of the wastes exist. The purpose of the value stream map is to identify those activities that add value and those that create waste, with the latter being targets for elimination."

The general process for value stream mapping is given in Figure 14.6. Although the process appears simple, it is wrought with detail and specifics and it requires walking the process, perhaps many times. This will be addressed later in the section on gemba walk.

The following subsection on current and future states is excerpted from Manos and Vincent (2012).

Current State

The value stream map can be considered a flowchart on steroids. While a flowchart is a simple visualization of a process, the value stream map is a bigger picture of the flow of a given product/service throughout the entire organization. Before beginning to map the process, it is important to plan how the process will be mapped. The following are considerations for value stream mapping:

1. *Collect information while walking the value stream. Common practice is to use a cross-functional team of individuals who operate within the value stream, including the value stream owner and supporting functions at various levels. Each individual should carry an A3-sized piece of paper and document the value stream and information while walking the product/service flow and information flow, from beginning to end of the value stream.*

2. *It is common practice to begin at the shipping end of the value stream, closest to the customer. This helps keep a customer focus throughout the entire value stream.*

3. *"Obtain all information directly from the value stream through observations, time study, inventory counting, and so on. Bring a stopwatch and*

physically count work-in-process (WIP). Do not rely on standard times and inventory levels established for the existing financial system, as this information is typically inaccurate.

4. *Draw the value stream manually during the mapping exercise. The team does not need to waste time while one plots on a computer and everyone else waits. Each person draws the value stream at the value stream. It is common practice to use butcher paper, markers, and sticky notes on a wall for the entire team to participate in constructing the value stream for discussion and clarifications. Sometimes things can be overlooked by a team, but a number of individuals performing the same task will likely catch information missed by a group.*

The common mapping format is for material flow on the bottom from left to right and information flow at the top from right to left (Figure 14.7).

The following are examples of information commonly gathered during the value stream mapping of the current state:

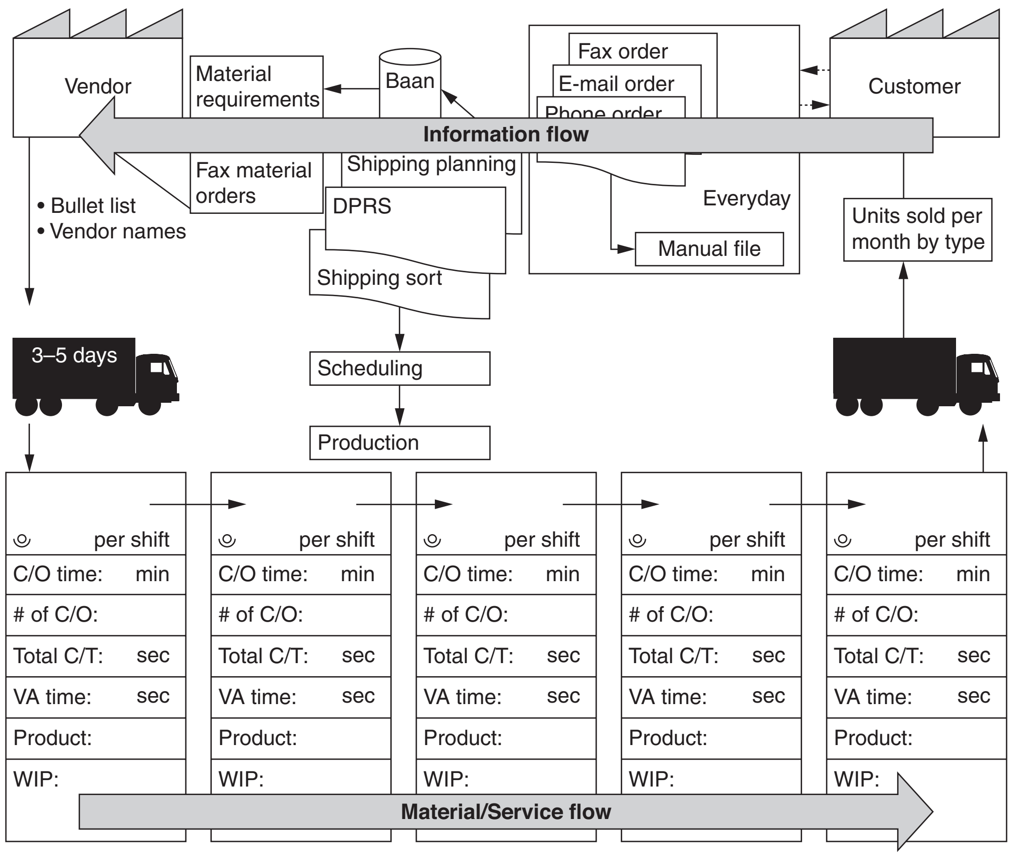


Figure 14.7 Basic structure of the value stream map.
Source: Manos and Vincent (2012).

- *Cycle time*
- *Changeover time*
- *Number of people*
- *Uptime/downtime*
- *WIP*
- *Inventory*
- *Packaging size*
- *Scrap/defect rate*
- *Total operating time (minus breaks)*
- *Value-added time versus non-value-added time*
- *Lead time*
- *Number of changeovers*

Figure 14.8 illustrates the basic structure of a value stream map, while many of the symbols are depicted in Figure 14.9.

Future State

With the current state of the value stream now mapped, it is time to look at what improvements are needed. Therefore, the lean value stream needs to be understood. Rother and Shook (2009) offer the following seven guidelines in Learning to See:

1. *Produce to the takt time*
2. *Create continuous flow wherever possible*
3. *Use supermarket pull systems to control production where continuous flow does not extend upstream*
4. *Send the customer schedule to only one production process (the pacemaker process)*
5. *Level the product/service mix evenly over time at the pacemaker process*
6. *Level the production volume by creating an "initial pull," releasing and withdrawing small, consistent increments of work at the pacemaker process*
7. *Develop the ability to make "every part every day" in processes upstream of the pacemaker process*

The intent is to brainstorm ideas for improving the current-state value stream and then reconstruct the value stream with the improvements in place (that is, the future-state value stream map).

Keep value stream mapping in perspective; it is simply a tool to visually show the current flow of product/service and information in the organization and to

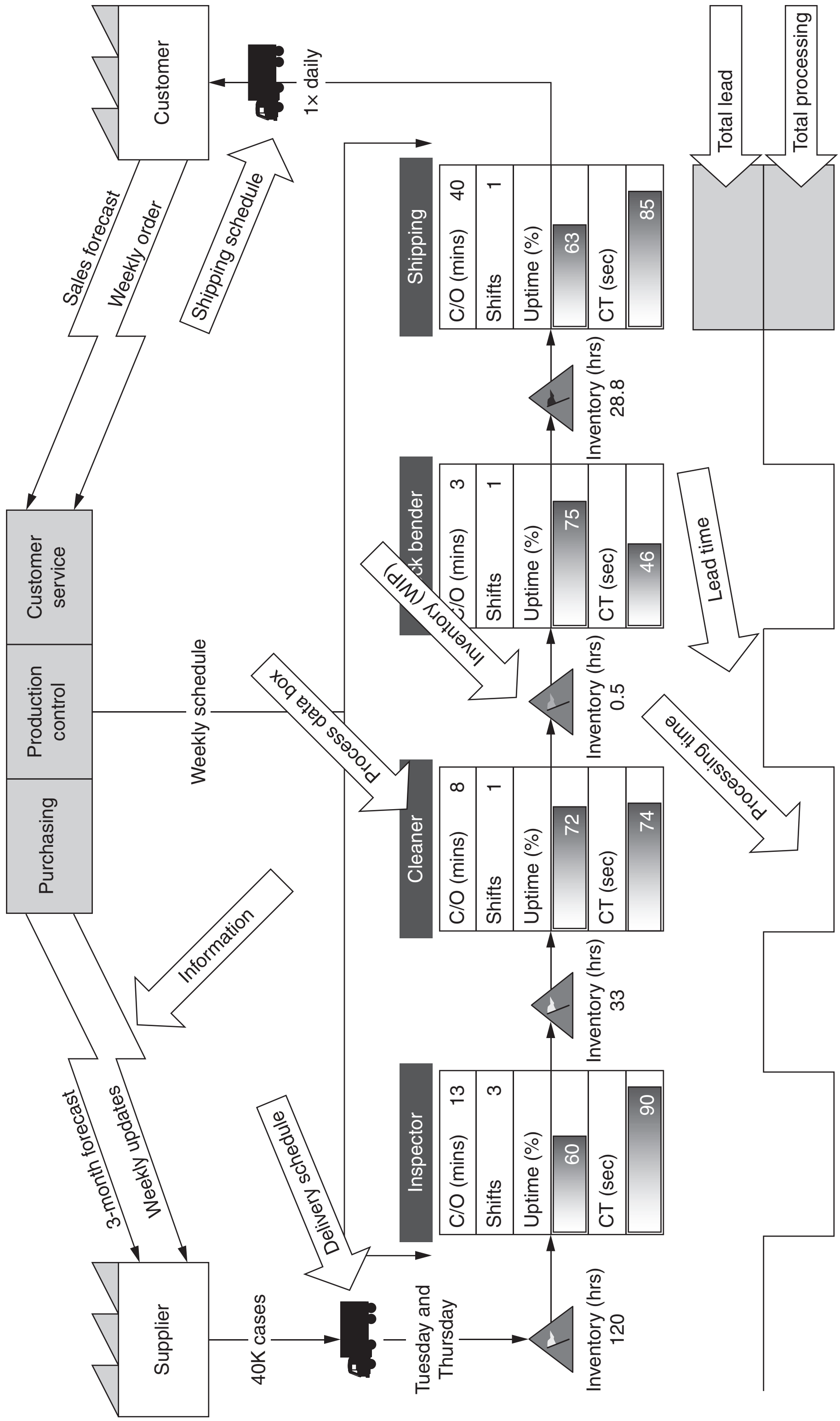


Figure 14.8 Value stream map components.
Source: Manos and Vincent (2012).

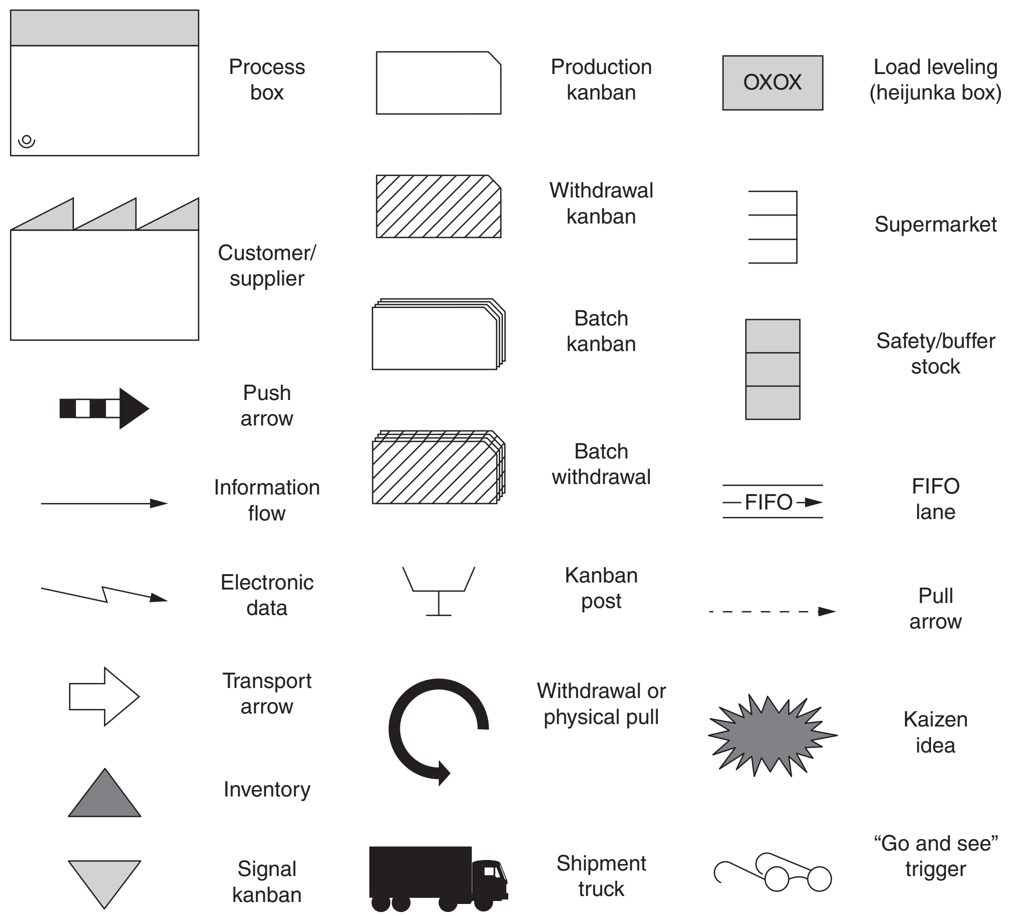


Figure 14.9 Common value stream mapping icons.
Source: Manos and Vincent (2012).

guide everyone in the organization through the analysis of the process to improve the flows and design improved value streams.

Work Instructions

A *work instruction* is something that describes how to perform a task. Regardless of how it is presented (that is, paper or electronic format, or something else), effective work instructions will identify at a minimum:

- All relevant header information that links the product or service to the work instruction
- All the parts required
- All the consumable raw materials

- All the tooling and fixtures required
- All the personnel required
- Potential safety hazards
- Potential opportunities to create defects
- Meaningful and properly annotated graphics and designs
- The exact sequence of steps required for assembling, manufacturing, producing, generating, and so on, the product or service governed by the work instruction

In addition to the above, a work instruction will contain a document number and revision number so that it can be properly identified. This is a critical piece of information so that proper work instruction documentation can be matched to the effectivity of engineering changes.

Furthermore, work instructions support lean activities through the creation of standardized work.

The question most often asked regarding work instructions is, "How much detail should be provided in a work instruction?" The answer to that question depends on how much variation an organization is willing or able to accept in the final output. Several key factors to consider are:

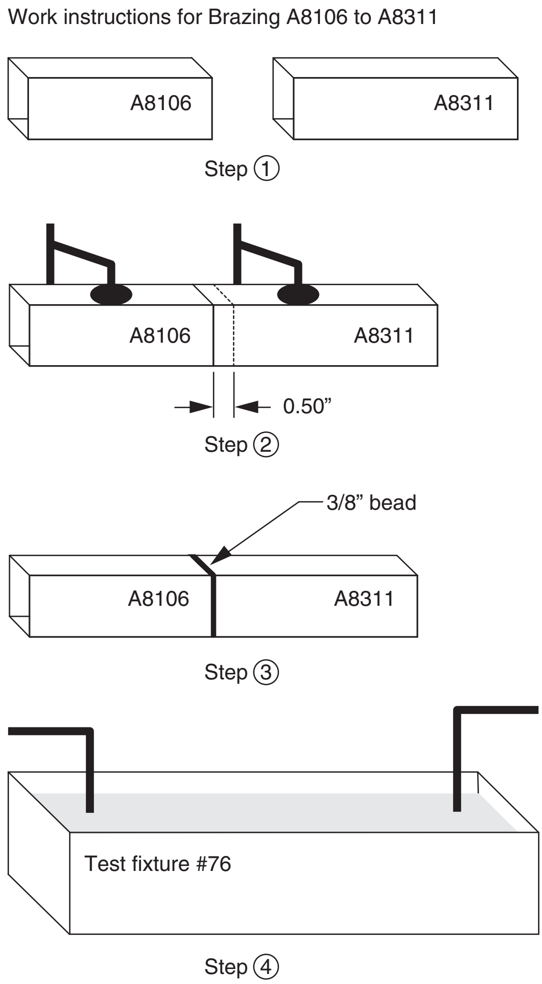
- How educated is the workforce that will be using the work instruction? The more educated, the less detail required. Consider "education" of the work force independent of a "trained" work force.
- How frequently will the workforce be using the work instruction? This has to be considered in conjunction with the first bullet. However, in general, the less frequently, the less detail, as long as the workforce has some training.
- How much variation can the organization tolerate in the output? The less tolerance, the more detail in the work instruction and vice versa.

There is no magic answer for writing work instructions. Each organization must address its own issues and strike a meaningful balance that it feels will work for it with regard to the above questions.

An example of written work instructions is illustrated in Figure 14.10.

Spaghetti Diagrams

Spaghetti diagrams refer to product flow paths in an organization. One example is shown in Figure 14.11. Spaghetti diagrams often depict very convoluted and circuitous paths. Consequently, these diagrams are very helpful in reducing unnecessary transportation and streamlining process product flow.



Work instructions for Brazing A8106 to A8311

Tools and parts list

- Type 44 brazer
- Clamping fixture #D22
- No. 24 brazing rod
- Test fixture #76
- Tubing parts A8106 and A8311

Step 1 Insert A8106 0.5” into A8311

Step 2 Clamp both parts

Step 3 Using #24 rod, place 3/8” bead at joint

Step 4 Water test in fixture #76

Step 5 Retouch as needed

Figure 14.10 Example of work instructions.

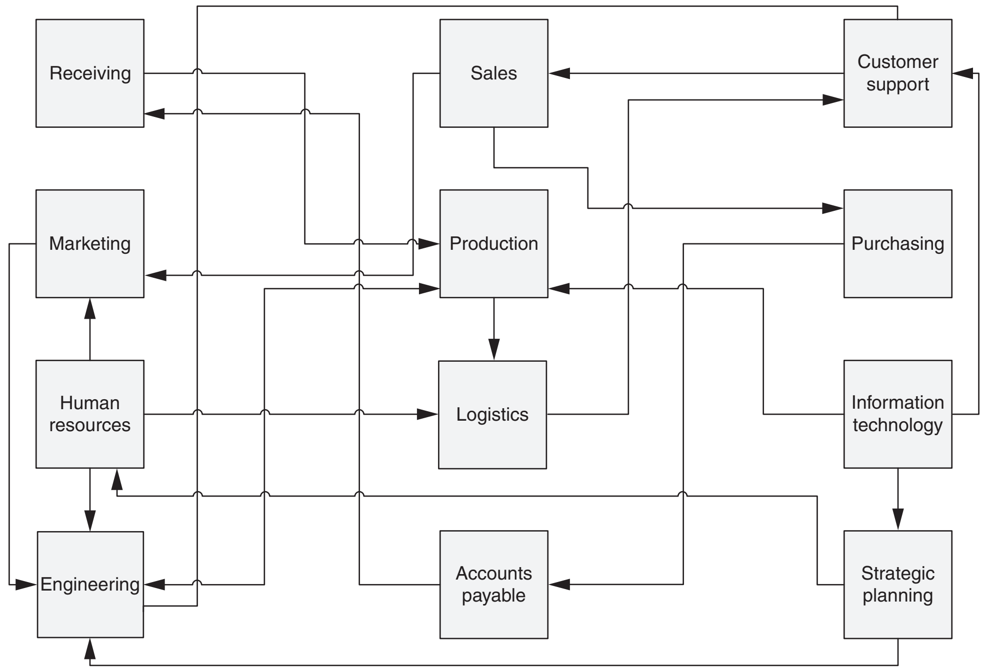


Figure 14.11 Example of a spaghetti diagram.

Circle Diagrams

A *circle diagram*, also known as a *hand-off map*, *relations diagram*, *or interrelationship digraph* (ID), is used to show linkages between various items. Mark Blazey, in his book *Insights to Performance Excellence*, uses these diagrams to show how the Baldrige Award criteria relate.

Figure 14.12 shows linkages for the criterion entitled "Process Effectiveness Outcomes" (Item 7.5). Although Figure 14.12 depicts item numbers, other relevant descriptors, such as departments, processes, and individuals, may be used. The direction of the arrows indicates the flow of products, services, information, and the like. Circle diagrams are highly useful in that they readily depict predecessor and successor relationships as well as potential bottlenecks. Too many inputs or outputs from any given descriptor around the circumference of the circle may indicate a limiting function. Examination of these specific descriptors might prove highly insightful.

Gemba Walk

A desk is a dangerous place from which to view the world.

—John le Carré

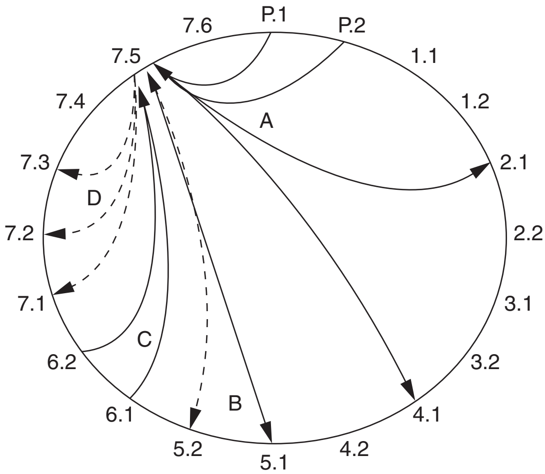


Figure 14.12 Example of a circle diagram.
Source: Used by permission from Mark L. Blazey, *Insights to Performance Excellence 2008: An Inside Look at the 2008 Baldrige Award Criteria* (Milwaukee: ASQ Quality Press, 2008), 222.

Gemba is a Japanese term that means "place of work" or "the place where the truth can be found." Google the term and you will see many different translations, all leading to the same intent. A gemba walk is also known as a *gemba visit*.

A gemba walk is a method of obtaining "voice of customer" information that requires the design team to visit and observe how the customer uses the product in his or her environment.

In the context of process analysis tools, a gemba walk is a critical and necessary step in learning how a process is actually accomplished. Otherwise, learning about a process is relegated to a conference room, a flip chart, and a half dozen conflicted subject matter experts arguing over how the process works.

Before beginning any gemba walk, it is important to establish a clear focus for the walk. The gemba walk is not the time to be critical, enforce policy, chastise, or solve problems. It is the time to learn, listen, and receive input.

If gemba walks are conducted on a regular basis independent of project improvement activities, it is important to consider the following:

- *Make the walks theme-based.* Change the focus of each gemba walk, such as housekeeping, up-to-date charts, safety, and so on.
- *Engage leaders.* Getting the leaders involved in the gemba walks helps them understand the processes and the value of the tool, and lets the organization know that gemba walks are sufficiently important to require their personal involvement.
- *Set expectations.* This sets the boundaries for what should and shouldn't be done on a gemba walk and how using gemba walks is beneficial.

- *Create a checklist.* The use of the checklist helps ensure that little is missed on the walk.
- *Publish what was learned.* This lets the entire organization know that the gemba walks are important. It also helps integrate their usage into the organization's culture.
- *Follow up on the process.* If processes are sufficiently important to receive a gemba walk, then the key learnings from the visit must be acted on. This should be built into the "gemba walk process."

Although the gemba walk appears to be a trivial technique, it is surprising how many process improvement activities are conducted without the benefit of a "look-and-see" visit to the workplace where the process is actually conducted. In essence, the improvement activity is conducted in absentia.

Chapter 15

Data Collection

Even if accurate data are available, they will be meaningless if they are not used correctly. The skill with which a company collects and uses data can make the difference between success and failure.

—Masaaki Imai

This chapter will cover many different aspects of data-related concepts and activities, including the types of data, the importance of understanding the different data measurement scales, various sampling designs and their uses, and effective approaches to collecting data.

TYPES OF DATA

> Define, classify, and distinguish between qualitative and quantitative data, and continuous and discrete data. (Evaluate)
>
> **Body of Knowledge V.B.1**

Qualitative and Quantitative Data

Data can be categorized into two types:

- Qualitative
- Quantitative

Qualitative data are data based on descriptive information rather than numerical information. This descriptive information is in terms of the characteristics, attributes, properties, and qualities of an object. For example, an object may be described as "heavy," "blue," "smooth," and so on. As such, these descriptions categorize. Qualitative data are sometimes referred to as *attribute* data. In addition, it is important to note that qualitative data are subjective data.

Qualitative data are usually collected in a free-form manner. This would include the use of observations, interviews, surveys with open-ended questions, and narrative data from social media and personal websites, among others.

Privacy issues aside, one of today's hottest topics is the collection of Big Data, which would include the collection of narrative (that is, qualitative) data from such sources as Twitter, Facebook, and other popular social media sites. As the data collection and analysis software matures, this data will yield valuable information to those who are able to mine it efficiently and effectively since it provides added dimension and insight into what was heretofore unknowable about customers.

Quantitative data are data that can be measured, verified, and manipulated. Quantitative data are also known as *numerical* data. For example, height, weight, volume, area, impedance, humidity, and so on, would be quantitative data. There is a special discussion regarding discrete data in the subsection below. Suffice it to say at this point that quantitative data can be discrete data as well.

Continuous and Discrete Data

Pyzdek and Keller (2010) define discrete data when they state, "Data are said to be discrete when they take on only a finite number of points that can be represented by the nonnegative integers." Discrete data are count data and are sometimes called *categorical* data. Herein lies the problem with the loose use of terminology. Categorical or attribute data are distinct from discrete data.

Pyzdek and Keller (2010) define continuous data when they state, "Data are said to be continuous when they exist on an interval, or on several intervals." Continuous data are also called *variables* data.

Continuous data result from measurement on a continuous scale such as length, weight, or temperature. These scales are called *continuous* because between any two values are an infinite number of other values. For example, between 1.537 inches and 1.538 inches, there are 1.5372, 1.5373, 1.53724, and so forth.

Effort should always be made to move from discrete to continuous measurements. There are several reasons for doing this:

- Control charts based on continuous data are more sensitive to process changes than those based on discrete data.
- When designed experiments are used for process improvement, changes in continuous data may be observed even though the discrete measurement hasn't changed.

EXAMPLE 15.1

Consider a stream of product that is being inspected. The inspector is reviewing the product and making notes on a data collection sheet. She labels the left column "Category/attribute" and the right column "Discrete data." In the left column, she writes either "non-reject" or "reject." In the right column, she writes either "0" for "non-reject" or "1" for "reject."

The two columns contain entirely different data. The right-hand column is scaled data (that is, transformed data) although the scale has no meaning.

- Continuous, or variables, data contain more information than discrete data. Note that variables data can be transformed into attribute data, but the reverse is not true. This can occur by grouping the variables data into attribute categories, which will transform to a nominal or ordinal scale.

  For example, a part with a measured value of seven inches exceeds the tolerance of five inches. Thus, the part would be considered a defective. However, because we measured the part, we know that it is not only defective, but by how much. We have taken a part measured on a continuous scale (that is, variables data) and transformed it to a nominal scale (that is, categorical/attribute data). The only information we now have about the part is that it is defective. Note that we initially measured the part, but did not record the length. We have lost valuable information. The various types of scales are discussed in detail in the next section.

MEASUREMENT SCALES

> Define and use nominal, ordinal, interval, and ratio measurement scales. (Apply)
>
> **Body of Knowledge V.B.2**

To avoid analysis errors, it is important to recognize the type of measurement scale used to collect data. There are four types of measurement scales:

- Nominal
- Ordinal
- Interval
- Ratio

Nominal scales classify data into categories with no order implied—for example, a list of equipment such as presses, drills, and lathes. When data are classified into two categories such as male/female, defective/non-defective, and so on, they are called *dichotomous*. Furthermore, the nominal scale does not really constitute a scale. In reality, it just labels, but does not scale along any dimension.

Ordinal scales refer to positions in a series, where order is important but precise differences between values aren't defined. An example might be the subjective ranking of three light bulbs as bright (third), brighter (second), and brightest (first). Unless the light emitted from each bulb were measured in lumens, the difference between the first and second bulbs and the second and the third bulbs would be unknown. Only the order of the bulbs would matter.

Interval scales have meaningful differences but no absolute zero, so ratios aren't useful. An example is temperature measured in °F, because 20 °F isn't twice as warm as 10 °F.

Ratio scales have meaningful differences, and an absolute zero exists. One example of a ratio scale is length in inches, because zero length is defined as having no length and 20 inches is twice as long as 10 inches. Other examples include weight and age.

The measurement scales along with other relevant information are given in Table 15.1. An important comment is necessary regarding this table. Note that the type of data for the ordinal scale is given as "Discrete." Many times ordinal data are treated or collected directly as discrete data, but manipulated as though they were continuous data and analyzed with parametric tests. Consider Example 15.2.

Table 15.1 A comparison of measurements scales and data types.

	Nominal	**Ordinal**	**Interval**	**Ratio**
=, ≠	✔	✔	✔	✔
<, >		✔	✔	✔
+, −			✔	✔
×, ÷				✔
Type of data	Qualitative	Qualitative	Quantitative	Quantitative
	Discrete	Discrete	Continuous	Continuous
Type of tests	Non-parametric	Non-parametric	Parametric	Parametric
Measures of central tendency	Mode	Mode	Mode	Mode
		Medium	Median	Median
			Mean	Mean
Measures of dispersion			Range	All
			Standard deviation	

EXAMPLE 15.2

An organization wants to survey its customers using a five-point Likert scale. The general construct of the scale looked like:

Very unsatisfied	Unsatisfied	Neutral	Satisfied	Very satisfied
1	2	3	4	5

Continued

Continued

Customers would match their feelings about the organization to the top categorical scale. The organization would transform the categorical data into discrete data using the bottom scale in the table above. These discrete data would be treated as though they were continuous data in order to compute a mean and standard deviation. For example, suppose the data computed to a mean of 3.25 and a standard deviation of 0.75.

From Example 15.2, we can see that transforming the data changes the measurement scale.

SAMPLING

Define and describe sampling concepts, including representative selection, homogeneity, bias, accuracy, and precision. Determine the appropriate sampling method (e.g., random, stratified, systematic, subgroup, block) to obtain valid representation in various situations. (Evaluate)

Body of Knowledge V.B.3

Sampling Concepts

Statistical sampling is a highly useful quality tool as long as we follow the underlying concepts:

• *Representative sample.* A sample that is representative of a targeted larger population. This means the characteristics of a representative sample reflect the characteristics of a targeted population from which the sample is drawn. True representative samples are difficult to accomplish. Consider a group of 32 children, of which 16 are male and 16 are female. A representative sample of size eight based on gender alone would be four males and four females. A nonrepresentative sample would be eight males. If we added ethnicity to the characteristics, finding a representative sample would become difficult.

The above describes a representative sample in qualitative terms. However, from a statistical point of view, a *representative sample* is a random sample that is drawn from a population such that observed values of the samples have the same distribution as the population. When samples are not representative of the populations, valid statistical inferences cannot be made.

• *Homogeneity.* This refers to the similarity or sameness of specified characteristic(s) of the members of the targeted population. Homogeneity of subpopulations is important in regard to stratified sampling, which is discussed in a subsection below.

- *Bias.* Bias can occur in the sampling in two ways: (1) some members of the targeted population are less likely to be included than others, and (2) the manner or process in which the sample is collected is nonrandom such that each member of the population does not have the same probability of being selected. Examples of the first case include when members self-select, when prescreening is performed on members, or through exclusion, which occurs when members move out of a study area. Furthermore, bias can be introduced if the sampling study takes too long to complete. This is because the starting and ending conditions may change during the course of the investigation.

  Although most statistics under- or overestimate population parameters, statistics computed from biased samples may significantly under- or overestimate the parameters. Other types of bias exist, and the reader is referred to sources more dedicated to this topic for details.

  A widely known and classic example of both biased and nonrepresentative sampling occurred in 1936 when the *American Literary Digest* magazine collected over two million postal surveys to predict that Alf Landon would defeat Franklin Roosevelt by a wide margin in the presidential election. Unfortunately, the sample was based on readers of the magazine and supplemented by records of registered automobile owners and telephone users. Unfortunately, the sample overrepresented wealthy individuals who were likely to vote for the Republican candidate.

- *Accuracy.* This refers to how close the sample statistic is to the population parameter. Accuracy is measured by the margin of error.

- *Precision.* This refers to how close estimates from different samples are to each other. Precision is inversely related to the standard error. Precision is great when the standard error is small, and vice versa.

- *Margin of error.* This refers to the maximum expected difference between the sample estimate of the population parameter and the true population parameter. The margin of error is equal to one-half of the width of the confidence interval.

- *Sampling frame.* This refers to a complete list of the members in the target population from which a sample is drawn.

- *Strata.* This refers to a mutually exclusive segment of the target population that is defined by one or more characteristics. In stratified sampling, all members of the targeted population are placed in a unique stratum, and the strata are mutually exclusive and collectively exhaustive of the population.

Types of Sampling

In this subsection, we will look at a variety of sampling methods, the first three of which fall under the category of probability sampling. The last one (that is, block) falls under the category of non-probability sampling. These sampling methods include:

- Random
- Stratified

- Systematic
- Block

Random Sampling. This method involves drawing a sample from a targeted population in a manner such that each member of the sample has an equal probability of being chosen. Further, every member is selected independently of every other member.

The basic procedure for random sampling follows:

1. Determine the sampling frame.
2. Assign each member of the population a number from 1 to *N*.
3. Generate a sample size of *n* different random numbers between 1 and *N*.
4. Select the members of the population associated with the *n* random numbers. This constitutes the sample.

This concept is illustrated in Figure 15.1.

Stratified Sampling. According to Anderson-Cook and Borror (2013), in stratified sampling "the population is grouped into subpopulations in which units in the same subpopulation are thought to be the same to each other or to one or more concomitant variables. The sample is drawn by randomly selecting units for the various subpopulations in numbers proportionate to the relative size of the subpopulations." (Note: A concomitant variable is a variable that is observed, but not measured or used in the analysis of the data.)

In stratified sampling, all members of the targeted population are assigned to a unique stratum. Consequently, the strata are mutually exclusive and collectively exhaustive with regard to the targeted population.

In addition, the stratification variable(s) should create a heterogeneous set of strata, while the members within each stratum should be as homogeneous as possible. This concept is depicted in Figure 15.2. In this figure, the stratification variable is "size." This stratifies the members of the population according to small, medium, and large. Thus, the strata are heterogeneous. Within each stratum, the

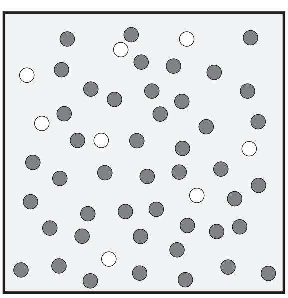


Figure 15.1 A graphical representation of random sampling.

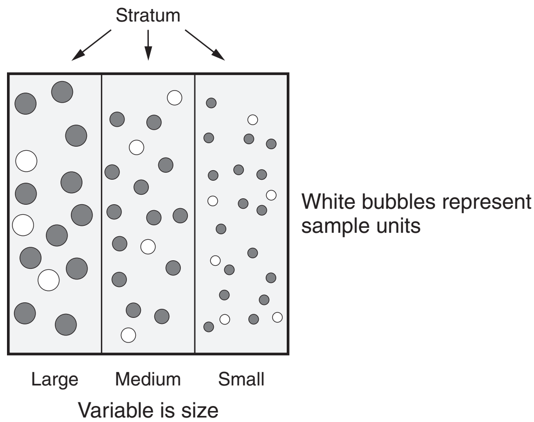


Figure 15.2 A graphical representation of stratified sampling.

sizes are either all small, medium, or large. Therefore, they are homogeneous within each stratum.

In proportionate stratified sampling, the size of the sample drawn from each stratum is proportionate to the relative size of that stratum in the total population.

The basic procedure for random sampling follows:

1. Determine the sampling frame.
2. Select the stratification variable(s) and the number of strata, k.
3. Divide the entire population into k strata. Based on the classification variable, each member of the subpopulation is assigned to one of the k strata.
4. Determine the sample size of each stratum, n_i, $i = 1, \ldots, k$.
5. In each stratum, number the members from 1 to N_i, $i = 1, \ldots, k$ where N_i is the subpopulation size of ith stratum.
6. Randomly draw sample n_i from each stratum (subpopulation) N_i, $i = 1, \ldots, k$ such that

$$\sum_{i=1}^{k} n_i = n; \sum_{i=1}^{k} N_i = N$$

Systematic Sampling. This method involves sampling from an ordered population at a specified sampling interval, i. This concept is shown in Figure 15.3. Systematic sampling is fast and efficient, but issues arise if, for some reason, the sampling interval synchronizes with a natural periodicity in the ordered population. For example, if defects occurred every $i - 1$, i, $i + 1$ interval. The probability of this occurring can be minimized by randomly starting within the first interval. Systematic sampling is sometimes called *interval sampling*.

The basic procedure for random sampling follows:

1. Determine the sampling frame.
2. Assign each member of the population a number from 1 to N.
3. Determine the sampling interval, i. The sampling interval is defined by $i = N/n$ where n is the predetermined sample size. Round i to the nearest integer.
4. Generate a random number k between 1 and i. This will be used to create a random start.
5. Sample the following numbers: $k, k + i, k + 2i, k + 3i, \ldots, k + (n - 1)i$.

Subgroup Sampling. There is no sampling method or design known as "subgroup" sampling. The body of knowledge in this context is referring to *rational subgrouping*, which will be addressed in Chapter 27.

Block Sampling. This method of sampling refers to the situation whereby an initial item is chosen from a population in sequential order and, consequently, the items in the balance of a defined block are automatically chosen. Block sampling falls into the category of non-probability sampling and is sometimes called *judgment block sampling*. Block sampling is illustrated in Figure 15.4.

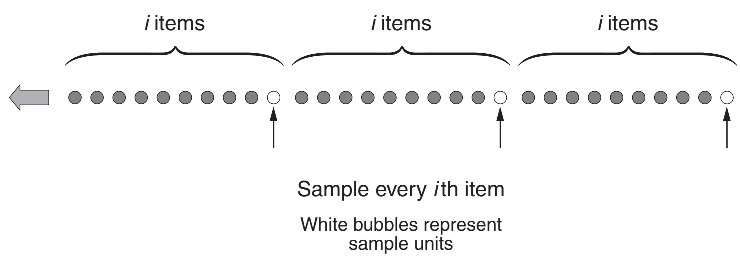


Figure 15.3 A graphical representation of systematic sampling.

Figure 15.4 A graphical representation of block sampling.

EXAMPLE 15.3

Consider an auditor who wants to review purchase orders. He has discovered the first 100 purchase orders filed by purchase order number. The purchase orders, along with supporting documentation, are kept in a five-drawer file cabinet in hanging folders with 20 folders per drawer. The auditor decides to select the second drawer from the top and pulls the twenty-third purchase order at random. As a result of this selection, the auditor has automatically selected purchase orders 21 through 40, which constitutes the block.

Edward Kisscorni, a CPA, has as an interesting comment to make on the use of block sampling and the taxpayer. According to Kisscorni (2009), "A judgment block sample is the least appropriate form of sampling because it violates the primary guiding principles of sampling. Normally the block of transactions being audited is not selected at random. Usually the transactions being audited are those most readily available. When a block sample is selected, it eliminates all the other transactions outside of the block. An invalid projection occurs if the intent is to project the results of the audit of the transactions in the block against the entire population including those transactions outside of the block. The taxpayer should be very careful before accepting any judgment block sampling."

Determining the Sample Size

A variety of factors can influence the sample size required. Some of these factors include:

- Precision and accuracy required
- Resources available
- Nature of the study, including time available to conduct it and geographical considerations
- Sampling design adopted
- Size and characteristics of the population
- Confidence and power required
- Methods of obtaining the sample

It should be apparent from the above list that there are many considerations that must be addressed when determining an appropriate sample size other than simply plugging numbers into a formula.

However, researchers have developed their own general rules of thumb for determining sample sizes:

- As many as possible

- For correlational, experimental design, and causal-comparative studies: 30 per group
- Population size < 100: sample the entire population
- Population size ~ 500: sample 50% of the population
- Population size ~ 1500: sample 20% of the population
- Population size > 5000: sample 400 of the population

Remember, these are rules of thumb and caution is in order. However, for the purpose of being able to make statistical statements, readers are encouraged to calculate sample sizes. When the population is a known quantity, the following formula can be used:

$$n = \frac{N}{1 + Ne^2}$$

where

N = Population

n = Sample size

e = Margin of error

Alternately, the formulas identified in Chapter 21 may be used.

What Makes a Good Sampling Design?

Many of the aspects of a good sampling design have been discussed previously. Let's pull them together in a meaningful list:

- Samples may be drawn randomly from the targeted population
- The sample is representative of the population from which it is drawn
- The design is economical and efficient
- Given the sampling design, it is sufficiently easy to conduct
- Standard error is minimal
- Bias is minimal
- Statistical assumptions are not violated
- Inferences can be made from the sample about the population with the desired level of confidence
- Its selection criteria are objective, not subject to data collector's bias

We've seen that the above characteristics may change significantly depending on the sampling design. Table 15.2 summarizes the advantages and disadvantages of the sampling methods discussed in this subsection.

Table 15.2 A comparison of statistical sampling methods.

Method	Advantages	Disadvantages	Comment
Random	• Easy to conduct and understand • Requires minimum knowledge about the population • Human bias eliminated • High probability of achieving a representative sample • Most accurate in generalizing about population	• Hard to do with large samples (that is, identifying all members of the population can be difficult) • Could be high cost	• Can be done with or without replacement • Population must be homogeneous
Stratified	• Higher sample accuracy • Higher sample precision	• Identifying all members of the population can be difficult • Identifying all members of a subgroup can be difficult • May be difficult to select stratification variables	• There are both proportion and nonproportional stratified sampling. • Random sample is applied within each stratum • Each stratum is a homogeneous group
Systematic	• Economical • Less time-consuming • Easy to conduct • If the order of the units is related to the characteristic of interest, this method will increase the representativeness of the sample	• Not all members of the population have an equal chance of being included (that is, decreased representativeness) • If the order of the units produces a cyclical pattern, this method may reduce the representativeness of the sample • Creates a risk if there is a natural periodicity in the data	• Useful in survey sampling • Sample size typically known in advance to determine sampling interval • Frequently used with a random start
Block	• Reduced cost • Reduced time involved • Knowledge-based sampling	• Lacks objectivity and introduces bias • Sample may not be representative • Individual-based	• Also known as judgmental block sampling • Used in auditing

DATA COLLECTION PLANS AND METHODS

> Develop and implement data collection plans that include data capture and processing tools, e.g., check sheets, data coding, data cleaning (imputation techniques). Avoid data collection pitfalls by defining the metrics to be used or collected, ensuring that collectors are trained in the tools and understand how the data will be used, and checking for seasonality effects. (Analyze)
>
> **Body of Knowledge V.B.4**

The plural of anecdote *is not* data.

—Unknown

Defining an Operational Definition

At the outset of any data collection activity, operational definitions should be established for each data element that will be collected. This is to ensure that the proper data elements will be collected and the opportunity for error will be minimized.

In the context of data collection and metric development, an operational definition is a clear, concise, and unambiguous statement that provides a unified understanding of the data for all involved before the data are collected or the metric is developed. It answers "Who collects the data, how are the data collected, what data are collected, where are the sources of the data, and when are the data collected? " Additional "who," "how," "what," "where," and "when" answers may be required.

When the data are used to construct a metric, the operational definition further defines the formula that will be used and each term used in the formula. As with data collection, the operational definition also delineates who provides the metrics and who is answerable to its results, how the metric is to be displayed or graphed, where the metric is displayed, and when it is available. Again, additional "who," "how," "what," "where," and "when" answers may be required.

Finally, the operational definition provides an interpretation of the metric, such as "up is good." While providing an interpretation of a metric seems trivial, this is not always the case.

For example, consider "employee attrition rate." How should this be interpreted? Is down good? Yes? How far down is "good"? Some authorities suggest that some amount of attrition is acceptable because it permits new thinking to enter the organization. A high level of attrition, however, can bleed the organization of institutional knowledge and memory.

In this example, the attrition rate metric may have boundaries in which:

- Above the upper boundary "down is good."
- Below the lower boundary "up is good."

Data Integrity and Reliability

Although data validity and reliability were discussed in the context of voice of the customer in Chapter 10, this subsection will address hands-on issues regarding data. As quality or Lean Six Sigma professionals, we have been taught to address the issue of data accuracy and integrity from the statistical viewpoint. However, this section will address most of the elements of data integrity and reliability by discussing why data accuracy is important, the numerous causes of poor data accuracy, and various techniques for improvement and the placement of effective data collection points (Kubiak 2008).

Importance of Data Accuracy

Poor data accuracy undermines two key precepts in a quality-focused or Lean Six Sigma organization: fact-based and data-driven decision making. When faced with data that have been rendered useless because of data accuracy issues, decision makers must revert to relying on intuition.

This might provide the naysayer with reason and justification to state, "This Lean Six Sigma approach won't work here." Remember, Lean Six Sigma drives cultural changes in organizations, and accurate data are necessary to drive and sustain real improvement.

In addition, some organizations might be required to submit highly accurate data because of regulatory or reporting requirements, or because of contractual requirements. This is particularly true in industries such as aerospace, in which data, in addition to product, are a deliverable. In fact, most organizations have processes that deliver only data to other organizations.

Consider the example of payroll data regularly reported to the Internal Revenue Service. The consequences associated with poor data accuracy can be quite troublesome and have a strong negative impact on the individual who is depending on the data.

Minimizing Poor Data Accuracy

There are many causes of poor data accuracy. Table 15.3 includes a few of the most significant causes. Consider how these issues relate to your own experience and create your own list so you can minimize the impact of poor data accuracy in your organization.

Table 15.4 lists data collection techniques used to minimize the occurrence of poor data accuracy or its impact.

Remember, processes are measurable and generate data. Treat poor data accuracy as you would any other quality defect: use the plethora of tools available to you as a quality or Lean Six Sigma professional to ferret out root cause. Chart data defects as you would product defects. Make them available for all to see.

Table 15.3 Common causes of poor data accuracy.

Issue	Comment
Multiple points of entry	When data entry is centralized to a single individual, entry errors are usually minimized over time due to the experience of the individual and, often, the ability of the individual to contact the originator of the data for clarifications. When data originators become responsible for their own data entry, data accuracy usually diminishes, particularly when frequency of data entry decreases or there is high turnover among the data originators.
Limited use of data validation	Such validation would include the presence or absence of data, data ranges, inconsistent data, and other logical checks. Surprisingly, computer technology has not completely eliminated this cause.
Batching input versus real-time input/ synchronization of system updates	This scenario usually results in instantaneous differences, but the differences are usually reconciled with time. Consider the example of an individual inquiring about a checking account balance through an ATM, a bank teller, or online. Because of different cut-off and system update times, the individual will see up to three different account balances.
Multiple means of data correction access to upstream systems	When multiple data correction access points are present, the opportunity to have synchronization issues increases. For example, suppose there are three upstream systems prior to the current system. Label them 4, 3, 2, and 1 for the current system. If corrections are made to 4, then 3, 2, and 1 are updated. However, if corrections are made to 2, only 2 and 1 are updated.
Unclear directions	This issue becomes increasingly important when multiple points of data entry exist.
Lack of training	This issue becomes increasingly important when multiple points of data entry exist.
Ambiguous terminology	Ask an operations manager for monthly data and you get monthly data according to the calendar month. Ask for the same data from an accountant and you get monthly data by fiscal month. Different months' lengths yield different data.
Manual versus automated means of entry	Automated data entry is usually preferred over manual data entry, particularly if you accept the premise that people are fallible.
Rounding	Improper rounding, for example, when rounding up is appropriate instead of rounding down. Also, the cumulative effects of rounding can generate errors. For example, computers store and calculate using numbers to 16, 32, or even 64 digits. An individual using numbers rounded to four digits in the same calculations can arrive at a result that is substantially different than the results arrived at by a computer.
Order of calculations	Consider $\frac{36}{(1/9)}$. If you rewrite this equation as $36\left(\frac{9}{1}\right)$ and multiply through, you obtain 324. If you divide out 1/9 and round to four decimals, you obtain 0.1111. Dividing 36/0.1111, you obtain 324.0324. Different order, different answers.

Continued

Table 15.3 *Continued.*

Issue	**Comment**
Calculation errors	This is simply an arithmetic error or an error in the specification of the formula to be used. For example, I worked at two organizations that computed "associate turnover" incorrectly.
Inadequate measurement systems	This is an issue commonly known among Six Sigma professionals who have been taught to examine the adequacy of the measurement system first. Examples include Kappa studies and gage repeatability and reproducibility studies.
Units of measure not defined	This type of error occurs when the wrong use of measure is assumed or used, such as "foot" instead of "inch," "yard" instead "foot," or "box" instead of "each." The last example can significantly impact inventory levels.
Failing to use the proper system of measurement	On September 23, 1999, after a 286-day journey from Earth to Mars, NASA lost the $125 million Mars Climate Orbiter after the spacecraft entered orbit 100 kilometers closer than planned. NASA used the metric system, while its partner Lockheed Martin, the organization that designed the navigation system, used English units.
Field inconsistencies/ truncation among systems or data fields	Moving an *n* character field to an *n* – 1 character field or smaller can result in field truncation, which would impact all calculations that use it. For example, 100,789 is moved into a three-character numeric field. The result would be 789, not 100,789.
Similarities of characters	Several letters and numbers can easily be confused, particularly when handwriting is poor or when reading other than the top sheet of a no carbon required form. Common examples include: 0 O (oh) 4 H 8 B 1 l (el) 5 S 9 g (gee) or q 2 Z and occasionally Q 6 G 3 B 7 Z Examples of possible confusion between letters: c-e, C-O, E-F, m-n, M-N, O-Q, P-R, r-v, u-v, U-V, v-w, V-W, v-y
Copying errors	Duplicates of duplicates can result in image degradation and impact the readability of the copy. Also, copy truncation occurs when the image being copied is close to the edge of the paper. Try this with your credit card statement sometime. Ensure that the purchase amounts are adjacent to the edge of the copier glass.

Table 15.4 Useful data collection techniques.

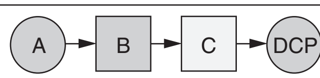
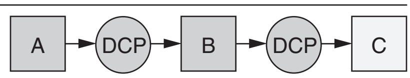
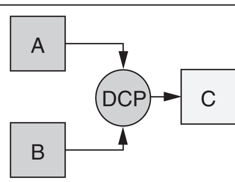
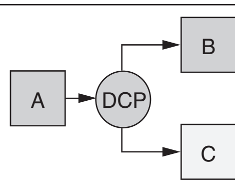
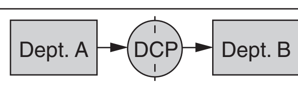
Technique	Comment
Walk the process first	Always walk the process first so everything that follows in this table has meaning and applicability.
Make the data collection process simple	Minimize the need to perform calculations and maximize the use of direct data collection. Direct data collection occurs when you take a reading from an instrument and log or enter that piece of data. In this case, a simple data transference has occurred.
Define collection points	Table 13.4 identifies typical and useful data collection points.
Use check sheets and checklists	These are simple but very effective tools and often overlooked. Perhaps they are too simple.
Bridge computer gaps	Some organizations still have processes requiring individuals to extract data from one computer report and enter it into a database residing on a different computer. In some cases, it is the same computer.
Minimize "other"	What is "other?" When collecting nominal or categorical data, it is often the last, and sometimes the largest, bar on a Pareto chart. A substantial amount of data classified as "other" usually indicates that data originators either don't understand clearly how to classify the data, or that categories are not mutually exclusive. The opportunity exists to classify data into multiple categories. As a result, data originators might disagree with one another, and the measurement system becomes inadequate.
Limit options	The more categories of data there are, the greater the possibility of a data collection error. This technique should be balanced against the previous "minimize other" technique.
Establish the rules	This technique is best illustrated by an example. Consider an organization that receives payments stating that any receipt received after 3:00 p.m. will be credited on the next business day.
Address timing and sequencing	This technique is important when a data collection point is inserted to collect data from multiple input streams. In this case, be particularly aware of the possibility of generating a bottleneck by waiting for a specific data input stream. Otherwise, it might be necessaryto collect the data at another point downstream. If this is not possible, rebalancing the upstream processes might be required.
Time-stamp data	This technique facilitates traceability of activities and events through time-based data and allows for more thorough root cause analysis.
Red-tag a unit of product or service	Track a specified unit of product or service through the entire process to understand how it is used to generate data or how data are collected from it.
Chart data accuracy	Remember, data collection is part of the process and therefore subject to charting techniques such as control charts and run charts.

Common Data Collection Points

Five common data collection points are summarized in Table 15.5 and are as follows:

- *End of process*. Traditionally, organizations have collected data to measure process performance at the very end when everything is said and done, and it's usually too late to take corrective action.
- *In process*. Over time, organizations that are more progressive have realized that data should be collected upstream and during the process. Data collection at critical subprocesses allows corrective action to be taken earlier, when change is still an option. This move facilitates more effective process management and allows the organizations to better gauge the overall health of their processes.
- *Points of convergence*. This occurs when multiple streams of processes come together. Collecting data at this point makes sense because it helps organizations understand how each input stream affects the downstream subprocesses.
- *Points of divergence*. This is nothing more than a flip of the points of convergence. Keep in mind that a decision point is a specialized point of divergence with two distinct outputs.

Table 15.5 Common data collection points (DCPs).

Point	Example
End of process	
In process	
Points of convergence	
Points of divergence	
Across functional boundaries	

Decision points involving "yes or no" decisions are useful data collection points, particularly when the "yes" path is the desired path and the "no" path is the undesired path.

Understanding the percentage of products or services taking the "no" path becomes an important piece of data necessary for driving process improvement.

- *Across functional boundaries*. These are nothing more than process hand-off points. However, responsibilities and accountability usually change across functional boundaries.

Collecting data at these focus points helps organizations understand how each functional department supports or detracts from process efficiency and effectiveness. It also helps minimize organizational finger-pointing with data and fact.

Accuracy Is Essential

Data accuracy is essential for managing and improving processes and achieving sustained results. Without it, decision making is, at best, uninformed.

Consequently, poor data accuracy can adversely impact an organization. Data defects, like product defects, should be charted and improved using the variety of tools at the disposal of the quality or Lean Six Sigma professional.

Data Cleaning

Data cleaning is the process of detecting and correcting and possibly removing inaccurate data from a set of data.

Ideally, inaccurate data are prevented from ever entering a system or database. However, inevitably, bad data occur and they must be addressed. Once data have already entered a system, the only way of detecting them is usually through manual means, which means human intervention.

If the data are qualitative in nature, this may prove to be a very difficult task as no general pattern to the data may be readily apparent, and the data will likely be in a narrative or free-form format.

However, if the data are quantitative, they may be sorted, graphed, or analyzed in some manner to discern outliers or others unwanted aspects of the data. For example, if outliers are present and discovered, they can be properly assessed to determine if they are legitimate and should be kept or can be appropriately discarded without unduly impacting underlying patterns in the data. Perhaps "spaces" are present in lieu of "zeroes," but "spaces" cannot be handled properly. In this case human intervention can replace the "spaces" with "zeroes" before data processing takes place. This process of replacing the missing values with data (that is, zeroes) is known as *imputation*.

Coding the Data

Most organizations collect large volumes of data, sometimes far more than they are able to analyze effectively. Check sheets are often used to record the presence or absence of particular features. Some data are collected manually by having people record counts or readings on check sheets or forms. In some cases, it is helpful to code data to simplify the recording process. For example, suppose several measurements of shaft diameters are to be recorded. The tolerance limits for shaft diameters are set at 1.5353 to 1.5358. The person doing the recording might read three values as 1.5355, 1.5354, and 1.5356 but record the last two digits only as 55, 54, and 56 for simplicity. Sometimes, it is useful to code data by using an algebraic transformation. Suppose a set of data has mean μ and standard deviation σ. A new set of data may be formed by using the formula $y = ax + b$. That is, each element of the new set is formed by multiplying an element of the original set by a and then adding b. The mean μ_y and standard deviation σ_y of the new set are given by

$$\mu_y = a\mu + b$$

$$\sigma_y = |a|\sigma$$

Increasingly, data are automatically transmitted from a measuring device to a database. A contoured part may be clamped into a fixture and have dozens of readings made and recorded instantaneously. Potential advantages of this automatic gauging approach include improved precision as well as reduction of labor, cycle time, error rates, and costs. When considering automated inspection, attention must be paid to the possibility of high initial costs, including the possibility of part redesign to adapt it to the constraints of the measurement system. If the measured values are fed directly into the database, care must be taken to make certain that the communication link is reliable and free of noise. It is also useful to have a graphic output so operators can quickly spot erroneous data visually.

Collecting Seasonal Data

When the data exhibit a seasonal effect, special consideration must be given to the data collection time frame. Figure 15.5 depicts a plot of data that exhibits a seasonal effect. In this figure, multiple seasons are shown and the seasonal effects are relatively equal.

Figure 15.5 also shows three different data collection ranges that might correspond to three different data collection plans. The first data collection plan for data range 1 shows a very small time frame in which the data are collected. Notice that this data range captures a limited portion of the seasonal cycle and is not representative of the entire seasonal effect. Consequently, this would not be an appropriate data collection plan.

The second data range depicts data being collected over one full seasonal cycle. This plan would capture all the data in a cycle and would be appropriate as a minimum data collection plan.

However, the third data collection plan encompasses multiple seasonal cycles and would be ideal if the data were available and the cost of data collection were acceptable. The presence of multiple cycles would permit us to explore variability with cycles.

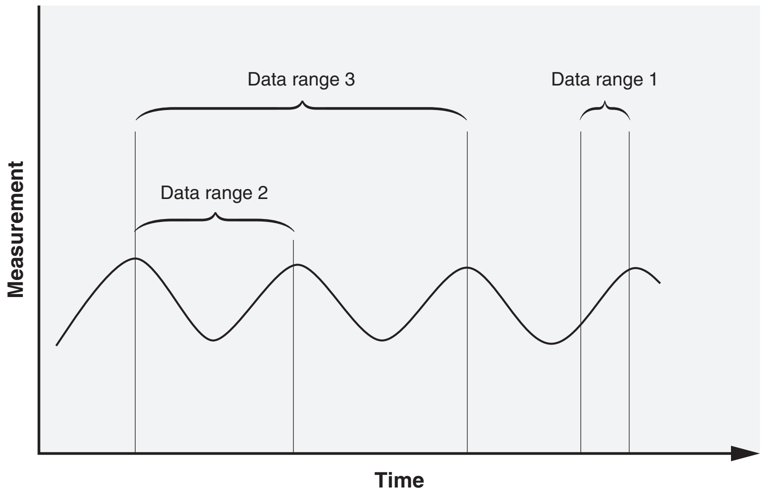


Figure 15.5 Collecting seasonal data.

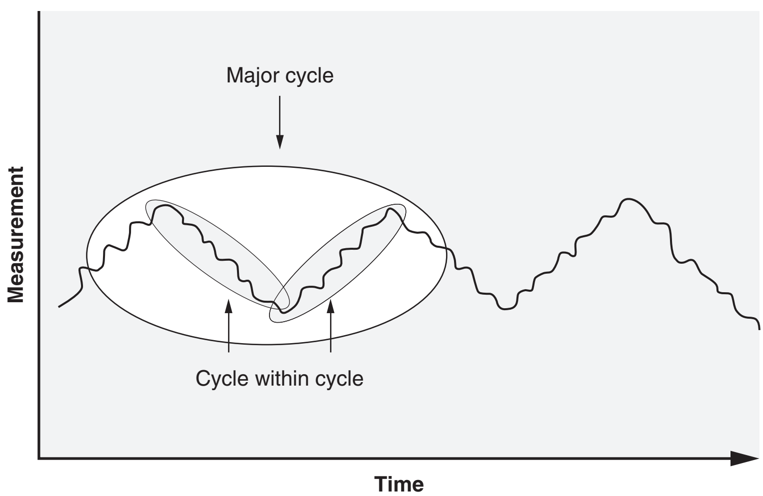


Figure 15.6 A seasonal effect within a seasonal effect.

Another important aspect of seasonality effects is that there is the possibility that multiple cycles might exist and be embedded within each other, as shown in Figure 15.6. This is why it is important to understand the nuances of the data before undertaking a major data collection effort.

Table 15.6 A summary of data collection strategies.

Collection strategy	DMAIC steps	Summary	Related strategies
1. Observational studies	D	• The relevance of the sample to the current study. Are there key differences in time, product, or process that may change patterns? • The quality of the data. How were the units selected? Have all potentially relevant inputs been gathered? • How the data can be used to advance the study without drawing conclusions that are not justifiable, given the lack of established causality.	2, 3, 4, 7
2. Monitoring techniques	A, C	• Selection of rational subgroups or individual observations to assess process stability • Action must be taken immediately (for example, when the monitoring technique is a control chart)	1, 3, 4, 5
3. Process capability	A, C	Selection of rational subgroups or individual observations to assess process stability	1, 2, 5
4. Measurement assessment	M	Assessment measurement of system capability and stability	1, 2 ,5 , 6
5. Sampling	A, C	Intentional selection of subset of population	2, 3, 4, 7
6. Design of experiments	I	Active manipulation of factors to establish causality	4
7. Complementary data and information	D, A	Qualitative, brainstorming, expert opinion, and informal assessment	1, 5

Data Collection Strategies

Anderson-Cook and Borror (2013) identify seven data collection strategies, shown in Table 15.6. Notice that that each strategy is linked to one or more DMAIC phases. Furthermore, Anderson-Cook and Borror emphasize that the strategies are not mutually exclusive. Table 15.7 identifies the same strategies, but includes advantages and disadvantages of each.

Table 15.7 Advantages and disadvantages of the data collection strategies.

Collection strategy	Advantages	Disadvantages
1. Observational studies	• Conveniently available data • Usually a large amount of data from which to sample	• Assuming a pattern observed in a convenience sample will be present in the larger population • Assuming causation between a set of inputs and its effect on the responses • Missing a lurking variable that wasn't measured and is driving change in both the input values and the response • Sampling from a portion of the total population, which might give a biased view of the relationship.
2. Monitoring techniques	• If used properly, these techniques can be highly beneficial to the organization • Can signal the need for swift action to be taken	• It is easy to misuse or select the wrong monitoring technique (for example, when the monitoring technique is a control chart) • Can easily fall into disuse
3. Process capability	• If used properly, this technique can be very effective in driving process improvement • Can provide estimates of short- and long-term variation	• It is easy to misuse or select the wrong process capability index • Much confusion exists around the use of this technique and associated indices such that the use of these indices might be a hard sell
4. Measurement assessment	• Provides insight into true process variation • Provides insight into the stability, bias, and linearity of the measurement system	• It is easy to misuse or select the wrong measurement assessment tool • Results may be difficult to interpret
5. Sampling	• Helps minimize costs • Can provide estimates of population parameters	• It is easy to select the wrong sampling technique • Once implemented, it is easy for production personnel to shortcut the sampling plan
6. Design of experiments	• Helps define the critical X's • Establishes the $Y = f(X)$ relationship	• It is easy to misuse or select the wrong experiment design • Can be expensive • May impact production
7. Complementary data and information	• Easy to teach and use • A little effort goes a long way	• Some techniques are subjective in nature

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Chapter 16

Measurement Systems

This chapter will address the basics of measurement systems analysis. Both variables and attributes analyses will be included, and an example of each will be demonstrated using Minitab. In addition, a firm foundation is provided of measurement systems analysis terminology, which is supplemented with ample use of graphics to solidify key points.

MEASUREMENT SYSTEM ANALYSIS (MSA)

> Use gauge repeatability and reproducibility (R&R) studies and other MSA tools (e.g., bias, correlation, linearity, precision to tolerance, percent agreement) to analyze measurement system capability. (Evaluate)
>
> **Body of Knowledge V.C.1**

Figure 16.1 attempts to depict how the terminology in this section fits together. As you read this section, please keep this figure in mind, as it will help provide perspective and aid in the interpretation of the concepts discussed.

Terminology

In order to analyze a measurement system, it is important to understand several key concepts and terminology, as well as how they relate:

- *Bias.* This is a systematic difference between the mean of the measurement results and a reference or true value. Bias, also known as *accuracy*, may be either positive or negative. For example, if one measures the length of 10 pieces of rope that range from one to 10 feet multiple times, each time computing the average and always concluding that the length of each average is two inches shorter than the true length, then the individual exhibits a bias of two inches. An organization's calibration process is used to control bias in a measurement system. The concept of bias is depicted in Figure 16.2. The components of bias include:

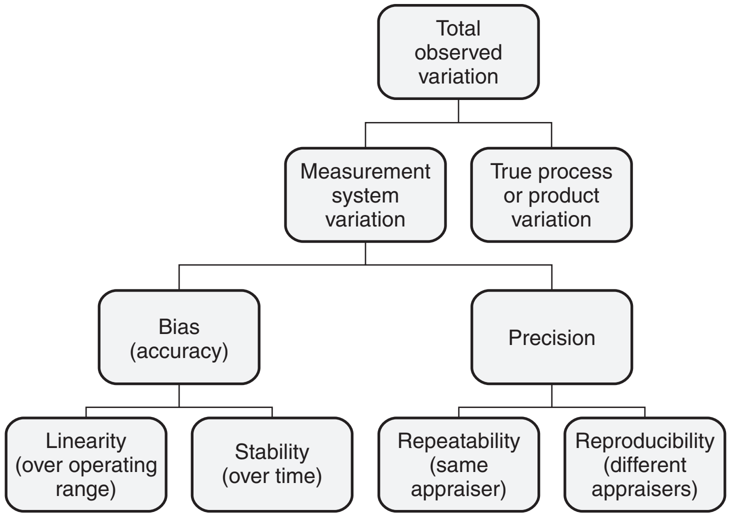

Figure 16.1 Measurement system hierarchy.

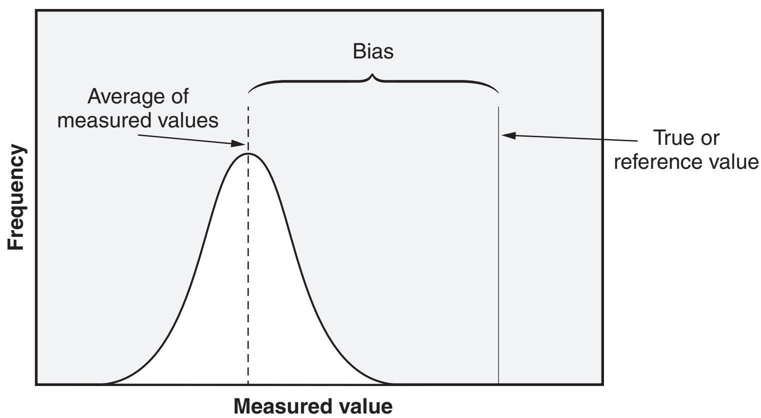

Figure 16.2 A graphical representation of bias.

– *Linearity.* This is the difference in bias through the operating range of measurements. A measurement system that has good linearity will have a constant bias no matter the magnitude of measurement. In the previous example, the range of measurement was from one foot to 10 feet, with a constant linear bias of two inches. Figure 16.3 provides an example of linear bias. By contrast, Figure 16.4 illustrates nonlinear and varying linear bias. Notice that, in this example, the varying linear bias grows over the measurement range.

– *Stability (of a measurement system).* This represents the change in bias of a measurement system over time and usage when that

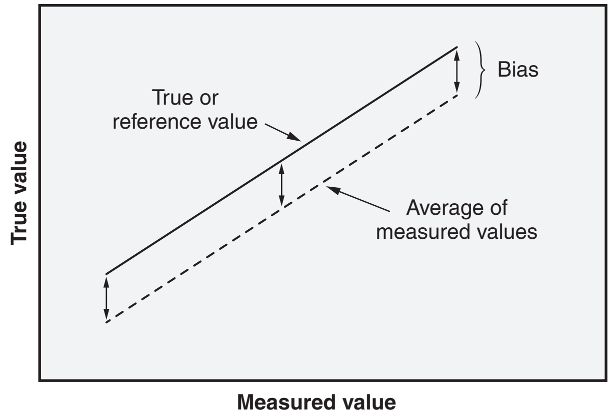

Figure 16.3 Example of a linear bias.

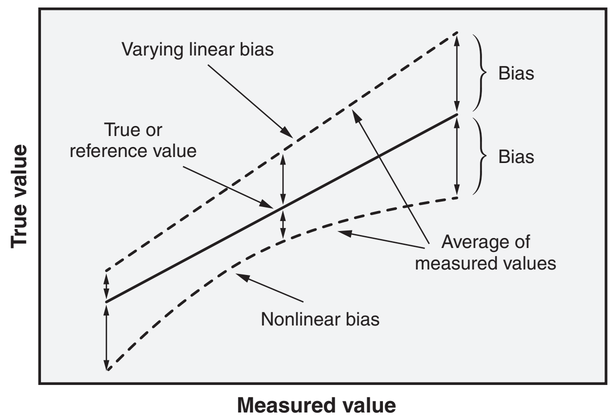

Figure 16.4 Examples of a nonlinear and varying linear bias.

system is used to measure a master part or standard. Thus, a stable measurement system is one in which the variation is in statistical control, which is typically demonstrated in practice using control charts. The control charts can be used in conjunction with an organization's calibration process to help establish the calibration recall time frame for measurement instrumentation. The concept of stability is illustrated in Figure 16.5.

- *Precision.* This is the closeness of agreement between randomly selected individual measurement results or test results. It is this aspect of measurement that addresses repeatability or consistency when an identical item is measured several times. Note that precision depends only on the distribution of random errors and does not relate to the reference or true value. Figure 16.6 illustrates the relationship between

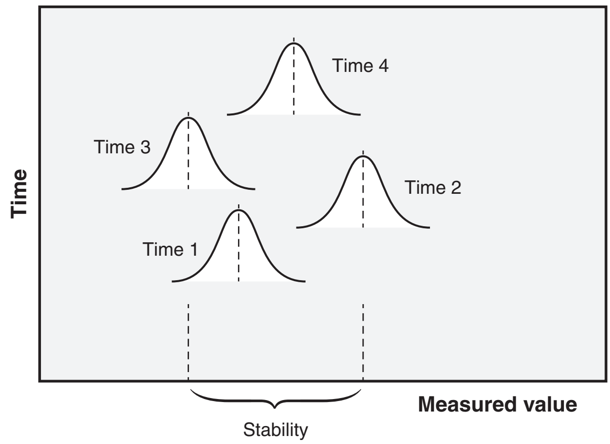

Figure 16.5 A graphical representation of stability.

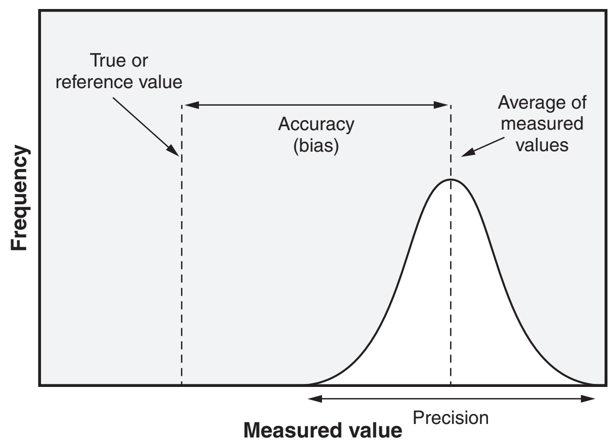

Figure 16.6 Precision and accuracy (bias).

accuracy and precision. Figure 16.7 depicts a more traditional view of the relationship between accuracy and precision. This table shows us how we can be accurate, but not precise, and vice versa.

The components of precision include:

– *Repeatability.* This is the precision under conditions where independent measurement results are obtained with the same method on identical measurement items by the same appraiser (that is, operator) using the same equipment within a short period of time. Although misleading, repeatability is often referred to as *equipment variation* (EV). It is also referred to as *within-system variation* when the conditions of measurement are fixed and defined

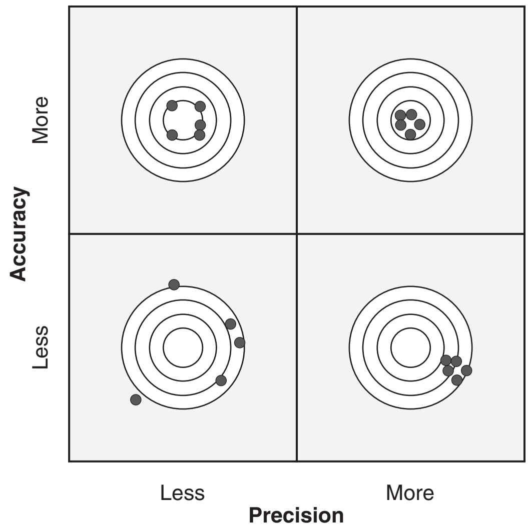

Figure 16.7 Accuracy versus precision.

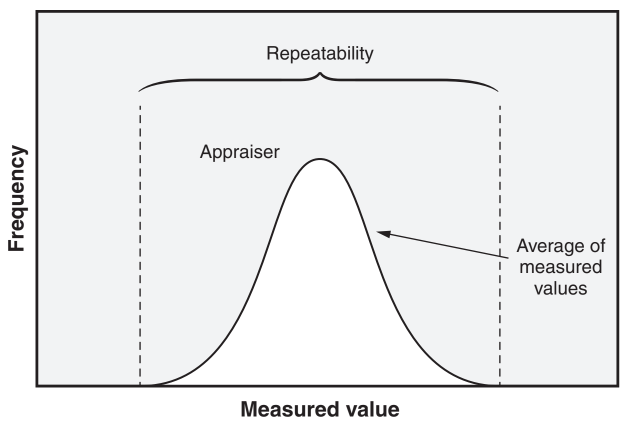

Figure 16.8 A graphical representation of repeatability.

(that is, equipment, appraiser, method, and environment). This concept is presented in Figure 16.8.

– *Reproducibility.* This is the precision under conditions where independent measurement results are obtained with the same method on identical measurement items with different operators using different equipment. This concept is illustrated in Figure 16.9. Although misleading, reproducibility is often referred to as appraiser variation (*AV*). The term "appraiser variation" is used because it is common practice to have different operators with

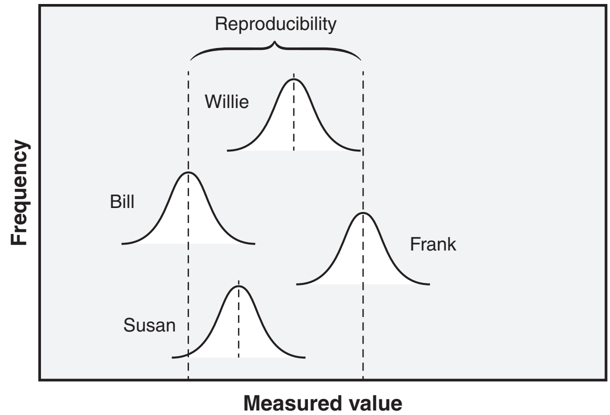

Figure 16.9 A graphical representation of reproducibility.

identical measuring systems. Reproducibility, however, can refer to any changes in the measurement system. For example, assume the same appraiser uses the same material, equipment, and environment, but uses two different measurement methods. The reproducibility calculation will show the variation due to a change in the method. It is also known as the *average variation between systems*, or *between-conditions variation* of measurement.

- *Discrimination.* This represents the measurement system's capability to detect and indicate small changes in the characteristic measured. Discrimination is also known as *resolution*. For example, a tape measure with one-inch gradations would be unable to distinguish between object lengths that fall in between the inch marks. Hence, we would say that the measurement system cannot properly discriminate between the objects. If an object to be measured is 2.5 inches, the measurement system (that is, tape measure) would produce a value of two or three inches depending on how the individual decides to round. Therefore, to measure an object that is 2.5 inches, a tape measure with finer gradations would be required.
- *Precision-to-tolerance ratio (PTR).* This is a measure of the capability of the measurement system. The smaller the PTR value, the better. It can be calculated by:

$$\text{PTR} = \frac{6\hat{\sigma}_{ms}}{\text{USL} - \text{LSL}}$$

where $\hat{\sigma}_{ms}$ is the estimated standard deviation of the total measurement system variability. Prior to the 1990s, the PTR formula was given by:

$$PTR = \frac{5.15\hat{\sigma}_{ms}}{USL - LSL}$$

where the 5.15 constant represents 5.15 standard deviations accounting for 99% of the measurement system variation.

Typically, PTR values less than 0.10 are considered good, while values greater than 0.30 require improvement in the measurement system.

- *Gage repeatability and reproducibility (GR&R) study.* This is one type of measurement system analysis done to evaluate the performance of a test method or measurement system. Such a study quantifies the capabilities and limitations of a measurement instrument, often estimating its repeatability and reproducibility. It typically involves multiple appraisers measuring a series of items multiple times.
- *Percent agreement.* This refers to the percentage of time in an attribute GR&R study that appraisers agree with (1) themselves (that is, repeatability), (2) other appraisers (that is, reproducibility), or (3) a known standard (that is, bias) when classifying or rating items using nominal or ordinal scales, respectively.

Guidelines for Determining Whether a Measurement System Is Acceptable

Four criteria are used to assess the adequacy of a measurement system when conducting a GR&R study. These include:

- Total percent GR&R in the percent study variation
- Percent tolerance
- Percent contribution
- Number of distinct categories (discrimination)

In general, if the total percent GR&R in the percent study variation is:

- Less than 10%, the measurement system is considered acceptable.
- Between 10% and 30%, inclusive, the measurement system is considered marginal and should be examined. Factors to be considered include the application, the cost of the measuring device, the cost of the repair, and any other factors deemed relevant.
- Greater than 30%, the measurement system is considered inadequate.

In general, if the percent tolerance is:

- Less than 10%, the measurement system is considered acceptable.
- Between 10% and 30%, inclusive, the measurement system is considered marginal and should be examined. Factors to be considered include the application, the cost of the measuring device, the cost of the repair, and any other factors deemed relevant.

- Greater than 30%, the measurement system is considered inadequate.

Notice that it is the same as the total percent GR&R in the percent study variation.

In general, if the percent tolerance is:

- Less than 1%, the measurement system is considered acceptable.
- Between 1% and 9%, inclusive, the measurement system is considered marginal and should be examined. Factors to be considered include the application, the cost of the measuring device, the cost of the repair, and any other factors deemed relevant.
- Greater than 9%, the measurement system is considered inadequate.

The last criterion refers to the number of distinct categories, which refers to the level of discrimination. The minimum number of levels that is recommended is five. Any measurement system with fewer than five levels should be reviewed for improvement.

The Automotive Industry Action Group (AIAG) established the above criteria. Therefore, unless your organization is part of this industry, you should consider the above criteria to be guidelines and dependent on the situation at hand.

Components of a Measurement System

Every measurement system consists of the following key components:

- Measurement instrument
- Appraiser(s) (commonly known as *operators*)
- Methods or procedures for conducting the measurement
- Environment

Each of the above components can introduce error into the measurement process.

Types of Measurement Systems Analysis

Measurement systems analysis can be categorized into two types:

- Variables
- Attributes

Three common methods fall into the variables category:

- AIAG method
- ANOVA method
- Control chart method

Each will be addressed separately in the following subsections.

AIAG Method. The Automotive Industry Action Group (AIAG) developed its own GR&R study methodology. Example 16.1 illustrates this methodology step by step.

EXAMPLE 16.1

Figure 16.10 is an example of how to conduct a GR&R study. It uses the data collection sheet published by the AIAG in *Measurement System Analysis* (1995). These steps provide a widely accepted procedure for analyzing measurement systems:

1. Label 10 parts with tags numbered 1–10 in such a way that the numbers on the tags aren't visible to the appraiser. Randomly place the parts on a work surface.
2. The first appraiser (we'll call her Susan) chooses a part, measures the designated dimension, announces the reading, and hands it to the recorder. The recorder looks at the number on the tag and records the reading announced by Susan in the appropriate column of row 1 (row labels are shown along the left margin).
3. Susan measures the remaining nine parts and announces the reading for each. The recorder enters the values in the columns of row 1 according to tag number.
4. The 10 parts are again randomly placed on the work surface with the tag numbers not visible. Frank, the second appraiser, measures each one. The recorder enters Frank's readings in row 6 of the form, which is the row for trial #1 for appraiser Frank.
5. Willie, the third appraiser, repeats the procedure, and the recorder enters these values in row 11.
6. Now it's Susan's turn to measure the 10 parts for trial #2. These readings are entered in row 2. Frank and Willie follow with their second set of readings, which are recorded in rows 7 and 12, respectively. (The appraiser order can be randomized if desired rather than Susan, then Frank, and then Willie as indicated here.)
7. Each appraiser completes the third trial of measurements, and these are recorded in rows 3, 8, and 13. This completes the data entry portion of the study. Figure 16.11 shows an example of the collected data.
8. Calculate the average and the range for each appraiser for each part tag number. The values go in rows 4 and 5, respectively, for Susan; 9 and 10, respectively, for Frank; and 14 and 15, respectively, for Willie.
9. Calculate the values for row 16 by averaging the values in rows 4, 9, and 14.
10. Calculate the values for the "Average/range" column for rows 4, 5, 9, 10, 14, and 15 by averaging the 10 values in their respective rows.
11. Calculate the value for the "Average/range" column of row 16 by averaging the 10 values in that row.
12. Calculate the value for the "Average/range" column of row 17 by using the formula given in that row.

Continued

Gage Repeatability and Reproducibility Data Collection Sheet													
		Part											
Row	**Appraiser/ Trial**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**Average/ Range**	**Appraiser/ Trial**
1	A1												A1
2	A2												A2
3	A3												A3
4	**Average A**												$\bar{X}_A$
5	**Range A**												$\bar{R}_A$
6	B1												B1
7	B2												B2
8	B3												B3
9	**Average B**												$\bar{X}_B$
10	**Range B**												$\bar{R}_B$
11	C1												C1
12	C2												C2
13	C3												C3
14	**Average C**												$\bar{X}_C$
15	**Range C**												$\bar{R}_C$
16	**Part Average**												$\bar{\bar{X}}_{Part}$
17	Maximum part average – Minimum part average =												R_{Part}
18	$(\bar{R}_A + \bar{R}_B + \bar{R}_C)$ / Number of appraisers =												$\bar{\bar{R}}$
19	$\bar{X}_{DIFF} = Max(\bar{X}_A, \bar{X}_B, \bar{X}_C) - Min(\bar{X}_A, \bar{X}_B, \bar{X}_C) =$												$\bar{X}_{DIFF}$
20	* $UCL_R = D_4\bar{\bar{R}} = (2.574)\bar{\bar{R}} =$												
21	* $LCL_R = D_3\bar{\bar{R}} = (0)\bar{\bar{R}} = 0$												
22	* $D_3 = 0; D_4 = 2.574$ for 3 trials.												

Figure 16.10 Blank gage repeatability and reproducibility data collection sheet.

Source: Adapted from AIAG. Used with permission of AIAG.

13. Calculate the value for the "Average/range" column of row 18 by using the formula given in that row.
14. Calculate the value for the "Average/range" column of row 19 by using the formula given in that row.
15. Calculate the UCL_R by using the formula shown in line 20 along with the proper control chart constant values from row 22.

Continued

Gage Repeatability and Reproducibility Data Collection Sheet													
		Part											
Row	Appraiser/ Trial	1	2	3	4	5	6	7	8	9	10	Average/ Range	Appraiser/ Trial
1	A1	0.29	-0.56	1.34	0.47	-0.80	0.02	0.59	-0.31	2.26	-1.36	**0.19**	A1
2	A2	0.41	-0.68	1.17	0.50	-0.92	-0.11	0.75	-0.20	1.99	-1.25	**0.17**	A2
3	A3	0.64	-0.58	1.27	0.64	-0.84	-0.21	0.66	-0.17	2.01	-1.31	**0.21**	A3
4	**Average A**	**0.45**	**-0.61**	**1.26**	**0.54**	**-0.85**	**-0.10**	**0.67**	**-0.23**	**2.09**	**-1.31**	0.19	$\bar{X}_A$
5	**Range A**	**0.35**	**0.12**	**0.17**	**0.17**	**0.12**	**0.23**	**0.16**	**0.14**	**0.27**	**0.11**	0.18	$\bar{R}_A$
6	B1	0.08	-0.47	1.19	0.01	-0.56	-0.20	0.47	-0.63	1.80	-1.68	**0.00**	B1
7	B2	0.25	-1.22	0.94	1.03	-1.20	0.22	0.55	0.08	2.12	-1.62	**0.12**	B2
8	B3	0.07	-0.68	1.34	0.20	-1.28	0.06	0.83	-0.34	2.19	-1.50	**0.09**	B3
9	**Average B**	**0.13**	**-0.79**	**1.16**	**0.41**	**-1.01**	**0.03**	**0.62**	**-0.30**	**2.04**	**-1.60**	**0.07**	$\bar{X}_B$
10	**Range B**	**0.18**	**0.75**	**0.40**	**1.02**	**0.72**	**0.42**	**0.36**	**0.71**	**0.39**	**0.18**	**0.51**	$\bar{R}_B$
11	C1	0.04	-1.38	0.88	0.14	-1.46	-0.29	0.02	-0.46	1.77	-1.49	**-0.22**	C1
12	C2	-0.11	-1.13	1.09	0.20	-1.07	-0.67	0.01	-0.56	1.45	-1.77	**-0.26**	C2
13	C3	-0.15	-0.96	0.67	0.11	-1.45	-0.49	0.21	-0.49	1.87	-2.16	**-0.28**	C3
14	**Average C**	**-0.07**	**-1.16**	**0.88**	**0.15**	**-1.33**	**-0.48**	**0.08**	**-0.50**	**1.70**	**-1.81**	**-0.25**	$\bar{X}_C$
15	**Range C**	**0.19**	**0.42**	**0.42**	**0.09**	**0.39**	**0.38**	**0.20**	**0.10**	**0.42**	**0.67**	**0.33**	$\bar{R}_C$
16	**Part Average**	**0.17**	**-0.85**	**1.10**	**0.37**	**-1.06**	**-0.19**	**0.45**	**-0.34**	**1.94**	**-1.57**	**0.00**	$\bar{\bar{X}}_{Part}$
17	Maximum part average − Minimum part average =											**3.51**	R_{Part}
18	$(\bar{R}_A + \bar{R}_B + \bar{R}_C)$ / Number of appraisers = $(0.18 + 0.51 + 0.33)/3 =$											**0.3417**	$\bar{\bar{R}}$
19	$\bar{X}_{DIFF} = Max(\bar{X}_A, \bar{X}_B, \bar{X}_C) - Min(\bar{X}_A, \bar{X}_B, \bar{X}_C) = 0.19 - (-0.25) =$											**0.4447**	$\bar{X}_{DIFF}$
20	$* \; UCL_R = D_4\bar{\bar{R}} = (2.574)(0.3417) = 0.8795$												
21	$* \; LCL_R = D_3\bar{\bar{R}} = (0)(0.3417) = 0$												
22	$* \; D_3 = 0; D_4 = 2.574$ for 3 trials.												

Figure 16.11 GR&R data collection sheet with data entered and calculations completed for Example 16.1.

Source: Adapted from AIAG. Used with permission of AIAG.

16. Compare each of the 10 R-values in row 5 to the UCL_R value calculated in row 20. Any *R*-value that exceeds the UCL_R should be circled. Repeat this for the *R*-values in rows 10 and 15. The circled *R*-values are significantly different from the others, and the cause of this difference should be identified and corrected. Once this has been done, the appropriate parts can be remeasured using the same appraiser, equipment, and so on, as for the original measurements. Recompute all appropriate values on the data collection sheet as necessary.

Continued

Continued

Recall that repeatability is the variation in measurements that occurs when the same measuring system—including equipment, material, and appraiser—is used. Repeatability, then, is reflected in the *R*-values as recorded in rows 5, 10, and 15 and summarized in row 17. Repeatability is often referred to as equipment variation, but the individual *R* averages may indicate differences between appraisers. In the example in Figure 16.11, R_A is smaller than R_B or R_C. This indicates that Susan has done a better job of getting the same answer upon repeated measurements of the same part than either Frank or Willie. Perhaps Susan has a better technique, more skill, or sharper eyesight than the others. Investigation of this issue may provide an opportunity for variation reduction.

Reproducibility is the variation that occurs between the overall average measurements for the three appraisers. It is reflected by the *X*-values in rows 4, 9, and 14 and summarized in the value of $\bar{X}_{\text{Diff}}$ in row 19. For example, had $\bar{X}_A$ and $\bar{X}_B$ been closer in value and $\bar{X}_C$ significantly different, then Willie's measurements would exhibit a bias. Again, further investigation would be necessary.

The next step in the study is to complete the GR&R report shown in Figure 16.12. A completed report based on the data from Figure 16.11 is shown in Figure 16.13. Each of the quantities in the "Measurement Unit Analysis" column will now be described:

- *Equipment variation* (EV) is an estimate of the standard deviation of the variation due to repeatability, and is sometimes denoted σ_E or σ_{rpt}.
- *Appraiser variation* (AV) is an estimate of the standard deviation of the variation due to reproducibility, and is sometimes denoted σ_A or σ_{rpd}.
- *Gage repeatability and reproducibility* (GR&R) is an estimate of the standard deviation of the variation due to the measurement system, and is sometimes denoted σ_M.
- *Part-to-part variation* (PV) is an estimate of the standard deviation of the variation due to part differences, and is sometimes denoted σ_P.
- *Total variation* (TV) is an estimate of the standard deviation of the total variation in the study, and is sometimes denoted σ_T.

The "Percent Total Variation" column in Figure 16.12 shows for each type of variation the percent of total variation it consumes. Sometimes, the right-hand column is based on the tolerance for the dimension. In this case, the value of *TV* is replaced by the tolerance.

ANOVA Method. The data depicted in Figure 16.11 can also be analyzed using the analysis of variance (ANOVA) methods (addressed in Chapter 21).

EXAMPLE 16.2

Open Minitab and:

1. Open the data file for Example 16.2.
2. Choose Stat > Quality Tools > Gage Study > Gage R&R Study (Crossed). Note: This is a crossed study since every appraiser will be measuring every part.

Continued

Gage Repeatability and Reproducibility Report									
Part No. and Name:			Gage Name:				Date:		
Characteristics:			Gage No.:				Performed by:		
Specifications:			Gage Type:						
From data sheet:		$\overline{\overline{R}} =$			$\overline{X}_{DIFF}$			R_P	

Measurement Unit Analysis			Percent Total Variation
Repeatability - Equipment Variation (EV) $EV = \overline{\overline{R}} \text{ x } K_1$ $= \_\_\_ \text{ x } \_\_\_$ $= \_\_\_\_$	Trials	K_1	$\%EV = 100(EV / TV)$ $= 100(\_\_\_/\_\_\_)$ $= \_\_\_\%$
	2	0.8862	
	3	0.5908	
Reproducibility - Appraiser Variation (AV) $AV = \sqrt{\left(\overline{X}_{DIFF} \text{ x } K_2\right)^2 - \left(EV^2/(nr)\right)}$ $= \sqrt{(\_\_ \text{ x } \_\_)^2 - (\_\_^2/(\_\_ \text{ x } \_\_))}$ $= \_\_\_\_$	Appraisers	K_2	$\%AV = 100(AV / TV)$ $= 100(\_\_\_/\_\_\_)$ $= \_\_\_\%$ n = number of parts r = number of trials
	2	0.7071	
	3	0.5231	
Gage Repeatability & Reproducibility (GRR) $GRR = \sqrt{EV^2 + AV^2}$ $= \sqrt{\_\_\_^2 + \_\_\_^2}$ $= \_\_\_\_$			$\%GRR = 100(GRR / TV)$ $= 100(\_\_\_/\_\_\_)$ $= \_\_\_\%$
	Parts	K_3	
	2	0.7071	
	3	0.5231	
Part Variation (PV) $PV = R_{Part} \text{ x } K_3$ $= \_\_\_ \text{ x } \_\_\_$ $= \_\_\_\_$	4	0.4467	$\%PV = 100(PV / TV)$ $= 100(\_\_\_/\_\_\_)$ $= \_\_\_\%$
	5	0.4030	
	6	0.3742	
	7	0.3534	
Total Variation (TV) $TV = \sqrt{GRR^2 + PV^2}$ $= \sqrt{\_\_\_^2 + \_\_\_^2}$ $= \_\_\_\_$	8	0.3375	$ndc = 1.41(PV / GRR)$ $= 1.41(\_\_\_/\_\_\_)$ $= \_\_\_$ ndc = number of distinct categories
	9	0.3249	
	10	0.3146	

Figure 16.12 Blank gage repeatibility and reproducibility report.

Source: Adapted from AIAG. Used with permission of AIAG.

3. Select or enter "Part" for *Part numbers*.
4. Select or enter "Operator" for *Operators*.
5. Select or enter "Measurement" for *Measurement data*.
6. Choose "ANOVA" for the *Method of analysis*.
7. Select "Options."
8. For the *Study variation*, enter the value of 5.15.

Continued

Gage Repeatability and Reproducibility Report									
Part No. and Name:			Gage Name:				Date:		
Characteristics:			Gage No.:				Performed by:		
Specifications:			Gage Type:						
From data sheet:		$\bar{\bar{R}} =$	0.3417		$\bar{X}_{DIFF}$	0.4447		R_{Part}	3.51

Measurement Unit Analysis			Percent Total Variation
Repeatability - Equipment Variation (EV) $EV = \bar{\bar{R}} \text{ x } K_1$ $= (0.3417)(0.5908)$ $= 0.2019$	Trials	K_1	$\%EV = 100(EV/TV)$ $= 100(0.2019/1.1458)$ $= 17.62\%$
	2	0.8862	
	3	0.5908	
Reproducibility - Appraiser Variation (AV) $AV = \sqrt{\left(\bar{X}_{DIFF} \text{ x } K_2\right)^2 - \left(EV^2/(nr)\right)}$ $= \sqrt{\left((0.4447)(0.5231)\right)^2 - \left((0.2019)^2/(10)(3)\right)}$ $= 0.2296$	Appraisers	K_2	$\%AV = 100(AV/TV)$ $= 100(0.2296/1.1458)$ $= 20.04\%$ n = number of parts r = number of trials
	2	0.7071	
	3	0.5231	
Gage Repeatability & Reproducibility (GRR) $GRR = \sqrt{EV^2 + AV^2}$ $= \sqrt{(0.2019)^2 + (0.2296)^2}$ $= 0.3058$			$\%GRR = 100(GRR/TV)$ $= 100(0.3058/1.1458)$ $= 26.69\%$
	Parts	K_3	
	2	0.7071	
	3	0.5231	
Part Variation (PV) $PV = R_{Part} \text{ x } K_3$ $= (3.51)(0.3146)$ $= 1.1042$	4	0.4467	$\%PV = 100(PV/TV)$ $= 100(1.1042/1.1458)$ $= 96.37\%$
	5	0.4030	
	6	0.3742	
	7	0.3534	
Total Variation (TV) $TV = \sqrt{GRR^2 + PV^2}$ $= \sqrt{(0.3058)^2 + (1.1042)^2}$ $= 1.1458$	8	0.3375	$ndc = 1.41(PV/GRR)$ $= 1.41(1.1042/0.3058)$ $= 5.09 \Rightarrow 5$ ndc = number of distinct categories
	9	0.3249	
	10	0.3146	

Figure 16.13 GR&R report with calculations for Example 16.1.
Source: Adapted from AIAG. Used with permission of AIAG.

9. For the *Process tolerance,* enter the value 8 for the *Upper spec—Lower spec*. Note: Entering a process tolerance will provide us with a column of data titled "% Tolerance" that shows the percentage of the tolerance used by the measurement system.
10. Close out each dialog box by clicking *OK*.

Continued

Figure 16.14 represents the results of an ANOVA using Minitab. The first source table indicates that the interaction term (that is, parts by operators) is not significant. Thus, the term is removed, resulting in the second source table.

Gage R&R Study - ANOVA Method

Two-Way ANOVA Table With Interaction

Source	DF	SS	MS	F	P
Part	9	88.3619	9.81799	492.291	0.000
Operator	2	3.1673	1.58363	79.406	0.000
Part * Operator	**18**	**0.3590**	**0.01994**	**0.434**	**0.974**
Repeatability	60	2.7589	0.04598		
Total	89	94.6471			

First source table

Interaction term

α to remove interaction term = 0.25

Two-Way ANOVA Table Without Interaction

Source	DF	SS	MS	F	P
Part	9	88.3619	9.81799	245.614	0.000
Operator	2	3.1673	1.58363	39.617	0.000
Repeatability	78	3.1179	0.03997		
Total	89	94.6471			

Second source table

Gage R&R

Source	VarComp	**%Contribution** (of VarComp)
Total Gage R&R	0.09143	**7.76**
Repeatability	0.03997	3.39
Reproducibility	0.05146	4.37
Operator	0.05146	4.37
Part-To-Part	1.08645	**92.24**
Total Variation	1.17788	100.00

Third source table

Process tolerance = 8

Source	**StdDev (SD)**	Study Var (5.15 × SD)	**%Study Var** (%SV)	**%Tolerance** (SV/Toler)
Total Gage R&R	0.30237	1.55721	**27.86**	**19.47**
Repeatability	0.19993	1.02966	18.42	12.87
Reproducibility	0.22684	1.16821	20.90	14.60
Operator	0.22684	1.16821	20.90	14.60
Part-To-Part	1.04233	5.36799	**96.04**	**67.10**
Total Variation	1.08530	5.58929	100.00	69.87

Fourth source table

Number of Distinct Categories = 4 ← Discrimination

Figure 16.14 Gage R&R Study—ANOVA method: source tables for Example 16.2.

Continued

Continued

The third source table provides the variances for each component. Notice that the GR&R variation (that is, %Contribution) is relatively small with respect to the part-to-part variance. Most of the variation is due to the difference between parts. This is what we look (and hope) for in a measurement systems analysis.

The last source table in Figure 16.14 depicts how much each component contributes to the total variation. The "SD" column provides the standard deviation for each component. Compare this column with the "Measurement Unit Analysis" column in Figure 16.13. The numbers are similar though not exact.

The "%Study Var" column shows the percentage contribution of each component. The Total Gage R&R is 27.86%. While it is in the high marginal range, there may be some opportunity for improvement. Compare these numbers with the "Percent Total Variation" column in Figure 16.13. Again, the numbers are similar but not exact.

Although three of the criteria for this study are either acceptable or marginal, the number of distinct categories is 4. Thus, the measurement system is unacceptable due to inadequate discrimination.

Figure 16.15 depicts what is called the "components of variation." Furthermore, it combines, in one graph, the three percentage criteria discussed previously. This graph confirms the discussion above.

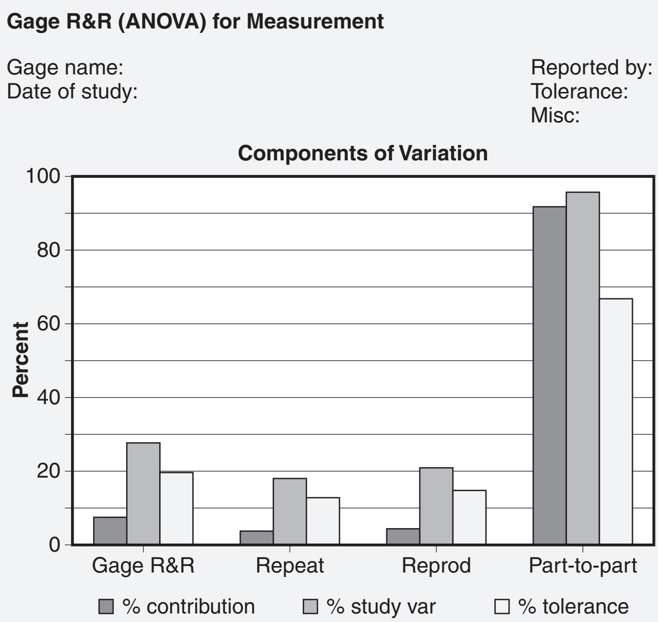

Figure 16.15 Gage R&R study—ANOVA method: components of variation for Example 16.2.

With the ready availability of statistical analysis software, the ANOVA method represents a quick and easy way to analyze data obtained from a GR&R study.

Control Chart Method. Another method for analyzing GR&R data is with $\bar{X}$ and R control charts.

For a more detailed discussion on the use of the control chart method, see Pyzdek and Keller (2010).

EXAMPLE 16.3

Open Minitab and:

1. Open the data file for Example 16.3.
2. Choose Stat > Quality Tools > Gage Study > Gage R&R Study (Crossed). Note: This is a crossed study since every appraiser will be measuring every part.
3. Select or enter "Part" for *Part numbers*.
4. Select or enter "Operator" for *Operators*.
5. Select or enter "Measurement" for *Measurement data*.
6. Choose "Xbar and R" for the *Method of analysis*.
7. Select "Options."
8. For the *Study variation,* enter the value of 5.15.
9. For the *Process tolerance,* enter the value 8 for the *Upper spec—Lower spec*. Note: Entering a process tolerance will provide us with a column of data titled "% Tolerance" that shows the percentage of the tolerance used by the measurement system.
10. Close out each dialog box by clicking *OK*.

Figure 16.16 provides the source tables for this method. Notice the similarities and differences between these two tables and the last two tables in Figure 16.14 as well as the "Percent Total Variation" column from Figure 16.13. The answers are, indeed, similar though not exact.

Note that, in contrast to Example 16.2, this method determines the number of distinct categories to be equal to five. However, for this particular data set, the ANOVA method would be more appropriate since it addressed the part–operator interaction.

Figure 16.17 is particularly interesting. Let's disregard the one out-of-control point on the range chart for Frank for the moment. Notice that each $\bar{X}$ chart is out of control. This is actually desirable since the R chart is based on repeatability error. If the $\bar{X}$ chart is in control, this would mean that part-to-part variation would be less than the repeatability variation. In other words, the part-to-part variation would be lost within the repeatability variation.

Continued

Continued

```
Gage R&R Study - XBar/R Method

                                 %Contribution
Source               VarComp    (of VarComp)
Total Gage R&R       0.09350            7.12
  Repeatability      0.04075            3.10
  Reproducibility    0.05275            4.02
Part-To-Part         1.21982           92.88
Total Variation      1.31332          100.00

Process tolerance = 8
                                       Study Var  %Study Var  %Tolerance
Source               StdDev (SD)  (5.15 × SD)       (%SV)  (SV/Toler)
Total Gage R&R           0.30578      1.57478       26.68       19.68
  Repeatability          0.20186      1.03959       17.61       12.99
  Reproducibility        0.22968      1.18287       20.04       14.79
Part-To-Part             1.10445      5.68793       96.37       71.10
Total Variation          1.14600      5.90191      100.00       73.77

Number of Distinct Categories = 5 <------------ Discrimination
```

Figure 16.16 Minitab session window output of the R&R study–$\bar{X}$ and R method: source tables for Example 16.3.

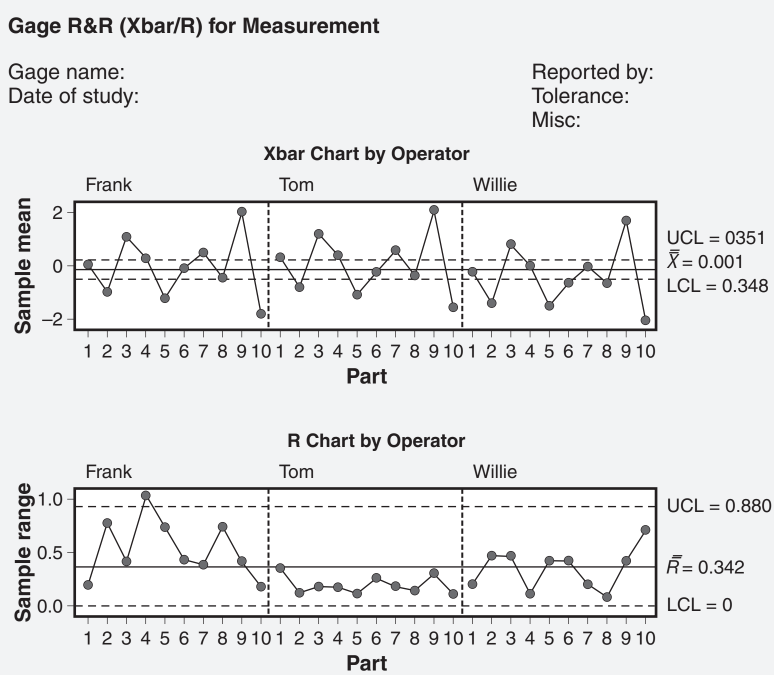

Figure 16.17 Graphical results of the gage R&R study–$\bar{X}$ and R method: $\bar{X}$ and R control charts by operators (appraisers) for Example 16.3.

Now we will discuss two *attributes* methods, using Minitab as the basis for the examples. The first is the *attribute agreement analysis*. This is an important type of analysis used to assess the level of agreement among appraisers grading on an ordinal scale. It is divided into four parts:

- Each appraiser versus standard
- Between appraisers
- All appraisers versus standard
- Within appraiser—how consistent is each appraiser within himself or herself?

The second method is the *attribute gage study—analytic method*. It is used to assess bias and repeatability in the measurement system.

Attribute Agreement Analysis

If three people say you are an ass, put on a bridle.

–Spanish proverb

Of course, the opening quote immediately above assumes an adequate measurement system.

Before we conduct an example of an attribute agreement analysis, let's introduce this subsection with two excerpts from *How Doctors Think* by Groopman (2007) that will definitely erode your confidence in the medical profession for years to come:

Excerpt 1:

Dr. E. James Potchen at Michigan State University in East Lansing has studied performance in reading chest x-rays. More than one hundred certified radiologists were assessed. These studies at Michigan State used a series of sixty chest x-rays that included duplicates of some of the films. When the radiologists were asked, "Is the film normal?" they disagreed among themselves an average of 20 percent of the time. This is called "interobserver variability." When a single radiologist reread on a later day the same sixty films, he contradicted his earlier analysis from 5 to 10 percent of the time. This is called "intraobserver variability."

One film of the sixty was of a patient who was missing his left clavicle. Presenting such a chest x-ray was meant to assess performance in noticing what was not on the film rather than merely searching for a positive finding—an exercise that points out our natural preference for focusing on positive data and ignoring the negative, as James Lock emphasized. Remarkably, 60 percent of the radiologists failed to identify the missing clavicle. When clinical data were added to the exercise, informing the radiologist that the sixty chest x-rays were obtained as part of an "annual physical examination," which primary care doctors perform in order to screen for serious diseases like lung cancer, 58 percent of the radiologists still missed it and scored the film as normal. However, when they were told that *the chest x-rays were obtained as part of a series of studies to find a cancer, then 83 percent of the radiologists identified the missing bone. This highlighted that a specific clinical cue can substantially improve performance, because the radiologist is systematically searching with attention to a particular condition, rather than relying on a flash impression.*

Excerpt 2:

Using a microscope, thirteen pathologists read 1001 specimens obtained from biopsies of the cervix, and repeated the readings later. On average, each pathologist agreed with himself only 89 percent of the time, and with a panel of senior pathologists only 87 percent of the time. With the patients who actually had an abnormal cervix, the doctors who reconsidered their own earlier conclusions agreed with their first readings only 68 percent of the time; the senior pathologists concurred with their juniors in only 51 percent of the cases. While the pathologists generally did well on distinguishing clearly cancerous tissues from clearly normal tissues, they did less well in identifying precancerous lesions.

While no details were provided regarding how the actual studies were conducted, these excerpts certainly highlight the importance and value of the attribute agreement analysis.

Let's consider the following example.

EXAMPLE 16.4

Open Minitab and:

1. Open the data file for Example 16.4.
2. Choose Stat > Quality Tools > Attribute Agreement Analysis.
3. Select or enter "Rating" for *Attribute column*.
4. Select or enter "Samples" for *Samples*.
5. Select or enter "Appraiser" for *Appraisers*.
6. Select enter "Attribute" for *Known standard/attribute*.
7. Check the box for *Categories of the attribute data are ordered*.
8. Close out the dialog box by clicking *OK*.

The data in Table 16.1 show the results of five new appraisers grading the written portion of a 12th-grade essay test. The appraisers graded on a five-point scale: –2, –1, 0, 1, 2. In addition, each appraiser grade for a given test will be compared with a known standard grade for that test. Each appraiser graded each of the 15 tests. $\alpha = 0.05$ will be used.

Figure 16.18 illustrates the Minitab session window for this example. The output is typically divided into four parts:

Continued

Table 16.1 Attribute agreement analysis—data for Example 16.4.

Simpson	1	2	2	Duncan	8	0	0
Montgomery	1	2	2	Hayes	8	0	0
Holmes	1	2	2	Simpson	9	–1	–1
Duncan	1	1	2	Montgomery	9	–1	–1
Hayes	1	2	2	Holmes	9	–1	–1
Simpson	2	–1	–1	Duncan	9	–2	–1
Montgomery	2	–1	–1	Hayes	9	–1	–1
Holmes	2	–1	–1	Simpson	10	1	1
Duncan	2	–2	–1	Montgomery	10	1	1
Hayes	2	–1	–1	Holmes	10	1	1
Simpson	3	1	0	Duncan	10	0	1
Montgomery	3	0	0	Hayes	10	2	1
Holmes	3	0	0	Simpson	11	–2	–2
Duncan	3	0	0	Montgomery	11	–2	–2
Hayes	3	0	0	Holmes	11	–2	–2
Simpson	4	–2	–2	Duncan	11	–2	–2
Montgomery	4	–2	–2	Hayes	11	–1	–2
Holmes	4	–2	–2	Simpson	12	0	0
Duncan	4	–2	–2	Montgomery	12	0	0
Hayes	4	–2	–2	Holmes	12	0	0
Simpson	5	0	0	Duncan	12	–1	0
Montgomery	5	0	0	Hayes	12	0	0
Holmes	5	0	0	Simpson	13	2	2
Duncan	5	–1	0	Montgomery	13	2	2
Hayes	5	0	0	Holmes	13	2	2
Simpson	6	1	1	Duncan	13	2	2
Montgomery	6	1	1	Hayes	13	2	2
Holmes	6	1	1	Simpson	14	–1	–1
Duncan	6	1	1	Montgomery	14	–1	–1
Hayes	6	1	1	Holmes	14	–1	–1
Simpson	7	2	2	Duncan	14	–1	–1
Montgomery	7	2	2	Hayes	14	–1	–1
Holmes	7	2	2	Simpson	15	1	1
Duncan	7	1	2	Montgomery	15	1	1
Hayes	7	2	2	Holmes	15	1	1
Simpson	8	0	0	Duncan	15	1	1
Montgomery	8	0	0	Hayes	15	1	1
Holmes	8	0	0				

Continued

```
Attribute Agreement Analysis for Rating

Each Appraiser vs Standard

Assessment Agreement

Appraiser    # Inspected   # Matched   Percent         95 % CI
Duncan                15           8     53.33   (26.59,  78.73)  ◄—— 4
Hayes                 15          13     86.67   (59.54,  98.34)  ◄—— 3
Holmes                15          15    100.00   (81.90, 100.00) }
Montgomery            15          15    100.00   (81.90, 100.00) } ◄— 1
Simpson               15          14     93.33   (68.05,  99.83)  ◄—— 2

# Matched: Appraiser's assessment across trials agrees with the known
           standard.


Fleiss' Kappa Statistics

Appraiser    Response     Kappa   SE Kappa         Z   P(vs > 0)
Duncan       -2         0.58333   0.258199   2.25924      0.0119
             -1         0.16667   0.258199   0.64550      0.2593
             0          0.44099   0.258199   1.70796      0.0438
             1          0.44099   0.258199   1.70796      0.0438
             2          0.42308   0.258199   1.63857      0.0507
             Overall    0.41176   0.130924   3.14508      0.0008
Hayes        -2         0.62963   0.258199   2.43855      0.0074
             -1         0.81366   0.258199   3.15131      0.0008
             0          1.00000   0.258199   3.87298      0.0001
             1          0.76000   0.258199   2.94347      0.0016
             2          0.81366   0.258199   3.15131      0.0008
             Overall    0.82955   0.134164   6.18307      0.0000
Holmes       -2         1.00000   0.258199   3.87298      0.0001
             -1         1.00000   0.258199   3.87298      0.0001
             0          1.00000   0.258199   3.87298      0.0001
             1          1.00000   0.258199   3.87298      0.0001
             2          1.00000   0.258199   3.87298      0.0001
             Overall    1.00000   0.131305   7.61584      0.0000
Montgomery   -2         1.00000   0.258199   3.87298      0.0001
             -1         1.00000   0.258199   3.87298      0.0001
             0          1.00000   0.258199   3.87298      0.0001
             1          1.00000   0.258199   3.87298      0.0001
             2          1.00000   0.258199   3.87298      0.0001
             Overall    1.00000   0.131305   7.61584      0.0000
Simpson      -2         1.00000   0.258199   3.87298      0.0001
             -1         1.00000   0.258199   3.87298      0.0001
             0          0.81366   0.258199   3.15131      0.0008
             1          0.81366   0.258199   3.15131      0.0008
             2          1.00000   0.258199   3.87298      0.0001
             Overall    0.91597   0.130924   6.99619      0.0000
```

Figure 16.18 Minitab session window output for Example 16.4.

Continued

Kendall's Correlation Coefficient

Appraiser	Coef	SE Coef	Z	P
Duncan	**0.87506**	0.192450	4.49744	0.0000
Hayes	0.94871	0.192450	4.88016	0.0000
Holmes	1.00000	0.192450	5.14667	0.0000
Montgomery	1.00000	0.192450	5.14667	0.0000
Simpson	0.96629	0.192450	4.97151	0.0000

Between Appraisers

Assessment Agreement

# Inspected	# Matched	Percent	95 % CI
15	6	40.00	(16.34, 67.71)

Matched: All appraisers' assessments agree with each other.

Fleiss' Kappa Statistics

Response	Kappa	SE Kappa	Z	P(vs > 0)
-2	0.680398	0.0816497	8.3331	0.0000
-1	0.602754	0.0816497	7.3822	0.0000
0	0.707602	0.0816497	8.6663	0.0000
1	0.642479	0.0816497	7.8687	0.0000
2	0.736534	0.0816497	9.0207	0.0000
Overall	**0.672965**	0.0412331	16.3210	0.0000

Kendall's Coefficient of Concordance

Coef	Chi - Sq	DF	P
0.966317	67.6422	14	0.0000

All Appraisers vs Standard

Assessment Agreement

# Inspected	# Matched	Percent	95 % CI
15	6	40.00	(16.34, 67.71)

Matched: All appraisers' assessments agree with the known standard.

Figure 16.18 *Continued.*

Continued

```
Fleiss' Kappa Statistics

Response      Kappa   SE Kappa         Z   P(vs > 0)
-2         0.842593   0.115470    7.2971      0.0000
-1         0.796066   0.115470    6.8941      0.0000
0          0.850932   0.115470    7.3693      0.0000
1          0.802932   0.115470    6.9536      0.0000
2          0.847348   0.115470    7.3383      0.0000
Overall    0.831455   0.058911   14.1136      0.0000


Kendall's Correlation Coefficient

    Coef     SE Coef         Z        P
0.958012   0.0860663   11.1090   0.0000

* NOTE * Single trial within each appraiser. No percentage of assessment
         agreement within appraiser is plotted.

Attribute Agreement Analysis
```

Figure 16.18 *Continued.*

- *Each appraiser versus standard*. The percent agreement table shows that both Holmes and Montgomery agreed with the standard on 15 out of 15 assessments, followed by Simpson with 14 out of 15, then Hayes with 13 out of 15, and Duncan trailing with 8 out of 15. With the exception of Duncan, all appraisers did well against the known standard. These results are confirmed with Fleiss's kappa statistics and Kendall's correlation coefficients. Notice Duncan's overall kappa was only 0.41176 while his correlation coefficient was the lowest at 0.87506. Figure 16.19 depicts in graphical form these same results using confidence intervals. It is obvious that Duncan requires additional training on how to grade essay tests, and both Hayes and Simpson could stand a little brushing up as well.
- *Between appraisers*. Fleiss's kappa statistic assumes the level of disagreement among scores is the same (that is, the difference between a score of –2 and –1 is the same as the difference between a score of –2 and 2). Therefore, Fleiss's overall kappa statistic reflects a value that indicates an unacceptable measurement system (that is, 0.672965). However, since Kendall's coefficient of concordance considers the difference between scores when the order of the scale is maintained, a high value is obtained (that is, 0.966317). This value is driven primarily by the fact that four of the five appraisers agreed on a minimum of 13 out of 15 trials.
- *All appraisers versus standard*. For reasons similar to "Between appraisers," Fleiss's overall kappa statistic is given as 0.831455, while Kendall's coefficient of concordance is 0.958012.
- *Within appraiser*. This table was not generated because each appraiser assessed each test only once.

Continued

Continued

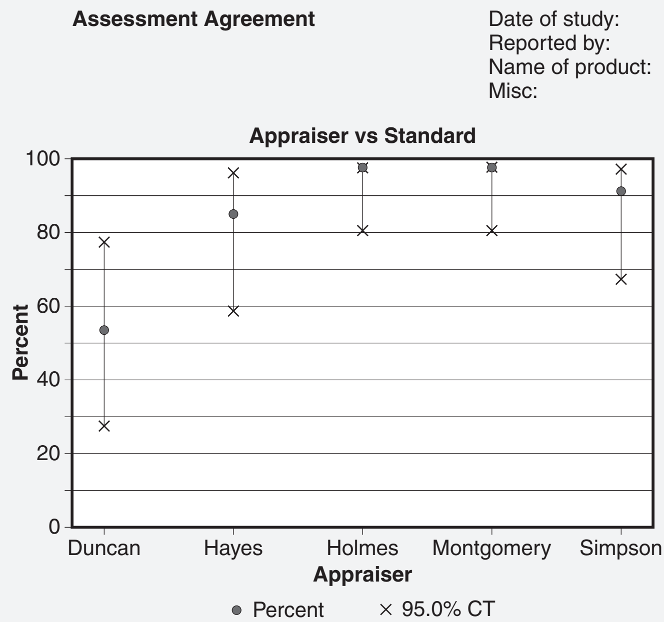

Figure 16.19 Graphical results of the attribute agreement analysis for Example 16.4.

See Kubiak (2012) for details regarding when to use Kappa statistics or either of Kendall's coefficients.

Attribute Gage Study—Analytic Method. An *attribute gage study* is used to determine the amount of bias and repeatability of a measurement system when the response is an attribute variable (that is, accept or reject). In such studies, reference values are known for each part and are selected at equal intervals across the entire range of the gage.

Depending on the particular attribute gage study method chosen (for example, regression method or AIAG method), additional requirements may be placed on the characteristics of the parts, the number of parts per trial, and the number of parts used in the study. Therefore, for additional information when constructing an attribute gage study, the reader is referred to the tutorial or help files of the specific software used. Also, the reader will find the second edition of the AIAG *Measurement Systems Analysis Reference Manual* useful in understanding the underlying theory.

It should now be clear that whenever a process improvement project is undertaken, the measurement system should be one of the first things analyzed, for the following reasons:

- All data from processes are, in effect, filtered through measurement systems.

- Reducing measurement system variation often represents a cost-effective way to reduce the total observed variation.
- Time, cost, and energy spent on improving a process without ensuring an adequate underlying measurement system could very well be counterproductive.

EXAMPLE 16.5

Open Minitab and:

1. Open the data file for Example 16.5.
2. Choose Stat > Quality Tools > Attribute Gage Study (Analytic Method).
3. Select or enter “Part Number” for *Part numbers*.
4. Select or enter “Reference” for *Reference values*.
5. Click the *Summarized counts* radial button.
6. Select or enter the value 20 for *Number of trials*.
7. Click the *Lower limit* radial button.
8. Select or enter the value –0.020.
9. Select “Options.”
10. Click the *AIAG method* radial button.
11. Close out each dialog box by clicking *OK*.

Table 16.2 shows the summarized data of 20 trials for each of 10 parts. For each part, a reference value has been provided and the number of acceptances has been captured. The AIAG method will be used.

Table 16.2 Attribute gage study—data for Example 16.5.

Part number	Reference	Acceptances
1	–0.050	0
2	–0.045	1
3	–0.040	2
4	–0.035	5
5	–0.030	8
6	–0.025	12
7	–0.020	17
8	–0.015	20
9	–0.010	20
10	–0.005	20

Continued

Continued

Figure 16.20 depicts the results of the attribute gage study analysis. The (adjusted) repeatability was determined to be 0.0458060, while the bias in the measurement system was 0.0097955. The *t*-test indicates that the bias is significantly different from zero, thus indicating that bias is present in the measurement system.

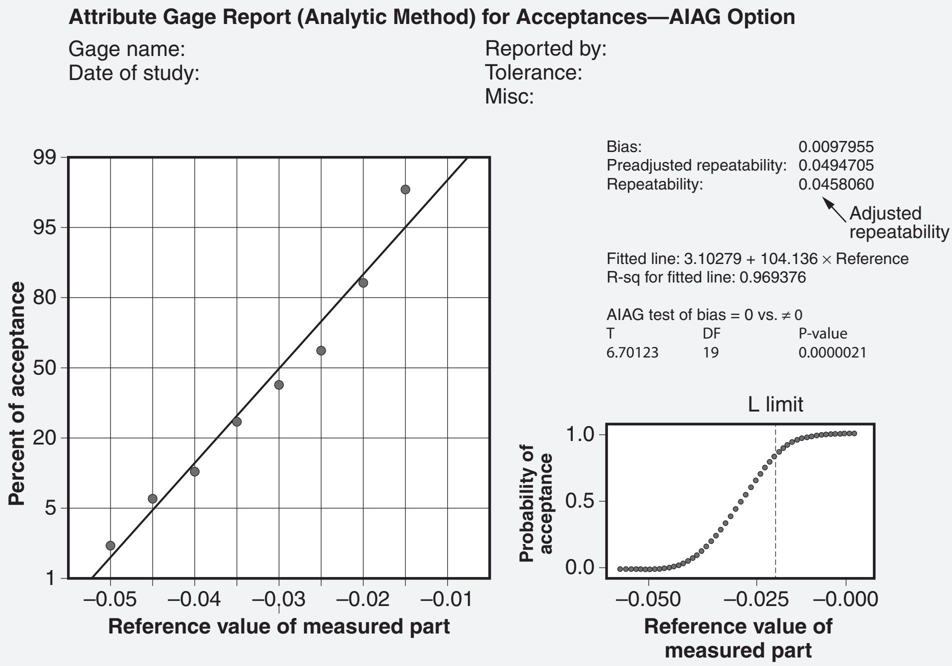

Figure 16.20 Graphical results of the attribute gage analysis for Example 16.5.

MEASUREMENT SYSTEMS ACROSS THE ORGANIZATION

Identify how measurement systems can be applied to marketing, sales, engineering, research and development (R&D), supply chain management, and customer satisfaction data. (Understand)

Body of Knowledge V.C.2

Measurement systems are widely found throughout most functions of an enterprise. Let's consider the following examples:

- *Human resources.* Though they are not often thought of as measurement systems in the context of Six Sigma, performance appraisal/evaluation systems are widely used throughout most organizations. More often than not, these systems are wrought with bias and lack repeatability. Managers, or *appraisers* as we call them, frequently must use ill-defined criteria to assess an employee's performance, usually on an annual basis. Such measurement systems can have a profound and deep impact on an individual and his or her career. In some instances, a single (annual) data point is used to determine whether to terminate an employee. In some organizational structures, employees may have multiple managers/appraisers who provide input that might be highly polarized. This discussion of a performance appraisal/evaluation system quickly raises questions regarding assessor agreement such as manager repeatability, each manager to standard, and all managers to standard. These terms should be familiar as they form the basis for the attribute agreement analysis discussed previously.
- *Marketing and sales/research and development.* These functional organizations traditionally have used employee surveys to gauge levels of customer satisfaction, dissatisfaction, and loyalty, as well as customer needs, wants, and other behaviors. Survey results are often fed to research and development functions to support and justify changes to product features and capabilities. Therefore, it is desirable that such surveys are highly valid and reliable and well defined. More than once, unreliable surveys have wreaked havoc on organizations. Consider the American automobile industry in the early 1970s. These companies continued to build large, gas-guzzling automobiles, missing the changing trend in consumer taste toward smaller, more fuel-efficient vehicles.
- *Quality engineering.* This function is often responsible for conducting various GR&R studies similar to those discussed previously. In addition, departments within this function usually are accountable for equipment calibration.
- *Supply chain management.* Probably the most common measurement system found in a supply chain management function is that for evaluating supplier performance and/or issuing supplier report cards. Supplier performance may be assessed inaccurately for a variety of reasons, such as the inability to determine a defect, misclassification of defects, inaccurate data entry, or even poor definitions of what constitutes a defect or late delivery. Some organizations include subjective components of measurement for determining levels of service or the ease of doing business with the supplier. Measuring supplier performance is critical in many organizations, as such performance provides the foundation for determining rework and repair cost allocations and reimbursements and occasionally percentage of profit sharing.

These are just a few examples of where measurement systems are routinely found in organizations. Their widespread use makes it imperative that they be sound and adequate to support their intended purpose. Without adequate measurement systems, organizational decision making may result in completely erroneous decisions that send an organization down a treacherous path. Alternatively, the basis for decision making deteriorates and decision makers must rely on gut feeling or guesswork. Either way, the outcomes are undesirable.

METROLOGY

> Define and describe elements of metrology, including calibration systems, traceability to reference standards, and the control and integrity of measurement devices and standards. (Understand)
>
> **Body of Knowledge V.C.3**

The purpose of a gage traceability document is to show the relationship between a given measurement system and the standards maintained by the United States Government at the National Institute of Standards and Technology (NIST). The traceability document quantifies the uncertainty involved in the various measurement transfers, from NIST calibration down to a measurement made on a part.

Measurement error is a source of variation. As such, it should be analyzed whenever a reduction in process variation is sought. It often happens that what was perceived as excess process variation was excess variation in the measurement system. When reducing the total observed process variation, improving the measurement system is often the most cost-effective strategy.

The causes of measurement error can be placed in the usual cause-and-effect categories:

- Machine (equipment)
  - Lack of accuracy
  - Lack of precision
  - Gage instability over time
- Methods (procedures)
  - Wrong tool specified
  - Improper procedure specified
  - Failure to use specified tool or procedure
- Man or woman (appraiser)

  - Lack of training
  - Lack of physical ability
  - Lack of motivation
- Mother Nature (environment)
  - Temperature, humidity, atmospheric pressure
  - Lighting, noise conditions
  - Vibration
  - Electronic emission
- Materials (parts)
  - Instability over time
  - Lack of homogeneity
  - Obsolescence
- Measurement
  - Measurement device not calibrated
  - Measurement device not "tared out"
  - Parallax effect

Throughout this book, we refer to the *7M's*. However, notice that this list contains only six "M's." The missing M is *management*, which we have framed in terms of the root cause categories based on the remaining six M's.

Whenever measuring equipment is used, a calibration system should be in place to help ensure that the measurement system does its job. The main thrust of a calibration system consists of a schedule that requires each piece of measuring equipment to be calibrated on a regular basis.

The time intervals between calibrations are based on the type of equipment and its uses. Calibration systems usually start with a relatively short interval for new equipment, with provisions for changing interval lengths depending on experience.

If, for instance, an instrument has been in for several successive calibrations with no adjustment needed, the calibration interval may be extended. On the other hand, instruments frequently requiring adjustment may need shorter intervals.

Some organizations set calibration quality goals of, for example, 95%, meaning that 95% of the instruments scheduled for calibration need no adjustment.

Each instrument must have a label that plainly displays its due date. Some organizations install a procedure to notify users of pending calibration.

In addition, a record system is required to document calibration activity. A system must also be in place for maintaining the calibration standards and equipment. This system should include a schedule for verifying that the standards are usable by checking them against recognized master standards.

Chapter 17

Basic Statistics

We live in a world of dispersion.

—Kaoru Ishikawa

This chapter will address basic statistical concepts, including distinguishing between a population and a sample, measures of central tendency and dispersion, the central limit theorem and its practical application, various descriptive statistics, the difference between descriptive and inferential statistics, numerous graphical methods and how to interpret them, and, of course, how to draw valid conclusions.

BASIC STATISTICAL TERMS

> Define and distinguish between population parameters and sample statistics, e.g., proportion, mean, standard deviation. (Apply)
>
> **Body of Knowledge V.D.1**

In a statistical study, the word "population" refers to the entire set of items under discussion. For example, the population might be the set of parts shipped to customer B last Friday. When discussing "population," it must be placed into proper context. What does this mean? Friday's "population" would actually be a sample if we defined the population as the set of parts shipped in a week that included Friday. Similarly, the week would be a sample if we defined the population as the set of parts shipped in the month that included the week we just labeled as a population. And so on.

When it is not feasible to measure a characteristic on each item in a population, a statistical study will randomly select a sample from the population, measure each item in the sample, and analyze the resulting data. The analysis of the sample data produces *statistics*. Examples of sample statistics include sample mean, sample median, sample standard deviation, and so on. These sample statistics are used to estimate the corresponding population *parameters*.

Table 17.1 Commonly used symbols for parameters and statistics.

	Population (Parameters)	Sample (Statistics)
Sample size	N	n
Mean	μ	$\bar{X}$ or $\bar{x}$
Standard deviation	σ	S or s or $\hat{\sigma}$
Median	None	$\tilde{X}$ or $\tilde{x}$
Mode	None	None
Proportion	p	$\hat{p}$

Traditionally, Latin letters are used to denote sample statistics and Greek letters are used to denote population parameters. However, the area of "quality," of which Lean Six Sigma is a subset, has been grossly inconsistent in its use of such nomenclature. Consequently, it is common to see a wide variety of symbols used for the same meaning. Table 17.1 shows commonly used symbols for the parameters and statistics used in this book.

The Greek letter μ is pronounced, "mew." The Greek letter σ is a lowercase sigma. Note in Table 17.1 that some parameters and statistics have yet to be assigned symbols.

CENTRAL LIMIT THEOREM

> Explain the central limit theorem and its significance in the application of inferential statistics for confidence intervals, hypothesis tests, and control charts. (Understand)
>
> **Body of Knowledge V.D.2**

Generally, it is advantageous to have normally distributed data. However, when data are not normally distributed, it useful to know about the central limit theorem. This theorem states that:

The distribution of sample averages will tend toward a normal distribution as the sample size, n, *approaches infinity.*

In other words, the central limit theorem guarantees at least approximate normality for the distribution of sample averages, even if the population from which the sample is drawn is not normally distributed.

Suppose $X_1, X_2, \ldots, X_n$ is a random sample taken from a distribution with mean μ and variance σ^2. If n is sufficiently large, then the sample mean, $\bar{X}$, follows approximately a normal distribution with $\mu_{\bar{X}} = \mu$ and variance $\sigma_{\bar{X}}^2 = \sigma^2 / n$, where σ^2 is from the population from which the random samples were drawn.

Notice that the definition does not state that the underlying distribution from which the sample is drawn must be normally distributed. The true form of the distribution does not have to be normally distributed as long as the sample size is sufficiently large. There have been several recommended cutoffs for large n including $n > 30$.

However, if the underlying distribution is normal, then the large n requirement is not necessary. In this case, the sampling distribution of $\bar{X}$ will also follow a normal distribution with mean $\mu_{\bar{X}} = \mu$ and variance $\sigma^2_{\bar{X}} = \sigma^2 / n$.

The central limit theorem has important practical applications in inferential statistics. *Inferential statistics* uses sample data to draw statistical conclusions about the population from which the sample was drawn. Inferential statistics is divided into two major areas:

- *Confidence intervals.* The *confidence interval* provides boundaries for an unknown parameter of a population with a specified degree of confidence that the parameter falls within the interval. A practical application of the central limit theorem is that as the sample size increases, the standard deviation of the sample mean decreases. Therefore, larger samples yield more accurate estimates of the population mean than smaller samples. Thus, confidence intervals become smaller with larger sample sizes. See Chapter 21 for details on confidence intervals.

- *Hypothesis testing.* Fundamentally, *hypothesis testing* is a test of significance and tests whether events occur by chance or not. Statistically speaking, a sample is drawn from a population and a statistic is computed from that sample. Let's say it is a mean. The hypothesis tests whether the mean occurred by chance at some specified level of significance.

  Let's construct a two-sided test for a sample mean:

  $$H_0 : \mu = \mu_0$$
  $$H_1 : \mu \neq \mu_0$$

  The above test would create an interval identical to the interval created by the confidence interval. Thus, the same practical application of the central limit theorem would apply as well. See Chapter 21 for details on hypothesis testing.

The central limit theorem also applies to the use of control charts. The plotted point on the $\bar{X}$ chart is the mean of a rational subgroup of size n. It follows directly from the central limit theorem that the distribution of the sample sizes will be approximately normal as the sample size increases.

The concept of the central limit theorem is illustrated in Figure 17.1. Note that the parent distribution in column "a" is a normal distribution. Consequently, all sampling distributions in this column are normal distributions, even for small values of n. The remaining columns contain distributions that depart significantly from normal. Consequently, higher values of n are required before normality

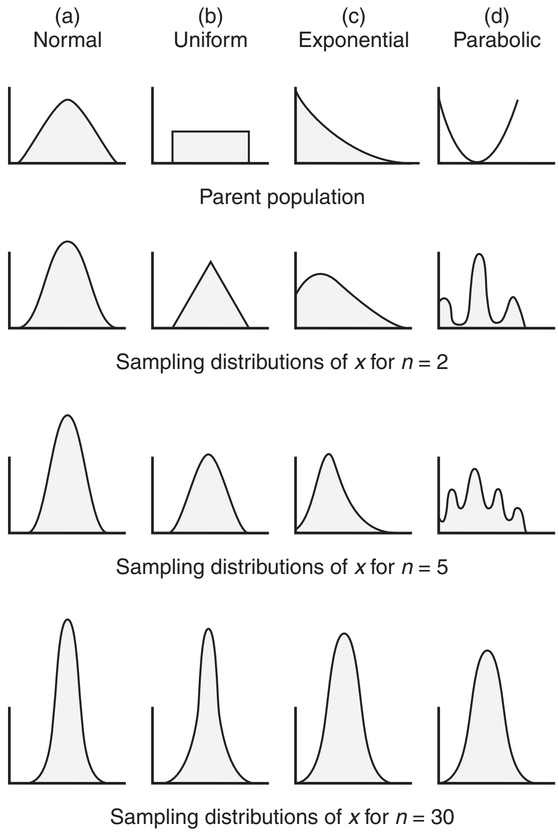

Figure 17.1 Examples of the impact of the CLT when increasing the sample size.
Source: http://imgarcade.com/1/central-limit-theorem-graph.

becomes apparent. Notice that as n grows larger, the variance of the sampling distribution grows smaller, which is consistent with the central limit theorem.

DESCRIPTIVE STATISTICS

> Calculate and interpret measures of dispersion and central tendency. (Evaluate)
>
> **Body of Knowledge V.D.3**

As presented, raw data provides little insight into underlying patterns, and limits our ability to draw meaningful conclusions. This is where descriptive statistics becomes useful.

Descriptive statistics is the collection of tools and techniques for displaying and summarizing data. There are many ways in which to display and summarize data. Therefore, it is important that one selects the best way to facilitate an understanding of the data. In some cases, multiple statistics and presentations or views are required to gain the most insight from the data.

This subsection will provide the basic descriptive statistics, while the next subsection will provide multiple graphical methods for viewing data. Combined, these two sections should provide a comprehensive understanding of most data sets.

Measures of Central Tendency

Three common ways for quantifying the centrality of a population or sample include the:

- *Mean.* Although there are many different types of statistical means, the most commonly dealt with in Lean Six Sigma is the arithmetic mean or arithmetic average of a data set.
- *Median.* The *median* is the middle value of an ordered data set. If the data are composed of an odd number of data points, the median is the middle value of the ordered data set. If the data are composed of an even number of data points, the median is the average of the two middle values of the ordered data set.
- *Mode.* The *mode* is the most frequently found value in a data set. Note: There may be more than one mode present.

EXAMPLE 17.1

Consider the data given in Table 17.2 that represent the diameters taken from a drilling operation. Compute all three measures of central tendency.

Because we are going to compute the median and the mode, the first thing we are going to do is to order the data set as shown in Figure 17.2.

From Figure 17.2, we see that the total of the diameters is 3.333. Thus,

$$\text{Mean} = \frac{3.333}{27} = 0.123$$

From Figure 17.2, we see that there are an odd number of data points in the ordered set of 27. Thus, the median occurs at point 14:

$$\text{Median} = 0.123$$

From Figure 17.2, we see that the most frequently found value in the data set is 0.123. Thus,

$$\text{Mode} = 0.123$$

Continued

Continued

Table 17.2 Data for Examples 17.1, 17.2, 17.3, 17.4, 17.5, 17.8, and 17.9.

Number	Diameter	Number	Diameter	Number	Diameter
1	0.127	10	0.125	19	0.126
2	0.125	11	0.121	20	0.121
3	0.123	12	0.123	21	0.124
4	0.123	13	0.122	22	0.121
5	0.120	14	0.125	23	0.124
6	0.124	15	0.124	24	0.122
7	0.126	16	0.122	25	0.126
8	0.122	17	0.123	26	0.125
9	0.123	18	0.123	27	0.123

Number	Sorted diameter
1	0.120
2	0.121
3	0.121
4	0.121
5	0.122
6	0.122
7	0.122
8	0.122
9	**0.123**
10	**0.123**
11	**0.123**
12	**0.123**
13	**0.123**
14	**0.123**
15	**0.123**
16	0.124
17	0.124
18	0.124
19	0.124
20	0.125
21	0.125
22	0.125
23	0.125
24	0.126
25	0.126
26	0.126
27	0.127
Total	**3.333**

Minimum = 0.120

Mode = 0.123

Median = 0.123

Maximum = 0.127

Mean = 0.123

Figure 17.2 Sorted data for Example 17.1

Measures of Dispersion

Although there are many ways to determine the dispersion of data, the two most common ways used in Lean Six Sigma include the:

- *Standard deviation.* The standard deviation is best described in its computational forms:
  - *Population.*

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$$

where N is the population size and μ is the population mean.

  - *Sample.*

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{X})^2}{n-1}}$$

where n is the sample size and $\bar{X}$ is the sample mean.

- *Range.* The *range* is simply the highest value minus the lowest value.

EXAMPLE 17.2

Consider the data given in Table 17.2 that represent the diameters taken from a drilling operation. Compute the standard deviations for the data assuming first that it is a population and second that it is a sample. Compute the range of the data.

Assuming the data represent a population, the standard deviation is given by:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}} = \sqrt{\frac{\sum_{i=1}^{27} (x_i - 3.333)^2}{27}} = 0.00175.$$

Assuming the data represent a sample, the standard deviation is given by:

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{X})^2}{n-1}} = \sqrt{\frac{\sum_{i=1}^{27} (x_i - 3.333)^2}{27-1}} = 0.00178.$$

Note: Readers may obtain different answers depending on the manner in which calculations are performed, numbers are rounded, number of significant digits used, and/or software used.

The *range* of the data is the maximum value minus the minimum value. From Figure 17.2, we see the maximum and minimum values are:

Range = Max – Min = 0.127 – 0.120 – 0.007.

Other Useful Descriptive Statistics

Other useful descriptive statistics include the following:

- *Kurtosis. Kurtosis* measures the degree to which a set of data is peaked or flat. When data are plotted as a histogram, a negative skewness value is depicted as a flatter distribution and a positive skewness value is depicted as a more peaked distribution. The normal distribution is considered the baseline for measuring kurtosis. The kurtosis value for the normal distribution is zero. The computation formula for kurtosis is complex and better left to a spreadsheet or statistical software program. Figure 17.3 provides a pictorial representation of kurtosis.
- *Skewness. Skewness* measures the degree to which a set of data is not symmetrical. A skewness value may be negative, zero, or positive. When data are plotted as a histogram, a negative skewness value is depicted with the tail of the distribution pointing toward the left, a positive skewness value is depicted with the tail pointing toward the right, and a zero skewness value indicates a symmetrical distribution. The computation formula for skewness is complex and better left to a spreadsheet or statistical software program. Figure 17.3 provides a pictorial representation of skewness.
- *Maximum.* As evident by the name, this is the maximum value in the sample.
- *Minimum.* As evident by the name, this is the minimum value in the sample.

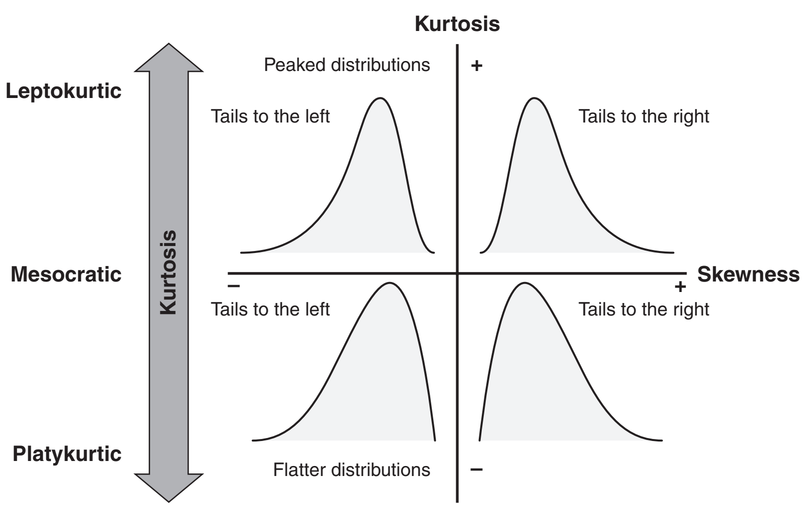

Figure 17.3 A pictorial representation of kurtosis and skewness.

- *First quartile (Q_1).* This is the point at which 25% of the data are less than or equal to this value. The first quartile does not have to be a value in the data set. See the subsection below on how to compute quartiles.
- *Second quartile (Q_2).* This is the point at which 50% of the data are less than or equal to this value. It is also known as the *median*. The second quartile or median does not have to be a value in the data set.
- *Third quartile (Q_3).* This is the point at which 75% of the data are less than or equal to this value. The third quartile does not have to be a value in the data set. See the subsection below on how to compute quartiles.
- *Fourth quartile.* This value is not computed as it is the maximum value in the data set.
- *Interquartile range (IQR).* The *interquartile range* is the third quartile minus the first quartile. It is considered a measure of dispersion. Again, this IQR does not have to be a value in the data set.

Most of the above have been included simply because they are provided as optional statistics for many statistical software packages.

EXAMPLE 17.3

Using the data given in Table 17.2, generate a histogram and compute the kurtosis and skewness of the data. Use Excel or other statistical software package.

Using Minitab, Figure 17.4 was generated. Notice the tail in the histogram in this figure is skewed right and the distribution is somewhat flat. This is consistent with the calculated values of kurtosis = –0.61 and skewness = 0.10.

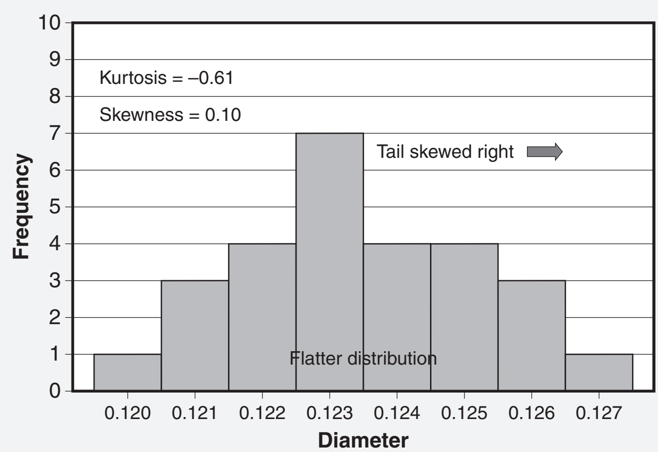

Figure 17.4 Histogram of data for Example 17.3.

Determining Quartiles

There are numerous methods and little agreement regarding how to calculate quartiles. Therefore, we will focus on a few selected approaches for doing so.

Method 1

1. Order the data set from smallest to largest.
2. Determine the median of the data set as described in the "Measures of Central Tendency" subsection above. This determination will result in two data sets: (1) a lower half and (2) an upper half.
3. To find the first quartile, find the median of the lower half as described in the "Measures of Central Tendency" subsection above. Note: Do not include the median in the lower-half data set from step 2.
4. To find the third quartile, find the median of the upper half as described in the "Measures of Central Tendency" subsection above. Note: Do not include the median in the upper-half data set from step 2.

Method 2

1. Order the data set from smallest to largest.
2. Determine the median of the data set as described in the "Measures of Central Tendency" subsection above. This determination will result in two data sets: (1) a lower half and (2) an upper half.
3. If the median found in step 2 is a datum (that is, a value in the ordered set), include the median in the two data sets. To find the first quartile, find the median of the lower half as described in the "Measures of Central Tendency" subsection above.
4. To find the third quartile, find the median of the upper half as described in the "Measures of Central Tendency" subsection above.

Method 3

1. Order the data set from smallest to largest.
2. Compute the first quartile as:

$$w = \frac{n + 1\text{th}}{4}$$

item, where n is the number of ordered data points. If the computed item number is an integer, count from the beginning of the ordered set up to the item number. The value in the ordered set corresponding to the item number is the first quartile. If the computed item is not an integer, the quartile must be computed by linear interpolation. The following method should be used:

a. w is formatted as "$y.z$" where y is an integer value of w and z is the decimal portion of w.

b. $Q_1 = x_y + z(x_{y+1} - x_y)$. This quartile is found by linear interpolation. However, if $y = w$, then $z = 0$ and $Q_1 = x_y$ and interpolation is not necessary.

c. $x_j = j$th observation of the ordered list of sample data.

3. Compute the third quartile as:

$$w = 3\left(\frac{n+1\text{th}}{4}\right)$$

item, where n is the number of ordered data points. If the computed item number is an integer, count from the beginning of the ordered set up to the item number. The value in the ordered set corresponding to the item number is the third quartile. If the computed item is not an integer, the quartile must be computed by linear interpolation.

a. w is formatted as "$y.z$" where y is an integer value of w and z is the decimal portion of w.

b. $Q_3 = x_y + z(x_{y+1} - x_y)$. This quartile is found by linear interpolation. However, if $y = w$, then $z = 0$ and $Q_3 = x_y$ and interpolation is not necessary.

c. $x_j = j$th observation of the ordered list of sample data.

Method 3 is the method used by the Minitab software.

EXAMPLE 17.4

Compute the first quartile, third quartile, and interquartile range (IQR) for each of the three methods just discussed using the data given in Table 17.2. The comparison of all three methods is shown in Figure 17.5.

Method 1:

From a previous example, we know that the median of the data given in Table 17.2 is item 14, which has a value of 0.123. This is removed from both the upper and lower sets of data as shown under Method 1 in Figure 17.5. The medians for both the upper and lower halves are found as discussed in the previous subsection. Thus, we have $Q_1 = 0.122$; $Q_3 = 0.125$; $IQR = 0.003$.

Method 2:

From a previous example, we know that the median of the data given in Table 17.2 is item 14, which has a value 0.123. Since the median is a datum in the ordered set, it is added to both the upper and lower halves. The medians for both the upper and lower halves are found as discussed in the previous subsection. In this case there is an even number of data items in each set. This requires us to average the seventh and eighth items in the lower set and the twentieth and twenty-first items in the upper set. Thus, we have $Q_1 = 0.122$; $Q_3 = 0.125$; $IQR = 0.003$.

Continued

Continued

Item	Sorted diameter		Method 1		Method 2	Method 3	Item	
1	0.120		0.120		0.120	0.120	1	
2	0.121		0.121		0.121	0.121	2	
3	0.121		0.121		0.121	0.121	3	
4	0.121		0.121		0.121	0.121	4	
5	0.122		0.122		0.122	0.122	5	
6	0.122		0.122		0.122	0.122	6	
7	0.122	Upper half	**0.122**	Upper half	**0.122**	**0.122**	**7**	← $Q_1 = \left(\frac{n+1}{4}\right)^{th}$ item
8	0.122		0.122		**0.122**	0.122	8	
9	0.123		0.123		0.123	0.123	9	
10	0.123		0.123		0.123	0.123	10	
11	0.123		0.123		0.123	0.123	11	
12	0.123		0.123		0.123	0.123	12	
13	0.123		0.123		0.123	0.123	13	
14	0.123				0.123	0.123	14	
15	0.123		0.123		0.123	0.123	15	
16	0.124		0.124		0.124	0.124	16	
17	0.124		0.124		0.124	0.124	17	
18	0.124		0.124		0.124	0.124	18	
19	0.124		0.124		0.124	0.124	19	
20	0.125		0.125		**0.125**	0.125	20	
21	0.125	Lower half	**0.125**	Lower half	**0.125**	**0.125**	**21**	← $Q_3 = 3\left(\frac{n+1}{4}\right)^{th}$ item
22	0.125		0.125		0.125	0.125	22	
23	0.125		0.125		0.125	0.125	23	
24	0.126		0.126		0.126	0.126	24	
25	0.126		0.126		0.126	0.126	25	
26	0.126		0.126		0.126	0.126	26	
27	0.127		0.127		0.127	0.127	27	

Figure 17.5 Three methods for determining first and third quartiles.

Method 3:

Computing Q_1 from the ordered data set, we have:

$$Q_1 = \left(\frac{n+1}{4}\right)^{th} = \left(\frac{27+1}{4}\right) = 7\text{th item.}$$

The seventh item of the ordered data list is 0.122. Thus, $Q_1 = 0.122$.

Computing Q_3 from the ordered data set, we have:

$$Q_3 = 3\left(\frac{n+1}{4}\right)^{th} = 3\left(\frac{27+1}{4}\right) = 3\left(\frac{28}{4}\right) = 21\text{st item}$$

The seventh item of the ordered data list is 0.125. Thus, $Q_1 = 0.125$.

Now computing the IQR we have:

$$IQR = Q_3 - Q_1 = 0.125 - 0.122 = 0.003$$

Notice that there were no decimal portions of the Q_1 and Q_3 values. Had there been, we would have to interpolate between the 7th and 8th items and the 21st and 22nd items, respectively, to find the appropriate quartile values.

Notice that in this example, all three methods produced the same results. This is not always the case and is simply a function of the data.

GRAPHICAL METHODS

Construct and interpret diagrams and charts, e.g., box-and-whisker plots, scatter diagrams, histograms, normal probability plots, frequency distributions, cumulative frequency distributions. (Evaluate)

Body of Knowledge V.D.4

In the previous subsection, we discussed descriptive statistics. In this subsection, we will discuss a variety of graphical methods for depicting data. Graphical methods are quite appealing because they provide an instant representation of the data while giving the viewer an almost complete understanding with little room for misinterpretation.

Frequency Distribution

A *frequency distribution* is a tabular display of data in mutually exclusive and collectively exhaustive classes or intervals that summarizes the number of occurrences in each class. Additionally, classes must be continuous and of equal width. A frequency distribution table may also show the absolute and cumulative absolute frequency and the relative and cumulative relative frequency. The relative and cumulative relative frequencies can be converted into percentages by multiplying by 100. Generally, a frequency distribution table contains between five and 20 classes, though this is considered a guideline.

To fully understand the frequency distribution table, some additional terminology is required:

- *Class* or *class interval.* This is the range in which data in a frequency distribution table are placed or binned. Frequency distributions have a predetermined number of classes. A class is also known as a *cell.*
- *Class limits.* A class has an upper and lower class limit. The *upper class limit* is the largest data value that can go into a given class. Similarly, the *lower class limit* is the smallest data value that can go into a given class. Class limits have the same number of decimals as the data values.
- *Class mark.* This is the average value of the class limits. This concept becomes more useful when discussing the concept of measures of central tendency with respect to grouped data.
- *Class boundaries.* The class boundaries allow us to create mutually exclusive and adjacent classes without gaps between classes. They

have one decimal place more than the original raw data and typically end in “5.” For example, if the data are whole numbers, then:

– Lower boundary = Lower class limit – 0.5

– Upper boundary = Upper class limit + 0.5

Notice that the interval formed by the class boundaries is larger than the class interval. In fact, the total range of all the class boundaries is greater than the total range of the raw data.

In order to calculate the class boundaries, we can employ the following approach:

1. Calculate the range of the data.
2. Specify the number of classes or cells desired. This will be discussed in detail in the subsection on histograms.
3. Compute the class width using the formula

$$W = \frac{\text{Range}}{\text{Number of cells}}$$

4. Round W as appropriate.
5. Set your lower class boundary for your first class.
6. Compute your upper class boundary for your first class by adding W to your lower class boundary.
7. Set your lower class boundary for your second class equal to your upper class boundary for your first class.
8. Compute your upper class boundary for your second class by adding W to your lower class boundary.
9. Continue this process until you establish an upper class boundary that includes your largest data value.

EXAMPLE 17.5

Using the data given in Table 17.2, construct a frequency distribution table using four class intervals.

1. We know that the range of data is 0.007 from a previous example.
2. The problem statement sets the number of class intervals to four.

Continued

Continued

3. Compute the class width:

$$W = \frac{0.007}{4} = 0.0018$$

4. We'll round W up to 0.002.
5. The smallest data value in our sample is 0.120. Therefore, we will set the lower class boundary of the first class equal to (0.120 – 0.0005) or 0.1195.
6. The first upper class boundary is the lower class boundary plus the class width:

$$0.1195 + 0.002 = 0.1215.$$

7. The lower class boundary for the second class is 0.1215.
8. The upper class boundary for the second class is 0.1235.
9. The remainder of the class boundaries are shown in Table 17.3 along with the completed frequency distribution table.

In general, we are not likely to develop a frequency distribution table with just four classes. Table 17.3 was completed to illustrate the process for developing the frequency distribution table.

Figure 17.6 has been developed to help illustrate all of the above concepts and terminology in one diagram using the data from Example 17.6.

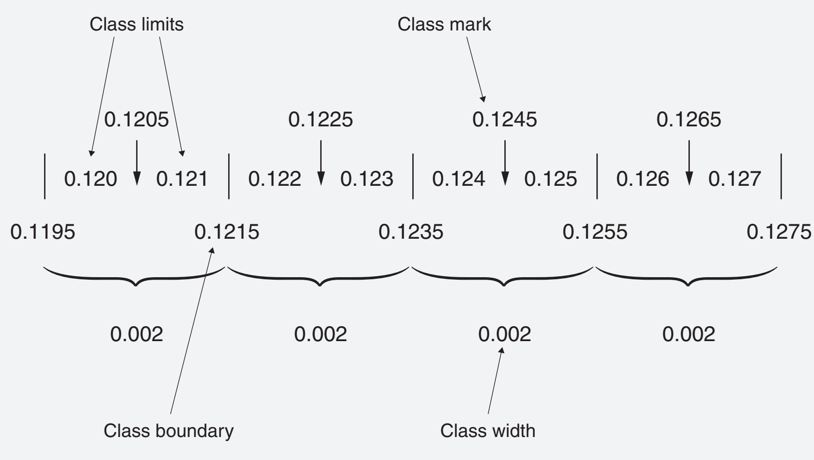

Figure 17.6 Understanding frequency distribution terminology.

Table 17.3 Example of a frequency distribution using the data given in Table 17.2.

<table>
<tr><th>Cell width</th><th>Lower class boundary</th><th>Upper class boundary</th><th>Tally</th><th>Data</th><th>Class interval</th><th>Class mark</th><th>Absolute frequency</th><th>Cumulative absolute frequency</th><th>Relative frequency</th><th>Cumulative relative frequency</th></tr>
<tr><td>0.002</td><td>0.1195</td><td>0.1215</td><td>||||</td><td>0.120, 0.121, 0.121, 0.121</td><td>0.120–0.121</td><td>0.1205</td><td>4</td><td>4</td><td>0.15</td><td>0.15</td></tr>
<tr><td>0.002</td><td>0.1215</td><td>0.1235</td><td>|||||||||||</td><td>0.122, 0.122, 0.122, 0.122, 0.123, 0.123, 0.123, 0.123, 0.123, 0.123, 0.123</td><td>0.122–0.123</td><td>0.1225</td><td>11</td><td>15</td><td>0.41</td><td>0.56</td></tr>
<tr><td>0.002</td><td>0.1235</td><td>0.1255</td><td>||||||||</td><td>0.124, 0.124, 0.124, 0.124, 0.125, 0.125, 0.125, 0.125</td><td>0.124–0.125</td><td>0.1245</td><td>8</td><td>23</td><td>0.30</td><td>0.85</td></tr>
<tr><td>0.002</td><td>0.1255</td><td>0.1275</td><td>||||</td><td>0.126, 0.126, 0.126, 0.127</td><td>0.126–0.127</td><td>0.1265</td><td>4</td><td>27</td><td>0.15</td><td>1.00</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>Total</td><td>27</td><td></td><td></td></tr>
</table>

Continued

Histogram

A *histogram* is a pictorial representation of data, the concepts of which are rooted in the frequency distribution. It is one of the seven basic tools of quality. Figure 17.4 provided an example of a histogram for the data given in Table 17.2. The figure was generated using Minitab, which uses its own set of rules and methods for determining the number of classes.

The advantage of using a histogram lies with its immediate visual feedback. With a brief review of Figure 17.4 we can quickly learn that the underlying distribution of the data is skewed right and tends to be flat. The central tendency of the distribution is at or about 0.123. There is a single mode at 0.123. The data are quite compact and there are no outliers. Thus, the range can be readily determined.

A key disadvantage of the histogram is that it groups data into classes. Consequently, the raw data are lost.

Typically, software is readily available to construct histograms. However, it is most likely that each set of software constructs a histogram differently or provides the user with a plethora of options for calculating the number of class intervals. Consider the following examples:

- *2 to the k rule*. Select the first integer value of k such that $2^k > n$ where n is the number of observations. Consider the data given in Table 17.2. There were 27 data values. Thus, $2^5 > 27$, making $k = 5$.
- *Sturges's formula*.
  a. $k = 1 + 3.3\log_{10} n$. k is rounded to the nearest whole number. From Table 17.2, $k = 6.61$, rounded to the nearest whole number 7.
  b. $k = (\log_2 n) + 1$. This formula assumes an approximately normal distribution of the data and tends to perform poorly for $n < 30$. k is rounded to the nearest whole number. From Table 17.2, $k = 5.8$, rounded to 6.
- *Square root rule*. $k = \sqrt{n}$. From Table 17.2, $k = 5.2$, rounded to the nearest whole number 5. Microsoft Excel uses this formula.
- *Rice's rule*. $k = 2n^{1/3}$. From Table 17.2, $k = 6$. k is rounded to the nearest whole number.

In addition to the above formulas, there are rules of thumb such as those given by Pyzdek (2003):

- Sample size of 100 or less 7–10 classes
- Sample size of 101–200 11–15 classes
- Sample size of 201 or more 13–20 classes

Other authors have suggested the following rules of thumb:

- Sample size of 50 or less 5–7 classes
- Sample size of 50–100 8–10 classes
- Sample size of 101–250 10–15 classes
- Sample size of 251 or more 15–20 classes

If one were to continue to research the number of classes relative to the sample size, numerous formulas and rules of thumb would continue to surface. In general, there appears to be little to no agreement in this area. Recall that Figure 17.4 was generated using Minitab. Notice that there were eight classes in this figure.

Our recommendation is to become familiar with your preferred quality software and its method of producing histograms. If options are available for changing the number of classes, try it out. You may be pleased with what you get.

Unlike the frequency distribution table, the histogram need not have equal class widths. However, most authors recommend equal class widths and most quality software provides only this option. Unequal class widths may be a consideration depending on the data density, skewness of the data, and presence of outliers.

Box-and-Whisker Diagrams

The *box-and-whisker diagram* (also called a *box plot*), developed by Professor John Tukey of Princeton University, displays graphically the high and low values of the data, quartiles, the median, outliers, and interquartile range. Depending on the software being used, other descriptive statistics may be available for display as well. An example of a box plot is depicted in Figure 17.7. Figure 17.7 was constructed using the data from Table 17.2. However, an additional data point was added to depict the outlier at 0.116.

With a brief visual review of Figure 17.7 we quickly determine many of the descriptive statistics calculated previously. In addition, we notice that the data are compact, but skewed, and that an outlier is present.

Box-and-whisker diagrams are quite appealing and provide a lot of information about a data set in just a single diagram.

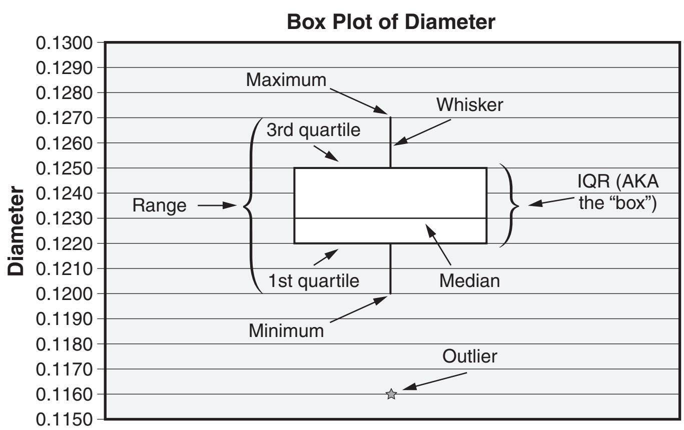

Figure 17.7 Basic components of a box-and-whisker diagram (box plot).

EXAMPLE 17.6

A stainless steel casting has a tight tolerance on the machined inside diameter (ID), which has been causing problems. The quality team has heard a number of proposed fixes. Some people believe the problem is caused by a slightly out-of-round condition on a cross-section of the casting. Others feel there is a taper, and still others insist the problem is too much part-to-part variation. The question is, which type of variation is giving the most trouble?

The team decides to measure the ID at three angles (12 o'clock, 2 o'clock, and 4 o'clock) at three locations along the bore (top, middle, and bottom) on five pieces. The data are shown in Table 17.4. Box plots created from these data are shown in Figure 17.8.

As we examine Figure 17.8, we notice that the within-part variation is quite small. However, the part-to-part variation is quite noticeable. For visual convenience, measurements within a part and from part-to-part have been connected with a mean connection line. This line helps enhance the two different types of variation.

Now that Figure 17.8 confirms the problem to be part-to-part variation, the team is no longer caught in the middle. It knows what the problem is and can now take the appropriate corrective action.

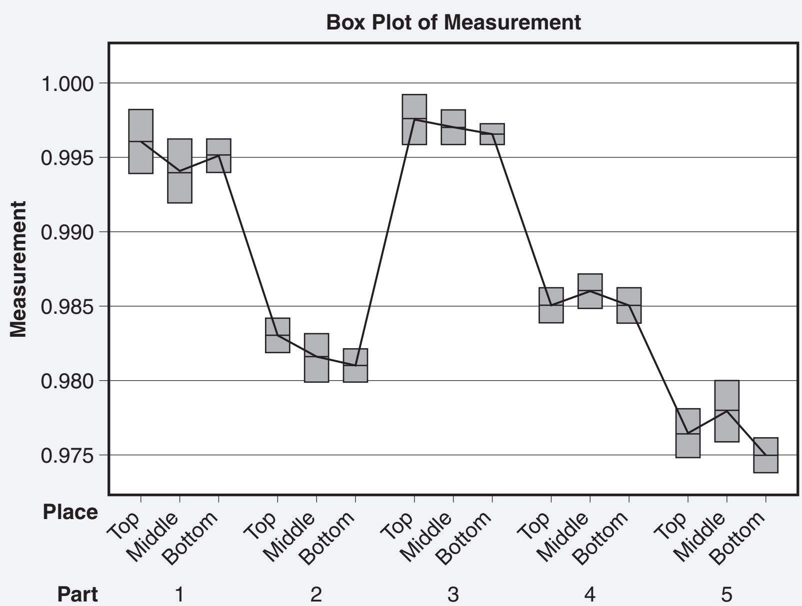

Figure 17.8 Multiple box plots for Example 17.6.

Continued

Continued

Table 17.4 Data for Example 17.6.

Part #1		Part #2		Part #3		Part #4		Part #5	
Place	**Measurement**	**Place**	**Measurement**	**Place**	**Measurement**	**Place**	**Measurement**	**Place**	**Measurement**
Top	0.998	Top	0.984	Top	0.998	Top	0.986	Top	0.975
Top	0.994	Top	0.982	Top	0.999	Top	0.985	Top	0.975
Top	0.996	Top	0.984	Top	0.996	Top	0.984	Top	0.978
Middle	0.992	Middle	0.982	Middle	0.998	Middle	0.987	Middle	0.980
Middle	0.996	Middle	0.980	Middle	0.998	Middle	0.986	Middle	0.976
Middle	0.994	Middle	0.983	Middle	0.996	Middle	0.985	Middle	0.980
Bottom	0.996	Bottom	0.981	Bottom	0.997	Bottom	0.986	Bottom	0.976
Bottom	0.994	Bottom	0.982	Bottom	0.997	Bottom	0.986	Bottom	0.974
Bottom	0.995	Bottom	0.980	Bottom	0.996	Bottom	0.984	Bottom	0.974

Scatter Diagrams

A *scatter diagram* is a plot of two variables, one on the y-axis and the other on the x-axis. The resulting graph allows visual examination for patterns to determine whether the variables show any relationship or if there is just random "scatter." This pattern, or lack thereof, aids in choosing the appropriate type of model for estimation. It is important to note that evidence of a pattern does not imply that a causal relationship exists between the variables. The scatter diagram is one of the seven basic tools of quality.

As mentioned above, we look for evidence of patterns. Such patterns may be linear or nonlinear. If the plotted points appear to be generally linear in nature, then we can measure the degree of linearity by using the sample correlation coefficient, denoted by the letter r. A positive correlation means that the line moves from the bottom left to the upper right. A negative correlation means that the line moves from the top left to the bottom right. If all the points fall exactly on a straight line that tips up on the right end, then $r = 1$. If all the points fall on a straight line that tips down on the right end, then $r = -1$. The value of r is always between –1 and +1, inclusive. This may be stated as $-1 \le r \le 1$. Therefore, the first step in analyzing a set of data for correlation is to construct a scatter diagram. If the points on this diagram tend to form a straight line, it makes sense to calculate the value of r.

The fact that a correlation exists does not imply that a cause-and-effect relationship exists. To conclude that a cause-and-effect relationship exists, we look to logic. When we obtain a high correlation coefficient, we must ask the question, "Does logic tell us that a cause-and-effect relationship exists or should exist?" On the other hand, when we obtain a low correlation coefficient, we can generally, but not always, conclude that a cause-and-effect relationship does not exist.

Keep in mind that the correlation coefficient is a statistic. As such, we are able to place a confidence interval around this statistic and make a probabilistic statement regarding its significance. Confidence intervals will be addressed in a later chapter.

EXAMPLE 17.7

For example, suppose an injection molding machine is producing parts with pitted surfaces. The following potential causes have been suggested:

- Mold pressure
- Coolant temperature
- Mold cooldown time
- Mold squeeze time

Values of each of these variables as well as the quality of the surface finish were collected on 10 batches. These data are shown in Table 17.5.

Four graphs have been plotted in Figure 17.9. In each graph, "Surface finish" is along the y-axis and one variable (that is, mold pressure, coolant temperature, mold

Continued

Table 17.5 Data for Example 17.7.

Batch number	Mold pressure, PSI	Coolant temperature, °F	Cooldown time, minutes	Squeeze time, minutes	Surface finish, microinches
1	220	102.5	14.5	0.72	37
2	200	100.8	16.0	0.91	30
3	410	102.6	15.0	0.90	40
4	350	101.5	16.2	0.68	32
5	490	100.8	16.8	0.85	27
6	360	101.4	14.8	0.76	35
7	370	102.5	14.3	0.94	43
8	330	99.8	16.5	0.71	23
9	280	100.8	15.0	0.65	32
10	400	101.2	16.6	0.96	30

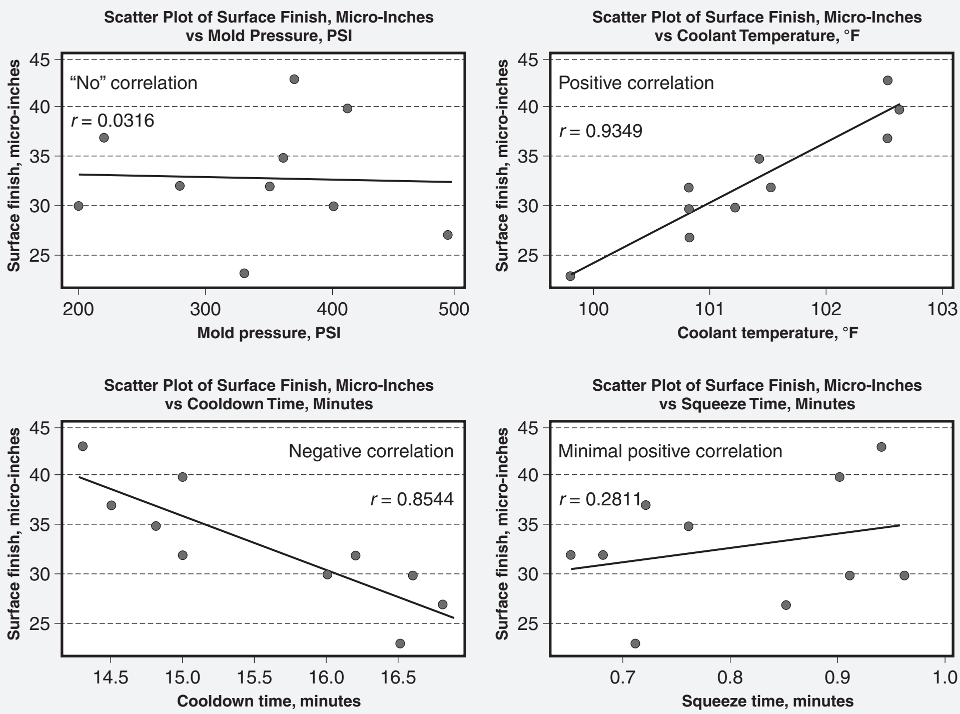

Figure 17.9 Scatter diagrams for Example 17.7.

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cooldown time, or mold squeeze time) is along the *x*-axis. Each data point on a graph represents one batch.

Let's briefly examine each of the graphs:

- *Scatter plot of surface finish versus mold pressure*. Notice the overall dispersion of the dot points. They are well dispersed and do not appear to present any noticeable pattern. The regression line for the variables is the black line in the middle of the graph. Notice that it is almost flat. This indicates that the *y*-value variation is almost completely unaffected by any change in the *x*-value. Consequently, there is little to no correlation present.
- *Scatter plot of surface finish versus coolant temperature*. The dot points are more tightly clustered together and clearly present a noticeable linear pattern. The addition of the regression line confirms a visually strong correlation coefficient.
- *Scatter plot of surface finish versus cooldown time*. The dot points in this graph are somewhat polarized in the lower and upper ends of the cooldown times, appearing to contribute to the correlation, which in this case is negative in nature.
- *Scatter plot of surface finish versus squeeze time*. Like the mold pressure, there is a lot of dispersion in the dot points, but a minor appearance of a linear pattern is present. The dot points are polarized and loosely clustered at the ends. An experienced eye might be able to determine the correlation to be minimally positive, but the aid of the regression line is certainly helpful.

Normal Probability Plot

Normal probability plots, also known as *Rankit* plots, *QQ* plots, *Quantile* plots, and *PP* plots, originally made use of normal probability graph paper. This special paper was used so that a random sample from a normally distributed population would plot as an approximately straight line. Today, advanced computer software has made the use of such paper all but obsolete.

Using the normal probability plot is similar to hypothesis testing. In this case, the null hypothesis is that the sample data come from a normal population, while the alternate hypothesis is that the data come from a nonnormal population.

The process for constructing a normal probability plot is straightforward. This process will use the "mean rank" approach as follows:

1. Collect and sort the data.
2. Generate a frequency distribution.
3. Generate a cumulative frequency distribution.
4. Compute the mean ranks.
5. Compute the probabilities. This is computed as:

$$\text{Probability} = \frac{\text{Mean rank}}{n+1}$$

6. Convert the probabilities from step 5 to percentages by multiplying by 100. This is computed as:

$$\left(\frac{\text{Mean rank}}{\text{n+1}}\right) \times 100.$$

It is important to note that normal probability plots are often presented in two distinct ways:

a. Data versus probability

b. Data versus percentage

Both plots are presented and interpreted in the same way. In "a," the probabilities are obtained from step 5. In "b," the percentages are obtained from step 6.

7. Note: This step is necessary only when normal probability plot paper is not available. This step permits regular quadrille graph paper to be used in lieu of normal probability plot paper.

Compute the normal score plot points using the inverse cumulative distribution function of the standard normal with the probability obtained from step 5. This is computed as:

$$\text{Normal score plot point } = \Phi^{-1}\left(\frac{\text{Mean rank}}{n+1}\right)$$

8. Generate the normal probability plot.

9. Analyze the plot and draw a conclusion.

The above approach is best illustrated by example.

EXAMPLE 17.8

Construct and interpret a normal probability plot using the data given in Table 17.2.
Using the process described above, we will construct Table 17.6.

1.–3. Figure 17.2 provides the sorted data. This allows us to easily construct columns A–C of Table 17.6. Column D is added for the convenience of performing row calculations.

Continued

Table 17.6 Mean rank method of computing normal scores—Example 17.8.

Column A	Column B	Column C	Column D	Column E	Column F	Column G	Column H
x	Frequency	Cumulative frequency (CF)	$n + 1$	Mean rank (MR)	MR/ $(n + 1)$	Mean rank %	Normal scores
0.120	1	1	28	1	0.036	3.6	–1.80274
0.121	3	4	28	3	0.107	10.7	–1.24187
0.122	4	8	28	6.5	0.232	23.2	–0.73181
0.123	7	15	28	12	0.429	42.9	0.18001
0.124	4	19	28	17.5	0.625	62.5	0.31864
0.125	4	23	28	21.5	0.768	76.8	0.73181
0.126	3	26	28	25	0.893	89.3	1.24187
0.127	1	27	28	27	0.964	96.4	1.80274

Continued

Table 17.7 Determining the mean ranks for Example 17.8.

x	Frequency	Data	Ranks occupied	Mean ranks
0.120	1	0.120	1	1
0.121	3	0.121, 0.121, 0.121	2, 3, 4	3
0.122	4	0.122, 0.122, 0.122, 0.122	5, 6, 7, 8	6.5
0.123	7	0.123, 0.123, 0.123, 0.123, 0.123, 0.123, 0.123	9, 10, 11, 12, 13, 14, 15	12
0.124	4	0.124, 0.124, 0.124, 0.124	16, 17, 18, 19	17.5
0.125	4	0.125, 0.125, 0.125, 0.125	20, 21, 22, 23	21.5
0.126	3	0.126, 0.126, 0.126	24, 25, 26	25
0.127	1	0.127	27	27

4. Compute the mean ranks. The computation of the mean ranks is shown in Table 17.7. The approach shown in this table could have easily been applied to computing median ranks as well.
5. Compute the probabilities. Note that "$n + 1$" is used in this determination. Thus,

$$\text{Probability} = \frac{\text{Mean rank}}{n+1}$$

   This is shown in Column F in Table 17.7.
6. Convert the probabilities from step 5 to percentages by multiplying by 100. This appears in Column G in the table.
7. Compute the normal score plot points using the inverse cumulative distribution function of the standard normal with the probabilities obtained from step 5. These values were computed using the NORM.S.INV function in Excel and are shown in Column H.
8. Generate the normal probability plot. See Figure 17.10. This figure was generated using Minitab.
9. Analyze the plot and draw a conclusion from Figure 17.10. Notice that the points form a straight line. Thus, we can conclude that the data come from a normal population.

If, for a moment, we assume the data were sampled from a controlled process, we can use the normal probability plot to estimate the percentage of product rejected by the process. Let's suppose the lower specification is set at 0.1205 and the upper specification is set at 0.1265. We can add these specifications to Figure 17.10 by drawing vertical lines from the *x*-axis at the specification points to the intersection points at the distribution fit line. At the points of intersection, draw horizontal lines to the *y*-axis to determine the percentages.

Continued

Continued

From this analysis, we can easily determine that approximately 5% of the product will be rejected below the lower specification limit and approximately 4% will be rejected above the upper specification limit.

When normal probability plot paper is not available, the percentages can be normalized using step 7 described in the approach above. This produced a plot like that shown in Figure 17.11. Notice that the graph is very similar except that the *y*-axis is now normal or *z*-scores and the *y*-axis is now uniformly spaced.

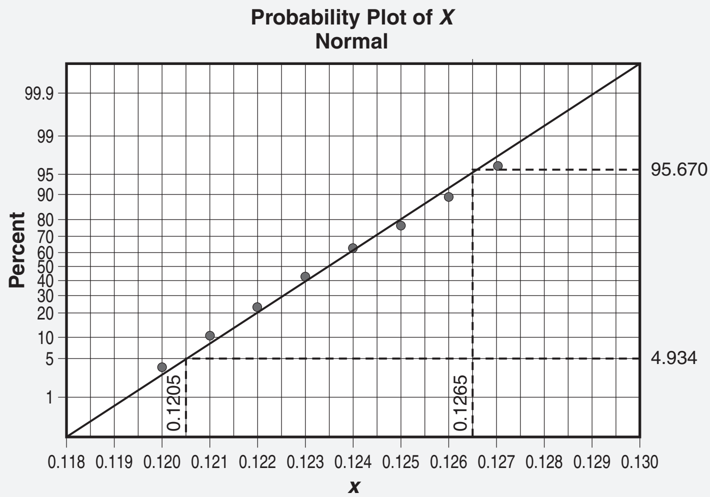

Figure 17.10 Normal probability plot for Example 17.8.

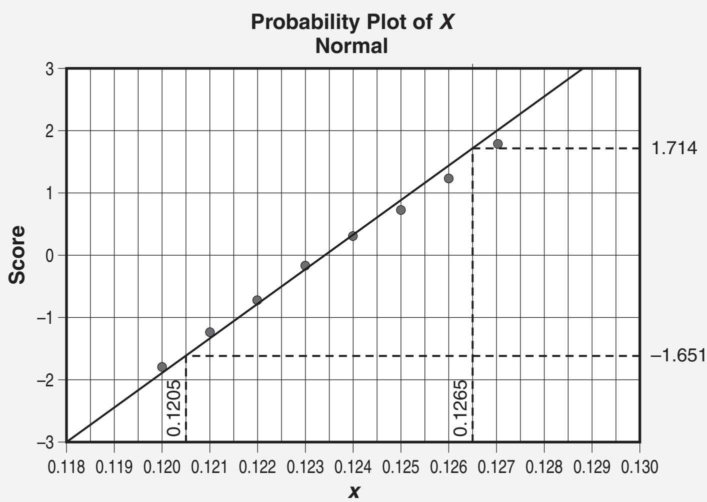

Figure 17.11 Normal probability plot using normal scores and standard graph paper.

The concept of the normal probability plots can be extended to testing hypotheses regarding populations beyond the normal distribution. These situations usually involve data transformations and will not be discussed here. However, some statistical software packages (for example, Minitab) will accommodate these situations.

Note: The normal probability scores found in Appendix 25 are based on the following formula:

$$U_i = \begin{cases} 1 - U_n & \text{for } i = 1 \\ \dfrac{i - 0.3175}{n - 0.365} & \text{for } i = 2, 3, \ldots, n-1 \\ 0.5^{(1/n)} & \text{for } i = n \end{cases}$$

Without the aid of a statistical test such as the Anderson-Darling or something similar, the interpretation of the normal probability plot can be somewhat subjective. It would not be unexpected to have two highly experienced Black Belts interpret the same plot and reach opposing conclusions regarding normality.

Usually, such disagreement occurs regarding how much departure from the distribution fit line there must be in the tails before one can conclude that the sample did not come from a population with a normal distribution.

In this subsection, we will look at various normal probability plots generated for specific distributions with defined parameters to determine what we might be able to learn about the population distribution. Five thousand points for each distribution were generated using Minitab. Parameters were chosen to obtain a specific shape and skewness and hence specific effect on each normal probability plot. The parameters for each of eleven distributions are given in Table 17.8. The statistics obtained from these randomly generated distributions are given in in Table 17.9.

Therefore, we will be working backwards, so to speak. However, we will have each population distribution side by side with its associated normal probability plot to facilitate the learning process.

In order to interpret normal probability plots consistently, it is important to keep several conventions in mind. These include the following:

- The definition of left and right skewness as given in Figure 17.2. Note that when we interpret distributions as skewed using the convention defined by Figure 17.2, left and right skewness is reflected in normal probability plots as shown in Figure 17.12.
- In order to maintain the conventions as noted in the first bullet above, the normal probability plots must be plotted with the percent, probability, or normal score on the y-axis. When normal probability plots are plotted in this manner, Figure 17.2, Figure 17.12, and Figure 17.13 hold.

Table 17.8 Parameters for Figures 17.11 through 17.25 and 17.27.

	Parameters								
Distribution	**Mean**	**Standard deviation**	**Shape (1)**	**Shape (2)**	**Location**	**Scale**	**Threshold**	**Degrees of freedom (1)**	**Degrees of freedom (2)**
Standard normal	0	1							
Wide normal	0	20							
Student's *t*								25	
Beta			2	2					
Cauchy					0	2			
Chi-square								25	
Exponential						1	0		
Lognormal					0	1	0		
F (20, 20)								20	20
F (1000, 1000)								1000	1000
Weibull			100			5	0		
Laplace					0	1			
Loglogistic					0	1	0		

Table 17.9 Statistics for Figures 17.12 through 17.25 and Figure 17.27.

Distribution	Mean	Standard deviation	*N*	Anderson-Darling	*p*-value ($\alpha = 0.05$)	Skewness	Kurtosis
Standard normal	0.001633	0.9718	5000	0.318	0.536	–0.02	–0.12
Wide normal	0.2726	19.787	5000	0.29	0.613	–0.03	–0.01
Student's *t*	0.01468	1.03423	5000	0.86	0.027	–0.08	0.23
Beta	0.4793	0.2227	5000	18.11	< 0.005	0.03	–0.87
Cauchy	–1.464	115.9	5000	1616.03	< 0.005	–24.78	1706.78
Chi-square	24.95	7.012	5000	19.35	< 0.005	0.55	0.33
Exponential	1.011	0.9991	5000	227.42	< 0.005	1.96	5.38
Lognormal	1.653	2.074	5000	473.07	< 0.005	4.12	27.78
F (20, 20)	1.1098	0.5389	5000	106.69	< 0.005	1.50	3.59
F (1000, 1000)	1.0031	0.0618	5000	1.80	< 0.005	0.19	0.09
Weibull	4.9722	0.06351	5000	54.18	< 0.005	–1.03	1.85
Laplace	–0.00696	1.41162	5000	56.77	< 0.005	–0.06	3.02
Loglogistic	4.111	12.82	5000	116.36	< 0.005	6.7941	53.1692

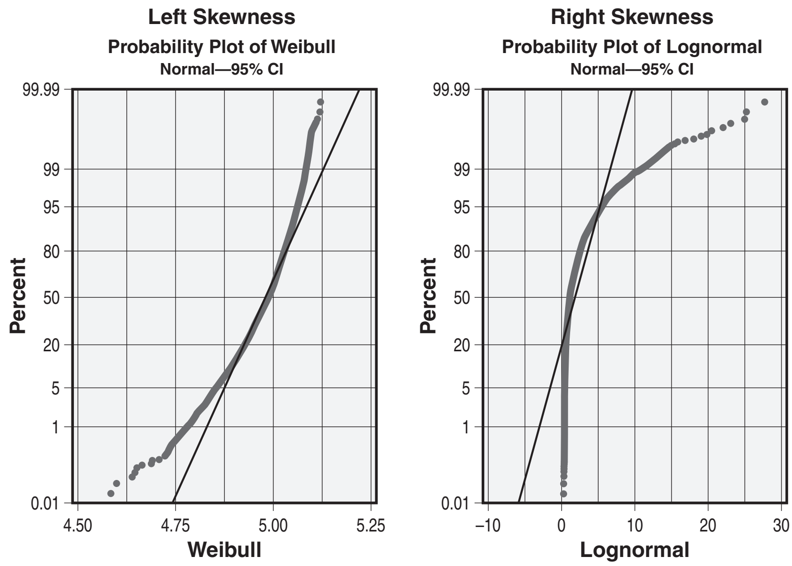

Figure 17.12 An illustration of left and right skewness in a normal probability plot.

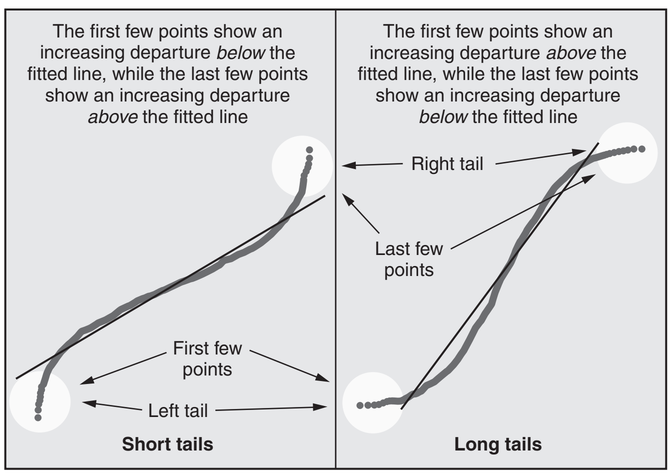

Figure 17.13 Direction of the tail bend on the normal probability plot tails for short- and long-tailed distribution.

Let's consider the following eleven distributions and their normal probability plots:

- *Standard normal distribution* ($\mu = 0$, $\sigma = 1$). Since this is the standard normal distribution, it plots as a straight line, as shown in the right-hand side of Figure 17.14. Since all the distributions were randomly generated, we do expect to see some minor perturbations, especially toward the tails of the distributions. A visual review of this plot by an experienced Black Belt would likely confirm it as normal. The Anderson-Darling test for normality and associated *p*-value shown in Table 17.9 would lead one to conclude normality as well.
- *Wide normal distribution* ($\mu = 0$, $\sigma = 20$). This distribution is a normal distribution with a large standard deviation, thus giving it slightly heavier tails. Heavy tails in a distribution tends to produce outliers at a rate greater than in a standard normal distribution. If we compare both right-hand sides of Figures 17.14 and 17.15, we will note some differences in the last few points. The last few points of Figure 17.15 tend to depart from the fitted distribution line at a slightly greater rate than the corresponding points in Figure 17.14. If compared to Figure 17.14, some experienced Black Belts might suggest that this normal probability plot is not from a normal distribution. However, when viewed alone, others may conclude that it is. Table 17.9 indicates that it passed the Anderson-Darling test for normality. This distribution is more platykurtic than the standard normal in Figure 17.14, but it is not readily apparent due to the matter of scale.
- *Student's* t-*distribution* (25 degrees of freedom). From Figure 17.13, we might determine that the normal probability plot shown in

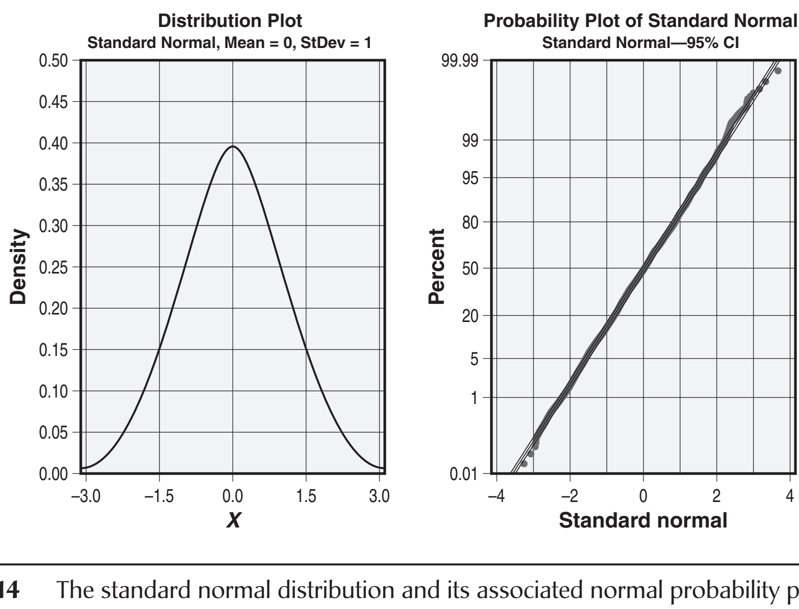

Figure 17.14 The standard normal distribution and its associated normal probability plot.

Figure 17.16 is a long-tailed distribution. It seems to be a flip of the normal probability plot shown in Figure 17.15. Similarly to the discussion regarding the wide normal distribution above, we should expect experienced Black Belts to debate whether this plot is from a normal distribution. We can see that it is slightly skewed left based on the first few data points. From Table 17.9 we know that it is negatively (left) skewed and leptokurtic. Furthermore, it did not pass the

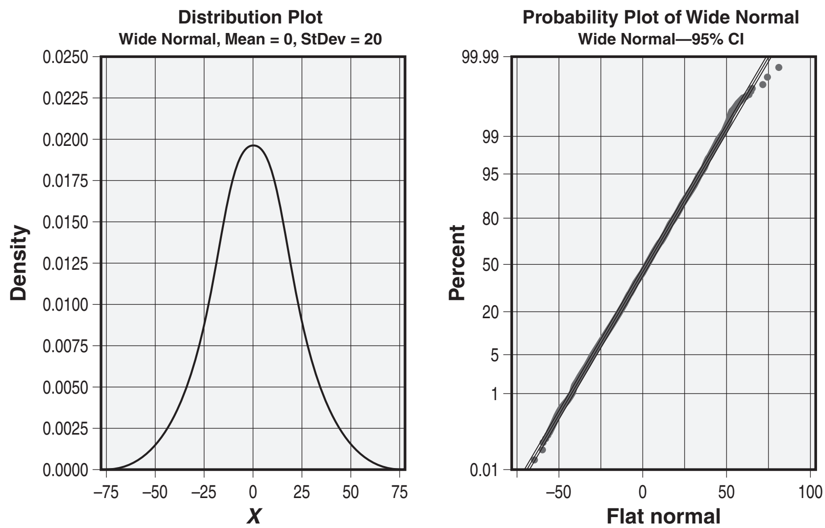

Figure 17.15 The wide normal distribution and its associated normal probability plot.

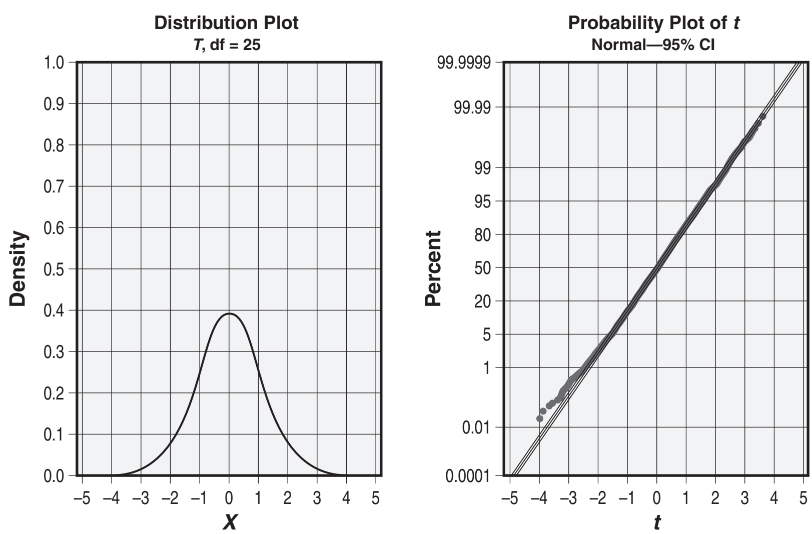

Figure 17.16 The Student's *t*-distribution and its associated normal probability plot.

Anderson-Darling test at $\alpha = 0.05$. Notice that the p-value is 0.027. From a practical viewpoint, this could be considered a judgment call.

- *Beta distribution* ($Shape_1 = 2$, $Shape_2 = 2$). The normal probability plot shown in Figure 17.17 appears as a symmetrical reverse S-curve that is bounded between 0 and 1, inclusive. This is indicative of mode and short tails, as shown in the left-hand side of Figure 17.13. If we look at Table 17.9, we see that the distribution statistics show that the beta is highly symmetric and slightly flatter than the standard normal distribution.
- *Cauchy distribution* (Location = 0, Scale = 2). Figure 17.13 indicates that the right-hand side of Figure 17.18 comes from a long-tailed distribution. The almost vertical slope of the fitted distribution line indicates a highly peaked (that is, leptokurtic) distribution. Also, notice that the outliers on the normal probability plot indicate a heavy-tailed distribution. The statistics shown in Table 17.9 are quite large for this distribution, and it clearly did not pass the Anderson-Darling test for normality. The skewness value is –24.78, which is not noticeable as a matter of scale.
- *Chi-square* (25 degrees of freedom). We know from Figure 17.12 that this distribution, depicted in Figure 17.19, is skewed right and from Figure 17.13 that it has a short left tail and long right tail. Clearly, the normal probability plot departs significantly from the line, indicating that the plot is not from a normal distribution. This is confirmed by the Anderson-Darling test from Table 17.9.

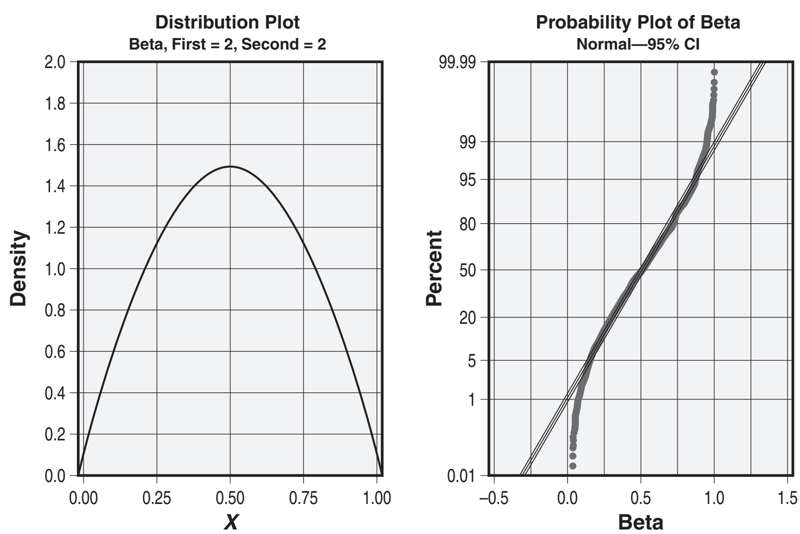

Figure 17.17 The beta distribution and its associated normal probability plot.

- *Exponential distribution* (Scale = 1, Threshold =0). From Figure 17.12, the distribution illustrated in Figure 17.20 is skewed right, and from Figure 17.13 it has a short left tail that is bounded by zero and a long right tail. The curvature in the normal probability plot would indicate that the data did not come from a normal distribution.

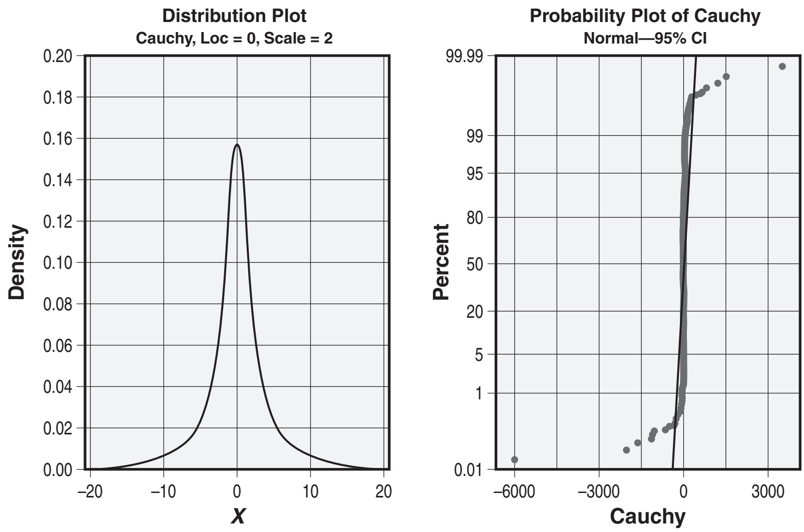

Figure 17.18 The Cauchy distribution and its associated normal probability plot.

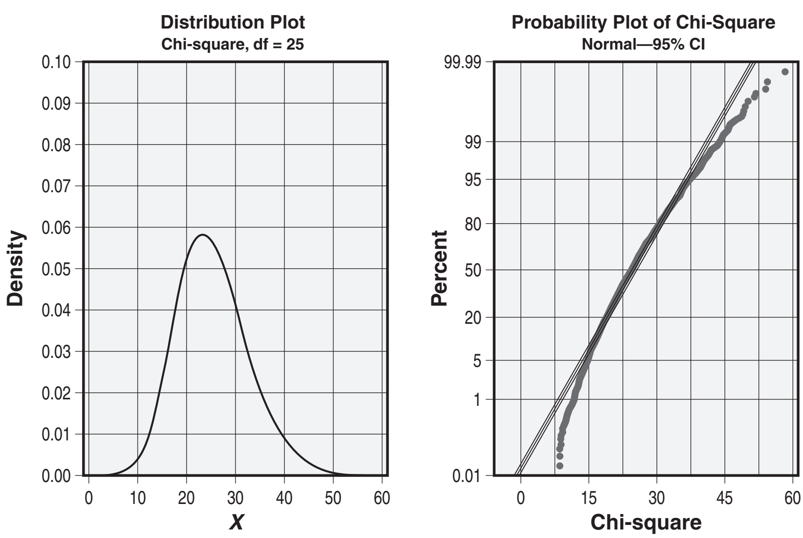

Figure 17.19 The chi-square distribution and its associated normal probability plot.

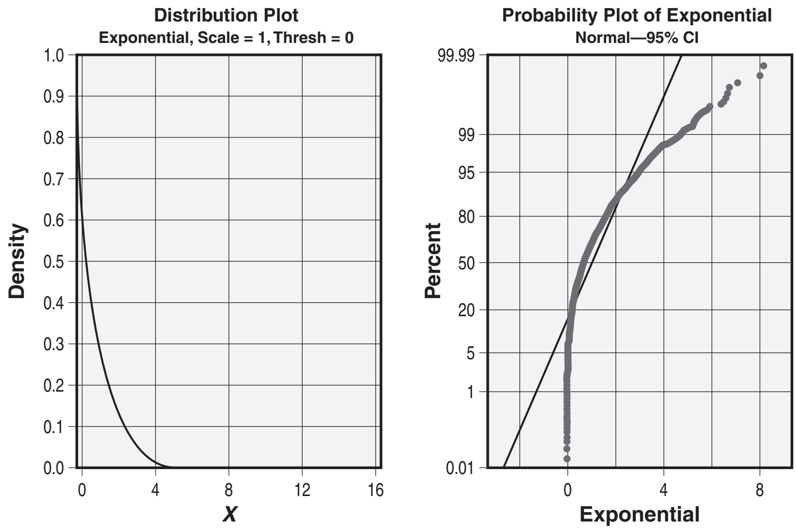

Figure 17.20 The exponential distribution and its associated normal probability plot.

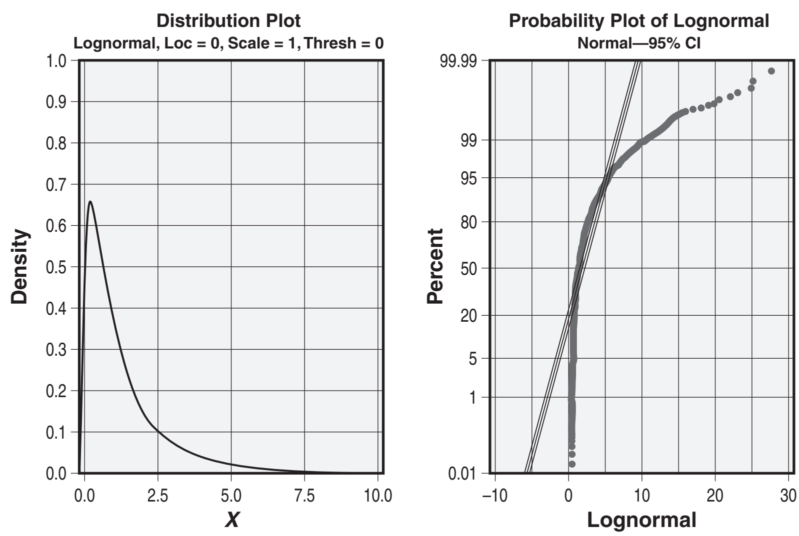

Figure 17.21 The lognormal distribution and its associated normal probability plot.

- *Lognormal distribution* (Location = 0, Scale = 1, Threshold = 0). The normal probability plot shown in Figure 17.21 is very similar to Figure 17.20 and is bounded by zero on the left. All that we can say about the normal probability plot is that the data did not come from a normal distribution. The normal probability plots for Figures 17.20 and 17.21 look very similar, yet one was generated by an exponential distribution and the other by a lognormal distribution.

- *F* (20 df, 20 df). The normal probability plot for Figure 17.22 indicates a right-skewed distribution with a short left tail that approaches zero, and a long right tail. The slope of the fitted distribution line indicates a highly peaked distribution. Clearly, the data did not come from a normal distribution.
- *F* (1000 df, 1000 df). The normal probability plot for Figure 17.23 may cause some concern for experienced Black Belts. At first glance, most

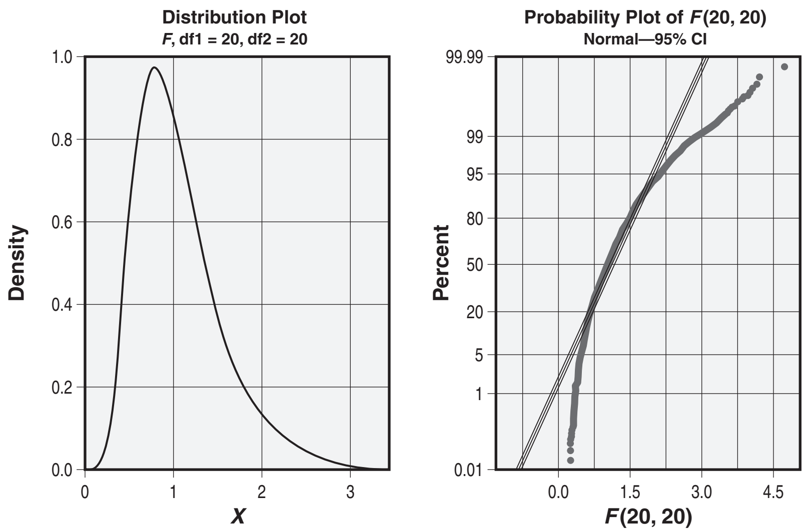

Figure 17.22 The F (20, 20) distribution and its associated normal probability plot.

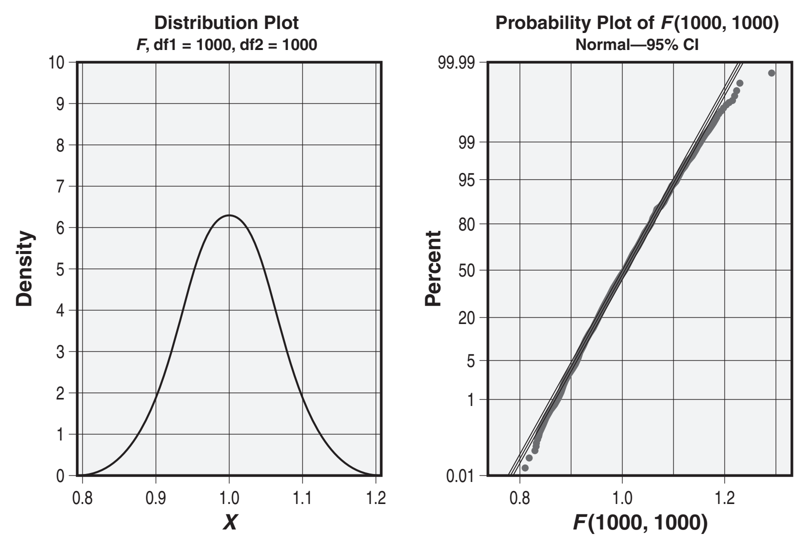

Figure 17.23 The F (1000, 1000) distribution and its associated normal probability plot.

of the points appear to plot quite well along a straight line. The last point does seem to be an outlier. However, upon closer review, there are points departing from the fitted distribution line beginning prior to 0.9 and just after 1.1. The normal probability plot does not come from a normal distribution. The statistics given in Table 17.9 confirm this. It is interesting to note that the theoretical distribution shown on the left-hand side in Figure 17.23 appears to be normal although it is an $F_{1000,1000}$ distribution.

- *Weibull distribution* (Shape = 100, Scale = 5, Threshold = 0). The normal probability plot shown in Figure 17.24 is left skewed and bounded by 5.1 with a short tail on the right and a long tail on the left. The curvature to the normal probability plot indicates that the data did not come from a normal distribution. From a practical viewpoint, Figure 17.24 is almost the mirror image of Figures 17.20 and 17.21.
- *Laplace distribution* (Location = 0, Scale = 1). This is an interesting normal probability plot. It comes from a distribution that is long and heavy in the tails (see Figure 17.25). The distribution is skewed both left and right about its mean. Notice that the left half of the normal probability plot is skewed left and the right half is skewed right. I hope that no Black Belt would conclude from the normal probability plot that the data come from a normal distribution. However, had a Black Belt generated a histogram such as that shown in Figure 17.26 and interpreted it without the benefit of an Anderson-Darling or similar test, he or she may have accepted the data as having come from a normal distribution.

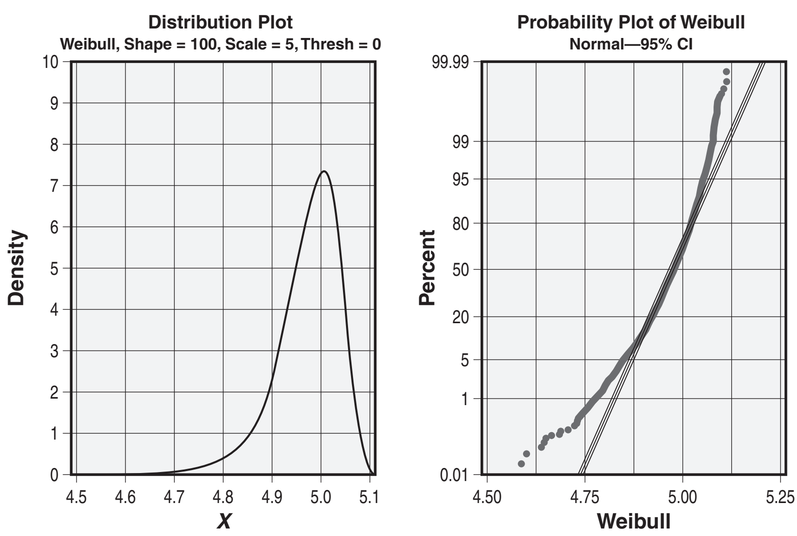

Figure 17.24 The Weibull distribution and its associated normal probability plot.

The normal probability plot in Figure 17.25 brings up an opportunity to learn something else about interpreting normal probability plots. Consider the bottom curve in Figure 17.25. Had this curve been flipped such that it appeared identical to the top curve, then we would have been able to conclude that the data came from a bimodal distribution.

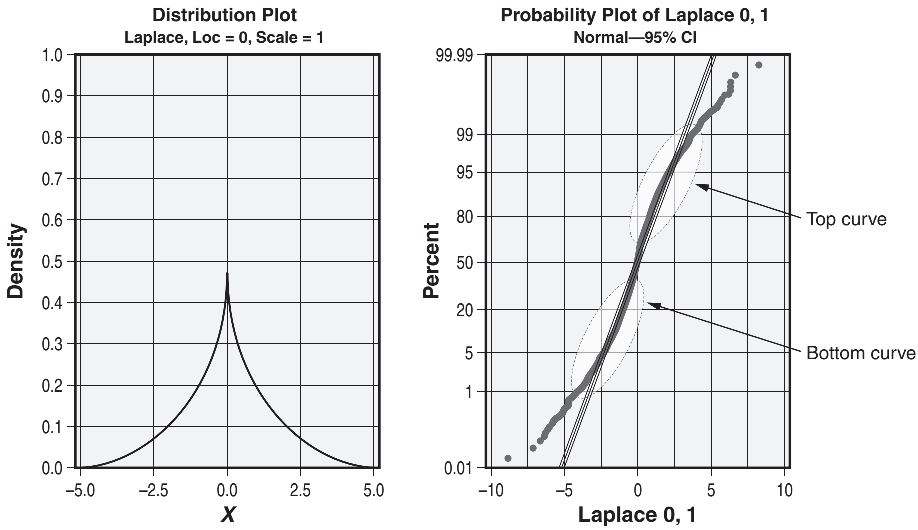

Figure 17.25 The Laplace distribution and its associated normal probability plot.

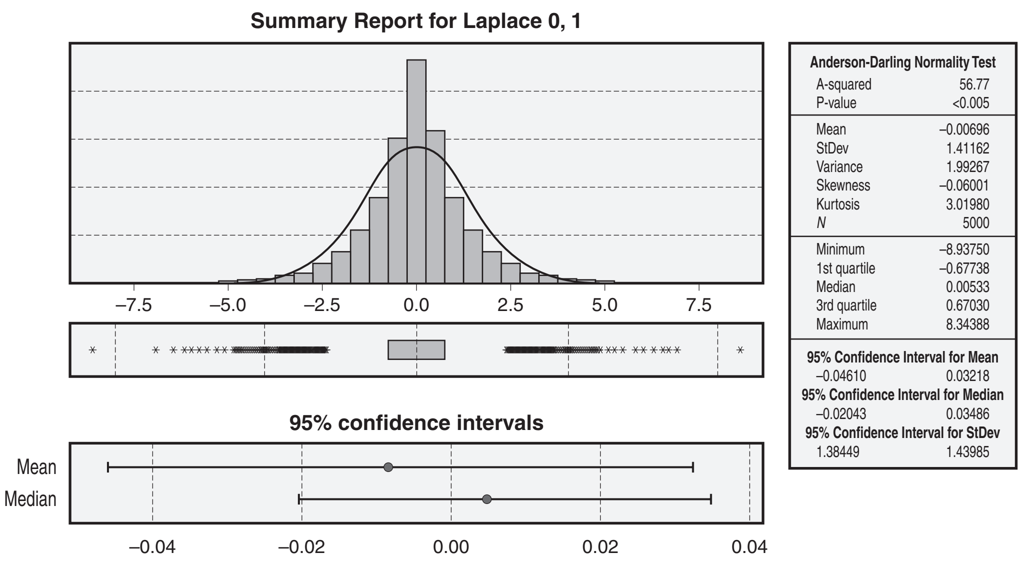

Figure 17.26 A histogram of the Laplace distribution shown in Figure 17.25.

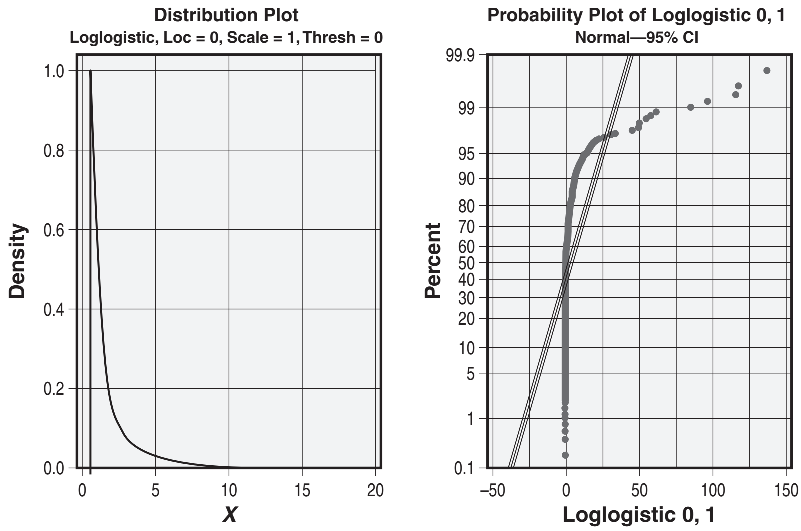

Figure 17.27 The loglogistic distribution and its associated normal probability plot.

More specifically, bimodal distributions exhibit two distinct peaks on the normal probability plot on the same side of the line.

- *Loglogistic distribution* (Location = 0, Scale = 1, Threshold = 0). Figure 17.27 exhibits characteristics we've seen in several of the normal probability plots previously. You may have begun to notice by now that the hard bend in the normal probability plots appears to indicate a ski-slope type of distribution function. The harder the bend, the deeper the slope. The gentler the bend, the gentler the slope. Review Figures 17.19 through 17.24 to convince yourself.

Normal probability plots aren't always easy to interpret. They require experience and occasionally statistical support via descriptive statistics and hypothesis testing. We've seen how various distributions can be disguised as a normal distribution and how easy it is to draw a wrong conclusion. All it takes is a moment or two to complement the normal probability plot with additional graphics and statistics.

In addition, we've learned a little about interpreting the normal probability plot and about what type of distribution may have generated your data. If your data are not normal and your software supports it, you may be able to run probability plots for other than normal distributions.

Stem-and-Leaf Diagram

The stem-and-leaf diagram is not part of the 2015 Body of Knowledge and will not be on the examination. However, it is part of the 2014 Green Belt Body of Knowledge. Consequently, it is being retained since Black Belts are commonly expected to train Green Belts.

A *stem-and-leaf diagram* is a graphical method for displaying quantitative data in a format similar to a histogram while maintaining all or nearly all of the raw data. A stem-and-leaf diagram is also known as a *stem-and-leaf plot* or *stem-and-leaf display*.

The stem-and-leaf diagram is useful for small sets of data that are easy to sort manually, but it quickly becomes unwieldy for large data sets. Pyzdek (2003) recommends data sets smaller than 200. However, the size of the data set is irrelevant when computer software is involved.

The process for developing a stem-and-leaf diagram is quite straightforward:

1. Gather and sort the data in ascending order.
2. Determine the stem and the leaf. As described by Tague (2005),

   Decide which digits in the data are changing. Out of this group, choose the two or three digits on the left that are the most significant. Of these, the digit on the right will be the leaf, and the one or two on the left will be the stem.

   Note: It is possible that some stems will not have any leaves depending on the nature of the data set.
3. Round all data in accordance with your selection of stem and leaf above.
4. Draw a vertical line on a page.
5. On the left side of the line, write the stems in ascending order down the page.
6. On the right side of the line, match the leaves to each stem (regardless of duplicates) in ascending order from left to right across the page. Do not use any digits to the right of the leaf.
7. Add a key or legend to indicate how to read the stem-and-leaf diagram. A common example of the key or legend is "6|3 = 63."

It is often a convention in stem-and-leaf diagrams to create separate stems for leaf digits 0–4 and 5–9. Consider Example 17.10.

EXAMPLE 17.9

Using the data in Table 17.2, construct a stem-and-leaf diagram.

For this example, we will use Minitab. We will select the leaf to be the digit in the fourth decimal place, and the stem will be everything in the third decimal place and to the left of it. The output from Minitab is shown in Figure 17.28.

The key or legend indicates that the leaf is the fourth decimal place and is in a different format than presented above. Notice that the leaves form a histogram. In reality, there are no leaves. Minitab filled in zeroes in this instance to provide a sense of count.

In addition, the parentheses surrounding the 7 in the first column of the fourth row indicates that the median is located in that row.

Continued

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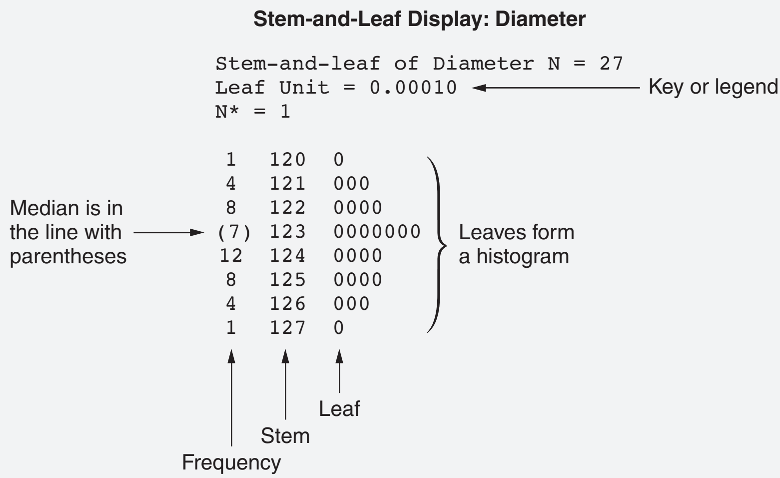

Figure 17.28 Stem-and-leaf diagram for Example 17.9.

EXAMPLE 17.10

Using the data in Table 17.10, construct a stem-and-leaf diagram.

For this example, we will use Minitab. We will select the leaf to be the digit in the second decimal place and the stem will be everything in the first decimal place. The

Table 17.10 Data for Example 17.10.

Number	Data	Sorted data	Number	Data	Sorted data	Number	Data	Sorted data
1	0.18	0.11	10	0.22	0.24	19	0.26	0.39
2	0.24	0.17	11	0.37	0.24	20	0.38	0.42
3	0.21	0.18	12	0.24	0.25	21	0.54	0.42
4	0.17	0.18	13	0.42	0.26	22	0.19	0.44
5	0.36	0.19	14	0.33	0.33	23	0.24	0.47
6	0.34	0.19	15	0.48	0.34	24	0.42	0.48
7	0.19	0.21	16	0.56	0.36	25	0.44	0.54
8	0.25	0.22	17	0.47	0.37	26	0.11	0.55
9	0.18	0.24	18	0.55	0.38	27	0.39	0.56

Continued

Continued

output from Minitab is shown in Figure 17.29. Note that there are only five rows of output and that each stem has only one row of leaves.

Now let's create another stem-and-leaf diagram using the same data, only this time we will create a stem for leaf digits 0–4 and a stem for leaf digits 5–9 using Minitab. This display is shown in Figure 17.30. Notice that there are now 10 rows in this figure compared to those in Figure 17.29.

Stem-and-Leaf Display: Data

```
Stem-and-leaf of Data N = 27
Leaf Unit = 0.010

 6    1   178899
13    2   1244456
(6)   3   346789
 8    4   22478
 3    5   456
```

Figure 17.29 Stem-and-leaf diagram for Example 17.10.

Stem-and-Leaf Display: Data

```
Stem-and-leaf of Data N = 27
Leaf Unit = 0.010

 1    1   1
 6    1   78899
11    2   12444
13    2   56
(2)   3   34
12    3   6789
 8    4   224
 5    4   78
 3    5   4
 2    5   56
```

Figure 17.30 Alternate view of a stem-and-leaf diagram for Example 17.10.

In other instances, a stem-and-leaf diagram may have to deal with negative numbers or the need to round numbers. Neither is difficult to deal with.

In the case of negative numbers, just place a negative before the stem unit.

The degree of rounding depends on the determination of the stem and leaf. In some instances, it may be necessary to round the entire stem. Consider Example 17.11.

EXAMPLE 17.11

Using the data in Table 17.11, construct a stem-and-leaf diagram.

For this example, we will use Minitab. We will select the leaf to be the units digit and the stem will be the tens digit. Before entering the data into Minitab, we will round the data as shown in Table 17.12. Several of the positive integers in the rounded data column do not have a tens digit. This means the tens digit will be treated as zero. Furthermore, the raw data was lacking stems 3 and 4. Minitab completed the range of stems by adding these values without leaves. The completed stem-and-leaf diagram is given in Figure 17.31.

Table 17.11 Data for Example 17.11.

Number	Data
1	–24.78957
2	–13.46
3	–3.3
4	4.34
5	5.6
6	5.787
7	17.83
8	25.8
9	53.7

Table 17.12 Rounded data for Example 17.11.

Number	Data	Rounded data
1	–24.78957	–25
2	–13.46	–13
3	–3.3	–3
4	4.34	4
5	5.6	6
6	5.787	6
7	17.83	18
8	25.8	26
9	53.7	54

Continued

Continued

Stem-and-Leaf Display: Rounded Data

```
Stem-and-leaf of Rounded Data N = 9
Leaf Unit = 1.0

 1    –2   5
 2    –1   3
 3    –0   3
(3)    0   466
 3     1   8
 2     2   6
 1     3
 1     4
 1     5   4
```

Figure 17.31 Stem-and-leaf diagram for Example 17.11.

VALID STATISTICAL CONCLUSIONS

Distinguish between descriptive and inferential statistical studies. Evaluate how the results of statistical studies are used to draw valid conclusions. (Evaluate)

Body of Knowledge V.D.5

One of the results of current technology is that we're drowning in data and thirsting for information. Statistical studies provide tools for squeezing information out of the data. The two principal types of statistical studies are usually called *descriptive (enumerative)* and *inferential (analytical)*.

Descriptive studies use techniques such as those in the previous sections to present data in an understandable format. A long column of numbers is rendered more meaningful when the mean, the median, the mode, and the standard deviation are known. A histogram, scatter diagram, or other graphic squeezes additional information from the data.

Inferential studies analyze data from a sample to infer properties of the population from which the sample was drawn. This is especially important in those situations in which population data are unavailable or infeasible to obtain.

Suppose, for instance, that one needs to know the mean diameter produced by a particular tooling setup on a lathe. The population would consist of all the parts that could be produced by that setup, which could take weeks or possibly years to acquire. Instead, a sample is selected, its properties studied, and a reasonable inference about the population mean diameter is obtained.

In this case, the mean of the sample would provide an approximation for the population mean. The amount of variation in the sample would provide information about how close that approximation is to the actual population mean.

Inferential studies produce statements involving probabilities, confidence levels, and so forth, and are covered in Chapter 21. Consequently, inferential studies can help us draw meaningful and practical conclusions when conducted properly. For example, was the study or experiment planned properly? Was a sufficient sample size taken in a randomized manner? Was the proper statistical methodology or test applied? When these and other related questions are answered affirmatively, then we can draw valid conclusions based on our statistical studies.

Chapter 18

Probability

A thousand probabilities do not make one fact.

—Italian proverb

This chapter will address the basic concepts of probability theory, as well as both the discrete and continuous statistical distributions commonly used in Lean Six Sigma. Numerous examples will be presented to help solidify the concepts presented.

BASIC CONCEPTS

> Describe and apply probability concepts, e.g., independence, mutually exclusive events, addition and multiplication rules, conditional probability, complementary probability, joint occurrence of events. (Apply)
>
> **Body of Knowledge V.E.1**

Probability is defined in two ways:

- *Classic definition*. If an event, say A, can occur in m ways out of a possible n equally likely ways, then

$$P(A) = \frac{m}{n}$$

- *Relative frequency definition*. The proportion that an event, say A, occurs in a series of repeated experiments as the number of those experiments approaches infinity. According to this definition, probability is a limiting value that may or may not be possible to quantify.

Almost all statistical activities are based on the rules of probability. This section lists and illustrates those rules, providing the groundwork for the study of probability distributions, confidence intervals, and hypothesis testing.

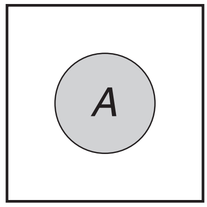

Figure 18.1 Venn diagram illustrating the probability of event A.

> **EXAMPLE 18.1**
>
> What is the probability that the card is a club (that is, P(♣))?
>
> There are 13 clubs in a deck of 52 cards; therefore, we have
>
> P(♣) = 13/52 = 0.25.

The probability that a particular event occurs is a number between 0 and 1, inclusive. For example, if a lot consisting of 100 parts has four defectives, we would say the probability of randomly drawing a defective is 0.04, or 4%. Symbolically, this is written P(defective) = 0.04.

The word "random" implies that each part has an equal chance of being drawn. If the lot had no defectives, the probability would be 0, or 0%. If the lot had 100 defectives, the probability would be 1, or 100%.

The probability that event A will occur can be illustrated with a *Venn diagram*, as shown in Figure 18.1. The area in the rectangle is called the universe and has a probability of 1. The shaded area inside the circle represents the probability that event A will occur and is denoted by P(A).

Complementary Rule of Probability

The complementary rule of probability states that the probability that event A will occur is 1 – P(event A does not occur). Stated symbolically,

$$P(A) = 1 - P(\text{not } A).$$

Some texts use other symbols for "not A," including $-A$, $\sim A$, A-prime, and $\overline{A}$. The Venn diagram illustrating the complementary rule is shown in Figure 18.2, where the probability that A will occur is shown as the area inside the circle, and P(not A) is the shaded area.

Addition Rule of Probability

We can see that P(♣ or ♠) = P(♣) + P(♠). Mathematically, we have

$$P(A \text{ or } B) = P(A) + P(B).$$

Alternately, using the appropriate nomenclature, we have

$$P(A \cup B) = P(A) + P(B).$$

where $\cup$ is the symbol for "union" or "addition."

Note that this formula applies only when A and B are independent of one another.

The Venn diagram for the addition rule of probability is shown in Figure 18.3. Note that events A and B cannot occur simultaneously. Two events that can't occur simultaneously are called *mutually exclusive* or *disjoint*. This is demonstrated by the fact that the two circles do not intersect.

This concept of probability is best illustrated through a series of examples.

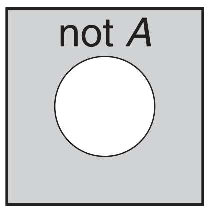

Figure 18.2 Venn diagram illustrating the complementary rule of probability.

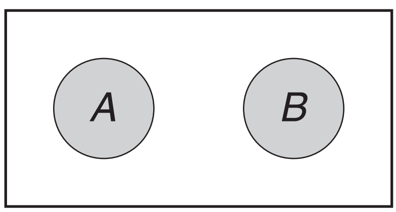

Figure 18.3 Venn diagram illustrating the addition rule of probability with independent events A and B.

EXAMPLE 18.2

Suppose a card is randomly selected from a standard 52-card deck. What is the probability that the card is either a club (♣) or a spade (♠)?

There are 26 cards that are either clubs or spades in the deck; therefore, we have

$$P(\clubsuit \text{ or } \spadesuit) = 13/52 + 13/52 = 26/52 = 0.5.$$

EXAMPLE 18.3

Now let's consider the probability of selecting either a king (K) or a club (♣). Using the addition rule, we have

$$\begin{aligned} P(K \text{ or } ♣) &= P(K) + P(♣) \\ &= 4/52 + 13/52 \\ &= 17/52 \\ &= 0.33. \end{aligned}$$

However, this is incorrect because there are only 16 cards that are either kings or clubs (that is, 13 clubs, which includes the king of clubs plus the *K*♦, the *K*♥, and the *K*♠). (Note: The drawing of a card that is both a king and a club is called the *joint occurrence of events*.) The reason the addition rule doesn't work here is that the two events (that is, drawing a king and drawing a club) can occur simultaneously. Hence, they are not independent. We'll denote the probability that A and B both occur as $P(A \cap B)$. This leads to a generalized version of the addition rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

where $\cap$ is the symbol for "intersection."

This generalized rule is always valid. For the previous example, we have

$$P(K \cap ♣) = 1/52 = 0.019$$

since there is only one card that is both a K and a ♣.

Now let's consider the probability of selecting either a king (K) or a club (♣).

$$\begin{aligned} P(K \cup ♣) &= P(K) = P(♣) - P(K \cap ♣) \\ &= 4/52 + 13/52 - 1/52 \\ &= 16/52 \\ &= 4/13 \\ &= 0.31. \end{aligned}$$

The Venn diagram illustrating the general version of the *addition rule* is shown in Figure 18.4. The circles in the figure are not disjoint since they clearly overlap. When circle A and circle B are added together, the intersection of the two circles will be counted twice. Therefore, we must subtract the intersection (that is, $P(A \cap B)$) to get the actual area of both circles.

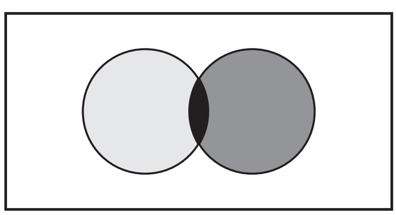

Figure 18.4 Venn diagram illustrating the general version of the addition rule of probability.

Contingency Tables

Contingency tables provide an effective way of arranging attribute data while allowing us to readily determine relevant probabilities. The following scenario best illustrates this statement: Suppose each part in a lot is one of four colors (red, yellow, green, or blue) and one of three sizes (small, medium, or large). A tool that displays these attributes is the contingency table shown in Table 18.1.

Each part belongs in exactly one column, and each part belongs in exactly one row, so each part belongs in exactly one of the 12 cells (that is, four colors times three sizes). When the columns and rows are totaled, the table becomes the one shown in Table 18.2.

Using the data in Table 18.2, we'll illustrate how to use a contingency table through the following examples.

Table 18.1 Example of a contingency table.

Size	**Red**	**Yellow**	**Green**	**Blue**
Small	16	21	14	19
Medium	12	11	19	15
Large	18	12	21	14

Table 18.2 Contingency table for Examples 18.4–18.11.

Size	**Red**	**Yellow**	**Green**	**Blue**	**Totals**
Small	16	21	14	19	70
Medium	12	11	19	15	57
Large	18	12	21	14	65
Totals	**46**	**44**	**54**	**48**	**192**

EXAMPLE 18.4

Find the probability that a part selected at random is red.

Since the size of the part is irrelevant in this example, we find that the total number of red parts from Table 18.2 is 46 and the total number of parts is 192. Therefore, we have

$$P(\text{red}) = \frac{46}{192} = 0.24$$

EXAMPLE 18.5

Find the probability that a part selected at random is small.

Since the color of the part is irrelevant in this example, we find that the total number of small parts from Table 18.2 is 70 and the total number of parts is 192. Therefore, we have

$$P(\text{small}) = \frac{70}{192} = 0.36$$

EXAMPLE 18.6

Find the probability that a part selected at random is both red *and* small.

From Table 18.2, we can see that the number of parts that are both red *and* small is 16. Therefore, we have

$$P(\text{red and small}) = \frac{16}{192} = 0.08$$

EXAMPLE 18.7

Find the probability that a part selected at random is red *or* small.

Using the addition rule of probability and the previous examples, we can formulate the following equation:

$$\begin{aligned} P(\text{red or small}) &= P(\text{red}) + P(\text{small}) - P(\text{red and small}) \\ &= \frac{46}{192} + \frac{70}{192} - \frac{16}{192} \\ &= \frac{100}{192} \\ &= 0.521 \end{aligned}$$

EXAMPLE 18.8

Find the probability that a part selected at random is red *or* yellow.

Using the addition rule of probability and Table 18.2, we can formulate the following equation:

Continued

Continued

$$\begin{aligned}P(\text{red or yellow}) &= P(\text{red}) + P(\text{yellow})\\ &= \frac{46}{192} + \frac{44}{192}\\ &= \frac{90}{192}\\ &= 0.47\end{aligned}$$

Note that parts cannot be both red and yellow simultaneously so that

$$P(\text{red and yellow}) = P(\text{red} \cap \text{yellow}) = 0$$

Conditional Probability

In the case of two events, *A* and *B*, conditional probability asks the question, "What is the probability of *B* occurring given that *A has* occurred?" This situation is formalized mathematically with the following definition of conditional probability:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}; P(A) \neq 0$$

Let's continue with the data from the previous example.

EXAMPLE 18.9

Now suppose that a part selected at random is known to be green. Find the probability that the selected part is large given that it is green.

Using the previous formula, we are looking for:

$$P(\text{large} \mid \text{green}) = \frac{P(\text{green} \cap \text{large})}{P(\text{green})}$$

We know from Table 18.2 that

$$P(\text{green}) = \frac{54}{192} = 0.28$$

and we also know that

$$\begin{aligned}P(\text{green and large}) &= P(\text{green} \cap \text{large})\\ &= \frac{21}{192}\\ &= 0.11\end{aligned}$$

Continued

Continued

Therefore, we can readily calculate

$$
\begin{aligned}
P(\text{large}|\text{green}) &= \frac{P(\text{green} \cap \text{large})}{P(\text{green})} \\
&= \frac{\frac{21}{192}}{\frac{54}{192}} \\
&= \frac{21}{54} \\
&= 0.39
\end{aligned}
$$

Alternately, we could have determined the probability directly from Table 18.2 by noting that of the 54 green parts, 21 of them are large. Thus, we would have computed the following directly:

$$P(\text{large}\,|\,\text{green}) = \frac{21}{54} = 0.39$$

EXAMPLE 18.10

Find the following probabilities using Table 18.2:

P(small|red); P(red|small); P(red|green)

Using the formulas for conditional probability and reading directly from Table 18.2:

$$P(\text{small}\,|\,\text{red}) = \frac{P(\text{small} \cap \text{red})}{P(\text{red})} = \frac{16/192}{46/192} = \frac{16}{46} = 0.35$$

$$P(\text{red}\,|\,\text{small}) = \frac{P(\text{red} \cap \text{small})}{P(\text{small})} = \frac{16/192}{70/192} = \frac{16}{70} = 0.23$$

$$P(\text{red}\,|\,\text{green}) = \frac{P(\text{red} \cap \text{green})}{P(\text{green})} = \frac{0/192}{54/192} = \frac{0}{54} = 0$$

Independent and Dependent Events

Two events A and B are said to be *independent* if and only if the occurrence of B has no effect on the occurrence of A (or vice versa). This can be stated mathematically as

EXAMPLE 18.11

Recall from Example 18.5 that

$$P(\text{small}) = \frac{70}{192} = 0.36$$

and from Example 18.10 that

$$P(\text{small}|\text{red}) = \frac{P(\text{small} \cap \text{red})}{P(\text{red})} = \frac{16/192}{46/192} = \frac{16}{46} = 0.35$$

Since

$$P(\text{small}) \neq P(\text{small}|\text{red})$$

we can conclude that the two are dependent events.

$$P(A \mid B) = P(A)$$

$$P(B \mid A) = P(B)$$

Similarly, events A and B are dependent if and only if this formula does not hold. In this case, we have:

$$P(A \mid B) \neq P(A)$$

$$P(B \mid A) \neq P(B)$$

Consider Example 18.11 using Table 18.2 along with some of the problems we solved previously.

Mutually Exclusive Events

Two events A and B are said to be mutually exclusive if both events cannot occur at the same time. Remember our previous example for which we sought P(red $\cap$ green) only to recognize from our contingency table (Table 18.2) that a part cannot be both red and green at the same time. Hence, event A (that a part is red) and event B (that a part is green) are mutually exclusive.

Multiplication Rule of Probabilities

Recall the previous formula for conditional probability. If we rearrange this formula, we will obtain the multiplication rule of probabilities:

$$P(A \cap B) = P(A) \times P(B \mid A)$$

$$P(A \cap B) = P(B) \times P(A \mid B)$$

Table 18.3 summarizes the key probability equations and concepts described in this section.

Table 18.3 Summary of the rules of probability.

<table>
<tr><td colspan="3">Axioms of probability
$0 \le \mathrm{P(Event)} \le 1$
$\sum_{i=1}^{n} \mathrm{P}(E_i) = 1$</td></tr>
<tr><td colspan="3">Complementary rule
$\mathrm{P}(A) = 1 - \mathrm{P}(A')$</td></tr>
<tr><td colspan="3" align="center">For two events A and B</td></tr>
<tr><td colspan="3">Conditional probability
$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$ for $\mathrm{P}(B) > 0$
$\mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)}$ for $\mathrm{P}(A) > 0$</td></tr>
<tr><td></td><td>Independent</td><td>Dependent</td></tr>
<tr><td>Events</td><td>$\mathrm{P}(A \mid B) = \mathrm{P}(A)$
$\mathrm{P}(B \mid A) = \mathrm{P}(B)$</td><td>$\mathrm{P}(A \mid B) \ne \mathrm{P}(A)$
$\mathrm{P}(B \mid A) \ne \mathrm{P}(B)$</td></tr>
<tr><td>Addition rule</td><td>$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B)$</td><td>$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$</td></tr>
<tr><td>Multiplication rule</td><td>$\mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$</td><td>$\mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B \mid A)$
$= \mathrm{P}(B) \times \mathrm{P}(A \mid B)$</td></tr>
<tr><td></td><td>Mutually exclusive</td><td>Not mutually exclusive</td></tr>
<tr><td></td><td>$\mathrm{P}(A \cap B) = 0$</td><td>$\mathrm{P}(A \cap B) \ne 0$</td></tr>
</table>

EXAMPLE 18.12

Let's consider a scenario whereby a box holds 129 parts, of which six are defective. We'll use this situation and study several examples to understand how multiplication rules work.

A part is randomly drawn from the box and placed in a fixture. A second part is then drawn from the box. What is the probability that the second part is defective?

The probability can't be determined directly unless the outcome of the first draw is known. In other words, probabilities associated with successive draws depend on the outcome of previous draws.

Let

D_1 denote the event that the first part is defective

D_2 denote the event that the second part is defective

G_1 denote the event that the first part is not defective (that is, good)

G_2 denote the event that the second part is not defective (that is, good)

Continued

Continued

There are two mutually exclusive events that can result in a defective part for the second draw: (1) good on first draw and defective on second or (2) defective on first draw and defective on second.

Using the notation we just defined, we can write these two events as ($G_1 \cap D_2$) and ($D_1 \cap D_2$). Note that we are looking for the probability that both of these events occur (that is, $P(G_1 \cap D_2) + P(D_1 \cap D_2)$). Now we can readily write the probability equations that will help us solve the problem:

$$P(G_1 \cap D_2) = P(G_1) \times P(D_2 \mid G_1)$$
$$P(D_1 \cap D_2) = P(D_1) \times P(D_2 \mid D_1)$$
$$P(D_2) = P(G_1 \cap D_2) + P(D_1 \cap D_2)$$

Filling in the numbers, we have

$$\left(\frac{123}{129}\right)\left(\frac{6}{128}\right) = 0.045$$
$$\left(\frac{6}{129}\right)\left(\frac{5}{128}\right) = 0.002$$
$$0.045 + 0.002 = 0.047$$

Therefore, the probability that the second part is defective is 0.047.

EXAMPLE 18.13

Given the same scenario as in Example 18.12, what is the probability of selecting only one defective?

There are two mutually exclusive events that can result in drawing only one defective part: (1) good on first draw and defective on second or (2) defective on first draw and good on second.

Using our notation, we can write these two events as

$$P(G_1 \cap D_2) = P(G_1)P(D_2 \mid G_1)$$
$$P(D_1 \cap G_2) = P(D_1)P(G_2 \mid D_1)$$
$$= P(G_1 \cap D_2) + P(D_1 \cap G_2)$$

Filling in the numbers, we have

$$\left(\frac{123}{129}\right)\left(\frac{6}{128}\right) = 0.045$$
$$\left(\frac{6}{129}\right)\left(\frac{123}{128}\right) = 0.045$$
$$0.045 + 0.045 = 0.090$$

Therefore, the probability of selecting only one defective is 0.090.

Permutations and Combinations

Permutations and *combinations* are basic concepts that deal with ordered and unordered arrangement and selection of things.

Permutations and combinations are not part of the Black Belt Body of Knowledge and hence will not be part of the certification examination. However, since they are part of the Green Belt Body of Knowledge, they are being introduced here so that Black Belts can become familiar with the topic.

The Factorial Nomenclature. The factorial notation is used extensively in calculating permutations and combinations (see Example 18.14). For example, we often need to compute $n!$. $n!$ is given by

$$
\begin{aligned}
n! &= \prod_{k=1}^{n} k \\
&= (n)(n-1)!(n-2)!\ldots(n-(n-2))(n-(n-1)) = (n)(n-1)!(n-2)!\ldots(2)(1) \\
&= \begin{cases} 1 & n = 0 \\ n \times (n-1)! & n > 0 \end{cases}
\end{aligned}
$$

Permutations. A *permutation* is an arrangement of a set of objects with regard to the order of the arrangement (see Example 18.15).

Consider the letters A and B. AB and BA represent two permutations because order is considered important. Thus, AB and BA are distinct.

The various notations for a permutation of n objects taken r at a time is given by

$$P(n,r) = {}_nP_r = P_r^n = \frac{n!}{(n-r)!}$$

Combinations. By contrast to a permutation, a *combination* addresses the *selection* of objects without regard to the order in which they are selected (see Example 18.16). Recall that permutations, in contrast, focus on the arrangement of objects with regard to the order in which they are arranged.

EXAMPLE 18.14

Compute 5!.

From the formulas given above, we know that

$$
\begin{aligned}
5! &= 5 \times 4 \times 3 \times 2 \times 1 \\
&= 120
\end{aligned}
$$

EXAMPLE 18.15

How many ways can three numbers (1, 2, 3) be arranged when order is considered important?

From reading the problem statement, we know that this calls for the use of a permutation. Therefore, arranging the three numbers three at a time we have:

1, 2, 3

1, 3, 2

2, 3, 1

2, 1, 3

3, 1, 2

3, 2, 1

Using the formula for a permutation, we obtain

$$\begin{aligned} P(3,3) &= \frac{3!}{(3-3)!} \\ &= \frac{3!}{0!} \\ &= \frac{6}{1} \\ &= 6 \end{aligned}$$

Alternately, there are three ways to choose the first number, two ways to choose the second number, and one way to choose the third number. Thus, the total number of ways

$$\begin{aligned} &= 3 \times 2 \times 1 \\ &= 6 \end{aligned}$$

EXAMPLE 18.16

How many ways can a gold medal, silver medal, and bronze medal be given to eight players?

From the problem statement, we know that this calls for the use of a permutation because order is considered important. Therefore, we must determine the number of ways three distinct medals can be arranged among eight players. Using the formula for permutations, we have

$$\begin{aligned} P(8,3) &= \frac{8!}{(8-3)!} \\ &= \frac{8 \times 7 \times 6 \times 5!}{5!} \\ &= 336 \end{aligned}$$

Continued

Continued

Alternately, we could approach the problem by noting that eight players have a chance to win a gold medal, seven players can win a silver medal, and six players can win a bronze medal. Thus, we have

$$= 8 \times 7 \times 6$$
$$= 336$$

Consider the letters *A* and *B*. *AB* and *BA* represent one combination because order is *not* considered important. Hence *AB* and *BA* are the same.

The various notations for a combination of *n* objects taken *r* at a time is given by

$$C(n,r) = {}_nC_r = C_r^n = \frac{{}_nP_r}{r!} = \frac{n!}{r!(n-r)!}$$

Notice that a combination is also a permutation divided by the number, *r*, taken at a time. In this context, *r* is the number of redundancies. Therefore, we could write

$$C(n,r) = \frac{\text{Number of permutations}}{\text{Number of redundancies}}$$

EXAMPLE 18.17

How many ways can three numbers (1, 2, 3) be arranged when order is *not* considered important?

From reading the problem statement, we know that this calls for the use of a combination. Therefore, we will arrange three numbers three at a time.

1, 2, 3

1, 3, 2

2, 3, 1

2, 1, 3

3, 1, 2

3, 2, 1

However, since order is not important, the above six ways are indistinguishable from one another and hence are all the same. Thus, we use the combination to calculate the number of ways:

Continued

$$
\begin{aligned}
C(3,3) &= \frac{3!}{3!(3-3)!} \\
&= \frac{3!}{3!0!} \\
&= \frac{6}{6\times 1} \\
&= \frac{6}{6} \\
&= 1
\end{aligned}
$$

EXAMPLE 18.18

How many ways can three tin cans be given to eight people?

Since the tin cans are indistinguishable from one another, we know this is a combination problem. Thus, we have

$$
\begin{aligned}
C(8,3) &= \frac{8!}{3!(8-3)!} \\
&= \frac{8!}{3!5!} \\
&= \frac{8\times 7\times 6\times 5!}{3!5!} \\
&= \frac{8\times 7\times 6}{3!} \\
&= \frac{8\times 7\times 6}{6} \\
&= 56
\end{aligned}
$$

Alternately, we could have looked at it as a permutation since there are eight ways we can distribute the first can, seven ways we can distribute the second can, and six ways we can distribute the third can. If we label the three cans as 1, 2, and 3 as in Example 18.17, there are six ways we can arrange the three cans, but these ways are redundant and hence all the same. Thus, there is really only one way. Recall that

$$
\begin{aligned}
C(n,r) &= \frac{{}_nP_r}{r!} \\
&= \frac{\text{Number of permutations}}{\text{Number of redundancies}} \\
&= \frac{336}{6} \\
&= 56
\end{aligned}
$$

DISTRIBUTIONS

> Describe, interpret, and use various distributions, e.g., normal, Poisson, binomial, chi square, Student's t, F, hypergeometric, bivariate, exponential, lognormal, Weibull. (Evaluate)
>
> **Body of Knowledge V.E.2**

This section will describe and interpret key and commonly used distributions in Lean Six Sigma. Pertinent information about each distribution is summarized in Table 18.4.

Normal Distribution

If X is a normal random variable with expected value μ (that is, E (X) = μ) and finite variance σ^2, then the probability density function for X is defined as

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}; -\infty \le x \le \infty$$

and the random variable Z is defined as

$$Z = \frac{X-\mu}{\sigma}.$$

When E (Z) = 0 and the Var (Z) = 1, then Z is distributed as a standard normal variable. The probability density function for the standard normal distribution is given as

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}; -\infty \le x \le \infty.$$

We will use the concept of the standard normal distribution extensively in later sections.

It is important to remember that the area between any two points under the standard normal curve can be determined from statistical tables such as those in Appendices 12 and 13. This can best be understood with a few examples.

Table 18.4 Summary of formulas, means, and variances of commonly used distributions.

Distribution	Formula	Mean	Variance
Normal	$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ for $-\infty\le x\le\infty$	μ	σ^2
Poisson	$f(x)=\frac{e^{-\lambda}\lambda^x}{x!}$ for $x=0,1,2...$	λ	λ
Binomial	$f(x)=\frac{n!}{x!(n-x)!}p^x(1-p)^{n-x}$ for $x=0,1,\ldots n$	np	$np(1-p)$
Chi-square	$f(x)=\frac{e^{\frac{-x}{2}}x^{\left(\frac{\nu}{2}\right)-1}}{2^{\frac{\nu}{2}}\Gamma\left(\frac{\nu}{2}\right)}$ for $x\ge 0, \nu=$ degrees of freedom	ν	2ν
t	$f(x)=\frac{1}{\sqrt{\pi\nu}}\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-(\nu+1)/2}$ for $-\infty\le x\le\infty$	0; for $\nu>1$	$\frac{\nu}{\nu-2}$; for $\nu>2$
F	$f(x)=\frac{\Gamma\left(\frac{\nu_1+\nu_2}{2}\right)\left(\frac{\nu_1}{\nu_2}\right)^{\nu_1/2}x^{(\nu_1/2)-1}}{\Gamma\left(\frac{\nu_1}{2}\right)\Gamma\left(\frac{\nu_2}{2}\right)\left[\left(\frac{\nu_1}{\nu_2}\right)x+1\right]^{\frac{(\nu_1+\nu_2)}{2}}}$ for $0\le x\le\infty, \nu_1>0, \nu_2>0$	$\frac{\nu_2}{\nu_2-2}$ for $\nu_2>2$	$\frac{2\nu_2^2(\nu_1+\nu_2-2)}{\nu_1(\nu_2-2)^2(\nu_2-4)}$ for $\nu_2>4$
Hypergeometric	$f(x)=\frac{\binom{S}{x}\binom{N-S}{n-x}}{\binom{N}{n}}$ for $x=\max\{0,n+S-N\}$ to $\min\{S,n\}$	np	$np(1-p)\left(\frac{N-n}{N-1}\right)$

Continued

Table 18.4 *Continued.*

Distribution	Formula	Mean	Variance
Bivariate normal	$f_{XY}(x,y)=\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}e^{\left\{\frac{-1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2}-\frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}+\frac{(y-\mu_Y)^2}{\sigma_Y^2}\right]\right\}}$ for $\left\{\begin{array}{l}-\infty<x<\infty\\ -\infty<y<\infty\\ -\infty<\mu_X<\infty\\ -\infty<\mu_Y<\infty\\ \sigma_X>0\\ \sigma_Y>0\\ -1\le\rho\le1\end{array}\right\}$	Conditional: $\mu_{x\vert y=y_0}=\mu_x+\rho\sigma_x\frac{(y-\mu_y)}{\sigma_y}$ $\mu_{y\vert x=x_0}=\mu_y+\rho\sigma_y\frac{(x-\mu_x)}{\sigma_x}$	Conditional: $\sigma_{x\vert y=y_0}=\sigma_x\sqrt{1-\rho^2}$ $\sigma_{y\vert x=x_0}=\sigma_y\sqrt{1-\rho^2}$
Exponential	$f(x)=\lambda e^{-\lambda x}$ for $0\le x\le\infty;\lambda>0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Lognormal	$f(x)=\frac{1}{x\sigma\sqrt{2\pi}}e^{\left[\frac{-2}{2\sigma^2}(\ln(x)-\mu)^2\right]}$ for $x>0$	$e^{\mu+\frac{\sigma^2}{2}}$	$e^{2\mu+\sigma^2}\left(e^{\sigma^2}-1\right)$
Weibull	$f(x)=\frac{\beta}{\alpha}\left(\frac{x-\gamma}{\alpha}\right)^{\beta-1}e^{-\left(\frac{x-\gamma}{\alpha}\right)^{\beta}}$ for $x>0,\alpha>0,\beta>0,\gamma\ge0$	$\gamma+\alpha\Gamma\left(1+\frac{1}{\beta}\right)$	$\alpha^2\left[\Gamma\left(1+\frac{2}{\beta}\right)-\left[\Gamma\left(1+\frac{1}{\beta}\right)\right]^2\right]$

EXAMPLE 18.19

Find the area under the standard normal curve between +1.22 standard deviations and +2.06 standard deviations.

The easiest way to do this is to find the cumulative area to the left of 2.06 and subtract the cumulative area to the left of 1.22. To do this, we can use Appendix 13, which is the cumulative standard normal distribution.

Using this table, we obtain the following:

Area to the left of 2.06 = 0.9803

Area to the left of 1.22 = 0.8888

Area between 1.22 and 2.06 = 0.0915

Therefore, the area under the curve between the two values is 0.0915, which approximates the value shown in Figure 18.5.

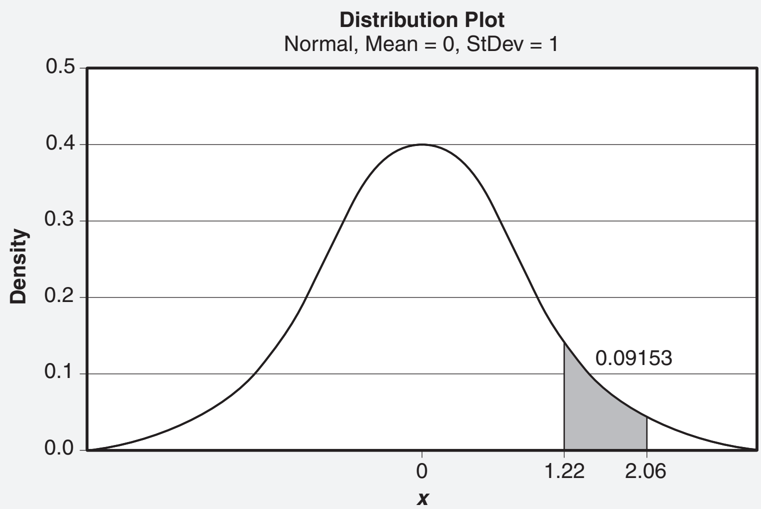

Figure 18.5 Standard normal distribution for Example 18.19.

For any data set that is normally distributed and for which the mean and the standard deviation are known or can be estimated, this technique can be used to determine the probability of falling between any two specified standard normal values or z-scores using

$$z = \frac{x - \mu}{\sigma}.$$

EXAMPLE 18.20

A product fill operation produces net weights that are normally distributed. Assume a mean of 8.06 ounces and standard deviation of 0.37 ounces. Estimate the percentage of the containers that have a net weight between 7.9 and 8.1 ounces.

To estimate the number of containers, we must first calculate the *z*-scores for 7.9 and 8.1 ounces. Filling in the numbers, we have

$$z = \frac{7.9 - 8.06}{0.37} = -0.4324$$
$$z = \frac{8.1 - 8.06}{0.37} = 0.1081.$$

The easiest way to do this is to find the cumulative area to the left of 0.1081 and subtract the cumulative area to the left of –0.4324. To do this, we can use Appendix 13, which is the cumulative standard normal distribution.

However, the tables do not provide us with the exact *z* indices for which to enter the table so we'll need to perform linear interpolation. Further, one of our *z* numbers is negative and the table does not provide negative numbers.

Fortunately, due to the symmetry of the normal distribution, we can state that

$$F(-z) = 1 - F(z)$$

where $F(z)$ is the cumulative standard normal distribution.

Using this relationship, we can enter the table at the positive *z* value to find the associated probability and subtract that probability from 1.

Pulling the above concepts all together we have the following:

Area to the left of .1081 = 0.5430
Area to the left of –0.4324 = 1 – Area to the left of 0.4324
= 1 – 0.6673
= 0.3327.

Subtracting 0.5430 and 0.3327 gives us the following area:

Area between –0.4324 and 0.1081 = 0.2103

Therefore, 21.03% of the containers have a net weight between 7.9 and 8.1 ounces. Put another way, the probability that a randomly selected container will have a net weight between 7.9 and 8.1 is approximately 0.2103. This value agrees with Figure 18.6.

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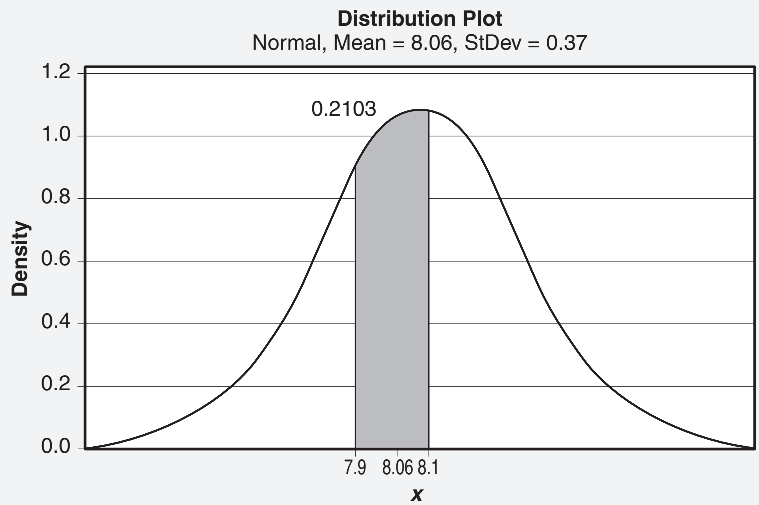

Figure 18.6 Standard normal distribution for Example 18.20.

Poisson Distribution

The Poisson distribution is important to the Black Belt because it is used to model the number of defects per unit of time (or distance) and is the basis for the control limit formulas for the *c* and *u* control charts. The Poisson has also been used to analyze such diverse phenomena as the number of phone calls received per day and the number of alpha particles emitted per minute. The formula for the probability mass function is

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!}; \lambda > 0; x = 0, 1, 2, \ldots$$

Lambda (λ) is the *shape parameter* and represents the average number of events in a given time interval. The mean and variance of the Poisson distribution are both λ.

EXAMPLE 18.21

The number of defects per shift has a Poisson distribution with $\lambda = 4.2$. A histogram plot of this distribution is shown in Figure 18.7. Find the probability that the second shift produces fewer than two defects.

Continued

Continued

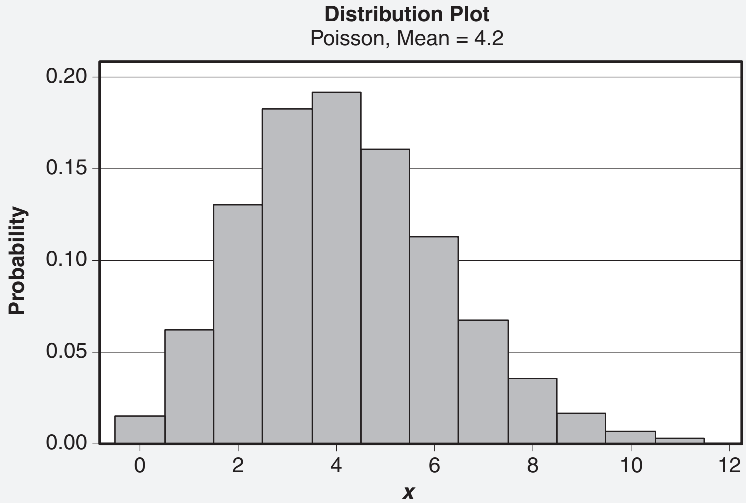

Figure 18.7 Poisson distribution with mean λ = 4.2 for Example 18.21.

From the problem statement, we seek

$$P(x < 2) = P(x = 0) + P(x = 1).$$

Filling in the numbers and using the Poisson table in Appendix 10, we have:

$$P(X = 0) = \frac{e^{4.2}4.2^0}{0!} = 0.0150$$

$$P(X = 1) = \frac{e^{4.2}4.2^1}{1!} = 0.0630.$$

Adding these two equations together, we get

$$P(X < 2) = 0.0780$$

Therefore, the probability that the second shift produces fewer than two defects is 0.0780.

Binomial Distribution

The binomial distribution represents a sequence of n Bernoulli trials.

If X is a random variable that equals the number of trials that result in a success, then X is binomially distributed with parameters p and n. The probability mass function for the binomial distribution is

$$P(X = x) = \frac{n!}{x!(n-x)!}p^x(1-p)^{n-x}; x = 0,1,2,\ldots; 0 < p < 1$$

The symbology relevant to the binomial distribution follows:

n = Sample size or number of trials

x = Number of successes in the sample

p = Probability of a success for each trial

Key assumptions regarding the binomial include the following:

- Each trial is independent
- Each trial can result in only one outcome: success or failure
- $0 < p < 1$ where p is constant for each trial

The term "success" can be defined in whatever way is deemed appropriate. For example, a "success" might mean that a part sampled is nonconforming (that is, we have succeeded in finding a defect).

Let's look at the formula for a moment. The term, $p^x(1 - p)^{n-x}$, from the binomial formula represents the probability of obtaining x successes in n trials while the term

$$\frac{n!}{x!(n-x)!}$$

represents the number of ways in which the x success in n trials can occur. With this brief background, we'll now look at an example of how the binomial distribution can be used.

EXAMPLE 18.22

A sample of size six is randomly selected from a batch with 14.28% nonconforming units. A plot of this distribution is shown in Figure 18.8. Find the probability that the sample has exactly two nonconforming units.

From the problem statement, we know that

$n = 6$

$x = 2$

$p = 0.1428$

Filling in the numbers, we have

$$\begin{aligned} P(X=2) &= \frac{6!}{2!(6-2)!}(0.1428)^2(1-0.1428)^{6-2} \\ &= \frac{6\times5\times4!}{2!4!}(0.1428)^2(0.8572)^4 \\ &= (15)(0.0204)(0.5399) \\ &= 0.1651 \end{aligned}$$

Continued

Continued

Thus, the probability that the sample contains exactly two nonconforming units is 0.1651, which is equal to the value shown in Figure 18.8.

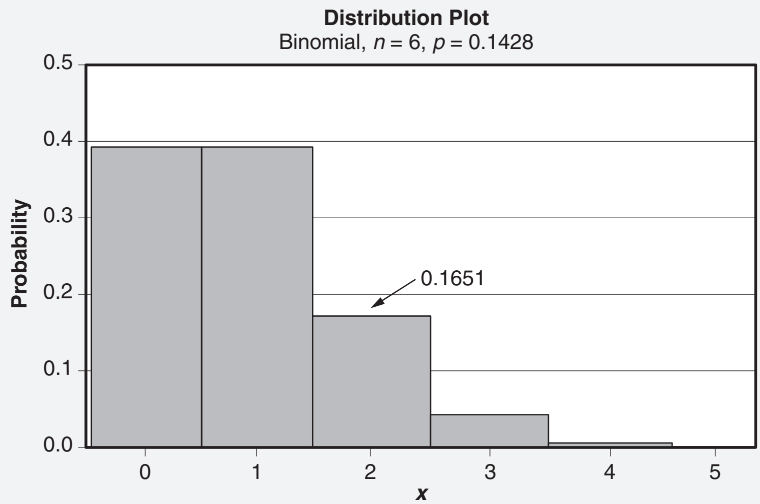

Figure 18.8 Binomial distribution with $n = 6$ and $p = 0.1428$ for Example 18.22.

Chi Square Distribution

If we obtain a random sample $x_1, x_2, \ldots, x_n$ of size n from a population that is normally distributed with mean μ and finite variance σ^2, the random variable

$$X^2 = \frac{(n-1)s^2}{\sigma^2}$$

is distributed as a chi-square (χ^2) distribution with $n - 1$ degrees of freedom where s^2 is the sample variance. The formula for the (χ^2) will be useful later when we discuss hypothesis testing and confidence intervals.

The probability density function for the chi-square is given by

$$f(x) = \frac{e^{-x/2} x^{(v/2)-1}}{2^{v/2}\Gamma\left(\frac{v}{2}\right)} \text{ for } x \geq 0$$

where ν is the number of the degrees of freedom.

Figure 18.9 illustrates the effects of various degrees of freedom on the chi-square distribution.

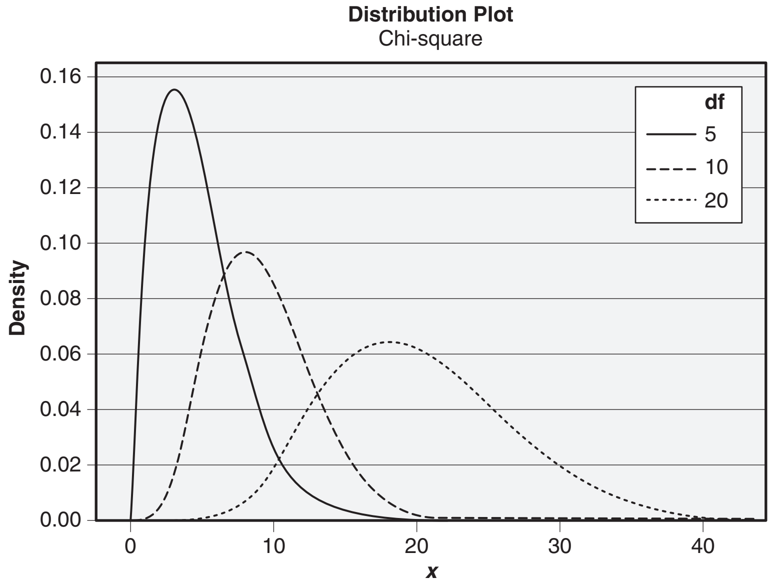

Figure 18.9 Example of a chi-square distribution with various degrees of freedom.

EXAMPLE 18.23

When using the $\bar{X}$ and *R* chart to monitor a process, we rely on the *R*-chart to track sample variation. This example illustrates how the chi-square distribution can be used, especially when the data are sparse.

A process with a normal distribution has been stable with $\sigma = 0.006$. Three samples from the current run have the following values:

Sample 1: 4.293, 4.275, 4.282

Sample 2: 4.305, 4.298, 4.296, 4.298

Sample 3: 4.301, 4.296

Do these data indicate that σ exceeded its historic value of 0.006 for any of the samples? Assume that we want to answer the question at a significance level of 0.05. The significance level is discussed in detail in the hypothesis testing section.

Using the data from this example, we can calculate sample statistics, and from Appendix 15, we can determine the test statistics values. These are summarized in Table 18.5. Upon reviewing this table, we find that none of the calculated chi-square values exceed the table values; we can conclude that the data do not exceed the historic value of σ.

Continued

Continued

Table 18.5 Chi-square table for Example 18.23.

Sample number	n	s	Chi-square statistic	Degrees of freedom	Chi-square table value
1	3	0.00907	5.474	2	5.991
2	4	0.00395	1.299	3	7.815
3	2	0.00354	0.347	1	3.842

t-Distribution (Also Known as Student's *t*-Distribution)

If we obtain a random sample $X_1, X_2, \ldots, X_n$ of size n from a population that is normally distributed with unknown mean μ and finite variance σ^2, then T is distributed as a t-distribution with $n - 1$ degrees of freedom:

$$T = \frac{\overline{x} - \mu}{\frac{s}{\sqrt{n}}}.$$

The probability density function for the t-distribution is given by

$$f(x) = \frac{1}{\sqrt{\pi\nu}} \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2} \quad \text{for } -\infty \leq x \leq \infty$$

where ν is the degrees of freedom.

Recall that when we reviewed the normal distribution previously, we assumed that σ was a known value. In general, this is unlikely since σ is a parameter of a population. Of course, when it is unknown, we must resort to estimating it. This is where the t-distribution becomes useful. In practical applications of this formula, μ is often assumed to be a known value, say μ_0, and σ is estimated by s, determined from the sample.

Typically, the t-distribution is used when $n < 30$, but more importantly when σ is unknown. The t-distribution is flatter than the normal distribution and has longer tails. As the degrees of freedom increase, the t-distribution approaches the normal distribution. Figure 18.10 illustrates the effects of various degrees of freedom on the t-distribution.

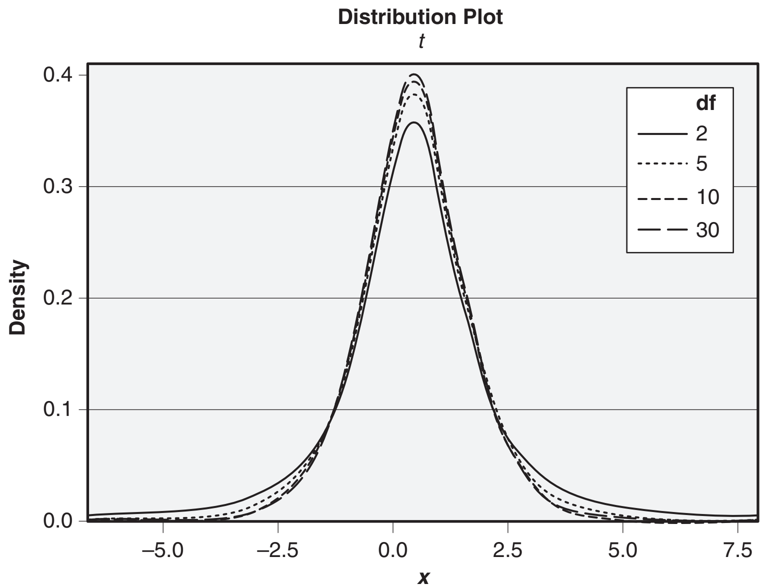

Figure 18.10 Example of a t-distribution with various degrees of freedom.

EXAMPLE 18.24

One of the length specifications for a masonry product requires that the mean length be at most 7.0 inches. The receiving inspection department's policy is to reject a lot if the sample at a 95% confidence level indicates that this length specification is violated. The following sample was taken:

7.2, 6.2, 7.5, 7.7, 7.8, 6.2

The mean of the above data is $\bar{X} = 7.1$ and the standard deviation is s = 0.73. If we assume a normal distribution, we can use the t-distribution table in Appendix 14 since n is less than 30. We enter Appendix 14 at row $\nu = n - 1 = 6 - 1 = 5$ for $\alpha = 0.05$ and find our critical t-value to be 2.015.

$$t = \frac{7.1 - 7.0}{\frac{0.73}{\sqrt{6}}} = 0.34$$

Since the calculated t-value doesn't exceed the table value, the batch cannot be rejected.

F-Distribution

If X and Y are two random variables distributed as χ^2 with ν_1 and ν_2 degrees of freedom, respectively, then the random variable

$$F = \frac{\dfrac{X}{v_1}}{\dfrac{Y}{v_2}}$$

is distributed as an F-distribution with degrees of freedom $\nu_1 = n_1 - 1$ in the numerator and degrees of freedom $\nu_2 = n_2 - 1$ in the denominator.

The probability density function for the F-distribution is given by

$$f(x) = \frac{\Gamma\left(\frac{\nu_1+\nu_2}{2}\right)\left(\frac{\nu_1}{\nu_2}\right)^{\nu_1/2} x^{(\nu_1/2)-1}}{\Gamma\left(\frac{\nu_1}{2}\right)\Gamma\left(\frac{\nu_2}{2}\right)\left[\left(\frac{\nu_1}{\nu_2}\right)x+1\right]^{\frac{(\nu_1+\nu_2)}{2}}} \text{ for } 0 < x < \infty, \nu_1 > 0, \nu_2 > 0.$$

This formula for the F-distribution will be useful later when we discuss hypothesis testing and confidence intervals. Figure 18.11 illustrates the effects of varying the degrees of freedom in both the numerator and the denominator.

A beneficial key property of the F-distribution follows:

$$F_{1-\alpha,\nu_1,\nu_2} = \frac{1}{F_{\alpha,\nu_2,\nu_1}}.$$

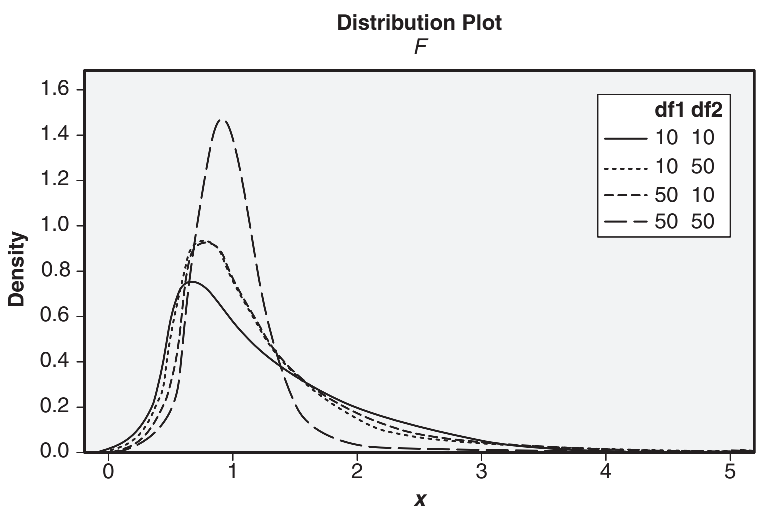

Figure 18.11 Example of an F-distribution with various degrees of freedom.

EXAMPLE 18.25

A quality team is deciding between two dynamometers for purchase. One question is whether the two models have the same standard deviation. Each dynamometer manufacturer will provide the team with twelve identical samples of each of its machines for destructive testing. This will enable the team to calculate

$$F = \frac{s_1^2}{s_2^2}.$$

If the standard deviations are the same, the F-value should be approximately 1. The team decides to assume the standard deviations are different unless they are in the 95% range of 1. The values of F from the $F_{0.025}$ and $F_{0.975}$ tables will provide this level of confidence. Appendices 17 and 22 are used to locate the critical F-values, 0.29 and 3.47, at 11 and 11 degrees of freedom. If the calculated F-value is less than 0.29 or larger than 3.47, the quality team will be convinced that the two dynamometers have different standard deviations. The F-distribution for this example is depicted in Figure 18.12.

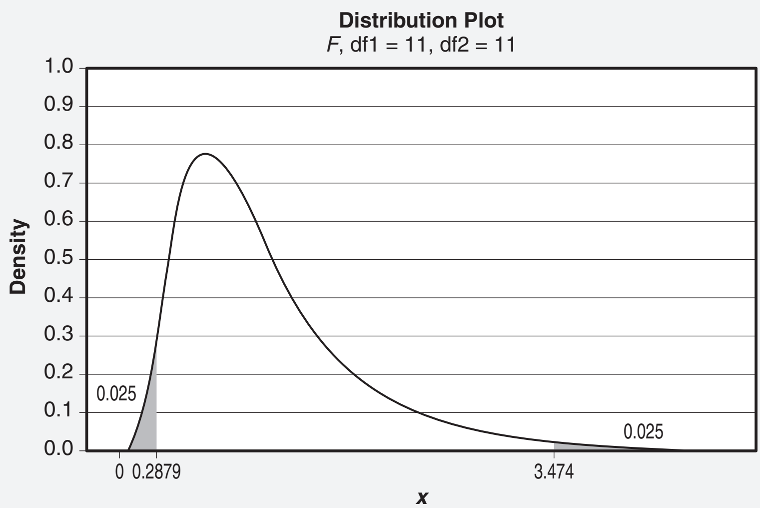

Figure 18.12 Probability density function for the F-distribution for Example 18.25.

If an α value table (that is, upper percentage points) is available, this reciprocal relationship is useful for determining the (1 – α) values (that is, lower percentage points). Of course, the reverse is true as well.

Hypergeometric Distribution

The hypergeometric distribution is useful when sampling from a finite population without replacement (that is, a sample is drawn from a population, inspected, and set aside, and another sample is drawn from the remaining population). Whereas in binomial sampling the probability of incurring a success (for example, defect) remains constant from sample to sample, the probability of incurring a success in hypergeometric sampling changes from sample to sample (that is, sampling without replacement).

If X represents the number of successes in a sample, then X is distributed as a hypergeometric random variable. The probability density function for the hypergeometric distribution is given as

$$f(x)=\frac{\binom{S}{x}\binom{N-S}{n-x}}{\binom{N}{n}} \text{ for } x=\max\{0,n+S-N\} \text{ to } \min\{S,n\}.$$

The elements in this formula are defined as follows:

N = The finite population size

S = The number of objects in the total population classified as successes

$N - S$ = The number of objects in the total population classified as failures

n = The sample size

x = The number of objects in the sample classified as successes
$n - x$ = The number of objects in the sample classified as failures

EXAMPLE 18.26

A batch of 100 parts, of which 25 are defective, has been received for inspection. Assuming sampling without replacement, what is the probability of obtaining three defects in a sample of size 10? Figure 18.13 illustrates the distribution of this example.

Before we go too far, let's first define a "success" as the finding of a defect. Then from the problem statement, we can readily see that

$N = 100$

$S = 25$

$N - S = 75$

$n = 10$

$x = 3$

$n - x = 7$

Continued

Continued

Filling in the numbers, we have

$$P(X=3) = \frac{\binom{25}{3}\binom{75}{7}}{\binom{100}{10}}$$
$$= \frac{(2300)(1.98\times10^9)}{1.73\times10^{13}}$$
$$= 0.264.$$

Therefore, the probability of obtaining exactly three defects in a sample of 10 is 0.264.

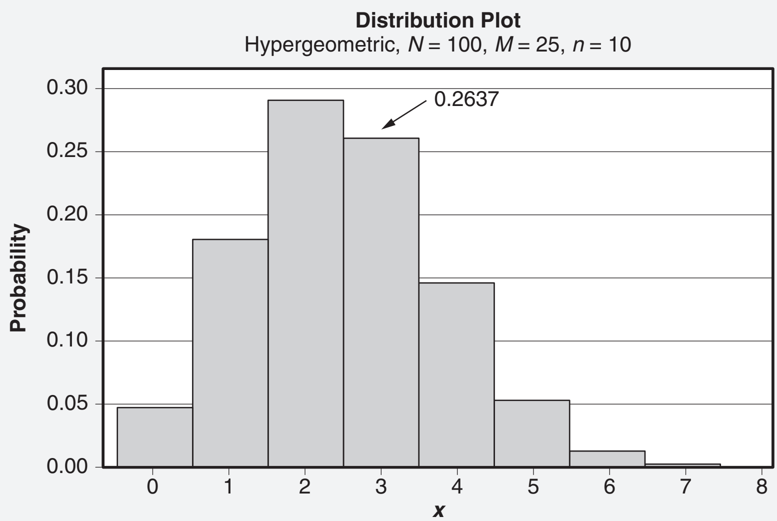

Figure 18.13 Hypergeometric distribution for Example 18.26.

Bivariate Normal Distribution

If there are two random variables of interest that are dependent or vary jointly (for example, length and weight), each of which is normally distributed, the joint probability density function of the variables is distributed as a bivariate normal distribution. The bivariate normal distribution is

$$f_{XY}(x,y;\mu_X,\mu_Y,\sigma_X,\sigma_Y,\rho)=\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}e^{\left\{\frac{-1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2}-\frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}+\frac{(y-\mu_Y)^2}{\sigma_Y^2}\right]\right\}}$$

$$\text{for}\begin{Bmatrix}-\infty<x<\infty\\ -\infty<y<\infty\\ -\infty<\mu_X<\infty\\ -\infty<\mu_Y<\infty\\ \sigma_X>0\\ \sigma_Y>0\\ -1\le\rho\le 1\end{Bmatrix}.$$

and ρ is the coefficient of correlation. Figure 18.14 provides an illustration of a typical bivariate normal distribution.

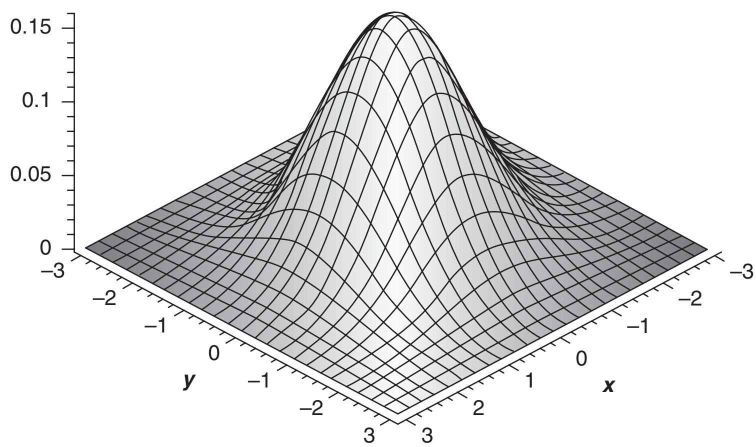

Figure 18.14 Example of a bivariate normal distribution.

EXAMPLE 18.27

The dimensions for a batch of part A are normally distributed. Likewise for part B. Using capability analysis, it is possible to calculate the percentage of the population that meets the specification for dimension A. Similarly, it is possible to calculate the percentage of the population that meets the specification for dimension B. A more important question is, "What percentage of the population are good parts (that is, meet the specifications for both dimension A and dimension B)." The distribution in this case is a *bivariate normal distribution*. Unfortunately, its analysis requires the use of specialized computer software.

Exponential Distribution

Recall that the Poisson distribution was used to model the number of defects per unit of time (or distance). By contrast, the exponential distribution allows us to model the time between defects and is a key distribution in the study of reliability engineering. Also, note that while the Poisson is a discrete distribution, the exponential is a continuous distribution.

If X is a random variable that equals the distance (or time) between successive defects from a Poisson process with mean $\lambda > 0$, then X is exponentially distributed as

$$f(x) = \lambda e^{-\lambda x} \quad \text{for } 0 \le x \le \infty.$$

The cumulative exponential distribution is given as

$$\begin{aligned} P(X < x) = F(x) &= \int_0^t \lambda e^{-\lambda x}\, dx \\ &= 1 - e^{-\lambda t}. \end{aligned}$$

EXAMPLE 18.28

Customers arrive at the self-checkout counter at the supermarket at a rate of five per hour. What is the probability that a customer arrives in less than 15 minutes? Figure 18.15 illustrates the distribution of this example, and it includes an additional curve to demonstrate how the value of λ affects the shape of the distribution.

Before we can begin to solve this problem, we need to ensure that all time units are made consistent. Thus, we need to convert the 15-minute time interval to 0.25 hours.

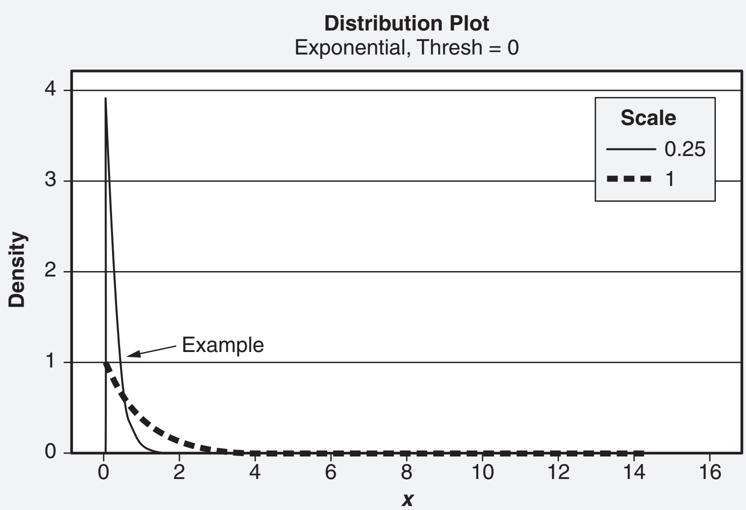

Figure 18.15 Exponential distribution for Example 18.28.

Continued

Continued

From the cumulative exponential distribution formula, we can determine that

$$\begin{aligned} P(X < 0.25) &= 1 - e^{-(5)(0.25)} \\ &= 1 - e^{-1.25} \\ &= 1 - 0.2865 \\ &= 0.7135. \end{aligned}$$

Therefore, the probability that a customer arrives in less than 15 minutes is 0.7135.

Lognormal Distribution

If a variable Y has a normal distribution with mean μ and finite variance σ^2, then the variable $X = e^Y$ has a lognormal distribution. This distribution has applications in modeling life spans for products that degrade over time. The probability density function for the lognormal distribution is defined as:

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{\left[\frac{-1}{2\sigma^2}(\ln(x)-\mu)^2\right]} \text{ for } x > 0.$$

EXAMPLE 18.29

Suppose the lifetime of a plasma television is distributed as lognormal with mean (that is, location parameter) μ = 10 hours and standard deviation (that is, scale parameter) σ = 2 hours. What is the probability that the lifetime of the plasma television exceeds 1000 hours? Figure 18.16 illustrates the distribution for this example.

$$\begin{aligned} P(X > 1000) &= 1 - P(X \le 1000) \\ &= 1 - P(e^Y \le 1000) \\ &= 1 - P(Y \le \ln(1000)) \\ &= \Phi\left(\frac{\ln(1000) - 10}{2}\right) \\ &= \Phi\left(\frac{6.9078 - 10}{2}\right) \\ &= \Phi(-1.55) \\ &= 1 - \Phi(1.55) \\ &= 1 - 0.9394 \\ &= 0.0606 \end{aligned}$$

Continued

Continued

Therefore, the probability that the lifetime of the plasma television exceeds 1000 hours is 0.0606.

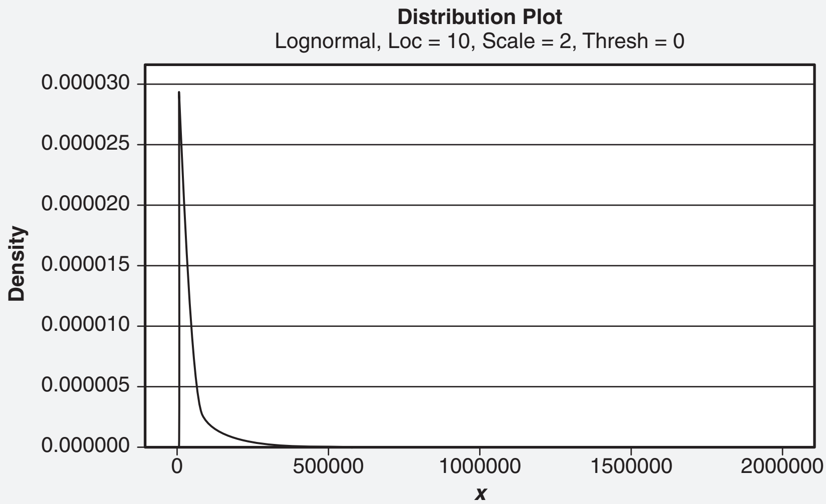

Figure 18.16 Lognormal distribution for Example 18.29.

Weibull Distribution

The Weibull distribution is a specialized form of the gamma distribution and is highly useful in the area of reliability. The general form of the Weibull distribution is given by

$$f(x) = \frac{\beta}{\alpha}\left(\frac{x-\gamma}{\alpha}\right)^{\beta-1} e^{-\left(\frac{x-\gamma}{\alpha}\right)^{\beta}} \quad \text{for } x > 0, \alpha > 0, \beta > 0, \gamma \geq 0$$

where

α = Scale parameter; $\alpha > 0$

β = Shape parameter; $\beta > 0$

γ = Location parameter; $-\infty < \gamma < \infty$

The cumulative distribution function of the Weibull is given by

$$F(x) = 1 - e^{-\left(\frac{x-\gamma}{\alpha}\right)^{\beta}}.$$

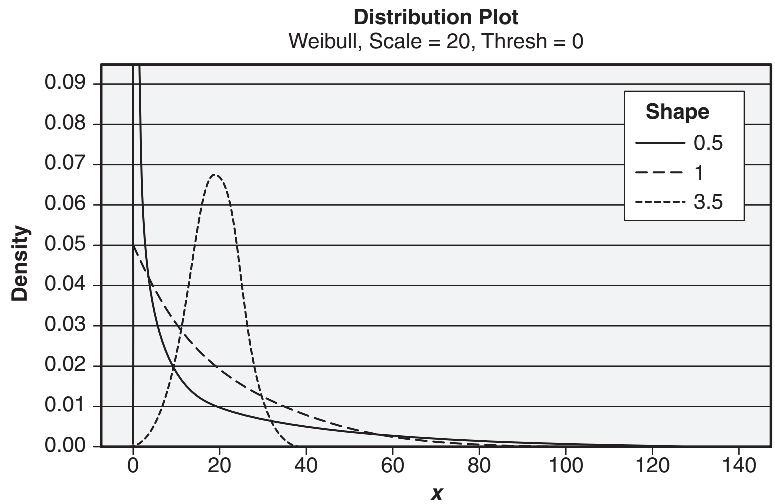

Figure 18.17 Example of a Weibull function for various values of the shape parameter.

This form is highly useful for solving reliability problems.

Although the Weibull distribution is actually a three-parameter distribution, it is sometimes referred to in the literature as a two-parameter distribution because the location parameter is assumed to be equal to zero.

The elegance of the Weibull function is that it takes on many shapes depending on the value of β, as shown in Figure 18.17. For example, when:

- $0 < \beta < 1$, the Weibull has a decreasing failure rate and can be used to model "infant mortality."
- $\beta = 1$, the Weibull is reduced to the exponential distribution with a constant failure rate.
- $\beta > 1$, the Weibull has an increasing failure rate and can be used to model "wear-out."
- $\beta = 3.5$, the Weibull is approximately the normal distribution, has an increasing failure rate, and can be used to model "wear-out."

As a result, the Weibull distribution is one of the fundamental distributions in the study of reliability engineering.

EXAMPLE 18.30

The time to failure of a belt on an engine is distributed as a Weibull random variable with $\beta = 0.5$ and $\alpha = 10{,}000$ hours. Determine the probability that the belt lasts at least 15,000 hours.

From the above problem, we are seeking to determine

$$P(x > 15{,}000).$$

Using the cumulative distribution function from above,

$$P(x > 15{,}000) = 1 - F(15{,}000) = e^{-\left[\left(\frac{15000}{10000}\right)^{1/2}\right]} = 0.2938.$$

In other words, 29.38% of all the belts will last at least 15,000 hours.

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Chapter 19

Process Capability

Total conformance to specifications comes only by aiming at continuous process improvement.

—Donald J. Wheeler and David S. Chambers

This chapter will address the area of process capability and its related concepts. Some aspects of process capability are controversial and even hotly debated in the literature. In some instances, concepts are even contradictory in nature. Consequently, it is important to understand each author's notation when reviewing the literature. We have tried to simplify the concepts to the degree possible while still adhering to the requirements of the body of knowledge.

PROCESS CAPABILITY INDICES

> Define, select, and calculate C_p and C_{pk}. (Evaluate)
>
> **Body of Knowledge V.F.1**

Process capability is defined as the inherent variability of the characteristics of a process. It represents the performance of the process over a period of stable operations. Process capability is expressed as 6σ. A process is said to be *capable* when the output always conforms to the process specifications.

The *process capability index* is a dimensionless number that is used to represent the ability to meet customer specification limits. The basics process capability indices commonly used are C_p, C_{pk}, and C_{pm}.

The C_p index describes process capability in relation to the specified tolerance of a characteristic divided by the natural process variation for a process in a state of statistical control. The formula is given by

$$C_p = \frac{USL - LSL}{6\sigma}$$

where

LSL = Lower specification limit

USL = Upper specification limit

6σ = The natural process variation

σ = Estimate of short-term variation obtained from a control chart, for example,

$$\frac{\bar{R}}{d_2}.$$

In simpler terms,

$$C_p = \frac{\text{Voice of the customer}}{\text{Voice of the process}} = \frac{\text{VOC}}{\text{VOP}} = \frac{\text{Tolerance}}{\text{Natural process variation}} = \frac{\text{USL} - \text{LSL}}{6\sigma}.$$

C_p values greater than or equal to 1 indicate that the process is technically capable. A C_p value equal to 2 is said to represent 6σ performance.

However, the C_p index is limited in its use since it does not address the centering of a process relative to the specification limits. For that reason, the C_{pk} was developed. It is defined as

$$C_{pk} = \min(C_{pk_L}, C_{pk_U})$$

where

C_{pk_L} is the lower process capability index and is given by

$$C_{pk_L} = \frac{\bar{x} - \text{LSL}}{3\sigma}.$$

C_{pk_U} is the upper process capability index and is given by

$$C_{pk_U} = \frac{\text{USL} - \bar{x}}{3\sigma}.$$

and

$\bar{X}$ = Process average

LSL = Lower specification limit

USL = Upper specification limit

σ = Process standard deviation

3σ = Half of the natural process variation

Notice that the C_{pk} determines the proximity of the process average to the nearest specification limit. Also, note that at least one specification limit must be stated in order to compute a C_{pk} value.

C_{pk} values of 1.33 or 1.67 are commonly set as goals since they provide some room for the process to drift in relation to the nominal setting.

Note that when the process average is obtained from a control chart, $\bar{x}$ is replaced by $\bar{\bar{x}}$.

The C_{pm}, also known as the *process capability index of the mean*, is an index that accounts for the location of the process average relative to a target value and is defined as

$$C_{pm} = \frac{(USL - LSL)/2}{3\sqrt{\sigma^2 + (\mu - T)^2}} = \frac{C_p}{\sqrt{1 + \frac{(\mu - T)^2}{\sigma^2}}}$$

where

μ = Process average

LSL = Lower specification limit

USL = Upper specification limit

σ = Process standard deviation

T = Target value (typically the center of the tolerance)

When the process average and the target value are equal, the C_{pm} equals the C_{pk}. When the process average drifts from the target value, the C_{pm} is less than the C_{pk}.

Note: There are other methods to compute the C_{pm}. For example, when the target value is at the midpoint between the specifications, Minitab computes the C_{pm} as follows:

$$C_{pm} = \frac{USL - LSL}{6\sqrt{\frac{\sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(x_{ij} - T\right)^2}{\sum_{i=1}^{k} n_i - 1}}}$$

where

LSL = Lower specification limit

USL = Upper specification limit

T = Target value (typically the center of the tolerance)

n_i = Sample size of *i*th subgroup

k = Number of subgroups

x_{ij} = Individual data element

The above method computes the standard deviation in a manner similar to the long-term computation of the variance described elsewhere in this chapter. Thus, the C_{pm} computed in this manner acts as a process performance index rather than a process capability index. Therefore, a cautionary warning regarding its use is warranted.

EXAMPLE 19.1

A batch process produces high fructose corn syrup with a specification on the dextrose equivalent (DE) of 6.00 to 6.15. The DEs are normally distributed, and a control chart shows that the process is stable. The standard deviation of the process obtained from the control chart is

$$\frac{\bar{R}}{d_2} = 0.035.$$

The process mean from the chart is 6.05. Determine the C_p and C_{pk}.

Using the formula for the C_p, we have

$$C_p = \frac{\text{Tolerance}}{6\sigma} = \frac{6.15 - 6.00}{6(0.035)} = 0.71.$$

Since the value of the C_p is less than 1, the process is deemed "not capable."

Now we must determine the C_{pk} using the formulas previously given:

$$C_{pk_L} = \frac{\bar{x} - LSL}{3\sigma} = \frac{6.05 - 6.00}{3(0.035)} = 0.48$$

$$C_{pk_U} = \frac{USL - \bar{x}}{3\sigma} = \frac{6.15 - 6.05}{3(0.035)} = 0.95$$

$$C_{pk} = \mathbf{min}((C_{pk_L}, C_{pk_U}) = \mathbf{min}(0.48, 0.95) = 0.48.$$

A C_{pk} of 0.48 indicates that the process is even less capable relative to the lower specification limit.

EXAMPLE 19.2

The data given in Table 19.1 were obtained from a control chart used to monitor the diameter of a cable. The lower specification is 0.5 and the upper specification is 0.6. A target value has been identified as 0.55. Determine the C_p, C_{pk}, and C_{pm} values.

Now that we know how to compute the indices from Example 19.1, we will use Minitab in this example.

The data for this example are in the first two columns and first five rows of the worksheet. To begin, we:

1. Choose Stat > Quality tools > Capability analysis > Normal.
2. Select "Subgroups across rows of" for the "Data are arranged as."
3. Click in the box below and select all 20 columns of data.
4. Enter 0.5 for the "Lower spec."
5. Enter 0.6 for the "Upper spec."
6. Select "Options."
7. Enter "0.55" for the target value and click OK.

Continued

Table 19.1 Cable diameter data for Example 19.2.

Subgroup																			
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
0.529	0.543	0.493	0.559	0.545	0.607	0.577	0.546	0.527	0.557	0.538	0.544	0.558	0.560	0.541	0.572	0.543	0.521	0.550	0.536
0.550	0.557	0.534	0.519	0.588	0.532	0.526	0.560	0.545	0.559	0.557	0.550	0.548	0.533	0.534	0.556	0.544	0.532	0.544	0.554
0.555	0.559	0.527	0.562	0.544	0.562	0.546	0.530	0.513	0.529	0.517	0.562	0.532	0.538	0.544	0.560	0.541	0.524	0.545	0.569
0.541	0.581	0.511	0.551	0.561	0.542	0.557	0.564	0.557	0.539	0.521	0.540	0.570	0.567	0.537	0.520	0.526	0.544	0.571	0.531
0.559	0.551	0.565	0.530	0.573	0.549	0.548	0.514	0.525	0.591	0.568	0.537	0.567	0.557	0.574	0.578	0.518	0.523	0.527	0.534

Continued

Continued

Click OK again to produce the graphical analysis shown in Figure 19.1.

From Figure 19.1, we can read directly the required values. The C_p = 0.85, C_{pk} = 0.79, and C_{pm} = 0.85. These results indicate that the process is not capable and improvement is required. Notice that the process mean and the target value are almost identical. Hence, the C_{pk} and the C_{pm} are equal to the two decimal places shown.

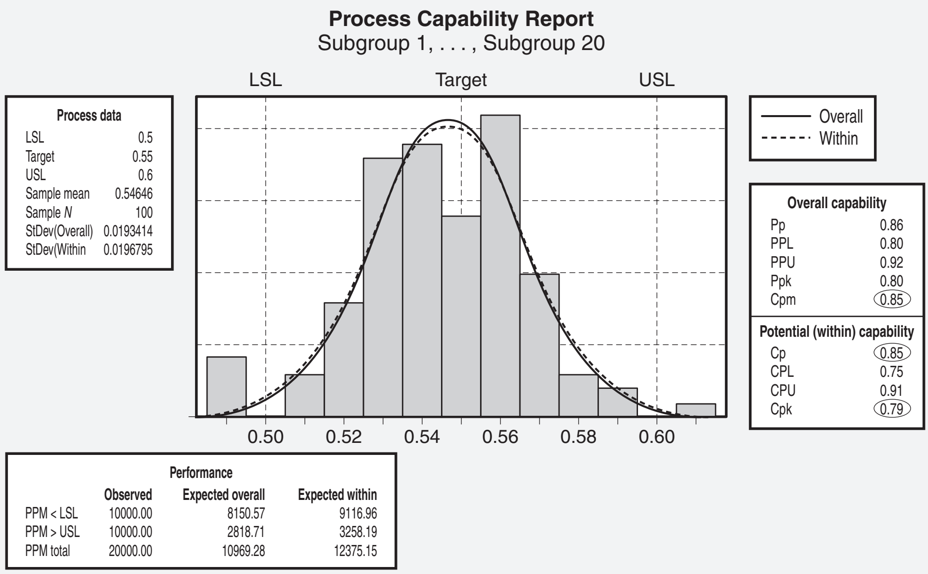

Figure 19.1 Process capability analysis for Example 19.2.

PROCESS PERFORMANCE INDICES

Define, select, and calculate P_p, P_{pk}, C_{pm} and process sigma. (Evaluate)

Body of Knowledge V.F.2

Process performance is defined as a statistical measure of the outcome of a characteristic from a process that *may not have been demonstrated to be in a state of statistical control*. Use this measure cautiously since it may contain a component of variability from special causes of an unpredictable nature. It differs from process capability because a *state of statistical control is not required*. This is an important distinction.

The *process performance index* is a dimensionless number that is used to represent the ability to meet specification limits on a characteristic of interest. The index compares the variability of the characteristic to the specification limits. Three basic process performance indices are P_p, P_{pk}, and P_{pm}. Note: C_{pm} was moved to section V.F.1 above as it more comparable to C_p and C_{pk} than it is to P_p and P_{pk}. P_{pm} was added in this subsection for completeness purposes.

The P_p is computed as

$$P_p = \frac{USL - LSL}{6s}$$

where

USL is the upper specification limit

LSL is the lower specification limit

s = Estimate of long-term variation obtained from a sample of size n. s is defined as:

$$s = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

6s is the natural process variation

Similarly to the C_{pk}, the P_{pk} is computed as

$$P_{pk} = \min(P_{pk_L}, P_{pk_U})$$

where

P_{pk_L} is the lower process capability index and is given by

$$P_{pk_L} = \frac{\bar{x} - LSL}{3s}$$

P_{pk_U} is the upper process capability index and is given by

$$P_{pk_U} = \frac{USL - \bar{x}}{3s}$$

and

$\bar{x}$ = Sample average

LSL = Lower specification limit

USL = Upper specification limit

s = Sample standard deviation

$3s$ = Half of the natural process variation

Notice that the P_{pk} determines the proximity of the process average to the nearest specification limit. Also, note that at least one specification limit must be stated in order to compute a P_{pk} value.

Process performance indices tend to be a bit larger than their corresponding process capability indices since they are often based on smaller sample sizes.

The P_{pm} index is analogous to the C_{pm}. It is computed as follows:

$$P_{pm} = \frac{(USL - LSL)/2}{3\sqrt{s^2 + (\bar{x} - T)^2}} = \frac{P_p}{\sqrt{1 + \frac{(\bar{x} - T)^2}{s^2}}}.$$

GENERAL PROCESS CAPABILITY STUDIES

> Describe and apply elements of designing and conducting process capability studies relative to characteristics, specifications, sampling plans, stability and normality. (Evaluate)
>
> **Body of Knowledge V.F.3**

Conducting a Process Capability Study

The purpose of a capability study is to determine whether a process is capable of meeting customer specifications and to devise a course of action if it cannot.

Although authors differ on the specific steps of a process capability study, the general elements of any study include the following:

1. Select a quality characteristic of a specific process for study. Ideally, a capability study should be performed for every critical quality characteristic. In practice, people familiar with the process are usually able to identify the few characteristics that merit a full capability study, typically characteristics that have shown to be difficult to hold to specification.

2. Confirm the measurement system used to acquire the process data. Large measurement system variation can obscure the "true" process variation and will produce an unreliable estimate of process capability.

3. Gather the data. Ideally, the data are readily available and are being tracked on a control chart. If not, collect data from the process—preferably randomly and with consideration given to rational subgrouping. Although the literature differs greatly on the appropriate number of subgroups, approximately 25–30 uniformly sized subgroups is usually sufficient. Subgroup sizes typically range from three to five. The exact size of the subgroup varies because the underlying

distribution may not be normal. The farther it departs from normality, the greater the subgroup size required.

4. Verify process stability and ensure that the process is in control. This can easily be accomplished using control charts. If the process is out of control, identify and eliminate special cause variation.
5. Verify that the individual data are normally distributed. Plot the data in a histogram and test for normality, which can be verified using a variety of statistical tests such as Anderson-Darling, Ryan-Joiner, Shapiro-Wilk, or Kolmogorov-Smirnov. If normality is not present and is required, it can be restored using the techniques described elsewhere in this chapter.
6. Obtain the process specifications. The process specifications may be either one- or two-sided specifications.
7. Determine process capability indices and interpret them. Apply the previously described methods. If the process is not capable, work to improve it.
8. Update the process control plan. Once the process has achieved a desired level of capability, update your control plan to ensure that the gains are maintained. If no control plan is present, create one.

Sampling Plans

Process capability indices are *point estimates* and should be presented along with their confidence intervals. Confidence intervals will be addressed in a later chapter.

Crossley (2008) provides a method for determining the lower confidence limit (LCL) on the C_{pk}. While his method doesn't directly provide a sample size in advance of sampling, it does provide information regarding the sufficiency of the sample size.

Crossley provides the LCL as

$$\mathrm{LCL} - C_{pk} - Z_{\alpha}\sqrt{\frac{1}{9n} + \frac{C_{pk}^2}{2n-2}}.$$

EXAMPLE 19.3

For $\alpha = 0.10$, $n = 45$, and a calculated $C_{pk} = 1.75$, compute the LCL C_{pk}.

Applying the above equation and filling in the numbers, we have

$$\mathrm{LCL} = 1.75 - 1.28\sqrt{\frac{1}{9(45)} + \frac{1.75^2}{2(45)-2}} = 1.75 - 0.25 = 1.50.$$

The above example can be interpreted as meaning that if you obtained a $C_{pk} = 1.75$ from a sample size of $n = 45$, then there is 90% confidence that the true C_{pk} is not lower than 1.50.

EXAMPLE 19.4

Using the data from Example 19.3, we enter the 90% Confidence Level Table of Appendix 34 and select the row for $n = 45$. Slide across this row until we reach the column for LCL $C_{pk} = 1.50$. At the intersection of the row and column, read the cell value. It should read $C_{pk} = 1.7470$ which rounds to $C_{pk} = 1.75$. This value agrees with our calculated C_{pk} from Exercise 19.3.

Notice that we are using Z_α instead of $Z_{\alpha/2}$. This is because we are dealing with a one-sided confidence limit. We are not concerned with how high the C_{pk} can go. We are only concerned that it does not go below a specified value.

To ease the burden of calculation, Appendix 34 has been added. The table works as follows:

1. Select your desired sample size, n, down the first column on the left.
2. Select your minimum acceptable LCL C_{pk} across the top row.
3. Determine the calculated C_{pk} the sample must achieve to ensure the LCL C_{pk}. This is where the row and column cells intersect.

PROCESS CAPABILITY FOR ATTRIBUTES DATA

Calculate the process capability and process sigma level for attributes data. (Apply)

Body of Knowledge V.F.4

According to Bothe (1997), "the capability measures for processes generating attribute data are formulated to allow a direct comparison to the corresponding measures for processes characterized with variable data." Although Bothe describes four attribute measures, we'll discuss the two most common:

$$\text{Equivalent } P_{pk}^{\%} = \text{Equivalent } P_{pk} \text{ based on percent}$$

$$\text{Equivalent } P_{p}^{\%} = \text{Equivalant } P_{p} \text{ based on percent}$$

As stated elsewhere in this chapter, processes must be in the state of statistical control before we can calculate meaningful process capability indices. One of the best means of demonstrating statistical control is by the use of control charts. It is through control charts that we will develop a procedure for determining both capability measures and the process sigma level derived from attributes data.

Consider the following procedure based on the use of the p-chart:

1. Determine $\bar{p}$, the percent nonconforming, from the process under investigation.

2. Convert the percent nonconforming to a fraction nonconforming.
3. Find the corresponding *Z*-value in the standard normal table for the associated fraction nonconforming where the fraction nonconforming represents the area under the right tail.
4. The Z-value represents the process sigma value.
5. Calculate

$$P_{pk} = \frac{\text{Process sigma level}}{3} = \frac{Z}{3}.$$

6. Calculate the P_p. The calculation of P_p is based on the assumption that half of the nonconforming parts are outside both specifications. Therefore, half of the fraction nonconforming should appear in both tails of the standard normal distribution. Consequently, the *Z*-value will be higher. Calculate

$$P_p = \frac{\text{Process sigma level}}{3} = \frac{Z}{3}.$$

It should be noted that the procedure outlined above is used for determining process performance indices for attributes data, not process capability indices. The reason for this is that the mean values obtained for entering the standard normal tables are calculated based on the total data, not subgroups. Consequently, standard deviations are as well. Since the standard deviations are computed in this manner, they are considered long term. Therefore, we must use the process performance indices.

EXAMPLE 19.5

Consider a stable process monitored by a *p*-chart. The process average is $\bar{p} = 0.01$. Compute P_{pk}, P_p, and the corresponding process sigma levels.

P_{pk} and the Process Sigma Level

From step 3 above, enter Appendix 13 at the fraction nonconforming of 0.99 (that is, 1 – 0.01). We do this because Appendix 13 is a table for the cumulative standard normal distribution.

Since Appendix 13 doesn't provide the exact value in the table, it will be necessary to perform linear interpolation to determine the process sigma level:

$$\text{Process sigma level} = 2.32 + \left(\frac{0.99 - 0.9898}{0.9901 - 0.9898}\right)(2.33 - 2.32) = 2.327$$

The corresponding P_{pk} is

$$P_{pk} = \frac{2.327}{3} = 0.776.$$

Continued

Continued

P_p and the Process Sigma Level

From step 3 above, enter Appendix 13 at the fraction nonconforming of 0.995 (that is, 1 – 0.005). We do this because Appendix 13 is a table for the cumulative standard normal distribution.

Since Appendix 13 doesn't provide the exact value in the table, it will be necessary to perform linear interpolation to determine the process sigma level:

$$\text{Process sigma level} = 2.57 + \left(\frac{0.9950 - 0.9949}{0.9951 - 0.9949}\right)(2.58 - 2.57) = 2.575$$

The corresponding P_p is

$$P_p = \frac{2.327}{3} = 0.858.$$

For other attributes charts such as the *np*-chart, the above procedure holds. Obtain $n\overline{p}$ from the centerline. Divide it by the constant sample size n to obtain $\overline{p}$. Then apply the procedure above.

When the *u*-charts or the *c*-harts are being used, we must determine the fraction nonconforming by other means.

Although control charts will be discussed in detail in a later chapter, a few basic concepts are required to proceed further.

The *c*-chart is a chart of nonconformities per unit based on a constant sample size, whereas the *u*-chart is a chart of nonconformities per unit based on a variable sample size.

The underlying distribution that governs both the *u*-chart and the *c*-chart is the Poisson distribution. Recall that the formula for the Poisson is given by

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!}; \lambda > 0; x = 0,1,2,\ldots.$$

This means that the probability that there are zero nonconformities per unit is

$$P(X = 0) = \frac{\lambda^{x}e^{-\lambda}}{x!} = \frac{\lambda^{0}e^{-\overline{u}}}{0!} = e^{-\overline{u}}.$$

Consequently, the probability that the number of nonconformities is non-zero is

$$1 - P(X = 0) = P(X > 0) = 1 - e^{-\overline{u}}.$$

The above formula provides the fraction nonconforming for the *u*-chart. Once the fraction nonconforming is determined, the procedure identified for *p*-charts can be followed.

For the c-chart, set

$$\overline{u} = \frac{\overline{c}}{n}$$

and proceed as just discussed.

PROCESS CAPABILITY FOR NON-NORMAL DATA

> Identify non-normal data and determine when it is appropriate to use Box-Cox or other transformation techniques. (Apply)
>
> **Body of Knowledge V.F.5**

If data are not normal and normality is required, the following approaches—all of which involve fairly complex calculations and are usually performed with the aid of statistical software packages—may be taken:

- Find another known distribution that fits the data—exponential, chi-square, Weibull, and F distributions are among the possibilities. A statistics software package is helpful in choosing the function and parameters that will provide the best fit for the data.
- Use nonlinear regression to fit a curve to the data, and apply numerical integration to find the areas of the "tails" beyond the specification limits.
- Transform the data to produce a second variable that is normally distributed; various normalizing transformations are available. Among the most used are the Box-Cox and Johnson transformations. These transformations introduce a new variable that will have a "more normal" distribution:
  – The Box-Cox transforms the input data, denoted by Y, using $W = Y^{\lambda}$, where λ is any number typically between –5 and 5. The trick, of course, is to choose λ that produces a curve that is as close to normal as possible. Statistics software packages will use sample data to recommend a range for λ, often with a confidence interval. The user may choose a value in the range and observe the resulting curve, eventually choosing the value that best fits the situation. The package typically generates values for the capability indices. Unfortunately, the Box-Cox transformation is limited to positive data values and often does not provide a suitable transformation. Also, it assumes that data are in subgroups.

– The Johnson transformation is chosen from three different functions within the Johnson system. As a result, this approach usually finds an appropriate transformation. It does not assume data are in subgroups.

An alternative to transforming nonnormal data is to assume the data fit a specific distribution. One favored distribution is the Weibull because by varying its parameters the distribution can take on a variety of different shapes.

EXAMPLE 19.6

The source of this example is Bower (2001). The data for this example are temperature data and are given in Table 19.2. A quick normal probability plot of the data can confirm the data are nonnormal. The Anderson-Darling statistic value is 1.347 with a p-value < 0.005. This is illustrated in Figure 19.2. This conclusion is supported by a graphical summary of the data shown in Figure 19.3.

Given the data, we might want to assume they fit a specific distribution. In this case, we will assume it fits the Weibull distribution. Instead of working the numbers by hand, we will choose to let Minitab estimate the parameters of the Weibull distribution for us by selecting the following sequence:

1. Choose Stat > Quality Tools > Capability Analysis > Nonnormal.

Table 19.2 Temperature data for Example 19.6.

13.81	13.05	13.29	22.33	4.40	10.70	3.42	10.04
17.64	8.19	14.48	4.95	6.19	15.37	11.37	7.79
10.41	3.35	5.82	5.82	10.01	3.78	13.93	6.14
11.31	1.87	13.07	8.04	8.85	5.30	10.87	4.65
9.81	4.58	2.52	15.20	4.59	8.73	9.57	16.15
12.09	21.21	7.29	10.44	15.22	5.57	4.67	17.95
8.46	4.41	10.33	6.18	2.36	2.97	7.95	7.15
9.69	3.79	9.75	4.34	8.60	6.13	4.20	17.19
6.26	15.43	6.28	8.37	13.64	4.09	6.03	7.12
2.39	7.70	6.60	12.66	7.58	9.39	4.14	9.60
9.72	7.93	2.44	9.54	8.25	11.32	11.67	7.61
10.05	8.07	6.38	5.99	12.99	7.25	4.97	5.43
8.20	5.16	6.87	9.74	17.37	12.93	8.58	4.00
9.59	4.53	15.25	9.38	13.00	4.91	8.14	4.48
8.47	8.05	3.02	10.38	7.02	18.20	8.97	11.62

Source: Bower (2001).

Continued

2. Select "Weibull" from the "Fit Distribution" drop-down menu.
3. Enter "1" into "Lower Spec."
4. Click "OK" to produce the analysis.

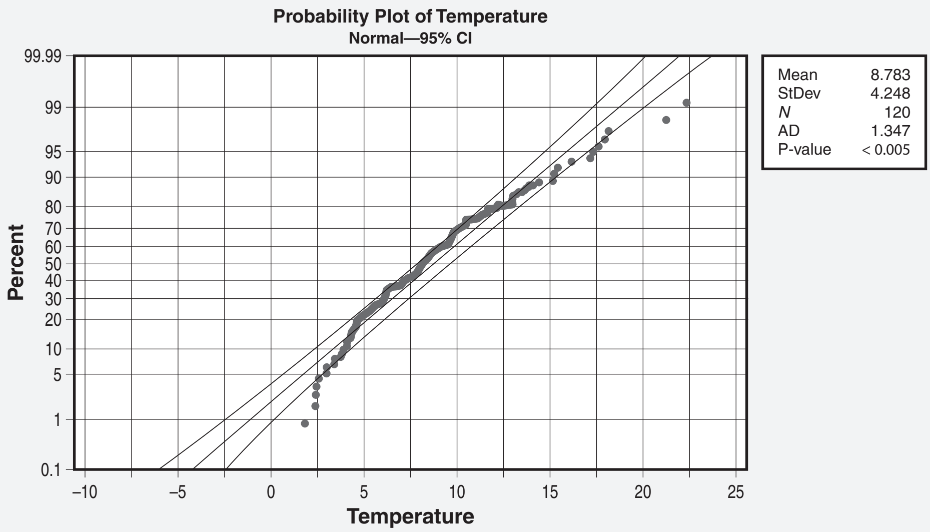

Figure 19.2 Normal probability plot of temperature data shown in Table 19.2.

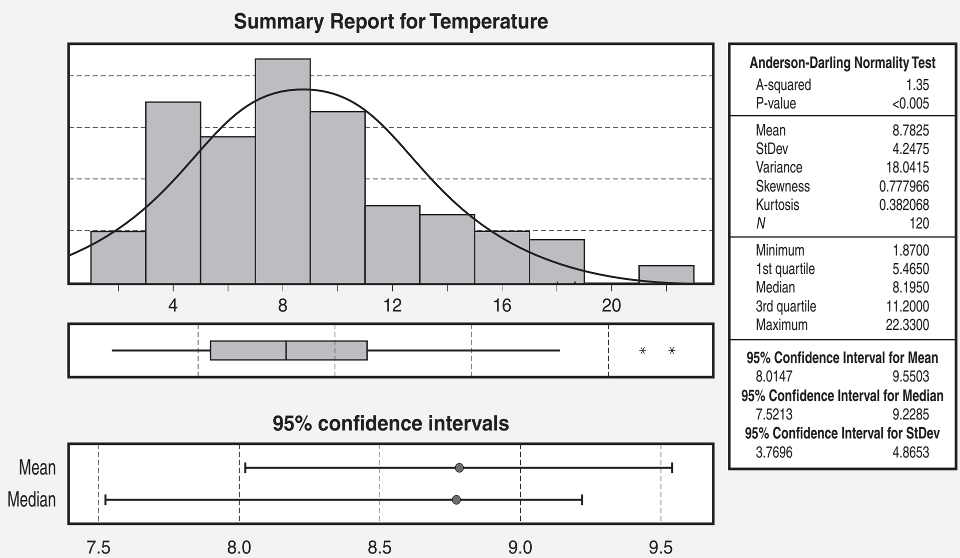

Figure 19.3 A graphical summary of the temperature data shown in Table 19.2.

Continued

Continued

The above will yield Figure 19.4 with fitted parameters. This figures illustrates a plot of the data with a Weibull plot overlay and shape parameter β = 2.214 and scale parameter α = 9.946. A corresponding Weibull probability plot illustrated in Figure 19.5 shows that we cannot reject the null hypothesis that the data fit a Weibull distribution. The Anderson-Darling statistic is 0.450 and the *p*-value > 0.250.

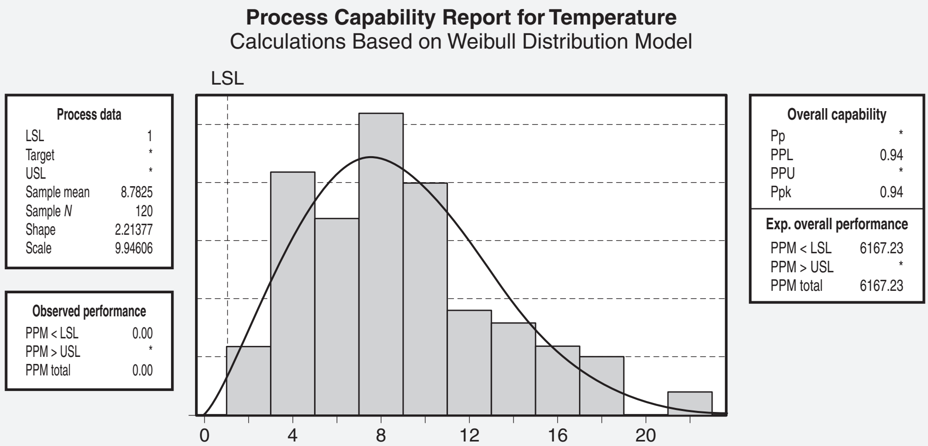

Figure 19.4 Nonnormal process capability analysis based on the Weibull distributiona.

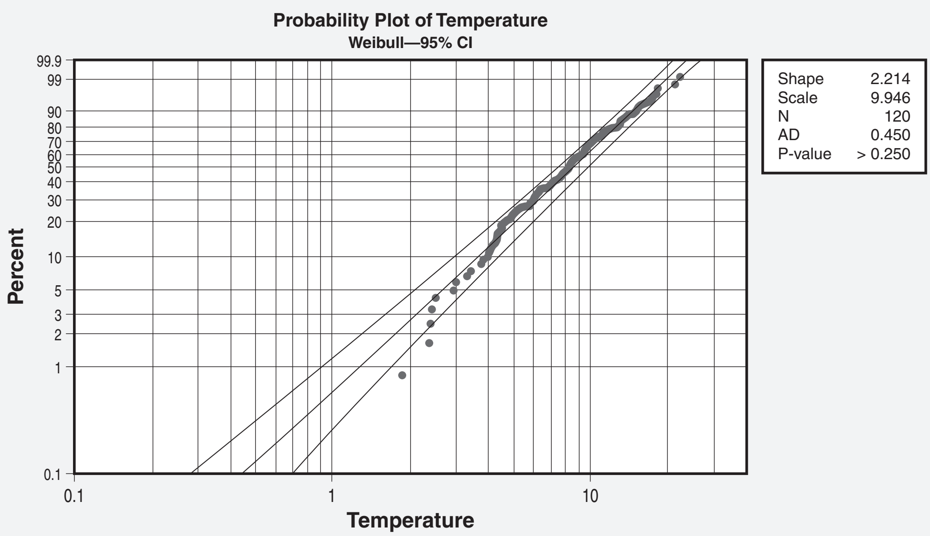

Figure 19.5 Weibull probability plot with fitted parameters.

Although the above example assumed the use of the Weibull distribution when the data were nonnormal, any other continuous distribution deemed appropriate could have been used as well.

PROCESS PERFORMANCE VS. SPECIFICATION

> Distinguish between natural process limits and specification limits. Calculate process performance metrics, e.g., percent defective, parts per million (PPM), defects per million opportunities (DPMO), defects per unit (DPU), throughput yield, rolled throughput yield (RTY). (Evaluate)
>
> **Body of Knowledge V.F.6**

Natural Process and Specification Limits

As mentioned previously, *natural process limits* are known by several names in the literature: *natural process variation*, *normal process variation*, and *natural tolerance*. In all cases, these terms refer to the $\pm 3\sigma$ limits (that is, 6σ spread) around a process average. Such limits include 99.73% of the process variation and are said to be the "voice of the process." Walter Shewhart originally proposed the $\pm 3\sigma$ limits as an economic trade-off between looking for special causes for points outside the control limits when no special causes existed and not looking for special causes when they did exist.

In contrast to natural process limits, *specification limits* are customer determined and are used to define acceptable levels of process performance. Said to be the "voice of the customer," specification limits may be one-sided (that is, upper or lower) or two-sided (that is, both upper and lower). The difference between the upper and lower specification limits is known as the *tolerance*.

The C_p index brings together the concepts of "voice of the process" and "voice of the customer." Recall that the C_p is defined as

$$C_p = \frac{\text{Voice of the customer}}{\text{Voice of the process}} = \frac{\text{VOC}}{\text{VOP}} = \frac{\text{Tolerance}}{\text{Natural process variation}} = \frac{\text{USL} - \text{LSL}}{6\sigma}.$$

Process Performance Metrics

Percentage Defective. The percentage defective is simply defined by the following ratio:

$$\text{Percentage defective} = \frac{\text{Total number of defective units}}{\text{Total number of units}} \times 100\%.$$

EXAMPLE 19.7

Consider a process in which the output is normally distributed with a mean of 0 and a standard deviation of 1. Specifications are set at ±3σ. The fraction defective for the process is shown in the shaded areas beyond ±3σ in Figure 19.6. The total fraction defective is the sum of the shaded areas, or 0.0027. Therefore, the percentage defective is 0.27%.

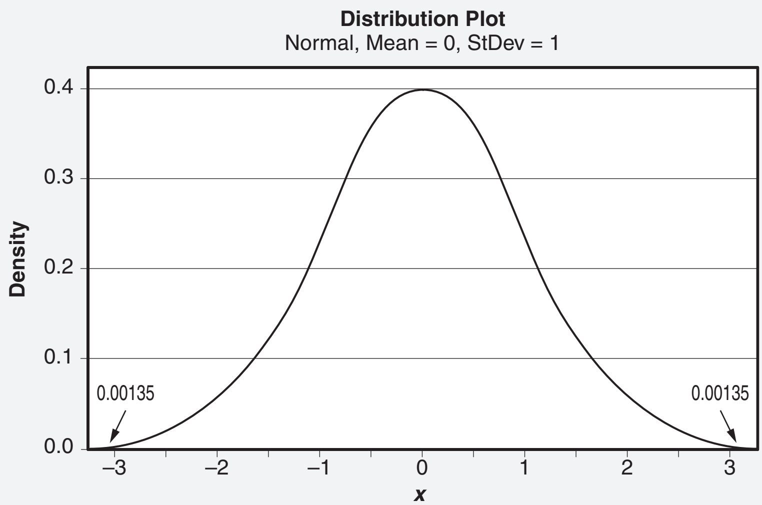

Figure 19.6 The fraction defective for the process given in Example 19.7.

Of course, a defective unit is any unit containing one or more defects. Note that the term

$$\frac{\text{Total number of defective units}}{\text{Total number of units}}$$

is known as the *fraction defective*.

Parts per Million (ppm). In a typical quality setting, the *parts per million* (ppm) metric usually indicates the number of times a defective part will occur in a million parts produced. By contrast, the DPMO metric reflects the number of defects occurring in a million opportunities. At this point, it is important to note that some authors state that the PPM and DPMO metrics are identical. However, if we follow the definitions above, this would only be true when the number of opportunities for a defect per unit or part is 1.

Additional confusion tends to surround the ppm metric due perhaps to a laxness in the terminology applied. In the Six Sigma context, ppm is also referred to as the ppm defect rate. Similarly, 3.4 ppm is often stated as 3.4 defects per million parts. However, in both examples, when we say *defects*, we are really referring to *defectives*.

EXAMPLE 19.8

Parts per million is also used to refer to contaminants. In a particular product, there are 0.23 grams of insect parts per 25 kilograms. What is the equivalent ppm?

This is a simple problem, but we must first convert everything to a common unit as follows:

$$\text{ppm} = \left(\frac{0.23 \text{ grams}}{25 \text{ kilograms}}\right)\left(\frac{1 \text{ kilogram}}{1000 \text{ grams}}\right) \times 1{,}000{,}000 = 9.2.$$

Finally, in the more traditional scientific context, ppm may simply refer to the various ratios of components in, say, a mixture. For example, the oxygen component of air is approximately 209,000 ppm. In this case, the idea of "defective" isn't a consideration.

Appendix 28 illustrates the linkages between multiple metrics, including ppm, sigma level, percentage in specification, and percentage defective. The familiar 3.4 ppm corresponds to a 6 sigma level of quality, assuming a $\pm 1.5\sigma$ shift in the mean. Both the sigma level of a process and the $\pm 1.5\sigma$ shift in the mean will be addressed in a later section.

Defects per Unit (DPU). The *defects per unit* (DPU) metric is a measure of capability for discrete (attributes) data defined by the following:

$$\text{Defects per unit (DPU)} = \frac{\text{Total number of defects}}{\text{Total number of units}}.$$

Defects Per Million Opportunities (DPMO). The *defects per million opportunities* (DPMO) metric is a measure of capability for discrete (attributes) data found by

$$\text{DPMO} = \frac{\text{Total number of defects}}{\text{Total number of opportunities}} \times 1{,}000{,}000.$$

The DPMO metric is important because it allows you to compare different types of products. Developing a meaningful DPMO metric scheme across multiple product

EXAMPLE 19.9

For example, a process produces 40,000 pencils. Three types of defects occurred. The number of occurrences of each defect type is (1) blurred printing: 36, (2) too long: 118, (3) rolled ends: 11. The total number of defects is 165.

Applying the DPU formula above yields the following:

$$\text{DPU} = \frac{\text{Total number of defects}}{\text{Total number of units}} = \frac{165}{40{,}000} = 0.004125.$$

EXAMPLE 19.10

Continuing with the pencil example, let's calculate the number of opportunities. First, determine the number of ways each defect can occur on each item. For this product, blurred printing occurs in only one way (the pencil slips in the fixture), so there are 40,000 opportunities for this defect to occur.

There are three independent places where dimensions are checked, so there are (3) (40,000) = 120,000 opportunities for this dimensional defect. Rolled ends can occur at the top and the bottom of the pencil, so there are (2) (40,000) = 80,000 opportunities for this defect to occur. Thus, the total number of opportunities for defects is

$$40{,}000 + 120{,}000 + 80{,}000 = 240{,}000.$$

Likewise, the total number of opportunities per unit is

$$1 + 3 + 2 = 6.$$

Applying the DPMO formula, we can readily determine the DPMO metric:

$$\text{DPMO} = \frac{\text{Total number of defects}}{\text{Total number of opportunities}} \times 1{,}000{,}000 = \frac{165{,}000{,}000}{240{,}000} = 687.5.$$

lines, however, can be very time-consuming because it is necessary to accurately determine the number of ways (or opportunities) a defect can occur per unit or part. This can be an enormous task, particularly when dealing with highly complex products and subassemblies, or even paperwork.

Throughput Yield (TPY)

Use of Input and Output. The traditional concept of yield for a process was defined as

$$\text{Process yield} = \frac{\text{Output}}{\text{Input}} \times 100\%.$$

Unfortunately, this definition overstates yield because it fails to reflect rework embedded within the process. The traditional concept is shown in Figure 19.7. Notice that reworked parts flow back into the main process stream and are counted as product that passed and was acceptable the first time. This method of counting inflates the numerator in the calculations and, consequently, overstates the yield as mention previously.

The throughput yield (TPY) concept accounts for the rework embedded within the process. Thus, it removes it from the calculation. This concept is illustrated in Figure 19.8. Notice the difference between the two figures. In Figure 19.8, rework units are not returned to the main process flow. Hence, they are not counted as part of the output.

Throughput yield is also known as *first-pass yield* (FPY) or *first-time yield* (FTY).

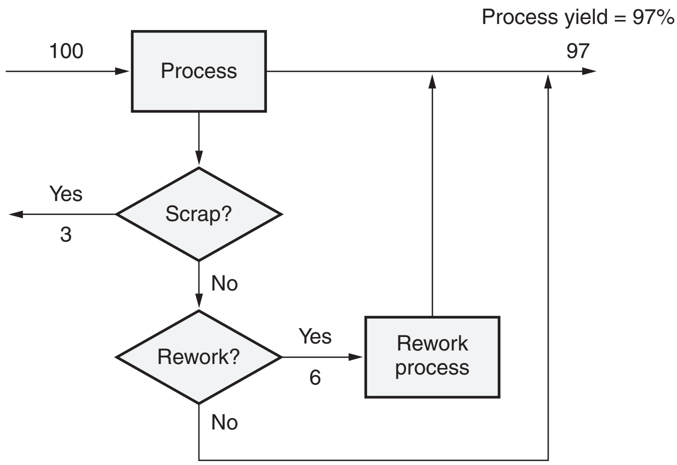

Figure 19.7 An illustration of the traditional yield calculation.

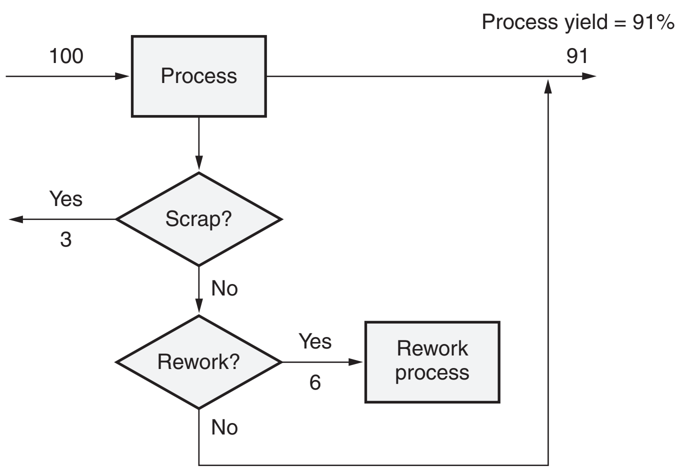

Figure 19.8 An illustration of the throughput yield calculation.

EXAMPLE 19.11

One hundred units enter a process. Three units are scrapped and six units are reworked. Compute the yield using the traditional approach.

Recall that this approach only accounts for scrap:

$$\text{Process yield} = \frac{100-3}{100} \times 100\% = \frac{97}{100} \times 100\% = 97\%.$$

EXAMPLE 19.12

One hundred units enter a process. Three units are scrapped and six units are reworked. Compute the yield using the throughput yield approach.

Recall that this approach accounts for scrap and rework:

$$\text{Process yield} = \frac{100-3-6}{100} \times 100\% = \frac{91}{100} \times 100\% = 91\%.$$

This represents six percentage points lower than the traditional calculation. Such a disparity might mean the difference between improvement action being taken or not.

EXAMPLE 19.13

Recall the pencil example from Example 19.9 above. The DPU from the example was 0.004125.

Applying the above formula, we have

$$\text{TPY} = e^{=0.004125} = 0.9959$$

On average, slightly more than 1 in 1000 pencils will have a defect.

Use of the DPU. Another way of determining the TPY is to calculate it by first determining the DPU of the process and the application of the following formula:

$$\text{TPY} = e^{-\text{DPU}}$$

Rolled Throughput Yield (RTY). The *rolled throughput yield* (RTY) metric represents the percentage of units of product passing defect free through an entire process. It is determined by multiplying throughput yields from each subprocess of the total process as follows:

$$\text{RTY} = \prod_{i=1}^{n} \text{TPY}_i$$

where

n = Number of subprocesses

TPY_i = Throughput yield of the ith subprocess.

It should be clear from the above formula that the multiplicative effect will work against the RTY. Consequently, it is advantageous to have the TPY of each subprocess as high as possible.

A significant advantage of using the RTY metric is that it provides a more complete view of the process. Subprocess yields that run high aren't likely to garner the attention necessary to drive improvement. Often, it is only when the total process yield becomes visible does real action occur.

EXAMPLE 19.14

The concept of the RTY is best illustrated by Figure 19.9, which depicts an overall process comprising four subprocesses. Suppose the TPY of each subprocess is 0.95. Then the RTY is easily computed as

$$\text{RTY} = \prod_{i=1}^{4} \text{FPY}_i = (0.95)(0.95)(0.95)(0.95) = 0.81 \text{ or } 81\%.$$

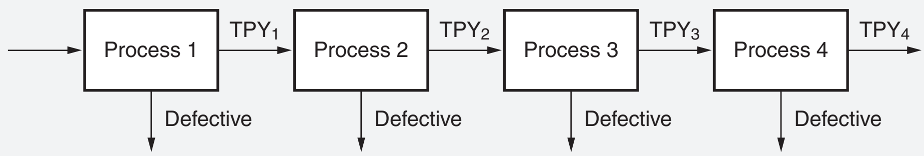

Figure 19.9 An illustration of the rolled throughput yield concept.

EXAMPLE 19.15

Recall the TPY example from Example 19.13. The TPY was 0.9959. Suppose an overall process comprises four subprocesses. Further, assume the TPY for each subprocess is 0.9959. Then the RTY is easily computed as

$$\text{RTY} = \prod_{i=1}^{4} \text{FPY}_i = (0.9959)(0.9959)(0.9959)(0.9959) = 0.9837 \text{ or } 98.37\%.$$

Although individual subprocess yields are very high, the total process yield has dropped.

SHORT-TERM AND LONG-TERM CAPABILITY

Describe and use appropriate assumptions and conventions when only short-term data or only long-term data are available. Interpret the relationship between short-term and long-term capability. (Evaluate)

Body of Knowledge V.F.7

The Difference between Short-Term and Long-Term Capability

Significant confusion abounds, and contradictory statements pepper the literature with regard to the meaning and determination of short-term and long-term capability and variability.

We will try to put forth the most common understanding of short-term and long-term capability and related concepts.

Short-Term Capability

Short-term capability encapsulates the following concepts:

- C_p, C_{pk}, and C_{pm} fall into this category and are known as process capability indices. They are sometime known as *within-subgroup*, *short-term*, or *potential* process capability indices.
- The process must be in statistical control for these indices to be used.
- These indices are derived using an estimate of short-term variation. Short-term variation is within-subgroups variation and is obtained from control charts. Examples from various control charts include:
  - $\bar{X}$ and R chart: $\dfrac{\bar{R}}{d_2}$
  - $\bar{X}$ and s chart: $\dfrac{\bar{s}}{c_4}$
  - XmR chart: $\dfrac{\overline{MR}}{d_2}$
- It is assumed that the individual process data elements are distributed approximately normally. When this assumption does not hold, normality can be restored statistically using the techniques described previously.
- The specification limits are based on customer requirements—ideally, all specifications are based on customer requirements.

Long-Term Capability

- P_p, P_{pk}, and P_{pm} fall into this category and are known as *process performance indices.*
- The process does not need be in statistical control for these indices to be used. However, if the process is not in statistical control, the indices are of little value.
- These indices are derived using an estimate of long-term variation. Long-term variation is known as *total* or *overall variation* (which includes both within- and between-subgroup variation) and is obtained as follows:

$$\sqrt{\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{n-1}}.$$

- It is assumed that the individual process data elements are distributed approximately normally. When this assumption does not hold, normality can be restored statistically using the techniques described previously.
- The specification limits are based on customer requirements—ideally, all specifications are based on customer requirements.

Relationship between Long-Term and Short-Term Capability

This is a difficult issue to address, as the literature is replete with inconsistencies. Further, the whole issue of long-term and short-term capability inevitably transforms into a discussion of long-term and short-term variability that includes the $\pm 1.5\sigma$ shift. Some authors say that:

- Long-term variability = Short-term variability $\pm 1.5\sigma$ shift. This interpretation is based on the underlying assumption of Six Sigma, which is that a process will drift or shift $\pm 1.5\sigma$ in the long term. When this drift is taken into account, 6σ process performance equates to 3.4 ppm; otherwise, a 4.5σ process performance equates to 3.4 ppm. See Appendix 28.
- Short-term variability = Long-term variability $\pm 1.5\sigma$ shift. A variation of this equation was used in the ASQ Black Belt Course. However, our previous discussions on long-term variability demonstrated that long-term variability was greater than short-term variability even without the consideration of the $\pm 1.5\sigma$ shift. If we follow the premise that the components of long-term variation are within-subgroup variation and between-subgroup variation, while short-term variation is considered to be only within-subgroup variation, then the equation above cannot hold true.

Regardless of which is correct, ensure that you are consistent with your interpretation when addressing this issue internally with your colleagues or externally with your customers or suppliers.

The 1.5 Sigma Shift

The $\pm 1.5\sigma$ concept was introduced by Motorola based on Motorola's observations of its processes. Six Sigma practitioners should not blindly accept the shift unless the shift is applicable to their process as well. In fact, their processes may shift, but at a constant other than $\pm 1.5\sigma$. However, this can't be known until it is proven with data.

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Part VI
Analyze

Chapter 20	Measuring and Modeling Relationships between Variables
Chapter 21	Hypothesis Testing
Chapter 22	Failure Mode and Effects Analysis (FMEA)
Chapter 23	Additional Analysis Methods

The purpose of this phase is to analyze and establish optimal performance settings for each *X* and verify root causes. Common tools used in this phase are shown in Table VI.1.

Table VI.1 Common tools used in *analyze* phase.

Affinity diagrams	*Histograms*	Qualitative analysis
ANOVA	*Hypothesis testing*	Regression
Basic statistics	*Interrelationship digraphs*	Reliability modeling
Brainstorming	Linear programming	Root cause analysis
Cause-and-effect diagrams	Linear regression	*Run charts*
Components of variation	Logistic regression	*Scatter diagrams*
Design of experiments (DOE)	*Meeting minutes*	Shop audits
Exponentially weighted moving average charts	*Multi-vari studies*	*Simulation*
Failure mode and effects analysis (FMEA)	Multiple regression	*Suppliers–inputs–process–outputs–customers (SIPOC)*
Force-field analysis	Multivariate tools	*Tollgate review*
Gap analysis	Nonparametric tests	*Tree diagrams*
General linear models (GLMs)	Preventive maintenance	Waste analysis
Geometric dimensioning and tolerancing (GD&T)	*Process capability analysis*	$Y=f(X)$
	Project management	
	Project tracking	

Note: Tools shown in *italic* are used in more than one phase.

Chapter 20

Measuring and Modeling Relationships between Variables

CORRELATION COEFFICIENT

> Calculate and interpret the correlation coefficient and its confidence interval, and describe the difference between correlation and causation. (Evaluate)
>
> **Body of Knowledge VI.A.1**

Note: Serial correlation will not be tested on the CSSBB exam.

The *sample correlation coefficient* r is a statistic that measures the degree of linear relationship between two sets of numbers and is computed as

$$r = \frac{s_{xy}}{\sqrt{s_{xx}s_{yy}}} = \frac{\sum_{i=1}^{n}(x_i - \overline{x})^2(y_i - \overline{y})^2}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2\sum_{i=1}^{n}(y_i - \overline{y})^2}} = \frac{\sum_{i=1}^{n}x_i y_i - \frac{\left(\sum_{i=1}^{n}x_i\right)\left(\sum_{i=1}^{n}y_i\right)}{n}}{\sqrt{\left(\sum_{i=1}^{n}x_i^2 - \frac{\left(\sum_{i=1}^{n}x_i\right)^2}{n}\right)\left(\sum_{i=1}^{n}y_i^2 - \frac{\left(\sum_{i=1}^{n}y_i\right)^2}{n}\right)}},$$

where $-1 \le r \le 1$.

Also, r is used to estimate the population correlation coefficient ρ. Correlation values of –1 and +1 represent perfect linear agreement between two independent and dependent variables. When r is 0, there is no linear relationship. If r is positive, the relationship between the variables is said to be positive; hence, when x increases, y increases. If r is negative, the relationship is said to be negative; hence, when x increases, y decreases. Figure 20.1 provides examples of different types of correlations.

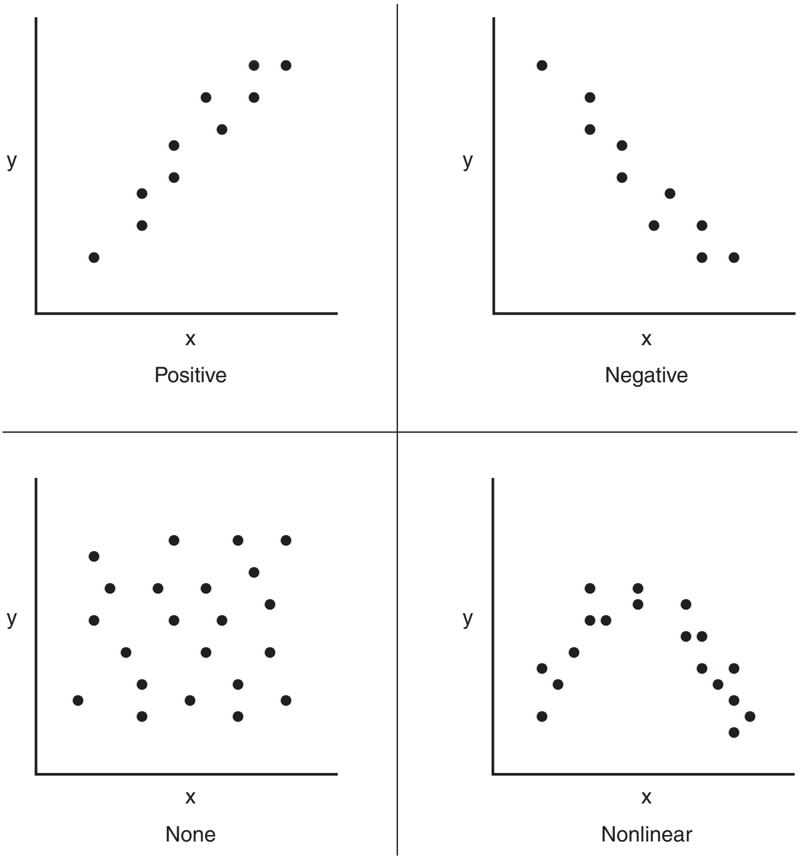


Figure 20.1 Examples of different types of correlations.

EXAMPLE 20.1

Compute the correlation coefficient from the data given in the first two columns of Table 20.1.

Table 20.1 Data for Example 20.1.

	x	y	x^2	xy	y^2
	10	2	100	20	4
	15	3	225	45	9
	20	5	400	100	25
	15	4	225	60	16
Total	60	14	950	225	54

Continued

Continued

For convenience in computing the correlation coefficient, additional columns and totals have been added. From Table 20.1, we find that

$$\sum_{i=1}^{n} x_i = 60,$$
$$\sum_{i=1}^{n} y_i = 14,$$
$$\sum_{i=1}^{n} x_i^2 = 950,$$
$$\sum_{i=1}^{n} x_i y_i = 225,$$
$$\sum_{i=1}^{n} y_i^2 = 54,$$
$$n = 4.$$

Filling in the formula from earlier, we have

$$r = \frac{225 - \frac{(60)(14)}{4}}{\sqrt{\left(950 - \frac{60^2}{4}\right)\left(54 - \frac{14^2}{4}\right)}} = \frac{15}{\sqrt{(50)(5)}} = 0.9489.$$

Another related value is the coefficient of determination, denoted r^2 where $r^2 \leq 1$. The *coefficient of determination* is the square of the correlation coefficient and represents the amount of the variation in y that is explained by the fitted simple linear equation. Similarly, the *adjusted coefficient of determination* (r^2_{adj}) is used with multiple regression because, unlike the R^2 statistic, the adjusted coefficient will increase when variables are added to the model that act to decrease the mean square error. r^2_{adj} is given by

$$r^2_{adj} = 1 - \left(1 - r^2\right)\left[\frac{n-1}{n-(p+1)}\right],$$

where p is the number of coefficients fitted in the regression equation.

Correlation versus Causality

Correlation differs from causality. It is necessary to understand this important distinction in order to use regression analysis effectively. Causality implies correlations, but correlation does not imply causality.

EXAMPLE 20.2

Empirical data demonstrate a strong correlation between height and weight in males. However, gaining weight does not cause one to grow taller. Hence, there is no causal relationship.

EXAMPLE 20.3

Test the hypotheses that

$$H_0 : \rho = 0$$
$$H_A : \rho \neq 0$$

Use the data and result from Example 20.1. Assume $\alpha = 0.10$.

Computing the T we have

$$T = \frac{0.9489\sqrt{4-2}}{\sqrt{1-(0.9489)^2}} = 4.2519$$

From Appendix 14, we find that $t_{\alpha/2,n-2} = t_{0.05,2}$ =2.920. Therefore, we reject the null hypothesis that $\rho = 0$ and conclude that $\rho \neq 0$ at the 0.10 significance level.

Hypothesis Test for the Correlation Coefficient

Recall that r is a sample statistic used to estimate ρ, the population correlation coefficient. As such, it will be useful to test the hypotheses

$$H_0 : \rho = 0$$
$$H_A : \rho \neq 0$$

As you might expect, such a test has been nicely defined. The test statistic for the hypotheses is given by

$$T = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

where T has the t-distribution with $n - 2$ degrees of freedom.

Confidence Interval for the Correlation Coefficient

An approximation for the $100(1 - \sigma)\%$ confidence interval for ρ is given by

$$\tanh(\operatorname{arctanh}(r) - \frac{z_{\alpha/2}}{\sqrt{n-3}}) \leq \rho \leq \tanh(\operatorname{arctanh}(r) + \frac{z_{\alpha/2}}{\sqrt{n-3}})$$

EXAMPLE 20.4

Construct a 100(1 – α)% confidence interval for ρ. Assume $\alpha = 0.10$, $r = 0.95$, and $n = 28$.

Using Appendix 12, we find that $z_{\alpha/2} = z_{0.05} = 1.645$. Filling in the formula from earlier, we have

$$\tanh\left(\operatorname{arctanh}(0.95) - \frac{1.645}{\sqrt{28-3}}\right) \le \rho \le \tanh\left(\operatorname{arctanh}(0.95) + \frac{1.645}{\sqrt{28-3}}\right)$$

$$\tanh\left(1.8318 - \frac{1.645}{5}\right) \le \rho \le \tanh\left(1.8318 + \frac{1.645}{5}\right)$$

$$\tanh(1.5028) \le \rho \le \tanh(2.1608)$$

Applying the identity for the tanh(u), we obtain

$$\frac{e^{1.5028} - e^{-1.5028}}{e^{1.5028} + e^{-1.5028}} \le \rho \le \frac{e^{2.1608} - e^{-2.1608}}{e^{2.1608} + e^{-2.1608}}$$

$$0.9057 \le \rho \le 0.9738$$

where

$$\tanh(u) = \frac{e^{u} - e^{-u}}{e^{u} + e^{-u}}$$

LINEAR REGRESSION

Calculate and interpret regression analysis, and apply and interpret hypothesis tests for regression statistics. Use the regression model for estimation and prediction, analyze the uncertainty in the estimate, and perform a residuals analysis to validate the model. (Evaluate)

Body of Knowledge VI.A.2

Regression analysis is a technique that typically uses continuous predictor variable(s), also called *regressor variables*, to predict the variation in a continuous response variable. Regression analysis uses the method of least squares to determine the values of the linear regression coefficients and the corresponding model.

The *method of least squares* is a technique for estimating a parameter that minimizes the sum of the squared differences between the observed and the predicted values derived from the model. These differences are known as *residuals* or *experimental errors*. Experimental errors associated with individual observations are assumed to be independent and normally distributed. It should be noted that analysis of variance, regression analysis, and analysis of covariance are all based on the method of least squares.

The *linear regression equation* is a mathematical function or model that indicates the linear relationship between a set of predictor variables and a response variable. *Linear regression coefficients* are numbers associated with each predictor variable in a linear regression equation that tell how the response variable changes with each unit increase in the predictor variable. The model also gives some sense of the degree of linearity present in the data.

Simple Linear Regression

When only one predictor variable is used, regression analysis is often referred to as simple linear regression. A simple linear regression model commonly uses a linear regression equation expressed as

$$E[Y|x] = \beta_0 + \beta_1 x + \varepsilon$$

where

γ = the response for a given x value

x = the value of the predictor variable

β_0 and β_1 are the linear regression coefficients

ε = the random error term

$V[Y|x] = \sigma^2$

$\varepsilon_1 \approx N(0,\sigma^2)$

β_0 is often called the *intercept* and β_1 the *slope*.

Applying the method of least squares provides us with a line fitted to the data so that the sum of the squared residuals

$$\sum_{i=1}^{n} \varepsilon_i$$

is minimized. That fitted line takes the form of

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

where

$\hat{y}$ is the response for a given x value

x is the value of the predictor variable

$\hat{\beta}_0$ and $\hat{\beta}_1$ are the linear regression coefficients and

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$
$$\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}}$$

and

$$S_{XY} = \sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n}; S_{XX} = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}$$

Special note: Statistical notation varies greatly among authors. b_0 and b_1 are often used for $\hat{\beta}_0$ and $\hat{\beta}_1$, respectively.

Figure 20.2 illustrates the relationships that exist among the concepts just discussed.

Of course, guessing is not the most desirable way to determine the most appropriate straight line fit of the data. A more appropriate technique is the method of least squares.

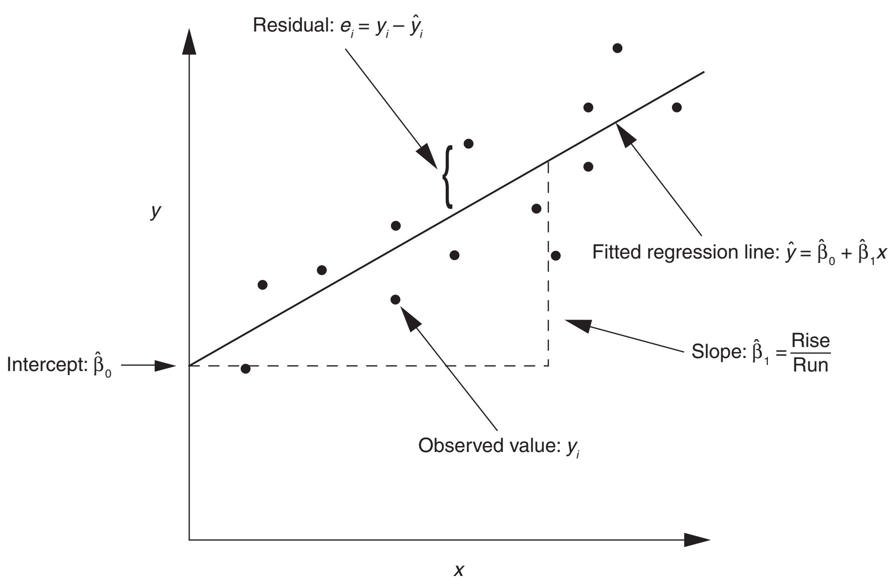


Figure 20.2 Graphical depiction of regression concepts.

EXAMPLE 20.5

Suppose a relationship is suspected between the ambient temperature *T* of a paint booth and paint viscosity *V*. If the relationship holds, then it may be possible to control temperature (predictor variable) in order to control viscosity (response variable).

Recall the data given in the first two columns of Table 20.1. The first step is to plot the data on a scatter diagram to determine whether it is reasonable to approximate them with a straight line. Although a straight line can't be drawn through these four points, the trend looks linear, as shown in Figure 20.3.

The next step is to find the linear equation that best fits the data. We might guess at the following two equations:

1. $V = 1 + 0.2T$
2. $V = -3 + 0.5T$

These lines, along with the data from Table 20.1, are plotted in Figure 20.4.

Which of the lines fits the data better? To answer this question, we first construct a table for each equation comparing the actual value of *V* with the value *V′* predicted by the equation for each value of *T*. Doing so yields the values shown in Table 20.2.

After inspecting these tables closely, it would appear that the first equation fits the data better because the values predicted by the equation seem to come closer to the actual data values. But is there another equation that fits them even better? To properly address this question, it is necessary to have a basis for judging goodness of fit. A method frequently employed is called "least squares." For each line in Table 20.2, the value of residual $\varepsilon = V - V'$ is computed and squared. These values are shown in Table 20.3.

Thus, the equation with the least value of $\Sigma\varepsilon^2$ is the better-fitting equation. The tables confirm the supposition that the first equation fits the data better.

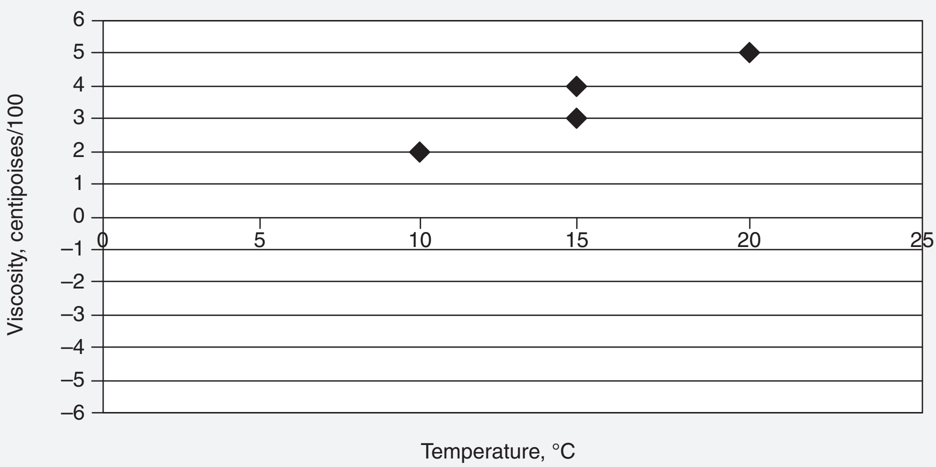


Figure 20.3 Scatter diagram developed from the data given in Table 20.1.

Continued

Continued

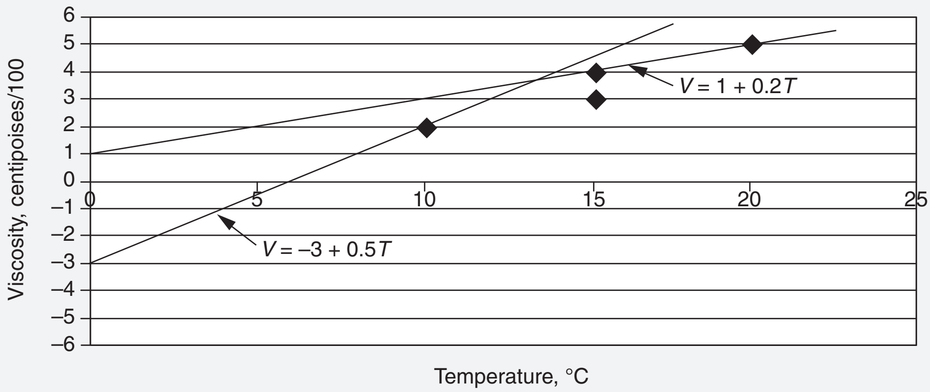


Figure 20.4 Scatter diagram from Figure 20.3 with two proposed lines.

Table 20.2 Computed values for the proposed lines in Figure 20.4.

V′ = 1 + 0.2T				**V′ = –3 + 0.5T**		
T	***V***	***V′***		***T***	***V***	***V′***
10	2	3		10	2	2
15	3	4		15	3	4.5
20	5	5		20	5	7
15	4	4		15	4	4.5

Table 20.3 Computed values for the proposed lines given in Figure 20.4 with residual values added.

V′ = 1 + 0.2*T*						**V′ = –3 + 0.5*T***				
T	***V***	***V′***	***e***	***e²***		***T***	***V***	***V′***	***e***	***e²***
10	2	3	–1	1		10	2	2	0	0
15	3	4	–1	1		15	3	4.5	–1.5	2.25
20	5	5	0	0		20	5	7	–2	4
15	4	4	0	0		15	4	4.5	–0.5	0.25
			$\sum e_i^2$	**2**					$\sum e_i^2$	**6.5**

Method of Least Squares

A straight line has the form $y = b_0 + b_1 x$. Formulas for finding b_0 and b_1 for the best-fitting line are given by

$$b_1 = \frac{S_{xy}}{S_{xx}}$$

$$b_0 = \bar{y} - b_1\bar{x}$$

The supporting formulas are repeated here for convenience:

$$S_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}$$

$$S_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n}$$

$$S_{yy} = \sum_{i=1}^{n} y_i^2 - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n}$$

where n is the number of data points.

EXAMPLE 20.6

Recalling the data given in Table 20.1 and replacing x for T and y for V, we have

$$S_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} = 950 - \frac{60^2}{4} = 50$$

$$S_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n} = 225 - \frac{(60)(14)}{4} = 15$$

$$S_{yy} = \sum_{i=1}^{n} y_i^2 - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n} = 54 - \frac{14^2}{4} = 5$$

Solving for the coefficients, we obtain

$$b_1 = \frac{S_{xy}}{S_{xx}} = \frac{15}{50} = 0.3$$

$$b_0 = \bar{y} - b_1\bar{x} = 3.5 - (0.3)(15) = -1$$

Continued

Continued

Thus, the best-fitting regression equation is given by

$$V = -1 + 0.3T \text{ or } y = -1 + 0.3x$$

and where

$$\sum_{i=1}^{n} e_i^2 = 0.5$$

is minimized, as shown in Table 20.4.

Table 20.4 Residual values for the least squares regression line from Example 20.6.

V′ = 1 + 0.3*T*				
T	*V*	*V′*	*e*	e^2
10	2	2	0	0
15	3	3.5	–0.5	0.25
20	5	5	0	0
15	4	3.5	0.5	0.25
			$\sum e_i^2$	**0.5**

Confidence Interval for the Regression Line

As you might expect, confidence intervals can be calculated about the mean response at a given value of $x = x_0$. The $100(1 - \alpha)\%$ confidence interval is given by

$$\hat{y}_0 - t_{\alpha/2,n-2} s_\varepsilon \sqrt{\frac{1}{n} + \frac{(x_o - \overline{x})^2}{S_{XX}}} \le E\left[Y \middle| x = x_0\right] \le \hat{y}_0 + t_{\alpha/2,n-2} s_\varepsilon \sqrt{\frac{1}{n} + \frac{(x_o - \overline{x})^2}{S_{XX}}}$$

where

$$S_e = \sqrt{\frac{\sum_{i=1}^{n} y_i^2 - n\overline{y}^2 - b_1 S_{XY}}{n-2}}$$

is known as the standard error of the estimate.

EXAMPLE 20.7

Compute the 95% confidence interval for the mean response using the data from Example 20.6 about $x_0 = 18$.

Using the results of Example 20.6 in conjunction with Table 20.1, we can compute

$$S_e = \sqrt{\frac{54-(4)(3.5)^2-(0.2)(15)}{2}} = 0.5$$
$$\hat{y} = -1 + 0.3x = -1 + (0.3)(18) = 4.4$$
$$\bar{x} = 15$$
$$\bar{y} = 3.5$$
$$t_{\alpha/2,n-2} = t_{0.025,2} = 4.303$$
$$n = 4$$

Filling in the numbers for the confidence interval formula given earlier, we have

$$4.4-(4.303)(0.5)\sqrt{\frac{1}{4}+\frac{(18-15)^2}{50}} \le E\left[Y \middle| x = x_0\right] \le 4.4+(4.303)(0.5)\sqrt{\frac{1}{4}+\frac{(18-15)^2}{50}}$$
$$2.9893 \le E\left[Y \middle| x = x_0\right] \le 5.8107$$

It is important to note that this confidence interval applies only to the interval obtained at $x = x_0$, not the entire range of x values.

Prediction Interval for a Future Observation

In addition to confidence intervals, we can calculate a prediction interval about a future observation, y_0, at $x = x_0$. The $100(1 - \alpha)\%$ prediction interval is given by

$$\hat{y}_0 - t_{\alpha/2,n-2} s_\varepsilon \sqrt{1+\frac{1}{n}+\frac{(x_o-\bar{x})^2}{S_{XX}}} \le y_0 \le \hat{y}_0 + t_{\alpha/2,n-2} s_\varepsilon \sqrt{1+\frac{1}{n}+\frac{(x_o-\bar{x})^2}{S_{XX}}}$$

EXAMPLE 20.8

Compute the 95% prediction interval for the next observation using the data from Example 20.7 about $x_0 = 18$.

Filling in the numbers obtained from Example 20.7, we have

$$4.4-(4.303)(0.5)\sqrt{1+\frac{1}{4}+\frac{(18-15)^2}{50}} \le y_0 \le 4.4+(4.303)(0.5)\sqrt{1+\frac{1}{4}+\frac{(18-15)^2}{50}}$$
$$1.8272 \le y_0 \le 6.9728.$$

Notice that the width of the prediction interval is much wider than the corresponding confidence interval at $x = x_0$.

Hypothesis Test for Regression Coefficient b_1

Recall that b_1 represents the slope of the regression equation. Although we can test the slope of the line against any value, we typically test to determine whether the slope is zero. In this case, the following hypothesis test is appropriate:

$$H_0: \beta_1 = 0$$
$$H_A: \beta_1 \neq 0$$

This hypothesis tests whether x can be considered a meaningful predictor of y. In other words, does a relationship exist between the two variables? The corresponding hypothesis test statistic is

$$t = \frac{b_1 - \beta_1}{s_{b_1}}$$

where

$$s_{b_1} = \sqrt{\frac{s_e^2}{S_{xx}}} = \frac{s_e}{\sqrt{S_{xx}}}$$

For a two-sided hypothesis test, the test statistic is compared to $t_{\alpha/2,n-2}$ and rejected if

$$t > t_{\alpha/2,n-2} \text{ or } t < -t_{\alpha/2,n-2}$$

EXAMPLE 20.9

Using the results obtained from the previous examples, test the following hypotheses at $\alpha = 0.05$:

$$H_0: \beta_1 = 0$$
$$H_A: \beta_1 \neq 0$$

Filling in the numbers from the test statistics, we obtain

$$t = \frac{b_1 - \beta_1}{s_{b_1}} = \frac{0.3 - 0}{\frac{0.5}{\sqrt{50}}} = 4.24$$

The critical values of $t_{\alpha/2,n-2}$ are −4.303 and 4.303. Since the test statistics values are within these values, we do not reject H_0.

Multiple Linear Regression

When multiple predictor variables are used, regression is referred to as *multiple linear regression*. For example, a multiple linear regression model with k predictor variables uses a linear regression equation expressed as

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + \varepsilon$$

where

Y is the response

$x_1, x_2, \ldots, x_k$ are the values of the predictor variables

$\beta_0, \beta_1, \beta_2, \ldots, \beta_k$ are the linear regression coefficients

ε is the random error term

The random error terms in regression analysis are assumed to be normally distributed with a constant variance. These assumptions can be readily checked through residual analysis or residual plots. See hypothesis tests for regression coefficients discussed earlier. Note: Other regression techniques are available when the input and/or output variables are not continuous.

Multicollinearity occurs when two or more predictor variables in a multiple regression model are correlated. Multicollinearity may cause the coefficient estimates and significance tests for each predictor variable to be underestimated. Note: The presence of multicollinearity does not reduce the predictive capability of the model—only calculations regarding individual predictor variables. However, including correlated variables does add to the overall model complexity and will likely increase the cost of data collection, perhaps needlessly.

Due to the tediousness of the calculations involved in computing the coefficients in multiple regression, software programs are often employed.

MULTIVARIATE TOOLS

> Use and interpret multivariate tools (e.g., factor analysis, discriminant analysis, multiple analysis of variance (MANOVA)) to investigate sources of variation. (Evaluate)
>
> **Body of Knowledge VI.A.3**

Multivariate analysis comprises a variety of tools used to understand and reduce the data dimensions by analyzing the data covariance structure. The following common tools are used to support multivariate analysis:

- *Principal components* are used "to form a smaller number of uncorrelated variables from a large set of data. The goal of principal components analysis is to explain the maximum amount of variance with the fewest number of principal components." See Minitab 15. Principal components have wide applicability in the social sciences and market research. This technique is not included in the Body of Knowledge.

- *Factor analysis* is used "to determine the underlying factors responsible for correlations in the data." See Minitab 15.
- *Discriminant analysis* is used "to classify observations into two or more groups if you have a sample with known groups. Discriminant analysis can also be used to investigate how variables contribute to group separation." See Minitab 15.
- *Multiple analysis of variance* (MANOVA) can be used to analyze both balanced and unbalanced experimental designs. MANOVA is used to "perform multivariate analysis of variance for balanced designs when you have more than one dependent variable. You can take advantage of the data covariance structure to simultaneously test the equality of means from different responses." See Minitab 15. The advantage of MANOVA over multiple one-way ANOVAs is that MANOVA controls the family error rate. By contrast, multiple one-way ANOVAs work to increase the alpha error rate.

Each of these tools will be explored through the use of examples.

EXAMPLE 20.10

Using the census data collected in Table 20.5, perform a principal components analysis using Minitab. Determine which factors contribute the greatest to the variability.

Setup

Choose *Stat > Multivariate > Principal Components*. Select all five data fields for *Variables*. Select "correlations" for type of matrix. Select "Scree plot" in the *Graphics* dialog box. The resulting analysis is depicted in Figures 20.5 and 20.6.

Analysis

The eigenvalue row in the top table of Figure 20.5 provides the variance of each component. For instance, population has a variance of 3.0289. Notice the second and third row. Population, school, and employment contribute 60.6%, 25.8%, and 11.4%, respectively, to variability. Taken together, the first two factors account for 86.4% of the variability, while the first three account for 97.8%. The scree plot shown in Figure 20.6 is simply a graphic depiction of eigenvalues. Conceptually, this plot is similar to a Pareto chart of variances. The principal component score for factor 1 (that is, population) is given by

$$\text{PC1 Score} = 0.558\textit{Population} + 0.313\textit{School} + 0.568\textit{Employment} \\ + 0.487\textit{Health} - 0.174\textit{Home}$$

Scores for the other factors can be determined in the same manner from the bottom table in Figure 20.5.

Continued

Table 20.5 Census data for Examples 20.10 and 20.11. Obtained from Minitab.

Population	School	Employ	Health	Home
5.935	14.2	2.265	2.27	2.91
1.523	13.1	0.597	0.75	2.62
2.599	12.7	1.237	1.11	1.72
4.009	15.2	1.649	0.81	3.02
4.687	14.7	2.312	2.50	2.22
8.044	15.6	3.641	4.51	2.36
2.766	13.3	1.244	1.03	1.97
6.538	17.0	2.618	2.39	1.85
6.451	12.9	3.147	5.52	2.01
3.314	12.2	1.606	2.18	1.82
3.777	13.0	2.119	2.83	1.80
1.530	13.8	0.798	0.84	4.25
2.768	13.6	1.336	1.75	2.64
6.585	14.9	2.763	1.91	3.17

```
Principal Component Analysis: Pop, School, Employ, Health, Home

Eigenanalysis of the Correlation Matrix

Eigenvalue   3.0289   1.2911   0.5725   0.0954   0.0121
Proportion    0.606    0.258    0.114    0.019    0.002
Cumulative    0.606    0.864    0.978    0.998    1.000


Variable        PC1      PC2      PC3      PC4      PC5
Pop           0.558    0.131   -0.008   -0.551    0.606
School        0.313    0.629    0.549    0.453   -0.007
Employ        0.568    0.004   -0.117   -0.268   -0.769
Health        0.487   -0.310   -0.455    0.648    0.201
Home         -0.174    0.701   -0.691   -0.015   -0.014
```

Figure 20.5 Example of a principal components analysis using the data given in Table 20.5.

Continued

Continued

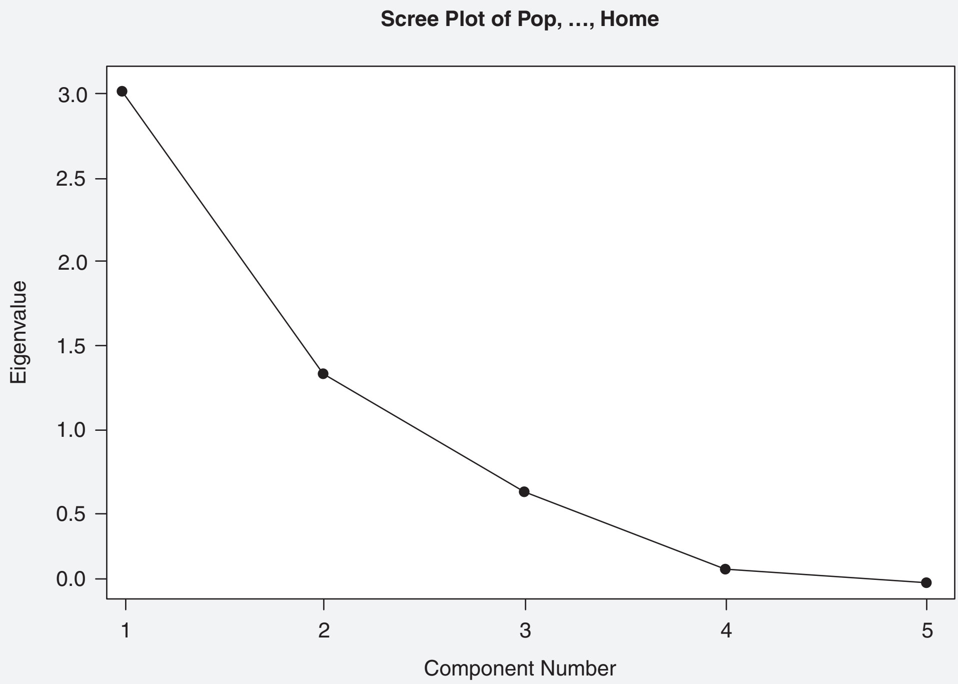


Figure 20.6 Scree plot for Examples 20.10 and 20.11.

EXAMPLE 20.11

Using the census data collected in Table 20.5, perform a factor analysis using Minitab.

Setup:

Choose *Stat > Multivariate > Factor Analysis*. Select all five data fields for *Variables*. Select "Scree plot" in the *Graphics* dialog box. The resulting analysis is depicted in Figures 20.6 and 20.7.

Analysis:

Although the five factors fit the data perfectly, we can see from the top table in Figure 20.7 that the last two factors (health and home) add little to the total variance. As a next step, we might consider performing two separate factor analyses—one for the first two components and one for the first three components.

Continued

Continued

Factor Analysis: Pop, School, Employ, Health, Home

Principal Component Factor Analysis of the Correlation Matrix

Unrotated Factor Loadings and Communalities

Variable	Factor1	Factor2	Factor3	Factor4	Factor5	Communality
Pop	0.972	0.149	-0.006	-0.170	0.067	1.000
School	0.545	0.715	0.415	0.140	-0.001	1.000
Employ	0.989	0.005	-0.089	-0.083	-0.085	1.000
Health	0.847	-0.352	-0.344	0.200	0.022	1.000
Home	-0.303	0.797	-0.523	-0.005	-0.002	1.000
Variance	3.0289	1.2911	0.5725	0.0954	0.0121	5.0000
% Var	0.606	0.258	0.114	0.019	0.002	1.000

Factor Score Coefficients

Variable	Factor1	Factor2	Factor3	Factor4	Factor5
Pop	0.321	0.116	-0.011	-1.782	5.511
School	0.180	0.553	0.726	1.466	-0.060
Employ	0.327	0.004	-0.155	-0.868	-6.988
Health	0.280	-0.272	-0.601	2.098	1.829
Home	-0.100	0.617	-0.914	-0.049	-0.129

Figure 20.7 Example of a factor analysis using the data given in Table 20.5.

EXAMPLE 20.12

Using the salmon data collected in Table 20.6, perform a discriminant analysis using Minitab to be able to classify a newly caught salmon as being from Alaska or Canada.

Setup

Choose *Stat* > *Multivariate* > *Discriminant Analysis*. Select "Salmon Origin" for *Groups*. Select "Freshwater Marine" for *Predictors*. The resulting analysis is depicted in Figure 20.8.

Analysis

The probability of correctly classifying an Alaskan salmon is 88%, while the corresponding probability for a Canadian salmon is 98%. A newly caught fish can be identified by using the linear discriminant functions provided in Figure 20.8. The function providing the highest discriminant value would be used. The two functions are

$$\text{Alaska} = -100.68 + 0.37\textit{Freshwater} + 0.38\textit{Marine}$$
$$\text{Canadian} = -95.14 + 0.50\textit{Freshwater} + 0.33\textit{Marine}$$

Continued

Table 20.6 Salmon data for Example 20.12. Obtained from Minitab.

Salmon origin	Freshwater	Marine	Salmon origin	Freshwater	Marine
Alaska	108	368	Canada	129	420
Alaska	131	355	Canada	148	371
Alaska	105	469	Canada	179	407
Alaska	86	506	Canada	152	381
Alaska	99	402	Canada	166	377
Alaska	87	423	Canada	124	389
Alaska	94	440	Canada	156	419
Alaska	117	489	Canada	131	345
Alaska	79	432	Canada	140	362
Alaska	99	403	Canada	144	345
Alaska	114	428	Canada	149	393
Alaska	123	372	Canada	108	330
Alaska	123	372	Canada	135	355
Alaska	109	420	Canada	170	386
Alaska	112	394	Canada	152	301
Alaska	104	407	Canada	153	397
Alaska	111	422	Canada	152	301
Alaska	126	423	Canada	136	438
Alaska	105	434	Canada	122	306
Alaska	119	474	Canada	148	383
Alaska	114	396	Canada	90	385
Alaska	100	470	Canada	145	337
Alaska	84	399	Canada	123	364
Alaska	102	429	Canada	145	376
Alaska	101	469	Canada	115	354
Alaska	85	444	Canada	134	383
Alaska	109	397	Canada	117	355
Alaska	106	442	Canada	126	345
Alaska	82	431	Canada	118	379
Alaska	118	381	Canada	120	369
Alaska	105	388	Canada	153	403

Continued

Table 20.6 *Continued.*

Salmon origin	Freshwater	Marine	Salmon origin	Freshwater	Marine
Alaska	121	403	Canada	150	354
Alaska	85	451	Canada	154	390
Alaska	83	453	Canada	155	349
Alaska	53	427	Canada	109	325
Alaska	95	411	Canada	117	344
Alaska	76	442	Canada	128	400
Alaska	95	426	Canada	144	403
Alaska	87	402	Canada	163	370
Alaska	70	397	Canada	145	355
Alaska	84	511	Canada	133	375
Alaska	91	469	Canada	128	383
Alaska	74	451	Canada	123	349
Alaska	101	474	Canada	144	373
Alaska	80	398	Canada	140	388
Alaska	95	433	Canada	150	339
Alaska	92	404	Canada	124	341
Alaska	99	481	Canada	125	346
Alaska	94	491	Canada	153	352
Alaska	87	480	Canada	108	339

```
Discriminant Analysis: SalmonOrigin versus Freshwater, Marine

Linear Method for Response: SalmonOrigin


Predictors: Freshwater, Marine


Group     Alaska     Canada
Count         50         50
```

Figure 20.8 Example of a discriminant analysis using the data given in Table 20.6.

Continued

Continued

```
Summary of classification

                   True Group
Put into Group    Alaska   Canada
Alaska              44        1
Canada               6       49
Total N             50       50
N correct           44       49
Proportion        0.880    0.980

N = 100           N Correct = 93          Proportion Correct = 0.930


Squared Distance Between Groups

          Alaska    Canada
Alaska   0.00000   8.29187
Canada   8.29187   0.00000


Linear Discriminant Function for Groups

              Alaska   Canada
Constant     -100.68   -95.14
Freshwater      0.37     0.50
Marine          0.38     0.33


Summary of Misclassified Observations

                                                      Squared
Observation      True Group   Pred Group     Group   Distance    Probability
          1**       Alaska       Canada    Alaska      3.544          0.428
                                           Canada      2.960          0.572
          2**       Alaska       Canada    Alaska     8.1131          0.019
                                           Canada     0.2729          0.981
         12**       Alaska       Canada    Alaska     4.7470          0.118
                                           Canada     0.7270          0.882
         13**       Alaska       Canada    Alaska     4.7470          0.118
                                           Canada     0.7270          0.882
         30**       Alaska       Canada    Alaska      3.230          0.289
                                           Canada      1.429          0.711
         32**       Alaska       Canada    Alaska      2.271          0.464
                                           Canada      1.985          0.536
         71**       Canada       Alaska    Alaska      2.045          0.948
                                           Canada      7.849          0.052
```

Figure 20.8 *Continued.*

EXAMPLE 20.13

Using the plastic film data collected in Table 20.7, perform a MANOVA using Minitab to determine the optimum levels for extruding plastic film.

Setup:

Choose *Stat* > *ANOVA* > *Balanced MANOVA*. Select "Tear," "Gloss," and "Opacity" for *Responses*. Select "Extrusion" and "Additive" for *Model*. Select "Matrices (hypothesis, error, partial correlations)" and "Eigen analysis" in the *Results* dialog box. The resulting analysis is depicted in Figure 20.9.

Analysis:

The *p*-values from the four statistical tests indicate that "Extrusion" and "Additive" are significant at $\alpha = 0.05$, while the interaction term "Extrusion * Additive" is not

Table 20.7 Plastic film data for Example 20.13. Obtained from Minitab.

Tear	Gloss	Opacity	Extrusion	Additive
6.5	9.5	4.4	1	1
6.2	9.9	6.4	1	1
5.8	9.6	3.0	1	1
6.5	9.6	4.1	1	1
6.5	9.2	0.8	1	1
6.9	9.1	5.7	1	2
7.2	10.0	2.0	1	2
6.9	9.9	3.9	1	2
6.1	9.5	1.9	1	2
6.3	9.4	5.7	1	2
6.7	9.1	2.8	2	1
6.6	9.3	4.1	2	1
7.2	8.3	3.8	2	1
7.1	8.4	1.6	2	1
6.8	8.5	3.4	2	1
7.1	9.2	8.4	2	2
7.0	8.8	5.2	2	2
7.2	9.7	6.9	2	2
7.5	10.1	2.7	2	2
7.6	9.2	1.9	2	2

Continued

```
ANOVA: Tear, Gloss, Opacity versus Extrusion, Additive

MANOVA for Extrusion
s = 1    m = 0.5    n = 6.0

                        Test                DF
Criterion          Statistic       F   Num   Denom       P
Wilks'               0.38186   7.554     3      14   0.003
Lawley-Hotelling     1.61877   7.554     3      14   0.003
Pillai's             0.61814   7.554     3      14   0.003
Roy's                1.61877


SSCP Matrix for Extrusion

            Tear    Gloss   Opacity
Tear       1.740   -1.504    0.8555
Gloss     -1.504    1.301   -0.7395
Opacity    0.855   -0.739    0.4205


SSCP Matrix for Error

            Tear     Gloss   Opacity
Tear       1.764    0.0200    -3.070
Gloss      0.020    2.6280    -0.552
Opacity   -3.070   -0.5520    64.924


Partial Correlations for the Error SSCP Matrix

              Tear      Gloss    Opacity
Tear       1.00000    0.00929   -0.28687
Gloss      0.00929    1.00000   -0.04226
Opacity   -0.28687   -0.04226    1.00000


EIGEN Analysis for Extrusion


Eigenvalue   1.619   0.00000   0.00000
Proportion   1.000   0.00000   0.00000
Cumulative   1.000   1.00000   1.00000


Eigenvector          1         2          3
Tear            0.6541    0.4315     0.0604
Gloss          -0.3385    0.5163     0.0012
Opacity         0.0359    0.0302    -0.1209
```

Figure 20.9 Example of MANOVA using the data given in Table 20.7.

Continued

```
MANOVA for Additive
s = 1    m = 0.5    n = 6.0

                       Test                    DF
Criterion         Statistic       F   Num   Denom       P
Wilks'              0.52303   4.256     3      14   0.025
Lawley-Hotelling    0.91192   4.256     3      14   0.025
Pillai's            0.47697   4.256     3      14   0.025
Roy's               0.91192


SSCP Matrix for Additive

            Tear    Gloss   Opacity
Tear      0.7605   0.6825     1.931
Gloss     0.6825   0.6125     1.732
Opacity   1.9305   1.7325     4.901


EIGEN Analysis for Additive


Eigenvalue   0.9119   0.00000   0.00000
Proportion   1.0000   0.00000   0.00000
Cumulative   1.0000   1.00000   1.00000


Eigenvector         1         2         3
Tear          -0.6330    0.4480   -0.1276
Gloss         -0.3214   -0.4992   -0.1694
Opacity       -0.0684    0.0000    0.1102


MANOVA for Extrusion*Additive
s = 1    m = 0.5    n = 6.0

                       Test                    DF
Criterion         Statistic       F   Num   Denom       P
Wilks'              0.77711   1.339     3      14   0.302
Lawley-Hotelling    0.28683   1.339     3      14   0.302
Pillai's            0.22289   1.339     3      14   0.302
Roy's               0.28683


SSCP Matrix for Extrusion*Additive

              Tear     Gloss   Opacity
Tear      0.000500   0.01650   0.04450
Gloss     0.016500   0.54450   1.46850
Opacity   0.044500   1.46850   3.96050
```

Figure 20.9 *Continued.*

Continued

Continued

```
EIGEN Analysis for Extrusion*Additive


Eigenvalue   0.2868   0.00000   0.00000
Proportion   1.0000   0.00000   0.00000
Cumulative   1.0000   1.00000   1.00000


Eigenvector         1         2         3
Tear          -0.1364    0.1806    0.7527
Gloss         -0.5376   -0.3028   -0.0228
Opacity       -0.0683    0.1102   -0.0000
```

Figure 20.9 *Continued.*

significant. The partial correlations matrix indicates that the two partial correlations (gloss-tear and opacity-gloss) are small. Therefore, a univariate ANOVA for the two responses may be acceptable. The eigenvalue analysis provides insight into how the mean responses differ among the levels of the model terms. Only the first eigenvalues for both "Extrusion" and "Additive" are non-zero and therefore have any meaning. The highest absolute value of each eigenvalue for each of the first eigenvectors indicates that "Tear" has the greatest difference in means between the two levels. The second highest is for "Gloss," followed by "Opacity."

Multivariate tools are more complicated than many of the other quantitative tools in the Six Sigma professional's arsenal. Although they are powerful and easily implemented with statistical software, the interpretation of the results is more subjective and caution is required.

MULTI-VARI STUDIES

(Not included in the Body of Knowledge)

Multi-vari analysis is a powerful graphical technique for viewing multiple sources of process variation. Sources of variation are categorized into families of related causes and quantified to reveal the largest causes.

Typical categories are positional, cyclical, and temporal. *Positional variation* is often called within-part variation and refers to variation of a characteristic on the same product. *Cyclical variation* covers part-to-part variation. *Temporal variation* occurs as change over time. For example, a study of the variation in hardness of a metal part might involve collection of hardness data at different locations on the part for positional variation. The study might find average hardness on consecutive parts to detect patterns of cyclical variation. It might also involve the average hardness of parts selected from production on several days, providing information on the temporal variation.

Six Sigma professionals find multi-vari studies useful in their constant battle to reduce variation. This is because once the predominant type of variation has been identified, the list of possible causes is greatly reduced.

EXAMPLE 20.14 (MINI CASE STUDY)

A cast stainless steel part has a tight tolerance on its machined inner diameter because a piston that moves inside must make a good seal. The part is a closed-end cylinder with a cast "ear" on one side and is illustrated in Figure 20.10. The problem is that the pistons are leaking. Machinists consider this a nightmare design because the walls are too thin, the depth is too great, and the dead end adds to the problems. A team has been assigned to work on the problem. One team member thinks the lathe chucks are clamping too tightly and squeezing the part out of round. Another member wants to change the tooling. Many of the ideas have been tried in the past with little improvement.

To initiate a multi-vari study, a data collection scheme that will capture types of variation must be designed. The data can be displayed on graphs that will aid in identifying the largest type of variation. To cover the within-part variation, the inside diameters (IDs) were measured at the top, middle, and bottom, as indicated by the section lines T, M, and B in Figure 20.11. Additional within-part variation was measured by checking for out-of-round conditions. To detect this out-of-round variation, the ID was measured at three different angles: 12 o'clock, 2 o'clock, and 4 o'clock, with 12 o'clock at the ear shown on the top.

To capture the variation over time, five pieces were selected at approximately equal time intervals during a shift. All measurements were obtained with a dial bore gage and recorded on the data sheet shown in Figure 20.11. The measurement results from five parts from one shift are shown in Table 20.8. What can be learned by looking at these numbers? Very little. Plotting the numbers on a graph, as shown in Figure 20.12, does point out an interesting pattern, however.

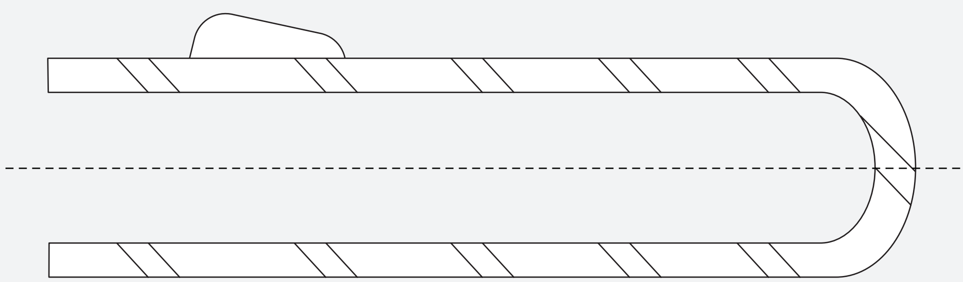
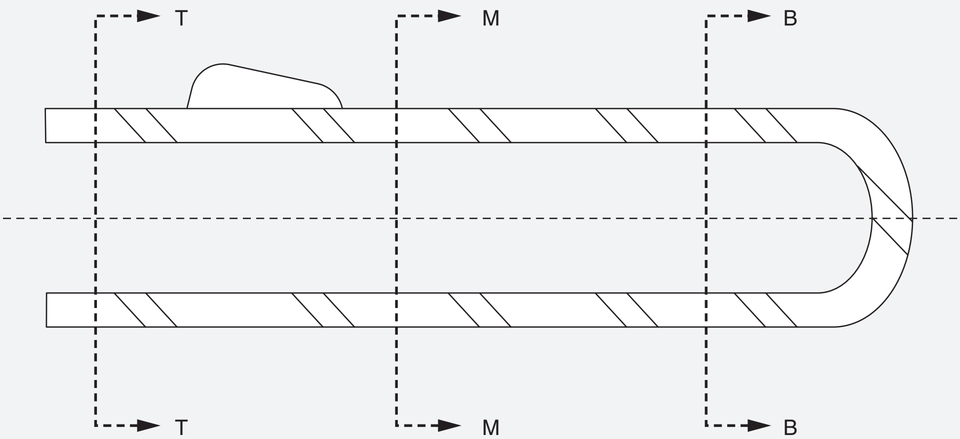
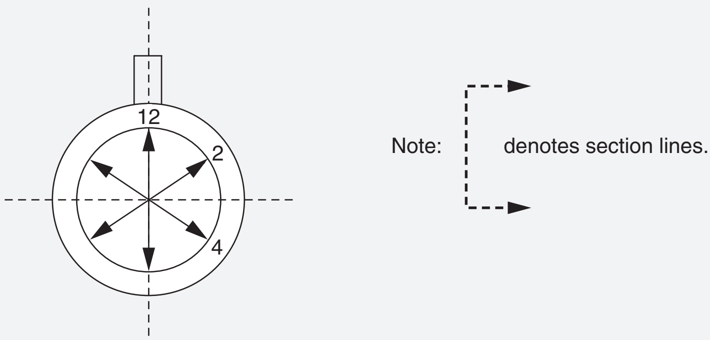


Figure 20.10 Stainless steel casting with critical ID.

Continued





The 12 o'clock measurement is the diameter from 12 o'clock to 6 o'clock.
The 2 o'clock measurement is the diameter from 2 o'clock to 8 o'clock.
The 4 o'clock measurement is the diameter from 4 o'clock to 10 o'clock.

Angle	Section T-T	Section M-M	Section B-B
12 o'clock			
2 o'clock			
4 o'clock			

Figure 20.11 Data collection sheet

Continued

Table 20.8 Casting data for Example 20.14.

	Part #1			**Part #2**			**Part #3**			**Part #4**			**Part #5**		
Angle	***T***	***M***	***B***	***T***	***M***	***B***	***T***	***M***	***B***	***T***	***M***	***B***	***T***	***M***	***B***
12	0.998	0.992	0.996	0.984	0.982	0.981	0.998	0.998	0.997	0.986	0.987	0.986	0.975	0.980	0.976
2	0.994	0.996	0.994	0.982	0.980	0.982	0.999	0.998	0.997	0.985	0.986	0.986	0.976	0.976	0.974
4	0.996	0.994	0.995	0.984	0.983	0.980	0.996	0.996	0.996	0.984	0.985	0.984	0.978	0.980	0.974

Continued

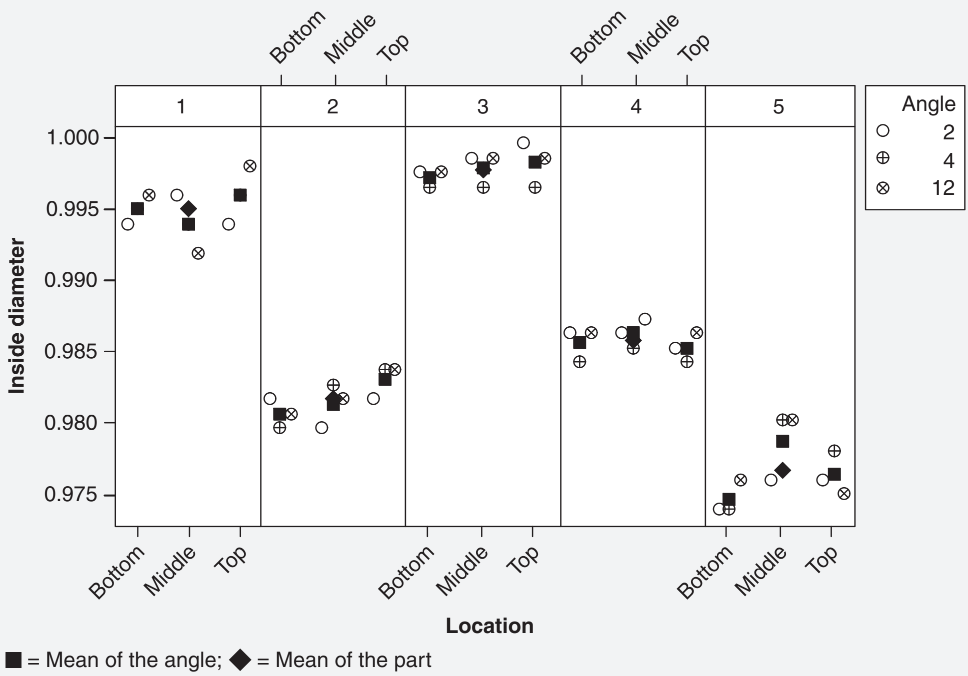


Figure 20.12 Multi-vari chart of data from Table 20.8.

A visual review of Figure 20.12 may help answer the question, Which type of variation is most dominant: out-of-round, top-to-bottom, or part-to-part? Some help may come from drawing a different type of line for each type of variation, as shown in Figure 20.13. In this figure it is fairly clear that the most significant variation is caused by part-to-part differences; therefore, the team brainstormed possible causes for part-to-part variation. People with ideas on how to reduce other variation, such as out-of-round, were asked to hold those ideas until later. It was finally hypothesized that the part-to-part variation was caused by the casting process, and a new foundry that could do precision casting instead of investment casting was located. The new batch of precision cast parts arrived, and again five parts were selected from a shift. The machined parts were measured, and the results are shown in Table 20.9. Notice that a more discriminating measurement system was introduced for these castings. A graph of the data appears in Figure 20.14.

Does it appear that the part-to-part variation has been reduced? Assuming that the part-to-part variation remains relatively small, which type of variation is now the most dominant? Figure 20.14 suggests that the out-of-round variation is now dominant.

The team then began to discuss possible causes for the out-of-round variation. At this point some team members returned to the theory that chuck pressure was causing the part to be squeezed into an out-of-round contour. A round hole was then machined, and when the chuck pressure was released, the part snapped back to a round contour that left an out-of-round hole. They suggested a better pressure regulator for the air

Continued

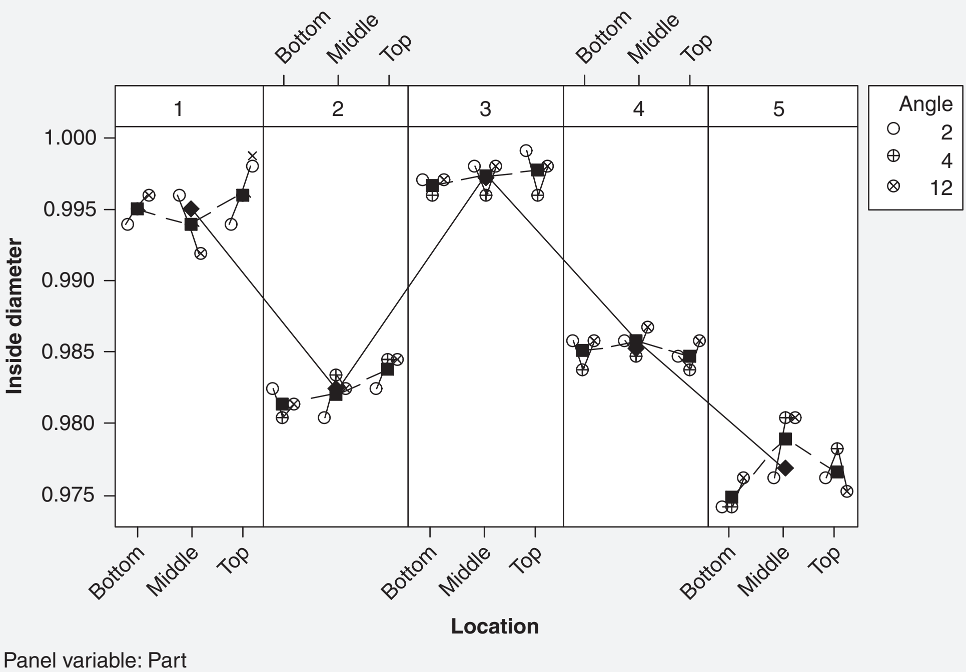


Figure 20.13 Multi-vari chart of data from Table 20.8 with the means of each factor connected.

line feeding the pneumatic chuck. One observant person who focused on the top of the hole, where the out-of-round should be most prominent, noticed that sometimes the 12 o'clock dimension is the longest (as in part #4) and sometimes another dimension is longer (see the numbers in Table 20.9). Since the parts are always chucked with the same orientation, the chuck pressure could not be the cause. The team members wondered if there might be a pattern relating the various orientations and the diameters, so they placed orientation numbers by each dot, as shown in Figure 20.15, and tried to find a pattern. Unfortunately, no pattern was discovered and the relationship seemed to be random.

So the question is, what could be causing random differences in diameters? At this point, a second-shift operator mentioned the fact that after the ID is cut, a burnishing tool is passed across the surface. He said, "I don't think we get all the machining chips out before we burnish, and some of those chips may be burnished into the surface, causing small diameter differences." He suggested that the parts be removed from the chuck and pressure-washed before the burnish operation. Some team members stated that this had been tried a year ago with no noticeable improvement. Others pointed out, however, that at that time the large part-to-part variation would have masked reduction in the minor variation now being seen. The team agreed that the pressure wash should be tried again. The resulting measurements are shown in Table 20.10.

Continued

Table 20.9 Casting data for Example 20.14 with precision parts.

	Part #1			Part #2			Part #3			Part #4			Part #5		
Angle	***T***	***M***	***B***	***T***	***M***	***B***	***T***	***M***	***B***	***T***	***M***	***B***	***T***	***M***	***B***
12	0.9835	0.9845	0.9805	0.9815	0.9850	0.9830	0.9825	0.9820	0.9825	0.9845	0.9840	0.9830	0.9815	0.9810	0.9840
2	0.9820	0.9810	0.9825	0.9850	0.9825	0.9810	0.9850	0.9845	0.9845	0.9810	0.9825	0.9820	0.9845	0.9825	0.9815
4	0.9850	0.9830	0.9835	0.9830	0.9815	0.9845	0.9820	0.9830	0.9825	0.9825	0.9815	0.9810	0.9835	0.9835	0.9820

Continued

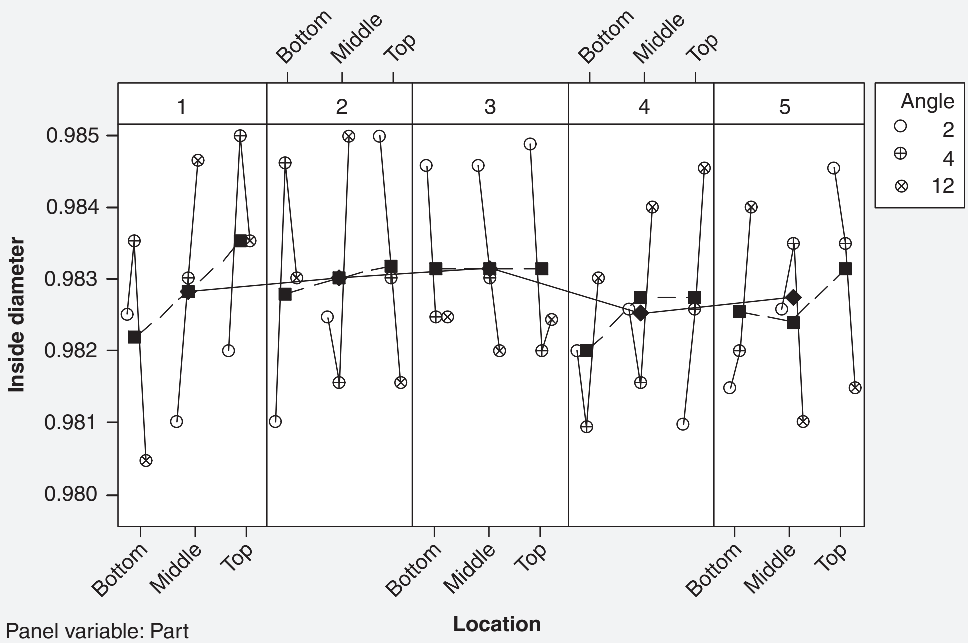


Figure 20.14 Multi-vari chart of data from Table 20.9 with the means of each factor connected.

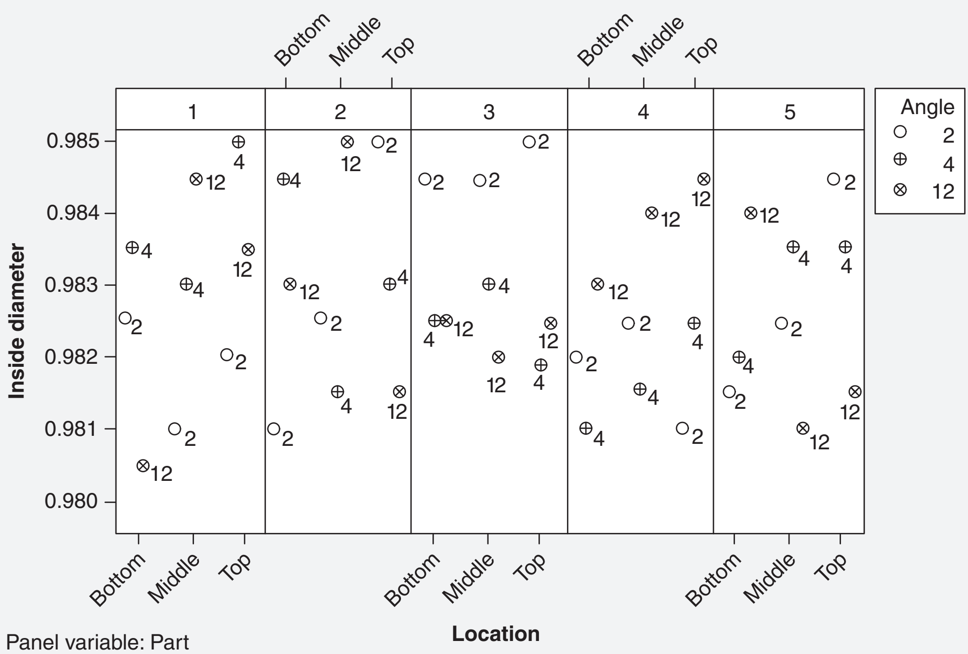


Figure 20.15 Multi-vari chart of data from Table 20.9.

Continued

Table 20.10 Casting data for Example 20.14 after pressure wash.

	Part #1			Part #2			Part #3			Part #4			Part #5		
Angle	***T***	***M***	***B***	***T***	***M***	***B***	***T***	***M***	***B***	***T***	***M***	***B***	***T***	***M***	***B***
12	0.9825	0.9825	0.9815	0.9830	0.9830	0.9815	0.9830	0.9820	0.9820	0.9820	0.9825	0.9820	0.9835	0.9830	0.9820
2	0.9835	0.9830	0.9825	0.9840	0.9830	0.9825	0.9835	0.9825	0.9820	0.9830	0.9825	0.9825	0.9840	0.9830	0.9825
4	0.9820	0.9820	0.9820	0.9832	0.9820	0.9820	0.9825	0.9825	0.9812	0.9820	0.9820	0.9820	0.9835	0.9830	0.9820

Continued

Continued

Plotting these data in Figure 20.16 shows that the out-of-round variation has been noticeably reduced. One of the team members noticed a remaining pattern of variation and suggested a tooling change that reduced the variation even further.

Hopefully, the power of multi-vari analysis is evident from the examples. This simple graphical technique can be quickly applied and potential components of variation readily identified. However, it should be emphasized that this approach is not a substitute for performing an analytical investigation to determine actual statistical significance.

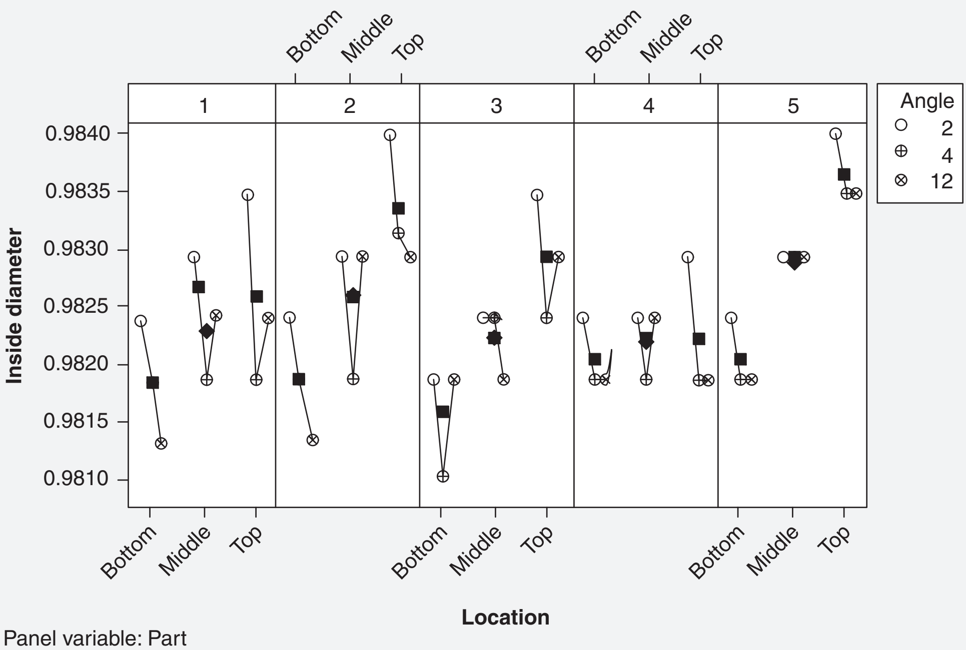


Figure 20.16 Multi-vari chart of data from Table 20.10.

Chapter 21

Hypothesis Testing

TERMINOLOGY

> Define and interpret the significance level, power, type I, and type II errors of statistical tests. (Evaluate)
>
> **Body of Knowledge VI.B.1**

Every hypothesis test uses samples to infer properties of a population on the basis of an analysis of the sampling information. Therefore, there is some chance that although the analysis is flawless, the conclusion may be incorrect. These sampling errors are not errors in the usual sense, because they can't be corrected (without using 100% sampling with no measurement errors).

Two types of errors can be made when testing hypotheses:

- *Type I error.* This error occurs when we reject the null hypothesis when it is true. We refer to the P(type I error) = P(rejecting H_0 when H_0 is true) = α. A type I error is also known as an α *error* and an *error of the first kind*. The P(type I error) is also known as α-*value*, *producer's risk*, *level of significance*, and *significance level* (see the note accompanying the definition of *p*-value).
- *Type II error.* This error occurs when we fail to reject the null hypothesis when it is false. We refer to the P(type II error) = P(not rejecting H_0 when H_0 is false) = β. A type II error is also known as a β *error* and an *error of the second kind*. The P(type II error) is also known as β-*value* or *consumer's risk*.

Figure 21.1 illustrates the relationship between type I and type II errors.

Concepts related to type I and type II errors include *p*-value and power. These terms are defined as:

- p-*value*. This is the *smallest* level of significance leading to the rejection of the null hypothesis. (Note: In Minitab, we look for a *p*-value less than our stated α-value in order to reject the null hypothesis. Some

		Nature (true condition)	
		H_0 is true	H_0 is false
Conclusion	H_0 is rejected	Type I error P (Type I) = a	**Correct decision**
	H_0 is not rejected	**Correct decision**	Type II error P (Type II) = b

Figure 21.1 Four outcomes assoicated with statistical hypotheses.

authors have used a level of significance and/or significance level synonymous with *p*-value.)

- *Power.* This is the probability of rejecting the null hypothesis when the alternative is true. We write this as P(rejecting $H_0 \mid H_0$ is not true) = $1 - \beta$.

The stronger the power of a hypothesis test, the more sensitive it is to detecting small differences.

STATISTICAL VS. PRACTICAL SIGNIFICANCE

> Define, compare, and interpret statistical and practical significance. (Evaluate)
>
> **Body of Knowledge VI.B.2**

In some situations it may be possible to detect a statistically significant difference between two populations when the difference has no practical importance. For example, suppose that a test is devised to determine whether there is a statistically significant difference in the surface finish when a lathe is operated at 400 rpm and at 700 rpm. If large sample sizes are used, it may be possible to determine that the 400 rpm population has a tiny but statistically significant improved surface. However, if both speeds produce surface finishes capable of meeting the specifications, it may be better to go with the faster speed because of its associated increase in throughput. Thus, we must ask ourselves, what does it take to translate a statistically significant difference into a practical significant difference? How much time, effort, cost, and energy are required? Is it worth it? For example, can we just change the setting on the machine to achieve 700 rpm, or is it necessary to purchase another costly machine?

SAMPLE SIZE

> Calculate sample size for common hypothesis tests: equality of means and equality of proportions. (Apply)
>
> **Body of Knowledge VI.B.3**

In order to make a statistically significant decision, it is important that a hypothesis test and confidence intervals use the proper sample size.

Table 21.1 incorporates some of the most commonly used sample size formulas for the following:

- Means
- Difference between means
- Proportions
- Difference between proportions

EXAMPLE 21.1

A Six Sigma professional wants to determine with 95% confidence whether the mean diameter of a fiberoptic cable differs by 0.5 mm. From past experience it is known that $\alpha = 2$ mm. Assume $\alpha = 0.05$.

From Appendix 12, we know that $z_{\alpha/2} = 1.96$. Also, we have

$\alpha = 0.05$

$\sigma = 2$

$E = 0.5$

We will use the following formula from Table 21.1 for our problem:

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2.$$

Filling in the numbers, we obtain

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2 = \left[\frac{(1.96)(2)}{0.5}\right]^2 = 61.47 \Rightarrow 62.$$

Thus, a sample size of 62 is required. The practice is to round the sample size up to the next whole number.

Table 21.1 Sample size formulas. Always round n up to the next highest integer.

Parameter	Sample size formula	Purpose
μ	$n=\left(\dfrac{z_{\alpha/2}\sigma}{E}\right)^2$	Estimating the mean μ of a normal distribution with known variance σ^2. Replace $z_{\alpha/2}$ with z_α for a one-sided alternative hypothesis.
p	$n=\left(\dfrac{z_{\alpha/2}}{E}\right)^2\hat{p}(1-\hat{p})$	Estimating the proportion p with an initial estimate $\hat{p}$. Replace $z_{\alpha/2}$ with z_α for a one-sided alternative hypothesis.
μ	$n=\dfrac{(z_{\alpha/2}+z_\beta)^2\sigma^2}{E^2}; E=\mu-\mu_0$	Detecting a difference of size E from the mean μ from a normal distribution with known variance σ^2. Replace $z_{\alpha/2}$ with z_α for a one-sided alternative hypothesis.
p	$n=\dfrac{z_{\alpha/2}\sqrt{p_0(1-p_0)}+z_\beta\sqrt{p(1-p)}}{p-p_0}$	Detecting a difference in the true population proportion μ.
$\mu_1-\mu_2$	$n=n_1=n_2=\dfrac{(z_{\alpha/2}+z_\beta)^2(\sigma_1^2+\sigma_2^2)}{E^2}$	Detecting a true difference of size E between population means from normal distributions. Replace $z_{\alpha/2}$ with z_α for a one-sided alternative hypothesis.
$\mu_1-\mu_2$	$n=n_1=n_2=\dfrac{(z_{\alpha/2})^2(\sigma_1^2+\sigma_2^2)}{E^2}$	Detecting a difference of size E between population means from normal distributions and known variances σ_1^2 and σ_2^2 when estimating $\mu_1-\mu_2$ by $\bar{X}_1-\bar{X}_2$. Replace $z_{\alpha/2}$ with z_α for a one-sided alternative hypothesis.
p_1-p_2	$n=n_1=n_2=\dfrac{\left[z_{\alpha/2}\sqrt{(p_1+p_2)\dfrac{(q_1+q_2)}{2}}+z_\beta\sqrt{p_1q_1+p_2q_2}\right]^2}{(p_1-p_2)^2}$ $q=1-p$	Detecting a difference between two proportions. Replace $z_{\alpha/2}$ with z_α for a one-sided alternative hypothesis.

EXAMPLE 21.2

A Six Sigma professional wants to determine with 95% confidence whether the true proportion defective in a lot of fiberoptic cable is no larger than 0.05. From past experience, we estimate p using $\hat{p} = 0.15$. Assume $\alpha = 0.05$.

From Appendix 12, we know that $z_{\alpha/2} = 1.96$. Also, we have

$\alpha = 0.05$

$E = 0.05$

We will use the following formula from Table 21.1 for our problem:

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 \hat{p}(1-\hat{p}).$$

Filling in the numbers, we obtain

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 \hat{p}(1-\hat{p}) = \left(\frac{1.96}{0.05}\right)^2 (0.15)(1-0.15) = 196.92 \Rightarrow 197.$$

Thus, a sample size of 197 is required. However, it might be more practical to sample an even 200 units of fiber optic cable.

All of the sample size formulas given in Table 21.1 reflect a two-sided alternative hypothesis. However, it is easy to obtain the one-sided alternative hypothesis formula by simply substituting $z_{\alpha/2}$ with z_α. Note that z_α is *not* changed.

Also note that the formulas given in Table 21.1 will most likely generate non-integer sample sizes. When this occurs, always round up to the next whole number to ensure statistically valid conclusions based on the stated α and β values.

POINT AND INTERVAL ESTIMATES

Define and distinguish between confidence and prediction intervals. Define and interpret the efficiency and bias of estimators. Calculate tolerance and confidence intervals. (Evaluate)

Body of Knowledge VI.B.4

Unbiased Estimators

An *unbiased estimator* is a point estimate of a parameter such that the expected value of the estimate equals the parameter. Mathematically, we write this as

$$E(\hat{\theta}) = \theta.$$

EXAMPLE 21.3

The sample variance is written as

$$s^2 = \frac{\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2}{n-1}$$

Maybe you have wondered why it isn't written as

$$s^2 = \frac{\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2}{n}$$

The reason is that

$$E(s^2) = E\left[\frac{\sum_{i-1}^{n}\left(X_i - \bar{X}\right)^2}{n-1}\right] = \sigma^2$$

The

$$E(s^2) = E\left[\frac{\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2}{n}\right] \neq \sigma^2$$

Readers who are interested in the specific mathematics of why this is true are encouraged to read a more advanced book on statistics.

Efficient Estimators

Efficiency is one measure of an estimator. If we have an unbiased estimator, T, the efficiency of the estimator is given by

$$e(T) = \frac{1/\Psi(\theta)}{\mathrm{var}(T)}$$

such that $e(T) \le 1$ and where $\psi(\theta)$ is known as the *Fisher transformation*. When $e(T) = 1$ for all values of the parameter θ, the estimator is said to be efficient.

Prediction Intervals

A *prediction interval* is similar to a confidence interval, but it is based on the predicted value that is likely to contain the values of future observations. It will be

EXAMPLE 21.4

A six sigma professional wants to determine the 95% prediction interval on the 26th observation of a piece of fiberoptic cable. The 25 observations yielded $\bar{X}$ = 15; s = 2.

From Appendix 14, we know that $t_{\alpha/2} = 2.064$. Also, we have

$\alpha = 0.05$

$n = 25$

$\bar{X} = 15$

$s = 2$

Filling in the numbers, we obtain

$$15-(2.064)(2)\sqrt{1+\frac{1}{25}} \le X_{26} \le 15+(2.064)(2)\sqrt{1+\frac{1}{25}}$$

$$10.7902 \le X_{26} \le 19.2093.$$

wider than the confidence interval because it contains bounds on individual observations rather than a bound on the mean of a group of observations. The 100(1 – α)% prediction interval for a single future observation from a normal distribution is given by

$$\bar{X}-t_{\alpha/2}s\sqrt{1+\frac{1}{n}} \le X_{n+1} \le \bar{X}+t_{\alpha/2}s\sqrt{1+\frac{1}{n}}.$$

Tolerance Intervals

A *tolerance interval* (also known as a *statistical tolerance interval*) is an interval estimator determined from a random sample so as to provide a level of confidence that the interval covers at least a specified proportion of the sampled population. The lower tolerance limit (LTL) and the upper tolerance limit (UTL) are given by

$$\text{LTL: } \bar{X}-\text{K}s$$

$$\text{UTL: } \bar{X}+\text{K}s$$

where

$\bar{X}$ = Sample mean

s = Sample standard deviation

K = Tolerance factor found in Appendixes 26 and 27

Like confidence intervals, tolerance intervals may be one-sided or two-sided.

EXAMPLE 21.5

A 15-piece sample from a process has a mean of $\bar{X} = 10.821$ and a standard deviation of $s = 0.027$. Find the tolerance interval so that there is 95% confidence that it will contain 99% of the population.

From Appendix 27, we know that K = 3.867. Also, we have

$\alpha = 0.05$

$n = 15$

$\bar{X} = 10.821$

$s = 0.027$

Filling in the numbers, we obtain

$$\text{LTL: } \bar{X} - \text{K}s = 10.821 - (3.867)(0.027) = 10.7166$$

$$\text{UTL: } \bar{X} + \text{K}s = 10.821 + (3.867)(0.027) = 10.9254$$

Point Estimates and Confidence Intervals

A *point estimate* is a statistic used to estimate a parameter of a population. Examples include the sample mean and sample proportion defective. However, point estimates provide no information regarding how close the estimate is to the actual population parameter. In other words, how well does $\bar{X}$, the sample mean, estimate μ, the population mean?

To address this question, we turn our attention to one type of interval estimate known as a confidence interval. A *confidence interval* is defined mathematically as

$$P(\text{lower limit} \leq \textit{parameter} \leq \text{upper limit}) = 1 - \alpha$$

Both the lower and upper limits are determined by the parameter and the sample data and are referred to as *confidence limits*. $(1 - \alpha)$ is known as the *confidence level*. Confidence intervals may be one-sided (for example, lower or upper) or two-sided.

In simpler terms, a confidence interval represents the degree of certainty to which the interval contains the true population parameter.

Table 21.2 provides seven commonly used confidence intervals for means.

Table 21.3 provides three commonly used confidence intervals for variances or ratios of variances.

Table 21.4 provides two commonly used confidence intervals for proportions.

Table 21.2 Confidence intervals for means.

Test name	Assumption	100(1 – α)% confidence interval
z-test for a population mean	Variance known, normal	$\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
z-test for two population means	Variances known and equal, normal	$\left(\bar{X}_1 - \bar{X}_2\right) - z_{\alpha/2}\sigma\sqrt{\frac{1}{n_1}+\frac{1}{n_2}} \le (\mu_1 - \mu_2) \le \left(\bar{X}_1 - \bar{X}_2\right) + z_{\alpha/2}\sigma\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$
z-test for two population means	Variances known and unequal, normal	$\left(\bar{X}_1 - \bar{X}_2\right) - z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}} \le (\mu_1 - \mu_2) \le \left(\bar{X}_1 - \bar{X}_2\right) + z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}$
t-test for a population mean	Variance unknown, normal	$\bar{X} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} \le \mu \le \bar{X} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}$ $s = \sqrt{\frac{\left(X_i - \bar{X}\right)^2}{n-1}}$
t-test for two population means	Variances unknown and equal, normal	$\left(\bar{X}_1 - \bar{X}_2\right) - t_{\alpha/2,n_1+n_2-2}s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}} \le (\mu_1 - \mu_2) \le \left(\bar{X}_1 - \bar{X}_2\right) + t_{\alpha/2,n_1+n_2-2}s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$ $s_p = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$

Continued

Table 21.2 *Continued.*

Test name	Assumption	100(1 – α)% confidence interval
t-test for population means	Variances unknown and unequal, normal	$\left(\bar{X}_1 - \bar{X}_2\right) - t_{\alpha/2,\nu}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \le (\mu_1 - \mu_2) \le \left(\bar{X}_1 - \bar{X}_2\right) + t_{\alpha/2,\nu}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$ $\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}{\frac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \frac{\left(s_2^2/n_2\right)^2}{n_2 - 1}}$ Round υ down to the next whole number
t-test for population means (paired *t* test)	Each data element in one population is paired with one data element in the other population, normal	$\bar{D} - t_{\alpha/2,n-1}s_D/\sqrt{n} \le \mu_D \le \bar{D} + t_{\alpha/2,n-1}s_D/\sqrt{n}$ $s_D = \sqrt{\frac{\sum_{i=1}^{n}\left(D_i - \bar{D}\right)^2}{n-1}}$ $\bar{D}$ = Average of the n differences $D_1, D_2, \ldots, D_n$

Table 21.3 Confidence intervals for variances.

Test name	Assumption	100(1 – α)% confidence interval
χ^2 test for one population variance	Sample is drawn from a normal population	$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}$
F test for two population variances	Both samples are taken from a normal distribution	$\frac{s_1^2}{s_2^2} f_{1-\alpha/2,\nu_2,\nu_1} \le \frac{\sigma_1^2}{\sigma_2^2} \le \frac{s_1^2}{s_2^2} f_{\alpha/2,\nu_2,\nu_1}$

EXAMPLE 21.6

Suppose an estimate is needed for the average coating thickness for a population of 1000 circuit boards received from a supplier. Rather than measure the coating thickness on all 1000 boards, one might randomly select a sample of 36 for measurement. Suppose that the average coating thickness on these 36 boards is 0.003, and the standard deviation of the 36 coating measurements is 0.0005. The standard deviation is assumed known from past experience. Determine the 95% confidence interval for the true mean.

From Appendix 12, we know that $z_{\alpha/2}$ =1.96. Also, we have

$\alpha = 0.05$

$\bar{X} = 0.003$

$\sigma = 0.0005$

$n = 36$

We will use the following formula from Table 21.2 for our problem:

$$\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

Filling in the numbers, we obtain

$$0.003-(1.96)\frac{0.0005}{\sqrt{36}} \le \mu \le 0.003+(1.96)\frac{0.0005}{\sqrt{36}}$$

$$0.00284 \le \mu \le 0.00316.$$

Thus, the 95% confidence interval for the mean is (0.00284, 0.00316).

Table 21.4 Confidence intervals for proportions.

Test name	Assumption	100(1 – α)% confidence interval
z-test for a given proportion p_0	Same size, $n \geq 30$	$\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq p \leq \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ $\hat{p} = \frac{X}{n}$; X is the number of observations of interest in the respective sample
z-test of the equality of two proportions	Same size, $n \geq 30$	$(\hat{p}_1 - \hat{p}_2) - z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \leq (p_1 - p_2) \leq (\hat{p}_1 - \hat{p}_2) + z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ $\hat{p}_1 = \frac{X_1}{n_1}; \hat{p}_2 = \frac{X_2}{n_2}; X_1, X_2$ are the numbers of observations of interest in their respective samples

EXAMPLE 21.7

Let's assume the same data given in Example 21.6 with the exception that the variance is not known and our sampling process yields $s = 0.1$. Determine the 95% confidence interval for the true variance.

From Appendix 15, we know that $\chi^2_{\alpha/2,n-1} = 53.203$; $\chi^2_{1-\alpha/2,n-1} = 20.569$. Also, we have

$\alpha = 0.05$

$\overline{X} = 0.003$

$s = 0.1$

$n = 36$

We will use the following formula from Table 25.3 for our problem:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}.$$

Filling in the numbers, we obtain

$$\frac{(35)(0.1)^2}{53.203} \leq \sigma^2 \leq \frac{(35)(0.1)^2}{20.569}$$

$$0.00658 \leq \sigma^2 \leq 0.01702.$$

Thus, the 95% confidence interval for the variance is (0.00658, 0.01702).

EXAMPLE 21.8

Let's assume the same data given in Example 21.6 with the exception that our sampling process yields four defectives. Determine the 95% confidence interval for the true proportion.

From Appendix 12, we know that $z_{\alpha/2} = 1.96$. Also, we have

$$\alpha = 0.05,$$

$$\hat{p} = \frac{4}{36} = 0.1111,$$

$$n = 36.$$

We will use the following formula from Table 25.4 for our problem:

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

Filling in the numbers, we obtain

$$0.1111-(1.96)\sqrt{\frac{(0.1111)(1-0.1111)}{36}} \le p \le 0.1111+(1.96)\sqrt{\frac{(0.1111)(1-0.1111)}{36}}$$

$$0.00844 \le p \le 0.21376.$$

Thus, the 95% confidence interval for the proportion is (0.00844, 0.21376).

TESTS FOR MEANS, VARIANCES, AND PROPORTIONS

Use and interpret the results of hypothesis tests for means, variances, and proportions. (Evaluate)

Body of Knowledge VI.B.5

Hypothesis Tests

To ensure that hypothesis tests are carried out properly, it is useful to have a well-defined process for conducting them:

1. Specify the parameter to be tested
2. State the null and alternative hypotheses (H_0, H_A)
3. State the α value
4. Determine the test statistic

5. Define the rejection criteria
6. Compute the critical values
7. Compute the test statistic
8. State the conclusion of the test

Note that the null hypothesis is either rejected or not rejected. It is never accepted.

Many Six Sigma professionals like to conduct hypothesis tests using a two-sided alternative hypothesis. On the surface this may seem correct. However, one should recognize that the job of the Six Sigma professional is to improve things. This suggests a one-sided hypothesis test. Of course, the direction of the alternative hypothesis depends on which direction represents "better." This does not mean that we should abandon the two-sided alternative hypothesis; it just means that we need to think more carefully about which alternative hypothesis is more appropriate.

Table 21.5 provides seven commonly used hypothesis tests for means.

Table 21.6 provides three commonly used confidence intervals for variances or ratios of variances.

Table 21.7 provides two commonly used confidence intervals for proportions.

EXAMPLE 21.9

We have just received a lot of 1000 circuit boards. The supplier is expected to coat the boards to a thickness of 0.0025 with a standard deviation of 0.0005. The standard deviation is assumed known from past experience. After randomly sampling 36 boards, we compute a mean thickness of 0.003. Has there been a change in the mean thickness?

1. The parameter to be tested is μ.
2. $H_0 : \mu = \mu_0$; $H_A : \mu \neq \mu_0$
3. $\alpha = 0.05$.
4. $z_0 = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$
5. Reject H_0 if $z_0 < -z_{\alpha/2}$ or $z_0 > z_{\alpha/2}$
6. From Appendix 12, $z_{0.05} = 1.645$
7. $z_0 = \dfrac{0.003 - 0.0025}{0.0005/\sqrt{36}} = 6$
8. Since the test statistic falls in the reject region, we reject the null hypothesis and conclude that there has been a change in the mean thickness of the coating on the circuit boards.

Table 21.5 Hypothesis tests for means.

Test name	Assumption	Test statistic	Null hypothesis	Alternative hypotheses	Rejection criteria
z-test for a population mean	Variance known, normal	$z_0 = \dfrac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$	$\mu = \mu_0$	$\mu \neq \mu_0$ $\mu > \mu_0$ $\mu < \mu_0$	$\lvert z_0 \rvert > z_{\alpha/2}$ $z_0 > z_\alpha$ $z_0 < -z_\alpha$
z-test for two population means	Variances known and equal, normal	$z_0 = \dfrac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}$ $\delta = \mu_1 - \mu_2$	$\mu_1 - \mu_2 = \delta$	$\mu_1 - \mu_2 \neq \delta$ $\mu_1 - \mu_2 > \delta$ $\mu_1 - \mu_2 < \delta$	$\lvert z_0 \rvert > z_{\alpha/2}$ $z_0 > z_\alpha$ $z_0 < -z_\alpha$
z-test for two population means	Variances known and unequal, normal	$z_0 = \dfrac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$ $\delta = \mu_1 - \mu_2$	$\mu_1 - \mu_2 = \delta$	$\mu_1 - \mu_2 \neq \delta$ $\mu_1 - \mu_2 > \delta$ $\mu_1 - \mu_2 < \delta$	$\lvert z_0 \rvert > z_{\alpha/2}$ $z_0 > z_\alpha$ $z_0 < -z_\alpha$
t-test for a population mean	Variance unknown, normal	$t_0 = \dfrac{\bar{X} - \mu_0}{s / \sqrt{n}}$ $s = \sqrt{\dfrac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}}$	$\mu = \mu_0$	$\mu \neq \mu_0$ $\mu > \mu_0$ $\mu < \mu_0$	$\lvert t_0 \rvert > t_{\alpha/2,n-1}$ $t_0 > t_{\alpha,n-1}$ $t_0 < -t_{\alpha,n-1}$

Continued

Table 21.5 *Continued*

Test name	Assumption	Test statistic	Null hypothesis	Alternative hypotheses	Rejection criteria
t-test for two population means	Variances unknown and equal, normal	$t_0 = \dfrac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{s_p\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}$ $\delta = \mu_1 - \mu_2$ $s_p = \dfrac{(n_1 - 1)s_1^1 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$	$\mu_1 - \mu_2 = \delta$	$\mu_1 - \mu_2 \neq \delta$ $\mu_1 - \mu_2 > \delta$ $\mu_1 - \mu_2 < \delta$	$\lvert t_0 \rvert > t_{\alpha/2, n_1+n_2-2}$ $t_0 > t_{\alpha, n_1+n_2-2}$ $t_0 < -t_{\alpha, n_1+n_2-2}$
t-test for population means	Variances unknown and unequal, normal	$t_0 = \dfrac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$ $\delta = \mu_1 - \mu_2$ $\nu = \dfrac{\left(\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}\right)^2}{\dfrac{(s_1^2/n_1)^2}{n_1 - 1} + \dfrac{(s_2^2/n_2)^2}{n_2 - 1}}$ Round ν down to the next whole number	$\mu_1 - \mu_2 = \delta$	$\mu_1 - \mu_2 \neq \delta$ $\mu_1 - \mu_2 > \delta$ $\mu_1 - \mu_2 < \delta$	$\lvert t_0 \rvert > t_{\alpha/2, \nu}$ $t_0 > t_{\alpha, \nu}$ $t_0 < -t_{\alpha, \nu}$

Continued

Table 21.5 *Continued.*

Test name	Assumption	Test statistic	Null hypothesis	Alternative hypotheses	Rejection criteria
t-test for population means (paired t-test)	Each data element in one population is paired with one data element in the other population, normal	$t_0 = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta}{s_D / \sqrt{n}}$ $s_D = \sqrt{\dfrac{\sum_{i=1}^{n}(D_i - \bar{D})^2}{n-1}}$ $\bar{D}$ = Average of the n differences D_1, D_2, …, D_n	$\mu_D = \delta$	$\mu_D \neq \delta$ $\mu_D > \delta$ $\mu_D < \delta$	$\lvert t_0 \rvert > t_{\alpha/2,n-1}$ $t_0 > t_{\alpha,n-1}$ $t_0 < -t_{\alpha,n-1}$

Table 21.6 Hypothesis tests for variances or ratios of variances.

Test name	Assumption	Test statistic	Null hypothesis	Alternative hypotheses	Rejection criteria
χ^2 test between a sample variance and an assumed population variance σ_0^2	Sample is drawn from a normal population	$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2}$	$\sigma^2 = \sigma_0^2$	$\sigma^2 \neq \sigma_0^2$ $\sigma^2 > \sigma_0^2$ $\sigma^2 < \sigma_0^2$	$\chi_0^2 > \chi^2_{\alpha/2,n-1}$ or $\chi_0^2 < \chi^2_{1-\alpha/2,n-1}$ $\chi_0^2 > \chi^2_{\alpha,n-1}$ $\chi_0^2 > \chi^2_{1-\alpha,n-1}$
F-test for two population variances	Both samples are taken from a normal distribution	$F_0 = \frac{s_1^2}{s_2^2}$ $f_{1-\alpha,\nu_1,\nu_2} = \frac{1}{f_{\alpha,\nu_2,\nu_1}}$	$\sigma_1^2 = \sigma_2^2$	$\sigma_1^2 \neq \sigma_2^2$ $\sigma_1^2 > \sigma_2^2$ $\sigma_1^2 < \sigma_2^2$	$F_0 > f_{\alpha/2,n_1-1,n_2-1}$ or $F_0 > f_{1-\alpha/2,n_1-1,n_2-1}$ $F_0 > f_{\alpha,n_1-1,n_2}$ $F_0 > f_{1-\alpha,n_1-1,n_2-1}$
χ^2 test between a population variance and an assumed variance σ_0^2	Sample is taken from a normal distribution	$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2}$	$\sigma^2 = \sigma_0^2$	$\sigma^2 \neq \sigma_0^2$ $\sigma^2 > \sigma_0^2$ $\sigma^2 < \sigma_0^2$	$\chi_0^2 > \chi^2_{\alpha/2,n-1}$ or $\chi_0^2 < \chi^2_{1-\alpha/2,n-1}$ $\chi_0^2 > \chi^2_{\alpha,n-1}$ $\chi_0^2 > \chi^2_{1-\alpha,n-1}$

Table 21.7 Hypothesis tests for proportions.

Test name	Assumption	Test statistic	Null hypothesis	Alternative hypotheses	Rejection criteria
z-test for a given proportion p_0	Same size, $n \geq 30$	$z_0 = \dfrac{p - p_0}{\sqrt{\dfrac{p_0(1-p_0)}{n}}}$	$p = p_0$	$p \neq p_0$ $p > p_0$ $p < p_0$	$\lvert z_0 \rvert > z_{\alpha/2}$ $z_0 > z_\alpha$ $z_0 < -z_\alpha$
z-test of the equality of two proportions	Same size, $n \geq 30$	$z_0 = \dfrac{p_1 - p_2}{\sqrt{P(1-P)\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}}$ $P = \dfrac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$	$p_1 = p_2$	$p_1 \neq p_2$ $p_1 > p_2$ $p_1 < p_2$	$\lvert z_0 \rvert > z_{\alpha/2}$ $z_0 > z_\alpha$ $z_0 < -z_\alpha$

EXAMPLE 21.10

Recently, a Six Sigma Black Belt project was undertaken to reduce the variance of the coating on circuit boards from 0.0005. We have just received a lot of 1000 circuit boards. After randomly sampling 36 boards, we compute a mean thickness of 0.003 and a variance of 0.00025. Has the variance been reduced ?

1. The parameter to be tested is σ^2.
2. $H_0 : \sigma^2 = \sigma_0^2$; $H_A : \sigma^2 < \sigma_0^2$
3. $\alpha = 0.05$
4. $\chi_0^2 = \dfrac{(n-1)s^2}{\sigma_0^2}$
5. Reject H_0 if $\chi_0^2 < \chi_{1-\alpha,n-1}^2$
6. From Appendix 15, $\chi_{0.95,35}^2 = 22.465$
7. $\chi_0^2 = \dfrac{(35)(0.00025)}{0.0005} = 17.5$
8. Since the test statistic falls in the reject region, we reject the null hypothesis and conclude that there has been a reduction in the variance of the coating thickness.

EXAMPLE 21.11

Let's assume the same data given in Example 21.9 with the exception that our sampling process yields two defectives. We want to determine whether our recent improvement efforts have been successful in reducing the proportion defective from 0.10. Has the proportion defective been reduced?

1. The parameter to be tested is p.
2. $H_0 : p = p_0$; $H_A : p < p_0$
3. $\alpha = 0.05$
4. $z_0 = \dfrac{p - p_0}{\sqrt{\dfrac{p_0(1-p_0)}{n}}}$
5. Reject H_0 if $z_0 < -z_\alpha$
6. From Appendix 15, $-z_{0.05} = -1.645$
7. $z_0 = \dfrac{0.0556 - 0.10}{\sqrt{\dfrac{(0.10)(1-0.10)}{36}}} = -0.888$
8. Since the test statistic does not fall in the reject region, we do not reject the null hypothesis and conclude there is not sufficient evidence to conclude that there has been a change in the proportion defective.

Figures 21.2–21.4 provide a simplified flowchart to help determine which hypothesis test to use.

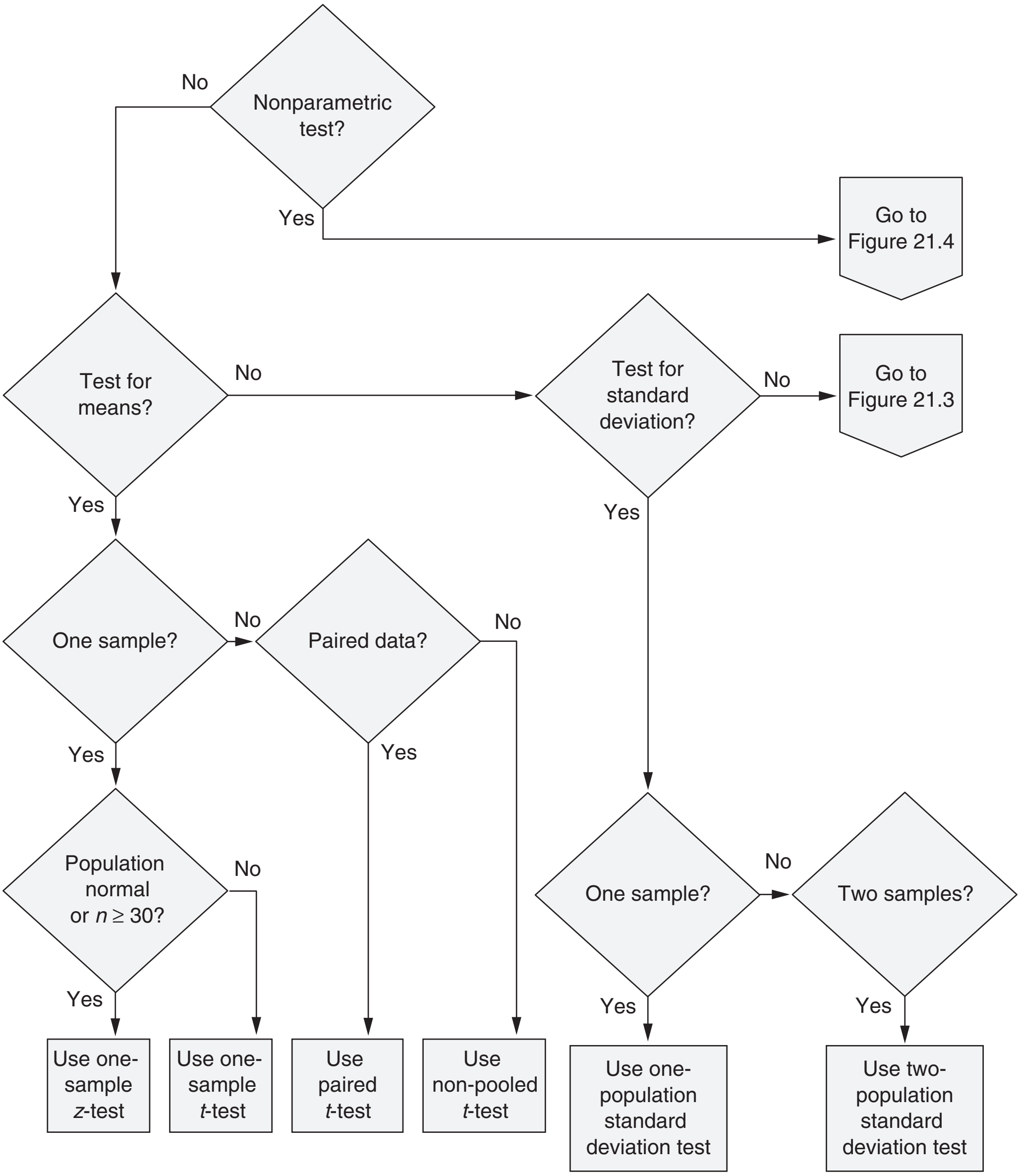


Figure 21.2 Hypothesis test flowchart (part 1).

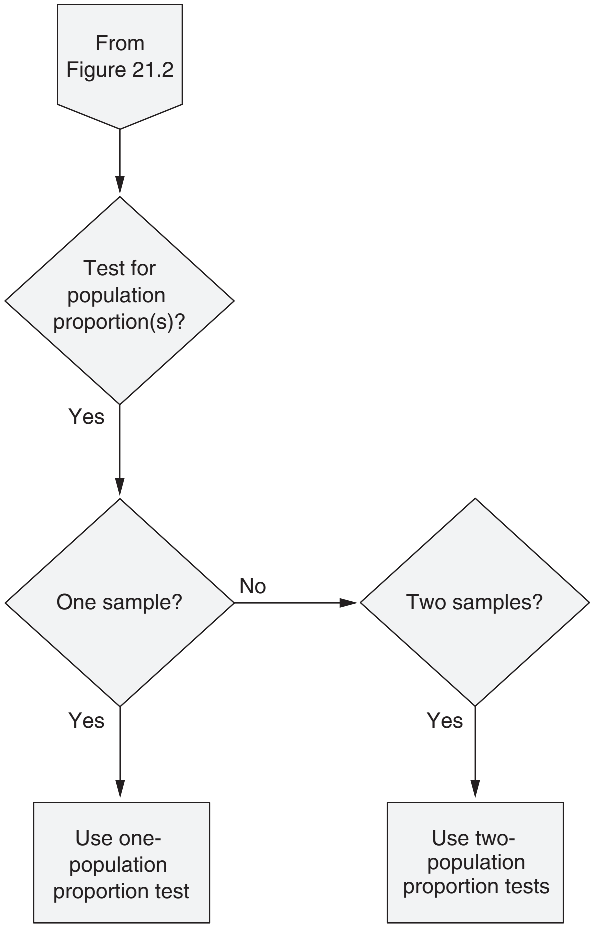


Figure 21.3 Hypothesis test flowchart (part 2).

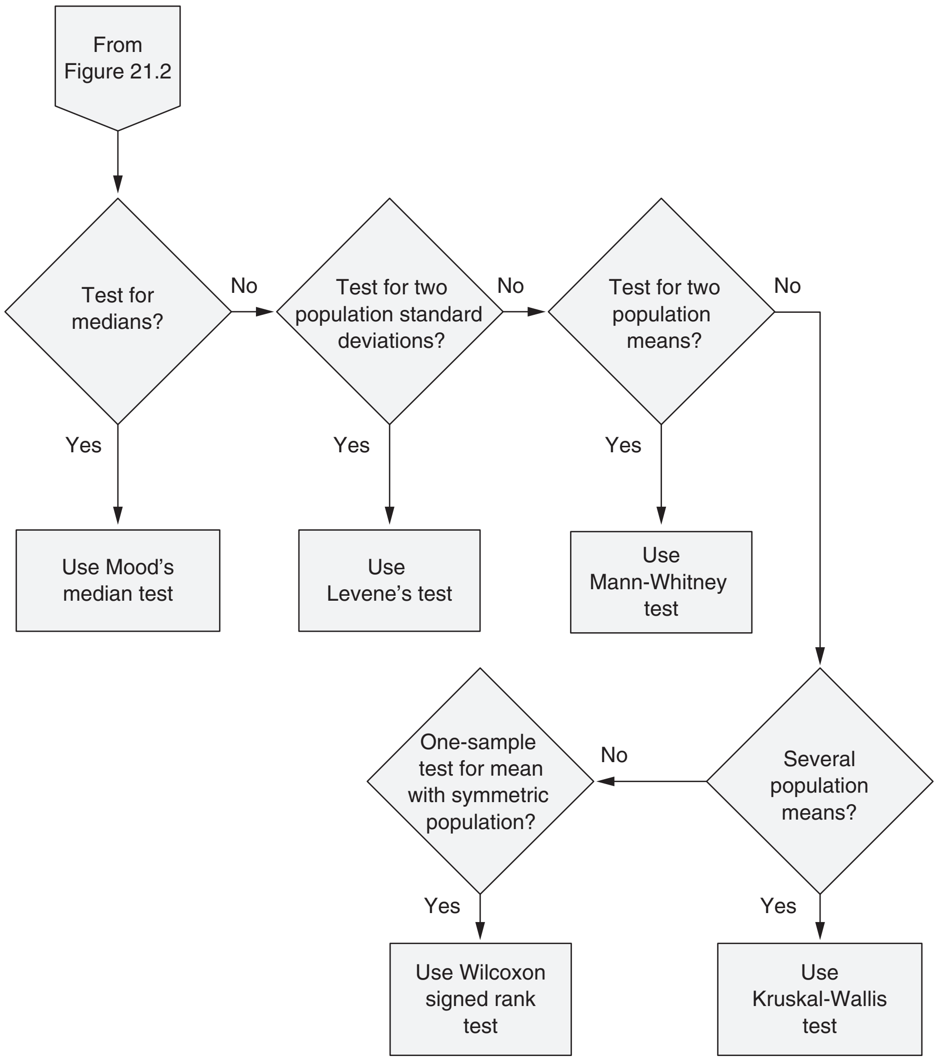


Figure 21.4 Hypothesis test flowchart (part 3).

ANALYSIS OF VARIANCE (ANOVA)

> Select, calculate, and interpret the results of ANOVAs. (Evaluate)
>
> **Body of Knowledge VI.B.6**

The ANOVA procedures extend the hypothesis tests for means that were discussed in the previous section to the means of more than two populations.

One-Way ANOVA

The one-way ANOVA procedure is used to determine whether the data from three or more populations formed by the treatment options from a single-factor designed experiment indicate that the population means are different. A treatment is a specific factor level. The hypotheses being tested are

$H_0 : \mu_1 = \mu_2 = \ldots = \mu_k$

H_A : Not all means are equal

From these hypotheses, we know that there are k factor levels.

Essentially, the idea behind ANOVA is to determine whether the variation due to the treatments is sufficiently large compared with the experimental error (that is, within-treatment variation). Although the null hypothesis is stated in terms of means, this determination is made through the use of the F-test, described previously.

To ensure that the ANOVA is carried out properly, it is useful to have a well-defined process for conducting it:

1. State the null and alternative hypotheses (H_0, H_A)
2. State the α value
3. Determine the test statistic
4. Define the rejection criteria
5. Compute the critical values
6. Construct the source table
7. Compute the test statistics
8. State the conclusion of the test using the ANOVA source table.

Table 21.8 illustrates an ANOVA source table where

k = Number of treatments

n = Number of readings per treatment

$N = kn$ = Total number of readings

Table 21.8 Example of a one-way ANOVA source table.

Source of variation	Sum of squares	Degrees of freedom	Mean squares	*F*-statistic
Between treatments	SS_B	$k-1$	$MS_B = \dfrac{SS_B}{k-1}$	$F_0 = \dfrac{MS_B}{MS_E}$
Within treatments (error)	SS_E	$N-k$	$MS_E = \dfrac{SS_E}{N-k}$	
Total	$\mathbf{SS_T}$	$\mathbf{N-1}$		

$$\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij} = \text{Total of all responses}$$

$$SS_T = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{..}^2}{N}$$

$$SS_{\text{Treatments}} = \sum_{i=1}^{k} \frac{y_i^2}{n} - \frac{y_{..}^2}{N}$$

$$SS_E = SS_T - SS_{\text{Treatments}}$$

$$y_{ij} = i\text{th treatment, } j\text{th measurement}$$

The underlying assumptions of ANOVA are that the data are independent, are normally distributed, and are of equal variance.

EXAMPLE 21.12

A process can be run at any of three temperatures: 180 °F, 200 °F, and 220 °F. Using the data in Table 21.9 and assuming $\alpha = 0.05$, determine whether temperature significantly affects the moisture content.

Table 21.9 Moisture content data for Example 21.12.

	Temperature		
Level	**180**	**200**	**220**
1	10.8	11.4	14.3
2	10.4	11.9	12.6
3	11.2	11.6	13.0
4	9.9	12.0	14.2

Continued

Continued

From the problem statement, we can determine

$$\alpha = 0.05$$

$$k = 3$$

$$n = 4$$

$$N = kn = 12$$

$$\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij} = 143.3$$

$$SS_T = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{..}^2}{N} = 1732.3 - \frac{(143.3)^2}{12} = 1732.27 - 1711.24 = 21.03$$

$$SS_{\text{Treatments}} = \sum_{i=1}^{k} \frac{y_i^2}{n} - \frac{y_{..}^2}{N} = 1728.93 - 1711.24 = 17.69$$

$$SS_E = SS_T - SS_{\text{Treatments}} = 21.03 - 17.69 = 3.34$$

Let's complete the ANOVA steps:

1. $H_0 = \mu_1 = \mu_2 = \ldots = \mu_k$; H_A = Not all means are equal
2. $\alpha = 0.05$
3. F-statistic given in Table 21.8
4. Reject H_0 if $F_0 > F_{\alpha,k-1,N-k}$
5. From Appendix 21, the critical value is $F_{0.05,2,9} = 4.26$
6. See Table 21.10 for the completed ANOVA source table
7. From Table 21.10, the test statistic is 23.92
8. Since the test statistic, 23.92, exceeds the critical value, 4.26, we reject the null hypothesis and conclude that temperature affects the moisture content

Table 21.10 Completed one-way ANOVA source table for the data given in Table 21.9.

Source of variation	Sum of squares	Degrees of freedom	Mean squares	*F*-statistic
Between treatments	17.69	2	8.85	23.92
Within treatments (error)	3.34	9	0.37	
Total	**21.03**	**11**		

Table 21.11 Example of a two-way ANOVA source table.

Source of variation	Sum of squares	Degrees of freedom	Mean squares	*F*-statistic
Factor A	SS_A	$a - 1$	$MS_A = \frac{SS_A}{a-1}$	$F_0 = \frac{MS_A}{MS_E}$
Factor B	SS_B	$b - 1$	$MS_B = \frac{SS_B}{b-1}$	$F_0 = \frac{MS_B}{MS_E}$
Interaction of factors A and B	SS_{AB}	$(a - 1)(b - 1)$	$MS_{AB} = \frac{SS_{AB}}{(a-1)(b-1)}$	$F_0 = \frac{MS_{AB}}{MS_E}$
Error	SS_E	$(N - ab)$	$MS_E = \frac{SS_E}{N-ab}$	
Total	$\boldsymbol{SS_T}$	$\boldsymbol{N - 1}$		

Two-Way ANOVA

The two-way ANOVA test can be used in the analysis of two-factor experiments. It differs from the one-way ANOVA procedure only in the details of the ANOVA table, so this discussion will focus on construction of the table.

In two-factor experiments, the effects to be analyzed are the main effects of factor A, the main effects of factor B, and the interaction between them (as will be discussed in Chapter 24). The two-way ANOVA source table is given in Table 21.11.

The structure of the table is similar to that of the one-way ANOVA table. The *F*-statistics in the last column are compared with the critical values in an *F*-table to determine whether there is a statistically significant effect due to factor A, factor B, or the interaction between them. Formulas for the sum of squares values are similar to those for one-way ANOVA. The equations for the two-way ANOVA are not provided. Though they are similar to the equations for the one-way ANOVA, they are tedious to apply. As a result, most two-way ANOVAs are computed using readily available statistical software.

As with the one-way ANOVA, the *F*-statistic measures the ratio between the effect and the experimental error. If the variation due to the effect is sufficiently large relative to the experimental error, the effect is considered statistically significant at a specified α significance level.

GOODNESS-OF-FIT (CHI SQUARE) TESTS

> Define, select, and interpret the results of these tests. (Evaluate)
>
> **Body of Knowledge VI.B.7**

χ^2 Goodness-of-Fit Test

The purpose of the χ^2 goodness-of-fit test is to test the hypotheses

H_0 = The data follow a specified distribution

H_A = The data do not follow a specified distribution

To conduct the χ^2 goodness-of-fit test:

1. State the null and alternative hypotheses (H_0, H_A).
2. Arrange a random sample of size n into a frequency histogram of k class intervals.
3. Determine O_i = Observed frequency in the ith class interval.
4. Determine E_i = Expected frequency in the ith class interval using the hypothesis distribution. Note: It is generally accepted that the expected frequencies should be no less than five. If they are less, adjacent class intervals should be combined until the recommended minimum of five is obtained.
5. State the α value.
6. Compute the test statistic $\chi_0^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$
7. Compute the critical value. Reject H_0 if $\chi_0^2 > \chi^2_{\alpha,k-p-1}$. The value of p is the number of parameters estimated. k is the number of class intervals or terms in the test statistic.
8. State the conclusion of the test.

EXAMPLE 21.13

Table 21.12 provides historical data of defect types for a consumer product. Each defective product has been categorized into one of four defect types. Percentages of each defect type have been computed from the historical data. Additionally, current data have been randomly selected and are also depicted in Table 21.12. Is the distribution that produced the current defect data the same as the distribution that produced the historical data?

1. H_0 = The data follow a specified distribution; H_A = The data do not follow a specified distribution. Alternatively, we would state H_0 = The distribution has not changed; H_A = The distribution has changed.

Continued

Continued

Table 21.12 Historical data of defect types along with current data from a randomly selected week for Example 21.13.

Defect type	Percent of defectives	Current data
Paint run	16	27
Paint blistered	28	60
Decal crooked	42	100
Door cracked	14	21
Total	**100**	**208**

2. See the arrangement of data in Table 21.13.
3. See column D in Table 21.13.
4. See column E in Table 21.13. This column is calculated by multiplying each probability in column C by the total of column D.
5. $\alpha = 0.05$.
6. $\chi_0^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} = 5.33$. This is given in the total of column G.
7. From Appendix 15, we have $\chi^2_{\alpha,k-p-1} = \chi^2_{\alpha,4-0-1} = \chi^2_{0.05,3} = 7.815$. Since $\chi_0^2 < 7.815$, do not reject the null hypothesis. The value of p is zero since no parameters were estimated.
8. Since we failed to reject the null hypothesis, we conclude that there is insufficient evidence to conclude that the distribution producing the current data has changed.

Table 21.13 Goodness-of-fit table for Example 21.13.

A	B	C	D	E	F	G
Defect type	**Percent of defectives**	**Probability of defective**	**Observed frequency**	**Expected frequency**	**(O)bserved – (E)xpected**	$\frac{(O-E)^2}{E}$
Paint run	16	0.16	27	33.28	–6.28	1.19
Paint blistered	28	0.28	60	58.24	1.76	0.05
Decal crooked	42	0.42	100	87.36	12.64	1.83
Door cracked	14	0.14	21	29.12	–8.12	2.26
Total	**100**	**1.00**	**208**	**208**		**5.33**

CONTINGENCY TABLES

> Select, develop, and use contingency tables to determine statistical significance. (Evaluate)
>
> **Body of Knowledge VI.B.8**

Contingency tables provide us with the ability to determine whether items classified into two or more categories act independently of one another.

A contingency table analysis allows us to test the hypotheses

H_0 = The classifications are independent

H_A = The classifications are dependent

To conduct a two-way contingency table analysis:

1. Determine the variables of classification.
2. State the null and alternative hypotheses (H_0, H_A).
3. Arrange the data according to the general form given in Table 21.14. This provides the observed frequencies O_{ij}.
4. Determine the expected frequencies E_{ij} as follows:

$$E_{ij} = \frac{r_i c_j}{N}$$

where

$$r_i = \sum_{j=1}^{c} O_{ij}$$

$$c_j = \sum_{i=1}^{r} O_{ij}$$

$$N = \sum_{i=1}^{r} \sum_{j=1}^{c} O_{ij}$$

5. State the α value.
6. Compute the test statistic $\chi_0^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}}$.
7. Compute the critical value. Reject H_0 if $\chi_0^2 > \chi^2_{\alpha,(r-1)(c-1)}$.
8. State the conclusion of the test.

Table 21.14 The general form of a two-way contingency table.

		Columns (c)
		Classification 2
Rows (r)	Classification 1	$\begin{matrix} O_{11} & O_{12} & \cdots & O_{1c} \\ O_{21} & O_{22} & \cdots & O_{2c} \\ \vdots & \vdots & \vdots & \vdots \\ O_{r1} & O_{r2} & \cdots & O_{rc} \end{matrix}$

EXAMPLE 21.14

Suppose there are three product families called red, blue, and yellow, and the four defect types shown in Table 21.15. Determine whether the classifications of product type and defect type are statistically independent.

1. Product classification and defect type.
2. H_0 = The classifications are independent; H_A = The classifications are not independent.
3. See Table 21.15 for the arrangement of the data and the associated observed frequencies.
4. The expected frequencies E_{ij} are given in Table 21.16.
5. $\alpha = 0.05$.
6. Compute the test statistic:

$$\chi_0^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij}-E_{ij}\right)^2}{E_{ij}} = \frac{(20-23.65)^2}{23.65}+\frac{(54-59.40)^2}{59.40}+\ldots+\frac{(17-16.18)^2}{16.18}=1.70$$

7. Reject H_0 if $\chi_0^2 > \chi^2_{\alpha,(r-1)(c-1)} = \chi^2_{0.05,6} = 12.592$.
8. Since $\chi_0^2 < 12.592$, we reject the null hypothesis and conclude that the two classifications are dependent.

Table 21.15 Observed frequencies of defectives for Example 21.14.

	Observed frequencies		
Defect type	**Red**	**Blue**	**Yellow**
Paint run	20	54	34
Paint blistered	35	71	50
Decal crooked	48	88	66
Door cracked	10	23	17

Continued

Continued

Table 21.16 Computation of the expected frequencies for Example 21.14.

	Observed frequencies			
Defect type	**Red**	**Blue**	**Yellow**	**Total**
Paint run	20	54	34	108
Paint blistered	35	71	50	156
Decal crooked	48	88	66	202
Door cracked	10	23	17	50
Total	**113**	**236**	**167**	**516**

	Expected frequencies			
Defect type	**Red**	**Blue**	**Yellow**	**Total**
Paint run	23.65	49.40	34.95	108
Paint blistered	34.16	71.35	50.49	156
Decal crooked	44.24	92.39	65.38	202
Door cracked	10.95	22.87	16.18	50
Total	**113**	**236**	**167**	**516**

NON-PARAMETRIC TESTS

Understand the importance of Kruskal-Wallis and Mann-Whitney tests and when they should be used. (Understand)

Body of Knowledge VI.B.9

When the underlying distribution of the data is not known, we must turn to non-parametric statistics to conduct hypothesis testing. Non-parametric tests tend to be less powerful than their corresponding parametric tests. Table 21.17 compares common parametric tests with their analogous non-parametric tests. Table 21.18 provides the details around each test depicted in Table 21.17. Let's examine each of the four tests in Table 21.18.

Mood's Median Test

(Note: This test is not included in the Body of Knowledge)

Mood's median test is used to determine whether there is sufficient evidence to conclude that k medians from samples of size n_j are equal. This test permits the use of unequal sample sizes. The hypotheses tested are

$$H_0 : \tilde{\mu}_1 = \tilde{\mu}_2 = \ldots,\ \tilde{\mu}_k$$

$$H_A : \tilde{\mu}_i \neq \tilde{\mu}_j \text{ for at least one pair } (i,j)$$

The procedure for using Mood's median test is as follows:

1. State the null and alternative hypotheses (H_0, H_A).
2. Collect a random sample of size n_i for each of the k populations.
3. Compute the overall median from the total sample size N, where

$$N = \sum_{i=1}^{k} n_i.$$

4. Construct a table with one column showing the number of readings above the overall median and another showing the number of readings below the overall median for each category. Construct this table using the observed values. There are k categories, one for each population. Half the readings that are equal to the median (that is, ties) should be counted in the "above" column and half in the "below" column. Note: A review of the literature indicates that authors handle ties in different ways. You may try to replicate this example using Minitab, only to find that the results differ. Generally, this difference would be attributed to the way ties are handled.
5. Construct a table with one column showing the number of readings above the sample median and another showing the number of readings below the sample median for each category. This table will provide the expected values.

Table 21.17 Comparison of parametric and non-parametric hypothesis tests.

Parametric	Non-parametric
Two-sample t-test	Mann-Whitney test
One-way ANOVA	Kruskal-Wallis test
One-way ANOVA	Mood's median test
Bartlett test	Levene's test

Table 21.18 Common non-parametric hypothesis tests.

Test name	Assumption	Test statistic	Null hypothesis	Alternative hypotheses	Rejection criteria
Mood's median test		$M=\sum_{i=1}^{k}\frac{(o_{AMi}-e_{AMi})^2}{e_{AMi}}+\sum_{i=1}^{k}\frac{(o_{BMi}-e_{BMi})^2}{e_{BMi}}$ AM = Number above median BM = Number below median	$\tilde{\mu}_1=\tilde{\mu}_2=,\ldots,\tilde{\mu}_K$	$\tilde{\mu}_i \neq \tilde{\mu}_j$ for at least one pair (i,j)	$H>\chi^2_{\alpha,k-1}$
Levene's test		$L=\frac{(N-k)\sum_{j=1}^{k}n_j(\bar{Y}_{.j}-\bar{Y}_{..})^2}{(k-1)\sum_{j=1}^{k}\sum_{i=1}^{n_j}(Y_{ij}-\bar{Y}_{i.})^2}$ k = Number of subgroups n_j = Sample size of the jth subgroup $N=\sum_{j=1}^{k}n_j$	$\sigma_1=\sigma_2=,\ldots,\sigma_k$	$\sigma_i \neq \sigma_j$ for at least one pair (i,j)	$L>F_{\alpha,k-1,N-k}$

Continued

Table 21.18 *Continued.*

Test name	Assumption	Test statistic	Null hypothesis	Alternative hypotheses	Rejection criteria
Kruskal-Wallis rank sum test of k populations	Each sample size should be at least 5	$H=\left[\frac{12}{N(N+1)}\sum_{j=1}^{k}\frac{R_j^2}{n_j}\right]-3(N+1)$ k = Number of subgroups n_j = Sample size of the jth subgroup $N=\sum_{j=1}^{k}n_j$ R_j = Sum of the ranks for the jth subgroup	$\mu_1=\mu_2=,\ldots,\mu_K$	$\mu_i \neq \mu_j$ for at least one pair (i,j)	$H>\chi^2_{\alpha,k-1}$
Mann-Whitney test (aka Wilcoxon rank-sum test)	Distributions have the same shape and variance	$W_2=\frac{(n_1+n_2)(n_1+n_2+1)}{2}-W_1$	$\mu_1=\mu_2$	$\mu_1 \neq \mu_2$ $\mu_1 > \mu_2$ $\mu_1 < \mu_2$	W_1 or $W_2 \neq W_{\alpha,n_1,n_2}$ $W_2 \leq W_{\alpha,n_1,n_2}$ $W_1 \leq W_{\alpha,n_1,n_2}$

6. Compute the test statistic

$$\chi_0^2 = \sum_{i=1}^{k} \frac{\left(O_{iA} - E_{iA}\right)^2}{E_{iA}} + \sum_{i=1}^{k} \frac{\left(O_{iB} - E_{iB}\right)^2}{E_{iB}},$$

   where the subscripts A and B denote above and below the median, respectively.

7. State the α value.
8. Compute the critical value. Reject H_0 if $\chi_0^2 > \chi_{\alpha,k-1}^2$.
9. State the conclusion of the test.

EXAMPLE 21.15

Three machines produce plastic parts with a critical location dimension. A sample size of at least 10 is collected from each machine. The dimension is measured on each part. Does Mood's median test permit rejection at the 95% significance level of the hypothesis that the median dimensions of the three machines are the same? The data are given in Table 21.19.

Table 21.19 Data for Mood's median test in Example 21.15.

Machine 1	Machine 2	Machine 3
6.48	6.42	6.46
6.42	6.46	6.48
6.47	6.41	6.45
6.48	6.47	6.41
6.49	6.45	6.47
6.47	6.46	6.44
6.48	6.42	6.47
6.46	6.46	6.42
6.45	6.46	6.43
6.46	6.48	6.47
	6.47	6.41
	6.43	
	6.48	
	6.47	
$n_1 = 10$	$n_2 = 14$	$n_3 = 11$

Continued

Continued

1. $H_0 : \tilde{\mu}_1 = \tilde{\mu}_2 = \ldots, \tilde{\mu}_k;\ H_A : \tilde{\mu}_1 = \tilde{\mu}_i \neq \tilde{\mu}_j$ for at least one pair (*i,j*)
2. See the data given in Table 21.19.
3. Overall median is 6.46.
4. See the top half of Table 21.20.
5. See the bottom half of Table 21.20.
6. $\chi_0^2 = \sum_{i=1}^{k} \frac{(O_{iA} - E_{iA})^2}{E_{iA}} + \sum_{i=1}^{k} \frac{(O_{iB} - E_{iB})^2}{E_{iB}} = 1.964$
7. $\alpha = 0.05$
8. Using Appendix 15, compute the critical value. Reject H_0 if

   $\chi_0^2 > \chi_{\alpha,k-1}^2 = \chi_{0.05,2}^2 = 5.991.$
9. Since $\chi_0^2 = 1.964 < \chi_{0.05,2}^2 = 5.991,$ do not reject the null hypothesis. Thus, we conclude that the medians are drawn from the same population.

Table 21.20 Computation of the expected frequencies for Example 21.15.

	Observed frequencies	
Machine	**Number above median**	**Number below median**
1	7	3
2	7	7
3	4.5	6.5

	Expected frequencies	
Machine	**Number above median**	**Number below median**
1	5	5
2	7	7
3	5.5	5.5

Levene's Test

(Note: This test is not included in the Body of Knowledge)

Levene's test is used to determine whether there is sufficient evidence to conclude that k variances from samples of size n_j are equal. This test permits the use of unequal sample sizes. It is an alternative to Bartlett's test when the underlying data are nonnormal. The hypotheses tested are

$H_0 : \sigma_1 = \sigma_2 = \ldots = \sigma_k$

$H_A : \sigma_i \neq \sigma_j$ for at least one pair (i, j)

Levene's original work used the mean in the test statistic as follows:

- $Y_{ij} = \left|X_{ij} - \bar{X}_{i.}\right|$, where $\bar{X}_i$ is the mean of the *i*th subgroup and is best when the underlying data follow a symmetrical distribution.

The test statistic was later modified to include both the median and the trimmed mean:

- $Y_{ij} = \left|X_{ij} - \tilde{X}_{i.}\right|$, where $\tilde{X}_{i.}$ is the median of the *i*th subgroup and is best when the underlying data are skewed.
- $Y_{ij} = \left|X_{ij} - \bar{X}'_{i.}\right|$, where $\bar{X}'_i$ is the 10% trimmed mean of the *i*th subgroup and is best when the underlying data follow a heavy-tailed distribution such as the Cauchy.

The procedure for using Levene's test is as follows:

1. State the null and alternative hypotheses (H_0, H_A).
2. Collect a random sample of size n_j for each of the k populations.
3. Compute the median for each sample.
4. Compute $Y_{ij} = \left|X_{ij} - \tilde{X}_{i.}\right|$.
5. Compute the test statistic:

$$L = \frac{(N-k)\sum_{j=1}^{k} n_j\left(\bar{Y}_{.j} - \bar{Y}_{..}\right)^2}{(k-1)\sum_{j=1}^{k}\sum_{i=1}^{n_j}\left(Y_{ij} - \bar{Y}_{i.}\right)^2}$$

6. State the α value.
7. Compute the critical value. Reject H_0 if $L > F_{\alpha,k-1,N-k}$.
8. State the conclusion of the test.

EXAMPLE 21.16

Using the data given in Table 21.21, determine whether the variances are equal.

1. $H_0 : \sigma_1 = \sigma_2 = \ldots = \sigma_k$; $H_A : \sigma_i \neq \sigma_j$ for at least one pair (i, j).
2. See the data given in Table 21.21.
3. See the median computed at the bottom of Table 21.22.
4. Compute $Y_{ij} = \left|X_{ij} - \tilde{X}_{i.}\right|$. See Table 21.22.
5. Using Tables 21.22–21.25, compute the test statistic:

$$L = \frac{(N-k)\sum_{j=1}^{k} n_j \left(\bar{Y}_{.j} - \bar{Y}_{..}\right)^2}{(k-1)\sum_{j=1}^{k}\sum_{i=1}^{n_u}\left(Y_{ij} - \bar{Y}_{i.}\right)^2}$$

6. $\alpha = 0.05$.
7. Using Appendix 21, compute the critical value. Reject H_0 if $L > F_{\alpha,k-1,N-k} = F_{0.05,2,27} = 3.35$.
8. Since $L > F_{0.05,2,27}$, we reject the null hypothesis and conclude that the variances are different.

Table 21.21 Data for Levene's test for Example 21.16.

Sample 1	Sample 2	Sample 3
27	19	10
26	23	12
22	20	15
20	27	18
15	21	14
8	23	16
7	17	8
24	18	18
27	26	15
12	25	9

Continued

Table 21.22 Levene's test for Example 21.16.

	X_{i1}	X_{i2}	X_{i3}	$\tilde{X}_1$	$\tilde{X}_2$	$\tilde{X}_3$	Y_{i1}	Y_{i2}	Y_{i3}
	27	19	10	21	22	14.5	6	3	4.5
	26	23	12	21	22	14.5	5	1	2.5
	22	20	15	21	22	14.5	1	2	0.5
	20	27	18	21	22	14.5	1	5	3.5
	15	21	14	21	22	14.5	6	1	0.5
	8	23	16	21	22	14.5	13	1	1.5
	7	17	8	21	22	14.5	14	5	6.5
	24	18	18	21	22	14.5	3	4	3.5
	27	26	15	21	22	14.5	6	4	0.5
	12	25	8	21	22	14.5	9	3	6.5
$\sum_{i=1}^{n_j} X_{ij}$	**188**	**219**	**134**						
$\tilde{X}_{i.}$	**21**	**22**	**14.5**						

Table 21.23 Levene's test for Example 21.16 (continued).

	Y_{i1}	Y_{i2}	Y_{i3}	$\bar{Y}_{i1}$	$\bar{Y}_{i2}$	$\bar{Y}_{i3}$	$Y_{i1}-\bar{Y}_{i1}$	$Y_{i2}-\bar{Y}_{i2}$	$Y_{i3}-\bar{Y}_{i3}$
	6	3	4.5	6.4	2.9	3.0	–0.4	0.1	1.5
	5	1	2.5	6.4	2.9	3.0	–1.4	–1.9	–0.5
	1	2	0.5	6.4	2.9	3.0	–5.4	–0.9	–2.5
	1	5	3.5	6.4	2.9	3.0	–5.4	2.1	0.5
	6	1	0.5	6.4	2.9	3.0	–0.4	–1.9	–2.5
	13	1	1.5	6.4	2.9	3.0	6.6	–1.9	–1.5
	14	5	6.5	6.4	2.9	3.0	7.6	2.1	3.5
	3	4	3.5	6.4	2.9	3.0	–3.4	1.1	0.5
	6	4	0.5	6.4	2.9	3.0	–0.4	1.1	–2.5
	9	3	6.5	6.4	2.9	3.0	2.6	0.1	3.5
$\sum_{i=1}^{n_j} X_{ij}$	**64**	**29**	**30**						
$\bar{Y}_{i.}$	**6.4**	**2.9**	**3.0**						
$\bar{Y}_{..} = \frac{\sum_{j=1}^{k}\sum_{i=1}^{n_j} Y_{ij}}{N}$	**4.1**								

Continued

Continued

Table 21.24 Levene's test for Example 21.16 (continued).

	$Y_{i1}-\bar{Y}_{i1}$	$Y_{i2}-\bar{Y}_{i2}$	$Y_{i3}-\bar{Y}_{i3}$	$(Y_{i1}-\bar{Y}_{i1})^2$	$(Y_{i2}-\bar{Y}_{i2})^2$	$(Y_{i3}-\bar{Y}_{i3})^2$
	–0.4	0.1	1.5	0.16	0.01	2.25
	–1.4	–1.9	–0.5	1.96	3.61	0.25
	–5.4	–0.9	–2.5	29.16	0.81	6.25
	–5.4	2.1	0.5	29.16	4.41	0.25
	–0.4	–1.9	–2.5	0.16	3.61	6.25
	6.6	–1.9	–1.5	43.56	3.61	2.25
	7.6	2.1	3.5	57.76	4.41	12.25
	–3.4	1.1	0.5	11.56	1.21	0.25
	–0.4	1.1	–2.5	0.16	1.21	6.25
	2.6	0.1	3.5	6.76	0.01	12.25
$\sum_{i=1}^{n_j}\left(Y_{ij}-\bar{Y}_{i.}\right)^2$				**180.40**	**22.90**	**48.50**
$\sum_{j=1}^{k}\sum_{i=1}^{n_j}\left(Y_{ij}-\bar{Y}_{i.}\right)^2$						**251.80**

Table 21.25 Levene's test for Example 21.16 (continued).

	$\bar{Y}_{.j}$	$\bar{Y}_{..}$	$\bar{Y}_{.j}-\bar{Y}_{..}$	$(\bar{Y}_{.j}-\bar{Y}_{..})^2$	n_j	$n_j(\bar{Y}_{.j}-\bar{Y}_{..})^2$
	6.4	4.1	2.3	5.29	10	52.9
	2.9	4.1	–1.2	1.44	10	14.4
	3.0	4.1	–1.1	1.21	10	12.1
$\sum_{j=1}^{k} n_j\left(\bar{Y}_{.j}-\bar{Y}_{..}\right)^2$						**79.4**

Kruskal-Wallis Test

The Kruskal-Wallis test is used to determine whether there is sufficient evidence to conclude that k means from samples of size n_j are equal. This test permits the use of unequal sample sizes. The hypotheses tested are

$H_0 : \mu_1 = \mu_2 = , \ldots , \mu_k$

$H_A : \mu_i \neq \mu_j$ for at least one pair (i, j)

The procedure for using the Kruskal-Wallis test is as follows:

1. State the null and alternative hypotheses (H_0, H_A).
2. Collect a random sample of size *n*j for each of the *k* populations.
3. Compute the ranks and average ranks where ties exist using the format shown in Table 21.27 for combined samples.
4. Compute the test statistic $\mathrm{H} = \left[\frac{12}{N(N+1)}\sum_{j=1}^{k}\frac{R_j^2}{n_j}\right] - 3(N+1)$.
5. State the α value.
6. Compute the critical value. Reject H_0 if $\mathrm{H} > \chi^2_{\alpha,k-1}$.
7. State the conclusion of the test.

EXAMPLE 21.17

Random samples of a population of fourth graders from three school districts were given a math test. The scores are listed in Table 21.26. Assuming the populations have the same shape, determine whether all means are equal.

1. $H_0 : \mu_1 = \mu_2 = , \ldots , \mu_k$; $H_A : \mu_i \neq \mu_j$ for at least one pair (*i, j*).
2. See the data given in Table 21.26.

Table 21.26 Data for Kruskal-Wallis test for Example 21.17.

District 1	District 2	District 3
104	100	103
106	106	111
111	103	108
108	102	113
110	101	112
104	102	109
107	106	107

Continued

3. See Table 21.27 for the ranks and average ranks.
4. Using Table 21.28 to assist in the calculations, compute the test statistic:

$$\mathrm{H}=\left[\frac{12}{N(N+1)}\sum_{j=1}^{k}\frac{R_j^2}{n_j}\right]-3(N+1)=10.36$$

5. $\alpha = 0.05$.
6. Using Appendix 15, compute the critical value. Reject H_0 if
   $\mathrm{H} > \chi^2_{\alpha,k-1} = \chi^2_{0.05,2} = 5.991.$
7. Since $\mathrm{H} > \chi^2_{\alpha,k-1} = \chi^2_{0.05,2} = 5.991,$ we reject the null hypothesis and conclude that the means are different.

Table 21.27 Determining ranks for Example 21.17.

Combined samples	Rank	Average rank
100	1	1
101	2	2
102	3, 4	3.5
103	5, 6	5.5
104	7, 8	7.5
106	9, 10, 11	10
107	12, 13	12.5
108	14, 15	14.5
109	16	16
110	17	17
111	18, 19	18.5
112	20	20
113	21	21

Continued

Continued

Table 21.28 Kruskal-Wallis test for Example 21.17.

District 1	District 1 rank	District 2	District 2 rank	District 3	District 3 rank	
104	7.5	100	1	103	5.5	
106	10	106	10	111	18.5	
111	18.5	103	5.5	108	14.5	
108	14.5	102	3.5	113	21	
110	17	101	2	112	20	
104	7.5	102	3.5	109	16	
107	12.5	106	10	107	12.5	
Total rank	**87.5**		**35.5**		**108.0**	
Ranks squared	**7656.3**		**1260.3**		**11,664.0**	
n_j	7		7		7	
$\frac{R_j^2}{n_j}$	**1094**		**180**		**1666**	
$\sum_{j=1}^{k}\frac{R_j^2}{n_j}$						**2940**

Mann-Whitney Test

The Mann-Whitney test is used to determine whether there is sufficient evidence to conclude that two means from samples of size n_j are equal. This test permits the use of unequal sample sizes. The hypotheses tested are

$H_0 : \mu_1 = \mu_2$

$H_A : \mu_1 \neq \mu_2$ or $\mu_1 > \mu_2$ or $\mu_1 < \mu_2$

The procedure for using the Mann-Whitney test is as follows:

1. State the null and alternative hypotheses (H_0, H_A).
2. Collect a random sample of size n_j for each of the two populations.
3. Compute the ranks and average ranks where ties exist using the format shown in Table 21.30 for combined samples.

4. Let W_1 be the sum of the ranks in the smaller sample.
5. Compute the test statistic $W_2 = \frac{(n_1+n_2)(n_1+n_2+1)}{2} - W$, if W_2 is necessary for the test.
6. State the α value.
7. Compute the critical value. Reject H_0 if

$$W_1 \text{ or } W_2 \neq W_{\alpha,n_1,n_2}$$
$$W_2 \leq W_{\alpha,n_1,n_2}$$
$$W_1 \leq W_{\alpha,n_1,n_2}$$

8. State the conclusion of the test.

EXAMPLE 21.18

Steel parts are painted by two shifts. A sample from each shift is inspected, and the number of paint bubbles is recorded for each part. The data are given in Table 21.29. Assuming the two populations have the same shape, determine whether the first shift produces more bubbles than the second shift.

1. $H_0: \mu_1 = \mu_2$; $H_A: \mu_1 < \mu_2$.
2. See the data given in Table 21.29.

Table 21.29 Data for Mann-Whitney test for Example 21.18.

Shift 1	Shift 2
12	8
11	7
13	9
12	8
9	7
10	8
9	8
	9
	9

Continued

Continued

3. Compute the ranks and average ranks where ties exist using the format shown in Table 21.30.
4. Let W_1 be the sum of the ranks in the smaller sample.
5. Using Table 21.31, we obtain $W_1 = 88.0$.
6. $\alpha = 0.05$.
7. Using Appendix 29, compute the critical value. Reject H_0 if $W_1 \le W_{\alpha,n_1,n_2} = W_{0.05,7,9} = 37$.
8. Since W_1 is greater than the critical value, we do not reject the null hypothesis and conclude that there is not sufficient evidence to conclude that the second shift produces fewer bubbles.

Table 21.30 Determining ranks for Example 21.18.

Combined samples	Rank	Average rank
7	1, 2	1.5
8	3, 4, 5, 6	4.5
9	7, 8, 9, 10, 11	9
10	12	12
11	13	13
12	14, 15	14.5
13	16	16

Table 21.31 Mann-Whitney test for Example 21.18.

Shift 1	Rank for shift 1	Shift 2
12	14.5	8
11	13	7
13	16	9
12	14.5	8
9	9	7
10	12	8
9	9	8
		9
		9
Total ranks	**88.0**	

Chapter 22

Failure Mode and Effects Analysis (FMEA)

> Describe the purpose and elements of FMEA, including risk priority number (RPN), and evaluate FMEA results for processes, products, and services. Distinguish between design FMEA (DFMEA) and process FMEA (PFMEA), and interpret their results. (Evaluate)
>
> **Body of Knowledge VI.C**

This chapter will:

- Address various types of failure mode and effects analysis tools
- Address when each type should be used
- Outline and explain the details of the FMEA document form
- Provide useful tips for conducting process FMEAs.

The FMEA Tool

Today's world is fraught with risk. A *failure mode and effects analysis* (FMEA) is a prevention-based risk management tool that focuses the user or team on systematically:

- Identifying and anticipating potential failures
- Identifying potential causes for the failures
- Prioritizing failures
- Taking action to reduce, mitigate, or eliminate failures

The real value of the FMEA is reflected in its use as a long-term living document. It is essential that the document is owned and updated as changes are made to the design or the process.

FMEA was first developed and used by reliability engineers in the 1950s to study malfunctions of military systems and has proven to be a worthy and valuable technique. Subsequently, it has become commonplace in just about every Lean Six Sigma practitioner's tool kit. As common as the tool is, however, it is often used incorrectly. Users who invest significant time and effort in the tool often do not reap all it has to offer.

Types of FMEAs. According to Stamatis (2003), there are four types of FMEAs: system, design, process, and service. A brief overview of each type follows:

1. *System FMEA*. An analysis process used to identify and evaluate the relative risk associated with a system, subsystem, or component usually associated with a particular system hardware design. The focus is on minimizing costs and failure effects on the system while maximizing the system reliability and maintainability.

2. *Design FMEA (DFMEA)*. An analysis process used to identify and evaluate the relative risk associated with a particular hardware design. The focus is on minimizing costs and failure effects on the design while maximizing the design reliability and maintainability.

   A specific variation of DFMEA is the *machine failure mode and effects analysis* (MFMEA). This analysis follows a specific hierarchical model that, according to Stamatis (2003) "divides the machine into subsystems, assemblies, and lowest replaceable units." Stamatis provides the following bullet breakdown:

   - System level—generic machine
   - Subsystem level—electrical, mechanical, controls
   - Assembly level—Fixtures and tools, material handling, drives
   - Component level

   For a more detailed discussion regarding MFMEAs, see Stamatis (2003).

3. *Process FMEA (PFMEA)*. An analysis process used to identify and evaluate the relative risks associated with a particular process design. Keeping the 6 M's in mind when conducting the PFMEA is an excellent way of capturing failures at each process step. The focus is on minimizing process failures and cost while maximizing process performance and productivity.

4. *Service FMEA*. An analysis process used to identify and evaluate the relative risks associated with a particular service. Since "service" is nothing more than a process in a nonmanufacturing environment, using the 6 M's when conducting the service FMEA is an excellent way of capturing failures. The focus of the service FMEA is to minimize process failures and cost while maximizing process performance and productivity. Consequently, it will be conducted like a PFMEA.

Figure 22.1 provides an example of an FMEA document format. All FMEAs are similar—with the exception of the first column.

The first column of the system and design FMEA uses a system, subsystem, product, assembly, subassembly, or part. By contrast, the process and service FMEAs use a process. Therefore, the first column contains the process steps.

May be a product, assembly, subassembly, or part

Initial development of the FMEA									**Improvement activities**		**Post-improvement activities**				
Process step/ input	Potential failure mode	Potential failure effects	SEV	Potential causes	OCC	Current controls	DET	RPN	Actions recom-mended	Resp	Actions taken	SEV	OCC	DET	RPN
1	2	3	4	5	6	7	8	9	10	11	12	13			

DET = Detection
OCC = Occurence
PFMEA = Process failure mode and effects analysis
Resp = Responsible
RPN = Risk priority number
SEV = Severity

Figure 22.1 FMEA document general format.

Thus, a process map, cause-and-effect matrix, suppliers, inputs, process, outputs, and customers (SIPOC) diagram, value stream map, cause-and-effect diagram, or something similar, usually feed these types.

The PFMEA Document. Of the primary types of FMEA documents, the Lean Six Sigma practitioner likely will deal with the PFMEA the most. The PFMEA document's columns include:

1. *Process step.* Identify the process step and input under investigation. Each step is identified sequentially. If the PFMEA is fed from a cause-and-effect matrix, only high-value steps might be listed.

2. *Potential failure mode.* Identify all the ways a failure can occur at this process step.

3. *Potential failure effects.* Identify all the effects each failure mode produces, including the effects on the customer. Use a new line for each failure effect. Table 22.1 demonstrates the many-to-many relationships that exist across the document columns for any given step.

   Stamatis (2003) introduces a column in his general FMEA format document between columns 3 and 4 for the use of "critical characteristics" and denotes it by the inverted delta symbol, ∇. This additional column accommodates the needs of the automotive industry. See Stamatis (2003) for details.

Table 22.1 Many-to-many relationships between key columns in a PFMEA document.

Process step/input	Potential failure mode	Potential failure effects	Potential causes	Current controls
n	1	1	1	1
				2
				3
			2	4
				5
				6
		2	3	7
				8
				9
			4	10
				11
				12
	2	3	5	13
				14
				15
			6	16
				17
				18
		4	7	19
				20
				21
			8	22
				23
				24
	3	5	9	25
$n + 1$	4	6	10	26

4. *Severity*. Quantify the severity of the impact of the failure effect. The scale for severity ranges from "no effect" on the low end to "safety hazard"—up to and including "loss of life without warning"—on the high end. Also, the effect can be expressed in monetary damages, as well as destruction and delays. All scales must be described in the context of the FMEA situation (see Table 22.2).

5. *Potential causes*. Identify all root causes leading to the failure. If root causes are unknown at the time the FMEA is conducted, it may be necessary to divert from the FMEA temporarily and conduct a root cause analysis using the variety of quality tools available.
6. *Occurrence*. Quantify the frequency of occurrence of the failure mode. The scale for occurrence ranges from "highly unlikely" on the low end to "highly likely" on the high end. Some users, teams, and organizations will go to great lengths to provide absolute definitions for the frequency of occurrence. For example, the Automotive Industry Action Group (1995b), *Potential Failure Mode and Effects Analysis (FMEA) Reference Manual,* stated that an occurrence entry value of 1 designates a possible failure rate ≤ 0.01 per thousand vehicles/items, and an entry value of 10 designates a possible failure rate ≥ 100 per thousand vehicles/items. The occurrence scale generally will translate to a rate or even a probability (see Table 22.3).
7. *Current controls*. Identify all the existing controls and procedures, including inspections and tests, that prevent the cause of the failure mode. Include a standard operating procedure number, if available.
8. *Detection*. Quantify the ability to detect the failure at a specific process step (that is, not at a previous or subsequent step, but at the step under consideration). The scale for detection ranges from "almost certain" on the low end to "not possible" on the high end (see Table 22.4).

Table 22.2 Severity scale example.

Severity value	Descriptor	Description
1	None	No effect
3	Minimal	Greater than $1000 and up to $100,000 in damages
7	Moderate	Greater than $100,000, but less than $1 million in damages
10	Extreme	Loss of life without warning or greater than $1 million in damages

Table 22.3 Occurrence scale example.

Occurrence value	Descriptor	Description
1	Highly unlikely	1 in 10,000
3	Unlikely	1 in 1000
7	Likely	1 in 100
10	Highly likely	1 in 10

Table 22.4 Detection scale example.

Detection value	Descriptor	Description
1	Almost certain	P(detection) 0.95
3	Likely	0.50 ≤ P(detection) < 0.95
7	Possible	0 < P(detection) < 0.50
10	Not possible	P(detection) = 0

9. *Risk priority number (RPN)*. Determine the multiplicative effect (that is, RPN = Severity value × Occurrence value × Detection value) of values assigned to columns 4, 6, and 8, respectively. Although teams generally work the highest RPN values first, they may set additional prioritization criteria, such as working any line item on the FMEA where the severity value is at the highest level, the detection value is at its highest, or any value is at its highest.
10. *Actions recommended*. Recommend actions for reducing the severity of the impact, frequency of occurrence, or the ability to improve detection.
11. *Responsible*. Identify who is responsible for the actions recommended. If more than one individual is identified, a lead should be specified as responsible.
12. *Actions taken*. List the actions taken and completed, and include the completion date.
13. *Severity, occurrence, detection, and RPN*. Identify new severity, occurrence, and detection values, and compute the new RPN value. These have the same meaning as items 4, 6, 8, and 9, respectively. However, these values reflect the actions taken in item 12. Ideally, one or more of these values will be reduced by the actions taken, resulting in a lower RPN value. If the value is not reduced, the actions taken were ineffective.

Developing the Scales. For FMEAs to be successful, a team must seriously consider the scales it will use to assign values to each component of the RPN.

Some authors advocate using a 10-point scale. One issue with this is that it tends to promote debate as to whether to assign an item a 2 versus a 3, or a 5 versus a 6. In such instances, the overall influence on the RPN value may be minimal, yet the team wastes significant time and energy debating values that are close together.

In contrast, other authors advocate using scales skewed and sparse in terms of assignable values. For example, instead of selecting a 1-to-10 scale, some teams will choose 1, 3, 7, and 10 scales, or something similar. The benefit of this type of scale is that it minimizes meaningless debate on close values and forces the team to discuss how to assign values. Further, it bounds the RPN values between one

EXAMPLE 22.1

Figure 22.2 provides an example of a PFMEA for two failure modes for the first step of a process. Notice that the severity and detection values are high in the first row. The detection control is ineffective, and the recommended actions attempt to address both of these issues. The recommended actions were taken, but only affected the detection value.

After applying the revised values, the new RPN value is 10. This represents a significant drop from the previous RPN value of 300.

Although the severity value remains at a value of 10, the occurrence value dropped from a 3 to a 1. Therefore, the team felt that because the failure is "highly unlikely" to occur, it will "almost certainly" be detected when it does. Consequently, the team decided not to pursue any additional improvements. Of course, this might be debatable.

In the second row, no current detection control mechanism exists. Therefore, the team must default to the highest value on the scale, which is a 10 in this case. This results in an RPN value of 490.

Thus, a simple action recommended is to separate the parts. This makes the revised detection value a 3 and the resulting RPN value a 21.

Process step/ input	Potential failure mode	Potential failure effects	SEV	Potential causes	OCC	Current controls	DET	RPN	Actions recom-mended	Resp.	Actions taken	SEV	OCC	DET	RPN
1	Part not installed	Device does not work	10	Process step skipped	3	SOP 123: process routing sheet	10	300	Modify program to halt production	T. Kubiak 06/17/14	Program modified to detect missing parts	10	1	1	10
1	Wrong part installed	Device overheats	7	Parts commingled in bin	7	None	10	490	Place different parts in different bins	T. Kubiak 06/17/14	Parts sorted and new bins added	7	1	3	21

DET = Detection
OCC = Occurence
PFMEA = Process failure mode and effects analysis
Resp = Responsible
RPN = Risk priority number
SEV = Severity
SOP = Standard operating procedure

Figure 22.2 FMEA for Example 22.1.

and 1000 (inclusive), which is convenient and easy to understand. Regardless of the scales used, they should be well defined, consistent, and clearly understood by each team member.

Useful Tips. More often than not, conducting an FMEA work session can be time-consuming, tiresome for all, and, in many cases, less than productive. Here are tips to make the sessions more meaningful:

1. *Establish team norms*. Team norms keep teams moving forward. Use them to the fullest.
2. *Keep work sessions to a reasonable length of time*. Though this might be included in the team norms, it requires special consideration. The FMEA work sessions can be quite intense and focused, and draining on the participants. Thus, this must be taken into consideration when setting the duration of the session. A work session of two or three hours is generally reasonable.
3. *Establish an FMEA owner*. I believe "No owner, no FMEA." An FMEA without an owner has no support and, consequently, either is done once, haphazardly, or not done at all.
4. *Use subject matter experts (SMEs)*. Some organizations actually use the "who's available method" (WAM). This is nonsense. SMEs understand the process or the component. Use them. Anything else is foolish.
5. *Use a professional facilitator*. Although team norms are helpful, a professional facilitator or individual who is exceptionally skilled in facilitation is occasionally necessary to keep the team moving forward. Using a facilitator should not be viewed as a negative by the team or a negative trait of the team leader. A team leader who recognizes the need for a facilitator demonstrates the ability of building a strong, cohesive team capable of producing results. Such an individual definitely shows his or her leadership abilities.
6. *Create meaningful scales*. Meaningful scales, in conjunction with proper facilitation, can keep the team moving forward. If thoughtful consideration is given to the development of the scales before beginning the FMEA, it is unlikely there will be a need to change the scales during the development of the FMEA. Unforeseen circumstances do arise, however, so there may be those rare times when it is necessary. An example of a scale that sometimes requires changing during the course of the FMEA is the *occurrence* scale. Usually, the need to do so arises because the scale was insufficient in the first place, and a situation arose that the scale could not accommodate.
7. *Establish a trigger*. Triggers should be used to initiate an update to an FMEA or the creation of a new one. When no trigger exists, FMEAs are at the mercy of someone's thought or memory, and that's a shame.
8. *Limit the ability of participants to review past decisions for current steps*. When columns 2 and 3, and/or 5, and 7 are identical, teams have a propensity to want to carry previous scores forward to the current step, based on the rationale that the current step wording is the same. Teams want to do this without regard to what has transpired between steps. This is usually done for convenience and is not necessarily supported by logic.

   For example, there is no reason to believe that identical failure modes and failure effects occurring in process steps 2 and 22 should have the same severity scores. If the scores are the same, the sameness should be supported by logic, not convenience.

9. *Remember that all decisions reflect the current step.* This is an important aspect of the FMEA and must be constantly reinforced to all team members. It is especially important when it comes to scoring the detection value because team members have a propensity to look ahead for detection potential.
10. *Write modes, effects, controls, and causes in clear and meaningful ways.* These should be written in clear, succinct, and unambiguous language. Otherwise, thoughts are lost from work session to work session. Try to avoid using adjectives and adverbs.
11. *Complete each step in a given work session to the extent possible.* Although this requires the participants to be flexible in the termination time of the work session, this helps maintain continuity of thought. If the termination time of the work session is unreasonable (for example, strays too far from tip number 2), seek to find a reasonable stopping point and make appropriate notes that will help the team pick up quickly at the next work session.
12. *Minimize the duration between work sessions.* If the duration between work sessions is too long, momentum is lost, and team members begin to forget their discussion points. This tip is particularly important when the FMEA is being developed for the first time.
13. *Identify root causes.* Ideally, root causes should be identified. Otherwise, the actions taken to reduce the high-value RPNs will likely be ineffective because they will be addressing symptoms.
14. *Score appropriately.* It is important to score as the situation dictates. Some individuals and teams are reluctant to score at the extreme values of the scales. For example, if it's not possible to detect a failure at a given step, a score of 10 should be given. Some individuals and teams resist scoring a 10 and fight to score a 7. It's unclear whether this is human nature or something else. Regardless, if the situation calls for a 10, a 10 must be given. Such situations should be addressed up front and included in the team norms, if necessary.
15. *Support FMEAs with additional quality tools.* It should be evident from this list of tips that additional quality tools can be beneficial when conducting FMEAs. For example, cause-and-effect diagrams, root cause analysis, and brainstorming can all play valuable roles in completing an FMEA.

Up-front work. FMEAs should be owned, considered living documents, and updated appropriately. They require intensive work up front on the part of the team, but their value is almost immeasurable in terms of providing a positive impact on quality. The tips provided are born from experience and, if applied, will ensure the effectiveness of your FMEA.

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Chapter 23

Additional Analysis Methods

GAP ANALYSIS

> Analyze scenarios to identify performance gaps, and compare current and future states using predefined metrics. (Analyze)
>
> **Body of Knowledge VI.D.1**

A *gap analysis* is a tool used to identify a performance difference between a current state and a desired or future state using metrics implied by the states. The desired or future state may be set by recognizing potential performance determined through activities such as benchmarking. Gap analyses are performed at multiple levels:

- *Business level.* A business may compare its performance directly with that of competitors or the general average industry performance. Gaps at this level are usually financial in nature.

  These gaps often cascade down to subordinate entities of an enterprise. For example, a $10M profit gap might be allocated as a gap of $2M to each of five business units.

EXAMPLE 23.1

Current state: Profit for FY 2016 = $4.26M
Future state: Profit for FY 2017 = $5M

EXAMPLE 23.2

Current state: Sales for FY 2016 = $18.1M
Future state: Sales for FY 2017 = $21.5M

EXAMPLE 23.3

Current state: Representatives in 11 cities
Future state: Representatives in 16 cities by 3Q 2017

Continued

Continued

EXAMPLE 23.4

Current state: 20K books circulated in academic year 2016
Future state: 22K books circulated in academic year 2017

EXAMPLE 23.5

Current state: Utility cost: $975K in calendar year 2016
Future state: Utility cost: $850K in calendar year 2017

EXAMPLE 23.6

Current state: FY 2016 square feet leased 100K
Future state: FY 2017 square feet leased 0

- *Process level.* The current performance of a process might be compared with the cost, cycle time, and quality characteristics of other processes or at performance levels required to remain competitive.

  These gaps may also cascade to lower entities. For instance, parts of a process cycle time reduction may be allocated to various subprocesses.

EXAMPLE 23.7

Current state: June volume 253 units
Future state: July volume 270 units

EXAMPLE 23.8

Current state: Standard deviation 0.006
Future state: Standard deviation 0.004 by 2Q 2017

EXAMPLE 23.9

Current state: Leaking packages 0.2% in August 2016
Future state: Leaking packages 0.05% in December 2016

EXAMPLE 23.10

Current state: Labor content: 0.13 hours per unit
Future state: Labor content: 0.10 hours per unit by 3Q 2017

EXAMPLE 23.11

Current state: External failures FY 2016 = 6
Future state: External failures FY 2017 = 0

- *Product level*. Gaps at this level are usually identified through differences in features and capabilities, cost, quality, or generally by critical-to-*X* perspectives.

EXAMPLE 23.12

Current state: Switch has 99% reliability @ 10K cycles
Future state: Switch has 99% reliability @ 20K cycles by 2/17

EXAMPLE 23.13

Current state: Invoice register doesn't show credit status
Future state: Invoice register shows credit status by 1Q 2017

EXAMPLE 23.14

Current state: Fleet mpg for model year 2016 = 22 mpg
Future state: Fleet mpg for model year 2019 = 35 mpg

If the gap between the current state and the future state is sufficiently large, a series of intermediate steps or milestones may be required, with more achievable gaps between each one. However, it is important to recognize that gaps are seldom stationary. Therefore, it may be necessary to account for an increasing gap over time, particularly when future states are set relative to competitor performance. Gap analysis may be stated in terms of external conditions.

EXAMPLE 23.15

Current state: FY 2016 Profit = $18.6M
Future state: FY 2017 Profit = $21.5M (If customer A builds 100,000 units)
FY 2017 Profit = $23M (If customer A builds 150,000 units)
FY 2017 Profit = $25M (If customer A builds 200,000 units)

EXAMPLE 23.16

Current state: Market share FY 2016 = 31%
Future State: Market share FY 2017 = 35% (If there are 100K housing starts)
Market share FY 2017 = 38% (If there are 140K housing starts)
Market share FY 2017 = 40% (If there are 160K housing starts)

EXAMPLE 23.17

Current state: Phosphates in effluent = 2.1 ppm June 2016
Future state: Phosphates in effluent < 1.6 ppm in June 2017 (If rain total < 2")
Phosphates in effluent < 1.8 ppm in June 2017 (If 2" < rain < 3")
Phosphates in effluent < 2.0 ppm in June 2017 (If rain > 3")

ROOT CAUSE ANALYSIS

> Define and describe the purpose of root cause analysis, recognize the issues involved in identifying a root cause, and use various tools (e.g., 5 whys, Pareto charts, fault tree analysis, cause and effect diagrams) to resolve chronic problems. (Analyze)
>
> **Body of Knowledge VI.D.2**

Solving a process problem means identifying the root cause and eliminating it. The ultimate test of whether the root cause has been eliminated is the ability to toggle the problem on and off by removing and reintroducing the root cause.

A number of tools for identifying root causes are discussed in the following paragraphs.

5 Whys

The *5 whys* is a technique used to drill down through the layers of cause and effect to the root cause. It consists of looking at an undesired result and asking, Why did this occur? When the question is answered, the next question is, And why did that occur? and so on. The number five is, of course, arbitrary.

EXAMPLE 23.18

Undesired result: A customer is not satisfied.

"Why is the customer not satisfied?" *Because the order arrived late*.

"Why did the order arrive late?" *Because it was shipped late*.

"Why was it shipped late?" *Because final assembly wasn't completed on time*.

"Why wasn't final assembly completed on time?" *Because a part was received from the paint line late*.

"Why was the part from the paint line late?" *Because it arrived late from the fabrication department*.

"Why was it late from the fabrication department?" *Because the fabrication machine was running another part for the same order*.

"Why was the fabrication machine running another part for the same order?" *Because that machine always runs these two parts*.

"Could they be on separate machines simultaneously when a delivery date is imminent?" *Um, yes*.

There is not one single root cause for every problem. Sometimes it is necessary to look at several causes and work on them one at a time, as Example 23.19 illustrates.

EXAMPLE 23.19

A problem-solving team at a trucking company is given the following gap analysis:

Current state: Number of customer complaints, FY 2016 = 238

Future state: Reduce the number by 25% for FY 2018

The team asks the first *why*: Why did the customers complain?

They decide to look at the complaint data for the 2016 FY:

Customer complaint	**Frequency**
Damaged goods	28
Missing goods	18
Delivered to wrong address	22
Late delivery	152
Wet goods	11
Incomplete shipment	3
Attempt to deliver after hours	4
Total	238

Since late delivery is the most frequent cause, the team asks the second *why*: Why did we have so many late deliveries?

They study the 152 late delivery events:

Reasons for late delivery	**Frequency**
Delayed departure	6
Driver illness	2
Paperwork has wrong address	7
Truck mechanical problem	127
Traffic accident	2
Flooding/storms	4
GPS failure	4
Total	152

The third *why*: Why did we have so many truck mechanical problems?

The team looks further at the "mechanical problem" events and produces the following list:

Continued

Mechanical problems	Frequency
Engine failure	5
Drive train failure	4
Electrical system failure	76
Exhaust system failure	11
Tire failure	18
Total	127

Fourth *why*: Why did we have so many electrical system failures?

Further analysis of the electrical systems failures:

Reasons for electrical system failure	Frequency
Battery failure	1
Headlight failure	63
Tail/brake light failure	10
Alternator failure	2
Total	76

To recapitulate, 41% of late deliveries are caused by a failed headlight.

Why do we have so many headlight failures?

The team studies the current headlight's life cycle. They find that after 1800 hours of service, there is a 50/50 chance that the bulb will fail. They also discover that the headlight repair procedure consists of having the driver pull off the road, call the nearest company depot, and wait for a mechanic to arrive. Depending on the weather, location, and so on, the headlight may be repaired on site or the truck may be towed to the nearest depot. Either way, enough time often elapses to cause a late delivery.

The team brainstormed several alternatives:

1. Train operators to change headlights
2. Provide a spare headlight that could be switched on when one fails
3. Establish a non-depot repair network to greatly increase the availability of repair people
4. Install a different type of headlight

A team member locates a more expensive headlight system that will run 20,000 hours before reaching the 50/50 failure point. In addition, these bulbs cast a reddish light when they have fewer than 200 hours of useful remaining life, so they can usually be changed prior to departure and without delaying delivery. Unfortunately, the new headlight requires a step-up transformer to supply the voltage needed. The team decides to recommend that all trucks in the system be fitted with the new transformer and headlights.

Continued

Continued

> Management agrees to retrofit 100 vehicles (10% of the fleet) with the new system for a one-year trial. At the end of the year, none of the 100 vehicles had a failed headlight. Management then decided to retrofit the entire fleet and to specify to their tractor supplier that all new units be equipped with the new headlight system. Customer complaints decreased by 22% during the following year. The team had a restrained celebration because they had not met the 25% decrease that had been their original goal. They set about analyzing the complaint types for further improvement.

Cause-and-Effect Diagram

A *cause-and-effect diagram* (also called the *Ishikawa diagram* or *fishbone diagram*) traditionally divides causes into several generic categories. In use, a large empty diagram is often drawn on a whiteboard or flip chart, as shown in Figure 23.1. Notice that the main branches of the diagram are the 7 M's, which were discussed in Chapter 16. An undesirable end effect is specified, and a brainstorming session is usually held to identify potential causes, sub-causes, and so on. The participants in the session should include people with a working knowledge of the process as well as subject matter experts. For example, suppose a circuit board plating operation is producing defects. After a few steps of brainstorming, the diagram might look like Figure 23.2. The beauty of a team-built cause-and-effect diagram is that the thought process of an individual is stimulated by others. A group will usually be much more creative than the same people working independently. Sometimes an idea from one person causes others to think along a new track. This may result in putting branches on branches as Figure 23.2 illustrates.

Some teams have found it useful to use a *flowchart* as a starting point rather than the traditional fishbone diagram, as shown in Figure 23.3.

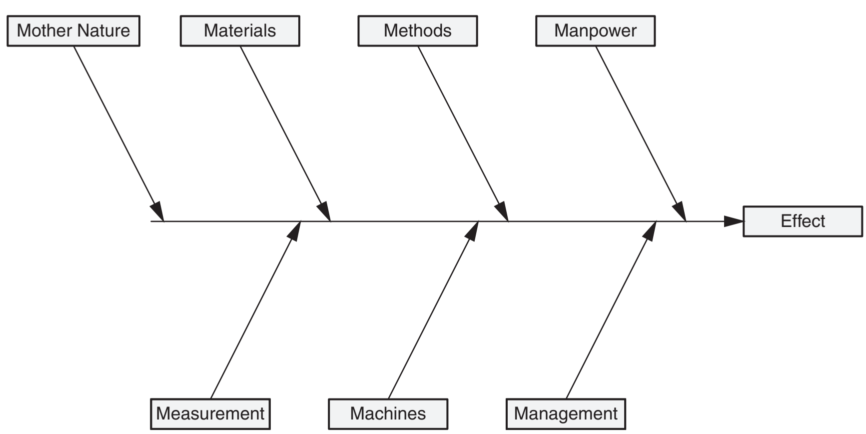


Figure 23.1 Example of a blank cause-and-effect diagram.

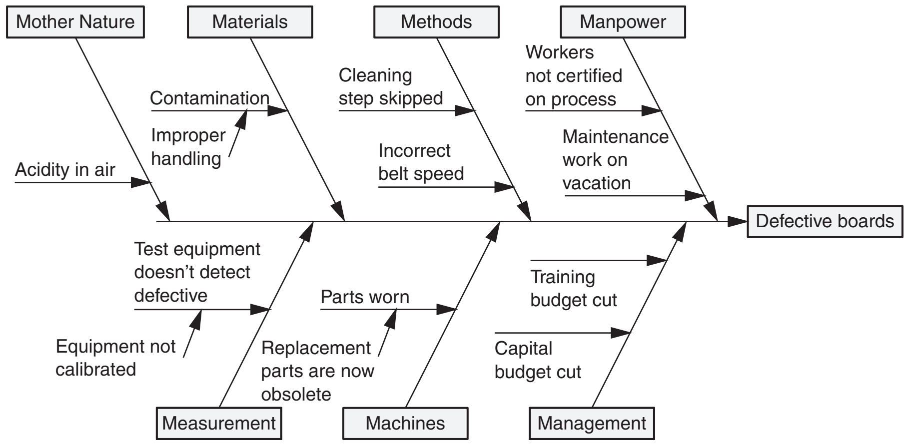


Figure 23.2 Example of a cause-and-effect diagram after a few brainstorming steps.

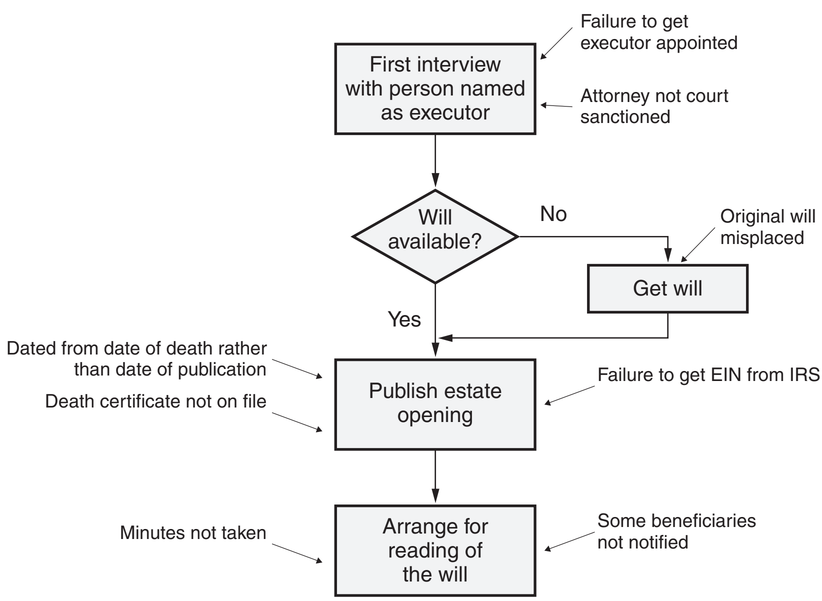


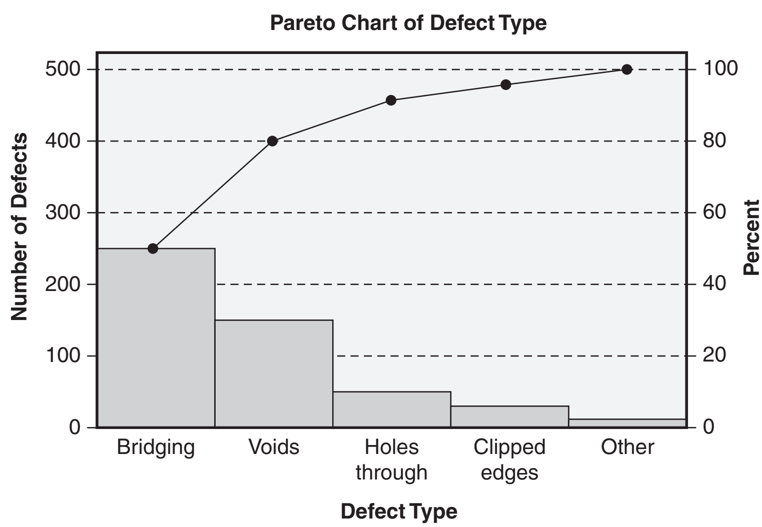
Figure 23.3 Causes for errors when opening an estate.

Pareto Chart

An excellent tool for seeking out and prioritizing potential root causes is the *Pareto chart* or diagram. An example of a Pareto chart for defect types is illustrated in Figure 23.4. Let's take a moment to analyze this chart. Notice that a list of mutually exclusive categories is given along the *x*-axis. Each defect type is graphed, and the height of each bar is proportional to the number of defects. The graph contains two *y*-axes. The left *y*-axis identifies the number of defects and is associated with the bar heights. The right *y*-axis represents the percentage of defects. The cumulative percentage of each defect type is graphed above the bars. The purpose of the Pareto chart is to separate the "vital few" causes from the "trivial many." This is often reflected in what is called the 80/20 rule. Figure 23.4 illustrates that the first two bars (that is, *bridging* and *voids*) represent 80% of the defects. The remaining 20% is spread across the three subsequent bars. Notice that the last bar is identified as "other." Because there are many types of minor defects that are few in quantity, they are collected in this category. There is no real need to identify them by specific type since they make up the trivial many and will not be investigated.

Pareto charts can and are often decomposed into lower-level charts. For example, an investigation into the *bridging* bar in Figure 23.4 may determine that bridging can be further subdivided into additional meaningful categories.

One drawback of Figure 23.4 is that it assumes all defects have an equal impact. However, if we quantify the cost of correcting each defect, we can weight each quantity of defect type by the cost of correction. This weight could very well yield a Pareto chart completely different from Figure 23.4.



Number of Defects	250	150	50	35	15
Percent	50.0	30.0	10.0	7.0	3.0
Cum %	50.0	80.0	90.0	97.0	100.0

Figure 23.4 Example of a Pareto chart for defects.

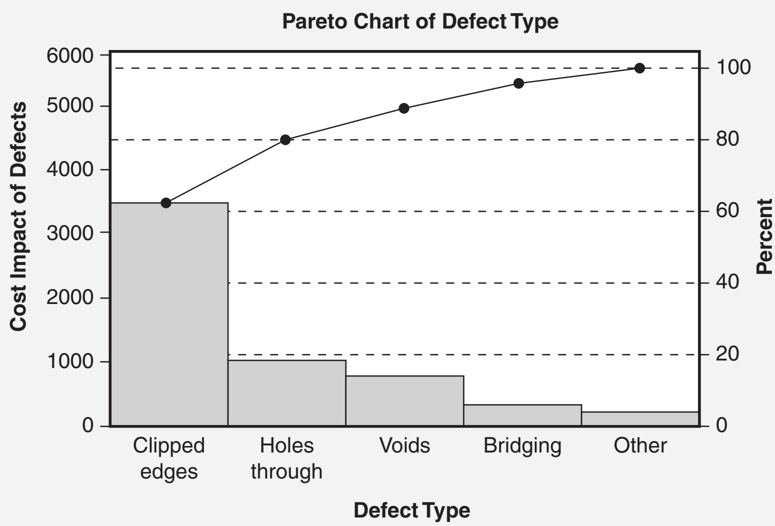
EXAMPLE 23.20

Let's consider the data given in Table 23.1. Applying the cost data in this table to Figure 23.4 yields the revised Pareto chart depicted in Figure 23.5, which illustrates a completely different picture and dictates a different course of investigation and action than Figure 23.4.

The Pareto principle says that the most important things should be treated differently than the least important things. To put it another way, the Pareto principle requires that the majority of our resources should be expended on the critical few rather than the trivial many. Consider the case of stockroom management. The basic requirements are (1) always have material on hand when it is needed and (2) don't maintain excess inventory. An elaborate software system with predictive capability, alarm messages,

Table 23.1 Cost to correct each defect type.

Defect type	Cost to correct ($)
Bridging	1.00
Voids	5.00
Holes through	25.00
Clipped edges	100.00
Other	10.00



Number of Defects	3500	1000	750	250	150
Percent	61.9	17.7	13.3	4.4	2.7
Cum %	61.9	79.6	92.9	97.3	100.0

Figure 23.5 Example of a Pareto chart for defects weighted by the cost to correct.

Continued

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sensors on all bins, automatic ordering, and careful monitoring may be established for all stock items. Another alternative is to use this elaborate, resource-intensive system for the few important items, in this case perhaps the most expensive and critical items. The inventories of the trivial many less expensive items can be allowed to be larger in number because they have less impact on the bottom line. The stockroom for an appliance manufacturer may have a month's supply of fasteners but only a carefully calculated three-day supply of electric motors. The medical community has long taught this principle for situations when a limited staff confronts a large group of people with injuries. They don't treat everyone equally nor do they say, "The vast majority of injuries are sprained ankles, so we'll treat those first." Instead, they use *triage*, dividing the group into three subgroups based on the severity of their injuries. They use the majority of their resources on the group that will survive only if they receive immediate care. The Pareto principle provides guidance in many areas.

Fault Tree Analysis

Once a failure has been identified as requiring additional study, a *fault tree analysis* (FTA) can be performed. Basic symbols used in FTA are borrowed from the electronics and logic fields. The fundamental symbols are the AND gate and the OR gate. Each of these has at least two inputs and a single output. Another key gate is the voting OR gate. In this gate, the output occurs if and only if *k* or more of the input events occur, where *k* is specified, usually on the gate symbol. Common FTA gate symbols are depicted in Figure 23.6.

The output for the AND gate occurs if and only if all inputs occur. The output for the OR gate occurs if and only if at least one input occurs. Rectangles are typically used for labeling inputs and outputs. The failure mode being studied is sometimes referred to as the "top" or "head" event. An FTA helps the user consider underlying causes for a failure mode and study relationships between various failures.

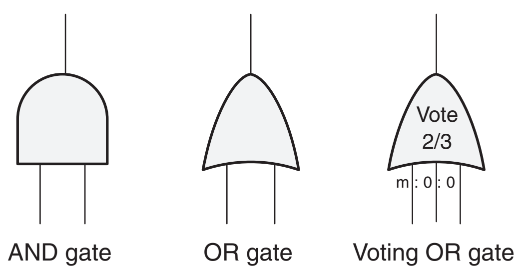

Figure 23.6 Basic fault tree analysis symbols.

EXAMPLE 23.21

The failure being studied is the stoppage of agitation in a tank before mixing is complete. This becomes the top event. Further team study indicates this will happen if any of the following occurs:

- Power loss due to both the external power source and the backup generator failing.
- Timer shuts off too soon because it is set incorrectly or has a mechanical failure.
- Agitator motor fails because it is overheated, or a fuse or capacitor fails.
- Agitator power train fails because both belts A and B break or the clutch fails or the transmission fails.

This test is symbolized by the FTA diagram shown in Figure 23.7.

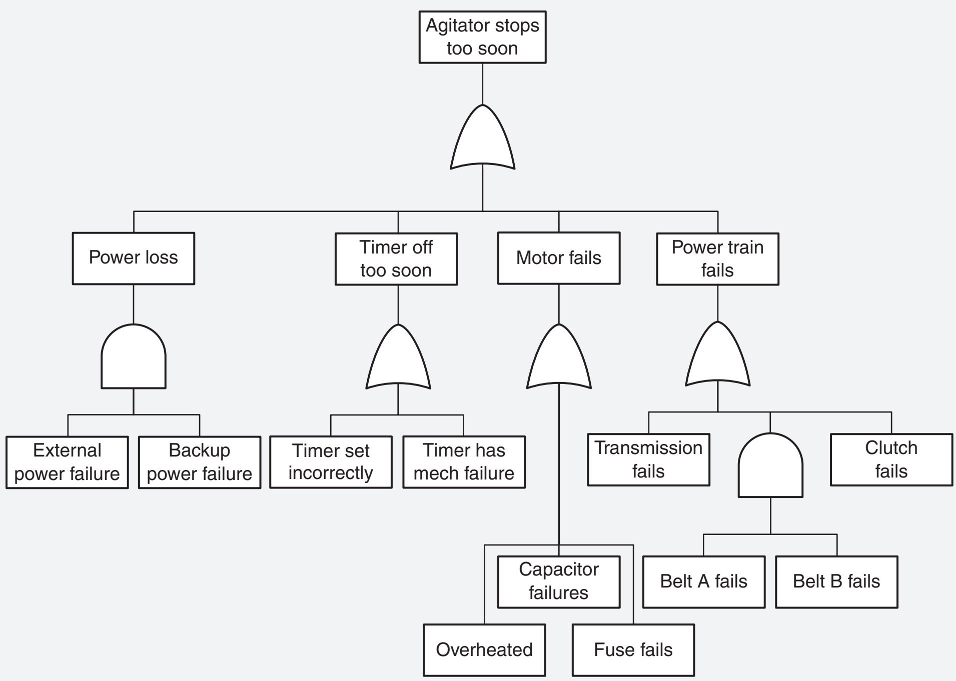

Figure 23.7 Example of stoppage of agitation in a tank.

WASTE ANALYSIS

> Identify and interpret the seven classic wastes (overproduction, inventory, defects, over-processing, waiting, motion, transportation) and resource under-utilization. (Analyze)
>
> **Body of Knowledge VI.D.3**

Finding and reducing waste has great potential for impact on the bottom line. Lean proponents put great emphasis on this effort. In fact, the National Institute of Standards and Technology (NIST) through its Manufacturing Extension Partnership (MEP) defines *lean* as "A systematic approach to identifying and eliminating waste (non-value-added activities) through continuous improvement by flowing the product at the pull of the customer in pursuit of perfection."

They define *non-value-added activities* as "Anything a final customer wouldn't want to pay for."

Shoichiro Toyoda, president of Toyota, says that waste is "anything other than the minimum amount of equipment, materials, parts, space, and worker's time which are absolutely necessary to add value to the product."

The difficult part about eliminating waste is it is so often hidden. Identifying hidden waste requires careful study of all aspects of an organization's processes.

NIST identifies eight sources of waste:

- *Overproduction*. Producing sooner or faster or in greater quantity than is needed by the next operation.

EXAMPLE 23.22

A punch press produces 20 parts for use elsewhere in the plant. On the first workday of each month it runs a one-month supply of part A, on the second workday of the month it runs a one-month supply of part B, and so on. This is overproduction and uses resources for investment and care for hundreds of parts. What if the schedule changes mid-month? What if a rush order arrives? Why doesn't the press run a one-week supply of part A during the first two hours on Monday and a one-week supply of part B during the second two hours on Monday, and so on? The usual reason is that the changeover time required to switch from part A to part B would not permit this. So overproduction is often caused by excess changeover time. As a result, much effort has been expended recently to reduce this time. The goal is sometimes referred to as *single-minute exchange of die* (SMED). If it could be sufficiently reduced, the press could run a one-day supply of part A in the first 20 minutes of the day, a one-day supply of part B in the second 20 minutes, and so on. One advantage of this would be a more agile

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or nimble operation. If twice as many of part C were required, it would not need to be ordered a full month in advance. Instead, the press could react on a much shorter time frame. This would allow the plant to react to customer orders more quickly. Analogous discussions could be held for document production in an office environment. Is it necessary to produce a month's supply of a document or form?

- *Inventory.* Excessive supply of parts or materials. Some accounting systems consider inventory an asset, but in light of the need to store, protect, locate, handle, and retrieve it, it may be considered a liability by lean professionals.

EXAMPLE 23.23

Organizations with assembly lines sometimes arrange with vendors of standard parts to go through their assembly line before a shift starts and stock it with the number of parts needed for that shift. This obviates the necessity of running the parts through the stockroom with the associated picking and moving operations

- *Defects.* When defects occur, additional material, labor, and machine time may be required in addition to a possible increase in warranty costs and the potential for product recalls.
- *Overprocessing.* This represents effort that does not add value.

EXAMPLE 23.24:

A company found that parts waiting to be painted were rusting. An oil dip tank was installed to prevent rust. Of course, a de-oiling dip tank was also needed because oily parts don't paint well. The oil/degrease steps were processing waste. Processing waste (some call it *overprocessing*) is one of the most difficult types of waste to detect.

- *Waiting.* Most workers can identify with this issue. Waiting for equipment, instructions, materials, and so forth, does not add value to a product or service.
- *Motion.* Movement by persons or equipment that does not add value. This can usually be improved by altering workplace layout whether behind a receptionist desk or on a production floor.

- *Transportation.* Movement of parts and materials around the plant or facility, usually due to poor facility layout. In some plants the decision as to where to place a newly arrived machine is largely based on where room can be found. This makes for a lot of movement of parts and materials that could be reduced with a saner layout.
- *People.* Failing to make use of all an employee has to offer in terms of ideas and creativity in making continuous improvement. This is often considered the eighth waste.

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Part VII

Improve

Chapter 24 Design of Experiments (DOE)
Chapter 25 Lean Methods
Chapter 26 Implementation

The purpose of this phase is to identify, select, pilot, if necessary, and implement process improvement solutions. Common tools used in this phase are shown in Table VII.1.

Table VII.1 Common tools used in *improve* phase.

Activity network diagrams	*Hypothesis testing*	*Project management*
Analysis of variance (ANOVA)	*Measurement systems analysis (MSA)*	*Project tracking*
Brainstorming	*Meeting minutes*	Reliability analysis
Control charts	Mixture experiments	Response surface methodology (RSM)
D-optimal designs	*Multi-vari studies*	Risk analysis
Design of experiments (DOE)	*Multivoting*	*Simulation*
Evolutionary operations (EVOP)	*Nominal group technique*	Statistical tolerancing
Failure mode and effects analysis (FMEA)	*Pareto charts*	Taguchi designs
Fault tree analysis (FTA)	*Project evaluation and review technique (PERT)*	Taguchi robustness concepts
Flowchart/process mapping (to be)	Pilot	*Tollgate review*
Gantt charts	*Prioritization matrix*	*Value stream maps*
Histograms	*Process capability analysis*	Work breakdown structure
	Process sigma	*Y=f(X)*

Note: Tools shown in *italic* are used in more than one phase.

Chapter 24

Design of Experiments (DOE)

Statistical design of experiments refers to the process of planning the experiment so that appropriate data will be collected and analyzed by statistical methods, resulting in valid and objective conclusions. The statistical approach to experimental design is necessary if we wish to draw meaningful conclusions from the data.

—Douglas C. Montgomery

This chapter will address a significant amount of terminology behind design of experiments, the principles of designing experiments, planning experiments, one-factor experiments, full factorial experiments, and two-level fractional factorial experiments. Numerous detailed examples are provided to help the reader understand the mechanics of the experiments. Some experiments are done both manually and with software.

TERMINOLOGY

> Define basic DOE terms, e.g., independent and dependent variables, factors and levels, response, treatment, error, nested. (Understand)
>
> **Body of Knowledge VII.A.1**

Design of experiments (DOE) is a statistical approach to designing and conducting experiments such that the experiment provides the most efficient and economical methods of determining the effect of a set of independent variables on a response variable. Knowledge of this relationship permits the experimenter to optimize a process and predict a response variable by setting the factors at specific levels. In other words, the objective of a designed experiment is to generate knowledge about a product or process and establish the mathematical relationship $Y = f(X)$, where X is a list of independent variables and Y is the dependent variable.

Figure 24.1 provides a visual concept of design of experiments. We have a process that uses a machine to control the surface finish of a product. The machine has three dials on the face of it. One dial controls the feed rate, one controls the speed rate, and another the coolant temperature. Each dial can be set at either a

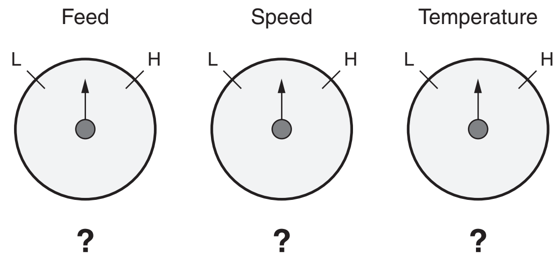

Figure 24.1 A visual concept of design of experiments.

low or high setting. The question is, "How should each dial be set to provide the smoothest surface finish?"

To better understand the concept of DOE, we will first provide a few fundamental definitions:

- An *experimental design* is a formal plan that details the specifics for conducting an experiment, such as which responses, factors, levels, blocks, treatments, and tools are to be used.
- An *effect* is the relationship between a factor and a response variable. Note that there can be many factors and response variables. Specific types include main effect, dispersion effect, and interaction effect. We will deal specifically with main and interaction effects.
- The *response variable* is the output variable that shows the observed results or value of an experimental treatment. It is sometimes known as the *dependent variable* or y *variable*. There may be multiple response variables in an experimental design.
- The *observed value* is a particular value of a response variable determined as a result of a test or measurement.
- A *factor* is an independent variable or assignable cause that may affect the responses, and of which different levels are included in the experiment. Factors are also known as *explanatory variables*, *predictor variables*, or *input variables*.
- A *noise factor* is an independent variable that is difficult or too expensive to control as part of standard experimental conditions. In general, it is not desirable to make inferences on noise factors, but they are included in an experiment to broaden the conclusions regarding control factors.
- A *level* is the setting or assignment of a factor at a specific value.
- The *design space* is the multidimensional region of possible treatment combinations formed by the selected factors and their levels.
- *Experimental error* is the variation that occurs in the response variable beyond that accounted for by the factors, blocks, or other assignable

source in conduct of an experiment. Usually it is truncated to just "error."

- An *experimental unit* is the smallest entity receiving a particular treatment that yields a value of the response variable.
- A *treatment* is the specific setting or combination of factor levels for an experimental unit.
- An *experimental run* is a single performance of the experiment for a specific set of treatment combinations.
- "n" represents the number of experimental runs that will be conducted in a given experiment and is computed as

  $$n = L^F$$

  where

  n = Number of runs

  L = Number of levels

  F = Number of factors

- According to Montgomery (2009), "by a *factorial design*, we mean that in each complete trial or replication of the experiment all possible combinations of the levels of the factors are investigated."
- A *nested design* is where the levels of one factor, say factor B, are nested under the levels of another factor, say factor A, and the levels of B are different for every level of A. This concept is best illustrated graphically as shown in Figure 24.2. Note the uniqueness of the numbering in Figure 24.2. In this figure, each subordinate level box is assigned to one unique higher-level box. To further illustrate, let's assume factor A is suppliers, factor B is ovens, and factor C is batches. Supplier 1 owns oven 1, which produced batches 1, 2, and 3, whereas suppler 2 owns oven 4, which produced batches 10, 11, and 12. These concepts are best illustrated by an example.

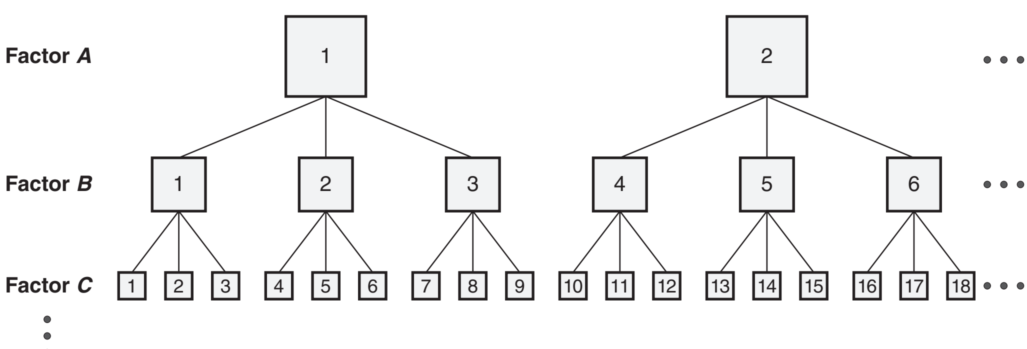

Figure 24.2 A typical structure of a nested design.

EXAMPLE 24.1

Suppose a machine operator can adjust the feed, speed, and coolant temperature of a process. Further, she wishes to find the settings that will produce the smoothest surface finish. The feed, speed, and coolant temperature are called the *factors*, or *independent variables*. The surface finish is called the *response*, or *dependent variable*, because its value depends on the values of the independent variables through the mathematical relationship that will be determined by the experiment. There may be additional independent variables, such as the hardness of the material or humidity of the room, that have an effect on the dependent variable. In this example, hardness and humidity will be considered noise factors since they are difficult or expensive to control with regard to this experiment.

The experimenter has decided that three factors will be tested at two levels each:

- Feed (F): 0.01 and 0.04 inches/revolution
- Speed (S): 1300 and 1800 revolutions/minute
- Coolant temperature (T): 100 °F and 140 °F

The experimenter has decided to conduct a full factorial experiment in order to generate the maximum amount of process knowledge. A full factorial experiment (discussed later in this chapter) tests all possible combinations of levels and factors, using one run for each combination. The number of runs in this experiment will be $n = 2^3 = 8$ since there are three factors at two levels each.

To ease the data collection process, the experimenter has developed a sheet listing those eight runs with room for recording five readings of the response variable for each run, as shown in Table 24.1.

The factor–level combinations are also called *treatments*, and the additional factor–level combinations within the experiment are called *repeated measures*. Repeated measures for this example are shown in Table 24.2. This is discussed in detail below.

Table 24.1 A 2^3 full factorial data collection sheet for Example 24.1.

Run	Feed	Speed	Temperature	y_1	y_2	y_3	y_4	y_5	Average response
1	0.01	1300	100						
2	0.01	1300	140						
3	0.01	1800	100						
4	0.01	1800	140						
5	0.04	1300	100						
6	0.04	1300	140						
7	0.04	1800	100						
8	0.04	1800	140						

Continued

Continued

Table 24.2 A 2^3 full factorial data collection sheet with data entered for Example 24.1.

Run	Feed	Speed	Temperature	y_1	y_2	y_3	y_4	y_5	Average response
1	0.01	1300	100	10.1	10.0	10.2	9.8	9.9	
2	0.01	1300	140	3.0	4.0	3.0	5.0	5.0	
3	0.01	1800	100	6.5	7.0	5.3	5.0	6.2	
4	0.01	1800	140	1.0	3.0	3.0	1.0	2.0	
5	0.04	1300	100	5.0	7.0	9.0	8.0	6.0	
6	0.04	1300	140	4.0	7.0	5.0	6.0	8.0	
7	0.04	1800	100	5.8	6.0	6.1	6.2	5.9	
8	0.04	1800	140	3.1	2.9	3.0	2.9	3.1	

DESIGN PRINCIPLES

Define and apply DOE principles, e.g., power, sample size, balance, repetition, replication, order, efficiency, randomization, blocking, interaction, confounding, resolution. (Apply)

Body of Knowledge VII.A.2

DOE is an effective tool for determining causal relationships between independent and response variables. However, considerable attention must be given to planning and conducting an experiment. Therefore, it is important to identify and define critical concepts underlying experimental designs:

- Power and sample size
- Repeated measures (repetition)
- Replication
- Confounding
- Order
- Randomization

- Blocking
- Main effect
- Interaction effect
- Balanced design
- Resolution
- Source table

Each of these concepts will be discussed in detail in the following subsections.

Power and Sample Size

The concepts of power and sample size were discussed in Chapter 21. Recall that the more power a test has, the greater the sensitivity of that test to detect small differences. Power increases as the sample size increases and can be determined before actually conducting an experiment. Also, as the power of the test increases, the probability of a type II error, β, decreases.

Repeated Measures (Repetition)

A *repeated measure*, also known as *repetition*, is the measurement of a response variable more than once under similar conditions. Repeated measures allow one to determine the inherent variability in the measurement system. Repetition occurs when each run is conducted n times in a row. This is in contrast to replication, whereby the entire experiment is repeated n times in a row. Table 24.3 provides an example of repeated measures.

Table 24.3 Example of repeated measures (repetition).

Run	Feed	Speed	Temperature	Responses		
1	–	–	–	10.0	10.0	10.0
2	–	–	+	4.0	6.0	4.0
3	–	+	–	6.0	6.0	6.0
4	–	+	+	2.0	2.0	1.0
5	+	–	–	7.0	5.0	5.0
6	+	–	+	6.0	6.0	4.0
7	+	+	–	6.0	5.0	7.0
8	+	+	+	3.0	3.0	4.0

Replication

Replication occurs when the entire experiment is performed more than once for a given set of independent variables. Each repetition of the experiment is called a *replicate*. Replication differs from repeated measures in that it is a repeat of the entire experiment for a given set of independent variables, not just a repeat of measurements for the same run. Replication increases the precision of the estimates of the effects in an experiment. It is more effective when all elements contributing to the experimental error are included. In some cases, replication may be limited to repeated measures under essentially the same conditions. Replication reflects the sources of variability both within and between runs and adds degrees of freedom to the experiment. Table 24.4 contrasts replication versus repetition (repeated measures).

Table 24.4 Replication versus repetition.

Repetition

Run	Feed	Speed	Temperature	Responses		
1	–	–	–	10.0	10.0	10.0
2	–	–	+	4.0	6.0	4.0
3	–	+	–	6.0	6.0	6.0
4	–	+	+	2.0	2.0	1.0
5	+	–	–	7.0	5.0	5.0
6	+	–	+	6.0	6.0	4.0
7	+	+	–	6.0	5.0	7.0
8	+	+	+	3.0	3.0	4.0

Replicate

Run	Feed	Speed	Temperature	Responses		
1	–	–	–	11.0	9.0	10.0
2	–	–	+	3.0	4.0	4.0
3	–	+	–	7.0	6.0	6.0
4	–	+	+	3.0	2.0	5.0
5	+	–	–	6.0	6.0	8.0
6	+	–	+	6.0	7.0	6.0
7	+	+	–	5.0	7.0	6.0
8	+	+	+	2.0	4.0	3.0

Replicate

Confounding

Confounding, or *aliasing*, occurs when factors or interactions are not distinguishable from one another. Confounding occurs when using fractional factorial designs because such designs do not include every possible combination of factors. However, higher-order interactions are generally assumed not statistically significant. (Note: In some processes, such as chemical processes, higher-order interactions must be considered.) Fortunately, we can choose which factors or interactions are confounded in order to permit the estimation of these higher-order terms.

Order

Order refers to the sequence in which the runs of an experiment will be conducted. Generally, we talk about two types of order:

- *Standard order* shows what the order of the runs in an experiment would be if the experiment were done in Yates's order. To understand Yates's order, look at the second, third, and fourth columns of Table 24.3. Notice the pattern of the lows and highs. The pattern is "cut in half" as we move from left to right.
- *Run order* shows what the order of the runs in an experiment would be if the experiment were run in random order. Random order works to spread the effects of noise variables.

Randomization

Randomization is used to assign treatments to experimental units so that each unit has an equal chance of being assigned a particular treatment, thus minimizing the effect of variation from uncontrolled noise factors. A *completely randomized design* is one in which the treatments are assigned at random to the full set of experimental units. No blocks are involved in a completely randomized design.

Blocking

A *block* is a collection of experimental units more homogeneous than the full set of experimental units. Blocks are usually selected to allow for special causes in addition to those introduced as factors to be studied. These special causes may be avoidable within blocks, thus providing a more homogeneous experimental subspace. *Blocking* refers to the method of including blocks in an experiment in order to broaden the applicability of the conclusions or minimize the impact of selected assignable causes. Randomization of the experiment is restricted and occurs within blocks. The *block effect* is the resulting effect from a block in an experimental design. Existence of a block effect generally means that the method of blocking was appropriate and that an assignable cause has been found.

An experimental design consisting of *b* blocks with *t* treatments assigned via randomization to the experimental units within each block is known as a *randomized block design* and is used to control the variability of experimental units.

EXAMPLE 24.2

Table 24.2 depicts eight treatments with five repeat measures per treatment, thus producing 40 tests. If it is not possible to run all 40 tests under the same conditions, the experimenter may decide to use blocking. For example, if the 40 tests must be spread over two shifts, the experimenter would be concerned about the impact the shift difference could have on the results. In this experiment, the coolant temperature is probably the most difficult to adjust, so the experimenter may be tempted to perform all the 100 °F runs during the first shift and the 140 °F runs during the second shift. The obvious problem here is that what appears to be the impact of the change in coolant temperature may in part reflect the impact of the change in shift. A better approach would be to randomly select the runs to be performed during each shift. This is called a *randomized block design*. For example, the random selection might put runs 1, 4, 5, and 8 in the first shift and runs 2, 3, 6, and 7 in the second shift.

Once the data are collected, as shown in Table 24.2, the next step is to find the average of the five repeated responses for each run. These averages are shown in Table 24.5.

Table 24.5 A 2^3 full factorial data collection sheet with data entered for Example 24.1.

Run	Feed	Speed	Temperature	y_1	y_2	y_3	y_4	y_5	Average response
1	0.01	1300	100	10.1	10.0	10.2	9.8	9.9	10.0
2	0.01	1300	140	3.0	4.0	3.0	5.0	5.0	4.0
3	0.01	1800	100	6.5	7.0	5.3	5.0	6.2	6.0
4	0.01	1800	140	1.0	3.0	3.0	1.0	2.0	2.0
5	0.04	1300	100	5.0	7.0	9.0	8.0	6.0	7.0
6	0.04	1300	140	4.0	7.0	5.0	6.0	8.0	6.0
7	0.04	1800	100	5.8	6.0	6.1	6.2	5.9	6.0
8	0.04	1800	140	3.1	2.9	3.0	2.9	3.1	3.0

For completely randomized designs, no stratification of the experimental units is made. In the randomized block design, the treatments are randomly allotted within each block; that is, the randomization is restricted. Similar to the randomized block design, a *randomized block factorial design* is a design run in a randomized block design where each block includes a complete set of factorial combinations.

Main Effect

A *main effect* is the impact or influence of a single factor on the mean of the response variable. A *main effects plot* is a graphical depiction that gives the average responses at the various levels of individual factors.

EXAMPLE 24.3

Refer to the data given in Table 24.5. The first step in calculating the main effects, sometimes called the *average main effects,* is to average the results for each level of each factor. This is accomplished by averaging the results of the four runs for that level. For example, $F_{0.01}$ (feed at the 0.01 inches/minute level) is calculated by averaging the result of the four runs in which feed was set at the 0.01 level.

Table 24.6 shows the calculations for the effect of the feed rate. Table 24.7 shows the calculations for the effect of the speed rate. Table 24.8 shows the calculations for the effect of the coolant temperature. Figure 24.3 graphically depicts the main effects. Notice that the feed rate in the first panel in Figure 24.3 is flat. This indicates it has no effect on the surface finish.

Table 24.6 Effect of feed rate.

Run	Feed	Average response
1	–	10.0
2	–	4.0
3	–	6.0
4	–	2.0
5	+	7.0
6	+	6.0
7	+	6.0
8	+	3.0

$$T_{100} = \frac{(10.0+4.0+6.0+2.0)}{4} = 5.5$$

$$T_{140} = \frac{(7.0+6.0+6.0+3.0)}{4} = 5.5$$

Table 24.7 Effect of speed rate.

Run	Speed	Average response
1	–	10.0
2	–	4.0
3	+	6.0
4	+	2.0
5	–	7.0
6	–	6.0
7	+	6.0
8	+	3.0

$$S_{1300} = \frac{(10.0+4.0+7.0+6.0)}{4} = 6.75$$

$$S_{1800} = \frac{(6.0+2.0+6.0+3.0)}{4} = 4.25$$

Continued

Continued

Table 24.8 Effect of coolant temperature.

Run	Temperature	Average response
1	–	10.0
2	+	4.0
3	–	6.0
4	+	2.0
5	–	7.0
6	+	6.0
7	–	6.0
8	+	3.0

$$T_{100} = \frac{(10.0+6.0+7.0+6.0)}{4} = 7.25$$

$$T_{140} = \frac{(4.0+2.0+6.0+3.0)}{4} = 3.75$$

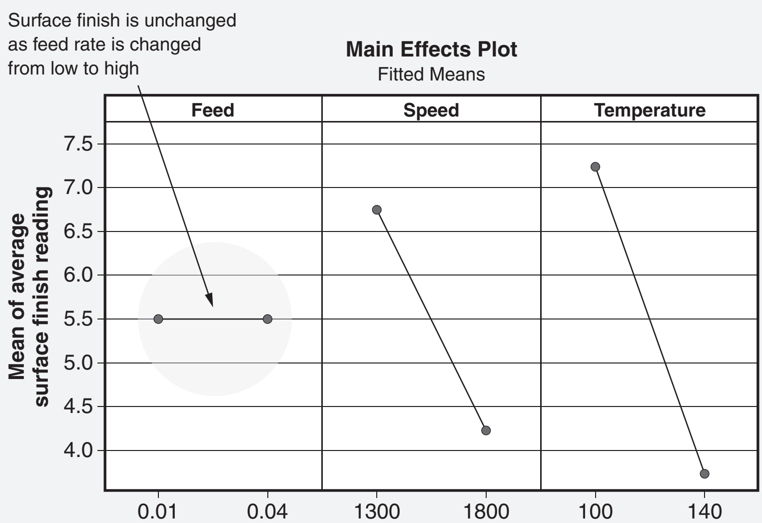

Figure 24.3 Graph of the main effects for Example 24.3.

A better surface finish is the one with the lowest response. Therefore, the experimenter would choose the level of each factor that produces the lowest result. The lowest response for speed is when the factor is set at 1800 revolutions/minute. Similarly, the lowest response occurs with a coolant temperature of 140 °F. What feed rate should be recommended? Since both $F_{0.01}$ and $F_{0.04}$ are 5.5, the feed rate doesn't impact surface finish in this range. A feed rate of 0.04 would be appropriate since it will result in a faster operation.

EXAMPLE 24.4

A quick review of the results from Example 24.3 reveals the impact of the main effects:

$$F_{0.04} - F_{0.01} = 5.5 - 5.5 = 0$$

$$S_{1800} - S_{1300} = 4.25 - 6.75 = -2.50$$

$$T_{140} - T_{100} = 3.75 - 7.25 = -3.50$$

Factors with the greatest difference between the "high" and "low" results have the greatest impact on the quality characteristic of interest (*surface finish* in the above example). The impact of the main effect is determined by subtracting the average response at the low level from the average response at the high level result for the factor.

The larger the absolute value of the main effect, the more influence that factor has on the quality characteristic. It is possible that the perceived difference between the high and low results is not statistically significant. This would occur if the experimental error is so large that it would be impossible to determine whether the difference between the high and low values is due to a real difference in the dependent variable or due to experimental error.

For analysis of data from an experiment, the null hypothesis is that changing the factor level does not make a statistically significant difference in the dependent variable. The α risk is the probability that the analysis will show that there is a significant difference when there is no difference. The β risk is the probability that the analysis will show that there is no significant difference when one exists. The power of the experiment is defined as $1 - \beta$, so the higher the power of the experiment, the lower the β risk. In general, a higher number of replications or a larger sample size provides a more precise estimate of experimental error, which in turn reduces the β risk.

Interaction Effect

An *interaction effect* occurs when the influence of one factor on the response variable depends on one or more other factors. Existence of an interaction effect means that the factors cannot be changed independently of one another. An *interaction plot* is a graphical depiction that gives the average responses at the combinations of levels of two distinct factors.

Interaction effects may be determined by replacing each high level with "+" and each low level with "–," as shown in Table 24.9.

Balanced Design

In a *balanced design*, all treatment combinations have the same number of observations (that is, in a factorial design, all factors are run at the same number of low and high levels).

Table 24.9 A 2^3 full factorial design using the + and – format.

Run	F	S	T	$F \times S$	$F \times T$	$S \times T$	$F \times S \times T$
1	–	–	–				
2	–	–	+				
3	–	+	–				
4	–	+	+				
5	+	–	–				
6	+	–	+				
7	+	+	–				
8	+	+	+				

EXAMPLE 24.5

To find an entry in the $F \times S$ column, multiply the entries in the F and S columns, using the multiplication rule: if the signs are the same, the result is positive; otherwise, the result is negative. Complete the other interaction columns the same way. To complete the $F \times S \times T$ column, multiply the $F \times S$ column by the T column. The completed interaction table is shown in Table 24.10.

The two- and three-factor interaction effect calculations for Table 24.10 are given in Table 24.11 along with their associated impact. The corresponding two-factor interaction plots are shown in Figure 24.4. The presence of interactions indicates that the main effects aren't additive.

Table 24.10 The completed interaction table for Example 24.5.

	Main effects			Two-factor interactions			Three-factor interaction	
Run	F	S	T	$F \times S$	$F \times T$	$S \times T$	$F \times S \times T$	**Average response**
1	–	–	–	+	+	+	–	10.0
2	–	–	+	+	–	–	+	4.0
3	–	+	–	–	+	–	+	6.0
4	–	+	+	–	–	+	–	2.0
5	+	–	–	–	–	+	+	7.0
6	+	–	+	–	+	–	–	6.0
7	+	+	–	+	–	–	–	6.0
8	+	+	+	+	+	+	+	3.0

Continued

Table 24.11 Two- and three-factor interaction effect calculations for Example 24.5.

Interaction effect calculations	Impact
$(F \times S)^{+} = \dfrac{10.0+4.0+6.0+3.0}{4} = 5.75$	5.75 – 5.25 = 0.50
$(F \times S)^{-} = \dfrac{6.0+2.0+7.0+6.0}{4} = 5.25$	
$(F \times T)^{+} = \dfrac{10.0+6.0+6.0+3.0}{4} = 6.25$	6.25 – 4.75 = 1.50
$(F \times T)^{-} = \dfrac{4.0+2.0+7.0+6.0}{4} = 4.75$	
$(S \times T)^{+} = \dfrac{10.0+2.0+7.0+3.0}{4} = 5.5$	5.5 – 5.5 = 0.0
$(S \times T)^{-} = \dfrac{4.0+6.0+6.0+6.0}{4} = 5.5$	
$(F \times S \times T)^{+} = \dfrac{4.0+6.0+7.0+3.0}{4} = 5.0$	5.0 – 6.0 = –1.0
$(F \times S \times T)^{-} = \dfrac{10.0+2.0+6.0+6.0}{4} = 6.0$	

Also, there are several things to note about this figure. This is a complete two-factor interaction graphic. Previously, Minitab provided the user with a choice of either the upper right half or the entire graphic as shown here. Minitab now provides either the lower left half or the full graphic. Our previous edition provided the upper right half. The newest version of Minitab does not provide an "upper right half only" option. The only way to obtain it is to select the full graphic. Why point this out if we can draw the same conclusions with either the upper right half or the lower left half? The upper right half graphic allows the user to obtain an estimate of the interaction effect impact assuming a meaningful scale for the graphs is chosen. The major tick marks in Figure 24.4 are 0.5 units apart. Notice that the right end of the speed–feed interaction graph lines are separated by 0.5 vertically. This is equivalent to the value of the impact. Similarly, the temperature–feed interaction graph lines are separated by 1.5 vertically, which is equivalent to the value of its impact.

If we review Figure 24.4 from a total perspective, we notice the following:

- Both the speed–temperature and temperature–speed lines are parallel, indicating that no interaction is present.
- In the top half of the chart, we notice that the speed–feed and temperature–feed interaction lines are crossed. The crossed lines indicate that some degree of interaction is present. The greater the degree of the cross, the greater the interaction present.

Continued

Continued

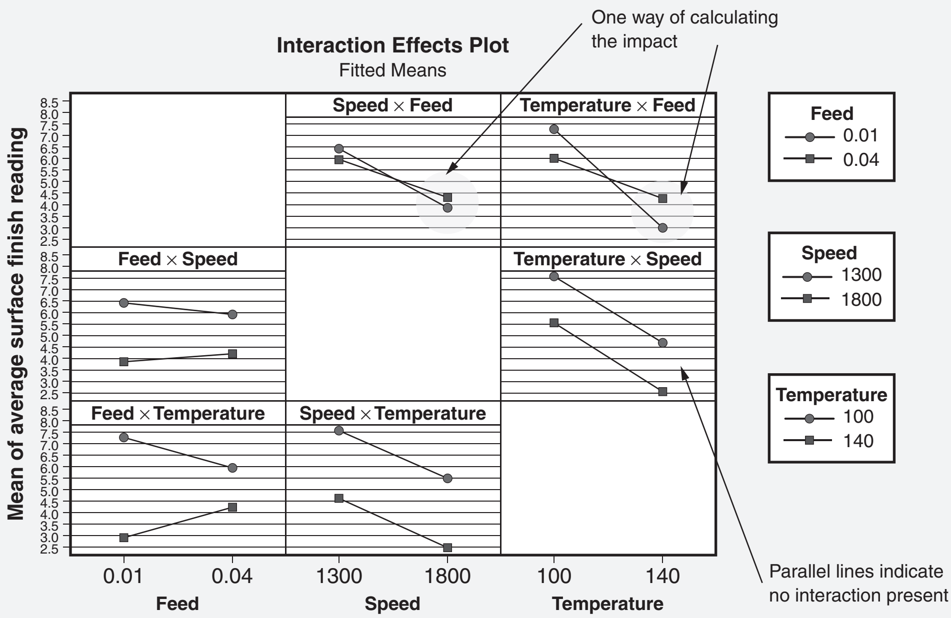

Figure 24.4 Graph of the two-factor interaction effects for Example 24.5.

- In the bottom half of the chart, we notice the feed–speed and feed–temperature interaction lines are not parallel to one another. In fact, if the lines are extended, they would cross at some point beyond the graph. Because they are not parallel, this indicates that some degree of interaction is present. The greater the degree of nonparallelism, the greater the degree of interaction. However, for reasons discussed above, we prefer the use of the top right half of the graph.

Table 24.9 represents a balanced design. This table contains three factors and eight runs. Each of the three factors is run four times at the high level and four times at the low level.

Resolution

In the context of experimental design, *resolution* refers to the level of confounding in a fractional factorial design. The resolution of a design is generally one more than the smallest-order interaction with which a main effect is confounded. For example, if factor A in a three-factor design is confounded with interaction $B \times C$, then the resolution is III. Resolutions are numbered with Roman numerals. Common resolution designs follow:

- Resolution II
  - Main effects are confounded with other main effects
- Resolution III
  - No main effects are confounded with another main effect
  - Main effects are confounded with two-factor interactions
  - Two-factor interactions may be confounded with other two-factor interactions
- Resolution IV
  - No main effects are confounded with another main effect
  - No main effects are confounded with two-factor interactions
  - Main effects are confounded with three-factor interactions
  - Two-factor interactions may be confounded with other two-factor interactions
- Resolution V
  - No main effects are confounded with another main effect
  - No main effects are confounded with two-factor interactions
  - No main effects are confounded with three-factor interactions
  - Main effects are confounded with four-factor interactions
  - No two-factor interactions are confounded with other two-factor interactions
  - Two-factor interactions may be confounded with three-factor interactions

Higher levels of resolution can also be achieved depending on the specific fractional factorial design. However, the issue of resolution is based on addressing the issue of confounding.

Some questions an experimenter must ask when developing a fractional factorial design with regard to confounding include:

- Where does the confounding exist?
- Can I live with the confounding? Is the confounding in higher-order interactions that can be assumed negligible?
- Does the confounding exist in lower-order interactions that have an impact?
- Can I remove the confounding through a carefully designed fractional factorial?
- Can I minimize the impact of confounding through a carefully designed factorial?

It is clear that thoughtful consideration along with subject matter expertise is required before designing fractional factorial designs.

EXAMPLE 24.6

Table 24.12 shows a 2^4 (two levels for four factors) full factorial design. Table 24.13 illustrates a one-half fraction of the 2^4 full factorial design with interaction columns added. Note that there are six two-factor interactions, four three-factor interactions, and one four-factor interaction. Also note that factor *A* is confounded with the *BCD* interaction because they have the same (+/–) pattern. All of the confounded or aliased pairs are shown in Table 24.13.

The big advantage of this particular fractional factorial is that, although there is confounding, main effects are confounded with three-factor interactions only. Since three-factor interactions are often small, the confounding of main effects is usually minor. Of course, if a three-factor interaction is significant, as it often is in chemical reactions, it will be missed with this design. Another downside of this design is that two-factor interactions are confounded with each other (for example, *AB* with *CD*). This means that an accurate picture of two-factor interactions is not possible.

Table 24.14 depicts an alternate or complementary one-half fraction of the same 2^4 factorial design given in Table 24.12. In fact, the confounded (or aliased) pairs are the same as Table 24.13. However, notice that the header row is a bit different. A negative sign precedes some of the interactions, and the signs in corresponding columns are reversed from Table 24.13. This is because the defining relation for this design is negative. The discussion of the defining relation is beyond the scope of this book. Readers interesting in pursuing this topic further are referred to Montgomery (2009).

Table 24.12 2^4 full factorial design without interactions shown for Example 2.6.

Run	*A*	*B*	*C*	*D*
1	–	–	–	–
2	+	–	–	–
3	–	+	–	–
4	+	+	–	–
5	–	–	+	–
6	+	–	+	–
7	–	+	+	–
8	+	+	+	–
9	–	–	–	+
10	+	–	–	+
11	–	+	–	+
12	+	+	–	+
13	–	–	+	+
14	+	–	+	+
15	–	+	+	+
16	+	+	+	+

Continued

Continued

Table 24.13 One-half fraction of a 2^4 factorial design given in Table 24.12.

Confounded (aliased) pairs

Runs	*A*	*B*	*C*	*D*	*AB*	*AC*	*AD*	*BC*	*BD*	*CD*	*ABC*	*ABD*	*ACD*	*BCD*	*ABCD*
1	−	−	−	−	+	+	+	+	+	+	−	−	−	−	+
4	+	+	−	−	+	−	−	−	−	+	−	−	+	+	+
6	+	−	+	−	−	+	−	−	+	−	−	+	−	+	+
7	−	+	+	−	−	−	+	+	−	−	−	+	+	−	+
10	+	−	−	+	−	−	+	+	−	−	+	−	−	+	+
11	−	+	−	+	−	+	−	−	+	−	+	−	+	−	+
13	−	−	+	+	+	−	−	−	−	+	+	+	−	−	+
16	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+

Table 24.14 Alternate (complementary) one-half fraction of a 2^4 factorial design given in Table 24.12.

Confounded (aliased) pairs

Runs	*A*	*B*	*C*	*D*	*–AB*	*–AC*	*–AD*	*BC*	*BD*	*CD*	*–ABC*	*–ABD*	*–ACD*	*BCD*	*–ABCD*
2	+	−	−	−	−	−	−	+	+	+	+	+	+	−	−
3	−	+	−	−	−	+	+	−	−	+	+	+	−	+	−
5	−	−	+	−	+	−	+	−	+	−	+	−	+	+	−
8	+	+	+	−	+	+	−	+	−	−	+	−	−	−	−
9	−	−	−	+	+	+	−	+	−	−	−	+	+	+	−
12	+	+	−	+	+	−	+	−	+	−	−	+	−	−	−
14	+	−	+	+	−	+	+	−	−	+	−	−	+	−	−
15	−	+	+	+	−	−	−	+	+	+	−	−	−	+	−

In today's world, experimental designs are frequently generated with the use of computer software. However, the application of such software requires more than a passing understanding of the concept of resolution. Unfortunately, a more detailed coverage of the topic of resolution is beyond the scope of this book. Readers interested in pursuing the topic in more depth should consult Montgomery (2009) or Box et al. (2005).

Source Table

A *source table* is a computational table used to analyze an experimental design. The typical structure of a source table is shown in Table 24.15. The table is five columns wide and the length depends on the number of factors. The table is built moving from left to right. Column 1 is used for identifying the factors, interactions, blocks, and so on. Column 2 is used for sums of squares. Formulas for sums of squares for common designs are provided later in this chapter. Column 3 is used for computing the degrees of freedom corresponding to each sum of squares in Column 2. As with the formulas, the degrees of freedoms are provided for common designs later in the chapter. Column 4 is determined by dividing column 2 (sum of squares) by column 3 (degrees of freedom). Column 5 is determined by dividing each factor or interaction by the error term in column 4. Ultimately, the resulting terms in column 5 will be used to compare critical F values obtained from the appropriate appendices to determine whether each factor or interaction term is significant.

Typically, terms related to error, such as *sums of squares* and *degrees of freedom*, are calculated by subtraction. More specifically:

Table 24.15 The typical structure of a source table.

	Sources of variation	**Sum of squares**	**Degrees of freedom**	**Mean squares**	**F_0 statistic**
List of factors	Factor A	SS_A	$a - 1$	$MS_A = \dfrac{SS_A}{a-1}$	$F_0 = \dfrac{MS_A}{MS_E}$
	Factor B	SS_B	$b - 1$	$MS_B = \dfrac{SS_B}{b-1}$	$F_0 = \dfrac{MS_B}{MS_E}$
List of interactions	Interaction AB	SS_{AB}	$(a - 1)(b - 1)$	$MS_{AB} = \dfrac{SS_{AB}}{(a-1)(b-1)}$	$F_0 = \dfrac{MS_{AB}}{MS_E}$
Error term	Error	SS_E	$ab(n - 1)$	$MS_E = \dfrac{SS_E}{ab(n-1)}$	
Total term	Total	SS_T	$abn - 1$		

Table 24.16 Applying the DOE terminology—full factorial and balanced design.

Factor | Factor level at low setting | Factor level at high setting

Run	Feed	Speed	Temperature	Average response
1	–	–	–	10.0
2	–	–	+	4.0
3	–	+	–	6.0
4	–	+	+	2.0
5	+	–	–	7.0
6	+	–	+	6.0
7	+	+	–	6.0
8	+	+	+	3.0

Experiment (one replicate)

Average response values based on repeated measures

n = Number of runs | Experimental run | Treatment (treatment combinations)

$$SS_E = SS_T - SS_A - SS_B - SS_{A\times B}$$

$$DF_E = DF_T - DF_A - DF_B - DF_{A\times B}$$

Of course, the above equations should be generalized to account for multiple factors, interactions, blocks, and so on.

Pulling It All Together

The above two sections introduced a substantial amount of terminology and concepts. Several figures and tables were also introduced to help pull many of these concepts together. We will now present one additional table. Table 24.16 includes many of the terms discussed above in one simple table.

PLANNING EXPERIMENTS

> Plan and evaluate DOEs by determining the objective, selecting appropriate factors, responses, and measurement methods, and choosing the appropriate design. (Evaluate)
>
> **Body of Knowledge VII.A.3**

When preparing to conduct an experiment, the first consideration is, "What question are we seeking to answer?" In Example 24.1, the objective was to find the combination of process settings that minimizes the surface finish reading. Examples of other experimental objectives follow:

- Find the combination of mail and media ads that produces the most sales
- Find the cake recipe that produces the most consistent taste in the presence of oven temperature variation
- Find the combination of valve dimensions that produces the greatest linear output

Sometimes the objective derives from a question. For example, "What's causing the excess variation in hardness at the rolling mill?" could generate the objective "Identify the factors responsible for the hardness variation and find the settings that minimize it." The objective must be related to the enterprise's goals and objectives. It must also be measurable, so the next step is to establish an appropriate measurement system. The measurement system must be reasonably simple and easy to operate.

In simpler terms, the primary objective of an experimental design is to:

- Identify which independent variables explain the most variation in the response variable
- Determine what levels of the independent variables minimize or maximize the response variable

Once the objective and a measurement system have been determined, the factors and levels are selected. People with the most experience and knowledge about the product or process should be consulted to determine what factors and what levels of each factor are important to achieve the objective. The list of factors and the levels for each factor should flow from their input.

The next step is to choose the appropriate design. The selection of the design may be constrained by such things as affordability and time available. At this stage, some experimenters establish a budget of time, cost, and other resources that may be used to reach the objective. If production equipment and personnel must be used, how much time is available? How much product and other consumables are available? Typically, 20% to 40% of the available budget should be allocated to the first experiment because it seldom meets the objective and, in fact, often raises as many questions as it answers. Examples of such new questions include:

- What if an additional level had been used for factor *A*?
- What if a different factor had been included instead of factor *B*?
- What if an additional factor had been included?

Therefore, rather than designing a massive experiment involving many variables and levels, it is usually best to begin with more modest screening designs. A *screening design* is an experiment intended to identify a subset of the collection of factors for subsequent study. An example of a screening design is the *Plackett-Burman* design, which is used when there are many factors to study. These designs study only the main effects and are primarily used as a screening design before applying other types of designs. In general, Plackett-Burman designs are two-level, but three-, five-, and seven-level designs are available. They allow for efficient estimation of the main effects and assume that interacting terms can be initially ignored.

The activities outlined in the earlier discussion should not be left to happenstance. Instead, they should result from the development of an experimental plan. An *experimental plan* is the assignment of treatments to each experimental unit and the time order in which the treatments are to be applied. Planning an experiment requires several design considerations:

- Noise factors (discussed previously)
- Confounding (discussed previously)
- Blocking, randomization, and resolution (all of which were discussed previously)
- The cost to run the experiment
- Time considerations—limited time might be available to conduct the experiment

Montgomery (2005) offers guidelines for conducting experiments:

1. *Recognition and statement of the problem*. As discussed earlier, this step might start with a question, but it must eventually be translated into a statistical problem.
2. *Selection of the response variable*. People most familiar with the process should be consulted.
3. *Choice of factors, levels, and range*. As with guideline 2, the subject matter experts who are familiar with the process should be consulted.
4. *Choice of experimental design*. Choosing an experimental design requires consideration of an experimental plan (discussed previously).
5. *Performing the experiment*. It is critical that the experimenter understand the conditions under which the experiment was conducted. Ideally, these conditions should match those conditions under which the process is routinely performed.
6. *Statistical analysis of the data*. The data must be carefully analyzed to ensure statistical validity.
7. *Conclusions and recommendations*. Statistical conclusions and recommendations must be translated into terms that management and operations people can understand, particularly if they are expected to support and/or fund process changes.

The next few sections discuss various designs.

ONE-FACTOR EXPERIMENTS

> Design and conduct completely randomized, randomized block, and Latin square designs, and evaluate their results. (Evaluate)
>
> **Body of Knowledge VII.A.4**

This section addresses the following designs:

- One-way ANOVA completely randomized
- Randomized complete block design (RCBD)
- Latin square design

For the reader's convenience, the statistical models, source tables, and definitions for the sums of squares have been summarized in three tables. This allows the reader to more readily compare and contrast each model.

In addition, a "dot" notation is commonly used in many DOE formulas. For readers who may be unfamiliar with this notation, please see Table 24.17. This

Table 24.17 Using the "dot" notation in matrices and in summations.

Sum across "j"

Sum across "i"

y_{11}	y_{12}	.	.	.	y_{1n}	$\sum_{j=1}^{n} y_{ij} = y_{1.}$
y_{21}	y_{22}				y_{2n}	$\sum_{j=1}^{n} y_{ij} = y_{2.}$
.	.				.	.
.	.				.	.
.	.				.	.
y_{n1}	y_{n2}	.	.	.	y_{nn}	
$\sum_{i=1}^{n} y_{ij} = y_{.1}$	$\sum_{i=1}^{n} y_{ij} = y_{.2}$					$\sum_{j=1}^{n} y_{.j} = \sum_{i=1}^{n} y_{i.} = \sum_{i=1}^{n}\sum_{j=1}^{n} y_{ij} = y_{..}$

table illustrates how the "dot" notation is simply a shorthand notation that uses subscripts to show completed summations. This notation is particularly convenient when multiple subscripts are being used concurrently.

Completely Randomized Design

A *one-factor* (that is, one-way) is a single-factor analysis of variance experiment.

The top portions of Tables 24.18–24.20 provide the statistical model, the source table, and the definition of the sums of squares for the one-way ANOVA, respectively. See Example 21.12.

Randomized Complete Block Design

In a *randomized complete block design* (RCBD), each block contains all treatments, and randomization is restricted within blocks.

The middle portions of Tables 24.18–24.20 provide the statistical model, the source table, and the definition of the sums of squares for the RCBD.

Typically, such designs are run using the software. However, the following example will be done manually.

EXAMPLE 24.7

This example is taken from Montgomery (2009). A medical device manufacturer produces vascular grafts. These grafts are produced by extruding resin along with a lubricant into tubes. The engineer suspects batch-to-batch variation. Consequently, the developer investigates the effects of four different levels of extrusion pressure using an RCBD design with the batches considered as blocks. The data for this example are shown in Table 24.21. Table 24.22 is similar to Table 24.21, but column and row totals have been added for treatment and block totals, respectively. In addition, the computations for the sum of squares for both blocks and treatments are shown. Table 24.23 is used for computing the sum of squares for the total. Each cell that held an original data element is now replaced by the square of that original data element. The bottom right corner shown in bold is $y^2 = 193999.3$. The computation for the sum of squares for the total is shown.

At this point, we're ready to transfer our sums of squares to the source table shown in Table 24.24 and begin to complete the table. There are four treatments. Therefore, there are three degrees of freedom. Similarly, there are six blocks. Therefore, there are five degrees of freedom. There are $4 \times 6 - 1$ or 23 degrees of freedom for the total. The degrees of freedom for the error term are obtained by subtraction. The remainder of the table is completed using the procedure described previously.

Assuming $\alpha = 0.05$, we must compare our F_0 value for the treatments to $F_{0.05,3,15}$. From Appendix 21, the F-value is 3.29. Since $8.1 > 3.29$, we reject the null hypothesis and conclude there is a difference in the mean extrusion pressure. For the blocks (batches), we must compare our F_0 value for the treatments to $F_{0.05,5,15}$. From Appendix 21, the F-value is 2.90. Since $5.3 > 2.90$, we reject the null hypothesis and conclude that the blocks (batches) are significant and affect the extrusion pressure.

Continued

Table 24.18 Statistical models for common experimental designs.

Name	Model	Hypotheses	Assumptions
One-way ANOVA (completely randomized design with fixed effects)	$y_{ij} = \mu + \tau_i + \varepsilon_{ij} \begin{cases} i = 1, 2, \ldots, a \\ j = 1, 2, \ldots, n \end{cases}$ y_{ij} = jth observation of the ith treatment μ = overall mean τ_i = effect of the ith treatment ε_{ij} = error term	$H_0 : \mu_1 = \mu_2 = \ldots = \mu_a$ $H_A : \mu_i \neq \mu_j$ for at least one pair (i, j) or $H_0 : \tau_1 = \tau_2 = \ldots = \tau_a = 0$ $H_A : \tau_i \neq 0$ for at least one i	$\varepsilon_{ij} : NID(0, \sigma^2)$ and the variance, σ^2, of each factor is constant for all levels
Randomized complete block design (RCBD)	$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij} \begin{cases} i = 1, 2, \ldots, a \\ j = 1, 2, \ldots, b \end{cases}$ y_{ij} = jth observation of the ith treatment jth block μ = overall mean τ_i = effect of the ith treatment β_j = effect of the jth block ε_{ij} = error term	$H_0 : \mu_1 = \mu_2 = \ldots = \mu_a$ $H_A : \mu_i \neq \mu_j$ for at least one pair (i, j) or $H_0 : \tau_1 = \tau_2 = \ldots = \tau_a = 0$ $H_A : \tau_i \neq 0$ for at least one i	$\varepsilon_{ij} : NID(0, \sigma^2)$

Continued

Table 24.18 *Continued.*

Name	Model	Hypotheses	Assumptions
Latin square design ($p \times p$)	$y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk} \begin{cases} i = 1,2,\ldots,p \\ j = 1,2,\ldots,p \\ k = 1,2,\ldots,p \end{cases}$ y_{ijk} = observation of the ith row, kth column for the jth treatment μ = overall mean α_i = effect of the ith row τ_j = effect of the jth treatment β_k = effect of the kth column ε_{ij} = error term	$H_0 : \mu_1 = \mu_2 = \ldots = \mu_p$ $H_A : \mu_i \neq \mu_j$ for at least one pair (i, j) or $H_0 : \tau_1 = \tau_2 = \ldots = \tau_p = 0$ $H_A : \tau_j \neq 0$ for at least one j	$\varepsilon_{ij} : NID(0, \sigma^2)$

Note: $NID(0, \sigma^2)$ means normal and independently distributed with mean 0 and variance σ^2.
Source: Montgomery (2009).

Table 24.19 Source tables for the models given in Table 24.18.

Sources of variation	Sum of squares	Degrees of freedom	Mean squares	F_0 statistic
Between treatments	$SS_{\text{Treatments}}$	$a-1$	$MS_B = \dfrac{SS_B}{a-1}$	$F_0 = \dfrac{MS_B}{MS_E}$
Error (within treatments)	SS_E (By subtraction	$N-a$	$MS_E = \dfrac{SS_E}{N-a}$	
Total	SS_T	$N-1$		

One-way ANOVA source table (completed randomized design with fixed effects)

Sources of variation	Sum of squares	Degrees of freedom	Mean squares	F statistic
Treatments	$SS_{\text{Treatments}}$	$a-1$	$MS_{\text{Treatments}} = \dfrac{SS_{\text{Treatments}}}{a-1}$	$F_0 = \dfrac{MS_{\text{Treatments}}}{MS_E}$
Blocks	SS_{Blocks}	$b-1$	$MS_{\text{Blocks}} = \dfrac{SS_{\text{Blocks}}}{b-1}$	
Error	SS_E	$(a-1)(b-1)$	$MS_E = \dfrac{SS_E}{(a-1)(b-1)}$	
Total	SS_T	$N-1$		

Randomized complete block design (RCBD) source table

Sources of variation	Sum of squares	Degrees of freedom	Mean squares	F statistic
Treatments	$SS_{\text{Treatments}}$	$p-1$	$MS_{\text{Treatments}} = \dfrac{SS_{\text{Treatments}}}{p-1}$	$F_0 = \dfrac{MS_{\text{Treatments}}}{MS_E}$
Rows	SS_{Rows}	$p-1$	$MS_{\text{Rows}} = \dfrac{SS_{\text{Rows}}}{p-1}$	
Columns	SS_{Columns}	$p-1$	$MS_{\text{Columns}} = \dfrac{SS_{\text{Columns}}}{p-1}$	
Error	SS_E (By subtraction)	$(p-2)(p-1)$	$MS_E = \dfrac{SS_E}{(p-2)(p-1)}$	
Total	SS_T	(p^2-1)		

Latin square design source table

Source: Montgomery (2009).

Table 24.20 Sums of squares for the models given in Table 28.10.

Statistical model	Sums of squares (SS)
One-way ANOVA (completely randomized design with fixed effects)	$SS_T = \sum_{i=1}^{a}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{..}^2}{N}$ $SS_{\text{Treatments}} = \sum_{i=1}^{a} \frac{y_{i}^2}{n} - \frac{y_{..}^2}{N}$ $SS_E = SS_T - SS_{\text{Treatments}}$
Randomized complete block design (RCBD)	$SS_T = \sum_{i=1}^{a}\sum_{j=1}^{b} y_{ij}^2 - \frac{y_{..}^2}{N}$ $SS_{\text{Treatments}} = \sum_{i=1}^{a} \frac{y_{i.}^2}{b} - \frac{y_{..}^2}{N}$ $SS_{\text{Blocks}} = \sum_{j=1}^{b} \frac{y_{.j}^2}{a} - \frac{y_{..}^2}{N}$ $SS_E = SS_T - SS_{\text{Treatments}} - SS_{\text{Blocks}}$
Latin square design ($p \times p$)	$SS_T = \sum_{i=1}^{p}\sum_{j=1}^{p}\sum_{k=1}^{p} y_{ijk}^2 - \frac{y_{...}^2}{N}$ $SS_{\text{Treatments}} = \sum_{i=1}^{p} \frac{y_{.j.}^2}{p} - \frac{y_{...}^2}{N}$ $SS_{\text{Rows}} = \sum_{i=1}^{p} \frac{y_{i..}^2}{p} - \frac{y_{...}^2}{N}$ $SS_{\text{Columns}} = \sum_{k=1}^{p} \frac{y_{..k}^2}{p} - \frac{y_{...}^2}{N}$ $SS_E = SS_T - SS_{\text{Treatments}} - SS_{\text{Rows}} - SS_{\text{Columns}}$

Source: Montgomery (2009).

Table 24.21 Data for Example 24.7.

Pressure	1	2	3	4	5	6
8500	90.3	89.2	98.2	93.9	87.4	97.9
8700	92.5	89.5	90.6	94.7	87.0	95.8
8900	85.5	90.8	89.6	86.2	88.0	93.4
9100	82.5	89.5	85.6	87.4	78.9	90.7

Continued

Continued

Table 24.22 Computation of SS_{Blocks} and $SS_{\text{Treatments}}$.

	Block (batches of resin)						
Pressure	**1**	**2**	**3**	**4**	**5**	**6**	**Treatment totals**
8500	90.3	89.2	98.2	93.9	87.4	97.9	556.9
8700	92.5	89.5	90.6	94.7	87.0	95.8	550.1
8900	85.5	90.8	89.6	86.2	88.0	93.4	533.5
9100	82.5	89.5	85.6	87.4	78.9	90.7	514.6
Block totals	350.8	359.0	364.0	362.2	341.3	377.8	**2155.1**

$$SS_{\text{Blocks}} = \frac{(350.8)^2 + (359.0)^2 + (364.0)^2 + (362.2)^2 + (341.3)^2 + (377.8)^2}{6} - \frac{(2155.1)^2}{24} = 192.3$$

$$SS_{\text{Treatments}} = \frac{(556.9)^2 + (550.1)^2 + (533.5)^2 + (514.6)^2}{4} - \frac{(2155.1)^2}{24} = 178.2$$

Table 24.23 Computation of SS_T.

	Block (batches of resin)						
Pressure	**1**	**2**	**3**	**4**	**5**	**6**	**Treatment totals**
8500	8154.1	7956.6	9643.2	8817.2	7638.8	9584.4	51794.4
8700	8556.3	8010.3	8208.4	8968.1	7569.0	9177.6	50489.6
8900	7310.3	8244.6	8028.2	7430.4	7744.0	8723.6	47481.1
6806.3	6806.3	8010.3	7327.4	7638.8	6225.2	8226.5	44234.3
Block totals	30826.8	32221.8	33207.1	32854.5	29177.0	35712.1	**193999.3**

$$SS_T = 193999.3 - \frac{(2155.1)^2}{24} = 193999.3 - 193519.0 = 480.3$$

Table 24.24 Source table for Example 24.7.

Source of variation	Sum of squares	Degrees of freedom	Mean squares	F_0 statistic
Treatments (pressure)	178.2	3	59.4	8.1
Blocks (batches)	192.3	5	38.5	5.3
Error	109.8	15	7.3	
Total	**480.3**	**23**		

Latin Square

A *Latin square* design is generally used to eliminate two block effects by balancing out their contributions. Table 24.25 illustrates examples of common Latin square designs up to order 5. Notice in these designs that each letter occurs exactly once in each row and column. As a result, these designs place restrictions on randomization. A basic assumption of a Latin square design is that these block factors do not interact with the factor of interest or with each other. The design is particularly useful (when the assumptions are valid) for minimizing the amount of experimentation.

Table 24.25 Examples of Latin squares from each main class up to order 5.

Latin square size ($n \times n$)	Example 1	Example 2
2×2	$\begin{bmatrix} A & B \\ B & A \end{bmatrix}$	$\begin{bmatrix} B & A \\ A & B \end{bmatrix}$
3×3	$\begin{bmatrix} A & B & C \\ B & C & A \\ C & A & B \end{bmatrix}$	$\begin{bmatrix} A & C & B \\ B & A & C \\ C & B & A \end{bmatrix}$
4×4	$\begin{bmatrix} A & B & C & D \\ B & A & D & C \\ C & D & A & B \\ D & C & B & A \end{bmatrix}$	$\begin{bmatrix} A & B & C & D \\ B & D & A & C \\ C & A & D & B \\ D & C & B & A \end{bmatrix}$
5×5	$\begin{bmatrix} A & B & C & D & E \\ B & C & E & A & D \\ C & E & D & B & A \\ D & A & B & E & C \\ E & D & A & C & B \end{bmatrix}$	$\begin{bmatrix} A & B & C & D & E \\ B & D & A & E & C \\ C & E & D & B & A \\ D & A & E & C & B \\ E & C & B & A & D \end{bmatrix}$

EXAMPLE 24.8

A process can be run at 180 °F, 200 °F, or 220 °F. What might an experimenter do to determine whether temperature significantly affects the moisture content?

Table 24.26 shows a 3 by 3 Latin square design that will be used as the basis for this example.

Table 24.27 depicts a design similar to Table 24.26, but data are now included. Note the linkage of the data to the Latin letters. This linkage will be become important later as we determine the sum of squares of the moisture content.

Continued

Table 24.26 The design is chosen and the Latin letter assignments are given for Example 24.8.

	Temperature		
Day	**180 °F**	**200 °F**	**220 °F**
1	A	B	C
2	B	C	A
3	C	A	B

Table 24.27 The moisture content responses are linked to the Latin letter assignments.

	Temperature		
Day	**180 °F**	**200 °F**	**220 °F**
1	A = 10.8	B = 11.4	C = 14.3
2	B = 10.4	C = 11.9	A = 12.6
3	C = 11.2	A = 11.6	B = 13.0

Table 24.28 Computation of SS_{Rows} (day) and SS_{Columns} (machine).

	Temperature					
Day	**180 °F**	**200 °F**	**220 °F**	**Total**	**Total squared**	**Average total squared**
1	10.8	11.4	14.3	36.5	1332.25	
2	10.4	11.9	12.6	34.9	1218.01	
3	11.2	11.6	13.0	35.8	1281.64	
Total	32.4	34.9	39.9	**107.2**	3831.90	
Total squared	1049.76	1218.01	1592.01	3859.78		**1286.59 (columns)**
Average total squared					**1277.30 (rows)**	

$$SS_{\text{Rows}} = \frac{1332.25+1218.01+1281.64}{3} - \frac{(107.2)^2}{9} = 1277.30 - 1276.87 = 0.43$$

$$SS_{\text{Columns}} = \frac{1049.76+1218.01+1592.01}{3} - \frac{(107.2)^2}{9} = 1286.59 - 1276.87 = 9.72$$

Tables 24.28 through 24.30 have the original data matrix as their basis but have been expanded to allow for additional column and/or row computations. Furthermore, the computations for sum of squares have been added to these tables for completeness purposes. All that is required is to transfer the sum of squares to the source table given in Table 24.31. Table 24.31 is completed using the procedure described previously.

Continued

Table 24.29 Computation of $SS_{\text{Moisture content}}$.

Latin letter	Temperature 180 °F	200 °F	220 °F	Total	Total squared
A	10.8	11.6	12.6	35.0	1225.00
B	10.4	11.4	13.0	34.8	1211.04
C	11.2	11.9	14.3	37.4	1398.76
Total	32.4	34.9	39.9	**107.2**	3834.80
Average total squared					**1278.27 (moisture content)**

$$SS_{\text{Moisture}} = \frac{1225.00 + 1211.04 + 1398.76}{3} - \frac{(107.2)^2}{9} = 1278.27 - 1276.87 = 1.40$$

Table 24.30 Computation of SS_T.

Day	Temperature 180 °F	200 °F	220 °F	Total squared
1	116.64	129.96	204.49	451.09
2	108.16	141.61	158.76	408.53
3	125.44	134.56	169.00	429.00
Total squared	350.24	406.13	532.25	**1288.62**

$$SS_T = (116.64 + 129.96 + \ldots + 169.00) - \frac{(107.2)^2}{9} = 1288.62 - 1276.87 = 11.75$$

Table 24.31 Source table for Example 24.8.

Source of variation	Sum of squares	Degrees of freedom	Mean squares	F_0 statistic
Treatments (moisture)	1.40	2	0.70	7.00
Rows (day)	0.43	2	0.22	2.15
Columns (temperature)	9.72	2	4.86	48.60
Error	0.20	2	0.10	
Total	**11.75**	**8**		

Continued

Continued

Assuming $\alpha = 0.05$, we must compare our F_0 value to $F_{0.05,2,2}$. From Appendix 21, the F-values are 19.0 since the same degrees of freedom are involved in all three comparisons. Since 7.00 < 19.0, we conclude that there is no significant difference in the mean moisture content. Row or day is not significant since 2.15 < 19.00. Temperature is significant since 48.60 > 19.00. Thus, we can conclude that temperature affects moisture content.

TWO-LEVEL FRACTIONAL FACTORIAL EXPERIMENTS

Design, analyze, and interpret these types of experiments, and describe how confounding can affect their use. (Evaluate)

Body of Knowledge VII.A.5

A *factorial design* is an experimental design consisting of all possible treatments formed from two or more factors, with each factor being studied at two or more levels. When all combinations are run, the interaction effects as well as the main effects can be estimated. By contrast, a *fractional factorial design* is an experimental design consisting of a subset (fraction) of the factorial design. Typically, the fraction is a simple proportion of the full set of possible treatment combinations. For example, half-fractions, quarter-fractions, and so forth, are common. While fractional factorial designs require fewer runs, some degree of confounding occurs.

Full factorial experiments (described in the next section) require a large number of runs, especially if several factors or several levels are involved. Recall that the formula for number of runs in a full factorial experiment is

$$n = L^F$$

where

n = Number of runs

L = Number of levels

F = Number of factors

For example, an experiment with eight two-level factors has $2^8 = 256$ runs, and an experiment with five three-level factors has $3^5 = 243$ runs. As you can see, full factorial experiments require significant numbers of runs, which can be very costly and time-consuming.

Furthermore, if the experiment is testing the effect of various factors on product quality in a manufacturing process, the tests typically must be run sequentially rather than simultaneously. A full factorial experiment with several factors and/or levels may require that a piece of production equipment be taken out of production for a considerable amount of time. Because of the extensive resource requirements of full factorial experiments, fractional factorial experimental designs were developed.

EXAMPLE 24.9

Consider the 2^{4-1} fractional factorial design and data given in Table 24.32. This is a resolution IV design. Analyze this design to determine which factors are significant. Assume $\alpha = 0.05$.

The following illustrates the analysis of this design:

- Figure 24.5 provides the Minitab session window for the initial model. All four main effects and three of the six two-way interactions are in the model. Since there are zero degrees of freedom for the error term, we are unable to obtain *F*-values and *p*-values in the source table.

  Notice from the alias table at the bottom of Figure 24.5 that *AB* is aliased with *CD*, *AC* with *BD*, and *BC* with *AD*. This is expected with a resolution IV design.

  Also, notice from the coefficients table that main effect *D* has the smallest coefficient among the main effects.

- Figure 24.6 provides a Pareto chart of the effects for the initial model. Main and interaction effects falling to the left of the heavy dotted line (31.05 value) are considered not significant. In this case, all terms except *A* are considered not to be significant.

- Figure 24.7 provides the main effects plot. Notice that the plots for main effects *B*, *C*, and *D* are relatively flat, thus suggesting they are relatively unimportant to

Table 24.32 A one-half 2^{4-1} fractional factorial in Minitab format for Example 24.9.

StdOrder	RunOrder	CenterPt	Blocks	A	B	C	D	Response
3	1	1	1	–1	1	–1	1	5
8	2	1	1	1	1	1	1	87
2	3	1	1	1	–1	–1	1	83
6	4	1	1	1	–1	1	–1	67
1	5	1	1	–1	–1	–1	–1	10
4	6	1	1	1	1	–1	–1	75
7	7	1	1	–1	1	1	–1	25
5	8	1	1	–1	–1	1	1	20

Continued

```
Factorial Regression: Response versus A, B, C, D

Analysis of Variance

Source                  DF   Adj SS    Adj MS  F-Value  P-Value
Model                    7  8424.00   1203.43        *        *
  Linear                 4  8081.00   2020.25        *        *
    A                    1  7938.00   7938.00        *        *
    B                    1    18.00     18.00        *        *
    C                    1    84.50     84.50        *        *
    D                    1    40.50     40.50        *        *
  2-Way Interactions     3   343.00    114.33        *        *
    A*B                  1    18.00     18.00        *        *
    A*C                  1   144.50    144.50        *        *
    B*C                  1   180.50    180.50        *        *
Error                    0        *         *
Total                    7  8424.00
```

Source table

- No *F* or *p*-values
- Zero degrees of freedom for error term

```
Model Summary

S     R-sq  R-sq(adj)  R-sq(pred)
*  100.00%          *           *

Coded Coefficients

                              SE
Term       Effect    Coef   Coef  T-Value  P-Value   VIF
Constant            46.50      *        *        *
A           63.00   31.50      *        *        *  1.00
B           3.000   1.500      *        *        *  1.00
C           6.500   3.250      *        *        *  1.00
D           4.500   2.250      *        *        *  1.00
A*B         3.000   1.500      *        *        *  1.00
A*C        -8.500  -4.250      *        *        *  1.00
B*C         9.500   4.750      *        *        *  1.00
```

Coefficients table

```
Regression Equation in Uncoded Units

Response = 46.50 + 31.50 A + 1.500 B + 3.250 C + 2.250 D + 1.500 A*B - 4.250 A*C
+ 4.750 B*C

Alias Structure

Factor Name

A      A
B      B
C      C
D      D

Aliases

I + ABCD
A + BCD
B + ACD
C + ABD
D + ABC
AB + CD
AC + BD
BC + AD
```

List of aliases (confounding terms)

Resolution IV

Figure 24.5 Session window for the initial model.

Continued

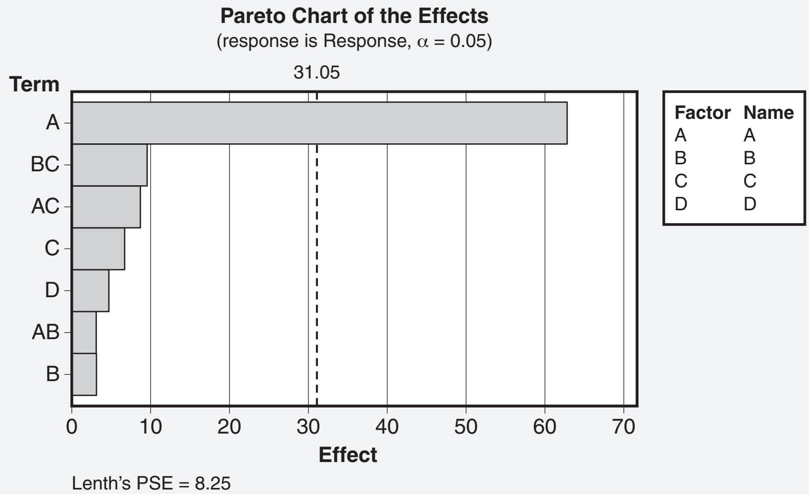

Figure 24.6 Pareto chart of effects for the initial model.

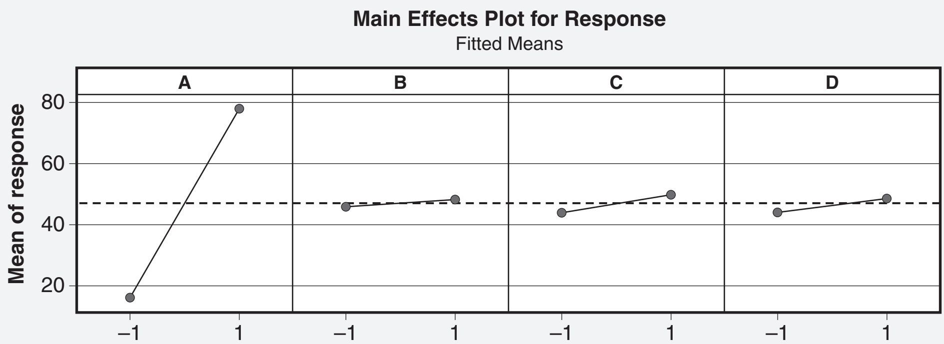

Figure 24.7 Main effects plot for the initial model.

the model. Figure 24.6 supports this conclusion. However, the source table in Figure 24.5, due to the lack of sufficient degrees of freedom for the error term, did not support it.

- Figure 24.8 provides the interaction plot. All of the interactions are small. Compare Figure 24.8 to Figure 24.6.
- Although we know where the model is heading, let's reduce the model a bit, moving a few terms out of it and into the error term. We'll rerun the model again. However, this time we'll put factors *A*, *B*, and *C* and two-way interactions *AC* and *BC* into the model.

Continued

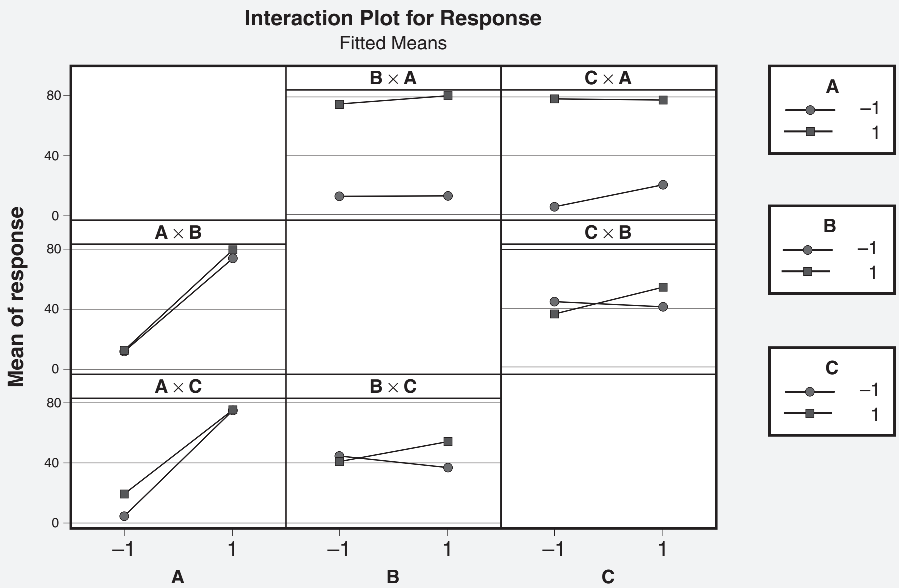

Figure 24.8 Interaction plot for the initial model.

- The session window for this reduced model is depicted in Figure 24.9. Although this isn't much, there are now two degrees of freedom for the error term. All factor and interaction terms in the model exceed the alpha value except factor *A*. Only factor *A* is significant, as shown in Figure 24.10, and the main effects plot as illustrated in Figure 24.11 is unchanged. Similarly, the interaction effects plot in Figure 24.12 is unchanged. (Note: It is good practice to always generate a source table, Pareto chart and both main and interaction plots when analyzing a DOE.)
- Based on the analysis above, we can reduce the model further. In this iteration, we will drop all terms except factor *A*. Figure 24.13 provides the session window for the final model with only factor *A*. The source table indicates that the error term now has six degrees of freedom. Factor *A* is significant. Figure 24.14 confirms the significance of factor *A*. The adjusted R^2 is high.

In addition to the above analysis, there are several key assumptions about residuals that should be analyzed. These assumptions are:

- Independent (uncorrelated)
- Identically distributed normal with mean 0 *and* constant variance, σ^2

Consequently, Minitab provides another set of charts worth producing when analyzing DOEs. There are a set of four charts (Figure 24.15) that can used to draw conclusions regarding the basic underlying assumptions about residuals:

Continued

```
Factorial Regression: Response versus A, B, C

Analysis of Variance

Source                 DF   Adj SS   Adj MS  F-Value  P-Value
Model                   5  8365.50  1673.10    57.20    0.017
  Linear                3  8040.50  2680.17    91.63    0.011
    A                   1  7938.00  7938.00   271.38    0.004
    B                   1    18.00    18.00     0.62    0.515
    C                   1    84.50    84.50     2.89    0.231
  2-Way Interactions    2   325.00   162.50     5.56    0.153
    A*C                 1   144.50   144.50     4.94    0.156
    B*C                 1   180.50   180.50     6.17    0.131
Error                   2    58.50    29.25
Total                   7  8424.00

Model Summary

      S    R-sq  R-sq(adj)  R-sq(pred)
5.40833  99.31%     97.57%      88.89%

Coded Coefficients

Term      Effect   Coef  SE Coef  T-Value  P-Value   VIF
Constant          46.50     1.91    24.32    0.002
A          63.00  31.50     1.91    16.47    0.004  1.00
B           3.00   1.50     1.91     0.78    0.515  1.00
C           6.50   3.25     1.91     1.70    0.231  1.00
A*C        -8.50  -4.25     1.91    -2.22    0.156  1.00
B*C         9.50   4.75     1.91     2.48    0.131  1.00

Regression Equation in Uncoded Units

Response = 46.50 + 31.50 A + 1.50 B + 3.25 C - 4.25 A*C + 4.75 B*C

Alias Structure

Factor  Name

A       A
B       B
C       C
D       D

Aliases

I + ABCD
A + BCD
B + ACD
C + ABD
AC + BD
BC + AD
```

Source table

- *F* and *p*-values now
- 2 degrees of freedom for error term

Coefficients table

List of aliases (confounding terms)

Resolution IV

Figure 24.9 Session window for the reduced model.

Continued

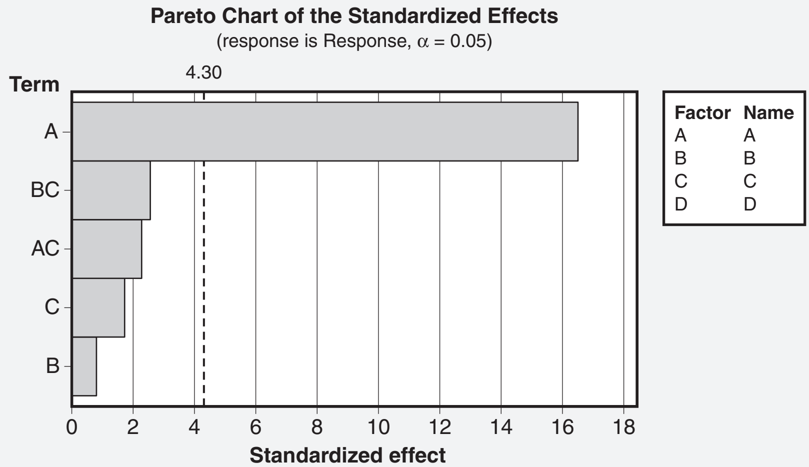

Figure 24.10 Pareto chart of effects for the reduced model.

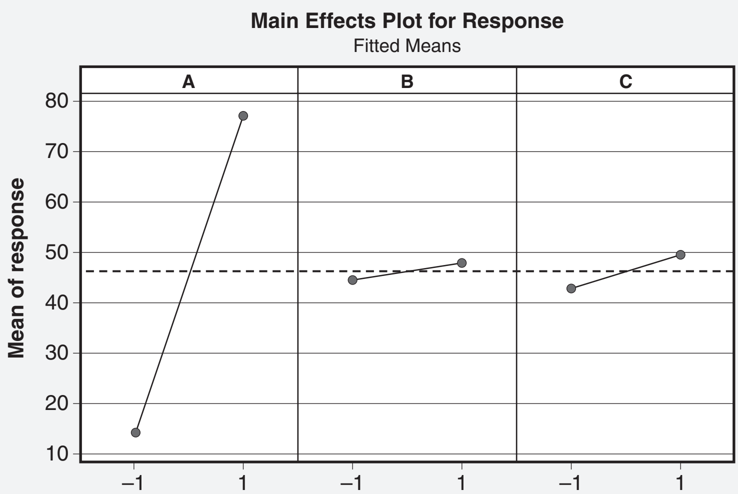

Figure 24.11 Main effects plot for the reduced model.

- *Normal probability plot*. This chart looks for residuals to fall along a straight line, thus indicating they are normally distributed. In this case, there appears to be some departure from normality. This conclusion is being made from a visual inspection of a graph but should be tested statistically by subjecting the residuals to the Anderson-Darling test for normality or a similar test. However, some departure from normality doesn't cause much of a problem, as there is a general robustness toward a violation of this assumption.

Continued

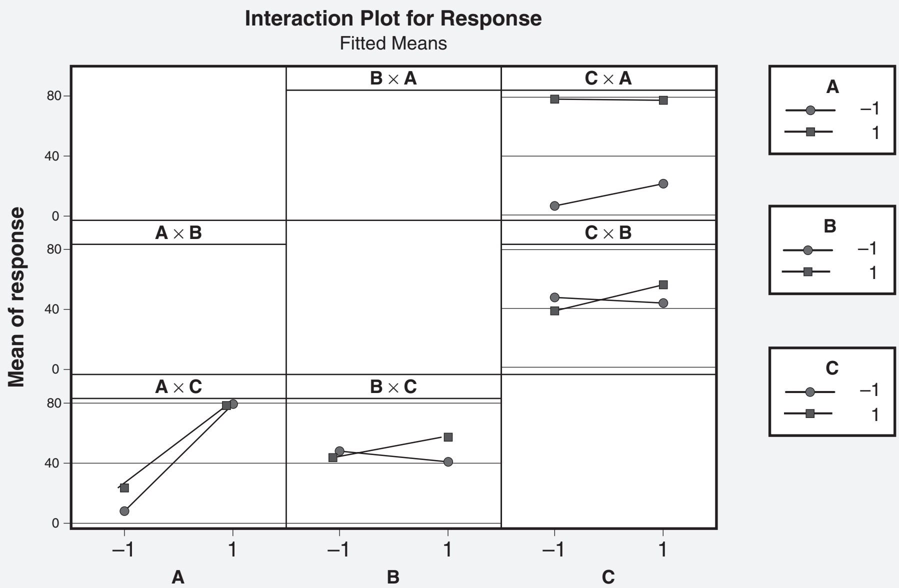

Figure 24.12 Interaction plot for the reduced model.

- *Histogram*. This chart serves the same purpose as the normal probability plot. One can sometimes draw the same conclusion from the histogram as from the normal probability plot.
- *Residuals versus fitted values*. This chart should look random on both sides of the zero line. No discernable pattern should be present. This chart is useful in detecting outliers. In addition, this chart will detect a violation of the assumption of constant variance (that is, homoscedasticity) of the residuals. This will be noticeable if the spread of the residual values increases as the fitted values increase. In the example, no noticeable pattern is evident.

  Montgomery (2009) states that "If the assumption of the homogeneity of variances is violated, the *F*-test is only slightly affected in the balanced (equal sized in all treatments) fixed effects model. However, in unbalanced designs or in cases where one variance is very much larger than the others, the problem is more serious."
- *Residuals versus order of the data*. This chart should present a random pattern on both sides of the zero line and is used to detect whether the residuals are uncorrelated with each other. The presence of a pattern would suggest the presence of autocorrelation. In the example, no noticeable pattern is evident.

Continued

Factorial Regression: Response versus A

```
Analysis of Variance

Source     DF  Adj SS   Adj MS  F-Value  P-Value
Model       1  7938.0  7938.00    98.00    0.000
  Linear    1  7938.0  7938.00    98.00    0.000
    A       1  7938.0  7938.00    98.00    0.000
Error       6   486.0    81.00
Total       7  8424.0
```

Source table

- *F* and *p*-values now
- 6 degrees of freedom for error term

Model Summary

```
S    R-sq  R-sq(adj)  R-sq(pred)
9  94.23%                 89.74%
```

Model summary

Coded Coefficients

```
Term      Effect     Coef  SE Coef  T-Value  P-Value   VIF
Constant            46.50     3.18    14.61    0.000
A          63.00    31.50     3.18     9.90    0.000  1.00
```

Coefficients table

Regression Equation in Uncoded Units

Response = 46.50 + 31.50 A

Regression equation

```
Alias Structure

Factor  Name

A       A
B       B
C       C
D       D
```

Aliases

```
I + ABCD
A + BCD
```

List of aliases (confounding terms)

Resolution IV

Figure 24.13 Session window for the final model.

Continued

Continued

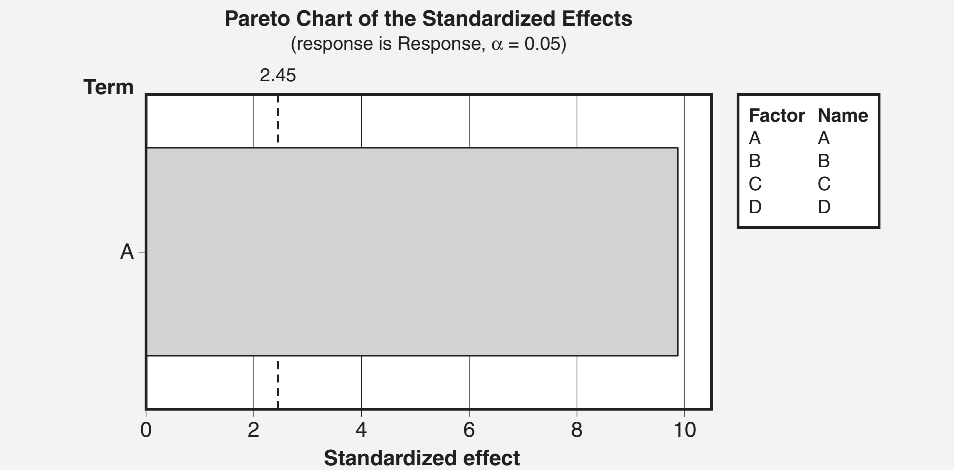

Figure 24.14 Pareto chart of effects for the final model.

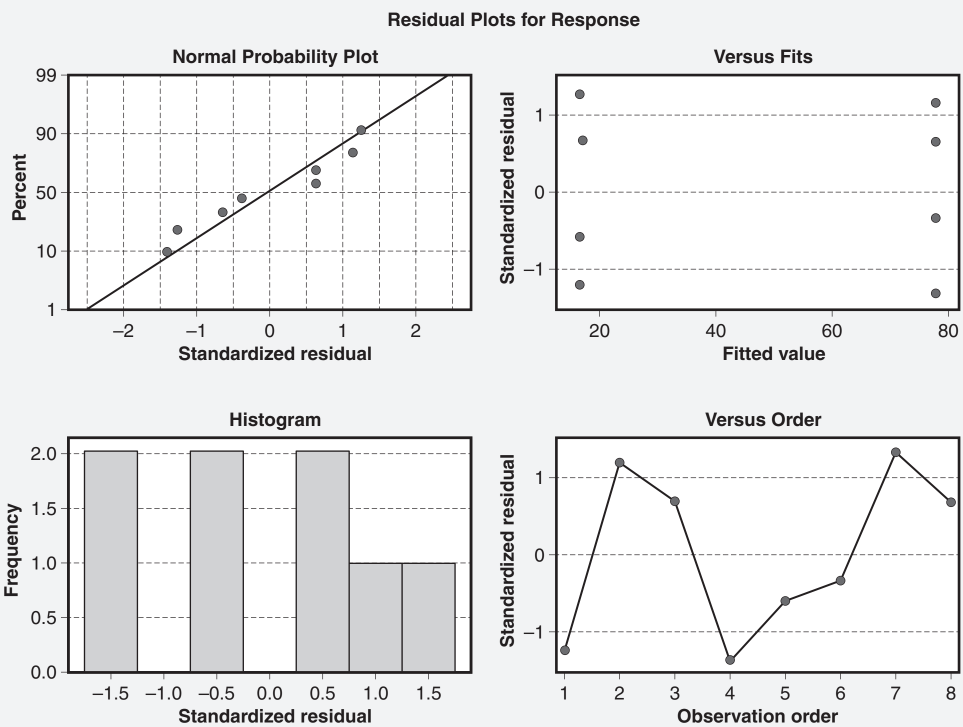

Figure 24.15 Residual analysis plots for the final model.

In general, DOEs can withstand some degree of violation of the underlying assumptions. However, when too much violation is present, various transformations can be employed to restore normality, eliminate or minimize autocorrelation, and equalize variances. When such situations occur, consult with your Master Black Belt.

FULL FACTORIAL EXPERIMENTS

Design, conduct, and analyze these types of experiments. (Evaluate)

Body of Knowledge VII.A.6

According to Mathews (2005), "In general, regardless of the number of classification variables, the ANOVA technique can be used to analyze multi-way classification problems." However, this section will be limited to a discussion on 2^2 full factorial experiments.

Table 24.33 provides the statistical model, source table, and the definition of the sums of squares for a full factorial design.

Table 24.33 Relevant tables for a two-way full factorial design.

Name	Model	Hypotheses	Assumption
Two-way ANOVA (completely randomized design with fixed effects)	$y_{ijk} = \mu + \tau_i + (\tau\beta)_{ij} + \varepsilon_{ijk} \begin{cases} i = 1,2,\ldots,a \\ j = 1,2,\ldots,b \\ k = 1,2,\ldots,n \end{cases}$ y_{ijk} = Observation in the ith row, jth column μ = Overal mean τ_i = Effect of the ith level of row factor A β_j = Effect of the jth level of column factor B $(\tau\beta)_{ij}$ = Effect of the interaction of τ_i on β_j e_{ijk} = Error term	$H_0 : \tau_1 = \tau_2 = \ldots = \tau_a = 0$ $H_A : \tau_i \neq 0$ for at least one i $H_0 : \beta_1 = \beta_2 = \ldots = \beta_b = 0$ $H_A : \beta_j \neq 0$ for at least one j $H_0 : (\tau\beta)_{ij} = 0$ for all i, j $H_A : (\tau\beta)_{ij} \neq 0$ for at least one i, j	$\varepsilon_{ijk} \sim NID(0, \sigma^2)$and the variance σ^2 of each factor is constant for all levels

Statistical model for the two-way ANOVA

Continued

Table 24.33 *Continued.*

Source of variation	Sum of squares	Degrees of freedom	Mean squares	F_0 statistic
Factor A	SS_A	$a - 1$	$MS_A = \frac{SS_A}{a-1}$	$F_0 = \frac{MS_A}{MS_E}$
Factor B	SS_B	$b - 1$	$MS_B = \frac{SS_B}{b-1}$	$F_0 = \frac{MS_B}{MS_E}$
Interaction AB	SS_{AB}	$(a - 1)(b - 1)$	$MS_{AB} = \frac{SS_{AB}}{(a-1)(b-1)}$	$F_0 = \frac{MS_{AB}}{MS_E}$
Error	SS_E By subtraction	By subtraction	$MS_E = \frac{SS_E}{ab(n-1)}$	
Total	SS_T	$abn - 1$		

Two-way ANOVA source table

Continued

Table 24.33 *Continued.*

Statistical model	Sums of squares (SS)
Two-way ANOVA (completely randomized design with fixed effects)	$SS_T = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{n} y_{ijk}^2 - \frac{y_{\ldots}^2}{abn}$ $SS_A = \frac{1}{bn}\sum_{i=1}^{a} y_{i..}^2 - \frac{y_{\ldots}^2}{abn}$ $SS_B = \frac{1}{an}\sum_{j=1}^{b} y_{.j.}^2 - \frac{y_{\ldots}^2}{abn}$ $SS_{AB} = \frac{1}{n}\sum_{i=1}^{a}\sum_{j=1}^{b} y_{ij.}^2 - \frac{y_{\ldots}^2}{abn} - SS_A - SS_B$ $SS_E = SS_T - SS_A - SS_B - SS_{AB}$

Sums of squares for the two-way ANOVA

Source: Montgomery (2009).

EXAMPLE 24.10

A 2^2 full factorial experiment (that is, two levels of acidity and two levels of bromine) with three replications has been established. The three replications will provide us with an estimate of experimental error. The order of the 12 tests has been randomized. The data table for the manual approach is shown in Table 24.34. Determine the effect that acidity and bromine have on nitrogen oxide (NOx) emissions. Assume $\alpha = 0.05$.

To facilitate the use of the equation in Table 24.33, we will expand Table 24.34. The result is Tables 24.35 and 24.36. We are able to apply the formulas directly to these tables. Note that the data cells in Table 24.36 are the squares of those given in Table 24.35.

Notice that the interaction term in Table 24.35 computes to a negative value. Since the sum of squares, by definition, cannot be negative, we set the sum of squares for the interaction term to zero.

Now we can complete the source table for this example as shown in Table 24.37 and compare our F_0 values to critical F values. From our source table, we have:

- Acidity: $F_{0.05, 1, 9} = 5.12$; since $0.16 < 5.12$, we must conclude that acidity has no effect on nitrogen oxide emissions.
- Bromine: $F_{0.05, 1, 9} = 5.12$; since $0.17 < 5.12$, we must conclude that acidity has no effect on nitrogen oxide emissions.

Table 24.34 Data table for Example 24.10.

		Bromine	
		–	+
Acidity	–	41.3	20.8
		14.6	50.0
		18.9	27.6
	+	20.6	28.7
		46.5	23.6
		31.2	40.2

Continued

Table 24.35 Computation of SS_{Acidity}, SS_{Bromine}, and $SS_{\text{Interaction}}$.

		Bromine			
		–	+	**Subtotal**	**Sum of squares**
Acidity	–	41.3	20.8		
		14.6	50.0		
		18.9	27.6		
	Subtotal	**74.8**	**98.4**	**173.2**	**29998.24**
	+	20.6	28.7		
		46.5	23.6		
		31.2	40.2		
	Subtotal	**98.3**	**92.5**	**190.8**	**36404.64**
	Total	**173.1**	**190.9**	**364.0**	**66402.88**
	Sum of squares	**29963.61**	**36442.81**	**66406.42**	

$$SS_{\text{Acidity}} = \frac{66402.88}{6} - \frac{(364.00)^2}{12} = 25.81$$

$$SS_{\text{Bromine}} = \frac{66406.42}{6} - \frac{(364.00)^2}{12} = 26.40$$

$$SS_{Interaction} = \frac{(74.8)^2 + (98.4)^2 + (98.3)^2 + (92.5)^2}{4} - 25.81 - 26.40 - \frac{(364.00)^2}{12} = -2719.37$$

$$SS_{Interaction} \Rightarrow 0.00$$

Continued

Continued

Table 24.36 Computation of SS_T and SS_{Error}.

		Bromine		
		–	+	Total (squared)
Acidity	–	1705.69	432.64	
		213.16	2500.00	
		357.21	761.76	
	Subtotal	**2276.06**	**3694.40**	**5970.46**
	+	424.36	823.69	
		2162.25	556.96	
		973.44	1616.04	
	Subtotal	**3560.05**	**2996.69**	**6556.74**
	Total (squared)	**5836.11**	**6691.09**	**12527.20**

$$SS_T = (2276.06 + 3694.40 + 3560.05 + 2996.69) - \frac{(364.00)^2}{12}$$
$$= 12527.20 - 11041.33 = 1485.87$$
$$SS_{Error} = SS_T - SS_{Acidity} - SS_{Bromine} = 1485.87 - 25.81 - 26.40 = 1433.66$$

Table 24.37 Source table for Example 24.10.

Source of variation	Sum of squares	Degrees of freedom	Mean squares	F_0 statistic
Acidity	25.81	1	25.81	0.16
Bromine	26.40	1	26.40	0.17
Error	1433.66	9	159.30	
Total	**1485.87**	**11**		

EXAMPLE 24.11

A 2^2 full factorial experiment (that is, two levels of acidity and two levels of bromine) with three replications has been established. The three replications will provide us with an estimate of experimental error. The order of the 12 tests has been randomized. The data table for the manual approach is shown in Table 24.34. Determine the effect that acidity and bromine have on nitrogen oxide (NOx) emissions. Assume $\alpha = 0.05$.

Unlike the previous example, we will solve this example using the Minitab software. The same data for this example will be used as in Example 24.10. However, it has been formatted differently to be compatible with Minitab. The data are shown in Table 24.38.

Because Minitab did not provide a two-way ANOVA choice (other than a balanced ANOVA), we elected to solve this example using the *general linear model* (GLM) approach. The GLM approach provides the greatest flexibility to solving many DOE models.

The session window for this analysis is depicted in Figure 24.16. A review of the source table indicates that neither factor is significant. Furthermore, the interaction term is not present. Recall, from the previous example, that we computed a negative interaction term that we set to zero.

Otherwise, only minor computational differences were noted between the two examples.

Figure 24.17 illustrates the Pareto chart of effects. Again, no terms are found to be significant. Although an interaction term was plotted on the Pareto chart, no interaction effect chart was plotted. When such conflicts arise, always resort to the source table in the session window.

Table 24.38 Full factorial with replicates in Minitab format for Example 24.11.

StdOrder	RunOrder	CenterPt	Blocks	Acidity	Bromine	Response
10	1	1	1	1	–1	20.6
5	2	1	1	–1	–1	41.3
12	3	1	1	1	1	28.7
2	4	1	1	1	–1	46.5
9	5	1	1	–1	–1	14.6
8	6	1	1	1	1	23.6
7	7	1	1	–1	1	20.8
6	8	1	1	1	–1	31.2
1	9	1	1	–1	–1	18.9
11	10	1	1	–1	1	50.0
4	11	1	1	1	1	40.2
3	12	1	1	–1	1	27.6

Continued

```
General Linear Model: Response versus Acidity, Bromine

Method

Factor coding (-1, 0, +1)

Factor Information

Factor    Type  Levels  Values
Acidity  Fixed       2  -1, 1
Bromine  Fixed       2  -1, 1

Analysis of Variance

Source         DF   Seq SS  Contribution   Adj SS  Adj MS  F-Value  P-Value
  Acidity       1    25.81         1.74%    25.81   25.81     0.16    0.697
  Bromine       1    26.40         1.78%    26.40   26.40     0.17    0.693
Error           9  1433.65        96.49%  1433.65  159.29
  Lack-of-Fit   1    72.03         4.85%    72.03   72.03     0.42    0.534
  Pure Error    8  1361.62        91.64%  1361.62  170.20
Total          11  1485.87       100.00%

Model Summary

      S   R-sq  R-sq(adj)    PRESS  R-sq(pred)
12.6212  3.51%      0.00%  2548.71       0.00%

Coefficients

Term       Coef  SE Coef          95% CI  T-Value  P-Value   VIF
Constant  30.33     3.64  (22.09, 38.58)     8.33    0.000
Acidity
  -1      -1.47     3.64  (-9.71, 6.78)     -0.40    0.697  1.00
Bromine
  -1      -1.48     3.64  (-9.73, 6.76)     -0.41    0.693  1.00

Regression Equation

Response = 30.33 - 1.47 Acidity_-1 + 1.47 Acidity_1 - 1.48
Bromine_-1 + 1.48 Bromine_1
```

Figure 24.16 Session window for Example 24.11 using the GLM.

Figure 24.18 provides a graphic of the main effects plot for this example. The effects appear to be significant, but don't be fooled by the scaling.

The residual analysis plots in Figure 24.19 appear to indicate a departure from normality. However, a test of the residuals using the Anderson-Darling test reveals that A-squared = 0.62 with a p-value of 0.079. Therefore, we cannot reject the null hypothesis that the residuals are normally distributed.

A review of the "residuals versus fits" and "residuals versus order" charts does not present any type of pattern.

Continued

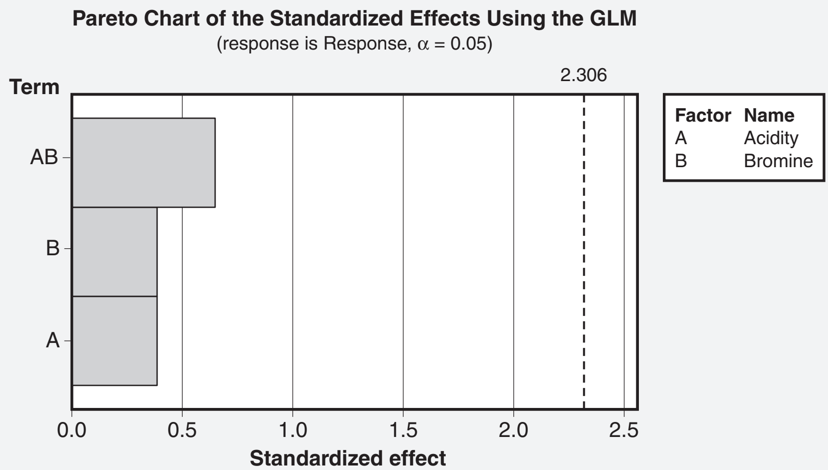

Figure 24.17 Pareto chart of effects.

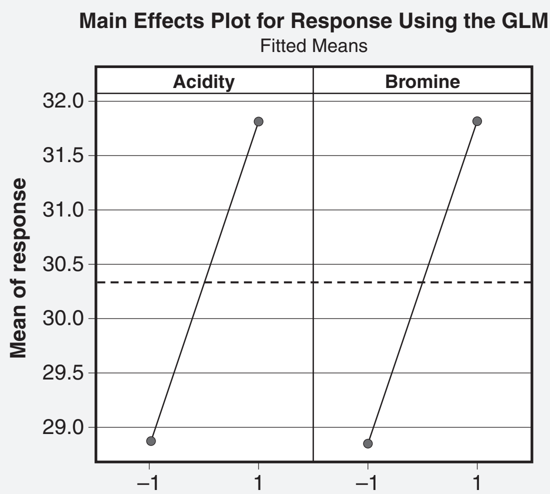

Figure 24.18 Main effects plot.

Continued

Continued

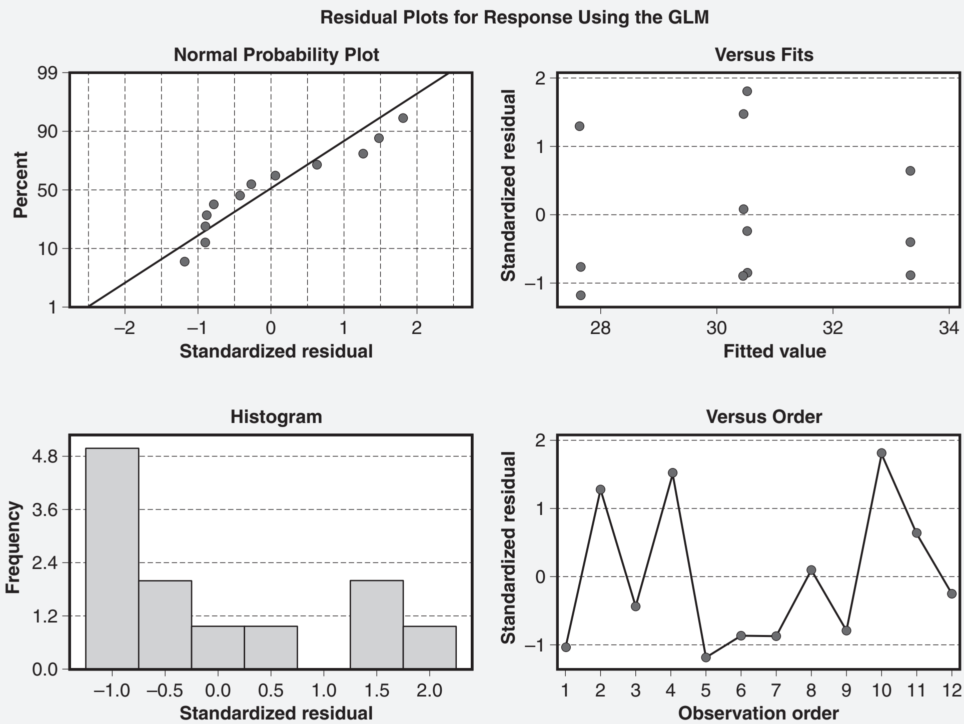

Figure 24.19 Residual analysis plots.

It should be clear from this and the preceding example that conducting DOEs requires careful planning and consideration as well as time. Furthermore, one needs to look at all aspects of the DOE during the analysis phase, such as the Pareto chart, main effects plot, interactions plot, and residuals analysis plots, and, if necessary, test the residuals for normality. Plots are useful, but it is easy to draw incorrect conclusions from them at times.

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Chapter 25

Lean Methods

WASTE ELIMINATION

> Select and apply tools and techniques for eliminating or preventing waste, e.g., pull systems, kanban, 5S, standard work, poka-yoke. (Analyze)
>
> **Body of Knowledge VII.B.1**

Over the years a great deal of effort has gone into improving *value-added activities*—those activities that impact the form or function of the product. For instance, improvements in machine operations, copiers, word processors, communications, and many other processes enhance the final product the customer receives. Lean thinking focuses on *non-value-added* activities—those activities that occur in every enterprise but do not add value for the customer. Some of these activities can be eliminated, and some can be simplified, improved, combined, and so forth. Tools for identifying and eliminating or reducing waste of all kinds are the topic of this section.

Kanban

A *kanban* is a system that signals the need to replenish stock or materials or to produce more of an item. Kanban is also known as a "pull" approach. Kanban systems need not be elaborate, sophisticated, or even computerized to be effective. Toyota's Taiichi Ohno developed the concept of a kanban system after visiting a U.S. supermarket. He observed that there were two boxes of each type of canned goods, one box placed behind the other on the shelf. When the front box was emptied and the rear box was pulled forward to replace it, that signaled them to bring in a full box and place it behind the box that was currently on the front of the shelf.

A system is best controlled when material and information flow into and out of the process in a smooth and rational manner. If process inputs arrive before they are needed, unnecessary confusion, inventory, and costs generally occur. If process outputs are not synchronized with downstream processes, the result is often delays, disappointed customers, and associated costs. A kanban system may be used to simplify and improve resupply procedure.

EXAMPLE 25.1

In a typical two-bin kanban arrangement on an assembly line, as the first bin is emptied, the user signals resupply personnel. The signal is usually visual and may involve displaying a card that came with the bin, turning on a light, or just showing the empty bin. The resupply employee gathers the information on supplies needed and replenishes the bins.

Sometimes, the bins are resupplied from a stockroom, although often it is from a closer supply point, sometimes referred to as a *supermarket*. In some cases, bins are replenished directly by an outside vendor. The entire string of events occurs routinely, often with no paperwork. The result is a smoother flow requiring less inventory. A properly administered kanban system improves system control by ensuring timely movement of products and information. It reduces waste involved with ordering, waiting for parts, and paperwork.

Pull Systems

In a *pull system*, the customer order process withdraws the needed items from a supermarket, and the supplying process produces product to replenish what was withdrawn. Also known as a *kanban* system, a pull system is the opposite of the traditional push system.

From the definition of the pull system, it can be seen that many manufacturing enterprises traditionally operate in a push system. This refers to the fact that raw materials and subassemblies are pushed through the production process with the anticipation that they will be needed by customers.

The main question is how large should the kanbans be? When the kanbans are too small, purchase orders are placed more often and stock-outs occur with greater frequency. When the kanbans are too large, work-in-progress inventory (one of the waste categories) builds up and incurs carrying costs that may include a tax burden as well.

A kanban can be a designated area on the floor, space on a shelf, a rack, a bin, or any place that is convenient to draw from and resupply. In some situations a card is placed below or behind the last item in a container. This kanban card is used by the resupply person as a reminder of the item to be restocked. The word *kanban* is translated as "cardboard," so the phrase *kanban card* is a little redundant.

EXAMPLE 25.2

An appliance manufacturer receives an order for 50 model XYZs. The warehouse ships the appliances to the customer, which leaves a blank space "open kanban" in the warehouse. This signals the assembly department to assemble 50 appliances to replenish those that have been sold. As the assembly department draws down the inventory of various subassemblies and purchased parts, the empty kanbans signal primary operations and purchasing agents to replenish those items.

It should be noted that some products do not fit the kanban pull system well. This is particularly true of seasonal products. An air conditioner manufacturer may sell 95% of its products in a three-month span. If it were to operate on a strict pull system it would be necessary to greatly expand manufacturing capacity—capacity that would not be needed the other nine months of the year. Instead, the company will likely spend most of the year building inventory that it hopes to sell during the rush months.

5S

5S, or *five S*, derives its name from the five Japanese terms beginning with "s" utilized to create a workplace suited for visual control and lean production. Collectively, 5S is about how to create a workplace that is visibly organized, free of clutter, neatly arranged, and sparkling clean. English-speaking authors have tried to come up with five "s" words that roughly translate as the Japanese words. Each "s" of 5S is defined as follows:

- *Seiri (sort* or *sifting).* To separate needed tools, parts, and instructions from unneeded materials and to remove the latter
- *Seiton (set in order)*. To neatly arrange and identify parts and tools for ease of use
- *Seiso (shine*, also *sanitize* or *scrub).* To conduct a cleanup campaign
- *Seiketsu (standardize)*. To conduct seiri, seiton, and seiso at frequent, indeed daily, intervals to maintain a workplace in perfect condition
- *Shitsuke (sustain* or *self-discipline).* To form the habit of always following the first four S's

A process is impacted by its environment, as is the ability of personnel to respond to process change. Improvements in the general state of the work area, including access to hand tools, aid in process control. Especially critical is the cleanliness, lighting, and general housekeeping status for any area where measurements are conducted since process control data are filtered through the measurement system.

The 5S system is often a starting place for implementing lean operations, so it is important that it be done properly and periodically. Remember shitsuke, the fifth "s," which essentially calls for performing the first four S's on an ongoing basis. The weak point of a 5S event is thinking of it as an event, like spring cleaning. It is instead to be thought of as part of daily activities. An effective way to make this happen is to have the workers in an area draw up a schedule with rotating 5S responsibilities, such as "Monday Bill sweeps the entryway."

Standard Work

Standard work is a concept whereby each work activity is organized around human motion to minimize waste. Each work activity is precisely described, including specifying cycle time, takt time, task sequence, and the minimum inventory of

EXAMPLE 25.3

An assembly line worker installs a wiring harness that is connected to various components at subsequent stations. Through conversations with fellow assemblers, it was discovered that if the wiring harness was always placed in a certain configuration, downstream work was simplified. Subsequently, a change was made to the product routing sheets to incorporate the wiring harness change to ensure that all assembly line workers install wiring harnesses in the same configuration. This change to the product routing sheets was supplemented with training.

parts on hand required to conduct the activity. Standard work is also known as *standardized work*.

Finding better ways of producing an ever more consistent product (or delivering an ever more consistent service) is the essence of process control. Standard work contributes to this effort by ensuring that product or services flowing into a process have minimal variation and that there is a reduction in the variation caused by the process. In addition, work is performed the same way every time. One of the problems with producing standard work documentation is that people are constantly finding better ways to do things, and the standard work document must be constantly updated.

Some organizations have found that rotating jobs helps everyone become acquainted with additional standard work opportunities.

Poka-Yoke

Poka-yoke is a term that means to mistake-proof a process by building safeguards into the system that avoid or immediately find errors. A poka-yoke device is one that prevents incorrect parts from being made, assembled, or stored, or that easily identifies a flaw or error. The term comes from the Japanese terms *poka*, which means "error," and *yokeru*, which means "to avoid." Poka-yoke is also known as *error-proofing*, *foolproofing*, and *mistake-proofing*. Poka-yoke activities enable process control by automatically eliminating a source of variation.

Poka-yoke methods are also helpful in reducing the occurrence of rare events.

EXAMPLE 25.4

Several people place documents in four separate trays depending on the document type. A kaizen team discovered that a person sorts the trays at the end of the day because about 5% of the documents are in the wrong tray, even though signs clearly state the document type. The team recommended printing the documents on different colored paper and also printing the signs on the corresponding paper color. This reduced the number of documents placed in the wrong tray to 0.7% and made the sorting job much easier. Note that the poka-yoke activity did not completely prevent errors, but the color-coding scheme did permit ready and immediate identification of an error.

EXAMPLE 25.5

A manufacturer found that about 1 in 2000 of its assemblies shipped was missing one of its 165 components. A poka-yoke technique was used to eliminate this defect. The manufacturer now uses a bar code on each component and scans the serial number and bar code of each component as it is added to the assembly. The software is written so that the printer at the shipping department will not print a shipping label if any component is missing or otherwise incorrect.

EXAMPLE 25.6

Another poka-yoke solution involves the selection of the correct part from a rack of bins with similar contents. As the product reaches the workstation, its bar code is read. Light beams crisscross the front of the bins. If the operator reaches into the wrong bin, the conveyor stops until appropriate corrections have been made.

EXAMPLE 25.7

A newly assigned press operator placed finished parts on a pallet in the incorrect orientation. The next operator didn't notice the error; thus, several hundred products were spoiled. A fixture for the pallet now makes it impossible to stack the parts in an incorrect orientation.

With the increase in flexible work assignments and cross-trained personnel, poka-yoke becomes even more important.

CYCLE-TIME REDUCTION

Use various tools and techniques for reducing cycle time, e.g., continuous flow, single-minute exchange of die (SMED), heijunka (production leveling). (Analyze)

Body of Knowledge VII.B.2

Cycle time is the time required to complete one cycle of an operation. It is useful to find ways to reduce cycle time and cycle time variation. Reducing variation in cycle time makes a system more predictable. Sometimes, the cycle time variation can be analyzed by studying the cycle times of sub-activities instead. For example, suppose the activity consists of using a word processor to modify a standard bid form. Sub-activities might include inserting client information, listing the proposed budget, detailing alternatives, and so on. The total time to prepare the bid might vary a great deal, while the time required to accomplish each sub-activity should show less variation. The activities performed should be continually studied in an effort to eliminate non-value-added components and find better and faster ways to complete the value-added components. One successful technique for accomplishing these goals goes by many names: *kaizen methods*, *kaizen blitz*, *rapid continuous improvement* (RCI), and similar titles. The usual procedure is to form a small team that is given a process to improve and a limited time frame, often only a few days. The team should include the people who perform the targeted activity, outsiders who can provide a fresh perspective, and people authorized to approve changes. The team observes the process and raises questions about its various parts, including:

- Why is that stored there? Is there a better place to put it?
- Why do things in that order?
- Would a different table height work better?
- Could your supplier (internal or external) provide a better service?
- Does your supplier know what you need?
- Do you know what your customer needs?
- Are you providing your customer (internal or external) with the best possible service?
- Should parts of this activity be performed by the customer or the supplier?
- Are there steps that can be eliminated?
- Is there enough light, fresh air, and so on, to do the job efficiently?
- Would another tool, software package, and so on, be more helpful?
- Are tools conveniently and consistently stored?
- Can the distance the person and/or product moves be reduced?
- Should this activity be moved closer to the supplier or customer?
- How many of these items should be kept on hand?
- Would it help to do this activity in less space?

In other words, the team questions everything about the process and its environment. Kaizen activity usually results in making several small improvements. In many situations the team actually implements a change and studies the result before making a recommendation.

Another useful time-based measure is *takt time*, which is sometimes confused with cycle time. However, the discussion immediately following should alleviate any confusion.

EXAMPLE 25.8

If 285 units are to be produced in a shift consisting of 27,000 seconds,

$$\text{Takt time} = \frac{27{,}000}{285} \approx 94.7 \text{ seconds/unit}$$

EXAMPLE 25.9

If the number of units scheduled is increased to 312 from 285, the takt time is reduced to 86.5 from 94.7 seconds. Adjustments to cycle times, by adding people or equipment, may be necessary.

Takt time is determined by customer demand. It is defined as

$$\text{Takt time} = \frac{\text{Time available}}{\text{Units required}}$$

In Example 25.8 the system must average one unit approximately every 95 seconds. To meet this demand rate, the cycle time for each process must be less than or equal to 95 seconds. So, to meet demand, the basic relationship between cycle time and takt time is

$$\text{Cycle time} \leq \text{Takt time}$$

Takt time is recalculated whenever the production schedule changes (Example 25.9).

It is important to note that if cycle time for every operation in a complete process can be reduced to equal or less than the takt time, products can be made in single-piece flow, or simply, a batch size of one. Achieving a batch size of one serves as the basis for continuous flow manufacturing, discussed in the next section.

Continuous Flow Manufacturing

Continuous flow manufacturing, commonly referred to as CFM, is a method in which items are produced and moved from one processing step to the next one piece at a time. Each process makes only the one piece that the next process needs, and the transfer batch size is one. CFM is sometimes known as *one-piece flow* or *single-piece flow*.

The traditional manufacturing wisdom, however, has been to study the marketplace to obtain a forecast of sales of various products. This forecast is used as a basis for orders that are issued to suppliers and to departments responsible for fabrication and assembly. This is referred to as a "push" system. One major problem with this strategy is that if the forecast is imperfect, products are produced that customers don't want, and/or products that customers want are not available.

A second major problem with the forecast-based strategy is the increasing expectation by customers for exactly the product they want, exactly when they want it. These two problems have led to a response by the manufacturing community that is sometimes called *mass customization*. As illustrated by the automotive industry, a vehicle ordered by a customer with a choice between dozens of options, with perhaps hundreds of possible combinations, cannot be accurately forecast. Instead, the customer order initiates the authorization to build the product. This is referred to as a "pull" system, because the pull of the customer instead of the push of the forecast activates the system.

Rather than producing batches of identical products, a pull-oriented organization produces a mix of products with the mix of features that customers order. In the ideal pull system, the receipt of the customer order initiates orders for the component parts to be delivered to the assembly line at scheduled times. The mixture of features of the components as they continuously flow to and through the line results in exactly the product the customer needs. Making this happen in a reasonable amount of time would have been unthinkable only a few years ago.

If a system were to achieve the state of perfection, each activity would move a component through the value stream so that it arrives at the next activity at the time it is needed. Achieving and maintaining this state may require a great deal of flexibility in allocating resources to various activities. Cross-training of personnel is essential. The resulting flexibility and system nimbleness would permit reduction of work-in-progress. Excessive cycle times are a barrier to CFM.

Reducing Changeover Time

Lean thinking is built on timely satisfaction of customer demand. This means there must be a system for quickly responding to changes in customer requirements. In metal-forming industries, for example, it was common practice to produce hundreds or even thousands of a particular part before changing the machine's dies and then producing hundreds of another part. This often led to vast inventories of work-in-progress and the associated waste. These procedures were "justified" because changeover time of dies took several hours. *Changeover time* is the time between the last good piece off the current run and the first good piece off the next run. This includes the time required to modify a system or workstation, which involves both teardown time for the existing condition and setup time for the new condition.

A system used to reduce changeover time and improve timely response to demand is based on a method known as *single-minute exchange of die* (SMED). SMED is a series of techniques pioneered by Shigeo Shingo to facilitate changeovers of production machinery in less than 10 minutes. The long-term objective is always zero setup, in which changeovers are instantaneous and do not interfere in any way with continuous flow. Setup in a single minute is not required but used as a reference. SMED is also known as *rapid exchange of tooling and dies* (RETAD).

The initial application of SMED often requires considerable resources in special staging tables, die storage areas, and so on. SMED is based on the concept that activities done while the machine is down are referred to as *internal activities*, whereas *external activities* are performed in preparation or follow-up to the die change. Shingo's method is to move as many activities from internal to external

EXAMPLE 25.10

A procedure for changing cameras requires several steps:

1. Shoot last good picture with camera A.
2. Remove camera A and its power supply and place in storage cupboard.
3. Remove type A tripod.
4. Remove type A lighting and reflectors.
5. Install type B lighting and reflectors.
6. Install type B tripod. Measure distance to subject with tape measure.
7. Locate camera B in cupboard and install it and its power supply.
8. Shoot first good picture with camera B.

A team working to reduce changeover time designed a fitting so both cameras could use the same tripod. By purchasing extra cables, the team was able to avoid moving power supplies. More-flexible lighting reflectors were designed so that one set would work for both cameras. Tape was placed on the floor to mark the location of the tripod feet, thus avoiding the use of a tape measure.

Of course, another alternative would be to obtain a more versatile camera that would not need to be changed. However, such an approach might be cost prohibitive.

as possible. A useful technique is to make a video recording of a typical changeover and have the user identify internal activities that can be converted to external activities. Positioning correct tooling, equipment, and manpower should all be done in external time.

The Shingo methodology can be summarized as follows:

- Identify and classify internal and external activities
- Make a video of the changeover process
- Separate internal and external activities
- Convert internal to external activities
- Apply engineering changes to convert any remaining internal activities to external activities as appropriate
- Minimize external activity time

Heijunka (Production Leveling)

Reducing variation is always a desirable goal. Variation in production scheduling can be a source of problems, so leveling production is a valid goal. Another goal is the satisfaction of the customer, that is, providing the customer what is needed when it is needed. These two goals can be contradictory because the customer's needs may be quite variable, and variation in production causes problems. Heijunka is a technique for meeting variable customer requirements while reducing variation in the production schedule. It has two forms, volume leveling and product leveling.

Volume Leveling. A customer needs 100 units per week, so the supplier decides to produce 20 units per day. On Monday the customer orders 15. The supplier ships 15 and puts the remaining five in heijunka storage. On Tuesday the customer orders 25 and the supplier ships the 20 produced that day plus the five in storage. On Wednesday the customer orders 18, so the supplier ships 18 and puts two units in storage. On Thursday the customer orders 19 and the supplier ships 19 and puts one additional unit in storage. On Friday the customer orders 23 units. The supplier ships the 20 produced that day plus the three that are in storage. Thus, although the customer's orders varied, the supplier was able to maintain a constant 20 units per week. The week's activity is summarized in Table 25.1.

The alert reader will ask, "What if the customer orders 23 on Monday?" The solution is to have a certain amount of heijunka storage before the week starts. The amount required will depend on experience, projections, and so on, but will not be more than one week's customer requirements, in this case 100 units. This would cover the case where the Monday order is 100. Actually, a heijunka storage amount of 80 would be sufficient because Monday's production of 20 units would bring the total to 100.

The first reaction to this arrangement is that it increases inventory, and of course it does. So the heijunka technique presents a trade-off between leveling production and maintaining a low inventory. Table 25.2 illustrates the situation in which the customer front-loads the orders during the week.

Product Leveling. In a mixed model scenario, the supplier ships more than one product type to the customer. If the customer needs 38 of part A and 62 of part B each week, the supplier might decide to produce 8 part A units and 12 part B units on Monday, Wednesday, and Friday. and produce 7 part A units and 13 part B units on Tuesday and Thursday. Table 25.3 depitcts a possible week.

In the real world the exact number of each product that a customer will order is usually not known, but the general approach has been shown to reduce variability in day-to-day production. The daily schedules are sometimes displayed in a *heijunka box*, which may be a matrix drawn on a display board as shown in Figure 25.1.

Table 25.1 Heijunka volume leveling example.

Customer order		Supplier reaction	Heijunka storage
Monday	15	Ship 15, put 5 in H storage	5
Tuesday	25	Ship 25 by drawing 5 from H	0
Wednesday	18	Ship 18, put 2 in H storage	2
Thursday	19	Ship 19, put 1 in H storage	3
Friday	23	Ship 23 by drawing 3 from H	0

Table 25.2 Another heijunka volume leveling example.

Customer order		Supplier reaction	Heijunka storage
Monday	50	Ship 50 by drawing 30 from H	50
Tuesday	50	Ship 50 by drawing 30 from H	20
Wednesday	0	Ship 0, put 20 in H storage	40
Thursday	0	Ship 0, put 20 in H storage	60
Friday	0	Ship 0, put 20 in H storage	80
Initial heijunka storage = 80			

Table 25.3 Heijunka product leveling example.

Customer order			Supplier reaction				Heijunka storage	
	A	B		A		B	A	B
Monday	5	13	Store	3	Draw	1	33	49
Tuesday	8	10	Draw	1	Store	3	32	52
Wednesday	15	25	Draw	7	Draw	13	25	39
Thursday	10	12	Draw	3	Store	1	22	40
Friday	0	2	Store	8	Store	10	30	50
Heijunka storage at start of week: A = 30, B = 50								

Mon	Tue	Wed	Thur	Fri
A	A	A	A	A
A	A	A	A	A
A	A	A	B	A
B	B	B	B	B
B	C	B	C	C
C	C	C	C	C
C	D	C	D	C
	E		E	

Figure 25.1 A heijunka box displaying daily schedules.

KAIZEN

> Define and distinguish between kaizen and kaizen blitz and describe when to use each method. (Apply)
>
> **Body of Knowledge VII.B.3**

Kaizen is a Japanese term made famous by Masaaki Imai in his book *Kaizen: The Key to Japan's Competitive Success* (1986). It refers to a mind-set in which all employees are responsible for making continuous incremental improvements to the functions they perform. The aggregate effect of this approach is cost-effective and practical improvements that have instant buy-in by those who use them.

Kaizen is a term that has come to mean gradual unending improvement by doing little things better and setting and achieving increasingly higher standards. A *kaizen event or kaizen blitz* is usually implemented as a small, intensive event or project over a relatively short duration, such as a week. It is advantageous to use the kaizen approach in the following situations:

- The responsibility for implementation of change as a result of the kaizen activity lies mostly within the team, and the risk of failure is small
- Projects are time bounded and results must be demonstrated quickly
- Projects are clearly defined

EXAMPLE 25.11

A kaizen team observed that fan blades were retrieved from large boxes that the operator was required to move around to accommodate model changes. A kaizen team designed a "Christmas tree" with arms holding the appropriate fan blades, which the assemblers could easily reach from their workstation. The assembly line supplier was responsible for assuring that the next blade on the tree was the right one for the next product on the line.

EXAMPLE 25.12

An overhead chain assembly line is populated by 35 workers. Each person can stop the line when he or she sees an opportunity for improvement. A clock that is visible to all is set at 12:00 at the start of each shift and runs only when the line is stopped for these improvements. Management policy is that the clock should run about 30–40 minutes each shift because, "Only when the clock is running are we solving problems and making things better. . . . We need around 30–40 minutes of improvements each shift."

- Improvement opportunities are readily identifiable, such as excess waste

Process control can be enhanced through kaizen events that reduce non-value-added activities. The resulting work is more productive and permits a closer process focus by all involved.

Kaizen, then, is really an approach to process and enterprise management that recognizes the importance of each employee's creative ability and interest in continuous improvement. This approach may be difficult to instill in organizations with a history of top-down management. Barriers include an adversarial attitude and the feeling that "It's not my job to make my job better." In some situations this may be overcome by demonstrating successful kaizen blitz events.

A *kaizen blitz* is performed by a team in a short amount of time, usually a few days. The team focuses on a specific work area with the intent of making low-cost improvements that are easy to implement and are often installed during the blitz. The team should consist of people from the work area plus others from similar work areas, technical support, supervisory personnel with decision-making responsibilities, and outsiders who can see with "fresh eyes." The blitz begins with some training and team-building activities and quickly proceeds to the work area, where each function is observed and explained in detail. Teams may split into sub-teams assigned to specific tasks. At some point the entire team hears reports and recommendations from the sub-teams and reaches a consensus on changes to be implemented. In some cases these changes can be carried out immediately, and in others a schedule of future implementation efforts is prepared. This schedule should include deadlines and persons responsible. The team sometimes makes a report to the next higher level of management, which provides recognition. Some

EXAMPLE 25.13

A team is assigned to perform a blitz event on an assembly line for an appliance manufacturer. The team consists of the departmental foreman, a group leader, three line workers (including one from a similar line elsewhere in the plant), one engineer with assembly line responsibilities, a technician, and a guest from a nearby factory. After introductory training, communication of the ground rules and expectations, and a discussion of previous kaizen events, the team proceeds to the observation phase, which takes about three hours. Special attention is given to material handling and resupply, availability of tools, and motion by individual workers. The team then splits into sub-teams. Two of the sub-teams will work on specific resupply problems, one will work with workplace organization and clutter, and a fourth will examine the need for better flow of subassemblies to the line.

The results of this blitz event included better design and positioning of material racks, with the stockroom responsible for making sure that the next item on the rack is the next item needed on the line. (Note: This is especially important when the schedule calls for a model change.) New routes for the resupply personnel were defined. Designated areas for subassemblies were signed and painted. Standard work issues were resolved and documented so that some assemblers now had slightly modified jobs that will make downstream functions more efficient. Material storage racks were redesigned.

organizations have found these reports and the time spent to prepare for them to be unnecessary as blitz events become routine and a part of the standard way of operating. A side benefit of blitz events is that employees begin to see that their input is valuable and valued, which may lead to more support for the incremental kaizen mind-set.

In summary, the term *kaizen* is used to describe the gradual but continuous improvement process that extends over time. It has been found to improve processes, products, and morale. *Kaizen blitz*, on the other hand, describes a short-term event designed to make quick and easy improvements and set the stage for the mind-set required for the long-term kaizen approach.

OTHER IMPROVEMENT TOOLS AND TECHNIQUES

> Identify and describe how other process improvement methodologies are used, e.g., theory of constraints (TOC), overall equipment effectiveness (OEE). (Understand)
>
> **Body of Knowledge VII.B.4**

The *theory of constraints* (TOC) is a problem-solving methodology that focuses on the weakest link in a chain of processes. Usually, the *constraint* is the slowest process. Flow rate through the system cannot increase unless the rate at the constraint increases. The TOC lists five steps to system improvement:

1. *Identify*. Find the process that limits the effectiveness of the system. If throughput is the concern, the constraint will often have work-in-progress (WIP) awaiting action.
2. *Exploit*. Use kaizen or other methods to improve the throughput of the constraining process, such as:
   - Enlarge kanban size so there is plenty of WIP ahead of the constraint so it never has to wait on upstream processes.
   - Ensure that no defective parts are in this WIP so the constraint only processes good parts.
   - Adjust the schedule so the constraint always runs, even during breaks and lunch. Use overtime.
   - Reduce changeover time.
   - Keep maintenance interruptions to a minimum.
   - Send some work through other processes.
3. *Subordinate*. Adjust (or subordinate) the rates of other processes in the chain to match that of the constraining process.

- If the constraint equipment needs maintenance, it should have highest priority—maintenance of other units should be subordinated to that of the constraint.

4. *Elevate*. If the system rate needs further improvement, the constraining process may require extensive revision (or elevation). This could mean investment in additional equipment or new technology.
5. *Repeat*. If these steps have improved the process to the point where it is no longer the constraint, the system rate can be further improved by repeating these steps with a new constraint.

The strength of the TOC is that it employs a systems approach, emphasizing that improvements to individual processes will not improve the rate of the system unless they improve the performance of the constraining process.

Drum-Buffer-Rope

The "subordinate" step requires further explanation. To illustrate how to manage operations, Goldratt (1997) uses a squad of soldiers walking in single file as an analogy for a string of production processes (see Figure 25.2). As the first soldier moves forward, that soldier receives unprocessed material, the fresh ground. Each succeeding soldier performs another process by walking on that same ground. As the last soldier passes over the ground, it becomes finished goods. Thus, the individual processes are moving over fixed material rather than the other way around. *Lead time* is the time it takes for the squad to pass a certain point. If each soldier moves as fast as possible, the lead time tends to lengthen, with the slower soldiers falling behind and holding up those behind them, since passing is not permitted.

The system constraint is the slowest soldier. The ground can't be processed faster than this soldier can move. This soldier sets the drumbeat for the entire system. To avoid lengthening the lead time, a rope connects the lead soldier to the slowest soldier (see Figure 25.3).

The length of the rope is the size of the *buffer.* When the lead soldier, marching faster than the constraint, gets far enough ahead, the rope becomes taut and signals the soldier to stop. In this way, maximum lead time and WIP are fixed.

If a soldier behind the slowest soldier happens to drop his rifle, he'll fall behind a little but will be able to catch up since he is not the slowest soldier. This is analogous to a minor process problem at one station. If a soldier in front of the

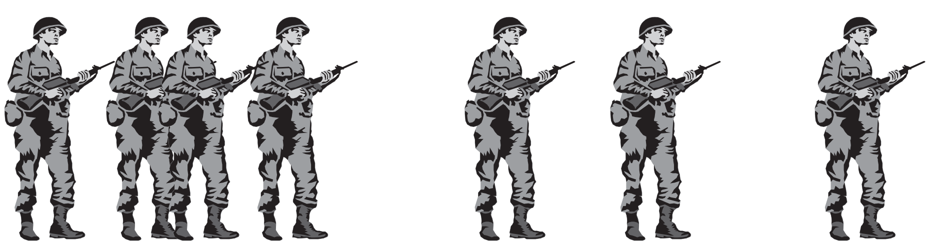

Figure 25.2 The drum-buffer-rope subordinate step analogy—no rope.

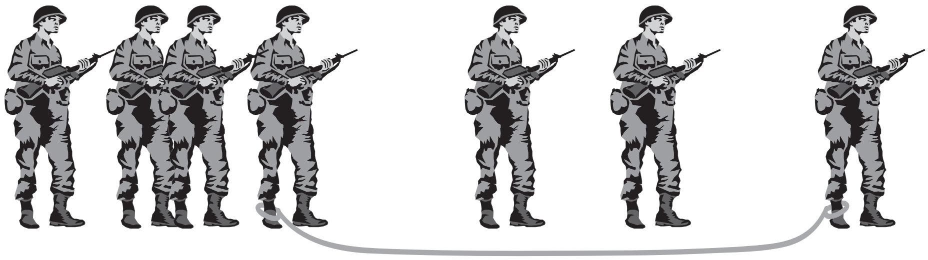

Figure 25.3 The drum-buffer-rope subordinate step analogy—with rope.

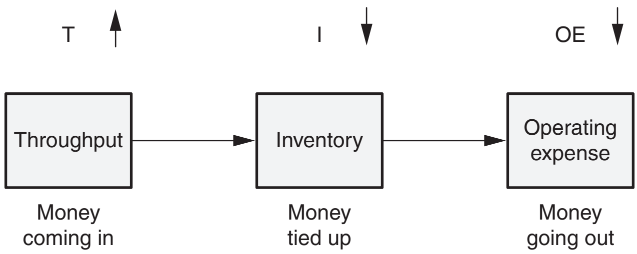

Figure 25.4 The interdependence of throughput, inventory, and operating expense measures.

slowest soldier drops her rifle, the squad will not have to stop unless the slowest soldier catches up with the one in front of him. So, if the squad has a high tendency to drop rifles, the rope must be longer. The length of the rope is the size of the buffer. In summary, to avoid long lead times and excess WIP, all system processes should be slowed down (via the rope) to the speed of the slowest process (the drum), and the amount of WIP (or buffer) is determined by the dependability of the individual processes.

The TOC as described by Goldratt has significant impact on three key interdependent operating measures of a business. Briefly, these are as described by Dettmer (2007):

- Throughput (designated as T)—the rate at which the entire system generates money through sales
- Investment (designated as I)—the money the system invests in things it intends to sell
- Operating expense (designated as OE)—the money the system spends turning inventory into throughput

Figure 25.4 illustrates the relationships between these operating measures. Notice that as throughput increases, the other two measures decrease. This makes sense because as constraints are removed, throughput increases. Increased throughput means less investment tied up in the system. Less investment, along with a higher throughput, means that less money (that is, operating expense) is required to turn inventory into throughput (or output as the case may be).

Overall Equipment Effectiveness (OEE)

OEE provides a metric for evaluating processes and process improvement efforts. The basic formulas are:

$$\text{OEE} = \text{Availability} \times \text{Performance} \times \text{Quality}$$

where

$$\text{Availability} = \frac{\text{Actual operating time}}{\text{Planned production time}}$$

$$\text{Performance} = \frac{\text{Total pieces/Actual operating time}}{\text{Ideal run time}}$$

$$\text{Quality} = \frac{\text{Number of good pieces}}{\text{Total number of pieces}}$$

and

Planned production time = Shift length – Breaks, lunch, and so on

Actual operating time = Planned production time – Downtime for setup, maintenance, and so on

Ideal run time = Designed maximum cycle time

EXAMPLE 25.14

A machine scheduled for a 10-hour shift minus 80 minutes for breaks and a meal is down for one hour and 10 minutes for maintenance. It is designed to process 300 pieces per hour. During the shift, it makes 1783 good parts and 91 bad parts. The OEE calculation follows (all times in minutes):

Planned production time = 600 – 80 = 520

Actual operating time = 520 – 70 = 450

Ideal run time = 5 parts/min

$$\text{Availability} = \frac{450}{520} \approx 0.865$$

$$\text{Performance} = \frac{\frac{1874}{450}}{5} \approx 0.833$$

$$\text{Quality} = \frac{1783}{1874} \approx 0.951$$

$\text{OEE} \approx (0.865)(0.833)(0.951) \approx 0.685$

The OEE number is a good metric for making comparisons between two identical machines making identical parts. It can also be used to compare a machine or process

Continued

Continued

before and after implementation of a quality improvement. Care must be used in comparing machines or processes that aren't very similar because the factors that are multiplied together can be influenced by maintenance differences, which may not be comparable. In addition, the "ideal run time" is not always calculated the same way.

The best place to do an initial OEE calculation is at the constraint step in a process, as defined in the previous section on theory of constraints. This is because improvements there improve the whole process. When seeking to improve the OEE value, examine the values of the three factors of availability, performance, and quality, and determine which one(s) can be cost-effectively raised. Conduct root cause analysis on the reasons for downtime in the *availability* factor, reduced cycle time in the *performance* factor, and defect level in the case of the *quality* factor.

Chapter 26

Implementation

> Develop plans for implementing proposed improvements, including conducting pilot tests or simulations, and evaluate results to select the optimum solution. (Evaluate)
>
> **Body of Knowledge VII.C**

Once the root causes have been identified, the next step should be to identify a potential set of solutions. These should be considered tentative until they are evaluated. When evaluating a set of solutions, it is important that specific criteria be established. Examples of such criteria include cost of implementation, ease of implementation, maintainability, reliability, organizational acceptance, customer impact, impact to the bottom line, and so forth. Whatever criteria are chosen, they should be relevant, well defined, and a result of team consensus.

Figure 26.1 depicts a matrix of potential solutions and the associated evaluation criteria. Also depicted are weights associated with each criterion. For each potential solution, a rank can be assigned to each cell, with the best being the highest number and the worst being the lowest number. Decimal values indicate ties between two rank values. For instance, if multiple criteria are tied for a specific rank, each would be assigned the average rank value. Once each cell has

		Potential solution				
	Weight value	**A**	**B**	**C**	**D**	**E**
Percent process improvement	0.25	5	1	2.5	2.5	4
Cost of implementation	0.20	2	3	5	4	1
Reliability	0.15	5	3.5	2	1	3.5
Maintainability	0.10	4	5	1.5	1.5	3
Installation resources required	0.10	2	5	4	1	3
Adverse impact	0.20	3	2	4	5	1
		3.60	**2.78**	**3.28**	**2.83**	**2.53**

Figure 26.1 Example of a ranking matrix with criteria weights shown.

been assigned a value, the corresponding weight value is multiplied by each of the assigned values and totaled. The potential solution with the highest weighted value becomes the solution of focus for implementation.

However, one big question remains: how are the various cell values determined? These values may be obtained in various ways. Following is a list of common and very useful approaches that yield highly actionable insight:

- *Pilot run.* A short, limited production run usually terminated by elapsed time or quantity produced. A pilot run provides the team with valuable information regarding how an improved process will run since the pilot represents actual production. It also provides significant insight into issues or problems that were overlooked. Hence, an additional opportunity is available to correct outstanding problems prior to full-scale implementation.

- *Simulation.* Usually performed with computer software, and can be either deterministic or probabilistic in design. A *deterministic* simulation provides little information other than to ensure that process flows are correct and that the general design of the simulation matches the operational environment. A *probabilistic*, or *stochastic*, simulation provides insight into the statistical and operational measures associated with the process being simulated. However, careful attention must be given to interpreting the statistical output to ensure that parameters are being estimated properly. Overall, simulation provides the ability to "implement" solutions off-line but draw conclusions regarding how the process improvement will operate online. When conducting simulations, caution is warranted. Simulations must be both verified and validated before use.

- *Model.* A smaller physical representation developed on a specified scale. Unlike simulations, models simply provide a three-dimensional miniature version of whatever is being modeled and are a very effective means of communicating an improvement alternative since they play to the visual sense.

- *Prototype.* A representation of the actual product in terms of form, fit, and functionality, with perhaps some limitations on functions and/or features. Prototypes provide the team with the ability to experiment with a product by adjusting a wide range of both product and process parameters, usually at a lower cost.

Once the team settles on a particular solution, it must then establish an implementation strategy. A first step in this approach might be the development of a force-field analysis. Recall that a force-field analysis is useful for identifying those forces that both support and oppose the process implementation. These forces can be integrated into the implementation strategy to ensure that those forces that support the implementation are retained or strengthened, and those that oppose it are eliminated or mitigated.

Although many authors propose various improvement implementation strategies and methodologies, they are usually offered in a one-size-fits-all approach.

EXAMPLE 26.1

A plastic recycling process grinds plastic waste, heats it to a liquid state, and pours it into rectangular molds approximately $2 \times 8 \times 84$ inches. The material is allowed to solidify in the molds. Then the molds containing the material are submerged in a water bath for cooling. The resulting product is somewhat unstable, tending to warp out of spec. The improvement team believes that the water bath develops hot spots around portions of the material that are more dense, thus retaining more heat. This results in different cooling rates and different final temperatures, which causes warping. The team believes that a flow of cool water should be passed through the bath container to provide a more stable cooling process. Before they implement this belief they consider each of the four testing techniques listed above.

Pilot run: The team could construct a set of troughs with water pumps and chillers and run several days of production through the new system. The resultant material could be checked for warp.

Simulation: The team would use heat transfer equations for various plastic densities and various water temperatures. They would use these formulas to predict the water temperature rise at the hot spots and the associated cooling rates. By executing the resulting software they could predict final temperature at various locations on the finished product by knowing the density of the material in that neighborhood. They could then check the predicted temperatures against measured values to validate the simulation. The next step would be to run the software under the assumption of flowing chilled water to predict final temperatures at various density points. This simulation could also be used to predict the best chiller output temperature, flow rate, and time in bath. Such predictions must be viewed with caution but they can provide assistance when deciding whether to embark on extensive capital investment.

Model: The team could build a 1/8-scale model of the proposed system and run water at various temperatures over miniature slabs in miniature molds. They could collect warpage data for comparison.

Prototype: The team could rig up a special trough with a water pump and chiller and measure the amount of warp in the finished product. Various water temperatures could be tried. The amount of warp in the finished product could be measured. If the prototype is constructed in a lab environment, consideration must be given to differences between lab conditions and the production environment.

Implementation of any initiative or process improvement must be performed with care and due consideration of the organization's culture. With respect to this idea, consider the following framework, which can be tailored to a specific organization's needs during an improvement implementation:

- *Infrastructure*. This includes everything necessary to ensure a successful implementation. Examples include changes to hardware or software, training, acquisition of talent, organizational changes, facilities, and so forth. The infrastructure requirements will vary greatly from project to project. Communication is one aspect whose importance is frequently underestimated, so the remainder of this chapter is devoted to this topic.

EXAMPLE 26.2

The board of directors of a hospital has decided to build a new building on the south side of town rather than continue to update the current building, parts of which are 100 years old. It is the only hospital in a town of 30,000, and they begin their implementation plan with a force-field analysis as shown in Figure 26.2.

The team forms a plan based on the force-field analysis with the following elements:

1. Request that the medical association make its support known by providing speakers at service clubs and publishing a letter of support in the local newspaper.
2. Assure the north side residents that the ambulance service will continue to be housed in the fire station on the north side.
3. Request that the nursing school inform the public of the advantages to student nurses of having access to the latest in facilities and equipment.
4. Request that residents send letters of support for the move. These letters will be combined, summarized, and provided to the regulatory entities.
5. Request that residents who must travel elsewhere for treatment provide testimonies of support to be used by hospital staff as they communicate with various stakeholders.
6. Assure the members of the Chamber of Commerce that the move will be incremental over a period of eight years, which will allow businesses to adjust plans accordingly.

Driving forces	Resisting forces
Doctors' association Nursing school Most residents Residents who currently must travel to other towns for treatment	Residents on north side of town Regulatory entities Taxpayers Chamber of Commerce

Figure 26.2 Example of a force-field analysis.

- *Communication plan*. A communication plan should be specific with regard to:
  - *Who communicates it*. It is important that communication start at the top; however, the message needs to cascade down through all levels of management until it reaches those who need to know. This cascading of communication cannot be emphasized enough as it creates and reinforces a sense of urgency. Furthermore, it

reinforces the message that management believes it is important to implement the process improvement.

– *What is communicated*. What should be communicated is the "who," "what," "when," "where," and "why." If each of these questions is addressed, then a clear and concise message will be delivered. No communication should raise more questions than it answers. And it should not assume an existing knowledge of the improvement implementation on the part of the recipient of the communication. What is to be communicated must remain consistent throughout. Thus, the development of "canned" presentations, a script, and so forth, are beneficial in maintaining that consistency.

– *When it is communicated*. The timeliness of a communication is, at best, tricky, and often more than one communication is required. Communications may be timed for release just prior to the phases of the implementation schedule. Occasionally, it is prudent to issue an overarching, up-front communication and support it with phased communications throughout the implementation. Regardless of the approach chosen, there are three important considerations:

1. Multiple communications may be necessary
2. Communications should reinforce and progress logically
3. Timely communications should occur before a phase is executed, not after

- *Where it is communicated*. This speaks to the geographic, demographic, and organizational areas of the communication. For example, the communication may be intended for the engineering function located in North Carolina.
- *How it is communicated*. All communication media are not created equal. An effective communication plan should recognize that most organizations have a variety of communication media available to them. Choosing the right medium for the right audience is essential. Examples of media include newsletters, e-mail, video, staff meetings, town halls, net meetings, telephone broadcasts, and so forth. It is important to keep in mind that for any given audience, multiple delivery media may be appropriate.
- *To whom it is communicated*. Without a doubt, it is critical to communicate to all who might be impacted by the improvement implementation. This includes not only those responsible for carrying out the process, but also upstream suppliers and downstream customers.
- *Competition for time, energy, and resources*. As with most organizations, the team's improvement initiative is not the only critical activity in the organization that demands resources. Therefore, it is essential to outline the requirements for implementation in advance and to secure

approval by management. An actual document requiring "sign-off" is helpful in this regard.

- *Management commitment*. By now, "management commitment" has become cliché. In terms of the framework, this item is pivotal. Nothing will be accomplished without it. Management must be visible, active, and engaged to ensure a successful implementation. Lip service won't cut it. If there is no management commitment, the only assurance the team has is that it will fail.

Part VIII

Control

The purpose of this phase is to transition the improved process back to the process owner and to establish and deploy a control plan to ensure that gains in performance are maintained. Common tools used in this phase are shown in Table VIII.1.

Table VIII.1 Common tools used in *control* phase.

5S	Lessons learned	Standard operating procedures (SOPs)
Basic statistics	Measurement systems analysis (MSA)	Standard work
Communication plan	Meeting minutes	*Statistical process control (SPC)*
Continuing process measurements	Mistake-proofing/ poka-yoke	*Tollgate review*
Control charts	Pre-control	Total productive maintenance
Control plan	*Project management*	Training plan deployment
Data collection	*Project tracking*	Visual factory
Data collection plan	*Run charts*	Work instructions
Kaizen	Six Sigma storyboard	
Kanban		

Note: Tools shown in *italic* are used in more than one phase.

Chapter 27

Statistical Process Control

Effective SPC is 10% statistics and 90% management action.

—John Hradesky

This chapter will address the basic concepts of statistical process control (SPC) including the objectives of SPC, the selection of critical process characteristics for control chart monitoring, the ever-important aspect of rational subgrouping, the selection and use of variables and attributes control charts, and the recognition and interpretation of out-of-control conditions. Numerous examples and control charts are presented, as well as several flowcharts to aid in the selection of control charts. In addition, several summary tables have been provided to assist the reader for examination and desktop reference purposes.

OBJECTIVES

> Explain the objectives of SPC, including monitoring and controlling process performance, tracking trends, runs, and reducing variation within a process. (Understand)
>
> **Body of Knowledge VIII.A.1**

Control Chart Objectives

Control charts are statistically based graphical tools used to monitor the behavior of a process. Walter A. Shewhart developed them in the mid-1920s while working at Bell Laboratories. More than 90 years later, control charts continue to serve as the foundation for statistical process control.

The goal of any quality activity is to meet the needs of the customer. Statistical process control (SPC) consists of a set of tools and activities that contribute to this goal through the following objectives:

- *Quantifying the variation of a process.* Fundamental to the development of any control chart is the quantification of the variation. Control chart limits are traditionally set at $\pm 3\sigma$ and are different from specifications. In addition, this is where studying the process capability is advantageous.
- *Centering a process.* Also fundamental to the development of any control chart is the quantification of the process average. It is important to note that the process average may not be equal to the nominal value defined by the specifications. In addition, this is where studying the process capability is advantageous.
- *Improving product and process design.* SPC is beneficial for improving the process design because it provides quantitative feedback regarding the process average and process variation. However, additional tools and methodologies are often required, such as Lean Six Sigma, DFSS, process capability studies, and machine capabilities, among others.
- *Tracking trends and runs in the process.* Rules have been developed to recognize statistically out-of-control conditions on control charts. Such conditions include trends and runs, which generally occur for specific reasons.
- *Determining when or when not to take action on a process.* This is a key point. Control charts illustrate two types of variation: common cause and special cause. We take action on special cause variation when we are trying to maintain control on a process. We do not take action on common cause variation unless we are trying to improve the process.
- *Making statistically valid decisions.* Control charts are all about making statistically based decisions. They are based on sound statistical theory, not gut feel.

The SPC tools achieve these objectives by collecting and analyzing data. Figure 27.1 depicts SPC as a machine that uses data as input and has information as its output. The SPC tools squeeze the information out of the raw data both graphically and statistically.

A wide variety of control charts have been developed over the years. However, we will focus on those that are the most popular in both manufacturing and transactional environments. These include the $\bar{X}$ and R chart, $\bar{X}$ and s chart, XmR chart, p-chart, np-chart, c-chart, u-chart, and several short-run charts.

These and other SPC tools are available in various SPC software packages and in some spreadsheets such as Microsoft Excel.

Special Cause versus Common Cause Variation

Every process has variation. The sources of process variation can be divided into two categories: special and common.

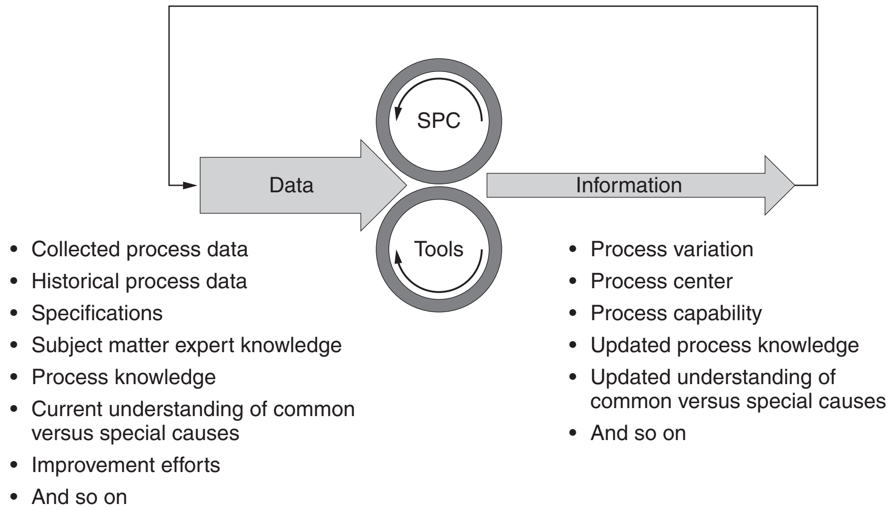

Figure 27.1 Function of SPC and related tools.

Common causes are those that are inherent to the process and generally are not controllable by process operators. Examples of common causes include variation in raw materials, variation in ambient temperature and humidity, variation in electrical or pneumatic sources, and variation within equipment, such as worn bearings.

In the case of service processes, common causes typically include variation in the input data, variations in customer load, and variation in computer operations. Some authors refer to common cause variation as *natural variation*.

Special causes of variation include unusual events that the operator, when properly alerted, can usually remove or adjust. Examples include large changes in raw materials, and broken equipment. Special causes are sometimes called *assignable causes*. Note: Some authors argue that one cannot refer to special causes as assignable causes until after the cause is known (that is, until after the cause is assigned to a special cause).

A principal problem in process management is the separation of special and common causes. If the process operator tries to adjust a process in response to common cause variation, the result is usually more variation rather than less. This is sometimes called *overadjustment* or *overcontrol*. If a process operator fails to respond to the presence of a special cause of variation, this cause is likely to produce additional process variation. This is referred to as *underadjustment* or *undercontrol*.

One of the main objectives of control charts is to help the process operator recognize the presence of special causes so that appropriate action can be taken in a timely manner. Control charts are discussed in detail in the following paragraphs.

Basic Structure of a Control Chart

Constructing control charts is straightforward and, more often than not, aided by computer software designed specifically for this purpose. Minitab and JMP, among others, are commonly used.

Figure 27.2 illustrates the general form for a control chart. Its critical components are:

1. *X-axis*. This axis represents the time order of subgroups. Subgroups represent samples of data taken from a process. It is critical that the integrity of the time dimension be maintained when plotting control charts.
2. *Y-axis*. This axis represents the measured value of the quality characteristic under consideration when using variables charts. When attributes charts are used, this axis is used to quantify defectives or defects.
3. *Centerline*. The centerline represents the process average.
4. *Control limits*. Control limits typically appear at ±3σ from the process average.
5. *Zones*. The zones represent the distance between each standard deviation and are useful when discussing specific out-of-control rules.
6. *Rational subgroups*. The variation within subgroups should be as small as possible so it's easier to detect subgroup-to-subgroup variation.

Notice that there are no specification limits present on the control chart in Figure 27.2. This is by design, not by accident. The presence of specification limits on control charts could easily lead to inaction, particularly when a process is out of control but within specification.

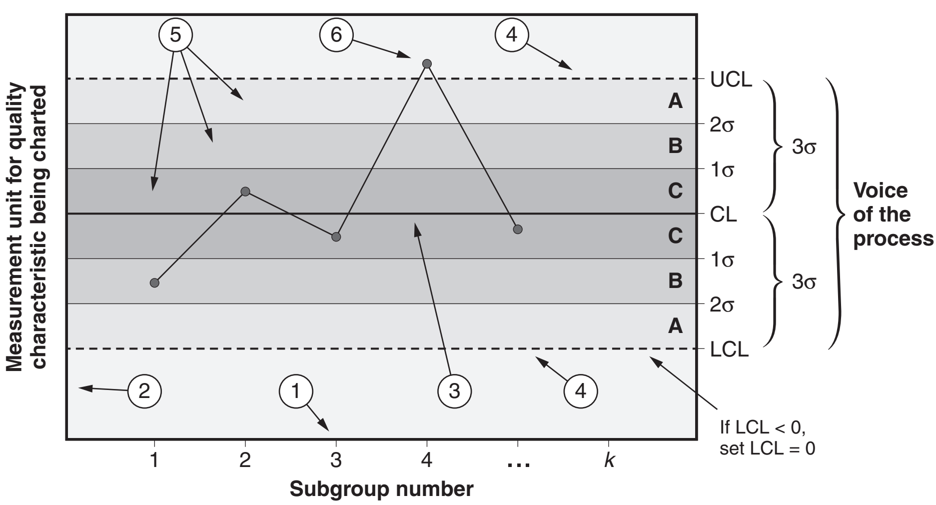

Figure 27.2 Structure of a typical control chart.

SELECTION OF VARIABLES

> Identify and select critical process characteristics for control chart monitoring. (Apply)
>
> **Body of Knowledge VIII.A.2**

When a control chart is to be used, a variable must be selected for monitoring. Sometimes that variable is the most critical dimension on the product. In some cases the variable of choice is a "leading indicator" of special causes—one that detects special causes before others do. Contractual requirements with a customer sometimes specify the variable(s) to be monitored via control chart. If the root cause of the special variation is known, an input variable may be monitored. Often, the variable to be monitored is the one that is the most difficult to hold, as determined by capability analyses. It is possible to monitor several variables on separate control charts, especially if computerized charting is employed. Ultimately, the selection of the variable to be charted depends on experience and judgment.

RATIONAL SUBGROUPING

> Define and apply the principle of rational subgrouping. (Apply)
>
> **Body of Knowledge VIII.A.3**

The method used to select samples for a control chart must be logical, or "rational." In the case of the $\bar{X}$ and R chart, it is desirable that the $\bar{X}$ chart detects a process shift, while the R chart should capture only common cause variation. That means there should be a high probability of variation between successive samples while the variation within the sample is kept low. Therefore, to minimize the within-sample variation, samples frequently consist of parts that are produced successively by the same process. The next sample is chosen somewhat later so that any process shifts that have occurred will be displayed on the chart as between-sample variation.

Choosing the rational subgroup requires care to make sure the same process is producing each item. For example, suppose a candy-making process uses 40 pistons to deposit 40 gobs of chocolate on a moving sheet of waxed paper in a 5×8 array, as shown in Figure 27.3. How should rational subgroups of size five be selected? Choosing the first five chocolates in each row, as indicated in Figure 27.3, would have the five elements of the sample being produced by five different processes (the five different pistons). A better choice would be to select the upper left-hand chocolate in five consecutive arrays, as shown in Figure 27.4,

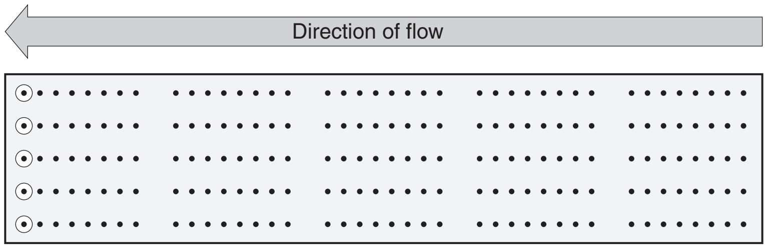

Figure 27.3 Conveyor belt in chocolate-making process.

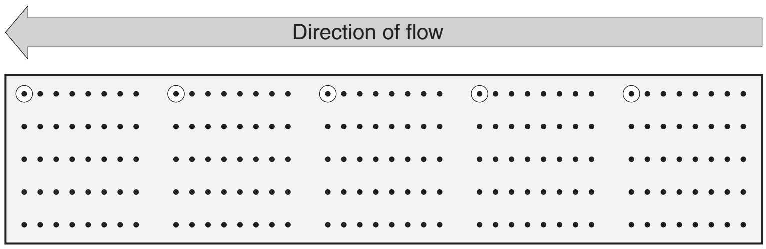

Figure 27.4 Conveyor belt in chocolate-making process with rational subgroup choice.

because the same piston forms them all. This rational subgrouping scheme minimizes the within-subgroup variation, thus allowing subgroup-to-subgroup variation to become more prominent.

The choice of sample size depends to some extent on the resources available to do the measuring. However, the larger the sample size, the more sensitive the chart.

CONTROL CHART SELECTION

> Select and use control charts in various situations: $\bar{X} - R$, $\bar{X} - s$, individual and moving range (ImR), p, np, c, u, short-run SPC, and moving average. (Apply)
>
> **Body of Knowledge VIII.A.4**

Types of Control Charts

Control charts can be categorized into two types: variables and attributes. Charts fall into the *variables* category when the data to be plotted result from measurement on a variable or continuous scale. *Attributes* charts are used for count data in which each data element is classified in one of two categories, such as good or bad.

Generally, variables charts are preferred over attributes charts because the data contain more information and are typically more sensitive to detecting process shifts than attributes charts.

It should be noted that measurement data collected for use with a variables control chart can be categorized and applied to an attributes control chart. For example, consider five temperature readings. Each reading can be classified as being within specification (for example, good) or out of specification (for example, bad). Had each reading been classified as attribute data (for example, either within or out of specification) at the time of collection and the actual measurement value not recorded, the attribute data could not be transformed for use on a variables control chart.

Variables Charts

The $\bar{X}$ and R chart is the flagship of variables control charts. It actually comprises two charts: the $\bar{X}$ chart, used to depict the process average, and the R chart, used to depict process variation using subgroup ranges.

The $\bar{X}$ and s chart is another variables control chart. With this chart, the sample standard deviation, s_i, for each subgroup is used to indicate process variation instead of the range. The standard deviation is a better measure of variation when the sample size is large (approximately 10 or larger).

The individuals and moving range chart, XmR, or ImR, is a useful chart when data are expensive to obtain or occur at a rate too slow to form rational subgroups. As the name implies, individual data points are plotted on one chart while the moving range (for example, absolute value of the difference between successive data points) is plotted on the moving range chart.

Figure 27.5 provides a flowchart for selecting the proper control chart based on the sample size.

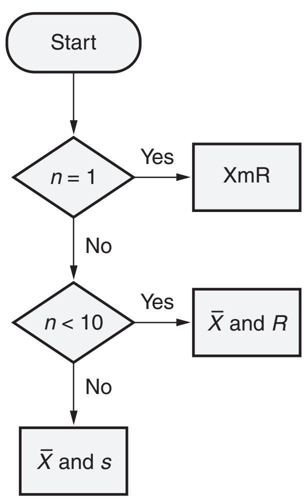

Figure 27.5 Selecting a variables control chart based on the subgroup size.

Appendix 3 provides useful combinations of variables control charts, and all necessary control chart constants are identified in Appendixes 4 and 5.

Control Limits. When first calculating control limits, it is prudent to collect as much data as practical. Many authors suggest at least 25–30 subgroups. The data are plotted and the charts reviewed for out-of-control conditions. If out-of-control conditions are found, root causes should be investigated and the associated data removed.

Consequently, a revised set of control limits should be computed. If it is not possible to determine root causes, the associated data should remain in the calculation for the limits. In this situation, it is likely the out-of-control condition was a statistical anomaly and really due to common cause variation.

Because control limits are calculated based on data from the process, they represent the *voice of the process* (VOP). Typically, they are set at $\pm 3\sigma$. The upper control limit is designated as *UCL* while the lower control limit is designated as *LCL*. The difference between the UCL and the LCL constitutes a 6σ spread. This spread is known as the VOP and is a necessary value when determining process capability (see Chapter 19).

Remember, $\pm 3\sigma$ represents 99.73% of the data. The probability of a point falling outside the limits is only 0.27%. Shewhart felt these limits represented an economical trade-off between the consequences of looking for a special cause that doesn't exist and not looking for one when it does exist. This being said, users are free to set limits at values other than $\pm 3\sigma$. However, caution is recommended since the out-of-control rules given later in this chapter would no longer apply.

It is important to note that calculating the LCL on the process variation chart could result in a negative value. If this should occur, the LCL is artificially set to zero because it is not possible to have negative ranges or standard deviations. Likewise, when an LCL computes to a negative value on attributes charts, the LCL is artificially set to zero because, again, it is not possible to have a negative percentage defective or negative defect counts.

The examples in the following sections use fewer samples for simplicity. Subgroup sizes are generally held constant with variables control charts. However, subgroup sizes may vary with specific attributes control charts.

Control limits for the $\overline{X}$ and R chart are given by the following formulas:

$\overline{X}$ and R Chart. The $\overline{X}$ and R chart is typically used when the sample size of each subgroup is approximately < 10 since the range is a better estimate of dispersion for smaller sample sizes.

Control limits for the $\overline{X}$ and R chart are given by the following formulas.

$$\overline{X}_i = \frac{\sum_{j=1}^{n} x_{ij}}{n} = \text{Average of the } i\text{th subgroup; plot points}$$

$$\overline{\overline{X}} = \frac{\sum_{i=1}^{k} \overline{X}_i}{k} = \text{Centerline of the } \overline{X} \text{ chart}$$

$$R_i = \text{Range of the } i\text{th subgroup; plot points}$$

$$\bar{R} = \frac{\sum_{i=1}^{k} R_i}{k} = \text{Centerline of the } R \text{ chart}$$

x_{ij} = jth individual data measurement within the ith subgroup

k = Number of subgroups

n = Constant sample size of each subgroup

$$\text{LCL}_{\bar{X}} = \bar{\bar{X}} - A_2\bar{R} \qquad \text{UCL}_{\bar{X}} = \bar{\bar{X}} + A_2\bar{R}$$

$$\text{LCL}_R = D_3\bar{R} \qquad \text{UCL}_R = D_4\bar{R}$$

The values of A_2, D_3, and D_4 are found in Appendix 4 for various values of n.

EXAMPLE 27.1

Data are collected in a face-and-plunge operation done on a lathe. The dimension being measured is the groove inside diameter (ID), which has a tolerance of 7.125 ± 0.010. Four parts are measured every hour. These values have been entered in Table 27.1. Determine the centerlines and control limits for the $\bar{X}$ and R chart.

Filling in the numbers from the earlier formulas, we have

$$\bar{\bar{X}} = 7.12565$$

$$\bar{R} = 0.0037$$

$$\text{LCL}_{\bar{X}} = 7.12565 - (0.729)(0.0037) = 7.122953$$

$$\text{UCL}_{\bar{X}} = 7.12565 + (0.729)(0.0037) = 7.128347$$

$$\text{LCL}_R = D_3\bar{R} = (0)(0.0037) = 0$$

$$\text{UCL}_R = D_4\bar{R} = (2.282)(0.0037) = 0.008443$$

The values of A_2, D_3, and D_4 were found in Appendix 4 for the value of $n = 4$.

The completed $\bar{X}$ and R chart is given in Figure 27.6. Both charts are in statistical control. Only minor computational differences are noted between the Minitab control limits and those given above that were computed using Excel.

Table 27.1 Data for Example 25.1 and Example 27.2.

7:00 a.m.	8:00 a.m.	9:00 a.m.	10:00 a.m.	11:00 a.m.	12:00 p.m.	1:00 p.m.	2:00 p.m.	3:00 p.m.	4:00 p.m.
7.127	7.125	7.123	7.127	7.128	7.125	7.126	7.126	7.127	7.128
7.123	7.126	7.129	7.127	7.125	7.125	7.123	7.126	7.129	7.123
7.123	7.121	7.129	7.124	7.126	7.127	7.123	7.127	7.128	7.122
7.126	7.122	7.124	7.125	7.127	7.128	7.125	7.128	7.129	7.124

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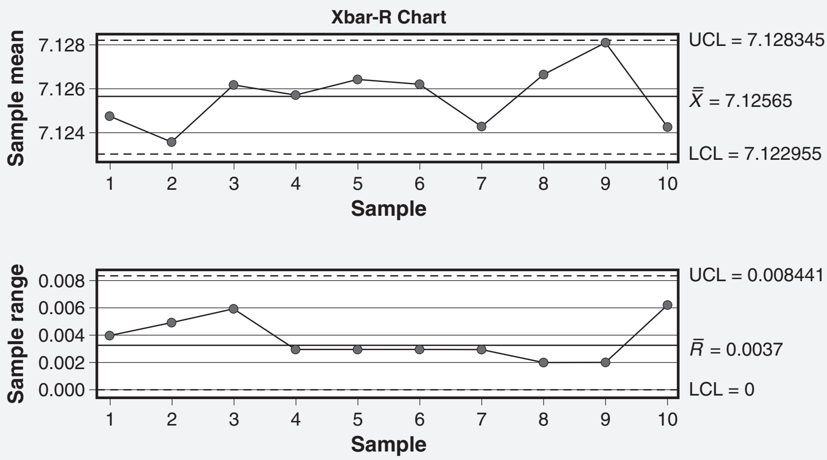

Figure 27.6 $\bar{X}$ and R chart for Example 27.1.

$\bar{X}$ and s Chart. The $\bar{X}$ and s chart is typically used when the sample size of each subgroup is approximately greater than or equal to 10 since the standard deviation is a better estimate of dispersion for larger sample sizes.

Control limits for an $\bar{X}$ and s chart are given by the following formulas:

$$\bar{X}_i = \frac{\sum_{j=1}^{n} x_{ij}}{n} = \text{Average of the } i\text{th subgroup; plot points}$$

$$\bar{\bar{X}} = \frac{\sum_{i=1}^{k} \bar{X}_i}{k} = \text{Centerline of the } \bar{X} \text{ chart}$$

$$s_i = \text{Standard deviation of the } i\text{th subgroup; plot points}$$

$$\bar{s} = \frac{\sum_{i=1}^{k} s_i}{k} = \text{Centerline of the } s \text{ chart}$$

x_{ij} = jth individual data measurement within the ith subgroup

k = Number of subgroups

n = Constant sample size of each subgroup

$$\text{LCL}_{\bar{X}} = \bar{\bar{X}} - A_3\bar{s} \qquad \text{UCL}_{\bar{X}} = \bar{\bar{X}} + A_3\bar{s}$$

$$\text{LCL}_s = B_3\bar{s} \qquad \text{UCL}_s = B_4\bar{s}$$

EXAMPLE 27.2

Using the same data provided in Table 27.1, construct an $\bar{X}$ and s chart.

Filling in the numbers from the earlier formulas, we have

$$\bar{\bar{X}} = 7.12565$$

$$\bar{s} = 0.001798$$

$$\mathrm{LCL}_{\bar{X}} = 7.12565 - (1.6278)(0.001798) = 7.122723$$

$$\mathrm{UCL}_{\bar{X}} = 7.12565 + (1.6278)(0.001798) = 7.128577$$

$$\mathrm{LCL}_s = B_3\bar{s} = (0)(0.001798) = 0$$

$$\mathrm{UCL}_s = B_4\bar{s} = (2.266)(0.001798) = 0.004074$$

The values of A_3, B_3, and B_4 were found in Appendix 4 for the value of $n = 4$.

The completed $\bar{X}$ and s chart is given in Figure 27.7. Both charts are in statistical control. In this example there were no computational differences noted between Minitab and Excel.

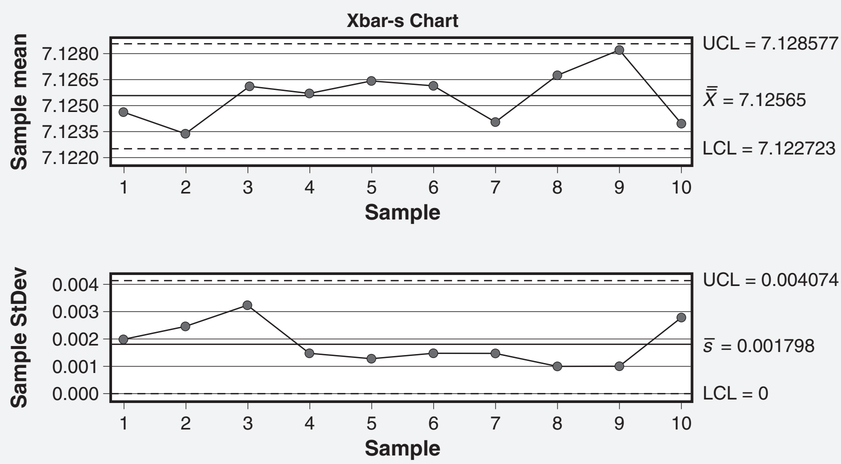

Figure 27.7 $\bar{X}$ and s chart for Example 27.2.

The values of A_3, B_3, and B_4 are found in Appendix 4 for various values of n.

Individuals and Moving Range (XmR, ImR) Chart. Recall that larger sample sizes produce charts that are more sensitive. In some situations, however, a sample size of one must be used. Examples include very slow processes or processes in which the measurement is very expensive to obtain, such as with destructive tests. If the sample size is one, an individuals and moving range (also known as *ImR* or *XmR*) chart is appropriate. The moving range is calculated by taking the absolute value of the difference between each measurement and the previous one. For this reason, the moving range chart has one less point plotted than the corresponding individuals chart.

Control limits for the individuals and moving range chart are given by the following formulas:

$$\bar{X} = \frac{\sum_{i=1}^{n} x_i}{n} = \text{Average of the individual data measurements; centerline of the } X \text{ chart}$$

$$mR_i = |x_i - x_{i-1}|;\ i = 2, 3, \ldots n;\ \text{plot points}$$

$$\overline{mR} = \frac{\sum_{i=2}^{n} mR_i}{n-1} = \text{Centerline for the } mR \text{ chart}$$

x_i = Individual data measurements; plot points

n = Number of individual data measurements

$$\text{LCL}_X = \bar{X} - E_2\overline{mR} \qquad \text{UCL}_X = \bar{X} + E_2\overline{mR}$$

$$\text{LCL}_{mR} = D_3\overline{mR} \qquad \text{UCL}_{mR} = D_4\overline{mR}$$

EXAMPLE 27.3

Using the data provided in Table 27.2, construct an XmR chart.

Filling in the numbers from the earlier formulas, we have

$$\bar{X} = 288.3$$

$$\overline{mR} = 2.89$$

$$\text{LCL}_X = 288.3 - (2.660)(2.89) = 280.61$$

$$\text{UCL}_X = 288.3 + (2.660)(2.89) = 295.99$$

$$\text{LCL}_{mR} = (0)(2.89) = 0$$

$$\text{UCL}_{mR} = (3.267)(2.89) = 9.44$$

Several of the calculations for this example are shown in Table 27.3. The values of E_2, D_3, and D_4 are found in Appendix 4 for n = 2. The completed individuals and moving range chart is given in Figure 27.8. Both charts demonstrate statistical control. Only minor computational differences are noted for the control limits for the individuals chart between Minitab and the Excel above.

Continued

Table 27.2 Data for Example 27.3—individuals and moving range chart.

Reading	Individual data element
1	290
2	288
3	285
4	290
5	291
6	287
7	284
8	290
9	290
10	288

Table 27.3 Calculations for Example 27.3—individuals and moving range chart.

Reading	Individual data element	Difference	Absolute value of the difference
1	290		
2	288	–2	2
3	285	–3	3
4	290	5	5
5	291	1	1
6	287	–4	4
7	284	–3	3
8	290	6	6
9	290	0	0
10	288	–2	2
Average	**288.3**	**Moving range**	**2.89**

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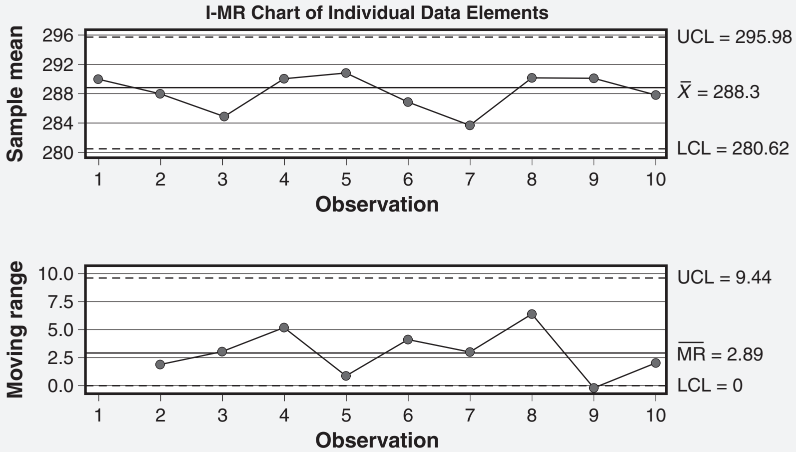

Figure 27.8 ImR chart for Example 27.3.

A summary of the formulas used for constructing the variables control charts is provided in Table 27.4.

Attributes Charts

Attributes charts are used for count data in which each data element is classified in one of two categories, such as good or bad. The *p*-charts and *np*-charts are used to plot proportion defective and number of defectives, respectively. The *c*-charts and *u*-charts are used to plot counts of defects and defects per unit, respectively.

Both *p*-charts and *np*-charts are based on the binomial distribution. Thus, it is assumed that the probability of a defective remains constant from subgroup to subgroup. Similarly, the *c*-charts and *u*-charts are based on the Poisson distribution. Therefore, the probability of the occurrence of a defect is assumed constant from subgroup to subgroup. When constructing attributes charts, it is important to keep these assumptions in mind to ensure statistical validity of the results.

Additionally, users of attributes charts should note that the *np*-chart and the *c*-chart require constant samples sizes. The *p*-chart and *u*-chart, however, permit variable sample sizes. As a result, the control limits for *p*-charts and *u*-charts will often appear ragged. These concepts are summarized in Figure 27.9.

Finally, readers should understand the differing terminology that surrounds these charts. For the most part, quality and Lean Six Sigma professionals tend to use the terms "defects" and "nonconformities" interchangeably. Consequently, we have used them interchangeably in this book.

However, there is a technical distinction between these two terms that is worth noting. ANSI Z1.4-2003 (R2013), which is in accordance with ISO 3534-2:2006(E), provides the following definitions:

Table 27.4 Formulas for calculating the centerline and control limits for variables charts.

Average chart				Individuals chart			
Chart	**Centerline**	**LCL**	**UCL**	**Chart**	**Centerline**	**LCL**	**UCL**
$\bar{X}$	$\bar{\bar{X}}$	$\text{LCL}_{\bar{X}} = \bar{\bar{X}} - A_2\bar{R}$	$\text{UCL}_{\bar{X}} = \bar{\bar{X}} + A_2\bar{R}$	X	$\bar{X}$	$\text{LCL}_X = \bar{X} - E_2\overline{mR}$	$\text{UCL}_X = \bar{X} + E_2\overline{mR}$
R	$\bar{R}$	$\text{LCL}_R = D_3\bar{R}$	$\text{UCL}_R = D_4\bar{R}$	mR	$\overline{mR}$	$\text{LCL}_{mR} = D_3\overline{mR}$	$\text{UCL}_{mR} = D_4\overline{mR}$
$\bar{X}$	$\bar{\bar{X}}$	$\text{LCL}_{\bar{X}} = \bar{\bar{X}} - A_3\bar{s}$	$\text{UCL}_{\bar{X}} = \bar{\bar{X}} + A_3\bar{s}$				
s	$\bar{s}$	$\text{LCL}_s = B_3\bar{s}$	$\text{UCL}_s = B_4\bar{s}$				

Average chart:

$$\bar{X}_i = \frac{\sum_{j=1}^{n} x_{ij}}{n}$$ = Average of the ith subgroup; plot points

$$\bar{\bar{X}} = \frac{\sum_{i=1}^{k} \bar{X}_i}{k}$$ = Centerline of the $\bar{X}$ chart

R_i = Range of the ith subgroup; plot points

$$\bar{R} = \frac{\sum_{i=1}^{k} R_i}{k}$$ = Centerline of the R chart

s_i = Standard deviation of the ith subgroup; plot points

$$\bar{s} = \frac{\sum_{i=1}^{k} s_i}{k}$$ = Centerline of the s chart

x_{ij} = j individual data measurements within i subgroups

k = Number of subgroups

n = Sample size of each subgroup

Individuals chart:

$$\bar{X} = \frac{\sum_{i=1}^{n} x_i}{n}$$ = Centerline of the X chart

$mR_i = |x_i - x_{i-1}|$ for $i = 2,3,\ldots,n$; plot points

$$\overline{mR} = \frac{\sum_{i=2}^{n} mR_i}{n-1}$$ = Centerline for the mR chart

x_i = Individual data measurements; plot points

n = Number of individual data measurements

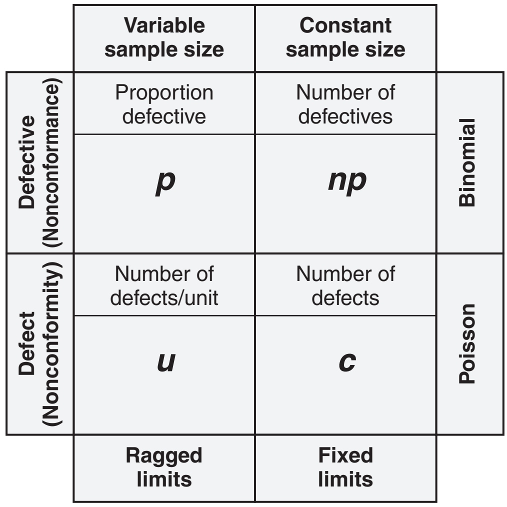

Figure 27.9 Characteristics of attributes control charts.

- *Defect*. A departure of a quality characteristic from its intended level or state that occurs with a severity sufficient to cause an associated product or service not to satisfy intended normal, or foreseeable, usage requirements.
- *Nonconformity*. A departure of a quality characteristic from its intended level or state that occurs with severity sufficient to cause an associated product or service not to meet a specification requirement.

ISO 3534-2:2006(E) adds two important notes to the definition of defects:

- *Note 1*. The distinction between the concepts *defect* and *nonconformity* is important as it has legal connotations, particularly those associated with product liability issues. Consequently, the term "defect" should be used with extreme caution.
- *Note 2*. The intended use by the customer can be affected by the nature of the information, such as operating or maintenance instructions, provided by the customer.

Based on the above definitions and supporting notes, one might argue that we can generally align a nonconformity to occurring within the organization and a defect to occurring outside the organization (that is, external customer).

Note: All the examples presented in the attributes charts section that follows use the same data. However, they are presented in different contexts. Only the sample sizes change as necessary to fit the particular type of chart being studied. This is done to help the reader grasp the underlying concepts of each chart more readily and to simultaneously compare and contrast the charts. We think this approach will be more beneficial to our readers than presenting each chart with entirely different data.

***p*-Chart.** Recall that the *p*-chart is used when the sample size varies. We use this chart to plot the proportion or percentage of defectives. Note: The control limits on this chart appear ragged because they reflect each subgroup's individual sample size.

The formulas for the centerline and the upper and lower control limits for the *p*-chart are:

$$p_i = \frac{D_i}{n_i} = \text{Plot points}$$

$$\bar{p} = \frac{\sum_{i=1}^{k} D_i}{\sum_{i=1}^{k} n_i}$$

$$D_i = \text{Number of defective units in the } i\text{th subgoup}$$

$$k = \text{Number of subgroups}$$

$$n_i = \text{Sample size of the } i\text{th subgroup}$$

$$\text{LCL}_p = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}} \qquad \text{UCL}_p = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$$

EXAMPLE 27.4

A test was conducted to determine the presence of the Rh factor in 13 samples of donated blood. The results of the test are given in Table 27.5. Construct a *p*-chart.

Using Excel and the formulas above, we obtain the results shown in Table 27.6 and immediately below:

$$\bar{p} = \frac{189}{1625} = 0.1163$$

$$\text{LCL}_p = 0.1163 - 3\sqrt{\frac{(0.1164)(1-0.1163)}{n_i}} = 0.1163 - 3\sqrt{\frac{(0.1163)(0.8837)}{n_i}}$$

$$\text{UCL}_p = 0.1163 + 3\sqrt{\frac{(0.1163)(1-0.1163)}{n_i}} = 0.1163 + 3\sqrt{\frac{(0.1163)(0.8837)}{n_i}}$$

The completed *p*-chart is given in Figure 27.10 and demonstrates statistical control. Notice that the values for the control limits in Figure 27.10 are actually the control limits for the last plot point only. Compare the last plot point for both the LCL and UCL from Table 27.6 and *the LCL and UCL* values given in Figure 27.10.

Continued

Table 27.5 Data for Example 27.4—*p*-chart.

Date	Number of defectives (D_i)	Sample size (n_i)
8-Sep	12	116
9-Sep	14	126
10-Sep	18	112
11-Sep	13	134
12-Sep	17	121
13-Sep	15	119
14-Sep	15	138
15-Sep	16	109
16-Sep	11	111
17-Sep	14	125
18-Sep	13	129
19-Sep	14	144
20-Sep	17	141

Table 27.6 Calculations for Example 27.4—*p*-chart.

Date	Number of defectives (D_i)	Sample size (n_i)	Plot points (p_i)	Centerline ($\bar{p}$)	LCL	UCL
8-Sep	12	116	0.1034	0.1163	0.0270	0.2056
9-Sep	14	126	0.1111	0.1163	0.0306	0.2020
10-Sep	18	112	0.1607	0.1163	0.0254	0.2072
11-Sep	13	134	0.0970	0.1163	0.0332	0.1994
12-Sep	17	121	0.1405	0.1163	0.0289	0.2037
13-Sep	15	119	0.1261	0.1163	0.0281	0.2045
14-Sep	15	138	0.1087	0.1163	0.0344	0.1982
15-Sep	16	109	0.1468	0.1163	0.0242	0.2084
16-Sep	11	111	0.0991	0.1163	0.0250	0.2076
17-Sep	14	125	0.1120	0.1163	0.0303	0.2023
18-Sep	13	129	0.1008	0.1163	0.0316	0.2010
19-Sep	14	144	0.0972	0.1163	0.0362	0.1964
20-Sep	17	141	0.1206	0.1163	**0.0353**	**0.1973**
Total	**189**	**1625**				
***p*-bar**	**0.1163**					

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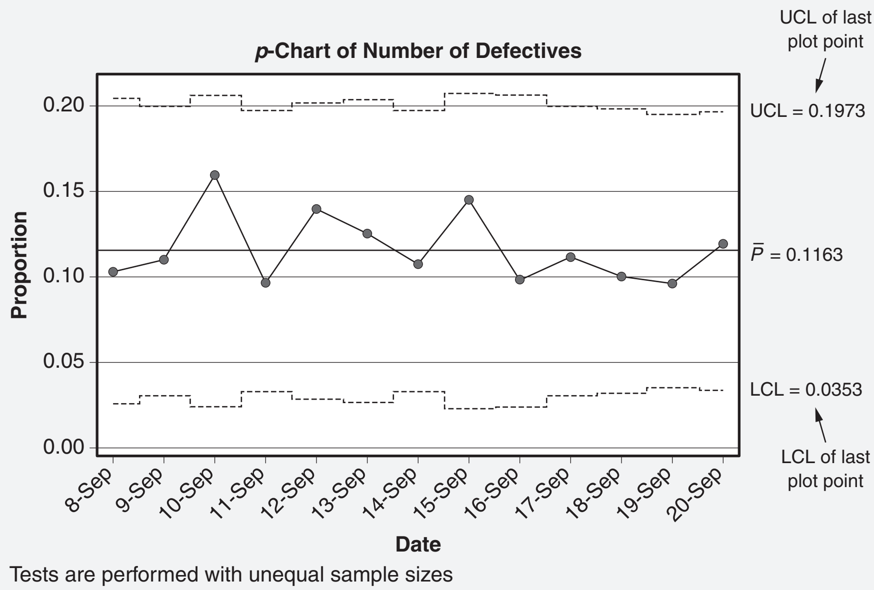

Figure 27.10 *p*-chart for Example 27.4.

***np*-Chart.** Recall that the *np*-chart is used when the sample size is constant. We use this chart to plot the number of defectives.

The formulas for the centerline and the upper and lower control limits for the *np*-chart are:

$$\bar{p} = \frac{\sum_{i=1}^{k} D_i}{\sum_{i=1}^{k} n_i}$$

$n\bar{p}$ = Centerline for the chart

D_i = Number of defective units in the ith subgoup; plot points

k = Number of subgroups

n = Constant sample size for each subgroup

$$\text{LCL}_{np} = n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})} \qquad \text{UCL}_{np} = n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})}$$

EXAMPLE 27.5

Packages containing 1000 light bulbs are randomly selected, and all 1000 bulbs are light-tested. The results of the tests are given in Table 27.7. Construct an *np*-chart.

Using Excel and the formulas above, we obtain the results shown in Table 27.8 and immediately below:

$$\bar{p} = \frac{189}{1625} = 0.1163$$

$$n\bar{p} = (125)(01163) = 14.54$$

The completed *np*-chart is given in Figure 27.11 and demonstrates statistical control. Only a minor difference for the LCL between Minitab and Excel is apparent.

Table 27.7 Data for Example 27.5—*np*-chart.

Date	Number of defectives (D_i)	Sample size (n)
8-Sep	12	125
9-Sep	14	125
10-Sep	18	125
11-Sep	13	125
12-Sep	17	125
13-Sep	15	125
14-Sep	15	125
15-Sep	16	125
16-Sep	11	125
17-Sep	14	125
18-Sep	13	125
19-Sep	14	125
20-Sep	17	125

Continued

Continued

Table 27.8 Calculations for Example 27.5—*np*-chart.

Date	Number of defectives (D_i)	Sample size (n)	Centerline (p bar)	LCL	UCL
8-Sep	12	125	14.54	3.78	25.29
9-Sep	14	125	14.54	3.78	25.29
10-Sep	18	125	14.54	3.78	25.29
11-Sep	13	125	14.54	3.78	25.29
12-Sep	17	125	14.54	3.78	25.29
13-Sep	15	125	14.54	3.78	25.29
14-Sep	15	125	14.54	3.78	25.29
15-Sep	16	125	14.54	3.78	25.29
16-Sep	11	125	14.54	3.78	25.29
17-Sep	14	125	14.54	3.78	25.29
18-Sep	13	125	14.54	3.78	25.29
19-Sep	14	125	14.54	3.78	25.29
20-Sep	17	125	14.54	3.78	25.29
Total	**189**	**1625**			
$\bar{p}$	**0.1163**				
$n\bar{p}$	**14.54**				

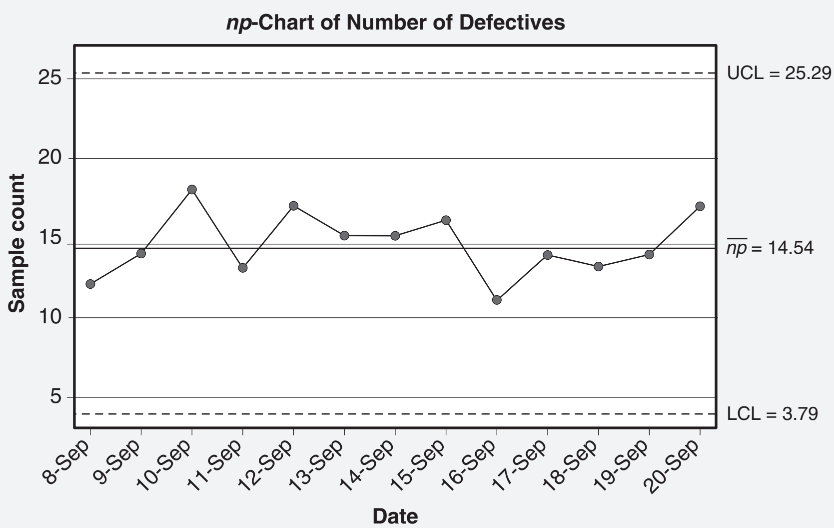

Figure 27.11 *np*-chart for Example 27.5.

***c*-Chart.** Recall that the *c*-chart is used when the sample size is constant. We use this chart to plot the number of defects.

The formulas for the centerline and the upper and lower control limits for the *c*-chart are:

$$\bar{c} = \frac{\sum_{i=1}^{k} c_i}{k}$$

$$c_i = \text{Number of defects in the } i\text{th subgroup; plot points}$$

$$k = \text{Number of subgroups}$$

$$n = \text{Constant sample size of each subgroup}$$

$$\text{LCL}_c = \bar{c} - 3\sqrt{\bar{c}} \qquad \text{UCL}_c = \bar{c} + 3\sqrt{\bar{c}}$$

EXAMPLE 27.6

Panes of glass are inspected for defects such as bubbles, scratches, chips, inclusions, waves, and dips. The results of the inspection are given in Table 27.9. Construct a *c*-chart.

Using Excel and the formulas above, we obtain the results shown in Table 27.10 and immediately below:

$$\bar{c} = \frac{189}{13} = 14.54$$

$$\text{LCL}_c = 14.54 - 3\sqrt{14.54} = 3.10 \qquad \text{UCL}_c = 14.54 + 3\sqrt{14.54} = 25.98$$

Table 27.9 Data for Example 27.6—*c*-chart.

Date	Number of defects (c_i)	Sample size (n)
8-Sep	12	125
9-Sep	14	125
10-Sep	18	125
11-Sep	13	125
12-Sep	17	125
13-Sep	15	125
14-Sep	15	125
15-Sep	16	125
16-Sep	11	125
17-Sep	14	125
18-Sep	13	125
19-Sep	14	125
20-Sep	17	125

Continued

Continued

The completed *c*-chart is given in Figure 27.12 and demonstrates statistical control.

Table 27.10 Calculations for Example 27.6—*c*-chart.

Date	Number of defects—plot points (c_i)	Sample size (n)	Centerline ($\bar{c}$)	LCL	UCL
8-Sep	12	125	14.54	3.10	25.98
9-Sep	14	125	14.54	3.10	25.98
10-Sep	18	125	14.54	3.10	25.98
11-Sep	13	125	14.54	3.10	25.98
12-Sep	17	125	14.54	3.10	25.98
13-Sep	15	125	14.54	3.10	25.98
14-Sep	15	125	14.54	3.10	25.98
15-Sep	16	125	14.54	3.10	25.98
16-Sep	11	125	14.54	3.10	25.98
17-Sep	14	125	14.54	3.10	25.98
18-Sep	13	125	14.54	3.10	25.98
19-Sep	14	125	14.54	3.10	25.98
20-Sep	17	125	14.54	3.10	25.98
Total	**189**				
$\bar{c}$	**14.54**				

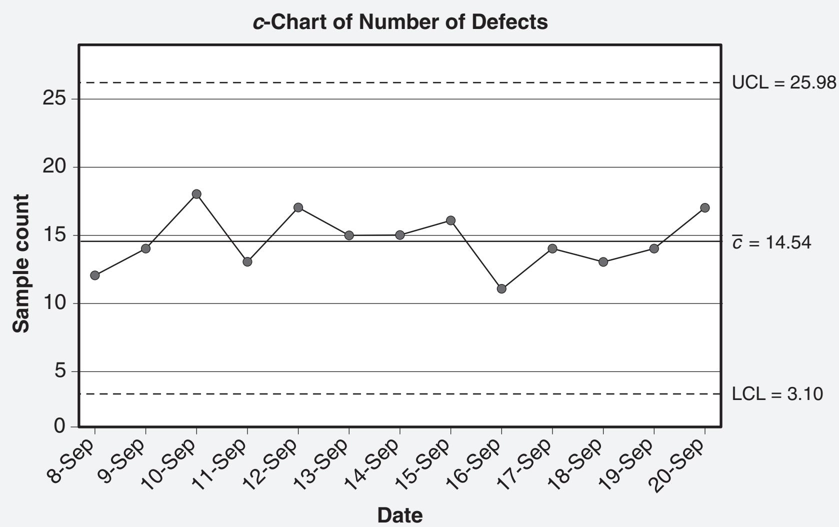

Figure 27.12 *c*-chart for Example 27.6.

***u*-Chart.** Recall that the *u*-chart is used when the sample size varies. We use this chart to plot the number of defects per unit. Note: The control limits on the chart appear ragged because they reflect each subgroup's individual sample size.

The formulas for the centerline and the upper and lower control limits for the *u*-chart are:

$$\bar{u} = \frac{\sum_{i=1}^{k} c_i}{\sum_{i=1}^{k} n_i} = \text{Centerline for the chart}$$

$$u_i = \frac{c_i}{n_i} = \text{Number of defects per unit in the } i\text{th subgroup; plot points}$$

$$c_i = \text{Number of defects in the } i\text{th subgroup; plot points}$$

$$k = \text{Number of subgroups}$$

$$n_i = \text{Sample size of the } i\text{th subgroup}$$

$$\text{LCL}_u = \bar{u} - 3\sqrt{\frac{\bar{u}}{n_i}} \qquad \text{UCL}_u = \bar{u} + 3\sqrt{\frac{\bar{u}}{n_i}}$$

EXAMPLE 27.7

Construct a *u*-chart using the data given in Table 27.11.

Table 27.11 Data for Example 27.7—*u*-chart.

Date	Number of defects (c_i)	Sample size (n_i)
8-Sep	12	116
9-Sep	14	126
10-Sep	18	112
11-Sep	13	134
12-Sep	17	121
13-Sep	15	119
14-Sep	15	138
15-Sep	16	109
16-Sep	11	111
17-Sep	14	125
18-Sep	13	129
19-Sep	14	144
20-Sep	17	141

Continued

Using Excel and the formulas given above, we obtain the results shown in Table 27.12 and immediately below:

$$\bar{u} = \frac{189}{1625} = 0.1163$$

$$\text{LCL}_u = 0.1163 - 3\sqrt{\frac{0.1163}{n_i}} \qquad \text{UCL}_u = 0.1163 + 3\sqrt{\frac{0.1163}{n_i}}$$

The completed u-chart is given in Figure 27.13 and demonstrates statistical control. Notice that the values for the control limits in Figure 27.13 are actually the control limits for the last plot point only. Compare the last plot point for both the LCL and UCL from Table 27.12 and the LCL and UCL values given in Figure 27.13.

Table 27.12 Calculations for Example 27.7—u-chart.

Date	Number of defectives (c_i)	Sample size (n_i)	Plot points (c_i/n_i)	Centerline ($\bar{u}$)	LCL	UCL
8-Sep	12	116	0.1034	0.1163	0.0213	0.2113
9-Sep	14	126	0.1111	0.1163	0.0252	0.2074
10-Sep	18	112	0.1607	0.1163	0.0196	0.2130
11-Sep	13	134	0.0970	0.1163	0.0279	0.2047
12-Sep	17	121	0.1405	0.1163	0.0233	0.2093
13-Sep	15	119	0.1261	0.1163	0.0225	0.2101
14-Sep	15	138	0.1087	0.1163	0.0292	0.2034
15-Sep	16	109	0.1468	0.1163	0.0183	0.2143
16-Sep	11	111	0.0991	0.1163	0.0192	0.2134
17-Sep	14	125	0.1120	0.1163	0.0248	0.2078
18-Sep	13	129	0.1008	0.1163	0.0262	0.2064
19-Sep	14	144	0.0972	0.1163	0.0310	0.2016
20-Sep	17	141	0.1206	0.1163	**0.0301**	**0.2025**
Total	**189**	**1625**				
$\bar{u}$	**0.1163**					

Continued

Continued

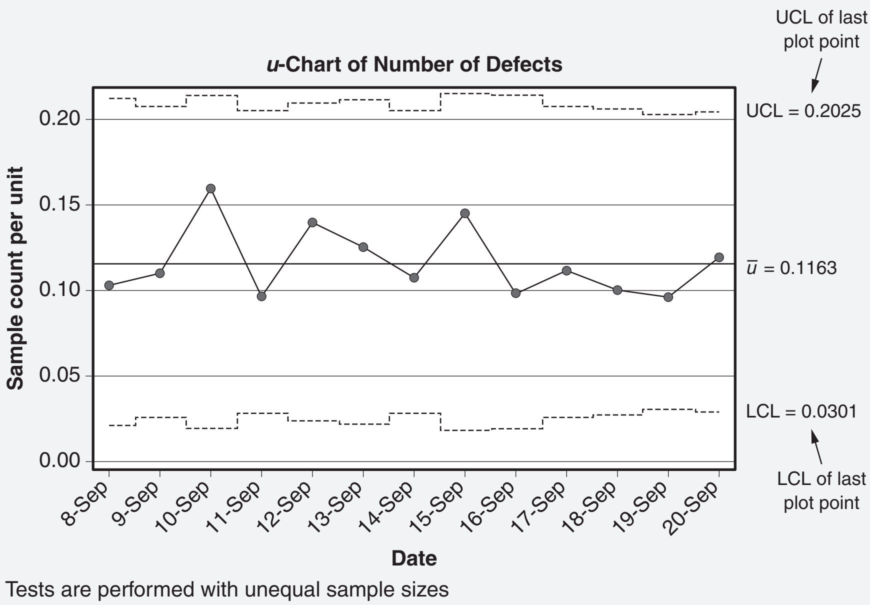

Figure 27.13 *u*-chart for Example 27.7.

A summary of the formulas used for constructing the attributes control charts is provided in Table 27.13.

Short-Run Control Charts

Short-run control charts should be considered when data are collected infrequently or aperiodically. They may be used with historical target or target values, attributes or variables data, and individual or subgrouped averages.

While the subject of short-run control charts is not overly difficult, it can be particularly confusing due to the various nomenclatures and terminologies used by authors. For this reason, the following figure and table have been prepared:

- *Figure 27.14*. This flowchart will aid the reader in determining what type of chart is most appropriate and will link to a comprehensive set of examples.
- *Table 27.14*. This table contains all the formulas needed to construct the short-run charts discussed in this subsection. Furthermore, an attempt has been made to provide the reader with alternate names of charts so that recognition is possible among the various sources and references.

Table 27.13 Formulas for calculating the centerline and control limits for attributes charts.

Chart	Centerline	LCL	UCL	Plot point
p	$\overline{p}=\dfrac{\sum_{i=1}^{k} D_i}{\sum_{i=1}^{k} n_i}$	$\text{LCL}_p=\overline{p}-3\sqrt{\dfrac{\overline{p}(1-\overline{p})}{n_i}}$	$\text{UCL}_p=\overline{p}+3\sqrt{\dfrac{\overline{p}(1-\overline{p})}{n_i}}$	$p_i=\dfrac{D_i}{n_i}$
np	$n\overline{p}=n\dfrac{\sum_{i=1}^{k} D_i}{\sum_{i=1}^{k} n_i}$	$\text{LCL}_{np}=n\overline{p}-3\sqrt{n\overline{p}(1-\overline{p})}$	$\text{UCL}_{np}=n\overline{p}+3\sqrt{n\overline{p}(1-\overline{p})}$	D_i
c	$\overline{c}=\dfrac{\sum_{i=1}^{k} c_i}{k}$	$\text{LCL}_c=\overline{c}-3\sqrt{\overline{c}}$	$\text{UCL}_c=\overline{c}+3\sqrt{\overline{c}}$	c_i
u	$\overline{u}=\dfrac{\sum_{i=1}^{k} c_i}{\sum_{i=1}^{k} n_i}$	$\text{LCL}_u=\overline{u}-3\sqrt{\dfrac{\overline{u}}{n_i}}$	$\text{UCL}_u=\overline{u}+3\sqrt{\dfrac{\overline{u}}{n_i}}$	$u_i=\dfrac{c_i}{n_i}$

Notes:

1. n = Constant sample size of each subgroup
2. n_i = Sample size of the ith subgroup
3. k = Number of subgroups
4. D_i = Number of *defective* units in the ith subgroup
5. c_i = Number of *defects* in the ith subgroup
6. u_i = Number of *defects*/unit in the ith subgroup

The focus of the next subsection will be on selecting the appropriate chart or pair of charts and computing the corresponding statistics to be plotted. Since formulas are provided and the construction of most tables and charts is fundamentally similar, an example table will be developed for plotting the $\overline{Z}$, W, $\overline{Z}^*$, and W^* charts only. Little attention will be given to the actual plotting of the data.

Key Considerations When Using a Short-Run Chart. Griffith (1996) recommends the following rules when dealing with short-run control charts:

- Focus on the process, not the parts. Traditional (or product) control charts should be used to monitor parts.
- Be sure the process stream is the same. For example, parts may vary in size, shape, and material, but the process remains the same.

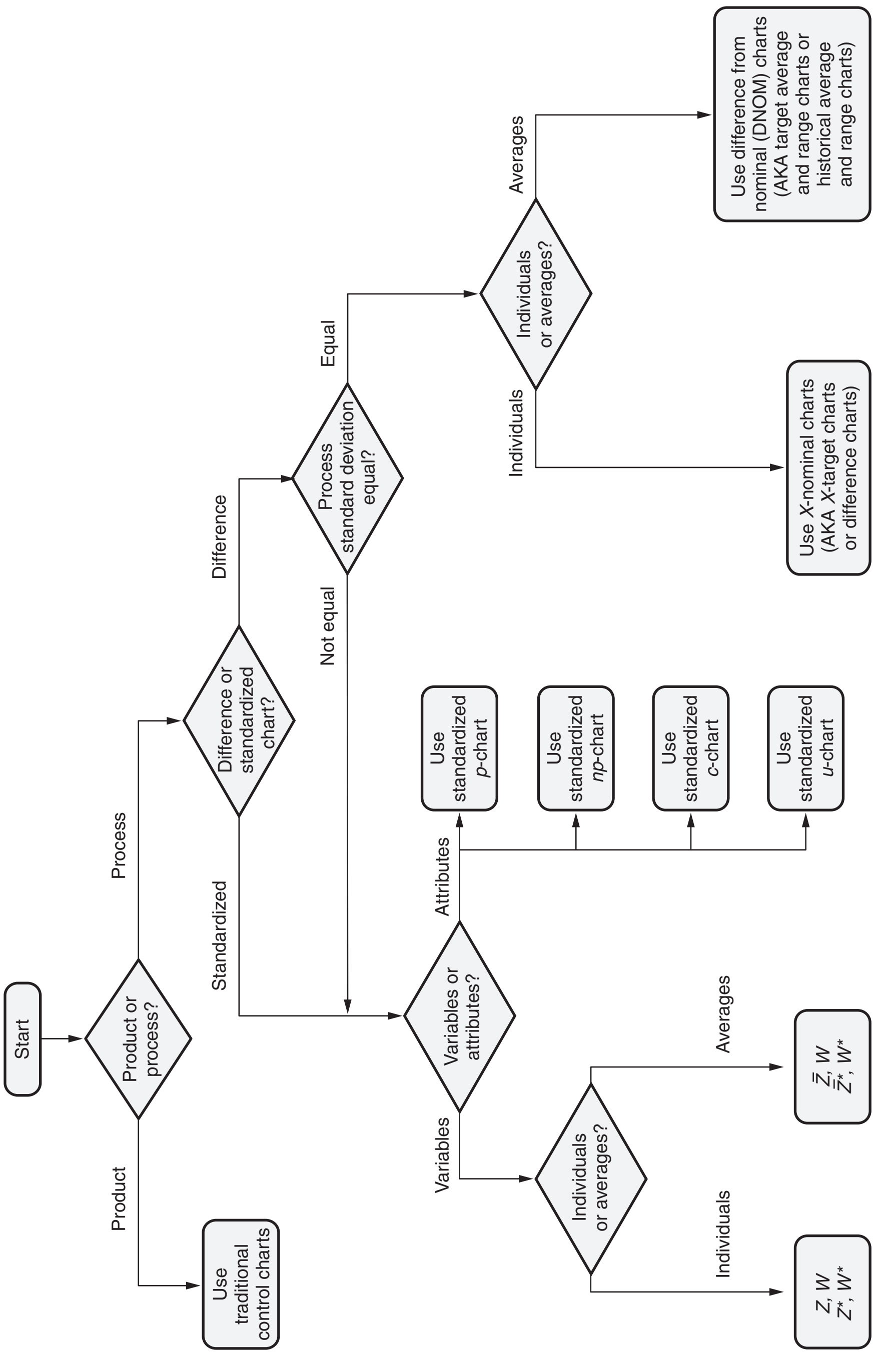

Figure 27.14 Short-run SPC control charts decision flowchart.

Table 27.14 Summary of formulas for short-run SPC charts.

	Charts for non-subgrouped data					
	Paired charts		**Paired charts**		**Paired charts**	
Chart type	Difference (*X*-nominal, *X*-target)	mR	Z	W	Z^*	W^*
Plot points	$X - Nom$	mR	$Z = \frac{X - Nom}{Sigma(X)}$	$W = mR_W$	$Z^* = \frac{X - Nom}{\overline{R}}$	$W^* = mR_W$
CL	0.0	$\overline{mR}$	0.0	d_2	0.0	1.0
UCL	$+A_3\overline{mR}$	$D_4\overline{mR}$	+3.0	$d_2 + 3d_3$	$+A_3$	D_4
LCL	$-A_3\overline{mR}$	$D_3\overline{mR}$	–3.0	0	$-A_3$	D_3

Where *Nom* = *Nominal* or *Target* and $Sigma(X) = \frac{\overline{R}}{d_2}$ and $n = 2$.

See Appendix 4 for A_3, D_3, and D_4 values.

See Appendix 6 for d_2 and d_3 values.

Continued

Table 27.14 *Continued.*

Charts for subgrouped data						
	Paired charts		**Paired charts**		**Paired charts**	
Chart type	Difference	R	$\bar{Z}$	W	$\bar{Z}^*$	W^*
Plot points	$\bar{X} - Nom$	R	$\bar{Z} = \dfrac{\bar{X} - Nom}{Sigma(\bar{X})}$	$W = \dfrac{R}{Sigma(X)}$	$\bar{Z}^* = \dfrac{\bar{X} - Nom}{\bar{R}}$	$W^* = \dfrac{R}{\bar{R}}$
CL	0.0	$\bar{\bar{R}}$	0.0	d_2	0.0	1.0
UCL	$+A_2\bar{\bar{R}}$	$D_4\bar{\bar{R}}$	+3.0	$d_2 + 3d_3$	$+A_2$	D_4
LCL	$-A_2\bar{\bar{R}}$	$D_3\bar{\bar{R}}$	–3.0	$d_2 - 3d_3$	$-A_2$	D_3

Where *Nom* = *Nominal* or *Target* and $Sigma(\bar{X}) = \dfrac{\bar{R}}{d_2\sqrt{n}}$ and $n = 2$.

See Appendix 4 for A_3, D_3, and D_4 values.

See Appendix 6 for d_2 and d_3 values.

Standardized attributes charts				
Chart type	p	np	c	u
Plot points	$Z_i = \dfrac{p_i - \bar{p}}{\sqrt{\bar{p}(1-\bar{p})/n}}$	$Z_i = \dfrac{np_i - n\bar{p}}{\sqrt{n\bar{p}(1-\bar{p})/n}}$	$Z_i = \dfrac{c_i - \bar{c}}{\sqrt{\bar{c}}}$	$Z_i = \dfrac{u_i - \bar{u}}{\sqrt{\bar{u}/n}}$
CL	0.0	0.0	0.0	0.0
UCL	+3.0	+3.0	+3.0	+3.0
LCL	–3.0	–3.0	–3.0	–3.0

- Look for generic families of product made by the process.
- Use coded data. This permits different parts, dimensions, tolerances, and so on, to be used together.
- Use the 20 subgroups minimum rule. The immediately preceding bullets all work to help meet this requirement.
- Variation between different parts aggregated on a chart must be the same. Statistically, we call this *homogeneity of variance*. Various tests are available to determine homogeneity of variance. When homogeneity of variance is not present in the data, the reader will note from Figure 27.14 that standardized charts must be used.

Also, it is important to point out that the measurement system should be sufficiently granular to detect variation. This is a particularly important consideration when dealing with parts of varying size, shape, and other measured characteristics.

A special word of caution is in order when interpreting short-run charts. Remember, these types of charts reflect multiple part numbers. Subsequently, the short-run chart may be in control but not the specific part numbers that comprise it. Undoubtedly, the reader will notice we are frequently working with coded data (that is, differences from target or historical target for averages and ranges). Therefore, it is critical that the data for each part be coded with its own target or historical target. This point cannot be emphasized enough.

Constructing a Short-Run Chart

1. Determine the historical average and historical range for each product. The historical average may also be a target or nominal value.
2. Collect a set of data—$x_1, x_2, \ldots, x_n$—for a number of different products that follow the same process.
3. Compute the average and range for each subgroup.
4. Compute sigma (X) or sigma ($\bar{X}$), depending on the set of charts to be developed.
5. Compute $\bar{Z}$, W, $\bar{Z}^*$, and W^* depending on the set of charts to be developed.
6. Compute the centerline for the $\bar{Z}$ moving average chart using the appropriate formulas given in the "Charts for Subgrouped Data" section of Table 27.14.
7. Compute the LCL and the UCL for the W chart, using the appropriate formulas given in the "Charts for Subgrouped Data" section of Table 27.14.
8. Compute the centerline for the $\bar{Z}^*$ chart using the appropriate formulas given in the "Charts for Subgrouped Data" section of Table 27.14.
9. Compute the LCL and the UCL for the W^* chart, using the appropriate formulas given in the "Charts for Subgrouped Data" section of Table 27.14.

10. Plot the data, centerline, LCL, and UCL for both of the chosen pairs of charts (that is, steps 6 and 7, or 8 and 9).
11. Interpret results and take action accordingly.

EXAMPLE 27.8

The data shown in Table 27.15 are taken from Wheeler (1991). Three different products from the same processes are included in the data. The data collected by product in time order by subgroup are given in columns E–I, respectively. A subgroup size of five

Table 27.15 Data for Example 27.8—Short-run chart.

Column A	Column B	Column C	Column D	Column E	Column F	Column G	Column H	Column I
Date	Product	Historical average/ Target/ Nominal value	Historical average range for 10 previous subgroups	X_1	X_2	X_3	X_4	X_5
11–Jul	1407	9.5	10.50	16	11	13	7	10
12–Jul	1407	9.5	10.50	14	15	15	10	16
15–Jul	1404	4.5	4.10	4	0	6	4	4
16–Jul	1404	4.5	4.10	2	4	5	8	5
17–Jul	1404	4.5	4.10	5	8	3	5	3
18–Jul	1404	4.5	4.10	6	6	5	3	5
19–Jul	1404	4.5	4.10	4	5	4	2	8
22–Jul	1407	9.5	10.50	5	7	7	8	15
23–Jul	1408	8.5	7.90	0	9	9	7	0
24–Jul	1408	8.5	7.90	5	7	7	12	8
25–Jul	1404	4.5	4.10	4	2	7	2	2
26–Jul	1404	4.5	4.10	3	6	4	2	6
29–Jul	1407	9.5	10.50	3	10	16	6	10
30–Jul	1408	8.5	7.90	9	7	8	8	13
31–Jul	1407	9.5	10.50	5	15	9	5	10
1–Aug	1408	8.5	7.90	8	8	8	10	10
2–Aug	1407	9.5	10.50	11	15	12	8	14
5–Aug	1404	4.5	4.10	3	2	5	4	4
6–Aug	1404	4.5	4.10	4	5	5	4	8
7–Aug	1404	4.5	4.10	2	3	1	4	1

Continued

has been used. The product, historical average, and historical range are given in columns B–D, respectively. Using the procedure given above, compute the plot points for a $\bar{Z} - W$ chart and a $\bar{Z}^* - W^*$ chart. Also provide the centerlines and LCL and UCL limits for each pair of charts.

Table 27.16 has been developed using Excel, Table 27.15, and the formulas given in Table 27.14.

If we review Table 27.16, we will see that the average and range for each subgroup have been computed and are given in columns J and K, respectively.

Sigma (X) is given in column L, and sigma ($\bar{X}$) is given in column M. These values were computed using $d_2 = 2.326$ for $n = 5$. Note: The historical average range given in column D has been used to compute these values for each product.

Again, using the formulas given in Table 27.14, the values for $\bar{Z}$, W, $\bar{Z}^*$, and W^* are computed as reflected in columns N–Q, respectively.

Let's review the $\bar{Z}$ chart. From Table 27.14, the centerline is 0, the LCL = –3.0 and the UCL = 3.0.

Using the same table for the W chart, we know that the centerline is $d_2 = 2.326$ and $d_3 = 0.8641$ for $n = 5$ from Appendix 6. This makes the LCL = –0.2663, which we set to 0.0, and the UCL = 4.9183.

Continuing with the $\bar{Z}^*$ chart, the centerline is 0 with the LCL at $-A_2$ and the UCL at $+A_2$. For $n = 5$, we have $A_2 = 0.577$ from Appendix 4. Therefore, the LCL = –0.577 and UCL = 0.577.

The centerline for the W^* chart is 1.0, while the LCL = D_3 and the UCL = D_4. For $n = 5$, $D_3 = 0$ and $D_4 = 2.114$. Therefore, the LCL = 0 and the UCL = 2.114.

The centerlines and control limits are summarized in Table 27.16.

Continued

Continued

Table 27.16 Calculations for Example 27.8—short-run chart.

Column A	Column B	Column C	Column D	Column E	Column F	Column G	Column H	Column I	Column J	Column K	Column L	Column M	Column N	Column O	Collumn P	Column Q
Date	Product	Historical average/ Target/ Nominal value	Historical average range for 10 previous subgroups	X_1	X_2	X_3	X_4	X_5	$\bar{X}$	R	σ_X	$\sigma_{\bar{X}}$	$\bar{Z}$	W	$\bar{Z}^*$	W^*
11–Jul	1407	9.5	10.50	16	11	13	7	10	11.4	9	4.51	2.02	0.94	2.0	0.18	0.9
12–Jul	1407	9.5	10.50	14	15	15	10	16	14.0	6	4.51	2.02	2.23	1.3	0.43	0.6
15–Jul	1404	4.5	4.10	4	0	6	4	4	3.6	6	1.76	0.79	–1.14	3.4	–0.22	1.5
16–Jul	1404	4.5	4.10	2	4	5	8	5	4.8	6	1.76	0.79	0.38	3.4	0.07	1.5
17–Jul	1404	4.5	4.10	5	8	3	5	3	4.8	5	1.76	0.79	0.38	2.8	0.07	1.2
18–Jul	1404	4.5	4.10	6	6	5	3	5	5.0	3	1.76	0.79	0.63	1.7	0.12	0.7
19–Jul	1404	4.5	4.10	4	5	4	2	8	4.6	6	1.76	0.79	0.13	3.4	0.02	1.5
22–Jul	1407	9.5	10.50	5	7	7	8	15	8.4	10	4.51	2.02	–0.54	2.2	–0.10	1.0
23–Jul	1408	8.5	7.90	0	9	9	7	0	5.0	9	3.40	1.52	–2.30	2.6	–0.44	1.1
24–Jul	1408	8.5	7.90	5	7	7	12	8	7.8	7	3.40	1.52	–0.46	2.1	–0.09	0.9
25–Jul	1404	4.5	4.10	4	2	7	2	2	3.4	5	1.76	0.79	–1.40	2.8	–0.27	1.2
26–Jul	1404	4.5	4.10	3	6	4	2	6	4.2	4	1.76	0.79	–0.38	2.3	–0.07	1.0
29–Jul	1407	9.5	10.50	3	10	16	6	10	9.0	13	4.51	2.02	–0.25	2.9	–0.05	1.2
30–Jul	1408	8.5	7.90	9	7	8	8	13	9.0	6	3.40	1.52	0.33	1.8	0.06	0.8
31–Jul	1407	9.5	10.50	5	15	9	5	10	8.8	10	4.51	2.02	–0.35	2.2	–0.07	1.0
1–Aug	1408	8.5	7.90	8	8	8	10	10	8.8	2	3.40	1.52	0.20	0.6	0.04	0.3
2–Aug	1407	9.5	10.50	11	15	12	8	14	12.0	7	4.51	2.02	1.24	1.6	0.24	0.7
5–Aug	1404	4.5	4.10	3	2	5	4	4	3.6	3	1.76	0.79	–1.14	1.7	–0.22	0.7
6–Aug	1404	4.5	4.10	4	5	5	4	8	5.2	4	1.76	0.79	0.89	2.3	0.17	1.0
7–Aug	1404	4.5	4.10	2	3	1	4	1	2.2	3	1.76	0.79	–2.92	1.7	–0.56	0.7
$d_2 =$	2.326															
$d_3 =$	0.8641											**UCL**	3.0	4.9183	0.557	2.114
$A_2 =$	0.577											**Centerline**	0.0	2.326	0.0	1.0
$D_3 =$	0											**LCL**	–3.0	0.0	–0.557	0.0
$D_4 =$	2.114															

Moving Average and Moving Range (MAMR) Chart. The moving average and moving range (MAMR) chart may be suitable in the following situations:

- When data are collected periodically or when it takes time to produce a single item
- When it may be desirable to dampen the effects of overcontrol
- When it may be necessary to detect smaller shifts in the process than with a comparable Shewhart chart

Key Considerations When Using an MAMR Chart. The user of an MAMR chart must address the following:

- *Selection of a moving average length*. The overall sensitivity of the chart in detecting process shifts is affected by the selection of the moving average length. Generally, the longer the length, the less sensitive the chart is to detecting shifts. However, specific selection of the length should be made with consideration of the out-of-control detection rules being used. Wheeler (1995) presents and compares a series of power function curves (that is, probability of detecting a shift in the process average) for *n*-point moving averages to various "chart for individuals" detection rules. When the moving average length becomes a practical consideration, the reader is encouraged to consult a more rigorous source on this topic. Note: Minitab allows the user to set the size of the moving average length.
- *Selection of a method for estimating* σ. Wheeler (1995) suggests two methods:
  - Average moving range ($\bar{R}$)
  - Median moving range ($\tilde{R}$)

  Although the use of the average moving range is more popular, variability present in the data may suggest the use of the dispersion statistics. However, Wheeler (1995) computes control limits by using a variety of dispersion statistics (for example, range, median moving range, standard deviation) and concludes that "there is no practical difference between any of the sets of limits."

The following two considerations are particularly important when using moving average charts:

- *Rational subgrouping*. As with any control chart, consideration of rational subgrouping remains paramount. Example 27.9 assumes a rational subgroup of one with a moving average length of three. If statistical and technical considerations were appropriate for a rational subgroup of five, the average of each subgroup would constitute a point in the moving average of length five. Note: Minitab allows the user to set the subgroup size.

- *Interpretation of the charts*. By nature of their construction, points on moving average charts and moving range charts do not represent independent subgroups. Hence, these points are correlated. While single points exceeding the control limits may still be used as out of control, other tests such as zone run tests may lead to false conclusions. Some software packages such as Minitab recognize this and limit the out-of-control tests on the moving range charts to the following:
  - One point more than three sigma from the centerline
  - Nine points in a row on the same side of the centerline
  - Six points in a row, all increasing or all decreasing
  - Fourteen points in a row, alternating up and down

Constructing an MAMR Chart. Use the following steps when constructing an MAMR chart:

1. Collect a set of data: $x_1, x_2, \ldots, x_n$.
2. Specify the length of the moving average (for example, 2, 3, . . .).
3. Calculate the moving averages.
4. Calculate the moving ranges.
5. Calculate the centerline of the moving average chart where the centerline equals $\bar{X}_{MA}$.
6. Calculate the centerline of the moving range chart where the centerline equals $\overline{mR}$.
7. Compute the LCL and the UCL for the moving range chart using $\text{LCL} = D_3\overline{mR}$ and $\text{UCL} = D_4\overline{mR}$, where D_3 and D_4 are constants found in Appendix 4.
8. Compute the LCL and the UCL for the moving average chart, using $\bar{X}_{MA} \pm A_2\overline{mR}$, where A_2 is a constant found in Appendix 4.
9. Plot the data, centerline, LCL, and UCL for both the moving average and moving range charts.
10. Interpret the results and take action accordingly.

EXAMPLE 27.9

The data shown in Table 27.17 are taken from Griffith (1996). Develop an MAMR chart with a moving average length of three.

Using Excel and the procedure given above, the results are shown in Table 27.18. Since a moving average length of three was specified in the problem statement, Table 27.18 shows two fewer plot points. The completed charts are given in Figures 27.15 and 27.16.

Continued

Table 27.17 Data for Example 27.9—MAMR chart.

Column A	Column B
Number	X_i
1	8.0
2	8.5
3	7.4
4	10.5
5	9.3
6	11.1
7	10.4
8	10.4
9	9.0
10	10.0
11	11.7
12	10.3
13	16.2
14	11.6
15	11.5
16	11.0
17	12.0
18	11.0
19	10.2
20	10.1
21	10.5
22	10.3
23	11.5
24	11.1
	Average

Let's first analyze Figure 27.15. We will note that this chart does not demonstrate statistical control. However, we will also note that the mean and the control limits do not agree with those presented in Table 27.18. This is because this chart plotted all 24 points and used the corresponding data in the calculations for the moving average and control limits, whereas this was not the case in Table 27.18. Contrast the moving average chart in Figure 27.15 to the corresponding moving range chart in Figure 27.16. Both charts are moving average charts of length three, but both calculated their averages and control

Continued

Table 27.18 Calculations for Example 27.9—MAMR chart.

Column A	Column B	Column C	Column D	Column E	Column F	Column G	Column H	Column I	Column J	Column K	Column L	Column M
Number	X_i	Moving average	Moving range	A_2	D_3	D_4	Average moving average	Average moving range	LCL_{MA}	UCL_{MA}	LCL_{MR}	UCL_{MR}
1	8.0											
2	8.5											
3	7.4	7.97	1.10	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
4	10.5	8.80	3.10	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
5	9.3	9.07	3.10	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
6	11.1	10.30	1.80	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
7	10.4	10.27	1.80	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
8	10.4	10.63	0.70	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
9	9.0	9.93	1.40	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
10	10.0	9.80	1.40	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
11	11.7	10.23	2.70	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
12	10.3	10.67	1.70	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
13	16.2	12.73	5.90	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
14	11.6	12.70	5.90	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
15	11.5	13.10	4.70	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
16	11.0	11.37	0.60	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
17	12.0	11.50	1.00	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
18	11.0	11.33	1.00	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
19	10.2	11.07	1.80	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
20	10.1	10.43	0.90	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
21	10.5	10.27	0.40	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
22	10.3	10.30	0.40	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
23	11.5	10.77	1.20	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
24	11.1	10.97	1.20	1.023	0.000	2.574	10.65	1.99	8.61	12.68	0.00	5.12
	Average	**10.65**	**1.99**									

Continued

Continued

limits differently. This is something readers must be aware of when using statistical software packages over which they have little control.

The moving range chart illustrated in Figure 27.16 depicts an out-of-control condition. However, only a minor computational difference between Minitab and Excel for the UCL is apparent.

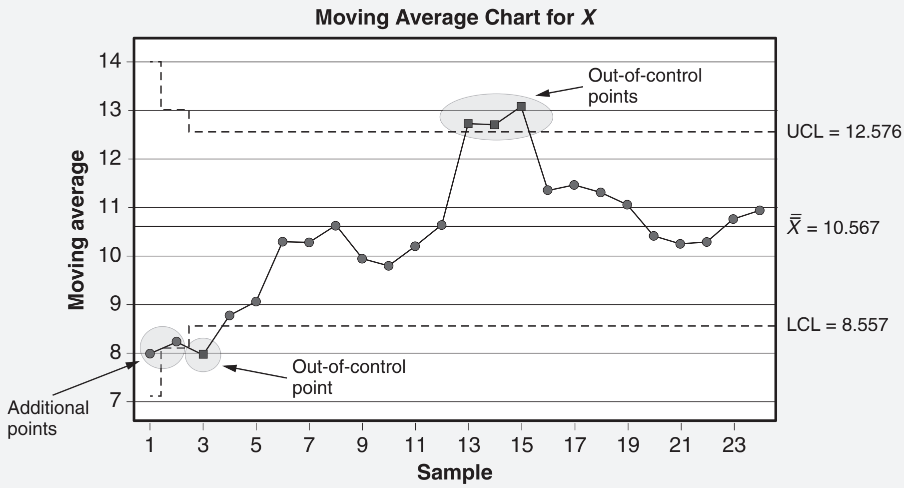

Figure 27.15 Moving average chart of length three for Example 27.9.

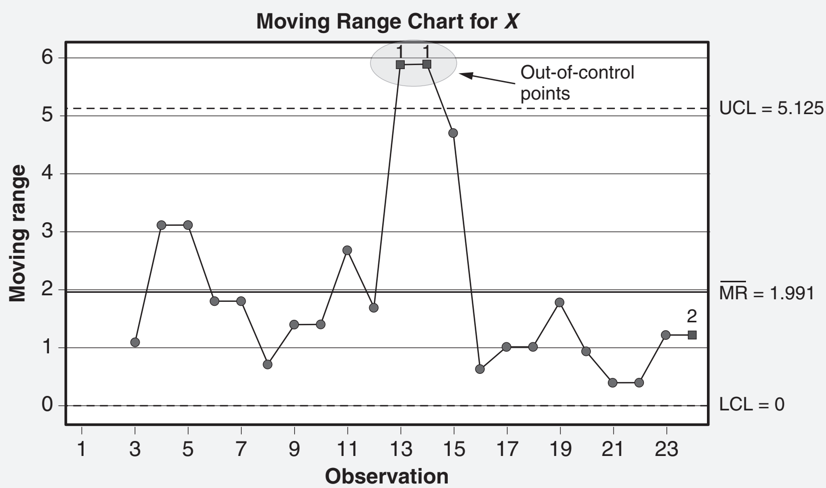

Figure 27.16 Moving range chart of length three for Example 27.9.

CONTROL CHART ANALYSIS

> Interpret control charts and distinguish between common and special causes using rules for determining statistical control. (Analyze)
>
> **Body of Knowledge VIII.A.5**

Rules for Determining Statistical Control

A control chart that has not triggered any out-of-control condition is considered stable and predictable, and operating in a state of statistical control. The variation depicted on the chart is due to common cause variation.

Points falling outside the limits or that meet any of the out-of-control conditions described below are attributed to special cause variation. Such points, regardless of whether they constitute "good" or "bad" occurrences, should be investigated immediately while the cause-and-effect relationships and individual memories are fresh, and access to documentation for process changes is readily available.

As the time between the out-of-control event and the beginning of the investigation increases, the likelihood of determining root causes diminishes greatly. Hence, the motto "time is of the essence" is most appropriate.

Finding root cause for out-of-control conditions may be frustrating and time-consuming, but the results are worthwhile. Ideally, root causes for good out-of-control conditions are incorporated into the process, while root causes for bad out-of-control conditions are removed.

Now a word of caution: adjusting a process when it is not warranted by out-of-control conditions constitutes process tampering. This usually results in destabilizing a process, causing it to spiral out of control.

Recall that when variable charts are being used, the chart used to measure process variation (for example, $\overline{R}$, $\overline{s}$, and $\overline{mR}$) should be reviewed first. Out-of-control conditions on this chart constitute changes in within-subgroup variation. Remember, from rational subgrouping, we would like the variation of subgroups on this chart to be as small as possible because this measure of dispersion is used to compute the control limits on the corresponding process average chart (for example, $\overline{X}$ and X). The tighter the control limits on the process average chart, the easier it is to detect subgroup-to-subgroup variation.

Commonly used rules or tests have been devised to detect out-of-control conditions. The specific rules vary. Conceptually, however, they are similar. The eight rules used by the software package Minitab are given in Table 27.19.

Generally, all of these rules apply to variables control charts, while a subset of them applies to attributes and short-term control charts. If you are using Minitab to develop control charts, only those that apply to the specific control charts will be available for selection. Figures 27.17 through 27.24 illustrate the out-of-control conditions for rules 1–8, respectively, in Table 27.19. The out-of-control condition

Table 27.19 Interpreting control chart out-of-control conditions used by Minitab.

		Chart type							
		Variables			Attributes				Short-run
Out-of-control condition	**Possible reasons**	$\bar{X}$ and R	$\bar{X}$ and s	XmR	p	np	c	u	MAMR
One point more than 3σ from the centerline (either side)	*Freaks*: error in plotting, calculation error, breakdown of facilities, extraneous causes	✔	✔	✔	✔	✔	✔	✔	✔
Nine points in a row on the same side of the centerline (Note: References to 6–9 points in a row on the same side of the centerline appear in the literature.)	A shift in the process mean has likely occurred	✔	✔	✔	✔	✔	✔	✔	✔
Six points in a row, all increasing or all decreasing	*Trend*: tool wear, skill improvement, deteriorating maintenance	✔	✔	✔	✔	✔	✔	✔	✔
Fourteen points in a row, alternating up and down	Alternating cause systems are present, such as two suppliers, and so on. Unlike stratification, the subgroups are homogeneous	✔	✔	✔	✔	✔	✔	✔	✔
Two out of three points more than 2σ from the centerline (same side)	Early warning of a potential process shift	✔	✔	✔					
Four out of five points more than 1σ from the centerline (same side)	Early warning of a potential process shift	✔	✔	✔					
Fifteen points in a row within 1σ of the centerline (either side)	*Stratification*: two or more different cause systems are present in every subgroup	✔	✔	✔					
Eight points in a row more than 1σ from the centerline (either side)	*Mixture*: two different operators being used, two machines, and so on.	✔	✔	✔					

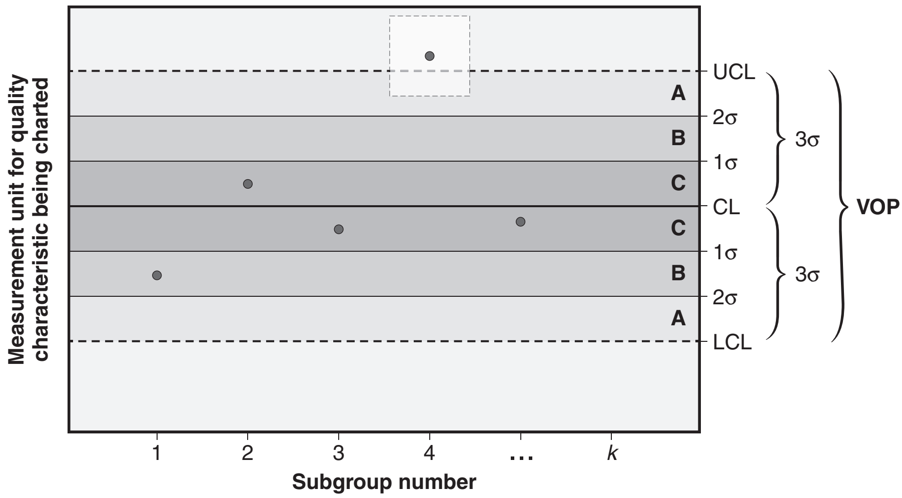

Figure 27.17 Example of rule 1 out-of-control condition.

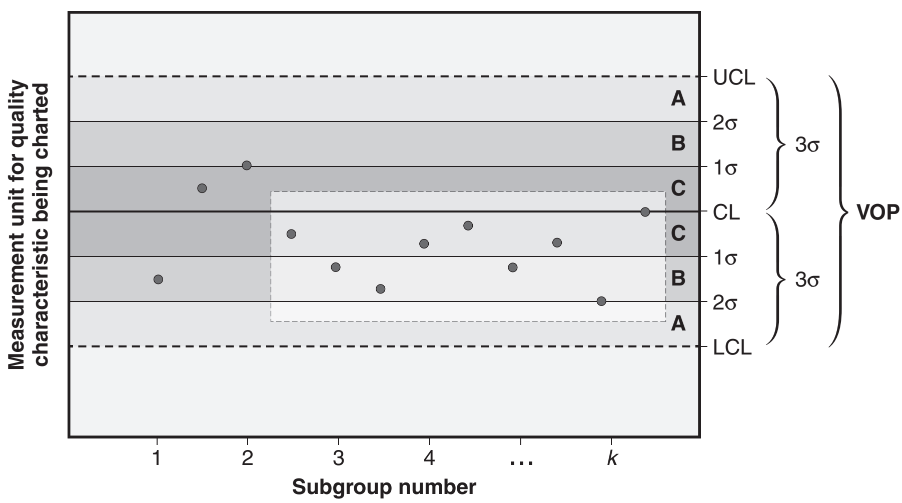

Figure 27.18 Example of rule 2 out-of-control condition.

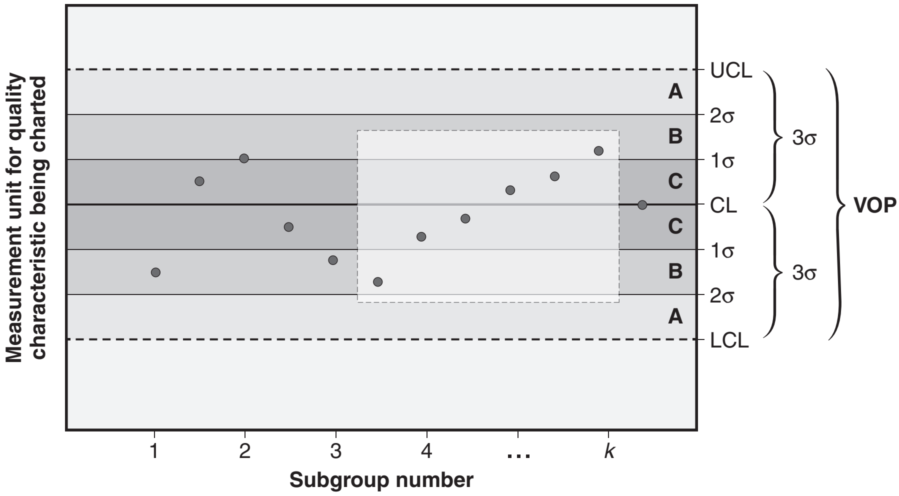

Figure 27.19 Example of rule 3 out-of-control condition.

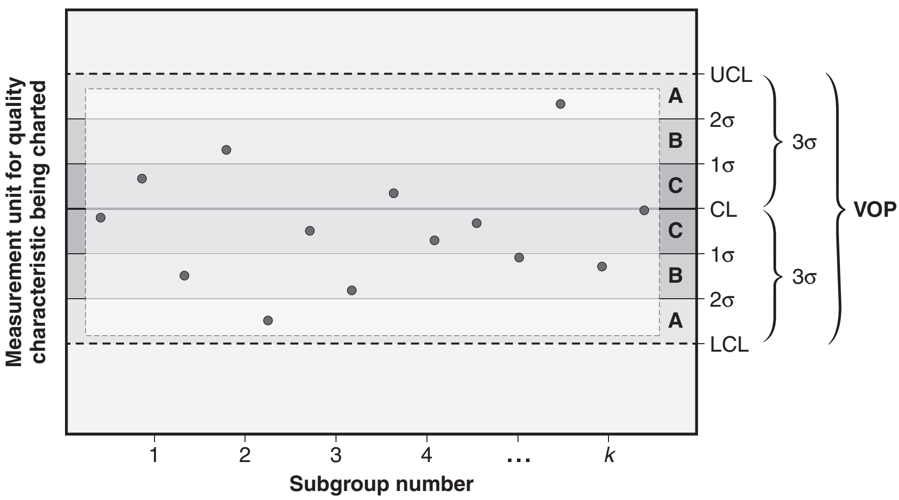

Figure 27.20 Example of rule 4 out-of-control condition.

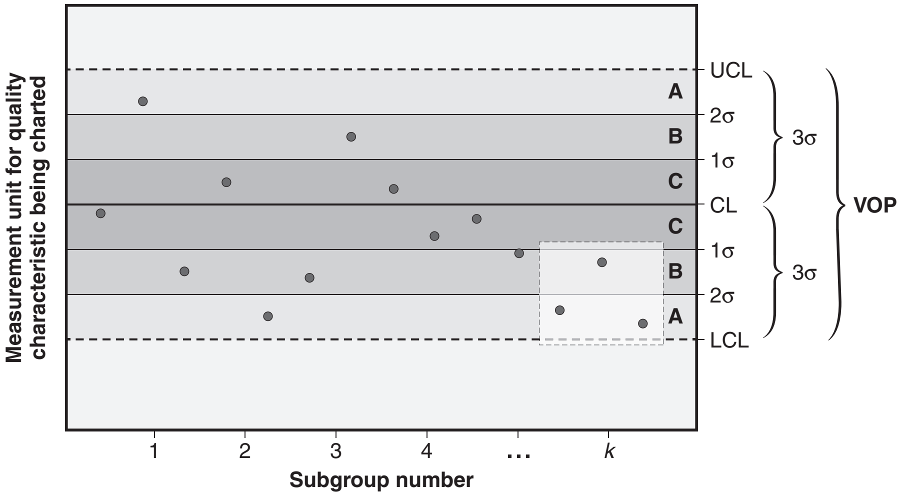

Figure 27.21 Example of rule 5 out-of-control condition.

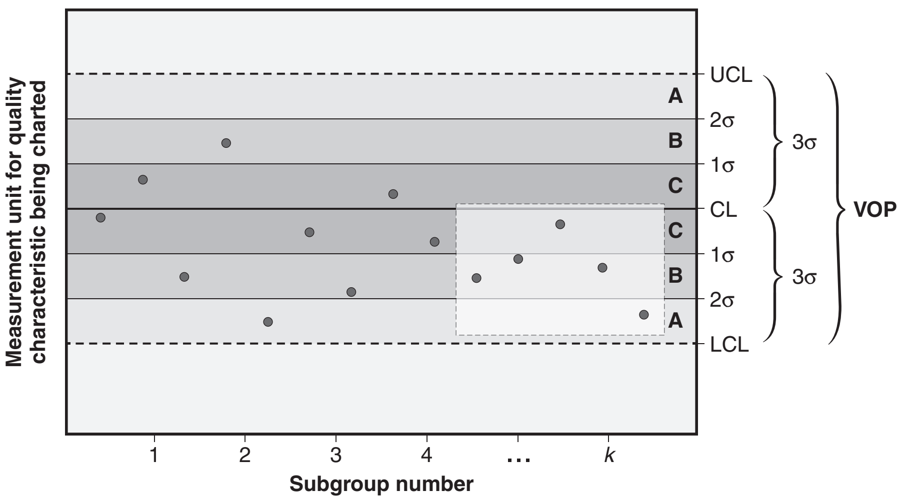

Figure 27.22 Example of rule 6 out-of-control condition.

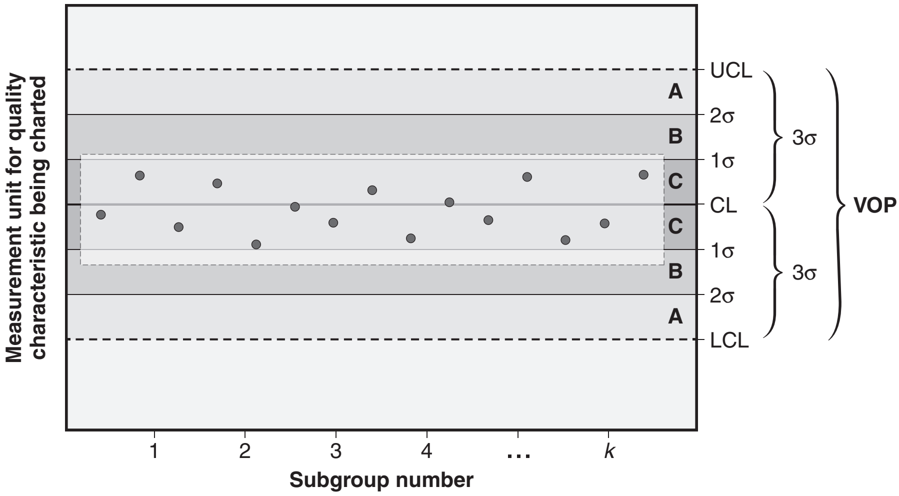

Figure 27.23 Example of rule 7 out-of-control condition.

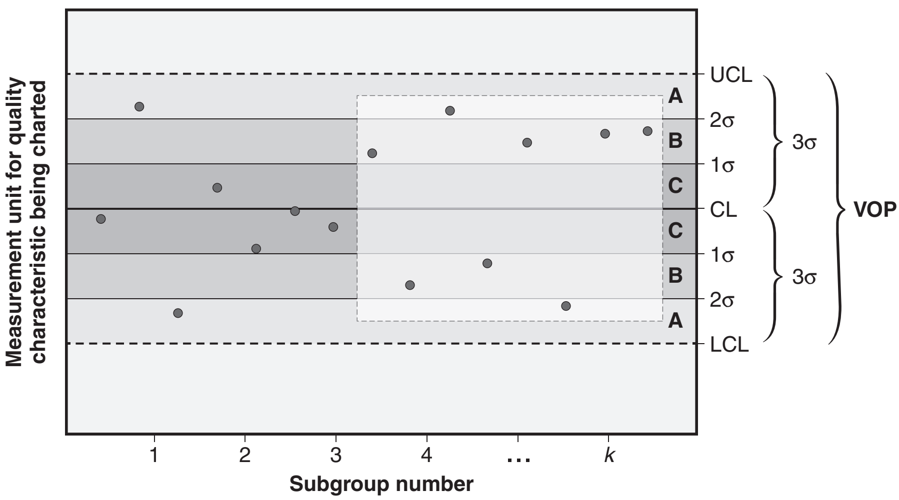

Figure 27.24 Example of rule 8 out-of-control condition.

found in each figure is enclosed in a dotted-lined box. Note that the subgroups along the horizontal axis are for illustrative purposes only and are not intended to align to each plot point.

The probabilities associated with the above out-of-control conditions occurring are both similar in value and relatively small. With the exception of points exceeding the control limits, most out-of-control conditions are subtle and would likely go unnoticed without the aid of a computerized control chart.

Interpreting Control Charts

As stated previously, specific rules have been devised to determine when out of-control conditions occur. These rules have been designed to permit us to detect early changes in a process, thus allowing us to take systematic action to discover the root cause of the variation or permit adjustment or other actions on the process before serious damage occurs.

Control charts operating in control are stable and predictable, and operating under the influence of common cause variation. Those operating with out-of-control conditions present are under the influence of special cause variation.

A control chart is relatively easy to develop and use, and it can be a highly effective statistical tool when selected properly and used correctly. Its selection and use alone, however, is not sufficient. When so indicated, control charts must be acted on in a timely manner so that root causes may be identified and removed from the process.

One last thing: when in doubt, avoid tampering with the process.

Chapter 28

Other Controls

Once a process has been improved, the next, and sometimes most difficult, challenge is to hold the gains. One reason this is so difficult is that it often requires people to change the way they do things. Two tools for holding the gains are total productive maintenance and visual controls.

TOTAL PRODUCTIVE MAINTENANCE (TPM)

> Define the elements of TPM and describe how it can be used to consistently control the improved process. (Understand)
>
> **Body of Knowledge VIII.B.1**

Total productive maintenance (TPM) is a methodology pioneered by Nippondenso (a member of the Toyota group) that works to ensure that every machine in a production process is always able to perform its required tasks so that production is never interrupted. TPM maximizes equipment effectiveness by using a preventive maintenance program throughout the life of the equipment.

In any situation where mechanical devices are used, the working state of those devices has an impact on the control of the process. If equipment deteriorates even subtly, the process output may be affected, often in unsuspected ways. For lean systems to work, all equipment must be ready to quickly respond to customer needs. This requires a system that foresees maintenance needs and takes appropriate action. A TPM system uses historical data, manufacturer's recommendations, reports by alert operators, diagnostic tests, and other techniques to schedule maintenance activity so machine downtime can be minimized. TPM goes beyond keeping everything running. An effective TPM system includes continuous improvement initiatives as it seeks more effective and efficient ways to predict and diagnose maintenance-related problems.

An important precept of TPM is the move toward ownership of the equipment and work area by the people that work there. Machine operators become responsible for lubrication, routine adjustments, and repair of machines. This means entrusting and enabling workers to perform functions previously carried out by maintenance personnel. A significant advantage to TPM is the tendency for production personnel to suggest improvements in workstation layout and operation.

EXAMPLE 28.1

A punch press operation experienced a stoppage to repair damage caused by the failure of a die spring. The repair entailed reworking the die, a process that sometimes takes two weeks. When the die was disassembled, it was determined that there were actually two failed die springs. It was speculated that a single spring failed and the machine continued to operate until the extra stress put on the other springs caused a second spring to fail. The operators who were monitoring a particular dimension with an SPC chart noticed a slight jump in the range chart for that dimension an hour or so before the second die spring broke. The next time that shift occurred on the range chart, they called maintenance and said, "We think we have a broken die spring." The maintenance person said, "No you don't; the die would have crashed." After some discussion, the maintenance person agreed to disassemble the die. A single broken spring was found inside. It was replaced, and the machine was back online in less than an hour.

EXAMPLE 28.2

The resolver on a pick-and-place robot was malfunctioning, causing sporadic part defects due to incorrect positioning of parts in a machine. A TPM team investigated the problem and determined that the malfunction was due to a buildup of contaminant located in and around the machine. As a result of this key learning, the team modified the robot maintenance schedule without creating any additional downtime.

If these suggestions are implemented, workers begin to recognize the connection between well-organized and maintained equipment and quality products. The performance of maintenance tasks by machine operators is called *autonomous maintenance*. Beginning steps for implementing autonomous maintenance include:

1. Be alert to paycheck/incentive issues. Little enthusiasm will be generated by a change that decreases take-home pay. For example, the worker may be paid at a lower rate while working on her machine than she would if maintenance personnel were working on it.

2. Make sure training, tools, and supplies are accessible. Don't ask people to perform tasks they aren't equipped to do.

3. Be alert for multi-shift tensions as in "I left this machine immaculate, now look at it . . . ." Cross-shift meetings to discuss standards will usually help. Some processes benefit from a log listing data from each shift.

4. Begin small, perhaps with daily cleaning and lubrication.

5. Determine if there are daily measurements that would help predict machine malfunction. Could a tachometer, ammeter, or recording thermostat be installed on equipment? If so, how is the output to be interpreted?

6. Be alert for, and celebrate, innovations made by workers.
7. Help maintenance department personnel understand that they are reserved for bigger things. Any slack time generated can be filled with training.

As TPM begins to work, it is possible to expand its scope to include:

- *Condition based maintenance.* The scheduling of maintenance tasks based on operating conditions.

  Examples:

  – Spot welding tips must be dressed or changed more frequently when used on galvanized steel.

  – Filters must be changed more frequently in summer.

  – If the patient weighs more than 300 lbs, the amps to the motor must be monitored.

- *Periodic/preventive maintenance*. The scheduling of maintenance tasks based on expected lifetime. Especially useful when multiple items are more efficiently maintained rather than one at a time, for example, light bulbs.

- *Reliability-based maintenance.* The scheduling of maintenance based on reliability data, especially if a failure is very costly.

  Examples:

  – A bearing that usually fails between 1800 and 2600 hours, causing a major shutdown, might be replaced on the weekend after it has had 1500 hours of service.

  – A timing belt that has a high probability of failure at 70,000 miles might be scheduled for replacement at 60,000 miles.

- *Maintenance analytics*. The scheduling of maintenance tasks based on data from sensing devices. For example, vibration sensors and noise sensors can sometimes detect a signature signal that precedes failure. An increase in amperage drawn could imply a failing motor or failure in its load.

In addition to using TPM to minimize equipment downtime, it can also be used to maintain process control by recognizing it as one component of total process variation. When we discuss measurement systems analysis, we look at total process variation as

$$\sigma^2_{\text{Measurement}} + \sigma^2_{\text{Process}} = \sigma^2_{\text{Total}}$$

However, we can further decompose $\sigma^2_{\text{Process}}$ as follows:

$$\sigma^2_{\text{Process}}$$

Having isolated the variation component due to TPM, it can be monitored and improved using a variety of tools and techniques discussed in other chapters.

VISUAL CONTROLS

> Define the elements of visual controls (e.g. pictures of correct procedures, color-coded components, indicator lights), and describe how they can help control the improved process. (Understand)
>
> **Body of Knowledge VIII.B.2**

Visual controls, sometimes known as the *visual factory*, are approaches and techniques that permit one to visually determine the status of a system, factory, or process at a glance and prevent or minimize process variation. To some degree, it can be viewed as a minor form of mistake-proofing. Some examples of visual controls include:

- Signage
- Product line identification
- Color-coded items such as parts, bins, racks, documentation, walkways, and so forth
- Schedule boards
- Conspicuous posting of performance indicators

Visual controls work to provide a constant focus and attention on the process. This level of attention can help stabilize variation at the improved level of a process. Examples of visual controls are shown in Figures 28.1 through 28.8.

Target	**358**
Completed	**316**
% completed	**88%**
% time used	**82%**

Figure 28.1 Scoreboard showing real-time performance data.

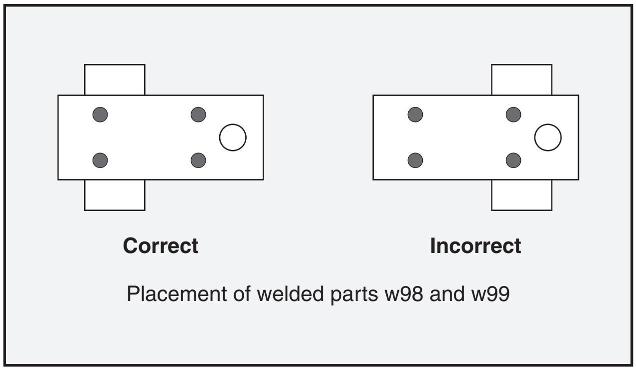

Figure 28.2 Visual work instructions.

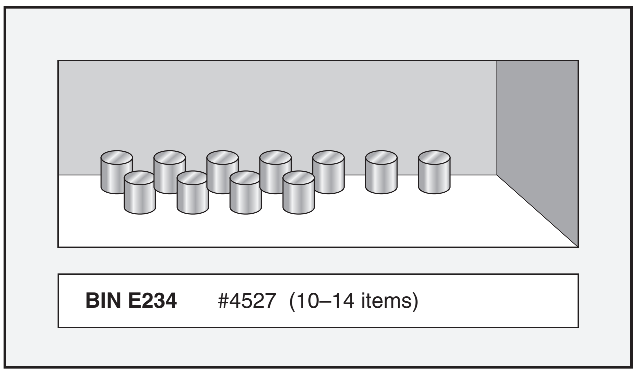

Figure 28.3 Visual stock control.

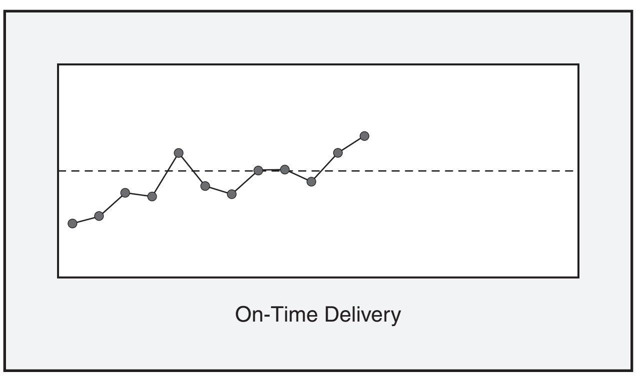

Figure 28.4 Performance data.

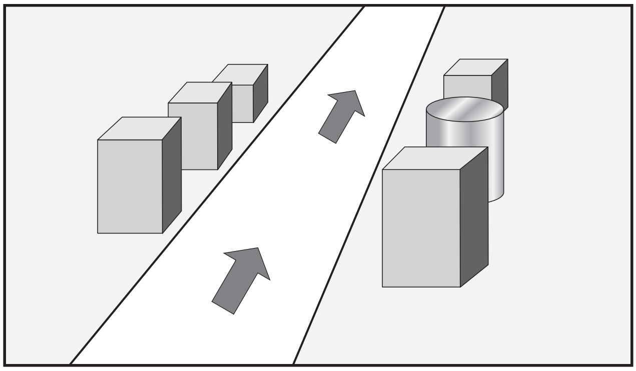

Figure 28.5 Floor marking.

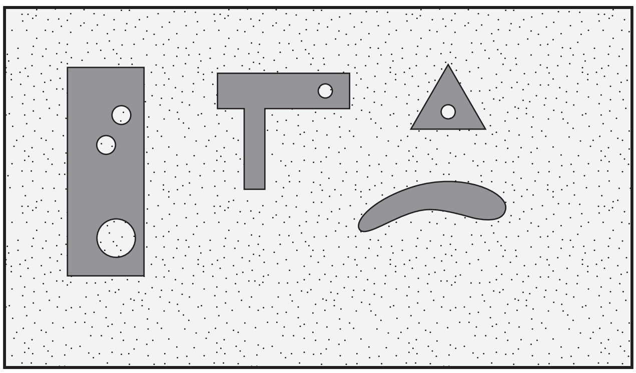

Figure 28.6 Tool board with templates.

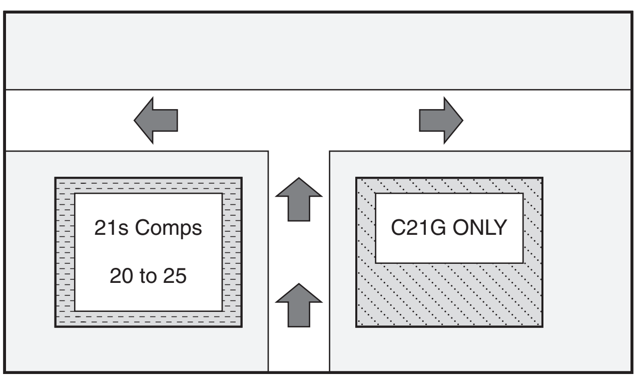

Figure 28.7 Floor-marked storage areas.

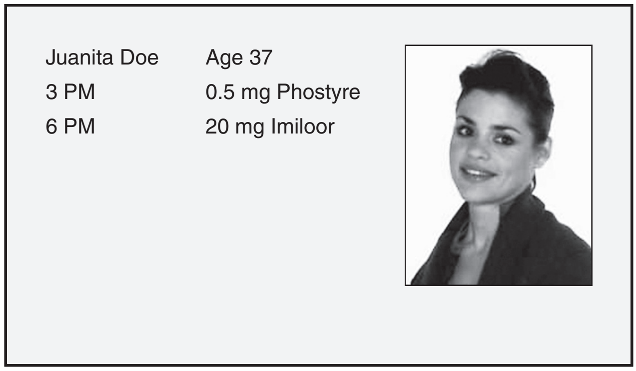

Figure 28.8 Patient's photo on medication chart.

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Chapter 29

Maintain Controls

MEASUREMENT SYSTEM REANALYSIS

> Review and evaluate measurement system capability as process capability improves, and ensure that measurement capability is sufficient for its intended use. (Evaluate)
>
> **Body of Knowledge VIII.C.1**

There are many reasons why a reanalysis of the measurement system is required; examples include change in process, change in customer specifications, new measuring devices, equipment calibration issues, and so on. Of course, most of these reasons are fairly obvious. One reason that is not so obvious is the impact that continuous process improvement has on the measurement system. The total observed variation in a process is made up of two components: process variation and measurement variation. We usually write this as

$$\sigma^2_{\text{Total observed}} = \sigma^2_{\text{Process}} + \sigma^2_{\text{Measurement system}}$$

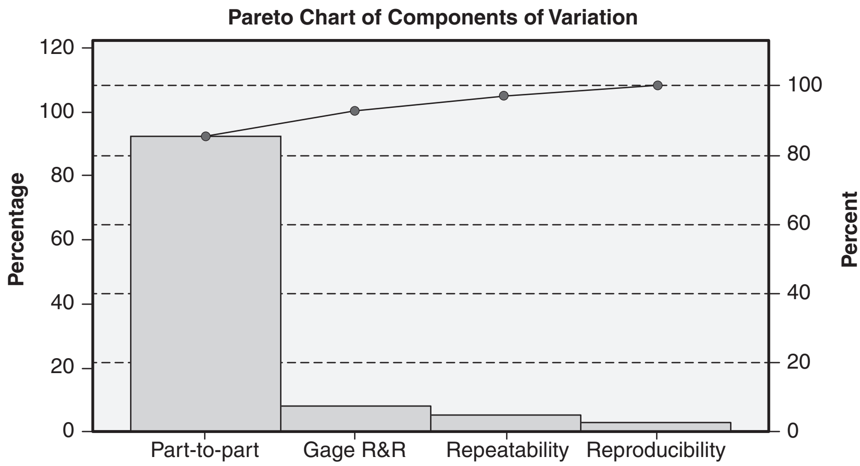
Hopefully, process improvement efforts result in reduced process variation. This, in turn, results in reduced total observed variation. However, the measurement system variation remains unchanged. As a result, measurement system variation increases as a percentage of total observed variation. Depending on the percentage value, the current measurement system variation could become unacceptably large.

Measurement variation can be calculated by performing a gage repeatability and reproducibility (GR&R) analysis, as described in Chapter 16. The following guidelines are typically used to determine the acceptability of a measurement system:

- < 10% error—The measurement system is considered acceptable.
- 10% to 30% error—The measurement system may be considered acceptable on the basis of importance of application, cost of gage, cost of repairs, and so on. This is a gray area and depends on the situation at hand.

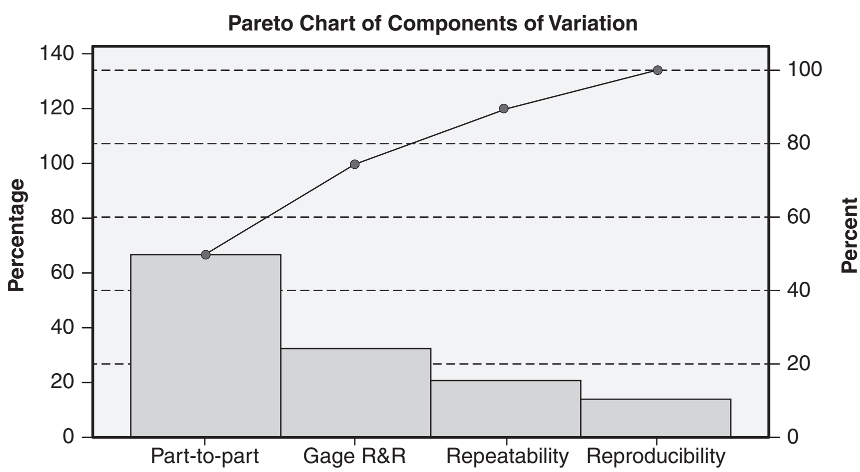
- > 30% error—The measurement system is considered unacceptable, and improvement is necessary.

Figure 29.1 depicts an acceptable measurement system using the above percentage guidelines, whereas Figure 29.2 depicts one that is unacceptable.



Components of variation percentage	92	8	5	3
Percent	85.2	7.4	4.6	2.8
Cum %	85.2	92.6	97.2	100.0

Figure 29.1 Example of an acceptable level of variation due to the measurement system.



Components of variation percentage	66	34	21	13
Percent	49.3	25.4	15.7	9.7
Cum %	49.3	74.6	90.3	100.0

Figure 29.2 Example of an unacceptable level of variation due to the measurement system.

EXAMPLE 29.1

An initial GR&R study determined the measurement system percentage to be 12%. In addition, the 10:1 ratio rule was met. Thus, the measurement system was judged acceptable. However, after a series of improvement projects, the part-to-part process variation was reduced by 75%. A reanalysis of the measurement system now showed that the GR&R increased to 35% and that the 10:1 ratio rule was still satisfied since the measuring devices and specification limits did not change. As a result, the organization must determine whether the measurement system is acceptable or needs improvement on the basis of intended use, customer specification, and cost of implementation.

Other rules or standards exist for determining the acceptability of measurement system variation:

- *10:1 ratio rule*. This rule states that the measurement system discrimination should be no greater than one-tenth of the smaller of the process variation or the specification tolerance.
- *4:1 ratio (25%) rule*. This rule states that measurement uncertainty should be no greater than 25% of the specification tolerance.

In Example 29.1, any future projects aimed at reducing process variation will most likely fail to produce the desired results. This is due to the large amount of measurement variation. The measurement system does not have adequate resolution within the process control limits and will not be able to detect the changes in the process variation. Future activities in this area will need to address the adequacy of the measurement system before any project to further reduce process variation is undertaken.

CONTROL PLAN

Develop a control plan to maintain the improved process performance, enable continuous improvement, and transfer responsibility from the project team to the process owner. (Apply)

Body of Knowledge VIII.C.2

A *control plan* is a living document that identifies critical input and output variables and associated activities that must be performed to maintain control of the variation of processes, products, and services in order to minimize deviation from their preferred values.

Control plans are formulated during the control phase of DMAIC and are intended to ensure that process improvement gains are not lost over time. A control plan is needed because, despite our best quality improvement efforts, some failure modes may occur. The control plan is designed to:

- Monitor the process
- Detect and diagnose potential failures
- Define the appropriate reaction to each failure type

Each of these purposes is discussed in more detail in the following paragraphs.

Monitoring

The control plan should specify a sampling plan that experience has shown to be effective in detecting changes to the process. The plan should identify gages or fixtures to be used. If measurements are to be made, the plan should cross-reference a control plan for monitoring GR&R for the gage involved.

Detection and Diagnosis

The control plan should identify indicators of process change. These could be patterns on a control chart, unusual sounds, signals on a control panel, or some other criteria. To the extent possible, the plan should list possible causes.

Reaction

The control plan should list the steps to be taken when a process change has been detected. This serves as an aid to the responsible personnel during what is often a stressful time. The plan should be quite specific; for example, the reaction to bulb #18 going to red should be to throw switch #4 and change filter #3A, or when seven successive points plot above the mean, the parting tool should be examined. The reaction section should cover requirements for containment and inspection of products suspected of having defects. It should also discuss disposition of parts found to be defective. Some control plans prescribe a more intense sampling protocol after certain corrective actions have been taken. Examples of control plans featuring some of these requirements are shown in Figure 29.3 and Figure 29.4

Notice that the two plans in Figure 29.3 identify the individual responsible for taking the actions defined in the control plan, using the basic premise that things get done when accountability is assigned.

A control plan is defined as a living document. It is designed to be maintained. Common triggers for updating control charts include:

- Process changes
- Specification changes
- Measurement technology changes
- Organizational changes
- Personnel turnover
- Defined period reviews

Control Plan for Process XYZ					
Activity/ step	**Input**	**Specification characteristic to be controlled**	**Control method**	**Control description**	**Responsibility**
1	Copper cable	Cross-section diameter	Measure per SOP 481-4.7	$\bar{X}$ and R control chart Out-of-control condition: Do not complete operation, notify production engineer immediately	Operator
2	Resistor	Resistance	Measure per SOP 596-2.4	$\bar{X}$ and R control chart Out-of-control condition: Pull lot from production, notify supplier, draw new lot	Production engineer

Control Plan for Process XYZ						
Control factor	**Specification or tolerance**	**Measurement technique**	**Sample size**	**Sampling method/ frequency**	**Reaction plan**	**Responsibility**
Cable diameter	0.5 mm ± 0.05 mm	Caliper, measure per SOP	7	$\bar{X}$ and R control chart, random sampling with a probability of 0.25	Out-of-control condition: Do not complete operation, notify production engineer immediately	Operator
Cable length	12.0 mm ± 0.10 mm	GKRD-45 optical reader, measure per SOP	5	$\bar{X}$ and R control chart, random sampling with a probability of 0.10	Out-of-control condition: Pull lot from production, notify supplier, draw new lot	Production engineer

Figure 29.3 Example of two different formats for control plans.

To reiterate, control plans hold the gains achieved from process improvement activities. A weak or inadequate control plan is a key failure mode for DMAIC projects. To this point, organizations with existing process or quality audit functions may want to consider adding control plans to their list of audit items.

Process	Specification	Inspection equipment See QR18 for GRR	Sampling plan	Recording medium	Discrepancy	Response action
Face and plunge	Face FIM ≤ 0.0008	CMM (Pgm #442)	1 every 50 parts	CMM printout	FIM > 0.0008	Stop production. Quarantine last 50 parts for inspection on CMM. Segregate any that fail flatness specification. Remove facing insert and replace. Inspect next 5 parts on CMM to make certain FIM = 0.0008 specification is met. If not, call supervisor.
Face and plunge	Slot width 0.250 ± 0.004	Caliper #Q955	5 consecutive parts once every 100	$\bar{X}$ and R chart #LA61	Point above UCL or below LCL on $\bar{X}$ 8 points one side of mean on $\bar{X}$	Stop production. Quarantine last 100 parts for inspection. Segregate any that fail width specification. Remove and replace insert on plunge tool. Inspect next 5 parts. If mean falls outside control limits, call supervisor.
					Point above UCL on R 8 points above mean on R	Stop production, call quality engineer.
Face and plunge	Slot depth 0.15 ± 0.005	Caliper #Q955	5 consecutive parts once every 100	$\bar{X}$ and R chart #LA62	Point above UCL or below LCL on $\bar{X}$	Stop production. Quarantine last 100 parts for inspection. Segregate any that fail depth specification. Remove and replace insert on plunge tool. Inspect next 5 parts. If mean falls outside control limits, call supervisor.

Figure 29.4 Another control plan format.

Chapter 30

Sustain Improvements

LESSONS LEARNED

> Document the lessons learned from all phases of a project and identify how improvements can be replicated and applied to other processes in the organization. (Apply)
>
> **Body of Knowledge VIII.D.1**

Organizational Memory

Many organizations find that they solve the same problems more than once. This is often because of poor documentation, poor communication, internal employee churn, or external attrition. Moreover, the organization's memory may exist only in the minds of employees. When a problem recurs or occurs in a slightly different form, it is often solved again, particularly if the people who worked on it before are not involved. This problem is only exacerbated by organizational downsizing, whereby the bottom line may receive some short-term benefit, but the organization has been crippled due to a significant loss of its intellectual capital and organizational memory.

The recognition of the need to maintain an organization's memory has given rise to the concept of *knowledge management*. While the definition of knowledge management remains vague, it can be characterized by its requirements to identify an organization's knowledge base and intellectual capital, maintain this information, and distribute it, make it accessible, or otherwise communicate it to those in the organization who have the need to use it.

Current technologies allow organizations to maintain and distribute information rather easily. The difficulty lies in capturing it. With respect to Six Sigma, the opportunity to capture project information and related lessons learned can easily be embedded into the control phase of the methodology. Organizations can readily establish the requirement that no project can be closed out without first capturing relevant project information and lessons learned in some permanent repository. Once captured, project information can be searched and compared against future project opportunities such that decisions can be made to charter a project, replicate a past success, or even kick-start new or innovative thinking on an existing project.

The organization must recognize that projects generate knowledge and that knowledge represents a critical resource for future projects. One approach to documentation of knowledge generated by a project is to specify a standard format for each project's final report. A suggested final report form is shown in Figure 30.1. These final reports should then be put in a searchable format for use by future project teams.

Having this information in a searchable format also provides the opportunity for future cross-project study and analysis, which could have implications for training, resource management, and project selection.

Project ID ________ **Final Report** Date ________

Six Sigma Project

Problem statement: ______________________________________

____

Team members: ____

____

Department/division: ____________________________________

Process: ____

Key steps toward problem solution: _________________________

____

Analysis tools that were most useful: _______________________

____

____

____

Corrective action implemented: ____________________________

____

Updates made to SOP: ___________________________________

____

Updates made to work instructions: _________________________

____

Updates made to control plan: _____________________________

____

Figure 30.1 Possible format for final report.

DOCUMENTATION

> Develop or modify documents including standard operating procedures (SOPs), work instructions, and control plans to ensure that the improvements are sustained over time. (Apply)
>
> **Body of Knowledge VIII.D.2**

A Six Sigma project often requires changes in processes and the way they are controlled. If these changes are not reflected in the appropriate documents, they will likely not be sustained. When this happens, the gains achieved by the project are lost. This, unfortunately, is a fairly common occurrence. Therefore, the format for a Six Sigma final report should include these updates, as shown in Figure 30.1.

Many organizations have found that documented standard operating procedures (SOPs) and work instructions (known in the lean manufacturing literature as "standard work") help reduce process variation. The purpose of these documents is to make certain that the activity is performed the same way over time. This is especially important when multi-skilled, cross-trained personnel move into a variety of positions.

The development and updating of these documents must involve the people who perform the work. The documentation process is begun by listing the major steps. Succeeding iterations examine the steps from the previous one and break them into smaller sub-steps. This process can continue until further deconstruction is not useful.

Two main considerations include:

- *Documents must be kept current*. In an era of continuous improvement, processes may be continuously changing. As a result, it is possible that different documentation releases for the same documentation may be in use simultaneously. Therefore, it is vital that employees always have access to the appropriate documentation based on the effective date of the change. It is easy to see how the complexity of managing such a documentation system can grow exponentially. With today's web-based and other technologies, documentation configuration management systems can be designed and developed to ensure that the proper documentation is available for the right process on the right part at the right time.

- *Multiple documentation formats exist*. The right choice depends on how the documentation is to be used, by whom, and at what skill level. However, documentation best practices suggest developing documents that are color-coded, rely heavily on graphics, illustrations, and photographs, and are light on words. Photographs of acceptable and unacceptable product and procedures have been found to be very useful. Some organizations have had success with the use of video for depicting complex operations. As always, the level of detail provided in any set of documentation should reflect the skills and education levels of the personnel doing the actual work and the degree to which variation must be controlled. Robust processes usually require less-detailed documentation than non-robust processes.

The best and most up-to-date SOPs, work instructions, and control plans will not be impactful if they are not followed. A considerable training effort is often required, as discussed in the next section of this chapter.

TRAINING FOR PROCESS OWNERS AND STAFF

> Develop and implement training plans to ensure consistent execution of revised process methods and standards to maintain process improvements. (Apply)
>
> **Body of Knowledge VIII.D.3**

A critical element for ensuring that the gains of an improved process are maintained is the training of the personnel responsible for executing the process. Training is often overlooked or viewed as an unnecessary cost or imposition on the time of already overworked employees.

Training comes in two main forms:

- *Initial*. This type is used for existing employees who have been responsible for executing the old process and will continue to be responsible for executing the new process. It also addresses the needs of employees who are new to the process. Furthermore, it may be conducted off the job or on the job, or both. The correct combination depends on the skill levels required, process complexity, and experience levels, among other factors. Initial training serves to calibrate employees and minimize variation in how the process is performed. If possible, the "starting tomorrow we're doing it this way" approach should be avoided. Instead, the rationale and methods for the change should be explained, preferably by the people closest to the process.

- *Recurring*. This type is used to minimize deterioration in process performance over time. Deterioration usually occurs as employees become comfortable with the new process and fail to recognize minor changes or tweaks to the process either through carelessness, poor documentation, or perhaps even a system that permits obsolete documentation to remain in the process stream. In addition, on-the-job training is another factor that often adversely impacts process performance. In a manner similar to the children's game of "telephone" that permits the description, interpretation, and understanding of an initial whispered message to transform itself into something entirely different after it has passed through a series of children trying to communicate the original message, so too does on-the-job training. Specific facts, details, and nuances are frequently left out as they pass from worker to worker. Recurring training restores process execution to its original design. The frequency of recurring training should be determined on the basis of process metrics and employee performance. Such training may be offered at specified intervals or conducted as required to serve the needs of underperforming employees.

When developing process-related training plans, important considerations include:

- Providing employees with the minimum skills and information needed to perform the functions required by the position. This goes beyond "how to turn on the screw gun." It includes how to read and interpret documentation and safety precautions, how to supply required data, and other skills as needed.
- Providing employees with additional skills and information that will ensure a broader view of what the position accomplishes for the enterprise. If the position entails installing 200 bimetallic oven controls each day, the employee should know how a bimetallic control works, which metals can be used, and perhaps a little history regarding earlier models.
- Providing employees with cross-training for additional functions. This often involves training employees on the processes immediately before and after the process for which they are responsible.

Other, larger training plan considerations beyond the immediate process include:

- Providing employees with opportunities for further education outside the enterprise on topics not directly related to the current organization needs. This is done under the assumption that exposure to ideas outside the box is valuable.
- Providing incentives and requirements that motivate employees to continue to seek education and training opportunities.
- Providing employees with experiences that demonstrate the need the organization has for their ideas for continuous improvement.
- Providing employees with the opportunity to help formulate a customized annual training plan. Such a plan should be routinely maintained, be supported by readily accessible records, and monitor progress toward plan completion.

These considerations should help employees make positive contributions toward the ability of the organization to sustain process improvements over the long term.

ONGOING EVALUATION

> Identify and apply tools (e.g., control charts, control plans) for ongoing evaluation of the improved process, including monitoring leading indicators, lagging indicators, and additional opportunities for improvement. (Apply)
>
> **Body of Knowledge VIII.D.4**

Once a process has been improved, it must be monitored to ensure that the gains are maintained and to determine when additional improvements are required. The control plan, as discussed in Chapter 29, is the key tool for maintaining the gains achieved by the project. It typically specifies the use of these tools:

- *Control charts*. These are used to monitor the stability of the process, determine when a special cause is present, and decide when to take appropriate action. The choice of a particular control chart depends on the nature of the process. When out-of-control conditions occur, action is required to restore stability. Control charts represent the voice of the process.
- *Process capability studies*. These studies provide the opportunity to understand how the voice of the process (that is, control limits) compares with the voice of the customer (that is, specifications) and help determine whether the process average must be shifted or re-centered or the variation reduced.
- *Process metrics*. This includes a wide variety of in-process and end-of-process metrics that measure the overall efficiency and effectiveness of the process. Examples include cycle times, takt time, takt rate, work-in-progress, backlog, defect rates, rework rates, and scrap rates.
- GR&R studies that monitor the measurement systems used by the process.

There is a need for a periodic formal review of completed projects to determine whether further improvements are appropriate. Criteria that are useful in making this determination include the following:

- Leading indicators
  - Have the customer requirements changed?

- Will they pay a higher price for an improved product?
- Are there other customers that could be served if we had an improved product?
- Are there changes in the regulatory environment that require process improvements?
- Are there changes in the marketplace that require process improvements?
- Has the market for the product reached maturity, indicating level or tapering demand?

- Lagging indicators
  - Have low-priced competitors taken market share?
  - Do data collected via the control plan indicate that process improvement would be easily achievable?
  - Has demand for the product decreased?
- Other indicators
  - Is the product still in alignment with enterprise goals?
  - Are new technologies/materials/processes available that might simplify improvement?

Taken collectively, these tools and methods help us gauge the overall health of a process and provide triggers for reevaluating a process for further improvement.

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Part IX

Design for Six Sigma (DFSS) Framework and Methodologies

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Chapter 31

Common DFSS Methodologies

> Identify and describe DMADV (define, measure, analyze, design, and validate) and DMADOV (define, measure, analyze, design, optimize, and validate). (Understand)
>
> **Body of Knowledge IX.A**

In addition to the two design for Six Sigma (DFSS) methodologies required by the Body of Knowledge, this chapter will address two other DFSS practices. Together, the four methodologies identified constitute the most common DFSS approaches used by practitioners today.

While DMAIC may be traditionally viewed as the foundation for Lean Six Sigma, its application is primarily limited to improving existing processes; it does little to address the design of new products or processes.

Fortunately, several additional structured methodologies exist. Several of the common methodologies are depicted in Figure 31.1. Each of these methodologies has its usefulness as long as the nuances of each are fully understood.

Table 31.1 has been created to provide a loose comparison between DMAIC and the DFSS methodologies. These comparisons are not exact, but do provide

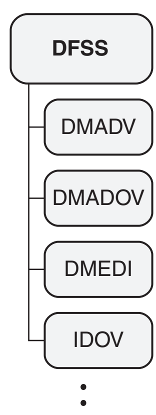

Figure 31.1 The DFSS family of methodologies.

Table 31.1 Comparing DMAIC and DFSS methodologies.

<table>
<tr><th>DFSS</th><th colspan="4"></th></tr>
<tr><th>DMAIC</th><th>DMADV</th><th>DMADOV</th><th>DMEDI</th><th>IDOV</th></tr>
<tr><td>Define</td><td>Define</td><td>Define</td><td>Define</td><td rowspan="2">Identify</td></tr>
<tr><td>Measure</td><td>Measure</td><td>Measure</td><td>Measure</td></tr>
<tr><td>Analyze</td><td>Analyze</td><td>Analyze</td><td>Explore*</td><td rowspan="2">Design</td></tr>
<tr><td rowspan="3">Improve</td><td rowspan="2">Design</td><td>Design</td><td rowspan="2">Develop</td></tr>
<tr><td>Optimize</td><td>Optimize</td></tr>
<tr><td>Verify**</td><td>Verify**</td><td>Implement</td><td>Validate***</td></tr>
<tr><td>Control</td><td></td><td></td><td></td><td></td></tr>
</table>

* Loosely aligned

** Often confused with "Validate"

*** Often confused with "Verify"

Table 31.2 Common tools used in DFSS.

Analytic hierarchical process (AHP)	Portfolio architecting
Axiomatic design	Process simulation
Critical parameter management	Pugh matrix
Design for X (DFX)	Quality function deployment (QFD)
DMADOV	Robust product design
DMADV	Statistical tolerancing
FMEA	Systematic design
IDOV	Tolerance design
Porter's five forces analysis	TRIZ

a sense of how the methodologies align. This table may be beneficial for readers who may be required to use multiple methodologies or find themselves in need of selecting one suitable for a particular project.

Table 31.2 identifies some of the commonly used tools in DFSS, including the DFSS methodologies shown in Figure 31.1. However, this table also lists other tools that are useful regardless of the DFSS methodology chosen. Examples include axiomatic design and TRIZ.

DMADV

DMADV is a well-recognized DFSS methodology and an acronym for *define-measure-analyze-design-verify*. Note that the ASQ Black Belt Body of Knowledge replaces "verify" with "validate." The difference is likely because, although different, "verify" and "validate" are often used synonymously.

Let's look at each letter of DMADV:

- *Define.* This phase is the same as in DMAIC and was addressed in Part IV.
- *Measure.* This phase is the same as in DMAIC and was addressed in Part V.
- *Analyze.* This phase is the same as in DMAIC and was addressed in Part VI.
- *Design*. Quite simply, this means carrying out the process of designing a product or process. Many organizations have well-established policies and procedures for their respective design processes. One valuable Lean Six Sigma technique that supports the design process is quality function deployment (QFD), the details of which were discussed in Chapter 10. Additional tools useful in this phase include pilot runs, simulations, prototypes, and models.
- *Verify.* This phase is directed at ensuring that the design output meets the design requirements and specifications, and is performed on the final product or process.

Verification means the design meets customer requirements and ensures that it yields the correct product or process. By contrast, *validation* speaks to the effectiveness of the design process itself and is intended to ensure that it is capable of meeting the requirements of the final product or process.

Both verification and validation are necessary functions in any design process. As such, this suggests that DMADV might be more appropriately named DMADVV.

DMADOV

The basic difference between DMADV and DMADOV is that "O" for "optimize" has been added. This may sound like a trivial observation, but many organizations' design processes do not include this refinement action. They are intended solely to produce a minimally workable product or process. DMADOV forces attention on the need to optimize the design.

Additional tools useful in this phase include design of experiments, *response surface methodology* (RSM), and *evolutionary operations* (EVOP). These methods help the design team establish and refine design parameters.

Note: The following two DFSS methodologies are included because, together with the above practices, they constitute the most common approaches used by practitioners today. They have been included for completeness purposes. However, they will not be included on the ASQ certification examination.

DMEDI

The *define–measure–explore–develop–implement* (DMEDI) methodology is appropriate where the limitations of DMAIC have been reached and customer requirements or expectations have not been met, or a quantum leap in performance is required.

Unlike DMAIC, DMEDI is not suitable for kaizen events, and it is generally more resource intensive, with projects taking considerably longer.

Let's explore each phase of DMEDI:

- *Define.* This phase is the same as in DMAIC.
- *Measure.* Unlike *measure* in DMAIC, this phase must emphasize the voice of the customer and the gathering of critical customer requirements since no baseline data are available. The quality function deployment (QFD) tool is valuable during this phase.
- *Explore.* This phase focuses on creating a high-level conceptual design of a new process that meets the critical customer requirements identified in the measure phase. Ideally, multiple designs are developed that provide options to be considered.
- *Develop.* The options developed in the explore phase are considered, and the "optimal" one is chosen based on its ability to meet customer requirements. Detailed designs are developed.
- *Implement.* During this phase, the new process may be simulated to verify its ability to meet customer requirements. In addition, a pilot is conducted, controls are established and put in place, and the new process is transferred back to the process owner.

IDOV

Unlike most of our other DFSS methodologies, IDOV contains only four phases and stands for *identify–design–optimize–validate*. Woodford ("Design for Six Sigma—IDOV Methodology") states that the IDOV phases parallel the original MAIC phases.

However, a review of his descriptions does not confirm this. Table 31.1 has been modified to conform to the descriptions of each phase he provides.

Let's explore each phase of IDOV:

- *Identify.* This phase links the design to the voice of the customer via the customer, product, and technical requirements, establishes the business case, identifies CTQ variables, and conducts a competitive analysis. Quality function deployment will be a particularly useful tool during this phase.

- *Design.* This phase focuses on identifying functional requirements, deploying CTQs, developing multiple design concepts, and selecting the best-fit concept.
- *Optimize.* This phase focuses on developing a detailed design for the best-fit design chosen in the previous phase, predicting design performance, and optimizing the design. Additional work in this phase includes error-proofing, statistical tolerancing, prototype development, and optimization of cost.
- *Validate (verify).* During this phase, the prototype is tested and validated, and reliability, risks, and performance are assessed. Note: The acronym for IDOV is not consistently defined. The "V" is used about equally as often for *validate* as it is for *verify.* Also, in many cases, when it is used as *verify* in the acronym, it is explicitly described as *validate* when the details of each letter are broken out.

Munro (2008) describes each phase of IDOV in more detail on pages 42–43 of *The Certified Six Sigma Green Belt Handbook,* First Edition.

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Chapter 32

Design for *X* (DFX)

> Describe design constraints, including design for cost, design for manufacturability (producibility), design for test, and design for maintainability. (Understand)
>
> **Body of Knowledge IX.B**

Traditionally, the design process was conducted almost in isolation, without regard to the downstream functions that might be impacted by that design. Once the design was complete, it was handed off to manufacturing, production, or the assembly function, whose job was to translate the design from concept into reality. Unfortunately, many design functions failed to address the capability of these downstream organizations. This frequently resulted in escalating production costs and numerous engineering change orders that created a significant divergence between the "as designed" and the "as built" configurations.

This scenario led to the development of the design-by-team approach known as *concurrent engineering* or *simultaneous engineering*. For the first time, stakeholders in the design function had a voice in how a product or process was designed. With fully cross-functional design teams in place, organizations could begin improving the efficiency and effectiveness of designs to benefit both the organization and the customer.

The concept behind *design for X* (DFX), where *X* is a variable such as cost, manufacturability, producibility, testability, maintainability, and so forth, is simply that "X" becomes the focus of or constraint to the design process. Let's look at some of the more familiar "X's."

Design to Cost

The need for design to cost (DTC) (also known as *design for cost*) usually arises when an organization establishes a fixed design budget in an effort to become more fiscally responsible, or perhaps a major customer has dictated what it is willing to spend. As a result, the design focus is shifted from "what works" to "what works within budget." The disciplines of value engineering and value analysis are keystones under a DTC scenario. This section will consider cost analysis from two viewpoints: *purchase cost* and *life cycle costs*.

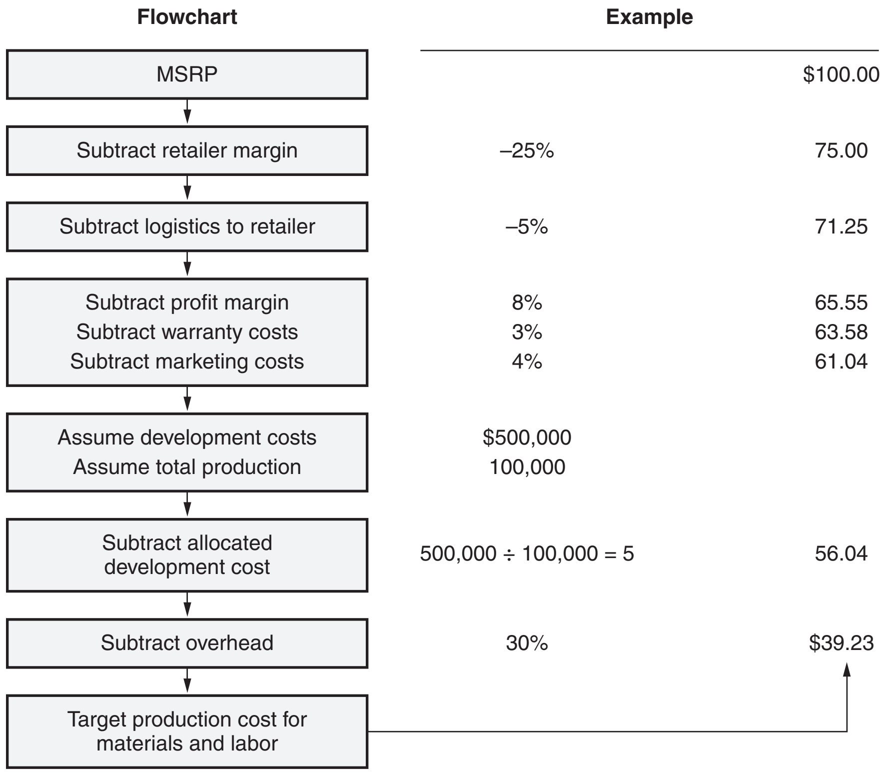

Figure 32.1 Flowchart for arriving at target cost.

Purchase Cost Issues. In the first months after a new product is introduced, its unique features or technology may determine customers' purchase decision. However, as a product matures, competing products tend to match technology and features. At that point cost becomes more of a determining factor in customers' decisions. Common cost-reduction efforts focus on labor and processes but often ignore cost-reduction potential related to the design cycle. Once a product and the processes that will be used to produce it have been determined, the cost has been largely determined. Trying to make major cost reductions after that point is generally futile. Therefore, significant cost reductions must start in the design stage. Figure 32.1 shows a procedure for arriving at a target production cost. Design for cost considerations during the design stage include:

1. Knowledge of the market and what customers want
2. Recognition of the trade-off between features and costs to help reduce creeping elegance
3. Exploration of design alternatives that lower costs

Life Cycle Cost (LCC) Issues. If the product will be used over a considerable period of time, the customer may look beyond the purchase price. Additional costs to be considered include:

		Option A	Option B
Acquisition costs (Purchase + Logistics)	$	1,000,000	1,000,000
Other capital costs (structural and utility changes)		+20,000	+15,000
Training costs (10 years / Operators + Maintenance)		+20,000	+10,000
Support costs (utilities, spare parts, software updates)		+5,000	+4,000
Disposal cost		0	0
Reclamation value		–100,000	–80,000
Total cost	**$**	**945,000**	**949,000**

Figure 32.2 Life cycle cost (LCC) example.

- Shipping
- Other capital costs (structural changes, wiring, plumbing, and so on)
- Training costs
- Support costs (utilities, software, and so on)
- Disposal costs
- Reclamation value

Figure 32.2 illustrates how these considerations help when comparing two competing products such as two machine tools, two diagnostic imaging machines or two point-of-sale systems. In the example, time value of money is not considered.

Design for Manufacturing/Design for Producibility/Design for Assembly

Although some degree of difference exists between the three terms, they are grouped together because the underlying DFX issues remain the same. Under these scenarios, the design team must recognize that limitations might exist with respect to manufacturing, producing, or assembling a product. Common examples include designs calling for:

- Equipment not currently available.
- Methods unfamiliar to the manufacturing or assembly workforce. An example might be requiring an automated method of soldering when the assembly workforce is skilled only in manual soldering.
- Expensive fixtures.
- Specialized tooling.
- Workplace redesign.

- Operators/assembly workers to use asymmetrical motions.
- Limited accessibility.
- Obsolete or hard-to-find parts.
- Use of exotic material when unnecessary.
- Special manufacturing, operator, or assembly worker skills.
- Tolerances beyond that which can be achieved given the available equipment or processes.

The best way to make certain that adequate provision for manufacturability and assembly has been made is to have manufacturing/assembly expertise present on the design team. Some guidelines to improve design for manufacturing (DFM)/ design for assembly (DFA) (also called design for manufacture and assembly [DFMA]) include:

1. Minimize part count. If two parts are in contact, ask these questions:
   - Do these parts need to move relative to each other?
   - Do these parts need to be separated for access or maintenance?
   - Do these parts need to be made of different material?
   - If none of the answers is yes, could one part suffice?
2. Avoid right/left parts. Design parts so they can be used on either side.
3. Use off-the-shelf parts, standard components, and purchased parts.
4. Use modular design where possible. If there is a family of similar products, consider using a common design for parts where possible.

Design for Test

The consideration given during the design stage to improve the testing of a product is called *design for test* (DFT) or *design for testability*.

Testing may be an integral part of ensuring quality, or it may be demanded by the customer. In such cases designs must accommodate an assemble-test-assemble type of production assembly process rather than rely entirely on functional tests of the finished product. This "no fault forward" approach can greatly decrease detection/repair of faults in final inspection. Such considerations are common with complex electromechanical equipment, including avionics and mass-market consumer electronics products. Historically, much of the DFT literature has referred to the testing of printed circuit boards. The goal is to design a board so that every function can be tested. It is important to verify that each circuit board can handle unusual input conditions without failing. To pose a simple example, how does the arithmetic section handle division by zero? It sometimes requires extra circuitry to allow input of the scenario being tested. So, the specifications for a board include the types of tests that the board needs to pass. The design team is responsible for designing the board to facilitate those tests. Software design teams

EXAMPLE 31.1

Suppose a truck manufacturer wishes to test the braking system on each truck before releasing it for shipment. The trucks have a redundant braking system with dual master cylinders. If the system controlled by one master cylinder fails, the other will stop the truck. When testing the braking system, however, the driver has no way of determining whether the redundancy functions correctly. To thoroughly test this feature, the braking system could be designed with additional equipment that permits disabling each master cylinder in turn (but not both at the same time!) so that the redundant function can be verified. The manufacturer might recommend that vehicle users perform this check as part of a regular preventive maintenance schedule.

have followed this example and designed software that permits testing for specified conditions. In some cases this requires additional code to facilitate the testing protocol.

Other branches of manufacturing could emulate the electronics industry. Why not design a port on a valve that would permit easy pressurization for leak tests? Why not include a simple thermocouple that would facilitate testing for heat transfer? The extra cost could be more than offset by the savings in the testing function, especially in situations where 100% testing is required.

Design for Maintainability

Customer loyalty depends on long-term satisfaction. For many products, this means that the ability to perform routine maintenance must be considered during the design process. Cars that require a floor jack to remove spark plugs or a contortionist to replace a shock absorber do not inspire customer confidence in other aspects of the product. Industrial products that require minimal downtime for repair can be of paramount interest, particularly when the customer is concerned with managing a product's or process's overall availability and life cycle costs.

Some guidelines to improve maintainability are:

1. Use standard parts and fasteners.
2. Don't require special tools to perform maintenance.
3. Design self-diagnosis and/or automated testing.
4. Routine maintenance should be at easily accessible points.
5. Minimize time to repair.
6. Design safety into the product and modules rather than depending on warning labels.
7. Permit visual inspection.

Accessibility and safety for all maintenance procedures helps reduce costs and increases availability for the user. Some firms have successfully employed 3D virtual reality systems to test the degree of difficulty for various maintenance operations. This permits design modifications prior to the construction of the first model.

Customers of complex systems may list *maintainability metrics* as part of the specifications for a product. For example, a specification might state, "Hydraulics system must be replaceable in 20 minutes."

Design for Robustness

In this scenario, designs must accommodate increased variation in the inputs while still producing the required output within defined specifications. For example, the impedance of a supplier's resistors is known to follow a normal distribution with a rather large variance. The design should be capable of accepting parts with such variation and still produce a tight output distribution.

In addition to being robust to variation in the inputs, a design needs to function when the operating environment varies. For example, some washing machines will be installed in garages, others in boiler rooms. Some products must function in salt air. Some operators will use liquid detergent, others will use powder. The design team needs to establish criteria in a host of areas as it moves through the design cycle. Product designs may need to consider variation in conditions of storage, transportation, and installation as well as operating conditions.

The concept of robustness is discussed in detail in the next chapter, but basically, design for robustness assumes uncertainty in the inputs, environment, operating conditions, and other variables. A robust design is able to function when these uncertainties stay within a stated range. For example, many automobile engines can function when the alcohol content of the fuel varies between 0% and 15%. The alcohol content for flex-fuel vehicles can range from 0% to 85%. These vehicles are robust to a wider range of variation in the alcohol content. If a vehicle engine functions in an ambient temperature ranging from –30 °F to 120 °F, that vehicle is robust to ambient temperature variation in that range. If the vehicle door latch functions with a slight nudge and also with a strong slam, the latch is robust to a range in operator temperament. As discussed in the next chapter, it is necessary to define the word "function" in each situation.

Chapter 33

Robust Designs

> Describe the elements of robust product design, tolerance design, and statistical tolerancing. (Apply)
>
> **Body of Knowledge IX.C**

Functional Requirements

The concepts of functional requirements are perhaps best understood with an example. A producer of black rubber provides door and window seals for the automotive industry. For many years the automobile designers have specified the hardness, toughness, and elasticity of the material, the geometry of the cross section, and other specifications. As might be expected, the rubber company has struggled to meet the specifications of its customers.

More recently, the designers have been specifying such things as "will not leak under a 20 mph wind and two inches of rain per hour for two hours." Such a statement is known as a *functional requirement*. It defines how the product is to perform and under what conditions.

Many companies have decided that their suppliers probably know more about the products they supply than they do and are probably in a better position to design a product that can be manufactured efficiently.

The use of functional requirements is sometimes referred to as "black box" engineering because the customer says, in essence, "I am not going to specify the design details; here's how I want it to function." Of course, the customer typically puts some constraints on the design, perhaps providing the space available, the relationship to other components, and so on. When specifications include functional and nonfunctional requirements, it is referred to as a "gray box" design.

Noise Strategies

Noise factors were introduced in Chapter 24. Figure 33.1 illustrates a response curve with three acceptable values for the input variable P. Regardless of the value of P chosen, a certain amount of variation or noise can be expected. This noise in the input variable is represented by the normal curves sketched on the horizontal axis. Figure 33.2 illustrates the impact on the output variable Q when $P = P_1$.

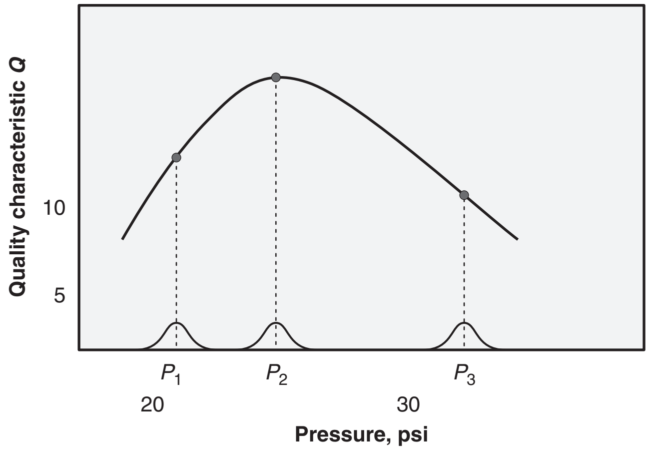

Figure 33.1 Nonlinear response curve with input noise.

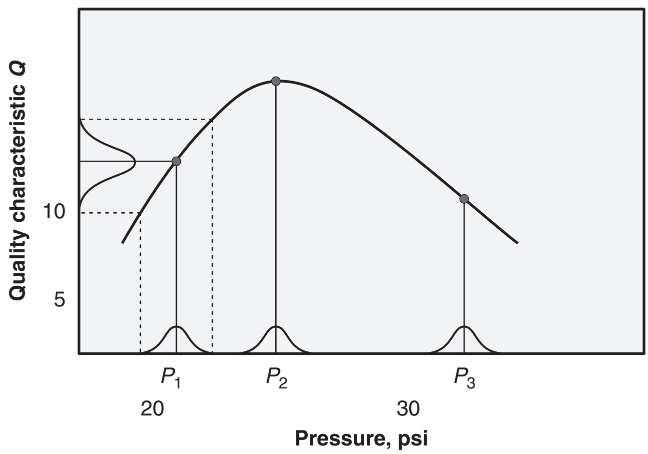

Figure 33.2 Nonlinear response curve showing the impact on Q of input noise at P_1.

Figure 33.3 shows the impact of noise at each of the three values of P. It is clear that the amount of variation in Q depends on the location of P. In fact, the location of P that minimizes "transmitted noise" is the area where the response curve is flattest—P_2 in this example.

The noise strategy, then, is to conduct a designed experiment or set of experiments, as shown in Chapter 28, and when nonlinear response curves are found, locate the flattest part of the curve, as there will be a point along this section that will minimize transmitted noise.

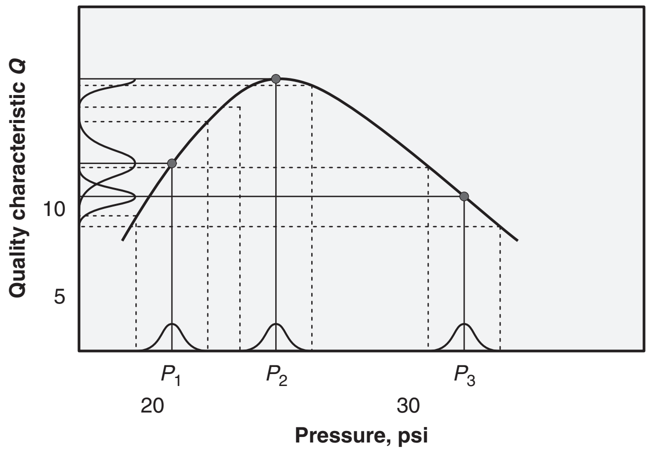

Figure 33.3 Nonlinear response curve showing the impact on Q of input noise at P_1, P_2, and P_3.

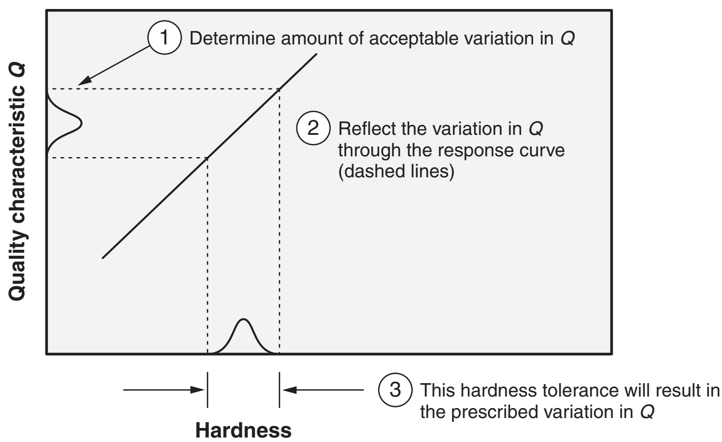

Figure 33.4 Using a response curve to determine tolerance.

Tolerance Design

Tolerance design uses the concept of transmitted noise to help determine product tolerance. For example, suppose a fabricator who uses steel to produce parts finds that the hardness of the steel impacts a quality characteristic of a formed part. As a result, a hardness tolerance is applied to the steel. What should the tolerance be? The amount of acceptable variation in the quality characteristic due to hardness is reflected back through the response curve to determine the amount of acceptable variation in hardness (see Figure 33.4). If a designed experiment determines that it is possible to operate at a relatively flat spot on the response curve relating hardness to the quality characteristic, the hardness tolerance can be looser than it would otherwise have to be. Therefore, tolerance design is considered a cost-saving technique since loosening tolerances on a specification generally reduces costs.

Statistical Tolerancing

In this section, we will address two methods for determining tolerance:

- Conventional tolerances
- Statistical tolerances

Conventional Tolerances. The conventional or traditional way to determine tolerance involves a situation in which several parts are stacked together, as shown in Figure 33.5.

Statistical Tolerances. If the processes producing the lengths of parts A, B, and C are capable and generate normal distributions, the tolerances on these parts are directly related to the standard deviations. But, unlike tolerances, standard deviations are not additive. Therefore, we must add the variances as follows:

$$\sigma^2_{\text{Stack}} = \sigma^2_A + \sigma^2_B + \sigma^2_C$$

$$\sigma_{\text{Stack}} = \sqrt{\sigma^2_A + \sigma^2_B + \sigma^2_C}$$

If we recognize that the worst-case scenario that occurs with the tolerance is exactly 6 standard deviations in width (that is, ± 3 deviations), we can now write

$$\text{Nominal}_{\text{Total}} = \text{Nominal}_A + \text{Nominal}_B + \text{Nominal}_C$$

and

$$\frac{T}{3} = \sqrt{\left(\frac{T_A}{3}\right)^2 + \left(\frac{T_B}{3}\right)^2 + \left(\frac{T_C}{3}\right)^2} \quad \text{or}$$

$$T_{\text{Stack}} = \sqrt{T^2_A + T^2_B + T^2_C}$$

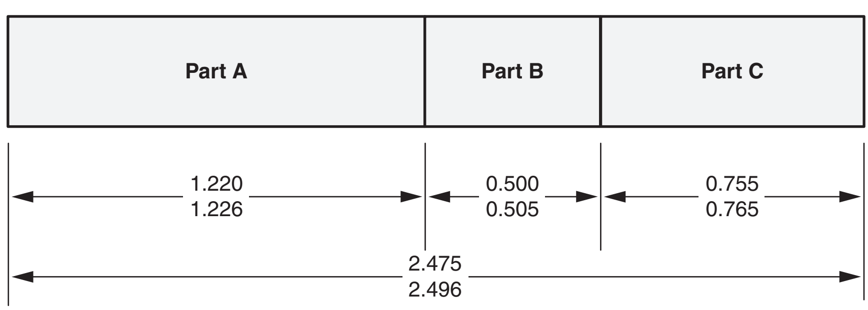

Figure 33.5 Conventional stack tolerance dimensioning.

EXAMPLE 33.1

The method to compute the overall tolerance on the length is as follows:

1. Use the sum of the minimum lengths of the three components as the lower tolerance limit:

$$\text{Lower tolerance limit} = 1.220 + 0.500 + 0.755 = 2.475$$

2. Use the sum of the maximum lengths of the three components as the upper tolerance limit:

$$\text{Upper tolerance limit} = 1.226 + 0.505 + 0.765 = 2.496$$

Therefore, the tolerance on the overall length is the sum of the tolerances on the individual parts:

$$\text{Tolerance on the stack} = 0.006 + 0.005 + 0.010 + 0.021 = \pm 0.0105$$

Applying the appropriate numerical values, we have

$$\text{Nominal}_{\text{Total}} = 1.223 + 0.5025 + 0.76 = 2.4855$$

and

$$T_{\text{Stack}} = \sqrt{(0.003)^2 + (0.0025)^2 + (0.005)^2} = \sqrt{0.00004025} \approx \pm 0.0063$$

Given this, we might want to inquire how the statistical tolerances might change in the previous equation such that we still achieve the tolerance of the stack obtained using the conventional approach. If we assume, for the moment, that each of the statistical tolerances is equal, we obtain the equation

$$T_{\text{Stack}} = \sqrt{3T_{\text{Part}}^2}$$

Solving for T_{Part}, we get

$$T_{\text{Part}} = \frac{T_{\text{Stack}}}{\sqrt{3}}$$

Substituting the numbers, we get

$$T_{\text{Part}} = \frac{0.0105}{\sqrt{3}} \approx 0.0061$$

Note that the tolerance of the part ($T_{\text{Part}} \approx 0.0061$) is greater than any of the part tolerances (that is, 0.003, 0.0025, and 0.005) using the conventional approach.

Thus, by using a statistical tolerance approach, the tolerance of the individual parts can be increased, giving the designer greater latitude.

Tolerance and Process Capability

If a process capability study has been completed, the process mean and standard deviation can be used to calculate tolerances. The formula used depends on the number of standard deviations to be included in the tolerance interval. The traditional number is 6 (that is, ±3).

EXAMPLE 33.2

A painting process produces coatings with a thickness of 0.0005 and a standard deviation of 0.00002. What should the tolerance limits be for this process?

$$0.0005 \pm 3 \times 0.00002$$

$$0.0005 \pm 0.00006$$

$$(0.00044, 0.00056)$$

Tolerance Intervals

See Chapter 21, section VI.B.4.

The Taguchi Loss Function

Conventionally, quality costs have been based on conformance to specifications, using scrap and rework costs. Therefore, tolerances become go/no-go limits, where any part meeting specifications is as good as any other part that does. Dr. Genichi Taguchi argued that parts near the nominal dimension have more value than others that are in specification but farther away from the nominal value. Thus, the conventional way to interpret specification or tolerances assumes that all parts that meet the spec are good and all that do not are bad. In fact, all that meet the spec are usually considered equally good and all that do not are considered equally bad. This interpretation is shown by the loss curve in Figure 33.6, where LSL and USL refer to the lower and upper spec limits. This loss curve, shown as a heavy dashed line, is a step function with the vertical "riser" located at the spec limits and at a height equal to the scrap or rework cost. This is sometimes referred to as *goal post analysis* because a football kicked between goal posts is worth the same number of points, regardless of how close it is to the posts.

Taguchi maintained that any deviation from nominal makes the product less valuable. This loss increases as the dimension gets farther from the nominal or "target" value. The loss that is incurred for dimensions that are in spec may not occur immediately because they will be sold to the customer at full price. Taguchi believed that the product would assemble and function in a less than perfect way, however, and therefore would be less valuable. The loss may appear in the form of a less useful product and therefore a less satisfied customer. Over time, these losses represent decrease in sales, market share, and so on.

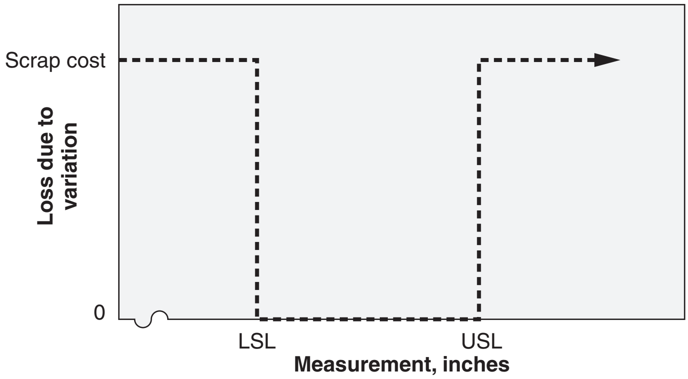

Figure 33.6 Conventional interpretation of loss.

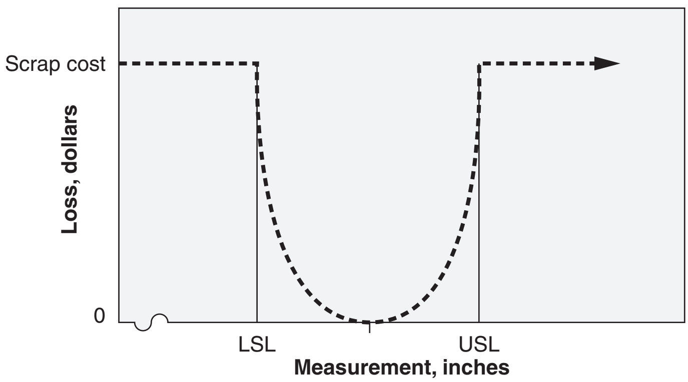

Figure 33.7 Taguchi loss function.

The exact shape of the loss function varies with product and within a product. Taguchi adapted a second-degree function as a generic approximation. This loss function is illustrated in Figure 33.7.

The heavy dashed curve is a second-degree curve, a parabola, of the form $y = (x - a)^2$. If a dimension is exactly at nominal, the part has zero loss. The more the dimension deviates from nominal, the greater the loss until it violates the spec limits, at which point the loss levels off at the scrap (or rework) cost.

The formula for the loss function is

$$L = \frac{c\sigma^2}{d^2}$$

where

L = Unit loss due to variation

EXAMPLE 33.3

A part with specs 1.455–1.465 that costs $6 to scrap is made on a process with $\sigma = 0.0015$. A proposal for process improvement will reduce σ to 0.001 but will cost $8000 per year. Is this expenditure justified if annual production is 100,000 units?

Solution: The current unit loss is

$$L = \frac{6(0.0015)^2}{0.005^2} \approx 0.54$$

This means that each piece produced has a $0.54 decrease in value due to variation.

The current annual loss is 100,000(0.54) ≈ $54,000.

If the proposed improvement is implemented, the unit loss will be

$$L = \frac{6(0.001)^2}{0.005^2} \approx 0.24$$

The annual loss with the proposed improvement is 100,000(0.24) ≈ $24,000.

This would seem to justify the $8000 investment.

Note: The accounting community does not necessarily agree with Taguchi's loss function, so some difficulty can be expected in justifying this analysis. If, however, a manager has a budget for continuous improvement, this type of analysis can help prioritize projects.

c = Scrap cost

σ = Standard deviation

$$d = \frac{\text{Tolerance}}{2}$$

The loss formula assumes that the process is centered, that is, the process mean coincides with the center of the specification. If this is not the case, the formula needs to be modified to the following general loss function.

Unit loss due to variation:

$$L = \frac{c\left[\sigma^2 + \left(\bar{\bar{x}} - t\right)^2\right]}{d^2}$$

where

$\bar{\bar{X}}$ = The process mean

t = Target, center of the specification

EXAMPLE 33.4

The following data have been collected for a product dimension:

It is normally distributed with $\bar{\bar{x}} = 23.7$, $\sigma = 2.11$, tolerance: 25 ± 5.

Scrap cost is \$3.55 per unit, annual production is 2000 units.

a. Using the standard capability analysis approach, calculate the annual scrap loss.

b. Using the general loss function, calculate annual scrap loss.

c. Two proposals for improving the process have been made:

   i. Build a new form tool to center the process, annual cost = \$650.

   ii. Purchase a new cam to reduce σ to 1.5, annual cost \$650

Which, if either, should be implemented?

Solution:

a. $$Z_U = \frac{USL - \bar{\bar{x}}}{\sigma} = \frac{30 - 23.7}{2.11} \approx 2.99$$

$$Z_L = \frac{\bar{\bar{x}} - LSL}{\sigma} = \frac{23.7 - 20}{2.11} \approx 1.75$$

From a *Z*-table, approximately 0.14% of production will exceed UCL and approximately 4.01% of production will be less than LCL.

Therefore, approximately $0.0014 \times 2000 \approx 2.8$ parts will exceed UCL and approximately $0.0401 \times 2000 \approx 80.2$ parts will be less than LCL.

So the annual scrap loss is $83 \times \$3.55 \approx \$294.65$.

b. Unit loss due to variation is

$$L = \frac{c\left[\sigma^2 + \left(\bar{\bar{x}} - t\right)^2\right]}{d^2} = \frac{3.55\left[2.11^2 + (23.7 - 25)^2\right]}{5^2} \approx 0.87$$

Annual loss due to variation $\approx 0.87 \times 2000 \approx \$1740$.

c.

   i. If the process is centered, unit loss will be

$$L = \frac{3.55\left[2.11^2 + (23.7 - 25)^2\right]}{5^2} \approx 0.63$$

Annual loss due to variation will be about $0.63 \times 2000 \approx \$1260$.

If this proposal is implemented, the annual loss due to variation will be reduced from \$1740 to \$1260. This proposal should not be implemented.

   ii. If $\sigma = 1.5$, unit loss is

$$L = \frac{3.55\left[1.5^2 + (23.7 - 25)^2\right]}{5^2} \approx 0.56$$

Annual loss due to variation is about $0.56 \times 2000 \approx \$1120$.

If his proposal is implemented, the annual loss due to variation will be reduced from \$1740 to \$1120. This proposal should not be implemented.

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Chapter 34

Retained Knowledge

This chapter is not part of the current Body of Knowledge and will not be on the examination. It was in the previous Body of Knowledge and we thought the material should not be lost. Therefore, we elected to retain the material and place it at the end of this edition.

Planning may be subdivided into three categories: strategic, tactical, and operational (that is, day-to-day) activities. This chapter deals with strategic and tactical planning. *Strategic planning* typically looks at a three- to five-year horizon, while *tactical planning* covers one to three years.

STRATEGIC

> Describe how Porter's five forces analysis, portfolio architecting, and other tools can be used in strategic design and planning. (Understand)

In general, a strategic planning process must do three things:

- Study the current state (that is, the business environment in which we operate)
- Envision the ideal future state (that is, where we'd like to be in three to five years)
- Plan the path (that is, how we'll get from here to there)

Strategic planning must be a priority of top management. For strategic planning to be successful, senior management must be actively engaged in the process. It is not something that can be delegated. Further, successful strategic planning must create a line of sight from strategic to tactical to operational activities. This does not mean that individuals executing activities at the lowest level of the organization must understand the details of the strategic plan. However, it does mean that their actions are traceable back to the strategic plan itself.

Various strategic planning models have been proposed, as described in the following subsections.

Porter's Five Forces

In 1979, Michael Porter listed five forces that affect the success of an enterprise:

- *The bargaining power of customers.* This force represents the ability and buying power of your customers to drive prices down.
- *The bargaining power of suppliers.* This force represents the ability and power of your suppliers to drive up prices of their products.
- *The threat of new entrants.* This force represents the ease with which new competitors can enter the market and drive prices down.
- *The threat of substitute products.* This force represents the extent to which different products or services can be substituted for your own.
- *The intensity of competitive rivalry.* This force represents the strength of the competition in your industry.

An organization should analyze each of these forces as it forms its strategic plan. It should also continually monitor each force because such forces can be highly dynamic, and changes in any one of these forces can require a change in strategy.

Portfolio Architecting

Portfolio architecting is the process of determining which products to produce. As a strategic planning tool, it can be used to add to a current product line or to develop an entirely new family of products. The strategic advantage of this approach is that the product family will have common modules that will make manufacturing and inventory more efficient and permit "mass customization." The steps in portfolio architecting are as follows:

1. Study the basic physical/chemical/biological principles involved (for example, how does powder coating work at the molecular level?)
2. Use the basic principles studied in step 1 to outline a family of products (for example, powder coating spray guns, curing systems, turnkey systems)
3. List the modules required for each product
4. Form a matrix with products composing the rows and function modules composing the columns (see Figure 34.1)
5. Study the matrix to determine which products to build

Steps 3–5 are repeated in an iterative fashion, adjusting modules, adding or deleting products, and redesigning modules until a final combination is determined.

	Module						
Product	**A**	**B**	**C**	**D**	**E**	**F**	**G**
Systems for custom auto shops	Y	Y		Y		Y	Y
Systems for education and training	Y	Y		Y			
Home systems	Y	Y		Y		Y	
Commercial turnkey systems	Y	Y	Y	Y	Y	Y	
Systems for coating wood	Y		Y	Y	Y	Y	Y
Systems for using thermosets	Y	Y	Y		Y		Y
Systems for curing other finishes		Y		Y	Y	Y	
Stand-alone spray guns		Y		Y			

Figure 34.1 Example of a product family matrix.

Hoshin Planning

Hoshin planning is a very tightly controlled planning and execution protocol that requires goal-setting beginning at the highest level. Each lower level of the organization is required to form objectives and goals that support those of the next higher level.

The following are the basic steps in the hoshin planning process at each level of the hierarchy:

- Statement of the organization's goals (occasionally referred to as *objectives*)
- Strategies that support achievement of the goals
- Time-based tactics that support achievement of each strategy
- Development of metrics that measure the progress toward achieving each strategy
- Assignment of accountability for strategies and tactics
- Regular reviews to assess actual performance of the plan
- Establishment of recovery plans or the adjustment of strategies and tactics as necessary

A key advantage of an effectively implemented hoshin planning process is that strategies are deployed and aligned both horizontally and vertically throughout an organization. See Kubiak (2012) for a detailed discussion on hoshin planning.

TACTICAL

Describe and use the theory of inventive problem-solving (TRIZ), systematic design, critical parameter management, and Pugh analysis in designing products or processes. (Apply)

Tactical plans are designed to support strategic plans. This section describes some tools for executing tactical plans.

TRIZ

TRIZ is an acronym for the Russian words *teorija rezbenija izobretaltelshih zadach*, meaning "theory of inventive problem solving." TRIZ provides another approach for solving difficult design problems. According to Rantanen and Domb (2002), the following are core concepts of this theory:

- *Contradiction.* Many problems have an inherent contradiction. For example, "How many stages should an air conditioner compressor have?" The more stages, the more efficiently it can respond to varying demand. However, additional stages are more expensive. The contradiction is that additional stages are both good and bad. Solving the problem consists of resolving the contradiction—preferably by removing it. Can a single-stage compressor become more efficient so that additional stages aren't needed? Can a multistage compressor be designed that is less expensive to build?
- *Resources.* TRIZ procedures require the analysis of items, information, energy, or material properties to determine which of them can be useful in resolving the contradiction.
- *Ideality.* In the beginning stages of problem solving, the ideal solution should be defined. In the case of the air conditioner, an ideal solution might be a single, infinitely variable stage.
- *Patterns of evolution.* Studies have shown that innovation tends to follow certain patterns. One such pattern is the move toward a more macro perspective, integrating a device or idea with others to form a new device or idea. An example would be the integration of the hand lawnmower with the small gasoline engine to form the power mower.
- *Inventive principles.* Extracting 40 inventive principles from a worldwide review of patents resulted in engineering hints aimed at creating inventive and patentable solutions to specific problems.

A Russian engineer, Genrich Altshuller, developed eight engineering system laws. According to Breyfogle (2003), Altshuller's laws are:

- *Law of completeness of the system.* Systems are derived from synthesis of separate parts into a functional system.
- *Law of energy transfer in the system.* Shafts, gears, magnetic fields, and charged particles are the heart of many inventive problems.
- *Law of increasing ideality.* Function is created with minimum complexity, which can be considered a ratio of system usefulness to its harmful effects. The ideal system has the desired outputs with no harmful effects, that is, no machine, just the function(s).
- *Law of harmonization.* Transferring energy more efficiently.
- *Law of uneven development of parts.* Not all parts evolve at the same pace. The least will limit the overall system.
- *Law of transition to a super system.* The solution system becomes the subsystem of a larger system.
- *Law of transition from macro to micro.* The use of physically smaller solutions (for example, electronic tubes to chips).
- *Law of increasing substance-field involvement.* Viewing and modeling systems as composed of two substances interacting through a field.

Axiomatic Design

Axiomatic design, yet another aid to the designer, divides the universe of the designer into four domains:

- *Customer domain.* Lists the features the customer wants
- *Functional domain.* Lists the various ways the product must work to meet the items in the customer domain
- *Physical domain.* Lists the parameters in the design that satisfy the items in the functional domain
- *Process domain.* Lists the processes that produce the product with the parameters listed in the physical domain

These domains are used sequentially in the design process:

- Concept design uses the customer domain to form the functional domain.
- Product design uses the functional domain to form the physical domain.
- Process design uses the physical domain to form the process domain.

There are two fundamental axioms of good design:

- A good design can be partitioned into sections so that changes in one section have minimal effect on the other sections.
- A good design requires a minimal amount of information to describe it.

Systematic Design

Systematic design refers to the current trend toward applying design principles to the design function itself. Some organizations are constructing a value stream map for the processes involved, finding waste, excess waiting, unneeded processing, and so forth. The result is a design process that is more efficient and better able to respond to changes in the marketplace. Improving the design process has a positive impact on the ability to produce and execute tactical plans.

Critical Parameter Management

In a manufacturing process, critical dimensions are carefully monitored by the use of SPC charts or other data collection and analysis techniques. Tactical planning must determine which information is critical to the success of the plan and establish a process for data collection and analysis. The tactical plan must also detail appropriate responses to changes in these critical parameters. This process is called *critical parameter management.*

Pugh Analysis

Pugh analysis, also called a *decision matrix* or *Pugh matrix*, is appropriate to use when a single option must be selected from several and when multiple criteria are to be used. Thus, it can have frequent applications in the design of products and processes. The matrix is formed by listing the criteria in the first column and identifying the options across the top of the remaining columns, as shown in Figure 34.2.

Next, the team assigns each criterion a weight, as shown in Figure 34.3. The more important the criterion, the higher the number assigned.

A baseline option is established, and for each criterion, all other options are ranked with regard to how they compare to the baseline. The value –1 is used for worse than the baseline, 0 is used when the criterion–option combination is the same as the baseline, and +1 is used for better than the baseline (see Figure 34.4). Of course, finer scales may be used. For example, –2, –1, 0, 1, and 2 may be more appropriate.

The final step is to multiply criteria weights by each cell value and calculate column totals. See Figure 34.5.

From Figure 34.5, we see that option D results in the highest positive score against the baseline. Although option D may not be the option ultimately selected, it is the relative score of each criterion that will generate discussion and debate regarding which option is most appropriate.

	Option					
Criteria	**A**	**B**	**C**	**D**	**E**	**F**
Profitable						
No new equipment required						
No new training required						
Doesn't require (work unknown)						
Doesn't require more floor space						

Figure 34.2 First step in forming a Pugh matrix.

		Option					
Criteria	**Weight**	**A**	**B**	**C**	**D**	**E**	**F**
Profitable	5						
No new equipment required	3						
No new training required	2						
Doesn't require (work unknown)	4						
Doesn't require more floor space	1						

Figure 34.3 Second step in forming a Pugh matrix.

		Option					
Criteria	**Weight**	**A**	**B**	**C**	**D**	**E**	**F**
Profitable	5	1	0	B	1	–1	1
No new equipment required	3	–1	0	B	1	0	1
No new training required	2	–1	1	B	1	–1	–1
Doesn't require (work unknown)	4	0	1	B	1	1	1
Doesn't require more floor space	1	–1	–1	B	–1	–1	0
	Total						

Figure 34.4 Third step in forming a Pugh matrix.

	Option						
Criteria	**Weight**	**A**	**B**	**C**	**D**	**E**	**F**
Profitable	5	1	0	B	1	–1	1
No new equipment required	3	–1	0	B	1	0	1
No new training required	2	–1	1	B	1	–1	–1
Doesn't require (work unknown)	4	0	1	B	1	1	1
Doesn't require more floor space	1	–1	–1	B	–1	–1	0
	Total	**–1**	**5**		**13**	**–4**	**10**

Figure 34.5 Final step in forming a Pugh matrix.

Part X

Appendices

Appendix 1
ASQ Code of Ethics (May 2005)

FUNDAMENTAL PRINCIPLES

ASQ requires its members and certification holders to conduct themselves ethically by:

1. Being honest and impartial in serving the public, their employers, customers, and clients.
2. Striving to increase the competence and prestige of the quality profession, and
3. Using their knowledge and skill for the enhancement of human welfare.

Members and certification holders are required to observe the tenets set forth below:

Relations with the Public

Article 1—Hold paramount the safety, health, and welfare of the public in the performance of their professional duties.

Relations with Employers and Clients

Article 2—Perform services only in their areas of competence.

Article 3—Continue their professional development throughout their careers and provide opportunities for the professional and ethical development of others.

Article 4—Act in a professional manner in dealings with ASQ staff and each employer, customer, or client.

Article 5—Act as faithful agents or trustees and avoid conflict of interest and the appearance of conflicts of interest.

Relations with Peers

Article 6—Build their professional reputation on the merit of their services and not compete unfairly with others.

Article 7—Assure that credit for the work of others is given to those to whom it is due.

Appendix 2A

ASQ Six Sigma Black Belt Certification Body of Knowledge (2015)

The topics in this Body of Knowledge include additional detail in the form of subtext explanations and the cognitive level at which test questions will be written. This information will provide guidance for the candidate preparing to take the exam. The subtext is not intended to limit the subject matter or be all-inclusive of what might be covered in an exam. It is meant to clarify the type of content to be included in the exam. The descriptor in parentheses at the end of each entry refers to the maximum cognitive level at which the topic will be tested. A complete description of cognitive levels is provided at the end of this document.

I. **Organization-wide Planning and Deployment (Questions 12)**

A. ***Organization-wide considerations***

1. *Fundamentals of six sigma and lean methodologies.* Define and describe the value, foundations, philosophy, history, and goals of these approaches, and describe the integration and complementary relationship between them. (Understand)

2. *Six sigma, lean, and continuous improvement methodologies.* Describe when to use six sigma instead of other problem-solving approaches, and describe the importance of aligning six sigma objectives with organizational goals. Describe screening criteria and how such criteria can be used for the selection of six sigma projects, lean initiatives, and other continuous improvement methods. (Apply)

3. *Relationships among business systems and processes.* Describe the interactive relationships among business systems, processes, and internal and external stakeholders , and the impact those relationships have on business systems. (Understand)

4. *Strategic planning and deployment for initiatives*. Define the importance of strategic planning for six sigma projects and lean initiatives. Demonstrate how hoshin kanri (X-matrix), portfolio analysis, and other tools can be used in support of strategic deployment of these projects. Use feasibility studies, SWOT analysis (strengths, weaknesses, opportunities, and threats), PEST analysis (political, economic, social, and technological) and contingency planning and business continuity planning to enhance strategic planning and deployment. (Apply)

B. ***Leadership***

1. *Roles and responsibilities*. Describe the roles and responsibilities of executive leadership, champions, sponsors, process owners, master black belts, black belts, and green belts in driving six sigma and lean initiatives. Describe how each group influences project deployment in terms of providing or managing resources, enabling changes in organizational structure, and supporting communications about the purpose and deployment of the initiatives. (Understand)

2. *Organizational roadblocks and change management*. Describe how an organization's structure and culture can impact six sigma projects. Identify common causes of six sigma failures, including lack of management support and lack of resources. Apply change management techniques, including stakeholder analysis, readiness assessments, and communication plans to overcome barriers and drive organization-wide change. (Apply)

II. **Organizational Process Management and Measures (10 Questions)**

A. ***Impact on stakeholders.*** Describe the impact six sigma projects can have on customers, suppliers, and other stakeholders. (Understand)

B. ***Benchmarking.*** Define and distinguish between various types of benchmarking, e.g., best practices, competitive, collaborative, breakthrough. Select measures and performance goals for projects resulting from benchmarking activities. (Apply)

C. ***Business measures***

1. *Performance measures*. Define and describe balanced scorecard, key performance indicators (KPIs), customer loyalty metrics, and leading and lagging indicators. Explain how to create a line of sight from performance measures to organizational strategies. (Analyze)

2. *Financial measures*. Define and use revenue growth, market share, margin, net present value (NPV), return on investment (ROI), and cost-benefit analysis (CBA). Explain the difference between hard cost measures (from profit and loss statements) and soft cost benefits of cost avoidance and reduction. (Apply)

III. **Team Management (18 Questions)**

A. ***Team formation***

1. *Team types and constraints*. Define and describe various teams, including virtual, cross-functional, and self-directed. Determine what team type will work best for a given a set of constraints, e.g., geography, technology availability, staff schedules, time zones. (Apply)

2. *Team roles and responsibilities*. Define and describe various team roles and responsibilities for leader, facilitator, coach, and individual member. (Understand)

3. *Team member selection criteria*. Describe various factors that influence the selection of team members, including the ability to influence, openness to change, required skills sets, subject matter expertise, and availability. (Apply)

4. *Team success factors*. Identify and describe the elements necessary for successful teams, e.g., management support, clear goals, ground rules, timelines. (Apply)

B. ***Team facilitation***

1. *Motivational techniques*. Describe and apply techniques to motivate team members. Identify factors that can demotivate team members and describe techniques to overcome them. (Apply)

2. *Team stages of development*. Identify and describe the classic stages of team development: forming, storming, norming, performing, and adjourning. (Apply)

3. *Team communication*. Describe and explain the elements of an effective communication plan, e.g., audience identification, message type, medium, frequency. (Apply)

4. *Team leadership models*. Describe and select appropriate leadership approaches (e.g., direct, coach, support, delegate) to ensure team success. (Apply)

C. ***Team dynamics***

1. *Group behaviors*. Identify and use various conflict resolution techniques (e.g., coaching, mentoring, intervention) to overcome negative group dynamics, including dominant and reluctant participants, groupthink, rushing to finish, and digressions. (Evaluate)

2. *Meeting management*. Select and use various meeting management techniques, including using agendas, starting on time, requiring pre-work by attendees, and ensuring that the right people and resources are available. (Apply)

3. *Team decision-making methods*. Define, select, and use various tools (e.g., consensus, nominal group technique, multi-voting) for decision-making. (Apply)

D. ***Team training***

1. *Needs assessment*. Identify the steps involved to implement an effective training curriculum: identify skills gaps, develop learning objectives, prepare a training plan, and develop training materials. (Understand)

2. *Delivery*. Describe various techniques used to deliver effective training, including adult learning theory, soft skills, and modes of learning. (Understand)

3. *Evaluation*. Describe various techniques to evaluate training, including evaluation planning, feedback surveys, pre-training and post-training testing. (Understand)

IV. **Define (20 questions)**

A. ***Voice of the customer***

1. *Customer identification*. Identify and segment customers and show how a project will impact both internal and external customers. (Apply)
2. *Customer data collection*. Identify and select appropriate data collection methods (e.g., surveys, focus groups, interviews, observations) to gather voice of the customer data. Ensure the data collection methods used are reviewed for validity and reliability. (Analyze)
3. *Customer requirements*. Define, select, and apply appropriate tools to determine customer needs and requirements, including critical-to-X (CTX when 'X' can be quality, cost, safety, etc.), CTQ tree, quality function deployment (QFD), supplier, input, process, output, customer (SIPOC) and Kano model. (Analyze)

B. ***Business case and project charter***

1. *Business case*. Describe business case justification used to support projects. (Understand)
2. *Problem statement*. Develop a project problem statement and evaluate it in relation to baseline performance and improvement goals. (Evaluate)
3. *Project scope*. Develop and review project boundaries to ensure that the project has value to the customer. (Analyze)
4. *Goals and objectives*. Identify SMART (specific, measureable, actionable, relevant and time bound) goals and objectives on the basis of the project?s problem statement and scope. (Analyze)
5. *Project performance measurements*. Identify and evaluate performance measurements (e.g., cost, revenue, delivery, schedule, customer satisfaction) that connect critical elements of the process to key outputs. (Analyze)
6. *Project charter review*. Explain the importance of having periodic project charter reviews with stakeholders. (Understand)

C. ***Project management (PM) tools.*** Identify and use the following PM tools to track projects and document their progress. (Evaluate)

1. Gantt charts
2. Toll-gate reviews
3. Work breakdown structure (WBS)

4. RACI Model (responsible, accountable, consulted and informed)

D. ***Analytical tools.*** Identify and use the following analytical tools throughout the DMAIC cycle. (Apply)

1. Affinity diagrams
2. Tree diagrams,
3. Matrix diagrams,
4. Prioritization matrices,
5. Activity network diagrams

V. **Measure (25 Questions)**

A. ***Process characteristics***

1. *Process flow metrics*. Identify and use process flow metrics (e.g., work in progress (WIP), work in queue (WIQ), touch time, takt time, cycle time, throughput) to determine constraints. Describe the impact that ?hidden factories? can have on process flow metrics. (Analyze)
2. *Process analysis tools*. Select, use and evaluate various tools, e.g., value stream maps, process maps, work instructions, flowcharts, spaghetti diagrams, circle diagrams, gemba walk. (Evaluate)

B. ***Data collection***

1. *Types of data*. Define, classify, and distinguish between qualitative and quantitative data, and continuous and discrete data. (Evaluate)
2. *Measurement scales*. Define and use nominal, ordinal, interval, and ratio measurement scales. (Apply)
3. *Sampling*. Define and describe sampling concepts, including representative selection, homogeneity, bias, accuracy, and precision. Determine the appropriate sampling method (e.g., random, stratified, systematic, subgroup, block) to obtain valid representation in various situations. (Evaluate)
4. *Data collection plans and methods*. Develop and implement data collection plans that include data capture and processing tools, e.g., check sheets, data coding, data cleaning (imputation techniques). Avoid data collection pitfalls by defining the metrics to be used or collected, ensuring that collectors are trained in the tools and understand how the data will be used, and checking for seasonality effects. (Analyze)

C. ***Measurement systems***

1. *Measurement system analysis (MSA)*. Use gauge repeatability and reproducibility (R&R) studies and other MSA tools (e.g., bias, correlation, linearity, precision to tolerance, percent agreement) to analyze measurement system capability. (Evaluate)

2. *Measurement systems across the organization*. Identify how measurement systems can be applied to marketing, sales, engineering, research and development (R&D), supply chain management, and customer satisfaction data. (Understand)

3. *Metrology*. Define and describe elements of metrology, including calibration systems, traceability to reference standards, and the control and integrity of measurement devices and standards. (Understand)

D. ***Basic statistics***

1. *Basic statistical terms*. Define and distinguish between population parameters and sample statistics, e.g., proportion, mean, standard deviation. (Apply)

2. *Central limit theorem*. Explain the central limit theorem and its significance in the application of inferential statistics for confidence intervals, hypothesis tests, and control charts. (Understand)

3. *Descriptive statistics*. Calculate and interpret measures of dispersion and central tendency. (Evaluate)

4. *Graphical methods*. Construct and interpret diagrams and charts, e.g., box-and-whisker plots, scatter diagrams, histograms, normal probability plots, frequency distributions, cumulative frequency distributions. (Evaluate)

5. *Valid statistical conclusions*. Distinguish between descriptive and inferential statistical studies. Evaluate how the results of statistical studies are used to draw valid conclusions. (Evaluate)

E. ***Probability***

1. *Basic concepts*. Describe and apply probability concepts, e.g., independence, mutually exclusive events, addition and multiplication rules, conditional probability, complementary probability, joint occurrence of events. (Apply)

2. *Distributions*. Describe, interpret, and use various distributions, e.g., normal, Poisson, binomial, chi square, Student?s t, F, hypergeometric, bivariate, exponential, lognormal, Weibull. (Evaluate)

F. ***Process capability***

1. *Process capability indices*. Define, select, and calculate Cp and Cpk. (Evaluate)

2. *Process performance indices*. Define, select, and calculate Pp, Ppk, Cpm, and process sigma. (Evaluate)

3. *General process capability studies*. Describe and apply elements of designing and conducting process capability studies relative

to characteristics, specifications, sampling plans, stability and normality. (Evaluate)

4. *Process capability for attributes data*. Calculate the process capability and process sigma level for attributes data. (Apply)
5. *Process capability for non-normal data*. Identify non-normal data and determine when it is appropriate to use Box-Cox or other transformation techniques. (Apply)
6. *Process performance vs. specification*. Distinguish between natural process limits and specification limits. Calculate process performance metrics, e.g., percent defective, parts per million (PPM), defects per million opportunities (DPMO), defects per unit (DPU), throughput yield, rolled throughput yield (RTY). (Evaluate)
7. *Short-term and long-term capability*. Describe and use appropriate assumptions and conventions when only short-term data or only long-term data are available. Interpret the relationship between short-term and long-term capability. (Evaluate)

VI. **Analyze (22 Questions)**

A. ***Measuring and modeling relationships and variables***

1. *Correlation coefficient*. Calculate and interpret the correlation coefficient and its confidence interval, and describe the difference between correlation and causation. (Evaluate)
2. *Linear regression*. Calculate and interpret regression analysis, and apply and interpret hypothesis tests for regression statistics. Use the regression model for estimation and prediction, analyze the uncertainty in the estimate, and perform a residuals analysis to validate the model. (Evaluate)
3. *Multivariate tools*. Use and interpret multivariate tools (e.g., factor analysis, discriminant analysis, multiple analysis of variance (MANOVA)) to investigate sources of variation. (Evaluate)

B. ***Hypothesis testing***

1. *Terminology*. Define and interpret the significance level, power, type I, and type II errors of statistical tests. (Evaluate)
2. *Statistical vs. practical significance*. Define, compare, and interpret statistical and practical significance. (Evaluate)
3. *Sample size*. Calculate sample size for common hypothesis tests: equality of means and equality of proportions. (Apply)
4. *Point and interval estimates*. Define and distinguish between confidence and prediction intervals. Define and interpret the efficiency and bias of estimators. Calculate tolerance and confidence intervals. (Evaluate)

5. *Tests for means, variances, and proportions*. Use and interpret the results of hypothesis tests for means, variances, and proportions. (Evaluate)
6. *Analysis of variance (ANOVA)*. Select, calculate, and interpret the results of ANOVAs. (Evaluate)
7. *Goodness-of-fit (chi square) tests*. Define, select, and interpret the results of these tests. (Evaluate)
8. *Contingency tables*. Select, develop, and use contingency tables to determine statistical significance. (Evaluate)
9. *Non-parametric tests*. Understand the importance of the Kruskal-Wallis and Mann-Whitney tests and when they should be used. (Understand)

C. ***Failure mode and effects analysis (FMEA).*** Describe the purpose and elements of FMEA, including risk priority number (RPN), and evaluate FMEA results for processes, products, and services. Distinguish between design FMEA (DFMEA) and process FMEA (PFMEA), and interpret their results. (Evaluate)

D. ***Additional analysis methods***

1. *Gap analysis*. Analyze scenarios to identify performance gaps, and compare current and future states using predefined metrics. (Analyze)
2. *Root cause analysis*. Define and describe the purpose of root cause analysis, recognize the issues involved in identifying a root cause, and use various tools (e.g., 5 whys, Pareto charts, fault tree analysis, cause and effect diagrams) to resolve chronic problems. (Analyze)
3. *Waste analysis*. Identify and interpret the seven classic wastes (overproduction, inventory, defects, over-processing, waiting, motion, transportation) and resource under-utilization. (Analyze)

VII. **Improve (21 Questions)**

A. ***Design of experiments (DOE)***

1. *Terminology*. Define basic DOE terms, e.g., independent and dependent variables, factors and levels, response, treatment, error, nested. (Understand)
2. *Design principles*. Define and apply DOE principles, e.g., power, sample size, balance, repetition, replication, order, efficiency, randomization, blocking, interaction, confounding, resolution. (Apply)
3. *Planning experiments*. Plan and evaluate DOEs by determining the objective, selecting appropriate factors, responses, and measurement methods, and choosing the appropriate design. (Evaluate)

4. *One-factor experiments*. Design and conduct completely randomized, randomized block, and Latin square designs, and evaluate their results. (Evaluate)

5. *Two-level fractional factorial experiments*. Design, analyze, and interpret these types of experiments, and describe how confounding can affect their use. (Evaluate)

6. *Full factorial experiments*. Design, conduct, and analyze these types of experiments. (Evaluate)

B. ***Lean methods***

1. *Waste elimination*. Select and apply tools and techniques for eliminating or preventing waste, e.g., pull systems, kanban, 5S, standard work, poka-yoke. (Analyze)

2. *Cycle-time reduction*. Use various tools and techniques for reducing cycle time, e.g., continuous flow, single-minute exchange of die (SMED), heijunka (production leveling). (Analyze)

3. *Kaizen*. Define and distinguish between kaizen and kaizen blitz and describe when to use each method. (Apply)

4. *Other improvement tools and techniques*. Identify and describe how other process improvement methodologies are used, e.g., theory of constraints (TOC), overall equipment effectiveness (OEE). (Understand)

C. ***Implementation.*** Develop plans for implementing proposed improvements, including conducting pilot tests or simulations, and evaluate results to select the optimum solution. (Evaluate)

VIII. **Control (15 Questions)**

A. ***Statistical process control (SPC)***

1. *Objectives*. Explain the objectives of SPC, including monitoring and controlling process performance, tracking trends, runs, and reducing variation within a process. (Understand)

2. *Selection of variables*. Identify and select critical process characteristics for control chart monitoring. (Apply)

3. *Rational subgrouping*. Define and apply the principle of rational subgrouping. (Apply)

4. *Control chart selection*. Select and use control charts in various situations: RX-, sX-, individual and moving range (ImR), p, np, c, u, short-run SPC, and moving average. (Apply)

5. *Control chart analysis*. Interpret control charts and distinguish between common and special causes using rules for determining statistical control. (Analyze)

B. ***Other controls***

1. *Total productive maintenance (TPM).* Define the elements of TPM and describe how it can be used to consistently control the improved process. (Understand)

2. *Visual controls (formerly "visual factory").* Define the elements of visual controls (e.g., pictures of correct procedures, color-coded components, indicator lights), and describe how they can help control the improved process. (Understand)

C. ***Maintain controls***

1. *Measurement system reanalysis.* Review and evaluate measurement system capability as process capability improves, and ensure that measurement capability is sufficient for its intended use. (Evaluate)

2. *Control plan.* Develop a control plan to maintain the improved process performance, enable continuous improvement, and transfer responsibility from the project team to the process owner. (Apply)

D. ***Sustain improvements***

1. *Lessons learned.* Document the lessons learned from all phases of a project and identify how improvements can be replicated and applied to other processes in the organization. (Apply)

2. *Documentation.* Develop or modify documents including standard operating procedures (SOPs), work instructions, and control plans to ensure that the improvements are sustained over time. (Apply)

3. *Training for process owners and staff.* Develop and implement training plans to ensure consistent execution of revised process methods and standards to maintain process improvements. (Apply)

4. *Ongoing evaluation.* Identify and apply tools (e.g., control charts, control plans) for ongoing evaluation of the improved process, including monitoring leading indicators, lagging indicators, and additional opportunities for improvement. (Apply)

IX. **Design For Six Sigma (DFSS) Framework and Methodologies (7 Questions)**

A. ***Common DFSS methodologies.*** Identify and describe DMADV (define, measure, analyze, design, and validate) and DMADOV (define, measure, analyze, design, optimize, and validate). (Understand)

B. ***Design for X (DFX).*** Describe design constraints, including design for cost, design for manufacturability (producibility), design for test, and design for maintainability. (Understand)

C. ***Robust designs.*** Describe the elements of robust product design, tolerance design, and statistical tolerancing. (Understand)

LEVELS OF COGNITION—BASED ON BLOOM'S TAXONOMY—REVISED (2001)

In addition to content specifics, the subtext for each topic in this BOK also indicates the intended complexity level of the test questions for that topic. These levels are based on "Levels of Cognition" (from Bloom's Taxonomy—Revised, 2001) and are presented below in rank order, from least complex to most complex.

Remember

Recall or recognize terms, definitions, facts, ideas, materials, patterns, sequences, methods, principles, etc.

Understand

Read and understand descriptions, communications, reports, tables, diagrams, directions, regulations, etc.

Apply

Know when and how to use ideas, procedures, methods, formulas, principles, theories, etc.

Analyze

Break down information into its constituent parts and recognize their relationship to one another and how they are organized; identify sublevel factors or salient data from a complex scenario.

Evaluate

Make judgments about the value of proposed ideas, solutions, etc., by comparing the proposal to specific criteria or standards.

Create

Put parts or elements together in such a way as to reveal a pattern or structure not clearly there before; identify which data or information from a complex set is appropriate to examine further or from which supported conclusions can be drawn.

Appendix 2B

ASQ Six Sigma Black Belt Certification Body of Knowledge (2007)

The topics in this Body of Knowledge include additional detail in the form of subtext explanations and the cognitive level at which the questions will be written. This information will provide useful guidance for both the Examination Development Committee and the candidates preparing to take the exam. The subtext is not intended to limit the subject matter or be all-inclusive of what might be covered in an exam. It is meant to clarify the type of content to be included in the exam. The descriptor in parentheses at the end of each entry refers to the maximum cognitive level at which the topic will be tested. A more complete description of cognitive levels is provided at the end of this document.

I. **Enterprise-Wide Deployment [9 Questions]**

A. ***Enterprise-wide view***

1. *History of continuous improvement*. Describe the origins of continuous improvement and its impact on other improvement models. (Remember)
2. *Value and foundations of Six Sigma*. Describe the value of Six Sigma, its philosophy, history, and goals. (Understand)
3. *Value and foundations of Lean.* Describe the value of Lean, its philosophy, history, and goals. (Understand)
4. *Integration of Lean and Six Sigma.* Describe the relationship between Lean and Six Sigma. (Understand)
5. *Business processes and systems.* Describe the relationship among various business processes (design, production, purchasing, accounting, sales, etc.) and the impact these relationships can have on business systems. (Understand)
6. *Six Sigma and Lean applications.* Describe how these tools are applied to processes in all types of enterprises: manufacturing, service, transactional, product and process design, innovation, etc. (Understand)

B. ***Leadership***

1. *Enterprise leadership responsibilities.* Describe the responsibilities of executive leaders and how they affect the deployment of Six Sigma in terms of providing resources, managing change, communicating ideas, etc. (Understand)

2. *Organizational roadblocks.* Describe the impact an organization's culture and inherent structure can have on the success of Six Sigma, and how deployment failure can result from the lack of resources, management support, etc.; identify and apply various techniques to overcome these barriers. (Apply)

3. *Change management.* Describe and use various techniques for facilitating and managing organizational change. (Apply)

4. *Six Sigma projects and kaizen events.* Describe how projects and kaizen events are selected, when to use Six Sigma instead of other problem-solving approaches, and the importance of aligning their objectives with organizational goals. (Apply)

5. *Six Sigma roles and responsibilities.* Describe the roles and responsibilities of Six Sigma participants: Black Belt, Master Black Belt, Green Belt, Champion, process owners, and project sponsors. (Understand)

II. **Organizational Process Management and Measures [9 Questions]**

A. ***Impact on stakeholders.*** Describe the impact Six Sigma projects can have on customers, suppliers, and other stakeholders. (Understand)

B. ***Critical to x (CTx) requirements.*** Define and describe various CTx requirements (critical to quality (CTQ), cost (CTC), process (CTP), safety (CTS), delivery (CTD), etc.) and the importance of aligning projects with those requirements. (Apply)

C. ***Benchmarking.*** Define and distinguish between various types of benchmarking, including best practices, competitive, collaborative, etc. (Apply)

D. ***Business performance measures.*** Define and describe various business performance measures, including balanced scorecard, key performance indicators (KPIs), the financial impact of customer loyalty, etc. (Understand)

E. ***Financial measures.*** Define and use financial measures, including revenue growth, market share, margin, cost of quality (COQ), net present value (NPV), return on investment (ROI), cost-benefit analysis, etc. (Apply)

III. **Team Management [16 Questions]**

A. ***Team formation.***

1. *Team types and constraints.* Define and describe various types of teams (e.g., formal, informal, virtual, cross-functional, self-directed, etc.), and determine what team model will work best for a given situation. Identify constraining factors including geography, technology, schedules, etc. (Apply)

2. *Team roles.* Define and describe various team roles and responsibilities, including leader, facilitator, coach, individual member, etc. (Understand)

3. *Team member selection.* Define and describe various factors that influence the selection of team members, including required skills sets, subject matter expertise, availability, etc. (Apply)

4. *Launching teams.* Identify and describe the elements required for launching a team, including having management support; establishing clear goals, ground rules, and timelines; and how these elements can affect the team's success. (Apply)

B. ***Team facilitation***

1. *Team motivation.* Describe and apply techniques that motivate team members and support and sustain their participation and commitment. (Apply)

2. *Team stages.* Facilitate the team through the classic stages of development: forming, storming, norming, performing, and adjourning. (Apply)

3. *Team communication.* Identify and use appropriate communication methods (both within the team and from the team to various stakeholders) to report progress, conduct milestone reviews, and support the overall success of the project. (Apply)

C. *Team dynamics.* Identify and use various techniques (e.g., coaching, mentoring, intervention, etc.) to overcome various group dynamic challenges, including overbearing/dominant or reluctant participants, feuding and other forms of unproductive disagreement, unquestioned acceptance of opinions as facts, groupthink, floundering, rushing to accomplish or finish, digressions, tangents, etc. (Evaluate)

D. *Time management for teams.* Select and use various time management techniques including publishing agendas with time limits on each entry, adhering to the agenda, requiring pre-work by attendees, ensuring that the right people and resources are available, etc. (Apply)

E. *Team decision-making tools.* Define, select, and use tools such as brainstorming, nominal group technique, multi-voting, etc. (Apply)

F. *Management and planning tools.* Define, select, and apply the following tools: affinity diagrams, tree diagrams, process decision program charts (PDPC), matrix diagrams, interrelationship digraphs, prioritization matrices, and activity network diagrams. (Apply)

G. *Team performance evaluation and reward.* Measure team progress in relation to goals, objectives, and other metrics that support team success, and reward and recognize the team for its accomplishments. (Analyze)

IV. **Define [15 Questions]**

A. ***Voice of the customer***

1. *Customer identification.* Segment customers for each project and show how the project will impact both internal and external customers. (Apply)

2. *Customer feedback.* Identify and select the appropriate data collection method (surveys, focus groups, interviews, observation, etc.) to gather customer feedback to better understand customer needs, expectations, and requirements. Ensure that the instruments used are reviewed for validity and reliability to avoid introducing bias or ambiguity in the responses. (Apply)

3. *Customer requirements.* Define, select, and use appropriate tools to determine customer requirements, such as CTQ flow-down, quality function deployment (QFD), and the Kano model. (Apply)

B. ***Project charter***

1. *Problem statement.* Develop and evaluate the problem statement in relation to the project's baseline performance and improvement goals. (Create)

2. *Project scope.* Develop and review project boundaries to ensure that the project has value to the customer. (Analyze)

3. *Goals and objectives.* Develop the goals and objectives for the project on the basis of the problem statement and scope. (Apply)

4. *Project performance measures.* Identify and evaluate performance measurements (e.g., cost, revenue, schedule, etc.) that connect critical elements of the process to key outputs. (Analyze)

C. ***Project tracking.*** Identify, develop, and use project management tools, such as schedules, Gantt charts, toll-gate reviews, etc., to track project progress. (Create)

V. **Measure [26 Questions]**

A. ***Process characteristics***

1. *Input and output variables.* Identify these process variables and evaluate their relationships using SIPOC and other tools. (Evaluate)

2. *Process flow metrics.* Evaluate process flow and utilization to identify waste and constraints by analyzing work in progress (WIP), work in queue (WIQ), touch time, takt time, cycle time, throughput, etc. (Evaluate)

3. *Process analysis tools.* Analyze processes by developing and using value stream maps, process maps, flowcharts, procedures, work instructions, spaghetti diagrams, circle diagrams, etc. (Analyze)

B. ***Data collection***

1. *Types of data.* Define, classify, and evaluate qualitative and quantitative data, continuous (variables) and discrete (attributes) data, and convert attributes data to variables measures when appropriate. (Evaluate)

2. *Measurement scales.* Define and apply nominal, ordinal, interval, and ratio measurement scales. (Apply)

3. *Sampling methods.* Define and apply the concepts related to sampling (e.g., representative selection, homogeneity, bias, etc.). Select and use appropriate sampling methods (e.g., random sampling, stratified sampling, systematic sampling, etc.) that ensure the integrity of data. (Evaluate)

4. *Collecting data.* Develop data collection plans, including consideration of how the data will be collected (e.g., check sheets, data coding techniques, automated data collection, etc.) and how it will be used. (Apply)

C. ***Measurement systems***

1. *Measurement methods.* Define and describe measurement methods for both continuous and discrete data. (Understand)

2. *Measurement systems analysis.* Use various analytical methods (e.g., repeatability and reproducibility (R&R), correlation, bias, linearity, precision to tolerance, percent agreement, etc.) to analyze and interpret measurement system capability for variables and attributes measurement systems. (Evaluate)

3. *Measurement systems in the enterprise.* Identify how measurement systems can be applied in marketing, sales, engineering, research and development (R&D), supply chain management, customer satisfaction, and other functional areas. (Understand)

4. *Metrology.* Define and describe elements of metrology, including calibration systems, traceability to reference standards, the control and integrity of standards and measurement devices, etc. (Understand)

D. ***Basic statistics***

1. *Basic terms.* Define and distinguish between population parameters and sample statistics (e.g., proportion, mean, standard deviation, etc.) (Apply)

2. *Central limit theorem.* Describe and use this theorem and apply the sampling distribution of the mean to inferential statistics for confidence intervals, control charts, etc. (Apply)

3. *Descriptive statistics.* Calculate and interpret measures of dispersion and central tendency, and construct and interpret frequency distributions and cumulative frequency distributions. (Evaluate)

4. *Graphical methods.* Construct and interpret diagrams and charts, including box-and-whisker plots, run charts, scatter diagrams, histograms, normal probability plots, etc. (Evaluate)

5. *Valid statistical conclusions.* Define and distinguish between enumerative (descriptive) and analytic (inferential) statistical studies, and evaluate their results to draw valid conclusions. (Evaluate)

E. ***Probability***

1. *Basic concepts.* Describe and apply probability concepts such as independence, mutually exclusive events, multiplication rules, complementary probability, joint occurrence of events, etc. (Apply)

2. *Commonly used distributions.* Describe, apply, and interpret the following distributions: normal, Poisson, binomial, chi square, Student's *t*, and *F* distributions. (Evaluate)

3. *Other distributions.* Describe when and how to use the following distributions: hypergeometric, bivariate, exponential, lognormal, and Weibull. (Apply)

F. ***Process capability***

1. *Process capability indices.* Define, select, and calculate C_p and C_{pk} to assess process capability. (Evaluate)

2. *Process performance indices.* Define, select, and calculate P_p, P_{pk}, and C_{pm} to assess process performance. (Evaluate)

3. *Short-term and long-term capability.* Describe and use appropriate assumptions and conventions when only short-term data or attributes data are available and when long-term data are available. Interpret the relationship between long-term and short-term capability. (Evaluate)

4. *Process capability for non-normal data.* Identify non-normal data and determine when it is appropriate to use Box-Cox or other transformation techniques. (Apply)

5. *Process capability for attributes data.* Calculate the process capability and process sigma level for attributes data. (Apply)

6. *Process capability studies.* Describe and apply elements of designing and conducting process capability studies, including identifying characteristics and specifications, developing sampling plans, and verifying stability and normality. (Evaluate)

7. *Process performance vs. specification.* Distinguish between natural process limits and specification limits, and calculate process

performance metrics such as percent defective, parts per million (PPM), defects per million opportunities (DPMO), defects per unit (DPU), process sigma, rolled throughput yield (RTY), etc. (Evaluate)

VI. **Analyze [24 Questions]**

A. ***Measuring and modeling relationships between variables***

1. *Correlation coefficient.* Calculate and interpret the correlation coefficient and its confidence interval, and describe the difference between correlation and causation. (Analyze)

   Note: Serial correlation will not be tested.

2. *Regression.* Calculate and interpret regression analysis, and apply and interpret hypothesis tests for regression statistics. Use the regression model for estimation and prediction, analyze the uncertainty in the estimate, and perform a residuals analysis to validate the model. (Evaluate)

   Note: Models that have non-linear parameters will not be tested.

3. *Multivariate tools.* Use and interpret multivariate tools such as principal components, factor analysis, discriminant analysis, multiple analysis of variance (MANOVA), etc., to investigate sources of variation. (Analyze)

4. *Multi-vari studies.* Use and interpret charts of these studies and determine the difference between positional, cyclical, and temporal variation. (Analyze)

5. *Attributes data analysis.* Analyze attributes data using logit, probit, logistic regression, etc., to investigate sources of variation. (Analyze)

B. **Hypothesis testing**

1. *Terminology.* Define and interpret the significance level, power, type I, and type II errors of statistical tests. (Evaluate)

2. *Statistical vs. practical significance.* Define, compare, and interpret statistical and practical significance. (Evaluate)

3. *Sample size.* Calculate sample size for common hypothesis tests (e.g., equality of means, equality of proportions, etc.). (Apply)

4. *Point and interval estimates.* Define and distinguish between confidence and prediction intervals. Define and interpret the efficiency and bias of estimators. Calculate tolerance and confidence intervals. (Evaluate)

5. *Tests for means, variances, and proportions.* Use and interpret the results of hypothesis tests for means, variances, and proportions. (Evaluate)

6. *Analysis of variance (ANOVA).* Select, calculate, and interpret the results of ANOVAs. (Evaluate)

7. *Goodness-of-fit (chi square) tests.* Define, select, and interpret the results of these tests. (Evaluate)

8. *Contingency tables.* Select, develop, and use contingency tables to determine statistical significance. (Evaluate)

9. *Non-parametric tests.* Select, develop, and use various non-parametric tests, including Mood's Median, Levene's test, Kruskal-Wallis, Mann-Whitney, etc. (Evaluate)

C. ***Failure mode and effects analysis (FMEA).*** Describe the purpose and elements of FMEA, including risk priority number (RPN), and evaluate FMEA results for processes, products, and services. Distinguish between design FMEA (DFMEA) and process FMEA (PFMEA), and interpret results from each. (Evaluate)

D. ***Additional analysis methods***

1. *Gap analysis.* Use various tools and techniques (gap analysis, scenario planning, etc.) to compare the current and future state in terms of pre-defined metrics. (Analyze)

2. *Root cause analysis.* Define and describe the purpose of root cause analysis, recognize the issues involved in identifying a root cause, and use various tools (e.g., the 5 whys, Pareto charts, fault tree analysis, cause and effect diagrams, etc.) for resolving chronic problems. (Evaluate)

3. *Waste analysis.* Identify and interpret the 7 classic wastes (overproduction, inventory, defects, over-processing, waiting, motion, and transportation) and other forms of waste such as resource under-utilization, etc. (Analyze)

VII. **Improve [23 Questions]**

A. ***Design of experiments (DOE)***

1. *Terminology.* Define basic DOE terms, including independent and dependent variables, factors and levels, response, treatment, error, etc. (Understand)

2. *Design principles.* Define and apply DOE principles, including power and sample size, balance, repetition, replication, order, efficiency, randomization, blocking, interaction, confounding, resolution, etc. (Apply)

3. *Planning experiments.* Plan, organize, and evaluate experiments by determining the objective, selecting factors, responses, and measurement methods; choosing the appropriate design, etc. (Evaluate)

4. *One-factor experiments.* Design and conduct completely randomized, randomized block, and Latin square designs, and evaluate their results. (Evaluate)

5. *Two-level fractional factorial experiments.* Design, analyze, and interpret these types of experiments, and describe how confounding affects their use. (Evaluate)

6. *Full factorial experiments.* Design, conduct, and analyze full factorial experiments. (Evaluate)

B. ***Waste elimination.*** Select and apply tools and techniques for eliminating or preventing waste, including pull systems, kanban, 5S, standard work, poka-yoke, etc. (Analyze)

C. ***Cycle-time reduction.*** Use various tools and techniques for reducing cycle time, including continuous flow, single-minute exchange of die (SMED), etc. (Analyze)

D. ***Kaizen and kaizen blitz.*** Define and distinguish between these two methods and apply them in various situations. (Apply)

E. ***Theory of constraints (TOC).*** Define and describe this concept and its uses. (Understand)

F. ***Implementation.*** Develop plans for implementing the improved process (i.e., conduct pilot tests, simulations, etc.), and evaluate results to select the optimum solution. (Evaluate)

G. ***Risk analysis and mitigation.*** Use tools such as feasibility studies, SWOT analysis (strengths, weaknesses, opportunities, and threats), PEST analysis (political, environmental, social, and technological), and consequential metrics to analyze and mitigate risk. (Apply)

VIII. **Control [21 Questions]**

A. ***Statistical process control (SPC)***

1. *Objectives.* Define and describe the objectives of SPC, including monitoring and controlling process performance, tracking trends, runs, etc., and reducing variation in a process. (Understand)

2. *Selection of variables.* Identify and select critical characteristics for control chart monitoring. (Apply)

3. *Rational subgrouping.* Define and apply the principle of rational subgrouping. (Apply)

4. *Control chart selection.* Select and use the following control charts in various situations:

   $\bar{X} - R$, $\bar{X} - s$, individual and moving range (ImR), p, np, c, u, short-run SPC, and moving average. (Apply)

5. *Control chart analysis.* Interpret control charts and distinguish between common and special causes using rules for determining statistical control. (Analyze)

B. ***Other control tools***

1. *Total productive maintenance (TPM).* Define the elements of TPM and describe how it can be used to control the improved process. (Understand)

2. *Visual factory.* Define the elements of a visual factory and describe how they can help control the improved process. (Understand)

C. ***Maintain controls***

1. *Measurement system re-analysis.* Review and evaluate measurement system capability as process capability improves, and ensure that measurement capability is sufficient for its intended use. (Evaluate)

2. *Control plan.* Develop a control plan for ensuring the ongoing success of the improved process including the transfer of responsibility from the project team to the process owner. (Apply)

D. ***Sustain improvements***

1. *Lessons learned.* Document the lessons learned from all phases of a project and identify how improvements can be replicated and applied to other processes in the organization. (Apply)

2. *Training plan deployment.* Develop and implement training plans to ensure continued support of the improved process. (Apply)

3. *Documentation.* Develop or modify documents including standard operating procedures (SOPs), work instructions, etc., to ensure that the improvements are sustained over time. (Apply)

4. *Ongoing evaluation.* Identify and apply tools for ongoing evaluation of the improved process, including monitoring for new constraints, additional opportunities for improvement, etc. (Apply)

IX. **Design for Six Sigma (DFSS) Frameworks and Methodologies [7 Questions]**

A. ***Common DFSS methodologies.*** Identify and describe these methodologies. (Understand)

1. DMADV (define, measure, analyze, design, and validate)

2. DMADOV (define, measure, analyze, design, optimize, and validate)

B. ***Design for X (DFX).*** Describe design constraints, including design for cost, design for manufacturability and producibility, design for test, design for maintainability, etc. (Understand)

C. ***Robust design and process.*** Describe the elements of robust product design, tolerance design, and statistical tolerancing. (Apply)

D. ***Special design tools***

1. *Strategic.* Describe how Porter's five forces analysis, portfolio architecting, and other tools can be used in strategic design and planning. (Understand)
2. *Tactical.* Describe and use the theory of inventive problem-solving (TRIZ), systematic design, critical parameter management, and Pugh analysis in designing products or processes. (Apply)

SIX LEVELS OF COGNITION BASED ON BLOOM'S TAXONOMY—REVISED (2001)

In addition to *content* specifics, the subtext for each topic in this BOK also indicates the intended *complexity level* of the test questions for that topic. These levels are based on "Levels of Cognition" (from Bloom's Taxonomy—Revised, 2001) and are presented below in rank order, from least complex to most complex.

Remember

Recall or recognize terms, definitions, facts, ideas, materials, patterns, sequences, methods, principles, etc.

Understand

Read and understand descriptions, communications, reports, tables, diagrams, directions, regulations, etc.

Apply

Know when and how to use ideas, procedures, methods, formulas, principles, theories, etc.

Analyze

Break down information into its constituent parts and recognize their relationship to one another and how they are organized; identify sublevel factors or salient data from a complex scenario.

Evaluate

Make judgments about the value of proposed ideas, solutions, etc., by comparing the proposal to specific criteria or standards.

Create

Put parts or elements together in such a way as to reveal a pattern or structure not clearly there before; identify which data or information from a complex set is appropriate to examine further or from which supported conclusions can be drawn.

Appendix 2C

ASQ Six Sigma Black Belt Certification Body of Knowledge (2001)

The topics in this Body of Knowledge include additional detail in the form of subtext explanations and the cognitive level at which the questions will be written. This information will provide useful guidance for both the Examination Development Committee and the candidates preparing to take the exam. The subtext is not intended to limit the subject matter or be all-inclusive of what might be covered in an exam. It is meant to clarify the type of content to be included in the exam. The descriptor in parentheses at the end of each entry refers to the maximum cognitive level at which the topic will be tested. A more complete description of cognitive levels is provided at the end of this document.

I. **Enterprise-Wide Deployment (9 Questions)**

A. ***Enterprise view***

1. *Value of Six Sigma.* Understand the organizational value of Six Sigma, its philosophy, goals, and definition. (Comprehension)
2. *Business systems and processes.* Understand and distinguish interrelationships between business systems and processes. (Comprehension)
3. *Process inputs, outputs, and feedback.* Describe how process inputs, outputs, and feedback of the system impact the enterprise system as a whole. (Comprehension)

B. ***Leadership***

1. *Enterprise leadership.* Understand leadership roles in the deployment of Six Sigma (e.g., resources, organizational structure). (Comprehension)
2. *Six Sigma roles and responsibilities.* Understand the roles/responsibilities of Black Belt, Master Black Belt, Green Belt, Champion, executive, process owners. (Comprehension)

C. ***Organizational goals and objectives.*** Understanding key drivers for business; understand key metrics/scorecards

1. *Linking projects to organizational goals.* Describe the project selection process including knowing when to use Six Sigma improvement methodology (DMAIC) as opposed to other problem-solving tools, and confirm link back to organizational goals. (Comprehension)

2. *Risk analysis.* Describe the purpose and benefit of strategic risk analysis (e.g., strengths, weaknesses, opportunities, threats (SWOT), scenario planning), including the risk of optimizing elements in a project or process resulting in suboptimizing the whole. (Comprehension)

3. *Closed-loop assessment/knowledge management.* Document the objectives achieved and manage the lessons learned to identify additional opportunities. (Comprehension)

D. ***History of organizational improvement/foundations of Six Sigma.*** Understand origin of continuous improvement tools used in Six Sigma (e.g., Deming, Juran, Shewhart, Ishikawa, Taguchi). (Comprehension)

II. **Business Process Management (9 Questions)**

A. ***Process vs. functional view***

1. *Process elements.* Understand process components and boundaries. (Analysis)

2. *Owners and stakeholders.* Identify process owners, internal and external customers, and other stakeholders. (Analysis)

3. *Project management and benefits.* Understand the difference between managing projects and maximizing their benefits to the business. (Analysis)

4. *Project measures.* Establish key performance metrics and appropriate project documentation. (Analysis)

B. ***Voice of the customer***

1. *Identify customer.* Segment customers as applicable to a particular project; list specific customers impacted by project within each segment; show how a project impacts internal and external customers; recognize the financial impact of customer loyalty. (Analysis)

2. *Collect customer data.* Use various methods to collect customer feedback (surveys, focus groups, interviews, observation, etc.) and understand the strengths and weaknesses of each approach; recognize the key elements that make surveys, interviews, and other feedback tools effective; review questions for integrity (bias, vagueness, etc.). (Application)

3. *Analyze customer data.* Use graphical, statistical, and qualitative tools to understand customer feedback. (Analysis)

4. *Determine critical customer requirements.* Translate customer feedback into strategic project focus areas using QFD or similar tools, and establish key project metrics that relate to the voice of the customer and yield process insights. (Analysis)

[Note: The analysis of QFD matrices is covered in section X.A.]

C. ***Business results***

1. *Process performance metrics.* Calculate DPU, RTY, and DPMO sigma levels; understand how metrics propagate upward and allocate downward; compare and contrast capability, complexity, and control; manage the use of sigma performance measures (e.g., PPM, DPMO, DPU, RTY, COPQ) to drive enterprise decisions. (Analysis)
2. *Benchmarking.* Understand the importance of benchmarking. (Knowledge)
3. *Financial benefits.* Understand and present financial measures and other benefits (soft and hard) of a project; understand and use basic financial models (e.g., NPV, ROI); describe, apply, evaluate, and interpret cost of quality concepts, including quality cost categories, data collection, reporting, etc. (Application)

III. **Project Management (15 Questions)**

A. ***Project charter and plan***

1. *Charter/plan elements.* Compare, select, and explain elements of a project's charter and plan. (Analysis)
2. *Planning tools.* Plan the project using tools such as Gantt chart, PERT chart, planning trees, etc. (Application)
3. *Project documentation.* Create data-driven and fact-driven project documentation using spreadsheets, storyboards, phased reviews, management reviews, presentations to the executive team, etc. (Synthesis)
4. *Charter negotiation.* Create and negotiate the charter, including objectives, scope, boundaries, resources, project transition, and project closure. (Analysis)

B. ***Team leadership***

1. *Initiating teams.* Know the elements of launching a team and why they are important: clear purpose, goals, commitment, ground rules, roles and responsibilities of team members, schedules, support from management, and team empowerment. (Application)
2. *Selecting team members.* Select team members who have appropriate skills sets (e.g., self-facilitation, technical/subject-matter expertise), and create teams with appropriate numbers of members and representation. (Application)
3. *Team stages.* Facilitate the stages of team evolution, including forming, storming, norming, performing, adjourning, and recognition. (Application)

C. ***Team dynamics and performance***

1. *Team-building techniques.* Recognize and apply the basic steps in team building: goals, roles and responsibilities, introductions, and both stated and hidden agendas. (Synthesis)
2. *Team facilitation techniques.* Apply coaching, mentoring, and facilitation techniques to guide a team and overcome problems such as overbearing, dominant, or reluctant participants; the unquestioned acceptance of opinions as facts; group-think; feuding; floundering; the rush to accomplishment; attribution; discounts and plops; digressions and tangents; etc. (Application)
3. *Team performance evaluation.* Measure team progress in relation to goals, objectives, and metrics that support team success. (Analysis)
4. *Team tools.* Define, select, and apply team tools such as nominal group technique, force field analysis, multivoting, conversion/diversion. (Application)

D. ***Change agent***

1. *Managing change.* Understand and apply techniques for facilitating or managing organizational change through change agent methodologies. (Application)
2. *Organizational roadblocks.* Understand the inherent structures of an organization (e.g., its cultures and constructs) that present basic barriers to improvement; select and apply techniques to overcome them. (Application)
3. *Negotiation and conflict resolution techniques.* Define, select, and apply tools such as consensus techniques, brainstorming, effort/impact, multivoting, interest-based bargaining to help conflicting parties (e.g., departments, groups, leaders, staff) recognize common goals and how to work together to achieve them. (Application)
4. *Motivation techniques.* Define, select, and apply techniques that support and sustain team member participation and commitment. (Application)
5. *Communication.* Use effective and appropriate communication techniques for different situations to overcome organizational barriers to success. (Application)

E. ***Management and Planning Tools.*** Define, select, and use 1) affinity diagrams, 2) interrelationship digraphs, 3) tree diagrams, 4) prioritization matrices, 5) matrix diagrams, 6) process decision program charts (PDPC), and 7) activity network diagrams. (Application)

IV. **Six Sigma Improvement Methodology and Tools—Define (9 Questions)**

A. ***Project scope.*** Determine project definition/scope using Pareto charts, top-level process (macro) maps, etc. (Synthesis)

B. ***Metrics.*** Establish primary and consequential metrics (quality, cycle time, cost). (Analysis)

C. ***Problem statement.*** Develop problem statement, including baseline and improvement goals. (Synthesis)

V. **Six Sigma Improvement Methodology and Tools—Measure (30 Questions)**

A. ***Process analysis and documentation***

1. *Tools.* Develop and review process maps, written procedures, work instructions, flowcharts, etc. (Analysis)

2. *Process inputs and outputs.* Identify process input variables and process output variables, and document their relationships through cause and effect diagrams, relational matrices, etc. (Evaluation)

B. ***Probability and statistics***

1. *Drawing valid statistical conclusions.* Distinguish between enumerative (descriptive) and analytical (inferential) studies, and distinguish between a population parameter and a sample statistic. (Evaluation)

2. *Central limit theorem and sampling distribution of the mean.* Define the central limit theorem and understand its significance in the application of inferential statistics for confidence intervals, control charts, etc. (Application)

3. *Basic probability concepts.* Describe and apply concepts such as independence, mutually exclusive, multiplication rules, complementary probability, joint occurrence of events, etc. (Application)

C. ***Collecting and summarizing data***

1. *Types of data.* Identify, define, classify and compare continuous (variables) and discrete (attributes) data, and recognize opportunities to convert attributes data to variables measures. (Evaluation)

2. *Measurement scales.* Define and apply nominal, ordinal, interval, and ratio measurement scales. (Application)

3. *Methods for collecting data.* Define and apply methods for collecting data such as check sheets, coding data, automatic gaging, etc. (Evaluation)

4. *Techniques for assuring data accuracy and integrity.* Define and apply techniques for assuring data accuracy and integrity such as random sampling, stratified sampling, sample homogeneity, etc. (Evaluation)

5. *Descriptive statistics.* Define, compute, and interpret measures of dispersion and central tendency, and construct and interpret

frequency distributions and cumulative frequency distributions. (Evaluation)

[Note: Measures of the geometric and harmonic mean will not be tested.]

6. *Graphical methods.* Depict *relationships* by constructing, applying and interpreting diagrams and charts such as stem-and-leaf plots, box-and-whisker plots, run charts, scatter diagrams, etc., and depict *distributions* by constructing, applying and interpreting diagrams such as histograms, normal probability plots, Weibull plots, etc. (Evaluation)

D. ***Properties and applications of probability distributions***

1. *Distributions commonly used by Black Belts.* Describe and apply binomial, Poisson, normal, chi-square, Student's *t*, and *F* distributions. (Evaluation)
2. *Other distributions.* Recognize when to use hypergeometric, bivariate, exponential, lognormal, and Weibull distributions. (Application)

E. ***Measurement systems***

1. *Measurement methods.* Describe and review measurement methods such as attribute screens, gauge blocks, calipers, micrometers, optical comparators, tensile strength, titration, etc. (Comprehension)
2. *Measurement system analysis.* Calculate, analyze, and interpret measurement system capability using repeatability and reproducibility, measurement correlation, bias, linearity, percent agreement, precision/tolerance (P/T), precision/total variation (P/TV), and use both ANOVA and control chart methods for non-destructive, destructive, and attribute systems. (Evaluation)
3. *Metrology.* Understand traceability to calibration standards, measurement error, calibration systems, control and integrity of standards and measurement devices. (Comprehension)

F. ***Analyzing process capability***

1. *Designing and conducting process capability studies.* Identify, describe, and apply the elements of designing and conducting process capability studies, including identifying characteristics, identifying specifications/tolerances, developing sampling plans, and verifying stability and normality. (Evaluation)
2. *Calculating process performance vs. specification.* Distinguish between natural process limits and specification limits, and calculate process performance metrics such as percent defective. (Evaluation)
3. *Process capability indices.* Define, select, and calculate C_p, C_{pk}, and assess process capability. (Evaluation)

4. *Process performance indices.* Define, select, and calculate P_p, P_{pk}, C_{pm}, and assess process performance. (Evaluation)
5. *Short-term vs. long-term capability.* Understand the assumptions and conventions appropriate when only short-term data are collected and when only attributes data are available; understand the changes in relationships that occur when long-term data are used; interpret relationships between long-term and short-term capability as it relates to technology and/or control problems. (Evaluation)
6. *Non-normal data transformations (process capability for non-normal data).* Understand the cause of non-normal data and determine when it is appropriate to transform. (Application)
7. *Process capability for attributes data.* Compute sigma level and understand its relationship to P_{pk}. (Application)

VI. **Six Sigma Improvement Methodology and Tools—Analyze (23 Questions)**

A. ***Exploratory data analysis***

1. *Mutli-vari studies.* Use multi-vari studies to interpret the difference between positional, cyclical, and temporal variation; design sampling plans to investigate the largest sources of variation; create and interpret multi-vari charts. (Application)
2. *Measuring and modeling relationships between variables*
   a. *Simple and multiple least-squares linear regression.* Calculate the regression equation; apply and interpret hypothesis tests for regression statistics; use the regression model for estimation and prediction, and analyze the uncertainty in the estimate. (Models that have non-linear parameters will not be tested.) (Evaluation)
   b. *Simple linear correlation.* Calculate and interpret the correlation coefficient and its confidence interval; apply and interpret a hypothesis test for the correlation coefficient; understand the difference between correlation and causation. (Serial correlation will not be tested.) (Evaluation)
   c. *Diagnostics.* Analyze residuals of the model. (Analysis)

B. ***Hypothesis testing***

1. *Fundamental concepts of hypothesis testing*
   a. *Statistical vs. practical significance.* Define, compare, and contrast statistical and practical significance. (Evaluation)
   b. *Significance level, power, type I and type II errors.* Apply and interpret the significance level, power, type I and type II errors of statistical tests. (Evaluation)

c. *Sample Size.* Understand how to calculate sample size for any given hypothesis test. (Application)

2. *Point and interval estimation.* Define and interpret the efficiency and bias of estimators; compute, interpret and draw conclusions from statistics such as standard error, tolerance intervals, and confidence intervals; understand the distinction between confidence intervals and prediction intervals. (Analysis)
3. *Tests for means, variances, and proportions.* Apply hypothesis tests for means, variances, and proportions, and interpret the results. (Evaluation)
4. *Paired-comparison tests.* Define, determine applicability, and apply paired-comparison parametric hypothesis tests, and interpret the results. (Evaluation)
5. *Goodness-of-fit tests.* Define, determine applicability, and apply chi-square tests and interpret the results. (Evaluation)
6. *Analysis of variance (ANOVA).* Define, determine applicability, and apply ANOVAs and interpret the results. (Evaluation)
7. *Contingency tables.* Define, determine applicability, and construct a contingency table and use it to determine statistical significance. (Evaluation)
8. *Non-parametric tests.* Define, determine applicability, and construct various non-parametric tests including Mood's Median, Levene's test, Kruskal-Wallis, Mann-Whitney, etc. (Analysis)

VII. **Six Sigma Improvement Methodology and Tools—Improve (22 Questions)**

A. ***Design of experiments (DOE)***

1. *Terminology.* Define independent and dependent variables, factors and levels, response, treatment, error, and replication. (Comprehension)
2. *Planning and organizing experiments.* Describe and apply the basic elements of experiment planning and organizing, including determining the experiment objective, selecting factors, responses, and measurement methods, choosing the appropriate design, etc. (Evaluation)
3. *Design principles.* Define and apply the principles of power and sample size, balance, replication, order, efficiency, randomization and blocking, interaction, and confounding. (Application)
4. *Design and analysis of one-factor experiments.* Construct these experiments such as completely randomized, randomized block and Latin square designs, and apply computational and graphical

methods to analyze and evaluate the significance of results. (Evaluation)

5. *Design and analysis of full-factorial experiments.* Construct these experiments and apply computational and graphical methods to analyze and evaluate the significance of results. (Evaluation)

6. *Design and analysis of two-level fractional factorial experiments.* Construct experiments (including Taguchi designs) and apply computational and graphical methods to analyze and evaluate the significance of results; understand limitations of fractional factorials due to confounding. (Evaluation)

7. *Taguchi robustness concepts.* Apply Taguchi robustness concepts and techniques such as signal-to-noise ratio, controllable and noise factors, and robustness to external sources of variability. (Analysis)

8. *Mixture experiments.* Construct these experiments and apply computational and graphical methods to analyze and evaluate the significance of results. (Analysis)

B. ***Response surface methodology***

1. *Steepest ascent/descent experiments.* Construct these experiments and apply computational and graphical methods to analyze the significance of results. (Analysis)

2. *Higher-order experiments.* Construct experiments such as CCD, Box-Behnken, etc., and apply computational and graphical methods to analyze the significance of results. (Analysis)

C. ***Evolutionary operations (EVOP).*** Understand the application and strategy of EVOP. (Comprehension)

VIII. **Six Sigma Improvement Methodology and Tools—*Control* (15 Questions)**

A. ***Statistical process control***

1. *Objectives and benefits.* Understand objectives and benefits of SPC (e.g., controlling process performance, distinguishing special from common causes). (Comprehension)

2. *Selection of variable.* Select critical characteristics for monitoring by control chart. (Application)

3. *Rational subgrouping.* Define and apply the principle of rational subgrouping. (Application)

4. *Selection and application of control charts.* Identify, select, construct and apply the following types of control charts: $\bar{X} - R$, $\bar{X} - s$, individual and moving range (ImR/XmR), median, p, np, c, and u. (Application)

5. *Analysis of control charts.* Interpret control charts and distinguish between common and special causes using rules for determining statistical control. (Analysis)

6. *PRE-control.* Define and explain PRE-control and perform PRE-control calculations and analysis. (Analysis)

B. ***Advanced statistical process control.*** Understand appropriate uses of short-run SPC, EWMA, CuSum, and moving average. (Comprehension)

C. ***Lean tools for control.*** Apply appropriate lean tools (e.g., 5S, visual factory, kaizen, kanban, poka-yoke, total productive maintenance, standard work) as they relate to the control phase of DMAIC. (Application)

[Note: The use of lean tools in other areas of DMAIC is covered in section IX.C.]

D. ***Measurement system re-analysis.*** Understand the need to improve measurement system capability as process capability improves; evaluate the use of control measurement systems (e.g., attributes, variables, destructive), and ensure that measurement capability is sufficient for its intended use. (Evaluation)

IX. **Lean Enterprise (9 Questions)**

A. ***Lean concepts***

1. *Theory of constraints.* Describe the theory of constraints. (Comprehension)

2. *Lean thinking.* Describe concepts such as value, value chain, flow, pull, perfection, etc. (Comprehension)

3. *Continuous flow manufacturing (CFM).* Describe the concept CFM. (Comprehension)

4. *Non-value-added activities.* Identify these activities in terms inventory, space, test inspection, rework, transportation, storage, etc. (Application)

5. *Cycle-time reduction.* Describe how cycle-time reduction can be used to identify defects and non-value-added activities using kaizen-type methods to reduce waste of space, inventory, labor, and distance. (Comprehension)

B. ***Lean tools.*** Define, select, and apply tools such as visual factory, kanban, poka-yoke, standard work, SMED, etc., in areas outside of DMAIC-Control. (Application)

[Note: The use of lean tools in DMAIC-Control is covered in section VIII.C.]

C. ***Total productive maintenance (TPM).*** Understand the concept of TPM. (Comprehension)

X. **Design for Six Sigma (DFSS) (9 Questions)**

A. ***Quality function deployment (QFD).*** Analyze a completed QFD matrix. (Analysis)

B. ***Robust design and process***

1. *Functional requirements.* Understand functional requirements of a design. (Comprehension)
2. *Noise strategies.* Develop a robust design using noise strategies. (Application)
3. *Tolerance design.* Understand the concepts of tolerance design and statistical tolerancing. (Analysis)
4. *Tolerance and process capability.* Calculate tolerances using process capability data. (Analysis)

C. ***Failure mode and effects analysis (FMEA).*** Understand the terminology, purpose, and use of scale criteria (RPN) and be able to apply it to a process, product, or service; understand the distinction between and interpret data associated with DFMEA and PFMEA. (Analysis)

D. ***Design for X (DFX).*** Understand design constraints such as design for cost, design for manufacturability and producibility, design for test, design for maintainability, etc. (Comprehension)

E. ***Special design tools.*** Understand the concept of special design tools such as the theory of inventive problem-solving (TRIZ), axiomatic design (conceptual structure robustness), etc. (Knowledge)

SIX LEVELS OF COGNITION BASED ON BLOOM'S TAXONOMY (1956)

In addition to *content* specifics, the subtext detail also indicates the intended *complexity level* of the test questions for that topic. These levels are based on "Levels of Cognition" (from Bloom's Taxonomy, 1956) and are presented below in rank order, from least complex to most complex.

Knowledge Level

(Also commonly referred to as recognition, recall, or rote knowledge.) Being able to remember or recognize terminology, definitions, facts, ideas, materials, patterns, sequences, methodologies, principles, etc.

Comprehension Level

Being able to read and understand descriptions, communications, reports, tables, diagrams, directions, regulations, etc.

Application Level

Being able to apply ideas, procedures, methods, formulas, principles, theories, etc., in job-related situations.

Analysis

Being able to break down information into its constituent parts and recognize the parts' relationship to one another and how they are organized; identify sublevel factors or salient data from a complex scenario.

Synthesis

Being able to put parts or elements together in such a way as to show a pattern or structure not clearly there before; identify which data or information from a complex set is appropriate to examine further or from which supported conclusions can be drawn.

Evaluation

Being able to make judgments regarding the value of proposed ideas, solutions, methodologies, etc., by using appropriate criteria or standards to estimate accuracy, effectiveness, economic benefits, etc.

Appendix 3

Control Chart Combinations for Measurement Data

Averages				Natural process limits			
Chart	Center line	LCL	UCL	Chart	Center line	UNPL	LNPL
$\bar{X}$	$\bar{\bar{X}}$	$\bar{\bar{X}} - A_2\bar{R}$	$\bar{\bar{X}} + A_2\bar{R}$	I	$\bar{\bar{X}}$	$\bar{\bar{X}} - E_2\bar{R}$	$\bar{\bar{X}} + E_2\bar{R}$
R	$\bar{R}$	$D_3\bar{R}$	$D_4\bar{R}$	R	$\bar{R}$	$D_3\bar{R}$	$D_4\bar{R}$
$\bar{X}$	$\bar{\bar{X}}$	$\bar{\bar{X}} - A_4\tilde{R}$	$\bar{\bar{X}} + A_4\tilde{R}$	I	$\bar{\bar{X}}$	$\bar{\bar{X}} - E_5\tilde{R}$	$\bar{\bar{X}} + E_5\tilde{R}$
R	$\tilde{R}$	$D_5\tilde{R}$	$D_6\tilde{R}$	R	$\tilde{R}$	$D_5\tilde{R}$	$D_6\tilde{R}$
$\bar{X}$	$\bar{\bar{X}}$	$\bar{\bar{X}} - A_1\bar{\sigma}_{RMS}$	$\bar{\bar{X}} + A_1\bar{\sigma}_{RMS}$	I	$\bar{\bar{X}}$	$\bar{\bar{X}} - E_1\bar{\sigma}_{RMS}$	$\bar{\bar{X}} + E_1\bar{\sigma}_{RMS}$
σ_{RMS}	$\bar{\sigma}_{RMS}$	$B_3\bar{\sigma}_{RMS}$	$B_4\bar{\sigma}_{RMS}$	σ_{RMS}	$\bar{\sigma}_{RMS}$	$B_3\bar{\sigma}_{RMS}$	$B_4\bar{\sigma}_{RMS}$
$\bar{X}$	$\bar{\bar{X}}$	$\bar{\bar{X}} - A_5\tilde{\sigma}_{RMS}$	$\bar{\bar{X}} + A_5\tilde{\sigma}_{RMS}$	I	$\bar{\bar{X}}$	$\bar{\bar{X}} - E_4\tilde{\sigma}_{RMS}$	$\bar{\bar{X}} + E_4\tilde{\sigma}_{RMS}$
σ_{RMS}	$\tilde{\sigma}_{RMS}$	$B_9\tilde{\sigma}_{RMS}$	$B_{10}\tilde{\sigma}_{RMS}$	σ_{RMS}	$\tilde{\sigma}_{RMS}$	$B_9\tilde{\sigma}_{RMS}$	$B_{10}\tilde{\sigma}_{RMS}$
$\bar{X}$	$\bar{\bar{X}}$	$\bar{\bar{X}} - A_3\bar{s}$	$\bar{\bar{X}} + A_3\bar{s}$	I	$\bar{\bar{X}}$	$\bar{\bar{X}} - E_3\bar{s}$	$\bar{\bar{X}} + E_3\bar{s}$
s	$\bar{s}$	$B_3\bar{s}$	$B_4\bar{s}$	s	$\bar{s}$	$B_3\bar{s}$	$B_4\bar{s}$
$\bar{X}$	$\bar{\bar{X}}$	$\bar{\bar{X}} - A_{10}\tilde{s}$	$\bar{\bar{X}} + A_{10}\tilde{s}$	I	$\bar{\bar{X}}$	$\bar{\bar{X}} - E_6\tilde{s}$	$\bar{\bar{X}} + E_6\tilde{s}$
s	$\tilde{s}$	$B_9\tilde{s}$	$B_{10}\tilde{s}$	s	$\tilde{s}$	$B_9\tilde{s}$	$B_{10}\tilde{s}$
$\bar{X}$	$\bar{\bar{X}}$	$\bar{\bar{X}} - A_7\sqrt{\bar{s}^2}$ $\approx \bar{\bar{X}} - A\sqrt{\bar{s}^2}$	$\bar{\bar{X}} + A_7\sqrt{\bar{s}^2}$ $\approx \bar{\bar{X}} + A\sqrt{\bar{s}^2}$	—	—	—	—
s	$\sqrt{\bar{s}^2}$	$B_7\sqrt{\bar{s}^2}$ $\approx B_5\sqrt{\bar{s}^2}$	$B_8\sqrt{\bar{s}^2}$ $\approx B_6\sqrt{\bar{s}^2}$	—	—	—	—
$\bar{X}$	$\bar{\bar{X}}$	$\bar{\bar{X}} - A_7\sqrt{\bar{s}^2}$ $\approx \bar{\bar{X}} - A\sqrt{\bar{s}^2}$	$\bar{\bar{X}} + A_7\sqrt{\bar{s}^2}$ $\approx \bar{\bar{X}} + A\sqrt{\bar{s}^2}$	—	—	—	—
s^2	$\bar{s}^2$	$B_{11}\bar{s}^2$	$B_{12}\bar{s}^2$	—	—	—	—
$\tilde{X}$	$\bar{\tilde{X}}$	$\bar{\tilde{X}} - A_6\bar{R}$	$\bar{\tilde{X}} + A_6\bar{R}$	I	$\bar{\tilde{X}}$	$\bar{\tilde{X}} - E_2\bar{R}$	$\bar{\tilde{X}} + E_2\bar{R}$
R	$\bar{R}$	$D_3\bar{R}$	$D_4\bar{R}$	R	$\bar{R}$	$D_3\bar{R}$	$D_4\bar{R}$
$\tilde{X}$	$\bar{\tilde{X}}$	$\bar{\tilde{X}} - A_9\tilde{R}$	$\bar{\tilde{X}} + A_9\tilde{R}$	I	$\bar{\tilde{X}}$	$\bar{\tilde{X}} - E_5\tilde{R}$	$\bar{\tilde{X}} + E_5\tilde{R}$
R	$\tilde{R}$	$D_5\tilde{R}$	$D_6\tilde{R}$	R	$\tilde{R}$	$D_5\tilde{R}$	$D_6\tilde{R}$

Averages				Natural process limits			
Chart	Center line	LCL	UCL	Chart	Center line	UNPL	LNPL
$\bar{X}$	Nominal	Nominal $- A\sigma_x$	Nominal $+ A\sigma_x$	I	Nominal	Nominal $- 3\sigma_x$	Nominal $+ 3\sigma_x$
R	$d_2\sigma_x$	$D_1\sigma_x$	$D_2\sigma_x$	R	$d_2\sigma_x$	$D_1\sigma_x$	$D_2\sigma_x$
$\bar{X}$	Nominal	Nominal $- A\sigma_x$	Nominal $+ A\sigma_x$	I	Nominal	Nominal $- 3\sigma_x$	Nominal $+ 3\sigma_x$
σ_{RMS}	$c_2\sigma_x$	$B_1\sigma_X$	$B_2\sigma_X$	σ_{RMS}	$c_2\sigma_x$	$B_1\sigma_X$	$B_2\sigma_X$
$\bar{X}$	Nominal	Nominal $- A\sigma_x$	Nominal $+ A\sigma_x$	I	Nominal	Nominal $- 3\sigma_x$	Nominal $+ 3\sigma_x$
s	$c_4\sigma_x$	$B_5\sigma_X$	$B_6\sigma_X$	s	$c_4\sigma_x$	$B_5\sigma_X$	$B_6\sigma_X$
$\tilde{X}$	Nominal	Nominal $- A_8\sigma_x$	Nominal $+ A_8\sigma_x$	I	Nominal	Nominal $- 3\sigma_x$	Nominal $+ 3\sigma_x$
R	$d_2\sigma_x$	$D_1\sigma_x$	$D_2\sigma_x$	R	$d_2\sigma_x$	$D_1\sigma_x$	$D_2\sigma_x$
—	—	—	—	I	$\bar{X}$	$\bar{X} - E_2 m\bar{R}$	$\bar{X} + E_2 m\bar{R}$
—	—	—	—	R	$m\bar{R}$	$D_3 m\bar{R}$	$D_4 m\bar{R}$
—	—	—	—	I	$\bar{X}$	$\bar{X} - E_5 m\tilde{R}$	$\bar{X} + E_5 m\tilde{R}$
—	—	—	—	R	$m\tilde{R}$	$D_5 m\tilde{R}$	$D_6 m\tilde{R}$

Notes:

When subgroups are involved, assume k subgroups of size n.

$\bar{R}$ is the average range of k subgroups of size n.

$\tilde{R}$ is the median range.

$m\bar{R}$ is the average moving range.

$m\tilde{R}$ is the median range of all the subgroup ranges.

σ_{RMS} is the average root mean squared deviation and computed as $\sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n}}$.

"Nominal" may be construed as given, known, target, and historical or past values.

See Appendix 6 for various ways of computing σ_X.

Appendix 4

Control Chart Constants

n	A	A_1	A_2	A_3	A_4	A_5	A_6	A_8	A_9	A_{10}	B_1	B_2	B_3	B_4	B_5	B_6	n
2	2.121	3.760	1.881	2.659	2.224	4.447				3.143	0.000	1.843	0.000	3.266	0.000	2.606	2
3	1.732	2.394	1.023	1.954	1.091	2.547	1.187	2.010	1.265	2.082	0.000	1.858	0.000	2.568	0.000	2.276	3
4	1.500	1.880	0.729	1.628	0.758	1.951				1.689	0.000	1.808	0.000	2.266	0.000	2.088	4
5	1.342	1.596	0.577	1.427	0.594	1.638	0.691	1.607	0.712	1.465	0.000	1.757	0.000	2.089	0.000	1.964	5
6	1.225	1.410	0.483	1.287	0.495	1.438				1.313	0.026	1.711	0.030	1.970	0.029	1.874	6
7	1.134	1.277	0.419	1.182	0.429	1.297	0.509	1.377	0.520	1.201	0.104	1.672	0.118	1.882	0.113	1.806	7
8	1.061	1.175	0.373	1.099	0.380	1.190				1.114	0.167	1.638	0.185	1.815	0.178	1.752	8
9	1.000	1.094	0.337	1.032	0.343	1.107	0.412	1.223	0.419	1.044	0.219	1.609	0.239	1.761	0.232	1.707	9
10	0.949	1.028	0.308	0.975	0.314	1.039				0.985	0.261	1.584	0.284	1.716	0.277	1.669	10
11	0.905	0.973	0.285	0.927	0.290	0.982	0.350	1.110	0.356	0.936	0.299	1.561	0.322	1.678	0.314	1.637	11
12	0.866	0.925	0.266	0.886	0.270	0.933				0.893	0.330	1.542	0.354	1.646	0.346	1.609	12
13	0.832	0.884	0.249	0.850	0.253	0.891	0.308	1.026	0.312	0.856	0.359	1.523	0.381	1.619	0.374	1.585	13
14	0.802	0.848	0.235	0.817	0.239	0.854				0.823	0.384	1.506	0.407	1.593	0.399	1.563	14
15	0.775	0.816	0.223	0.789	0.226	0.821	0.276	0.958	0.280	0.794	0.406	1.492	0.428	1.572	0.420	1.544	15
16	0.750	0.788	0.212	0.763	0.215						0.427	1.477	0.448	1.552	0.441	1.526	16
17	0.728	0.762	0.203	0.739	0.206						0.445	1.466	0.466	1.534	0.458	1.511	17
18	0.707	0.738	0.194	0.718	0.197						0.461	1.455	0.482	1.518	0.475	1.496	18
19	0.688	0.717	0.187	0.698	0.189						0.477	1.443	0.496	1.504	0.490	1.483	19
20	0.671	0.697	0.180	0.680	0.182						0.490	1.434	0.510	1.490	0.503	1.471	20
21	0.655	0.679	0.173	0.663	0.176						0.504	1.423	0.523	1.477	0.517	1.459	21
22	0.640	0.662	0.167	0.647	0.170						0.517	1.414	0.535	1.465	0.529	1.448	22
23	0.626	0.647	0.162	0.633	0.164						0.528	1.406	0.545	1.455	0.539	1.438	23
24	0.612	0.632	0.157	0.619	0.159						0.539	1.398	0.555	1.445	0.549	1.429	24
25	0.600	0.619	0.153	0.606	0.155						0.544	1.395	0.564	1.436	0.558	1.421	25

Notes: Only odd values of n are given for A_6, A_8, and A_9. Otherwise, the median for an even n is simply the average of the two middle values of the subgroup.

See Appendix 5 for values of A_7, B_7, and B_8.

n	B_9	B_{10}	B_{11}	B_{12}	D_1	D_2	D_3	D_4	D_5	D_6	E_1	E_2	E_3	E_4	E_5	E_6	n
2		3.864	0.000	5.243	0.000	3.686	0.000	3.267	0.000	3.863	5.317	2.660	3.760	6.289	3.145	4.444	2
3		2.733	0.000	4.000	0.000	4.358	0.000	2.574	0.000	2.744	4.146	1.772	3.385	4.412	1.889	3.606	3
4		2.351	0.000	3.449	0.000	4.698	0.000	2.282	0.000	2.375	3.760	1.457	3.256	4.115	1.517	3.378	4
5		2.145	0.000	3.121	0.000	4.918	0.000	2.114	0.000	2.179	3.568	1.290	3.191	3.663	1.329	3.275	5
6	0.031	2.008	0.000	2.897	0.000	5.078	0.000	2.004	0.000	2.054	3.454	1.184	3.153	3.521	1.214	3.215	6
7	0.120	1.913	0.000	2.732	0.204	5.204	0.076	1.924	0.077	1.967	3.378	1.109	3.127	3.432	1.134	3.178	7
8	0.188	1.839	0.000	2.604	0.388	5.306	0.136	1.864	0.139	1.901	3.323	1.054	3.109	3.367	1.075	3.151	8
9	0.242	1.782	0.000	2.500	0.547	5.393	0.184	1.816	0.188	1.850	3.283	1.010	3.095	3.322	1.029	3.132	9
10	0.287	1.735	0.000	2.414	0.687	5.469	0.223	1.777	0.227	1.809	3.251	0.975	3.084	3.286	0.992	3.115	10
11	0.324	1.695	0.000	2.342	0.811	5.535	0.256	1.744	0.260	1.773	3.226	0.945	3.076	3.257	0.961	3.106	11
12	0.357	1.661	0.000	2.279	0.923	5.594	0.283	1.717	0.288	1.744	3.205	0.921	3.069	3.233	0.935	3.093	12
13	0.384	1.631	0.000	2.225	1.025	5.647	0.307	1.693	0.312	1.719	3.188	0.899	3.063	3.212	0.913	3.086	13
14	0.409	1.604	0.000	2.177	1.118	5.696	0.328	1.672	0.333	1.697	3.174	0.881	3.058	3.195	0.894	3.080	14
15	0.431	1.582	0.000	2.134	1.203	5.741	0.347	1.653	0.352	1.678	3.161	0.864	3.054	3.181	0.877	3.074	15
16			0.000	2.095	1.282	5.782	0.363	1.637	0.368	1.660	3.150	0.849	3.050		0.862		16
17			0.000	2.061	1.356	5.820	0.378	1.622	0.383	1.645	3.141	0.836	3.047		0.848		17
18			0.000	2.029	1.424	5.856	0.391	1.609	0.397	1.631	3.133	0.824	3.044		0.835		18
19			0.000	2.000	1.489	5.890	0.403	1.597	0.409	1.618	3.125	0.813	3.042		0.824		19
20			0.027	1.973	1.549	5.921	0.415	1.585	0.420	1.606	3.119	0.803	3.040		0.814		20
21			0.051	1.949	1.596	5.960	0.423	1.577	0.428	1.598	3.113	0.794	3.038		0.804		21
22			0.074	1.926	1.659	5.979	0.434	1.566	0.440	1.585	3.107	0.786	3.036		0.796		22
23			0.095	1.905	1.710	6.006	0.443	1.557	0.449	1.576	3.102	0.778	3.034		0.787		23
24			0.115	1.885	1.759	6.031	0.452	1.548	0.457	1.568	3.098	0.770	3.033		0.780		24
25			0.134	1.866	1.806	6.056	0.459	1.541	0.465	1.560	3.094	0.763	3.032		0.773		25

Notes: Only odd values of n are given for A_6, A_8, and A_9. Otherwise, the median for an even n is simply the average of the two middle values of the subgroup.

See Appendix 5 for values of A_7, B_7, and B_8.

Appendix 5

Constants for A_7, B_7, and B_8

n	k	A_7	B_7	B_8
2	2	2.3937	0.0000	2.9409
2	3	2.3025	0.0000	2.8289
2	4	2.2568	0.0000	2.7727
2	5	2.2294	0.0000	2.7391
2	6	2.2112	0.0000	2.7167
2	7	2.1982	0.0000	2.7008
2	8	2.1885	0.0000	2.6888
2	9	2.1809	0.0000	2.6796
2	10	2.1749	0.0000	2.6722
2	11	2.1700	0.0000	2.6661
2	12	2.1659	0.0000	2.6611
2	13	2.1625	0.0000	2.6569
2	14	2.1595	0.0000	2.6532
2	15	2.1569	0.0000	2.6501
2	16	2.1547	0.0000	2.6473
2	17	2.1527	0.0000	2.6449
2	18	2.1510	0.0000	2.6427
2	19	2.1494	0.0000	2.6408
2	20	2.1480	0.0000	2.6391
2	21	2.1467	0.0000	2.6375
2	22	2.1456	0.0000	2.6361
2	23	2.1445	0.0000	2.6348
2	24	2.1435	0.0000	2.6336
2	25	2.1426	0.0000	2.6325
3	2	1.8426	0.0000	2.4213
3	3	1.8054	0.0000	2.3724
3	4	1.7869	0.0000	2.3480
3	5	1.7758	0.0000	2.3335

n	k	A_7	B_7	B_8
3	6	1.7685	0.0000	2.3238
3	7	1.7632	0.0000	2.3170
3	8	1.7593	0.0000	2.3118
3	9	1.7563	0.0000	2.3078
3	10	1.7538	0.0000	2.3046
3	11	1.7518	0.0000	2.3020
3	12	1.7502	0.0000	2.2998
3	13	1.7488	0.0000	2.2980
3	14	1.7476	0.0000	2.2964
3	15	1.7465	0.0000	2.2950
3	16	1.7456	0.0000	2.2938
3	17	1.7448	0.0000	2.2928
3	18	1.7441	0.0000	2.2918
3	19	1.7435	0.0000	2.2910
3	20	1.7429	0.0000	2.2902
3	21	1.7424	0.0000	2.2896
3	22	1.7419	0.0000	2.2889
3	23	1.7415	0.0000	2.2884
3	24	1.7411	0.0000	2.2879
3	25	1.7407	0.0000	2.2874
4	2	1.5635	0.0000	2.1762
4	3	1.5422	0.0000	2.1464
4	4	1.5315	0.0000	2.1316
4	5	1.5252	0.0000	2.1228
4	6	1.5210	0.0000	2.1169
4	7	1.5180	0.0000	2.1127
4	8	1.5157	0.0000	2.1096
4	9	1.5140	0.0000	2.1072

Note: Values are based on *k* subgroups of size *n*.

n	k	A_7	B_7	B_8
4	10	1.5125	0.0000	2.1052
4	11	1.5114	0.0000	2.1036
4	12	1.5105	0.0000	2.1023
4	13	1.5096	0.0000	2.1012
4	14	1.5090	0.0000	2.1002
4	15	1.5084	0.0000	2.0994
4	16	1.5078	0.0000	2.0987
4	17	1.5074	0.0000	2.0980
4	18	1.5070	0.0000	2.0974
4	19	1.5066	0.0000	2.0969
4	20	1.5063	0.0000	2.0965
4	21	1.5060	0.0000	2.0961
4	22	1.5057	0.0000	2.0957
4	23	1.5054	0.0000	2.0953
4	24	1.5052	0.0000	2.0950
4	25	1.4955	0.0000	2.0815
5	2	1.3841	0.0000	2.0258
5	3	1.3699	0.0000	2.0049
5	4	1.3628	0.0000	1.9945
5	5	1.3585	0.0000	1.9883
5	6	1.3557	0.0000	1.9842
5	7	1.3537	0.0000	1.9812
5	8	1.3522	0.0000	1.9790
5	9	1.3510	0.0000	1.9773
5	10	1.3501	0.0000	1.9759
5	11	1.3493	0.0000	1.9748
5	12	1.3486	0.0000	1.9739
5	13	1.3481	0.0000	1.9731
5	14	1.3476	0.0000	1.9724
5	15	1.3472	0.0000	1.9718
5	16	1.3469	0.0000	1.9713
5	17	1.3466	0.0000	1.9709
5	18	1.3463	0.0000	1.9705
5	19	1.3461	0.0000	1.9701
5	20	1.3458	0.0000	1.9698
5	21	1.3456	0.0000	1.9695
5	22	1.3455	0.0000	1.9692
5	23	1.3453	0.0000	1.9690
5	24	1.3451	0.0000	1.9687

n	k	A_7	B_7	B_8
5	25	1.3450	0.0000	1.9685
6	2	1.2557	0.0296	1.9215
6	3	1.2453	0.0294	1.9056
6	4	1.2401	0.0293	1.8977
6	5	1.2371	0.0292	1.8930
6	6	1.2350	0.0291	1.8899
6	7	1.2335	0.0291	1.8876
6	8	1.2324	0.0291	1.8859
6	9	1.2316	0.0291	1.8846
6	10	1.2309	0.0290	1.8836
6	11	1.2303	0.0290	1.8827
6	12	1.2299	0.0290	1.8820
6	13	1.2295	0.0290	1.8814
6	14	1.2291	0.0290	1.8809
6	15	1.2288	0.0290	1.8804
6	16	1.2286	0.0290	1.8800
6	17	1.2284	0.0290	1.8797
6	18	1.2282	0.0290	1.8794
6	19	1.2280	0.0290	1.8791
6	20	1.2278	0.0290	1.8789
6	21	1.2277	0.0290	1.8786
6	22	1.2275	0.0290	1.8784
6	23	1.2274	0.0290	1.8783
6	24	1.2273	0.0290	1.8781
6	25	1.2272	0.0289	1.8779
7	2	1.1577	0.1153	1.8438
7	3	1.1497	0.1145	1.8311
7	4	1.1458	0.1141	1.8247
7	5	1.1434	0.1138	1.8209
7	6	1.1418	0.1137	1.8184
7	7	1.1407	0.1136	1.8166
7	8	1.1398	0.1135	1.8153
7	9	1.1392	0.1134	1.8142
7	10	1.1386	0.1134	1.8134
7	11	1.1382	0.1133	1.8127
7	12	1.1378	0.1133	1.8121
7	13	1.1375	0.1133	1.8116
7	14	1.1373	0.1132	1.8112
7	15	1.1370	0.1132	1.8109

Note: Values are based on k subgroups of size n.

n	k	A_7	B_7	B_8
7	16	1.1369	0.1132	1.8105
7	17	1.1367	0.1132	1.8103
7	18	1.1365	0.1132	1.8100
7	19	1.1364	0.1132	1.8098
7	20	1.1363	0.1131	1.8096
7	21	1.1361	0.1131	1.8094
7	22	1.1360	0.1131	1.8093
7	23	1.1359	0.1131	1.8091
7	24	1.1359	0.1131	1.8090
7	25	1.1358	0.1131	1.8088
8	2	1.0798	0.1818	1.7830
8	3	1.0734	0.1808	1.7724
8	4	1.0702	0.1802	1.7671
8	5	1.0683	0.1799	1.7640
8	6	1.0670	0.1797	1.7619
8	7	1.0661	0.1795	1.7604
8	8	1.0654	0.1794	1.7593
8	9	1.0649	0.1793	1.7584
8	10	1.0645	0.1793	1.7577
8	11	1.0641	0.1792	1.7571
8	12	1.0638	0.1791	1.7567
8	13	1.0636	0.1791	1.7563
8	14	1.0634	0.1791	1.7559
8	15	1.0632	0.1790	1.7556
8	16	1.0630	0.1790	1.7554
8	17	1.0629	0.1790	1.7551
8	18	1.0628	0.1790	1.7549
8	19	1.0627	0.1790	1.7547
8	20	1.0626	0.1789	1.7546
8	21	1.0625	0.1789	1.7544
8	22	1.0624	0.1789	1.7543
8	23	1.0623	0.1789	1.7542
8	24	1.0622	0.1789	1.7541
8	25	1.0622	0.1789	1.7539
9	2	1.0157	0.2354	1.7337
9	3	1.0105	0.2342	1.7247
9	4	1.0078	0.2336	1.7202
9	5	1.0063	0.2332	1.7175
9	6	1.0052	0.2330	1.7157

n	k	A_7	B_7	B_8
9	7	1.0045	0.2328	1.7145
9	8	1.0039	0.2327	1.7135
9	9	1.0035	0.2326	1.7128
9	10	1.0031	0.2325	1.7122
9	11	1.0028	0.2325	1.7117
9	12	1.0026	0.2324	1.7113
9	13	1.0024	0.2324	1.7109
9	14	1.0022	0.2323	1.7106
9	15	1.0021	0.2323	1.7104
9	16	1.0020	0.2322	1.7102
9	17	1.0018	0.2322	1.7100
9	18	1.0017	0.2322	1.7098
9	19	1.0016	0.2322	1.7096
9	20	1.0016	0.2322	1.7095
9	21	1.0015	0.2321	1.7094
9	22	1.0014	0.2321	1.7093
9	23	1.0014	0.2321	1.7091
9	24	1.0013	0.2321	1.7091
9	25	1.0013	0.2321	1.7090
10	2	0.9619	0.2798	1.6927
10	3	0.9575	0.2785	1.6849
10	4	0.9553	0.2779	1.6810
10	5	0.9540	0.2775	1.6787
10	6	0.9531	0.2772	1.6771
10	7	0.9525	0.2770	1.6760
10	8	0.9520	0.2769	1.6752
10	9	0.9516	0.2768	1.6745
10	10	0.9513	0.2767	1.6740
10	11	0.9511	0.2766	1.6736
10	12	0.9509	0.2766	1.6732
10	13	0.9507	0.2765	1.6729
10	14	0.9506	0.2765	1.6727
10	15	0.9504	0.2765	1.6725
10	16	0.9503	0.2764	1.6723
10	17	0.9502	0.2764	1.6721
10	18	0.9501	0.2764	1.6719
10	19	0.9501	0.2764	1.6718
10	20	0.9500	0.2763	1.6717
10	21	0.9499	0.2763	1.6716

Note: Values are based on *k* subgroups of size *n.*

n	k	A_7	B_7	B_8
10	22	0.9499	0.2763	1.6715
10	23	0.9498	0.2763	1.6714
10	24	0.9498	0.2763	1.6713
10	25	0.9497	0.2763	1.6712
11	2	0.9159	0.3173	1.6579
11	3	0.9121	0.3160	1.6510
11	4	0.9102	0.3153	1.6476
11	5	0.9091	0.3149	1.6455
11	6	0.9083	0.3147	1.6442
11	7	0.9078	0.3145	1.6432
11	8	0.9074	0.3143	1.6425
11	9	0.9071	0.3142	1.6419
11	10	0.9068	0.3141	1.6414
11	11	0.9066	0.3141	1.6411
11	12	0.9064	0.3140	1.6408
11	13	0.9063	0.3140	1.6405
11	14	0.9062	0.3139	1.6403
11	15	0.9060	0.3139	1.6401
11	16	0.9059	0.3139	1.6399
11	17	0.9059	0.3138	1.6397
11	18	0.9058	0.3138	1.6396
11	19	0.9057	0.3138	1.6395
11	20	0.9057	0.3138	1.6394
11	21	0.9056	0.3137	1.6393
11	22	0.9056	0.3137	1.6392
11	23	0.9055	0.3137	1.6391
11	24	0.9055	0.3137	1.6390
11	25	0.9054	0.3137	1.6390
12	2	0.8759	0.3495	1.6279
12	3	0.8726	0.3482	1.6218
12	4	0.8710	0.3475	1.6187
12	5	0.8700	0.3472	1.6169
12	6	0.8693	0.3469	1.6156
12	7	0.8688	0.3467	1.6148
12	8	0.8685	0.3466	1.6141
12	9	0.8682	0.3465	1.6136
12	10	0.8680	0.3464	1.6132
12	11	0.8678	0.3463	1.6129
12	12	0.8677	0.3462	1.6126

n	k	A_7	B_7	B_8
12	13	0.8675	0.3462	1.6124
12	14	0.8674	0.3461	1.6122
12	15	0.8673	0.3461	1.6120
12	16	0.8673	0.3461	1.6118
12	17	0.8672	0.3460	1.6117
12	18	0.8671	0.3460	1.6116
12	19	0.8671	0.3460	1.6115
12	20	0.8670	0.3460	1.6114
12	21	0.8670	0.3460	1.6113
12	22	0.8669	0.3459	1.6112
12	23	0.8669	0.3459	1.6111
12	24	0.8668	0.3459	1.6111
12	25	0.8668	0.3459	1.6110
13	2	0.8408	0.3776	1.6017
13	3	0.8378	0.3763	1.5962
13	4	0.8364	0.3756	1.5934
13	5	0.8355	0.3753	1.5917
13	6	0.8349	0.3750	1.5906
13	7	0.8345	0.3748	1.5898
13	8	0.8342	0.3747	1.5892
13	9	0.8340	0.3746	1.5888
13	10	0.8338	0.3745	1.5884
13	11	0.8336	0.3744	1.5881
13	12	0.8335	0.3743	1.5879
13	13	0.8334	0.3743	1.5877
13	14	0.8333	0.3743	1.5875
13	15	0.8332	0.3742	1.5873
13	16	0.8331	0.3742	1.5872
13	17	0.8331	0.3742	1.5871
13	18	0.8330	0.3741	1.5869
13	19	0.8330	0.3741	1.5869
13	20	0.8329	0.3741	1.5868
13	21	0.8329	0.3741	1.5867
13	22	0.8328	0.3741	1.5866
13	23	0.8328	0.3740	1.5865
13	24	0.8328	0.3740	1.5865
13	25	0.8327	0.3740	1.5864
14	2	0.8095	0.4024	1.5785
14	3	0.8069	0.4011	1.5735

Note: Values are based on k subgroups of size n.

n	*k*	A_7	B_7	B_8
14	4	0.8056	0.4004	1.5710
14	5	0.8049	0.4001	1.5695
14	6	0.8044	0.3998	1.5684
14	7	0.8040	0.3996	1.5677
14	8	0.8037	0.3995	1.5672
14	9	0.8035	0.3994	1.5668
14	10	0.8033	0.3993	1.5664
14	11	0.8032	0.3992	1.5662
14	12	0.8031	0.3992	1.5659
14	13	0.8030	0.3991	1.5657
14	14	0.8029	0.3991	1.5656
14	15	0.8028	0.3990	1.5654
14	16	0.8027	0.3990	1.5653
14	17	0.8027	0.3990	1.5652
14	18	0.8026	0.3989	1.5651
14	19	0.8026	0.3989	1.5650
14	20	0.8026	0.3989	1.5649
14	21	0.8025	0.3989	1.5649
14	22	0.8025	0.3989	1.5648
14	23	0.8025	0.3988	1.5647
14	24	0.8024	0.3988	1.5647
14	25	0.8024	0.3988	1.5646
15	2	0.7815	0.4244	1.5578

n	*k*	A_7	B_7	B_8
15	3	0.7792	0.4231	1.5532
15	4	0.7781	0.4225	1.5509
15	5	0.7774	0.4221	1.5495
15	6	0.7769	0.4219	1.5486
15	7	0.7766	0.4217	1.5479
15	8	0.7763	0.4216	1.5475
15	9	0.7761	0.4215	1.5471
15	10	0.7760	0.4214	1.5468
15	11	0.7759	0.4213	1.5465
15	12	0.7758	0.4213	1.5463
15	13	0.7757	0.4212	1.5461
15	14	0.7756	0.4212	1.5460
15	15	0.7755	0.4211	1.5458
15	16	0.7755	0.4211	1.5457
15	17	0.7754	0.4211	1.5456
15	18	0.7754	0.4210	1.5455
15	19	0.7753	0.4210	1.5455
15	20	0.7753	0.4210	1.5454
15	21	0.7753	0.4210	1.5453
15	22	0.7752	0.4210	1.5453
15	23	0.7752	0.4210	1.5452
15	24	0.7752	0.4209	1.5452
15	25	0.7752	0.4209	1.5452

Note: Values are based on *k* subgroups of size *n.*

Appendix 6

Factors for Estimating σ_X

n	c_2	c_4	d_2	d_3	d_4	n
2	0.5642	0.7979	1.128	0.8525	0.954	2
3	0.7236	0.8862	1.693	0.8884	1.588	3
4	0.7979	0.9213	2.059	0.8798	1.978	4
5	0.8407	0.9400	2.326	0.8641	2.257	5
6	0.8686	0.9515	2.534	0.8480	2.472	6
7	0.8882	0.9594	2.704	0.8332	2.645	7
8	0.9027	0.9650	2.847	0.8198	2.791	8
9	0.9139	0.9693	2.970	0.8078	2.915	9
10	0.9227	0.9727	3.078	0.7971	3.024	10
11	0.9300	0.9754	3.173	0.7873	3.121	11
12	0.9359	0.9776	3.258	0.7785	3.207	12
13	0.9410	0.9794	3.336	0.7704	3.285	13
14	0.9453	0.9810	3.407	0.7630	3.356	14
15	0.9490	0.9823	3.472	0.7562	3.422	15
16	0.9523	0.9835	3.532	0.7499	3.482	16
17	0.9551	0.9845	3.588	0.7441	3.538	17
18	0.9576	0.9854	3.640	0.7386	3.591	18
19	0.9599	0.9862	3.689	0.7335	3.640	19
20	0.9619	0.9869	3.735	0.7287	3.686	20
21	0.9638	0.9876	3.778	0.7272	3.730	21
22	0.9655	0.9882	3.819	0.7199	3.771	22
23	0.9670	0.9887	3.858	0.7159	3.811	23
24	0.9684	0.9892	3.895	0.7121	3.847	24
25	0.9695	0.9896	3.931	0.7084	3.883	25

Note: σ_x may be estimated from k subgroups of size n:

$$\frac{\bar{s}_{RMS}}{c_2} \qquad \frac{\bar{s}}{c_4}$$

$$\frac{\bar{R}}{d_2} \qquad \frac{\overline{mR}}{d_2}$$

$$\frac{\tilde{R}}{d_4}$$

Appendix 7

Control Charts Count Data

Chart	Centerline	LCL	UCL	Plot point
p	$\bar{p}=\frac{\sum_{i=1}^{k}D_i}{\sum_{i=1}^{k}n_i}$	$LCL_p=\bar{p}-3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$	$UCL_p=\bar{p}+3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$	$p_i=\frac{D_i}{n_i}$
np	$n\bar{p}=n\frac{\sum_{i=1}^{k}D_i}{\sum_{i=1}^{k}n_i}$	$LCL_{np}=n\bar{p}-3\sqrt{n\bar{p}(1-\bar{p})}$	$UCL_{np}=n\bar{p}+3\sqrt{n\bar{p}(1-\bar{p})}$	D_i
c	$\bar{c}=\frac{\sum_{i=1}^{k}c_i}{k}$	$LCL_c=\bar{c}-3\sqrt{\bar{c}}$	$UCL_c=\bar{c}+3\sqrt{\bar{c}}$	c_i
u	$\bar{u}=\frac{\sum_{i=1}^{k}c_i}{\sum_{i=1}^{k}n_i}$	$LCL_u=\bar{u}-3\sqrt{\frac{\bar{u}}{n_i}}$	$UCL_u=\bar{u}+3\sqrt{\frac{\bar{u}}{n_i}}$	$u_i=\frac{c_i}{n_i}$

Notes:

1. n = Constant sample size of each subgroup
2. n_i = Sample size of the ith subgroup
3. k = Number of subgroups
4. D_i = Number of *defective* (nonconforming) units in the ith subgroup
5. c_i = Number of *defects* in the ith subgroup
6. u_i = Number of *defects*/unit in the ith subgroup

Appendix 8

Binomial Distribution Table

$$\Pr(X = x) = f(n; x, p) = \binom{n}{k} p^k (1-p)^{n-k}$$

		Probability of *x* occurrences in a sample of size *n*																				
n	***x***	**0.05**	**0.10**	**0.15**	**0.20**	**0.25**	**0.30**	**0.35**	**0.40**	**0.45**	**0.50**	**0.55**	**0.60**	**0.65**	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	***n***	***x***
1	**0**	0.9500	0.9000	0.8500	0.8000	0.7500	0.7000	0.6500	0.6000	0.5500	0.5000	0.4500	0.4000	0.3500	0.3000	0.2500	0.2000	0.1500	0.1000	0.0500	**1**	**0**
	1	0.0500	0.1000	0.1500	0.2000	0.2500	0.3000	0.3500	0.4000	0.4500	0.5000	0.5500	0.6000	0.6500	0.7000	0.7500	0.8000	0.8500	0.9000	0.9500		**1**
2	**0**	0.9025	0.8100	0.7225	0.6400	0.5625	0.4900	0.4225	0.3600	0.3025	0.2500	0.2025	0.1600	0.1225	0.0900	0.0625	0.0400	0.0225	0.0100	0.0025	**2**	**0**
	1	0.0950	0.1800	0.2550	0.3200	0.3750	0.4200	0.4550	0.4800	0.4950	0.5000	0.4950	0.4800	0.4550	0.4200	0.3750	0.3200	0.2550	0.1800	0.0950		**1**
	2	0.0025	0.0100	0.0225	0.0400	0.0625	0.0900	0.1225	0.1600	0.2025	0.2500	0.3025	0.3600	0.4225	0.4900	0.5625	0.6400	0.7225	0.8100	0.9025		**2**
3	**0**	0.8574	0.7290	0.6141	0.5120	0.4219	0.3430	0.2746	0.2160	0.1664	0.1250	0.0911	0.0640	0.0429	0.0270	0.0156	0.0080	0.0034	0.0010	0.0001	**3**	**0**
	1	0.1354	0.2430	0.3251	0.3840	0.4219	0.4410	0.4436	0.4320	0.4084	0.3750	0.3341	0.2880	0.2389	0.1890	0.1406	0.0960	0.0574	0.0270	0.0071		**1**
	2	0.0071	0.0270	0.0574	0.0960	0.1406	0.1890	0.2389	0.2880	0.3341	0.3750	0.4084	0.4320	0.4436	0.4410	0.4219	0.3840	0.3251	0.2430	0.1354		**2**
	3	0.0001	0.0010	0.0034	0.0080	0.0156	0.0270	0.0429	0.0640	0.0911	0.1250	0.1664	0.2160	0.2746	0.3430	0.4219	0.5120	0.6141	0.7290	0.8574		**3**
4	**0**	0.8145	0.6561	0.5220	0.4096	0.3164	0.2401	0.1785	0.1296	0.0915	0.0625	0.0410	0.0256	0.0150	0.0081	0.0039	0.0016	0.0005	0.0001	0.0000	**4**	**0**
	1	0.1715	0.2916	0.3685	0.4096	0.4219	0.4116	0.3845	0.3456	0.2995	0.2500	0.2005	0.1536	0.1115	0.0756	0.0469	0.0256	0.0115	0.0036	0.0005		**1**
	2	0.0135	0.0486	0.0975	0.1536	0.2109	0.2646	0.3105	0.3456	0.3675	0.3750	0.3675	0.3456	0.3105	0.2646	0.2109	0.1536	0.0975	0.0486	0.0135		**2**
	3	0.0005	0.0036	0.0115	0.0256	0.0469	0.0756	0.1115	0.1536	0.2005	0.2500	0.2995	0.3456	0.3845	0.4116	0.4219	0.4096	0.3685	0.2916	0.1715		**3**
	4	0.0000	0.0001	0.0005	0.0016	0.0039	0.0081	0.0150	0.0256	0.0410	0.0625	0.0915	0.1296	0.1785	0.2401	0.3164	0.4096	0.5220	0.6561	0.8145		**4**
5	**0**	0.7738	0.5905	0.4437	0.3277	0.2373	0.1681	0.1160	0.0778	0.0503	0.0313	0.0185	0.0102	0.0053	0.0024	0.0010	0.0003	0.0001	0.0000	0.0000	**5**	**0**
	1	0.2036	0.3281	0.3915	0.4096	0.3955	0.3602	0.3124	0.2592	0.2059	0.1563	0.1128	0.0768	0.0488	0.0284	0.0146	0.0064	0.0022	0.0005	0.0000		**1**
	2	0.0214	0.0729	0.1382	0.2048	0.2637	0.3087	0.3364	0.3456	0.3369	0.3125	0.2757	0.2304	0.1811	0.1323	0.0879	0.0512	0.0244	0.0081	0.0011		**2**
	3	0.0011	0.0081	0.0244	0.0512	0.0879	0.1323	0.1811	0.2304	0.2757	0.3125	0.3369	0.3456	0.3364	0.3087	0.2637	0.2048	0.1382	0.0729	0.0214		**3**
	4	0.0000	0.0005	0.0022	0.0064	0.0146	0.0284	0.0488	0.0768	0.1128	0.1563	0.2059	0.2592	0.3124	0.3602	0.3955	0.4096	0.3915	0.3281	0.2036		**4**
	5	0.0000	0.0000	0.0001	0.0003	0.0010	0.0024	0.0053	0.0102	0.0185	0.0313	0.0503	0.0778	0.1160	0.1681	0.2373	0.3277	0.4437	0.5905	0.7738		**5**

$$\Pr(X = x) = f(n; x, p) = \binom{n}{k} p^k (1-p)^{n-k}$$

		Probability of *x* occurrences in a sample of size *n*																				
n	**x**	**0.05**	**0.10**	**0.15**	**0.20**	**0.25**	**0.30**	**0.35**	**0.40**	**0.45**	**0.50**	**0.55**	**0.60**	**0.65**	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	**n**	**x**
6	**0**	0.7351	0.5314	0.3771	0.2621	0.1780	0.1176	0.0754	0.0467	0.0277	0.0156	0.0083	0.0041	0.0018	0.0007	0.0002	0.0001	0.0000	0.0000	0.0000	**6**	**0**
	1	0.2321	0.3543	0.3993	0.3932	0.3560	0.3025	0.2437	0.1866	0.1359	0.0938	0.0609	0.0369	0.0205	0.0102	0.0044	0.0015	0.0004	0.0001	0.0000		**1**
	2	0.0305	0.0984	0.1762	0.2458	0.2966	0.3241	0.3280	0.3110	0.2780	0.2344	0.1861	0.1382	0.0951	0.0595	0.0330	0.0154	0.0055	0.0012	0.0001		**2**
	3	0.0021	0.0146	0.0415	0.0819	0.1318	0.1852	0.2355	0.2765	0.3032	0.3125	0.3032	0.2765	0.2355	0.1852	0.1318	0.0819	0.0415	0.0146	0.0021		**3**
	4	0.0001	0.0012	0.0055	0.0154	0.0330	0.0595	0.0951	0.1382	0.1861	0.2344	0.2780	0.3110	0.3280	0.3241	0.2966	0.2458	0.1762	0.0984	0.0305		**4**
	5	0.0000	0.0001	0.0004	0.0015	0.0044	0.0102	0.0205	0.0369	0.0609	0.0938	0.1359	0.1866	0.2437	0.3025	0.3560	0.3932	0.3993	0.3543	0.2321		**5**
	6	0.0000	0.0000	0.0000	0.0001	0.0002	0.0007	0.0018	0.0041	0.0083	0.0156	0.0277	0.0467	0.0754	0.1176	0.1780	0.2621	0.3771	0.5314	0.7351		**6**
7	**0**	0.6983	0.4783	0.3206	0.2097	0.1335	0.0824	0.0490	0.0280	0.0152	0.0078	0.0037	0.0016	0.0006	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	**7**	**0**
	1	0.2573	0.3720	0.3960	0.3670	0.3115	0.2471	0.1848	0.1306	0.0872	0.0547	0.0320	0.0172	0.0084	0.0036	0.0013	0.0004	0.0001	0.0000	0.0000		**1**
	2	0.0406	0.1240	0.2097	0.2753	0.3115	0.3177	0.2985	0.2613	0.2140	0.1641	0.1172	0.0774	0.0466	0.0250	0.0115	0.0043	0.0012	0.0002	0.0000		**2**
	3	0.0036	0.0230	0.0617	0.1147	0.1730	0.2269	0.2679	0.2903	0.2918	0.2734	0.2388	0.1935	0.1442	0.0972	0.0577	0.0287	0.0109	0.0026	0.0002		**3**
	4	0.0002	0.0026	0.0109	0.0287	0.0577	0.0972	0.1442	0.1935	0.2388	0.2734	0.2918	0.2903	0.2679	0.2269	0.1730	0.1147	0.0617	0.0230	0.0036		**4**
	5	0.0000	0.0002	0.0012	0.0043	0.0115	0.0250	0.0466	0.0774	0.1172	0.1641	0.2140	0.2613	0.2985	0.3177	0.3115	0.2753	0.2097	0.1240	0.0406		**5**
	6	0.0000	0.0000	0.0001	0.0004	0.0013	0.0036	0.0084	0.0172	0.0320	0.0547	0.0872	0.1306	0.1848	0.2471	0.3115	0.3670	0.3960	0.3720	0.2573		**6**
	7	0.0000	0.0000	0.0000	0.0000	0.0001	0.0002	0.0006	0.0016	0.0037	0.0078	0.0152	0.0280	0.0490	0.0824	0.1335	0.2097	0.3206	0.4783	0.6983		**7**
8	**0**	0.6634	0.4305	0.2725	0.1678	0.1001	0.0576	0.0319	0.0168	0.0084	0.0039	0.0017	0.0007	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	**8**	**0**
	1	0.2793	0.3826	0.3847	0.3355	0.2670	0.1977	0.1373	0.0896	0.0548	0.0313	0.0164	0.0079	0.0033	0.0012	0.0004	0.0001	0.0000	0.0000	0.0000		**1**
	2	0.0515	0.1488	0.2376	0.2936	0.3115	0.2965	0.2587	0.2090	0.1569	0.1094	0.0703	0.0413	0.0217	0.0100	0.0038	0.0011	0.0002	0.0000	0.0000		**2**
	3	0.0054	0.0331	0.0839	0.1468	0.2076	0.2541	0.2786	0.2787	0.2568	0.2188	0.1719	0.1239	0.0808	0.0467	0.0231	0.0092	0.0026	0.0004	0.0000		**3**
	4	0.0004	0.0046	0.0185	0.0459	0.0865	0.1361	0.1875	0.2322	0.2627	0.2734	0.2627	0.2322	0.1875	0.1361	0.0865	0.0459	0.0185	0.0046	0.0004		**4**
	5	0.0000	0.0004	0.0026	0.0092	0.0231	0.0467	0.0808	0.1239	0.1719	0.2188	0.2568	0.2787	0.2786	0.2541	0.2076	0.1468	0.0839	0.0331	0.0054		**5**
	6	0.0000	0.0000	0.0002	0.0011	0.0038	0.0100	0.0217	0.0413	0.0703	0.1094	0.1569	0.2090	0.2587	0.2965	0.3115	0.2936	0.2376	0.1488	0.0515		**6**
	7	0.0000	0.0000	0.0000	0.0001	0.0004	0.0012	0.0033	0.0079	0.0164	0.0313	0.0548	0.0896	0.1373	0.1977	0.2670	0.3355	0.3847	0.3826	0.2793		**7**
	8	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0002	0.0007	0.0017	0.0039	0.0084	0.0168	0.0319	0.0576	0.1001	0.1678	0.2725	0.4305	0.6634		**8**

$$\Pr(X = x) = f(n; x, p) = \binom{n}{k} p^k (1-p)^{n-k}$$

		Probability of *x* occurrences in a sample of size *n*																				
n	***x***	**0.05**	**0.10**	**0.15**	**0.20**	**0.25**	**0.30**	**0.35**	**0.40**	**0.45**	**0.50**	**0.55**	**0.60**	**0.65**	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	***n***	***x***
9	**0**	0.6302	0.3874	0.2316	0.1342	0.0751	0.0404	0.0207	0.0101	0.0046	0.0020	0.0008	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**9**	**0**
	1	0.2985	0.3874	0.3679	0.3020	0.2253	0.1556	0.1004	0.0605	0.0339	0.0176	0.0083	0.0035	0.0013	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000		**1**
	2	0.0629	0.1722	0.2597	0.3020	0.3003	0.2668	0.2162	0.1612	0.1110	0.0703	0.0407	0.0212	0.0098	0.0039	0.0012	0.0003	0.0000	0.0000	0.0000		**2**
	3	0.0077	0.0446	0.1069	0.1762	0.2336	0.2668	0.2716	0.2508	0.2119	0.1641	0.1160	0.0743	0.0424	0.0210	0.0087	0.0028	0.0006	0.0001	0.0000		**3**
	4	0.0006	0.0074	0.0283	0.0661	0.1168	0.1715	0.2194	0.2508	0.2600	0.2461	0.2128	0.1672	0.1181	0.0735	0.0389	0.0165	0.0050	0.0008	0.0000		**4**
	5	0.0000	0.0008	0.0050	0.0165	0.0389	0.0735	0.1181	0.1672	0.2128	0.2461	0.2600	0.2508	0.2194	0.1715	0.1168	0.0661	0.0283	0.0074	0.0006		**5**
	6	0.0000	0.0001	0.0006	0.0028	0.0087	0.0210	0.0424	0.0743	0.1160	0.1641	0.2119	0.2508	0.2716	0.2668	0.2336	0.1762	0.1069	0.0446	0.0077		**6**
	7	0.0000	0.0000	0.0000	0.0003	0.0012	0.0039	0.0098	0.0212	0.0407	0.0703	0.1110	0.1612	0.2162	0.2668	0.3003	0.3020	0.2597	0.1722	0.0629		**7**
	8	0.0000	0.0000	0.0000	0.0000	0.0001	0.0004	0.0013	0.0035	0.0083	0.0176	0.0339	0.0605	0.1004	0.1556	0.2253	0.3020	0.3679	0.3874	0.2985		**8**
	9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0003	0.0008	0.0020	0.0046	0.0101	0.0207	0.0404	0.0751	0.1342	0.2316	0.3874	0.6302		**9**
10	**0**	0.5987	0.3487	0.1969	0.1074	0.0563	0.0282	0.0135	0.0060	0.0025	0.0010	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**10**	**0**
	1	0.3151	0.3874	0.3474	0.2684	0.1877	0.1211	0.0725	0.0403	0.0207	0.0098	0.0042	0.0016	0.0005	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000		**1**
	2	0.0746	0.1937	0.2759	0.3020	0.2816	0.2335	0.1757	0.1209	0.0763	0.0439	0.0229	0.0106	0.0043	0.0014	0.0004	0.0001	0.0000	0.0000	0.0000		**2**
	3	0.0105	0.0574	0.1298	0.2013	0.2503	0.2668	0.2522	0.2150	0.1665	0.1172	0.0746	0.0425	0.0212	0.0090	0.0031	0.0008	0.0001	0.0000	0.0000		**3**
	4	0.0010	0.0112	0.0401	0.0881	0.1460	0.2001	0.2377	0.2508	0.2384	0.2051	0.1596	0.1115	0.0689	0.0368	0.0162	0.0055	0.0012	0.0001	0.0000		**4**
	5	0.0001	0.0015	0.0085	0.0264	0.0584	0.1029	0.1536	0.2007	0.2340	0.2461	0.2340	0.2007	0.1536	0.1029	0.0584	0.0264	0.0085	0.0015	0.0001		**5**
	6	0.0000	0.0001	0.0012	0.0055	0.0162	0.0368	0.0689	0.1115	0.1596	0.2051	0.2384	0.2508	0.2377	0.2001	0.1460	0.0881	0.0401	0.0112	0.0010		**6**
	7	0.0000	0.0000	0.0001	0.0008	0.0031	0.0090	0.0212	0.0425	0.0746	0.1172	0.1665	0.2150	0.2522	0.2668	0.2503	0.2013	0.1298	0.0574	0.0105		**7**
	8	0.0000	0.0000	0.0000	0.0001	0.0004	0.0014	0.0043	0.0106	0.0229	0.0439	0.0763	0.1209	0.1757	0.2335	0.2816	0.3020	0.2759	0.1937	0.0746		**8**
	9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0005	0.0016	0.0042	0.0098	0.0207	0.0403	0.0725	0.1211	0.1877	0.2684	0.3474	0.3874	0.3151		**9**
	10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0003	0.0010	0.0025	0.0060	0.0135	0.0282	0.0563	0.1074	0.1969	0.3487	0.5987		**10**

Appendix 9

Cumulative Binomial Distribution Table

$$\Pr(X \le x) = F(n; x, p) = \sum_{k=0}^{x} \binom{n}{x} p^k (1-p)^{n-k}$$

		Probability of ≤*x* occurrences in a sample of size *n*																				
n	***x***	**0.05**	**0.10**	**0.15**	**0.20**	**0.25**	**0.30**	**0.35**	**0.40**	**0.45**	**0.50**	**0.55**	**0.60**	**0.65**	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	***n***	***x***
1	**0**	0.9500	0.9000	0.8500	0.8000	0.7500	0.7000	0.6500	0.6000	0.5500	0.5000	0.4500	0.4000	0.3500	0.3000	0.2500	0.2000	0.1500	0.1000	0.0500	**1**	**0**
	1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000		**1**
2	**0**	0.9025	0.8100	0.7225	0.6400	0.5625	0.4900	0.4225	0.3600	0.3025	0.2500	0.2025	0.1600	0.1225	0.0900	0.0625	0.0400	0.0225	0.0100	0.0025	**2**	**0**
	1	0.9975	0.9900	0.9775	0.9600	0.9375	0.9100	0.8775	0.8400	0.7975	0.7500	0.6975	0.6400	0.5775	0.5100	0.4375	0.3600	0.2775	0.1900	0.0975		**1**
	2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000		**2**
3	**0**	0.8574	0.7290	0.6141	0.5120	0.4219	0.3430	0.2746	0.2160	0.1664	0.1250	0.0911	0.0640	0.0429	0.0270	0.0156	0.0080	0.0034	0.0010	0.0001	**3**	**0**
	1	0.9928	0.9720	0.9393	0.8960	0.8438	0.7840	0.7183	0.6480	0.5748	0.5000	0.4253	0.3520	0.2818	0.2160	0.1563	0.1040	0.0608	0.0280	0.0073		**1**
	2	0.9999	0.9990	0.9966	0.9920	0.9844	0.9730	0.9571	0.9360	0.9089	0.8750	0.8336	0.7840	0.7254	0.6570	0.5781	0.4880	0.3859	0.2710	0.1426		**2**
	3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000		**3**
4	**0**	0.8145	0.6561	0.5220	0.4096	0.3164	0.2401	0.1785	0.1296	0.0915	0.0625	0.0410	0.0256	0.0150	0.0081	0.0039	0.0016	0.0005	0.0001	0.0000	**4**	**0**
	1	0.9860	0.9477	0.8905	0.8192	0.7383	0.6517	0.5630	0.4752	0.3910	0.3125	0.2415	0.1792	0.1265	0.0837	0.0508	0.0272	0.0120	0.0037	0.0005		**1**
	2	0.9995	0.9963	0.9880	0.9728	0.9492	0.9163	0.8735	0.8208	0.7585	0.6875	0.6090	0.5248	0.4370	0.3483	0.2617	0.1808	0.1095	0.0523	0.0140		**2**
	3	1.0000	0.9999	0.9995	0.9984	0.9961	0.9919	0.9850	0.9744	0.9590	0.9375	0.9085	0.8704	0.8215	0.7599	0.6836	0.5904	0.4780	0.3439	0.1855		**3**
	4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000		**4**
5	**0**	0.7738	0.5905	0.4437	0.3277	0.2373	0.1681	0.1160	0.0778	0.0503	0.0313	0.0185	0.0102	0.0053	0.0024	0.0010	0.0003	0.0001	0.0000	0.0000	**5**	**0**
	1	0.9774	0.9185	0.8352	0.7373	0.6328	0.5282	0.4284	0.3370	0.2562	0.1875	0.1312	0.0870	0.0540	0.0308	0.0156	0.0067	0.0022	0.0005	0.0000		**1**
	2	0.9988	0.9914	0.9734	0.9421	0.8965	0.8369	0.7648	0.6826	0.5931	0.5000	0.4069	0.3174	0.2352	0.1631	0.1035	0.0579	0.0266	0.0086	0.0012		**2**
	3	1.0000	0.9995	0.9978	0.9933	0.9844	0.9692	0.9460	0.9130	0.8688	0.8125	0.7438	0.6630	0.5716	0.4718	0.3672	0.2627	0.1648	0.0815	0.0226		**3**
	4	1.0000	1.0000	0.9999	0.9997	0.9990	0.9976	0.9947	0.9898	0.9815	0.9688	0.9497	0.9222	0.8840	0.8319	0.7627	0.6723	0.5563	0.4095	0.2262		**4**
	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000		**5**

$$\Pr(X \le x) = F(n; x, p) = \sum_{k=0}^{x} \binom{n}{x} p^k (1-p)^{n-k}$$

		Probability of ≤*x* occurrences in a sample of size *n*																				
n	**x**	**0.05**	**0.10**	**0.15**	**0.20**	**0.25**	**0.30**	**0.35**	**0.40**	**0.45**	**0.50**	**0.55**	**0.60**	**0.65**	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	**n**	**x**
6	**0**	0.7351	0.5314	0.3771	0.2621	0.1780	0.1176	0.0754	0.0467	0.0277	0.0156	0.0083	0.0041	0.0018	0.0007	0.0002	0.0001	0.0000	0.0000	0.0000	**6**	**0**
	1	0.9672	0.8857	0.7765	0.6554	0.5339	0.4202	0.3191	0.2333	0.1636	0.1094	0.0692	0.0410	0.0223	0.0109	0.0046	0.0016	0.0004	0.0001	0.0000		**1**
	2	0.9978	0.9842	0.9527	0.9011	0.8306	0.7443	0.6471	0.5443	0.4415	0.3438	0.2553	0.1792	0.1174	0.0705	0.0376	0.0170	0.0059	0.0013	0.0001		**2**
	3	0.9999	0.9987	0.9941	0.9830	0.9624	0.9295	0.8826	0.8208	0.7447	0.6563	0.5585	0.4557	0.3529	0.2557	0.1694	0.0989	0.0473	0.0159	0.0022		**3**
	4	1.0000	0.9999	0.9996	0.9984	0.9954	0.9891	0.9777	0.9590	0.9308	0.8906	0.8364	0.7667	0.6809	0.5798	0.4661	0.3446	0.2235	0.1143	0.0328		**4**
	5	1.0000	1.0000	1.0000	0.9999	0.9998	0.9993	0.9982	0.9959	0.9917	0.9844	0.9723	0.9533	0.9246	0.8824	0.8220	0.7379	0.6229	0.4686	0.2649		**5**
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000		**6**
7	**0**	0.6983	0.4783	0.3206	0.2097	0.1335	0.0824	0.0490	0.0280	0.0152	0.0078	0.0037	0.0016	0.0006	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	**7**	**0**
	1	0.9556	0.8503	0.7166	0.5767	0.4449	0.3294	0.2338	0.1586	0.1024	0.0625	0.0357	0.0188	0.0090	0.0038	0.0013	0.0004	0.0001	0.0000	0.0000		**1**
	2	0.9962	0.9743	0.9262	0.8520	0.7564	0.6471	0.5323	0.4199	0.3164	0.2266	0.1529	0.0963	0.0556	0.0288	0.0129	0.0047	0.0012	0.0002	0.0000		**2**
	3	0.9998	0.9973	0.9879	0.9667	0.9294	0.8740	0.8002	0.7102	0.6083	0.5000	0.3917	0.2898	0.1998	0.1260	0.0706	0.0333	0.0121	0.0027	0.0002		**3**
	4	1.0000	0.9998	0.9988	0.9953	0.9871	0.9712	0.9444	0.9037	0.8471	0.7734	0.6836	0.5801	0.4677	0.3529	0.2436	0.1480	0.0738	0.0257	0.0038		**4**
	5	1.0000	1.0000	0.9999	0.9996	0.9987	0.9962	0.9910	0.9812	0.9643	0.9375	0.8976	0.8414	0.7662	0.6706	0.5551	0.4233	0.2834	0.1497	0.0444		**5**
	6	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9994	0.9984	0.9963	0.9922	0.9848	0.9720	0.9510	0.9176	0.8665	0.7903	0.6794	0.5217	0.3017		**6**
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000		**7**
8	**0**	0.6634	0.4305	0.2725	0.1678	0.1001	0.0576	0.0319	0.0168	0.0084	0.0039	0.0017	0.0007	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	**8**	**0**
	1	0.9428	0.8131	0.6572	0.5033	0.3671	0.2553	0.1691	0.1064	0.0632	0.0352	0.0181	0.0085	0.0036	0.0013	0.0004	0.0001	0.0000	0.0000	0.0000		**1**
	2	0.9942	0.9619	0.8948	0.7969	0.6785	0.5518	0.4278	0.3154	0.2201	0.1445	0.0885	0.0498	0.0253	0.0113	0.0042	0.0012	0.0002	0.0000	0.0000		**2**
	3	0.9996	0.9950	0.9786	0.9437	0.8862	0.8059	0.7064	0.5941	0.4770	0.3633	0.2604	0.1737	0.1061	0.0580	0.0273	0.0104	0.0029	0.0004	0.0000		**3**
	4	1.0000	0.9996	0.9971	0.9896	0.9727	0.9420	0.8939	0.8263	0.7396	0.6367	0.5230	0.4059	0.2936	0.1941	0.1138	0.0563	0.0214	0.0050	0.0004		**4**
	5	1.0000	1.0000	0.9998	0.9988	0.9958	0.9887	0.9747	0.9502	0.9115	0.8555	0.7799	0.6846	0.5722	0.4482	0.3215	0.2031	0.1052	0.0381	0.0058		**5**
	6	1.0000	1.0000	1.0000	0.9999	0.9996	0.9987	0.9964	0.9915	0.9819	0.9648	0.9368	0.8936	0.8309	0.7447	0.6329	0.4967	0.3428	0.1869	0.0572		**6**
	7	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9993	0.9983	0.9961	0.9916	0.9832	0.9681	0.9424	0.8999	0.8322	0.7275	0.5695	0.3366		**7**
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000		**8**

$$\Pr(X \le x) = F(n; x, p) = \sum_{k=0}^{x} \binom{n}{x} p^k (1-p)^{n-k}$$

		Probability of ≤*x* occurrences in a sample of size *n*																				
n	**x**	**0.05**	**0.10**	**0.15**	**0.20**	**0.25**	**0.30**	**0.35**	**0.40**	**0.45**	**0.50**	**0.55**	**0.60**	**0.65**	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**	**n**	**x**
9	**0**	0.6302	0.3874	0.2316	0.1342	0.0751	0.0404	0.0207	0.0101	0.0046	0.0020	0.0008	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**9**	**0**
	1	0.9288	0.7748	0.5995	0.4362	0.3003	0.1960	0.1211	0.0705	0.0385	0.0195	0.0091	0.0038	0.0014	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000		**1**
	2	0.9916	0.9470	0.8591	0.7382	0.6007	0.4628	0.3373	0.2318	0.1495	0.0898	0.0498	0.0250	0.0112	0.0043	0.0013	0.0003	0.0000	0.0000	0.0000		**2**
	3	0.9994	0.9917	0.9661	0.9144	0.8343	0.7297	0.6089	0.4826	0.3614	0.2539	0.1658	0.0994	0.0536	0.0253	0.0100	0.0031	0.0006	0.0001	0.0000		**3**
	4	1.0000	0.9991	0.9944	0.9804	0.9511	0.9012	0.8283	0.7334	0.6214	0.5000	0.3786	0.2666	0.1717	0.0988	0.0489	0.0196	0.0056	0.0009	0.0000		**4**
	5	1.0000	0.9999	0.9994	0.9969	0.9900	0.9747	0.9464	0.9006	0.8342	0.7461	0.6386	0.5174	0.3911	0.2703	0.1657	0.0856	0.0339	0.0083	0.0006		**5**
	6	1.0000	1.0000	1.0000	0.9997	0.9987	0.9957	0.9888	0.9750	0.9502	0.9102	0.8505	0.7682	0.6627	0.5372	0.3993	0.2618	0.1409	0.0530	0.0084		**6**
	7	1.0000	1.0000	1.0000	1.0000	0.9999	0.9996	0.9986	0.9962	0.9909	0.9805	0.9615	0.9295	0.8789	0.8040	0.6997	0.5638	0.4005	0.2252	0.0712		**7**
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9992	0.9980	0.9954	0.9899	0.9793	0.9596	0.9249	0.8658	0.7684	0.6126	0.3698		**8**
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000		**9**
10	**0**	0.5987	0.3487	0.1969	0.1074	0.0563	0.0282	0.0135	0.0060	0.0025	0.0010	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**10**	**0**
	1	0.9139	0.7361	0.5443	0.3758	0.2440	0.1493	0.0860	0.0464	0.0233	0.0107	0.0045	0.0017	0.0005	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000		**1**
	2	0.9885	0.9298	0.8202	0.6778	0.5256	0.3828	0.2616	0.1673	0.0996	0.0547	0.0274	0.0123	0.0048	0.0016	0.0004	0.0001	0.0000	0.0000	0.0000		**2**
	3	0.9990	0.9872	0.9500	0.8791	0.7759	0.6496	0.5138	0.3823	0.2660	0.1719	0.1020	0.0548	0.0260	0.0106	0.0035	0.0009	0.0001	0.0000	0.0000		**3**
	4	0.9999	0.9984	0.9901	0.9672	0.9219	0.8497	0.7515	0.6331	0.5044	0.3770	0.2616	0.1662	0.0949	0.0473	0.0197	0.0064	0.0014	0.0001	0.0000		**4**
	5	1.0000	0.9999	0.9986	0.9936	0.9803	0.9527	0.9051	0.8338	0.7384	0.6230	0.4956	0.3669	0.2485	0.1503	0.0781	0.0328	0.0099	0.0016	0.0001		**5**
	6	1.0000	1.0000	0.9999	0.9991	0.9965	0.9894	0.9740	0.9452	0.8980	0.8281	0.7340	0.6177	0.4862	0.3504	0.2241	0.1209	0.0500	0.0128	0.0010		**6**
	7	1.0000	1.0000	1.0000	0.9999	0.9996	0.9984	0.9952	0.9877	0.9726	0.9453	0.9004	0.8327	0.7384	0.6172	0.4744	0.3222	0.1798	0.0702	0.0115		**7**
	8	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9995	0.9983	0.9955	0.9893	0.9767	0.9536	0.9140	0.8507	0.7560	0.6242	0.4557	0.2639	0.0861		**8**
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9990	0.9975	0.9940	0.9865	0.9718	0.9437	0.8926	0.8031	0.6513	0.4013		**9**
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000		**10**

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Appendix 10

Poisson Distribution Table

$$\Pr(X = x) = f(x;\lambda) = \frac{\lambda^x}{x!}e^{-\lambda}$$

	Probability of x occurrences													
λ	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	λ
0.05	0.9512	0.0476	0.0012	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.05**
0.10	0.9048	0.0905	0.0045	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.10**
0.15	0.8607	0.1291	0.0097	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.15**
0.20	0.8187	0.1637	0.0164	0.0011	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.20**
0.25	0.7788	0.1947	0.0243	0.0020	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.25**
0.30	0.7408	0.2222	0.0333	0.0033	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.30**
0.35	0.7047	0.2466	0.0432	0.0050	0.0004	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.35**
0.40	0.6703	0.2681	0.0536	0.0072	0.0007	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.40**
0.45	0.6376	0.2869	0.0646	0.0097	0.0011	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.45**
0.50	0.6065	0.3033	0.0758	0.0126	0.0016	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.50**
0.55	0.5769	0.3173	0.0873	0.0160	0.0022	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.55**
0.60	0.5488	0.3293	0.0988	0.0198	0.0030	0.0004	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.60**
0.65	0.5220	0.3393	0.1103	0.0239	0.0039	0.0005	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.65**
0.70	0.4966	0.3476	0.1217	0.0284	0.0050	0.0007	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.70**
0.75	0.4724	0.3543	0.1329	0.0332	0.0062	0.0009	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.75**
0.80	0.4493	0.3595	0.1438	0.0383	0.0077	0.0012	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.80**
0.85	0.4274	0.3633	0.1544	0.0437	0.0093	0.0016	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.85**
0.90	0.4066	0.3659	0.1647	0.0494	0.0111	0.0020	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.90**
0.95	0.3867	0.3674	0.1745	0.0553	0.0131	0.0025	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	**0.95**
1.00	0.3679	0.3679	0.1839	0.0613	0.0153	0.0031	0.0005	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	**1.00**
1.10	0.3329	0.3662	0.2014	0.0738	0.0203	0.0045	0.0008	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	**1.10**
1.20	0.3012	0.3614	0.2169	0.0867	0.0260	0.0062	0.0012	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	**1.20**
1.30	0.2725	0.3543	0.2303	0.0998	0.0324	0.0084	0.0018	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	**1.30**
1.40	0.2466	0.3452	0.2417	0.1128	0.0395	0.0111	0.0026	0.0005	0.0001	0.0000	0.0000	0.0000	0.0000	**1.40**
1.50	0.2231	0.3347	0.2510	0.1255	0.0471	0.0141	0.0035	0.0008	0.0001	0.0000	0.0000	0.0000	0.0000	**1.50**
1.60	0.2019	0.3230	0.2584	0.1378	0.0551	0.0176	0.0047	0.0011	0.0002	0.0000	0.0000	0.0000	0.0000	**1.60**
1.70	0.1827	0.3106	0.2640	0.1496	0.0636	0.0216	0.0061	0.0015	0.0003	0.0001	0.0000	0.0000	0.0000	**1.70**
1.80	0.1653	0.2975	0.2678	0.1607	0.0723	0.0260	0.0078	0.0020	0.0005	0.0001	0.0000	0.0000	0.0000	**1.80**
1.90	0.1496	0.2842	0.2700	0.1710	0.0812	0.0309	0.0098	0.0027	0.0006	0.0001	0.0000	0.0000	0.0000	**1.90**
2.00	0.1353	0.2707	0.2707	0.1804	0.0902	0.0361	0.0120	0.0034	0.0009	0.0002	0.0000	0.0000	0.0000	**2.00**
2.10	0.1225	0.2572	0.2700	0.1890	0.0992	0.0417	0.0146	0.0044	0.0011	0.0003	0.0001	0.0000	0.0000	**2.10**
2.20	0.1108	0.2438	0.2681	0.1966	0.1082	0.0476	0.0174	0.0055	0.0015	0.0004	0.0001	0.0000	0.0000	**2.20**
2.30	0.1003	0.2306	0.2652	0.2033	0.1169	0.0538	0.0206	0.0068	0.0019	0.0005	0.0001	0.0000	0.0000	**2.30**
2.40	0.0907	0.2177	0.2613	0.2090	0.1254	0.0602	0.0241	0.0083	0.0025	0.0007	0.0002	0.0000	0.0000	**2.40**
2.50	0.0821	0.2052	0.2565	0.2138	0.1336	0.0668	0.0278	0.0099	0.0031	0.0009	0.0002	0.0000	0.0000	**2.50**
2.60	0.0743	0.1931	0.2510	0.2176	0.1414	0.0735	0.0319	0.0118	0.0038	0.0011	0.0003	0.0001	0.0000	**2.60**
2.70	0.0672	0.1815	0.2450	0.2205	0.1488	0.0804	0.0362	0.0139	0.0047	0.0014	0.0004	0.0001	0.0000	**2.70**

$$\Pr(X = x) = f(x;\lambda) = \frac{\lambda^x}{x!}e^{-\lambda}$$

	Probability of *x* occurrences													
λ	13	14	15	16	17	18	19	20	21	22	23	24	25	λ
0.05	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.05**
0.10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.10**
0.15	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.15**
0.20	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.20**
0.25	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.25**
0.30	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.30**
0.35	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.35**
0.40	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.40**
0.45	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.45**
0.50	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.50**
0.55	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.55**
0.60	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.60**
0.65	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.65**
0.70	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.70**
0.75	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.75**
0.80	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.80**
0.85	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.85**
0.90	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.90**
0.95	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**0.95**
1.00	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**1.00**
1.10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**1.10**
1.20	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**1.20**
1.30	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**1.30**
1.40	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**1.40**
1.50	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**1.50**
1.60	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**1.60**
1.70	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**1.70**
1.80	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**1.80**
1.90	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**1.90**
2.00	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**2.00**
2.10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**2.10**
2.20	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**2.20**
2.30	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**2.30**
2.40	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**2.40**
2.50	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**2.50**
2.60	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**2.60**
2.70	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**2.70**

$$\Pr(X = x) = f(x;\lambda) = \frac{\lambda^x}{x!}e^{-\lambda}$$

	Probability of *x* occurrences													
λ	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	λ
2.80	0.0608	0.1703	0.2384	0.2225	0.1557	0.0872	0.0407	0.0163	0.0057	0.0018	0.0005	0.0001	0.0000	**2.80**
2.90	0.0550	0.1596	0.2314	0.2237	0.1622	0.0940	0.0455	0.0188	0.0068	0.0022	0.0006	0.0002	0.0000	**2.90**
3.00	0.0498	0.1494	0.2240	0.2240	0.1680	0.1008	0.0504	0.0216	0.0081	0.0027	0.0008	0.0002	0.0001	**3.00**
3.10	0.0450	0.1397	0.2165	0.2237	0.1733	0.1075	0.0555	0.0246	0.0095	0.0033	0.0010	0.0003	0.0001	**3.10**
3.20	0.0408	0.1304	0.2087	0.2226	0.1781	0.1140	0.0608	0.0278	0.0111	0.0040	0.0013	0.0004	0.0001	**3.20**
3.30	0.0369	0.1217	0.2008	0.2209	0.1823	0.1203	0.0662	0.0312	0.0129	0.0047	0.0016	0.0005	0.0001	**3.30**
3.40	0.0334	0.1135	0.1929	0.2186	0.1858	0.1264	0.0716	0.0348	0.0148	0.0056	0.0019	0.0006	0.0002	**3.40**
3.50	0.0302	0.1057	0.1850	0.2158	0.1888	0.1322	0.0771	0.0385	0.0169	0.0066	0.0023	0.0007	0.0002	**3.50**
3.60	0.0273	0.0984	0.1771	0.2125	0.1912	0.1377	0.0826	0.0425	0.0191	0.0076	0.0028	0.0009	0.0003	**3.60**
3.70	0.0247	0.0915	0.1692	0.2087	0.1931	0.1429	0.0881	0.0466	0.0215	0.0089	0.0033	0.0011	0.0003	**3.70**
3.80	0.0224	0.0850	0.1615	0.2046	0.1944	0.1477	0.0936	0.0508	0.0241	0.0102	0.0039	0.0013	0.0004	**3.80**
3.90	0.0202	0.0789	0.1539	0.2001	0.1951	0.1522	0.0989	0.0551	0.0269	0.0116	0.0045	0.0016	0.0005	**3.90**
4.00	0.0183	0.0733	0.1465	0.1954	0.1954	0.1563	0.1042	0.0595	0.0298	0.0132	0.0053	0.0019	0.0006	**4.00**
4.10	0.0166	0.0679	0.1393	0.1904	0.1951	0.1600	0.1093	0.0640	0.0328	0.0150	0.0061	0.0023	0.0008	**4.10**
4.20	0.0150	0.0630	0.1323	0.1852	0.1944	0.1633	0.1143	0.0686	0.0360	0.0168	0.0071	0.0027	0.0009	**4.20**
4.30	0.0136	0.0583	0.1254	0.1798	0.1933	0.1662	0.1191	0.0732	0.0393	0.0188	0.0081	0.0032	0.0011	**4.30**
4.40	0.0123	0.0540	0.1188	0.1743	0.1917	0.1687	0.1237	0.0778	0.0428	0.0209	0.0092	0.0037	0.0013	**4.40**
4.50	0.0111	0.0500	0.1125	0.1687	0.1898	0.1708	0.1281	0.0824	0.0463	0.0232	0.0104	0.0043	0.0016	**4.50**
4.60	0.0101	0.0462	0.1063	0.1631	0.1875	0.1725	0.1323	0.0869	0.0500	0.0255	0.0118	0.0049	0.0019	**4.60**
4.70	0.0091	0.0427	0.1005	0.1574	0.1849	0.1738	0.1362	0.0914	0.0537	0.0281	0.0132	0.0056	0.0022	**4.70**
4.80	0.0082	0.0395	0.0948	0.1517	0.1820	0.1747	0.1398	0.0959	0.0575	0.0307	0.0147	0.0064	0.0026	**4.80**
4.90	0.0074	0.0365	0.0894	0.1460	0.1789	0.1753	0.1432	0.1002	0.0614	0.0334	0.0164	0.0073	0.0030	**4.90**
5.00	0.0067	0.0337	0.0842	0.1404	0.1755	0.1755	0.1462	0.1044	0.0653	0.0363	0.0181	0.0082	0.0034	**5.00**
5.10	0.0061	0.0311	0.0793	0.1348	0.1719	0.1753	0.1490	0.1086	0.0692	0.0392	0.0200	0.0093	0.0039	**5.10**
5.20	0.0055	0.0287	0.0746	0.1293	0.1681	0.1748	0.1515	0.1125	0.0731	0.0423	0.0220	0.0104	0.0045	**5.20**
5.30	0.0050	0.0265	0.0701	0.1239	0.1641	0.1740	0.1537	0.1163	0.0771	0.0454	0.0241	0.0116	0.0051	**5.30**
5.40	0.0045	0.0244	0.0659	0.1185	0.1600	0.1728	0.1555	0.1200	0.0810	0.0486	0.0262	0.0129	0.0058	**5.40**
5.50	0.0041	0.0225	0.0618	0.1133	0.1558	0.1714	0.1571	0.1234	0.0849	0.0519	0.0285	0.0143	0.0065	**5.50**
5.60	0.0037	0.0207	0.0580	0.1082	0.1515	0.1697	0.1584	0.1267	0.0887	0.0552	0.0309	0.0157	0.0073	**5.60**
5.70	0.0033	0.0191	0.0544	0.1033	0.1472	0.1678	0.1594	0.1298	0.0925	0.0586	0.0334	0.0173	0.0082	**5.70**
5.80	0.0030	0.0176	0.0509	0.0985	0.1428	0.1656	0.1601	0.1326	0.0962	0.0620	0.0359	0.0190	0.0092	**5.80**
5.90	0.0027	0.0162	0.0477	0.0938	0.1383	0.1632	0.1605	0.1353	0.0998	0.0654	0.0386	0.0207	0.0102	**5.90**
6.00	0.0025	0.0149	0.0446	0.0892	0.1339	0.1606	0.1606	0.1377	0.1033	0.0688	0.0413	0.0225	0.0113	**6.00**
6.10	0.0022	0.0137	0.0417	0.0848	0.1294	0.1579	0.1605	0.1399	0.1066	0.0723	0.0441	0.0244	0.0124	**6.10**
6.20	0.0020	0.0126	0.0390	0.0806	0.1249	0.1549	0.1601	0.1418	0.1099	0.0757	0.0469	0.0265	0.0137	**6.20**
6.30	0.0018	0.0116	0.0364	0.0765	0.1205	0.1519	0.1595	0.1435	0.1130	0.0791	0.0498	0.0285	0.0150	**6.30**
6.40	0.0017	0.0106	0.0340	0.0726	0.1162	0.1487	0.1586	0.1450	0.1160	0.0825	0.0528	0.0307	0.0164	**6.40**

$$\Pr(X = x) = f(x; \lambda) = \frac{\lambda^x}{x!} e^{-\lambda}$$

	Probability of *x* occurrences													
λ	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	λ
2.80	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**2.80**
2.90	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**2.90**
3.00	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**3.00**
3.10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**3.10**
3.20	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**3.20**
3.30	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**3.30**
3.40	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**3.40**
3.50	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**3.50**
3.60	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**3.60**
3.70	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**3.70**
3.80	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**3.80**
3.90	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**3.90**
4.00	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**4.00**
4.10	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**4.10**
4.20	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**4.20**
4.30	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**4.30**
4.40	0.0005	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**4.40**
4.50	0.0006	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**4.50**
4.60	0.0007	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**4.60**
4.70	0.0008	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**4.70**
4.80	0.0009	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**4.80**
4.90	0.0011	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**4.90**
5.00	0.0013	0.0005	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**5.00**
5.10	0.0015	0.0006	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**5.10**
5.20	0.0018	0.0007	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**5.20**
5.30	0.0021	0.0008	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**5.30**
5.40	0.0024	0.0009	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**5.40**
5.50	0.0028	0.0011	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**5.50**
5.60	0.0032	0.0013	0.0005	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**5.60**
5.70	0.0036	0.0015	0.0006	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**5.70**
5.80	0.0041	0.0017	0.0007	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**5.80**
5.90	0.0046	0.0019	0.0008	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**5.90**
6.00	0.0052	0.0022	0.0009	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**6.00**
6.10	0.0058	0.0025	0.0010	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**6.10**
6.20	0.0065	0.0029	0.0012	0.0005	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**6.20**
6.30	0.0073	0.0033	0.0014	0.0005	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**6.30**
6.40	0.0081	0.0037	0.0016	0.0006	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**6.40**

$$\Pr(X = x) = f(x;\lambda) = \frac{\lambda^x}{x!}e^{-\lambda}$$

	Probability of *x* occurrences													
λ	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	λ
6.50	0.0015	0.0098	0.0318	0.0688	0.1118	0.1454	0.1575	0.1462	0.1188	0.0858	0.0558	0.0330	0.0179	**6.50**
6.60	0.0014	0.0090	0.0296	0.0652	0.1076	0.1420	0.1562	0.1472	0.1215	0.0891	0.0588	0.0353	0.0194	**6.60**
6.70	0.0012	0.0082	0.0276	0.0617	0.1034	0.1385	0.1546	0.1480	0.1240	0.0923	0.0618	0.0377	0.0210	**6.70**
6.80	0.0011	0.0076	0.0258	0.0584	0.0992	0.1349	0.1529	0.1486	0.1263	0.0954	0.0649	0.0401	0.0227	**6.80**
6.90	0.0010	0.0070	0.0240	0.0552	0.0952	0.1314	0.1511	0.1489	0.1284	0.0985	0.0679	0.0426	0.0245	**6.90**
7.00	0.0009	0.0064	0.0223	0.0521	0.0912	0.1277	0.1490	0.1490	0.1304	0.1014	0.0710	0.0452	0.0263	**7.00**
7.10	0.0008	0.0059	0.0208	0.0492	0.0874	0.1241	0.1468	0.1489	0.1321	0.1042	0.0740	0.0478	0.0283	**7.10**
7.20	0.0007	0.0054	0.0194	0.0464	0.0836	0.1204	0.1445	0.1486	0.1337	0.1070	0.0770	0.0504	0.0303	**7.20**
7.30	0.0007	0.0049	0.0180	0.0438	0.0799	0.1167	0.1420	0.1481	0.1351	0.1096	0.0800	0.0531	0.0323	**7.30**
7.40	0.0006	0.0045	0.0167	0.0413	0.0764	0.1130	0.1394	0.1474	0.1363	0.1121	0.0829	0.0558	0.0344	**7.40**
7.50	0.0006	0.0041	0.0156	0.0389	0.0729	0.1094	0.1367	0.1465	0.1373	0.1144	0.0858	0.0585	0.0366	**7.50**
7.60	0.0005	0.0038	0.0145	0.0366	0.0696	0.1057	0.1339	0.1454	0.1381	0.1167	0.0887	0.0613	0.0388	**7.60**
7.70	0.0005	0.0035	0.0134	0.0345	0.0663	0.1021	0.1311	0.1442	0.1388	0.1187	0.0914	0.0640	0.0411	**7.70**
7.80	0.0004	0.0032	0.0125	0.0324	0.0632	0.0986	0.1282	0.1428	0.1392	0.1207	0.0941	0.0667	0.0434	**7.80**
7.90	0.0004	0.0029	0.0116	0.0305	0.0602	0.0951	0.1252	0.1413	0.1395	0.1224	0.0967	0.0695	0.0457	**7.90**
8.00	0.0003	0.0027	0.0107	0.0286	0.0573	0.0916	0.1221	0.1396	0.1396	0.1241	0.0993	0.0722	0.0481	**8.00**
8.10	0.0003	0.0025	0.0100	0.0269	0.0544	0.0882	0.1191	0.1378	0.1395	0.1256	0.1017	0.0749	0.0505	**8.10**
8.20	0.0003	0.0023	0.0092	0.0252	0.0517	0.0849	0.1160	0.1358	0.1392	0.1269	0.1040	0.0776	0.0530	**8.20**
8.30	0.0002	0.0021	0.0086	0.0237	0.0491	0.0816	0.1128	0.1338	0.1388	0.1280	0.1063	0.0802	0.0555	**8.30**
8.40	0.0002	0.0019	0.0079	0.0222	0.0466	0.0784	0.1097	0.1317	0.1382	0.1290	0.1084	0.0828	0.0579	**8.40**
8.50	0.0002	0.0017	0.0074	0.0208	0.0443	0.0752	0.1066	0.1294	0.1375	0.1299	0.1104	0.0853	0.0604	**8.50**
8.60	0.0002	0.0016	0.0068	0.0195	0.0420	0.0722	0.1034	0.1271	0.1366	0.1306	0.1123	0.0878	0.0629	**8.60**
8.70	0.0002	0.0014	0.0063	0.0183	0.0398	0.0692	0.1003	0.1247	0.1356	0.1311	0.1140	0.0902	0.0654	**8.70**
8.80	0.0002	0.0013	0.0058	0.0171	0.0377	0.0663	0.0972	0.1222	0.1344	0.1315	0.1157	0.0925	0.0679	**8.80**
8.90	0.0001	0.0012	0.0054	0.0160	0.0357	0.0635	0.0941	0.1197	0.1332	0.1317	0.1172	0.0948	0.0703	**8.90**
9.00	0.0001	0.0011	0.0050	0.0150	0.0337	0.0607	0.0911	0.1171	0.1318	0.1318	0.1186	0.0970	0.0728	**9.00**
9.10	0.0001	0.0010	0.0046	0.0140	0.0319	0.0581	0.0881	0.1145	0.1302	0.1317	0.1198	0.0991	0.0752	**9.10**
9.20	0.0001	0.0009	0.0043	0.0131	0.0302	0.0555	0.0851	0.1118	0.1286	0.1315	0.1210	0.1012	0.0776	**9.20**
9.30	0.0001	0.0009	0.0040	0.0123	0.0285	0.0530	0.0822	0.1091	0.1269	0.1311	0.1219	0.1031	0.0799	**9.30**
9.40	0.0001	0.0008	0.0037	0.0115	0.0269	0.0506	0.0793	0.1064	0.1251	0.1306	0.1228	0.1049	0.0822	**9.40**
9.50	0.0001	0.0007	0.0034	0.0107	0.0254	0.0483	0.0764	0.1037	0.1232	0.1300	0.1235	0.1067	0.0844	**9.50**
9.60	0.0001	0.0007	0.0031	0.0100	0.0240	0.0460	0.0736	0.1010	0.1212	0.1293	0.1241	0.1083	0.0866	**9.60**
9.70	0.0001	0.0006	0.0029	0.0093	0.0226	0.0439	0.0709	0.0982	0.1191	0.1284	0.1245	0.1098	0.0888	**9.70**
9.80	0.0001	0.0005	0.0027	0.0087	0.0213	0.0418	0.0682	0.0955	0.1170	0.1274	0.1249	0.1112	0.0908	**9.80**
9.90	0.0001	0.0005	0.0025	0.0081	0.0201	0.0398	0.0656	0.0928	0.1148	0.1263	0.1250	0.1125	0.0928	**9.90**
10.00	0.0000	0.0005	0.0023	0.0076	0.0189	0.0378	0.0631	0.0901	0.1126	0.1251	0.1251	0.1137	0.0948	**10.00**

$$\Pr(X = x) = f(x;\lambda) = \frac{\lambda^x}{x!}e^{-\lambda}$$

	Probability of *x* occurrences													
λ	13	14	15	16	17	18	19	20	21	22	23	24	25	λ
6.50	0.0089	0.0041	0.0018	0.0007	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**6.50**
6.60	0.0099	0.0046	0.0020	0.0008	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**6.60**
6.70	0.0108	0.0052	0.0023	0.0010	0.0004	0.0001	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**6.70**
6.80	0.0119	0.0058	0.0026	0.0011	0.0004	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**6.80**
6.90	0.0130	0.0064	0.0029	0.0013	0.0005	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**6.90**
7.00	0.0142	0.0071	0.0033	0.0014	0.0006	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**7.00**
7.10	0.0154	0.0078	0.0037	0.0016	0.0007	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**7.10**
7.20	0.0168	0.0086	0.0041	0.0019	0.0008	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	**7.20**
7.30	0.0181	0.0095	0.0046	0.0021	0.0009	0.0004	0.0001	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	**7.30**
7.40	0.0196	0.0104	0.0051	0.0024	0.0010	0.0004	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	**7.40**
7.50	0.0211	0.0113	0.0057	0.0026	0.0012	0.0005	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	**7.50**
7.60	0.0227	0.0123	0.0062	0.0030	0.0013	0.0006	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	**7.60**
7.70	0.0243	0.0134	0.0069	0.0033	0.0015	0.0006	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	**7.70**
7.80	0.0260	0.0145	0.0075	0.0037	0.0017	0.0007	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	**7.80**
7.90	0.0278	0.0157	0.0083	0.0041	0.0019	0.0008	0.0003	0.0001	0.0001	0.0000	0.0000	0.0000	0.0000	**7.90**
8.00	0.0296	0.0169	0.0090	0.0045	0.0021	0.0009	0.0004	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	**8.00**
8.10	0.0315	0.0182	0.0098	0.0050	0.0024	0.0011	0.0005	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	**8.10**
8.20	0.0334	0.0196	0.0107	0.0055	0.0026	0.0012	0.0005	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	**8.20**
8.30	0.0354	0.0210	0.0116	0.0060	0.0029	0.0014	0.0006	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	**8.30**
8.40	0.0374	0.0225	0.0126	0.0066	0.0033	0.0015	0.0007	0.0003	0.0001	0.0000	0.0000	0.0000	0.0000	**8.40**
8.50	0.0395	0.0240	0.0136	0.0072	0.0036	0.0017	0.0008	0.0003	0.0001	0.0001	0.0000	0.0000	0.0000	**8.50**
8.60	0.0416	0.0256	0.0147	0.0079	0.0040	0.0019	0.0009	0.0004	0.0002	0.0001	0.0000	0.0000	0.0000	**8.60**
8.70	0.0438	0.0272	0.0158	0.0086	0.0044	0.0021	0.0010	0.0004	0.0002	0.0001	0.0000	0.0000	0.0000	**8.70**
8.80	0.0459	0.0289	0.0169	0.0093	0.0048	0.0024	0.0011	0.0005	0.0002	0.0001	0.0000	0.0000	0.0000	**8.80**
8.90	0.0481	0.0306	0.0182	0.0101	0.0053	0.0026	0.0012	0.0005	0.0002	0.0001	0.0000	0.0000	0.0000	**8.90**
9.00	0.0504	0.0324	0.0194	0.0109	0.0058	0.0029	0.0014	0.0006	0.0003	0.0001	0.0000	0.0000	0.0000	**9.00**
9.10	0.0526	0.0342	0.0208	0.0118	0.0063	0.0032	0.0015	0.0007	0.0003	0.0001	0.0000	0.0000	0.0000	**9.10**
9.20	0.0549	0.0361	0.0221	0.0127	0.0069	0.0035	0.0017	0.0008	0.0003	0.0001	0.0001	0.0000	0.0000	**9.20**
9.30	0.0572	0.0380	0.0235	0.0137	0.0075	0.0039	0.0019	0.0009	0.0004	0.0002	0.0001	0.0000	0.0000	**9.30**
9.40	0.0594	0.0399	0.0250	0.0147	0.0081	0.0042	0.0021	0.0010	0.0004	0.0002	0.0001	0.0000	0.0000	**9.40**
9.50	0.0617	0.0419	0.0265	0.0157	0.0088	0.0046	0.0023	0.0011	0.0005	0.0002	0.0001	0.0000	0.0000	**9.50**
9.60	0.0640	0.0439	0.0281	0.0168	0.0095	0.0051	0.0026	0.0012	0.0006	0.0002	0.0001	0.0000	0.0000	**9.60**
9.70	0.0662	0.0459	0.0297	0.0180	0.0103	0.0055	0.0028	0.0014	0.0006	0.0003	0.0001	0.0000	0.0000	**9.70**
9.80	0.0685	0.0479	0.0313	0.0192	0.0111	0.0060	0.0031	0.0015	0.0007	0.0003	0.0001	0.0001	0.0000	**9.80**
9.90	0.0707	0.0500	0.0330	0.0204	0.0119	0.0065	0.0034	0.0017	0.0008	0.0004	0.0002	0.0001	0.0000	**9.90**
10.00	0.0729	0.0521	0.0347	0.0217	0.0128	0.0071	0.0037	0.0019	0.0009	0.0004	0.0002	0.0001	0.0000	**10.00**

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Appendix 11

Cumulative Poisson Distribution Table

$$\Pr(X \le x) = F(x;\lambda) = \sum_{x=0}^{n} \frac{\lambda^x}{x!} e^{-\lambda}$$

	Probability of ≤ *x* occurrences													
λ	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	λ
0.05	0.9512	0.9988	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.05**
0.10	0.9048	0.9953	0.9998	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.10**
0.15	0.8607	0.9898	0.9995	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.15**
0.20	0.8187	0.9825	0.9989	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.20**
0.25	0.7788	0.9735	0.9978	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.25**
0.30	0.7408	0.9631	0.9964	0.9997	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.30**
0.35	0.7047	0.9513	0.9945	0.9995	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.35**
0.40	0.6703	0.9384	0.9921	0.9992	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.40**
0.45	0.6376	0.9246	0.9891	0.9988	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.45**
0.50	0.6065	0.9098	0.9856	0.9982	0.9998	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.50**
0.55	0.5769	0.8943	0.9815	0.9975	0.9997	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.55**
0.60	0.5488	0.8781	0.9769	0.9966	0.9996	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.60**
0.65	0.5220	0.8614	0.9717	0.9956	0.9994	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.65**
0.70	0.4966	0.8442	0.9659	0.9942	0.9992	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.70**
0.75	0.4724	0.8266	0.9595	0.9927	0.9989	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.75**
0.80	0.4493	0.8088	0.9526	0.9909	0.9986	0.9998	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.80**
0.85	0.4274	0.7907	0.9451	0.9889	0.9982	0.9997	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.85**
0.90	0.4066	0.7725	0.9371	0.9865	0.9977	0.9997	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.90**
0.95	0.3867	0.7541	0.9287	0.9839	0.9971	0.9995	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.95**
1.00	0.3679	0.7358	0.9197	0.9810	0.9963	0.9994	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**1.00**
1.10	0.3329	0.6990	0.9004	0.9743	0.9946	0.9990	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**1.10**
1.20	0.3012	0.6626	0.8795	0.9662	0.9923	0.9985	0.9997	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**1.20**
1.30	0.2725	0.6268	0.8571	0.9569	0.9893	0.9978	0.9996	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	**1.30**
1.40	0.2466	0.5918	0.8335	0.9463	0.9857	0.9968	0.9994	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	**1.40**
1.50	0.2231	0.5578	0.8088	0.9344	0.9814	0.9955	0.9991	0.9998	1.0000	1.0000	1.0000	1.0000	1.0000	**1.50**
1.60	0.2019	0.5249	0.7834	0.9212	0.9763	0.9940	0.9987	0.9997	1.0000	1.0000	1.0000	1.0000	1.0000	**1.60**
1.70	0.1827	0.4932	0.7572	0.9068	0.9704	0.9920	0.9981	0.9996	0.9999	1.0000	1.0000	1.0000	1.0000	**1.70**
1.80	0.1653	0.4628	0.7306	0.8913	0.9636	0.9896	0.9974	0.9994	0.9999	1.0000	1.0000	1.0000	1.0000	**1.80**
1.90	0.1496	0.4337	0.7037	0.8747	0.9559	0.9868	0.9966	0.9992	0.9998	1.0000	1.0000	1.0000	1.0000	**1.90**
2.00	0.1353	0.4060	0.6767	0.8571	0.9473	0.9834	0.9955	0.9989	0.9998	1.0000	1.0000	1.0000	1.0000	**2.00**
2.10	0.1225	0.3796	0.6496	0.8386	0.9379	0.9796	0.9941	0.9985	0.9997	0.9999	1.0000	1.0000	1.0000	**2.10**
2.20	0.1108	0.3546	0.6227	0.8194	0.9275	0.9751	0.9925	0.9980	0.9995	0.9999	1.0000	1.0000	1.0000	**2.20**
2.30	0.1003	0.3309	0.5960	0.7993	0.9162	0.9700	0.9906	0.9974	0.9994	0.9999	1.0000	1.0000	1.0000	**2.30**
2.40	0.0907	0.3084	0.5697	0.7787	0.9041	0.9643	0.9884	0.9967	0.9991	0.9998	1.0000	1.0000	1.0000	**2.40**
2.50	0.0821	0.2873	0.5438	0.7576	0.8912	0.9580	0.9858	0.9958	0.9989	0.9997	0.9999	1.0000	1.0000	**2.50**
2.60	0.0743	0.2674	0.5184	0.7360	0.8774	0.9510	0.9828	0.9947	0.9985	0.9996	0.9999	1.0000	1.0000	**2.60**
2.70	0.0672	0.2487	0.4936	0.7141	0.8629	0.9433	0.9794	0.9934	0.9981	0.9995	0.9999	1.0000	1.0000	**2.70**

$$\Pr(X \le x) = F(x;\lambda) = \sum_{x=0}^{n} \frac{\lambda^x}{x!} e^{-\lambda}$$

	Probability of ≤ *x* occurrences													
λ	13	14	15	16	17	18	19	20	21	22	23	24	25	λ
0.05	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.05**
0.10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.10**
0.15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.15**
0.20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.20**
0.25	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.25**
0.30	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.30**
0.35	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.35**
0.40	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.40**
0.45	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.45**
0.50	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.50**
0.55	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.55**
0.60	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.60**
0.65	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.65**
0.70	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.70**
0.75	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.75**
0.80	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.80**
0.85	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.85**
0.90	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.90**
0.95	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**0.95**
1.00	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**1.00**
1.10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**1.10**
1.20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**1.20**
1.30	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**1.30**
1.40	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**1.40**
1.50	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**1.50**
1.60	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**1.60**
1.70	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**1.70**
1.80	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**1.80**
1.90	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**1.90**
2.00	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**2.00**
2.10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**2.10**
2.20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**2.20**
2.30	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**2.30**
2.40	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**2.40**
2.50	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**2.50**
2.60	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**2.60**
2.70	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**2.70**

$$\Pr(X \le x) = F(x;\lambda) = \sum_{x=0}^{n} \frac{\lambda^x}{x!} e^{-\lambda}$$

	Probability of ≤ *x* occurrences													
λ	0	1	2	3	4	5	6	7	8	9	10	11	12	λ
2.80	0.0608	0.2311	0.4695	0.6919	0.8477	0.9349	0.9756	0.9919	0.9976	0.9993	0.9998	1.0000	1.0000	**2.80**
2.90	0.0550	0.2146	0.4460	0.6696	0.8318	0.9258	0.9713	0.9901	0.9969	0.9991	0.9998	0.9999	1.0000	**2.90**
3.00	0.0498	0.1991	0.4232	0.6472	0.8153	0.9161	0.9665	0.9881	0.9962	0.9989	0.9997	0.9999	1.0000	**3.00**
3.10	0.0450	0.1847	0.4012	0.6248	0.7982	0.9057	0.9612	0.9858	0.9953	0.9986	0.9996	0.9999	1.0000	**3.10**
3.20	0.0408	0.1712	0.3799	0.6025	0.7806	0.8946	0.9554	0.9832	0.9943	0.9982	0.9995	0.9999	1.0000	**3.20**
3.30	0.0369	0.1586	0.3594	0.5803	0.7626	0.8829	0.9490	0.9802	0.9931	0.9978	0.9994	0.9998	1.0000	**3.30**
3.40	0.0334	0.1468	0.3397	0.5584	0.7442	0.8705	0.9421	0.9769	0.9917	0.9973	0.9992	0.9998	0.9999	**3.40**
3.50	0.0302	0.1359	0.3208	0.5366	0.7254	0.8576	0.9347	0.9733	0.9901	0.9967	0.9990	0.9997	0.9999	**3.50**
3.60	0.0273	0.1257	0.3027	0.5152	0.7064	0.8441	0.9267	0.9692	0.9883	0.9960	0.9987	0.9996	0.9999	**3.60**
3.70	0.0247	0.1162	0.2854	0.4942	0.6872	0.8301	0.9182	0.9648	0.9863	0.9952	0.9984	0.9995	0.9999	**3.70**
3.80	0.0224	0.1074	0.2689	0.4735	0.6678	0.8156	0.9091	0.9599	0.9840	0.9942	0.9981	0.9994	0.9998	**3.80**
3.90	0.0202	0.0992	0.2531	0.4532	0.6484	0.8006	0.8995	0.9546	0.9815	0.9931	0.9977	0.9993	0.9998	**3.90**
4.00	0.0183	0.0916	0.2381	0.4335	0.6288	0.7851	0.8893	0.9489	0.9786	0.9919	0.9972	0.9991	0.9997	**4.00**
4.10	0.0166	0.0845	0.2238	0.4142	0.6093	0.7693	0.8786	0.9427	0.9755	0.9905	0.9966	0.9989	0.9997	**4.10**
4.20	0.0150	0.0780	0.2102	0.3954	0.5898	0.7531	0.8675	0.9361	0.9721	0.9889	0.9959	0.9986	0.9996	**4.20**
4.30	0.0136	0.0719	0.1974	0.3772	0.5704	0.7367	0.8558	0.9290	0.9683	0.9871	0.9952	0.9983	0.9995	**4.30**
4.40	0.0123	0.0663	0.1851	0.3594	0.5512	0.7199	0.8436	0.9214	0.9642	0.9851	0.9943	0.9980	0.9993	**4.40**
4.50	0.0111	0.0611	0.1736	0.3423	0.5321	0.7029	0.8311	0.9134	0.9597	0.9829	0.9933	0.9976	0.9992	**4.50**
4.60	0.0101	0.0563	0.1626	0.3257	0.5132	0.6858	0.8180	0.9049	0.9549	0.9805	0.9922	0.9971	0.9990	**4.60**
4.70	0.0091	0.0518	0.1523	0.3097	0.4946	0.6684	0.8046	0.8960	0.9497	0.9778	0.9910	0.9966	0.9988	**4.70**
4.80	0.0082	0.0477	0.1425	0.2942	0.4763	0.6510	0.7908	0.8867	0.9442	0.9749	0.9896	0.9960	0.9986	**4.80**
4.90	0.0074	0.0439	0.1333	0.2793	0.4582	0.6335	0.7767	0.8769	0.9382	0.9717	0.9880	0.9953	0.9983	**4.90**
5.00	0.0067	0.0404	0.1247	0.2650	0.4405	0.6160	0.7622	0.8666	0.9319	0.9682	0.9863	0.9945	0.9980	**5.00**
5.10	0.0061	0.0372	0.1165	0.2513	0.4231	0.5984	0.7474	0.8560	0.9252	0.9644	0.9844	0.9937	0.9976	**5.10**
5.20	0.0055	0.0342	0.1088	0.2381	0.4061	0.5809	0.7324	0.8449	0.9181	0.9603	0.9823	0.9927	0.9972	**5.20**
5.30	0.0050	0.0314	0.1016	0.2254	0.3895	0.5635	0.7171	0.8335	0.9106	0.9559	0.9800	0.9916	0.9967	**5.30**
5.40	0.0045	0.0289	0.0948	0.2133	0.3733	0.5461	0.7017	0.8217	0.9027	0.9512	0.9775	0.9904	0.9962	**5.40**
5.50	0.0041	0.0266	0.0884	0.2017	0.3575	0.5289	0.6860	0.8095	0.8944	0.9462	0.9747	0.9890	0.9955	**5.50**
5.60	0.0037	0.0244	0.0824	0.1906	0.3422	0.5119	0.6703	0.7970	0.8857	0.9409	0.9718	0.9875	0.9949	**5.60**
5.70	0.0033	0.0224	0.0768	0.1800	0.3272	0.4950	0.6544	0.7841	0.8766	0.9352	0.9686	0.9859	0.9941	**5.70**
5.80	0.0030	0.0206	0.0715	0.1700	0.3127	0.4783	0.6384	0.7710	0.8672	0.9292	0.9651	0.9841	0.9932	**5.80**
5.90	0.0027	0.0189	0.0666	0.1604	0.2987	0.4619	0.6224	0.7576	0.8574	0.9228	0.9614	0.9821	0.9922	**5.90**
6.00	0.0025	0.0174	0.0620	0.1512	0.2851	0.4457	0.6063	0.7440	0.8472	0.9161	0.9574	0.9799	0.9912	**6.00**
6.10	0.0022	0.0159	0.0577	0.1425	0.2719	0.4298	0.5902	0.7301	0.8367	0.9090	0.9531	0.9776	0.9900	**6.10**
6.20	0.0020	0.0146	0.0536	0.1342	0.2592	0.4141	0.5742	0.7160	0.8259	0.9016	0.9486	0.9750	0.9887	**6.20**
6.30	0.0018	0.0134	0.0498	0.1264	0.2469	0.3988	0.5582	0.7017	0.8148	0.8939	0.9437	0.9723	0.9873	**6.30**
6.40	0.0017	0.0123	0.0463	0.1189	0.2351	0.3837	0.5423	0.6873	0.8033	0.8858	0.9386	0.9693	0.9857	**6.40**
6.50	0.0015	0.0113	0.0430	0.1118	0.2237	0.3690	0.5265	0.6728	0.7916	0.8774	0.9332	0.9661	0.9840	**6.50**

$$\Pr(X \le x) = F(x;\lambda) = \sum_{x=0}^{n} \frac{\lambda^x}{x!} e^{-\lambda}$$

	Probability of ≤ *x* occurrences													
λ	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**λ**
2.80	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**2.80**
2.90	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**2.90**
3.00	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**3.00**
3.10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**3.10**
3.20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**3.20**
3.30	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**3.30**
3.40	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**3.40**
3.50	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**3.50**
3.60	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**3.60**
3.70	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**3.70**
3.80	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**3.80**
3.90	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**3.90**
4.00	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**4.00**
4.10	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**4.10**
4.20	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**4.20**
4.30	0.9998	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**4.30**
4.40	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**4.40**
4.50	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**4.50**
4.60	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**4.60**
4.70	0.9996	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**4.70**
4.80	0.9995	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**4.80**
4.90	0.9994	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**4.90**
5.00	0.9993	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**5.00**
5.10	0.9992	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**5.10**
5.20	0.9990	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**5.20**
5.30	0.9988	0.9996	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**5.30**
5.40	0.9986	0.9995	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**5.40**
5.50	0.9983	0.9994	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**5.50**
5.60	0.9980	0.9993	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**5.60**
5.70	0.9977	0.9991	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**5.70**
5.80	0.9973	0.9990	0.9996	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**5.80**
5.90	0.9969	0.9988	0.9996	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**5.90**
6.00	0.9964	0.9986	0.9995	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**6.00**
6.10	0.9958	0.9984	0.9994	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**6.10**
6.20	0.9952	0.9981	0.9993	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**6.20**
6.30	0.9945	0.9978	0.9992	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**6.30**
6.40	0.9937	0.9974	0.9990	0.9996	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**6.40**
6.50	0.9929	0.9970	0.9988	0.9996	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**6.50**

$$\Pr(X \le x) = F(x;\lambda) = \sum_{x=0}^{n} \frac{\lambda^x}{x!} e^{-\lambda}$$

	Probability of ≤ *x* occurrences													
λ	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	λ
6.60	0.0014	0.0103	0.0400	0.1052	0.2127	0.3547	0.5108	0.6581	0.7796	0.8686	0.9274	0.9627	0.9821	**6.60**
6.70	0.0012	0.0095	0.0371	0.0988	0.2022	0.3406	0.4953	0.6433	0.7673	0.8596	0.9214	0.9591	0.9801	**6.70**
6.80	0.0011	0.0087	0.0344	0.0928	0.1920	0.3270	0.4799	0.6285	0.7548	0.8502	0.9151	0.9552	0.9779	**6.80**
6.90	0.0010	0.0080	0.0320	0.0871	0.1823	0.3137	0.4647	0.6136	0.7420	0.8405	0.9084	0.9510	0.9755	**6.90**
7.00	0.0009	0.0073	0.0296	0.0818	0.1730	0.3007	0.4497	0.5987	0.7291	0.8305	0.9015	0.9467	0.9730	**7.00**
7.10	0.0008	0.0067	0.0275	0.0767	0.1641	0.2881	0.4349	0.5838	0.7160	0.8202	0.8942	0.9420	0.9703	**7.10**
7.20	0.0007	0.0061	0.0255	0.0719	0.1555	0.2759	0.4204	0.5689	0.7027	0.8096	0.8867	0.9371	0.9673	**7.20**
7.30	0.0007	0.0056	0.0236	0.0674	0.1473	0.2640	0.4060	0.5541	0.6892	0.7988	0.8788	0.9319	0.9642	**7.30**
7.40	0.0006	0.0051	0.0219	0.0632	0.1395	0.2526	0.3920	0.5393	0.6757	0.7877	0.8707	0.9265	0.9609	**7.40**
7.50	0.0006	0.0047	0.0203	0.0591	0.1321	0.2414	0.3782	0.5246	0.6620	0.7764	0.8622	0.9208	0.9573	**7.50**
7.60	0.0005	0.0043	0.0188	0.0554	0.1249	0.2307	0.3646	0.5100	0.6482	0.7649	0.8535	0.9148	0.9536	**7.60**
7.70	0.0005	0.0039	0.0174	0.0518	0.1181	0.2203	0.3514	0.4956	0.6343	0.7531	0.8445	0.9085	0.9496	**7.70**
7.80	0.0004	0.0036	0.0161	0.0485	0.1117	0.2103	0.3384	0.4812	0.6204	0.7411	0.8352	0.9020	0.9454	**7.80**
7.90	0.0004	0.0033	0.0149	0.0453	0.1055	0.2006	0.3257	0.4670	0.6065	0.7290	0.8257	0.8952	0.9409	**7.90**
8.00	0.0003	0.0030	0.0138	0.0424	0.0996	0.1912	0.3134	0.4530	0.5925	0.7166	0.8159	0.8881	0.9362	**8.00**
8.10	0.0003	0.0028	0.0127	0.0396	0.0940	0.1822	0.3013	0.4391	0.5786	0.7041	0.8058	0.8807	0.9313	**8.10**
8.20	0.0003	0.0025	0.0118	0.0370	0.0887	0.1736	0.2896	0.4254	0.5647	0.6915	0.7955	0.8731	0.9261	**8.20**
8.30	0.0002	0.0023	0.0109	0.0346	0.0837	0.1653	0.2781	0.4119	0.5507	0.6788	0.7850	0.8652	0.9207	**8.30**
8.40	0.0002	0.0021	0.0100	0.0323	0.0789	0.1573	0.2670	0.3987	0.5369	0.6659	0.7743	0.8571	0.9150	**8.40**
8.50	0.0002	0.0019	0.0093	0.0301	0.0744	0.1496	0.2562	0.3856	0.5231	0.6530	0.7634	0.8487	0.9091	**8.50**
8.60	0.0002	0.0018	0.0086	0.0281	0.0701	0.1422	0.2457	0.3728	0.5094	0.6400	0.7522	0.8400	0.9029	**8.60**
8.70	0.0002	0.0016	0.0079	0.0262	0.0660	0.1352	0.2355	0.3602	0.4958	0.6269	0.7409	0.8311	0.8965	**8.70**
8.80	0.0002	0.0015	0.0073	0.0244	0.0621	0.1284	0.2256	0.3478	0.4823	0.6137	0.7294	0.8220	0.8898	**8.80**
8.90	0.0001	0.0014	0.0068	0.0228	0.0584	0.1219	0.2160	0.3357	0.4689	0.6006	0.7178	0.8126	0.8829	**8.90**
9.00	0.0001	0.0012	0.0062	0.0212	0.0550	0.1157	0.2068	0.3239	0.4557	0.5874	0.7060	0.8030	0.8758	**9.00**
9.10	0.0001	0.0011	0.0058	0.0198	0.0517	0.1098	0.1978	0.3123	0.4426	0.5742	0.6941	0.7932	0.8684	**9.10**
9.20	0.0001	0.0010	0.0053	0.0184	0.0486	0.1041	0.1892	0.3010	0.4296	0.5611	0.6820	0.7832	0.8607	**9.20**
9.30	0.0001	0.0009	0.0049	0.0172	0.0456	0.0986	0.1808	0.2900	0.4168	0.5479	0.6699	0.7730	0.8529	**9.30**
9.40	0.0001	0.0009	0.0045	0.0160	0.0429	0.0935	0.1727	0.2792	0.4042	0.5349	0.6576	0.7626	0.8448	**9.40**
9.50	0.0001	0.0008	0.0042	0.0149	0.0403	0.0885	0.1649	0.2687	0.3918	0.5218	0.6453	0.7520	0.8364	**9.50**
9.60	0.0001	0.0007	0.0038	0.0138	0.0378	0.0838	0.1574	0.2584	0.3796	0.5089	0.6329	0.7412	0.8279	**9.60**
9.70	0.0001	0.0007	0.0035	0.0129	0.0355	0.0793	0.1502	0.2485	0.3676	0.4960	0.6205	0.7303	0.8191	**9.70**
9.80	0.0001	0.0006	0.0033	0.0120	0.0333	0.0750	0.1433	0.2388	0.3558	0.4832	0.6080	0.7193	0.8101	**9.80**
9.90	0.0001	0.0005	0.0030	0.0111	0.0312	0.0710	0.1366	0.2294	0.3442	0.4705	0.5955	0.7081	0.8009	**9.90**
10.00	0.0000	0.0005	0.0028	0.0103	0.0293	0.0671	0.1301	0.2202	0.3328	0.4579	0.5830	0.6968	0.7916	**10.00**

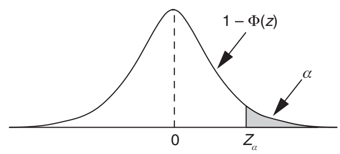
$$\Pr(X \le x) = F(x;\lambda) = \sum_{x=0}^{n} \frac{\lambda^x}{x!} e^{-\lambda}$$

		Probability of ≤ *x* occurrences												
λ	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	λ
6.60	0.9920	0.9966	0.9986	0.9995	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**6.60**
6.70	0.9909	0.9961	0.9984	0.9994	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**6.70**
6.80	0.9898	0.9956	0.9982	0.9993	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**6.80**
6.90	0.9885	0.9950	0.9979	0.9992	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**6.90**
7.00	0.9872	0.9943	0.9976	0.9990	0.9996	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**7.00**
7.10	0.9857	0.9935	0.9972	0.9989	0.9996	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**7.10**
7.20	0.9841	0.9927	0.9969	0.9987	0.9995	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**7.20**
7.30	0.9824	0.9918	0.9964	0.9985	0.9994	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**7.30**
7.40	0.9805	0.9908	0.9959	0.9983	0.9993	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**7.40**
7.50	0.9784	0.9897	0.9954	0.9980	0.9992	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**7.50**
7.60	0.9762	0.9886	0.9948	0.9978	0.9991	0.9996	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	**7.60**
7.70	0.9739	0.9873	0.9941	0.9974	0.9989	0.9996	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	**7.70**
7.80	0.9714	0.9859	0.9934	0.9971	0.9988	0.9995	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	**7.80**
7.90	0.9687	0.9844	0.9926	0.9967	0.9986	0.9994	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	**7.90**
8.00	0.9658	0.9827	0.9918	0.9963	0.9984	0.9993	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	**8.00**
8.10	0.9628	0.9810	0.9908	0.9958	0.9982	0.9992	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	**8.10**
8.20	0.9595	0.9791	0.9898	0.9953	0.9979	0.9991	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	**8.20**
8.30	0.9561	0.9771	0.9887	0.9947	0.9977	0.9990	0.9996	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	**8.30**
8.40	0.9524	0.9749	0.9875	0.9941	0.9973	0.9989	0.9995	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	**8.40**
8.50	0.9486	0.9726	0.9862	0.9934	0.9970	0.9987	0.9995	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	**8.50**
8.60	0.9445	0.9701	0.9848	0.9926	0.9966	0.9985	0.9994	0.9998	0.9999	1.0000	1.0000	1.0000	1.0000	**8.60**
8.70	0.9403	0.9675	0.9832	0.9918	0.9962	0.9983	0.9993	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	**8.70**
8.80	0.9358	0.9647	0.9816	0.9909	0.9957	0.9981	0.9992	0.9997	0.9999	1.0000	1.0000	1.0000	1.0000	**8.80**
8.90	0.9311	0.9617	0.9798	0.9899	0.9952	0.9978	0.9991	0.9996	0.9998	0.9999	1.0000	1.0000	1.0000	**8.90**
9.00	0.9261	0.9585	0.9780	0.9889	0.9947	0.9976	0.9989	0.9996	0.9998	0.9999	1.0000	1.0000	1.0000	**9.00**
9.10	0.9210	0.9552	0.9760	0.9878	0.9941	0.9973	0.9988	0.9995	0.9998	0.9999	1.0000	1.0000	1.0000	**9.10**
9.20	0.9156	0.9517	0.9738	0.9865	0.9934	0.9969	0.9986	0.9994	0.9998	0.9999	1.0000	1.0000	1.0000	**9.20**
9.30	0.9100	0.9480	0.9715	0.9852	0.9927	0.9966	0.9985	0.9993	0.9997	0.9999	1.0000	1.0000	1.0000	**9.30**
9.40	0.9042	0.9441	0.9691	0.9838	0.9919	0.9962	0.9983	0.9992	0.9997	0.9999	1.0000	1.0000	1.0000	**9.40**
9.50	0.8981	0.9400	0.9665	0.9823	0.9911	0.9957	0.9980	0.9991	0.9996	0.9999	0.9999	1.0000	1.0000	**9.50**
9.60	0.8919	0.9357	0.9638	0.9806	0.9902	0.9952	0.9978	0.9990	0.9996	0.9998	0.9999	1.0000	1.0000	**9.60**
9.70	0.8853	0.9312	0.9609	0.9789	0.9892	0.9947	0.9975	0.9989	0.9995	0.9998	0.9999	1.0000	1.0000	**9.70**
9.80	0.8786	0.9265	0.9579	0.9770	0.9881	0.9941	0.9972	0.9987	0.9995	0.9998	0.9999	1.0000	1.0000	**9.80**
9.90	0.8716	0.9216	0.9546	0.9751	0.9870	0.9935	0.9969	0.9986	0.9994	0.9997	0.9999	1.0000	1.0000	**9.90**
10.00	0.8645	0.9165	0.9513	0.9730	0.9857	0.9928	0.9965	0.9984	0.9993	0.9997	0.9999	1.0000	1.0000	**10.00**

Appendix 12

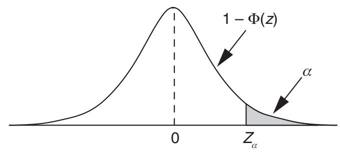
Standard Normal Distribution Table

$$\Pr(Z \geq z) = 1 - \Phi(Z \leq z) = \int_z^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\mu^2/2} d\mu$$



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09	*Z*
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641	**0.0**
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247	**0.1**
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859	**0.2**
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483	**0.3**
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121	**0.4**
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776	**0.5**
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451	**0.6**
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148	**0.7**
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867	**0.8**
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611	**0.9**
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379	**1.0**
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170	**1.1**
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985	**1.2**
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823	**1.3**
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681	**1.4**
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559	**1.5**
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455	**1.6**
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367	**1.7**
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294	**1.8**
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233	**1.9**
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183	**2.0**
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143	**2.1**
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110	**2.2**
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084	**2.3**
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064	**2.4**
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048	**2.5**

$$\Pr(Z \ge z) = 1 - \Phi(Z \le z) = \int_z^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\mu^2/2} d\mu$$

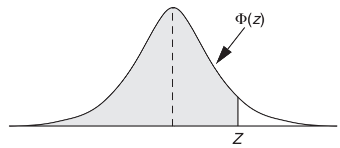


Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09	Z
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036	**2.6**
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026	**2.7**
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019	**2.8**
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014	**2.9**
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010	**3.0**
3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071	**3.1**
3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050	**3.2**
3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035	**3.3**
3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024	**3.4**
3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017	**3.5**
3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011	**3.6**
3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008	**3.7**
3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005	**3.8**
3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003	**3.9**
4.0	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00002	0.00002	0.00002	0.00002	**4.0**
4.1	0.00002	0.00002	0.00002	0.00002	0.00002	0.00002	0.00002	0.00002	0.00001	0.00001	**4.1**
4.2	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	**4.2**
4.3	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	0.00001	**4.3**
4.4	0.000005	0.000005	0.000005	0.000005	0.000004	0.000004	0.000004	0.000004	0.000004	0.000004	**4.4**
4.5	0.000003	0.000003	0.000003	0.000003	0.000003	0.000003	0.000003	0.000002	0.000002	0.000002	**4.5**
4.6	0.000002	0.000002	0.000002	0.000002	0.000002	0.000002	0.000002	0.000002	0.000001	0.000001	**4.6**
4.7	0.000001	0.000001	0.000001	0.000001	0.000001	0.000001	0.000001	0.000001	0.000001	0.000001	**4.7**
4.8	0.000001	0.000001	0.000001	0.000001	0.000001	0.000001	0.000001	0.000001	0.000001	0.000001	**4.8**
4.9	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**4.9**
5.0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**5.0**
5.1	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**5.1**
5.2	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**5.2**
5.3	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**5.3**
5.4	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**5.4**
5.5	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**5.5**
5.6	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**5.6**
5.7	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**5.7**
5.8	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**5.8**
5.9	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**5.9**
6.0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**6.0**

Appendix 13

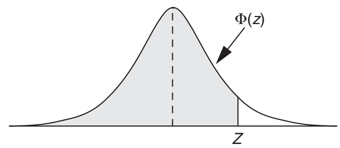
Cumulative Standard Normal Distribution Table

$$\Pr(Z \le z) = \Phi(Z \le z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\mu^2/2} d\mu$$



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09	Z
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359	**0.0**
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753	**0.1**
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141	**0.2**
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517	**0.3**
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879	**0.4**
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224	**0.5**
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549	**0.6**
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852	**0.7**
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133	**0.8**
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389	**0.9**
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621	**1.0**
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830	**1.1**
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015	**1.2**
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177	**1.3**
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319	**1.4**
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441	**1.5**
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545	**1.6**
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633	**1.7**
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706	**1.8**
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767	**1.9**
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817	**2.0**
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857	**2.1**
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890	**2.2**
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916	**2.3**
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936	**2.4**
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952	**2.5**

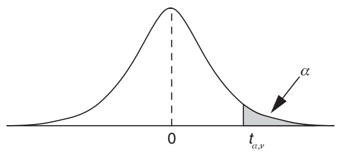
$$\Pr(Z \le z) = \Phi\left(Z \le z\right) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\mu^2/2} d\mu$$



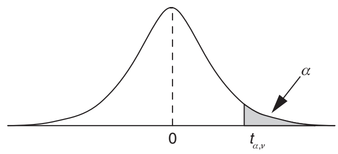
Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09	Z
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964	**2.6**
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974	**2.7**
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981	**2.8**
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986	**2.9**
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990	**3.0**
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929	**3.1**
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950	**3.2**
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965	**3.3**
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976	**3.4**
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983	**3.5**
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989	**3.6**
3.7	0.99989	0.99990	0.99990	0.99990	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992	**3.7**
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995	**3.8**
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997	**3.9**
4.0	0.99997	0.99997	0.99997	0.99997	0.99997	0.99997	0.99998	0.99998	0.99998	0.99998	**4.0**
4.1	0.99998	0.99998	0.99998	0.99998	0.99998	0.99998	0.99998	0.99998	0.99999	0.99999	**4.1**
4.2	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999	**4.2**
4.3	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999	**4.3**
4.4	0.999995	0.999995	0.999995	0.999995	0.999996	0.999996	0.999996	0.999996	0.999996	0.999996	**4.4**
4.5	0.999997	0.999997	0.999997	0.999997	0.999997	0.999997	0.999997	0.999998	0.999998	0.999998	**4.5**
4.6	0.999998	0.999998	0.999998	0.999998	0.999998	0.999998	0.999998	0.999998	0.999999	0.999999	**4.6**
4.7	0.999999	0.999999	0.999999	0.999999	0.999999	0.999999	0.999999	0.999999	0.999999	0.999999	**4.7**
4.8	0.999999	0.999999	0.999999	0.999999	0.999999	0.999999	0.999999	0.999999	0.999999	0.999999	**4.8**
4.9	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	**4.9**
5.0	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	**5.0**
5.1	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	**5.1**
5.2	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	**5.2**
5.3	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	**5.3**
5.4	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	**5.4**
5.5	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	**5.5**
5.6	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	**5.6**
5.7	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	**5.7**
5.8	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	**5.8**
5.9	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**5.9**
6.0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	**6.0**

Appendix 14

t-Distribution Table



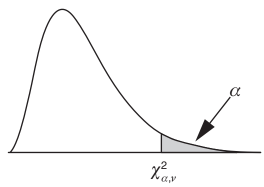
	α					
ν	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	ν
1	3.078	6.314	12.706	31.821	63.657	**1**
2	1.886	2.920	4.303	6.965	9.925	**2**
3	1.638	2.353	3.182	4.541	5.841	**3**
4	1.533	2.132	2.776	3.747	4.604	**4**
5	1.476	2.015	2.571	3.365	4.032	**5**
6	1.440	1.943	2.447	3.143	3.707	**6**
7	1.415	1.895	2.365	2.998	3.499	**7**
8	1.397	1.860	2.306	2.896	3.355	**8**
9	1.383	1.833	2.262	2.821	3.250	**9**
10	1.372	1.812	2.228	2.764	3.169	**10**
11	1.363	1.796	2.201	2.718	3.106	**11**
12	1.356	1.782	2.179	2.681	3.055	**12**
13	1.350	1.771	2.160	2.650	3.012	**13**
14	1.345	1.761	2.145	2.624	2.977	**14**
15	1.341	1.753	2.131	2.602	2.947	**15**
16	1.337	1.746	2.120	2.583	2.921	**16**
17	1.333	1.740	2.110	2.567	2.898	**17**
18	1.330	1.734	2.101	2.552	2.878	**18**
19	1.328	1.729	2.093	2.539	2.861	**19**
20	1.325	1.725	2.086	2.528	2.845	**20**
21	1.323	1.721	2.080	2.518	2.831	**21**
22	1.321	1.717	2.074	2.508	2.819	**22**
23	1.319	1.714	2.069	2.500	2.807	**23**



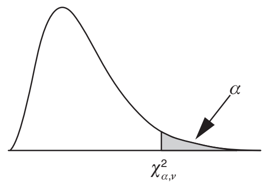
	α					
ν	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	ν
24	1.318	1.711	2.064	2.492	2.797	**24**
25	1.316	1.708	2.060	2.485	2.787	**25**
26	1.315	1.706	2.056	2.479	2.779	**26**
27	1.314	1.703	2.052	2.473	2.771	**27**
28	1.313	1.701	2.048	2.467	2.763	**28**
29	1.311	1.699	2.045	2.462	2.756	**29**
30	1.310	1.697	2.042	2.457	2.750	**30**
40	1.303	1.684	2.021	2.423	2.704	**40**
50	1.299	1.676	2.009	2.403	2.678	**50**
60	1.296	1.671	2.000	2.390	2.660	**60**
70	1.294	1.667	1.994	2.381	2.648	**70**
80	1.292	1.664	1.990	2.374	2.639	**80**
90	1.291	1.662	1.987	2.368	2.632	**90**
100	1.290	1.660	1.984	2.364	2.626	**100**
120	1.289	1.658	1.980	2.358	2.617	**120**
∞	1.282	1.645	1.960	2.326	2.576	∞

Appendix 15

Chi-Square Distribution Table



	α										
υ	**0.995**	**0.990**	**0.975**	**0.950**	**0.900**	**0.100**	**0.050**	**0.025**	**0.010**	**0.005**	υ
1	0.00004	0.00016	0.00098	0.00393	0.01579	2.7055	3.8415	5.0239	6.6349	7.8794	**1**
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597	**2**
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838	**3**
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860	**4**
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750	**5**
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548	**6**
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278	**7**
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955	**8**
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589	**9**
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188	**10**
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757	**11**
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300	**12**
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819	**13**
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319	**14**
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801	**15**
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267	**16**
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718	**17**
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156	**18**
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582	**19**
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997	**20**
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401	**21**
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796	**22**
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181	**23**
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559	**24**



	α										
υ	**0.995**	**0.990**	**0.975**	**0.950**	**0.900**	**0.100**	**0.050**	**0.025**	**0.010**	**0.005**	υ
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928	**25**
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290	**26**
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645	**27**
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993	**28**
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336	**29**
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672	**30**
35	17.192	18.509	20.569	22.465	24.797	46.059	49.802	53.203	57.342	60.275	**35**
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766	**40**
45	24.311	25.901	28.366	30.612	33.350	57.505	61.656	65.410	69.957	73.166	**45**
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490	**50**
55	31.735	33.570	36.398	38.958	42.060	68.796	73.311	77.380	82.292	85.749	**55**
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952	**60**
65	39.383	41.444	44.603	47.450	50.883	79.973	84.821	89.177	94.422	98.105	**65**
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215	**70**
75	47.206	49.475	52.942	56.054	59.795	91.061	96.217	100.839	106.393	110.286	**75**
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321	**80**
85	55.170	57.634	61.389	64.749	68.777	102.079	107.522	112.393	118.236	122.325	**85**
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299	**90**
95	63.250	65.898	69.925	73.520	77.818	113.038	118.752	123.858	129.973	134.247	**95**
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169	**100**

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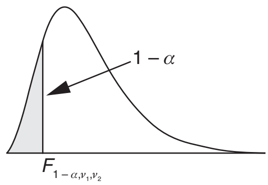
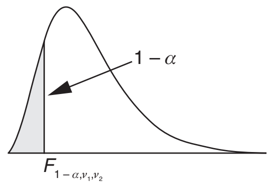
Appendix 16

F(0.99) Distribution Table



	Degrees of freedom for the numerator (ν_1)																	
(ν_2)	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	(ν_2)
1	0.00	0.01	0.03	0.05	0.06	0.07	0.08	0.09	0.09	0.10	0.10	0.11	0.11	0.11	0.12	0.12	0.12	**1**
2	0.00	0.01	0.03	0.06	0.08	0.09	0.10	0.12	0.12	0.13	0.14	0.14	0.15	0.15	0.16	0.16	0.16	**2**
3	0.00	0.01	0.03	0.06	0.08	0.10	0.12	0.13	0.14	0.15	0.16	0.17	0.17	0.18	0.18	0.19	0.19	**3**
4	0.00	0.01	0.03	0.06	0.09	0.11	0.13	0.14	0.16	0.17	0.18	0.18	0.19	0.20	0.20	0.21	0.21	**4**
5	0.00	0.01	0.04	0.06	0.09	0.11	0.13	0.15	0.17	0.18	0.19	0.20	0.21	0.21	0.22	0.23	0.23	**5**
6	0.00	0.01	0.04	0.07	0.09	0.12	0.14	0.16	0.17	0.19	0.20	0.21	0.22	0.22	0.23	0.24	0.24	**6**
7	0.00	0.01	0.04	0.07	0.10	0.12	0.14	0.16	0.18	0.19	0.20	0.22	0.23	0.23	0.24	0.25	0.25	**7**
8	0.00	0.01	0.04	0.07	0.10	0.12	0.15	0.17	0.18	0.20	0.21	0.22	0.23	0.24	0.25	0.26	0.26	**8**
9	0.00	0.01	0.04	0.07	0.10	0.13	0.15	0.17	0.19	0.20	0.22	0.23	0.24	0.25	0.26	0.26	0.27	**9**
10	0.00	0.01	0.04	0.07	0.10	0.13	0.15	0.17	0.19	0.21	0.22	0.23	0.24	0.25	0.26	0.27	0.28	**10**
11	0.00	0.01	0.04	0.07	0.10	0.13	0.15	0.17	0.19	0.21	0.22	0.24	0.25	0.26	0.27	0.28	0.28	**11**
12	0.00	0.01	0.04	0.07	0.10	0.13	0.15	0.18	0.20	0.21	0.23	0.24	0.25	0.26	0.27	0.28	0.29	**12**
13	0.00	0.01	0.04	0.07	0.10	0.13	0.16	0.18	0.20	0.22	0.23	0.24	0.26	0.27	0.28	0.29	0.29	**13**
14	0.00	0.01	0.04	0.07	0.10	0.13	0.16	0.18	0.20	0.22	0.23	0.25	0.26	0.27	0.28	0.29	0.30	**14**
15	0.00	0.01	0.04	0.07	0.10	0.13	0.16	0.18	0.20	0.22	0.24	0.25	0.26	0.27	0.28	0.29	0.30	**15**
16	0.00	0.01	0.04	0.07	0.10	0.13	0.16	0.18	0.20	0.22	0.24	0.25	0.26	0.28	0.29	0.30	0.31	**16**
17	0.00	0.01	0.04	0.07	0.10	0.13	0.16	0.18	0.20	0.22	0.24	0.25	0.27	0.28	0.29	0.30	0.31	**17**
18	0.00	0.01	0.04	0.07	0.10	0.13	0.16	0.18	0.21	0.22	0.24	0.26	0.27	0.28	0.29	0.30	0.31	**18**
19	0.00	0.01	0.04	0.07	0.10	0.13	0.16	0.19	0.21	0.23	0.24	0.26	0.27	0.28	0.29	0.30	0.31	**19**
20	0.00	0.01	0.04	0.07	0.10	0.14	0.16	0.19	0.21	0.23	0.24	0.26	0.27	0.29	0.30	0.31	0.32	**20**
21	0.00	0.01	0.04	0.07	0.10	0.14	0.16	0.19	0.21	0.23	0.25	0.26	0.27	0.29	0.30	0.31	0.32	**21**
22	0.00	0.01	0.04	0.07	0.11	0.14	0.16	0.19	0.21	0.23	0.25	0.26	0.28	0.29	0.30	0.31	0.32	**22**
23	0.00	0.01	0.04	0.07	0.11	0.14	0.16	0.19	0.21	0.23	0.25	0.26	0.28	0.29	0.30	0.31	0.32	**23**
24	0.00	0.01	0.04	0.07	0.11	0.14	0.16	0.19	0.21	0.23	0.25	0.26	0.28	0.29	0.30	0.31	0.32	**24**
25	0.00	0.01	0.04	0.07	0.11	0.14	0.17	0.19	0.21	0.23	0.25	0.27	0.28	0.29	0.31	0.32	0.33	**25**
26	0.00	0.01	0.04	0.07	0.11	0.14	0.17	0.19	0.21	0.23	0.25	0.27	0.28	0.29	0.31	0.32	0.33	**26**
27	0.00	0.01	0.04	0.07	0.11	0.14	0.17	0.19	0.21	0.23	0.25	0.27	0.28	0.30	0.31	0.32	0.33	**27**
28	0.00	0.01	0.04	0.07	0.11	0.14	0.17	0.19	0.21	0.23	0.25	0.27	0.28	0.30	0.31	0.32	0.33	**28**
29	0.00	0.01	0.04	0.07	0.11	0.14	0.17	0.19	0.21	0.23	0.25	0.27	0.28	0.30	0.31	0.32	0.33	**29**
30	0.00	0.01	0.04	0.07	0.11	0.14	0.17	0.19	0.22	0.24	0.25	0.27	0.29	0.30	0.31	0.32	0.33	**30**
40	0.00	0.01	0.04	0.07	0.11	0.14	0.17	0.20	0.22	0.24	0.26	0.28	0.29	0.31	0.32	0.33	0.34	**40**
60	0.00	0.01	0.04	0.07	0.11	0.14	0.17	0.20	0.22	0.24	0.26	0.28	0.30	0.31	0.33	0.34	0.35	**60**
100	0.00	0.01	0.04	0.07	0.11	0.14	0.17	0.20	0.23	0.25	0.27	0.29	0.31	0.32	0.34	0.35	0.36	**100**
∞	0.00	0.01	0.04	0.07	0.11	0.15	0.18	0.21	0.23	0.26	0.28	0.30	0.32	0.33	0.35	0.36	0.38	∞

Degrees of freedom for the denominator



	Degrees of freedom for the numerator																	
	(ν_1)																	
(ν_2)	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**40**	**60**	**100**	∞	(ν_2)
1	0.12	0.12	0.12	0.12	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.14	0.14	0.15	0.15	**1**
2	0.17	0.17	0.17	0.17	0.17	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.19	0.19	0.20	0.21	0.22	**2**
3	0.20	0.20	0.20	0.21	0.21	0.21	0.21	0.21	0.22	0.22	0.22	0.22	0.22	0.23	0.24	0.25	0.26	**3**
4	0.22	0.22	0.23	0.23	0.23	0.23	0.24	0.24	0.24	0.24	0.25	0.25	0.25	0.26	0.27	0.28	0.30	**4**
5	0.24	0.24	0.24	0.25	0.25	0.25	0.26	0.26	0.26	0.26	0.27	0.27	0.27	0.28	0.30	0.31	0.33	**5**
6	0.25	0.25	0.26	0.26	0.27	0.27	0.27	0.28	0.28	0.28	0.28	0.29	0.29	0.30	0.32	0.33	0.36	**6**
7	0.26	0.27	0.27	0.27	0.28	0.28	0.29	0.29	0.29	0.30	0.30	0.30	0.30	0.32	0.34	0.35	0.38	**7**
8	0.27	0.28	0.28	0.29	0.29	0.29	0.30	0.30	0.30	0.31	0.31	0.31	0.32	0.33	0.35	0.37	0.40	**8**
9	0.28	0.28	0.29	0.29	0.30	0.30	0.31	0.31	0.31	0.32	0.32	0.32	0.33	0.35	0.37	0.39	0.42	**9**
10	0.29	0.29	0.30	0.30	0.31	0.31	0.32	0.32	0.32	0.33	0.33	0.33	0.34	0.36	0.38	0.40	0.43	**10**
11	0.29	0.30	0.30	0.31	0.31	0.32	0.32	0.33	0.33	0.33	0.34	0.34	0.34	0.37	0.39	0.41	0.44	**11**
12	0.30	0.30	0.31	0.32	0.32	0.33	0.33	0.33	0.34	0.34	0.35	0.35	0.35	0.38	0.40	0.42	0.46	**12**
13	0.30	0.31	0.31	0.32	0.33	0.33	0.34	0.34	0.34	0.35	0.35	0.36	0.36	0.38	0.41	0.43	0.47	**13**
14	0.31	0.31	0.32	0.33	0.33	0.34	0.34	0.35	0.35	0.35	0.36	0.36	0.36	0.39	0.42	0.44	0.48	**14**
15	0.31	0.32	0.32	0.33	0.34	0.34	0.35	0.35	0.36	0.36	0.36	0.37	0.37	0.40	0.43	0.45	0.49	**15**
16	0.31	0.32	0.33	0.33	0.34	0.35	0.35	0.36	0.36	0.36	0.37	0.37	0.38	0.40	0.43	0.46	0.50	**16**
17	0.32	0.32	0.33	0.34	0.34	0.35	0.35	0.36	0.36	0.37	0.37	0.38	0.38	0.41	0.44	0.46	0.51	**17**
18	0.32	0.33	0.33	0.34	0.35	0.35	0.36	0.36	0.37	0.37	0.38	0.38	0.38	0.41	0.44	0.47	0.52	**18**
19	0.32	0.33	0.34	0.34	0.35	0.36	0.36	0.37	0.37	0.38	0.38	0.38	0.39	0.42	0.45	0.48	0.52	**19**
20	0.32	0.33	0.34	0.35	0.35	0.36	0.37	0.37	0.38	0.38	0.38	0.39	0.39	0.42	0.45	0.48	0.53	**20**
21	0.33	0.34	0.34	0.35	0.36	0.36	0.37	0.37	0.38	0.38	0.39	0.39	0.40	0.43	0.46	0.49	0.54	**21**
22	0.33	0.34	0.35	0.35	0.36	0.37	0.37	0.38	0.38	0.39	0.39	0.40	0.40	0.43	0.46	0.49	0.55	**22**
23	0.33	0.34	0.35	0.35	0.36	0.37	0.37	0.38	0.38	0.39	0.39	0.40	0.40	0.43	0.47	0.50	0.55	**23**
24	0.33	0.34	0.35	0.36	0.36	0.37	0.38	0.38	0.39	0.39	0.40	0.40	0.41	0.44	0.47	0.50	0.56	**24**
25	0.34	0.34	0.35	0.36	0.37	0.37	0.38	0.38	0.39	0.39	0.40	0.40	0.41	0.44	0.48	0.51	0.56	**25**
26	0.34	0.35	0.35	0.36	0.37	0.37	0.38	0.39	0.39	0.40	0.40	0.41	0.41	0.44	0.48	0.51	0.57	**26**
27	0.34	0.35	0.36	0.36	0.37	0.38	0.38	0.39	0.39	0.40	0.40	0.41	0.41	0.45	0.48	0.52	0.57	**27**
28	0.34	0.35	0.36	0.36	0.37	0.38	0.38	0.39	0.40	0.40	0.41	0.41	0.41	0.45	0.49	0.52	0.58	**28**
29	0.34	0.35	0.36	0.37	0.37	0.38	0.39	0.39	0.40	0.40	0.41	0.41	0.42	0.45	0.49	0.52	0.58	**29**
30	0.34	0.35	0.36	0.37	0.37	0.38	0.39	0.39	0.40	0.40	0.41	0.41	0.42	0.45	0.49	0.53	0.59	**30**
40	0.35	0.36	0.37	0.38	0.39	0.39	0.40	0.41	0.41	0.42	0.42	0.43	0.43	0.47	0.52	0.56	0.63	**40**
60	0.36	0.37	0.38	0.39	0.40	0.41	0.42	0.42	0.43	0.44	0.44	0.45	0.45	0.50	0.54	0.59	0.68	**60**
100	0.37	0.38	0.39	0.40	0.41	0.42	0.43	0.44	0.44	0.45	0.46	0.46	0.47	0.52	0.57	0.63	0.74	**100**
∞	0.39	0.40	0.41	0.42	0.43	0.44	0.45	0.46	0.47	0.48	0.48	0.49	0.50	0.55	0.62	0.70	1.00	∞

Degrees of freedom for the denominator

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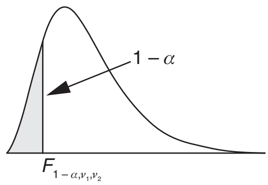
Appendix 17

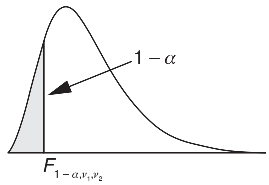
F(0.975) Distribution Table



Degrees of freedom for the numerator (ν_1)

Degrees of freedom for the denominator

(ν_2)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	(ν_2)
1	0.00	0.03	0.06	0.08	0.10	0.11	0.12	0.13	0.14	0.14	0.15	0.15	0.16	0.16	0.16	0.16	0.17	**1**
2	0.00	0.03	0.06	0.09	0.12	0.14	0.15	0.17	0.17	0.18	0.19	0.20	0.20	0.21	0.21	0.21	0.22	**2**
3	0.00	0.03	0.06	0.10	0.13	0.15	0.17	0.18	0.20	0.21	0.22	0.22	0.23	0.24	0.24	0.25	0.25	**3**
4	0.00	0.03	0.07	0.10	0.14	0.16	0.18	0.20	0.21	0.22	0.23	0.24	0.25	0.26	0.26	0.27	0.27	**4**
5	0.00	0.03	0.07	0.11	0.14	0.17	0.19	0.21	0.22	0.24	0.25	0.26	0.27	0.27	0.28	0.29	0.29	**5**
6	0.00	0.03	0.07	0.11	0.14	0.17	0.20	0.21	0.23	0.25	0.26	0.27	0.28	0.29	0.29	0.30	0.31	**6**
7	0.00	0.03	0.07	0.11	0.15	0.18	0.20	0.22	0.24	0.25	0.27	0.28	0.29	0.30	0.30	0.31	0.32	**7**
8	0.00	0.03	0.07	0.11	0.15	0.18	0.20	0.23	0.24	0.26	0.27	0.28	0.30	0.30	0.31	0.32	0.33	**8**
9	0.00	0.03	0.07	0.11	0.15	0.18	0.21	0.23	0.25	0.26	0.28	0.29	0.30	0.31	0.32	0.33	0.34	**9**
10	0.00	0.03	0.07	0.11	0.15	0.18	0.21	0.23	0.25	0.27	0.28	0.30	0.31	0.32	0.33	0.33	0.34	**10**
11	0.00	0.03	0.07	0.11	0.15	0.18	0.21	0.24	0.26	0.27	0.29	0.30	0.31	0.32	0.33	0.34	0.35	**11**
12	0.00	0.03	0.07	0.11	0.15	0.19	0.21	0.24	0.26	0.28	0.29	0.31	0.32	0.33	0.34	0.35	0.35	**12**
13	0.00	0.03	0.07	0.11	0.15	0.19	0.22	0.24	0.26	0.28	0.29	0.31	0.32	0.33	0.34	0.35	0.36	**13**
14	0.00	0.03	0.07	0.12	0.15	0.19	0.22	0.24	0.26	0.28	0.30	0.31	0.32	0.34	0.35	0.35	0.36	**14**
15	0.00	0.03	0.07	0.12	0.16	0.19	0.22	0.24	0.27	0.28	0.30	0.31	0.33	0.34	0.35	0.36	0.37	**15**
16	0.00	0.03	0.07	0.12	0.16	0.19	0.22	0.25	0.27	0.29	0.30	0.32	0.33	0.34	0.35	0.36	0.37	**16**
17	0.00	0.03	0.07	0.12	0.16	0.19	0.22	0.25	0.27	0.29	0.30	0.32	0.33	0.34	0.36	0.37	0.37	**17**
18	0.00	0.03	0.07	0.12	0.16	0.19	0.22	0.25	0.27	0.29	0.31	0.32	0.34	0.35	0.36	0.37	0.38	**18**
19	0.00	0.03	0.07	0.12	0.16	0.19	0.22	0.25	0.27	0.29	0.31	0.32	0.34	0.35	0.36	0.37	0.38	**19**
20	0.00	0.03	0.07	0.12	0.16	0.19	0.22	0.25	0.27	0.29	0.31	0.33	0.34	0.35	0.36	0.37	0.38	**20**
21	0.00	0.03	0.07	0.12	0.16	0.19	0.22	0.25	0.27	0.29	0.31	0.33	0.34	0.35	0.36	0.38	0.38	**21**
22	0.00	0.03	0.07	0.12	0.16	0.19	0.23	0.25	0.27	0.30	0.31	0.33	0.34	0.36	0.37	0.38	0.39	**22**
23	0.00	0.03	0.07	0.12	0.16	0.19	0.23	0.25	0.28	0.30	0.31	0.33	0.34	0.36	0.37	0.38	0.39	**23**
24	0.00	0.03	0.07	0.12	0.16	0.20	0.23	0.25	0.28	0.30	0.32	0.33	0.35	0.36	0.37	0.38	0.39	**24**
25	0.00	0.03	0.07	0.12	0.16	0.20	0.23	0.25	0.28	0.30	0.32	0.33	0.35	0.36	0.37	0.38	0.39	**25**
26	0.00	0.03	0.07	0.12	0.16	0.20	0.23	0.25	0.28	0.30	0.32	0.33	0.35	0.36	0.37	0.38	0.39	**26**
27	0.00	0.03	0.07	0.12	0.16	0.20	0.23	0.26	0.28	0.30	0.32	0.33	0.35	0.36	0.37	0.39	0.40	**27**
28	0.00	0.03	0.07	0.12	0.16	0.20	0.23	0.26	0.28	0.30	0.32	0.34	0.35	0.36	0.38	0.39	0.40	**28**
29	0.00	0.03	0.07	0.12	0.16	0.20	0.23	0.26	0.28	0.30	0.32	0.34	0.35	0.36	0.38	0.39	0.40	**29**
30	0.00	0.03	0.07	0.12	0.16	0.20	0.23	0.26	0.28	0.30	0.32	0.34	0.35	0.37	0.38	0.39	0.40	**30**
40	0.00	0.03	0.07	0.12	0.16	0.20	0.23	0.26	0.29	0.31	0.33	0.34	0.36	0.37	0.39	0.40	0.41	**40**
60	0.00	0.03	0.07	0.12	0.16	0.20	0.24	0.26	0.29	0.31	0.33	0.35	0.37	0.38	0.40	0.41	0.42	**60**
100	0.00	0.03	0.07	0.12	0.16	0.20	0.24	0.27	0.29	0.32	0.34	0.36	0.37	0.39	0.40	0.42	0.43	**100**
∞	0.00	0.03	0.07	0.12	0.17	0.21	0.24	0.27	0.30	0.32	0.35	0.37	0.39	0.40	0.42	0.43	0.44	∞



	Degrees of freedom for the numerator																	
	(ν_1)																	
(ν_2)	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**40**	**60**	**100**	**∞**	**(ν_2)**
1	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.19	0.19	0.20	**1**
2	0.22	0.22	0.22	0.23	0.23	0.23	0.23	0.23	0.23	0.24	0.24	0.24	0.24	0.25	0.25	0.26	0.27	**2**
3	0.25	0.26	0.26	0.26	0.26	0.27	0.27	0.27	0.27	0.27	0.28	0.28	0.28	0.29	0.30	0.31	0.32	**3**
4	0.28	0.28	0.28	0.29	0.29	0.29	0.30	0.30	0.30	0.30	0.30	0.31	0.31	0.32	0.33	0.34	0.36	**4**
5	0.30	0.30	0.30	0.31	0.31	0.31	0.32	0.32	0.32	0.32	0.33	0.33	0.33	0.34	0.36	0.37	0.39	**5**
6	0.31	0.32	0.32	0.32	0.33	0.33	0.33	0.34	0.34	0.34	0.34	0.35	0.35	0.36	0.38	0.39	0.42	**6**
7	0.32	0.33	0.33	0.34	0.34	0.34	0.35	0.35	0.35	0.36	0.36	0.36	0.36	0.38	0.40	0.41	0.44	**7**
8	0.33	0.34	0.34	0.35	0.35	0.36	0.36	0.36	0.37	0.37	0.37	0.37	0.38	0.40	0.41	0.43	0.46	**8**
9	0.34	0.35	0.35	0.36	0.36	0.37	0.37	0.37	0.38	0.38	0.38	0.39	0.39	0.41	0.43	0.45	0.47	**9**
10	0.35	0.35	0.36	0.37	0.37	0.37	0.38	0.38	0.39	0.39	0.39	0.40	0.40	0.42	0.44	0.46	0.49	**10**
11	0.36	0.36	0.37	0.37	0.38	0.38	0.39	0.39	0.39	0.40	0.40	0.40	0.41	0.43	0.45	0.47	0.50	**11**
12	0.36	0.37	0.37	0.38	0.38	0.39	0.39	0.40	0.40	0.41	0.41	0.41	0.41	0.44	0.46	0.48	0.51	**12**
13	0.37	0.37	0.38	0.38	0.39	0.40	0.40	0.40	0.41	0.41	0.42	0.42	0.42	0.44	0.47	0.49	0.53	**13**
14	0.37	0.38	0.38	0.39	0.40	0.40	0.41	0.41	0.41	0.42	0.42	0.42	0.43	0.45	0.48	0.50	0.54	**14**
15	0.37	0.38	0.39	0.39	0.40	0.41	0.41	0.41	0.42	0.42	0.43	0.43	0.43	0.46	0.49	0.51	0.55	**15**
16	0.38	0.39	0.39	0.40	0.40	0.41	0.41	0.42	0.42	0.43	0.43	0.44	0.44	0.46	0.49	0.52	0.55	**16**
17	0.38	0.39	0.40	0.40	0.41	0.41	0.42	0.42	0.43	0.43	0.44	0.44	0.44	0.47	0.50	0.52	0.56	**17**
18	0.39	0.39	0.40	0.41	0.41	0.42	0.42	0.43	0.43	0.44	0.44	0.44	0.45	0.47	0.50	0.53	0.57	**18**
19	0.39	0.40	0.40	0.41	0.42	0.42	0.43	0.43	0.44	0.44	0.44	0.45	0.45	0.48	0.51	0.54	0.58	**19**
20	0.39	0.40	0.41	0.41	0.42	0.42	0.43	0.43	0.44	0.44	0.45	0.45	0.46	0.48	0.51	0.54	0.59	**20**
21	0.39	0.40	0.41	0.42	0.42	0.43	0.43	0.44	0.44	0.45	0.45	0.46	0.46	0.49	0.52	0.55	0.59	**21**
22	0.40	0.40	0.41	0.42	0.42	0.43	0.44	0.44	0.45	0.45	0.45	0.46	0.46	0.49	0.52	0.55	0.60	**22**
23	0.40	0.41	0.41	0.42	0.43	0.43	0.44	0.44	0.45	0.45	0.46	0.46	0.47	0.49	0.53	0.56	0.60	**23**
24	0.40	0.41	0.42	0.42	0.43	0.43	0.44	0.45	0.45	0.46	0.46	0.46	0.47	0.50	0.53	0.56	0.61	**24**
25	0.40	0.41	0.42	0.42	0.43	0.44	0.44	0.45	0.45	0.46	0.46	0.47	0.47	0.50	0.54	0.56	0.62	**25**
26	0.40	0.41	0.42	0.43	0.43	0.44	0.45	0.45	0.46	0.46	0.47	0.47	0.47	0.50	0.54	0.57	0.62	**26**
27	0.40	0.41	0.42	0.43	0.44	0.44	0.45	0.45	0.46	0.46	0.47	0.47	0.48	0.51	0.54	0.57	0.63	**27**
28	0.41	0.41	0.42	0.43	0.44	0.44	0.45	0.45	0.46	0.46	0.47	0.47	0.48	0.51	0.55	0.58	0.63	**28**
29	0.41	0.42	0.42	0.43	0.44	0.44	0.45	0.46	0.46	0.47	0.47	0.48	0.48	0.51	0.55	0.58	0.63	**29**
30	0.41	0.42	0.43	0.43	0.44	0.45	0.45	0.46	0.46	0.47	0.47	0.48	0.48	0.51	0.55	0.58	0.64	**30**
40	0.42	0.43	0.44	0.45	0.45	0.46	0.47	0.47	0.48	0.48	0.49	0.49	0.50	0.53	0.57	0.61	0.67	**40**
60	0.43	0.44	0.45	0.46	0.47	0.47	0.48	0.49	0.49	0.50	0.51	0.51	0.52	0.55	0.60	0.64	0.72	**60**
100	0.44	0.45	0.46	0.47	0.48	0.49	0.49	0.50	0.51	0.51	0.52	0.53	0.53	0.57	0.63	0.67	0.77	**100**
∞	0.46	0.47	0.48	0.49	0.50	0.51	0.52	0.52	0.53	0.54	0.55	0.55	0.56	0.61	0.67	0.74	1.00	**∞**

Degrees of freedom for the denominator

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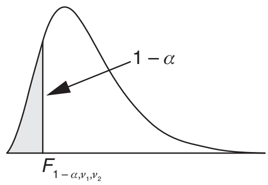
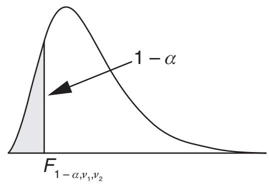
Appendix 18

F(0.95) Distribution Table



Degrees of freedom for the numerator (ν_1); Degrees of freedom for the denominator (ν_2)

(ν_2)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	(ν_2)
1	0.01	0.05	0.10	0.13	0.15	0.17	0.18	0.19	0.20	0.20	0.21	0.21	0.21	0.22	0.22	0.22	0.22	**1**
2	0.01	0.05	0.10	0.14	0.17	0.19	0.21	0.22	0.23	0.24	0.25	0.26	0.26	0.27	0.27	0.28	0.28	**2**
3	0.00	0.05	0.11	0.15	0.18	0.21	0.23	0.25	0.26	0.27	0.28	0.29	0.29	0.30	0.30	0.31	0.31	**3**
4	0.00	0.05	0.11	0.16	0.19	0.22	0.24	0.26	0.28	0.29	0.30	0.31	0.31	0.32	0.33	0.33	0.34	**4**
5	0.00	0.05	0.11	0.16	0.19	0.23	0.25	0.27	0.29	0.30	0.31	0.32	0.33	0.34	0.34	0.35	0.36	**5**
6	0.00	0.05	0.11	0.16	0.20	0.23	0.26	0.28	0.30	0.31	0.32	0.33	0.34	0.35	0.36	0.36	0.37	**6**
7	0.00	0.05	0.11	0.16	0.21	0.24	0.26	0.29	0.30	0.32	0.33	0.34	0.35	0.36	0.37	0.38	0.38	**7**
8	0.00	0.05	0.11	0.17	0.21	0.24	0.27	0.29	0.31	0.33	0.34	0.35	0.36	0.37	0.38	0.39	0.39	**8**
9	0.00	0.05	0.11	0.17	0.21	0.24	0.27	0.30	0.31	0.33	0.35	0.36	0.37	0.38	0.39	0.39	0.40	**9**
10	0.00	0.05	0.11	0.17	0.21	0.25	0.27	0.30	0.32	0.34	0.35	0.36	0.37	0.38	0.39	0.40	0.41	**10**
11	0.00	0.05	0.11	0.17	0.21	0.25	0.28	0.30	0.32	0.34	0.35	0.37	0.38	0.39	0.40	0.41	0.41	**11**
12	0.00	0.05	0.11	0.17	0.21	0.25	0.28	0.30	0.33	0.34	0.36	0.37	0.38	0.39	0.40	0.41	0.42	**12**
13	0.00	0.05	0.11	0.17	0.21	0.25	0.28	0.31	0.33	0.35	0.36	0.38	0.39	0.40	0.41	0.42	0.42	**13**
14	0.00	0.05	0.11	0.17	0.22	0.25	0.28	0.31	0.33	0.35	0.37	0.38	0.39	0.40	0.41	0.42	0.43	**14**
15	0.00	0.05	0.11	0.17	0.22	0.25	0.28	0.31	0.33	0.35	0.37	0.38	0.39	0.41	0.42	0.43	0.43	**15**
16	0.00	0.05	0.12	0.17	0.22	0.25	0.29	0.31	0.33	0.35	0.37	0.38	0.40	0.41	0.42	0.43	0.44	**16**
17	0.00	0.05	0.12	0.17	0.22	0.26	0.29	0.31	0.34	0.36	0.37	0.39	0.40	0.41	0.42	0.43	0.44	**17**
18	0.00	0.05	0.12	0.17	0.22	0.26	0.29	0.32	0.34	0.36	0.37	0.39	0.40	0.41	0.42	0.43	0.44	**18**
19	0.00	0.05	0.12	0.17	0.22	0.26	0.29	0.32	0.34	0.36	0.38	0.39	0.40	0.42	0.43	0.44	0.45	**19**
20	0.00	0.05	0.12	0.17	0.22	0.26	0.29	0.32	0.34	0.36	0.38	0.39	0.41	0.42	0.43	0.44	0.45	**20**
21	0.00	0.05	0.12	0.17	0.22	0.26	0.29	0.32	0.34	0.36	0.38	0.39	0.41	0.42	0.43	0.44	0.45	**21**
22	0.00	0.05	0.12	0.17	0.22	0.26	0.29	0.32	0.34	0.36	0.38	0.40	0.41	0.42	0.43	0.44	0.45	**22**
23	0.00	0.05	0.12	0.17	0.22	0.26	0.29	0.32	0.34	0.36	0.38	0.40	0.41	0.42	0.44	0.45	0.45	**23**
24	0.00	0.05	0.12	0.17	0.22	0.26	0.29	0.32	0.34	0.37	0.38	0.40	0.41	0.43	0.44	0.45	0.46	**24**
25	0.00	0.05	0.12	0.17	0.22	0.26	0.29	0.32	0.35	0.37	0.38	0.40	0.41	0.43	0.44	0.45	0.46	**25**
26	0.00	0.05	0.12	0.17	0.22	0.26	0.29	0.32	0.35	0.37	0.39	0.40	0.42	0.43	0.44	0.45	0.46	**26**
27	0.00	0.05	0.12	0.17	0.22	0.26	0.29	0.32	0.35	0.37	0.39	0.40	0.42	0.43	0.44	0.45	0.46	**27**
28	0.00	0.05	0.12	0.17	0.22	0.26	0.30	0.32	0.35	0.37	0.39	0.40	0.42	0.43	0.44	0.45	0.46	**28**
29	0.00	0.05	0.12	0.17	0.22	0.26	0.30	0.32	0.35	0.37	0.39	0.40	0.42	0.43	0.44	0.45	0.46	**29**
30	0.00	0.05	0.12	0.17	0.22	0.26	0.30	0.32	0.35	0.37	0.39	0.41	0.42	0.43	0.45	0.46	0.47	**30**
40	0.00	0.05	0.12	0.17	0.22	0.26	0.30	0.33	0.35	0.38	0.40	0.41	0.43	0.44	0.45	0.46	0.48	**40**
60	0.00	0.05	0.12	0.18	0.23	0.27	0.30	0.33	0.36	0.38	0.40	0.42	0.44	0.45	0.46	0.47	0.49	**60**
100	0.00	0.05	0.12	0.18	0.23	0.27	0.31	0.34	0.36	0.39	0.41	0.43	0.44	0.46	0.47	0.48	0.49	**100**
∞	0.00	0.05	0.12	0.18	0.23	0.27	0.31	0.34	0.37	0.39	0.42	0.44	0.45	0.47	0.48	0.50	0.51	∞



	Degrees of freedom for the numerator (ν_1)																	
(ν_2)	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**40**	**60**	**100**	**∞**	**(ν_2)**
1	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.25	0.25	0.26	**1**
2	0.28	0.28	0.29	0.29	0.29	0.29	0.29	0.30	0.30	0.30	0.30	0.30	0.30	0.31	0.32	0.32	0.33	**2**
3	0.32	0.32	0.32	0.33	0.33	0.33	0.33	0.33	0.34	0.34	0.34	0.34	0.34	0.35	0.36	0.37	0.38	**3**
4	0.34	0.35	0.35	0.35	0.36	0.36	0.36	0.36	0.36	0.37	0.37	0.37	0.37	0.38	0.40	0.41	0.42	**4**
5	0.36	0.36	0.37	0.37	0.38	0.38	0.38	0.38	0.39	0.39	0.39	0.39	0.39	0.41	0.42	0.43	0.45	**5**
6	0.38	0.38	0.38	0.39	0.39	0.40	0.40	0.40	0.40	0.41	0.41	0.41	0.41	0.43	0.44	0.46	0.48	**6**
7	0.39	0.39	0.40	0.40	0.41	0.41	0.41	0.42	0.42	0.42	0.42	0.43	0.43	0.44	0.46	0.48	0.50	**7**
8	0.40	0.40	0.41	0.41	0.42	0.42	0.42	0.43	0.43	0.43	0.44	0.44	0.44	0.46	0.48	0.49	0.52	**8**
9	0.41	0.41	0.42	0.42	0.43	0.43	0.43	0.44	0.44	0.44	0.45	0.45	0.45	0.47	0.49	0.51	0.53	**9**
10	0.41	0.42	0.43	0.43	0.44	0.44	0.44	0.45	0.45	0.45	0.46	0.46	0.46	0.48	0.50	0.52	0.55	**10**
11	0.42	0.43	0.43	0.44	0.44	0.45	0.45	0.45	0.46	0.46	0.46	0.47	0.47	0.49	0.51	0.53	0.56	**11**
12	0.43	0.43	0.44	0.44	0.45	0.45	0.46	0.46	0.47	0.47	0.47	0.48	0.48	0.50	0.52	0.54	0.57	**12**
13	0.43	0.44	0.44	0.45	0.46	0.46	0.46	0.47	0.47	0.48	0.48	0.48	0.48	0.51	0.53	0.55	0.58	**13**
14	0.44	0.44	0.45	0.46	0.46	0.47	0.47	0.47	0.48	0.48	0.48	0.49	0.49	0.51	0.54	0.56	0.59	**14**
15	0.44	0.45	0.45	0.46	0.46	0.47	0.47	0.48	0.48	0.49	0.49	0.49	0.50	0.52	0.54	0.57	0.60	**15**
16	0.44	0.45	0.46	0.46	0.47	0.47	0.48	0.48	0.49	0.49	0.49	0.50	0.50	0.53	0.55	0.57	0.61	**16**
17	0.45	0.46	0.46	0.47	0.47	0.48	0.48	0.49	0.49	0.50	0.50	0.50	0.51	0.53	0.56	0.58	0.62	**17**
18	0.45	0.46	0.46	0.47	0.48	0.48	0.49	0.49	0.50	0.50	0.50	0.51	0.51	0.54	0.56	0.59	0.62	**18**
19	0.45	0.46	0.47	0.47	0.48	0.49	0.49	0.49	0.50	0.50	0.51	0.51	0.51	0.54	0.57	0.59	0.63	**19**
20	0.46	0.46	0.47	0.48	0.48	0.49	0.49	0.50	0.50	0.51	0.51	0.51	0.52	0.54	0.57	0.60	0.64	**20**
21	0.46	0.47	0.47	0.48	0.49	0.49	0.50	0.50	0.51	0.51	0.51	0.52	0.52	0.55	0.58	0.60	0.64	**21**
22	0.46	0.47	0.48	0.48	0.49	0.49	0.50	0.50	0.51	0.51	0.52	0.52	0.52	0.55	0.58	0.61	0.65	**22**
23	0.46	0.47	0.48	0.48	0.49	0.50	0.50	0.51	0.51	0.52	0.52	0.52	0.53	0.55	0.58	0.61	0.65	**23**
24	0.47	0.47	0.48	0.49	0.49	0.50	0.50	0.51	0.51	0.52	0.52	0.53	0.53	0.56	0.59	0.61	0.66	**24**
25	0.47	0.47	0.48	0.49	0.50	0.50	0.51	0.51	0.52	0.52	0.52	0.53	0.53	0.56	0.59	0.62	0.66	**25**
26	0.47	0.48	0.48	0.49	0.50	0.50	0.51	0.51	0.52	0.52	0.53	0.53	0.53	0.56	0.59	0.62	0.67	**26**
27	0.47	0.48	0.49	0.49	0.50	0.50	0.51	0.52	0.52	0.52	0.53	0.53	0.54	0.57	0.60	0.63	0.67	**27**
28	0.47	0.48	0.49	0.49	0.50	0.51	0.51	0.52	0.52	0.53	0.53	0.54	0.54	0.57	0.60	0.63	0.68	**28**
29	0.47	0.48	0.49	0.50	0.50	0.51	0.51	0.52	0.52	0.53	0.53	0.54	0.54	0.57	0.60	0.63	0.68	**29**
30	0.47	0.48	0.49	0.50	0.50	0.51	0.52	0.52	0.53	0.53	0.54	0.54	0.54	0.57	0.61	0.64	0.69	**30**
40	0.48	0.49	0.50	0.51	0.52	0.52	0.53	0.53	0.54	0.54	0.55	0.55	0.56	0.59	0.63	0.66	0.72	**40**
60	0.50	0.51	0.51	0.52	0.53	0.54	0.54	0.55	0.55	0.56	0.57	0.57	0.57	0.61	0.65	0.69	0.76	**60**
100	0.51	0.52	0.52	0.53	0.54	0.55	0.56	0.56	0.57	0.57	0.58	0.58	0.59	0.63	0.68	0.72	0.80	**100**
∞	0.52	0.53	0.54	0.55	0.56	0.57	0.58	0.58	0.59	0.60	0.60	0.61	0.62	0.66	0.72	0.78	1.00	**∞**

Degrees of freedom for the denominator

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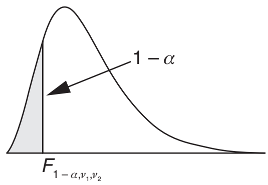
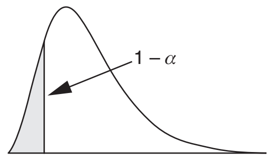
Appendix 19

F(0.90) Distribution Table



	Degrees of freedom for the numerator (ν_1)																	
(ν_2)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	(ν_2)
1	0.03	0.12	0.18	0.22	0.25	0.26	0.28	0.29	0.30	0.30	0.31	0.31	0.32	0.32	0.33	0.33	0.33	1
2	0.02	0.11	0.18	0.23	0.26	0.29	0.31	0.32	0.33	0.34	0.35	0.36	0.36	0.37	0.37	0.37	0.38	2
3	0.02	0.11	0.19	0.24	0.28	0.30	0.33	0.34	0.36	0.37	0.38	0.38	0.39	0.40	0.40	0.41	0.41	3
4	0.02	0.11	0.19	0.24	0.28	0.31	0.34	0.36	0.37	0.38	0.39	0.40	0.41	0.42	0.42	0.43	0.43	4
5	0.02	0.11	0.19	0.25	0.29	0.32	0.35	0.37	0.38	0.40	0.41	0.42	0.43	0.43	0.44	0.45	0.45	5
6	0.02	0.11	0.19	0.25	0.29	0.33	0.35	0.37	0.39	0.41	0.42	0.43	0.44	0.45	0.45	0.46	0.46	6
7	0.02	0.11	0.19	0.25	0.30	0.33	0.36	0.38	0.40	0.41	0.43	0.44	0.45	0.46	0.46	0.47	0.48	7
8	0.02	0.11	0.19	0.25	0.30	0.34	0.36	0.39	0.40	0.42	0.43	0.45	0.46	0.46	0.47	0.48	0.49	8
9	0.02	0.11	0.19	0.25	0.30	0.34	0.37	0.39	0.41	0.43	0.44	0.45	0.46	0.47	0.48	0.49	0.49	9
10	0.02	0.11	0.19	0.26	0.30	0.34	0.37	0.39	0.41	0.43	0.44	0.46	0.47	0.48	0.49	0.49	0.50	10
11	0.02	0.11	0.19	0.26	0.30	0.34	0.37	0.40	0.42	0.43	0.45	0.46	0.47	0.48	0.49	0.50	0.51	11
12	0.02	0.11	0.19	0.26	0.31	0.34	0.37	0.40	0.42	0.44	0.45	0.47	0.48	0.49	0.50	0.50	0.51	12
13	0.02	0.11	0.19	0.26	0.31	0.35	0.38	0.40	0.42	0.44	0.46	0.47	0.48	0.49	0.50	0.51	0.52	13
14	0.02	0.11	0.19	0.26	0.31	0.35	0.38	0.40	0.43	0.44	0.46	0.47	0.48	0.49	0.50	0.51	0.52	14
15	0.02	0.11	0.19	0.26	0.31	0.35	0.38	0.41	0.43	0.45	0.46	0.48	0.49	0.50	0.51	0.52	0.52	15
16	0.02	0.11	0.19	0.26	0.31	0.35	0.38	0.41	0.43	0.45	0.46	0.48	0.49	0.50	0.51	0.52	0.53	16
17	0.02	0.11	0.19	0.26	0.31	0.35	0.38	0.41	0.43	0.45	0.47	0.48	0.49	0.50	0.51	0.52	0.53	17
18	0.02	0.11	0.19	0.26	0.31	0.35	0.38	0.41	0.43	0.45	0.47	0.48	0.49	0.51	0.52	0.52	0.53	18
19	0.02	0.11	0.19	0.26	0.31	0.35	0.38	0.41	0.43	0.45	0.47	0.48	0.50	0.51	0.52	0.53	0.53	19
20	0.02	0.11	0.19	0.26	0.31	0.35	0.39	0.41	0.44	0.45	0.47	0.49	0.50	0.51	0.52	0.53	0.54	20
21	0.02	0.11	0.19	0.26	0.31	0.35	0.39	0.41	0.44	0.46	0.47	0.49	0.50	0.51	0.52	0.53	0.54	21
22	0.02	0.11	0.19	0.26	0.31	0.35	0.39	0.41	0.44	0.46	0.47	0.49	0.50	0.51	0.52	0.53	0.54	22
23	0.02	0.11	0.19	0.26	0.31	0.35	0.39	0.42	0.44	0.46	0.48	0.49	0.50	0.51	0.53	0.53	0.54	23
24	0.02	0.11	0.19	0.26	0.31	0.35	0.39	0.42	0.44	0.46	0.48	0.49	0.50	0.52	0.53	0.54	0.54	24
25	0.02	0.11	0.19	0.26	0.31	0.36	0.39	0.42	0.44	0.46	0.48	0.49	0.51	0.52	0.53	0.54	0.55	25
26	0.02	0.11	0.19	0.26	0.31	0.36	0.39	0.42	0.44	0.46	0.48	0.49	0.51	0.52	0.53	0.54	0.55	26
27	0.02	0.11	0.19	0.26	0.31	0.36	0.39	0.42	0.44	0.46	0.48	0.49	0.51	0.52	0.53	0.54	0.55	27
28	0.02	0.11	0.19	0.26	0.31	0.36	0.39	0.42	0.44	0.46	0.48	0.50	0.51	0.52	0.53	0.54	0.55	28
29	0.02	0.11	0.19	0.26	0.31	0.36	0.39	0.42	0.44	0.46	0.48	0.50	0.51	0.52	0.53	0.54	0.55	29
30	0.02	0.11	0.19	0.26	0.32	0.36	0.39	0.42	0.44	0.46	0.48	0.50	0.51	0.52	0.53	0.54	0.55	30
40	0.02	0.11	0.19	0.26	0.32	0.36	0.39	0.42	0.45	0.47	0.49	0.50	0.52	0.53	0.54	0.55	0.56	40
60	0.02	0.11	0.19	0.26	0.32	0.36	0.40	0.43	0.45	0.47	0.49	0.51	0.53	0.54	0.55	0.56	0.57	60
100	0.02	0.11	0.19	0.26	0.32	0.36	0.40	0.43	0.46	0.48	0.50	0.52	0.53	0.55	0.56	0.57	0.58	100
∞	0.02	0.11	0.19	0.27	0.32	0.37	0.40	0.44	0.46	0.49	0.51	0.53	0.54	0.56	0.57	0.58	0.59	∞

Degrees of freedom for the denominator



Degrees of freedom for the numerator (ν_1)

Degrees of freedom for the denominator (ν_2)

(ν_2)	18	19	20	21	22	23	24	25	26	27	28	29	30	40	60	100	∞	(ν_2)
1	0.33	0.33	0.34	0.34	0.34	0.34	0.34	0.34	0.34	0.34	0.35	0.35	0.35	0.35	0.36	0.36	0.37	**1**
2	0.38	0.38	0.39	0.39	0.39	0.39	0.39	0.40	0.40	0.40	0.40	0.40	0.40	0.41	0.42	0.42	0.43	**2**
3	0.41	0.42	0.42	0.42	0.43	0.43	0.43	0.43	0.43	0.44	0.44	0.44	0.44	0.45	0.46	0.47	0.48	**3**
4	0.44	0.44	0.44	0.45	0.45	0.45	0.46	0.46	0.46	0.46	0.46	0.47	0.47	0.48	0.49	0.50	0.51	**4**
5	0.46	0.46	0.46	0.47	0.47	0.47	0.48	0.48	0.48	0.48	0.48	0.49	0.49	0.50	0.51	0.52	0.54	**5**
6	0.47	0.47	0.48	0.48	0.49	0.49	0.49	0.49	0.50	0.50	0.50	0.50	0.50	0.52	0.53	0.55	0.56	**6**
7	0.48	0.49	0.49	0.49	0.50	0.50	0.50	0.51	0.51	0.51	0.51	0.52	0.52	0.53	0.55	0.56	0.58	**7**
8	0.49	0.50	0.50	0.50	0.51	0.51	0.52	0.52	0.52	0.52	0.53	0.53	0.53	0.55	0.56	0.58	0.60	**8**
9	0.50	0.50	0.51	0.51	0.52	0.52	0.52	0.53	0.53	0.53	0.54	0.54	0.54	0.56	0.58	0.59	0.61	**9**
10	0.51	0.51	0.52	0.52	0.53	0.53	0.53	0.54	0.54	0.54	0.54	0.55	0.55	0.57	0.59	0.60	0.63	**10**
11	0.51	0.52	0.52	0.53	0.53	0.54	0.54	0.54	0.55	0.55	0.55	0.55	0.56	0.58	0.60	0.61	0.64	**11**
12	0.52	0.52	0.53	0.53	0.54	0.54	0.55	0.55	0.55	0.56	0.56	0.56	0.56	0.58	0.60	0.62	0.65	**12**
13	0.52	0.53	0.53	0.54	0.54	0.55	0.55	0.56	0.56	0.56	0.56	0.57	0.57	0.59	0.61	0.63	0.66	**13**
14	0.53	0.53	0.54	0.54	0.55	0.55	0.56	0.56	0.56	0.57	0.57	0.57	0.58	0.60	0.62	0.64	0.66	**14**
15	0.53	0.54	0.54	0.55	0.55	0.56	0.56	0.56	0.57	0.57	0.57	0.58	0.58	0.60	0.62	0.64	0.67	**15**
16	0.53	0.54	0.55	0.55	0.56	0.56	0.56	0.57	0.57	0.58	0.58	0.58	0.59	0.61	0.63	0.65	0.68	**16**
17	0.54	0.54	0.55	0.55	0.56	0.56	0.57	0.57	0.58	0.58	0.58	0.59	0.59	0.61	0.63	0.65	0.69	**17**
18	0.54	0.55	0.55	0.56	0.56	0.57	0.57	0.58	0.58	0.58	0.59	0.59	0.59	0.62	0.64	0.66	0.69	**18**
19	0.54	0.55	0.55	0.56	0.57	0.57	0.58	0.58	0.58	0.59	0.59	0.59	0.60	0.62	0.64	0.66	0.70	**19**
20	0.54	0.55	0.56	0.56	0.57	0.57	0.58	0.58	0.59	0.59	0.59	0.60	0.60	0.62	0.65	0.67	0.70	**20**
21	0.55	0.55	0.56	0.57	0.57	0.58	0.58	0.58	0.59	0.59	0.60	0.60	0.60	0.63	0.65	0.67	0.71	**21**
22	0.55	0.56	0.56	0.57	0.57	0.58	0.58	0.59	0.59	0.60	0.60	0.60	0.61	0.63	0.66	0.68	0.71	**22**
23	0.55	0.56	0.56	0.57	0.58	0.58	0.59	0.59	0.59	0.60	0.60	0.60	0.61	0.63	0.66	0.68	0.72	**23**
24	0.55	0.56	0.57	0.57	0.58	0.58	0.59	0.59	0.60	0.60	0.60	0.61	0.61	0.64	0.66	0.68	0.72	**24**
25	0.55	0.56	0.57	0.57	0.58	0.58	0.59	0.59	0.60	0.60	0.61	0.61	0.61	0.64	0.66	0.69	0.73	**25**
26	0.56	0.56	0.57	0.58	0.58	0.59	0.59	0.60	0.60	0.60	0.61	0.61	0.62	0.64	0.67	0.69	0.73	**26**
27	0.56	0.56	0.57	0.58	0.58	0.59	0.59	0.60	0.60	0.61	0.61	0.61	0.62	0.64	0.67	0.69	0.73	**27**
28	0.56	0.57	0.57	0.58	0.58	0.59	0.59	0.60	0.60	0.61	0.61	0.62	0.62	0.64	0.67	0.70	0.74	**28**
29	0.56	0.57	0.57	0.58	0.59	0.59	0.60	0.60	0.61	0.61	0.61	0.62	0.62	0.65	0.68	0.70	0.74	**29**
30	0.56	0.57	0.58	0.58	0.59	0.59	0.60	0.60	0.61	0.61	0.62	0.62	0.62	0.65	0.68	0.70	0.75	**30**
40	0.57	0.58	0.59	0.59	0.60	0.60	0.61	0.61	0.62	0.62	0.63	0.63	0.64	0.66	0.70	0.72	0.77	**40**
60	0.58	0.59	0.60	0.60	0.61	0.62	0.62	0.63	0.63	0.64	0.64	0.65	0.65	0.68	0.72	0.75	0.81	**60**
100	0.59	0.60	0.61	0.61	0.62	0.63	0.63	0.64	0.64	0.65	0.65	0.66	0.66	0.70	0.74	0.77	0.84	**100**
∞	0.60	0.61	0.62	0.63	0.64	0.65	0.65	0.66	0.67	0.67	0.68	0.68	0.69	0.73	0.77	0.82	1.00	∞

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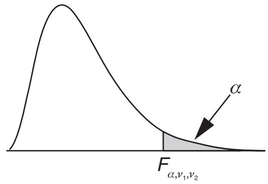
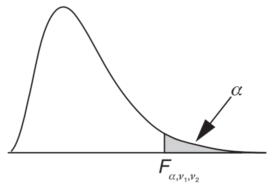
Appendix 20

F(0.10) Distribution Table



Degrees of freedom for the denominator

	Degrees of freedom for the numerator (ν_1)																	
(ν_2)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	(ν_2)
1	39.86	49.50	53.59	55.83	57.24	58.20	58.91	59.44	59.86	60.19	60.47	60.71	60.90	61.07	61.22	61.35	61.46	**1**
2	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38	9.39	9.40	9.41	9.41	9.42	9.42	9.43	9.43	**2**
3	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24	5.23	5.22	5.22	5.21	5.20	5.20	5.20	5.19	**3**
4	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.95	3.94	3.92	3.91	3.90	3.89	3.88	3.87	3.86	3.86	**4**
5	4.06	3.78	3.62	3.52	3.45	3.40	3.37	3.34	3.32	3.30	3.28	3.27	3.26	3.25	3.24	3.23	3.22	**5**
6	3.78	3.46	3.29	3.18	3.11	3.05	3.01	2.98	2.96	2.94	2.92	2.90	2.89	2.88	2.87	2.86	2.85	**6**
7	3.59	3.26	3.07	2.96	2.88	2.83	2.78	2.75	2.72	2.70	2.68	2.67	2.65	2.64	2.63	2.62	2.61	**7**
8	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56	2.54	2.52	2.50	2.49	2.48	2.46	2.45	2.45	**8**
9	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44	2.42	2.40	2.38	2.36	2.35	2.34	2.33	2.32	**9**
10	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35	2.32	2.30	2.28	2.27	2.26	2.24	2.23	2.22	**10**
11	3.23	2.86	2.66	2.54	2.45	2.39	2.34	2.30	2.27	2.25	2.23	2.21	2.19	2.18	2.17	2.16	2.15	**11**
12	3.18	2.81	2.61	2.48	2.39	2.33	2.28	2.24	2.21	2.19	2.17	2.15	2.13	2.12	2.10	2.09	2.08	**12**
13	3.14	2.76	2.56	2.43	2.35	2.28	2.23	2.20	2.16	2.14	2.12	2.10	2.08	2.07	2.05	2.04	2.03	**13**
14	3.10	2.73	2.52	2.39	2.31	2.24	2.19	2.15	2.12	2.10	2.07	2.05	2.04	2.02	2.01	2.00	1.99	**14**
15	3.07	2.70	2.49	2.36	2.27	2.21	2.16	2.12	2.09	2.06	2.04	2.02	2.00	1.99	1.97	1.96	1.95	**15**
16	3.05	2.67	2.46	2.33	2.24	2.18	2.13	2.09	2.06	2.03	2.01	1.99	1.97	1.95	1.94	1.93	1.92	**16**
17	3.03	2.64	2.44	2.31	2.22	2.15	2.10	2.06	2.03	2.00	1.98	1.96	1.94	1.93	1.91	1.90	1.89	**17**
18	3.01	2.62	2.42	2.29	2.20	2.13	2.08	2.04	2.00	1.98	1.95	1.93	1.92	1.90	1.89	1.87	1.86	**18**
19	2.99	2.61	2.40	2.27	2.18	2.11	2.06	2.02	1.98	1.96	1.93	1.91	1.89	1.88	1.86	1.85	1.84	**19**
20	2.97	2.59	2.38	2.25	2.16	2.09	2.04	2.00	1.96	1.94	1.91	1.89	1.87	1.86	1.84	1.83	1.82	**20**
21	2.96	2.57	2.36	2.23	2.14	2.08	2.02	1.98	1.95	1.92	1.90	1.87	1.86	1.84	1.83	1.81	1.80	**21**
22	2.95	2.56	2.35	2.22	2.13	2.06	2.01	1.97	1.93	1.90	1.88	1.86	1.84	1.83	1.81	1.80	1.79	**22**
23	2.94	2.55	2.34	2.21	2.11	2.05	1.99	1.95	1.92	1.89	1.87	1.84	1.83	1.81	1.80	1.78	1.77	**23**
24	2.93	2.54	2.33	2.19	2.10	2.04	1.98	1.94	1.91	1.88	1.85	1.83	1.81	1.80	1.78	1.77	1.76	**24**
25	2.92	2.53	2.32	2.18	2.09	2.02	1.97	1.93	1.89	1.87	1.84	1.82	1.80	1.79	1.77	1.76	1.75	**25**
26	2.91	2.52	2.31	2.17	2.08	2.01	1.96	1.92	1.88	1.86	1.83	1.81	1.79	1.77	1.76	1.75	1.73	**26**
27	2.90	2.51	2.30	2.17	2.07	2.00	1.95	1.91	1.87	1.85	1.82	1.80	1.78	1.76	1.75	1.74	1.72	**27**
28	2.89	2.50	2.29	2.16	2.06	2.00	1.94	1.90	1.87	1.84	1.81	1.79	1.77	1.75	1.74	1.73	1.71	**28**
29	2.89	2.50	2.28	2.15	2.06	1.99	1.93	1.89	1.86	1.83	1.80	1.78	1.76	1.75	1.73	1.72	1.71	**29**
30	2.88	2.49	2.28	2.14	2.05	1.98	1.93	1.88	1.85	1.82	1.79	1.77	1.75	1.74	1.72	1.71	1.70	**30**
40	2.84	2.44	2.23	2.09	2.00	1.93	1.87	1.83	1.79	1.76	1.74	1.71	1.70	1.68	1.66	1.65	1.64	**40**
60	2.79	2.39	2.18	2.04	1.95	1.87	1.82	1.77	1.74	1.71	1.68	1.66	1.64	1.62	1.60	1.59	1.58	**60**
100	2.76	2.36	2.14	2.00	1.91	1.83	1.78	1.73	1.69	1.66	1.64	1.61	1.59	1.57	1.56	1.54	1.53	**100**
∞	2.71	2.30	2.08	1.94	1.85	1.77	1.72	1.67	1.63	1.60	1.57	1.55	1.52	1.50	1.49	1.47	1.46	∞



	Degrees of freedom for the numerator (ν_1)																	
(ν_2)	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**40**	**60**	**100**	∞	(ν_2)
1	61.57	61.66	61.74	61.81	61.88	61.95	62.00	62.05	62.10	62.15	62.19	62.23	62.26	62.53	62.79	63.01	63.33	**1**
2	9.44	9.44	9.44	9.44	9.45	9.45	9.45	9.45	9.45	9.45	9.46	9.46	9.46	9.47	9.47	9.48	9.49	**2**
3	5.19	5.19	5.18	5.18	5.18	5.18	5.18	5.17	5.17	5.17	5.17	5.17	5.17	5.16	5.15	5.14	5.13	**3**
4	3.85	3.85	3.84	3.84	3.84	3.83	3.83	3.83	3.83	3.82	3.82	3.82	3.82	3.80	3.79	3.78	3.76	**4**
5	3.22	3.21	3.21	3.20	3.20	3.19	3.19	3.19	3.18	3.18	3.18	3.18	3.17	3.16	3.14	3.13	3.10	**5**
6	2.85	2.84	2.84	2.83	2.83	2.82	2.82	2.81	2.81	2.81	2.81	2.80	2.80	2.78	2.76	2.75	2.72	**6**
7	2.61	2.60	2.59	2.59	2.58	2.58	2.58	2.57	2.57	2.56	2.56	2.56	2.56	2.54	2.51	2.50	2.47	**7**
8	2.44	2.43	2.42	2.42	2.41	2.41	2.40	2.40	2.40	2.39	2.39	2.39	2.38	2.36	2.34	2.32	2.29	**8**
9	2.31	2.30	2.30	2.29	2.29	2.28	2.28	2.27	2.27	2.26	2.26	2.26	2.25	2.23	2.21	2.19	2.16	**9**
10	2.22	2.21	2.20	2.19	2.19	2.18	2.18	2.17	2.17	2.17	2.16	2.16	2.16	2.13	2.11	2.09	2.06	**10**
11	2.14	2.13	2.12	2.12	2.11	2.11	2.10	2.10	2.09	2.09	2.08	2.08	2.08	2.05	2.03	2.01	1.97	**11**
12	2.08	2.07	2.06	2.05	2.05	2.04	2.04	2.03	2.03	2.02	2.02	2.01	2.01	1.99	1.96	1.94	1.90	**12**
13	2.02	2.01	2.01	2.00	1.99	1.99	1.98	1.98	1.97	1.97	1.96	1.96	1.96	1.93	1.90	1.88	1.85	**13**
14	1.98	1.97	1.96	1.96	1.95	1.94	1.94	1.93	1.93	1.92	1.92	1.92	1.91	1.89	1.86	1.83	1.80	**14**
15	1.94	1.93	1.92	1.92	1.91	1.90	1.90	1.89	1.89	1.88	1.88	1.88	1.87	1.85	1.82	1.79	1.76	**15**
16	1.91	1.90	1.89	1.88	1.88	1.87	1.87	1.86	1.86	1.85	1.85	1.84	1.84	1.81	1.78	1.76	1.72	**16**
17	1.88	1.87	1.86	1.86	1.85	1.84	1.84	1.83	1.83	1.82	1.82	1.81	1.81	1.78	1.75	1.73	1.69	**17**
18	1.85	1.84	1.84	1.83	1.82	1.82	1.81	1.80	1.80	1.80	1.79	1.79	1.78	1.75	1.72	1.70	1.66	**18**
19	1.83	1.82	1.81	1.81	1.80	1.79	1.79	1.78	1.78	1.77	1.77	1.76	1.76	1.73	1.70	1.67	1.63	**19**
20	1.81	1.80	1.79	1.79	1.78	1.77	1.77	1.76	1.76	1.75	1.75	1.74	1.74	1.71	1.68	1.65	1.61	**20**
21	1.79	1.78	1.78	1.77	1.76	1.75	1.75	1.74	1.74	1.73	1.73	1.72	1.72	1.69	1.66	1.63	1.59	**21**
22	1.78	1.77	1.76	1.75	1.74	1.74	1.73	1.73	1.72	1.72	1.71	1.71	1.70	1.67	1.64	1.61	1.57	**22**
23	1.76	1.75	1.74	1.74	1.73	1.72	1.72	1.71	1.70	1.70	1.69	1.69	1.69	1.66	1.62	1.59	1.55	**23**
24	1.75	1.74	1.73	1.72	1.71	1.71	1.70	1.70	1.69	1.69	1.68	1.68	1.67	1.64	1.61	1.58	1.53	**24**
25	1.74	1.73	1.72	1.71	1.70	1.70	1.69	1.68	1.68	1.67	1.67	1.66	1.66	1.63	1.59	1.56	1.52	**25**
26	1.72	1.71	1.71	1.70	1.69	1.68	1.68	1.67	1.67	1.66	1.66	1.65	1.65	1.61	1.58	1.55	1.50	**26**
27	1.71	1.70	1.70	1.69	1.68	1.67	1.67	1.66	1.65	1.65	1.64	1.64	1.64	1.60	1.57	1.54	1.49	**27**
28	1.70	1.69	1.69	1.68	1.67	1.66	1.66	1.65	1.64	1.64	1.63	1.63	1.63	1.59	1.56	1.53	1.48	**28**
29	1.69	1.68	1.68	1.67	1.66	1.65	1.65	1.64	1.63	1.63	1.62	1.62	1.62	1.58	1.55	1.52	1.47	**29**
30	1.69	1.68	1.67	1.66	1.65	1.64	1.64	1.63	1.63	1.62	1.62	1.61	1.61	1.57	1.54	1.51	1.46	**30**
40	1.62	1.61	1.61	1.60	1.59	1.58	1.57	1.57	1.56	1.56	1.55	1.55	1.54	1.51	1.47	1.43	1.38	**40**
60	1.56	1.55	1.54	1.53	1.53	1.52	1.51	1.50	1.50	1.49	1.49	1.48	1.48	1.44	1.40	1.36	1.29	**60**
100	1.52	1.50	1.49	1.48	1.48	1.47	1.46	1.45	1.45	1.44	1.43	1.43	1.42	1.38	1.34	1.29	1.21	**100**
∞	1.44	1.43	1.42	1.41	1.40	1.39	1.38	1.38	1.37	1.36	1.35	1.35	1.34	1.30	1.24	1.18	1.00	∞

Degrees of freedom for the denominator

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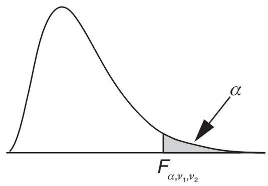
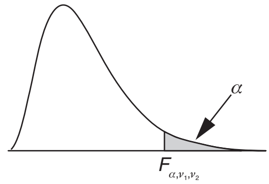
Appendix 21

F(0.05) Distribution Table



		Degrees of freedom for the numerator (ν_1)																	
	(ν_2)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	(ν_2)
Degrees of freedom for the denominator	1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	242.98	243.91	244.69	245.36	245.95	246.46	246.92	1
	2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.40	19.41	19.42	19.42	19.43	19.43	19.44	2
	3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.76	8.74	8.73	8.71	8.70	8.69	8.68	3
	4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.94	5.91	5.89	5.87	5.86	5.84	5.83	4
	5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.70	4.68	4.66	4.64	4.62	4.60	4.59	5
	6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.03	4.00	3.98	3.96	3.94	3.92	3.91	6
	7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.60	3.57	3.55	3.53	3.51	3.49	3.48	7
	8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.31	3.28	3.26	3.24	3.22	3.20	3.19	8
	9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.10	3.07	3.05	3.03	3.01	2.99	2.97	9
	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.94	2.91	2.89	2.86	2.85	2.83	2.81	10
	11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.82	2.79	2.76	2.74	2.72	2.70	2.69	11
	12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.72	2.69	2.66	2.64	2.62	2.60	2.58	12
	13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.63	2.60	2.58	2.55	2.53	2.51	2.50	13
	14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.57	2.53	2.51	2.48	2.46	2.44	2.43	14
	15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.51	2.48	2.45	2.42	2.40	2.38	2.37	15
	16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.46	2.42	2.40	2.37	2.35	2.33	2.32	16
	17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.41	2.38	2.35	2.33	2.31	2.29	2.27	17
	18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.37	2.34	2.31	2.29	2.27	2.25	2.23	18
	19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.34	2.31	2.28	2.26	2.23	2.21	2.20	19
	20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.31	2.28	2.25	2.22	2.20	2.18	2.17	20
	21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.28	2.25	2.22	2.20	2.18	2.16	2.14	21
	22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.26	2.23	2.20	2.17	2.15	2.13	2.11	22
	23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.24	2.20	2.18	2.15	2.13	2.11	2.09	23
	24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.22	2.18	2.15	2.13	2.11	2.09	2.07	24
	25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.20	2.16	2.14	2.11	2.09	2.07	2.05	25
	26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.18	2.15	2.12	2.09	2.07	2.05	2.03	26
	27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.17	2.13	2.10	2.08	2.06	2.04	2.02	27
	28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.15	2.12	2.09	2.06	2.04	2.02	2.00	28
	29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.14	2.10	2.08	2.05	2.03	2.01	1.99	29
	30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.13	2.09	2.06	2.04	2.01	1.99	1.98	30
	40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.04	2.00	1.97	1.95	1.92	1.90	1.89	40
	60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.95	1.92	1.89	1.86	1.84	1.82	1.80	60
	100	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.97	1.93	1.89	1.85	1.82	1.79	1.77	1.75	1.73	100
	∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.79	1.75	1.72	1.69	1.67	1.64	1.62	∞



Degrees of freedom for the numerator (ν_1)

Degrees of freedom for the denominator (ν_2)

(ν_2)	18	19	20	21	22	23	24	25	26	27	28	29	30	40	60	100	∞	(ν_2)
1	247.32	247.69	248.01	248.31	248.58	248.83	249.05	249.26	249.45	249.63	249.80	249.95	250.10	251.14	252.20	253.04	254.31	**1**
2	19.44	19.44	19.45	19.45	19.45	19.45	19.45	19.46	19.46	19.46	19.46	19.46	19.46	19.47	19.48	19.49	9.49	**2**
3	8.67	8.67	8.66	8.65	8.65	8.64	8.64	8.63	8.63	8.63	8.62	8.62	8.62	8.59	8.57	8.55	5.13	**3**
4	5.82	5.81	5.80	5.79	5.79	5.78	5.77	5.77	5.76	5.76	5.75	5.75	5.75	5.72	5.69	5.66	3.76	**4**
5	4.58	4.57	4.56	4.55	4.54	4.53	4.53	4.52	4.52	4.51	4.50	4.50	4.50	4.46	4.43	4.41	3.10	**5**
6	3.90	3.88	3.87	3.86	3.86	3.85	3.84	3.83	3.83	3.82	3.82	3.81	3.81	3.77	3.74	3.71	2.72	**6**
7	3.47	3.46	3.44	3.43	3.43	3.42	3.41	3.40	3.40	3.39	3.39	3.38	3.38	3.34	3.30	3.27	2.47	**7**
8	3.17	3.16	3.15	3.14	3.13	3.12	3.12	3.11	3.10	3.10	3.09	3.08	3.08	3.04	3.01	2.97	2.29	**8**
9	2.96	2.95	2.94	2.93	2.92	2.91	2.90	2.89	2.89	2.88	2.87	2.87	2.86	2.83	2.79	2.76	2.16	**9**
10	2.80	2.79	2.77	2.76	2.75	2.75	2.74	2.73	2.72	2.72	2.71	2.70	2.70	2.66	2.62	2.59	2.06	**10**
11	2.67	2.66	2.65	2.64	2.63	2.62	2.61	2.60	2.59	2.59	2.58	2.58	2.57	2.53	2.49	2.46	1.97	**11**
12	2.57	2.56	2.54	2.53	2.52	2.51	2.51	2.50	2.49	2.48	2.48	2.47	2.47	2.43	2.38	2.35	1.90	**12**
13	2.48	2.47	2.46	2.45	2.44	2.43	2.42	2.41	2.41	2.40	2.39	2.39	2.38	2.34	2.30	2.26	1.85	**13**
14	2.41	2.40	2.39	2.38	2.37	2.36	2.35	2.34	2.33	2.33	2.32	2.31	2.31	2.27	2.22	2.19	1.80	**14**
15	2.35	2.34	2.33	2.32	2.31	2.30	2.29	2.28	2.27	2.27	2.26	2.25	2.25	2.20	2.16	2.12	1.76	**15**
16	2.30	2.29	2.28	2.26	2.25	2.24	2.24	2.23	2.22	2.21	2.21	2.20	2.19	2.15	2.11	2.07	1.72	**16**
17	2.26	2.24	2.23	2.22	2.21	2.20	2.19	2.18	2.17	2.17	2.16	2.15	2.15	2.10	2.06	2.02	1.69	**17**
18	2.22	2.20	2.19	2.18	2.17	2.16	2.15	2.14	2.13	2.13	2.12	2.11	2.11	2.06	2.02	1.98	1.66	**18**
19	2.18	2.17	2.16	2.14	2.13	2.12	2.11	2.11	2.10	2.09	2.08	2.08	2.07	2.03	1.98	1.94	1.63	**19**
20	2.15	2.14	2.12	2.11	2.10	2.09	2.08	2.07	2.07	2.06	2.05	2.05	2.04	1.99	1.95	1.91	1.61	**20**
21	2.12	2.11	2.10	2.08	2.07	2.06	2.05	2.05	2.04	2.03	2.02	2.02	2.01	1.96	1.92	1.88	1.59	**21**
22	2.10	2.08	2.07	2.06	2.05	2.04	2.03	2.02	2.01	2.00	2.00	1.99	1.98	1.94	1.89	1.85	1.57	**22**
23	2.08	2.06	2.05	2.04	2.02	2.01	2.01	2.00	1.99	1.98	1.97	1.97	1.96	1.91	1.86	1.82	1.55	**23**
24	2.05	2.04	2.03	2.01	2.00	1.99	1.98	1.97	1.97	1.96	1.95	1.95	1.94	1.89	1.84	1.80	1.53	**24**
25	2.04	2.02	2.01	2.00	1.98	1.97	1.96	1.96	1.95	1.94	1.93	1.93	1.92	1.87	1.82	1.78	1.52	**25**
26	2.02	2.00	1.99	1.98	1.97	1.96	1.95	1.94	1.93	1.92	1.91	1.91	1.90	1.85	1.80	1.76	1.50	**26**
27	2.00	1.99	1.97	1.96	1.95	1.94	1.93	1.92	1.91	1.90	1.90	1.89	1.88	1.84	1.79	1.74	1.49	**27**
28	1.99	1.97	1.96	1.95	1.93	1.92	1.91	1.91	1.90	1.89	1.88	1.88	1.87	1.82	1.77	1.73	1.48	**28**
29	1.97	1.96	1.94	1.93	1.92	1.91	1.90	1.89	1.88	1.88	1.87	1.86	1.85	1.81	1.75	1.71	1.47	**29**
30	1.96	1.95	1.93	1.92	1.91	1.90	1.89	1.88	1.87	1.86	1.85	1.85	1.84	1.79	1.74	1.70	1.46	**30**
40	1.87	1.85	1.84	1.83	1.81	1.80	1.79	1.78	1.77	1.77	1.76	1.75	1.74	1.69	1.64	1.59	1.38	**40**
60	1.78	1.76	1.75	1.73	1.72	1.71	1.70	1.69	1.68	1.67	1.66	1.66	1.65	1.59	1.53	1.48	1.29	**60**
100	1.71	1.69	1.68	1.66	1.65	1.64	1.63	1.62	1.61	1.60	1.59	1.58	1.57	1.52	1.45	1.39	1.21	**100**
∞	1.60	1.59	1.57	1.56	1.54	1.53	1.52	1.51	1.50	1.49	1.48	1.47	1.46	1.39	1.32	1.24	1.00	∞

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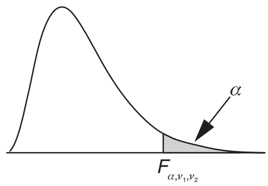
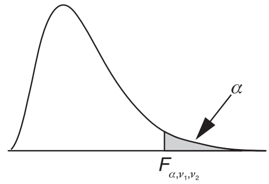
Appendix 22

F(0.025) Distribution Table



		Degrees of freedom for the numerator (ν_1)																	
Degrees of freedom for the denominator	(ν_2)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	(ν_2)
	1	647.79	799.50	864.16	899.58	921.85	937.11	948.22	956.66	963.28	968.63	973.03	976.71	979.84	982.53	984.87	986.92	988.73	1
	2	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40	39.41	39.41	39.42	39.43	39.43	39.44	39.44	2
	3	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42	14.37	14.34	14.30	14.28	14.25	14.23	14.21	3
	4	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90	8.84	8.79	8.75	8.71	8.68	8.66	8.63	8.61	4
	5	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62	6.57	6.52	6.49	6.46	6.43	6.40	6.38	5
	6	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52	5.46	5.41	5.37	5.33	5.30	5.27	5.24	5.22	6
	7	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76	4.71	4.67	4.63	4.60	4.57	4.54	4.52	7
	8	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30	4.24	4.20	4.16	4.13	4.10	4.08	4.05	8
	9	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96	3.91	3.87	3.83	3.80	3.77	3.74	3.72	9
	10	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72	3.66	3.62	3.58	3.55	3.52	3.50	3.47	10
	11	6.72	5.26	4.63	4.28	4.04	3.88	3.76	3.66	3.59	3.53	3.47	3.43	3.39	3.36	3.33	3.30	3.28	11
	12	6.55	5.10	4.47	4.12	3.89	3.73	3.61	3.51	3.44	3.37	3.32	3.28	3.24	3.21	3.18	3.15	3.13	12
	13	6.41	4.97	4.35	4.00	3.77	3.60	3.48	3.39	3.31	3.25	3.20	3.15	3.12	3.08	3.05	3.03	3.00	13
	14	6.30	4.86	4.24	3.89	3.66	3.50	3.38	3.29	3.21	3.15	3.09	3.05	3.01	2.98	2.95	2.92	2.90	14
	15	6.20	4.77	4.15	3.80	3.58	3.41	3.29	3.20	3.12	3.06	3.01	2.96	2.92	2.89	2.86	2.84	2.81	15
	16	6.12	4.69	4.08	3.73	3.50	3.34	3.22	3.12	3.05	2.99	2.93	2.89	2.85	2.82	2.79	2.76	2.74	16
	17	6.04	4.62	4.01	3.66	3.44	3.28	3.16	3.06	2.98	2.92	2.87	2.82	2.79	2.75	2.72	2.70	2.67	17
	18	5.98	4.56	3.95	3.61	3.38	3.22	3.10	3.01	2.93	2.87	2.81	2.77	2.73	2.70	2.67	2.64	2.62	18
	19	5.92	4.51	3.90	3.56	3.33	3.17	3.05	2.96	2.88	2.82	2.76	2.72	2.68	2.65	2.62	2.59	2.57	19
	20	5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.91	2.84	2.77	2.72	2.68	2.64	2.60	2.57	2.55	2.52	20
	21	5.83	4.42	3.82	3.48	3.25	3.09	2.97	2.87	2.80	2.73	2.68	2.64	2.60	2.56	2.53	2.51	2.48	21
	22	5.79	4.38	3.78	3.44	3.22	3.05	2.93	2.84	2.76	2.70	2.65	2.60	2.56	2.53	2.50	2.47	2.45	22
	23	5.75	4.35	3.75	3.41	3.18	3.02	2.90	2.81	2.73	2.67	2.62	2.57	2.53	2.50	2.47	2.44	2.42	23
	24	5.72	4.32	3.72	3.38	3.15	2.99	2.87	2.78	2.70	2.64	2.59	2.54	2.50	2.47	2.44	2.41	2.39	24
	25	5.69	4.29	3.69	3.35	3.13	2.97	2.85	2.75	2.68	2.61	2.56	2.51	2.48	2.44	2.41	2.38	2.36	25
	26	5.66	4.27	3.67	3.33	3.10	2.94	2.82	2.73	2.65	2.59	2.54	2.49	2.45	2.42	2.39	2.36	2.34	26
	27	5.63	4.24	3.65	3.31	3.08	2.92	2.80	2.71	2.63	2.57	2.51	2.47	2.43	2.39	2.36	2.34	2.31	27
	28	5.61	4.22	3.63	3.29	3.06	2.90	2.78	2.69	2.61	2.55	2.49	2.45	2.41	2.37	2.34	2.32	2.29	28
	29	5.59	4.20	3.61	3.27	3.04	2.88	2.76	2.67	2.59	2.53	2.48	2.43	2.39	2.36	2.32	2.30	2.27	29
	30	5.57	4.18	3.59	3.25	3.03	2.87	2.75	2.65	2.57	2.51	2.46	2.41	2.37	2.34	2.31	2.28	2.26	30
	40	5.42	4.05	3.46	3.13	2.90	2.74	2.62	2.53	2.45	2.39	2.33	2.29	2.25	2.21	2.18	2.15	2.13	40
	60	5.29	3.93	3.34	3.01	2.79	2.63	2.51	2.41	2.33	2.27	2.22	2.17	2.13	2.09	2.06	2.03	2.01	60
	100	5.18	3.83	3.25	2.92	2.70	2.54	2.42	2.32	2.24	2.18	2.12	2.08	2.04	2.00	1.97	1.94	1.91	100
	∞	5.02	3.69	3.12	2.79	2.57	2.41	2.29	2.19	2.11	2.05	1.99	1.94	1.90	1.87	1.83	1.80	1.78	∞



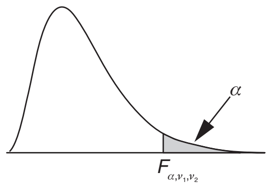
Degrees of freedom for the numerator (ν_1)

Degrees of freedom for the denominator (ν_2)

(ν_2)	18	19	20	21	22	23	24	25	26	27	28	29	30	40	60	100	∞	(ν_2)
1	990.35	991.80	993.10	994.29	995.36	996.35	997.25	998.08	998.85	999.56	1000.22	1000.84	1001.41	1005.60	1009.80	1013.17	1018.26	**1**
2	39.44	39.45	39.45	39.45	39.45	39.45	39.46	39.46	39.46	39.46	39.46	39.46	39.46	39.47	39.48	39.49	39.50	**2**
3	14.20	14.18	14.17	14.16	14.14	14.13	14.12	14.12	14.11	14.10	14.09	14.09	14.08	14.04	13.99	13.96	13.90	**3**
4	8.59	8.58	8.56	8.55	8.53	8.52	8.51	8.50	8.49	8.48	8.48	8.47	8.46	8.41	8.36	8.32	8.26	**4**
5	6.36	6.34	6.33	6.31	6.30	6.29	6.28	6.27	6.26	6.25	6.24	6.23	6.23	6.18	6.12	6.08	6.02	**5**
6	5.20	5.18	5.17	5.15	5.14	5.13	5.12	5.11	5.10	5.09	5.08	5.07	5.07	5.01	4.96	4.92	4.85	**6**
7	4.50	4.48	4.47	4.45	4.44	4.43	4.41	4.40	4.39	4.39	4.38	4.37	4.36	4.31	4.25	4.21	4.14	**7**
8	4.03	4.02	4.00	3.98	3.97	3.96	3.95	3.94	3.93	3.92	3.91	3.90	3.89	3.84	3.78	3.74	3.67	**8**
9	3.70	3.68	3.67	3.65	3.64	3.63	3.61	3.60	3.59	3.58	3.58	3.57	3.56	3.51	3.45	3.40	3.33	**9**
10	3.45	3.44	3.42	3.40	3.39	3.38	3.37	3.35	3.34	3.34	3.33	3.32	3.31	3.26	3.20	3.15	3.08	**10**
11	3.26	3.24	3.23	3.21	3.20	3.18	3.17	3.16	3.15	3.14	3.13	3.13	3.12	3.06	3.00	2.96	2.88	**11**
12	3.11	3.09	3.07	3.06	3.04	3.03	3.02	3.01	3.00	2.99	2.98	2.97	2.96	2.91	2.85	2.80	2.72	**12**
13	2.98	2.96	2.95	2.93	2.92	2.91	2.89	2.88	2.87	2.86	2.85	2.85	2.84	2.78	2.72	2.67	2.60	**13**
14	2.88	2.86	2.84	2.83	2.81	2.80	2.79	2.78	2.77	2.76	2.75	2.74	2.73	2.67	2.61	2.56	2.49	**14**
15	2.79	2.77	2.76	2.74	2.73	2.71	2.70	2.69	2.68	2.67	2.66	2.65	2.64	2.59	2.52	2.47	2.40	**15**
16	2.72	2.70	2.68	2.67	2.65	2.64	2.63	2.61	2.60	2.59	2.58	2.58	2.57	2.51	2.45	2.40	2.32	**16**
17	2.65	2.63	2.62	2.60	2.59	2.57	2.56	2.55	2.54	2.53	2.52	2.51	2.50	2.44	2.38	2.33	2.25	**17**
18	2.60	2.58	2.56	2.54	2.53	2.52	2.50	2.49	2.48	2.47	2.46	2.45	2.44	2.38	2.32	2.27	2.19	**18**
19	2.55	2.53	2.51	2.49	2.48	2.46	2.45	2.44	2.43	2.42	2.41	2.40	2.39	2.33	2.27	2.22	2.13	**19**
20	2.50	2.48	2.46	2.45	2.43	2.42	2.41	2.40	2.39	2.38	2.37	2.36	2.35	2.29	2.22	2.17	2.09	**20**
21	2.46	2.44	2.42	2.41	2.39	2.38	2.37	2.36	2.34	2.33	2.33	2.32	2.31	2.25	2.18	2.13	2.04	**21**
22	2.43	2.41	2.39	2.37	2.36	2.34	2.33	2.32	2.31	2.30	2.29	2.28	2.27	2.21	2.14	2.09	2.00	**22**
23	2.39	2.37	2.36	2.34	2.33	2.31	2.30	2.29	2.28	2.27	2.26	2.25	2.24	2.18	2.11	2.06	1.97	**23**
24	2.36	2.35	2.33	2.31	2.30	2.28	2.27	2.26	2.25	2.24	2.23	2.22	2.21	2.15	2.08	2.02	1.94	**24**
25	2.34	2.32	2.30	2.28	2.27	2.26	2.24	2.23	2.22	2.21	2.20	2.19	2.18	2.12	2.05	2.00	1.91	**25**
26	2.31	2.29	2.28	2.26	2.24	2.23	2.22	2.21	2.19	2.18	2.17	2.17	2.16	2.09	2.03	1.97	1.88	**26**
27	2.29	2.27	2.25	2.24	2.22	2.21	2.19	2.18	2.17	2.16	2.15	2.14	2.13	2.07	2.00	1.94	1.85	**27**
28	2.27	2.25	2.23	2.22	2.20	2.19	2.17	2.16	2.15	2.14	2.13	2.12	2.11	2.05	1.98	1.92	1.83	**28**
29	2.25	2.23	2.21	2.20	2.18	2.17	2.15	2.14	2.13	2.12	2.11	2.10	2.09	2.03	1.96	1.90	1.81	**29**
30	2.23	2.21	2.20	2.18	2.16	2.15	2.14	2.12	2.11	2.10	2.09	2.08	2.07	2.01	1.94	1.88	1.79	**30**
40	2.11	2.09	2.07	2.05	2.03	2.02	2.01	1.99	1.98	1.97	1.96	1.95	1.94	1.88	1.80	1.74	1.64	**40**
60	1.98	1.96	1.94	1.93	1.91	1.90	1.88	1.87	1.86	1.85	1.83	1.82	1.82	1.74	1.67	1.60	1.48	**60**
100	1.89	1.87	1.85	1.83	1.81	1.80	1.78	1.77	1.76	1.75	1.74	1.72	1.71	1.64	1.56	1.48	1.35	**100**
∞	1.75	1.73	1.71	1.69	1.67	1.66	1.64	1.63	1.61	1.60	1.59	1.58	1.57	1.48	1.39	1.30	1.00	∞

Appendix 23

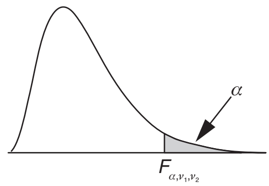
F(0.01) Distribution Table



Degrees of freedom for the numerator (ν_1)

Degrees of freedom for the denominator (ν_2)

(ν_2)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	(ν_2)
1	4052.18	4999.50	5403.35	5624.58	5763.65	5858.99	5928.36	5981.07	6022.47	6055.85	6083.32	6106.32	6125.86	6142.67	6157.28	6170.10	6181.43	1
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40	99.41	99.42	99.42	99.43	99.43	99.44	99.44	2
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23	27.13	27.05	26.98	26.92	26.87	26.83	26.79	3
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.45	14.37	14.31	14.25	14.20	14.15	14.11	4
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.96	9.89	9.82	9.77	9.72	9.68	9.64	5
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.79	7.72	7.66	7.60	7.56	7.52	7.48	6
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.54	6.47	6.41	6.36	6.31	6.28	6.24	7
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.73	5.67	5.61	5.56	5.52	5.48	5.44	8
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.18	5.11	5.05	5.01	4.96	4.92	4.89	9
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.77	4.71	4.65	4.60	4.56	4.52	4.49	10
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.46	4.40	4.34	4.29	4.25	4.21	4.18	11
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.22	4.16	4.10	4.05	4.01	3.97	3.94	12
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	4.02	3.96	3.91	3.86	3.82	3.78	3.75	13
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94	3.86	3.80	3.75	3.70	3.66	3.62	3.59	14
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.73	3.67	3.61	3.56	3.52	3.49	3.45	15
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.62	3.55	3.50	3.45	3.41	3.37	3.34	16
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.52	3.46	3.40	3.35	3.31	3.27	3.24	17
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.43	3.37	3.32	3.27	3.23	3.19	3.16	18
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.36	3.30	3.24	3.19	3.15	3.12	3.08	19
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.29	3.23	3.18	3.13	3.09	3.05	3.02	20
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31	3.24	3.17	3.12	3.07	3.03	2.99	2.96	21
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.18	3.12	3.07	3.02	2.98	2.94	2.91	22
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.14	3.07	3.02	2.97	2.93	2.89	2.86	23
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.09	3.03	2.98	2.93	2.89	2.85	2.82	24
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13	3.06	2.99	2.94	2.89	2.85	2.81	2.78	25
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09	3.02	2.96	2.90	2.86	2.81	2.78	2.75	26
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06	2.99	2.93	2.87	2.82	2.78	2.75	2.71	27
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03	2.96	2.90	2.84	2.79	2.75	2.72	2.68	28
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00	2.93	2.87	2.81	2.77	2.73	2.69	2.66	29
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.91	2.84	2.79	2.74	2.70	2.66	2.63	30
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.73	2.66	2.61	2.56	2.52	2.48	2.45	40
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.56	2.50	2.44	2.39	2.35	2.31	2.28	60
100	6.90	4.82	3.98	3.51	3.21	2.99	2.82	2.69	2.59	2.50	2.43	2.37	2.31	2.27	2.22	2.19	2.15	100
∞	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.25	2.18	2.13	2.08	2.04	2.00	1.97	∞



	Degrees of freedom for the numerator																	
	(ν_1)																	
(ν_2)	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**40**	**60**	**100**	**∞**	**(ν_2)**
1	6191.53	6200.58	6208.73	6216.12	6222.84	6228.99	6234.63	6239.83	6244.62	6249.07	6253.20	6257.05	6260.65	6286.78	6313.03	6334.11	6365.86	**1**
2	99.44	99.45	99.45	99.45	99.45	99.46	99.46	99.46	99.46	99.46	99.46	99.46	99.47	99.47	99.48	99.49	99.50	**2**
3	26.75	26.72	26.69	26.66	26.64	26.62	26.60	26.58	26.56	26.55	26.53	26.52	26.50	26.41	26.32	26.24	26.13	**3**
4	14.08	14.05	14.02	13.99	13.97	13.95	13.93	13.91	13.89	13.88	13.86	13.85	13.84	13.75	13.65	13.58	13.46	**4**
5	9.61	9.58	9.55	9.53	9.51	9.49	9.47	9.45	9.43	9.42	9.40	9.39	9.38	9.29	9.20	9.13	9.02	**5**
6	7.45	7.42	7.40	7.37	7.35	7.33	7.31	7.30	7.28	7.27	7.25	7.24	7.23	7.14	7.06	6.99	6.88	**6**
7	6.21	6.18	6.16	6.13	6.11	6.09	6.07	6.06	6.04	6.03	6.02	6.00	5.99	5.91	5.82	5.75	5.65	**7**
8	5.41	5.38	5.36	5.34	5.32	5.30	5.28	5.26	5.25	5.23	5.22	5.21	5.20	5.12	5.03	4.96	4.86	**8**
9	4.86	4.83	4.81	4.79	4.77	4.75	4.73	4.71	4.70	4.68	4.67	4.66	4.65	4.57	4.48	4.41	4.31	**9**
10	4.46	4.43	4.41	4.38	4.36	4.34	4.33	4.31	4.30	4.28	4.27	4.26	4.25	4.17	4.08	4.01	3.91	**10**
11	4.15	4.12	4.10	4.08	4.06	4.04	4.02	4.01	3.99	3.98	3.96	3.95	3.94	3.86	3.78	3.71	3.60	**11**
12	3.91	3.88	3.86	3.84	3.82	3.80	3.78	3.76	3.75	3.74	3.72	3.71	3.70	3.62	3.54	3.47	3.36	**12**
13	3.72	3.69	3.66	3.64	3.62	3.60	3.59	3.57	3.56	3.54	3.53	3.52	3.51	3.43	3.34	3.27	3.17	**13**
14	3.56	3.53	3.51	3.48	3.46	3.44	3.43	3.41	3.40	3.38	3.37	3.36	3.35	3.27	3.18	3.11	3.00	**14**
15	3.42	3.40	3.37	3.35	3.33	3.31	3.29	3.28	3.26	3.25	3.24	3.23	3.21	3.13	3.05	2.98	2.87	**15**
16	3.31	3.28	3.26	3.24	3.22	3.20	3.18	3.16	3.15	3.14	3.12	3.11	3.10	3.02	2.93	2.86	2.75	**16**
17	3.21	3.19	3.16	3.14	3.12	3.10	3.08	3.07	3.05	3.04	3.03	3.01	3.00	2.92	2.83	2.76	2.65	**17**
18	3.13	3.10	3.08	3.05	3.03	3.02	3.00	2.98	2.97	2.95	2.94	2.93	2.92	2.84	2.75	2.68	2.57	**18**
19	3.05	3.03	3.00	2.98	2.96	2.94	2.92	2.91	2.89	2.88	2.87	2.86	2.84	2.76	2.67	2.60	2.49	**19**
20	2.99	2.96	2.94	2.92	2.90	2.88	2.86	2.84	2.83	2.81	2.80	2.79	2.78	2.69	2.61	2.54	2.42	**20**
21	2.93	2.90	2.88	2.86	2.84	2.82	2.80	2.79	2.77	2.76	2.74	2.73	2.72	2.64	2.55	2.48	2.36	**21**
22	2.88	2.85	2.83	2.81	2.78	2.77	2.75	2.73	2.72	2.70	2.69	2.68	2.67	2.58	2.50	2.42	2.31	**22**
23	2.83	2.80	2.78	2.76	2.74	2.72	2.70	2.69	2.67	2.66	2.64	2.63	2.62	2.54	2.45	2.37	2.26	**23**
24	2.79	2.76	2.74	2.72	2.70	2.68	2.66	2.64	2.63	2.61	2.60	2.59	2.58	2.49	2.40	2.33	2.21	**24**
25	2.75	2.72	2.70	2.68	2.66	2.64	2.62	2.60	2.59	2.58	2.56	2.55	2.54	2.45	2.36	2.29	2.17	**25**
26	2.72	2.69	2.66	2.64	2.62	2.60	2.58	2.57	2.55	2.54	2.53	2.51	2.50	2.42	2.33	2.25	2.13	**26**
27	2.68	2.66	2.63	2.61	2.59	2.57	2.55	2.54	2.52	2.51	2.49	2.48	2.47	2.38	2.29	2.22	2.10	**27**
28	2.65	2.63	2.60	2.58	2.56	2.54	2.52	2.51	2.49	2.48	2.46	2.45	2.44	2.35	2.26	2.19	2.06	**28**
29	2.63	2.60	2.57	2.55	2.53	2.51	2.49	2.48	2.46	2.45	2.44	2.42	2.41	2.33	2.23	2.16	2.03	**29**
30	2.60	2.57	2.55	2.53	2.51	2.49	2.47	2.45	2.44	2.42	2.41	2.40	2.39	2.30	2.21	2.13	2.01	**30**
40	2.42	2.39	2.37	2.35	2.33	2.31	2.29	2.27	2.26	2.24	2.23	2.22	2.20	2.11	2.02	1.94	1.80	**40**
60	2.25	2.22	2.20	2.17	2.15	2.13	2.12	2.10	2.08	2.07	2.05	2.04	2.03	1.94	1.84	1.75	1.60	**60**
100	2.12	2.09	2.07	2.04	2.02	2.00	1.98	1.97	1.95	1.93	1.92	1.91	1.89	1.80	1.69	1.60	1.43	**100**
∞	1.93	1.90	1.88	1.85	1.83	1.81	1.79	1.77	1.76	1.74	1.72	1.71	1.70	1.59	1.47	1.36	1.00	**∞**

Degrees of freedom for the denominator

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Appendix 24

Median Ranks Table

	Sample of size *n*															
i	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	***i***
1	0.5000	0.2929	0.2063	0.1591	0.1294	0.1091	0.0943	0.0830	0.0741	0.0670	0.0611	0.0561	0.0519	0.0483	0.0452	**1**
2		0.7071	0.5000	0.3855	0.3136	0.2643	0.2284	0.2011	0.1797	0.1623	0.1480	0.1361	0.1259	0.1171	0.1095	**2**
3			0.7937	0.6145	0.5000	0.4214	0.3642	0.3207	0.2864	0.2588	0.2360	0.2169	0.2007	0.1867	0.1746	**3**
4				0.8409	0.6864	0.5786	0.5000	0.4402	0.3932	0.3553	0.3240	0.2978	0.2755	0.2564	0.2397	**4**
5					0.8706	0.7357	0.6358	0.5598	0.5000	0.4518	0.4120	0.3787	0.3504	0.3260	0.3048	**5**
6						0.8909	0.7716	0.6793	0.6068	0.5482	0.5000	0.4596	0.4252	0.3956	0.3698	**6**
7							0.9057	0.7989	0.7136	0.6447	0.5880	0.5404	0.5000	0.4652	0.4349	**7**
8								0.9170	0.8203	0.7412	0.6760	0.6213	0.5748	0.5348	0.5000	**8**
9									0.9259	0.8377	0.7640	0.7022	0.6496	0.6044	0.5651	**9**
10										0.9330	0.8520	0.7831	0.7245	0.6740	0.6302	**10**
11											0.9389	0.8639	0.7993	0.7436	0.6952	**11**
12												0.9439	0.8741	0.8133	0.7603	**12**
13													0.9481	0.8829	0.8254	**13**
14														0.9517	0.8905	**14**
15															0.9548	**15**
16																**16**
17																**17**
18																**18**
19																**19**
20																**20**
21																**21**
22																**22**
23																**23**
24																**24**
25																**25**
26																**26**
27																**27**
28																**28**
29																**29**
30																**30**

$$\text{Median rank}(i) = \begin{cases} 1 - \text{median rank}(n) & \text{for } i = 1 \\ \dfrac{i - 0.3175}{n + 0.365} & \text{for } i = 2, 3, \ldots, n-1 \\ 0.5^{1/n} & \text{for } i = n \end{cases}$$

	Sample of size *n*															
i	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	***i***
1	0.0424	0.0400	0.0378	0.0358	0.0341	0.0325	0.0310	0.0297	0.0285	0.0273	0.0263	0.0253	0.0245	0.0236	0.0228	**1**
2	0.1028	0.0969	0.0916	0.0869	0.0826	0.0788	0.0752	0.0720	0.0691	0.0663	0.0638	0.0615	0.0593	0.0573	0.0554	**2**
3	0.1639	0.1545	0.1461	0.1385	0.1317	0.1256	0.1199	0.1148	0.1101	0.1058	0.1017	0.0980	0.0946	0.0914	0.0883	**3**
4	0.2250	0.2121	0.2005	0.1902	0.1808	0.1724	0.1647	0.1576	0.1511	0.1452	0.1397	0.1346	0.1298	0.1254	0.1213	**4**
5	0.2861	0.2697	0.2550	0.2418	0.2299	0.2192	0.2094	0.2004	0.1922	0.1846	0.1776	0.1711	0.1651	0.1595	0.1542	**5**
6	0.3472	0.3272	0.3094	0.2934	0.2790	0.2660	0.2541	0.2432	0.2332	0.2240	0.2155	0.2077	0.2003	0.1935	0.1871	**6**
7	0.4083	0.3848	0.3639	0.3451	0.3281	0.3128	0.2988	0.2860	0.2743	0.2635	0.2535	0.2442	0.2356	0.2276	0.2201	**7**
8	0.4694	0.4424	0.4183	0.3967	0.3772	0.3596	0.3435	0.3288	0.3153	0.3029	0.2914	0.2807	0.2708	0.2616	0.2530	**8**
9	0.5306	0.5000	0.4728	0.4484	0.4263	0.4064	0.3882	0.3716	0.3564	0.3423	0.3293	0.3173	0.3061	0.2957	0.2859	**9**
10	0.5917	0.5576	0.5272	0.5000	0.4754	0.4532	0.4329	0.4144	0.3974	0.3817	0.3672	0.3538	0.3414	0.3297	0.3189	**10**
11	0.6528	0.6152	0.5817	0.5516	0.5246	0.5000	0.4776	0.4572	0.4384	0.4212	0.4052	0.3904	0.3766	0.3638	0.3518	**11**
12	0.7139	0.6728	0.6361	0.6033	0.5737	0.5468	0.5224	0.5000	0.4795	0.4606	0.4431	0.4269	0.4119	0.3978	0.3847	**12**
13	0.7750	0.7303	0.6906	0.6549	0.6228	0.5936	0.5671	0.5428	0.5205	0.5000	0.4810	0.4635	0.4471	0.4319	0.4177	**13**
14	0.8361	0.7879	0.7450	0.7066	0.6719	0.6404	0.6118	0.5856	0.5616	0.5394	0.5190	0.5000	0.4824	0.4659	0.4506	**14**
15	0.8972	0.8455	0.7995	0.7582	0.7210	0.6872	0.6565	0.6284	0.6026	0.5788	0.5569	0.5365	0.5176	0.5000	0.4835	**15**
16	0.9576	0.9031	0.8539	0.8098	0.7701	0.7340	0.7012	0.6712	0.6436	0.6183	0.5948	0.5731	0.5529	0.5341	0.5165	**16**
17		0.9600	0.9084	0.8615	0.8192	0.7808	0.7459	0.7140	0.6847	0.6577	0.6328	0.6096	0.5881	0.5681	0.5494	**17**
18			0.9622	0.9131	0.8683	0.8276	0.7906	0.7568	0.7257	0.6971	0.6707	0.6462	0.6234	0.6022	0.5823	**18**
19				0.9642	0.9174	0.8744	0.8353	0.7996	0.7668	0.7365	0.7086	0.6827	0.6586	0.6362	0.6153	**19**
20					0.9659	0.9212	0.8801	0.8424	0.8078	0.7760	0.7465	0.7193	0.6939	0.6703	0.6482	**20**
21						0.9675	0.9248	0.8852	0.8489	0.8154	0.7845	0.7558	0.7292	0.7043	0.6811	**21**
22							0.9690	0.9280	0.8899	0.8548	0.8224	0.7923	0.7644	0.7384	0.7141	**22**
23								0.9703	0.9309	0.8942	0.8603	0.8289	0.7997	0.7724	0.7470	**23**
24									0.9715	0.9337	0.8983	0.8654	0.8349	0.8065	0.7799	**24**
25										0.9727	0.9362	0.9020	0.8702	0.8405	0.8129	**25**
26											0.9737	0.9385	0.9054	0.8746	0.8458	**26**
27												0.9747	0.9407	0.9086	0.8787	**27**
28													0.9755	0.9427	0.9117	**28**
29														0.9764	0.9446	**29**
30															0.9772	**30**

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Appendix 25

Normal Scores Table

	n															
n	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	***n***
1	0.000	–0.545	–0.819	–0.998	–1.129	–1.231	–1.315	–1.385	–1.446	–1.499	–1.546	–1.588	–1.626	–1.662	–1.694	**1**
2		0.545	0.000	–0.291	–0.486	–0.630	–0.744	–0.838	–0.917	–0.985	–1.045	–1.098	–1.146	–1.189	–1.229	**2**
3			0.819	0.291	0.000	–0.198	–0.347	–0.466	–0.564	–0.647	–0.719	–0.783	–0.839	–0.890	–0.936	**3**
4				0.998	0.486	0.198	0.000	–0.150	–0.271	–0.371	–0.456	–0.531	–0.596	–0.655	–0.707	**4**
5					1.129	0.630	0.347	0.150	0.000	–0.121	–0.222	–0.309	–0.384	–0.451	–0.511	**5**
6						1.231	0.744	0.466	0.271	0.121	0.000	–0.102	–0.189	–0.265	–0.332	**6**
7							1.315	0.838	0.564	0.371	0.222	0.102	0.000	–0.087	–0.164	**7**
8								1.385	0.917	0.647	0.456	0.309	0.189	0.087	0.000	**8**
9									1.446	0.985	0.719	0.531	0.384	0.265	0.164	**9**
10										1.499	1.045	0.783	0.596	0.451	0.332	**10**
11											1.546	1.098	0.839	0.655	0.511	**11**
12												1.588	1.146	0.890	0.707	**12**
13													1.626	1.189	0.936	**13**
14														1.662	1.229	**14**
15															1.694	**15**
16																**16**
17																**17**
18																**18**
19																**19**
20																**20**
21																**21**
22																**22**
23																**23**
24																**24**
25																**25**
26																**26**
27																**27**
28																**28**
29																**29**
30																**30**

	n															
n	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**n**
1	–1.724	–1.751	–1.777	–1.801	–1.824	–1.846	–1.866	–1.885	–1.904	–1.921	–1.938	–1.954	–1.969	–1.984	–1.998	**1**
2	–1.266	–1.299	–1.331	–1.360	–1.388	–1.414	–1.438	–1.461	–1.483	–1.504	–1.524	–1.542	–1.561	–1.578	–1.595	**2**
3	–0.978	–1.017	–1.053	–1.087	–1.118	–1.148	–1.175	–1.201	–1.226	–1.249	–1.272	–1.293	–1.313	–1.332	–1.351	**3**
4	–0.755	–0.799	–0.840	–0.877	–0.912	–0.945	–0.976	–1.004	–1.032	–1.057	–1.082	–1.105	–1.127	–1.148	–1.169	**4**
5	–0.565	–0.614	–0.659	–0.701	–0.739	–0.775	–0.809	–0.840	–0.870	–0.898	–0.925	–0.950	–0.974	–0.997	–1.019	**5**
6	–0.393	–0.448	–0.497	–0.543	–0.586	–0.625	–0.662	–0.696	–0.728	–0.759	–0.787	–0.815	–0.840	–0.865	–0.888	**6**
7	–0.232	–0.293	–0.348	–0.399	–0.445	–0.488	–0.528	–0.565	–0.600	–0.633	–0.664	–0.693	–0.721	–0.747	–0.772	**7**
8	–0.077	–0.145	–0.206	–0.262	–0.313	–0.360	–0.403	–0.443	–0.481	–0.516	–0.549	–0.581	–0.610	–0.638	–0.665	**8**
9	0.077	0.000	–0.068	–0.130	–0.186	–0.237	–0.284	–0.328	–0.368	–0.406	–0.442	–0.475	–0.507	–0.537	–0.565	**9**
10	0.232	0.145	0.068	0.000	–0.062	–0.118	–0.169	–0.216	–0.260	–0.301	–0.339	–0.375	–0.409	–0.441	–0.471	**10**
11	0.393	0.293	0.206	0.130	0.062	0.000	–0.056	–0.107	–0.155	–0.199	–0.240	–0.278	–0.314	–0.348	–0.380	**11**
12	0.565	0.448	0.348	0.262	0.186	0.118	0.056	0.000	–0.051	–0.099	–0.143	–0.184	–0.223	–0.259	–0.293	**12**
13	0.755	0.614	0.497	0.399	0.313	0.237	0.169	0.107	0.051	0.000	–0.048	–0.092	–0.133	–0.172	–0.208	**13**
14	0.978	0.799	0.659	0.543	0.445	0.360	0.284	0.216	0.155	0.099	0.048	0.000	–0.044	–0.085	–0.124	**14**
15	1.266	1.017	0.840	0.701	0.586	0.488	0.403	0.328	0.260	0.199	0.143	0.092	0.044	0.000	–0.041	**15**
16	1.724	1.299	1.053	0.877	0.739	0.625	0.528	0.443	0.368	0.301	0.240	0.184	0.133	0.085	0.041	**16**
17		1.751	1.331	1.087	0.912	0.775	0.662	0.565	0.481	0.406	0.339	0.278	0.223	0.172	0.124	**17**
18			1.777	1.360	1.118	0.945	0.809	0.696	0.600	0.516	0.442	0.375	0.314	0.259	0.208	**18**
19				1.801	1.388	1.148	0.976	0.840	0.728	0.633	0.549	0.475	0.409	0.348	0.293	**19**
20					1.824	1.414	1.175	1.004	0.870	0.759	0.664	0.581	0.507	0.441	0.380	**20**
21						1.846	1.438	1.201	1.032	0.898	0.787	0.693	0.610	0.537	0.471	**21**
22							1.866	1.461	1.226	1.057	0.925	0.815	0.721	0.638	0.565	**22**
23								1.885	1.483	1.249	1.082	0.950	0.840	0.747	0.665	**23**
24									1.904	1.504	1.272	1.105	0.974	0.865	0.772	**24**
25										1.921	1.524	1.293	1.127	0.997	0.888	**25**
26											1.938	1.542	1.313	1.148	1.019	**26**
27												1.954	1.561	1.332	1.169	**27**
28													1.969	1.578	1.351	**28**
29														1.984	1.595	**29**
30															1.998	**30**

Appendix 26

Factors for One-Sided Tolerance Limits

	Confidence level								
	0.90			0.95			0.99		
	Proportion			Proportion			Proportion		
n	**0.90**	**0.95**	**0.99**	**0.90**	**0.95**	**0.99**	**0.90**	**0.95**	**0.99**
4	2.955	3.668	5.040	4.255	5.293	7.291	26.036	33.146	46.579
5	2.586	3.207	4.401	3.381	4.190	5.750	7.698	9.649	13.382
6	2.378	2.950	4.048	2.962	3.667	5.025	5.281	6.572	9.055
7	2.242	2.783	3.820	2.711	3.355	4.595	4.300	5.332	7.321
8	2.145	2.663	3.659	2.541	3.145	4.307	3.759	4.652	6.373
9	2.071	2.573	3.538	2.416	2.992	4.099	3.412	4.217	5.771
10	2.012	2.503	3.442	2.321	2.875	3.940	3.168	3.913	5.352
11	1.964	2.445	3.365	2.245	2.782	3.814	2.986	3.687	5.041
12	1.924	2.397	3.301	2.182	2.706	3.712	2.844	3.511	4.800
13	1.891	2.356	3.247	2.130	2.642	3.627	2.729	3.371	4.607
14	1.861	2.321	3.201	2.085	2.589	3.554	2.635	3.255	4.449
15	1.836	2.291	3.160	2.046	2.542	3.492	2.556	3.157	4.317
16	1.813	2.264	3.125	2.013	2.501	3.438	2.488	3.074	4.204
17	1.793	2.240	3.093	1.983	2.465	3.390	2.429	3.002	4.107
18	1.775	2.218	3.065	1.956	2.433	3.347	2.377	2.939	4.022
19	1.759	2.199	3.039	1.932	2.404	3.309	2.331	2.884	3.947
20	1.744	2.181	3.016	1.910	2.378	3.274	2.290	2.834	3.880
21	1.730	2.165	2.994	1.890	2.355	3.243	2.254	2.790	3.821
22	1.717	2.150	2.975	1.872	2.333	3.214	2.220	2.749	3.766
23	1.706	2.137	2.957	1.855	2.313	3.188	2.190	2.713	3.717
24	1.695	2.124	2.940	1.840	2.295	3.164	2.162	2.679	3.673
25	1.685	2.112	2.925	1.826	2.278	3.142	2.137	2.649	3.631
26	1.676	2.101	2.911	1.812	2.262	3.121	2.113	2.620	3.594
27	1.667	2.091	2.897	1.800	2.247	3.101	2.091	2.594	3.559
28	1.659	2.081	2.885	1.788	2.234	3.083	2.071	2.569	3.526
29	1.651	2.072	2.873	1.777	2.221	3.066	2.052	2.547	3.496
30	1.644	2.064	2.862	1.767	2.209	3.050	2.034	2.525	3.468

	Confidence level								
	0.90			0.95			0.99		
	Proportion			Proportion			Proportion		
n	**0.90**	**0.95**	**0.99**	**0.90**	**0.95**	**0.99**	**0.90**	**0.95**	**0.99**
31	1.637	2.056	2.851	1.758	2.197	3.035	2.018	2.506	3.441
32	1.630	2.048	2.841	1.748	2.186	3.021	2.002	2.487	3.416
33	1.624	2.041	2.832	1.740	2.176	3.008	1.987	2.469	3.393
34	1.618	2.034	2.823	1.732	2.166	2.995	1.973	2.453	3.371
35	1.613	2.027	2.814	1.724	2.157	2.983	1.960	2.437	3.350
36	1.608	2.021	2.806	1.716	2.148	2.972	1.948	2.422	3.330
37	1.602	2.015	2.799	1.709	2.140	2.961	1.936	2.408	3.312
38	1.598	2.009	2.791	1.702	2.132	2.950	1.924	2.394	3.294
39	1.593	2.004	2.784	1.696	2.124	2.941	1.914	2.381	3.277
40	1.588	1.999	2.777	1.690	2.117	2.931	1.903	2.369	3.261
41	1.584	1.994	2.771	1.684	2.110	2.922	1.894	2.358	3.245
42	1.580	1.989	2.765	1.678	2.104	2.913	1.884	2.346	3.231
43	1.576	1.984	2.759	1.673	2.097	2.905	1.875	2.336	3.217
44	1.572	1.980	2.753	1.668	2.091	2.897	1.867	2.326	3.203
45	1.569	1.976	2.748	1.662	2.085	2.889	1.858	2.316	3.190
46	1.565	1.971	2.742	1.658	2.079	2.882	1.850	2.306	3.178
47	1.562	1.967	2.737	1.653	2.074	2.875	1.843	2.297	3.166
48	1.558	1.964	2.732	1.648	2.069	2.868	1.835	2.288	3.154
49	1.555	1.960	2.727	1.644	2.063	2.861	1.828	2.280	3.143
50	1.552	1.956	2.723	1.640	2.058	2.855	1.821	2.272	3.133
51	1.549	1.953	2.718	1.636	2.054	2.848	1.815	2.264	3.122
52	1.546	1.949	2.714	1.632	2.049	2.842	1.808	2.257	3.113
53	1.543	1.946	2.710	1.628	2.045	2.837	1.802	2.249	3.103
54	1.541	1.943	2.706	1.624	2.040	2.831	1.796	2.242	3.094
55	1.538	1.940	2.702	1.621	2.036	2.826	1.791	2.235	3.085
56	1.536	1.937	2.698	1.617	2.032	2.820	1.785	2.229	3.076
57	1.533	1.934	2.694	1.614	2.028	2.815	1.780	2.222	3.068
58	1.531	1.931	2.690	1.610	2.024	2.810	1.774	2.216	3.060
59	1.528	1.929	2.687	1.607	2.020	2.805	1.769	2.210	3.052
60	1.526	1.926	2.684	1.604	2.017	2.801	1.764	2.204	3.044
61	1.524	1.923	2.680	1.601	2.013	2.796	1.759	2.199	3.037
62	1.522	1.921	2.677	1.598	2.010	2.792	1.755	2.193	3.029
63	1.520	1.918	2.674	1.595	2.007	2.787	1.750	2.188	3.022
64	1.518	1.916	2.671	1.593	2.003	2.783	1.746	2.183	3.016
65	1.516	1.914	2.668	1.590	2.000	2.779	1.741	2.178	3.009
66	1.514	1.911	2.665	1.587	1.997	2.775	1.737	2.173	3.003

n	Confidence level								
	0.90			0.95			0.99		
	Proportion			Proportion			Proportion		
	0.90	0.95	0.99	0.90	0.95	0.99	0.90	0.95	0.99
67	1.512	1.909	2.662	1.585	1.994	2.771	1.733	2.168	2.996
68	1.510	1.907	2.659	1.582	1.991	2.767	1.729	2.163	2.990
69	1.508	1.905	2.657	1.580	1.988	2.764	1.725	2.159	2.984
70	1.506	1.903	2.654	1.577	1.985	2.760	1.721	2.154	2.978
71	1.504	1.901	2.651	1.575	1.983	2.756	1.718	2.150	2.973
72	1.503	1.899	2.649	1.573	1.980	2.753	1.714	2.145	2.967
73	1.501	1.897	2.646	1.570	1.977	2.750	1.711	2.141	2.962
74	1.499	1.895	2.644	1.568	1.975	2.746	1.707	2.137	2.957
75	1.498	1.893	2.642	1.566	1.972	2.743	1.704	2.133	2.951
76	1.496	1.891	2.639	1.564	1.970	2.740	1.701	2.129	2.946
77	1.495	1.890	2.637	1.562	1.967	2.737	1.697	2.126	2.941
78	1.493	1.888	2.635	1.560	1.965	2.734	1.694	2.122	2.937
79	1.492	1.886	2.633	1.558	1.963	2.731	1.691	2.118	2.932
80	1.490	1.885	2.630	1.556	1.961	2.728	1.688	2.115	2.927
81	1.489	1.883	2.628	1.554	1.958	2.725	1.685	2.111	2.923
82	1.488	1.881	2.626	1.552	1.956	2.722	1.682	2.108	2.918
83	1.486	1.880	2.624	1.550	1.954	2.720	1.679	2.105	2.914
84	1.485	1.878	2.622	1.549	1.952	2.717	1.677	2.101	2.910
85	1.484	1.877	2.620	1.547	1.950	2.714	1.674	2.098	2.906
86	1.482	1.875	2.618	1.545	1.948	2.712	1.671	2.095	2.902
87	1.481	1.874	2.617	1.543	1.946	2.709	1.669	2.092	2.898
88	1.480	1.872	2.615	1.542	1.944	2.707	1.666	2.089	2.894
89	1.479	1.871	2.613	1.540	1.942	2.704	1.664	2.086	2.890
90	1.477	1.870	2.611	1.539	1.940	2.702	1.661	2.083	2.886
91	1.476	1.868	2.610	1.537	1.938	2.700	1.659	2.080	2.882
92	1.475	1.867	2.608	1.535	1.937	2.697	1.656	2.077	2.879
93	1.474	1.866	2.606	1.534	1.935	2.695	1.654	2.075	2.875
94	1.473	1.864	2.605	1.532	1.933	2.693	1.652	2.072	2.872
95	1.472	1.863	2.603	1.531	1.932	2.691	1.649	2.069	2.868
96	1.471	1.862	2.601	1.530	1.930	2.688	1.647	2.067	2.865
97	1.470	1.861	2.600	1.528	1.928	2.686	1.645	2.064	2.862
98	1.469	1.859	2.598	1.527	1.927	2.684	1.643	2.062	2.858
99	1.468	1.858	2.597	1.525	1.925	2.682	1.641	2.059	2.855
100	1.467	1.857	2.595	1.524	1.923	2.680	1.639	2.057	2.852
200	1.410	1.791	2.511	1.448	1.836	2.568	1.524	1.923	2.679
300	1.385	1.763	2.476	1.416	1.799	2.521	1.476	1.868	2.609

	Confidence level								
	0.90			0.95			0.99		
	Proportion			Proportion			Proportion		
n	**0.90**	**0.95**	**0.99**	**0.90**	**0.95**	**0.99**	**0.90**	**0.95**	**0.99**
400	1.371	1.747	2.455	1.397	1.777	2.493	1.448	1.836	2.568
500	1.361	1.736	2.441	1.385	1.762	2.475	1.430	1.814	2.540
600	1.354	1.727	2.430	1.375	1.752	2.461	1.416	1.799	2.521
700	1.349	1.721	2.422	1.368	1.744	2.451	1.406	1.787	2.505
800	1.344	1.716	2.416	1.362	1.737	2.442	1.397	1.777	2.493
900	1.340	1.712	2.411	1.358	1.732	2.436	1.390	1.769	2.483
1000	1.337	1.708	2.406	1.354	1.727	2.430	1.385	1.762	2.475

Appendix 27

Factors for Two-Sided Tolerance Limits

n	Confidence level 0.90			0.95			0.99		
	Proportion			Proportion			Proportion		
	0.90	**0.95**	**0.99**	**0.90**	**0.95**	**0.99**	**0.90**	**0.95**	**0.99**
4	4.166	4.930	6.420	5.368	6.353	8.274	9.397	11.121	14.483
5	3.494	4.141	5.407	4.274	5.065	6.614	6.611	7.834	10.230
6	3.131	3.713	4.856	3.712	4.402	5.757	5.336	6.328	8.277
7	2.901	3.443	4.507	3.368	3.997	5.232	4.612	5.473	7.165
8	2.742	3.255	4.265	3.135	3.722	4.876	4.147	4.922	6.449
9	2.625	3.117	4.086	2.967	3.522	4.617	3.822	4.538	5.948
10	2.535	3.010	3.948	2.838	3.371	4.420	3.582	4.254	5.578
11	2.463	2.925	3.837	2.737	3.251	4.264	3.397	4.034	5.292
12	2.404	2.856	3.747	2.655	3.153	4.137	3.249	3.860	5.064
13	2.355	2.798	3.671	2.586	3.073	4.032	3.129	3.717	4.878
14	2.313	2.748	3.607	2.529	3.005	3.943	3.029	3.599	4.723
15	2.277	2.706	3.552	2.480	2.946	3.867	2.944	3.498	4.592
16	2.246	2.669	3.503	2.437	2.895	3.801	2.871	3.412	4.479
17	2.218	2.636	3.461	2.399	2.851	3.743	2.808	3.337	4.380
18	2.194	2.607	3.423	2.366	2.812	3.691	2.752	3.271	4.294
19	2.172	2.581	3.389	2.336	2.776	3.645	2.703	3.212	4.218
20	2.152	2.558	3.358	2.310	2.745	3.604	2.659	3.160	4.149
21	2.134	2.536	3.331	2.285	2.716	3.566	2.619	3.113	4.088
22	2.118	2.517	3.305	2.264	2.690	3.532	2.584	3.071	4.032
23	2.103	2.499	3.282	2.243	2.666	3.501	2.551	3.032	3.981
24	2.089	2.483	3.261	2.225	2.644	3.473	2.521	2.996	3.935
25	2.077	2.468	3.241	2.208	2.624	3.446	2.494	2.964	3.892
26	2.065	2.454	3.223	2.192	2.606	3.422	2.469	2.934	3.853
27	2.054	2.441	3.206	2.178	2.588	3.399	2.445	2.906	3.817
28	2.044	2.429	3.190	2.164	2.572	3.378	2.424	2.881	3.783
29	2.034	2.418	3.175	2.151	2.557	3.358	2.403	2.857	3.752
30	2.025	2.407	3.161	2.140	2.543	3.340	2.385	2.834	3.722

	Confidence level								
	0.90			0.95			0.99		
	Proportion			Proportion			Proportion		
n	0.90	0.95	0.99	0.90	0.95	0.99	0.90	0.95	0.99
31	2.017	2.397	3.148	2.128	2.530	3.323	2.367	2.813	3.695
32	2.009	2.387	3.136	2.118	2.517	3.306	2.350	2.794	3.669
33	2.001	2.379	3.124	2.108	2.505	3.291	2.335	2.775	3.645
34	1.994	2.370	3.113	2.099	2.494	3.276	2.320	2.757	3.622
35	1.987	2.362	3.103	2.090	2.484	3.262	2.306	2.741	3.600
36	1.981	2.355	3.093	2.081	2.474	3.249	2.293	2.725	3.580
37	1.975	2.347	3.083	2.073	2.464	3.237	2.280	2.711	3.560
38	1.969	2.341	3.074	2.066	2.455	3.225	2.269	2.696	3.542
39	1.964	2.334	3.066	2.058	2.447	3.214	2.257	2.683	3.524
40	1.958	2.328	3.058	2.051	2.438	3.203	2.246	2.670	3.507
41	1.953	2.322	3.050	2.045	2.431	3.193	2.236	2.658	3.491
42	1.948	2.316	3.042	2.038	2.423	3.183	2.226	2.646	3.476
43	1.944	2.311	3.035	2.032	2.416	3.173	2.217	2.635	3.462
44	1.939	2.305	3.028	2.027	2.409	3.164	2.208	2.625	3.448
45	1.935	2.300	3.022	2.021	2.402	3.156	2.200	2.615	3.434
46	1.931	2.295	3.015	2.016	2.396	3.147	2.191	2.605	3.421
47	1.927	2.291	3.009	2.010	2.390	3.139	2.183	2.595	3.409
48	1.923	2.286	3.003	2.006	2.384	3.131	2.176	2.586	3.397
49	1.920	2.282	2.998	2.001	2.378	3.124	2.168	2.578	3.386
50	1.916	2.278	2.992	1.996	2.373	3.117	2.161	2.569	3.375
51	1.913	2.274	2.987	1.992	2.367	3.110	2.155	2.561	3.364
52	1.910	2.270	2.982	1.987	2.362	3.103	2.148	2.553	3.354
53	1.906	2.266	2.977	1.983	2.357	3.097	2.142	2.546	3.344
54	1.903	2.262	2.972	1.979	2.353	3.090	2.136	2.538	3.334
55	1.900	2.259	2.967	1.975	2.348	3.084	2.130	2.531	3.325
56	1.897	2.256	2.963	1.972	2.344	3.078	2.124	2.525	3.316
57	1.895	2.252	2.958	1.968	2.339	3.073	2.118	2.518	3.308
58	1.892	2.249	2.954	1.964	2.335	3.067	2.113	2.512	3.299
59	1.889	2.246	2.950	1.961	2.331	3.062	2.108	2.505	3.291
60	1.887	2.243	2.946	1.958	2.327	3.057	2.103	2.499	3.283
61	1.884	2.240	2.942	1.954	2.323	3.052	2.098	2.494	3.276
62	1.882	2.237	2.939	1.951	2.319	3.047	2.093	2.488	3.268
63	1.880	2.234	2.935	1.948	2.316	3.042	2.088	2.482	3.261
64	1.877	2.232	2.931	1.945	2.312	3.037	2.084	2.477	3.254
65	1.875	2.229	2.928	1.942	2.309	3.033	2.080	2.472	3.247
66	1.873	2.226	2.925	1.939	2.305	3.029	2.075	2.467	3.241

	Confidence level								
	0.90			0.95			0.99		
	Proportion			Proportion			Proportion		
n	**0.90**	**0.95**	**0.99**	**0.90**	**0.95**	**0.99**	**0.90**	**0.95**	**0.99**
67	1.871	2.224	2.921	1.937	2.302	3.024	2.071	2.462	3.234
68	1.869	2.221	2.918	1.934	2.299	3.020	2.067	2.457	3.228
69	1.867	2.219	2.915	1.931	2.296	3.016	2.063	2.453	3.222
70	1.865	2.217	2.912	1.929	2.293	3.012	2.059	2.448	3.216
71	1.863	2.214	2.909	1.926	2.290	3.008	2.056	2.444	3.210
72	1.861	2.212	2.906	1.924	2.287	3.004	2.052	2.439	3.204
73	1.859	2.210	2.903	1.922	2.284	3.001	2.048	2.435	3.199
74	1.857	2.208	2.900	1.919	2.282	2.997	2.045	2.431	3.193
75	1.856	2.206	2.898	1.917	2.279	2.994	2.042	2.427	3.188
76	1.854	2.204	2.895	1.915	2.276	2.990	2.038	2.423	3.183
77	1.852	2.202	2.893	1.913	2.274	2.987	2.035	2.419	3.178
78	1.851	2.200	2.890	1.911	2.271	2.984	2.032	2.415	3.173
79	1.849	2.198	2.888	1.909	2.269	2.980	2.029	2.412	3.168
80	1.848	2.196	2.885	1.907	2.266	2.977	2.026	2.408	3.163
81	1.846	2.195	2.883	1.905	2.264	2.974	2.023	2.405	3.159
82	1.845	2.193	2.880	1.903	2.262	2.971	2.020	2.401	3.154
83	1.843	2.191	2.878	1.901	2.259	2.968	2.017	2.398	3.150
84	1.842	2.189	2.876	1.899	2.257	2.965	2.014	2.394	3.145
85	1.840	2.188	2.874	1.897	2.255	2.962	2.012	2.391	3.141
86	1.839	2.186	2.872	1.895	2.253	2.960	2.009	2.388	3.137
87	1.838	2.184	2.870	1.894	2.251	2.957	2.006	2.385	3.133
88	1.836	2.183	2.868	1.892	2.249	2.954	2.004	2.382	3.129
89	1.835	2.181	2.866	1.890	2.247	2.952	2.001	2.379	3.125
90	1.834	2.180	2.864	1.889	2.245	2.949	1.999	2.376	3.121
91	1.833	2.178	2.862	1.887	2.243	2.947	1.996	2.373	3.117
92	1.831	2.177	2.860	1.885	2.241	2.944	1.994	2.370	3.114
93	1.830	2.176	2.858	1.884	2.239	2.942	1.992	2.368	3.110
94	1.829	2.174	2.856	1.882	2.237	2.939	1.989	2.365	3.107
95	1.828	2.173	2.854	1.881	2.236	2.937	1.987	2.362	3.103
96	1.827	2.171	2.852	1.879	2.234	2.935	1.985	2.360	3.100
97	1.826	2.170	2.851	1.878	2.232	2.932	1.983	2.357	3.096
98	1.824	2.169	2.849	1.876	2.231	2.930	1.981	2.355	3.093
99	1.823	2.167	2.847	1.875	2.229	2.928	1.979	2.352	3.090
100	1.822	2.166	2.846	1.874	2.227	2.926	1.977	2.350	3.087
200	1.764	2.097	2.754	1.798	2.137	2.808	1.865	2.217	2.912
300	1.740	2.068	2.842	1.767	2.100	2.922	1.820	2.163	3.080

	Confidence level								
	0.90			0.95			0.99		
	Proportion			Proportion			Proportion		
n	**0.90**	**0.95**	**0.99**	**0.90**	**0.95**	**0.99**	**0.90**	**0.95**	**0.99**
400	1.726	2.052	2.841	1.749	2.079	2.920	1.794	2.132	3.077
500	1.717	2.041	2.839	1.737	2.065	2.918	1.777	2.112	3.074
600	1.710	2.033	2.838	1.728	2.055	2.916	1.764	2.097	3.071
700	1.705	2.027	2.836	1.722	2.047	2.914	1.755	2.086	3.069
800	1.701	2.022	2.835	1.717	2.041	2.912	1.747	2.077	3.066
900	1.697	2.018	2.833	1.712	2.035	2.910	1.741	2.069	3.063
1000	1.694	2.014	2.832	1.709	2.031	2.908	1.736	2.063	3.060

Appendix 28

Equivalent Sigma Levels, Percent Defective, and ppm

With no sigma shift (centered)				With 1.5 sigma shift			
Sigma level	Percent in specification	Percent defective	ppm	Sigma level	Percent in specification	Percent defective	ppm
0.10	7.9656	92.0344	920344	0.10	2.5957	97.40426	974043
0.20	15.8519	84.1481	841481	0.20	5.2235	94.77650	947765
0.30	23.5823	76.4177	764177	0.30	7.9139	92.08606	920861
0.40	31.0843	68.9157	689157	0.40	10.6950	89.30505	893050
0.50	38.2925	61.7075	617075	0.50	13.5905	86.40949	864095
0.60	45.1494	54.8506	548506	0.60	16.6196	83.38043	833804
0.70	51.6073	48.3927	483927	0.70	19.7952	80.20480	802048
0.80	57.6289	42.3711	423711	0.80	23.1240	76.87605	768760
0.90	63.1880	36.8120	368120	0.90	26.6056	73.39444	733944
1.00	68.2689	31.7311	317311	1.00	30.2328	69.76721	697672
1.10	72.8668	27.1332	271332	1.10	33.9917	66.00829	660083
1.20	76.9861	23.0139	230139	1.20	37.8622	62.13784	621378
1.30	80.6399	19.3601	193601	1.30	41.8185	58.18148	581815
1.40	83.8487	16.1513	161513	1.40	45.8306	54.16937	541694
1.50	86.6386	13.3614	133614	1.50	49.8650	50.13499	501350
1.60	89.0401	10.9599	109599	1.60	53.8860	46.11398	461140
1.70	91.0869	8.9131	89131	1.70	57.8573	42.14274	421427
1.80	92.8139	7.1861	71861	1.80	61.7428	38.25720	382572
1.90	94.2567	5.7433	57433	1.90	65.5085	34.49152	344915
2.00	95.4500	4.5500	45500	2.00	69.1230	30.87702	308770
2.10	96.4271	3.5729	35729	2.10	72.5588	27.44122	274412
2.20	97.2193	2.7807	27807	2.20	75.7929	24.20715	242071
2.30	97.8552	2.1448	21448	2.30	78.8072	21.19277	211928
2.40	98.3605	1.6395	16395	2.40	81.5892	18.41082	184108
2.50	98.7581	1.2419	12419	2.50	84.1313	15.86869	158687
2.60	99.0678	0.9322	9322	2.60	86.4313	13.56867	135687
2.70	99.3066	0.6934	6934	2.70	88.4917	11.50830	115083
2.80	99.4890	0.5110	5110	2.80	90.3191	9.68090	96809

With no sigma shift (centered)				With 1.5 sigma shift			
Sigma level	Percent in specification	Percent defective	ppm	Sigma level	Percent in specification	Percent defective	ppm
2.90	99.6268	0.3732	3732	2.90	91.9238	8.07621	80762
3.00	99.7300	0.2700	2700	3.00	93.3189	6.68106	66811
3.10	99.8065	0.1935	1935	3.10	94.5199	5.48014	54801
3.20	99.8626	0.1374	1374	3.20	95.5433	4.45668	44567
3.30	99.9033	0.0967	967	3.30	96.4069	3.59311	35931
3.40	99.9326	0.0674	674	3.40	97.1283	2.87170	28717
3.50	99.9535	0.0465	465	3.50	97.7250	2.27504	22750
3.60	99.9682	0.0318	318	3.60	98.2135	1.78646	17865
3.70	99.9784	0.0216	216	3.70	98.6096	1.39035	13904
3.80	99.9855	0.0145	145	3.80	98.9276	1.07242	10724
3.90	99.9904	0.0096	96.2	3.90	99.1802	0.81976	8198
4.00	99.9937	0.0063	63.3	4.00	99.3790	0.62097	6210
4.10	99.9959	0.0041	41.3	4.10	99.5339	0.46612	4661
4.20	99.9973	0.0027	26.7	4.20	99.6533	0.34670	3467
4.30	99.9983	0.0017	17.1	4.30	99.7445	0.25551	2555
4.40	99.9989	0.0011	10.8	4.40	99.8134	0.18658	1866
4.50	**99.9993**	**0.0007**	**6.8**	4.50	99.8650	0.13499	1350
4.60	99.9996	0.0004	4.2	4.60	99.9032	0.09676	968
4.70	99.9997	0.0003	2.6	4.70	99.9313	0.06871	687
4.80	99.9998	0.0002	1.6	4.80	99.9517	0.04834	483
4.90	99.99990	0.00010	1.0	4.90	99.9663	0.03369	337
5.00	99.99994	0.00006	0.6	5.00	99.9767	0.02326	233
5.10	99.99997	0.00003	0.3	5.10	99.9841	0.01591	159
5.20	99.99998	0.00002	0.2	5.20	99.9892	0.01078	108
5.30	99.999988	0.000012	0.12	5.30	99.9928	0.00723	72.3
5.40	99.999993	0.000007	0.07	5.40	99.9952	0.00481	48.1
5.50	99.999996	0.000004	0.04	5.50	99.9968	0.00317	31.7
5.60	99.999998	0.000002	0.02	5.60	99.9979	0.00207	20.7
5.70	99.9999988	0.0000012	0.012	5.70	99.9987	0.00133	13.3
5.80	99.9999993	0.0000007	0.007	5.80	99.9991	0.00085	8.5
5.90	99.9999996	0.0000004	0.004	5.90	99.9995	0.00054	5.4
6.00	99.9999998	0.0000002	0.002	**6.00**	**99.9997**	**0.00034**	**3.4**

Appendix 29

Critical Values for the Mann-Whitney Test Table (One-Tail, Alpha = 0.05)

n_2	n_1																					
	4	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**
4	12																					
5	13	19																				
6	14	20	24																			
7	15	22	25	34																		
8	16	23	26	35	45																	
9	17	25	28	37	47	59																
10	18	26	29	39	49	61	83															
11	19	27	30	40	51	63	86	101														
12	20	29	32	42	53	66	90	105	121													
13	21	30	33	44	55	68	93	109	125	143												
14	22	32	34	45	57	71	96	112	130	148	167											
15	23	33	36	47	59	73	100	116	134	152	172	192										
16	24	35	37	49	62	75	103	120	138	157	177	198	220									
17	25	36	38	50	64	78	107	124	142	162	182	203	226	249								
18	26	37	40	52	66	80	110	128	147	166	187	209	232	256	281							
19	27	39	41	54	68	83	114	132	151	171	192	215	238	262	287	314						
20	28	40	42	55	70	85	117	136	155	176	197	220	244	269	294	321	349					
21	29	42	44	57	72	88	121	140	160	181	203	226	250	275	301	328	356	386				
22	30	43	45	59	74	90	124	143	164	185	208	231	256	281	308	336	364	394	424			
23	31	45	46	61	76	93	128	147	168	190	213	237	262	288	315	343	372	402	433	465		
24	32	46	48	62	78	95	131	151	172	195	218	243	268	294	322	350	380	410	442	474	508	
25	33	47	49	64	80	97	134	155	177	200	223	248	274	301	329	358	387	418	450	483	517	552

Note: $n_2 > n_1$

Appendix 30

Critical Values for the Mann-Whitney Test Table (One-Tail, Alpha = 0.01)

n_2	n_1																					
	4	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**
4	9																					
5	10	16																				
6	11	17	24																			
7	11	18	25	34																		
8	12	19	26	35	45																	
9	12	20	28	37	47	59																
10	13	21	29	39	49	61	74															
11	14	21	30	40	51	63	76	91														
12	14	22	32	42	53	66	79	94	109													
13	15	23	33	44	55	68	82	97	113	130												
14	16	24	34	45	57	71	85	100	116	134	152											
15	16	25	36	47	59	73	88	103	120	138	156	176										
16	17	26	37	49	62	75	90	106	123	141	161	181	202									
17	18	27	38	50	64	78	93	110	127	145	165	185	207	229								
18	18	28	40	52	66	80	96	113	131	149	169	190	212	235	259							
19	19	29	41	54	68	83	99	116	134	153	174	195	217	241	265	290						
20	19	30	42	55	70	85	102	119	138	157	178	200	222	246	271	297	324					
21	20	31	44	57	72	88	104	122	141	161	182	205	228	252	277	303	330	359				
22	21	32	45	59	74	90	107	126	145	165	187	209	233	257	283	310	337	366	395			
23	21	33	46	61	76	93	110	129	149	169	191	214	238	263	289	316	344	373	403	434		
24	22	34	48	62	78	95	113	132	152	173	196	219	243	269	295	322	351	380	411	442	475	
25	23	35	49	64	80	97	116	135	156	177	200	224	248	274	301	329	358	388	418	450	483	517

Note: $n_2 > n_1$

Appendix 31

Critical Values for the Mann-Whitney Test Table (Two-Tail, Alpha = 0.025)

n_2	n_1																					
	4	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**
4	11																					
5	11	18																				
6	12	19	26																			
7	13	20	28	37																		
8	14	21	29	39	49																	
9	15	22	31	40	51	63																
10	16	23	32	42	53	65	79															
11	16	25	34	44	56	68	82	96														
12	17	26	36	46	58	71	85	100	116													
13	18	27	37	48	60	74	88	103	119	137												
14	19	28	39	50	63	76	91	107	123	141	160											
15	20	30	40	52	65	79	94	110	127	145	165	185										
16	21	31	42	54	67	82	97	114	131	150	169	190	211									
17	22	32	44	56	70	85	100	117	135	154	174	195	217	240								
18	22	33	45	58	72	87	104	121	139	159	179	200	223	246	271							
19	23	34	47	60	75	90	107	124	143	163	184	205	228	252	277	303						
20	24	36	48	62	77	93	110	128	147	167	188	211	234	258	283	310	337					
21	25	37	50	64	79	96	113	132	151	172	193	216	240	264	290	317	344	373				
22	26	38	51	66	82	98	116	135	155	176	198	221	245	270	296	324	352	381	411			
23	27	39	53	68	84	101	119	139	159	180	203	226	251	276	303	330	359	389	419	451		
24	28	41	55	70	86	104	123	142	163	185	208	232	257	282	309	337	366	396	427	459	492	
25	29	42	56	72	89	107	126	146	167	189	213	237	262	289	316	344	374	404	436	468	502	536

Note: $n_2 > n_1$

Appendix 32

Critical Values for the Mann-Whitney Test Table (Two-Tail, Alpha = 0.005)

n_2	n_1																					
	4	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**
4	9																					
5	9	15																				
6	9	15	22																			
7	10	16	23	32																		
8	10	17	25	33	43																	
9	11	18	26	35	45	56																
10	11	18	27	36	47	58	70															
11	12	19	28	38	48	60	73	87														
12	12	20	29	39	50	62	75	90	105													
13	13	21	30	40	52	64	78	93	108	125												
14	13	22	31	42	54	67	81	95	111	128	146											
15	14	22	32	43	56	69	83	98	115	132	150	170										
16	14	23	34	45	57	71	86	101	118	136	155	174	195									
17	15	24	35	46	59	73	88	104	121	139	159	179	200	222								
18	15	25	36	48	61	75	91	107	125	143	163	183	205	227	251							
19	16	26	37	49	63	78	93	110	128	147	167	188	210	233	257	282						
20	16	27	38	51	65	80	96	113	131	151	171	192	215	238	262	288	314					
21	17	27	39	52	67	82	99	116	135	154	175	197	219	243	268	294	321	349				
22	17	28	40	54	69	84	101	119	138	158	179	201	224	249	274	300	327	355	385			
23	18	29	42	55	70	87	104	122	141	162	183	206	229	254	279	306	334	362	392	423		
24	18	30	43	57	72	89	106	125	145	166	187	210	234	259	285	312	340	369	399	430	463	
25	19	31	44	58	74	91	109	128	148	169	192	215	239	264	291	318	347	376	407	438	471	504

Note: $n_2 > n_1$

Appendix 33

Critical Values for the Wilcoxon Signed-Rank Test

	α			
Two-tail test	**0.100**	**0.050**	**0.020**	**0.010**
One-tail test	**0.050**	**0.025**	**0.010**	**0.005**
n				
4	0			
5	1	0		
6	2	1		
7	4	2	0	
8	6	4	1	
9	8	5	2	0
10	11	8	4	2
11	14	10	6	4
12	18	14	9	6
13	21	17	12	8
14	26	21	15	11
15	31	25	19	14
16	36	30	23	18
17	41	35	27	22
18	47	40	32	26
19	54	46	37	30
20	60	52	42	35
21	68	59	48	41
22	75	66	54	47
23	83	73	61	53
24	92	81	68	59
25	101	89	76	66

Note: In most cases, the critical value is a non-integer value for a given alpha value. Therefore, the table values have been rounded down to the nearest whole number. This results in a smaller, more conservative alpha value than shown in the column headers.

Appendix 34

Glossary of Lean Six Sigma and Related Terms

α—*See* alpha.

β—*See* beta.

C_p (process capability index)—An index describing process capability in relation to specified tolerance of a characteristic divided by the natural process variation for a process in a state of statistical control:

$$C_p = \frac{USL - LSL}{6\sigma}$$

where USL = upper specification limit, LSL = lower specification limit, and 6σ = the natural process variation. *See also* P_p (process performance index), a similar term, except that the process may not be in a state of statistical control; process capability index.

C_{pk} (minimum process capability index)—The smaller of the upper process capability index C_{pk_U} and the lower process capability index C_{pk_L}. The C_{pk} considers the mean of the process with respect to the upper and lower specifications. The C_{pk} can be converted to a sigma level using sigma level = $3C_{pk}$.

C_{pk_L} (lower process capability index; CPL)—An index describing process capability in relation to the lower specification limit. The index is defined as

$$C_{pk_L} = \frac{\bar{x} - LSL}{3\sigma}$$

where $\bar{x}$ = process average, LSL = lower specification limit, and 3σ = half of the process capability. Note: When the process average is obtained from a control chart, $\bar{x}$ is replaced by $\bar{\bar{x}}$.

C_{pk_U} (upper process capability index; CPU)—An index describing process capability in relation to the upper specification limit. The index is defined as

$$C_{pk_U} = \frac{USL - \bar{x}}{3\sigma}$$

where $\bar{x}$ = process average, USL = upper specification limit, and 3σ = half of the process capability. Note: When the process average is obtained from a control chart, $\bar{x}$ is replaced by $\bar{\bar{x}}$.

C_{pm} (process capability index of the mean)—An index that takes into account the location of the process average relative to a target value and is defined as

$$C_{pm} = \frac{(\text{USL} - \text{LSL})/2}{3\sqrt{\sigma^2 + (\mu - T)^2}}$$

where USL = upper specification limit, LSL = lower specification limit, σ = standard deviation, μ = expected value, and T = target value. When the process average and the target value are the same, the $C_{pm} = C_{pk}$. When the process average drifts from the target value, the $C_{pm} < C_{pk}$..

n—*See* sample size.

1 – α—*See* confidence level.

1 – β—*See* power.

ρ—The population correlation coefficient. *See also* correlation coefficient.

***p*–value**—The smallest level of significance leading to the rejection of the null hypothesis.

P_p (process performance index)—An index describing process performance in relation to specified tolerance:

$$P_p = \frac{\text{USL} - \text{LSL}}{6s}$$

s is used for standard deviation instead of σ since both random and special causes may be present. Note: A state of statistical control is not required. Also note the similarity to the formula for C_p.

P_{pk} (minimum process performance index)—The smaller of the upper process performance index P_{pk_U} and the lower process performance index P_{pk_L}. The P_{pk}, similar to the C_{pk}, considers the mean of the process with respect to the upper and lower specifications. The P_{pk} is known as the potential process capability and is used in the validation stage of a new product launch. A state of statistical control is not required.

P_{pk_L} (process performance index, lower; PPL)—An index describing process performance in relation to the lower specification limit. The index is defined as:

$$P_{pk_L} = \frac{\bar{x} - \text{LSL}}{3s}$$

where $\bar{x}$ = process average, LSL = lower specification limit, and 3σ = half of the process capability. Note: When the process average is obtained from a control chart, $\bar{x}$ is replaced by $\bar{\bar{x}}$.

P_{pk_U} (process performance index, upper; PPU)—An index describing process performance in relation to the upper specification limit. The index is defined as:

$$P_{pk_U} = \frac{\text{USL} - \bar{x}}{3s}$$

where $\bar{x}$ = process average, USL = upper specification limit, and 3σ = half of the process capability. Note: When the process average is obtained from a control chart, $\bar{x}$ is replaced by $\bar{\bar{x}}$.

r**—***See* correlation coefficient.

R**—***See* correlation coefficient.

r^2**—***See* coefficient of determination.

R^2**—***See* coefficient of determination.

r^2_{adj}**—***See* adjusted coefficient of determination.

R^2_{adj}**—***See* adjusted coefficient of determination.

t_k**—**This is the t statistic at lag k. The formula is:

$$t_k = \frac{r_k}{SE(r_k)}$$

where $SE(r_k)$ = standard error of the autocorrelation at lag k.

$Y = f(X)$**—**A foundational concept of Six Sigma, this equation represents the idea that process outputs are a function of process inputs.

5S (Five S)—A term that derives its name from the five terms beginning with "s" that are used to create a workplace suited for visual control and lean production. Collectively, 5S is about how to create a workplace that is visibly organized, free of clutter, neatly arranged, and sparkling clean. The 5S's are defined as follows: *seiri* (sort; also sifting) means to separate needed tools, parts, and instructions from unneeded materials and to remove the latter; *seiton* (set in order) means to neatly arrange and identify parts and tools for ease of use; *seiso* (shine; also sanitize or scrub) means to conduct a cleanup campaign; *seiketsu* (standardize) means to conduct seiri, seiton, and seiso at frequent, indeed daily, intervals to maintain a workplace in perfect condition; and *shitsuke* (sustain; also self-discipline) means to form the habit of always following the first four S's. *See also* individual listing for each of the 5S's.

5 whys—A persistent questioning technique to probe deeper to reach the root cause of a problem. *See also* 5Ws & 1H.

5Ws & 1H—Addressing the *who*, *what*, *where*, *when*, *why*, and *how* questions is a useful technique to help develop an objective and a concise statement of a problem. *See also* 5 whys.

6Ms—Typically the primary categories of the cause-and-effect diagram: machines, manpower, materials, measurements, methods, and Mother Nature. Using these categories as a structured approach provides some assurance that few causes will be overlooked. *See also* 7Ms.

7Ms—The 6Ms with the addition of management. *See also* 6Ms.

A

ABC—*See* activity-based costing.

acceptance control chart—A control chart intended primarily to evaluate whether the plotted measure can be expected to satisfy specified tolerances.

acceptance quality limit (AQL)—The quality level that is the worst tolerable product average when a continuing series of lots is submitted for acceptance sampling. Note the following: (1) This concept applies only when an acceptance sampling scheme with rules for switching and discontinuation is used. (2) Although individual lots with quality as bad as the AQL can be accepted with fairly high probability, the designation of an AQL does not suggest that this is a desirable quality level. (3) Acceptance sampling schemes found in standards with their rules for switching and discontinuation of sampling inspection are designed to encourage suppliers to have process averages consistently better than the AQL. If suppliers fail to do so, there is a high probability of being switched from normal inspection to tightened inspection, where lot acceptance becomes more difficult. Once on tightened inspection, unless corrective action is taken to improve product quality, it is very likely that the rule requiring discontinuation of sampling inspection pending such improvement will be invoked unless action is taken to improve the process. (4) The use of the abbreviation AQL to mean *acceptable quality level* is no longer recommended since modern thinking is that no fraction defective is really acceptable. Using "acceptance quality limit" rather than "acceptable quality level" indicates a technical value where acceptance occurs.

accuracy—The closeness of agreement between a test result or measurement result and the true or reference value.

ACF—*See* sample autocorrelation function.

action plan—*See* operational plan.

activity—An action of some type that requires a time duration for accomplishment.

activity network diagram (AND)—A tool used to illustrate a sequence of events or activities (nodes) and the interconnectivity of such nodes. It is used for scheduling and especially for determining the critical path through nodes. It is also known as an *arrow diagram*. The activity network diagram is one of the seven management and planning tools.

activity-based costing (ABC)—A cost allocation method that allocates overhead expenses to the activities based on the proportion of use, rather than proportion of costs.

actual cost of work performed (ACWP)—*See* actual value.

actual value (AV)—The actual cost incurred for work completed by a specific date. It is also known as the *actual cost of work performed* (ACWP).

ACWP—*See* actual value.

adjusted coefficient of determination (R^2_{adj})—A statistic that takes into consideration the number of predictor variables in the model and is helpful when comparing models with a different number of predictor variables. It is computed as follows:

$$R^2_{adj} = 1 - \left(\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\left(\sum_{i=1}^{n} y_i - \bar{y} \right)^2} \right) \left(\frac{n-1}{n-p-1} \right)$$

where:

n = Number of observations

y_i = ith observed response value

$\bar{y}_i$ = Mean observed response value

$\hat{y}_i$ = ith fitted value

p = Number of terms in the model including the constant term

Notice when $p = 2$ (that is, simple linear regression), R^2 equals R^2_{adj}. Sometimes R^2_{adj} is written as r^2_{adj}.

affinity diagram—A tool used to organize information and help achieve order out of the chaos that can develop in a brainstorming session. Large amounts of data, concepts, and ideas are grouped based on their natural relationship to one another. It is more a creative process than a logical process. Also known as the *KJ method*, the affinity diagram is one of the seven management and planning tools.

agile approach—*See* lean.

AIAG—Automotive Industry Action Group.

alias—An effect that is completely confounded with another effect due to the nature of the designed experiment. Aliases are the result of confounding, which may or may not be deliberate.

alpha (α)—(1) The maximum probability or risk of making a type I error when dealing with the significance level of a test. (2) The probability or risk of incorrectly deciding that a shift in the process mean has occurred when in fact the process has not changed (when referring to α in general or the p–value obtained in the test). (3) α is usually designated as producer's risk.

alpha (α) risk—*See* type I error.

alternative hypothesis—A hypothesis formulated from new information.

analysis of means (ANOM)—A statistical procedure for troubleshooting industrial processes and analyzing the results of experimental designs with factors at fixed levels. It provides a graphical display of data. Ellis R. Ott developed the procedure in 1967 because he observed that non-statisticians had difficulty understanding analysis of variance. Analysis of means is easier for quality practitioners to use because it is an extension of the control chart. In 1973, Edward G. Schilling further extended the concept, enabling analysis of means to be used with nonnormal distributions and attributes data where the normal approximation to the binomial distribution does not apply. This is referred to as *analysis of means for treatment effects*.

analysis of variance (ANOVA)—A basic statistical technique for analyzing experimental data. It subdivides the total variation of a data set into meaningful component parts associated with specific sources of variation in order to test a hypothesis on the parameters of the model or to estimate variance components. There are three model types: fixed, random, and mixed.

analytical thinking—Breaking down a problem or situation into discrete parts to understand how each part contributes to the whole.

AND—*See* activity network diagram.

Andon board—A visual device (usually lights) displaying status alerts that can easily be seen by those who should respond.

ANOM—*See* analysis of means.

ANOVA—*See* analysis of variance.

ANSI—American National Standards Institute.

APC—*See* automated process control.

appraisal costs—The costs associated with measuring, evaluating, or auditing products or services to ensure conformance to quality standards and performance requirements. These include costs such as incoming and source inspection/test of purchased material; in-process and final inspection/test; product, process, or service audit; calibration of measuring and test equipment; and the cost of associated supplies and materials.

AQL—*See* acceptance quality limit.

ARIMA—*See* autoregressive integrated moving average.

arrow diagram—*See* activity network diagram.

AS-9100—A standard for the aeronautics industry embracing the ISO 9001 standard.

ASQ—American Society for Quality.

assessment of training effectiveness—According to the Manpower Services Commission (MSC) "Glossary of Training Terms": "A general term for the process of ascertaining whether training is efficient or effective in achieving prescribed objectives. It covers both evaluation and validation."

assignable cause—A specifically identified factor that contributes to variation and is detectable. Eliminating assignable causes so that the points plotted on the control chart remain within the control limits helps achieve a state of statistical control. Note: Although assignable cause is sometimes considered synonymous with special cause, a special cause is assignable only when it is specifically identified. *See also* special cause.

assumption—A condition that must be true in order for a statistical procedure to be valid.

attribute data—Data that are categorized for analysis or evaluation. (Attribute data may involve measurements as long as they are used only to place given data in a category for further analysis or evaluation. Contrast with variables data.)

attrition—Refers to loss of people to jobs outside the organization.

autocorrelation—The internal correlation between members of series observations ordered in time.

automated process control (APC)—The application of automation to effect process control by maintaining manipulated variables at set point levels such that controlled variables are maintained at specified levels. Generally, automation allows for a greater number of manipulated variables to be controlled with greater speed and accuracy than possible by human intervention.

autonomation—*See* jidoka.

autoregressive integrated moving average (ARIMA)—According to the SAS Institute, Inc. "an ARIMA model predicts a value in a response time series as a linear combination of its own past values, past errors (also called *shocks* or *innovations*), and current and past values of other time series."

AV—*See* actual value.

avoidance, risk—Refers to the practice of eliminating the risk factor.

axiomatic design—A theory by Dr. Nam Pyo Suh that stresses each functional requirement be designed to achieve robustness without affecting other functional requirements.

B

BAC—*See* budget at completion.

baka-yoke—A term for a manufacturing technique for preventing mistakes by designing the manufacturing process, equipment, and tools so that an operation literally cannot be performed incorrectly. In addition to preventing incorrect operation, the technique usually provides a warning signal of some sort for incorrect performance.

balanced design—A design where all treatment combinations have the same number of observations. If replication in a design exists, it would be balanced only if the replication was consistent across all treatment combinations. In other words, the number of replicates of each treatment combination is the same.

balanced incomplete block design—Incomplete block design in which each block contains the same number k of different levels from the l levels of the principal factor arranged so that every pair of levels occurs in the same number l of blocks from the b blocks. Note: This design implies that every level of the principal factor will appear the same number of times in the experiment.

balanced scorecard—Translates an organization's mission and strategy into a comprehensive set of performance measures to provide a basis for strategic measurement and management, utilizing four balanced views: financial, customers, internal business processes, and learning and growth.

batch processing—Running large batches of a single product through the process at one time, resulting in queues awaiting the next steps in the process.

bathtub curve—A graphic representation of the relationship of the life of a product versus the probable failure rate. The curve contains three phases: early or infant failure (break-in), a stable rate during normal use, and wear-out. The bathtub curve is also known as the life-history curve.

BB—*See* Black Belt.

BCWP—*See* budgeted cost of work performed.

BCWS—*See* budgeted cost of work scheduled.

behavior, directive—Establishes a clear path to the goal by specifying what is to be done, how it is to be done, and who is going to do it.

behavior, supportive—Involves two-way communication and focuses on emotional and social support.

benchmark—An organization, a part of an organization, or a measurement that serves as a reference point or point of comparison.

benchmarking—An improvement process in which an organization measures its performance against that of best-in-class organizations (or others that are good performers), determines how those organizations achieved their performance levels, and uses the information to improve its own performance. Areas that can be benchmarked include strategies, operations, processes, and procedures.

benefit-cost analysis—A collection of the dollar value of benefits derived from an initiative divided by the associated costs incurred. A benefit-cost analysis is also known as a cost-benefit analysis.

benefit-cost ratio—The formula for computing a benefit-cost is:

$$\frac{\sum \text{NPV of all benefits anticipated}}{\sum \text{NPV of all costs anticipated}}.$$

Occasionally it is referred to as a *cost-benefit ratio*, which is the reciprocal.

best subsets—A very useful tool to help select a regression model among many other viable models. It is based on the concept that for a model comprising m

predictor variables, there are $2^m - 1$ possible subsets of predictor variables possible. Each model can be compared using such measures as VIF, Mallows' C_p, PRESS, R^2_{adj}, and squared error to aid in the model selection.

beta (β)—(1) The maximum probability or risk of making a type II error. (2) The probability or risk of incorrectly deciding that a shift in the process mean has not occurred when in fact the process has changed. (3) β is usually designated as consumer's risk.

beta (β) risk—*See* type II error.

bias—A systematic difference between the mean of the test result or measurement result and a true or reference value. For example, if one measures the lengths of 10 pieces of rope that range from 1 foot to 10 feet and always concludes that the length of each piece is 2 inches shorter than the true length, then the individual is exhibiting a bias of 2 inches. Bias is a component of accuracy.

Black Belt (BB)—A Six Sigma role associated with an individual who is typically assigned full time to train and mentor Green Belts as well as lead improvement projects using specified methodologies such as define, measure, analyze, improve, and control (DMAIC) and design for Six Sigma (DFSS).

block—A collection of experimental units more homogeneous than the full set of experimental units. Blocks are usually selected to allow for special causes, in addition to those introduced as factors to be studied. These special causes may be avoidable within blocks, thus providing a more homogeneous experimental subspace.

block diagram—A diagram that describes the operation, interrelationships, and interdependencies of components in a system. Boxes, or blocks (hence the name), represent the components; connecting lines between the blocks represent interfaces. There are two types of block diagrams: a functional block diagram, which shows a system's subsystems and lower-level products, their interrelationships, and interfaces with other systems, and a reliability block diagram, which is similar to the functional block diagram except that it is modified to emphasize those aspects influencing reliability.

block effect—An effect resulting from a block in an experimental design. Existence of a block effect generally means that the method of blocking was appropriate and that an assignable cause has been found.

blocking—Including blocks in an experiment in order to broaden the applicability of the conclusions or minimize the impact of selected assignable causes. The randomization of the experiment is restricted and occurs within blocks.

BPM—*See* business process management.

brainstorming—A problem-solving tool that teams use to generate as many ideas as possible that are related to a particular subject. Team members begin by offering all their ideas; the ideas are not discussed or reviewed until after the brainstorming session.

break-even point—Conducted as an analysis, it is a useful approach when there are multiple alternatives each of which may be economical under its own set of conditions. The break-even point is determined by altering one variable while holding all others constant. The variable generally represents some passage of time or number of units processed and can be calculated or determined graphically to identify where the alternatives are equal from an economical viewpoint. This point of equality is known as the break-even point.

budget at completion (BAC)—This is the total planned value for a given task at the end of the project.

budgeted cost of work performed (BCWP)—*See* earned value.

budgeted cost of work scheduled (BCWS)—*See* planned value.

bullwhip effect—Whenever an organization changes the schedule or the demand for product frequently, these perturbations are rippled upstream throughout the value chain and amplified along the way, sending the supply chain into turmoil. This is known as the bullwhip effect. Some organizations have minimized the bullwhip effect by batching changes and releasing the changes at less frequent intervals. This introduces stability into the value chain.

business process management (BPM)—A set of activities an organization implements to optimize its processes. Business process management encompasses both software tools and manual processes designed to assist organizations in achieving process optimization.

C

calibration—The comparison of a measurement instrument or system of unverified accuracy to a measurement instrument or system of known accuracy to detect any variation from the true value.

capability—The performance of a process demonstrated to be in a state of statistical control. *See also* process capability; process performance.

capability index—*See* process capability index.

causal factor—A variable that when changed or manipulated in some manner serves to influence a given effect or result.

cause-and-effect analysis—The process of identifying the likely causes of any outcome. Some cause-and-effect analyses produce an output like a fishbone diagram. Others produce a cause-and-effect tree with multiple sub-branches.

cause-and-effect diagram—A diagram resembling a fish skeleton that is used to illustrate the main causes and sub-causes leading to an effect. One of the seven basic tools of quality, the cause-and-effect diagram is also known as the *Ishikawa diagram* or the *fishbone diagram*.

central limit theorem (CLT)—A theorem that states that irrespective of the shape of the distribution of a population, the distribution of sample means is approximately normal when the sample size is large.

central tendency—The propensity of data collected on a process to concentrate around a value situated approximately midway between the lowest and highest values. Three common measures of central tendency include (arithmetic) mean, median, and mode. *See also* mean, median, mode.

centralization versus decentralization—In the context of organizational design refers to how much decision-making authority has been delegated to lower management levels.

Certified Six Sigma Black Belt (CSSBB)—An ASQ certification.

Certified Six Sigma Green Belt (CSSGB)—An ASQ certification.

CFM—*See* continuous flow manufacturing.

chaku-chaku—A term meaning "load-load" in a cell layout where a part is taken from one machine and loaded into the next.

champion—A Six Sigma role associated with a senior manager who ensures that his or her projects are aligned with the organization's strategic goals and priorities, provides the Six Sigma team with resources, removes organizational barriers for the team, participates in project tollgate reviews, and essentially serves as the team's backer. Although many organizations define the terms "champion" and "sponsor" differently, they are frequently used interchangeably. *See also* sponsor.

chance cause—*See* random cause.

chance variation—Variation due to chance causes. *See also* random variation.

change agent—According to Hutton (1994), a change agent is an individual who play a specific role in the planning and implementation of the change management process. They may be members of the organization or may be outsiders.

changeover time—The time interval between the last good piece off the current run and the first good piece off the next run.

characteristic—A property that helps differentiate between items of a given sample or population.

charter—A documented statement officially initiating the formation of a committee, team, project, or other effort in which a clearly stated purpose and approval are conferred.

check sheet—A simple data recording device typically used to collect the frequency of occurrences of nominal data by category. Whenever the user observes the occurrence of one of the categories, he or she places a mark next to the category. When the observation process is complete, final counts by category can then be obtained. The user custom designs the check sheet so that he or she can readily interpret the results. The check sheet is occasionally considered to be one of the seven basic tools of quality. Note: Check sheets are often confused with checklists. *See also* checklist.

checklist—A quality tool that is used for processes where there is a large human element. Examples are audit checklists and an airplane pilot's preflight checklist. Checklists serve as reminders and, depending on design, evidence that important items have been observed or appropriate actions have been taken. A well-designed checklist helps gather information to support findings and observations and serves as a guide and a place to record information. Note: Checklists are often confused with check sheets. *See also* check sheet.

churn—The movement of people to different job positions within the organization.

circle diagram—A tool used to show linkages between various items. The circle diagram is constructed by evenly placing item descriptors around a circle. Such item descriptors might include products, services, organizations, individuals, and so forth. Arrows are drawn from each item to other items where a flow exists. Circle diagrams are highly useful in that they readily depict predecessor and successor relationships as well as potential bottlenecks. Too many inputs or outputs from any given descriptor around the circumference of the circle may indicate a limiting function. A circle diagram is also known as a *hand-off map*.

closed-loop feedback system—When the manipulated variable is adjusted by the final control element, and that variable, in turn, affects the next measurement, the system is said to be a closed-loop feedback system.

CLT—*See* central limit theorem.

CMM—Capability maturity model.

CMMI—Capability maturity model integration.

coaching—A process by which a more experienced individual helps enhance the existing skills and capabilities that reside in a less experienced individual. Coaching is about listening, observing, and providing constructive, practical, and meaningful feedback. Typically, coaching is used on a one-to-one basis, or for a small group or team, and conducted at the job site or during the training process. During training, coaching helps the trainee translate the theoretical learning into applied learning while helping the trainee developing confidence in their newly developing knowledge and skills. Post training, coaches help projects stay on track and advance toward completion in a timely manner.

coefficient of determination (R^2)—A statistic that indicates how much variation in the response variable is accounted for by the regression model. $R^2 \le 1$. It is computed as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}_i\right)^2}$$

where:

n = Number of observations

y_i = ith observed response value

$\bar{y}_i$ = Mean observed response value

$\hat{y}_i$ = ith fitted value

Sometimes R^2 is written as r^2. The coefficient of determination is known as the *coefficient of multiple determination* when used in multiple regression.

coefficient of multiple determination—*See* coefficient of determination.

communication plan—A document that defines what will be communicated, to whom, how often, and by what means.

complete block—A block that accommodates a complete set of treatment combinations.

completely randomized design—A design in which the treatments are randomly assigned to the full set of experimental units. No blocks are involved.

completely randomized factorial design—A factorial design in which all the treatments are randomly assigned to the full set of experimental units. *See also* completely randomized design.

compliance—An affirmative indication or judgment that the supplier of a product or service has met the requirements of the relevant specifications, contract, or regulation; also the state of meeting the requirements.

components of variation (COV)—A statistical technique that allows us to separate process variation by the inputs. Components of variation is useful in the situation where our problem can be formulated as a nested hierarchal structure.

concurrent engineering—A way to reduce cost, improve quality, and shrink cycle time by simplifying a product's system of life-cycle tasks during the early concept stages. Concurrent engineering is a process to get all departments including engineering, purchasing, marketing, manufacturing, and finance to work on a new design at once to speed development. The emphasis is on upstream prevention versus downstream correction. Concurrent engineering is also known as *simultaneous engineering*.

confidence coefficient (1 – α)—*See* confidence level.

confidence interval—An estimate of the interval between two statistics that includes the true value of the parameter with some probability. This probability is called the *confidence level* of the estimate. Confidence levels typically used are 90%, 95%, and 99%. Either the interval contains the parameter or it does not.

confidence level (1 – α)—The probability that (1) the confidence interval described by a set of confidence limits actually includes the population parameter and (2) an interval about a sample statistic actually includes the population parameter. The confidence level is also known as the *confidence coefficient*.

confidence limits—The endpoints of the interval about the sample statistic that is believed, with a specified confidence level, to include the population parameter. *See also* confidence interval.

conflict resolution—A process for resolving disagreements in a manner acceptable to all parties.

confounding—Indistinguishably combining an effect with other effects or blocks. When done deliberately, higher-order effects are systematically aliased so as to allow estimation of lower-order effects. Sometimes, confounding results from inadvertent changes to a design during the running of an experiment or from poor planning of the design. This can diminish or even invalidate the effectiveness of the experiment.

consensus—Finding a proposal acceptable enough that all team members support the decision and no member opposes it.

constraint—A bottleneck or limitation of the throughput of a process.

constraint management—Pertains to identifying a constraint and working to remove or diminish it, while dealing with resistance to change.

consumer's risk (β)—The probability of acceptance when the quality level has a value stated by the acceptance sampling plan as unsatisfactory. Note: (1) such acceptance is a Type II error; (2) consumer's risk is usually designated as beta (β). Consumer's risk is also known as *beta* (β) *risk*, *beta* (β) *error*, *error of the second kind*, *type 2 error*, and *type II error*.

continuous flow manufacturing (CFM)—A method in which items are produced and moved from one processing step to the next, one piece at a time. Each process makes only the one piece that the next process needs, and the transfer batch size is one. CFM is sometimes known as *one-piece flow* or *single-piece flow*.

continuous variable—A variable whose possible values form an interval set of numbers such that between each two values in the set another member of the set occurs.

control chart—A chart that plots a statistical measure of a series of samples in a particular order to steer the process regarding that measure and to control and reduce variation. The control chart comprises the plotted points, a set of upper and lower control limits, and a centerline. Specific rules are used to determine when the control chart goes out of control. Note: (1) the order is either time or sample number order based, and (2) the control chart operates most effectively when the measure is a process characteristic correlated with an ultimate product or service characteristic. The control chart is one of the seven basic tools of quality.

control chart, acceptance—*See* acceptance control chart.

control limit—A line on a control chart used for judging the stability of a process. Note: (1) control limits provide statistically determined boundaries for the deviations from the centerline of the statistic plotted on a Shewhart control chart due to random causes alone; (2) control limits (with the exception of the acceptance control chart) are based on actual process data, not on specification limits; (3) other than points outside the control limits, "out-of-control" criteria can include runs, trends, cycles, periodicity, and unusual patterns within the control limits; (4) the calculation of control limits depends on the type of control chart.

control plan—A living document that identifies critical input or output variables and associated activities that must be performed to maintain control of the variation of processes, products, and services in order to minimize deviation from their preferred values.

controlled variable—In the context of automated process control, the controlled variable is a variable that must be maintained at a set or specified level. The control variable is also known as the *process variable* or simply the *measurement*. In Lean Six Sigma terminology, it is the "*Y*" variable.

controller—In the context of automated process control, the controller is considered the brain of the system since it compares the value of the signal to the set value (that is, nominal or desired value) and sends a signal to the final control element for action. Essentially, the controller makes a decision. The basic operation performed is "decide."

COPQ—*See* cost of poor quality.

COQ—*See* cost of quality.

corrective action—Action taken to eliminate the root cause(s) and symptom(s) of an existing deviation or nonconformity to prevent recurrence.

correlation coefficient (*r*)—A statistic that measures the degree of linear relationship between two sets of numbers and is computed as

$$r = \frac{s_{xy}}{\sqrt{s_{xx}s_{yy}}} = \frac{\sum_{i=1}^{n}\left(x_i-\bar{x}\right)^2\left(y_i-\bar{y}\right)^2}{\sqrt{\sum_{i=1}^{n}\left(x_i-\bar{x}\right)^2\sum_{i=1}^{n}\left(y_i-\bar{y}\right)^2}} = \frac{\sum_{i=1}^{n}x_iy_i-\frac{\left(\sum_{i=1}^{n}x_i\right)\left(\sum_{i=1}^{n}y_i\right)}{n}}{\sqrt{\left(\sum_{i=1}^{n}x_i^2-\frac{\left(\sum_{i=1}^{n}x_i\right)^2}{n}\right)\left(\sum_{i=1}^{n}y_i^2-\frac{\left(\sum_{i=1}^{n}y_i\right)^2}{n}\right)}}$$

where $-1 \le r \le 1$. Also, r is used to estimate the population correlation coefficient ρ. Correlation values of –1 and +1 represent perfect linear agreement between two independent and dependent variables. An $r = 0$ means there is no linear relationship at all. If r is positive, the relationship between the variables is said to be positive. Hence, when x increases, y increases. If r is negative, the relationship is said to be negative. Hence, when x increases, y decreases. The correlation coefficient is also known as the *sample correlation coefficient* and is sometimes written as *R*.

cost avoidance (budget impacting)—This type of cost avoidance eliminates or reduces items in the budget marked for future spending. For example, three engineers are budgeted for hire in the fourth quarter. The cost of the three engineers is not in the baseline, nor has any spending for these engineers occurred. Eliminating these planned expenses has an impact on the future budget and is thus cost avoidance.

cost avoidance (non-budget impacting)—This type of cost avoidance results from productivity or efficiencies gained in a process (i.e., reduction of non-value-added activities) without a headcount reduction. For example, the process cycle time that involved two workers was reduced by ten percent. Assuming the ten percent amounts to two hours per worker per week, the two workers save four hours per week. These four hours are allocated to other tasks.

cost avoidance savings—Are, by exclusion, everything that is not hard dollar savings. Cost avoidance savings are also known as soft dollar savings. There are two types of cost avoidance savings: budget impacting, and non-budget impacting.

cost of poor quality (COPQ)—The costs associated with the production of non-conforming material. These costs include both external and internal failure costs. *See also* external failure costs; internal failure costs.

cost of quality (COQ)—The costs specifically associated with the achievement or nonachievement of product or service quality, including all product or service requirements established by the organization and its contracts with customers and society. More specifically, quality costs are the total costs incurred by (1) investing in the prevention of nonconformances to requirements, (2) appraising a product or service for conformance to requirements, and (3) failing to meet requirements. These can then be categorized as prevention, appraisal, and failure. *See also* appraisal costs; failure costs; prevention costs.

cost performance index (CPI)—This is a dimensionless index used to measure the project's cost efficiency. It is calculated as

$$\mathrm{CPI} = \frac{\mathrm{EV}}{\mathrm{AC}} = \frac{\mathrm{BCWP}}{\mathrm{ACWP}}.$$

If the CPI is greater than 1, the project is under budget. If the CPI is equal to 1, the project is on budget. If the CPI is less than 1, the project is over budget and corrective action must be taken.

cost savings—For project savings to generate real cost savings, a prior baseline of spending must be established; the dollars must have been planned and in the budget; and the savings must affect the bottom line (that is, the profit and loss statement or balance sheet). Cost savings are also known as *hard dollar savings*.

cost variance (CV)—The cost variance is the difference between the amount budgeted for the work performed and the actual cost of the work performed. It is calculated as

$$\mathrm{CV} = \mathrm{EV} - \mathrm{AC}.$$

If the CV is positive, the project is under budget. If the CV is zero, the project is on budget. If the CV is negative, the project is over budget and corrective action must be taken.

cost-benefit analysis—*See* benefit-cost analysis.

COV—*See* components of variation.

CPI—*See* cost performance index.

CPL—*See* C_{pk_L} .

CPM—*See* critical path method.

CPU—*See* C_{pk_U}.

creativity—The mental capability to generate ideas.

critical path—The sequence of tasks that takes the longest time and determines a project's completion date.

critical path method (CPM)—An activity-oriented project management technique that uses arrow-diagramming methods to demonstrate both the time and the cost required to complete a project. It provides one time estimate: normal time.

critical success factor (CSF)—*See* key business drivers.

criticality—An indication of the consequences expected to result from a failure.

critical-to-quality (CTQ)—A characteristic of a product or service that is essential to ensure customer satisfaction.

cross-functional team—A group consisting of members from more than one department that is organized to accomplish a project.

CSF—*See* critical success factor.

CSSBB—*See* Certified Six Sigma Black Belt.

CSSGB—*See* Certified Six Sigma Green Belt.

CSSYB—Certified Six Sigma Yellow Belt.

CSSWB—Certified Six Sigma White Belt.

CTQ—*See* critical-to-quality.

culture—Westcott (2006) refers to culture as "the collective beliefs, values, attitudes, manners, customs, behaviors, and artifacts unique to an organization."

cumulative sum (CUSUM) control chart—A control chart on which the plotted value is the cumulative sum of deviations of successive samples from a target value. The ordinate of each plotted point represents the algebraic sum of the previous ordinate and the most recent deviations from the target.

customer loyalty—A term used to describe the behavior of customers—in particular, customers who exhibit a high level of satisfaction, conduct repeat business, or provide referrals and testimonials. It is the result of an organization's processes, practices, and efforts designed to deliver its services or products in ways that drive such behavior.

CUSUM—*See* cumulative sum (CUSUM) control chart.

CV—*See* cost variance.

cycle time—The time required to complete one cycle of an operation. If cycle time for every operation in a complete process can be reduced to equal takt time, products can be made in single-piece flow. *See also* takt time.

cycle-time reduction—A method for reducing the amount of time it takes to execute a process or build a specific product. Elimination of duplicate or unnecessary tasks also reduces the time that is necessary to execute a process. Improving hand-offs, which tend to be points of substantial discontinuity, also eliminates rework or duplicate efforts.

D

decentralization versus centralization—In the context of organizational design, refers to how much decision-making authority has been delegated to lower management levels.

decision matrix—*See* Pugh matrix.

defect—The nonfulfillment of a requirement related to an intended or specified use for a product or service. In simpler terms, a defect is anything not done correctly the first time. Note: The distinction between the concepts *defect* and *nonconformity* is important because it has legal connotations, particularly those associated with product liability issues. Consequently, the term "defect" should be used with extreme caution.

defective—A unit of product that contains one or more defects with respect to the quality characteristic(s) under consideration.

defects per million opportunities (DPMO)—A measure of capability for discrete (attribute) data found by dividing the number of defects by the opportunities for defects multiplied by a million. DPMO allows for comparison of different types of product. *See also* parts per million.

defects per unit (DPU)—A measure of capability for discrete (attribute) data found by dividing the number of defects by the number of units.

Deming cycle—*See* plan-do-study-act (PDSA) cycle.

Deming Prize—Award given annually to organizations that, according to the award guidelines, have successfully applied organization-wide quality control based on statistical quality control and will keep up with it in the future. Although the award is named in honor of W. Edwards Deming, its criteria are not specifically related to Deming's teachings. There are three separate divisions for the award: the Deming Application Prize, the Deming Prize for Individuals, and the Deming Prize for Overseas Companies. The award process is overseen by the Deming Prize Committee of the Union of Japanese Scientists and Engineers in Tokyo.

Deming wheel—*See* plan-do-study-act (PDSA) cycle.

dependability—The degree to which a product is operable and capable of performing its required function at any randomly chosen time during its specified operating time, provided that the product is available at the start of that period. (Non-operation-related influences are not included.) Dependability can be expressed by the ratio

$$\frac{\text{Time available}}{\text{Time available} + \text{Time required}}.$$

dependent events—Two events, *A* and *B*, are dependent if the probability of one event occurring is higher given the occurrence of the other event.

derating—The practice of using components at lower stress levels than those stated by the component specifications.

descriptive statistics—The collection of tools and techniques for displaying and summarizing data.

design failure mode and effects analysis (DFMEA)—An analysis process used to identify and evaluate the relative risk associated with a particular design.

design for Six Sigma (DFSS)—A structured methodology that focuses on designing new products or services with the intent of achieving Six Sigma quality levels. *See also* DMADV, DMADOV, DMEDI, and IDOV.

design for X (DFX)—An umbrella term whereby *X* is a variable that can assume multiple descriptions such as maintainability, manufacturability, producibility, quality, testability, or whatever component of design, logistics, production, quality, and so forth, of the product needs to be emphasized. The various components of the product are collectively referred to as the "ilities." Design for X involves the principle of designing products so that they are cost-effective and achieve the objectives set forth for whatever "ilities" are of concern.

design of experiments (DOE, DOX)—The arrangement in which an experimental program is to be conducted, including the selection of factor combinations and their levels. Note: The purpose of designing an experiment is to provide the most efficient and economical methods of reaching valid and relevant conclusions from the experiment. The selection of the design is a function of many considerations, such as the type of questions to be answered, the applicability of the conclusions, the homogeneity of experimental units, the randomization scheme, and the cost to run the experiment. A properly designed experiment will permit simple interpretation of valid results.

design review—Documented, comprehensive, and systematic examination of a design to evaluate its capability to fulfill the requirements for quality.

design space—The multidimensional region of possible treatment combinations formed by the selected factors and their levels.

design for cost (DFC)—*See* design-to-cost.

design to cost (DTC)—A set of tools, techniques, or methods used to set and achieve product cost goals through design trade-off analysis. Design to cost is also known as *design for cost*.

detection—The likelihood of finding a failure once it has occurred. Detection may be evaluated based on a 10-point scale. On the lowest end of the scale (1), it is assumed that a design control will detect a failure with certainty. On the highest end of the scale (10), it is assumed that a design control will not detect a failure when a failure occurs.

DFC—An acronym for design for cost. *See* design to cost.

DFM—An acronym for design for manufacturability. *See* design for X.

DFMaint—An acronym for design for maintainability. *See* design for X.

DFMEA—*See* design failure mode and effects analysis.

DFP—An acronym for design for producibility. *See* design for X.

DFQ—An acronym for design for quality. *See* design for X.

DFSS—*See* design for Six Sigma.

DFT—An acronym for design for testability. *See* design for X.

DFX—*See* design for X.

differences—Difference between data values for a specified lag value *k*. For example, for a given set of ordered data, there is a lag *k* between values *xt* and *xt+k*. Differencing is used to introduce stationarity to a time series. Data transformation can also be used to stabilize the variance.

discrete scale—A scale with only a set or sequence of distinct values. Examples include defects per unit, events in a given time period, types of defects, and number of orders on a truck.

discrete variable—A variable whose possible values form a finite or, at most, countably infinite set.

discriminant analysis—A statistical technique that is used "to classify observations into two or more groups if you have a sample with known groups. Discriminant analysis can also be used to investigate how variables contribute to group separation" (Minitab 15).

discrimination—The capability of the measurement system to detect and indicate small changes in the characteristic measured. Discrimination is also known as *resolution*. For example, a tape measure with gradations of one inch would be unable to distinguish between lengths of objects that fall in between the inch marks. Hence, we would say the measurement system could not properly discriminate between the objects. If an object to be measured is 2.5 inches, the measurement system (that is, tape measure) would produce a value of 2 or 3 inches depending on how the individual decided to round. Therefore, to measure an object that is 2.5 inches, a tape measure with finer gradations would be required.

dispersion effect—The influence of a single factor on the variance of the response variable.

dissatisfiers—Those features or functions that the customer or employee has come to expect, and if they are no longer present, the customer will be dissatisfied.

disturbance—In the context of automated process control, these variables cause the controlled variable to deviate from the set point. In Lean Six Sigma terminology, these are noise variables. A disturbance is also known as an *upset*.

DMADOV—A structured DFSS methodology. DMADOV stands for define, measure, analyze, design, optimize, and verify.

DMADV—A structured DFSS methodology. DMADV is an acronym for define, measure, analyze, design, and verify. Variations of DMADV exist. *See also* design for Six Sigma, DMADOV, DMEDI, IDOV.

DMAIC—A structured methodology that focuses on improving existing processes with the intent of achieving Six Sigma quality levels. DMAIC is an acronym for define, measure, analyze, improve, and control. *See also* design for Six Sigma, DMADV, DMADOV, DMEDI, IDOV.

DMEDI—A structured DFSS methodology. DMEDI stands for define, measure, explore, develop, and implement.

DOE—*See* design of experiments.

DOX—*See* design of experiments.

DPMO—*See* defects per million opportunities.

DPU—*See* defects per unit.

driving forces—In the context of a force-field analysis, driving forces are those forces that aid in achieving the objective.

DTC—*See* design to cost.

E

EAC—Estimate at completion

earned value (EV)—This is the approved budget for the work actually completed by a specific date. It is also known as the *budgeted cost of work performed* (BCWP).

education—Focuses on broadening an individual's knowledge base and expands thinking processes. Further, it helps employees understand concepts and accept increased job responsibilities, and prepares them for future jobs and leadership roles. Simply, education helps individuals learn how to think.

effect—A relationship between factor(s) and response variable(s) based on cause. Note that there can be many factors and response variables. Specific types include main effect, dispersion effect, and interaction effect.

efficiency—An unbiased estimator is considered to be more efficient than another unbiased estimator if it has a smaller variance.

eight types of waste—The seven types of waste plus an additional waste: people. The eighth category was added by Taiichi Ohno, as he saw the underutilization of employees (for example, brainpower, skills, experience, and talents) as a waste. *See also* seven types of waste.

eighty-twenty (80-20) rule—A term referring to the Pareto principle, which was first defined by J. M. Juran in 1950. The principle suggests that most effects come from relatively few causes; that is, 80% of the effects come from 20% of the possible causes. *See also* Pareto principle.

entity—An item that can be individually described and considered.

error—(1) Error in measurement is the difference between the indicated value and the true value of a measured quantity. (2) A fault resulting from defective judgment, deficient knowledge, or carelessness. It is not to be confused with measurement error, which is the difference between a computed or measured value and the true or reference value.

error detection—A hybrid form of error-proofing whereby a bad part can still be made but will be caught immediately, and corrective action will be taken to prevent another bad part from being produced. An error-detection device is used to spot the error and stop the process when a bad part is made. This is used when error-proofing is too expensive or not easily implemented.

error of the first kind—*See* type I error.

error of the second kind—*See* type II error.

error-proofing—The use of process or design features to prevent the acceptance or further processing of nonconforming products. *See also* foolproofing, mistake-proofing, and poka-yoke.

estimate at completion (EAC)—This can be simply an estimate or it can be calculated as:

1. Assuming cost performance for the remainder of the project will revert to what was originally planned:

   EAC = Approved budget for the entire task – Cost variance for work performed to date on task

   = BAC + AC – EV

2. Assuming cost performance for the remainder of the project will continue as it has for the work performed to date:

$$\text{EAC} = \frac{\text{BAC}}{\text{Cumulative CPI}}.$$

estimate to complete (ETC)—This can be simply an estimate or it can be calculated as ETC = BAC – AC to date.

ETC—*See* estimate to complete.

EV—*See* earned value.

evolutionary operations (EVOP)—The procedure of adjusting variables in a process in small increments in search of a more optimal point on the response surface.

EVOP—*See* evolutionary operations.

executive—*See* the section on Roles and Responsibilities in Chapter 2 for a detailed discussion.

expected value—The mean of a random variable.

experiential training—The focus of this form of training is on having the learner experience the effects typically encountered in a real-life situation. The experiential approach employs many types of structured experiences (for example, games, simulations, role plays) to facilitate learning. Typically, participants assume roles within a stated scenario designed to surface one or more learning points.

experimental design—A formal plan that details the specifics for conducting an experiment, such as which responses, factors, levels, blocks, treatments, and tools are to be used. *See also* design of experiments.

experimental error—Variation in the response variable beyond that accounted for by the factors, blocks, or other assignable sources in the conduct of the experiment.

experimental run—A single performance of the experiment for a specific set of treatment combinations.

experimental unit—The smallest entity receiving a particular treatment, subsequently yielding a value of the response variable.

explanatory variable—*See* predictor variable.

external customer—A person or organization that receives a product, service, or information but is not part of the organization supplying it. *See also* internal customer.

external failure costs—The failure costs occurring after delivery or shipment of the product, and during or after furnishing of a service, to the customer. Examples are the costs of processing customer complaints, customer returns, warranty claims, and product recalls.

F

facilitator—An individual responsible for creating favorable conditions that will enable a team to reach its purpose or achieve its goals by bringing together the necessary tools, information, and resources to get the job done.

factor—An independent variable or assignable cause that may affect the responses and of which different levels are included in the experiment. Factors are also known as *explanatory*, *independent*, *regressor*, or *predictor* variables.

factor analysis—A statistical technique used "to determine the underlying factors responsible for correlations in the data" (Minitab 15).

factor relationship diagram—A visual tool that assists in the planning of experiment designs. Typically, a factor relationship diagram is composed of two structures: the design structure and the unit structure. The design structure contains those factors that are deliberately manipulated during the experimental design. The unit structure is determined by how we manage noise and other un-manipulated sources of variation during the experimental design. In addition to these two structures, we have the inference space. Hild (2009) states, "The inference space represents the range of conditions over which the experiment's results can expect to repeat."

failure—The termination, due to one or more defects, of the ability of an item, product, or service to perform its required function when called upon to do so. A failure may be partial, complete, or intermittent.

failure analysis—The process of breaking down a failure to determine its cause and to put measures in place to prevent future problems.

failure costs—The costs resulting from product or services not conforming to requirements or customer/user needs. Failure costs are further divided into internal and external categories. *See also* external failure costs, internal failure costs.

failure mode—The type of defect contributing to a failure.

failure mode and effects analysis (FMEA)—A procedure in which each potential failure mode in every subitem of an item is analyzed to determine its effect on other subitems and on the required function of the item.

failure rate—The number of failures per unit time (for equal time intervals).

final control element—In the context of automated process control, the final control element, upon receiving direction from the controller, will take action to adjust the appropriate "manipulated variable." (that is, input variable). Examples of final control elements include control values, electric motors, conveyors, and variable speed pumps. The basic operation of the final control element is "action."

first-pass yield (FPY)—The percentage of units that complete a process and meet quality guidelines without being scrapped, rerun, retested, returned, or diverted into an offline repair area. FPY is calculated by dividing the units entering the process minus the defective units by the total number of units entering the process. First-pass yield is also known as the *quality rate* or *first-time yield* (FTY).

first-time yield (FTY)—*See* first-pass yield.

fishbone diagram—*See* cause-and-effect diagram.

fitness for use—A term used to indicate that a product or service fits the customer's or user's defined purpose for that product or service.

five S—*See* 5S.

five whys—*See* 5 whys.

flowchart—A basic quality tool that uses graphical representation for the steps in a process. Effective flowcharts include decisions, inputs, and outputs as well as the sequence of process steps. The flowchart is occasionally considered to be one of the seven basic tools of quality.

FMEA—*See* failure mode and effects analysis.

foolproofing—A method of making a product or process immune to errors on the part of the user or operator. Foolproofing is synonymous with error-proofing. *See also* error-proofing, mistake-proofing, poka-yoke.

force-field analysis—A technique for analyzing the forces that aid or hinder an organization in reaching an objective.

forecasting—A prediction of future values of a time series. For example, one might want to predict the next value, the next five values, or the next 10 values in the series.

forming—The first stage of team growth. *See also* stages of team growth.

fourteen (14) points—W. Edwards Deming's 14 management practices to help organizations increase their quality and productivity. They are as follows: (1) create constancy of purpose for improving products and services; (2) adopt a new philosophy; (3) cease dependence on inspection to achieve quality; (4) end the practice of awarding business on price alone; instead, minimize total cost by working with a single supplier; (5) improve constantly and forever every process for planning, production, and service; (6) institute training on the job; (7) adopt and institute leadership; (8) drive out fear; (9) break down barriers between staff areas; (10) eliminate slogans, exhortations, and targets for the workforce; (11) eliminate numerical quotas for the workforce and numerical goals for management; (12) remove barriers that rob people of pride of workmanship, and eliminate the annual rating or merit system; (13) institute a vigorous program of education and self-improvement for everyone; (14) put everybody in the company to work to accomplish the transformation.

FPY—*See* first-pass yield.

FTY—*See* first-pass yield.

funnel experiment—An experiment that demonstrates the effects of tampering. Marbles are dropped through a funnel in an attempt to hit a mark on a flat surface target below. The experiment shows that adjusting a stable process to compensate for an undesirable result or an extraordinarily good result will produce output that is worse than if the process had been left alone.

G

gage repeatability and reproducibility (GR&R) study—A type of measurement system analysis done to evaluate the performance of a test method or measurement system. Such a study quantifies the capabilities and limitations of a measurement instrument, often estimating its repeatability and reproducibility. It typically involves multiple operators measuring a series of measurement items multiple times.

Gantt chart—A type of bar chart used in process/project planning and control to display planned work and finished work in relation to time. Also called a *milestone chart*.

gap analysis—A technique that compares an organization's existing state with its desired state (typically expressed by its long-term plans) to help determine what needs to be done to remove or minimize the gap.

GB—*See* Green Belt.

GD&T—*See* geometric dimensioning and tolerancing.

gemba visit—A term that means "place of work" or "the place where the truth can be found." Still others may call it "the value proposition." A gemba visit is a method of obtaining voice of customer information that requires the design team to visit and observe how the customer uses the product in his or her environment.

general linear model (GLM)—A statistical tool that synthesizes many of the other common statistical techniques for analyzing both continuous and discrete data.

geometric dimensioning and tolerancing (GD&T)—A method to minimize production costs by considering the functions or relationships of part features in order to define dimensions and tolerances.

GLM—*See* general linear model.

goal—A statement of general intent, aim, or desire. It is the point toward which the organization (or individual) directs its efforts; goals are often non-quantitative.

governance—According to Bertin and Watson (2007), governance establishes the policy framework within which organization leaders will make strategic decisions to fulfill the organizational purpose as well as the tactical actions that they take at the level of operational management to deploy and execute the organization's guiding policy and strategic direction. Governance in the context of Lean Six Sigma deployment includes all the processes, procedures, rules, roles, and responsibilities associated with the strategic, tactical, and operational deployment of Lean Six Sigma. It also includes the authoritative, decision-making body charged with the responsibility of providing the required governance.

GR&R study—*See* gage repeatability and reproducibility (GR&R) study.

Green Belt (GB)—A Six Sigma role associated with an individual who retains his or her regular position within the firm but is trained in the tools, methods, and skills necessary to conduct Six Sigma improvement projects either individually or as part of larger teams.

H

hand-off map—*See* circle diagram.

hard dollar savings—*See* cost savings.

heijunka—The act of leveling the variety or volume of items produced at a process over a period of time. Heijunka is used to avoid excessive batching of product types and volume fluctuations, especially at a pacemaker process.

histogram—A plot of a frequency distribution in the form of rectangles (cells) whose bases are equal to the class interval and whose areas are proportional to the frequencies. A histogram is a graphic summary of variation in a set of data. The graphical nature of the histogram permits easier visualization of patterns of variation that are difficult to detect in a simple table of numbers. The histogram is one of the seven basic tools of quality.

hoshin kanri—A Japanese term referring to a top-down, bottom-up, systematic and structured strategic planning process that engages all levels of the organization while creating measurable and aligned goals that imbue the concept of continuous improvement through use of the plan–do–check–act cycle. (Note: some sources will cite the PDSA cycle.) Hoshin kanri is also known as *hoshin planning*, *policy deployment*, *policy management*, *policy control*, and *management by policy*.

hoshin planning—A term meaning "breakthrough planning." Hoshin planning is a strategic planning process in which an organization develops up to four vision statements that indicate where the company should be in the next five years. Company goals and work plans are developed on the basis of the vision statements. Periodic audits are then conducted to monitor progress. *See also* hoshin kanri.

house of quality—A diagram named for its house-shaped appearance that clarifies the relationship between customer needs and product features. It helps correlate market or customer requirements and analysis of competitive products with higher-level technical and product characteristics and makes it possible to bring several factors into a single figure. The house of quality is also known as *quality function deployment* (QFD).

hygiene factors—A term used by Frederick Herzberg to label "dissatisfiers." *See also* dissatisfiers.

hypothesis—A statement about a population or the form of a probability distribution to be tested.

hypothesis test—According to Montgomery (2006), a hypothesis test is any procedure used to test a statistical hypothesis.

I

ICC—*See* intra-class correlation coefficient.

ID—*See* interrelationship digraph.

ideation—According to Jonson (2005) it is "the creative process of generating, developing, and communicating new ideas, where an idea is understood as a basic element of thought that can be visual, concrete, or abstract."

IDOV—A structured methodology as an alternative to design for Six Sigma. IDOV is an acronym for identify, design, optimize, and validate. *See also* design for Six Sigma.

imprecision—A measure of precision computed as a standard deviation of the test or measurement results. Less precision is reflected by a larger standard deviation.

incomplete block—A block that accommodates only a subset of treatment combinations.

incomplete block design—A design in which the design space is subdivided into blocks in which there are insufficient experimental units available to run a complete replicate of the experiment.

in-control process—A condition where the existence of special causes is no longer indicated by a Shewhart control chart. This does not indicate that only random causes remain, nor does it imply that the distribution of the remaining values is normal (Gaussian). It indicates (within limits) a predictable and stable process.

independent events—Two events, A and B, are called independent if the probability that they both occur is the product of the probabilities of their individual occurrences. That is:

$$P(A \cap B) = P(A)P(B).$$

independent variable—*See* predictor variable.

inferential statistics—Techniques for reaching conclusions about a population on the basis of analysis of data from a sample.

inherent process variation—The variation in a process when the process is operating in a state of statistical control.

innovation—The ability to not only be creative, but to put idea into practice.

inspection—The process of measuring, examining, testing, gauging, or otherwise comparing the unit with the applicable requirements.

interaction effect—The effect for which the apparent influence of one factor on the response variable depends on one or more other factors. Existence of an interaction effect means that the factors cannot be changed independently of each other.

interaction plot—A plot providing the average responses at the combinations of levels of two distinct factors.

internal customer—The recipient, person, or department of another person's or department's output (product, service, or information) within an organization. *See also* external customer.

internal failure costs—The failure costs occurring prior to delivery or shipment of the product, or the furnishing of a service, to the customer. Examples are the costs of scrap, rework, re-inspection, retesting, material review, and downgrading.

internal rate of return (IRR)—The discount rate that causes the sum of NPV of the costs (negative cash flows) plus the sum of the NPV of the savings (positive cash flows) to equal zero.

interrelationship digraph (ID)—A tool that displays the cause-and-effect relationships between ideas or factors in a complex situation. It identifies meaningful categories from a mass of ideas and is useful when relationships are difficult to determine. Also known as a *relations diagram*, the interrelationship digraph is one of the seven management and planning tools.

interval scale—A quantitative scale in which meaningful differences exist, but where multiplication and division are not allowed since there is not absolute zero. An example is temperature measured in °F, because 20 °F isn't twice as warm as 10 °F.

intervention—An action taken by a leader or a facilitator to support the effective functioning of a team or work group.

intra-class correlation coefficient (ICC)—The intra-class correlation coefficient helps gauge the effectiveness of attribute (discrete) measurement systems and is used under the following conditions or assumptions: there is the need to classify things in order, rank, or scale, ranges are equally distributed such as –2, –1, 0, 1, 2, there are consequences for misclassifying, units must be independent, raters make their classifications independently of other raters, and the categories are mutually exclusive and collectively exhaustive. There are six forms of the intra-class correlation coefficient.

IRR—*See* internal rate of return.

Ishikawa diagram—*See* cause-and-effect diagram.

ISO—"Equal" (Greek). A prefix for a series of standards published by the International Organization for Standardization.

ISO 9000 series—A set of individual, but related, international standards and guidelines on quality management and quality assurance developed to help organizations effectively document the quality system elements to be implemented to maintain an efficient quality system. The standards were developed by the International Organization for Standardization, a specialized international agency for standardization composed of the national standards bodies of nearly 100 countries.

ISO 14000 series—A set of individual, but related, international standards and guidelines relevant to developing and sustaining an environmental management system.

ITIL—Information technology infrastructure library

J

jidoka—A method of autonomous control involving the adding of intelligent features to machines to start or stop operations as control parameters are reached, and to signal operators when necessary. Jidoka is also known as *autonomation*.

JIT—*See* just-in-time manufacturing.

job aids—Virtually any type of media that can either substitute for formal training and/or provide reinforcement or reference after training.

job analysis—A process for collecting information about duties, responsibilities, skills, and outcomes. Also, a description of the work environment may be included if it is unique or extraordinary in some manner. An example of such an environment might be a manufacturing clean room.

job description—A job analysis, task analysis, and skills analysis are components of the job description. Once completed, these three aspects will be used to draft a job description and a job specification. Job descriptions, according to Rae (1997) "are statements of the outline of the whole job and show the duties and responsibilities involved in the job." It also "details the skills, knowledge, and attitudes which are required by the individual in order to carry out the duties involved in the job."

job specification—A subset of the job description, the job specification, according to Rae (1997), defines "what the job holder should do and be capable of doing." Bee and Bee (1994) add that the job specification also includes carrying out the job to a prescribed standard. The greater the detail of the job specification, the easier it will be to determine performance gaps.

Juran trilogy—*See* quality trilogy.

just-in-time manufacturing (JIT manufacturing)—A material requirement planning system for a manufacturing process in which there is little or no manufacturing material inventory on hand at the manufacturing site and little or no incoming inspection.

K

kaikaku—A term meaning a breakthrough improvement in eliminating waste.

kaizen—A term that means gradual unending improvement by doing little things better and setting and achieving increasingly higher standards. The kaizen approach is usually implemented as a small, intensive event or project over a relatively short duration, such as a week.

kaizen blitz/event—An intense team approach to employ the concepts and techniques of continuous improvement in a short time frame (for example, to reduce cycle time, increase throughput).

kanban—A system that signals the need to replenish stock or materials or to produce more of an item. Kanban is also known as a "pull" approach. Kanban systems need not be elaborate, sophisticated, or even computerized to be effective. Taiichi Ohno of Toyota developed the concept of a kanban system after a visit to a U.S. supermarket.

Kano model—A representation of the three levels of customer satisfaction, defined as dissatisfaction, neutrality, and delight.

kappa (K)—Futrell (1995) defines kappa as "the proportion of agreement between rates after agreement by chance has been removed." Kappa techniques help gauge the effectiveness of attribute (discrete) measurement systems and are used under the following conditions or assumptions: there is the need to classify things in a nominal manner, some categories can be used more frequently than others, units must be independent, raters make their classifications independently of other raters, and the categories are mutually exclusive and collectively exhaustive. The formula for kappa is:

$$K = \frac{\mathrm{P}_{\text{Observed}} - \mathrm{P}_{\text{Chance}}}{1 - \mathrm{P}_{\text{Chance}}}$$

When multiple raters are involved, we must be concerned with two types of kappa statistics: K_{Overall} and K_{Category}. The overall kappa assesses the rater agreement across all of the categories, while the category kappa is used to compute individual kappa values for each category. This provides an indication regarding where each rater has trouble. These formulas are

$$K_{\text{Overall}} = 1 - \frac{nm^2 - \sum_{i=1}^{n}\sum_{j=1}^{k} x_{ij}^2}{nm(m-1)\sum_{j=1}^{k}\left(\overline{p}_j\right)\left(1-\overline{p}_j\right)}$$

$$K_{\text{Category}} = 1 - \frac{\sum_{i=1}^{n} x_{ij}\left(m - x_{ij}\right)}{nm(m-1)\left(\overline{p}_j\right)\left(1-\overline{p}_j\right)}$$

where

m = Number of raters

n = Number of units

k = Number of categories

$$\overline{p} = \frac{\text{Ratings within category}}{nm}$$

KBD—See key business drivers.

Kendall's coefficient of concordance—This coefficient measures the association between appraisers and the association within appraisers and is used with attributed (discrete) measurement system. Kendall's coefficient of concordance ranges between –1 and 1. A positive value of –1 indicates positive association. Similarly, a negative value indicates negative association. A value of zero indicates no agreement or association. The formula for computing Kendall's coefficient of concordance is given by:

$$W = \frac{12\sum_{i=1}^{N} R_i^2 - 3k^2 N(N+1)^2}{k^2 N(N+1) - k\sum_{j=1}^{k} T_j}$$

where:

N = The number of units

$\sum_{i=1}^{N} R_i$ = The sum of the squared sums of ranks for each of the ranked N units

k = The number of appraisers

T_j = Assigns the average of ratings to tied observation = $T_j = \sum_{i=1}^{g_j} \left(t_i^3 - t_i\right)$

t = The number of tied ranks in the ith grouping of ties

g_j = The number of groups of ties in the jth set of ranks

Kendall's correlation coefficient—This coefficient measures the association between all appraisers and the known standard and each appraiser and the known standard and is used in with attribute (discrete) measurement system. Kendall's correlations coefficient is also known as Kendall's rank-order correlation coefficient, Kendall's Tau correlation coefficient, and Kendall's Tau-b correlation coefficient. Kendall's correlation coefficient ranges between -1 and 1. A positive value of -1 indicates positive association. Similarly, a negative value indicates negative association. A value of zero indicates no agreement or association. The formula for computing Kendall's correlation coefficient is given by:

$$\tau = \frac{C - D}{\sqrt{\left[N_{\text{Total}}\left(N_{\text{Total}} - 1\right)(0.5) - T_X\right]}\sqrt{\left[N_{\text{Total}}\left(N_{\text{Total}} - 1\right)(0.5) - T_Y\right]}}$$

where:

C = Number of concordant pairs = $\sum_{i<k}\sum_{j<i} n_{ij} n_{ki}$

D = Number of discordant pairs = $\sum_{i<k}\sum_{j>i} n_{ij} n_{ki}$

n_{+i} = Number of observations in the ith row

n_{+j} = Number of observations in the jth column

n_{ij} = Number of observations in the cell in the ith row and jth column

n_{ki} = Number of observations in the cell in the kth row and ith column

N = Total number of observations

T_X = Number of pairs tied on $X = 0.5\sum_i n_{+i}\left(n_{+i} - 1\right)$

T_Y = Number of pairs tied on $Y = 0.5\sum_j n_{+j}\left(n_{+j} - 1\right)$

k = Number of appraisers

Kendall's rank-order correlation coefficient—*See* Kendall's correlation coefficient.

Kendall's tau correlation coefficient—*See* Kendall's correlation coefficient.

Kendall's tau-b correlation coefficient—*See* Kendall's correlation coefficient.

key business drivers (KBDs)—Those few things that must be done well for the organization to succeed. KBDs should reflect the organization's goals and are vital for its strategies to be successful. Key business drivers are also known as *critical success factors* (CSFs).

key performance indicator (KPI)—Quantifiable financial and nonfinancial measurements that reflect an organization's key business drivers (KBDs) and are usually long term.

KJ method—*See* affinity diagram.

KPI—*See* key performance indicator.

L

lag—The number of time units between differenced values.

latent defects—Are hidden defects in either material and/or workmanship that may cause a component to malfunction or fail, but is not discoverable through inspection.

law of unintended consequences—Named by sociologist, Robert K. Merton, is meant to describe outcomes not intended by purposeful actions. "Unintended consequences" is sometimes known as unanticipated consequences, unforeseen consequences, or even side effects.

LBQ—*See* Ljung-Box Q (statistics).

LCI—*See* learner-controlled instruction.

lean—A comprehensive approach complemented by a collection of tools and techniques that focus on reducing cycle time, standardizing work, and reducing waste. Lean is also known as "lean approach" or "lean thinking."

lean approach—*See* lean.

lean manufacturing—Applying the lean approach to improving manufacturing operations.

Lean Six Sigma—A fact-based, data-driven philosophy of improvement that values defect prevention over defect detection. It drives customer satisfaction and bottom-line results by reducing variation, waste, and cycle time, while promoting the use of work standardization and flow, thereby creating a competitive advantage. It applies anywhere variation and waste exist, and every employee should be involved. Note: In the first edition of *The Certified Six Sigma Black Belt Handbook*, this definition was attributed to Six Sigma. However, through experience and empirical evidence, it has become clear that Lean and Six Sigma are different sides of the same coin. Both concepts are required to effectively drive sustained breakthrough improvement. Subsequently, the definition is being changed to reflect the symbiotic relationship that must exist between Lean and Six Sigma to ensure lasting and positive change.

lean thinking—*See* lean.

learner-controlled instruction (LCI)—*See* self-directed learning.

learning—According to Rothwell (2008) "learning gives individuals the potential to get results."

level—The setting or assignment of a factor at a specific value.

linear programming—A constrained optimization technique comprising both a linear objective function and linear constraints.

linear regression coefficients—Numbers associated with each predictor variable in a linear regression equation that tell how the response variable changes with each unit increase in the predictor variable. The model also gives some sense of the degree of linearity present in the data. More specifically, when we talk about linear regression, we mean we are linear in the coefficients or parameters.

linear regression equation—A mathematical function or model that indicates the linear relationship between a set of predictor variables and a response variable.

linearity (general sense)—The degree to which a pair of variables follows a straight-line relationship. Linearity can be measured by the correlation coefficient.

linearity (measurement system sense)—The difference in bias through the range of measurement. A measurement system that has good linearity will have a constant bias no matter the magnitude of measurement. If one views the relation between the observed measurement result on the y-axis and the true value on the x-axis, an ideal measurement system would have a line of slope = 1. Linearity is a component of accuracy.

link function—A function that transforms probabilities of a response variable from the closed interval, (0,1), to a continuous scale that is unbounded. Once the transformation is complete, the relationship between the predictor and response variable can be modeled with linear regression.

Ljung-Box Q statistics—A statistic that tests the null hypothesis that autocorrelations up to lag $k = 0$. The statistic is computed as

$$Q_k = n(n+2)\sum_{m=1}^{k} \frac{r_m^2}{n-m}$$

where

n = Number of observations

r_m = Autocorrelation at lag m; $m = 1,2,\ldots,k$

k = Lag j; $k = 1,2,\ldots$

Q is distributed as $\chi^2_{\alpha,k}$ and tests the hypothesis of model adequacy. If the value of Q exceeds the chi-square value, the hypothesis is rejected.

logistic regression—A type of regression that is used when the response variable is discrete. Although there are many type of logistic regression, three common types include, binary, nominal, and ordinal.

M

MAD—*See* mean absolute deviation.

main effect—The influence of a single factor on the mean of the response variable.

main effects plot—The plot giving the average responses at the various levels of individual factors.

maintainability—The measure of the ability of an item to be retained or restored to a specified condition when maintenance is performed by personnel having specified skill levels, using prescribed procedures and resources, at each prescribed level of maintenance and repair.

Malcolm Baldrige National Quality Award (MBNQA)—An award established by Congress in 1987 to raise awareness of quality management and to recognize U.S. organizations that have implemented successful quality management systems. A Criteria for Performance Excellence is published each year. Three awards may be given annually in each of five categories: manufacturing businesses, service businesses, small businesses, education institutions, and healthcare organizations. The award is named after the late secretary of commerce Malcolm Baldrige, a proponent of quality management. The U.S. Commerce Department's National Institute of Standards and Technology manages the award, and ASQ administers it. The major emphasis in determining success is achieving results driven by effective processes.

Mallows's C_p—A statistic used to measure the goodness of a prediction. It is computed by using

$$\text{Mallows's } C_p = \frac{SSE_p}{MSE_m} - (n - 2p)$$

where

SSE_p = Sum of the squared error for the model under consideration

MSE_m = Mean squared error for the model with all predictors included

n = Number of observations

p = Number of terms in the model including the constant term

manipulated variable—In the context of automated process control, the manipulated variable is the variable used to maintain the controlled variable at the set point. In Lean Six Sigma terminology, it is the "X" variable. There can be many manipulated variables. In the event there are, the APC system becomes more complex. Examples of such variables include, concentration, temperature, pressure, and flow rate.

MANOVA—*See* multiple analysis of variance.

MAPE —*See* mean absolute percentage error.

margin—Refers to the difference between income and cost.

market share—An organization's market share of a particular product or service is that percentage of the dollar value that is sold relative to the total dollar value sold by all organizations in a given market.

Master Black Belt (MBB)—A Six Sigma role associated with an individual typically assigned full time to train and coach Black Belts to ensure improvement projects chartered are the right strategic projects for the organization. Master Black Belts are usually the authorizing body to certify lower-level belts within an organization.

materials review board (MRB)—A quality control committee or team usually employed in manufacturing or other materials-processing installations that has the responsibility and authority to deal with items or materials that do not conform to fitness-for-use specifications.

matrix chart—*See* matrix diagram.

matrix diagram—A tool that identifies the relationships that exist between groups of data. It can be also used to identify the strength of that relationship as well. The matrix diagram does not use a singular format. Instead, there are six different forms including the C, L, T, X, Y and roof-shaped format. The different shapes can be selected depending on the number of variables to be compared. The matrix diagram is one of the seven management and planning tools.

MBB—*See* Master Black Belt.

MBNQA—*See* Malcolm Baldrige National Quality Award.

mean—A measure of central tendency and the arithmetic average of all measurements in a data set.

mean absolute deviation (MAD)—A measure of accuracy of the fitted time series values, computed as

$$MAD = \sum_{t=1}^{n} \left| \left(Y_t - \hat{Y}_t \right) \right|$$

where

n = Number of observations

Y_t = Actual value at time t

$\hat{Y}_t$ = Fitted value at time t

mean absolute percentage error (MAPE)—A measure of accuracy of the fitted time series values, computed as

$$MAPE = \frac{\left(\sum_{t=1}^{n} \frac{\left| Y_t - \hat{Y}_t \right|}{Y_t} \right)(100)}{n} ; Y_t \neq 0$$

where

n = Number of observations

Y_t = Actual value at time t

$\hat{Y}_t$ = Fitted value at time t

MSD—*See* mean squared deviation.

mean squared deviation (MSD)—A measure of accuracy of the fitted time series values, computed as

$$MSD = \frac{\sum_{t=1}^{n} \left| Y_t - \hat{Y}_t \right|^2}{n}$$

where

n = Number of forecasted values

Y_t = Actual value at time t

$\hat{Y}_t$ = Forecast value at time t

mean time between failures (MTBF)—A basic measure of reliability for repairable items. The mean number of life units during which all parts of an item perform within their specified limits, during a particular measurement interval under stated conditions.

mean time to failure (MTTF)—A basic measure of system reliability for nonrepairable items. The total number of life units for an item divided by the total number of failures within that population, during a particular measurement interval under stated conditions.

mean time to repair (MTTR)—A basic measure of maintainability. The sum of corrective maintenance times at any specific level of repair, divided by the total number of failures within an item repaired at that level, during a particular interval under stated conditions.

measurement error—The difference between the actual value and the measured value of a quality characteristic.

median—The middle number or center value of a set of data when all the data are arranged in increasing order.

megaprojects—Projects that are so large they can't be managed effectively. Consequently, they are decomposed into smaller, more manageable projects that can be completed in shorter times with fewer resources. Megaprojects provide a focused way of advancing an organization's strategies. However, they require additional coordination among projects and functional groups to be successful.

mentoring—While coaching focuses on the individual as it relates to Lean Six Sigma, mentoring focuses on the individual from the career perspective. Mentors are usually experienced individuals (not necessarily in Lean Six Sigma) who have in-depth knowledge about the organization, as well as the individual (that is, mentee). Usually, they come from within the organization, though not necessarily the same department as their mentee. Their role is to help provide guidance, wisdom, and a possible road map to career advancement.

method of least squares—A technique for estimating a parameter that minimizes the sum of the squared differences between the observed and the predicted values derived from the model.

metrology—The science and practice of measurements.

mind mapping—A technique for creating a visual representation of a multitude of issues or concerns by forming a map of the interrelated ideas.

mistake-proofing—The use of process or design features to prevent manufacture of nonconforming product. It is typically engineering techniques that make the process or product sufficiently robust so as to avoid failing. *See also* error-proofing, foolproofing, and poka-yoke.

mitigation—Minimizing the impact of the risk.

mode—The value that occurs most frequently in a data set.

monitoring—Continuing to observe low risks where the cost of mitigation or avoidance is too high.

motivation, extrinsic—The satisfaction of either material or psychological needs that are applied by others or the organization through pre-action (incentive) or post-action (reward).

motivation, intrinsic—The qualities of work itself or of relationships, events, or situations that satisfy basic psychological needs (for example, achievement, power, affiliation, autonomy, responsibility, creativity, and self-actualization) in a self-rewarding process.

MRB—*See* materials review board.

MSC—*See* Manpower Services Commission.

MTBF—*See* mean time between failures.

MTTF—*See* mean time to failure.

MTTR—*See* mean time to repair.

muda—An activity that consumes resources but creates no value; the seven categories are correction, overprocessing, inventory, waiting, overproduction, internal transport, and motion. *See also* seven types of waste.

multicollinearity—Multicollinearity is said to occur when two or more predictor variables in a multiple regression model are correlated. Multicollinearity may cause the coefficient estimates and significance tests for each predictor variable to be underestimated and difficult to interpret.

multiphase planning—The process of dissecting the planning time frame into smaller time elements or phases to recognize or even celebrate significant events or accomplishments and to demonstrate progress.

multiple analysis of variance (MANOVA)—A statistical technique that can be used to analyze both balanced and unbalanced experimental designs. MANOVA is used to perform multivariate analysis of variance for balanced designs when there is more than one dependent variable. The advantage of MANOVA over multiple one-way ANOVAs is that MANOVA controls the family error rate. By contrast, multiple one-way ANOVAs work to increase the alpha error rate.

multiple linear regression—A linear regression equation in which multiple predictor variables are used.

multi-vari analysis—A graphical technique for viewing multiple sources of process variation. Different sources of variation are categorized into families of related causes and quantified to reveal the largest causes.

multivariate control chart—A variables control chart that allows plotting of more than one variable. These charts make use of the T^2 statistic to combine information from the dispersion and mean of several variables.

multi voting—A decision-making tool that enables a group to sort through a long list of ideas to identify priorities.

N

natural process variation—*See* voice of the process.

natural team—A work group responsible for a particular process.

needs analysis—*See* training needs analysis

net present value—A discounted cash flow technique for finding the present value of each future year's cash flow.

next operation as customer—The concept that the organization is composed of service/product providers and service/product receivers or "internal customers." *See also* internal customer.

NGT—*See* nominal group technique.

NIST—National Institute of Standards and Technology (U.S.).

noise factor—In robust parameter design, a noise factor is a predictor variable that is hard to control or is not desirable to control as part of the standard experimental conditions. In general, it is not desirable to make inference on noise factors, but they are included in an experiment to broaden the conclusions regarding control factors.

nominal group technique (NGT)—A technique similar to brainstorming that is used by teams to generate ideas on a particular subject. Team members are asked to silently come up with as many ideas as possible and write them down. Each member then shares one idea, which is recorded. After all the ideas are recorded, they are discussed and prioritized by the group.

nominal scale—A scale with unordered, labeled categories, or a scale ordered by convention. Examples include type of defect, breed of dog, and complaint category. Note: It is possible to count by category but not by order or measure.

nonconformance—*See* defective.

nonconformity—*See* defect.

nonlinear regression—In nonlinear regression as with linear regression, we are talking about the form of the coefficients or parameters. In the case of nonlinear regression, the coefficients are nonlinear. With linear regression, the parameters have a closed-form solution. However, with nonlinear regression they do not. Instead, the parameters must be solved for using iterative optimization techniques that may or may not converge.

non-value-added (NVA)—Tasks or activities that can be eliminated with no deterioration in product or service functionality, performance, or quality in the eyes of the customer. Generally, customers are unwilling to pay for non-value-added activities.

norming—The third stage of team growth. *See also* stages of team growth.

norms—Behavioral expectations, mutually agreed-upon rules of conduct, protocols to be followed, or social practices.

NPV—*See* net present value.

null hypothesis—A hypothesis that is formulated without considering any new information or formulated based on the belief that what already exists is true.

NVA—*See* non-value-added.

O

objective—A quantitative statement of future expectations and an indication of when the expectations should be achieved; it flows from goal(s) and clarifies what people must accomplish.

observation—The process of determining the presence or absence of attributes or measuring a variable. Also, the result of the process of determining the presence or absence of attributes or measuring a variable.

observational study—Analysis of data collected from a process without imposing changes on the process.

observed value—The particular value of a response variable determined as a result of a test or measurement.

occurrence—The likelihood of a failure occurring. Occurrence is evaluated using a 10-point scale. On the lowest end of the scale (1), it is assumed the probability of a failure is unlikely. On the highest end of the scale (10), it is assumed the probability of a failure is nearly inevitable.

OOC—*See* out of control.

operational plan—A set of short-term plans that support the achievement of an organization's tactical plans. The planning horizon for operational plans is often one year or less, but may vary depending on the nature of the organization. Operational plans are sometimes called *action plans*.

order—There are two types of order used by Minitab when creating experimental designs: (1) standard order (StdOrder) shows what the order of the runs in the experiment would be if the experiment were done in standard order (also called Yates's order), and (2) run order (RunOrder) shows what the order of the runs in the experiment would be if the experiment were run in random order.

ordinal scale—A scale with ordered, labeled categories. Note: (1) The borderline between ordinal and discrete scales is sometimes blurred. When subjective opinion ratings such as excellent, very good, neutral, poor, and very poor are coded (as numbers 1–5), the apparent effect is conversion from an ordinal to a discrete scale. However, such numbers should not be treated as ordinary numbers, because the distance between 1 and 2 may not be the same as that between 2 and 3, 3 and 4, and so forth. On the other hand, some categories that are ordered objectively according to magnitude—such as the Richter scale, which ranges from 0 to 8 according to the amount of energy release—could equally well be related to a discrete scale. (2) Sometimes, nominal scales are ordered by convention. An example is the blood groups *A*, *B*, and *O*, which are always stated in this order. It's the same case if different categories are denoted by single letters; they are then ordered by convention, according to the alphabet.

organizational dynamics—A field of study that deals with the behavioral nature of organizations, specifically with regard to the following elements: organizational culture, organizational structure and alignment, theories of motivation, leadership styles, group or team dynamics, conflict management and resolution, and change management. More importantly, organizational dynamics deals with the interaction of the above elements when they are all brought together.

out of control—A process is described as operating out of control when special causes are present.

P

parameter—A constant or coefficient that describes some characteristic of a population.

Pareto chart—A graphical tool based on the Pareto principle for ranking causes from most significant to least significant. It utilizes a vertical bar graph in which the bar height reflects the frequency or impact of causes. The graph is distinguished by the inclusion of a cumulative percentage line that identifies the vital few opportunities for improvement. It is also known as the *Pareto diagram*. The Pareto chart is one of the seven basic tools of quality.

Pareto diagram—*See* Pareto chart.

Pareto principle—An empirical rule named after the nineteenth-century economist Vilfredo Pareto that suggests that most effects come from relatively few causes; that is, about 80% of the effects come from about 20% of the possible causes. The Pareto principle is also known as the *eighty-twenty (80-20) rule*.

parts per billion (ppb)—The number of times an occurrence happens in one billion chances. In a typical quality setting it usually indicates the number of times a defective part will happen in a billion parts produced; the calculation is projected into the future on the basis of past performance. Parts per billion allows for comparison of different types of product. Two ppb corresponds to a Six Sigma level of quality assuming there is not a 1.5σ shift of the mean.

parts per million (ppm)—The number of times an occurrence happens in one million chances. In a typical quality setting it usually indicates the number of times a defective part will happen in a million parts produced; the calculation is often projected into the future on the basis of past performance. Parts per million allows for comparison of different types of product. A ppm of 3.4 corresponds to a Six Sigma level of quality assuming a 1.5σ shift of the mean.

payback period—The number of units it will take the results of a project or capital investment to recover the investment from net cash flows. (Units may refer to the amount of product, passage of time, or other meaningful component of measure.)

payout period—*See* payback period.

PDCA cycle—*See* plan–do–check–act (PDCA) cycle.

PDPC—*See* process decision program chart.

PDSA cycle—*See* plan–do–study–act (PDSA) cycle.

percent agreement—The percentage of time in an attribute GR&R study that appraisers agree with (1) themselves (that is, repeatability), (2) other appraisers (that is, reproducibility), or (3) a known standard (that is, bias) when classifying or rating items using nominal or ordinal scales, respectively.

performance—According to Rothwell (2008) "performance is the actual realization of that potential" (gained by learning).

performing—The fourth stage of team growth. *See also* stages of team growth.

PERT—*See* program evaluation and review technique.

PEST—*See* PEST analysis.

PEST analysis—An analysis similar to SWOT, the PEST analysis brings together four environmental scanning perspectives that serve as useful input into an organization's strategic planning process. PEST stands for *political*, *economic*, *social*, and *technological*.

PFMEA—*See* process failure mode and effects analysis.

pipeline—A term applied to where projects "reside" that have been selected and/or prioritized and are waiting to be assigned.

PIT—*See* process improvement team.

plan–do–check–act (PDCA) cycle—A four-step process for quality improvement. In the first step (plan), a plan to effect improvement is developed. In the second step (do), the plan is carried out, preferably on a small scale. In the third step (check), the effects of the plan are observed. In the last step (act), the results are studied to determine what was learned and what can be predicted. The plan–do–check–act cycle is sometimes referred to as the *Shewhart cycle*, after Walter A. Shewhart.

plan–do–study–act (PDSA) cycle—A variation on the plan–do–check–act (PDCA) cycle, with the variation indicating that additional study is required after a change is made. The PDSA cycle is attributed to W. Edwards Deming.

planned value (PV)—This is the approved budget for work scheduled for completion by a specific date. It is also known as the *budgeted cost of work scheduled* (BCWS).

point estimate—The single value used to estimate a population parameter. Point estimates are commonly referred to as the points at which the interval estimates are centered; these estimates give information about how much uncertainty is associated with the estimate.

poka-yoke—A term that means to mistake-proof a process by building safeguards into the system that avoids or immediately finds errors. A poka-yoke device prevents incorrect parts from being made or assembled and easily identifies a flaw or error. The term comes from the Japanese terms *poka*, which means "error," and *yokeru*, which means "to avoid." *See also* baka-yoke, error-proofing, foolproofing, mistake-proofing.

policy—A high-level overall statement embracing the general goals and acceptable practices of a group.

policy control—*See* hoshin kanri

policy deployment—*See* hoshin kanri

policy management—*See* hoshin kanri

PONC—*See* price of nonconformance.

population—The totality of items or units of material under consideration.

portfolio management—The process of managing both active (assigned) and inactive (pipeline) projects.

potential process capability—*See* P_{pk}.

power (1 – β)—The probability of rejecting the null hypothesis when the alternative is true. We write this as P(rejecting $H_0 | H_0$ is not true) = 1 – β. The stronger the power of a hypothesis test, the more sensitive the test is to detecting small differences.

ppb—*See* parts per billion.

PPL—*See* P_{pk_L}.

ppm—*See* parts per million.

PPU—*See* P_{pk_U}.

precision—The closeness of agreement among randomly selected individual measurements or test results. It is that aspect of measure that addresses repeatability or consistency when an identical item is measured several times.

precision-to-tolerance ratio (PTR)—A measure of the capability of the measurement system. It can be calculated by

$$PTR = \frac{5.15\hat{\sigma}_{ms}}{\text{USL} - \text{LSL}}$$

where $\hat{\sigma}_{ms}$ is the estimated standard deviation of the total measurement system variability. In general, reducing the PTR will yield an improved measurement system.

prediction interval—Similar to a confidence interval, it is based on the predicted value that is likely to contain the values of future observations. It will be wider than the confidence interval because it contains bounds on individual observations rather than a bound on the mean of a group of observations. The 100(1 – α)% prediction interval for a single future observation from a normal distribution is given by

$$\bar{X} - t_{\alpha/2} s\sqrt{1+\frac{1}{n}} \le X_{n+1} \le \bar{X} + t_{\alpha/2} s\sqrt{1+\frac{1}{n}}$$

prediction sum of squares—*See* PRESS.

predictor variable—A variable that can contribute to the explanation of the outcome of an experiment. A predictor variable is also known as an *independent variable*, *explanatory variable*, or *regressor variable*.

PRESS—A statistic is used to assess the fit of a regression model for a given set of observations that were not used to estimate the model's parameters. The smaller the PRESS value the better. The PRESS statistic is computed as:

$$PRESS = \sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2$$

where:

n = Number of observations

y_i = ith observed value

$\hat{y}_i$ = ith predicted value based on a model fitted to the remaining $n - 1$ points

prevention costs—The cost of all activities specifically designed to prevent poor quality in products or services. Examples are the cost of new product review, quality planning, supplier capability surveys, process capability evaluations, quality improvement team meetings, quality improvement projects, and quality education and training.

prevention versus detection—A phrase used to contrast two types of quality activities. Prevention refers to those activities designed to prevent nonconformances in products and services. Detection refers to those activities designed to find nonconformances already in products and services. Another phrase used to describe this distinction is "designing in quality versus inspecting in quality."

price of nonconformance (PONC)—The cost of not doing things right the first time.

principal components—Used "to form a smaller number of uncorrelated variables from a large set of data. The goal of principal components analysis is to explain the maximum amount of variance with the fewest number of principal components" (Minitab 15). Principal components have wide applicability in the social sciences and market research.

principles—Covey (1989) defines as "guidelines for human conduct that are proven to have enduring, permanent value. They're fundamental. They're essentially unarguable because they are self-evident. One way to quickly grasp the self-evident nature of principles is to simply consider the absurdity of attempting to live an effective life based on their opposites. I doubt that anyone would seriously consider unfairness, deceit, baseness, uselessness, mediocrity, or degeneration to be a solid foundation for lasting happiness and success."

prioritization matrix—A tool used to choose between several options that have many useful benefits, but where not all of them are of equal value. The choices are prioritized according to known weighted criteria and then narrowed down to the most desirable or effective one(s) to accomplish the task or problem at hand. The prioritization matrix is one of the seven management and planning tools.

process—A series of interrelated steps consisting of resources and activities that transform inputs into outputs and work together to a common end. A process can be graphically represented using a flowchart. A process may or may not add value.

process capability—The calculated inherent variability of a characteristic of a product. It represents the best performance of the process over a period of stable operations. Process capability is expressed as $6\hat{\sigma}$, where $\hat{\sigma}$ is the sample standard deviation (short-term component of variation) of the process under a state of statistical control. Note: $\hat{\sigma}$ is often shown as σ in most process capability formulas. A process is said to be "capable" when the output of the process always conforms to the process specifications.

process capability index—A single-number assessment of ability to meet specification limits on a quality characteristic of interest. This index compares the variability of the characteristic to the specification limits. Three basic process capability indices are C_p, C_{pk}, and C_{pm}. *See also* C_p, C_{pk}, C_{pm}.

process decision program chart (PDPC)—A tool that identifies all events that can go wrong and the appropriate countermeasures for these events. It graphically represents all sequences that lead to a desirable effect. It is one of the seven management and planning tools.

process failure mode and effects analysis (PFMEA)—An analysis process used to identify and evaluate the relative risks associated with a particular process.

process improvement team (PIT)—A natural work group or cross-functional team whose responsibility is to achieve needed improvements in existing processes. The lifespan of the team is based on the completion of the team purpose and specific tasks.

process management—The set of techniques and tools applied to a process to implement and improve process effectiveness, hold the gains, and ensure process integrity in fulfilling customer requirements.

process mapping—The flowcharting of a process in detail that includes the identification of inputs and outputs.

process owner—A Six Sigma role associated with an individual who coordinates the various functions and work activities at all levels of a process, has the authority or ability to make changes in the process as required, and manages the entire process cycle so as to ensure performance efficiency and effectiveness.

process performance—A statistical measure of the outcome of a characteristic from a process that may not have demonstrated to be in a state of statistical control. Note: Use this measure cautiously since it may contain a component of variability from special causes of unpredictable value. It differs from process capability because a state of statistical control is not required.

process performance index—A single-number assessment of ability to meet specification limits on a quality characteristic of interest. The index compares the variability of the characteristic with the specification limits. Three basic process performance indices are P_p, P_{pk}, and P_{pm}. The P_{pm} is analogous to the $C_{pm.}$ *See also* P_p, P_{pk}.

producer's risk (α)—The probability of nonacceptance when the quality level has a value stated by the acceptance sampling plan as acceptable. Note: (1) such nonacceptance is a type I error; (2) producer's risk is usually designated as alpha (α); (3) quality level could relate to fraction nonconforming and acceptable to AQL; (4) interpretation of producer's risk requires knowledge of the stated quality level. Producer's risk is also known as *alpha* (α) *risk*, *alpha* (α) *error*, *error of the first kind*, *type 1 error*, and *type I error*.

product identification—A means of marking parts with a label, etching, engraving, ink, or other means so that part numbers and other key attributes can be identified.

program evaluation and review technique (PERT)—An event-oriented project management planning and measurement technique that utilizes an arrow diagram or road map to identify all major project events and demonstrates the amount of time (critical path) needed to complete a project. It provides three time estimates: optimistic, most likely, and pessimistic.

program management—Occasionally, project management is used interchangeably with program management. However, program management is actually a higher-level activity often involving the management of multiple projects with common goal.

project—A project exists for producing results. Three components must be present for an activity to be defined as a project: scope, schedule, and resources.

project, active—Projects that have been assigned and are under way.

project, inactive—Projects that currently reside in the pipeline. These include both selected and prioritized projects.

project assignment—A prioritized project and a resource (that is, Black Belt or Black Belt candidate) are available to be matched. If the matching is successful, the resource is assigned to the project and the project gets under way.

project closure—A project has been terminated for one of two reasons: successful completion or no longer viable.

project identification—The delineation of clear and specific improvement opportunities that is required in the organization.

project life cycle—A typical project life cycle consists of five sequential phases in project management: concept, planning, design, implementation, and evaluation.

project management—The discipline of planning, organizing, and managing resources to bring about the successful completion of specific project goals and objectives, categorized into five component processes (sometimes called life cycle phases): initiating, planning, executing, controlling, and closing. The Project Management Institute notes that project management is the "application of knowledge, skills, tools, and techniques" to activities to meet the requirements of a specific project.

project plan—Documents that contain the details of why the project is to be initiated, what the project is to accomplish, when and where it is to be implemented, who will be responsible, how implementation will be carried out, how much it will cost, what resources are required, and how the project's progress and results will be measured.

project portfolio—The totality of the projects that have been selected and prioritized and waiting in the pipeline plus those that have been assigned and currently under way.

project prioritization—A selected project will be independently assessed against specific criteria in order to establish a rank score. Rank scores of other selected projects will be compared and a final ranking of the projects will be established. This final ranking will set the precedence and order for project assignment.

project qualification—Translation of an identified improvement opportunity into Lean Six Sigma charter format for recognizing its completeness and the ability to assess its potential as a viable project.

project reserve budget—In instances where doubt exists, it may be possible to establish a project reserve budget. Such budgets may be set as a percentage (for example, 10%, 15%, 20%) of the actual dollar budget. These budgets are typically set aside as contingency dollars to help mitigate project risk.

project selection—To select a project means that it has met the criteria of being a Lean Six Sigma project (that is, it has been qualified) and now must be judged to determine whether it is sufficiently worthy such that the organization is willing to invest time and resources to achieve the expected benefits of the project.

propagation of error—Also known as *transmission of error;* has application in two major areas of Lean Six Sigma: (1) response model analysis where the analysis will addresses how random noise in the x_i values affects the predicted value of y and (2) tolerance stack-up analysis where the analysis will address tolerance stack-up problems.

PTR—*See* precision-to-tolerance ratio.

Pugh matrix—Also known as a *decision matrix;* appropriate to use when a single option must be selected from several and when multiple criteria are to be used.

PV—*See* planned value.

Q

QFD—*See* quality function deployment.

quality assurance—The planned or systematic actions necessary to provide adequate confidence that a product or service will satisfy given needs.

quality control—The operational techniques and the activities that sustain a quality of product or service that will satisfy given needs; also the use of such techniques and activities.

quality council—A group within an organization that drives the quality improvement effort and usually has oversight responsibility for the implementation and maintenance of the quality management system; it operates in parallel with the normal operation of the business. A quality council is sometimes referred to as a *quality steering committee*.

quality function deployment (QFD)—A method used to translate voice of customer information into product requirements/CTQs and to continue deployment (for example, cascading) of requirements to parts and process requirements. Quality function deployment is also known as the *house of quality*. *See also* house of quality.

quality improvement—The actions taken throughout an organization to increase the effectiveness and efficiency of activities and processes in order to provide added benefits to both the organization and its customers.

quality loss function—A parabolic approximation of the quality loss that occurs when a quality characteristic deviates from its target value. The quality loss function is expressed in monetary units. The cost of deviating from the target increases as a quadratic function the farther the quality characteristic moves from the target. The formula used to compute the quality loss function depends on the type of quality characteristic used. Genichi Taguchi first introduced the quality loss function in this form.

quality management—The totality of functions involved in organizing and leading the effort to determine and achieve quality.

quality manual—A document stating the quality policy and describing the quality system of an organization.

quality planning—The activity of establishing quality objectives and quality requirements.

quality policy—Top management's formally stated intentions and direction for the organization pertaining to quality.

quality system—The organizational structure, procedures, processes, and resources needed to implement quality management.

quality trilogy—A three-pronged approach identified by J. M. Juran for managing for quality. The three legs are quality planning (developing the products and processes required to meet customer needs), quality control (meeting product and process goals), and quality improvement (achieving unprecedented levels of performance).

queue—A waiting line.

queueing theory—The mathematical study of queues. "Queueing" is thought to be the only word in the English language that includes five consecutive vowels. Alternate spelling of "queueing" is "queuing."

R

random cause—A source of process variation that is inherent in a process over time. Note: In a process subject only to random cause variation, the variation is predictable within statistically established limits. Random cause is also known as *chance cause* and *common cause*.

random sampling—The process of selecting units for a sample in such a manner that all combinations of units under consideration have an equal or ascertainable chance of being selected for the sample.

random variable—A variable whose value depends on chance.

random variation—Variation due to random cause.

randomization—The process used to assign treatments to experimental units so that each experimental unit has an equal chance of being assigned a particular treatment.

randomized block design—An experimental design consisting of b blocks with t treatments assigned via randomization to the experimental units within each block. This is a method for controlling the variability of experimental units. For the completely randomized design, no stratification of the experimental units is made. In the randomized block design, the treatments are randomly allotted within each block; that is, the randomization is restricted.

randomized block factorial design—A factorial design run in a randomized block design where each block includes a complete set of factorial combinations.

ratio scale—A scale where meaningful differences are shown, where absolute zero exists, and where multiplication and division are permitted. One example of a ratio scale is length in inches, because zero length is defined as having no length, and 20 inches is twice as long as 10 inches.

rational subgroup—A subgroup that is expected to be as free as possible from assignable causes (usually consecutive items). In a rational subgroup, variation is presumed to be only from random cause.

red bead experiment—An experiment developed by W. Edwards Deming to illustrate that it is impossible to put employees in rank order of performance for the coming year based on their performance during the past year because performance differences must be attributed to the system, not to the employees. Four thousand red and white beads in a jar (of which 20% are red) and six people are needed for the experiment. The participants' goal is to produce white beads, as the customer will not accept red beads. One person stirs the beads and then, blindfolded, selects a sample of 50 beads. That person hands the jar to the next person, who repeats the process, and so on. When all participants have their samples, the number of red beads for each is counted. The limits of variation between employees that can be attributed to the system are calculated. Everyone will fall within the calculated limits of variation that could arise from the system. The calculations will show that there is no evidence that one person will be a better performer than another in the future. The experiment shows that it would be a waste of management's time to try to find out why, say, John produced 4 red beads and Jane produced 15; instead, management should improve the system, making it possible for everyone to produce more white beads.

reengineering—The process of completely redesigning or restructuring a whole organization, an organizational component, or a complete process. It's a "start all over again from the beginning" approach, sometimes called a *breakthrough*. In terms of improvement approaches, reengineering is contrasted with incremental improvement (kaizen).

regression analysis—A technique that typically uses continuous predictor variable(s) (that is, regressor variables) to predict the variation in a continuous response variable. Regression analysis uses the method of least squares to determine the values of the linear regression coefficients and the corresponding model.

regressor variable—*See* predictor variable.

reliability—The probability that an item can perform its intended function for a specified interval under stated conditions. In the context of training, reliability refers to the consistency and stability of the measurement process over time. For example, if an individual's skills have not been enhanced between two designated time periods, a reliable measurement process would yield the same results for the same participant.

reliability growth—The improvement in product reliability over a period of time.

repeatability—The precision under conditions where independent measurement results are obtained with the same method on identical measurement items by the same appraiser (that is, operator) using the same equipment within a short period of time. Although misleading, repeatability is often referred to as equipment variation (EV). It is also referred to as within-system variation when the conditions of measurement are fixed and defined (that is, equipment, appraiser, method, and environment). Repeatability is a component of precision.

repeated measures—The measurement of a response variable more than once under similar conditions. Repeated measures allow one to determine the inherent variability in the measurement system. Repeated measures are also known as *duplication* or *repetition*.

replication—The performance of an experiment more than once for a given set of predictor variables. Each repetition of the experiment is called a *replicate*.

Replication differs from repeated measures in that it is a repeat of the entire experiment for a given set of predictor variables, not just a repeat of measurements on the same experiment. Note: Replication increases the precision of the estimates of the effects in an experiment. It is more effective when all elements contributing to the experimental error are included.

reproducibility—The precision under conditions where independent measurement results are obtained with the same method on identical measurement items with different operators using different equipment. Although misleading, reproducibility is often referred to as appraiser variation (*AV*). The term "appraiser variation" is used because it is common practice to have different operators with identical measuring systems. Reproducibility, however, can refer to any changes in the measurement system. For example, assume the same appraiser uses the same material, equipment, and environment, but uses two different measurement methods; the reproducibility calculation will show the variation due to change in methods. It is also known as the *average variation between systems* or *between-conditions variation of measurement*. Reproducibility is a component of precision.

resolution—(1) The smallest measurement increment that can be detected by the measurement system. (2) In the context of experimental design, resolution refers to the level of confounding in a fractional factorial design. For example, in a resolution III design, the main effects are confounded with the two-way interaction effects.

response variable—A variable that shows the observed results of an experimental treatment. It is sometimes known as the *dependent* variable or *y* variable. There may be multiple response variables in an experimental design.

restraining forces—In the context of a force-field analysis, restraining forces are those forces that hinder or oppose the objective.

RETAD—*See* single-minute exchange of die.

return on equity (ROE)—The net profit after taxes, divided by last year's tangible stockholders' equity, and then multiplied by 100 to provide a percentage.

return on investment (ROI)—An umbrella term for a variety of ratios measuring an organization's business performance, the ROI metric measures the effectiveness of an organization's ability to use its resources to generate income. It is computed as:

$$\mathrm{ROI} = \frac{\mathrm{Income}}{\mathrm{Cost}}(100\%).$$

In its most basic form, ROI indicates what remains from all money taken in after all expenses are paid.

revenue growth—The projected increase in income that will result from a project. This is calculated as the increase in gross income minus the cost. Revenue growth may be stated in dollars per year or as a percentage per year.

risk—The probability of not achieving a project goal or schedule, budget, or resource target because something did or didn't occur. Risk may be negative in the form of threats or positive in the form of opportunities.

risk, consumer's (β)—*See* consumer's risk.

risk, producer's (α)—*See* producer's risk.

risk assessment/management—The process of determining what risks are present in a situation and what actions might be taken to eliminate or mediate them.

robust designs—Products or processes that continue to perform as intended in spite of manufacturing variation and extreme environmental conditions during use.

robustness—The condition of a product or process design that remains relatively stable with a minimum of variation even though factors that influence operations or usage, such as environment and wear, are constantly changing.

ROE—*See* return on equity.

ROI—*See* return on investment.

rolled throughput yield (RTY)—The probability of a unit of product passing through an entire process defect-free. The rolled throughput yield is determined by multiplying first-pass yield from each subprocess of the total process.

root cause—A factor (that is, original cause) that, through a chain of cause and effect, causes a defect or nonconformance to occur. Root causes should be permanently eliminated through process improvement. Note: Several root causes may be present and may work either together or independently to cause a defect.

root cause analysis—A structured approach or process of identifying root (that is, original) causes. Many techniques and statistical tools are available for analyzing data to ultimately determine the root cause.

RTY—*See* rolled throughput yield.

run chart—A chart showing a line connecting consecutive data points collected from a process running over a period of time. The run chart is occasionally considered to be one of the seven basic tools of quality. Note: A trend is indicated when the series of collected data points head up or down.

run rates—A projection of future performance based on the assumption that the remainder of the project will continue as it has for the work performed to date. These estimates may be 12-month, year-end, or project-end projections.

S

sample—A group of units, portions of material, or observations taken from a larger collection of units, quantity of material, or observations that serves to provide information that may be used as a basis for making a decision concerning the larger quantity.

sample autocorrelation function (ACF)—The sample autocorrelation function is calculated as

$$r_k = \frac{\sum_{t=1}^{n} (x_{t-k} - \bar{x})(x_t - \bar{x})}{\sum_{t=1}^{n} (x_t - \bar{x})^2}$$

where

n = Number of observations

x_t = Value of x at time t

k = Lag j; k = 1,2,…

sample size (n)—The number of units in a sample. Note: In a multistage sample, the sample size is the total number of units at the conclusion of the final stage of sampling.

Sarbanes-Oxley Act of 2002 (SOX)—A legislative act enacted after a series of financial scandals that mandates strict requirements for financial accounting and for top management's responsibility and accountability in disclosing the financial status of their organization.

scatter diagram—A plot of two variables, one on the y-axis and the other on the x-axis. The resulting graph allows visual examination for patterns to determine if the variables show any relationship or if there is just random "scatter." This pattern, or lack thereof, aids in choosing the appropriate type of model for estimation. Note: Evidence of a pattern does not imply that a causal relationship exists between the variables. The scatter diagram is one of the seven tools of quality.

scatter plot—*See* scatter diagram.

schedule performance index (SPI)—This is a dimensionless index used to measure the project's schedule efficiency. It is calculated as

$$\text{SPI} = \frac{\text{EV}}{\text{PV}} = \frac{\text{BCWP}}{\text{BCWS}}.$$

If the SPI is greater than 1, the project is ahead of schedule. If the SPI is equal to 1, the project is on schedule. If the SPI is less than 1, the project is behind schedule and corrective action must be taken.

schedule variance (SV)—The schedule variance is the difference between the amount budgeted for the work performed and the planned cost of the work performed. It is calculated as

$$SV = EV - PC.$$

If the SV is positive, the project is ahead of schedule. If the SV is zero, the project is on schedule. If the SV is negative, the project is behind schedule and corrective action must be taken.

scribe—The member of a team assigned to record minutes of meetings.

SDL—*See* self-directed learning.

SDWT—*See* self-directed work team.

SEI—Software Engineering Institute.

seiban—The name of a management practice taken from the Japanese words *sei*, which means "manufacturing," and *ban*, which means "number." A seiban number is assigned to all parts, materials, and purchase orders associated with a particular customer, job, or project. This enables a manufacturer to track everything related to a particular product, project, or customer and facilitates setting aside inventory for specific projects or priorities. Seiban is an effective practice for project and build-to-order manufacturing.

seiketsu—One of the 5S's that means to conduct seiri, seiton, and seiso at frequent, indeed daily, intervals to maintain a workplace in perfect condition.

seiri—One of the 5S's that means to separate needed tools, parts, and instructions from unneeded materials and to remove the latter.

seiso—One of the 5S's that means to conduct a cleanup campaign.

seiton—One of the 5S's that means to neatly arrange and identify parts and tools for ease of use.

self-directed learning (SDL)—A learning method by which students learn without an instructor and at their own pace. This permits adult learners to build toward achieving the desired competency level in the needed knowledge or skill.

self-directed work team (SDWT)—A team that requires little supervision and manages itself and the day-to-day work it does; self-directed teams are responsible for whole work processes and schedules, with each individual performing multiple tasks.

self-sustaining department—One that must promote itself to the remainder of the organization with the hope that the organization will fund its services.

sensor/transmitter—In the context of automated process control, these components are often combined. They capture the measurement, convert it to a signal, and transmit it to the controller for interpretation. These are also called the *primary* and *secondary elements*. The basic operation performed is "measurement."

set point—In the context of automated process control, the set point is the specified value of the controlled variable. This variable is equivalent to the nominal value in a specification.

seven basic tools of quality—A set of both qualitative and quantitative tools that help organizations understand and improve. Unfortunately, there is no longer a unified set of seven tools that are universally agreed on, but most authors agree that five of the seven basic tools include the cause-and-effect diagram, control charts, histogram, Pareto chart, and scatter diagram. Additional tools suggested as part of the seven basic tools of quality include the check sheet, flowchart, graphs, run chart, trend chart, and stratification. Note: (1) graphs are sometimes included with control charts and are distinct at other times; (2) run charts are sometimes included with control charts and are distinct at other times; (3) run charts are sometimes synonymous with trend charts and are distinct at other times; (4) run charts are sometimes included with graphs and are distinct at other times. Some authors are known to have eight basic tools of quality. *See also* individual entries.

seven management and planning tools—The seven commonly recognized management and planning tools are the affinity diagram, tree diagram, process decision program chart (PDPC), matrix diagram, interrelationship diagram, prioritization matrices, and the activity network diagram. *See also* individual entries.

seven tools of quality—*See* seven basic tools of quality.

seven types of waste—Taiichi Ohno proposed value as the opposite of waste and identified seven categories: (1) defects, (2) overproduction (ahead of demand), (3) overprocessing (beyond customer requirements), (4) waiting, (5) unnecessary motions, (6) transportation, and (7) inventory (in excess of the minimum).

severity—An indicator of the degree of a failure should a failure occur. Severity can be evaluated based on a 10-point scale. On the lowest end of the scale (1), it is assumed a failure will have no noticeable effect. On the highest end of the scale (10), it is assumed a failure will impact safe operation or violate compliance with regulatory mandate.

Shewhart cycle—*See* plan–do–check–act (PDCA) cycle.

shitsuke—One of the 5S's that means to form the habit of always following the first four S's.

side effects—*See* law of unintended consequences.

simple linear regression—A linear regression equation in which one predictor variable is used

simulation—The act of modeling the characteristics or behaviors of a physical or abstract system.

simultaneous engineering—*See* concurrent engineering.

single-minute exchange of die (SMED)—A series of techniques pioneered by Shigeo Shingo for changeovers of production machinery in less than 10 minutes. The long-term objective is always zero setup, in which changeovers are instantaneous and do not interfere in any way with continuous flow. Setup in a single minute is not required but used as a reference. SMED is also known as *rapid exchange of tooling and dies* (RETAD).

SIPOC—A macro-level analysis of the suppliers, inputs, process, outputs, and customers.

Six Sigma—Can take on various definitions across a broad spectrum depending on the level of focus and implementation. (1) Philosophy—the philosophical perspective views all work as processes that can be defined, measured, analyzed, improved, and controlled (DMAIC). Processes require inputs and produce outputs. If you control the inputs, you will control the outputs. This is generally expressed as the $Y = f(X)$ concept. (2) Set of tools—Six Sigma as a set of tools includes all the qualitative and quantitative techniques used by the Six Sigma practitioner to drive process improvement through defect reduction and the minimization of variation. A few such tools include statistical process control (SPC), control charts, failure mode and effects analysis, and process mapping. There is probably little agreement among Six Sigma professionals as to what constitutes the tool set. (3) Methodology—this view of Six Sigma recognizes the underlying and rigorous approach known as DMAIC. DMAIC defines the steps a Six Sigma practitioner is expected to follow, starting with identifying the problem and ending with implementing long-lasting solutions. While DMAIC is not the only Six Sigma methodology in use, it is certainly the most widely adopted and recognized. (4) Metrics—in simple terms, six sigma quality performance means 3.4 defects per million opportunities (accounting for a 1.5-sigma shift in the mean).

skills analysis—The purpose of the skills analysis is to identify the specific skills required to perform the specific tasks identified by the task analysis. In many instances, it is likely that multiple skills will be required to perform a specific task. Examples of skills associated with the task example might be "ability to listen effectively" and "ability to write clearly."

slack time—The time an activity can be delayed without delaying the entire project; it is determined by calculating the difference between the latest allowable date and the earliest expected date.

SMART—SMART stands for *specific*, *measurable*, *achievable*, *relevant*, and *timely* and used generally in regard to the creation of goal statements.

SMED—*See* single-minute exchange of die.

soft dollar savings—*See* cost avoidance savings.

SOW—*See* statement of work.

SOX—*See* Sarbanes-Oxley Act of 2002.

spaghetti chart—A before-improvement chart of existing steps in a process and the many back-and-forth interrelationships (can resemble a bowl of spaghetti); used to see the redundancies and other wasted movements of people and material.

span of control—The number of subordinates a manager can effectively and efficiently manage.

SPC—*See* statistical process control.

special cause—A source of process variation other than inherent process variation. Note: (1) sometimes special cause is considered synonymous with assignable cause, but a special cause is assignable only when it is specifically identified; (2) a special cause arises because of specific circumstances that are not always present. Therefore, in a process subject to special causes, the magnitude of the variation over time is unpredictable. *See also* assignable cause.

specification—An engineering requirement used for judging the acceptability of a particular product/service based on product characteristics such as appearance, performance, and size. In statistical analysis, specifications refer to the document that prescribes the requirements within which the product or service has to perform.

specification limits—Limiting value(s) stated for a characteristic. *See also* tolerance.

sponsor—A member of management who oversees, supports, and implements the efforts of a team or initiative. Although many organizations define the terms "champion" and "sponsor" differently, they are frequently used interchangeably. *See also* champion.

SPI—*See* schedule performance index.

SQC—*See* statistical quality control.

stability (of a measurement system)—The change in bias of a measurement system over time and usage when that system is used to measure a master part or standard. Thus, a stable measurement system is one in which the variation is in statistical control, which is typically demonstrated through the use of control charts. Stability is a component of accuracy.

stable process—A process that is predictable within limits; a process that is subject only to random causes. (This is also known as a state of statistical control.) Note: (1) a stable process will generally behave as though the results are simple random samples from the same population; (2) this state does not imply that the random variation is large or small, within or outside specification limits, but rather that the variation is predictable using statistical techniques; (3) the process capability of a stable process is usually improved by fundamental changes that reduce or remove some of the random causes present and/or adjusting the mean toward the target value.

stages of team growth—The four development stages through which groups typically progress are *forming*, *storming*, *norming*, and *performing*. Knowledge of the stages helps team members accept the normal problems that occur on the path from forming a group to becoming a team.

stakeholders—The people, departments, and groups that have an investment or interest in the success or actions taken by the organization.

standard—A statement, specification, or quantity of material against which measured outputs from a process may be judged as acceptable or unacceptable.

standard error of the ACF—The standard error of the ACF is calculated as:

$$SE(r_k) = \sqrt{\frac{\left(1 + 2\sum_{m=1}^{k-1} r_m^2\right)}{n}}; SE(r_1) = \sqrt{\frac{1}{n}}$$

where

n = Number of observations

r_m = Autocorrelation at lag m; $m = 1,2,\ldots,k$

k = Lag j; $k = 1,2,\ldots$

standard work—A concept whereby each work activity is organized around human motion to minimize waste. Each work activity is precisely described and includes specifying cycle time, takt time, task sequence, and the minimum inventory of parts on hand required to conduct the activity. Standard work is also known as *standardized work*.

statement of work (SOW)—A description of the actual work to be accomplished. It is derived from the work breakdown structure and, when combined with the project specifications, becomes the basis for the contractual agreement on the project. The statement of work is also known as *scope of work*.

stationarity—The statistical characteristics of the mean, variance, and autocorrelation structure over time. Time series data are assumed stationary when the mean, variance, and autocorrelation structure are constant over the entire range of the data.

statistic—A quantity calculated from a sample of observations, most often to form an estimate of some population parameter.

statistical control—A process is considered to be in a state of statistical control if variations among the observed sampling results from it can be attributed to a constant system of chance causes.

statistical process control (SPC)—The use of statistical techniques such as control charts to reduce variation, increase knowledge about the process, and steer the process in the desired way. Note: (1) SPC operates most efficiently by controlling variation of process or in-process characteristics that correlate with a final product characteristic, and/or by increasing the robustness of the process against this variation; (2) a supplier's final product characteristic can be a process characteristic to the next downstream supplier's process.

statistical quality control (SQC)—The application of statistical techniques to control quality. The term "statistical process control" is often used interchangeably with "statistical quality control," although statistical quality control includes acceptance sampling as well as statistical process control.

statistical tolerance interval—An interval estimator determined from a random sample so as to provide a specified level of confidence that the interval covers at least a specified proportion of the sampled population.

storming—The second stage of team growth. *See also* stages of team growth.

strategic plan—A set of long-term plans generally focused on a three-to-five year planning horizon and designed to achieve an organization's goals and objectives.

strategic planning—A process to set an organization's long-range goals and identify the actions needed to reach those goals.

stratification—The layering of objects or data; also, the process of classifying data into subgroups based on characteristics or categories. Stratification is occasionally considered to be one of the seven tools of quality.

subsystem—A combination of sets, groups, and so on, that performs an operational function within a system and its major subdivision of the system.

supply chain—The series of processes and/or organizations that are involved in producing and delivering a product or service to the customer or user.

SV—*See* schedule variance.

SWOT—Strengths, weaknesses, opportunities, threats.

SWOT analysis—An assessment of an organization's key strengths, weaknesses, opportunities, and threats. It considers factors such as the organization's industry, competitive position, functional areas, and management.

system—A composite of equipment, skills, and techniques capable of performing or supporting an operational role, or both. A complete system includes all equipment, related facilities, material, software, services, and personnel required for its operation and support to the degree that it can be considered self-sufficient in its intended operating environment.

systems thinking—A problem-solving approach that expands the view of something under analysis to determine how it interacts with other elements of the system of which it is a part.

T

tactical plan—A set of mid-term plans that support the achievement of an organization's strategic plans. The planning horizon for tactical plans is often in the one- to three-year range, but may vary depending on the nature of the organization.

Taguchi methods—The American Supplier Institute's trademarked term for the quality engineering methodology developed by Genichi Taguchi. In this engineering approach to quality control, Taguchi calls for off-line quality control, online quality control, and a system of experimental designs to improve quality and reduce costs.

takt time—A term derived from the German word *taktzeit*, meaning "clock cycle." Takt time is the available production time divided by the rate of customer demand. Operating under takt time sets the production pace to customer demand.

task analysis—The purpose of the task analysis is to identify the specific tasks required to perform a particular job. Often, this is accomplished by direct observation of the individual performing the job. It is important that all tasks be identified regardless of whether or not they are observable at the time the task analysis is being performed. An example of a task might be "prepare minutes for all management meetings."

team—Two or more people who are equally accountable for the accomplishment of a purpose and specific performance goals; it is also defined as a small number of people with complementary skills who are committed to a common purpose.

team building—The process of transforming a group of people into a team and developing the team to achieve its purpose.

testing—A means of determining the capability of an item to meet specified requirements by subjecting the item to a set of physical, chemical, environmental, and operating actions and conditions.

theory of constraints (TOC)—Goldratt's theory deals with techniques and tools for identifying and eliminating the constraints (bottlenecks) in a process.

theory of knowledge—A belief that management is about prediction, and people learn not only from experience but also from theory. When people study a process and develop a theory, they can compare their predictions with their observations; profound learning results.

theory X and theory Y—A theory developed by Douglas McGregor that maintains that there are two contrasting assumptions about people, each of which is based on the manager's view of human nature. Theory X managers take a negative view and assume that most employees do not like work and try to avoid it. Theory Y managers take a positive view and believe that employees want to work, will seek and accept responsibility, and can offer creative solutions to organizational problems.

theory Z—A term coined by William G. Ouchi, theory Z refers to a Japanese style of management that is characterized by long-term employment, slow promotions, considerable job rotation, consensus-style decision making, and concern for the employee as a whole.

three-sixty-degree (360°) feedback process—An evaluation method that provides feedback from the perspectives of self, peers, direct reports, superiors, customers, and suppliers.

throughput time—The total time required (processing + queue time) from concept to launch, from order received to delivery, or from raw materials received to delivery to customer.

time series—A sequence of measurements of some quantity taken at different times, often at equally spaced intervals. Time series models often have the following components:

T_i = Long-term trend

C_i = Cyclical effect (due to business or economic downturns)

S_i = Seasonal effect

R_i = Residual or error effect

TOC—*See* theory of constraints.

tolerance—The difference between upper and lower specification limits.

tolerance design (Taguchi)—A rational grade limit for components of a system; determines which parts and processes need to be modified and to what degree it is necessary to increase their control capacity; a method for rationally determining tolerances.

tollgate review—A formal review process conducted by a champion who asks a series of focused questions aimed at ensuring that the team has performed diligently during the current phase. The end result of a tollgate is a "go" or "no-go" decision. The "go" decision allows the team to move forward to the next phase. If it is in the last phase, the "go" decision" brings about project closure. If the decision is "no-go," the team must remain in the phase or retreat to an earlier phase, or perhaps the project is terminated or suspended.

total productive maintenance (TPM)—A methodology pioneered by Nippondenso (a member of the Toyota group) that works to ensure that every machine in a production process is always able to perform its required tasks such that production is never interrupted. TPM maximizes equipment effectiveness by using a preventive maintenance program throughout the life of the equipment.

TPM—*See* total productive maintenance.

training—A subset of education that focuses on increasing proficiency in a skill. Therefore, training typically refers to skill-based instruction and addresses the specific skills employees need to perform a current job or task.

training need—Essentially, a performance gap between what people know or are able to do and what they should know or be able to do to perform their work competently. For example, consider the situation where a new piece of equipment is to be installed and an employee needs to know how to operate it.

training needs analysis—A diagnostic method to identify the gap between current performance and desired performance. A *needs analysis* is also known as a *needs assessment* or *training requirements analysis*.

training plan—A road map for meeting critical training requirements. All training plans, though, should be a result of a strategic business planning process and should provide a mechanism for ensuring that training and education build the organization's capabilities as well as enable employees to a make a positive contribution to the organization's success.

training requirements analysis—*See* training needs analysis.

training want—A training want may surface when an employee's proficiency is low in certain job tasks or behaviors. However, those tasks and behaviors may not necessarily be important to work performance outcomes. For example, consider a customer service employee who wants to learn more about an engineering topic because of a personal interest.

transfer, risk—Refers to moving the risk to another party or individual and allows them to assume the risk responsibility.

transmission of error—*See* propagation of error.

transmitter—*See* sensor/transmitter.

treatment—The specific setting or combination of factor levels for an experimental unit.

tree diagram—A tool that depicts the hierarchy of tasks and subtasks needed to complete an objective. The finished diagram resembles a tree. The tree diagram is one of the seven management and planning tools.

trend chart—A control chart in which the deviation of the subgroup average, $\bar{X}$, from an expected trend in the process level is used to evaluate the stability of a process. The trend chart is also known as the *trend control chart*. The trend chart is occasionally considered to be one of the seven basic tools of quality.

trend control chart—*See* trend chart.

TRIZ—A Russian acronym for "theory of inventive problem solving"; a systematic means of inventing and solving design conflicts. TRIZ involves three items to solve technical problems: (1) various tricks, (2) methods based on utilizing physical effects and phenomena, and (3) complex methods.

type 1 error—*See* type I error.

type 2 error—*See* type II error.

type I error—An error that occurs when we reject the null hypothesis when it is true. We refer to the P(type I error) = P(rejecting H_0 when H_0 is true) = α. A type I error is also known as an *alpha* (α) *error* and *error of the first kind*. The P(type I error) is also known as α – *value*, *producer's risk*, *level of significance*, and *significance level*.

type II error—An error that occurs when we fail to reject the null hypothesis when it is false. We refer to the P(type II error) = P(not rejecting H_0 when H_0 is false) = β. A type II error is also known as a *beta* (β) *error* and *error of the second kind*. The P(type II error) is also known as β – *value* and *consumer's risk*.

U

unanticipated consequences—*See* law of unintended consequences.

underwritten department—One that receives a budget and is expected to meet its objectives within the constraints of that budget.

unforeseen consequences—*See* law of unintended consequences.

upset—*See* disturbance.

V

VA—*See* value-added or value analysis.

valence—Is the value placed on achievement and reveals the "why" beneath the "what" is being asked of someone.

validation—Refers to the effectiveness of the design process itself and is intended to ensure the design process is capable of meeting the requirements of the final product or process.

validation, external—According to the Manpower Services Commission (MSC) "Glossary of Training Terms": "A series of tests and assessments designed to ascertain whether the behavioural objectives of an internally valid training programme were realistically based on an accurate initial identification of training needs in relations to the criteria of effectiveness adopted by the organization."

validation, internal—According to the Manpower Services Commission (MSC) "Glossary of Training Terms": "A series of test and assessments designed to ascertain whether a training programme has achieved the behavioural objective specified."

validity—In the context of training, a valid measurement process is one that measures what it is designed to measure. The most commonly used method for training programs is the content-validity approach. This is a highly subjective approach by which subject matter experts provide opinions regarding whether the process measures what is needed on the job.

value—The net difference between customer-perceived benefits and burdens; it is sometimes expressed as a ratio of benefits to burdens or a ratio of worth to cost.

value analysis—An analytical process that assumes a process, procedure, product, or service is of no value unless proved otherwise. Value analysis assigns a price to every step of a process and then computes the worth-to-cost ratio of that step. *See also* value-added.

value engineering (VE)—An engineering discipline responsible for analyzing the components and processes that create a product, with an emphasis on minimizing costs while maintaining standards required by the customer.

value stream—All activities, both value-added and non-value-added, required to bring a product from a raw material state into the hands of the customer, a customer requirement from order to delivery, or a design from concept to launch.

value stream analysis (VSA)—An analytical process designed to enhance the benefits of a value delivery system while reducing or eliminating all non-value-adding costs associated with value delivery.

value stream mapping—A technique for following the production path for a product or service from beginning to end while drawing a visual representation of every process in the material and information flows. Subsequently, a future state map is drawn of how value should flow.

value-added (VA)—The tasks or activities that convert resources into products or services consistent with customer requirements. The customer can be internal or external to the organization. Value-added activities add worth to the product or service from the customer's perspective and typically change form, fit, or function.

variables data—Data resulting from the measurement of a parameter or a variable. The resulting measurements may be recorded on a continuous scale.

variance inflation factor—Measures the increase in the variance of a regression coefficient when predictor variables are correlated. The formula for the VIF is given by:

$$VIF\left(\beta_j\right) = \frac{1}{1 - R_j^2}; j = 1, 2, \ldots, k$$

where: R_j^2 is the coefficient of multiple determination. It is determined by regressing the predictor variable, x_j, on the other remaining $k - 1$ predictor variables. Consequently, when R_j^2 is large, $VIF(\beta_j)$ is also large. This causes the variances of the coefficients to become inflated.

VE—*See* value engineering.

verification—Refers to the design meeting customer requirements and ensures that the design yields the correct product or process.

VIF—*See* variance inflation factor.

virtual team—A boundaryless team functioning without a commonly shared physical structure or physical contact, using technology to link the team members. Team members are typically remotely situated and affiliated with a common organization, purpose, or project.

visual controls—The collection of approaches and techniques that permit one to visually determine the status of a system, factory, or process at a glance and to prevent or minimize process variation. To some degree, it can be viewed as a minor form of mistake-proofing. *Visual controls* are sometimes referred to as the visual factory.

visual factory—*See* visual controls.

vital few, useful many—A phrase used by J. M. Juran to describe his use of the Pareto principle, which he first defined in 1950. (The principle was used much earlier in economics and inventory control methodologies.) The principle suggests that most effects come from relatively few causes; that is, 80% of the effects come from 20% of the possible causes. The 20% of the possible causes is referred to as the "vital few"; the remaining causes are referred to as the "useful many." When Juran first defined this principle, he was referring to the remaining causes as the "trivial many," but realizing that no problems are trivial in quality assurance, he changed it to "useful many."

VOC—*See* voice of the customer.

voice of the customer (VOC)—An organization's efforts to understand the customers' needs and expectations ("voice") and to provide products and services that truly meet them.

voice of the process (VOP)—The 6σ spread between the upper and lower control limits as determined from an in-control process. The VOP is also known as *natural process variation*.

VOP—*See* voice of the process.

VSA—*See* value stream analysis.

W

waste—Any activity that consumes resources but does not add value to the product or service a customer receives. Waste is also known as *muda*.

WB—*See* White Belt.

White Belt (WB)—A Six Sigma role associated with an individual who works on local problem-solving teams, but may not be part of a Six Sigma project team.

work breakdown structure—A hierarchical decomposition of the work content for a given project.

work group—A group composed of people from one functional area who work together on a daily basis and share a common purpose.

Y

YB—*See* Yellow Belt.

Yellow Belt (WB)—A Six Sigma role associated with an individual who participates as a project team member. They may review process improvements that support the project. Such individuals may have a hands-on role with the process being improved.

Please submit suggestions, additions, corrections, or deletions to authors@asq.org.

Appendix 35

Consolidated Acronym Glossary

5S (Five S)—seiri (sort; also sifting), seiton (set in order), seiso (shine; also sanitize or scrub), seiketsu (standardize), and shitsuke (sustain; also self-discipline).

5Ws & 1H—who, what, where, when, why, and how.

6Ms—machines, manpower, materials, measurements, methods, and Mother Nature.

7Ms—machines, manpower, materials, measurements, methods, Mother Nature and management.

ABC—activity-based costing.

ACF—(sample) autocorrelation function.

ACWP—actual cost of work performed.

AIAG—Automotive Industry Action Group.

AND—activity network diagram.

ANOM—analysis of means.

ANOVA—analysis of variance.

ANSI—American National Standards Institute.

APC—automated process control.

AQL—acceptance quality limit.

ARIMA—autoregressive integrated moving average.

AS-9100—a standard for the aeronautics industry embracing the ISO 9001 standard.

ASQ—American Society for Quality.

AV—actual value.

BAC—budget at completion.

BB—Black Belt.

BCWP—budgeted cost of work performed.

BCWS—budgeted cost of work scheduled.

BPM—business process management.

CFM—continuous flow manufacturing.

CLT—central limit theorem.

CMM—capability maturity model.

CMMI—capability maturity model integration

COPQ—cost of poor quality.

COQ—cost of quality.

COV—components of variation.

CPI—cost performance index.

CPL—lower process capability index.

CPM—critical path method.

CPU—upper process capability index.

CSF—critical success factors.

CSSBB—Certified Six Sigma Black Belt.

CSSGB—Certified Six Sigma Green Belt.

CSSYB—Certified Six Sigma Yellow Belt.

CSSWB—Certified Six Sigma White Belt.

CTQ—critical-to-quality.

CUSUM—cumulative sum (control chart).

CV—cost variance.

DFC—design for cost.

DFM—design for manufacturability or design for maintainability.

DFMEA—design failure mode and effects analysis.

DFP—design for producibility.

DFQ—design for quality.

DFSS—design for Six Sigma.

DFT—design for testability.

DFX—design for X.

DMADOV—define, measure, analyze, design, optimize, verify.

DMADV—define, measure, analyze, design, verify.

DMAIC—define, measure, analyze, improve, control.

DMEDI—define, measure, explore, develop, implement.

DOE—design of experiments.

DOX—design of experiments.

DPMO—defects per million opportunities.

DPU—defects per unit.

DTC—design to cost.

EAC—estimate at completion.

ETC—estimate to complete.

EV—earned value.

EVOP—evolutionary operations.

FMEA—failure mode and effects analysis.

FPY—first-pass yield.

FTY— first-time yield.

GB—Green Belt.

GD&T—geometric dimensioning and tolerancing.

GLM—general linear model.

GR&R—gage repeatability and reproducibility.

ICC—intra-class correlation coefficient.

ID—interrelationship digraph.

IDOV—identify, design, optimize, validate.

IRR—internal rate of return.

ISO—International Organization for Standardization.

ITIL—information technology infrastructure library

JIT—just-in-time.

KBD—key business driver.

KJ method—affinity diagram.

KPI—key performance indicator.

LBQ— Ljung-Box Q (statistics).

LCI—learner-controlled instruction.

MAD—mean absolute deviation.

MANOVA—multiple analysis of variance.

MAPE —mean absolute percentage error.

MBB—Master Black Belt.

MBNQA—Malcolm Baldrige National Quality Award.

MRB—materials review board.

MSC—Manpower Services Commission.

MSD—mean squared deviation.

MTBF—mean time between failures.

MTTF—mean time to failure.

MTTR—mean time to repair.

NGT—nominal group technique.

NIST—National Institute of Standards and Technology (U.S.).

NPV—net present value.

NVA—non-value-added.

OOC—out of control.

PDCA—plan–do–check–act.

PDPC—process decision program chart.

PDSA—plan–do–study–act.

PERT—program evaluation and review technique.

PEST—political, economic, social, technology.

PFMEA—process failure mode and effects analysis.

PIT—process improvement team.

PONC—price of nonconformance.

ppb—parts per billion.

PPL—lower process performance index.

ppm—parts per million.

PPU—upper process performance index.

PRESS—prediction sum of squares.

PTR—precision-to-tolerance ratio.

PV—planned value.

QFD—quality function deployment.

RETAD—single-minute exchange of die.

ROE—return on equity.

ROI—return on investment.

RTY—rolled throughput yield.

SDL—self-directed learning.

SDWT—self-directed work team.

SEI—Software Engineering Institute.

SIPOC—suppliers, inputs, process, outputs, customers.

SMART—specific, measurable, achievable, relevant, timely.

SMED—single-minute exchange of die.

SOW—statement of work.

SOX—Sarbanes-Oxley Act of 2002.

SPC—statistical process control.

SPI—schedule performance index.

SQC—statistical quality control.

SV—schedule variance.

SWOT—strengths, weaknesses, opportunities, threats.

TOC—theory of constraints.

TPM—total productive maintenance.

TRIZ—theory of inventive problem solving.

VA—value-added or value analysis.

VE—value engineering.

VIF—variance inflation factor.

VOC—voice of the customer.

VOP—voice of the process.

VSA—value stream analysis.

YB—Yellow Belt.

WB—White Belt.

Please submit suggestions, additions, corrections, or deletions to authors@asq.org.

Appendix 36

Glossary of Japanese Terms

baka-yoke—A term for a manufacturing technique for preventing mistakes by designing the manufacturing process, equipment, and tools so that an operation literally cannot be performed incorrectly. In addition to preventing incorrect operation, the technique usually provides a warning signal of some sort for incorrect performance.

chaku-chaku—A term meaning "load-load" in a cell layout where a part is taken from one machine and loaded into the next.

gemba visit—A term that means "place of work" or "the place where the truth can be found." Still others may call it "the value proposition." A gemba visit is a method of obtaining voice of customer information that requires the design team to visit and observe how the customer uses the product in his or her environment.

heijunka—The act of leveling the variety or volume of items produced at a process over a period of time. Heijunka is used to avoid excessive batching of product types and volume fluctuations, especially at a pacemaker process.

hoshin kanri—A Japanese term referring to a top-down, bottom-up, systematic and structured strategic planning process that engages all levels of the organization while creating measurable and aligned goals that imbue the concept of continuous improvement through use of the plan–do–check–act cycle. (Note: some sources will cite the PDSA cycle.) Hoshin kanri is also known as *hoshin planning*, *policy deployment*, *policy management*, *policy control*, and *management by policy*.

hoshin planning—A term meaning "breakthrough planning." Hoshin planning is a strategic planning process in which a company develops up to four vision statements that indicate where the company should be in the next five years. Company goals and work plans are developed on the basis of the vision statements. Periodic audits are then conducted to monitor progress. *See* hoshin kanri.

jidoka—A method of autonomous control involving the adding of intelligent features to machines to start or stop operations as control parameters are reached, and to signal operators when necessary. Jidoka is also known as *autonomation*.

kaikaku—A term meaning a breakthrough improvement in eliminating waste.

kaizen—A term that means gradual unending improvement by doing little things better and setting and achieving increasingly higher standards.

kaizen blitz/event—An intense team approach to employ the concepts and techniques of continuous improvement in a short time frame (for example, to reduce cycle time, increase throughput).

kanban—A system that signals the need to replenish stock or materials or to produce more of an item (also called a "pull" approach). The system was inspired by Tauchi Ohno's (Toyota) visit to a U.S. supermarket.

muda—An activity that consumes resources but creates no value; the seven categories are correction, overprocessing, inventory, waiting, overproduction, internal transport, and motion.

poka-yoke—A term that means to mistake-proof a process by building safeguards into the system that avoids or immediately finds errors. The term comes from the Japanese terms *poka*, which means "error," and *yokeru*, which means "to avoid."

seiban—The name of a management practice taken from the words *sei*, which means "manufacturing," and *ban*, which means "number." A seiban number is assigned to all parts, materials, and purchase orders associated with a particular customer, job, or project. This enables a manufacturer to track everything related to a particular product, project, or customer and facilitates setting aside inventory for specific projects or priorities. Seiban is an effective practice for project and build-to-order manufacturing.

seiketsu—One of the 5S's that means to conduct seiri, seiton, and seiso at frequent, indeed daily, intervals to maintain a workplace in perfect condition.

seiri—One of the 5S's that means to separate needed tools, parts, and instructions from unneeded materials and to remove the latter.

seiso—One of the 5S's that means to conduct a cleanup campaign.

seiton—One of the 5S's that means to neatly arrange and identify parts and tools for ease of use.

shitsuke—One of the 5S's that means to form the habit of always following the first four S's.

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